Calculus From the Ground Up

Limits, Insight, and Why It Works — From Functions to Differential Equations

Built for Understanding, Not Just Solving

June 2026

Preface

A different promise

Most calculus textbooks make you a quiet promise: memorize these formulas, practice these templates, and you will pass. This book takes a different path. Work through these ideas with pencil in hand, and you will *understand*. You will see why the derivative takes the form it does, where it came from, what a limit actually asserts, why the chain rule holds together geometrically, and why the area under a curve and the slope of a curve turn out to be two sides of a single fact. Beyond calculating answers, you will look at a symbol like $\mathrm{d}x$, or a phrase like "the limit as x approaches a," and see what is actually happening underneath.

That is a demanding promise to keep, and it asks more of you. Understanding cannot be poured into an unresisting mind; it has to be constructed from direct experience. So this book is built, brick by brick, from the ground up. We start with what a function truly represents, and we conclude, many chapters later, with differential equations – the language in which physics, biology, and economics write down how the world evolves. Every step in between is crafted to feel natural, as though you could have discovered it yourself while exploring a concrete problem.

Focus

The single sentence that organizes this entire book: *calculus is the mathematics of change and accumulation, and the bridge between them is the limit.* If you keep that sentence in view, every chapter will find its natural place.

There is a compelling reason to insist on understanding beyond intellectual satisfaction. Techniques acquired only by rote memorization evaporate. Within a year of finishing a procedural course, many students cannot integrate by parts or state the Mean Value Theorem, because nothing anchored the rules in place – they were an arbitrary list, and unanchored lists decay. But ideas you have explored visually and conceptually do not vanish in that way, because each one is supported by the geometric intuitions around it. If you forget the quotient rule, you can sketch a small nudge and rederive it in two lines. If you forget the formula for a Taylor polynomial, you can reconstruct the matching-derivative requirement that produced it. True understanding is never slower than memorization; it is the only thing that endures.

How this book is built: intuition, then picture, then proof

Every important idea in this book is introduced in the same three movements, and it will help you to know the rhythm in advance.

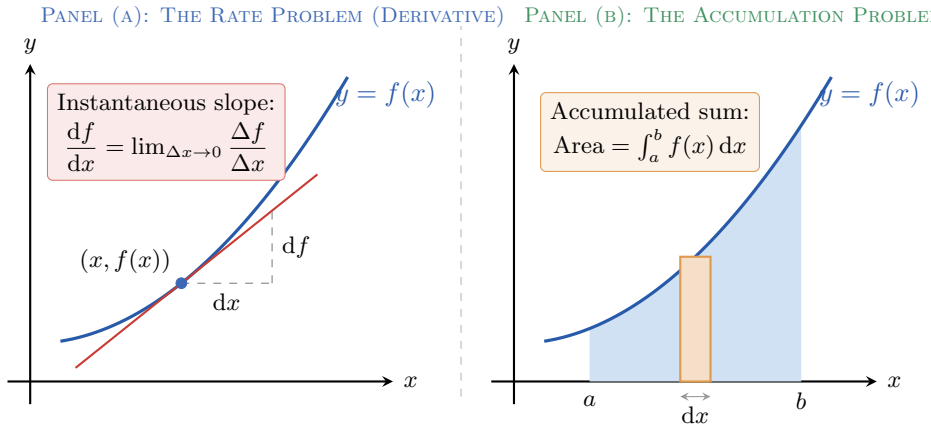


Figure 1: The two grand pillars of calculus. Left: The rate problem zooms into a single point to find the local steepness df/dx. Right: The accumulation problem sums up infinitely many thin slices of area $f(x)\,dx$ across an interval. The central discovery of calculus is that differentiation and integration invert each other.

First, intuition. Before any symbols, we ask what the idea *is* in plain human terms. What is a derivative? It is how fast something is changing right now – the speedometer reading, not the trip average. What is an integral? It is a total accumulated from infinitely many tiny contributions – the full distance built up from every instant's speed. We will state the idea the way you might explain it to a sharp friend over coffee, with no apology and no formula.

Second, the picture. Calculus was discovered with pictures, and it is still understood with pictures. A derivative is the slope of a tangent line. An integral is an area. A limit is a process of closing in. Almost every idea in this book has an exact geometric meaning, and we will draw it. You will see secant lines pivot down into a tangent; you will see rectangles thicken into a smooth region of area; you will see the graphs of Taylor polynomials creep outward to hug a curve. The figures are not decoration. They are arguments you can see.

Third, the proof – clear and complete, but never dry. Intuition can mislead, and pictures can hide special cases. So after we see why an idea should be true, we make it precise and we prove it. But a proof in this book is never a wall of symbols dropped on you from a height. We always explain the *idea* of the proof first – the one move that makes it work – and then we walk through the steps narrating as we go, the way a good teacher talks through a derivation at the board. A proof is just a careful retelling of why something has to be true. If you cannot feel why each line follows, the proof has failed you, and we will try not to fail you.

Intuition

You can read this book at two depths. On a first pass, read the intuition and study the pictures, and let the proofs wash over you to get the shape. On a second pass, slow down and follow every line of the derivations. Both passes are legitimate. Mathematics is rarely understood in one sitting, and nobody – not Newton, not you – absorbs it linearly. Circling back is not failure; it is the method.

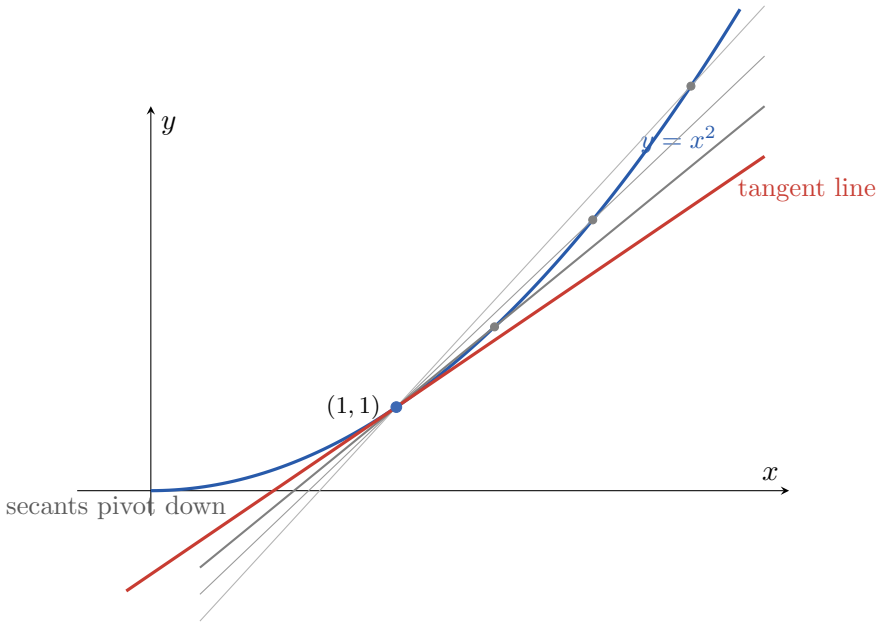


Figure 2: The core geometric intuition of the derivative. Each gray secant line cuts the curve at two points; as the second point slides toward the point of tangency, the secants pivot and close in on a single red line. That limiting line is the tangent, and its slope is the derivative. We will build this picture carefully in the chapters on limits and the derivative.

The colored boxes: a visual language

Throughout the book you will meet recurring colored boxes. Each one plays a fixed role, so once you learn the color, you know what kind of thinking is coming. Three of them carry the special spirit of this book, and they deserve introduction by name.

History

A **History** box tells you where an idea came from, and – just as important – what its inventors got right by instinct before anyone could justify it. Calculus was used brilliantly for a century and a half before it was made rigorous. Newton and Leibniz computed orbits and areas with "infinitesimals" – quantities supposed to be smaller than any positive number yet not quite zero – and they were savagely, correctly criticized for it (the philosopher George Berkeley mocked these vanishing quantities as "the ghosts of departed quantities"). And yet the answers were right. History boxes show you this drama: the right intuition arriving first, the rigorous foundation arriving later, with Cauchy and Weierstrass finally pinning down the limit in the 1800s. Knowing the story tells you which ideas were hard-won, and reassures you that your own confusion is in good company.

What is really happening

A **What is really happening** box pulls back the curtain on a symbol or a computation. What *is* that dx in $\int f(x)\,dx$ – a number, a ghost, a leftover? What does $\lim_{x\to a}$ actually claim, when nothing ever reaches a? What does e^x truly mean, and why does it equal its own

derivative? These boxes exist because notation is compressed thought, and the compression can hide the very thing you most need to see. We open the box and look inside.

Why the technique works

A **Why the technique works** box explains the underlying mechanism behind a method rather than merely cataloging steps. Every computational rule in calculus is a disguised truth about limits or about functions, and the disguise is usually thin once you know to look. Substitution, for example, is nothing more than the chain rule run backwards; integration by parts is the product rule run backwards. When you see *why* a technique works, it stops being a trick you have to remember and becomes a consequence you could have predicted. That is the difference between a magician and a physicist watching the same card trick.

The remaining boxes round out the conversation, and you will get to know them by sight:

- An **Intuition** box (green) states an idea in plain language before we make it precise.
- A **Common trap** box (red) flags a mistake almost everyone makes – a place where the intuition lies, or where the notation invites an error. Forewarned is forearmed.
- A **Walking question** box (orange) hands you a question to chew on away from the page, on a walk or in the shower. The best calculus thinking often happens when you are not staring at symbols. These have no answers printed; they are yours to turn over.
- A **Work it yourself** box (purple) asks you to compute one small thing by hand, right then, because some truths only land in the fingers. Reading about a derivative is not the same as taking one.
- An **Application** box (magenta) shows the idea doing real work in physics, biology, economics, or machine learning – where the abstraction earns its keep.
- A **Connection** box (teal) links the idea to other parts of mathematics, so the subject feels like one landscape rather than a list of topics.
- A **Why this matters** box (cyan) steps back to say why an idea is beautiful or central – the aesthetic and structural payoff.
- A **Focus** box (blue) marks the one thing in a section you should not lose.

Reading note

You do not have to read the boxes in any particular order relative to the main text, and you will not break anything by skipping one and coming back. They are asides in the original sense: voices leaning in from the margin. The main argument always stands on its own in the running prose.

The arc of the book: from functions to differential equations

It helps to see the whole journey before you take the first step, the way you might glance at a trail map before a long hike. Here is the territory, and why it is laid out the way it is.

We begin with functions and the objects calculus acts on. Calculus does not study numbers; it studies *functions* – rules that turn an input into an output, machines for relating one changing quantity to another. So before anything else we make sure you truly understand functions, and especially the two families that calculus cannot live without: the exponential functions (and the remarkable number e, which we will meet from three different directions – compound interest, a limit, and the function that is its own rate of change) and the trigonometric functions (which, properly understood, measure rotation and periodic oscillation alongside static geometry). These early chapters are about *meaning*: what these objects are, before we ever differentiate them.

Then we build the limit, the keystone of everything. The limit is the idea that makes the whole subject possible and the idea that took mathematicians the longest to get right. We approach it first through pure intuition and a long historical look at the centuries of productive confusion, and only then through the precise "ε–δ" definition that finally made it rigorous. We spend real time here because everything downstream – derivatives, integrals, series, all of it – is secretly a statement about limits. Rush the limit and the rest stays foggy; understand it and the rest comes into focus.

With limits in hand, we build the derivative. The derivative is the limit of those pivoting secant lines in Figure 2: instantaneous rate of change, the slope of the tangent. We learn what it means, then we earn the rules – the power rule, product and quotient rules, and the chain rule – by deriving each from the definition, so they are never arbitrary. Then we put the derivative to work: linear approximation, the Mean Value Theorem (the quiet theorem that secretly justifies half of calculus), finding maxima and minima, optimization, and l'Hôpital's rule.

Next we build the integral, and meet the great theorem. An integral is an accumulation – an area, a total, a sum of infinitely many infinitesimal pieces. We define it carefully as a limit of sums, and then we arrive at the summit of the first half of the book: the Fundamental Theorem of Calculus, which reveals that differentiation and integration are inverse operations. The slope problem and the area problem, which look completely unrelated, turn out to be the same problem read forwards and backwards. From there we develop the techniques of integration – substitution, parts, partial fractions – always with a "why it works" box exposing the rule underneath the trick.

Then we extend the reach: series, and then more than one variable. We learn to add up infinitely many terms (sequences and series), and we discover Taylor series, which let us rebuild familiar functions like e^x, $\sin x$, and $\cos x$ out of nothing but their derivatives at a single point – one of the most beautiful ideas in all of mathematics. Then we step up a dimension: functions of several variables, partial derivatives, the gradient, multiple integrals, and a first taste of vector calculus.

Finally we arrive at differential equations. This is where calculus turns back toward the world it came from. A differential equation is a sentence about how a quantity changes, written in the language of derivatives, and its solution is the quantity's whole life story. Population growth, radioactive decay, a swinging pendulum, a cooling cup of coffee, an electrical circuit – all of them are differential equations. Reaching them is the point of the climb.

Why this matters

Notice the shape of Figure 3. It is not a list of independent topics to be ticked off. It is a single structure where each stone holds up the next, and the limit is the keystone in the arch. This is why we refuse to rush the early chapters. A keystone set carelessly brings down everything resting on it.

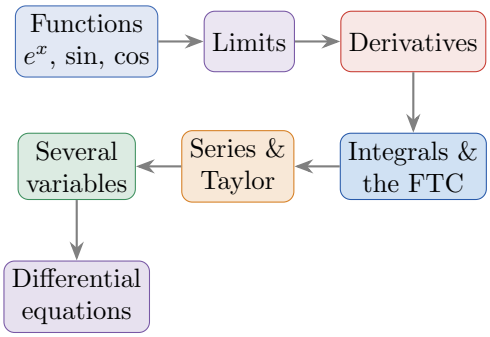


Figure 3: The arc of the book. Each idea rests on the one before it. The limit is the keystone: every arrow after it is, underneath, a statement about limits. The destination – differential equations – is where calculus describes how the world actually changes.

How to read this book

A few practical suggestions, earned from watching many people learn this subject.

Draw. This is the single most important piece of advice in the preface, and it is why it gets its own paragraph. Keep paper and pencil beside you and redraw the figures yourself – not trace them, draw them. Sketch the secant lines pivoting into the tangent. Shade the area under a curve and chop it into rectangles. Plot a function and guess its derivative before you compute it. The act of drawing forces you to decide what you actually believe about the shape, and it exposes vague understanding instantly. Mathematicians who look like they are "just thinking" are very often drawing in their heads, and they learned to do that by drawing on paper first.

Walking question

Here is your first walking question, to carry away from this preface. A car's speedometer reads 60 miles per hour at a single instant – say, exactly at noon. But speed is distance divided by time, and at a single *instant* no time passes and no distance is covered: the car is, in that frozen instant, nowhere going anywhere. So what could "60 miles per hour *right now*" possibly mean? Sit with that. The real resolution of this paradox is the derivative, and it is also, in miniature, the central idea of the whole first half of this book.

Ask "why" relentlessly, and refuse to be satisfied too early. When the book states a rule, stop and ask why it should be true before you read on. When it draws a picture, ask what would change if the curve bent the other way. When a proof makes a move, ask why that move and not another. This book is written to reward that interrogation – nearly every "why" you can think to ask is answered somewhere, often in a colored box – but the answer means far more if you have felt the question first. A question you have actually wrestled with is a hook the answer can hang on. An answer to a question you never asked just slides off.

Compute things by hand, especially when you would rather not. There is a kind of understanding that only arrives through your fingers. When a "Work it yourself" box asks you to take one derivative or evaluate one integral, do it then and there, even though you could read past it. The friction is the point.

Work it yourself

Try this one now, before you turn the page, with nothing but arithmetic. Take the function that squares its input, $f(x) = x^2$. Compute the average rate of change between $x = 1$ and a nearby point $x = 1 + h$ for shrinking h: that average rate is

$$\frac{f(1+h) - f(1)}{h} = \frac{(1+h)^2 - 1}{h} = \frac{2h + h^2}{h} = 2 + h.$$

Now plug in $h = 0.1$, then $h = 0.01$, then $h = 0.001$. You get 2.1, then 2.01, then 2.001. The numbers are visibly homing in on 2. That 2 is the slope of the red tangent line in Figure 2, and you just computed your first derivative – by watching a quotient close in on a value it never quite reaches. Hold on to the feeling of that "closing in." It is the limit, and the whole book turns on it.

Do the exercises. Every chapter ends with a set of them – some computational, some conceptual, and usually a proof or two. The computational ones build fluency; the conceptual ones build understanding; the proofs build the ability to be *sure*. You do not have to do all of them, but do not do none of them. Mathematics is not a spectator sport, and reading a worked example feels deceptively like doing one. It is not the same, in the way that watching someone swim is not the same as getting wet.

Common trap

The most common way to fail at calculus is to read it like a novel – smoothly, continuously, never stopping. It cannot be read that way. If you find yourself nodding along for three pages without pausing, you have almost certainly stopped understanding and started skimming. Real reading of mathematics is slow and full of stops: rereading a line, working a tiny case in the margin, drawing the picture. If a sentence does not make sense, do not push past it hoping it will clear up. Stop. Draw something. Try a number. The fog is information.

Pause

And when you get stuck – you will get stuck, everyone gets stuck – that is not a sign you are bad at this. Getting stuck is what doing mathematics feels like from the inside, for everyone, at every level. The difference between people who learn calculus and people who give up is almost never raw talent. It is whether, on getting stuck, they draw a picture, try a smaller case, reread the definition, and stay with the discomfort for five more minutes. Confusion is not the opposite of understanding. It is the doorway to it.

A word on rigor, and a word of welcome

You may have heard that calculus comes in two flavors: an easy "intuitive" version for engineers and a hard "rigorous" version for mathematicians, and that you must choose between them. This book rejects that choice. Intuition and rigor are not rivals; they are partners, and each is incomplete without the other. Intuition without rigor led Newton and Leibniz to brilliant results along with occasional nonsense, with no systematic way to distinguish them. Rigor without intuition is how calculus becomes a dry sequence of formal ε and δ manipulations that produce correct proofs of

theorems you cannot picture and do not believe. We will develop both side by side. We use intuition to discover and pictures to see, and we use rigor to be *certain* – to know, not merely to hope, that the geometric insight is sound. The history boxes will show you, again and again, that this is how the subject truly developed: the insight first, followed by the rigorous foundation.

History

The people who invented calculus did not understand it as well as you are about to. Newton and Leibniz, working independently in the 1660s and 1670s, had the central insights and the spectacular applications, but their foundations were genuinely shaky – they leaned on infinitesimal quantities nobody could define without contradiction. It took roughly two more centuries for Augustin-Louis Cauchy and Karl Weierstrass to replace those ghosts with the precise theory of limits you will learn in the early chapters. So if the foundations feel subtle, that is not your weakness. They *are* subtle – subtle enough to elude the greatest minds of the seventeenth and eighteenth centuries. You get to learn, in a few clear chapters, what cost humanity two hundred years to work out. That is the gift of being born later.

Connection

One last orientation. Although this book is about calculus, you will keep glimpsing the rest of mathematics through its windows. The limit connects to the deep structure of the real numbers (an appendix at the end pursues this). The derivative is a linear approximation, which ties calculus to linear algebra. Series connect to the theory of approximation and to complex analysis. Differential equations open onto all of physics. We will not chase every thread, but we will point at them, so you always feel that calculus is one province of a single connected country, not an island.

So: welcome. You are about to learn one of the most powerful and beautiful things human beings have ever figured out – a way of thinking that decoded the planets, underlies every bridge and circuit and economic model, and reveals a hidden unity beneath change and accumulation that nobody suspected was there. It was hard-won, and it is worth winning. Read slowly. Draw constantly. Ask why without mercy. Get stuck, and stay a little longer than is comfortable.

Let us begin from the ground up.

Exercises

These are not computational drills – the real exercises begin in Chapter 1. They are meant to fix the habits and the central image of this preface before you turn the page.

1. **The closing-in feeling.** Return to the "Work it yourself" box above, where the quotient $\dfrac{(1+h)^2-1}{h}$ simplified to $2+h$. Compute it once more, this time for $h=-0.1$, $h=-0.01$, and $h=-0.001$ – that is, approaching from *below* $x=1$. Do the values still close in on 2? What does the simplified form $2+h$ predict they must do? (Approaching a value from both sides and getting the same answer is exactly what a limit will demand of us.)

2. **Reading the picture.** Look carefully at Figure 2. Every gray secant passes through the same point $(1,1)$, yet each has a different slope. In one or two sentences, say in your own words

what is sliding, what is staying fixed, and what the red line is the limit *of*.

3. **The arc.** Without looking back, try to redraw Figure 3 from memory – the chain from functions all the way to differential equations. Which single box did the preface call the keystone, and why does every arrow after it secretly carry that same idea forward?

4. **Carry the question.** Write down, in one sentence, the speedometer paradox from the "Walking question" box: how can a car have a definite speed at a single frozen instant, when no time passes and no distance is covered in an instant? You are not asked to solve it – only to state it sharply. The whole first half of the book is the answer.

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Part I

Foundations: Before the Calculus

Chapter 1

What Calculus Is Really About

Before we touch a single rule, before we differentiate or integrate anything, I want you to understand what game we are playing and why anyone ever bothered to invent it. Calculus has a reputation as a thicket of formulas to be memorized and survived. That reputation is a shame, because underneath the symbols sits one of the sharpest ideas humans have ever had, and it is an idea you can *see*. This whole book rests on a promise: you will not be asked to take anything on faith. We build every notion from a picture and an intuition first, and only then make it precise. By the end you should feel not that you have learned a technique, but that you have understood a truth.

So let us begin where calculus itself began – with two questions that look completely unrelated, and a stunning discovery that they are secretly the same question asked from two directions.

1.1 Two questions that built a subject

1.1.1 The first question: how fast, right now?

Imagine a ball dropped from a tall building. You have a stopwatch and a measuring tape, and you record how far the ball has fallen at each instant. After 1 second it has fallen about 4.9 meters; after 2 seconds, about 19.6 meters; after 3 seconds, about 44.1 meters. Galileo found, by rolling balls down ramps to slow time down enough to measure, that the distance fallen grows like the *square* of the time:

$$s(t) = 4.9\,t^2 \quad \text{(meters, with } t \text{ in seconds)}.$$

Now here is a question a child might ask and a physicist must answer: *how fast is the ball going at the exact instant $t = 2$ seconds?*

This sounds innocent. It is not. Speed is distance divided by time – but at a single *instant* no time passes and no distance is covered. The arithmetic comes out to $0/0$, which means nothing. "Speed at an instant" seems to be a contradiction in terms. Yet your car's speedometer reads something definite at every instant, and the ball really is moving at a definite rate as it passes the second-floor window. Something genuine is being measured. Calculus exists, in large part, to say exactly what.

Intuition

You already know how to dodge the $0/0$ problem in practice: you measure speed over a *short* interval and call it the speed "at" the middle. Drive 60 miles in one hour and your average speed is 60 mph; but to know your speed *now* you would glance at how far you went in the last second, or the last tenth of a second. The shorter the interval, the more the average over it deserves to be called your speed *right now*. Calculus is what happens when you take that "shorter and shorter" seriously – all the way to the limit.

Let us actually do it. Average speed over an interval is

$$\text{average speed} = \frac{\text{change in distance}}{\text{change in time}}.$$

We want the speed near $t = 2$. Pick a small extra time h and look at the interval from $t = 2$ to $t = 2 + h$. The distance covered in that interval is $s(2 + h) - s(2)$, so

$$\text{average speed over } [2,\, 2 + h] = \frac{s(2 + h) - s(2)}{h}.$$

With $s(t) = 4.9\,t^2$ this is

$$\frac{4.9(2 + h)^2 - 4.9(2)^2}{h} = \frac{4.9\,(4 + 4h + h^2) - 4.9 \cdot 4}{h} = \frac{4.9\,(4h + h^2)}{h} = 4.9\,(4 + h) = 19.6 + 4.9\,h.$$

Look hard at that last expression. As long as $h \neq 0$ we were allowed to cancel it, and we are left with $19.6 + 4.9\,h$. Now watch what happens as h shrinks:

interval length h (s)	average speed $19.6 + 4.9h$ (m/s)
1	24.5
0.1	20.09
0.01	19.649
0.001	19.6049
0.0001	19.60049

The averages are not jumping around. They are homing in on a single number, 19.6 m/s, as cleanly as you could wish. We never set $h = 0$ – that would have given the meaningless $0/0$ – but we watched where the answer was *heading* as h approached 0. That destination, 19.6 m/s, is what we will mean by the instantaneous speed at $t = 2$. This act of watching where an expression heads, without ever arriving, is the **limit**, and it is the single technical idea on which all of calculus is built.

What is really happening

The fraction $\dfrac{s(2 + h) - s(2)}{h}$ is genuinely $\dfrac{0}{0}$ at $h = 0$ – there is no dodging that. What rescues us is that for every *nonzero* h, the fraction simplifies to the perfectly ordinary expression $19.6 + 4.9h$, which has no trouble at all at $h = 0$. The instantaneous speed is not the value of the bad fraction *at* $h = 0$ (it has none); it is the value the *good* expression is unmistakably approaching. A limit asserts: "no matter how close you demand the output be to 19.6, I can guarantee it by making h small enough." That is a statement about an entire neighborhood

of h's, not about the forbidden point itself. Holding those two things apart – the value at a point versus the value approached near it – is the whole art.

1.1.2 The second question: how much, in total?

Now flip the situation around. Suppose you do *not* know the distance fallen, but you do know the speed at every instant. From the table above we might guess (correctly) that the falling ball's speed grows linearly with time:

$$v(t) = 9.8\,t \quad \text{(meters per second)}.$$

The new question is the reverse of the first: *given the speed at every instant, how far has the ball travelled in total between $t = 0$ and $t = 2$?*

If the speed were constant – say a steady 20 m/s – this would be no harder than a grocery bill: distance is speed times time, $20 \times 2 = 40$ meters. But the speed is *changing* every instant, from 0 at the start to 19.6 at the end. There is no single speed to multiply by. So again we are stuck, and again the escape is to chop the problem into small pieces over which things barely change.

Intuition

Over a very short stretch of time the speed hardly changes, so over that short stretch "distance = speed × time" is *almost* exactly right. Slice the two seconds into many short intervals, on each one pretend the speed holds steady, multiply to get the little distance covered, and add up all the little distances. The thinner the slices, the better the pretense, and the closer the sum creeps to the true total distance. Once more it is "finer and finer, then take the limit."

Now look at what those little distances are. Distance over a short interval is (roughly) speed times the width of that interval – but "a height times a width" is the *area of a rectangle*. So adding up all those little distances is the same as adding up the areas of thin rectangles drawn under the speed graph. The total distance travelled *is the area under the speed curve*. A question about accumulated motion has transformed directly into a question about geometric area.

Let us add up the rectangles for $v(t) = 9.8\,t$ from 0 to 2, using n equal slices each of width $\Delta t = 2/n$. On the k-th slice we take the height to be the speed at its right edge, $t_k = 2k/n$. The area of the k-th rectangle is its height times its width, $(9.8 \cdot \frac{2k}{n}) \cdot \frac{2}{n}$. Summing all n slices gives

$$\sum_{k=1}^{n} \underbrace{9.8 \cdot \frac{2k}{n}}_{\text{height}} \cdot \underbrace{\frac{2}{n}}_{\text{width}} = \frac{9.8 \cdot 4}{n^2} \sum_{k=1}^{n} k.$$

To evaluate this sum, we use the classic identity for the sum of the first n integers: $\sum_{k=1}^{n} k = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}$. (You can see this geometrically by pairing the first and last numbers, $(1 + n) + (2 + n - 1) + \ldots$, giving n pairs each summing to $n + 1$, divided by 2). Substituting this into our total area gives:

$$\text{Total rectangle area} = \frac{39.2}{n^2} \cdot \frac{n(n+1)}{2} = 19.6 \cdot \frac{n+1}{n} = 19.6 \left(1 + \frac{1}{n}\right).$$

As $n \to \infty$, the excess fraction $\frac{1}{n} \to 0$, so the staircase sum approaches exactly 19.6 meters. And indeed, the area of the triangular region under the straight line $v(t) = 9.8t$ from $t = 0$ to $t = 2$ is

$\frac{1}{2} \times$ base $\times$ height $= \frac{1}{2}(2)(19.6) = 19.6$. Notice that $s(2) = 4.9 \cdot 2^2 = 19.6$: the area under the speed graph reproduces precisely the distance we started from in the first question (Figure 1.1). The two questions are already in conversation.

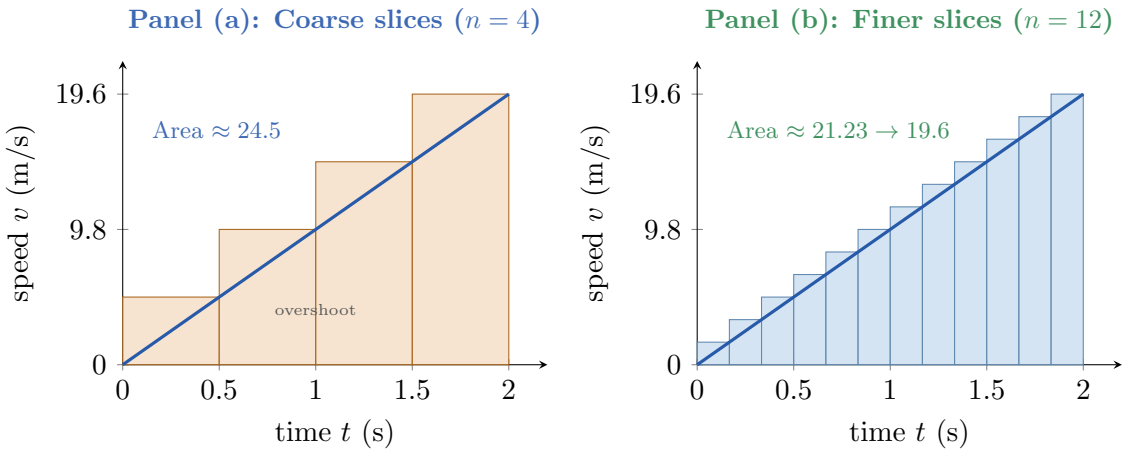


Figure 1.1: The total distance fallen between $t = 0$ and $t = 2$ equals the area under the speed graph $v(t) = 9.8t$. Panel (a) uses 4 coarse right-endpoint rectangles, overestimating the area due to visible stair-step wedges. Panel (b) uses 12 thinner slices, where the staircase clings tightly to the linear curve. As $n \to \infty$, the staircase error vanishes completely, converging to the exact area of 19.6 meters.

1.2 The two big ideas, named

Those two questions are the two pillars of calculus. Each has a name, and each is really a single idea wearing different clothes depending on the field that asks it.

Focus

The derivative answers "how fast, right now?" It is the *instantaneous rate of change* of one quantity with respect to another – and, drawn on a graph, it is the *slope of the curve at a point*. The process of finding it is *differentiation*.
The integral answers "how much, in total?" It is the *accumulation* of a changing quantity – and, drawn on a graph, it is the *area under the curve*. The process of finding it is *integration*.

The same two ideas appear everywhere, and recognizing them is half the battle. The derivative is velocity (rate of change of position), acceleration (rate of change of velocity), current (rate of flow of charge), marginal cost (rate of change of cost as you make one more), reaction rate, population growth rate, and the slope of any trend you have ever drawn through data. The integral is total distance from a speed, total charge from a current, total cost from a marginal cost, the work done by a varying force, the area of an irregular field, the mass of a rod whose density varies, and the probability piled up under a bell curve. Learn the two ideas once and you have learned them for every subject at once.

1.2.1 The derivative is the slope of a tangent

We computed the falling ball's instantaneous speed by shrinking a time interval. Let us now *see* that computation. Draw the graph of $s(t) = 4.9\,t^2$ with time across and distance up. Pick the point on the curve at $t = 2$. To estimate the speed there we used the interval from 2 to $2 + h$; on the graph, that amounts to drawing the straight line through the two curve points at $t = 2$ and $t = 2 + h$. A line through two points of a curve is called a **secant** line, and its slope – rise over run – is exactly

$$\text{slope of secant} = \frac{s(2+h) - s(2)}{(2+h) - 2} = \frac{s(2+h) - s(2)}{h},$$

the very same average-speed fraction as before. *Average speed is the slope of a secant line.* Now let $h \to 0$: the second point slides along the curve toward the first, the secant pivots, and in the limit it becomes the line that just grazes the curve at $t = 2$ – the **tangent** line. Its slope is the instantaneous speed, 19.6. The derivative *is* the slope of the tangent, and the tangent *is* the limit of secants (Figure 1.2).

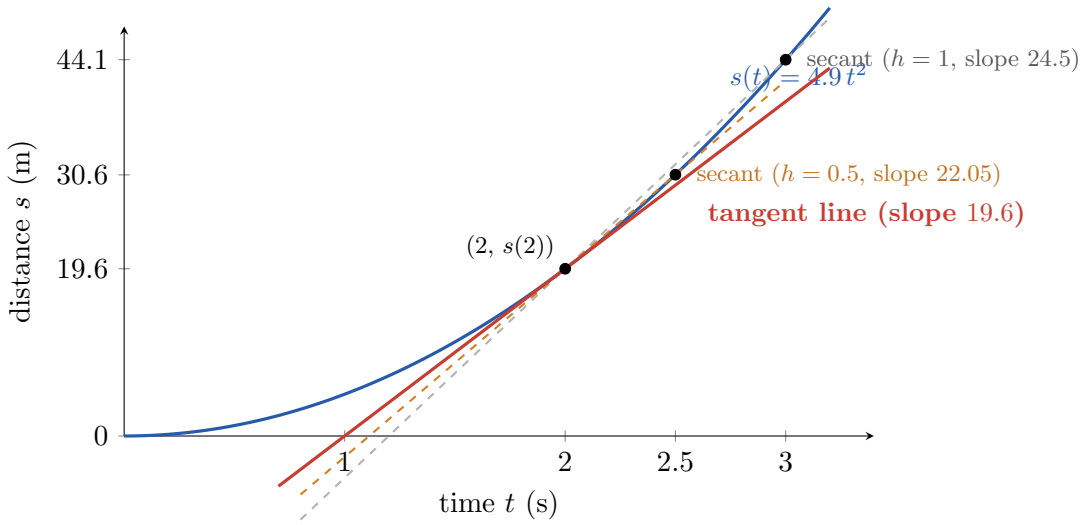


Figure 1.2: Secant to tangent convergence. The dashed lines join $(2, s(2))$ to $(2 + h, s(2 + h))$ for $h = 1$ and $h = 0.5$; their slopes represent the average speed over each shrinking interval. As $h \to 0$, the pivot point slides along the parabola toward $(2, 19.6)$ and the rotating secant freezes into the solid red tangent line, whose slope is the instantaneous speed 19.6 m/s.

Why the technique works

Why should "slope of the tangent" equal "rate of change"? Because slope *is* a rate of change – that is all slope ever was. The slope of a straight line is rise over run: how much the output changes per unit change in the input. A curve is not straight, so it has no single slope; but if you zoom in on one point of a smooth curve, it looks straighter and straighter, until under enough magnification it is indistinguishable from a line. That line is the tangent, and its slope is the rate of change of the curve *at that point*. The limit-of-secants construction is just the precise way to find the slope of a graph that refuses to be a single straight line. Every derivative you will ever compute is, at heart, this zoom-in.

Walking question

Picture a curve that goes up steeply, then levels off, then dips down – say a hill in profile. Walk along it in your mind and watch the tangent line tilt as you go: steep up, then flat at the crest, then tilting down the far side (Figure 1.3). Where is the tangent perfectly horizontal? What is special about that point on the hill? You have just discovered, with no formulas, why setting the derivative to zero finds the tops and bottoms – the peaks and valleys – of a curve.

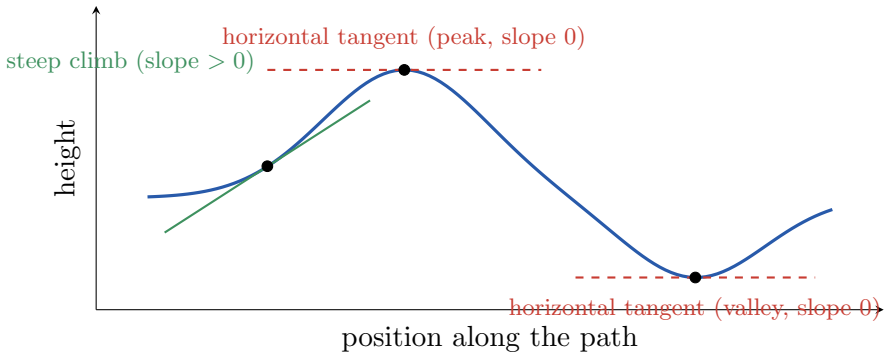


Figure 1.3: Walking the hill. On the climb the tangent tilts steeply upward (positive slope); at the crest and again at the bottom of the valley the tangent is perfectly horizontal (slope zero); on the descent the slope is negative. Those flat-tangent points identify the peaks and valleys – which is why, once we can compute derivatives, solving $f'(x) = 0$ pinpoints the extremes of any smooth curve.

1.2.2 The integral is the area under a curve

The accumulation question turned distance into area, and that picture is exactly the integral. Given any curve $y = f(x)$ sitting above the horizontal axis, the integral of f from $x = a$ to $x = b$ is the area of the region trapped between the curve and the axis over that span. We approximate it, as we did for the falling ball, by filling the region with thin rectangles: slice $[a, b]$ into many narrow strips, on each strip build a rectangle whose height is the value of f somewhere on that strip, sum the rectangle areas, and then let the strips become infinitely thin (Figure 1.4). The sum of rectangle areas is called a *Riemann sum*; the limit it approaches is the integral.

What is really happening

The symbol $\int_a^b f(x)\,\mathrm{d}x$ is a frozen photograph of this whole process. Leibniz wrote the integral sign $\int$ as a stretched letter "S" for *summa*, Latin for "sum," because an integral is the sum of the rectangle areas. Each rectangle has area $f(x)\,\mathrm{d}x$: a height $f(x)$ times an infinitely thin width $\mathrm{d}x$. So $\int_a^b f(x)\,\mathrm{d}x$ literally reads "sum, from a to b, of heights times infinitesimal widths." The notation is not a hurdle to clear – it is a sentence describing the picture. When you meet $\mathrm{d}x$ later it will not be a magic talisman; it is the ghost of a strip's width after you have let the strip shrink to nothing.

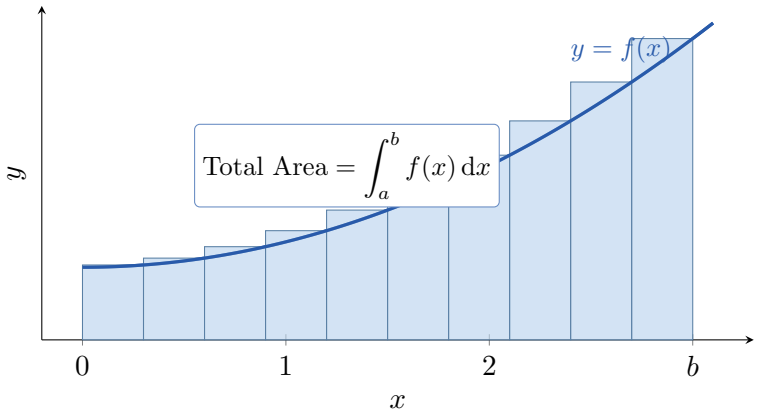


Figure 1.4: The integral as area. The shaded rectangles approximate the area under $y = f(x)$; as they get thinner their combined area converges to the exact area, written $\int_a^b f(x)\,dx$. The elongated "S" is Leibniz's sum sign, and dx is the vanishing width of a strip.

1.3 The central difficulty: the infinitely small and the infinitely many

Both pillars rest on the same precarious move. To get the derivative we divided two quantities that were both rushing to zero. To get the integral we added up infinitely many rectangles, each of vanishing area. In both cases we flirted with infinity – once as the infinitely small ($h \to 0$), once as the infinitely many ($n \to \infty$) – and somehow walked away with a finite, exact answer. *That* is the central difficulty calculus had to solve, and for two centuries it was solved with more boldness than rigor.

Intuition

Why is the infinite so hard to handle? Because our two trustiest tools break on it. "Rate = change in output over change in input" breaks when the change in input is zero: you get $0/0$. "Total = value times amount" breaks when there is no single value, only a changing one. The genius of calculus is to never actually *reach* the infinite. We keep the input change small but nonzero, or the rectangles many but finite, do ordinary finite arithmetic, and then ask where the answers are heading. The destination – the limit – is finite and exact, even though every step on the road still involves a real, harmless, nonzero quantity. We tame infinity by approaching it and reading off where the approach points, not by arriving.

Common trap

The seductive wrong move is to "just set $h = 0$." If you do that in $\dfrac{s(2+h) - s(2)}{h}$ before simplifying, you get $\dfrac{0}{0}$ and learn nothing; and morally, you have destroyed the very interval whose shrinking told you the rate. Equally tempting is to call an infinitely thin rectangle's area "zero," conclude that the total is a sum of zeros, and announce that all areas are 0. Both errors come from treating "infinitely small" as "equal to zero." It is not. An infinitesimal is smaller than any positive number you can name yet not zero – which is exactly why the early

pioneers could not say what it *was*. The cure, as we will see, is the limit, which lets us speak of the small without ever pretending it has become nothing.

1.3.1 How the pioneers got away with it

History

The two ideas are far older than the name "calculus." Around 250 BCE *Archimedes* found the area of a parabolic segment and approximated π by trapping a circle between inscribed and circumscribed polygons with ever more sides – a genuine limit argument, the "method of exhaustion." He had the integral's soul without its notation, and he was so wary of infinity that he re-proved each result by contradiction to avoid speaking of it directly.

The explosion came in the 1660s and 1670s, independently, with *Isaac Newton* in England and *Gottfried Wilhelm Leibniz* in Germany. Newton, with his eye on the physics of motion, thought of quantities as *flowing* in time and of derivatives as "fluxions" – velocities of flow. Leibniz, with the eye of a logician and notation-builder, thought in terms of infinitely small differences $\mathrm{d}x$ and infinite sums $\int$, and gave us the supple symbols $\frac{\mathrm{d}y}{\mathrm{d}x}$ and $\int y\,\mathrm{d}x$ that we still use because they *do our thinking for us*. Their followers fought a bitter, nationalistic priority war over who invented it first. The fair verdict: both did, independently, and we use Leibniz's notation with Newton's physical instincts.

What did they get right, and what was missing? They got the *answers* right, with uncanny reliability, and they grasped the two pillars and their inverse relationship with full clarity. What was missing was a coherent account of the infinitely small. Newton spoke of quantities "ultimately vanishing"; Leibniz of infinitesimals that were somehow nonzero when you divided by them and zero when you wished to discard them. It worked, but no one could say cleanly *why*.

History

The sharpest critic was the philosopher Bishop George *Berkeley*, who in 1734 skewered the infinitesimal in a pamphlet called *The Analyst*. Of those quantities that are first treated as nonzero (so we may divide by them) and then dropped as zero (so the answer comes out clean), he asked the unforgettable question: "May we not call them the *ghosts of departed quantities*?" He was right that the logic was incoherent, and wrong that the results were therefore suspect – the answers were correct. The gap between "correct answers" and "coherent reasons" stood open for nearly a century.

It was closed in the 1820s by Augustin-Louis *Cauchy* and, with full rigor, in the 1860s–70s by Karl *Weierstrass*, who replaced the mystical infinitesimal with the *limit*, defined in plain, finite language about how close you can force an output to be by controlling the input – the ε–δ definition you will meet in the next chapters. Weierstrass banished the ghosts: no infinitely small quantity appears in his definitions at all, only ordinary numbers and the word "approaches" made precise. Calculus could finally say exactly what it had always done correctly.

Why this matters

There is a moral here worth carrying through the whole book. For two hundred years the most powerful mathematics on Earth rested on a notion no one could define. That is not a scandal to be embarrassed by; it is a clue about how understanding actually grows. Intuition runs ahead, gets the right answers, and *drags* rigor along behind it until the foundations catch up. We will honor both halves: we will let intuition and pictures lead, and then we will make them precise. You are allowed to *see* that something is true before you can *prove* it – that is how Newton, Leibniz, and Archimedes worked too.

1.4 The central duality: differentiation and integration invert each other

We have two processes. Differentiation takes a quantity and returns its rate of change. Integration takes a rate and accumulates it into a total. Stated like that, they sound like opposites – and they are, in the precise sense that *each undoes the other*. This is the **Fundamental Theorem of Calculus**, the keystone of the subject, and the reason these two questions belong in one book rather than two.

You have already watched it happen. We started with position $s(t) = 4.9t^2$. Differentiating it (shrinking the time interval) gave the velocity $v(t) = 9.8\,t$. Then, starting from that velocity, integrating it (summing rectangles under its graph from 0 to 2) gave back 19.6 meters $= s(2)$. We went out by the derivative and came home by the integral. Accumulate a rate of change over an interval and you recover the total change of the original quantity. In symbols, with F a quantity and $f = F'$ its rate of change,

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

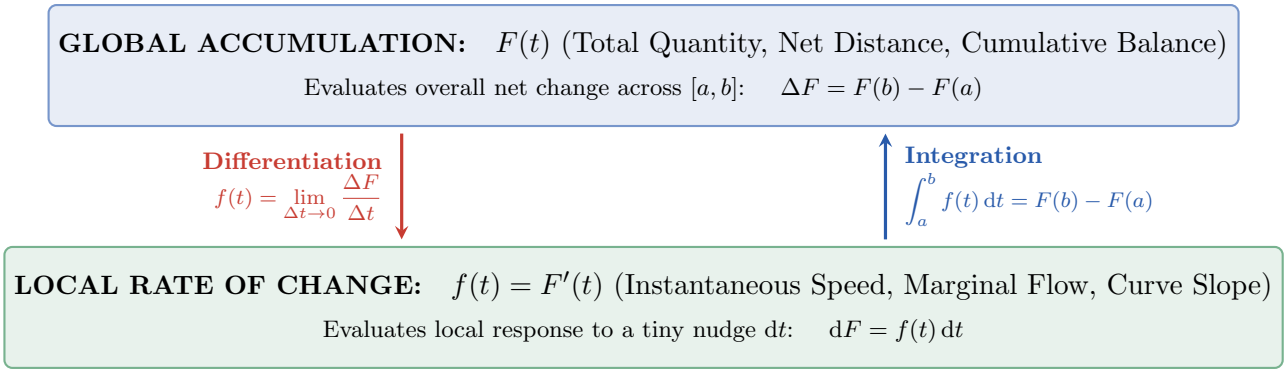


Figure 1.5: The Fundamental Theorem loop. Differentiation zooms into an infinitesimal window to measure the local rate of change $f(t) = F'(t)$. Integration reverses the operation, summing infinitely many infinitesimal contributions $f(t)\,\mathrm{d}t$ to reconstruct the global net change $F(b) - F(a)$. Cutting and reassembling are exact inverses.

Why the technique works

Why must accumulating a rate give back the total change? Strip away the symbols and the underlying mechanism is remarkably direct. Suppose $F(t)$ is your bank balance and $f(t) = F'(t)$ is the rate money flows in and out at each instant. Over a tiny slice of time $\mathrm{d}t$, the balance changes by (rate)×(time) $= f(t)\,\mathrm{d}t$. Add up all those tiny changes from a to b – that is the integral $\int_a^b f\,\mathrm{d}t$ – and of course you get the *net* change in the balance, $F(b) - F(a)$, because the total change is nothing but the sum of all the little changes along the way. Differentiation chops a total into its instantaneous pieces; integration glues the pieces back into a total. They are inverse because cutting and reassembling are inverse. The Fundamental Theorem is the precise statement that the gluing really does recover what the cutting took apart.

Why this matters

Pause to feel how surprising this ought to be. Tangent slopes and areas under curves have, on their face, nothing to do with each other – one is about steepness at a point, the other about accumulated bulk over a span. The geometry is unrelated; the questions arose in different centuries from different needs (astronomy and motion on one side, land area and volume on the other). That they turn out to be exact inverses is one of those moments where mathematics stops being a collection of tricks and reveals an underlying unity. When you truly grasp the Fundamental Theorem, hard integrals become easy: instead of summing infinitely many rectangles, you just ask "what quantity has *this* as its rate of change?" and read off the answer. The whole computational power of calculus flows from these two ideas being one.

Connection

This inverse pairing echoes structures you already know. Addition and subtraction undo each other; multiplication and division undo each other; squaring and square-rooting undo each other. Calculus adds one more pair to the list – differentiation and integration – but a far richer one, because the objects being paired are not numbers but entire *functions*, and the pairing ties together the local (slope at a point) with the global (area over a span). Much of higher mathematics, from differential equations to the divergence theorem of vector calculus to the topology of differential forms, is this single duality, generalized.

1.5 What this book promises

Here is the road we are on. We will spend the early chapters making sure we know what a *function* is, and what the special characters of calculus – the number e, the exponential e^x, the trigonometric functions – actually *mean*, beyond merely manipulating formulas. Then we will build the *limit* carefully, because everything rests on it, and we will see that the ε–δ definition is not bureaucratic fussiness but exactly the precise version of "heads toward." With limits in hand we will construct the derivative, learn to compute it, and understand *why* each rule (the product rule, the chain rule, and the rest) is forced upon us by the meaning of slope. We will do the same for the integral, prove the Fundamental Theorem that binds the two, and finally turn the whole machine loose on the equations that govern change itself – differential equations.

Focus

Three commitments for the road ahead. *First*, a picture before a formula, always: if you cannot see it, you do not yet understand it. *Second*, every rule comes with a reason – we will never ask you to memorize a procedure whose mechanism we have not exposed. *Third*, the limit is the hero of the story; every apparent paradox of the infinitely small and the infinitely many is resolved by it. Keep these three in mind and calculus stops being a list of tricks and becomes what it really is: a small number of deep ideas, seen clearly.

Work it yourself

Before you turn the page, do the falling-ball computation yourself, but at a different instant. Take $s(t) = 4.9\,t^2$ and find the instantaneous speed at $t = 3$ by forming $\dfrac{s(3+h) - s(3)}{h}$, simplifying until the h in the denominator cancels, and reading off the limit as $h \to 0$. You should get 29.4 m/s. Notice that in each case the speed came out as 9.8 times the time – you have just *derived*, by hand, that the derivative of $4.9\,t^2$ is $9.8\,t$. That is the whole subject in miniature, and you did it with nothing but algebra and the nerve to let h approach zero.

Exercises

Exercise 1.1 (Average to instantaneous)**.** A pebble dropped from rest falls $s(t) = 4.9\,t^2$ meters in t seconds. Compute the average speed over each interval $[1,\ 1+h]$ for $h = 1,\ 0.1,\ 0.01,\ 0.001$. Then simplify $\dfrac{s(1+h) - s(1)}{h}$ algebraically and take the limit as $h \to 0$. What is the instantaneous speed at $t = 1$? Confirm it equals $9.8 \cdot 1$.

Exercise 1.2 (A different curve)**.** Let $f(x) = x^2$. Form the secant slope $\dfrac{f(2+h) - f(2)}{h}$, simplify so the denominator's h cancels, and find the limit as $h \to 0$. This is the slope of the tangent to $y = x^2$ at $x = 2$. Repeat at a general point $x = a$ and show the tangent slope is $2a$.

Exercise 1.3 (Area by rectangles)**.** Estimate the area under $y = x$ from $x = 0$ to $x = 1$ by summing n rectangles of equal width $1/n$, using right-endpoint heights. Show the sum equals $\dfrac{1}{n^2} \cdot \dfrac{n(n+1)}{2} = \dfrac{n+1}{2n}$, and take $n \to \infty$. Check your limit against the area of the triangle from elementary geometry ($\frac{1}{2} \times 1 \times 1$).

Exercise 1.4 (Conceptual: instant versus interval)**.** Explain in a few sentences, to a curious friend who has not taken calculus, why "speed at a single instant" is not simply "distance divided by time" at that instant, and how the idea of a limit rescues the notion. Avoid all symbols.

Exercise 1.5 (Reading the notation)**.** In the expression $\displaystyle\int_0^3 f(x)\,\mathrm{d}x$, identify in words what each piece refers to in the rectangle picture: the sign $\int$, the limits 0 and 3, the height $f(x)$, and the width $\mathrm{d}x$. Why is the integral sign shaped like an "S"?

Exercise 1.6 (The inverse relationship, by hand)**.** The velocity of a cart is $v(t) = 6t$ (meters per second). Using n equal rectangles, show the area under v from 0 to 2 approaches 12 meters as $n \to \infty$. Separately, guess a position function $s(t)$ whose instantaneous rate of change is $6t$ and check that $s(2) - s(0) = 12$. You have just verified one instance of the Fundamental Theorem.

Exercise 1.7 (Where the tangent is flat)**.** Sketch (by hand or in your head) the graph of $y = x^2 - 4x + 3$. Using the result that the tangent slope at $x = a$ for $y = x^2$ is $2a$, argue that the slope of this parabola at $x = a$ is $2a - 4$. At what x is the tangent horizontal? Relate your answer to the lowest point of the parabola, and explain the connection to the "walk along the hill" picture (Figure 1.3) from this chapter.

Exercise 1.8 (Conceptual: taming infinity)**.** Berkeley mocked infinitesimals as "the ghosts of departed quantities." In your own words, explain what was logically uncomfortable about treating h as nonzero (to divide by it) and then as zero (to finish the calculation), and describe how the limit avoids ever doing both to the same quantity at the same time.

Exercise 1.9 (A proof in miniature)**.** Let $f(x) = c$ be a constant function. Using the secant-slope definition $\dfrac{f(x+h) - f(x)}{h}$, prove carefully that the tangent slope is 0 at every point. Then explain geometrically why this had to be so, and physically why "the rate of change of something that never changes is zero" is the only sensible answer.

Exercise 1.10 (Speed from a table)**.** A runner's position is recorded as 0, 4, 16, 36, 64 meters at times $t = 0, 1, 2, 3, 4$ seconds. Compute the average speed over each one-second interval. Are these average speeds, or instantaneous ones? What extra information (or limiting process) would you need to claim the runner's exact speed at $t = 2$? Discuss why a finite table can never quite pin down an instantaneous rate.

Chapter 2

Functions: The Language of Change

Calculus is the mathematics of change. But before we can study *how* things change, we need a precise way to say *what* is changing, and *what it depends on*. That language is the language of functions. Every derivative is the rate of change of a function; every integral is the accumulation of a function; every differential equation is a sentence whose unknown is a function. So this chapter is not a warm-up to be skimmed. It is the vocabulary of everything that follows. If you understand functions deeply — not as formulas to plug into, but as *rules that turn one quantity into another* — the rest of calculus will feel like a story you can already half-tell.

We will build the idea from the ground up: first the intuition (a function is a machine, a dependable rule), then the precise definition (domain, codomain, range), then the four ways we meet functions in the wild (formula, table, graph, words). From there we assemble the toolkit calculus leans on constantly: composition, inverses, symmetry, monotonicity, the families of polynomial, rational, and algebraic functions, and the art of transforming a graph by shifting, stretching, and reflecting. We end by asking the question the whole chapter is secretly about: *why are functions the objects calculus acts on?*

2.1 What a Function Really Is

Start with the everyday experience that functions are trying to capture: one thing depends on another. The cost of filling your tank depends on how many litres you pump. The distance a dropped stone has fallen depends on how long it has been falling. The temperature of your coffee depends on how long it has been sitting. In each case there is an *input* you can dial (litres, seconds) and an *output* that responds (cost, distance, temperature). A function is the exact bookkeeper of that dependence.

Intuition

Picture a machine with a funnel on top and a spout on the bottom. You drop a number in the top; a single number comes out the bottom. The machine is *deterministic*: drop in the same number twice and you get the same answer twice. That is the whole idea. A function is a reliable rule that assigns, to each allowed input, *exactly one* output.

The phrase "exactly one" is doing quiet but essential work. If a rule could send the input 3 sometimes

to 9 and sometimes to 10, it would be useless for prediction — and prediction is the entire point. So we demand single-valuedness: one input, one output, no exceptions.

Definition 2.1 (Function)**.** A *function* f from a set D to a set Y is a rule that assigns to each element $x \in D$ exactly one element $f(x) \in Y$. We write

$$f : D \to Y, \qquad x \mapsto f(x).$$

The set D is the *domain* (the allowed inputs). The set Y is the *codomain* (a set guaranteed to contain all the outputs). The actual set of outputs,

$$\text{range}(f) = \{f(x) : x \in D\} \subseteq Y,$$

is the *range* (or image).

Read the notation out loud so it stops being decoration. "f colon D arrow Y" says f *takes things from D and produces things in Y*. The second arrow, the little "$\mapsto$" with a bar, is read "maps to" and tells you the actual rule: x maps to $f(x)$. The first arrow is about *sets*; the second is about a single *element*.

Focus

Three sets travel with every function, and confusing them causes more grief than any other early mistake.

- **Domain** D: where inputs are allowed to live.
- **Codomain** Y: a roomy "target" set we promise the outputs fall inside. It is part of how the function is *declared*.
- **Range**: the outputs that actually occur. Always a subset of the codomain, often a strict subset.

For $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, the codomain is all of $\mathbb{R}$, but the range is only $[0, \infty)$: no real input ever produces a negative output.

2.1.1 The domain is part of the function

Here is a subtlety beginners routinely miss: *a function is its rule together with its domain.* The two formulas $g(x) = x$ on $\mathbb{R}$ and $h(x) = x^2/x$ are not the same function, even though x^2/x simplifies to x wherever it is defined. The second one is forbidden from eating $x = 0$, because division by zero is undefined, so its domain is $\mathbb{R} \setminus \{0\}$. Same rule almost everywhere, different domains, different functions. Calculus cares about this fiercely: the behavior of a function *near* a missing point is exactly the kind of thing a limit is built to examine.

When a function is handed to you as a bare formula with no domain stated, the convention is to use the *natural domain*: every real number for which the formula makes sense (no division by zero, no even roots of negatives, no logs of non-positive numbers, and so on).

Example 2.1 (Finding the natural domain)**.** Find the natural domain of

$$f(x) = \frac{\sqrt{x+2}}{x-3}.$$

Two requirements must both hold. First, the square root needs a non-negative argument: $x+2 \geq 0$, i.e. $x \geq -2$. Second, the denominator must not vanish: $x - 3 \neq 0$, i.e. $x \neq 3$. Intersecting the two conditions gives the natural domain

$$[-2, 3) \cup (3, \infty).$$

Notice how the domain is a *geometric* object on the number line — two intervals with a single point punched out — rather than a mere algebraic footnote.

Common trap

"$f(x) = x$" and "$f(x) = \dfrac{x^2}{x}$" are different functions because they have different domains. More generally, cancelling a factor can silently enlarge the domain. Algebra that "simplifies" $\frac{(x-1)(x+2)}{x-1}$ to $x + 2$ has quietly invited $x = 1$ back in — but the original expression was never defined there. Keep track of what you threw away; in calculus those punched-out points are where the interesting things (removable discontinuities, limits) live.

What is really happening

What is a function "really," underneath the machine metaphor? Mathematicians who wanted no metaphors at all settled on this: a function *is* its complete input-output ledger. Formally, f is the set of ordered pairs

$$\{(x, f(x)) : x \in D\},$$

subject to one rule — no two pairs share the same first coordinate (that is single-valuedness restated). So a function is literally a set of points: precisely the set of points you plot to draw its graph. This is why "the function" and "the graph of the function" are, at the deepest level, the same object. The graph is not a picture *of* the function; it *is* the function, drawn.

History

The word "function" enters mathematics with Leibniz in the 1670s, originally meaning a quantity (like a tangent length or a curvature) that varied along a curve. For the next century a "function" meant, in practice, a single analytic formula — Euler in 1748 defined it as "an analytic expression composed of a variable and constants." That formula-bound view cracked in the early 1800s when Fourier showed that wildly different formulas could describe the same curve, and that strange, piece-by-piece rules deserved to be functions too. Dirichlet in 1837 gave the modern liberation: y is a function of x if to each x there corresponds a definite y — *no formula required.* He even offered the deliberately formula-less example that is 1 on rationals and 0 on irrationals. That shift, from "function = formula" to "function = any single-valued rule," is one of the great quiet revolutions in mathematics, and it is exactly the generality calculus needs.

2.2 Graphs: Seeing a Function

If a function is a set of input-output pairs, we can *see* it by plotting each pair $(x, f(x))$ as a point in the plane, with the input along the horizontal axis and the output up the vertical axis. Sweep x

across the domain and the points trace a curve: the graph. The graph turns an abstract rule into geometry, and geometry is where intuition lives.

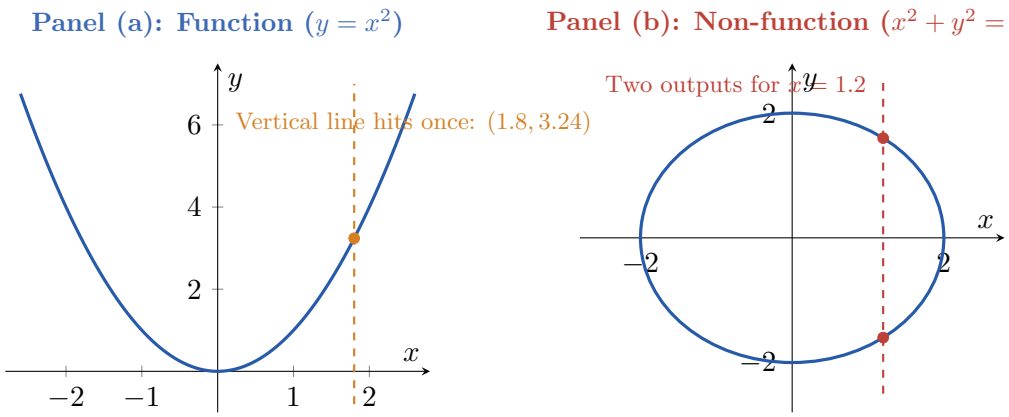


Figure 2.1: The Vertical Line Test in action. Panel (a): On the graph of $f(x) = x^2$, every vertical line crosses the curve at most once, guaranteeing that each input produces exactly one output. Panel (b): On the circle $x^2 + y^2 = 4$, the vertical line at $x = 1.2$ strikes twice (at $y \approx 1.6$ and $y \approx -1.6$), violating single-valuedness and proving the relation is not a single function of x.

Figure 2.1 also makes the single-valuedness rule visible, and gives us a quick diagnostic.

Focus

The Vertical Line Test. A curve in the plane is the graph of a function (of x) if and only if every vertical line crosses it *at most once*. A vertical line is the set of all points with a fixed input x; if it hit the curve twice, that single input would have two outputs, breaking the "exactly one" rule. So the circle $x^2 + y^2 = 1$ is *not* the graph of a function — the vertical line $x = 0$ stabs it at both $(0, 1)$ and $(0, -1)$.

Learning to read a graph pays off constantly, because calculus keeps asking you to read off rates and areas from pictures. A graph shows you at a glance where the function is positive or negative (above or below the x-axis), where it is rising or falling, where it peaks or bottoms out, how fast it climbs (steepness), and what it does far out toward the edges of the domain. Each of those visual features will get a precise calculus name in later chapters — zero, increasing, maximum, derivative, limit at infinity. The graph is the bridge between seeing and computing.

2.3 Four Ways to Represent a Function

The same function can show up wearing four different outfits, and a fluent reader of calculus can translate among them without thinking. Each representation makes some things obvious and hides others.

1. **A formula (symbolic).** $f(x) = x^2$. Compact, exact, perfect for algebra and for the differentiation rules to come. But a formula can hide behavior — you may not *see* the minimum until you think.

2. **A table (numerical).** A list of input-output pairs. This is what a thermometer log or an

experiment actually produces. Exact about the real data; silent about the values *between* the rows.

3. **A graph (visual).** The curve. Reveals shape, trend, symmetry, and speed at a glance; less good for reading off an exact value.
4. **Words (verbal).** "The output is the square of the input." How problems first arrive in the world. Often the hardest step in an applied problem is translating words into one of the other three forms.

Work it yourself

Take $f(x) = x^2$ and build all four views by hand. Words: *square the input.* Formula: $f(x) = x^2$. Table: compute f at $x = -2, -1, 0, 1, 2$ to get $4, 1, 0, 1, 4$. Graph: plot those five points and connect them with a smooth U. Now ask yourself which representation made the symmetry $f(-x) = f(x)$ jump out, which made the minimum at $x = 0$ obvious, and which would let you read off $f(1.5)$ most reliably. Different jobs, different outfits.

Table 2.1: A function as a table: $f(x) = x^2$ sampled at five inputs. The table is exact at these points and says nothing about what happens between them — that silence is precisely what a graph (or a limit) fills in.

x	-2	-1	0	1	2
$f(x)$	4	1	0	1	4

Application

In data science and machine learning, you almost never meet a function as a clean formula. You meet it as a *table*: rows of measured inputs and outputs (pixels and labels, features and prices). The central act of supervised learning is to *guess a formula or graph* that fits the table and generalizes to inputs you have not seen. The four representations are not academic — choosing which one to trust, and how to interpolate between table rows, is the daily work. Calculus enters the moment you ask "which way should I nudge the formula to fit the table better?" — that nudge is a derivative.

2.4 Composition: Functions in Series

Machines can be plumbed together: send the output of one into the input of the next. If g turns x into $g(x)$, and f then turns *that* into $f(g(x))$, the combined two-stage machine is the *composition* $f \circ g$.

Definition 2.2 (Composition)**.** Given $g : A \to B$ and $f : B \to C$, their composition $f \circ g : A \to C$ is defined by

$$(f \circ g)(x) = f\big(g(x)\big).$$

It is defined exactly at those x in the domain of g for which $g(x)$ lands in the domain of f. You apply g *first*, then f — read the notation from the inside out.

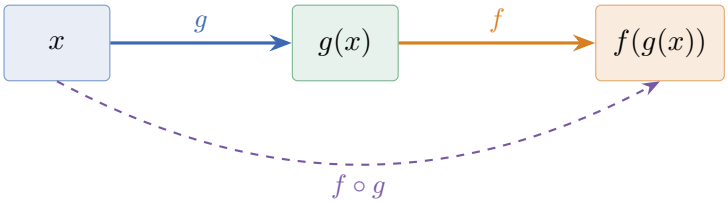


Figure 2.2: Composition is wiring two machines in series. The input x passes through g, and its output is immediately fed to f. The dashed arrow is the combined machine $f \circ g$.

Example 2.2 (Computing a composition)**.** Let $f(u) = \sqrt{u}$ and $g(x) = x^2 + 1$. Then

$$(f \circ g)(x) = f(g(x)) = \sqrt{x^2 + 1},$$

which is defined for *all* real x (since $x^2 + 1 \geq 1 > 0$ always). Reverse the order and you get a genuinely different function:

$$(g \circ f)(x) = g(f(x)) = (\sqrt{x})^2 + 1 = x + 1,$$

but now the domain is only $x \geq 0$, inherited from the inner $\sqrt{x}$. So even where $g \circ f$ and the bare formula $x + 1$ agree, the domain remembers the square root. **Order matters, and so does the inner domain.**

Common trap

$f \circ g$ is almost never the same as $g \circ f$. Composition is not commutative. Above, one branch gave $\sqrt{x^2 + 1}$ and the other gave $x + 1$ — not even close. Always apply the inner function first, and carry its domain restrictions along.

Why this matters

Why fuss over composition this early? Because the single most important rule in all of differential calculus — the *chain rule* — is the rule for differentiating a composition. "Most functions you will ever differentiate are compositions in disguise" is barely an exaggeration: $\sin(x^2)$, e^{3x}, $\sqrt{1 - x^2}$, $\ln(\cos x)$ are all $f(g(x))$ for some inner g. Learning to *see* the inner and outer machine now is learning to use the chain rule later. We will return to this exact picture when the chain rule arrives.

2.5 Inverses: Running the Machine Backwards

Sometimes you know the output and want the input. You measured the cost; how many litres did you pump? You know the area of a square; what was the side length? You want to *undo* the function. The undoing, when it exists, is the *inverse function* f^{-1}.

Intuition

If f is a machine that sends $x \mapsto y$, the inverse f^{-1} is the machine that sends $y \mapsto x$ — it runs the original backwards. Feeding a number through f and then through f^{-1} must return

you to where you started: $f^{-1}(f(x)) = x$. The inverse exactly cancels f.

But not every machine can be run backwards. Suppose $f(2) = 4$ and $f(-2) = 4$ (as for $f(x) = x^2$). Hand the inverse the output 4 and ask for the input — it cannot decide between 2 and -2. The inverse would have to be two-valued, which disqualifies it from being a function at all. The fix is to demand that f never sends two different inputs to the same output.

Definition 2.3 (One-to-one and inverse)**.** A function $f : D \to Y$ is *one-to-one* (injective) if distinct inputs give distinct outputs:

$$x_1 \neq x_2 \implies f(x_1) \neq f(x_2).$$

If f is one-to-one, its *inverse* f^{-1} has domain range(f) and is defined by the rule

$$f^{-1}(y) = x \quad \text{exactly when} \quad f(x) = y.$$

It satisfies $f^{-1}(f(x)) = x$ for all $x \in D$ and $f(f^{-1}(y)) = y$ for all y in the range.

Common trap

The notation f^{-1} means *inverse function*, not reciprocal. So $f^{-1}(x) \neq \dfrac{1}{f(x)}$. The reciprocal of $\sin x$ is $\csc x$; the inverse of $\sin x$ (on a suitable domain) is $\arcsin x$. Same superscript, completely different objects. Blame history for the clash, and read carefully.

There is a beautiful diagnostic for "one-to-one," the mirror image of the vertical line test.

Focus

The Horizontal Line Test. f is one-to-one — and so has an inverse — exactly when every *horizontal* line crosses its graph at most once. A horizontal line is a fixed output value y; if it hit the graph twice, two inputs would share that output, killing invertibility. $f(x) = x^2$ fails this test on all of $\mathbb{R}$ (the line $y = 4$ hits at $x = \pm 2$), which is exactly why $\sqrt{\cdot}$ is only the inverse of x^2 *after* we restrict to $x \geq 0$.

2.5.1 The graph of an inverse is a reflection

Here is the geometric punchline, and it is genuinely lovely. To invert a function you swap the roles of input and output — you swap x and y. On a graph, swapping the two coordinates is exactly reflecting across the diagonal line $y = x$. So the graph of f^{-1} is the mirror image of the graph of f in that diagonal.

Why the technique works

Why does inverting a function mean reflecting its graph across $y = x$? Trace it slowly. A point on the graph of f is a pair (a, b) with $b = f(a)$ — "input a, output b." The inverse reverses the roles: it takes input b to output a, so the corresponding point on the graph of f^{-1} is (b, a). The operation $(a, b) \mapsto (b, a)$ that turns one graph into the other is precisely the swap of horizontal and vertical coordinates. And swapping coordinates is a reflection across the line where the two coordinates are equal — the line $y = x$. So the geometry is not a trick

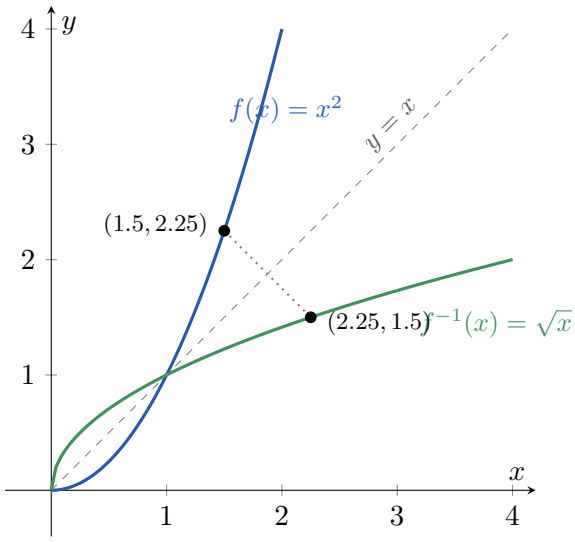


Figure 2.3: An inverse is a reflection across the diagonal line $y = x$. The point $(1.5, 2.25)$ on $f(x) = x^2$ (restricted to $x \geq 0$) is mirrored across $y = x$ to $(2.25, 1.5)$ on $f^{-1}(x) = \sqrt{x}$. Swapping input and output coordinates is geometrically identical to reflecting across the line of equality.

to memorize; it is the visible shadow of the algebra "swap input and output."

Example 2.3 (Finding an inverse algebraically)**.** Find the inverse of $f(x) = 2x + 3$. Set $y = 2x + 3$, then *solve for* x (this is the "swap input and output" step made concrete):

$$y = 2x + 3 \implies x = \frac{y - 3}{2}.$$

So $f^{-1}(y) = \dfrac{y-3}{2}$, or, renaming the input back to x, $f^{-1}(x) = \dfrac{x-3}{2}$. Sanity check by composing: $f^{-1}(f(x)) = \frac{(2x+3)-3}{2} = \frac{2x}{2} = x$. The machine and its reverse cancel, exactly as required.

2.6 Symmetry and Monotonicity: Reading a Function's Character

Two qualitative features of a function come up so often that they earn permanent names. Both are visible at a glance on the graph, and both will matter enormously once we are differentiating and integrating.

2.6.1 Even and odd: the two symmetries

Definition 2.4 (Even and odd functions)**.** A function f is *even* if $f(-x) = f(x)$ for every x in its domain, and *odd* if $f(-x) = -f(x)$ for every x in its domain.

Even means *mirror symmetry across the y-axis*: the left half of the graph is the reflection of the right half. Think $f(x) = x^2$ or $\cos x$. Odd means *point symmetry through the origin*: rotate the graph $180°$ about $(0, 0)$ and it lands on itself. Think $f(x) = x^3$ or $\sin x$. The names come from powers of x: x^{even} is an even function, x^{odd} is an odd function, which is the cleanest possible mnemonic.



Figure 2.4: Even functions are mirror-symmetric across the y-axis ($f(-x) = f(x)$); odd functions are rotationally symmetric through the origin ($f(-x) = -f(x)$).

Example 2.4 (Testing symmetry algebraically)**.** Classify $f(x) = x^4 - 3x^2$. Replace x by $-x$ and simplify:

$$f(-x) = (-x)^4 - 3(-x)^2 = x^4 - 3x^2 = f(x).$$

It equals the original, so f is *even*. Now try $g(x) = x^3 - x$:

$$g(-x) = (-x)^3 - (-x) = -x^3 + x = -(x^3 - x) = -g(x),$$

so g is *odd*. Most functions are neither: $h(x) = x^2 + x$ gives $h(-x) = x^2 - x$, which is neither $h(x)$ nor $-h(x)$.

Connection

Symmetry is more than visual balance — it performs substantial computational work. When you integrate an *odd* function over a symmetric interval $[-a, a]$, the left half exactly cancels the right half and the integral is 0, for free, with no antiderivative needed. For an *even* function the two halves are equal, so $\int_{-a}^{a} f = 2\int_{0}^{a} f$, halving the work. We will use these shortcuts in the integration chapters; they are gifts that symmetry hands you. Even and odd are also the building blocks of Fourier analysis, where every function is split into its even (cosine) and odd (sine) parts.

2.6.2 Increasing and decreasing

Definition 2.5 (Increasing and decreasing)**.** A function f is *increasing* on an interval I if larger inputs give larger outputs there:

$$x_1 < x_2 \implies f(x_1) < f(x_2) \quad \text{for all } x_1, x_2 \in I.$$

It is *decreasing* on I if $x_1 < x_2 \implies f(x_1) > f(x_2)$. A function that is one or the other throughout I is called *monotonic* on I.

On the graph, increasing means the curve goes *uphill* as you read left to right; decreasing means *downhill*. Most interesting functions do both, in different stretches: $f(x) = x^2$ falls on $(-\infty, 0]$ and rises on $[0, \infty)$, turning around at the bottom of the U.

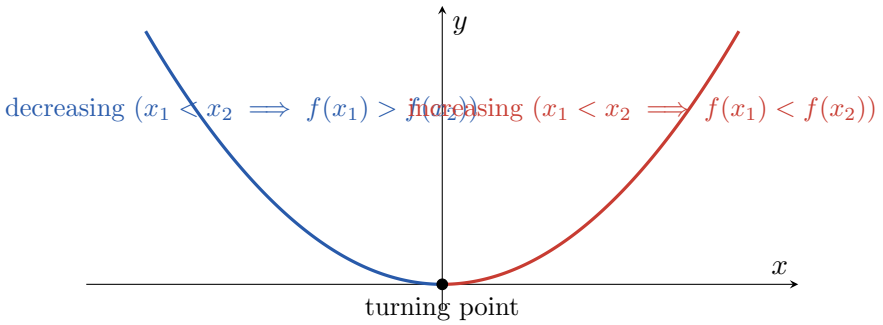


Figure 2.5: The parabola $f(x) = x^2$ decreases (downhill) on $(-\infty, 0]$ and increases (uphill) on $[0, \infty)$, turning around at the bottom of the U. The sign of the slope — and later, of the derivative — is exactly what flips at the turning point.

Connection

Notice the quiet link to invertibility from the previous section: a strictly monotonic function automatically passes the horizontal line test, so *every strictly increasing or strictly decreasing function has an inverse*. Monotonicity is the cleanest guarantee of invertibility we have. And in the chapters ahead, the sign of the *derivative* will be precisely the tool that detects where a function is increasing or decreasing — positive slope means uphill. So this informal idea graduates into a sharp, computable test.

2.7 Families of Functions

Calculus is easier when you recognize the species of function in front of you, because whole families share behavior. We build them up from the simplest.

2.7.1 Polynomials

A *polynomial* is built from x using only addition, subtraction, and multiplication — no division by x, no roots, no exotic operations. Concretely,

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where the coefficients a_i are constants and n (the highest power with $a_n \neq 0$) is the *degree*. The natural domain of every polynomial is all of $\mathbb{R}$: you can always add and multiply real numbers, so nothing is ever forbidden. Low degrees have familiar shapes — degree 0 is a horizontal line, degree 1 a slanted line, degree 2 a parabola, degree 3 an S-shaped cubic.

Why this matters

Polynomials are the friendliest functions in all of calculus, for three reasons. They are defined everywhere; they are smooth with no breaks or corners; and you can differentiate or integrate them with one trivial rule applied term by term. The deeper payoff comes much later: *Taylor's theorem* will show that a huge class of functions can be approximated, as closely as you like, by polynomials. So when we eventually want to understand $\sin x$ or e^x or $\ln x$ near a point,

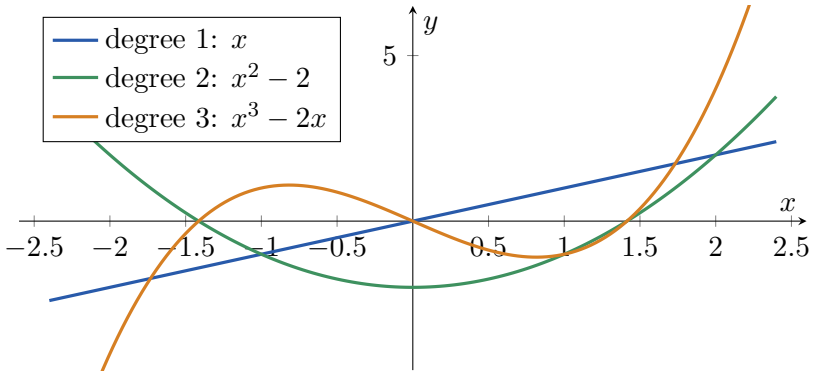


Figure 2.6: Polynomials of degree 1, 2, and 3. Degree controls the maximum number of turns: a degree-n polynomial can change direction at most $n - 1$ times, which is why the line never turns, the parabola turns once, and the cubic turns twice.

our strategy will be to replace them, locally, with a polynomial we already understand. Time spent on polynomials now is repaid then.

2.7.2 Rational functions

Allow division by polynomials and you get the *rational functions*: ratios

$$r(x) = \frac{p(x)}{q(x)}, \qquad p, q \text{ polynomials}, \quad q \not\equiv 0.$$

The name is the same "ratio" as in rational *numbers* (which are ratios of integers); here it is a ratio of polynomials. The new and important feature is that the domain now has holes: $r(x)$ is undefined wherever the denominator vanishes. Those forbidden points are where the most interesting calculus behavior lives — vertical asymptotes, where the function blows up toward $\pm\infty$.

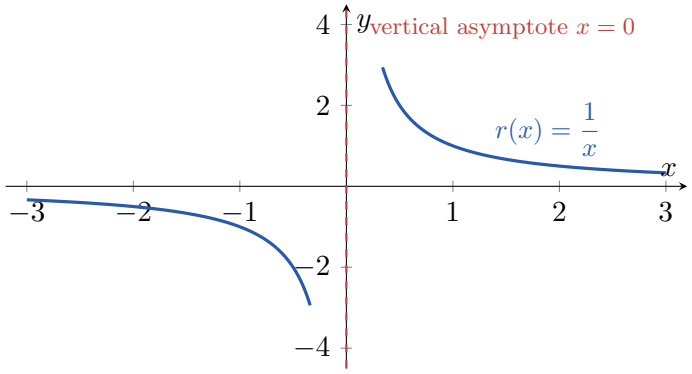


Figure 2.7: The rational function $r(x) = 1/x$. Its natural domain is $\mathbb{R} \setminus \{0\}$; as $x \to 0$ the value shoots off to $\pm\infty$, producing a vertical asymptote. Holes in the domain are where rational functions get dramatic.

2.7.3 Algebraic functions

Go one step further and allow roots — square roots, cube roots, fractional powers — in addition to the four arithmetic operations. Anything you can build from x with $+, -, \times, \div$ and roots is an *algebraic function*. Examples: $\sqrt{x}$, $\sqrt[3]{x^2+1}$, $\dfrac{x}{\sqrt{1-x^2}}$. The presence of even roots restricts the domain (you cannot take $\sqrt{\text{negative}}$ over the reals), which is why $\sqrt{x}$ lives only on $[0, \infty)$ and $\dfrac{x}{\sqrt{1-x^2}}$ only on $(-1, 1)$.

Focus

A useful hierarchy, each family containing the previous:

$$\text{polynomials} \subset \text{rational} \subset \text{algebraic}.$$

Every polynomial is a rational function (with denominator 1); every rational function is algebraic (no roots needed). Functions that are *not* algebraic at all — that can never be captured by finitely many arithmetic operations and roots — are called *transcendental*. These are the headline acts of the next chapters: e^x, $\ln x$, $\sin x$, $\cos x$. They "transcend" algebra, and calculus is the right tool to tame them.

2.8 Transformations: Reshaping a Graph

Once you know the graph of one function, you can get the graphs of a whole crowd of relatives for free — by shifting, stretching, and reflecting. Understanding transformations means you never start a graph from scratch; you start from a shape you know and *move* it. There are four basic moves, and the key to all of them is a single distinction: changes *outside* the function act on the *output* (y), and changes *inside* act on the *input* (x).

Focus

The four transformations. Start from $y = f(x)$ and let $c > 0$.

- **Vertical shift:** $f(x) + c$ moves the graph *up* by c; $f(x) - c$ moves it *down*. (Outside; behaves as you expect.)
- **Horizontal shift:** $f(x - c)$ moves the graph *right* by c; $f(x + c)$ moves it *left*. (Inside; behaves *opposite* to the sign — the famous surprise.)
- **Vertical stretch/reflect:** $a\, f(x)$ scales the output by $|a|$ (a stretch if $|a| > 1$, a squash if $|a| < 1$) and flips across the x-axis if $a < 0$. (Outside.)
- **Horizontal stretch/reflect:** $f(bx)$ scales the input axis by the factor $1/|b|$, so $|b| > 1$ *compresses* horizontally and $|b| < 1$ stretches it, with a flip across the y-axis if $b < 0$. (Inside; again the counterintuitive direction.)

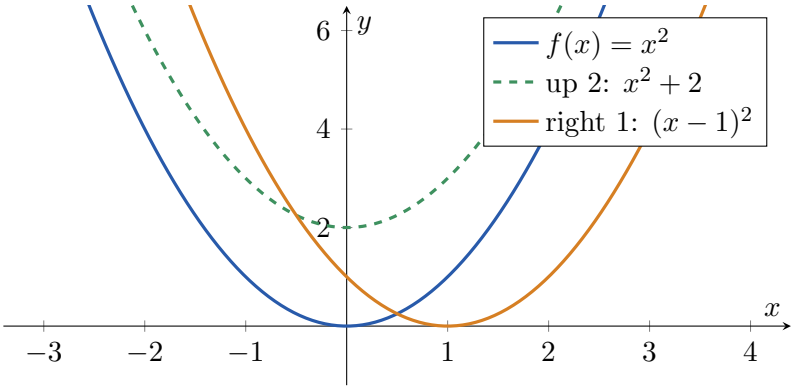


Figure 2.8: The base parabola x^2 (solid blue), shifted *up* by 2 via $x^2 + 2$ (an outside change), and shifted *right* by 1 via $(x-1)^2$ (an inside change). Note that $(x-1)^2$ moves right even though we *subtracted* — the inside surprise.

Why the technique works

Why does $f(x-c)$ shift the graph to the *right* when we *subtract*? It feels backwards, so let us pin down the mechanism instead of memorizing it. Call the new function $h(x) = f(x-c)$ and ask: where does h achieve the value that f achieved at some point x_0? It does so when its *inside* equals x_0, that is when $x - c = x_0$, i.e. at $x = x_0 + c$. So whatever f did at x_0, the new function h does at $x_0 + c$ — a position c units farther right. The graph's features all move right by c. The minus sign is there precisely because the input has to grow by c to "catch up" to the value the inside used to reach. Once you see it as "the input must increase to compensate," the direction stops being a paradox.

Example 2.5 (Reading a transformed graph)**.** Describe how to build the graph of $g(x) = -2(x+3)^2 + 1$ from the parabola $y = x^2$, taking the operations from the inside out:

1. $(x+3)^2$: inside change, shift *left* 3 units.
2. $2(x+3)^2$: outside multiply, stretch vertically by 2 (taller, thinner).
3. $-2(x+3)^2$: the minus reflects across the x-axis (the U now opens downward).
4. $-2(x+3)^2 + 1$: outside add, shift *up* 1 unit.

Result: a downward parabola with its peak (vertex) at $(-3, 1)$, twice as steep as the original. You read off the whole shape without plotting a single point.

Common trap

Order can matter when you stack transformations, because some of them do not commute. "Stretch then shift" generally differs from "shift then stretch." A safe rule: handle everything *inside* the function (affecting x) as one group and everything *outside* (affecting y) as another, and within each group respect the algebraic order of operations. When in doubt, test one point: a single tracked point will tell you immediately whether you applied the moves correctly.

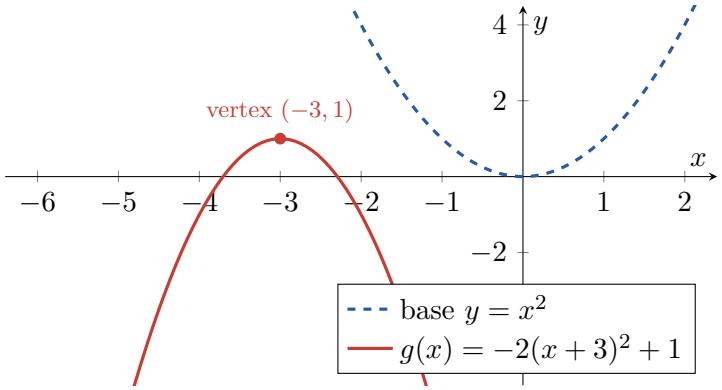


Figure 2.9: Building $g(x) = -2(x+3)^2 + 1$ from $y = x^2$: shift left 3, stretch vertically by 2, reflect across the x-axis, then shift up 1. The vertex lands at $(-3, 1)$ and the parabola opens downward.

2.9 Why Functions Are the Objects Calculus Acts On

We close by answering the question lurking under the whole chapter. Calculus has two central verbs — *differentiate* and *integrate* — and both take a function as input and return a function as output. Differentiation asks "how fast is this output changing as the input moves?" and hands you back the *rate-of-change function*, the derivative. Integration asks "how much has accumulated?" and hands you back the *accumulation function*, the integral. Neither question even makes sense for a lone number; both are questions about a *relationship* between a varying input and a responding output. That relationship is exactly what a function is.

What is really happening

Here is the deep reason functions are unavoidable. To talk about *change*, you need at least two quantities and a dependence between them: you cannot ask "how fast" without "how fast *relative to what*." The most economical object that packages "output depends on input" is a function. So the moment you decide to study change quantitatively, the function is not a choice — it is forced on you. Calculus did not pick functions for convenience; functions are the smallest accurate description of a varying world, and calculus is what you can *do* with them. Everything ahead — limits, derivatives, integrals, series, differential equations — is a verb whose noun is a function.

Walking question

On your next walk, pick something changing around you — your shadow's length as the afternoon wears on, your speed as you go up and down a hill, the loudness of traffic as a car passes. For each one, name the input and the output explicitly. Then ask the two calculus questions in plain words: *how fast is the output changing right now?* (that is a derivative) and *how much has accumulated so far?* (that is an integral). You will find that once you have named the function, the calculus questions almost ask themselves. That is the central habit of mind, and you can already practice it.

History

The historical rise of the function concept and the rise of calculus are inextricably braided together. Newton thought in terms of *fluents* — quantities flowing in time — and *fluxions*, their rates of flow; his "functions" were quantities changing with time, the very picture above. Leibniz, who gave us the word "function" and the notation $\mathrm{d}y/\mathrm{d}x$ and $\int$, thought in terms of relations between varying quantities along a curve. For both founders, the object under study was already a function in all but the modern name. The careful set-theoretic definition we gave at the start of this chapter came two centuries later, once mathematicians like Cauchy, Dirichlet, and Weierstrass insisted on making the foundations airtight. The intuition came first and worked brilliantly; the rigor came later and explained *why* it worked. That order — intuition leading, rigor following to secure the ground — is the rhythm of this entire book.

2.10 Exercises

Exercise 2.1 (Natural domains)**.** Find the natural domain of each function and express it in interval notation.

1. $f(x) = \dfrac{1}{x^2 - 4}$
2. $g(x) = \sqrt{9 - x^2}$
3. $h(x) = \dfrac{\sqrt{x+1}}{x - 2}$
4. $k(x) = \dfrac{1}{\sqrt{x} - 2}$ (watch for *two* forbidden situations)

Exercise 2.2 (Range from the graph)**.** For each function, determine the range, and justify it geometrically (what is the lowest/highest output, and is every value between achieved?).

1. $f(x) = x^2 - 4$ on $\mathbb{R}$.
2. $g(x) = \dfrac{1}{x}$ on $(0, \infty)$.
3. $h(x) = 3 - |x|$ on $\mathbb{R}$.

Exercise 2.3 (Composition and order)**.** Let $f(x) = x + 1$ and $g(x) = x^2$. Compute $(f \circ g)(x)$ and $(g \circ f)(x)$, show they are different functions, and find every x at which the two compositions happen to be *equal*.

Exercise 2.4 (Decomposing a composition)**.** Each of the following is a composition $f(g(x))$. Identify a natural inner function g and outer function f. (This is the skill the chain rule will depend on.)

1. $\sqrt{x^2 + 1}$
2. $(3x - 5)^{10}$
3. $\dfrac{1}{x^2 + 1}$

Exercise 2.5 (Finding inverses)**.** Find the inverse of each function on the indicated domain, state the domain of the inverse, and verify by composition that it undoes the original.

1. $f(x) = 3x - 7$ on $\mathbb{R}$.
2. $g(x) = x^2 + 1$ on $[0, \infty)$.
3. $h(x) = \dfrac{1}{x-2}$ on $(2, \infty)$.

Exercise 2.6 (Even, odd, or neither)**.** Classify each function as even, odd, or neither, showing the algebra $f(-x) = \ldots$ in each case.

1. $f(x) = x^4 + 5x^2 - 1$
2. $g(x) = x^5 - x$
3. $h(x) = x^3 + x^2$
4. $p(x) = \dfrac{x}{x^2+1}$

Exercise 2.7 (Transformations)**.** Starting from the base graph $y = x^2$, describe in words (inside-out) the sequence of transformations that produces each graph, and state the location of the vertex.

1. $y = (x-2)^2 + 3$
2. $y = -\frac{1}{2}(x+1)^2$
3. $y = 2(x-4)^2 - 5$

Exercise 2.8 (Conceptual: representations)**.** A function is given only by the table

x	0	1	2	3	4
$f(x)$	1	2	4	8	16

Propose a formula that fits all five rows. Explain clearly what the table can and cannot tell you: in particular, name two *different* functions that agree with the table at these five inputs but disagree at $x = 1.5$. What does this say about the limits of the numerical representation?

Exercise 2.9 (Proof: the vertical line test)**.** Prove the vertical line test carefully. That is, prove that a set S of points in the plane is the graph of a function $y = f(x)$ (for x ranging over some domain D) *if and only if* every vertical line meets S in at most one point. (Hint: a function is, by Definition 2.1, a single-valued rule; translate "single-valued" into a statement about the points $(x, y) \in S$.)

Exercise 2.10 (Proof: composition of one-to-one functions)**.** Suppose f and g are both one-to-one, and the composition $f \circ g$ is defined. Prove that $f \circ g$ is also one-to-one. (Hint: assume $(f \circ g)(x_1) = (f \circ g)(x_2)$ and peel off one function at a time, using injectivity of each.) Then explain in one sentence why this guarantees $f \circ g$ has an inverse, and state a formula for $(f \circ g)^{-1}$ in terms of f^{-1} and g^{-1}.

Exercise 2.11 (Proof: even/odd decomposition)**.** Let f be any function whose domain is symmetric about 0 (so $-x$ is in the domain whenever x is). Define

$$f_{\text{even}}(x) = \frac{f(x) + f(-x)}{2}, \qquad f_{\text{odd}}(x) = \frac{f(x) - f(-x)}{2}.$$

Prove that f_{even} is even, that f_{odd} is odd, and that $f = f_{\text{even}} + f_{\text{odd}}$. (This shows *every* such function splits uniquely into an even part plus an odd part — a fact at the heart of Fourier analysis.) As a check, carry out the decomposition for $f(x) = e^x$ and identify the two famous functions you obtain.

Chapter 3

Exponentials and the Number e

There is a number, written e, that sits at the center of calculus the way π sits at the center of geometry. It is about 2.71828, it never stops and never repeats, and the first time you meet it, it can feel like a magic constant that a teacher pulled from a hat. "Use base e," they say, "because it is natural." But *why* is it natural? Natural for whom? Natural for what?

This chapter is a slow, careful answer to that question. We are going to refuse to accept e as a gift. Instead we will *corner* it: we will ask a few very ordinary questions about growth, money, and slopes, and we will watch the same number $2.71828\ldots$ come walking out of every one of them, whether we wanted it to or not. By the end you should feel that e was not invented. It was discovered, the way a continent is discovered, because it was already there.

We start before e, with the thing it is the king of: exponentials.

3.1 What an Exponential Really Is

3.1.1 From repeated multiplication to a smooth curve

You first met powers as repeated multiplication. If a is a positive number,

$$a^1 = a, \qquad a^2 = a \cdot a, \qquad a^3 = a \cdot a \cdot a, \qquad a^n = \underbrace{a \cdot a \cdots a}_{n \text{ factors}}.$$

This is airtight when n is a positive whole number. You cannot multiply a by itself "$\frac{1}{2}$ times" or "-3 times" or "π times" by counting factors. So if we want a function $f(x) = a^x$ defined for *every* real number x – a smooth curve, rather than merely a discrete scatter of points over the integers – we have to *extend* the idea. The remarkable fact is that there is exactly one consistent way to do it.

The whole extension is forced by a single property we refuse to give up:

$$a^{x+y} = a^x \, a^y. \tag{3.1}$$

This is the law of exponents, and over the integers it is just bookkeeping: $a^{2+3} = a^2 a^3$ because 5 factors is 2 factors followed by 3 factors. We will *insist* that this law keeps holding for all real exponents, and that single insistence pins down everything else.

Intuition

Think of a^x as a machine that turns *addition into multiplication*. Slide the input x along by a fixed step Δx, and the output does not change by a fixed *amount* – it changes by a fixed *factor*. Walk one unit to the right and you multiply by a. Walk another unit and you multiply by a again. Exponentials are the functions for which equal strides cause equal *percentage* changes. That is the whole personality of the thing.

Let us see how (3.1) forces the rest of the definition, step by step.

Zero. Put $y = 0$ in (3.1): $a^{x+0} = a^x a^0$, i.e. $a^x = a^x\, a^0$. Dividing by a^x (which is positive, never zero) gives

$$a^0 = 1.$$

We did not *decree* that $a^0 = 1$ to be tidy; the exponent law leaves us no choice.

Negative exponents. Put $y = -x$: $a^{x+(-x)} = a^0 = 1$, so $a^x a^{-x} = 1$, hence

$$a^{-x} = \frac{1}{a^x}.$$

Again, forced. A negative exponent must mean a reciprocal, because that is the only thing consistent with the law.

Roots and fractions. What could $a^{1/2}$ be? By the law, $a^{1/2} \cdot a^{1/2} = a^{1/2+1/2} = a^1 = a$. So $a^{1/2}$ is a number whose square is a: it is $\sqrt{a}$. More generally $a^{1/n} = \sqrt[n]{a}$, and $a^{m/n} = \left(\sqrt[n]{a}\right)^m$. Every rational exponent is now determined.

Irrational exponents. This is the only delicate step, and it is the place where the integers quietly hand the baton to calculus. What is 2^π? We cannot reach π by counting or by fractions exactly. But we can *sneak up on it*. The number $\pi = 3.14159\ldots$ is the limit of the rational numbers

$$3, \quad 3.1, \quad 3.14, \quad 3.141, \quad 3.1415, \quad \ldots$$

and each of those has a perfectly good power: $2^3, 2^{3.1}, 2^{3.14}, \ldots$ These values march steadily toward a single limiting number, and *that* limit is what we define 2^π to be.

What is really happening

When you write 2^π, nothing is being "multiplied π times." What is really happening is a *limit*. We define

$$a^x \;=\; \lim_{\substack{r \to x \\ r \text{ rational}}} a^r,$$

the value the rational powers approach as their exponent closes in on x. The reason this works – the reason the rational powers do not wander off but settle on one number – is that a^r is *monotonic and continuous* in r, a fact guaranteed by the completeness of the real numbers (the same property that gives every length a number). So the exponential is the first place in this book where the smoothness of the real line is doing real labor under the surface. Calculus is already here, hiding inside an innocent-looking 2^x.

3.1.2 The shape of the graph

With a^x now defined for all real x, let us look at it. Take $a > 1$, say $a = 2$. The function climbs, gently at first and then ferociously. Take $0 < a < 1$, say $a = \frac{1}{2}$; then $a^x = (1/2)^x = 2^{-x}$ is just the

mirror image, decaying instead of growing. And $a = 1$ is the boring flat line $1^x = 1$.

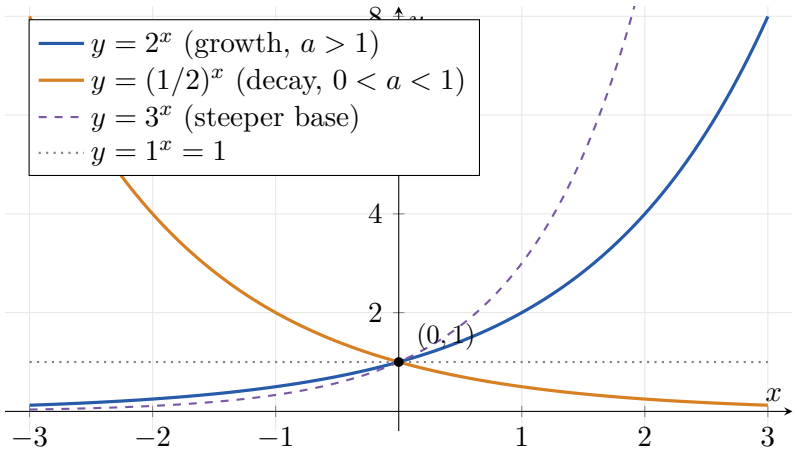


Figure 3.1: Every exponential $y = a^x$ passes through $(0, 1)$, stays strictly positive, and is either increasing ($a > 1$) or decreasing ($0 < a < 1$). Growth and decay are mirror images: $(1/2)^x$ is 2^x reflected across the y-axis.

Three features of Figure 3.1 are worth saying out loud, because they are the soul of the exponential and we will lean on all three.

- **It is always positive.** $a^x > 0$ for every x. The curve hugs the x-axis on one side but never touches or crosses it. (A positive number raised to any power stays positive; the x-axis is a horizontal asymptote.)
- **It always passes through** $(0, 1)$**.** Because $a^0 = 1$ for every base, the whole family pivots through the single point $(0, 1)$.
- **Equal steps multiply.** Step right by 1 and the height multiplies by a. This is constant *proportional* change, and it is what makes exponentials the law of anything that grows or shrinks at a rate set by its own current size: populations, money, radioactive atoms, cooling coffee, viral posts.

Common trap

An exponential function is a^x: the variable is upstairs, in the exponent. A *power* function is x^a: the variable is downstairs, in the base. They are completely different animals. x^2 is a gentle parabola; 2^x eventually leaves every power function in the dust. Confusing 2^x with x^2 – or differentiating one as if it were the other – is one of the most common early mistakes. Say it to yourself: "exponential, variable on top."

Walking question

Here is a famous gut-check. A lily pad doubles its area every day, and on day 30 it covers the whole pond. On which day was the pond *half* covered? Most people instinctively say day 15. Walk with that for a minute and feel why the answer is day 29 – and why "half" happens at the very last step. That single beat of delay, packed into the final doubling, is the entire emotional content of exponential growth.

Application

Why nature loves exponentials. Suppose a quantity changes at a rate proportional to how much of it there is right now. Money in an account earns interest in proportion to the balance. A bacterial colony makes new cells in proportion to the cells already present. A lump of uranium loses atoms in proportion to the atoms still intact. In symbols, the rate of change is k times the amount, and – as we will prove once we have derivatives in hand – the only functions obeying that are exponentials, $A(t) = A_0\,a^t$ for a suitable base. The exponential is not one model among many. It is *the* signature of self-proportional change, which is why it shows up everywhere from compound interest to carbon dating to the early phase of an epidemic.

3.2 Logarithms: Running the Exponential Backwards

3.2.1 The question a logarithm answers

Since a^x (for $a > 1$) is strictly increasing, it hits every positive height exactly once. That means it can be *undone*. Given a positive output y, there is exactly one input x with $a^x = y$. We give that input a name.

Definition 3.1 (Logarithm)**.** For a base $a > 0$ with $a \neq 1$, the **logarithm** $\log_a y$ is the exponent you must raise a to in order to get y:

$$\log_a y = x \quad \Longleftrightarrow \quad a^x = y, \qquad (y > 0).$$

In words: $\log_a y$ answers the question "a to the *what* gives y?"

So $\log_2 8 = 3$ because $2^3 = 8$, and $\log_{10} 1000 = 3$ because $10^3 = 1000$, and $\log_2 \frac{1}{4} = -2$ because $2^{-2} = \frac{1}{4}$. A logarithm is just an exponent wearing a different hat. The logarithm and the exponential are *inverse functions*: each undoes the other,

$$a^{\log_a y} = y \quad (y > 0), \qquad \log_a(a^x) = x \quad (\text{all } x).$$

Geometrically, inverse functions are reflections across the line $y = x$ – you swap the roles of input and output, which swaps the roles of the two axes.

3.2.2 Why logs turn multiplication into addition

Every algebraic property of logarithms is just the exponent law (3.1) read backwards. This is worth seeing once, slowly, because it is the reason logarithms were invented and the reason they still matter.

Why the technique works

The mechanism: a logarithm is an exponent, so adding exponents becomes adding logarithms. Take two positive numbers u and v. Write them in terms of the base: $u = a^p$ and $v = a^q$, which is to say $p = \log_a u$ and $q = \log_a v$. Now multiply:

$$u\,v = a^p\,a^q = a^{p+q}$$

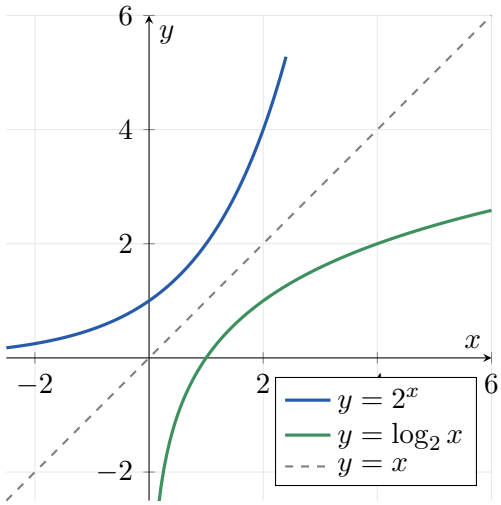


Figure 3.2: The logarithm is the exponential reflected across the dashed line $y = x$. The point $(0, 1)$ on the exponential becomes $(1, 0)$ on the logarithm; the vertical asymptote of the log (at $x = 0$) is the reflected image of the horizontal asymptote of the exponential.

by the exponent law. Read that last equation through the logarithm: the exponent that produces uv is $p + q$. Therefore

$$\boxed{\log_a(uv) = \log_a u + \log_a v.}$$

Multiplication downstairs became addition upstairs. The same one-line trick gives every other rule:

$$\log_a\left(\frac{u}{v}\right) = \log_a u - \log_a v, \qquad \log_a(u^n) = n \log_a u.$$

There is no separate "log algebra" to memorize. There is only the exponent law, seen from the other side.

History

This is not a curiosity – it changed history. In 1614 John Napier published the first table of logarithms, and Henry Briggs soon recast them in base 10. Before computers, multiplying two eight-digit numbers by hand was slow and error-prone. With a log table you could *look up* the logs, *add* them (easy), and look the answer back up. Multiplication had been demoted to addition. Kepler used logarithms to grind through the astronomical arithmetic behind his laws of planetary motion; Laplace later said logarithms, "by shortening the labours, doubled the life of the astronomer." The slide rule – the engineer's constant companion until the 1970s – is nothing but two logarithmic rulers sliding past each other, *adding lengths* to multiply numbers. Every Apollo mission was flown with help from a device whose whole principle is $\log(uv) = \log u + \log v$.

Focus

Changing base. Any one logarithm can be rebuilt from any other by a constant factor. If $a^x = y$, take $\log_b$ of both sides: $x \log_b a = \log_b y$, so

$$\log_a y = \frac{\log_b y}{\log_b a}.$$

The shape of the log curve never really changes with the base – only its vertical scale. That is exactly why, in a moment, we will be free to hunt for the *one* base that makes calculus cleanest, knowing every other base is just a rescaling of it.

3.3 The Central Question: A Base That Grows At Its Own Rate

Now we come to the heart of the chapter. We have a whole family of exponentials a^x, one for each base a. For calculus, the question that matters is: *how fast does a^x grow?* That is, what is its derivative?

Even before we have developed the rules of differentiation, we can compute the derivative of a^x from the very definition of slope, and what falls out is startling.

3.3.1 Differentiating a^x from scratch

The derivative is the limit of the slope of a tiny secant line. For $f(x) = a^x$:

$$f'(x) = \lim_{h \to 0} \frac{a^{x+h} - a^x}{h}.$$

We exploit the defining algebraic structure of the exponential: it converts additions in the input into multiplications in the output. Split $a^{x+h} = a^x a^h$, then factor a^x out of the entire difference quotient:

$$f'(x) = \lim_{h \to 0} \frac{a^x a^h - a^x}{h} = \lim_{h \to 0} a^x \cdot \frac{a^h - 1}{h} = a^x \cdot \underbrace{\lim_{h \to 0} \frac{a^h - 1}{h}}_{\text{call this } M(a)}.$$

The factor a^x slides straight out of the limit because it does not depend on h. What is left, $M(a) = \lim_{h \to 0}(a^h - 1)/h$, is just a *number* – it depends on the base a but not on x. So we have learned something remarkable already:

$$\frac{\mathrm{d}}{\mathrm{d}x} a^x \;=\; M(a) \cdot a^x. \tag{3.2}$$

Intuition

Read (3.2) out loud: *the derivative of an exponential is a constant multiple of itself.* Whatever the base, the rate of growth of a^x is proportional to its current height – exactly the "self-proportional" behavior we said exponentials embody. The constant of proportionality $M(a)$ is the only thing that changes from base to base. And $M(a)$ has a beautifully simple meaning: setting $x = 0$ in (3.2) gives $f'(0) = M(a) \cdot a^0 = M(a)$. So $M(a)$ is simply *the slope of a^x as it crosses the y-axis at the point $(0, 1)$.*

So every exponential is its own derivative *up to a constant factor* $M(a)$, the starting slope. And $M(a)$ is itself a limit of secant slopes through $(0,1)$: the quotient $(a^h-1)/h$ is the slope of the line joining $(0,1)$ to the nearby point (h,a^h) on the curve. As h shrinks, those secants pivot down onto the tangent at $(0,1)$, and their slopes settle on $M(a)$. Figure 3.3 shows the pivot in action.

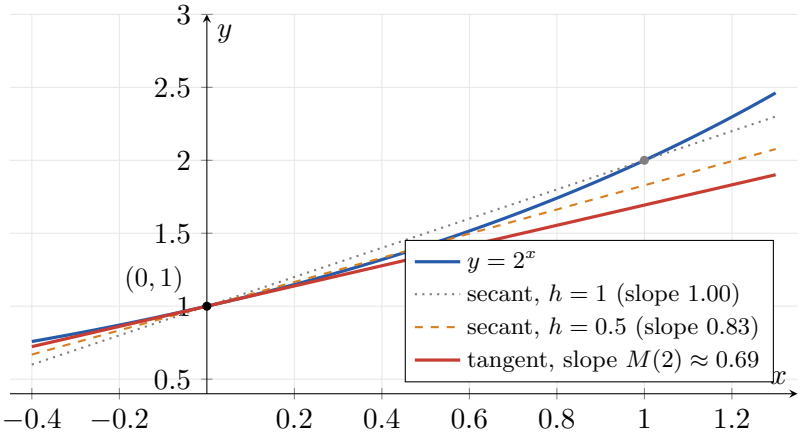


Figure 3.3: The starting slope $M(a)$ as a limit of secants, shown for $a = 2$. Each secant joins $(0,1)$ to a point $(h, 2^h)$ on the curve; as $h \to 0$ the secants pivot down onto the tangent at $(0,1)$, whose slope is $M(2) \approx 0.69$. The number $M(a)$ is nothing but the slope this pivoting line lands on.

This raises an irresistible question.

Focus

The central question of this chapter.

Is there a base a for which the fudge factor is exactly 1?

That is, a base whose curve a^x crosses $(0,1)$ at slope exactly 1, so that $M(a) = 1$ and the derivative of a^x is a^x *on the nose*, with no leftover constant? Such a function would be its own rate of change – the purest possible expression of self-proportional growth. *That base, if it exists, is e.*

3.3.2 Cornering the base by measuring starting slopes

Let us actually measure $M(a)$ for a few bases and watch the special base reveal itself. Numerically, with a small h, $M(a) \approx (a^h-1)/h$. Taking $h = 0.0001$:

base a	2	2.5	2.7	2.718	2.8	3
starting slope $M(a)$	0.693	0.916	0.993	1.0000	1.030	1.099

Look at the trend. At base 2 the curve crosses $(0,1)$ too *shallowly* – slope 0.69, less than 1. At base 3 it crosses too *steeply* – slope 1.10, more than 1. Somewhere in between, the slope passes through exactly 1. The table says it happens right around $a = 2.718$. We have trapped the special

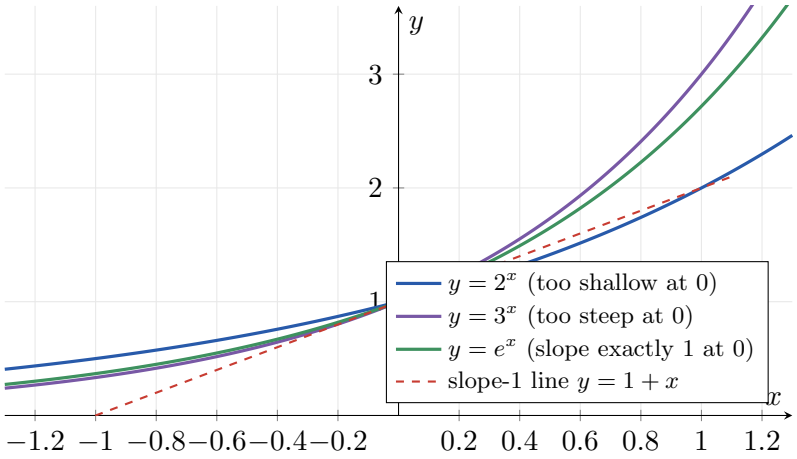


Figure 3.4: The hunt for the natural base. As x leaves 0, the curve 2^x rises too slowly and 3^x too fast. The base e is the one whose tangent at $(0, 1)$ is exactly the 45° line $y = 1 + x$ – it kisses that line rather than crossing it shallow or steep.

base between 2 and 3, and pinned it near 2.718, purely by asking the calculus question "which curve leaves $(0, 1)$ at 45°?"

This is our first sighting of e, and notice how we got it: not by decree, but by *demanding* a clean derivative. The number had to exist, because the starting slope $M(a)$ slides continuously from below 1 (at $a = 2$) to above 1 (at $a = 3$), so by the Intermediate Value Theorem it equals 1 somewhere in between. That "somewhere" is e.

Definition 3.2 (The number e, slope view)**.** e is the unique positive base for which the exponential e^x has slope 1 at $x = 0$; equivalently, the unique base making

$$\frac{\mathrm{d}}{\mathrm{d}x} e^x = e^x.$$

Up to a constant multiple, the functions that equal their own derivative are precisely Ce^x (with e^x being the unique one with $f(0) = 1$).

Why this matters

Pause on how special Definition 3.2 is. "A function that is its own derivative" sounds almost paradoxical – the curve's height *is* its steepness, everywhere, simultaneously. Picture it: wherever e^x is at height 5, it is climbing at rate 5; at height 100, climbing at rate 100. The function carries its own velocity inside its position. This is why e^x is the natural language of every process whose speed is set by its present state, and why differential equations – the mathematics of change – are written in base e and almost never in base 10.

3.4 Three Faces of e

We now have one definition of e (the slope-1 base). But e is famous precisely because it refuses to be just one thing. The same number arises from money, from slopes, and from a tidy infinite sum,

and the fact that these all agree is the real evidence that e is fundamental. We will meet *three faces* of e and show they are the same number.

3.4.1 First face: the compound-interest limit

Forget calculus for a moment and think about money. You invest \$1 at 100% annual interest. If interest is paid once, at year's end, you finish with $\$1 \times (1+1) = \$2$. Fair enough.

But suppose the bank compounds *twice* a year: it pays 50% at the half-year, and then 50% on the new, larger balance. After six months you have $1 \times (1+\frac{1}{2}) = 1.5$; after the full year, $1.5 \times (1+\frac{1}{2}) = (1+\frac{1}{2})^2 = 2.25$. By splitting the year you earned interest *on your interest*, and beat \$2.

Greed sets in. Why not compound monthly? Quarterly? Daily? Every second? If we split the year into n equal periods, each period pays interest $1/n$, and the year-end balance is

$$\left(1+\frac{1}{n}\right)^n.$$

Let us tabulate this as n grows.

compounding	n	$\left(1+\frac{1}{n}\right)^n$
yearly	1	2.00000
twice	2	2.25000
quarterly	4	2.44141
monthly	12	2.61304
daily	365	2.71457
hourly	8760	2.71813
each minute	525600	2.71828
$\to$ instant	∞	2.718281828…

Intuition

There are two opposing forces in $(1+1/n)^n$, and the suspense is which one wins. The base $1+1/n$ is shrinking toward 1, which alone would drag the whole thing down toward $1^\infty = 1$. But the exponent n is growing without bound, which alone would blow it up to infinity. They fight to a draw, and the draw is a finite number: about 2.718. Compounding more often helps, but with sharply diminishing returns – the jump from yearly to monthly is large, from monthly to "every second" almost nothing. There is a ceiling on greed, and the ceiling is e.

Definition 3.3 (The number e, compound-interest view)**.**

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n.$$

This limit really does converge – the values are increasing but bounded above (you can check $(1+1/n)^n < 3$ for all n), so by the completeness of the reals they close in on a definite limit. That limit is $e \approx 2.71828$.

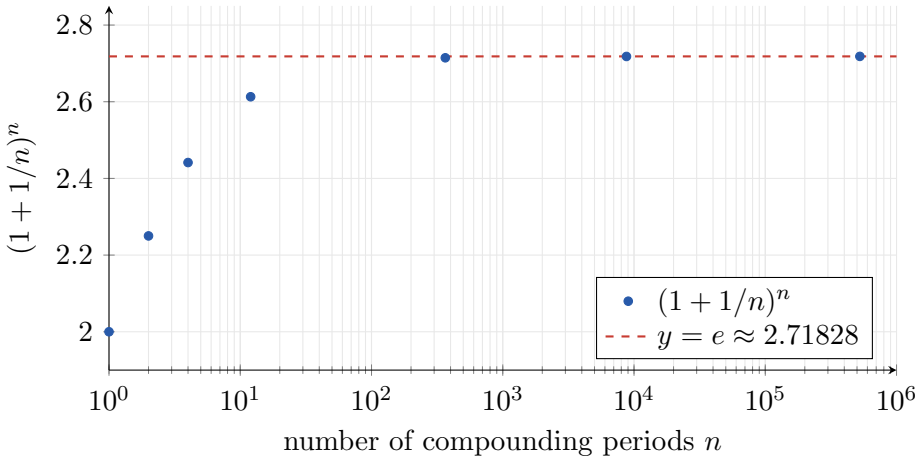


Figure 3.5: Compounding more and more finely. The balance climbs toward a hard ceiling: the horizontal line $y = e$. The horizontal axis is logarithmic, so even multiplying the compounding frequency by a thousand barely moves the dots once n is large.

Application

Continuous growth. Replace "100% for one year" with a rate r over time t, compounded n times. The balance is $\left(1 + \frac{rt}{n}\right)^n$, and as $n \to \infty$ this becomes e^{rt} (substitute $m = n/(rt)$ to recover Definition 3.3). This is the *continuous compounding* formula $A(t) = A_0 e^{rt}$, and it is the same formula that governs bacterial growth, radioactive decay (with $r < 0$), and Newton's law of cooling. The bank's idealized "interest paid every instant" is mathematically identical to a population reproducing continuously. e is what self-reinforcing growth looks like when you stop chopping time into pieces and let it flow.

3.4.2 Second face: the base whose derivative is itself

This is Definition 3.2, the slope-1 base. We have already cornered it numerically near 2.718 in Figure 3.4. The thrilling part is that it gives the *same* 2.718... as the money limit. Let us see why these two faces are secretly one.

Why the technique works

Why the slope base equals the compound-interest limit. Suppose a is the base with starting slope 1, so for tiny h,

$$\frac{a^h - 1}{h} \approx 1, \quad \text{i.e.} \quad a^h \approx 1 + h.$$

Now build up $a^1 = a$ by taking $h = 1/n$ and stacking n of these little steps. The slope-1 approximation $a^h \approx 1 + h$ with $h = 1/n$ reads $a^{1/n} \approx 1 + \frac{1}{n}$, and this gets better the larger n is (the smaller the step). Since $a = (a^{1/n})^n$,

$$a \;=\; \left(a^{1/n}\right)^n \;\approx\; \left(1 + \frac{1}{n}\right)^n.$$

As $n \to \infty$ the approximation tightens to equality, and the right-hand side is exactly Definition 3.3. So the base with slope 1 *is* $\lim_{n\to\infty}(1+1/n)^n$. The "money number" and the "slope number" were never two numbers. The fine-grained compounding $(1+1/n)^n$ is literally the process of building self-proportional growth out of tiny equal multiplicative steps – which is the very thing a derivative of 1 at the origin demands.

Work it yourself

Confirm the bridge with your own hands. Take the slope-1 condition $a^h \approx 1+h$ at $h = 0.001$, which says the special base should satisfy $a \approx (1+0.001)^{1000}$. Compute $(1.001)^{1000}$ on a calculator. You should get about 2.7169 – already closing in on 2.71828. Now do it again with $h = 0.0001$, i.e. $(1.0001)^{10000} \approx 2.7181$. Watch the digits of e assemble themselves as h shrinks. You are watching two definitions of e shake hands.

3.4.3 Third face: the infinite sum $\sum 1/n!$

The third face is the one a computer actually uses, and it is gorgeous. It says

$$e \;=\; 1+1+\frac{1}{2!}+\frac{1}{3!}+\frac{1}{4!}+\cdots \;=\; \sum_{n=0}^{\infty}\frac{1}{n!},$$

where $n! = 1\cdot 2\cdot 3\cdots n$ is the factorial and $0! = 1$ by convention. The terms shrink with breathtaking speed, because each one divides by the next integer:

$$1,\; 1,\; \tfrac{1}{2} = 0.5,\; \tfrac{1}{6} \approx 0.1667,\; \tfrac{1}{24} \approx 0.0417,\; \tfrac{1}{120} \approx 0.0083,\; \tfrac{1}{720} \approx 0.0014,\; \ldots$$

Add them up cumulatively:

terms used	2	3	4	5	6	7
partial sum	2.0000	2.5000	2.6667	2.7083	2.7167	2.7181

After only seven terms we already have 2.7181, correct to four places. Compare that with the compound-interest limit, which needed half a million periods to reach the same accuracy. *This* is why the series is the practical face: it converges with astonishing speed.

Where does the series come from? It is hiding inside the compound-interest limit. If you expand $(1+1/n)^n$ with the binomial theorem and let $n \to \infty$, each binomial term turns into a $1/k!$. Here is the mechanism.

Why the technique works

Why $(1+1/n)^n$ unfolds into $\sum 1/k!$. The binomial theorem gives

$$\left(1+\frac{1}{n}\right)^n = \sum_{k=0}^{n}\binom{n}{k}\frac{1}{n^k} = \sum_{k=0}^{n}\frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}\cdot\frac{1}{n^k}.$$

Look at the k-th term. The numerator $n(n-1)\cdots(n-k+1)$ is a product of k factors, each

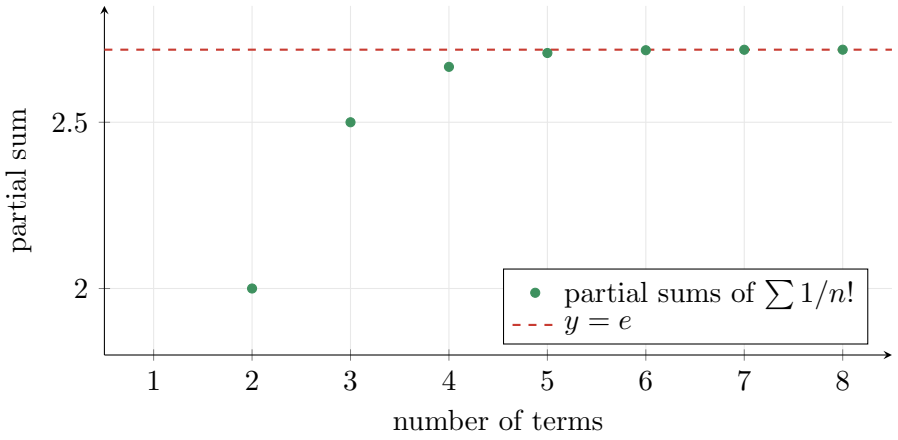


Figure 3.6: The series $\sum 1/n!$ races to e. Each new term is far smaller than the last, so the partial sums lock onto the limit after only a handful of steps – a stark contrast to the crawling convergence of $(1 + 1/n)^n$.

just a hair below n; dividing by n^k pairs them up as

$$\frac{n}{n} \cdot \frac{n-1}{n} \cdot \frac{n-2}{n} \cdots \frac{n-k+1}{n} = 1 \cdot \left(1 - \tfrac{1}{n}\right)\left(1 - \tfrac{2}{n}\right) \cdots \left(1 - \tfrac{k-1}{n}\right).$$

As $n \to \infty$ every one of those factors heads to 1, so the k-th term collapses to exactly $\dfrac{1}{k!}$. Summing over k,

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^{\infty} \frac{1}{k!}.$$

The series is not a third, separate definition pulled from nowhere; it is what the money limit *turns into* when you open it up term by term. Three faces, one number.

The series view will return with full force later in the book, when we discover that e^x itself has the series $\sum x^n/n!$ – and that this is exactly the function that equals its own derivative (differentiate term by term: $\frac{\mathrm{d}}{\mathrm{d}x}(x^n/n!) = x^{n-1}/(n-1)!$, and the sum reproduces itself, shifted). All three faces are then revealed as the same creature seen from three sides.

History

The symbol e and the number's deep role are due to Leonhard Euler, the most prolific mathematician in history, in the 1720s–1740s. The number had already surfaced – Jacob Bernoulli ran straight into the compound-interest limit around 1683 while studying exactly the interest problem above, and could only show it lay between 2 and 3 – but it was Euler who named it, computed it to many digits via the series $\sum 1/n!$, proved it irrational, and stitched it into the fabric of analysis. (The letter is probably just "e for exponential"; the charming story that Euler named it after himself is almost certainly false, since he was not one for vanity.) Euler also gave us the most beautiful equation in mathematics, $e^{i\pi} + 1 = 0$, which ties e to π, to i, and to the bare numbers 0 and 1 – but that is a story for the chapter on complex exponentials.

3.5 Why e Is Inevitable, and the Natural Log as Area

3.5.1 The natural logarithm

The logarithm to base e is so central that it gets its own name and symbol: the **natural logarithm**,

$$\ln x \;=\; \log_e x.$$

It is the inverse of e^x: $\ln(e^x) = x$ and $e^{\ln x} = x$ for $x > 0$. Everything we proved about logarithms holds, of course – $\ln(uv) = \ln u + \ln v$, and so on – but the natural log carries an extra gift inherited from its base. Because e^x is its own derivative, its inverse $\ln x$ has the cleanest possible derivative of any logarithm. We will prove later that

$$\frac{\mathrm{d}}{\mathrm{d}x}\ln x = \frac{1}{x},$$

with no messy constant out front. (For any other base, $\frac{\mathrm{d}}{\mathrm{d}x}\log_a x = \frac{1}{x\ln a}$ – the natural log is the one that washes the constant away, precisely because its base is the slope-1 base.)

3.5.2 A preview: the natural log is an area

That clean derivative $1/x$ opens a stunning back door to e, one that needs no exponents at all. Here is the idea, as a preview of the integral chapters to come.

Consider the curve $y = 1/x$ for $x > 0$ – a simple hyperbola sloping down. Define $L(x)$ to be the *area* under this curve from 1 to x:

$$L(x) = (\text{area beneath } y = 1/t \text{ from } t = 1 \text{ to } t = x).$$

This is a perfectly concrete geometric quantity: shade the region and measure it. Yet this innocent area function turns out to *be* the natural logarithm, $L(x) = \ln x$. And it secretly carries the exponent law: stretching the region horizontally by a factor reproduces $\ln(ab) = \ln a + \ln b$ as a statement about areas.

Why this matters

Sit with what just happened. We have now met e as a limit of products of money, as the base with slope 1, as a sum of reciprocal factorials, and now as the place where the area under a hyperbola hits 1. Four utterly different starting points – finance, slopes, infinite sums, geometry – and every road leads to $2.71828\ldots$. When a number is this insistent, refusing to stay confined to the problem that produced it, you are not looking at an accident of notation. You are looking at a load-bearing pillar of mathematics. *That* is why we call base e natural: it is the base the universe keeps choosing for itself.

Connection

This area-under-$1/t$ picture is more than a curiosity – it is the cleanest *logical* foundation for the whole subject. Notice the awkwardness we waded through in the opening section: to define a^x for irrational x we needed limits, and to differentiate it we needed the mysterious constant $M(a)$. The modern, airtight approach turns the whole story upside down. *First*

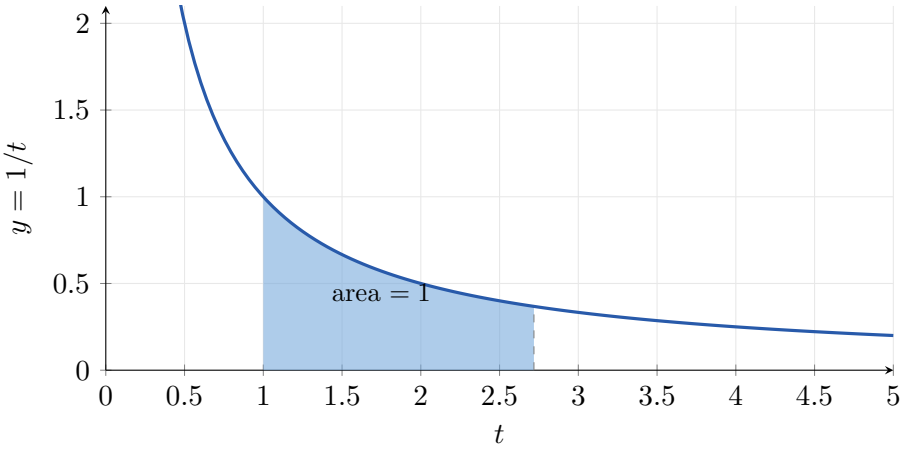


Figure 3.7: The natural logarithm as accumulated area under $y = 1/t$. The shaded region from $t = 1$ to $t = e$ has area exactly 1. This is yet another way to single out e: it is the upper limit at which the running area under $1/t$ first reaches 1, i.e. the x with $\ln x = 1$.

FACE 1 (GROWTH & FINANCE): $e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n \approx 2.7182818\ldots$ *Continuous compounding limit*

FACE 2 (CALCULUS & TANGENTS): $\frac{\mathrm{d}}{\mathrm{d}x} e^x = e^x \iff \lim_{h\to 0} \frac{e^h - 1}{h} = 1$ *Leaves* $(0, 1)$ *at* $45°$*, slope equals heigh*

FACE 3 (SERIES EXPANSION): $e = \sum_{k=0}^{\infty} \frac{1}{k!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \ldots$ *Rapid factorial convergence*

FACE 4 (GEOMETRIC ACCUMULATION): $\int_1^e \frac{1}{t}\,\mathrm{d}t = 1 \iff \ln(e) = 1$ *Hyperbolic area under* $1/t$ *reaching unit*

Figure 3.8: Four distinct doors opening onto the exact same mathematical constant $e \approx 2.71828\ldots$. Whether arrived at through continuous compound interest, the requirement that an exponential equal its own derivative, the binomial expansion into reciprocal factorials, or the natural accumulation under $y = 1/t$, the identical universal number appears.

define $\ln x$ as the area $\int_1^x \frac{1}{t}\,dt$ – which needs only integration, no irrational exponents at all. *Then* define e^x as its inverse, and *finally* define $a^x = e^{x \ln a}$. Every property we fought for – the exponent law, the derivative rule, the value of e – drops out as a clean theorem. We will walk that road in the chapters on the integral and the Fundamental Theorem. For now, the hyperbola in Figure 3.7 is quietly holding up everything in this chapter.

3.6 Worked Examples

Example 3.1 (Reading the exponent law backward)**.** Simplify $\dfrac{e^3 \cdot e^x}{e^{x-2}}$.

Solution. Combine the numerator with the law $e^a e^b = e^{a+b}$: the top is e^{3+x}. Dividing by e^{x-2} subtracts exponents:

$$\frac{e^{3+x}}{e^{x-2}} = e^{(3+x)-(x-2)} = e^{3+x-x+2} = e^5.$$

The x's cancel entirely. Whenever you see a ratio of exponentials with the same base, do not panic – just add and subtract the exponents.

Example 3.2 (Logs turn an equation inside out)**.** Solve $5 \cdot 2^x = 320$ for x.

Solution. First isolate the exponential: $2^x = 64$. Now ask the logarithm's question, "2 to the what is 64?" Since $64 = 2^6$, we read off $x = 6$. If the right-hand side were not a clean power of 2, we would take ln of both sides:

$$x \ln 2 = \ln 64 \quad \Longrightarrow \quad x = \frac{\ln 64}{\ln 2} = \log_2 64 = 6.$$

The logarithm is exactly the tool that brings the unknown *down* from the exponent where you cannot reach it.

Example 3.3 (Continuous vs. annual compounding)**.** You invest \$1000 at 6% per year for 10 years. Compare yearly compounding with continuous compounding.

Solution. Yearly: $A = 1000(1 + 0.06)^{10} = 1000(1.06)^{10} \approx 1000(1.7908) = \$1790.85$. Continuous: $A = 1000\, e^{0.06 \cdot 10} = 1000\, e^{0.6} \approx 1000(1.8221) = \$1822.12$. Compounding "every instant" instead of once a year earns an extra \$31 over the decade – real money, but a far cry from doubling. The gap is the diminishing return we saw in Figure 3.5: most of the benefit of compounding is captured already by compounding a few times a year.

Example 3.4 (How many terms of the series?)**.** Estimate e using the series through the $1/4!$ term, and bound the error.

Solution. The partial sum is

$$1 + 1 + \tfrac{1}{2} + \tfrac{1}{6} + \tfrac{1}{24} = 2 + 0.5 + 0.16667 + 0.04167 = 2.70833.$$

The true value is 2.71828, so the error is about 0.00995. The first omitted term is $1/5! = 1/120 \approx 0.00833$, and indeed the total error is just a touch more than that one term – because every *later* term is far smaller still. This is the hallmark of a fast series: the error is essentially the size of the first thing you dropped.

Example 3.5 (Estimating the starting slope of 3^x)**.** Estimate $M(3) = \lim_{h\to 0}(3^h-1)/h$ numerically, and say what it means.

Solution. With $h = 0.001$: $3^{0.001} = e^{0.001\ln 3} \approx 1+0.001(1.0986) = 1.0010986$, so $(3^{0.001}-1)/0.001 \approx 1.0986$. Thus $M(3) \approx 1.099$, matching our table. It means the curve 3^x crosses $(0,1)$ at slope 1.099 – a bit steeper than 45° – which is why 3 overshoots the magic base e. (And note $1.0986 \approx \ln 3$: the starting slope of a^x is precisely $\ln a$, a fact we will prove once we have the chain rule, since $a^x = e^{x\ln a}$.)

3.7 Exercises

Exercise 3.1. Using only the exponent law $a^{x+y} = a^x a^y$ and $a^0 = 1$, prove that $a^{-x} = 1/a^x$ and that $(a^x)^y = a^{xy}$ for integer y. State clearly where you use each law.

Exercise 3.2. Simplify without a calculator: (a) $\log_2 32$; (b) $\log_3 \frac{1}{27}$; (c) $\ln(e^4)$; (d) $e^{\ln 7}$; (e) $\log_5 1$. For each, say in one sentence what question the logarithm is answering.

Exercise 3.3. Solve for x: (a) $3^x = 81$; (b) $4\cdot 10^x = 4000$; (c) $e^{2x} = 5$ (leave the answer in terms of ln); (d) $\ln(x-1) = 2$.

Exercise 3.4. A culture of bacteria doubles every 3 hours and starts at 500 cells. (a) Write the population as $N(t) = 500\cdot 2^{t/3}$ and explain why the exponent is $t/3$. (b) How many cells after 12 hours? (c) Rewrite the model in the form $N(t) = 500\,e^{kt}$ by finding k, and interpret k as a continuous growth rate.

Exercise 3.5. Compute $(1+1/n)^n$ by hand for $n = 1, 2, 3, 4$ and confirm the sequence is increasing. Then compute $(1+1/n)^{n+1}$ for the same n and confirm *that* sequence is decreasing. (These two sequences squeeze e from below and above – a clean way to see the limit exists.)

Exercise 3.6. Using the series $e = \sum_{n=0}^{\infty} 1/n!$, compute the partial sum through $1/6!$ and compare with 2.718282. How many decimal places do you get right, and how does the error compare to the first omitted term $1/7!$?

Exercise 3.7 (Conceptual)**.** Explain in a short paragraph, without formulas, why a function whose derivative equals itself must grow exponentially. (Hint: at every height its steepness equals its height, so the bigger it gets, the faster it climbs – describe the runaway feedback.)

Exercise 3.8 (Conceptual)**.** Why is base e called "natural" while base 10 is merely convenient for humans? Give two distinct reasons drawn from this chapter – one about derivatives and one about the area under $1/t$.

Exercise 3.9. Show, using the change-of-base formula, that the graphs of $\log_2 x$, $\ln x$, and $\log_{10} x$ are all vertical rescalings of one another. By what factor must you stretch $\ln x$ to obtain $\log_{10} x$?

Exercise 3.10 (Proof)**.** Prove that the starting slope $M(a) = \lim_{h\to 0}(a^h - 1)/h$ satisfies $M(a^2) = 2M(a)$, and more generally $M(a^k) = k\,M(a)$ for positive integers k. (Hint: $a^{2h}-1 = (a^h-1)(a^h+1)$, and $a^h \to 1$ as $h \to 0$.) Explain why this is consistent with the claim that $M(a) = \ln a$.

Exercise 3.11 (Challenge)**.** The number e is irrational. Here is Fourier's argument. Suppose for contradiction $e = p/q$ for integers p, q. Multiply the series $e = \sum 1/n!$ by $q!$. Argue that $q!\,e$ would have to be an integer, but that the tail $\sum_{n=q+1}^{\infty} q!/n!$ is a positive number strictly less than 1 (compare it with a geometric series). Conclude that no such p, q exist. Fill in the comparison that bounds the tail below 1.

Chapter 4

Trigonometric and Inverse Functions

The functions in this chapter were not invented by people doing calculus. They were invented thousands of years before, by astronomers tracking the rotation of the heavens and surveyors dividing up land. Sine and cosine are the mathematics of *going around*: of anything that repeats, oscillates, rotates, or vibrates. A swinging pendulum, an alternating current, a sound wave, the seasons, the orbit of a planet, the position of a point on a spinning wheel – all of them, when you measure them carefully, turn out to be made of sines and cosines.

So before we can do calculus *with* these functions – before we can differentiate $\sin x$ or integrate $\cos x$ – we need to really understand what they *are*. Not as buttons on a calculator, and not as ratios in a right triangle (though we will see that picture too), but as something deeper and more useful: the coordinates of a point traveling around a circle. That single picture will explain every identity, every graph, every symmetry, and it will explain why we measure angles in those natural units called *radians*, and why doing so is the secret that makes calculus with these functions come out clean.

We will also meet the *inverse* trig functions, the functions that answer the question "what angle has this sine?" Inverses are subtle – a periodic function repeats, so it cannot be undone without care – and sorting out exactly how to undo a sine teaches a lesson about inverse functions in general that will pay off later.

Let us build it all from the circle up.

4.1 Angles: What Are We Really Measuring?

Start with the most basic question. An angle measures *how much you have turned*. Stand facing north, rotate to face east: you have turned through one quarter of a full turn. Rotate to face south: half a turn. The natural, dimensionless way to talk about turning is as a *fraction of a full revolution*.

Everything after that is just choosing a *unit* – a name for "one full turn" – and the choice is pure convention. The Babylonians, working in base 60 and tracking a sky that seemed to cycle once a year through roughly 360 days, declared one full turn to be 360 *degrees*. There is nothing sacred about 360; it is a culturally inherited number that happens to divide nicely ($360 = 2^3 \cdot 3^2 \cdot 5$, so halves, thirds, quarters, fifths, sixths, eighths, ninths, tenths, twelfths all come out whole). It is a fine unit for carpentry and navigation. It is a *terrible* unit for calculus, and by the end of this chapter you will feel in your bones exactly why.

Intuition

An angle is not a length and not an area. It is a *turn* – a fraction of a full rotation. Degrees and radians are just two different rulers laid against that same turn. Degrees call a full turn 360; radians call it 2π. Same turn, different number. The essential step is choosing the ruler that makes the rest of mathematics simplest, and that ruler is radians.

4.1.1 Radians: Measuring an Angle by the Arc It Cuts

Here is a more natural idea, one that uses no arbitrary number at all. Draw a circle of radius r and an angle at its center. That angle opens up onto an *arc* – a piece of the circle's edge. The bigger the angle, the longer the arc. In fact the arc length is exactly proportional to the angle. So why not let the arc *be* the measure of the angle?

There is one wrinkle: a bigger circle gives a longer arc for the same angle, so raw arc length depends on r. We fix that by dividing out the radius. Define:

$$\theta = \frac{\text{arc length}}{\text{radius}} = \frac{s}{r}.$$

This ratio is the same for every circle, because doubling r doubles s. The number you get is the angle measured in **radians**.

Definition 4.1 (Radian)**.** The radian measure of a central angle is the length of the arc it subtends divided by the radius of the circle: $\theta = s/r$. Equivalently, an angle of one radian subtends an arc whose length equals the radius.

Because $s = r\theta$, an angle of one radian is the angle for which the arc is *exactly as long as the radius*. Wrap one radius-length of string along the rim of the circle, and the angle it spans at the center is one radian – a bit under 60° (57.2958…°, to be precise).

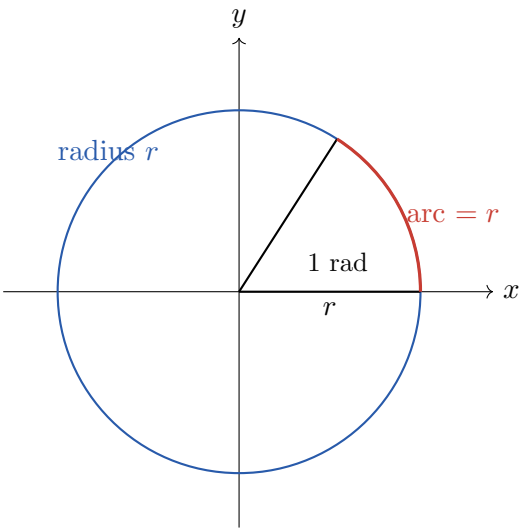


Figure 4.1: One radian: the central angle whose arc length equals the radius. Since the full circumference is $2\pi r$, a full turn is 2π radians.

How many radians in a full turn? The whole circumference has length $2\pi r$, so a full revolution measures $s/r = 2\pi r/r = 2\pi$ radians. That gives us the master conversion:

$$2\pi \text{ radians } = 360^\circ, \qquad \text{so} \qquad \pi \text{ radians} = 180^\circ, \qquad 1 \text{ rad} = \frac{180^\circ}{\pi}.$$

A handful of values worth knowing cold:

$$30^\circ = \tfrac{\pi}{6}, \quad 45^\circ = \tfrac{\pi}{4}, \quad 60^\circ = \tfrac{\pi}{3}, \quad 90^\circ = \tfrac{\pi}{2}, \quad 180^\circ = \pi, \quad 360^\circ = 2\pi.$$

What is really happening

When you write "$\theta = \pi/3$" you are not really writing an abstract angle – you are writing a *length*. You are saying: on the unit circle (radius 1), walk a distance of $\pi/3 \approx 1.047$ along the rim. The angle and the arc length are literally the same number when $r = 1$. This is why, throughout calculus, the variable inside sin and cos feels like a distance: in radians on the unit circle, it *is* one. Degrees hide this by inserting the arbitrary scaling factor $\pi/180$, which is exactly the factor that will later clutter up every derivative if you let it.

Two immediate payoffs of the $s = r\theta$ definition, both free:

Arc length. The length of an arc of radius r spanning angle θ (in radians) is simply $s = r\theta$. No factor of $\pi/180$, no fuss.

Sector area. A full disk has area πr^2 and spans 2π radians. A sector of angle θ is the fraction $\theta/(2\pi)$ of the disk, so its area is

$$A = \frac{\theta}{2\pi} \cdot \pi r^2 = \tfrac{1}{2} r^2 \theta.$$

Hold onto that formula $A = \frac{1}{2}r^2\theta$; we will use it in a clever sandwich argument later in the book to pin down the most important limit in trigonometry.

Walking question

A bicycle wheel has radius 0.35 m. As you ride, the wheel turns. Convince yourself that the distance the bike travels equals the arc length traced on the rim, and hence that if the wheel turns through θ radians the bike moves $0.35\,\theta$ meters. Now feel why "radians" is the natural rotational unit for anything that rolls: the angle, scaled by the radius, *is* the distance. What ugly conversion factor would appear if you insisted on degrees?

4.2 Sine and Cosine as Coordinates on the Unit Circle

Now the central idea of the whole chapter. Forget triangles for a moment. Draw the **unit circle**: the circle of radius 1 centered at the origin, the set of points with $x^2 + y^2 = 1$.

Start at the point $(1, 0)$ – three o'clock – and walk counterclockwise along the rim. After walking an arc-length t (equivalently, after turning through an angle of t radians, since $r = 1$), you arrive at some point on the circle. *That point's coordinates are the definition of cosine and sine.*

Definition 4.2 (Sine and cosine, circle definition)**.** Let t be a real number. Travel counterclockwise a distance t along the unit circle, starting from $(1, 0)$. The coordinates of the point you reach are

denoted $(\cos t,\ \sin t)$. That is, $\cos t$ is the horizontal coordinate and $\sin t$ is the vertical coordinate of the point at arc-length t on the unit circle.

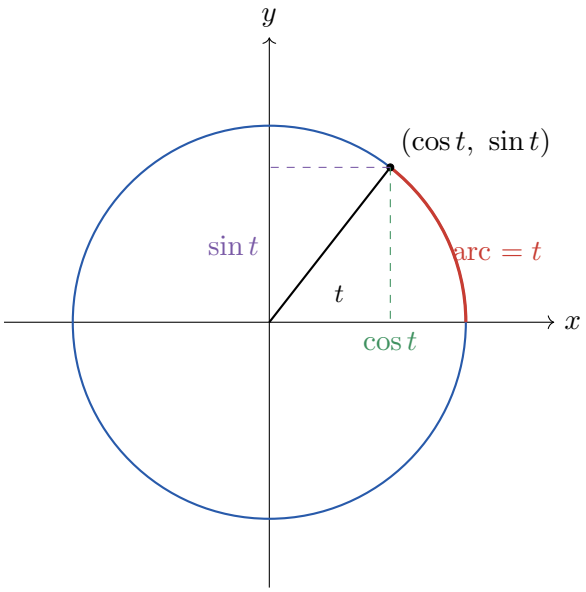


Figure 4.2: Cosine and sine are the coordinates of the point reached by walking arc-length t counterclockwise around the unit circle from $(1, 0)$. Cosine is horizontal, sine is vertical.

This definition improves upon the elementary "opposite over hypotenuse" formula in three fundamental ways. It applies to *every* real number t, rather than only acute angles between 0 and 90°: you can keep walking past the top of the circle, past the left side, all the way around and around. It makes the periodic symmetries transparent, as the next section shows. And it ties sin and cos directly to motion along a curve, which is exactly what derivatives are about – the payoff that matters for calculus.

The connection to the right-triangle picture is immediate. Look at Figure 4.2: the radius to the point, the dashed vertical segment, and the segment along the x-axis form a right triangle with hypotenuse 1. The horizontal leg is $\cos t$, the vertical leg is $\sin t$. So the familiar SOH-CAH-TOA ratios are just this picture scaled: in a right triangle with hypotenuse h, the side opposite angle t is $h \sin t$ and the side adjacent is $h \cos t$. The unit circle is the master triangle, hypotenuse normalized to 1.

Intuition

Picture a point riding counterclockwise around the unit circle at steady speed. Its *shadow on the x-axis* – the horizontal coordinate – traces out $\cos t$ as t increases. Its *shadow on the y-axis* traces out $\sin t$. Both shadows slosh back and forth between -1 and 1 forever. Sine and cosine are nothing more than the two shadows of uniform circular motion. Every wave you will ever meet is a shadow of something going around.

4.2.1 Everything That Follows Immediately

From this one definition, a cascade of facts falls out with almost no work. This is the sign of a good definition.

The Pythagorean identity. The point $(\cos t, \sin t)$ lies *on* the unit circle, and the unit circle is the set with $x^2 + y^2 = 1$. Therefore, for every real t,

$$\boxed{\cos^2 t + \sin^2 t = 1.}$$

This is not a theorem you prove with effort; it *is* the equation of the circle, with x renamed $\cos t$ and y renamed $\sin t$. (The notation $\cos^2 t$ is shorthand for $(\cos t)^2$, an old convention you simply have to memorize.)

Range. Both coordinates of a point on the unit circle lie between -1 and 1. So $-1 \le \sin t \le 1$ and $-1 \le \cos t \le 1$ for all t. The functions never escape that band.

Periodicity. Walking an extra full lap – an extra 2π of arc – brings you back to the same point. Hence

$$\sin(t + 2\pi) = \sin t, \qquad \cos(t + 2\pi) = \cos t.$$

The functions repeat with **period** 2π. This is why they model anything cyclic.

Special values. Read them straight off the circle. At $t = 0$ you are at $(1, 0)$, so $\cos 0 = 1,\ \sin 0 = 0$. A quarter turn $t = \pi/2$ lands you at the top $(0, 1)$: $\cos(\pi/2) = 0,\ \sin(\pi/2) = 1$. Half a turn $t = \pi$ puts you at $(-1, 0)$. Three-quarters $t = 3\pi/2$ at the bottom $(0, -1)$.

t	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin t$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos t$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0

Notice the elegant pattern: the sine row reads $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$, and the cosine row is the same thing backwards. The values at $\pi/6, \pi/4, \pi/3$ come from the two special triangles – the 30-60-90 and the 45-45-90 – which we will derive as an example below.

Work it yourself

Let us nail down $\sin(\pi/4)$ and $\cos(\pi/4)$ from the circle. At $t = \pi/4$ (a 45° turn) you are exactly on the diagonal line $y = x$, by symmetry: the point is equidistant from the two axes. So $\cos(\pi/4) = \sin(\pi/4)$; call the common value c. Pythagoras says $c^2 + c^2 = 1$, hence $2c^2 = 1$, so $c = 1/\sqrt{2} = \sqrt{2}/2 \approx 0.707$. You just computed a "table value" with nothing but the circle and Pythagoras – no calculator, no memorization.

4.3 The Other Four: Tangent and Friends

Sine and cosine are the parents; the remaining four trig functions are built from them by division. The most important is the **tangent**:

$$\tan t \;=\; \frac{\sin t}{\cos t}.$$

Geometrically, $\tan t$ is the *slope* of the radius line to the point $(\cos t, \sin t)$ – it is rise over run, $\sin t/\cos t$. It is also, beautifully, the length cut off on the vertical tangent line to the circle at $(1, 0)$, which is where the name comes from (*tangent* = touching). The remaining three are reciprocals, named for historical convenience:

$$\cot t = \frac{\cos t}{\sin t} = \frac{1}{\tan t}, \qquad \sec t = \frac{1}{\cos t}, \qquad \csc t = \frac{1}{\sin t}.$$

Each is undefined wherever its denominator hits zero. Tangent and secant blow up where $\cos t = 0$, i.e. at $t = \pm\pi/2, \pm 3\pi/2, \ldots$; cotangent and cosecant blow up where $\sin t = 0$, i.e. at $t = 0, \pm\pi, \pm 2\pi, \ldots$ Dividing the Pythagorean identity $\cos^2 + \sin^2 = 1$ through by $\cos^2 t$ gives a useful cousin:

$$1 + \tan^2 t = \sec^2 t,$$

and dividing instead by $\sin^2 t$ gives $1 + \cot^2 t = \csc^2 t$. These are the same circle equation wearing different clothes.

Common trap

$\tan t$ has period π, *not* 2π. After half a lap the point $(\cos t, \sin t)$ flips to its antipode $(-\cos t, -\sin t)$, and the two sign flips cancel in the ratio $\sin t/\cos t$. So $\tan(t + \pi) = \tan t$. Students who blindly assume every trig function repeats every 2π get tangent wrong. Always reason from the circle, not from a memorized rule.

4.4 Symmetries: Reading Identities Off the Circle

A common student fear is that trigonometry is a swamp of identities to memorize. It is not. Almost every identity is a geometric symmetry of the circle, *seen* rather than recalled. Here are the main ones, each as a picture.

Even and odd. Replace t by $-t$: instead of walking counterclockwise you walk clockwise the same distance. You land at the mirror image across the x-axis. Mirroring across the x-axis keeps the horizontal coordinate and flips the vertical one:

$$\cos(-t) = \cos t \quad \text{(even)}, \qquad \sin(-t) = -\sin t \quad \text{(odd)}.$$

Cosine is symmetric, sine is antisymmetric. One picture, two identities.

Supplementary angle ($\pi - t$). The point at angle $\pi - t$ is the mirror image of the point at angle t across the *y-axis*. Mirroring across the y-axis flips x and keeps y:

$$\cos(\pi - t) = -\cos t, \qquad \sin(\pi - t) = \sin t.$$

Co-function / quarter turn ($\frac{\pi}{2} - t$). This is the deep one, and it is the source of the prefix "co-" in cosine. The point at angle $\frac{\pi}{2} - t$ is the reflection of the point at angle t across the diagonal line $y = x$. Reflecting across $y = x$ *swaps* the coordinates. Therefore

$$\sin\left(\tfrac{\pi}{2} - t\right) = \cos t, \qquad \cos\left(\tfrac{\pi}{2} - t\right) = \sin t.$$

Cosine of an angle is the sine of its complement – "cosine" literally means "**co**mplement's **sine**." Rather than a separate rule to memorize, it is the $y = x$ reflection, seen once and never forgotten.

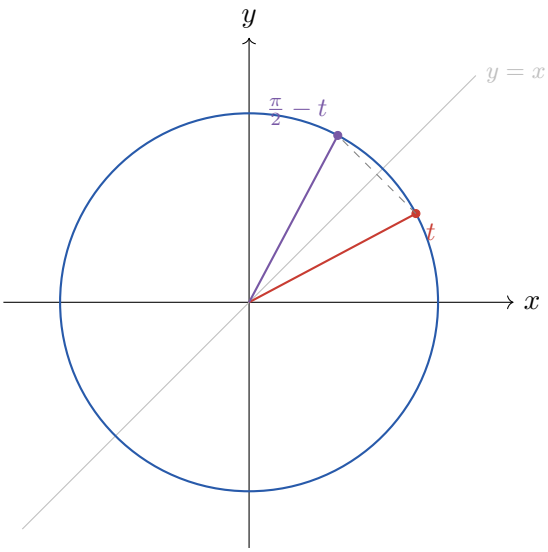


Figure 4.3: The co-function identity is a mirror. Reflecting the point for angle t across the line $y = x$ gives the point for angle $\frac{\pi}{2} - t$, and that reflection swaps the coordinates: sin and cos trade places.

Why the technique works

Why can we get away with *reading* identities off pictures instead of grinding algebra? Because sin and cos were *defined* as coordinates of a point on the circle, and a geometric transformation of the circle (a reflection, a rotation) does a predictable thing to coordinates. Reflect across the x-axis: $y \mapsto -y$, so $\sin \mapsto -\sin$. Rotate by a quarter turn, reflect across a diagonal – each motion is a simple coordinate operation. So every symmetry of the circle is automatically an identity, and the "identity" is just the bookkeeping of where the coordinates went. The algebra works because the geometry works; the picture is the proof.

4.4.1 The Addition Formulas

The one family of identities you cannot quite *see* in a single glance, but which everything downstream depends on, is the addition formulas:

$$\cos(a + b) = \cos a \cos b - \sin a \sin b, \qquad \sin(a + b) = \sin a \cos b + \cos a \sin b.$$

These say: if you know where angles a and b land, you can find where angle $a + b$ lands without walking the whole way. They are the engine behind the derivative of sine, so they earn a real derivation. The cleanest one is a rotation argument.

Focus

Deriving the addition formula by rotation. Rotating the plane counterclockwise by an angle a sends the point $(1,0)$ to $(\cos a, \sin a)$ and sends $(0,1)$ to $(-\sin a, \cos a)$ (a quarter turn ahead of the first). A rotation is linear, so it is completely determined by what it does to these two basis points: the point (x,y) goes to

$$(x\cos a - y\sin a, \quad x\sin a + y\cos a).$$

Now take the point at angle b, namely $(x,y) = (\cos b, \sin b)$, and rotate it by a. Geometrically that lands it at angle $a+b$, i.e. at $(\cos(a+b), \sin(a+b))$. Algebraically the formula above sends it to

$$(\cos b\cos a - \sin b\sin a, \quad \cos b\sin a + \sin b\cos a).$$

Matching the two coordinates gives exactly the cosine and sine addition formulas. The single idea "rotating by a then by b is the same as rotating by $a+b$" contains both identities at once.

Two consequences we will lean on. Setting $b = a$ gives the **double-angle** formulas:

$$\sin 2a = 2\sin a\cos a, \qquad \cos 2a = \cos^2 a - \sin^2 a = 1 - 2\sin^2 a = 2\cos^2 a - 1.$$

The last two forms (using $\cos^2 + \sin^2 = 1$) rearrange into the **half-angle** formulas, which are workhorses in integration:

$$\sin^2 a = \frac{1-\cos 2a}{2}, \qquad \cos^2 a = \frac{1+\cos 2a}{2}.$$

Keep these last two in your pocket; when you later integrate $\sin^2$ or $\cos^2$, these are how you do it.

History

The addition formulas are ancient. In effect they appear in Ptolemy's *Almagest* (c. 150 CE) as "Ptolemy's theorem" about chords of a circle – Greek astronomers worked not with our sine but with the *chord* of an angle, the straight-line distance across the arc, which equals $2\sin(\theta/2)$ on a unit circle. The word "sine" itself is a chain of mistranslations: the Sanskrit *jya* (chord/bowstring) became the Arabic *jiba*, written without vowels as *jb*; later Latin translators misread *jb* as *jaib*, meaning "bay" or "fold," and rendered it *sinus* – Latin for a bay or the fold of a garment. So every time you write sin, you are writing the Latin for "bay," by way of a twelfth-century reading error. The mathematics is older and clearer than its name.

4.5 The Graphs: Unrolling the Circle

We have been watching a point go around a circle. To graph $\sin t$ and $\cos t$ as functions of t, we *unroll* that circular motion: put t on the horizontal axis (how far around we have walked) and the coordinate on the vertical axis. The circle's repetition becomes a wave that rolls on forever.

Read the structure straight off the graph. The wave oscillates between -1 and 1 (the **amplitude** is 1). It completes one full cycle every 2π (the **period**). Sine starts at 0 climbing; cosine starts at 1 already at its peak. And the two curves are *the same shape shifted*: $\cos t = \sin(t + \frac{\pi}{2})$. Shifting sine to the left by a quarter period lands it exactly on cosine. (This is the co-function identity in motion:

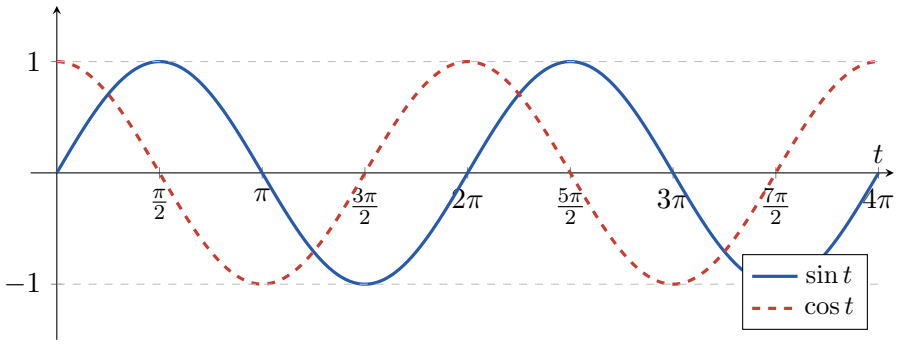


Figure 4.4: Sine and cosine, the unrolled shadows of circular motion. Each is a wave of amplitude 1 and period 2π. Cosine is sine shifted left by $\pi/2$ – the co-function identity, made visual.

using cos even, $\sin(t + \frac{\pi}{2}) = \sin(\frac{\pi}{2} - (-t)) = \cos(-t) = \cos t$.) More simply still: the cosine wave is the sine wave with a head start of a quarter turn, which is precisely what a quarter turn around the circle does.

Connection

A general sinusoid is written $y = A\sin(\omega t + \varphi) + k$. Each parameter is a geometric knob: A is the amplitude (vertical stretch), ω is the angular frequency (horizontal compression – the period is $2\pi/\omega$), φ is the phase (horizontal shift, the head start), and k is the vertical offset (the midline). Because sine and cosine differ only by a phase, *any* sinusoid can be written with either one. This four-knob family is exactly what describes simple harmonic motion in physics – a mass on a spring, $x(t) = A\cos(\omega t + \varphi)$ – and it is the building block of Fourier analysis, where astonishingly *every* periodic signal, however jagged, is a sum of these smooth waves.

The tangent graph looks entirely different, and the circle tells us why. As t climbs toward $\pi/2$, the point nears the top of the circle: $\sin t \to 1$ while $\cos t \to 0^+$, so $\tan t = \sin t/\cos t \to +\infty$. Just past $\pi/2$, cosine is negative and tangent leaps to $-\infty$. The result is a curve that races up to a vertical asymptote, vanishes off the top, and reappears from the bottom – repeating every π, not 2π.

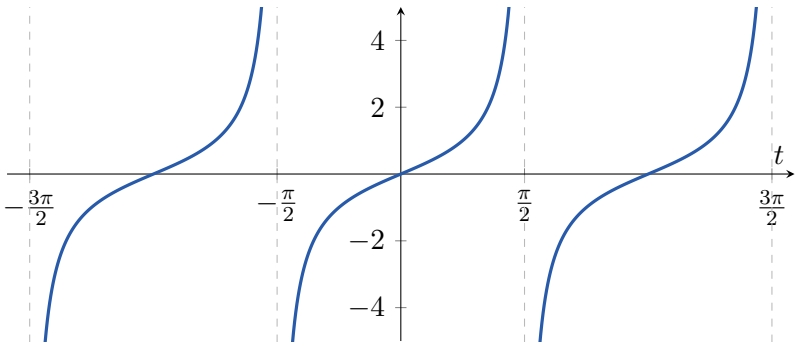


Figure 4.5: The tangent function $\tan t = \sin t/\cos t$. Vertical asymptotes occur where $\cos t = 0$. The function runs from $-\infty$ to $+\infty$ on each interval of width π, then repeats.

4.6 The Small-Angle Behavior: $\sin t \approx t$

This section hands the rest of the book its single most important fact about trig. Look again at the unit circle for a *small* angle t, just above $(1, 0)$.

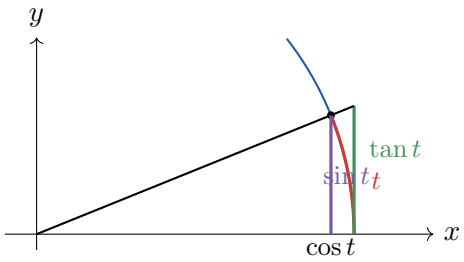


Figure 4.6: For a small angle t, the vertical segment $\sin t$, the arc t, and the tangent segment $\tan t$ are nearly equal, with $\sin t < t < \tan t$. As $t \to 0$ the three become indistinguishable, which forces $\sin t/t \to 1$.

Three lengths are racing to zero together as $t \to 0$: the vertical drop $\sin t$, the arc t along the rim, and the tangent segment $\tan t$. Geometrically they nest:

$$\sin t \;<\; t \;<\; \tan t \qquad (0 < t < \tfrac{\pi}{2}).$$

The first inequality says the straight vertical chord is shorter than the curved arc above it; the second compares two triangles against the sector. As the angle shrinks, the arc straightens out and these three lengths become indistinguishable. The most useful way to say this precisely is in terms of the ratio $\sin t/t$.

Focus

The fundamental trig limit.

$$\boxed{\lim_{t \to 0} \frac{\sin t}{t} = 1.}$$

In words: for small t (*in radians*), $\sin t$ and t are almost equal – their ratio rushes to 1. Equivalently, $\sin t \approx t$ when t is small – Figure 4.7 shows the curve and the line clinging together near the origin. This is the seed from which the entire calculus of trig functions grows; the derivative of sine, as you will see in the next chapter, is exactly this limit in disguise.

We can already see *why* it must be true, and even sketch the rigorous proof, using only the sector-area formula $A = \frac{1}{2}r^2\theta$ from the start of the chapter.

Focus

Why $\sin t/t \to 1$: the squeeze. Work on the unit circle ($r = 1$) with $0 < t < \pi/2$. Compare three areas in Figure 4.6:

- the inner triangle with vertices $(0, 0), (1, 0), (\cos t, \sin t)$ has area $\frac{1}{2}\sin t$;
- the circular sector of angle t has area $\frac{1}{2}t$ (using $A = \frac{1}{2}r^2\theta$ with $r = 1$);
- the outer triangle with the vertical tangent, vertices $(0, 0), (1, 0), (1, \tan t)$, has area

$\frac{1}{2}\tan t$.

The inner triangle sits inside the sector, which sits inside the outer triangle, so

$$\tfrac{1}{2}\sin t \;<\; \tfrac{1}{2}t \;<\; \tfrac{1}{2}\tan t.$$

Multiply by 2 and divide by $\sin t$ (positive here):

$$1 \;<\; \frac{t}{\sin t} \;<\; \frac{1}{\cos t}.$$

Flip everything (taking reciprocals reverses the inequalities):

$$\cos t \;<\; \frac{\sin t}{t} \;<\; 1.$$

As $t \to 0$, $\cos t \to 1$. The ratio $\sin t/t$ is trapped – *squeezed* – between something heading to 1 and the constant 1. It has no choice: $\lim_{t\to 0} \sin t/t = 1$. (The case $t < 0$ follows because $\sin t/t$ is even.)

Common trap

This limit equals 1 **only in radians**. If t is measured in degrees, then walking "t degrees" covers an arc of only $\frac{\pi}{180}t$, so $\sin(t^\circ)/t \to \pi/180 \approx 0.01745$, a needless ugly constant. Every derivative formula for trig functions would be polluted by that factor forever. *This is the real reason calculus uses radians.* Radians are the unit in which arc length equals angle, which is the unit in which $\sin t \approx t$, which is the unit in which the derivatives come out clean.

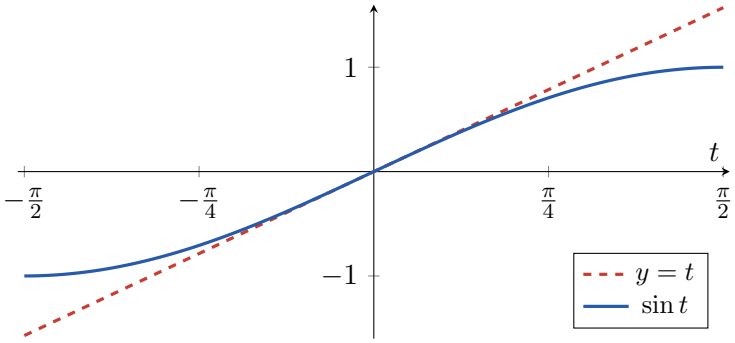


Figure 4.7: Near $t = 0$ the curve $\sin t$ hugs the line $y = t$: they share the same value and the same slope 1 at the origin, and only peel apart slowly. This is the visual content of $\sin t/t \to 1$, equivalently $\sin t \approx t$ for small t.

Why this matters

Pause on how much rides on $\sin t/t \to 1$. It is the statement that, up close, a sine wave and a straight line of slope 1 are the same thing – that the circle, examined under a microscope at $(1, 0)$, looks flat and vertical, rising one unit of height per one unit of arc. From this single "the curve is locally straight" fact will come $\frac{\mathrm{d}}{\mathrm{d}t}\sin t = \cos t$, the wave equation, the description of light and sound, and the Taylor series $\sin t = t - \frac{t^3}{6} + \cdots$ whose very first term is this

approximation $\sin t \approx t$. A great deal of physics is the sentence "for small angles, $\sin t \approx t$" taken seriously.

Application

The pendulum. A pendulum of length L obeys, exactly, $\ddot{\theta} = -\frac{g}{L}\sin\theta$ – a hard, nonlinear equation with no elementary solution. But for small swings, $\sin\theta \approx \theta$, and the equation *linearizes* to $\ddot{\theta} = -\frac{g}{L}\theta$, whose solution is a pure sinusoid with period $T = 2\pi\sqrt{L/g}$. That is why a pendulum clock keeps near-constant time regardless of how far it swings (as long as the swing is small): Galileo's observation, explained in one approximation. The whole of small-oscillation physics – springs, molecules, circuits, bridges – rests on replacing a curve by its tangent line, and $\sin\theta \approx \theta$ is the first and most famous instance.

A companion limit, derived the same way and needed alongside the first, is

$$\lim_{t\to 0}\frac{1-\cos t}{t} = 0.$$

Intuitively: near $t = 0$ the height $\sin t$ falls off the circle like t (first order), but the horizontal gap $1-\cos t$ falls off like $t^2/2$ (second order, much faster), so dividing it by t still leaves something heading to zero. We prove it with a clean algebraic trick – multiply by the conjugate:

$$\frac{1-\cos t}{t} = \frac{(1-\cos t)(1+\cos t)}{t(1+\cos t)} = \frac{1-\cos^2 t}{t(1+\cos t)} = \frac{\sin^2 t}{t(1+\cos t)} = \underbrace{\frac{\sin t}{t}}_{\to 1}\cdot\underbrace{\frac{\sin t}{1+\cos t}}_{\to 0/2=0}.$$

The first factor tends to 1 by our main limit; the second tends to $0/2 = 0$. Product: 0. Both limits will reappear, fused, the moment we differentiate sine.

4.7 Inverse Trigonometric Functions

We turn now from "given the angle, find the coordinate" to its reverse: "given the coordinate, find the angle." If you know that a point on the unit circle has height $\sin t = 0.5$, which t produced it? This is the job of the inverse trig functions – but to define them properly we must first confront a real obstacle.

4.7.1 The Obstacle: Periodic Functions Are Not One-to-One

A function has an inverse only if it is **one-to-one**: each output comes from exactly one input. Graphically, this is the horizontal line test – no horizontal line may cross the graph more than once, or else that output traces back to several inputs and "the inverse" is ambiguous.

Sine fails this test spectacularly. The horizontal line $y = 0.5$ slices through the sine graph infinitely often: $\sin t = 0.5$ at $t = \pi/6$, at $t = 5\pi/6$, at $t = \pi/6 + 2\pi$, and at infinitely many other places. "The angle whose sine is 0.5" has no single answer. A periodic function repeats, so by its nature it can never be globally invertible.

Intuition

The fix is blunt and a little ruthless: we *restrict* sine to a stretch of its domain on which it never repeats a value – a stretch where it is strictly increasing (or strictly decreasing) and sweeps through its full range exactly once. On that restricted piece the inverse exists and is unique. We are not inverting all of sine – that is impossible. We are inverting one carefully chosen, representative slice of it, and then living forever with the consequences of that choice.

4.7.2 Arcsine: Choosing the Principal Branch

For sine, the agreed-upon slice is the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$. Why this one? On it, sine climbs monotonically from $\sin(-\frac{\pi}{2}) = -1$ up to $\sin(\frac{\pi}{2}) = 1$, hitting every value in $[-1, 1]$ exactly once. It is the slice straddling the origin, it is symmetric, and it captures sine at its most natural – rising through zero with slope 1. This restricted, invertible sine has an inverse called the **arcsine**.

Definition 4.3 (Arcsine)**.** For $-1 \le x \le 1$, $\arcsin x$ (also written $\sin^{-1} x$) is the unique angle $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ with $\sin\theta = x$. Its domain is $[-1, 1]$ (the possible sine values) and its range is $[-\frac{\pi}{2}, \frac{\pi}{2}]$ (the chosen slice of angles). The name *arc*sine reflects exactly what it returns: the *arc* (angle) belonging to a given coordinate.

Common trap

The superscript notation $\sin^{-1} x$ is a genuine menace. It does **not** mean $1/\sin x$ – that is $\csc x$ or $(\sin x)^{-1}$. Here the "-1" marks *functional inverse*, not a reciprocal power. The clash with $\sin^2 x = (\sin x)^2$ is an unfortunate historical inconsistency in the same notation. When in doubt, write $\arcsin x$; it cannot be misread.

The graph of an inverse function is the *reflection* of the original across the line $y = x$ – because inverting a function swaps the roles of input and output, which swaps the axes. So arcsin is the restricted sine curve flipped over the diagonal.

4.7.3 Arccosine and Arctangent

The same surgery defines the other inverses; only the chosen slice differs, each picked to be a monotonic stretch covering the full range once.

Arccosine. Cosine *decreases* from 1 to -1 on $[0, \pi]$, sweeping its whole range once, so that is the slice. Define $\arccos x$ for $x \in [-1, 1]$ as the unique $\theta \in [0, \pi]$ with $\cos\theta = x$. Note its range starts at 0, not $-\frac{\pi}{2}$: a different function needs a different home interval. A pretty relation ties the two together. Since $\cos\theta = \sin(\frac{\pi}{2} - \theta)$ (the co-function identity), one can show

$$\arcsin x + \arccos x = \frac{\pi}{2} \qquad \text{for all } x \in [-1, 1].$$

The arcsine and arccosine of the same number are complementary angles – again the "co-" relationship, now read backwards through the inverses.

Arctangent. Tangent runs monotonically from $-\infty$ to $+\infty$ on the open interval $(-\frac{\pi}{2}, \frac{\pi}{2})$, so that is its slice. Define $\arctan x$ for *every* real x as the unique $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$ with $\tan\theta = x$. Because

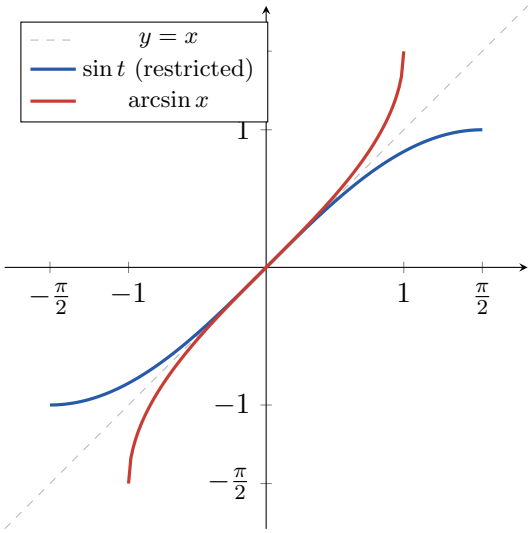


Figure 4.8: Restricted sine (on $[-\frac{\pi}{2}, \frac{\pi}{2}]$) and its mirror image across $y = x$, the arcsine. Reflecting a graph across $y = x$ is what inverting a function does: it swaps input and output.

tangent's range is all of $\mathbb{R}$, arctangent accepts every real input – a genuinely useful feature. Its graph has two horizontal asymptotes, at $y = \pm\frac{\pi}{2}$: feed in an enormous number and the angle approaches, but never reaches, a quarter turn.

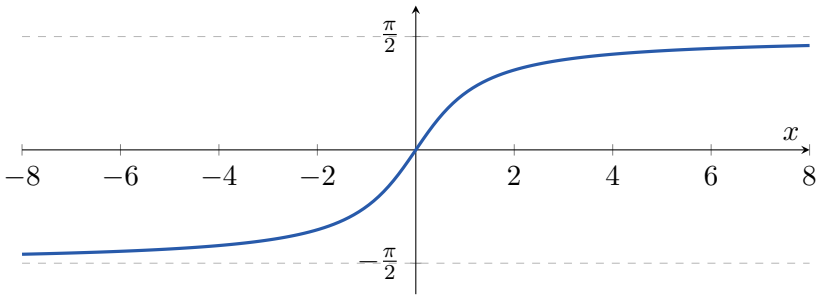


Figure 4.9: Arctangent: defined for all real x, with horizontal asymptotes at $\pm\pi/2$. It is the smooth, bounded "angle-of" function – which is why it appears everywhere a saturating or angle-recovering response is needed.

Inverse	Domain (inputs)	Range (outputs)	Chosen slice of the original
$\arcsin x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$	sine increasing through 0
$\arccos x$	$[-1, 1]$	$[0, \pi]$	cosine decreasing from 1
$\arctan x$	$(-\infty, \infty)$	$(-\frac{\pi}{2}, \frac{\pi}{2})$	tangent increasing through 0

What is really happening

What an inverse trig function "really" returns is a *location on the circle*, reported as the arc length from $(1, 0)$ – but *only the location within the chosen branch.* When your calculator says $\arcsin(0.5) = \pi/6$, it is not lying, but it is silently making a choice: it reports the one answer in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ and quietly discards the equally-valid $5\pi/6$ and all the $+2\pi k$ cousins. This is why

solving $\sin t = 0.5$ in full requires you to add back what the inverse threw away: $t = \pi/6 + 2\pi k$ *and* $t = 5\pi/6 + 2\pi k$. The inverse gives you one root; the symmetries of the circle give you the rest.

Why the technique works

Why must we restrict the domain at all, rather than just "allowing" a multivalued inverse? Because in calculus we need a genuine *function* – one output per input – so that we can graph it, differentiate it, and feed it into the chain rule. A function that returns infinitely many answers cannot have a single well-defined derivative at a point. By amputating sine down to one monotone branch we buy a clean, differentiable, single-valued inverse. The cost is that the inverse only ever reports angles in one limited window; we accept that cost because differentiability is worth more than completeness, and because the discarded answers are always recoverable from the periodicity and symmetry of the original.

4.7.4 Computing With Inverses: the Triangle Trick

A frequent and initially baffling task is to simplify a composition like $\cos(\arcsin x)$. The expression mixes an angle-finder with a coordinate-finder, but a right-triangle picture untangles it instantly.

Example 4.1 (Simplify $\cos(\arcsin x)$)**.** Let $\theta = \arcsin x$, so $\sin\theta = x$ and, by definition of the arcsine branch, $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. We want $\cos\theta$. Build a right triangle in which $\sin\theta = x$: opposite side x, hypotenuse 1 (so that opposite/hypotenuse $= x$). By Pythagoras the adjacent side is $\sqrt{1-x^2}$ (Figure 4.10). Then

$$\cos(\arcsin x) = \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \sqrt{1-x^2}.$$

We take the *positive* root because θ lies in $[-\frac{\pi}{2}, \frac{\pi}{2}]$, where cosine is never negative. The branch we chose for arcsine is exactly what resolves the sign ambiguity – another reason the restriction matters.

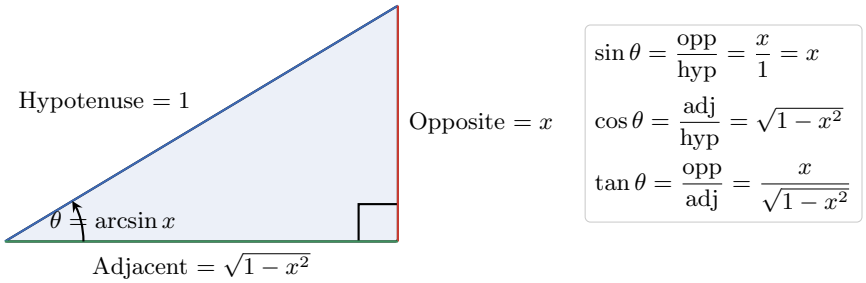


Figure 4.10: The right-triangle method for simplifying inverse trigonometric compositions. Setting $\theta = \arcsin x$ builds a reference right triangle with opposite side x and hypotenuse 1. The Pythagorean theorem supplies the missing adjacent side $\sqrt{1-x^2}$, allowing any trig function of $\arcsin x$ to be read off directly.

This little identity, $\cos(\arcsin x) = \sqrt{1-x^2}$, is not a curiosity: it is precisely what makes the derivative of arcsin come out to the clean $1/\sqrt{1-x^2}$ you will derive next chapter, and it is the substitution behind a whole family of integrals.

Work it yourself

Try the same trick on $\tan(\arccos x)$. Set $\theta = \arccos x$, so $\cos\theta = x$ with adjacent $= x$, hypotenuse $= 1$, hence opposite $= \sqrt{1-x^2}$. Then $\tan\theta =$ opposite/adjacent $= \sqrt{1-x^2}/x$. Sketch the triangle yourself and confirm each side; the picture does all the work.

4.8 Why Radians Make Calculus Clean: The Payoff

One thread has run through the whole chapter, and it is the most important practical takeaway. Why radians?

Trace the logic. We *defined* radians so that arc length equals angle ($s = r\theta$, and θ itself on the unit circle). That choice made the arc above $(1, 0)$ have length exactly t. That, in turn, forced the squeeze $\cos t < \frac{\sin t}{t} < 1$, hence $\sin t/t \to 1$, hence $\sin t \approx t$ for small t. And *that* approximation is, almost verbatim, the statement that the derivative of $\sin t$ at $t = 0$ equals 1:

$$\frac{\mathrm{d}}{\mathrm{d}t}\sin t\Big|_{t=0} = \lim_{t\to 0}\frac{\sin t - \sin 0}{t - 0} = \lim_{t\to 0}\frac{\sin t}{t} = 1.$$

With that, the next chapter's central results $\frac{\mathrm{d}}{\mathrm{d}t}\sin t = \cos t$ and $\frac{\mathrm{d}}{\mathrm{d}t}\cos t = -\sin t$ fall out clean, with no stray constants. Had we used degrees, every one of these would carry a factor of $\pi/180 \approx 0.01745$, multiplying and compounding through every higher derivative, every integral, every Taylor series. Radians are not a convention chosen for tidiness; they are the *unique* angle unit in which the calculus of trigonometry has no extraneous constant at all.

Why this matters

This is a recurring pattern in mathematics worth naming. The right *units* or the right *definitions* do not merely simplify the bookkeeping – they reveal that two things you thought were different are secretly the same. Radians reveal that angle and arc length are one quantity. The number e (you have met it, or will) is the base that reveals that an exponential and its own rate of growth are one. In both cases, choosing the "natural" unit makes a derivative equal to 1 exactly where it should, and the whole subject snaps into focus. When the formulas suddenly lose their clutter, it is usually a sign you have found the coordinates the problem wanted all along.

4.9 Exercises

Exercise 4.1 (Conversions and arc length)**.** (a) Convert to radians: 135°, 210°, 300°. (b) Convert to degrees: $\frac{5\pi}{6}$, $\frac{7\pi}{4}$, 3 radians (approximate). (c) A circular arc has radius 4 cm and subtends an angle of $\frac{2\pi}{3}$. Find its length and the area of the corresponding sector.

Exercise 4.2 (Reading the circle)**.** Without a calculator, give exact values of sin, cos, and tan at $t = \frac{2\pi}{3}$, $t = \frac{5\pi}{4}$, and $t = \frac{11\pi}{6}$. For each, state which quadrant the point lands in and use that to predict the signs *before* computing.

Exercise 4.3 (Identities from symmetry)**.** Using only the unit-circle picture (reflections and rotations), explain why each holds: (a) $\sin(\pi + t) = -\sin t$; (b) $\cos(\pi + t) = -\cos t$; (c) $\tan(\pi + t) = \tan t$; (d) $\sin(\frac{\pi}{2} + t) = \cos t$. Draw the relevant reflection or rotation for at least two of them.

Exercise 4.4 (Double and half angle)**.** (a) Starting from the addition formula for $\cos(a+b)$, derive $\cos 2a = 1 - 2\sin^2 a$. (b) Use the half-angle formula to find the exact value of $\cos^2(\pi/8)$, and hence $\cos(\pi/8)$.

Exercise 4.5 (The fundamental limit at work)**.** Evaluate each limit, reducing it to $\lim_{t\to 0} \sin t/t = 1$ (and its cousin $\frac{1-\cos t}{t} \to 0$): (a) $\lim\limits_{t\to 0} \dfrac{\sin 5t}{t}$; (b) $\lim\limits_{t\to 0} \dfrac{\sin 3t}{\sin 7t}$; (c) $\lim\limits_{t\to 0} \dfrac{\tan t}{t}$; (d) $\lim\limits_{t\to 0} \dfrac{1-\cos t}{t^2}$. (Hint for (a): write $\frac{\sin 5t}{t} = 5 \cdot \frac{\sin 5t}{5t}$.)

Exercise 4.6 (Inverse compositions)**.** Simplify to an algebraic expression in x (state any restrictions on x): (a) $\sin(\arccos x)$; (b) $\tan(\arcsin x)$; (c) $\cos(\arctan x)$; (d) $\sin(2\arctan x)$.

Exercise 4.7 (Why the branch matters)**.** (a) Is it true that $\arcsin(\sin t) = t$ for *every* real t? Test $t = \frac{3\pi}{4}$ and explain the result using the range of arcsin. (b) For which t *is* $\arcsin(\sin t) = t$ exactly true? (c) Sketch the function $g(t) = \arcsin(\sin t)$ over $[-2\pi, 2\pi]$ and describe its shape.

Exercise 4.8 (Solving trig equations completely)**.** Find *all* real solutions, using an inverse for one solution and the circle's symmetry for the rest: (a) $2\sin t - 1 = 0$; (b) $\cos t = -\frac{\sqrt{3}}{2}$; (c) $\tan t = 1$. Express each answer as a family $t = (\dots) + 2\pi k$ or $+\pi k$, $k \in \mathbb{Z}$.

Exercise 4.9 (Prove the squeeze, carefully)**.** Reproduce the proof that $\lim_{t\to 0^+} \sin t/t = 1$ by comparing the areas of the inner triangle, the sector, and the outer triangle on the unit circle. Then explain why the two-sided limit follows, by arguing that $f(t) = \sin t/t$ is an even function.

Exercise 4.10 (A modeling problem)**.** The number of daylight hours at a mid-northern latitude is modeled by

$$D(t) = 12 + 3\sin\!\left(\frac{2\pi}{365}(t - 80)\right),$$

where t is the day of the year. (a) What are the amplitude, period, and midline, and what do they mean physically? (b) On which day is daylight longest? (c) For roughly how many days per year does $D(t) > 13.5$? (Use an inverse sine and the symmetry of the curve.)

Exercise 4.11 (Conceptual: the missing constant)**.** Suppose a careless textbook *defines* sine using degrees, so that $S(x) = \sin(x^\circ)$ where x is a pure number of degrees. Using $\sin(x^\circ) = \sin\!\left(\frac{\pi}{180}x \text{ rad}\right)$ and the radian limit, compute $\lim_{x\to 0} S(x)/x$. Explain in one or two sentences why this answer makes degrees a poor choice for calculus, connecting it to the eventual formula for the derivative of S.

Chapter 5

The Idea and History of Limits

There is a single idea sitting underneath every formula in this book. It is not the derivative, and it is not the integral. It is older than both, and both are defined in terms of it. It is the *limit*.

A limit answers a question that sounds almost paradoxical: *what value is a quantity heading toward, even if it never actually gets there?* How can we talk about where something is going if it never arrives? And yet your intuition handles this kind of reasoning every day. When you watch a car decelerate toward a stop sign, you can tell it is heading for a complete stop long before its wheels stop turning. When you pour water into a glass and the level rises more and more slowly, you can see it approaching the brim. The limit is the mathematics of *approach* – of destinations inferred from the journey.

This chapter is about the *idea* of a limit and the long history of how mathematicians fought to understand it. The famous "epsilon–delta" definition can wait for the next chapter. Here we want something harder to get from a formula: we want you to *see* what a limit is, why calculus cannot live without it, and why it took roughly two hundred years and some of the sharpest minds in history to say precisely what it means. That long delay is not an embarrassment to be skipped over – it is the clue. The fact that brilliant people used calculus successfully for two centuries *before* they could define its central concept tells you that the intuition comes first and the rigor comes second – which is exactly how this book is organized, and exactly how you should learn it.

5.1 A quantity that approaches

Start with the most concrete possible picture. Consider the function

$$f(x) = \frac{x^2 - 1}{x - 1}.$$

There is something wrong with this function at exactly one point. If you try to evaluate it at $x = 1$, you get

$$f(1) = \frac{1 - 1}{1 - 1} = \frac{0}{0},$$

which is undefined. Division by zero is not allowed, and $0/0$ is not allowed twice over. So the function f has a *hole* at $x = 1$: it is defined everywhere else on the number line, but at $x = 1$ it simply has no value.

Now here is the question that started calculus. The function is undefined *at* $x = 1$, but what is it doing *near* $x = 1$? What value is it approaching as x creeps toward 1?

Work it yourself

Before reading on, build the table yourself with a calculator. Compute $f(x) = \dfrac{x^2 - 1}{x - 1}$ for $x = 0.9,\ 0.99,\ 0.999$ and then for $x = 1.1,\ 1.01,\ 1.001$. Watch what happens to the output. Do not simplify the formula first – plug the numbers in raw, the way a machine would.

Here is the table you should have produced. The values of x march in toward 1 from below (the left) and from above (the right), and we watch the outputs.

x (from left)	$f(x)$	x (from right)	$f(x)$
0.9	1.9	1.1	2.1
0.99	1.99	1.01	2.01
0.999	1.999	1.001	2.001
0.9999	1.9999	1.0001	2.0001

The pattern is unmistakable. As x approaches 1 from either side, $f(x)$ approaches 2. It never *equals* 2 in this table, and it never can equal anything at $x = 1$, because f is not defined there. But the outputs are clearly homing in on 2. We say:

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = 2.$$

Read this aloud as "the limit, as x approaches 1, of $(x^2 - 1)/(x - 1)$, equals 2."

Focus

The notation $\lim_{x \to a} f(x) = L$ is a statement about the *approach*, not about the destination point itself. It says: *as x gets close to a (but not equal to a), the outputs $f(x)$ get close to L.* Whether f is defined at a, and what value it might have there, is a separate question that the limit deliberately ignores.

Why does f approach 2? Algebra reveals the secret. As long as $x \neq 1$, we may factor and cancel:

$$f(x) = \frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1} = x + 1.$$

The cancellation is legal precisely *because* $x \neq 1$, so the factor $x - 1$ in the denominator is not zero and we are allowed to divide it out. For every x except 1, our complicated-looking $f(x)$ is just the simple line $x + 1$ wearing a disguise. And a line has no trouble at $x = 1$: the line $x + 1$ gives $1 + 1 = 2$ there. So as x approaches 1, the disguised function f approaches the same value the line would have, namely 2.

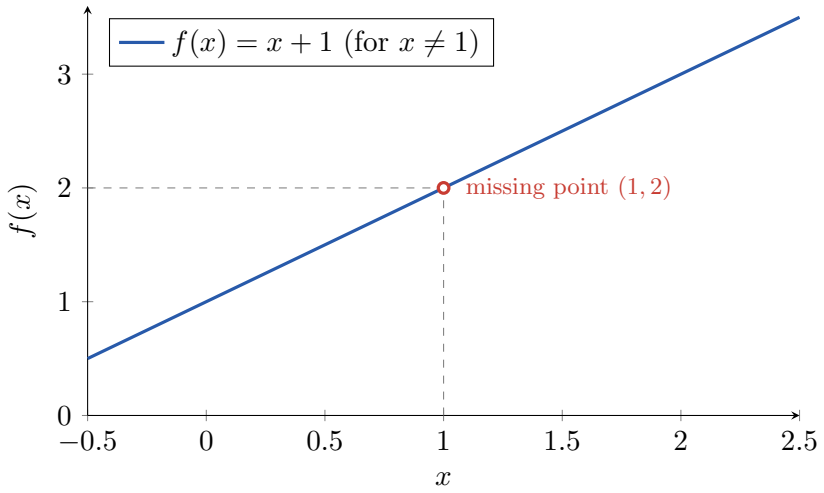


Figure 5.1: The graph of $f(x) = \dfrac{x^2 - 1}{x - 1}$ is the straight line $y = x + 1$ with a single point punched out at $(1, 2)$. The hollow circle marks the missing point. As x slides toward 1 along the line, the height slides toward 2 – regardless of the fact that the point itself is absent.

Intuition

A limit is what your eye does automatically when it follows a curve toward a point. If you trace the line in Figure 5.1 with your finger heading for $x = 1$, your finger arrives at height 2 – and the tiny hole does not stop you, because your finger is reading the *trend* of the curve, not the single missing pixel. The limit is the trend.

5.1.1 Why the hole is the whole point

You might shrug and say: this is a fuss over nothing. Just use the line $x + 1$ and forget the hole. But the hole is not a defect to be swept aside; it reveals the very reason limits are necessary. Look closely at what we are about to do throughout calculus.

When we measure the steepness of a curve at a point, we will compute the slope of a line through two points on the curve and then let those two points slide together. At the instant they coincide, the slope formula becomes

$$\frac{\text{rise}}{\text{run}} = \frac{0}{0},$$

because the two points are the same point: zero rise over zero run. When we measure the area under a curve, we will add up the areas of thin rectangles and let their width shrink to zero; in the limit each rectangle has zero width, and we are summing infinitely many zeros. Every central computation in calculus runs straight into a $0/0$ or an "infinity of zeros," an expression that is meaningless if you plug in the forbidden value directly. The limit is the tool that lets us extract a sensible, finite answer from these forbidden expressions by asking not "what is the value *at* the bad point?" but "what is the value *approaching* as we close in on the bad point?"

Why this matters

This is the deep reason the limit is the foundation. Calculus is the study of quantities at an *instant* – instantaneous speed, instantaneous slope, the area swept out at a precise moment. But an instant has no duration, and at an instant the naive formulas collapse to 0/0. The limit is how we make sense of "at an instant" without ever actually dividing by zero. It is the bridge from the smooth, continuous world we want to describe to the arithmetic of finite numbers we are allowed to do.

5.2 A limit that does not need a hole

It would be a mistake to think limits are only about repairing broken functions. Most of the time the limit of a function at a point is simply its value there, and that is the boring, reassuring case. If $g(x) = x^2 + 3$, then

$$\lim_{x \to 2} g(x) = 2^2 + 3 = 7,$$

and the outputs near $x = 2$ approach 7 for the unremarkable reason that g is a nice smooth function with $g(2) = 7$. There is no hole, no drama. The function approaches the value it actually has (Figure 5.2).

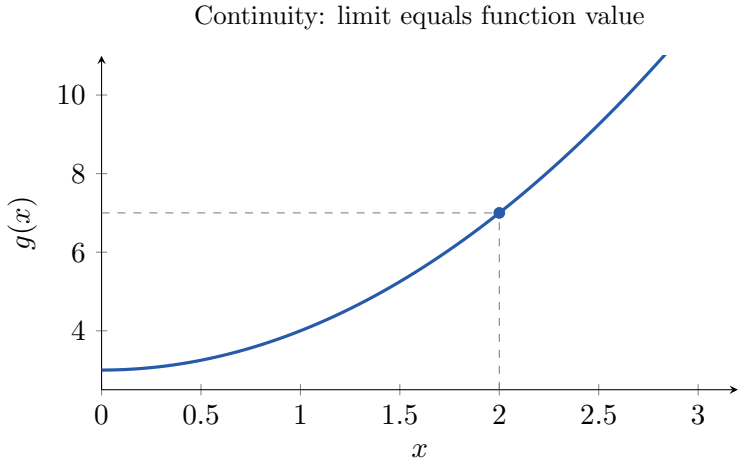


Figure 5.2: The graph of $g(x) = x^2 + 3$ has no hole at $x = 2$. As x slides toward 2 the height slides toward 7 – and the point $(2, 7)$ is genuinely *there* (filled dot), so here the limit and the value agree. Contrast this with the punched-out point of Figure 5.1.

Functions that behave this well – where the limit at a point equals the value at the point – are called *continuous*, and they get their own chapter. For now, just notice that the limit and the value can either *agree* (the continuous case, g above) or *disagree* (the hole case, f above, where the limit is 2 but the value does not exist). The limit concept earns its place precisely because it does not care which case you are in. It always asks the same question – where are the outputs heading? – and that question has an answer of 2 for f at $x = 1$ and an answer of 7 for g at $x = 2$, whether or not the function bothers to be defined and equal there.

What is really happening

When you write $\lim_{x\to a} f(x) = L$, you are making a claim about *infinitely many* values of f at once – all the values at points near a. The claim is: *you can force the outputs to be as close to L as you like, just by keeping x close enough to a.* The phrase "as close as you like" is the heart of it. It is not enough that the outputs get *somewhat* close; they must be squeezable arbitrarily close. We deliberately exclude $x = a$ itself from the conversation, which is why the value $f(a)$ – present, absent, or weird – has no vote. The next chapter turns "as close as you like" into the precise ε–δ machinery. Everything there is just this sentence made unambiguous.

5.3 When the approach disagrees with itself: one-sided limits

So far our examples approached the same value whether x came in from the left or from the right. That symmetry is not guaranteed, and watching it break is one of the most instructive things you can do.

Consider the *sign-step* function

$$H(x) = \begin{cases} 0 & x < 0, \\ 1 & x \geq 0. \end{cases}$$

This is the Heaviside step. It is flat at height 0 for all negative x, then jumps suddenly to height 1 at $x = 0$ and stays there. Ask the limit question at $x = 0$: what is H approaching?

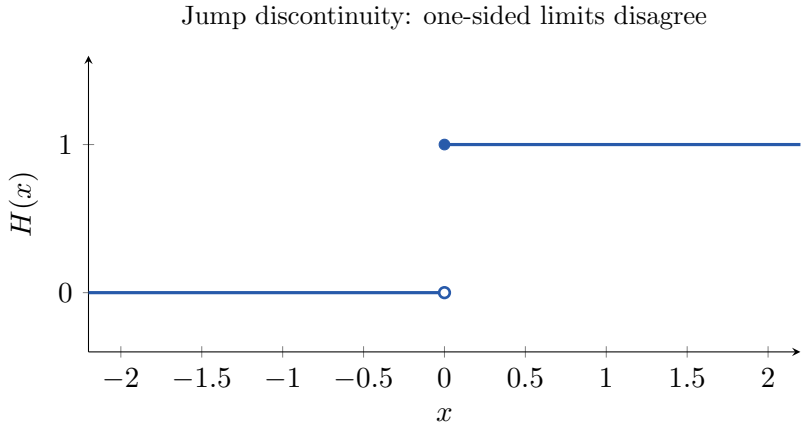


Figure 5.3: The Heaviside step $H(x)$. Approaching $x = 0$ from the left, the outputs sit at 0; approaching from the right, they sit at 1. The two approaches disagree, so there is no single value that H is heading toward.

Figure 5.3 shows the jump. If you walk in toward 0 from the negative side – $x = -0.1, -0.01, -0.001$ – every output is exactly 0. The function is approaching 0. But if you walk in from the positive side – $x = 0.1, 0.01, 0.001$ – every output is exactly 1. The function is approaching 1. The two sides give different answers. There is no single number that $H(x)$ is "heading toward" as $x \to 0$, because it depends entirely on which direction you came from.

We capture this with *one-sided limits*. The *left-hand limit* watches only x coming from below:

$$\lim_{x\to 0^-} H(x) = 0,$$

where the little minus sign on 0^- means "x approaches 0 through values less than 0." The *right-hand limit* watches only x coming from above:

$$\lim_{x\to 0^+} H(x) = 1,$$

with 0^+ meaning "through values greater than 0." Both one-sided limits exist and are perfectly definite. They simply disagree.

Focus

The ordinary (two-sided) limit $\lim_{x\to a} f(x)$ exists *only when both one-sided limits exist and are equal*. In symbols,

$$\lim_{x\to a} f(x) = L \quad\Longleftrightarrow\quad \lim_{x\to a^-} f(x) = L \text{ and } \lim_{x\to a^+} f(x) = L.$$

For the Heaviside step, the left limit is 0 and the right limit is 1; they disagree, so the two-sided limit $\lim_{x\to 0} H(x)$ *does not exist*. "Does not exist" is a legitimate, useful answer, not a failure.

Intuition

Think of two hikers approaching a canyon from opposite rims. The left hiker ends at the canyon floor; the right hiker ends on the far rim. If someone asks "where does the trail arrive?" there is no single answer – it depends which hiker you follow. A two-sided limit exists only when both hikers, coming from opposite directions, arrive at the *same* place. One-sided limits let us still describe each hiker's destination separately.

Application

One-sided limits are not a curiosity; they are how the physical world encodes *sudden events*. Flip a switch and the voltage jumps; the current "before" and "after" have different limits at the instant of the flip. In economics, a tax bracket creates a jump in the marginal rate: the rate just below the threshold and just above it are different one-sided limits. In probability, the cumulative distribution of a discrete variable steps up at each value, exactly like the Heaviside function. Whenever reality has a genuine jump, the two-sided limit fails and the one-sided limits carry the real information.

5.3.1 A more violent disagreement

A jump is a polite failure: each side still settles on a number. Limits can fail more dramatically. Consider $p(x) = 1/x$ near $x = 0$. From the right, as $x = 0.1, 0.01, 0.001$, the outputs are $10, 100, 1000$ – shooting off to $+\infty$. From the left, $x = -0.1, -0.01, -0.001$ give $-10, -100, -1000$ – plunging to $-\infty$. Neither one-sided limit settles on a finite number; the function blows up, as Figure 5.4 makes vivid. We write $\lim_{x\to 0^+} 1/x = +\infty$ and $\lim_{x\to 0^-} 1/x = -\infty$ as shorthand for "grows without bound," but you should hear those as descriptions of *how* the limit fails, not as ordinary numerical answers. We will treat such infinite behavior carefully in the chapter on asymptotes.

Infinite limits: blow-up near the pole

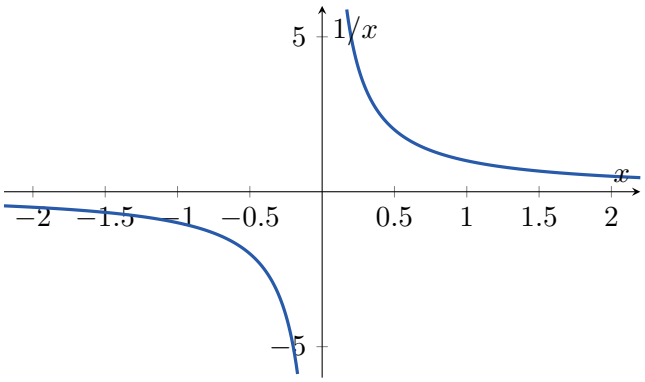


Figure 5.4: $p(x) = 1/x$. The right-hand branch races up toward $+\infty$ as $x \to 0^+$; the left-hand branch dives toward $-\infty$ as $x \to 0^-$. There is no finite value being approached from either side.

5.4 The historical crisis: a triumph built on sand

Now for the history, where the idea of a limit turns dramatic. Calculus was invented – twice, independently – in the late seventeenth century. Isaac Newton in England (working in the 1660s and 1670s, publishing later) and Gottfried Wilhelm Leibniz in Germany (publishing in the 1680s) each built a calculus of *infinitely small quantities*. Their methods were astonishingly powerful. Newton used his "fluxions" to derive the laws of planetary motion; Leibniz's elegant $\mathrm{d}y/\mathrm{d}x$ and $\int$ notation – which we still use – let problems that had stumped the ancients fall in a few lines. By any practical measure, calculus was a colossal success.

And yet at its logical core sat something nobody could justify.

History

Two inventors, one shaky foundation. Both Newton and Leibniz computed derivatives by the same essential maneuver. To find the slope of $y = x^2$, you let x change by a tiny amount o (Newton's notation) or $\mathrm{d}x$ (Leibniz's), compute the resulting change in y, and form the ratio. For $y = x^2$:

$$\frac{(x+o)^2 - x^2}{o} = \frac{x^2 + 2xo + o^2 - x^2}{o} = \frac{2xo + o^2}{o} = 2x + o.$$

Then – here is the suspicious step – you *discard* the leftover o to conclude that the slope is $2x$. But look at what just happened. In the third equality we *divided* by o, which is only legal if $o \neq 0$. Then in the last step we *threw away* the term o, which is only legal if $o = 0$. The quantity o had to be nonzero and zero in the same breath. Newton and Leibniz both sensed this was uncomfortable; neither could say cleanly why the answer $2x$ was nonetheless exactly right.

This was not a small loose thread. It was a contradiction at the foundation of the most celebrated mathematics of the age, and it did not escape notice. The most famous attack came from an unexpected direction: a bishop.

History

Berkeley and the "ghosts of departed quantities" (1734). George Berkeley, an Irish philosopher and Anglican bishop, was irritated that scientists who demanded rigorous proof for religious claims were happily using calculus whose own logic was incoherent. In a witheringly clever pamphlet titled *The Analyst*, he dissected the infinitesimal exactly as we did above. The infinitely small increments, he wrote, are neither finite quantities, nor quantities infinitely small, nor yet nothing. He coined the line that has haunted calculus ever since: they are

> *"the ghosts of departed quantities."*

His point was sharp and, on the logic available at the time, entirely correct. The quantity o is alive (nonzero) when we divide by it and dead (zero) when we discard it. Berkeley did not deny that calculus got right answers – he could see that it did – he denied that anyone understood *why*. He even suggested, mischievously, that the correct answers came from a lucky "compensation of errors."

Walking question

Take Berkeley's challenge seriously on your next walk. The computation $\frac{(x+o)^2-x^2}{o} = 2x + o$ is exact and innocent as long as $o \neq 0$. The whole crisis lives in a single move: at the end we replace $2x + o$ by $2x$. Ask yourself: *is that an approximation, or is it exact?* If it is an approximation, calculus is only approximately true – but it is not, the answers are exact. If it is exact, then in what sense is dropping a nonzero o legitimate? Hold that question. The limit is precisely the answer, and feeling the tension first will make the resolution land.

5.4.1 What the early masters got right – and what was missing

Do not read this history as "Newton and Leibniz were sloppy." They were not sloppy; they were *early*. Their physical and geometric intuition was nearly flawless. When Newton spoke of a quantity's "ultimate ratio," he described the ratio $2x + o$ *as* o *vanishes* – not its value at any fixed nonzero o, and not a meaningless ratio at $o = 0$, but the value the ratio is *heading toward*. Read that sentence again: Newton was reaching, in words, for exactly the limit concept. He wrote of "the limit to which they always approach, and to which they approach nearer than by any given difference, but never go beyond." That is almost a definition of a limit – in 1687.

History

The gap was language, not insight. What Newton and Leibniz lacked was not the right picture – they had it – but a way to talk about "approaching" that did not smuggle in an actual infinitely small number. They kept reaching for the destination (the limit) but describing the journey with a paradoxical object (the infinitesimal o or $\mathrm{d}x$ treated as a real number that is somehow both zero and not). The fix, when it finally came a century and a half later, changed almost none of the answers and all of the reasoning. It replaced the noun "infinitesimal" with the verb "approaches," made precise.

What is really happening

What is really happening in the $0/0$ derivative computation? *Nothing is ever divided by zero, and nothing nonzero is ever discarded.* The expression $\frac{(x+o)^2-x^2}{o}$ equals $2x+o$ for *every* nonzero o – this is exact algebra, no approximation. The slope is then *defined* as the limit of that exact expression as $o \to 0$, i.e. the number that $2x+o$ is heading toward, which is $2x$. We never set $o = 0$; we ask where $2x + o$ goes as o shrinks. The ghost is exorcised because we never needed it to be a number at all. "$o \to 0$" is a process, not a value. That single reinterpretation – from "a number infinitely close to zero" to "a process of getting closer to zero" – is the entire content of the limit revolution.

5.5 Cauchy and Weierstrass: rebuilding on rock

For about a hundred and forty years after Newton's *Principia*, calculus advanced spectacularly while its foundation remained, in Berkeley's phrase, ghostly. Euler computed with infinite sums and infinitesimals with breathtaking audacity and almost always got the right answer through sheer instinct. But "almost always" is the problem: as functions got stranger and infinite series got subtler, the instinct started producing genuine contradictions, and nobody could tell a trustworthy manipulation from a treacherous one without a firm definition to appeal to. The nineteenth century decided to fix this once and for all.

History

Cauchy puts the limit first (1820s). Augustin-Louis Cauchy, lecturing at the École Polytechnique in Paris, made a decisive choice in his *Cours d'analyse* (1821) and related works: he made the *limit* the primary concept and *defined* everything else – continuity, the derivative, the integral, convergence – in terms of it. He wrote, in words, that a variable's limit is the fixed value it approaches so that it ends up differing from it by "as little as one wishes." In this formulation, the infinitesimal is demoted: for Cauchy, "infinitely small" is no longer a mysterious number but simply a *variable whose limit is zero.* That re-definition quietly dissolves Berkeley's ghost. An infinitesimal is not a tiny fixed quantity; it is a quantity in the *act* of shrinking toward zero.

Cauchy's phrase "as little as one wishes" is the seed of everything. It contains a hidden challenge-and-response: *however* small a gap you want, I can guarantee the difference is smaller, provided the variable is far enough along. But Cauchy still leaned on intuitive, sometimes vague verbal phrasing – "approaches indefinitely," "successive values" – and his words occasionally hid subtle errors (his famous claim that a convergent sum of continuous functions is itself continuous is, as he stated it, false). The job was nearly done. The final tightening fell to Berlin.

History

Weierstrass makes it static and exact (1850s–1870s). Karl Weierstrass, a former schoolteacher who became the towering analyst of his era, removed the last trace of motion and mystery from the limit. Where Cauchy still spoke of a variable *approaching* – a word that suggests time and movement – Weierstrass replaced the whole idea with a frozen, purely arithmetic statement involving two fixed challenge numbers, traditionally ε and δ. There is

no flowing variable, no ghost, no "infinitely small": just a flat claim that for *every* tolerance $\varepsilon > 0$ you name, *there exists* a closeness $\delta > 0$ that delivers it. This is the celebrated ε–δ definition. With it, calculus finally rested on logic as solid as arithmetic itself, more than two centuries after Newton and Leibniz first used it.

Why this matters

Notice the direction of the centuries-long motion: from *dynamic* to *static*. Newton's quantities *flowed*; Cauchy's variables *approached*; Weierstrass's ε and δ just *sit there* in a logical relationship with no motion at all. This is a recurring pattern in mathematics – we tame a slippery dynamic intuition by reformulating it as a static statement about "for all … there exists …." You lose the vivid sense of movement, but you gain something priceless: a definition you can actually *prove things about*, because it no longer relies on the reader sharing your mental picture of motion.

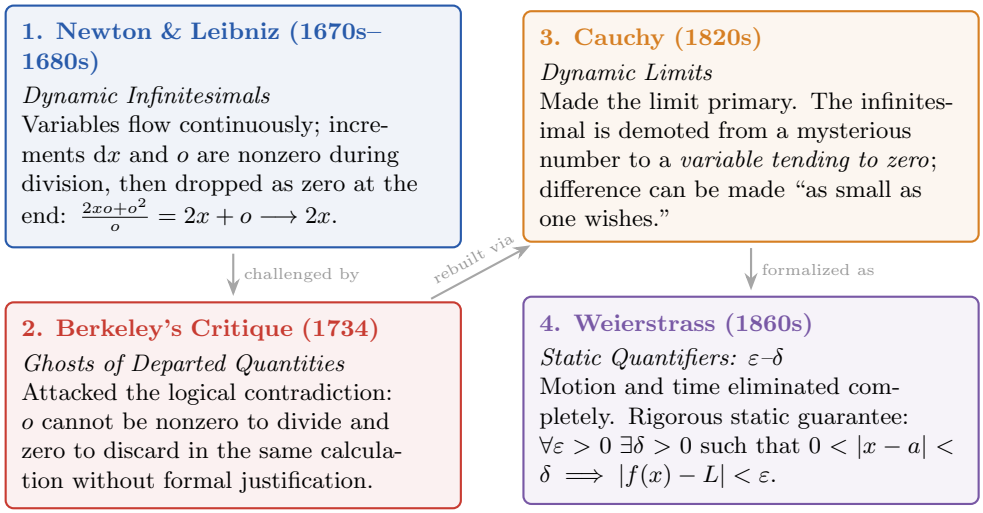


Figure 5.5: The two-century conceptual progression of the limit: from physical intuition of flowing infinitesimals to a completely static arithmetic statement.

History

A coda: the ghosts came back, legitimately. There is a satisfying twist. In the 1960s the logician Abraham Robinson built *nonstandard analysis*, a rigorous number system that genuinely contains infinitesimals – numbers smaller than every positive real yet not zero. In that framework Newton and Leibniz's original calculations are literally valid as written. So the infinitesimal was not wrong, only premature: it took three hundred years to build a logic solid enough to hold it. The mainstream of calculus, and this book, still uses the ε–δ limit, because it asks less of the foundations and is what you will meet everywhere. Still, the early masters' instinct was eventually vindicated in full.

5.5.1 Why intuition alone is insufficient

You might reasonably ask: if the intuitive idea – "outputs head toward L" – gets the right answers, why bother with the austere ε–δ machinery at all? Why did this need fixing?

The answer is that intuition, unaided, eventually lies to you. Consider how you would *prove* a basic fact such as "the limit of a sum is the sum of the limits," or decide a genuinely hard case where the function oscillates infinitely fast as it approaches the point. Intuition has no traction on these; it can only gesture. Worse, the nineteenth century discovered functions that violate every naive expectation – a function continuous at every point yet with a sharp corner at every point, so that "follow the curve with your finger" becomes meaningless. On such monsters, the only way to know what is true is to have a definition precise enough to *compute* with, one that does not secretly depend on the function being as tame as the pictures in this chapter.

Common trap

The most common misconception about limits, and the one the whole history is fighting, is the belief that $\lim_{x\to a} f(x)$ means "plug in $x = a$." Sometimes plugging in works (the continuous case), and that lucky coincidence breeds the bad habit. But the limit is fundamentally a statement about values *near* a with a *itself excluded*. Our very first example, $\frac{x^2-1}{x-1}$, has a perfectly good limit of 2 at a point where you *cannot* plug in at all. Train yourself to read $\lim_{x\to a}$ as "as x approaches a," never as "at $x = a$."

Why the technique works

Why does turning "approaches" into a static ε–δ statement actually work – what makes it the definitive formalization? It captures the exact operational content of "as close as you like." The phrase secretly contains a quantifier duel: a skeptic challenges you with a tolerance ("get within $\varepsilon = 0.001$ of L"), and you must respond with a guarantee ("keep x within δ of a and I promise you will"). If you can win this duel for *every* challenge ε – no matter how brutally small – then in every practical and provable sense the outputs are being forced toward L, and there is no wiggle room left for them to be heading anywhere else. The genius of Weierstrass's formulation is that it converts the vague "gets closer and closer" into a finite, checkable promise about two ordinary positive numbers. The next chapter is entirely devoted to playing this game until it is second nature.

5.6 A numerical caution: tables suggest, they do not prove

The table that opened this chapter was persuasive, and tables are a wonderful way to *guess* a limit. But a table can mislead, and seeing how is excellent training for why we eventually need proof rather than evidence.

Work it yourself

Tabulate $f(x) = \sin(\pi/x)$ at the seductive points $x = 1,\ \frac{1}{2},\ \frac{1}{3},\ \frac{1}{4},\ \ldots$, i.e. $x = 1/n$ for $n = 1, 2, 3, \ldots$. At every one of these points $\pi/x = n\pi$, so $\sin(\pi/x) = \sin(n\pi) = 0$. The table reads $0, 0, 0, 0, \ldots$ and screams "the limit as $x \to 0^+$ is 0." Now tabulate instead at

$x = 2,\ \frac{2}{5},\ \frac{2}{9},\ \ldots$, i.e. $x = 2/(4k+1)$. There $\pi/x = (4k+1)\pi/2$, an odd multiple of $\pi/2$, so the output is $+1$ every time. A second table, just as correct, screams "the limit is 1."

Both tables are arithmetically correct, and they contradict each other. The truth is that $\sin(\pi/x)$ oscillates between -1 and $+1$ faster and faster as $x \to 0^+$, never settling on anything: the limit *does not exist*, as the crowding oscillations of Figure 5.6 show. No finite table could ever reveal this for certain, because a table only samples finitely many points and the misbehavior hides in between them.

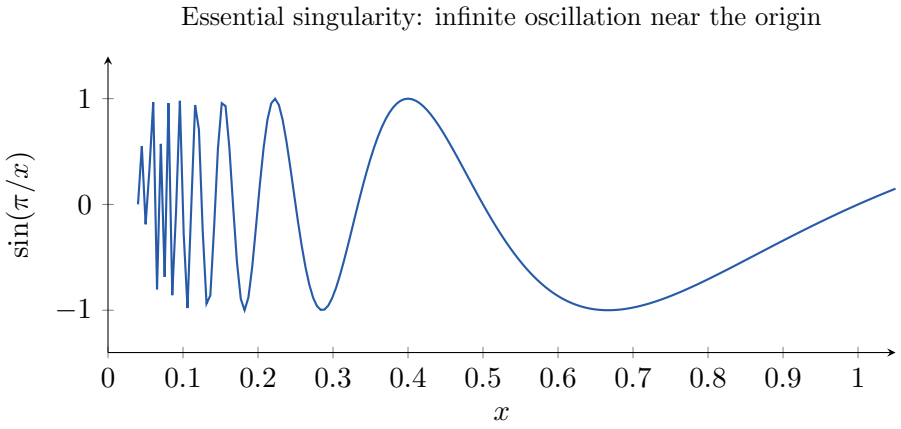


Figure 5.6: $\sin(\pi/x)$ near $x = 0$. The oscillations crowd together without bound as $x \to 0^+$, so the curve never homes in on a single height. A numerical table that happens to land on the zeros (or on the peaks) would falsely suggest a limit. Evidence is not proof.

Focus

The moral is not "never use tables." Tables and graphs are how you *form* the conjecture, and forming the right conjecture is most of the battle. The moral is that a finite amount of evidence cannot *settle* a statement about infinitely many points. That gap – between suggestive evidence and airtight proof – is exactly the gap the ε–δ definition was built to close. This is why the next chapter exists.

5.7 What a limit is, in one breath

Here is the whole chapter in a single paragraph, the kind you could recite on a walk.

A limit is the value a function's outputs *head toward* as the input is driven toward some point, deliberately ignoring what – if anything – happens *at* that point. It exists when the approach from the left and the approach from the right agree on a single destination, and it fails to exist when they disagree (a jump), or when the outputs run off to infinity, or when they oscillate forever without settling. Newton and Leibniz *used* this idea fluently while describing it with a paradoxical infinitesimal that Berkeley rightly ridiculed as a ghost; Cauchy made the limit the centerpiece and the infinitesimal merely "a variable tending to zero"; Weierstrass froze the whole thing into an exact statement about challenge numbers ε and δ with no motion and no ghosts. The intuition came first by two centuries; the rigor came to protect the intuition from the cases where it would otherwise lie.

Connection

Everything ahead is the limit in disguise. The *derivative* is the limit of a slope as two points merge (0/0 tamed). The *integral* is the limit of a sum of shrinking rectangles (infinitely many zeros tamed). *Continuity* is the statement that a function's limit equals its value. *Infinite series* are limits of partial sums. *Improper integrals* are limits of ordinary ones. If you truly own the idea in this chapter – a destination inferred from an approach, with the endpoint itself excluded – you own the spine of the entire subject, and each later technique becomes a special way of computing one particular limit.

5.8 Exercises

Exercise 5.1. Build a table of values for $g(x) = \dfrac{x^2 - 9}{x - 3}$ at $x = 2.9,\ 2.99,\ 2.999$ and $x = 3.1,\ 3.01,\ 3.001$. Conjecture $\lim_{x \to 3} g(x)$ from the table, then confirm your conjecture algebraically by factoring and cancelling. Explain in one sentence why the cancellation is legitimate even though $g(3)$ is undefined.

Exercise 5.2. For each function, decide from a graph or short table whether the two-sided limit at the given point exists, and if so what it is. If it does not exist, state both one-sided limits (or explain that the behavior is infinite or oscillatory).

(a) $\dfrac{|x|}{x}$ at $x = 0$.

(b) $\dfrac{x^3 - x}{x - 1}$ at $x = 1$.

(c) $\dfrac{1}{(x-2)^2}$ at $x = 2$.

Exercise 5.3. Consider the piecewise function

$$f(x) = \begin{cases} x + 1 & x < 2, \\ 5 & x = 2, \\ 3x - 3 & x > 2. \end{cases}$$

Compute $\lim_{x \to 2^-} f(x)$, $\lim_{x \to 2^+} f(x)$, and $\lim_{x \to 2} f(x)$. Then state the value $f(2)$. Which of these four numbers can disagree with the others, and why does that not contradict anything?

Exercise 5.4. (*Berkeley's challenge, made concrete.*) Carry out the early-calculus computation for $y = x^3$. That is, expand $\dfrac{(x + o)^3 - x^3}{o}$, simplify (dividing by o where $o \neq 0$), and then identify the value the resulting expression heads toward as $o \to 0$. State clearly which step requires $o \neq 0$ and which step requires $o \to 0$, and explain in your own words why no single number o is ever required to be both zero and nonzero.

Exercise 5.5. Explain, in plain prose and to a friend who has not taken calculus, why $\lim\limits_{x \to a} f(x)$ deliberately ignores the value $f(a)$. Give a specific example where the limit exists but the function is undefined at the point, and a second example where the limit exists, the function is defined at the point, but the two *disagree*.

Exercise 5.6. (*Tables can lie.*) Let $h(x) = \cos(\pi/x)$. Evaluate h at $x = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$ (that is, $x = 1/n$). Show that these values do not all agree, and use this to argue that $\lim_{x\to 0^+} h(x)$ does not exist. Then explain why *no* finite table could ever *prove* that a limit *does* exist, even when one does.

Exercise 5.7. The Heaviside function H has $\lim_{x\to 0^-} H(x) = 0$ and $\lim_{x\to 0^+} H(x) = 1$. Define a new function $S(x) = H(x) + H(-x)$. Work out $S(x)$ explicitly for $x < 0$, $x = 0$, and $x > 0$, then find both one-sided limits of S at 0 and decide whether $\lim_{x\to 0} S(x)$ exists. (This is a nice case where two discontinuous functions combine into something whose two-sided limit *does* exist.)

Exercise 5.8. (*Conceptual / historical.*) In two or three sentences each, describe the distinct contribution of (a) Newton and Leibniz, (b) Berkeley, (c) Cauchy, and (d) Weierstrass to our understanding of limits. Your answer to (a) should make clear what they got *right* intuitively, and your answer to (d) should explain what the shift from "approaches" to "for every ε there exists δ" actually buys us.

Exercise 5.9. (*Toward ε–δ.*) For $f(x) = 3x + 2$, we believe $\lim_{x\to 1} f(x) = 5$. Suppose a skeptic demands that the output be within 0.6 of 5, i.e. $|f(x) - 5| < 0.6$. Find an interval of x-values around 1 (of the form $|x - 1| < \delta$ for some $\delta > 0$) that guarantees this. Then redo it for a tolerance of 0.06. What is the pattern relating the tolerance to your δ? You have just played the ε–δ game by hand.

Part II

Limits and Continuity

Chapter 6

Limits: Intuition and the Epsilon-Delta Definition

In the previous chapter you met the *idea* of a limit. You watched values of a function pile up near a single number L as the input crept toward some point a, and you learned to write

$$\lim_{x \to a} f(x) = L$$

to record that fact. You also saw that the early calculus – Newton's "ultimate ratios," Leibniz's infinitesimals – ran on exactly this kind of intuition, and that for a century and a half nobody could say with full precision what the words actually *meant*. "Approaches." "Tends to." "Becomes infinitely close." These are pictures, not definitions. A picture cannot be argued with; a definition can.

This chapter is where the fog burns off. We take the intuitive phrase "$f(x)$ gets close to L as x gets close to a" and turn it, slowly and with no hand-waving left in it, into a single sentence – the *epsilon-delta definition*. Then we use that sentence to prove things: that specific limits really are what we think they are, and that the familiar limit laws (the limit of a sum is the sum of the limits, and so on) follow from the definition rather than standing as extra assumptions.

A warning, and a promise. The first time anyone sees the epsilon-delta definition it looks like a thicket of quantifiers built to keep students out. That reaction is normal and it is temporary. By the end of this chapter the definition will read like a transcript of a game you already know how to play – a game of *challenge and response*. The whole trick is to hear the quantifiers as moves by two players, and we will build that game up one piece at a time before we ever write the symbols.

6.1 Why intuition alone is not enough

You might reasonably ask: if the intuitive picture works, if everyone *agrees* that $\lim_{x \to 2}(3x + 1) = 7$, why bother making it precise? The answer is that intuition, left to itself, contradicts itself. It gives different answers to different people, and sometimes the wrong answer to everyone.

Intuition

Consider the sloppy slogan "the limit is the value the function gets close to." Now look at

$$f(x) = \frac{x^2 - 1}{x - 1}.$$

At $x = 1$ this is $\frac{0}{0}$, undefined. But for every $x \neq 1$ we can cancel: $f(x) = x + 1$. So as x heads toward 1, $f(x)$ heads toward 2. The function never *reaches* 2 at the relevant input – it is not even defined at $x = 1$ – yet 2 is unmistakably the limit. "Gets close to" has to mean something that does not depend on the value at a at all. The intuitive phrase does not say that out loud, so different readers fill the gap differently. Figure 6.1 shows the situation: a perfectly ordinary straight line with a single point punched out of it.

Removable discontinuity: approaching the hole

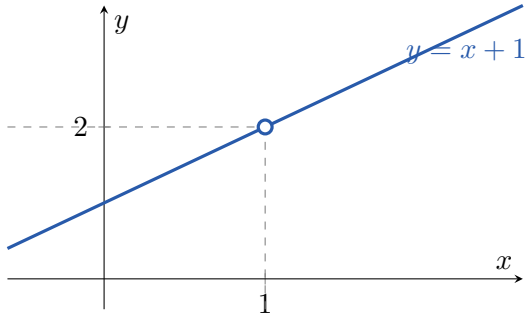


Figure 6.1: The graph of $f(x) = \frac{x^2-1}{x-1}$. For every $x \neq 1$ it agrees exactly with the line $y = x + 1$; only at $x = 1$ is it undefined, leaving the open hole. The limit asks where the graph is *heading* as $x \to 1$, which is plainly 2 – the very spot the hole sits over. The missing point is exactly what the limit is built to ignore.

Worse, the eighteenth-century notion of an "infinitely small" quantity $\mathrm{d}x$ – a number bigger than zero but smaller than every ordinary positive number – is, taken literally, self-contradictory in the ordinary real numbers. If $\mathrm{d}x > 0$ and $\mathrm{d}x$ is smaller than every positive number, then $\mathrm{d}x$ is smaller than $\mathrm{d}x/2$, which is absurd. The early users got correct answers *despite* this, by a kind of trained instinct for when the infinitesimals were safe to drop. What they lacked was a definition that a skeptic could check mechanically, without instinct.

History

The skeptic arrived. In 1734 Bishop George Berkeley published *The Analyst*, a withering attack on the foundations of the calculus, asking how a quantity $\mathrm{d}x$ could be treated as nonzero (so you can divide by it) and then as zero (so you can throw it away) in the same calculation. He called the vanished increments "the ghosts of departed quantities." Berkeley was not being obtuse; he was right that no one could yet answer him. It took until the 1820s for Augustin-Louis Cauchy to base calculus on limits rather than infinitesimals, and until Karl Weierstrass's lectures in the 1850s–1870s for the limit itself to be pinned down in the static, quantifier-based form – epsilon and delta – that we use today. The definition in this chapter is, almost word for word, Weierstrass's answer to Berkeley's ghost.

So the goal is a definition with three properties: it must (1) not refer to the value $f(a)$ at all, (2) capture "arbitrarily close" without any mystical infinitely-small numbers, using only ordinary positive reals, and (3) be checkable – a finite, mechanical test that settles whether a proposed L is the limit. Remarkably, one sentence does all three.

6.2 The challenge–response game

Let me describe the game first, with no symbols, and then we will name the pieces.

Two players sit across a table. Player One is the *Skeptic*; Player Two is the *Prover*. On the table is a claim: "As $x \to a$, $f(x) \to L$." The Skeptic does not believe it and wants to test it. Here is the single move each player gets, repeated as often as the Skeptic likes.

Focus

The move. The Skeptic names a *tolerance*: a positive number, a margin of error around L that they are willing to accept. Call it ε (epsilon, for "error"). The Skeptic says: "Fine. I want $f(x)$ to land within ε of L. Can you guarantee that?"
The Prover must respond with a *closeness*: a positive number δ (delta, for "distance"), a radius around the input a, and a promise: "Yes. As long as you keep x within δ of a (but not exactly at a), I guarantee $f(x)$ stays within your ε of L."

The Prover *wins a round* if the promise holds: every x in the punctured interval $(a-\delta,\ a+\delta)$, $x \neq a$, really does produce an $f(x)$ inside $(L-\varepsilon,\ L+\varepsilon)$. The Prover *wins the game* – and the limit really is L – if the Prover has a winning response *no matter what tolerance the Skeptic names*, including absurdly tiny ones like $\varepsilon = 10^{-100}$.

Pause on the asymmetry, because it is the whole point.

Intuition

The Skeptic moves *first*. The Skeptic gets to pick ε as small as they please, and only *then* does the Prover get to choose δ. This is exactly right. "$f(x)$ gets arbitrarily close to L" should mean: name any closeness you want for the *output*, and I can deliver it by squeezing the *input* hard enough. The Prover's δ is allowed to depend on the Skeptic's ε. Usually it must: the smaller the output tolerance, the tighter the Prover has to clamp the input. A winning strategy is a *rule* that turns any ε into a δ that works.

If even one stubborn tolerance defeats the Prover – one ε for which no δ keeps the function inside the band – then the limit is not L. That is how the game lets us *disprove* a wrong claim too, which we will use later.

Walking question

Before reading on, play a round in your head with $f(x) = 3x + 1$ near $a = 2$, claiming $L = 7$. The Skeptic says $\varepsilon = 0.6$: "land within 0.6 of 7, so between 6.4 and 7.6." What inputs x near 2 give outputs in that band? Since $3x + 1$ is between 6.4 and 7.6 exactly when x is between 1.8 and 2.2, the Prover can answer $\delta = 0.2$. Now ask yourself the move that matters: if the Skeptic had said $\varepsilon = 0.06$ instead, what δ would the Prover answer? Notice the pattern in

how δ shrinks with ε. That pattern *is* the proof.

6.2.1 Two pictures: a band and an interval

The game lives in a picture, and you should carry the picture with you forever. The Skeptic's ε is a horizontal *band* of height 2ε centered on the line $y = L$. The Prover's δ is a vertical *strip* of width 2δ centered on $x = a$. The Prover wins the round when the graph of f, restricted to that vertical strip, stays trapped inside the horizontal band – with the single point above $x = a$ exempted, since we never ask about $x = a$ itself. See Figure 6.2.

The challenge–response game: horizontal error band vs. vertical input strip

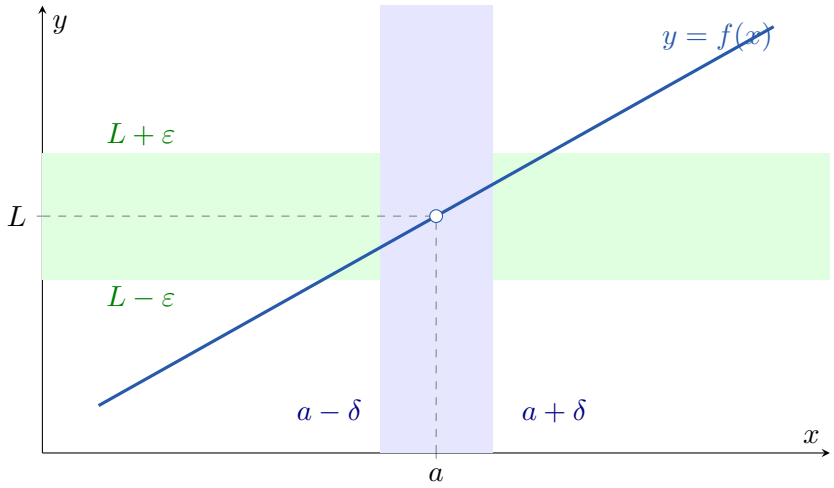


Figure 6.2: The challenge–response game as a picture. The Skeptic paints the horizontal band of half-height ε about $y = L$. The Prover must paint a vertical strip of half-width δ about $x = a$ *narrow enough* that the graph, inside the strip, never escapes the band. The open circle marks that $x = a$ is excluded: we care about the approach, not the value there.

What is really happening

What is the limit definition really asserting? Not that "$f(x)$ moves toward L" – "moves" smuggles in time and motion, which the static real numbers do not contain. It is asserting something purely about *containment of sets*. For every horizontal band around L, however thin, there exists a vertical strip around a, possibly very thin, such that the part of the graph over the punctured strip lies entirely inside the band. No motion, no infinitely small quantities, no reference to $f(a)$. Just: "for every band, there is a strip that fits inside it." Weierstrass's achievement was to see that this single containment statement captures everything the older mathematicians *meant* by "approaches," while saying nothing they could not check.

6.3 From the game to the definition

Now we translate, phrase by phrase. This is the moment to go slowly; if you internalize the translation, the definition will never again look foreign.

Game language	$\Longleftrightarrow$	**Symbol language**
"$f(x)$ lands within ε of L"	$\Longleftrightarrow$	$\lvert f(x) - L\rvert < \varepsilon$
"x is within δ of a, but not at a"	$\Longleftrightarrow$	$0 < \lvert x - a\rvert < \delta$
"the Skeptic may name any tolerance"	$\Longleftrightarrow$	for every $\varepsilon > 0$
"the Prover can find a closeness"	$\Longleftrightarrow$	there exists $\delta > 0$
"the promise holds for those x"	$\Longleftrightarrow$	$(0 < \lvert x-a\rvert < \delta)$ implies $(\lvert f(x)-L\rvert < \varepsilon)$

Two of these rows deserve a second look.

First, why $|f(x) - L| < \varepsilon$ and not $\leq \varepsilon$? Because "within ε" is a strict band, and – more importantly – it does not matter. Any tolerance the Skeptic can hit with $\leq$ they can also hit with $<$ by switching to a slightly larger ε, so the two versions of the definition are equivalent. We use strict inequalities throughout for clean algebra.

Second, and this is the subtle one: why the *left* inequality $0 < |x - a|$? The expression $|x - a|$ is zero precisely when $x = a$. Writing $0 < |x - a|$ forbids that single case. This is the formal incarnation of "we never ask about $x = a$ itself." It is what makes the limit blind to $f(a)$ – exactly the property we demanded in Section 6.1, the one that let $\frac{x^2-1}{x-1}$ have a limit of 2 at a point where it is undefined.

Stack the rows and you have the definition. Read it now not as a wall of symbols but as the game you just played.

Definition 6.1 (The limit, Weierstrass)**.** Let f be defined on some open interval containing a, except possibly at a itself. We say

$$\lim_{x \to a} f(x) = L$$

if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that, for all x,

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

Formal

Read the quantifiers in order, left to right, as alternating players:

$$\underbrace{\forall\, \varepsilon > 0}_{\text{Skeptic}} \ \underbrace{\exists\, \delta > 0}_{\text{Prover}} \ \underbrace{\forall x \left(0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon\right)}_{\text{the promise is kept}}.$$

The *order* is not decoration. δ comes after ε, so δ is allowed to be chosen in full knowledge of ε – it may, and usually must, depend on ε. If you secretly swapped the order to "$\exists \delta\, \forall \varepsilon$," you would be demanding one magic δ that works for all tolerances at once, which is almost never possible. The grammar encodes the strategy.

Common trap

The value at a is irrelevant – truly irrelevant. Three different functions can share the same limit at a: one undefined there, one defined with the "right" value L, and one defined with a deliberately wrong value. Because Definition 6.1 only ever quantifies over x with $|x-a|>0$, none of these differences at the single point a can change the limit. Beginners routinely "simplify" by plugging in $x=a$; that is a *shortcut* that happens to work for nice (continuous) functions, not the definition. We will see in the next chapter exactly when the shortcut is legal.

6.3.1 What it would take to be *false*

A definition you cannot negate is a definition you do not understand. The limit fails to be L when the Skeptic has a winning tolerance. Negating the quantifiers (each $\forall$ becomes $\exists$ and vice versa) gives:

Focus

$\lim_{x\to a} f(x) \neq L$ means: *there exists* some $\varepsilon>0$ such that *for every* $\delta>0$ there is at least one x with $0<|x-a|<\delta$ but $|f(x)-L|\geq\varepsilon$.
In words: there is one bad band the Prover can never fit into, no matter how thin a strip they try – every strip leaks at least one point of the graph out of that band.

We will put this to work in Section 6.7 to show that $\lim_{x\to 0}\sin(1/x)$ does not exist: there is a fixed band that every strip, however thin, fails to contain because the graph oscillates forever.

6.4 Worked epsilon–delta proofs: linear functions

Time to prove something. The structure of every epsilon-delta proof is the same, and once you see it twice it becomes a template:

1. **Scratch work (find δ).** Start from the goal $|f(x)-L|<\varepsilon$ and work *backwards*, unraveling it into a condition on $|x-a|$. This tells you what δ to pick. This step is exploration; it does not go in the final proof.
2. **The proof (verify δ works).** Announce the δ you found. Then, starting from $0<|x-a|<\delta$, run the algebra *forward* to reach $|f(x)-L|<\varepsilon$.

The scratch work is where the thinking happens; the proof is where you check it. Let me show both halves out loud the first time, then tighten up.

Example 6.1 (A linear limit)**.** Prove that $\lim_{x\to 2}(3x+1)=7$.

Scratch work. The output gap is

$$|f(x)-L| = |(3x+1)-7| = |3x-6| = 3|x-2|.$$

There it is: the output distance is exactly 3 times the input distance. We want $3|x-2| < \varepsilon$, i.e. $|x-2| < \varepsilon/3$. So if the Skeptic hands us ε, we should respond $\delta = \varepsilon/3$. (Now we know the answer; the proof below pretends we are seeing it for the first time and just verifies it.)

Proof. Let $\varepsilon > 0$ be given. Choose $\delta = \varepsilon/3$. Suppose $0 < |x-2| < \delta$. Then

$$|(3x+1)-7| = 3|x-2| < 3\delta = 3 \cdot \frac{\varepsilon}{3} = \varepsilon.$$

Thus $0 < |x-2| < \delta$ implies $|(3x+1)-7| < \varepsilon$. Since $\varepsilon > 0$ was arbitrary, $\lim_{x\to 2}(3x+1) = 7$. ■

Notice how short the actual proof is, and how completely the scratch work did the work. The Skeptic-to-Prover rule here is the function $\delta = \varepsilon/3$: hand it any tolerance and it returns a closeness that provably works.

Why the technique works

Why does $\delta = \varepsilon/3$ work, and why *that* number? Because a line of slope 3 magnifies horizontal distances into vertical ones by exactly the factor 3. If you want the vertical wandering held under ε, you must hold the horizontal wandering under $\varepsilon/3$ – you "pre-divide" by the magnification. For a general line $f(x) = mx + b$ with $m \neq 0$, the same reasoning gives $\delta = \varepsilon/|m|$: the steeper the line, the tighter you must clamp the input. This is your first glimpse of a theme that runs through all of calculus – *the local rate of change controls how output error responds to input error.* The number $|m|$ is, of course, the derivative of the line. We are already computing with derivatives; we just have not named them yet.

Work it yourself

Carry out the whole template yourself for $\lim_{x\to -1}(5-2x) = 7$. (Scratch: $|(5-2x)-7| = |-2x-2| = 2|x+1|$, so take $\delta = \varepsilon/2$.) Then do $\lim_{x\to 3}(7) = 7$ for the constant function and watch something curious happen: the output gap is $|7-7| = 0 < \varepsilon$ *automatically*, so *any* δ works – say $\delta = 1$. The Prover never even has to look at ε. A constant function is the one case where the Prover can answer before the Skeptic finishes speaking.

6.5 Worked epsilon–delta proofs: a quadratic

Lines are easy because the output gap factors cleanly as a constant times the input gap. For a curve, the magnification is not constant – it depends on where you are – and that forces one new idea: *control the wild factor first.*

Example 6.2 (A quadratic limit)**.** Prove that $\lim_{x\to 3} x^2 = 9$.

Scratch work. The output gap is

$$|x^2 - 9| = |(x-3)(x+3)| = |x-3|\,|x+3|.$$

One factor, $|x-3|$, is the input gap – small, under our control via δ. The other factor, $|x+3|$, is the troublemaker: near $x = 3$ it is about 6, but it is not constant, and we cannot let it run away. The fix is the standard two-step move.

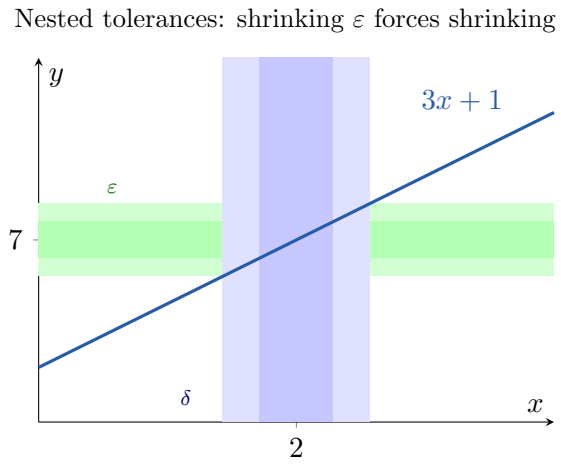


Figure 6.3: For the line $3x + 1$ at $a = 2$: shrink the green band (smaller ε) and the matching blue strip shrinks in lockstep, by the factor $\delta = \varepsilon/3$. The nested bands and strips make the rule $\varepsilon \mapsto \varepsilon/3$ visible.

Step one: tame the wild factor with a preliminary clamp. Decide in advance that δ will be *at most* 1. If $|x - 3| < 1$, then $2 < x < 4$, so

$$5 < x + 3 < 7, \quad \text{hence} \quad |x + 3| < 7.$$

We have boxed the troublemaker: as long as we keep x within 1 of 3, the factor $|x + 3|$ can never exceed 7. (Any correct upper bound would do; 7 is convenient.)

Step two: use the bound to size δ. Now, whenever $|x - 3| < 1$,

$$|x^2 - 9| = |x - 3|\,|x + 3| < 7\,|x - 3|.$$

To force this under ε it suffices to have $|x - 3| < \varepsilon/7$. We need *both* clamps at once – $|x - 3| < 1$ to validate the bound, and $|x - 3| < \varepsilon/7$ to land in the band – so we take the smaller of the two:

$$\delta = \min\Big(1,\ \frac{\varepsilon}{7}\Big).$$

Proof. Let $\varepsilon > 0$ be given. Choose $\delta = \min(1, \varepsilon/7)$. Suppose $0 < |x - 3| < \delta$. Since $\delta \le 1$, we have $|x - 3| < 1$, so $2 < x < 4$ and therefore $|x + 3| < 7$. Since also $\delta \le \varepsilon/7$, we have $|x - 3| < \varepsilon/7$. Combining,

$$|x^2 - 9| = |x - 3|\,|x + 3| < \frac{\varepsilon}{7} \cdot 7 = \varepsilon.$$

Hence $\lim_{x \to 3} x^2 = 9$. ■

Intuition

The min is not a trick; it is the natural answer to a genuine two-part demand. One part of the requirement ("keep the slope factor bounded") sets a ceiling on δ; the other part ("land inside the ε band") sets another. To satisfy both promises simultaneously you take the tighter one. Whenever you see $\delta = \min(\dots, \dots)$ in an epsilon-delta proof, read it as "I made two promises and I am keeping both."

Why the technique works

Why does the curve need a min but the line did not? Because the line's magnification factor was the *constant* 3 – there was nothing wild to tame, so one clamp sufficed. For x^2 the magnification near 3 is about 6 (the slope of x^2 at $x = 3$), but it varies as x moves, so we first pen x into a small region where we can put a hard ceiling on that varying factor, then shrink within the pen to hit the band. The deep moral: *the controllable input gap times a bounded "amplifier" gives a controllable output gap.* Every limit-of-a-product argument, including the product limit law below, is this same "small times bounded is small" idea wearing different clothes.

Common trap

Do not forget to bound the wild factor. A common wrong "proof" of $\lim_{x\to 3} x^2 = 9$ writes $|x^2 - 9| = |x - 3||x + 3|$ and then sets $\delta = \varepsilon/|x + 3|$. But δ is not allowed to depend on x – x is the variable we are quantifying over, not a fixed number you may plug into your choice of δ. The Prover must commit to a single δ depending only on ε (and on the fixed data a), *before* any particular x is examined. The preliminary clamp $|x - 3| < 1$ exists precisely so that we can replace the forbidden, x-dependent $|x + 3|$ by the fixed constant 7.

Work it yourself

Prove $\lim\limits_{x\to 2} x^2 = 4$ on your own. Clamp $|x - 2| < 1$ so that $1 < x < 3$ and $|x + 2| < 5$, then take $\delta = \min(1, \varepsilon/5)$. Compare with the $x = 3$ case: the amplifier ceiling changed from 7 to 5 because you are sitting at a less steep part of the parabola. The δ you are allowed to use literally tracks the steepness of the curve at the point – another preview of the derivative.

Let us record the general lesson as a small lemma, because it is the engine of all the later laws.

Lemma 6.1 (Small times bounded is small)**.** *Suppose that on some punctured interval around a we can write $f(x) - L = g(x)\,h(x)$, where $|g(x)| \le M$ for some constant $M > 0$ on that interval, and $\lim_{x\to a} h(x) = 0$. Then $\lim_{x\to a} f(x) = L$.*

Proof. Let $\varepsilon > 0$. Since $\lim_{x\to a} h(x) = 0$, there is $\delta_1 > 0$ with $0 < |x - a| < \delta_1 \Rightarrow |h(x)| < \varepsilon/M$. Let $\delta_2 > 0$ be a radius small enough that the bound $|g(x)| \le M$ holds for $0 < |x - a| < \delta_2$. Put $\delta = \min(\delta_1, \delta_2)$. Then for $0 < |x - a| < \delta$,

$$|f(x) - L| = |g(x)|\,|h(x)| < M \cdot \frac{\varepsilon}{M} = \varepsilon. \qquad \square$$

The quadratic proof was exactly this lemma with $g(x) = x + 3$ (bounded by $M = 7$ near 3) and $h(x) = x - 3$ (heading to 0). Hold this pattern; it returns.

6.6 The limit laws, and why they are theorems

In practice nobody computes $\lim_{x\to 3}(x^2 + 5x)$ from scratch with epsilon and delta. Instead they say: the limit of a sum is the sum of the limits, the limit of a product is the product of the limits, and so on – so the answer is $9 + 15 = 24$. These *limit laws* are what make limits usable. But they are not

axioms. Each one is a *theorem* that must be *earned* from Definition 6.1. The intuition (errors add, small times bounded is small) tells you the laws should be true; the epsilon-delta machine is what *makes* them true. Let us prove the two central ones and see the intuition become rigor.

Assume throughout that $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$.

Theorem 6.1 (Sum law)**.** $\lim\limits_{x\to a} \big(f(x) + g(x)\big) = L + M.$

Intuition

The idea before the proof. The total error of the sum, $|(f+g) - (L+M)|$, is at most the error of f plus the error of g (triangle inequality). If you want the *combined* error under ε, just demand each piece be under $\varepsilon/2$. Halve the tolerance, hand half to each function, collect their separate δ's, and use the tighter one. "Split the budget" is the entire strategy.

Proof. Let $\varepsilon > 0$. Apply the hypotheses with the tolerance $\varepsilon/2$:

- Since $\lim_{x\to a} f = L$, there is $\delta_1 > 0$ with $0 < |x-a| < \delta_1 \Rightarrow |f(x) - L| < \varepsilon/2$.
- Since $\lim_{x\to a} g = M$, there is $\delta_2 > 0$ with $0 < |x-a| < \delta_2 \Rightarrow |g(x) - M| < \varepsilon/2$.

Let $\delta = \min(\delta_1, \delta_2)$. For $0 < |x-a| < \delta$ *both* bounds hold at once, so by the triangle inequality

$$\big|\big(f(x)+g(x)\big) - (L+M)\big| = \big|\big(f(x)-L\big) + \big(g(x)-M\big)\big| \le |f(x)-L| + |g(x)-M| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Therefore $\lim_{x\to a}(f+g) = L + M$. □

Common trap

Why $\varepsilon/2$ and not ε? If you only forced each piece under ε, the triangle inequality would give you $|(f+g) - (L+M)| < \varepsilon + \varepsilon = 2\varepsilon$ – twice the target, a miss. Splitting the budget in half is what makes the two halves *add up* to exactly ε. The choice $\varepsilon/2$ is reverse-engineered from the triangle inequality's "+". This budget-splitting is one of the most reused tactics in all of analysis; it will reappear as "$\varepsilon/3$ tricks" and "ε/n tricks" whenever finitely many errors must be controlled together.

Theorem 6.2 (Product law)**.** $\lim\limits_{x\to a} \big(f(x)\, g(x)\big) = L\, M.$

Intuition

Products are subtler than sums because the errors interact. Write $f = L + u$ and $g = M + v$, where $u = f - L$ and $v = g - M$ are the (small) errors. Then

$$fg = (L+u)(M+v) = LM + Lv + Mu + uv,$$

so the product's error is

$$fg - LM = \underbrace{Lv}_{\text{const}\times\text{small}} + \underbrace{Mu}_{\text{const}\times\text{small}} + \underbrace{uv}_{\text{small}\times\text{small}}.$$

Every one of the three pieces is small: the first two are a fixed constant times something heading to 0, and the third is even smaller, a product of two vanishing errors. Make each of

the three under $\varepsilon/3$ and add. That is the proof; the rest is bookkeeping to turn "small" into actual inequalities.

Proof. Let $\varepsilon > 0$. We control the three pieces $|Lv|$, $|Mu|$, $|uv|$, where $u = f(x) - L$ and $v = g(x) - M$, aiming each below $\varepsilon/3$.

The cross term $|uv|$. Pick $\delta_1, \delta_2 > 0$ so that $|u| = |f(x) - L| < \sqrt{\varepsilon/3}$ for $0 < |x - a| < \delta_1$ and $|v| = |g(x) - M| < \sqrt{\varepsilon/3}$ for $0 < |x - a| < \delta_2$. Then $|uv| < \varepsilon/3$.

The term $|Mu|$. Pick $\delta_3 > 0$ so that $|u| < \dfrac{\varepsilon}{3(|M| + 1)}$ for $0 < |x - a| < \delta_3$. Then $|Mu| \le |M|\,|u| \le \dfrac{|M|}{|M| + 1} \cdot \dfrac{\varepsilon}{3} < \dfrac{\varepsilon}{3}$. (We divide by $|M| + 1$ rather than $|M|$ so the bound is safe even when $M = 0$.)

The term $|Lv|$. Symmetrically, pick $\delta_4 > 0$ so that $|v| < \dfrac{\varepsilon}{3(|L| + 1)}$ for $0 < |x - a| < \delta_4$, giving $|Lv| < \varepsilon/3$.

Now set $\delta = \min(\delta_1, \delta_2, \delta_3, \delta_4)$. For $0 < |x - a| < \delta$ all four conditions hold simultaneously, so by the triangle inequality

$$|f(x)g(x) - LM| = |Lv + Mu + uv| \le |Lv| + |Mu| + |uv| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

Hence $\lim_{x \to a}(fg) = LM$. □

Remark 6.1. The proof above is the direct, self-contained one, and it shows clearly where each term goes. A slicker route uses Lemma 6.1: since f has a limit it is bounded near a, and $fg - LM = f\,(g - M) + M\,(f - L)$ is then "bounded times small plus constant times small," each piece $\to 0$. Both proofs are really the same observation – *small errors, multiplied by quantities that stay bounded, stay small* – which is why we isolated that lemma earlier.

Why this matters

Look at what just happened. The sum law and product law – the rules you will use thousands of times to evaluate limits – are not separate facts to memorize. They are both consequences of two elementary truths about distance: the triangle inequality ($|p+q| \le |p| + |q|$), and "small times bounded is small." The entire arithmetic of limits rests on those two pillars. When a subject's working rules all collapse into two short principles, you have found bedrock. That is the kind of understanding this book is after: not a longer list of rules, but the short reason underneath the list.

For completeness, here are the remaining standard laws. The constant-multiple and difference laws are immediate from the two above (take g constant, or replace g by $-g$). The quotient law needs one extra idea – keeping the denominator away from zero – which we state and sketch.

Theorem 6.3 (The limit laws)**.** *If $\lim_{x \to a} f(x) = L$ and $\lim_{x \to a} g(x) = M$, then:*

1. *(Constant multiple)* $\lim\limits_{x \to a} c\,f(x) = cL$ *for any constant c.*
2. *(Sum/difference)* $\lim\limits_{x \to a} \big(f(x) \pm g(x)\big) = L \pm M$.
3. *(Product)* $\lim\limits_{x \to a} f(x)g(x) = LM$.

4. *(Quotient)* $\lim_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{L}{M}$, *provided* $M \neq 0$.

Sketch of the quotient law. It is enough, given the product law, to handle $\lim_{x\to a} \frac{1}{g(x)} = \frac{1}{M}$ when $M \neq 0$; then $\frac{f}{g} = f \cdot \frac{1}{g}$ follows. Compute the error:

$$\left|\frac{1}{g(x)} - \frac{1}{M}\right| = \frac{|M - g(x)|}{|g(x)|\,|M|}.$$

The numerator $|M - g(x)| \to 0$ – that is small. The danger is the $|g(x)|$ in the denominator: if it could be near 0, the fraction could blow up. So we first secure the denominator. Since $M \neq 0$, apply the limit of g with tolerance $|M|/2$: there is a radius within which $|g(x)| > |M|/2 > 0$. On that region the fraction is at most $\frac{2}{|M|^2}\,|M - g(x)|$ – once again "small times bounded" – and shrinking the radius further forces it under any ε. The structure is identical to the quadratic proof: clamp first to control the dangerous factor, then shrink to hit the band. □

Connection

The limit laws say the map "$f \mapsto \lim_{x\to a} f$" respects addition and multiplication: it sends sums to sums and products to products. In the language of later mathematics, taking the limit at a is a *homomorphism* from the algebra of functions having a limit at a into $\mathbb{R}$. The same structural fact – "the operation is linear and multiplicative" – recurs for the derivative (linear, with a product rule) and the integral (linear). Spotting which operations preserve which structures is one of the organizing instincts of all higher mathematics, and limits are where it starts.

6.7 When the game cannot be won: limits that do not exist

The definition's real power is that it can also certify *failure*. To show a limit does *not* exist, we play the Skeptic: we exhibit one tolerance ε that defeats every possible δ.

Example 6.3 (A jump)**.** The Heaviside step $H(x) = 0$ for $x < 0$ and $H(x) = 1$ for $x \geq 0$ has no limit at 0.

Argument. Suppose, for contradiction, some L were the limit. Take the Skeptic's tolerance $\varepsilon = \frac{1}{2}$. For *any* $\delta > 0$, the punctured interval $(-\delta, \delta) \setminus \{0\}$ contains both negative points (where $H = 0$) and positive points (where $H = 1$). For the promise to hold we would need both $|0 - L| < \frac{1}{2}$ and $|1 - L| < \frac{1}{2}$, i.e. L within $\frac{1}{2}$ of *both* 0 and 1. But 0 and 1 are a distance 1 apart, and no single point is within $\frac{1}{2}$ of both. So $\varepsilon = \frac{1}{2}$ defeats every δ; the limit cannot exist. ■

Example 6.4 (Infinite oscillation)**.** $\lim_{x\to 0} \sin\dfrac{1}{x}$ does not exist.

As $x \to 0$, the input $1/x$ races off to infinity, so $\sin(1/x)$ runs through its full range $[-1, 1]$ infinitely often – it never settles. No matter how thin a strip $(-\delta, \delta)$ the Prover proposes, that strip still contains inputs where $\sin(1/x) = 1$ and inputs where $\sin(1/x) = -1$ (because $1/x$ sweeps through infinitely many periods). With the Skeptic's tolerance $\varepsilon = 1$, no candidate L can be within 1 of both $+1$ and -1 at once. Every δ leaks; the limit fails to exist. Figure 6.5 shows the graph refusing to settle.

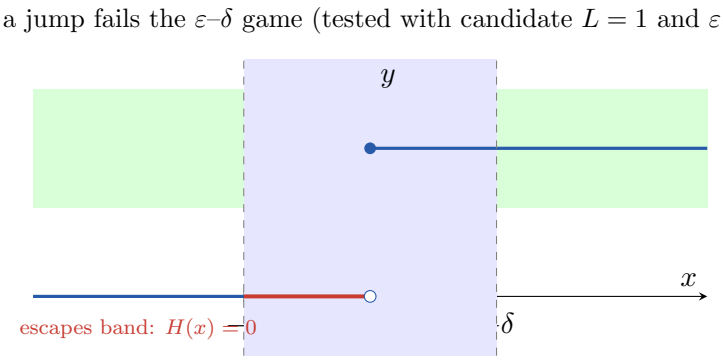


Figure 6.4: Negation of the limit at a jump. Testing candidate $L = 1$ with $\varepsilon = 0.4$ creates a horizontal band $[0.6, 1.4]$. Any proposed vertical strip $(-\delta, \delta)$ inevitably contains negative inputs (red segment) where $H(x) = 0$, lying entirely outside the band. No δ can ever succeed.

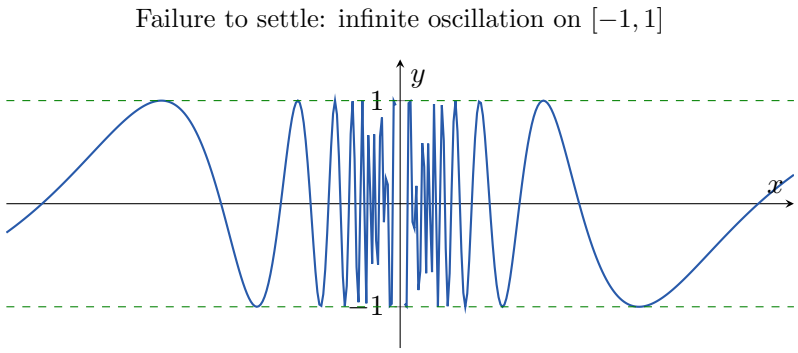


Figure 6.5: $y = \sin(1/x)$ near 0. The oscillations bunch up infinitely as $x \to 0$, forever sweeping the whole band from -1 to 1. No vertical strip around 0, however thin, can trap the graph inside any band thinner than the full height 2, so no limit exists. (Pixels cannot draw the infinitely many wiggles closest to 0; imagine them packed ever tighter.)

Connection

The jump in H fails for a "two-sided disagreement" reason: the left approach wants $L = 0$, the right wants $L = 1$. This motivates *one-sided limits*, written $\lim_{x\to a^-}$ and $\lim_{x\to a^+}$, where the strip is only the left or right half-interval. A two-sided limit exists exactly when both one-sided limits exist and agree – which they do not for H. We develop one-sided limits and the closely related idea of continuity in the chapters just ahead; here, note only that the epsilon-delta frame absorbs them with a single change (replace the symmetric strip by a half-strip).

6.8 Making the intuition and the formalism agree

We end where the chapter began: with the relationship between the picture and the sentence. They had better agree – a definition that contradicted the clear cases would be a bad definition, not a brave one. Check the handshake from both sides.

Formalism honors intuition. Every limit you would compute "by inspection" – $\lim_{x\to 2}(3x+1) = 7$, $\lim_{x\to 3} x^2 = 9$, $\lim_{x\to a} c = c$ – comes out the expected way under Definition 6.1, as we proved or sketched. The definition does not overturn your instincts for the easy cases; it ratifies them. What it adds is a referee for the *hard* cases, where instinct stalls: removable holes like $\frac{x^2-1}{x-1}$, jumps like H, oscillations like $\sin(1/x)$. There the picture alone gives no confident answer, and the definition decides.

Intuition tames the formalism. Conversely, you should never run the epsilon-delta machine blind. Before writing a single δ, look at the picture and *predict* the answer; the formal proof is then a verification, not a search in the dark. And the scratch-work phase – working backward from $|f(x) - L| < \varepsilon$ to find δ – is just the picture turned into algebra: "how tight a strip keeps the graph in this band?" The two are one activity in two notations.

What is really happening

Here is the cleanest way to hold the whole chapter in one thought. The intuitive limit and the epsilon-delta limit are not two different limits that happen to agree; they are the *same* statement told at two resolutions. "$f(x)$ approaches L" is the statement at human resolution – fast, visual, occasionally wrong at the edges. "$\forall\varepsilon\,\exists\delta\ldots$" is the identical statement at full resolution – slow, symbolic, never wrong. You learn both because you think in the first and prove in the second, and a working mathematician fluidly translates between them mid-sentence. When that translation becomes automatic, the epsilon-delta definition stops being an obstacle and becomes what it truly is: the exact written form of something you already see.

Walking question

A question to turn over on a walk. We insisted the order of quantifiers is "$\forall\varepsilon\,\exists\delta$." Imagine a stricter notion in which a single δ must work for *all* ε simultaneously: $\exists\delta\,\forall\varepsilon$. Which functions, if any, satisfy *that*? (Hint: it would force $|f(x) - L| < \varepsilon$ for every $\varepsilon > 0$ on a fixed punctured interval, hence $f(x) = L$ there.) You will discover that the strict version describes only locally-constant functions – which is exactly why the ordinary definition uses the looser, correct order. The grammar is doing real work.

6.9 Exercises

Exercise 6.1 (Linear, by the template)**.** Using the two-step template (scratch work, then proof), prove from Definition 6.1 that $\lim_{x\to 4}(2x-3)=5$. Identify the rule $\varepsilon \mapsto \delta$ explicitly and relate the constant in it to the slope of the line.

Exercise 6.2 (Negative slope)**.** Prove that $\lim_{x\to 1}(8-5x)=3$. Be careful that a negative slope does not change the size of the magnification: explain why $\delta=\varepsilon/5$ (not $\varepsilon/(-5)$) is the right response, and where the absolute value enters.

Exercise 6.3 (Quadratic with a clamp)**.** Prove that $\lim_{x\to 4} x^2 = 16$. State your preliminary clamp, derive the resulting ceiling on $|x+4|$, and write $\delta=\min(1,\varepsilon/9)$ (or justify a different constant if you clamp differently).

Exercise 6.4 (A reciprocal)**.** Prove that $\lim_{x\to 2}\dfrac{1}{x}=\dfrac{1}{2}$ directly from the definition. (Compute $|\frac{1}{x}-\frac{1}{2}| = \frac{|x-2|}{2|x|}$, clamp $|x-2|<1$ so that $1<x<3$ and hence $|x|>1$, then choose $\delta=\min(1,2\varepsilon)$.) Point out exactly where you used the clamp to keep the denominator away from 0.

Exercise 6.5 (Negating the definition)**.** Write out, with quantifiers in the correct order, the precise statement "$\lim_{x\to a} f(x)\neq L$." Then use it to prove that $\lim_{x\to 0}\dfrac{|x|}{x}$ does not exist. (Take $\varepsilon=1$ and show every δ contains points where the function is $+1$ and points where it is -1.)

Exercise 6.6 (The $\varepsilon/2$ budget, examined)**.** In the proof of the sum law we chose tolerance $\varepsilon/2$ for each function. **(a)** Redo the final inequality using $\varepsilon/3$ for each piece and confirm it still works (you land at $2\varepsilon/3<\varepsilon$). **(b)** Show that using ε for each piece *fails* to prove the claim, and pinpoint the exact step where the argument breaks. **(c)** Conclude that any split summing to at most ε works, and that $\varepsilon/2$ is simply the tidiest.

Exercise 6.7 (Constant multiple law)**.** Prove from the definition that $\lim_{x\to a} c\,f(x) = cL$ whenever $\lim_{x\to a} f(x)=L$. Treat $c=0$ separately and explain why that case needs no hypothesis on f at all.

Exercise 6.8 (Small times bounded, applied)**.** Use Lemma 6.1 to prove $\lim_{x\to 0} x\sin\dfrac{1}{x}=0$ even though $\sin\frac{1}{x}$ has no limit at 0. Identify g and h in the lemma, and the bound M. (This is a striking case: a wildly oscillating factor, multiplied by something heading to 0, is tamed completely.)

Exercise 6.9 (A removable hole)**.** Let $f(x)=\dfrac{x^2-4}{x-2}$ for $x\neq 2$ and $f(2)=100$. Prove that $\lim_{x\to 2} f(x)=4$ directly from the definition, and explain in one sentence why the absurd value $f(2)=100$ is invisible to the limit. Which inequality in Definition 6.1 is responsible for that invisibility?

Exercise 6.10 (Conceptual: order of quantifiers, proof)**.** Prove that if a single $\delta>0$ satisfied the condition "$0<|x-a|<\delta \Rightarrow |f(x)-L|<\varepsilon$" for *every* $\varepsilon>0$ at once, then $f(x)=L$ for all x in the punctured interval $0<|x-a|<\delta$. Conclude that the order $\forall\varepsilon\,\exists\delta$ in the definition is essential and cannot be reversed without trivializing it.

Exercise 6.11 (Proof: uniqueness of the limit)**.** Prove that a limit, if it exists, is unique: if $\lim_{x\to a} f(x)=L$ and $\lim_{x\to a} f(x)=L'$, then $L=L'$. (Suppose $L\neq L'$ and let $\varepsilon=|L-L'|/2>0$. Use the definition for each limit to trap $f(x)$ within ε of both L and L' simultaneously on a small punctured interval, and derive a contradiction from the triangle inequality.) This is the theorem that licenses the phrase "*the* limit."

Chapter 7

Computing Limits

In the last two chapters you learned what a limit *is*. You met the sharp-edged definition — for every challenge ε there is a response δ — and you saw that this machinery exists to pin down a single idea: *the value a function approaches as the input approaches a point, regardless of what happens at the point itself.* That was the foundation. It was also slow. Nobody wants to run an epsilon-delta argument every time they need a limit, any more than you want to prove the distributive law every time you multiply two numbers.

This chapter is about *fluency*. We will build a small toolbox — substitution, factoring, rationalizing, the squeeze theorem, one-sided limits, absolute values — that lets you compute almost any limit you will meet, quickly and correctly. But this is not a bag of tricks to memorize. Every tool here is *licensed* by something we already proved: the limit laws. Each technique is just a clever, legal rearrangement that turns a hard-looking limit into one you can read off by inspection. The whole art is this:

Focus

Computing a limit means **transforming the expression — using only moves the limit laws permit — until you reach a form where the answer is obvious.** The skill is not in the final step. It is in seeing which legal move clears the obstacle.

We start by recalling the laws, because everything else stands on them.

7.1 The Limit Laws: Our License to Compute

Suppose $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$, where L and M are finite real numbers. In the epsilon-delta chapter we proved the following facts, which drive everything that follows.

Theorem 7.1 (Limit Laws)**.** *With the hypotheses above, and c a constant:*

1. ***(Constant)*** $\lim_{x\to a} c = c$.
2. ***(Identity)*** $\lim_{x\to a} x = a$.
3. ***(Sum/Difference)*** $\lim_{x\to a} \big(f(x) \pm g(x)\big) = L \pm M$.

4. ***(Constant multiple)*** $\lim_{x\to a} c\,f(x) = c\,L$.
5. ***(Product)*** $\lim_{x\to a} f(x)\,g(x) = L\,M$.
6. ***(Quotient)*** $\lim_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{L}{M}$, provided $M \neq 0$.
7. ***(Power)*** $\lim_{x\to a} \big(f(x)\big)^n = L^n$ *for any positive integer* n, *and* $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{L}$ *when the root is defined.*

Read these slowly. They say something striking: *the limit of a combination is the combination of the limits.* Addition, multiplication, division — the limit passes straight through them. This is exactly why calculus is computable at all. If limits did not respect arithmetic, every single problem would be a fresh epsilon-delta battle.

Why the technique works

Why are we allowed to "pass the limit inside"? The intuition is short. If $f(x)$ is squeezing in on L and $g(x)$ is squeezing in on M, then their sum $f(x)+g(x)$ has no choice but to squeeze in on $L+M$: push f within half your tolerance of L and g within half your tolerance of M, and the sum lands within your full tolerance of $L+M$. That "split the tolerance in two" move is the whole proof of the sum law, and the product and quotient laws are the same spirit with a little more bookkeeping. The laws are not axioms we hope are true; they are theorems we earned with ε and δ. Having earned them, we may now use them freely and never look back.

There is one boundary on this freedom, and it is the single most important caveat in the chapter. **The quotient law requires $M \neq 0$.** The moment a denominator's limit is zero, the law goes silent — it tells you nothing. That silence is exactly where all the interesting work of this chapter lives. Most of our techniques are tricks for handling a $0/0$ that the quotient law refuses to touch.

Common trap

A limit law that does not apply is not a law that gives the answer 0, or ∞, or "undefined." It simply *does not apply.* When you hit $\frac{0}{0}$ you have not found an answer — you have found a question. The expression is called an *indeterminate form* precisely because the laws determine nothing. You must transform it first.

7.2 Direct Substitution: The First Thing to Try

Here is the happiest fact in limit computation. For the functions you build out of the everyday pieces — polynomials, roots, sines, cosines, exponentials, logs, and any sums, products, quotients, and compositions of these — the limit as $x \to a$ is just the value at a, *whenever that value makes sense.* You plug in. You are done.

Theorem 7.2 (Substitution for continuous functions)**.** *If f is continuous at a, then*

$$\lim_{x\to a} f(x) = f(a).$$

Every polynomial is continuous everywhere. Rational functions, roots, exp, log, sin, cos, *and their compositions are continuous everywhere they are defined.*

We will give continuity its own careful chapter. For now take the working definition: *f is continuous at a when the limit there equals the value there*, with no gap, no jump, and no hole. The theorem then looks almost circular — "the limit is the value when the limit is the value" — but the content is in the second sentence: *the standard building blocks really are continuous*, and that fact is itself a consequence of the limit laws.

Why the technique works

Why does substitution work for a polynomial? Watch the laws assemble it. Take $p(x) = 3x^2 - 5x + 1$ and $a = 2$. By the identity law $\lim x = 2$. By the power law $\lim x^2 = 2^2 = 4$. By the constant-multiple law $\lim 3x^2 = 12$ and $\lim 5x = 10$. By the sum/difference law the whole thing is $12 - 10 + 1 = 3$, which is exactly $p(2)$. Nothing here was a leap of faith — each step cited a law from Theorem 7.1. Substitution is not a shortcut that bypasses the laws; it is the laws, applied so routinely we stopped narrating them.

Example 7.1 (Substitution, straight ahead)**.**

$$\lim_{x \to 3} \frac{x^2 + 1}{x - 1} = \frac{3^2 + 1}{3 - 1} = \frac{10}{2} = 5.$$

The denominator's limit is $2 \neq 0$, so the quotient law applies and we simply evaluate. No drama.

Example 7.2 (Substitution through a composition)**.**

$$\lim_{x \to 0} \cos\big(e^x - 1\big) = \cos\big(e^0 - 1\big) = \cos(0) = 1.$$

The inside $e^x - 1 \to 0$, and cos is continuous at 0, so we evaluate through the composition.

Common trap

Substitution is the *first* thing to try, not the only thing. If plugging in gives a genuine number, you are finished. If it gives $\frac{0}{0}$, $\frac{\text{nonzero}}{0}$, $\frac{\infty}{\infty}$, or any other nonsense, **stop** — you have learned that the function is not continuous at a (or not even defined there), and you must reach for a different tool. Substitution failing is not failure; it is a diagnosis.

7.3 When Substitution Fails: The 0/0 Form

The interesting limits — the ones calculus actually needs — almost always arrive as $\frac{0}{0}$, for a structural reason. The derivative itself, which we are marching toward, is

$$\lim_{h \to 0} \frac{f(x + h) - f(x)}{h},$$

and as $h \to 0$ the top goes to $f(x) - f(x) = 0$ while the bottom goes to 0. *Every derivative is a 0/0 limit.* So learning to handle 0/0 is not a sideshow; it is the central engine of differential calculus, rehearsed here on simpler functions.

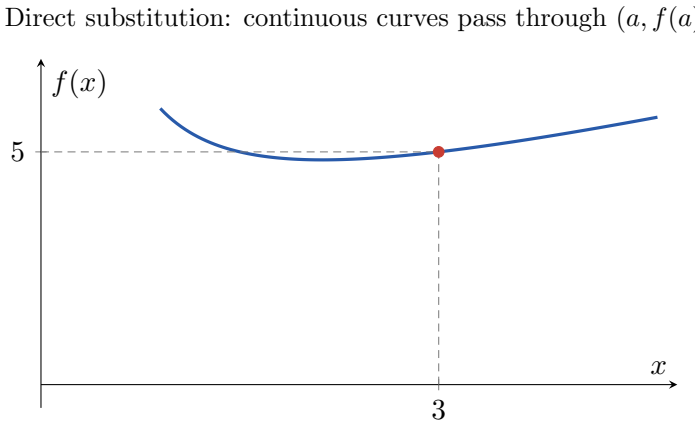


Figure 7.1: For $\dfrac{x^2+1}{x-1}$ near $x = 3$ the curve is smooth and unbroken. The limit is simply the height of the curve there: 5. This is the easy case — substitution and nothing more.

What is really happening

What is really happening in a $\frac{0}{0}$ limit? Both the numerator and the denominator are shrinking to zero, and the limit measures *how fast each shrinks compared to the other*. If the top heads to zero twice as fast as the bottom, the ratio approaches 2. If they vanish at exactly the same rate, you get a nonzero number; if the top wins, you get 0; if the bottom wins, you get a blow-up. The symbol $\frac{0}{0}$ hides all of this — it is "indeterminate" precisely because the bare zeros throw away the rate information. Our algebra tricks are nothing but ways to *recover the hidden rate* by canceling the common factor that is causing both to vanish.

The key structural fact is this. When a rational expression gives $\frac{0}{0}$ at $x = a$, it is because $(x - a)$ is a factor of *both* the numerator and the denominator. That shared factor is the culprit: it is what drives both to zero. Cancel it, and the limit becomes visible.

Why the technique works

Why is canceling a factor legitimate when the factor is zero in the limit? Here is the subtle, beautiful point. The functions

$$\frac{x^2-9}{x-3} \qquad \text{and} \qquad x+3$$

are *not the same function*: the first is undefined at $x = 3$, the second is perfectly happy there. But a limit as $x \to 3$ **never looks at** $x = 3$ **itself** — only at the values *near* it. For every $x \neq 3$ the two expressions are identical, because dividing $x^2 - 9 = (x-3)(x+3)$ by $(x-3)$ is legal whenever $x - 3 \neq 0$. Since the limit ignores the single point $x = 3$, the two functions have the same limit there. The cancellation is licensed not by the quotient law (which is silent) but by the very *definition* of a limit: limits care about the neighborhood, not the point. This is the deepest "why" in the chapter, so reread it until it feels obvious.

Canceling the zero-factor: exposing the continuous line

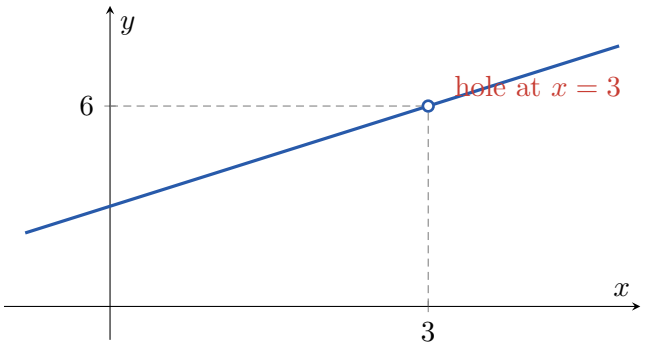


Figure 7.2: The graph of $\dfrac{x^2-9}{x-3}$ is the straight line $y = x + 3$ with a single *puncture* at $x = 3$. The function is undefined there, but the limit reads the height the line is heading for: 6. The missing point is invisible to the limit.

7.3.1 Technique 1: Factor and Cancel

The recipe writes itself: when substitution gives $\frac{0}{0}$ in a rational expression, factor top and bottom, cancel the common $(x - a)$, then substitute into what remains.

Example 7.3 (The canonical cancellation)**.**

$$\lim_{x\to 3} \frac{x^2-9}{x-3}.$$

Substitution gives $\frac{0}{0}$. Factor the numerator:

$$\frac{x^2-9}{x-3} = \frac{(x-3)(x+3)}{x-3} = x+3 \quad \text{for } x \neq 3.$$

Now substitution into $x + 3$ is legal: $\lim_{x\to 3}(x+3) = 6$.

Example 7.4 (Factoring a quadratic over a quadratic)**.**

$$\lim_{x\to 2} \frac{x^2-x-2}{x^2-4}.$$

At $x = 2$ both top and bottom vanish, so $(x - 2)$ divides both. Factor:

$$\frac{x^2-x-2}{x^2-4} = \frac{(x-2)(x+1)}{(x-2)(x+2)} = \frac{x+1}{x+2} \quad (x \neq 2).$$

Substitute: $\lim_{x\to 2} \dfrac{x+1}{x+2} = \dfrac{3}{4}$.

Example 7.5 (A cube; remember the factoring patterns)**.**

$$\lim_{x\to 1} \frac{x^3-1}{x-1}.$$

Use $x^3 - 1 = (x - 1)(x^2 + x + 1)$:

$$\frac{x^3-1}{x-1} = x^2+x+1 \quad (x \neq 1), \qquad \lim_{x\to 1}(x^2+x+1) = 3.$$

Work it yourself

Try this one before reading on:

$$\lim_{x\to -2}\frac{x^2+5x+6}{x+2}.$$

Substitution gives $\frac{0}{0}$, so factor: $x^2+5x+6=(x+2)(x+3)$. Cancel the $(x+2)$ to get $x+3$, then substitute $x=-2$ to find the limit is 1. Did you also notice that the answer is the *slope-like* number you'd get from the rate idea? The numerator and denominator both vanish at $x=-2$, and the ratio of their vanishing rates is exactly 1.

7.3.2 Technique 2: Rationalize

Square roots block factoring. The cure is the *conjugate*. Recall the difference-of-squares identity $(\sqrt{A}-\sqrt{B})(\sqrt{A}+\sqrt{B})=A-B$: multiplying a root expression by its conjugate turns the roots into ordinary subtraction, which usually exposes the hidden $(x-a)$ factor.

Why the technique works

Why does multiplying by the conjugate help? A subtraction of roots, $\sqrt{x+1}-2$, is opaque — you cannot see what factor is making it vanish. But $(\sqrt{x+1}-2)(\sqrt{x+1}+2)=(x+1)-4=x-3$ is transparent: the root is gone and the vanishing factor $(x-3)$ stares back at you. We multiply top and bottom by the same conjugate, so we have multiplied by 1 and changed nothing — but we have *relocated* the difficulty from the unreadable numerator to a readable one. It is the conjugate's job to convert "difference of roots" into "difference of the things under the roots."

Example 7.6 (Rationalizing the numerator)**.**

$$\lim_{x\to 3}\frac{\sqrt{x+1}-2}{x-3}.$$

Substitution gives $\frac{0}{0}$ (the top is $\sqrt{4}-2=0$). Multiply top and bottom by the conjugate $\sqrt{x+1}+2$:

$$\frac{\sqrt{x+1}-2}{x-3}\cdot\frac{\sqrt{x+1}+2}{\sqrt{x+1}+2}=\frac{(x+1)-4}{(x-3)\big(\sqrt{x+1}+2\big)}=\frac{x-3}{(x-3)\big(\sqrt{x+1}+2\big)}.$$

Now the $(x-3)$ cancels:

$$=\frac{1}{\sqrt{x+1}+2}\quad(x\neq 3),\qquad\lim_{x\to 3}\frac{1}{\sqrt{x+1}+2}=\frac{1}{\sqrt{4}+2}=\frac{1}{4}.$$

Example 7.7 (Conjugate on the denominator)**.**

$$\lim_{x\to 0}\frac{x}{\sqrt{x+4}-2}.$$

Multiply by the conjugate $\sqrt{x+4}+2$ over itself:

$$\frac{x}{\sqrt{x+4}-2}\cdot\frac{\sqrt{x+4}+2}{\sqrt{x+4}+2}=\frac{x\big(\sqrt{x+4}+2\big)}{(x+4)-4}=\frac{x\big(\sqrt{x+4}+2\big)}{x}=\sqrt{x+4}+2\quad(x\neq 0).$$

Then $\lim_{x\to 0}\big(\sqrt{x+4}+2\big)=2+2=4$.

7.3.3 Technique 3: Clear Complex Fractions

A fraction buried inside a fraction looks fierce but yields to the oldest trick in algebra: combine the little fractions over a common denominator, then simplify. These appear constantly in derivative computations.

Example 7.8 (A fraction of fractions)**.**

$$\lim_{x\to 2} \frac{\dfrac{1}{x} - \dfrac{1}{2}}{x-2}.$$

Combine the numerator over the common denominator $2x$:

$$\frac{1}{x} - \frac{1}{2} = \frac{2-x}{2x}.$$

So the whole expression is

$$\frac{2-x}{2x} \cdot \frac{1}{x-2} = \frac{-(x-2)}{2x(x-2)} = \frac{-1}{2x} \quad (x \neq 2),$$

and $\lim_{x\to 2} \dfrac{-1}{2x} = \dfrac{-1}{4}$. Notice the sign flip from $2-x = -(x-2)$ — a classic place to lose a minus.

Common trap

The recurring danger in all three techniques is **sign errors** during factoring, especially $a-b = -(b-a)$. When you cancel, watch which way the factor points. A single dropped minus sign turns a correct answer into its negative, and the algebra will look perfectly clean while it lies to you.

7.4 The Squeeze Theorem

Factoring and rationalizing handle algebraic obstructions. But some limits — the ones involving wild oscillation, or transcendental functions like sin near 0 — resist algebra entirely. For these we need a different idea, and it happens to be one of the most intuitive in analysis.

Intuition

Imagine a small fish swimming between two larger fish, one above and one below. If the top fish and the bottom fish both swim toward the same doorway and meet there, the little fish, trapped between them, has no choice — it must go through that doorway too. The squeeze theorem is exactly this. If a function is trapped between two others that converge to a common limit, it is forced to that same limit, no matter how erratically it wiggles in between.

Theorem 7.3 (Squeeze Theorem)**.** *Suppose that for all x near a (except possibly at a itself)*

$$g(x) \leq f(x) \leq h(x),$$

and that

$$\lim_{x\to a} g(x) = \lim_{x\to a} h(x) = L.$$

Then $\lim_{x\to a} f(x) = L$ as well.

Why the technique works

Why is the squeeze theorem true? An epsilon-delta argument makes it airtight, but the idea is the picture. Fix a tolerance ε. Since $g \to L$, eventually $g(x) > L - \varepsilon$. Since $h \to L$, eventually $h(x) < L + \varepsilon$. Once both of these hold (take the smaller of the two δ's), the sandwich $g \le f \le h$ forces

$$L - \varepsilon < g(x) \le f(x) \le h(x) < L + \varepsilon,$$

so $f(x)$ is within ε of L. That is precisely the definition of $\lim f = L$. The bread does the work; f is just along for the ride.

Example 7.9 (Taming an oscillation)**.**

$$\lim_{x \to 0} x^2 \sin \frac{1}{x}.$$

Substitution is hopeless: $\sin(1/x)$ oscillates infinitely fast as $x \to 0$ and has no limit. But sin of anything lives in $[-1, 1]$, so

$$-1 \le \sin \frac{1}{x} \le 1 \quad \Longrightarrow \quad -x^2 \le x^2 \sin \frac{1}{x} \le x^2.$$

Both ends, $-x^2$ and x^2, go to 0 as $x \to 0$. Squeezed between them, so does our function:

$$\lim_{x \to 0} x^2 \sin \frac{1}{x} = 0.$$

The x^2 factor is a clamp that crushes the wild oscillation down to zero.

Squeeze theorem: bounding curves clamp wild oscillations

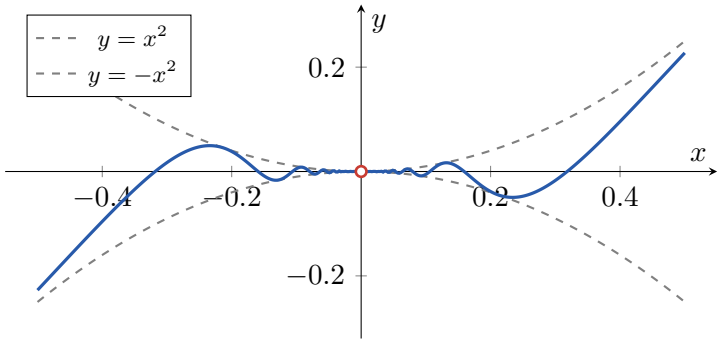


Figure 7.3: $y = x^2 \sin(1/x)$ oscillates faster and faster near 0, but it is pinned between the parabolas $y = \pm x^2$ (dashed). Since both clamps close on 0, the trapped curve is forced to 0.

7.4.1 The Crown Jewel: $\sin x / x \to 1$

We now use the squeeze theorem to prove the most important single limit in all of calculus:

$$\boxed{\lim_{x \to 0} \frac{\sin x}{x} = 1.}$$

This is the limit from which *every* derivative of a trig function flows. When you later learn that $\frac{\mathrm{d}}{\mathrm{d}x}(\sin x) = \cos x$, the engine inside that fact is this one limit. It is worth proving carefully.

Intuition

For a small angle x (in radians!), the arc of the unit circle and the straight chord beneath it are nearly the same length, and $\sin x$ is almost the arc itself. "Nearly" becomes "exactly in the limit." The claim $\sin x/x \to 1$ is the precise statement that, for tiny angles, *the sine of an angle is essentially the angle*. This is why radians matter: only in radians does arc length equal the angle, making the comparison clean.

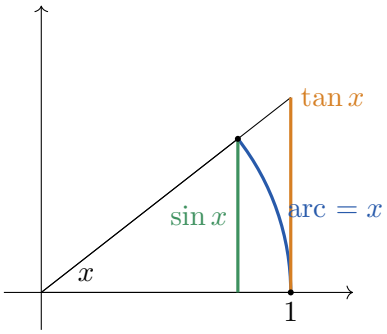


Figure 7.4: The unit circle, angle x in radians. The vertical green segment is $\sin x$, the bold blue arc has length x, and the orange segment up to the tangent line has length $\tan x$. Comparing the three areas gives $\sin x \le x \le \tan x$.

Focus

The proof, in one sentence: we trap x between $\sin x$ and $\tan x$ using three nested areas in the unit circle, divide through to isolate $\dfrac{\sin x}{x}$, and squeeze.

Setting up the inequality. Work in the unit circle (radius 1) and take a small positive angle x, with $0 < x < \frac{\pi}{2}$. Look at Figure 7.4. We compare three regions:

- the *small triangle* OAB with the chord, of area $\frac{1}{2}\sin x$;
- the *circular sector* OAB, of area $\frac{1}{2}x$;
- the *large triangle* OAT out to the tangent line, of area $\frac{1}{2}\tan x$.

Let us justify each area. The small triangle OAB has base $OA = 1$ along the x-axis and height equal to the vertical drop from B, which is $\sin x$; so its area is $\frac{1}{2} \cdot 1 \cdot \sin x = \frac{1}{2}\sin x$. The sector is the fraction $\frac{x}{2\pi}$ of the full disk of area $\pi(1)^2 = \pi$, giving area $\frac{x}{2\pi} \cdot \pi = \frac{1}{2}x$. The large triangle OAT has base $OA = 1$ and height $AT = \tan x$ (the tangent line meets the vertical at height $\tan x$), so its area is $\frac{1}{2}\tan x$.

Now read off the geometry: the small triangle sits *inside* the sector, which sits *inside* the large triangle. Hence

$$\underbrace{\tfrac{1}{2}\sin x}_{\text{triangle}} \;\le\; \underbrace{\tfrac{1}{2}x}_{\text{sector}} \;\le\; \underbrace{\tfrac{1}{2}\tan x}_{\text{triangle}}.$$

Multiply through by 2:

$$\sin x \;\le\; x \;\le\; \tan x = \frac{\sin x}{\cos x}.$$

Rearranging into a sandwich. Since $0 < x < \frac{\pi}{2}$ we have $\sin x > 0$, so we may divide the whole chain by $\sin x$ without flipping any inequalities:

$$1 \;\leq\; \frac{x}{\sin x} \;\leq\; \frac{1}{\cos x}.$$

Take reciprocals — which *does* reverse the inequalities, since all quantities are positive:

$$\cos x \;\leq\; \frac{\sin x}{x} \;\leq\; 1.$$

There it is: $\dfrac{\sin x}{x}$ is sandwiched between $\cos x$ and 1 (Figure 7.5).

Squeezing. As $x \to 0^+$, the lower bread $\cos x \to \cos 0 = 1$, and the upper bread is the constant 1. Both converge to 1, so by the squeeze theorem

$$\lim_{x \to 0^+} \frac{\sin x}{x} = 1.$$

The other side. We assumed $x > 0$. What about $x \to 0^-$? The function $\dfrac{\sin x}{x}$ is *even*: replacing x by $-x$ gives $\dfrac{\sin(-x)}{-x} = \dfrac{-\sin x}{-x} = \dfrac{\sin x}{x}$, unchanged. So the left-hand limit equals the right-hand limit. Both are 1, and therefore the two-sided limit is

$$\lim_{x \to 0} \frac{\sin x}{x} = 1. \qquad \blacksquare$$

The squeeze sandwich: $\cos x \leq \frac{\sin x}{x} \leq 1$ around $x = 0$

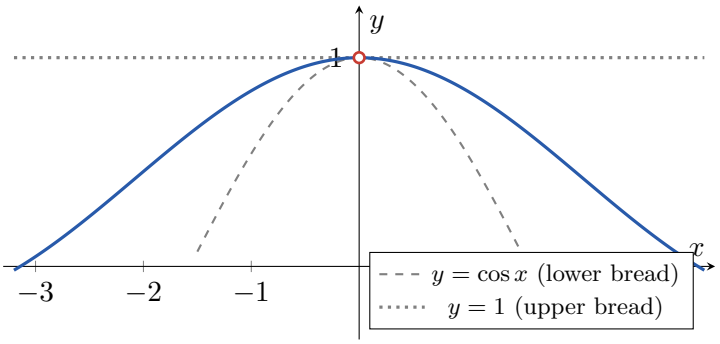


Figure 7.5: The function $y = \dfrac{\sin x}{x}$ (solid) is pinned between $y = \cos x$ (dashed, shown only near 0 where it is the lower bound) and the line $y = 1$ (dotted). As $x \to 0$ both breads close on 1, forcing the curve up to the punctured value 1. The point $x = 0$ is a removable hole: the formula is undefined there, but the limit is plainly 1.

History

The fact that $\sin x \approx x$ for small angles was known and *used* long before anyone could prove it with limits. Indian astronomers of the Kerala school (Madhava, around 1400) worked with series for sine that encode exactly this small-angle behavior, and Newton and Euler computed

fluently with $\sin x/x \to 1$ in the late 1600s and 1700s while treating the "x" as a genuine infinitesimal — a number smaller than any positive number yet not zero. Their instinct was flawless and their answers correct, but the foundation was shaky: what *is* an infinitesimal? It was Cauchy in the 1820s who recast such statements as rigorous limits, and the squeeze theorem — the area argument you just saw — is essentially his way of replacing the hand-wavy "$\sin x$ and x become equal" with a rigorous bound that anyone can check. The geometry was the ancients'; the rigor is the nineteenth century's.

Example 7.10 (A companion limit)**.** A close relative, proved with the same machinery and a trig identity:

$$\lim_{x\to 0} \frac{1-\cos x}{x} = 0.$$

Multiply by the conjugate $1+\cos x$:

$$\frac{1-\cos x}{x}\cdot\frac{1+\cos x}{1+\cos x} = \frac{1-\cos^2 x}{x(1+\cos x)} = \frac{\sin^2 x}{x(1+\cos x)} = \frac{\sin x}{x}\cdot\frac{\sin x}{1+\cos x}.$$

As $x\to 0$ the first factor $\to 1$ and the second $\to \frac{0}{2}=0$, so the product $\to 1\cdot 0 = 0$. Two of our tricks — conjugates and the $\sin x/x$ limit — working together.

Example 7.11 (Using $\sin x/x$ as a building block)**.**

$$\lim_{x\to 0} \frac{\sin 5x}{3x}.$$

The trick is to manufacture the form $\frac{\sin(\text{something})}{\text{same something}}$. Multiply and divide to make the argument of sine match its denominator:

$$\frac{\sin 5x}{3x} = \frac{\sin 5x}{5x}\cdot\frac{5x}{3x} = \frac{\sin 5x}{5x}\cdot\frac{5}{3}.$$

As $x\to 0$, $5x\to 0$ too, so $\dfrac{\sin 5x}{5x}\to 1$, and the limit is $1\cdot\dfrac{5}{3} = \dfrac{5}{3}$.

7.5 One-Sided Limits

So far we have let x approach a from both sides at once. But sometimes a function behaves differently coming from the left than from the right, and we need to look at each approach separately. These are *one-sided limits*.

Definition 7.1 (One-sided limits)**.** The **right-hand limit** $\lim\limits_{x\to a^+} f(x) = L$ means $f(x)$ approaches L as x approaches a through values *greater* than a. The **left-hand limit** $\lim\limits_{x\to a^-} f(x) = L$ uses values *less* than a. The epsilon-delta phrasing is identical to the two-sided one, except we restrict to $0 < x-a < \delta$ (right) or $-\delta < x-a < 0$ (left).

The connection between one-sided and two-sided limits is the whole reason we care about them, and it is clean:

Theorem 7.4 (Two-sided = both one-sided agreeing)**.** $\lim\limits_{x\to a} f(x) = L$ if and only if

$$\lim_{x\to a^-} f(x) = L \quad \textit{and} \quad \lim_{x\to a^+} f(x) = L.$$

That is: ***the two-sided limit exists exactly when both one-sided limits exist and are equal.***

Intuition

A two-sided limit is a meeting. For the function to "approach L as $x \to a$" it must approach L whether you sneak up from the left or from the right. If the two approaches head for different heights, there is no single number the function is closing in on — so the two-sided limit *does not exist*. This is the standard way to prove a limit fails to exist: show the two sides disagree.

Example 7.12 (A jump: the limit fails to exist)**.** The Heaviside step function

$$H(x) = \begin{cases} 0 & x < 0, \\ 1 & x \geq 0, \end{cases}$$

has, at $x = 0$,

$$\lim_{x \to 0^-} H(x) = 0, \qquad \lim_{x \to 0^+} H(x) = 1.$$

The two sides disagree, so $\lim_{x \to 0} H(x)$ **does not exist.** The graph jumps; no single height is being approached.

Step discontinuity: left limit $0 \neq$ right limit 1

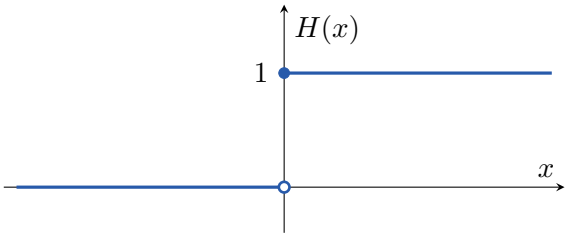


Figure 7.6: The Heaviside step. From the left, $H \to 0$; from the right, $H \to 1$. Because the two one-sided limits disagree, the two-sided limit at 0 does not exist.

Example 7.13 (Different formulas on each side)**.** Let

$$f(x) = \begin{cases} x^2 + 1 & x \leq 1, \\ 4 - 2x & x > 1. \end{cases}$$

From the left we use $x^2 + 1$: $\lim_{x \to 1^-} f(x) = 1^2 + 1 = 2$. From the right we use $4 - 2x$: $\lim_{x \to 1^+} f(x) = 4 - 2 = 2$. The sides agree, so $\lim_{x \to 1} f(x) = 2$ *exists*, regardless of the fact that the formula switches at $x = 1$ (Figure 7.7).

Walking question

Take a walk and picture this. A function can have a perfectly good two-sided limit at a point where it is built from *two different formulas*, as long as those formulas meet at the same height. And a function can *fail* to have a limit at a point where it is given by a single tidy formula — if that formula blows up or jumps. The limit cares only about the *heights being approached from each side*, never about how the function is written or what it does at the point itself. Can you sketch a function continuous everywhere except one jump, and one that is defined by

Piecewise agreement: two formulas meeting at the same height

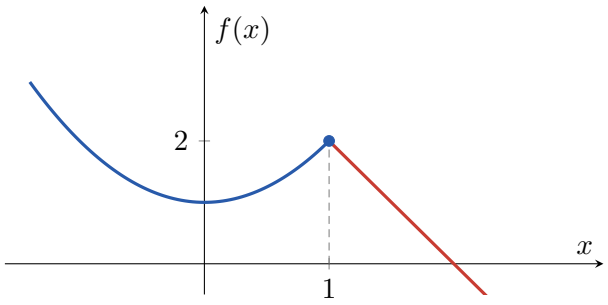


Figure 7.7: The piecewise f: the parabola x^2+1 (blue, for $x \le 1$) and the line $4-2x$ (red, for $x > 1$) meet at the same height 2 when $x = 1$. Both one-sided limits equal 2, so the two-sided limit exists even though the formula changes. Contrast this with the Heaviside jump, where the pieces fail to meet.

two formulas yet perfectly smooth?

7.6 Limits with Absolute Values

The absolute value $|x|$ is secretly a piecewise function in disguise:

$$|x| = \begin{cases} x & x \ge 0, \\ -x & x < 0, \end{cases}$$

and that hidden split is *exactly* the place where one-sided limits earn their keep. Whenever an absolute value sits on top of (or inside) an expression, the right move is almost always: **break it at the point where its inside changes sign, and take the limit from each side.**

Example 7.14 (The classic $|x|/x$)**.**

$$\lim_{x \to 0} \frac{|x|}{x}.$$

For $x > 0$, $|x| = x$, so $\frac{|x|}{x} = \frac{x}{x} = 1$. For $x < 0$, $|x| = -x$, so $\frac{|x|}{x} = \frac{-x}{x} = -1$. Therefore

$$\lim_{x \to 0^+} \frac{|x|}{x} = 1, \qquad \lim_{x \to 0^-} \frac{|x|}{x} = -1.$$

The two sides disagree, so the two-sided limit **does not exist**. The function is $+1$ on the right and -1 on the left — a jump straight through the origin (Figure 7.8).

What is really happening

What is the absolute value really doing to a limit? It *folds the negative side up*, which reverses the direction of approach on the left. So an expression that is heading one way as x increases through 0 from the right may head the opposite way from the left — the fold flips its sign. That is why absolute-value limits so often split into two unequal one-sided limits. The cure

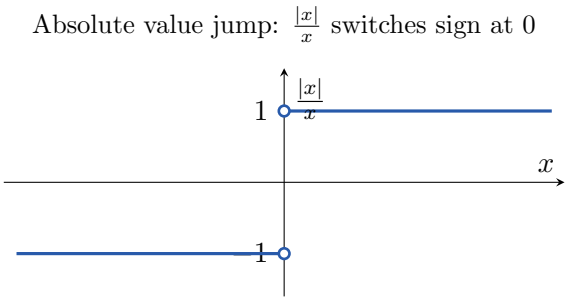


Figure 7.8: The sign function $\dfrac{|x|}{x}$ equals $+1$ to the right of 0 and -1 to the left, with the point $x = 0$ removed (the function is undefined there). The left and right limits, -1 and $+1$, disagree, so the two-sided limit at 0 does not exist.

is never to fight the absolute value head-on but to *unfold it* into its two cases and treat each side as the plain function it really is.

Example 7.15 (Absolute value not at zero)**.**

$$\lim_{x \to 3} \frac{|x-3|}{x-3}.$$

The inside $x - 3$ changes sign at $x = 3$. For $x > 3$, $x - 3 > 0$ so the ratio is $+1$; for $x < 3$, $x - 3 < 0$ so the ratio is -1. Right limit $= 1$, left limit $= -1$: the two-sided limit does not exist. (It is the previous example shifted three units to the right.)

Example 7.16 (When the absolute value does *not* cause trouble)**.**

$$\lim_{x \to 0} x\,|x|.$$

Here splitting is easy and the sides agree. For $x \geq 0$, $x|x| = x^2 \to 0$; for $x < 0$, $x|x| = -x^2 \to 0$. Both one-sided limits are 0, so the two-sided limit is 0. An absolute value does not automatically destroy a limit — it only matters when the two folded sides head to different places.

Common trap

Do not "simplify" $|x|$ to x and barge ahead. $|x| = x$ is true only for $x \geq 0$. On the left of the trouble point you must use $|x| = -x$, and forgetting the sign is the single most common error here. When you see an absolute value in a limit, your reflex should be: *find where the inside is zero, then split.*

7.7 One-Sided Limits and Infinite Blow-Ups

One-sided limits also let us describe what happens when a function blows up — the $\frac{\text{nonzero}}{0}$ situation that the quotient law also refuses to touch, but for a different reason than $\frac{0}{0}$.

Example 7.17 (Opposite infinities from each side)**.**

$$\lim_{x \to 0} \frac{1}{x}.$$

Substitution gives $\frac{1}{0}$ — not indeterminate, but undefined. Approach from the right ($x > 0$, tiny): $\frac{1}{x}$ is a huge positive number, so $\lim_{x\to 0^+} \frac{1}{x} = +\infty$. From the left ($x < 0$, tiny): $\frac{1}{x}$ is a huge *negative* number, so $\lim_{x\to 0^-} \frac{1}{x} = -\infty$. The two sides shoot off in opposite directions, so the two-sided limit does not exist.

Example 7.18 (Same infinity from both sides)**.**

$$\lim_{x\to 0} \frac{1}{x^2} = +\infty.$$

Here $x^2 > 0$ no matter which side you come from, so $\frac{1}{x^2} \to +\infty$ from *both* sides. We write $+\infty$ to describe the shared behavior, while remembering that "$= \infty$" is shorthand for "grows without bound," not a claim that the limit is a number.

Common trap

$\frac{\text{nonzero}}{0}$ and $\frac{0}{0}$ are completely different diagnoses. A nonzero number over zero *blows up* (vertical asymptote); there is nothing to cancel and no finite limit. A zero over zero is *indeterminate*; there is a hidden common factor and the limit may be any finite number. When substitution fails, your very first question is: *is the numerator also zero?* The answer routes you to entirely different techniques.

7.8 A Preview of Indeterminate Forms

We have spent this chapter conquering one indeterminate form, $\frac{0}{0}$, with algebra. Its siblings recur throughout calculus, and each demands its own tools, so they are worth naming now. An *indeterminate form* is a symbolic "shape" a limit can take where the limit laws give no answer — the value genuinely depends on the competing rates, not on the shape alone.

Focus

The seven classic indeterminate forms:

$$\frac{0}{0}, \quad \frac{\infty}{\infty}, \quad 0\cdot\infty, \quad \infty-\infty, \quad 1^\infty, \quad 0^0, \quad \infty^0.$$

Each is "indeterminate" because the answer can be *anything* depending on how fast the pieces move. The shape alone does not decide.

Why are these indeterminate while, say, $\frac{2}{0}$ or 0^5 are not? The difference is whether the symbolic form pins down the answer.

Intuition

Consider $0\cdot\infty$. One factor is shrinking to nothing, the other growing without bound — they are at war, and who wins depends on *how fast* each moves. Watch three skirmishes as $x \to 0^+$,

each of the form (something $\to 0$) $\cdot$ (something $\to \infty$):

$$x \cdot \frac{1}{x} \to 1, \qquad x \cdot \frac{1}{x^2} \to +\infty, \qquad x^2 \cdot \frac{1}{x} \to 0.$$

Same shape $0 \cdot \infty$, three different answers. That is the entire meaning of "indeterminate": the form is a question, never an answer. By contrast $\frac{2}{0}$ always blows up and 0^5 is always 0 — those forms *are* determined, so we do not call them indeterminate.

The cleanest power tool for the two leading forms, $\frac{0}{0}$ and $\frac{\infty}{\infty}$, is *l'Hôpital's rule*, which turns the limit of a ratio into the limit of the ratio of derivatives. It is wonderfully effective — and it is worthless until you have derivatives, which is why it waits for its own chapter much later. The forms $0 \cdot \infty$, $\infty - \infty$, and the three exponential shapes 1^∞, 0^0, ∞^0 are all handled by first *algebraically rewriting* them as $\frac{0}{0}$ or $\frac{\infty}{\infty}$ (the exponential ones by taking logs), and then applying the same machinery. The algebra you practiced in this chapter — factoring, rationalizing, clearing fractions — is the warm-up; l'Hôpital is the heavy artillery you get once derivatives are in hand.

Connection

There is a thread running from this chapter all the way to Taylor series near the end of the book. The reason $\frac{\sin x}{x} \to 1$ is that $\sin x = x - \frac{x^3}{6} + \cdots$; divide by x and the leading term is 1 with the rest vanishing. Every indeterminate $\frac{0}{0}$ is, underneath, a competition between the *leading terms* of two series. The algebra tricks of this chapter, l'Hôpital's rule, and Taylor expansion are three faces of the same idea: *a limit is decided by the lowest-order behavior of each piece.*

7.9 A Strategy, Start to Finish

Put the tools in order. Faced with $\lim_{x \to a} f(x)$:

1. **Try substitution.** Plug in a. If you get a finite number and the function is continuous there, you are done.
2. **Diagnose the failure.** If substitution gives $\frac{\text{nonzero}}{0}$, it is a blow-up — use one-sided limits to describe $\pm\infty$ and conclude the two-sided limit (usually) does not exist. If it gives $\frac{0}{0}$, it is indeterminate — proceed.
3. **Remove the common factor.** Factor and cancel; if there are roots, rationalize with the conjugate; if there are nested fractions, clear them. Then substitute into the simplified form.
4. **Squeeze if algebra fails.** For oscillations or trig limits like $\frac{\sin x}{x}$, trap the function between two others with a common limit.
5. **Split at sign changes.** For absolute values or piecewise formulas, take left and right limits separately and check whether they agree.
6. **Recognize the deeper forms.** If you are left with $\frac{\infty}{\infty}$, $0 \cdot \infty$, or an exponential shape, name the indeterminate form and remember l'Hôpital is coming.

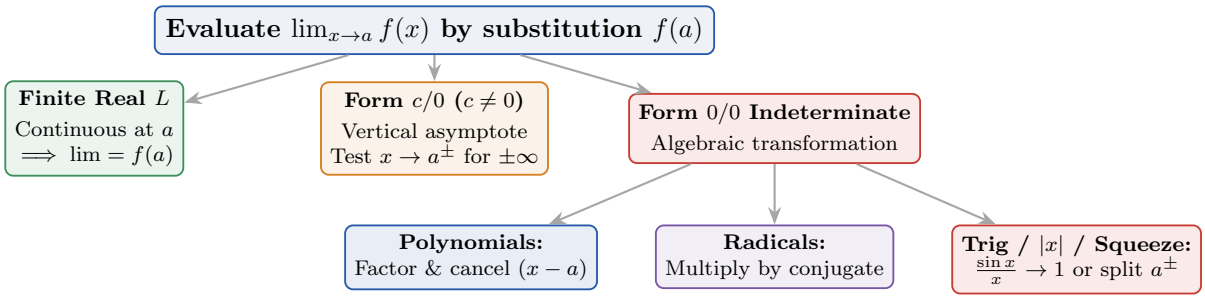


Figure 7.9: The limit computation decision roadmap: diagnose the initial form upon substitution, then deploy the matching algebraic or geometric tool to resolve the obstruction.

Example 7.19 (A limit that uses several tools)**.**

$$\lim_{x\to 0} \frac{\sqrt{1+x}-1}{\sin x}.$$

Substitution gives $\frac{0}{0}$. There is a root, so rationalize the numerator, and there is a sine, so plan to use $\frac{\sin x}{x}$. Multiply the numerator by its conjugate:

$$\frac{\sqrt{1+x}-1}{\sin x}\cdot\frac{\sqrt{1+x}+1}{\sqrt{1+x}+1} = \frac{(1+x)-1}{\sin x\,(\sqrt{1+x}+1)} = \frac{x}{\sin x\,(\sqrt{1+x}+1)}.$$

Now split the factors deliberately:

$$= \frac{x}{\sin x}\cdot\frac{1}{\sqrt{1+x}+1} = \frac{1}{\frac{\sin x}{x}}\cdot\frac{1}{\sqrt{1+x}+1}.$$

As $x \to 0$, $\frac{\sin x}{x} \to 1$ so its reciprocal $\to 1$, and $\sqrt{1+x}+1 \to 2$. The limit is

$$1\cdot\frac{1}{2} = \frac{1}{2}.$$

Rationalizing and the $\sin x/x$ limit, working side by side.

Why this matters

Notice the shape of what you have learned. Every technique in this chapter does the same thing: it *changes the appearance of an expression without changing the value it approaches*, until the answer becomes self-evident. Factoring, conjugates, common denominators, the squeeze sandwich, splitting at a sign change — these are not five unrelated tricks but five dialects of a single sentence: *a limit lives in the neighborhood of a point, not at the point, so I am free to rewrite the function on that neighborhood however I please.* That freedom, licensed by the limit laws and the definition of a limit itself, is the quiet engine of all of calculus. Once you feel it, computing limits stops being a chore and becomes a search for the right disguise to remove.

Exercises

Exercise 7.1 (Substitution)**.** Evaluate by direct substitution, justifying that it is legal:

$$\text{(a)}\ \lim_{x\to 2}\left(3x^2-4x+7\right), \qquad \text{(b)}\ \lim_{x\to 0}\frac{e^x+\cos x}{x+2}, \qquad \text{(c)}\ \lim_{x\to \pi}\sin\left(\frac{x}{2}\right).$$

Exercise 7.2 (Factor and cancel)**.** Evaluate:

$$\text{(a) } \lim_{x\to 4}\frac{x^2-16}{x-4}, \qquad \text{(b) } \lim_{x\to -1}\frac{x^2+3x+2}{x^2-1}, \qquad \text{(c) } \lim_{x\to 2}\frac{x^3-8}{x-2}.$$

Exercise 7.3 (Rationalize)**.** Evaluate:

$$\text{(a) } \lim_{x\to 0}\frac{\sqrt{9+x}-3}{x}, \qquad \text{(b) } \lim_{x\to 1}\frac{\sqrt{x}-1}{x-1}, \qquad \text{(c) } \lim_{x\to 7}\frac{\sqrt{x+2}-3}{x-7}.$$

Exercise 7.4 (Complex fractions)**.** Evaluate:

$$\text{(a) } \lim_{x\to 3}\frac{\frac{1}{x}-\frac{1}{3}}{x-3}, \qquad \text{(b) } \lim_{h\to 0}\frac{\frac{1}{(2+h)^2}-\frac{1}{4}}{h}.$$

Exercise 7.5 (Trig limits)**.** Using $\lim_{x\to 0}\frac{\sin x}{x}=1$ and $\lim_{x\to 0}\frac{1-\cos x}{x}=0$, evaluate:

$$\text{(a) } \lim_{x\to 0}\frac{\sin 7x}{2x}, \qquad \text{(b) } \lim_{x\to 0}\frac{\sin 3x}{\sin 5x}, \qquad \text{(c) } \lim_{x\to 0}\frac{\tan x}{x}.$$

Exercise 7.6 (Squeeze)**.** (a) Show $\lim_{x\to 0} x^4\cos\frac{1}{x^2}=0$ by squeezing. (b) Given that $1-\frac{x^2}{6}\le\frac{\sin x}{x}\le 1$ for small x, use the squeeze theorem to re-derive $\lim_{x\to 0}\frac{\sin x}{x}=1$.

Exercise 7.7 (One-sided and absolute value)**.** Evaluate each one-sided limit and decide whether the two-sided limit exists:

$$\text{(a) } \lim_{x\to 5}\frac{|x-5|}{x-5}, \qquad \text{(b) } \lim_{x\to 2^+}\frac{x-2}{|x-2|}, \qquad \text{(c) } \lim_{x\to 0}\frac{x^2}{|x|}.$$

Exercise 7.8 (Piecewise)**.** Let

$$f(x)=\begin{cases}2x+1 & x<1,\\ k & x=1,\\ x^2+2 & x>1.\end{cases}$$

Find $\lim_{x\to 1^-} f(x)$ and $\lim_{x\to 1^+} f(x)$. Does $\lim_{x\to 1} f(x)$ exist? Does the value of k affect your answer? Explain in one sentence why the limit ignores k.

Exercise 7.9 (Blow-ups)**.** Describe the behavior with one-sided limits, and state whether the two-sided limit exists:

$$\text{(a) } \lim_{x\to 2}\frac{1}{x-2}, \qquad \text{(b) } \lim_{x\to 2}\frac{1}{(x-2)^2}, \qquad \text{(c) } \lim_{x\to 1}\frac{x}{x^2-1}.$$

Exercise 7.10 (Conceptual)**.** Decide whether each statement is true or false, and justify in a sentence. (a) If $\lim_{x\to a} f(x)$ exists then f is defined at a. (b) If $f(a)$ is defined then $\lim_{x\to a} f(x)=f(a)$. (c) The form $\frac{0}{0}$ always has limit 1 because "anything over itself is 1." (d) If both one-sided limits exist, the two-sided limit exists.

Exercise 7.11 (Proof: a squeeze)**.** Prove using the squeeze theorem that if $|f(x)|\le M|x-a|$ for all x near a (with M a fixed constant), then $\lim_{x\to a} f(x)=0$. *Hint:* sandwich f between $-M|x-a|$ and $M|x-a|$.

Exercise 7.12 (Proof: the conjugate, in general)**.** Let $a > 0$. Prove that

$$\lim_{x \to a} \frac{\sqrt{x} - \sqrt{a}}{x - a} = \frac{1}{2\sqrt{a}}$$

by rationalizing. Explain in one sentence which limit law licenses the final substitution step, and why it is now legal when it was not before.

Chapter 8

Continuity and the Intermediate Value Theorem

There is a word we use loosely in everyday life and precisely in mathematics: *continuous*. We say a road runs continuously from one town to the next, meaning you can drive the whole way without ever being teleported across a gap. We say someone worked continuously for three hours, meaning no interruptions. Underneath both uses is the same picture: *no sudden jumps*. You can get from here to there by moving a little at a time.

Calculus is built on this picture. Almost every theorem you will meet later — that a function must hit every value between two of its outputs, that it must attain a highest and a lowest point on a closed interval, that the area under a curve makes sense, that a differentiable function is well behaved — rests on continuity. So before we can differentiate or integrate anything with confidence, we need to say, with total precision, what it means for a function to have "no jumps."

The right definition, oddly enough, does not mention jumps at all. It is phrased entirely in the language of limits you built in the previous chapters. By the end of this chapter you will see *why* the limit definition captures exactly the no-jump idea, you will be able to classify every way a function can fail to be continuous, and you will prove and use two of the great existence theorems of analysis: the Intermediate Value Theorem and the Extreme Value Theorem.

8.1 From "no jumps" to a definition

Start with the picture, and refuse to leave it until the definition feels forced upon us.

Imagine dragging a pencil along the graph of a function f. Continuity at a point a should mean: as your pencil approaches the vertical line $x = a$ from either side, the height of the curve approaches the height $f(a)$ of the point actually sitting above a. Nothing jumps; nothing is missing; the point you would *predict* by sliding in is exactly the point that is *there*.

That last sentence is the whole idea, and it splits into three separate promises, each of which can fail independently.

Intuition

For f to be continuous at a, three things must all hold:

1. $f(a)$ must *exist* — there has to be a point on the graph above a.
2. $\lim_{x\to a} f(x)$ must *exist* — the curve must approach a single definite height as you slide in from both sides.
3. These two numbers must *agree* — the height you slide toward equals the height that is actually there.

The first is about the point. The second is about the approach. The third demands that the approach and the point match. Continuity is precisely the demand that prediction and reality coincide.

Compressing all three promises into one line gives the definition.

Definition 8.1 (Continuity at a point)**.** A function f is **continuous at** a if

$$\lim_{x\to a} f(x) = f(a).$$

We mean this in the strongest sense: a is in the domain of f (so $f(a)$ is a real number), the limit on the left exists, and the two are equal. If f is continuous at every point of a set S, we say f is **continuous on** S.

This single equation quietly bundles all three promises. If $f(a)$ did not exist, the right side would be meaningless. If the limit did not exist, the left side would be meaningless. And if both existed but disagreed, the equation would simply be false. The equation $\lim_{x\to a} f(x) = f(a)$ asserts all three at once. That economy is exactly why mathematicians chose it.

What is really happening

Why does an equation about limits capture a *geometric* idea about jumps? Because a limit *is* the machinery of "sliding in." Recall what $\lim_{x\to a} f(x) = L$ really asserts: for every tolerance $\varepsilon > 0$, no matter how small, there is a window of inputs $0 < |x - a| < \delta$ on which f stays within ε of L. Continuity says we may take $L = f(a)$ *and drop the restriction* $x \neq a$. In one breath:

$$\text{for every } \varepsilon > 0 \text{ there is } \delta > 0 \text{ such that } |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

Read that aloud as a promise about jumps: "if you want my output to land within ε of $f(a)$, just keep your input within δ of a." A jump is exactly a failure of this promise — a height the output refuses to approach no matter how close the input gets. Continuity is the guarantee that small input changes force small output changes. There is no gap to teleport across.

Focus

There is an essential difference from the bare limit definition of the previous chapter. There we wrote $0 < |x - a| < \delta$, deliberately *excluding* the point $x = a$, because a limit is about the neighborhood, not the point itself. Continuity is the stronger statement that we may *include* $x = a$: the inequality $|x - a| < \delta$ now allows $x = a$, where it reads $|f(a) - f(a)| = 0 < \varepsilon$ for free. The limit ignores the point; continuity insists the point cooperate with its neighbors.

8.1.1 One-sided continuity and continuity on an interval

Sometimes a function only "slides in" nicely from one side — think of $\sqrt{x}$ at $x = 0$, where there are no inputs to the left at all. We handle this with one-sided limits.

Definition 8.2 (One-sided continuity)**.** f is **continuous from the right** at a if $\lim_{x \to a^+} f(x) = f(a)$, and **continuous from the left** at a if $\lim_{x \to a^-} f(x) = f(a)$. A function is continuous at an interior point exactly when it is continuous from both sides there.

This lets us say cleanly what it means for f to be continuous on a *closed* interval $[a, b]$: continuous at every interior point, continuous from the right at the left endpoint a, and continuous from the left at the right endpoint b. We only ever ask a function to behave with respect to inputs that actually exist. This convention will matter enormously when we state the Extreme Value Theorem, which needs a closed interval.

8.2 A gallery of discontinuities

The fastest way to understand continuity is to study how it breaks. Each of the three promises can fail, and the failures look different on a graph. Here is the zoo.

8.2.1 Removable discontinuity: the patchable hole

Here the limit exists — the curve clearly aims at a single height — but either $f(a)$ is undefined or it is defined "in the wrong place." The graph has a single missing or misplaced point. The defect is *removable*: redefining f at the one point a to equal the limit repairs continuity completely.

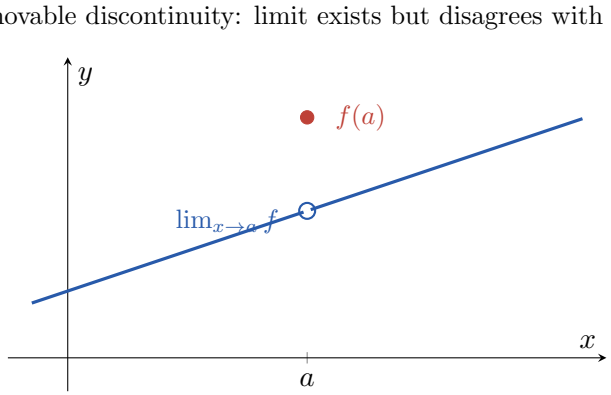


Figure 8.1: A removable discontinuity. The curve aims at the open circle, but the actual value $f(a)$ (filled dot) sits elsewhere. The limit exists; it just disagrees with $f(a)$. Move the dot down into the hole and continuity is restored.

Example 8.1 (The classic hole)**.** Consider

$$g(x) = \frac{x^2 - 1}{x - 1}.$$

At $x = 1$ the formula reads $0/0$, undefined, so promise (1) fails: $g(1)$ does not exist. But for every $x \neq 1$ we may cancel:

$$g(x) = \frac{(x-1)(x+1)}{x-1} = x + 1,$$

so $\lim_{x\to 1} g(x) = \lim_{x\to 1}(x+1) = 2$. The limit exists and equals 2; only the point is missing. Defining $g(1) = 2$ plugs the hole and makes g continuous everywhere.

Common trap

A removable discontinuity is still a genuine discontinuity. The function g above is *not* continuous at 1 as originally written — it has no value there, so the equation $\lim_{x\to 1} g(x) = g(1)$ cannot hold. People sometimes say "g is basically $x+1$." It *becomes* $x+1$ only after you supply the missing value. Until you do, there is a hole, and a hole is a discontinuity. Cancellation in a formula does not by itself mend the function; it tells you which value would mend it.

8.2.2 Jump discontinuity: the cliff

Here both one-sided limits exist but *disagree*. The curve approaches one height from the left and a different height from the right. There is a genuine cliff. No single redefinition of $f(a)$ can fix this, because there are two different heights competing and $f(a)$ can only be one number.

Jump discontinuity: one-sided limits exist but disagree

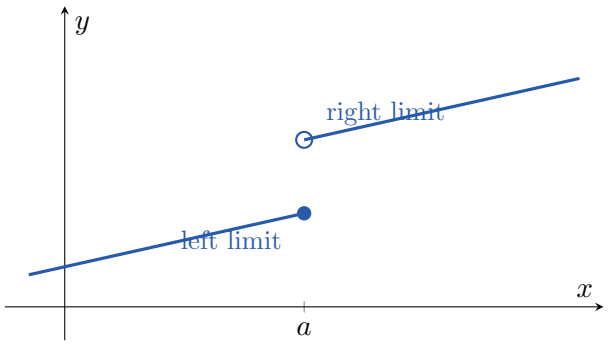


Figure 8.2: A jump discontinuity. The left-hand limit and right-hand limit both exist but are unequal, so the two-sided limit does not exist. The gap between the dots is the size of the jump.

Example 8.2 (A step function)**.** Let $H(x) = 0$ for $x < 0$ and $H(x) = 1$ for $x \geq 0$ (the Heaviside step, the idealized "switch turns on" function). Then $\lim_{x\to 0^-} H(x) = 0$ while $\lim_{x\to 0^+} H(x) = 1$. The one-sided limits disagree, so $\lim_{x\to 0} H(x)$ does not exist, and H is discontinuous at 0. It *is* continuous from the right there, since $\lim_{x\to 0^+} H(x) = 1 = H(0)$.

Application

Jump discontinuities are not pathological — they are everywhere in modeling. A light switch, a digital signal flipping from 0 to 1, the price of a stock the instant news breaks, the charge on a capacitor the moment a switch closes: all are modeled by jumps. This is exactly why engineers love the Heaviside function and its cousin, the Dirac delta. Calculus copes with

such functions by working on the pieces *between* jumps, where everything is continuous, and treating the jumps as boundaries. "Piecewise continuous" is the accurate description of most real-world signals.

8.2.3 Infinite discontinuity: the pole

Here a one-sided limit (or both) is infinite. The curve shoots off to $\pm\infty$ near a. There is a vertical asymptote. No finite value of $f(a)$ could possibly match an infinite approach.

Infinite discontinuity: vertical asymptote at the pole

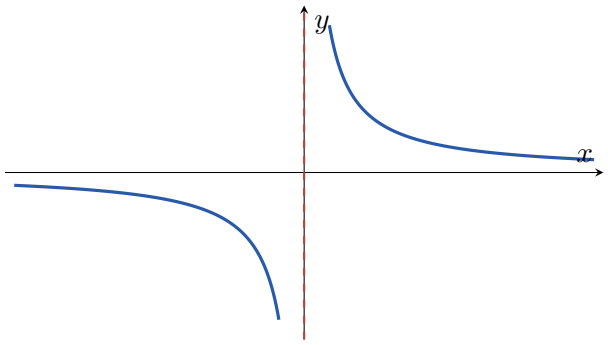


Figure 8.3: An infinite discontinuity: $f(x) = 1/x$ near $x = 0$. The left limit is $-\infty$, the right limit is $+\infty$. The dashed line is the vertical asymptote.

8.2.4 Oscillatory discontinuity: the blur

The subtlest failure: the function neither settles down nor blows up; it oscillates forever, infinitely fast, as $x \to a$. The limit fails not because of a jump or a pole but because there is no single height the values approach at all.

Example 8.3 (Endless wiggling)**.** Let $s(x) = \sin(1/x)$ for $x \neq 0$. As $x \to 0$, the argument $1/x$ races off to infinity, so $\sin(1/x)$ sweeps through every value in $[-1, 1]$ infinitely many times in any neighborhood of 0. There is no limiting height. No definition of $s(0)$ can rescue continuity, because the values do not approach *anything*.

Contrast its tamer relative $x\sin(1/x)$: the wild oscillation is throttled by the factor $x \to 0$, the wiggles get squeezed to zero amplitude, and $\lim_{x\to 0} x\sin(1/x) = 0$ by the Squeeze Theorem. Defining the value at 0 to be 0 makes *that* one continuous. The difference between a curable and an incurable wiggle is whether the amplitude dies out.

Walking question

Take a walk and sort the discontinuities by "how badly broken" they are. The removable hole is barely broken — one point out of place. The jump is broken but tame — two genuine one-sided limits. The pole is broken and unbounded. The oscillation is broken and not even *trying* to have a limit. Now ask yourself: which of these can a computer plotting 200 points

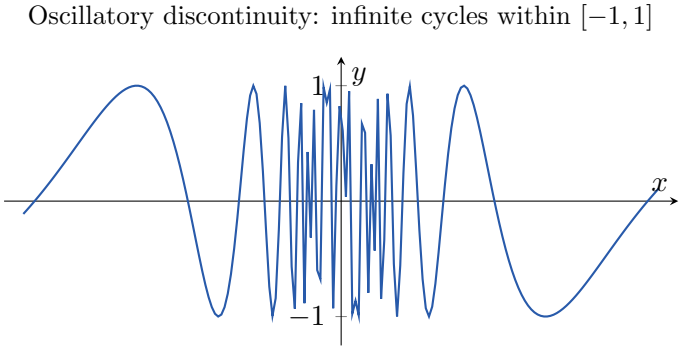


Figure 8.4: $\sin(1/x)$ near the origin. The oscillation never slows; the values fill the band $[-1, 1]$ densely as $x \to 0$, so no limit exists.

possibly miss? (Hint: think about whether your sample points might land on the "wrong" side of the trouble, or skip over a thin spike. This is why we prove things instead of trusting pictures.)

8.3 Building continuous functions

If we had to check the limit definition by hand for every function, calculus would be unbearable. Fortunately continuity is *hereditary*: continuous functions combine to make more continuous functions. We get to know a few atoms are continuous, then assemble everything else for free.

8.3.1 The atoms

The following are continuous everywhere on their natural domains; each follows directly from the limit laws of the previous chapters.

- Constant functions $f(x) = c$ and the identity $f(x) = x$. (For the identity, $\lim_{x \to a} x = a = f(a)$ — this is almost the definition of a limit.)
- Polynomials, everywhere on $\mathbb{R}$.
- Rational functions $p(x)/q(x)$, everywhere $q(x) \neq 0$.
- $\sin x$, $\cos x$, everywhere; $\exp x$ and, on $(0, \infty)$, $\ln x$.
- $\sqrt{x}$ on $[0, \infty)$ (continuous from the right at 0), and more generally x^r on its domain.

8.3.2 The combination laws

Theorem 8.1 (Continuity of combinations)**.** *Suppose f and g are continuous at a. Then so are*

$$f + g, \qquad f - g, \qquad fg, \qquad cf \ (c \in \mathbb{R}),$$

and f/g provided $g(a) \neq 0$.

Why the technique works

Why is this true with no new work at all? Because each statement is just a limit law rephrased. Continuity of f at a means $\lim_{x\to a} f(x) = f(a)$, and likewise for g. The sum law for limits says

$$\lim_{x\to a}\big(f(x)+g(x)\big) = \lim_{x\to a} f(x) + \lim_{x\to a} g(x) = f(a)+g(a) = (f+g)(a),$$

which is exactly the statement that $f+g$ is continuous at a. The product, difference, and quotient cases are the corresponding limit laws, read the same way. The proviso $g(a)\neq 0$ in the quotient is precisely the proviso in the limit quotient law: you may not divide by a quantity heading to zero. *Every* continuity combination law is a limit law wearing a different hat.

This single observation explains the whole list of atoms. A polynomial is built from the identity and constants by multiplication and addition; apply Theorem 8.1 finitely many times and continuity is guaranteed at every point. A rational function is a quotient of polynomials, continuous wherever the denominator is nonzero. You never have to return to ε and δ for these.

8.3.3 Composition: the most important law

The deepest of the combination laws concerns *chaining* functions, g then f. It will reappear as the engine of the chain rule in differentiation, so it is worth seeing clearly now.

Theorem 8.2 (Continuity of compositions)**.** *If g is continuous at a and f is continuous at $g(a)$, then the composition $f\circ g$, defined by $(f\circ g)(x) = f(g(x))$, is continuous at a.*

Intuition

Picture two machines in a row. The first, g, takes x near a and (being continuous) produces output near $g(a)$. The second, f, takes inputs near $g(a)$ and (being continuous there) produces outputs near $f(g(a))$. Feed a small wiggle in x through the pair: g converts it to a small wiggle near $g(a)$, and f converts that to a small wiggle near $f(g(a))$. Small in, small out, all the way through. The chain of two no-jump machines is itself a no-jump machine.

Proof. We use the ε–δ form of continuity. Let $\varepsilon>0$ be given; we must find $\delta>0$ so that $|x-a|<\delta$ forces $|f(g(x))-f(g(a))|<\varepsilon$.

Work from the outside in. Because f is continuous at the point $b := g(a)$, there is an $\eta>0$ (the tolerance f needs on *its* input) such that

$$|y-b|<\eta \implies |f(y)-f(b)|<\varepsilon.$$

Now hand that η to g. Because g is continuous at a, there is a $\delta>0$ such that

$$|x-a|<\delta \implies |g(x)-g(a)|<\eta.$$

Chain them: if $|x-a|<\delta$, then $|g(x)-b|<\eta$, and therefore $|f(g(x))-f(b)|<\varepsilon$. Since $f(b)=f(g(a))$, this says exactly $|f(g(x))-f(g(a))|<\varepsilon$. The required δ exists for every ε, so $f\circ g$ is continuous at a. □

Focus

Watch the direction of the argument. We pick the tolerance ε at the *final* output, pull it back through f to get a tolerance η on f's input, then pull *that* back through g to get a tolerance δ on the original input. Tolerances propagate backward, from the end of the pipeline to the start. This backward flow is the shape of nearly every chain-rule-style argument in calculus. Hold onto it; it returns when we differentiate compositions.

Example 8.4 (Reading off continuity at a glance)**.** The function $h(x) = \sqrt{x^2+1}$ is continuous on all of $\mathbb{R}$. Here $g(x) = x^2 + 1$ is a polynomial (continuous, and always $\geq 1 > 0$), and $f(u) = \sqrt{u}$ is continuous on $[0, \infty)$, which safely contains every value of g. By Theorem 8.2 the composition is continuous everywhere. Similarly $\sin(e^x)$, $e^{\cos x}$, and $\dfrac{1}{1+x^2}$ are continuous on all of $\mathbb{R}$ — each is a finite assembly of continuous atoms by addition, multiplication, division (with nonzero denominator), and composition. After a little practice you simply *see* it.

History

For nearly two centuries calculus thrived on this no-jump intuition without a precise definition. Newton spoke of quantities flowing; Euler manipulated functions as if they were always smooth. But "obvious" continuity hid traps. In the early nineteenth century Bernard Bolzano (1817) and then Augustin-Louis Cauchy (1821) finally pinned continuity to limits, roughly as in Definition 8.1. Bolzano even gave a correct proof of the Intermediate Value Theorem — decades ahead of his time, and largely ignored. The real shock came later: in 1872 Karl Weierstrass exhibited a function that is continuous *everywhere* yet differentiable *nowhere* — an unbroken curve so jagged it has a corner at every point. That monster proved continuity and smoothness are genuinely different ideas, and that intuition alone, however good, cannot replace a definition. The ε–δ machinery you are using is the tool that finally tamed the monsters.

8.4 The Intermediate Value Theorem

We now reach the first great payoff of continuity. It is a theorem you already believe in your bones, which is exactly why it is so instructive to see proved.

Intuition

Suppose a function is continuous on $[a, b]$, starts below some level N at $x = a$, and ends above N at $x = b$. To get from below N to above N *without jumping*, the graph must cross the line $y = N$ somewhere in between. A continuous pencil stroke that begins below a horizontal line and ends above it has to touch that line. That is the entire content of the Intermediate Value Theorem. The hard part is not believing it — it is proving that the word "continuous" is strong enough to force the crossing, using only the definition.

Theorem 8.3 (Intermediate Value Theorem, IVT)**.** *Let f be continuous on the closed interval $[a, b]$, and let N be any number strictly between $f(a)$ and $f(b)$. Then there exists at least one $c \in (a, b)$ with $f(c) = N$.*

Intermediate Value Theorem: continuous curve crossing $y = N$

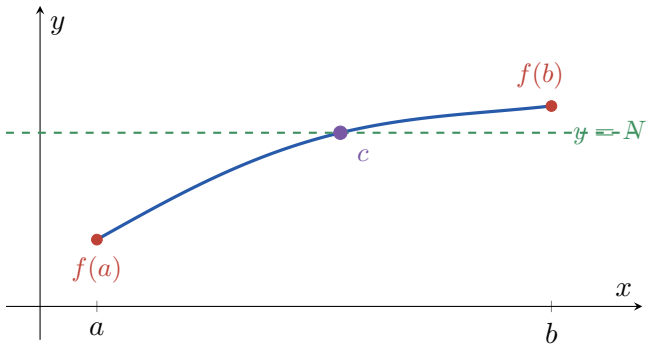


Figure 8.5: The Intermediate Value Theorem. A continuous curve running from height $f(a)$ up to height $f(b)$ must meet every intermediate level N at least once; here it crosses $y = N$ at the point c. For a wigglier curve there could be several crossings, but the theorem guarantees at least one.

8.4.1 The necessity of completeness alongside continuity

Before the proof, a warning that reveals what is really at stake. The IVT is *false* if we work over the rational numbers alone. Consider $f(x) = x^2 - 2$ on $[0, 2]$, restricted to rational inputs. It is "continuous" in the obvious sense, $f(0) = -2 < 0$ and $f(2) = 2 > 0$, yet there is *no rational* c with $f(c) = 0$ — the crossing happens at $\sqrt{2}$, which is not a rational number. The curve sneaks through a gap in the rationals without ever landing on a rational point.

What is really happening

The IVT is not really a theorem about continuity alone. It is a theorem about continuity *plus the absence of gaps in the real line*. That gaplessness has a name: **completeness**, the property that every nonempty set of reals that is bounded above has a least upper bound (a supremum). The rationals fail completeness — the set of rationals whose square is below 2 has no rational least upper bound — and the IVT fails right along with it. The real numbers were, in a sense, *built* to make theorems like the IVT true: $\mathbb{R}$ is the gapless completion of $\mathbb{Q}$. So when we prove the IVT, the decisive step will be the one place we invoke a least upper bound. Watch for it.

8.4.2 Proof by "the last moment it is still below"

There are two standard proofs. One repeatedly bisects the interval, trapping the crossing in shrinking subintervals. The other, which we give, captures the crossing in a single clean stroke using completeness. The idea is irresistible once you see it.

Proof. It is enough to handle the case $f(a) < N < f(b)$; the other case ($f(a) > N > f(b)$) follows by applying this one to $-f$ and $-N$. Define

$$S = \{\, x \in [a, b] : f(x) < N \,\}.$$

Think of S as the set of inputs at which the curve is still strictly below the target level. It is nonempty, since $a \in S$ (because $f(a) < N$), and it is bounded above by b. By the **completeness** of

$\mathbb{R}$, S has a least upper bound; call it $c = \sup S$. Intuitively, c is *the last moment the curve is still below N* — the exact place where "below" must hand off to "above." We claim $f(c) = N$, and we rule out the only two alternatives.

Suppose $f(c) < N$. Then $c \neq b$ (since $f(b) > N$), so $c < b$. Because f is continuous at c and $f(c)$ is strictly less than N, values of f stay below N on a whole little interval to the right of c: taking $\varepsilon = N - f(c) > 0$, continuity gives a $\delta > 0$ with $f(x) < f(c) + \varepsilon = N$ for all x within δ of c. So some points just to the *right* of c also lie in S. But then c was not an upper bound of S — contradiction. The curve cannot still be below N *and* have room to its right.

Suppose $f(c) > N$. Then $c \neq a$, so $c > a$. By the same continuity argument with $\varepsilon = f(c) - N > 0$, the values of f stay *above* N on a whole little interval around c, including a stretch to the *left* of c. On that stretch no point belongs to S, so $c - \delta$ would already be an upper bound for S — smaller than c, contradicting that c is the *least* upper bound.

Both alternatives are impossible, so the only survivor is $f(c) = N$. And c lies strictly between a and b because $f(a) < N < f(b)$ rules out the endpoints. □

Focus

Feel where each hypothesis earned its keep. *Completeness* produced the candidate point $c = \sup S$ — without it, S might have no least upper bound and the proof could not even begin. *Continuity* was used twice, once on each side, to show that the curve cannot be strictly below or strictly above N at c without contradicting how c was chosen. Drop continuity and a jump could leap over N; drop completeness and the crossing could fall into a gap. The theorem lives exactly at the meeting of these two ideas.

8.4.3 Root-finding: the IVT as an existence machine

The IVT is an *existence* theorem. It does not tell you *where* the crossing is, only that one exists. That modest-sounding guarantee is the foundation of every numerical root-finder.

Corollary 8.1 (Bolzano's sign-change test)**.** *If f is continuous on $[a, b]$ and $f(a)$ and $f(b)$ have opposite signs (one negative, one positive), then f has a root in (a, b): some c with $f(c) = 0$.*

This is just the IVT with $N = 0$: the value 0 lies between a negative number and a positive number, so it is attained.

Example 8.5 (A root of a quintic you cannot solve by formula)**.** Does $p(x) = x^5 + x - 1$ have a real root? There is no algebraic formula for the roots of a general quintic, but we do not need one. p is a polynomial, hence continuous everywhere. Compute the ends:

$$p(0) = 0 + 0 - 1 = -1 < 0, \qquad p(1) = 1 + 1 - 1 = 1 > 0.$$

Opposite signs across $[0, 1]$, so by Corollary 8.1 there is a root in $(0, 1)$ (Figure 8.6). Existence, established in two arithmetic lines and one theorem, with no formula in sight.

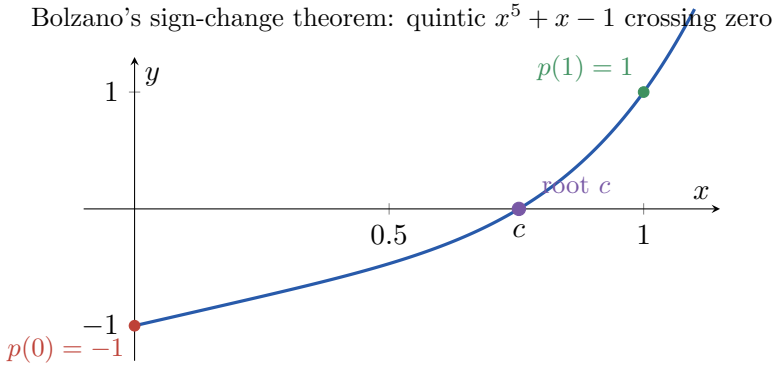


Figure 8.6: The quintic $p(x) = x^5 + x - 1$ on $[0, 1]$. Since $p(0) = -1 < 0$ and $p(1) = 1 > 0$, the continuous curve must cross the axis; the IVT guarantees a root c (here $c \approx 0.755$) even though no algebraic formula produces it.

Work it yourself

Pin that quintic root down by hand using *bisection* — the IVT applied over and over. We know the root lies in $[0, 1]$ since $p(0) < 0 < p(1)$. Test the midpoint:

$$p(0.5) = 0.5^5 + 0.5 - 1 = 0.03125 - 0.5 = -0.46875 < 0.$$

Same sign as $p(0)$, so the root is in $[0.5, 1]$. Now $p(0.75) = 0.2373 + 0.75 - 1 = -0.0127 < 0$, so the root is in $[0.75, 1]$. Next $p(0.875) \approx 0.5129 + 0.875 - 1 = 0.388 > 0$, so the root is in $[0.75, 0.875]$. Each step halves the interval and the IVT promises the root never escapes. Two more halvings put you near $x \approx 0.755$. You have computed a root of an unsolvable quintic with nothing but arithmetic and a theorem.

Why the technique works

Why is bisection guaranteed to succeed? Each step preserves the property "the endpoints have opposite signs" as an *invariant*. As long as that invariant holds, the IVT guarantees a root inside the current interval. Halving the interval shrinks its length to $(b - a)/2^n$ after n steps, which tends to 0, so the trapped root is pinned to arbitrary precision. The method cannot fail to converge, because at every stage a root is provably present in a provably shrinking cage. Bisection is the IVT turned into an algorithm.

8.4.4 Fixed points: where a function meets the diagonal

Here is a second use, and a striking one. A **fixed point** of g is an input that g leaves unchanged: $g(c) = c$. Geometrically, it is where the graph of g meets the diagonal line $y = x$.

Corollary 8.2 (A fixed-point theorem)**.** *If g is continuous and maps $[a, b]$ into itself — that is, $a \le g(x) \le b$ for all $x \in [a, b]$ — then g has a fixed point in $[a, b]$.*

Proof. Consider the auxiliary function $h(x) = g(x) - x$, which is continuous as a difference of continuous functions. At the left end, $g(a) \ge a$ gives $h(a) = g(a) - a \ge 0$. At the right end, $g(b) \le b$

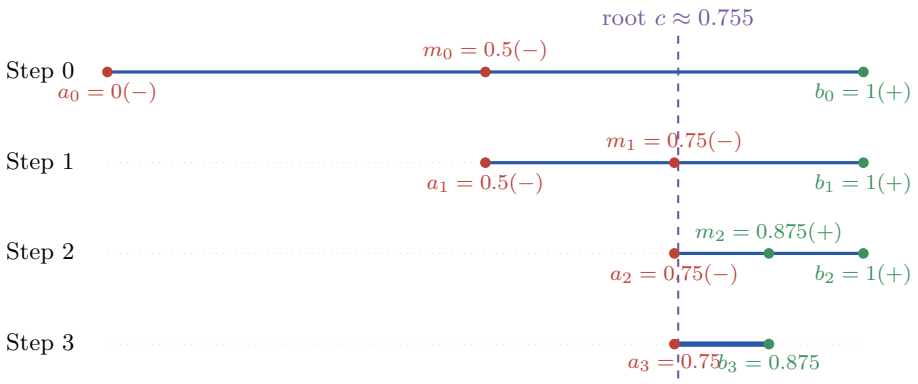


Figure 8.7: The bisection algorithm in action. At each stage the interval is bisected at its midpoint; sign evaluation determines which half preserves opposite signs, halving the cage around the root at every step.

gives $h(b) = g(b) - b \leq 0$. So h runs from a nonnegative value down to a nonpositive one. If either end is exactly 0 we have our fixed point outright; otherwise $h(a) > 0 > h(b)$, and the IVT hands us a c with $h(c) = 0$, i.e. $g(c) = c$. □

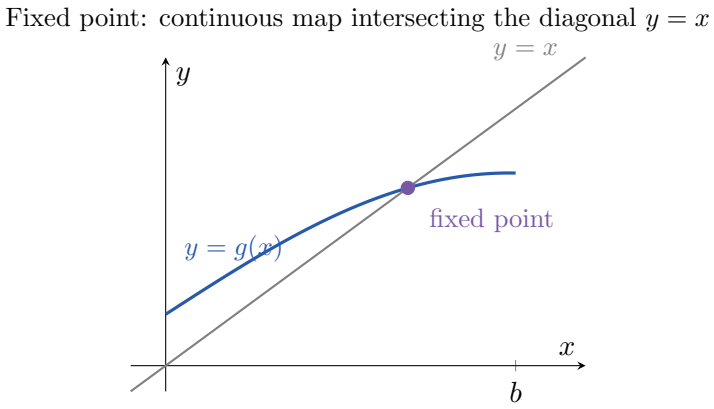


Figure 8.8: A fixed point is a crossing of $y = g(x)$ with the diagonal $y = x$. If a continuous g keeps its outputs inside $[a, b]$, its graph starts on or above the diagonal at a and ends on or below it at b, so it must cross.

Connection

This one-dimensional fixed-point theorem is the humble ancestor of a towering family. In higher dimensions it becomes Brouwer's Fixed-Point Theorem (every continuous map of a disk to itself has a fixed point), which underlies the existence of economic equilibria (Nash's theorem on equilibria leans on it) and of solutions to differential equations. The everyday image: stir a cup of coffee continuously, and when it settles, at least one molecule sits exactly where it started. Stretch and fold a paper map of a region and lay it back inside the same region — some point lands on top of where it began. All of it grows from the same seed: a continuous self-map cannot avoid the diagonal.

8.5 The Extreme Value Theorem

The IVT guarantees a function hits intermediate *values*. Our second great theorem guarantees it hits extreme ones — a genuine highest point and a genuine lowest point — provided we ask on a closed, bounded interval.

Theorem 8.4 (Extreme Value Theorem, EVT)**.** *If f is continuous on a closed bounded interval $[a, b]$, then f attains a maximum and a minimum there: there exist points $p, q \in [a, b]$ with*

$$f(q) \le f(x) \le f(p) \qquad \text{for all } x \in [a, b].$$

The word *attains* is the heart of it. It is not merely that the values stay bounded — it is that the largest value is actually *achieved* by some input p, and the smallest by some input q. The maximum is a value the function truly reaches, and not an unreached ceiling.

8.5.1 Why every hypothesis is necessary

This theorem is fragile: weaken any single condition and it collapses. Studying the failures shows exactly what each word is buying.

- **Drop "closed" (omit an endpoint).** On the *open* interval $(0, 1)$, $f(x) = x$ is continuous but attains neither a max nor a min: it gets arbitrarily close to 1 and to 0 without ever reaching them. The supremum 1 and infimum 0 exist as bounds but are never *achieved*, because the points that would achieve them were excluded from the domain.
- **Drop "bounded" (let the interval run to infinity).** On $[0, \infty)$, $f(x) = x$ is continuous but has no maximum — it grows without bound. A maximum cannot exist where the values escape to infinity.
- **Drop "continuous."** On $[0, 1]$, let $f(x) = 1/x$ for $x > 0$ and $f(0) = 0$. This is bounded below but shoots to $+\infty$ near 0, so it has no maximum despite living on a perfectly closed bounded interval. A single infinite discontinuity is enough to destroy the conclusion.

Extreme Value Theorem: global maximum and minimum on $[a, b]$

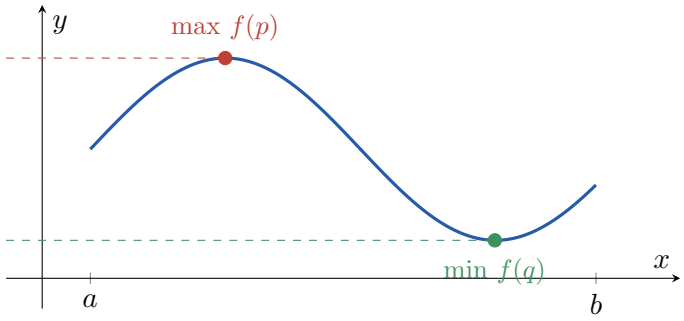


Figure 8.9: The Extreme Value Theorem. A continuous function on a closed bounded interval genuinely reaches a highest value $f(p)$ and a lowest value $f(q)$. The dashed lines mark the ceiling and floor, each *touched* by the curve.

Focus

The EVT is the silent partner behind all of optimization. Later, when we hunt for maxima and minima by setting derivatives to zero, we will be searching among *candidate* points — critical points and endpoints. But why should a maximum exist among them at all? Because the EVT promises one exists *somewhere*, and a short argument then shows that "somewhere" has to be a critical point or an endpoint. Without the EVT, we would be hunting for something that might not be there. It guarantees the prize is on the table before we start looking for it.

8.5.2 The shape of the argument

A complete proof leans on two facts about continuous functions on $[a, b]$ that are slightly beyond our scope to prove in full, but whose ideas come across clearly enough in outline. We sketch the route so the theorem does not feel like magic.

Intuition

Step 1: the function is bounded. First show f cannot run off to $\pm\infty$ on $[a, b]$. The mechanism: if f were unbounded, you could find inputs $x_1, x_2, \ldots$ in $[a, b]$ with $|f(x_n)| \to \infty$. Because $[a, b]$ is closed and bounded, these inputs must *accumulate* somewhere inside it — they cannot all spread out to infinity (the interval is finite) and cannot escape out an open end (there is none). Near that accumulation point, continuity forces the f-values to stay close to one finite number, flatly contradicting $|f(x_n)| \to \infty$. So f must be bounded.
Step 2: the bound is achieved. Let M be the least upper bound of the values $f(x)$ — it exists by completeness now that we know the values are bounded. By definition of least upper bound, there are inputs with f-values as close to M as we like; collect such inputs $x_1, x_2, \ldots$ with $f(x_n) \to M$. Again they accumulate at some point p in $[a, b]$, and continuity at p forces $f(p) = M$. So the supremum M is actually *attained* at p. Apply the same reasoning to $-f$ to attain the minimum.

What is really happening

Both steps secretly use the same precious property of $[a, b]$: *any infinite sequence of points in a closed bounded interval has a point it crowds up against, inside the interval.* This is the Bolzano–Weierstrass property, and it is what "closed and bounded" really buys you. "Bounded" keeps the points from escaping to infinity; "closed" keeps the accumulation point from leaking out through a missing endpoint. Strip away either and the accumulation point can vanish — which is precisely why the counterexamples on $(0, 1)$ and $[0, \infty)$ work. Continuity then does the final job: it transfers the accumulation of *inputs* into a statement about *outputs*, letting a limiting value be genuinely reached. The EVT is the marriage of compactness (closed + bounded) with continuity.

Connection

The phrase "closed and bounded" is the one-dimensional face of a concept that organizes all of analysis and topology: **compactness**. The grand statement, of which the EVT is a special case, is that *a continuous real-valued function on a compact set attains its maximum and minimum.* Compactness is exactly the abstraction of "no points can escape" that we needed

in both steps above. When you later meet compactness in a topology course, you can think back to this chapter: it is the property that makes the Extreme Value Theorem true, distilled to its essence and freed from the real line.

8.6 Why continuity is the right notion of "no jumps"

We close by stepping back. We *defined* continuity by a limit equation, but is that definition the right one — does it really capture "no jumps," nothing more and nothing less? The two big theorems are the evidence.

A function with a jump fails the IVT: the Heaviside step H leaps from 0 to 1 at the origin and skips every value strictly between, like $\frac{1}{2}$. So "jumpy" functions are exactly the ones that can dodge intermediate values, and continuous functions are exactly the ones that cannot. The IVT is, in effect, the theorem that says *our definition of continuity is strong enough to forbid jumps.* Conversely, our gallery showed that every way a function can fail Definition 8.1 — hole, jump, pole, oscillation — is visibly a break in the curve. The definition forbids breaks, and only breaks.

Why this matters

There is a deep economy here worth pausing on. We wanted a geometric idea — an unbroken curve — and we found we could capture it with a single algebraic equation, $\lim_{x\to a} f(x) = f(a)$, that mentions no pictures at all. Then that one humble equation, applied with the completeness of $\mathbb{R}$, forced two sweeping existence theorems: every intermediate value is hit (IVT), and extreme values are genuinely attained (EVT). This is the recurring miracle of analysis: a precise local condition (how a function behaves near each single point) propagates into powerful global conclusions (what the function must do across an entire interval). Local restraint, global consequence. The whole subject runs on this current, and continuity is where you first feel it.

Pause

Before the exercises, the through-line in one breath. Continuity at a means the limit equals the value, which is the precise form of "no jump." Continuity is inherited through sums, products, quotients (away from zeros), and compositions, so almost every function you write down is continuous on its natural domain. On the real line — gapless by completeness — continuity forces the Intermediate Value Theorem (every in-between value is attained, the engine of root-finding and fixed points) and, on closed bounded intervals, the Extreme Value Theorem (max and min are truly reached, the foundation of optimization). Every one of these rests on the same modest equation. If a step ever feels mysterious, return to the picture of a pencil that never leaves the page.

8.7 Exercises

Exercise 8.1 (Classify the discontinuity)**.** For each function, find every point of discontinuity on $\mathbb{R}$ and classify it as removable, jump, infinite, or oscillatory.

$$\text{(a) } \frac{x^2-9}{x-3}, \quad \text{(b) } \frac{1}{(x-2)^2}, \quad \text{(c) } \frac{|x|}{x}, \quad \text{(d) } \cos\Big(\frac{1}{x}\Big).$$

For any removable case, state the value that repairs it.

Exercise 8.2 (Make it continuous)**.** Find the value of k that makes

$$f(x) = \begin{cases} \dfrac{x^2-4}{x-2}, & x \neq 2, \\ k, & x = 2, \end{cases}$$

continuous at $x = 2$. Then do the same for $g(x) = \dfrac{\sqrt{x+1}-1}{x}$ at $x = 0$ (rationalize the numerator first).

Exercise 8.3 (Continuity by inheritance)**.** Without using ε and δ, explain why each function is continuous on the stated set, naming which combination or composition laws you use: (a) $\dfrac{\sin x}{x^2+1}$ on $\mathbb{R}$; (b) $\sqrt{4-x^2}$ on $[-2,2]$; (c) $e^{1/x}$ on $(0,\infty)$.

Exercise 8.4 (A root exists)**.** Show that the equation $\cos x = x$ has a solution in $\big[0, \frac{\pi}{2}\big]$. (Hint: apply the IVT to $h(x) = \cos x - x$ and check the endpoints.) Then use one bisection step to locate the solution in a smaller interval.

Exercise 8.5 (A fixed point in the wild)**.** Let $g(x) = \frac{1}{2}\cos x$. Show g maps the interval $[-1,1]$ into itself and is continuous, hence has a fixed point there. What equation does the fixed point satisfy?

Exercise 8.6 (An interval always has a midpoint value)**.** Suppose f is continuous on $[a,b]$. Prove there is a point $c \in [a,b]$ with

$$f(c) = \frac{f(a)+f(b)}{2}.$$

(The average of the two endpoint heights lies between them.) Is the same true for $\frac{f(a)+2f(b)}{3}$? Explain.

Exercise 8.7 (Every hypothesis matters)**.** Give an explicit example of a function that is continuous on the half-open interval $(0,1]$ and is *unbounded* there, showing the EVT genuinely needs the left endpoint. Then give a function continuous and bounded on $(0,1]$ that nonetheless attains no maximum.

Exercise 8.8 (The temperature on a great circle, a proof)**.** Temperature varies continuously along the equator. Prove there exist two diametrically opposite points on the equator with exactly the same temperature. Model the equator by $[0, 2\pi]$ with $T(0) = T(2\pi)$, and apply the IVT to the function $D(\theta) = T(\theta + \pi) - T(\theta)$ on $[0, \pi]$. (Hint: examine $D(0)$ and $D(\pi)$ and notice they have opposite signs unless one is already zero.)

Exercise 8.9 (A continuous bijection's inverse, conceptual)**.** Argue, using the IVT, that a continuous function on an interval that is one-to-one must be strictly monotonic (always increasing or always decreasing). (Hint: if it rose then fell, the IVT would force it to repeat a value, contradicting one-to-oneness.) This is the secret reason continuous invertible functions have continuous inverses — a fact we will use when differentiating $\ln x$ and $\arcsin x$.

Exercise 8.10 (Prove a combination law)**.** Using only Definition 8.1 and the limit laws, prove carefully that if f and g are continuous at a then fg is continuous at a. Then explain in one sentence why your proof breaks down for f/g when $g(a) = 0$.

Exercise 8.11 (Dirichlet's monster, a challenge)**.** Let $D(x) = 1$ if x is rational and $D(x) = 0$ if x is irrational. Show D is discontinuous at *every* real number. (Hint: in any interval, however small, there are both rationals and irrationals, so the values 0 and 1 both occur arbitrarily close to any point, and no single limit can exist.) This function has no points of continuity at all — the opposite extreme from the well-behaved functions of this chapter.

Chapter 9

Limits at Infinity and Asymptotes

So far our limits have asked a local question: as x creeps toward some finite landmark a, where does $f(x)$ head? In this chapter we step back and ask a global one. *Where does the function go in the long run?* Walk a mile to the right along the graph, then ten miles, then a thousand. Does the curve settle down onto a level highway? Does it shoot off to the heavens? Does it hug a tilted line forever without ever touching it?

This is the study of **end behavior**, and its natural language is the **limit at infinity**. It is also the study of **asymptotes** — the straight lines a curve approaches but (often) never reaches. Asymptotes are the skeleton of a graph. If you know the asymptotes, you already know the silhouette of the function before you plot a single careful point. And once you can read end behavior fluently, you can compare how fast different functions grow — which is the single most useful skill in all of applied mathematics, from sorting algorithms to the stability of bridges to the half-life of a drug.

9.1 What "Limit at Infinity" Really Means

When we wrote $\lim_{x\to a} f(x) = L$ for a finite a, the symbol $x \to a$ meant "x gets arbitrarily close to a." But ∞ is not a number, and x cannot get *close* to it — there is nothing to get close to. So we have to say precisely what the new symbol asserts.

What is really happening

The statement $\lim_{x\to\infty} f(x) = L$ does **not** say that x arrives anywhere. It is a promise about the long run:

No matter how tight a tolerance you demand around L, I can name a starting point far enough to the right so that, from there on, $f(x)$ stays inside that tolerance.

There is no last value of x, no destination. There is only the guarantee that *eventually and forever after*, the output is pinned near L. The whole content of "$x \to \infty$" is the word *eventually*.

Here is the precise version, the exact analogue of the ε–δ definition but with the role of "close to a" played by "far to the right."

Definition 9.1 (Limit at infinity)**.** We write $\lim_{x\to\infty} f(x) = L$ if: for every $\varepsilon > 0$ there exists a real number M such that

$$x > M \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

Likewise $\lim_{x\to-\infty} f(x) = L$ if for every $\varepsilon > 0$ there is an M with $x < M \implies |f(x) - L| < \varepsilon$.

Compare this with the finite-a definition. There, ε bought you a δ and you looked inside the window $0 < |x - a| < \delta$. Here, ε buys you a threshold M and you look at the entire tail $x > M$. The picture is not a shrinking window around a point; it is a horizontal corridor of half-width ε around the height L, and the claim is that the graph enters that corridor and never leaves.

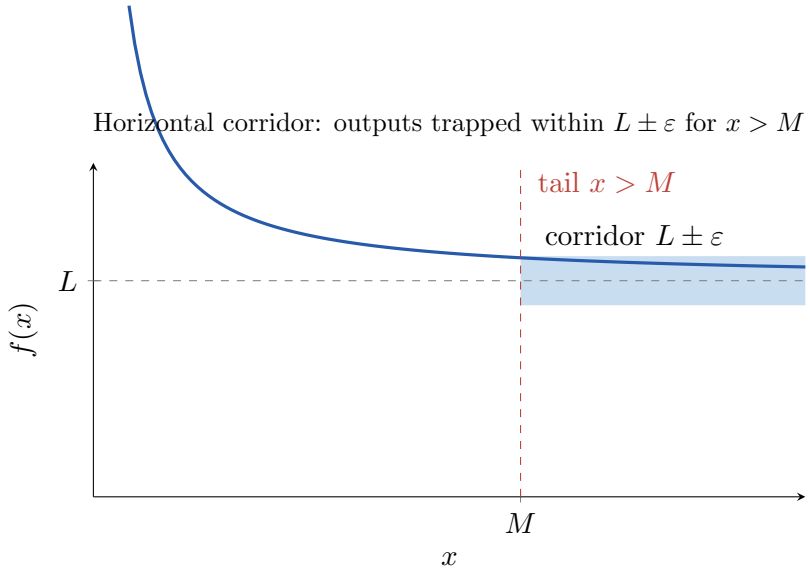


Figure 9.1: A limit at infinity. Once x passes the threshold M, the graph is trapped inside the corridor of half-width ε around the height L. Shrink ε and you may need a larger M, but such an M always exists.

Work it yourself

Show directly from Definition 9.1 that $\lim_{x\to\infty} \frac{1}{x} = 0$. Given $\varepsilon > 0$, we need $x > M \implies |1/x - 0| < \varepsilon$, i.e. $1/x < \varepsilon$ (for $x > 0$), i.e. $x > 1/\varepsilon$. So *choose* $M = 1/\varepsilon$. Then any $x > M$ satisfies $1/x < 1/M = \varepsilon$. Done. Notice how M depends on ε: demand a tolerance ten times tighter and your threshold moves ten times farther out. That is the whole mechanism.

9.1.1 Infinite limits: when the output runs away

There is a second, completely different way for a function to behave badly: the *output* can blow up. We already met this near vertical features like $1/x$ at $x = 0$. Let us pin down the language, because "$= \infty$" is another phrase that needs a careful reading.

Definition 9.2 (Infinite limit)**.** We write $\lim_{x\to a} f(x) = +\infty$ if for every $N > 0$ there is a $\delta > 0$ such that

$$0 < |x - a| < \delta \quad \Longrightarrow \quad f(x) > N.$$

That is: name any height N, however large, and the graph eventually climbs above it and stays above it as x closes in on a. The case $-\infty$ and the cases $x \to \pm\infty$ are defined by the obvious changes.

Common trap

The equals sign in $\lim f(x) = \infty$ is a **convenient lie**. There is no number ∞ that the limit equals. Strictly, the limit *does not exist* — we are just describing *how* it fails to exist: not by oscillating, not by jumping, but by growing without bound. Treat "$= \infty$" as one indivisible phrase meaning "increases beyond every bound," never as an arithmetic value you can add or divide.

History

The struggle to make these ideas rigorous is the great drama of 19th-century analysis. Newton and Leibniz computed with quantities that were "ultimately" vanishing or "infinitely" large and got correct answers by sure instinct, but Bishop Berkeley mocked the infinitesimals as "ghosts of departed quantities," and he had a point: nobody could say what they *were*. The repair came in stages. Bernard Bolzano (1810s) and especially Augustin-Louis Cauchy (1820s) recast limits as statements about getting arbitrarily close. Then Karl Weierstrass, lecturing in Berlin in the 1860s, supplied the final grammar — the ε, the δ, and (for our chapter) the threshold M — so that "tends to infinity" became a finite, checkable sentence with no ghosts in it. Definition 9.1 is Weierstrass's gift: a way to talk about the infinite using only finite quantities.

9.2 Horizontal Asymptotes: the Level Highway

A horizontal asymptote is simply a finite limit at infinity, read off the graph.

Definition 9.3 (Horizontal asymptote)**.** The line $y = L$ is a **horizontal asymptote** of f if $\lim_{x\to\infty} f(x) = L$ or $\lim_{x\to-\infty} f(x) = L$ (or both).

A few things worth saying out loud, because each contradicts a common belief.

- A graph **may cross its horizontal asymptote**, even infinitely often. The asymptote describes the *tail*, not a barrier. The function $f(x) = \dfrac{\sin x}{x}$ crosses $y = 0$ at every nonzero multiple of π yet still has $y = 0$ as a horizontal asymptote, because the crossings get ever smaller.
- A function can have **two different** horizontal asymptotes, one to the right and one to the left. We will meet $\arctan x$ and the logistic curve, which level off at different heights in the two directions.
- Having a horizontal asymptote is a statement about ends only. It says nothing about what happens in the middle, where the function may do anything.

9.2.1 Two asymptotes, one curve: the inverse tangent and the logistic

Some of the most important curves in science have a floor on one side and a ceiling on the other. The inverse tangent levels off at $\pi/2$ as $x \to +\infty$ and at $-\pi/2$ as $x \to -\infty$. The logistic function

Damped oscillation: $\frac{\sin x}{x}$ repeatedly crossing its asymptote $y = 0$

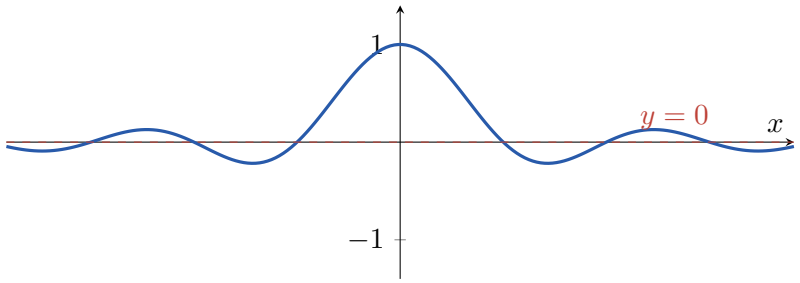


Figure 9.2: $f(x) = \sin(x)/x$ crosses its horizontal asymptote $y = 0$ at every nonzero multiple of π. The asymptote governs the *tail*, not a forbidden line. The wiggles decay because the amplitude $1/|x|$ shrinks.

$\sigma(x) = \dfrac{1}{1 + e^{-x}}$ — the workhorse of population models and of machine-learning classifiers — rises from the floor $y = 0$ to the ceiling $y = 1$. In both cases the graph is sandwiched between two horizontal highways.

Two distinct horizontal asymptotes: logistic sigmoidal saturation

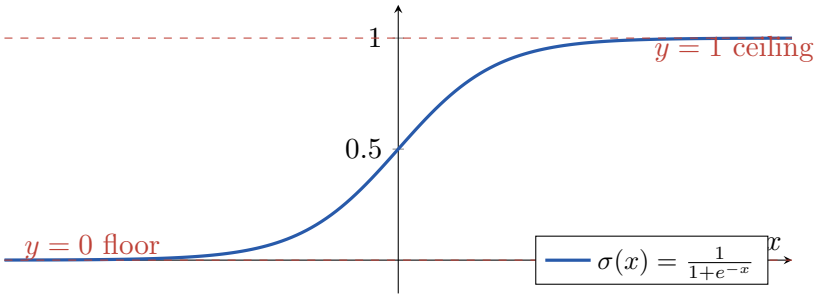


Figure 9.3: The logistic curve $\sigma(x)$ has two horizontal asymptotes: a floor at $y = 0$ (as $x \to -\infty$) and a ceiling at $y = 1$ (as $x \to +\infty$). Most saturating processes in nature look like this.

Application

Why classifiers love the ceiling and floor. A neural network outputs a raw score z that can be any real number, but a probability must live in $[0, 1]$. Passing z through σ squashes it: huge positive z saturates near 1 ("confident yes"), huge negative z saturates near 0 ("confident no"). The two horizontal asymptotes are precisely what make σ a sensible probability. The same shape models a population approaching its carrying capacity, a chemical reaction approaching completion, or a rumor saturating a town — anything that grows fast at first and then runs out of room.

9.3 Vertical Asymptotes: the Cliff

A horizontal asymptote happens because x ran off to infinity. A **vertical** asymptote is the opposite: x stays at a finite spot, but the *output* runs to infinity.

Definition 9.4 (Vertical asymptote)**.** The line $x = a$ is a **vertical asymptote** of f if at least one of the one-sided limits $\lim_{x\to a^+} f(x)$ or $\lim_{x\to a^-} f(x)$ is $+\infty$ or $-\infty$.

The usual culprit is a denominator that hits zero while the numerator does not. Near such a point the fraction has a fixed, nonzero amount on top divided by a vanishing amount on the bottom, and "fixed over tiny" is "huge." The sign on each side tells you whether the cliff goes up or down.

Work it yourself

Find the vertical asymptote of $f(x) = \dfrac{x+2}{x-3}$ and its two-sided behavior. The denominator vanishes at $x = 3$, where the numerator is $5 \neq 0$, so $x = 3$ is a vertical asymptote. Now test the signs. Just to the *right* of 3, say $x = 3.01$: top $\approx 5 > 0$, bottom $= 0.01 > 0$, ratio $\approx +500$, so $f \to +\infty$. Just to the *left*, $x = 2.99$: top ≈ 5, bottom $= -0.01 < 0$, ratio ≈ -500, so $f \to -\infty$. The curve dives down on the left of the cliff and rockets up on the right.

Common trap

A zero in the denominator does not guarantee a vertical asymptote. If the numerator vanishes there too, the factor may cancel and leave only a *hole*. Consider $g(x) = \dfrac{x^2-9}{x-3} = \dfrac{(x-3)(x+3)}{x-3}$. For $x \neq 3$ this equals $x + 3$, a perfectly tame line with a single missing point at $(3, 6)$. No cliff — just a puncture. Always factor and cancel *before* declaring an asymptote. Asymptote when the zero survives in the bottom; hole when it cancels.

Duality of asymptotes: vertical cliff ($x = 3$) and horizontal highway ($y = 1$)

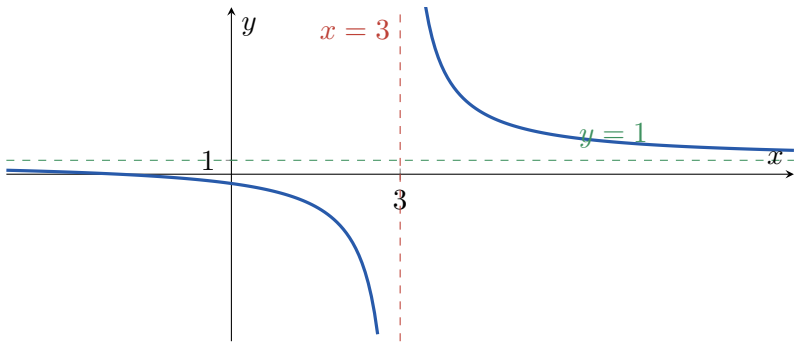


Figure 9.4: $f(x) = \frac{x+2}{x-3}$ has a vertical asymptote at $x = 3$ (output runs off) and a horizontal asymptote at $y = 1$ (input runs off). The two kinds of asymptote are dual: one freezes x and lets y escape, the other freezes the level and lets x escape.

9.4 Dominant-Term Reasoning for Rational Functions

How did we know the horizontal asymptote in Figure 9.4 was $y = 1$? There is a fast, intuitive method that works for every rational function and, once understood, frees you from memorizing a table of cases.

Intuition

When x is enormous, a polynomial is **ruled by its highest-degree term**. In $3x^2+50x+9000$, plug in $x = 10^6$: the $3x^2$ term is 3×10^{12}, while $50x = 5 \times 10^7$ and the constant 9000 are rounding error by comparison. The leading term is the elephant; the rest are fleas. So for end behavior, *keep only the leading term of the top and the leading term of the bottom, and simplify.*

Why the technique works

Why dominant-term reasoning is exact and rigorous. Take $f(x) = \dfrac{3x^2+50x+9000}{x^2-7}$. Divide top and bottom by x^2, the highest power present:

$$f(x) = \frac{3 + \dfrac{50}{x} + \dfrac{9000}{x^2}}{1 - \dfrac{7}{x^2}}.$$

This is not an approximation; it is an exact equality (for $x \neq 0$). Now every term with an x in the denominator $\to 0$ as $x \to \infty$, by the $1/x \to 0$ fact we proved by hand. So the whole thing tends to $\dfrac{3+0+0}{1-0} = 3$. The "keep the leading terms" shortcut is just this division, performed in your head. Dividing by the dominant power is the *mechanism*; the shortcut is its shadow.

The general rule falls out of comparing the degree of the numerator p with the degree of the denominator q. Let $\deg p = n$ with leading coefficient a, and $\deg q = m$ with leading coefficient b.

Focus

End behavior of $\dfrac{p(x)}{q(x)}$ as $x \to \pm\infty$.

- **Bottom wins** ($n < m$): the fraction $\to 0$. Horizontal asymptote $y = 0$. (Top grows slower; the big denominator crushes it.)
- **Tie** ($n = m$): the fraction $\to a/b$, the ratio of leading coefficients. Horizontal asymptote $y = a/b$.
- **Top wins** ($n > m$): the fraction $\to \pm\infty$. No horizontal asymptote. If $n = m+1$ exactly, there is instead a *slant* asymptote (next section).

Example 9.1 (All three cases at once)**.**

$$\frac{2x+1}{x^2+5} \to 0 \quad (n = 1 < m = 2), \qquad \frac{4x^2-x}{2x^2+3} \to \frac{4}{2} = 2 \quad (n = m = 2),$$

$$\frac{x^3}{x^2+1} \to \pm\infty \quad (n = 3 > m = 2).$$

For the middle one, the horizontal asymptote is $y = 2$ in *both* directions, since the leading coefficients do not care about the sign of x when both degrees are even and equal.

Common trap

When degrees *tie*, students sometimes answer "1" by reflex, as if the leading x's simply cancel. They do not: you keep the *coefficients*. In $\frac{4x^2-x}{2x^2+3}$ the limit is $4/2 = 2$, not 1. And the lower-degree terms ($-x$ and $+3$) are genuinely irrelevant to the limit — but only to the *limit*. They still bend the curve before it settles.

Walking question

Without computing, decide the end behavior of each, just by eyeing leading terms: $\dfrac{x^4 - 100x}{3x^4 + 1}$, $\dfrac{7x}{x^3 - x}$, $\dfrac{x^2 - 1}{x + 1}$. (Answers: $\to \frac{1}{3}$; $\to 0$; the last one cancels to $x - 1$, so it heads to $\pm\infty$ along a line — a slant asymptote in disguise.)

9.4.1 Roots and other non-polynomial tops

Dominant-term reasoning extends far beyond polynomials. Whenever you can identify the fastest-growing piece on top and on bottom, the same divide trick works — you just have to be careful about signs under square roots.

Example 9.2 (A square root that needs care)**.** Find $\lim_{x\to\pm\infty} \dfrac{\sqrt{4x^2+1}}{x+3}$. The dominant piece on top is $\sqrt{4x^2} = 2|x|$, and on the bottom it is x. Divide top and bottom by x. The subtlety: $\sqrt{4x^2+1}/x = \pm\sqrt{4+1/x^2}$, where the sign is $+$ if $x > 0$ and $-$ if $x < 0$, because $\sqrt{x^2} = |x|$. Hence

$$x \to +\infty: \quad \frac{\sqrt{4+1/x^2}}{1+3/x} \to \frac{\sqrt{4}}{1} = 2, \qquad x \to -\infty: \quad \frac{-\sqrt{4+1/x^2}}{1+3/x} \to -2.$$

Two different horizontal asymptotes, $y = 2$ and $y = -2$, born entirely from the $\sqrt{x^2} = |x|$ sign rule.

Common trap

$\sqrt{x^2} = |x|$, **not** x. For $x \to -\infty$ this single fact flips a sign and gives you the second asymptote. Forgetting it is the most common error in limits-at-infinity with radicals. When dividing by x inside a square root, you are really dividing by $\sqrt{x^2} = |x|$, so a minus sign appears on the negative side.

Example 9.3 (A sneaky difference of roots)**.** Find $\lim_{x\to\infty}\left(\sqrt{x^2+x} - x\right)$. As written this is "$\infty - \infty$," which decides nothing. Multiply by the conjugate over itself:

$$\sqrt{x^2+x} - x = \frac{(\sqrt{x^2+x} - x)(\sqrt{x^2+x} + x)}{\sqrt{x^2+x} + x} = \frac{x^2 + x - x^2}{\sqrt{x^2+x} + x} = \frac{x}{\sqrt{x^2+x} + x}.$$

Now divide top and bottom by x (using $\sqrt{x^2+x}/x = \sqrt{1+1/x}$ for $x > 0$):

$$= \frac{1}{\sqrt{1+1/x} + 1} \longrightarrow \frac{1}{\sqrt{1+1}} = \frac{1}{2}.$$

The two huge quantities differ by a finite gap of $\frac{1}{2}$ in the limit. The conjugate trick converts a hopeless $\infty - \infty$ into a clean ratio where dominant terms can speak.

9.5 Slant (Oblique) Asymptotes: the Tilted Highway

When the top of a rational function out-grows the bottom by exactly one degree, the curve does not flatten and does not merely blow up vertically. Instead it approaches a *slanted line*. The graph rides a tilted highway off to infinity.

Intuition

A slant asymptote is what polynomial *long division* sees. Dividing $p(x)$ by $q(x)$ when $\deg p = \deg q + 1$ gives a linear quotient $mx + b$ plus a proper remainder $r(x)/q(x)$:

$$\frac{p(x)}{q(x)} = \underbrace{mx + b}_{\text{the line}} + \underbrace{\frac{r(x)}{q(x)}}_{\to 0}.$$

The remainder term has a smaller-degree top than bottom, so by the "bottom wins" rule it vanishes at infinity. What is left in the long run is exactly the line $y = mx + b$. **The quotient of long division is the asymptote.**

Example 9.4 (Finding a slant asymptote by division)**.** Consider $f(x) = \dfrac{x^2 + 2x - 1}{x - 1}$. Degree 2 over degree 1: expect a slant. Long-divide:

$$x^2 + 2x - 1 = (x - 1)(x + 3) + 2, \qquad \text{so} \qquad f(x) = x + 3 + \frac{2}{x - 1}.$$

As $x \to \pm\infty$ the remainder $\dfrac{2}{x-1} \to 0$, so $f(x)$ approaches the line $y = x + 3$. The sign of $\dfrac{2}{x-1}$ also tells you *which side* of the line the curve rides: positive (above the line) for $x > 1$, negative (below) for $x < 1$. There is also a vertical asymptote at $x = 1$, where the remainder itself blows up.

Slant asymptote: long-division quotient $y = x + 3$ taming degree-2 rational curve

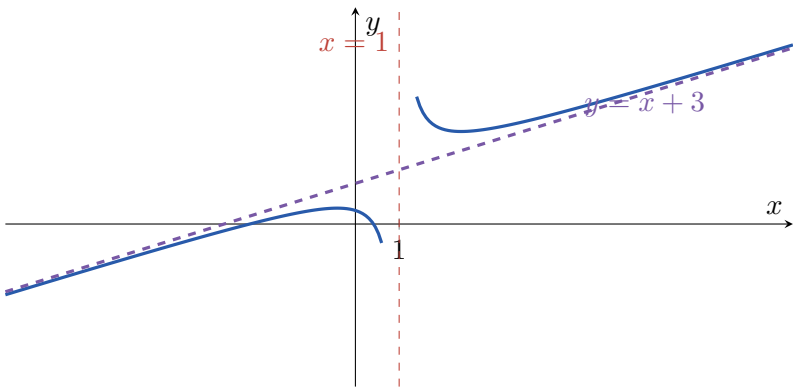


Figure 9.5: $f(x) = \frac{x^2+2x-1}{x-1}$ has a slant asymptote $y = x + 3$ and a vertical asymptote $x = 1$. Far out, the curve is indistinguishable from the tilted line; the gap between them is the remainder $2/(x - 1)$, which melts away.

Why the technique works

Why "the quotient is the asymptote" is an exact theorem. Two functions f and g are *asymptotic* if $f(x) - g(x) \to 0$ as $x \to \infty$ — their graphs become indistinguishable. Long division writes $f = (mx+b) + \text{remainder}$ with remainder $\to 0$, which is *exactly* the statement $f(x) - (mx+b) \to 0$. So the line $mx+b$ is asymptotic to f by definition. The same logic shows the asymptote need not be a line at all: if the top exceeds the bottom by two degrees, the quotient is a *parabola*, and the curve hugs that parabola at infinity. People rarely name these "parabolic asymptotes," but the mechanism is identical — whatever the polynomial quotient is, the curve becomes it.

Common trap

A graph has a slant asymptote **or** a horizontal asymptote, never both in the same direction — a line is either tilted or level. If $\deg p = \deg q$ you get a horizontal asymptote; if $\deg p = \deg q + 1$ you get a slant one; if the gap is 2 or more there is no straight-line asymptote at all. Always compare degrees *first* to know which question you are even asking.

9.6 Growth Rates Revisited: Who Wins the Long Race?

Dominant-term reasoning settled contests *within* the polynomials. But the deepest and most useful comparisons are *across* families: logarithms versus powers versus exponentials. These three grow at wildly different speeds, and the pecking order never changes once x is large enough.

Focus

The growth hierarchy (for large x).

$$\ln x \ \ll\ x^p\ (p>0)\ \ll\ a^x\ (a>1)\ \ll\ x!\ \ll\ x^x.$$

Here $f \ll g$ means $\dfrac{f(x)}{g(x)} \to 0$ as $x \to \infty$: f is utterly outrun by g. *Logs crawl, powers walk, exponentials sprint, factorials fly.* Every exponential, no matter how gentle its base (say 1.000001^x), eventually overtakes every power, no matter how steep (say x^{1000}).

The claim is important enough, and counterintuitive enough, to earn a proof rather than a slogan. Here is *why* the exponential always wins.

Intuition

Why exponentials beat powers: constant percentage versus constant amount. A power like x^2 grows by an ever-smaller *percentage* as x climbs — going from 100 to 101 barely budges x^2 in relative terms. An exponential like 2^x grows by a *fixed percentage* (100% per unit step) forever. Fixed-percentage growth compounds; eventually-shrinking-percentage growth cannot keep up. The race is not close: it is a rout, just delayed.

Theorem 9.1 (Exponential dominates every power)**.** *For every fixed $p > 0$ and every $a > 1$,*

$$\lim_{x\to\infty} \frac{x^p}{a^x} = 0.$$

Proof. The cleanest path is through logarithms, which turn the lopsided ratio into a manageable difference. Take logs of the ratio:

$$\ln\left(\frac{x^p}{a^x}\right) = p\ln x - x\ln a.$$

We will show this $\to -\infty$; then the ratio itself, being e to that power, $\to 0$. Factor out x:

$$p\ln x - x\ln a = x\left(\frac{p\ln x}{x} - \ln a\right).$$

Here $\ln a > 0$ is a fixed positive constant. So everything hinges on the term $\dfrac{\ln x}{x}$. If we can show $\dfrac{\ln x}{x} \to 0$, then the bracket tends to $0 - \ln a = -\ln a < 0$, while the factor $x \to +\infty$; a positive-going x times a negative-bounded bracket gives $-\infty$. So it remains only to prove the cornerstone fact: $\ln x/x \to 0$.

Cornerstone: $\ln x \ll x$. For $t \geq 1$ we have $\dfrac{1}{t} \leq \dfrac{1}{\sqrt{t}}$, so integrating from 1 to x,

$$\ln x = \int_1^x \frac{1}{t}\,\mathrm{d}t \leq \int_1^x \frac{1}{\sqrt{t}}\,\mathrm{d}t = 2\sqrt{x} - 2 \leq 2\sqrt{x}.$$

Therefore $\dfrac{\ln x}{x} \leq \dfrac{2\sqrt{x}}{x} = \dfrac{2}{\sqrt{x}} \to 0$. That bound — $\ln x \leq 2\sqrt{x}$ — is the whole secret: the logarithm is trapped beneath a mere square root, so it is hopelessly slow. Feeding this back, $\dfrac{p\ln x}{x} \to 0$, the bracket $\to -\ln a < 0$, the product $\to -\infty$, and the original ratio $\to 0$. □

What is really happening

What the logarithm trick is really doing. Exponentials and powers live on different scales, and dividing them directly is like comparing a sprinter's position to a walker's — both numbers race off to infinity and the ratio is a ∞/∞ blur. The logarithm *changes the ruler*: it converts multiplicative growth into additive growth, a^x into the straight line $x\ln a$, and x^p into the slow curve $p\ln x$. On this new ruler the contest is between a *line* and a *logarithm*, and the line grows dramatically faster. Taking logs is the standard move whenever you must compare two runaway quantities: it tames both at once.

Connection

Big-O and the speed of algorithms. Computer science lives on this hierarchy. An algorithm that does $\ln n$ work (binary search) is extremely fast; n or $n\ln n$ (a good sort) is fine; n^2 (a naive sort) is sluggish on big inputs; and 2^n (brute-forcing every subset) is hopeless past a few dozen items. The symbol "$f = O(g)$" means f grows *no faster than* a constant times g — a bounded ratio — and "$f = o(g)$" is exactly our $f \ll g$. The same limits that shape a curve's tail decide whether a program finishes before the sun burns out.

Work it yourself

Feel the rout with your own arithmetic. Compare x^{10} against 2^x at a few points. At $x = 10$: $x^{10} = 10^{10} = 10{,}000{,}000{,}000$ while $2^{10} = 1024$ — the power is ahead by seven orders of magnitude. At $x = 60$: $2^{60} \approx 1.15 \times 10^{18}$ while $60^{10} \approx 6.0 \times 10^{17}$ — the exponential has just pulled even. Past there, 2^x leaves x^{10} in the dust forever. The crossover is late, but it is inevitable. That delay is exactly why the behavior can appear settled for small inputs while the asymptotic truth is the exact opposite.

Example 9.5 (Putting the hierarchy to work)**.** Evaluate each by spotting the dominant family.

$$\lim_{x\to\infty} \frac{x^{100}}{e^x} = 0 \quad \text{(exp beats power)}, \qquad \lim_{x\to\infty} \frac{\ln x}{x^{0.001}} = 0 \quad \text{(power beats log)},$$

$$\lim_{x\to\infty} \frac{3^x + x^5}{3^x - 100} = \lim_{x\to\infty} \frac{1 + x^5/3^x}{1 - 100/3^x} = \frac{1+0}{1-0} = 1.$$

In the last one we divided by the dominant term 3^x, then used $x^5/3^x \to 0$ (exp beats power) and $100/3^x \to 0$. Same dominant-term strategy as for rational functions — only now the "leading term" is the fastest-growing *family*, not the highest power.

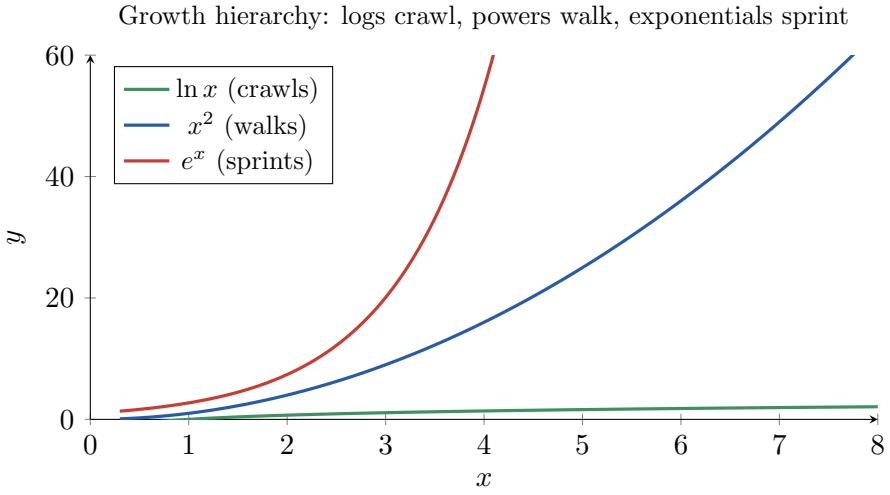


Figure 9.6: The growth hierarchy on one set of axes. For small x the order can look ambiguous, but e^x soon rockets past x^2, which long ago left $\ln x$ behind crawling along the floor. Extend the window and the gaps only widen.

9.7 Reading a Graph From Its Asymptotes

Put the pieces together. Given a rational function (or any function whose tails you can analyze), a short checklist lets you sketch an accurate silhouette before plotting careful points. Asymptotes are the scaffolding; the rest is filling in.

Focus

Asymptote checklist for $f(x) = p(x)/q(x)$.

1. **Factor** top and bottom. Cancel common factors — each cancellation is a *hole*, not an asymptote.
2. **Vertical asymptotes:** the surviving zeros of the denominator. Check the sign on each side to see if the cliff goes up or down.
3. **Horizontal/slant:** compare degrees. $n < m \Rightarrow y = 0$; $n = m \Rightarrow y = a/b$; $n = m+1 \Rightarrow$ slant (do the division); $n > m+1 \Rightarrow$ no line asymptote.
4. **Intercepts:** x-intercepts are the surviving zeros of the *numerator*; the y-intercept is $f(0)$ if defined.
5. **Sketch** the asymptotes as dashed lines first, then thread the curve through the intercepts and into the correct corners.

Example 9.6 (A full reading)**.** Sketch $f(x) = \dfrac{x^2-4}{x^2-1}$. *Factor:* $f(x) = \dfrac{(x-2)(x+2)}{(x-1)(x+1)}$ — no common factors, so no holes. *Vertical asymptotes:* $x = 1$ and $x = -1$. *Horizontal:* degrees tie ($n = m = 2$) with leading coefficients 1 and 1, so $y = 1$. *Intercepts:* zeros of the numerator give $x = \pm 2$; and $f(0) = \dfrac{-4}{-1} = 4$, the y-intercept. With two vertical cliffs, a level highway $y = 1$ at both ends, and the curve pinned through $(\pm 2, 0)$ and $(0, 4)$, the entire shape is forced. The middle branch has a local minimum at $(0, 4)$, staying strictly above the asymptote $y = 1$.

Asymptotic skeleton: reconstructing $f(x) = \frac{x^2-4}{x^2-1}$ from bounds and intercepts

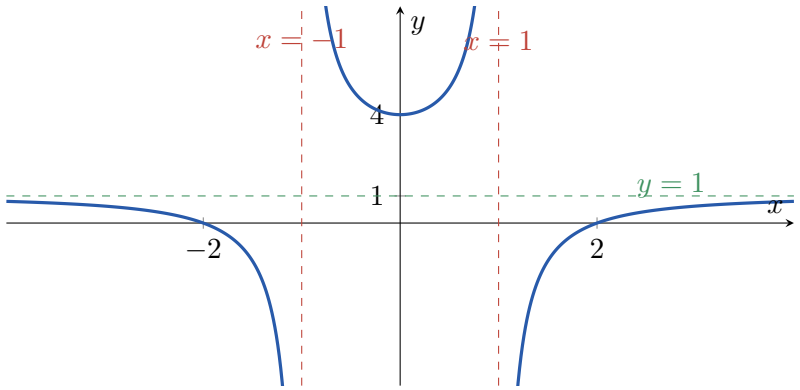


Figure 9.7: $f(x) = \frac{x^2-4}{x^2-1}$, read entirely from its asymptotes: verticals at $x = \pm 1$, a horizontal $y = 1$ at both ends, x-intercepts at ± 2, and the local minimum $(0, 4)$. The skeleton determines the silhouette.

Why this matters

Notice the economy. We never solved $f(x) = k$ for a single generic k, never built a table of twenty points. We asked three structural questions — where does the denominator die, how do the degrees compare, where are the zeros — and the graph fell out almost for free. This is the recurring pleasure of calculus: the right *questions* replace mountains of arithmetic. Asymptotes are the first place a beginner feels that leverage, and it is a feeling worth chasing

through every later chapter.

Connection

Asymptotes return as improper integrals. The vertical and horizontal asymptotes you sketch here become, a few chapters from now, the boundaries of *improper integrals* — areas that stretch out to infinity or pile up against a cliff. Whether such an infinite region has finite area is decided by exactly the growth-rate comparisons of this chapter: $\int_1^\infty x^{-2}\,dx$ converges because $1/x^2$ dies fast, while $\int_1^\infty x^{-1}\,dx$ diverges because $1/x$ dies too slowly. End behavior is not a side topic; it is the gatekeeper of which infinite sums and areas make sense.

9.8 Exercises

Exercise 9.1 (Warm-up limits)**.** Evaluate each limit, naming the dominant terms you used.

$$\text{(a) } \lim_{x\to\infty}\frac{5x^3-2x}{2x^3+7}, \quad \text{(b) } \lim_{x\to\infty}\frac{x+9}{x^2-4}, \quad \text{(c) } \lim_{x\to-\infty}\frac{6x^4+1}{1-3x^2}.$$

Exercise 9.2 (Two asymptotes from one radical)**.** Find both horizontal asymptotes of $f(x)=\dfrac{\sqrt{9x^2+2}}{4x-1}$, treating $x\to+\infty$ and $x\to-\infty$ separately. Explain, using $\sqrt{x^2}=|x|$, why the two answers differ in sign.

Exercise 9.3 (Conjugate trick)**.** Evaluate $\lim_{x\to\infty}\left(\sqrt{x^2+5x}-\sqrt{x^2+x}\right)$ by multiplying by the conjugate. (Answer: 2.) Then explain in one sentence why the naive answer "$\infty-\infty=0$" is wrong.

Exercise 9.4 (Hole versus asymptote)**.** For each function, state every vertical asymptote and every hole, with coordinates.

$$\text{(a) } \frac{x-2}{x^2-4}, \qquad \text{(b) } \frac{x^2+x-6}{x^2-9}, \qquad \text{(c) } \frac{x^2+1}{x^2-2x}.$$

Exercise 9.5 (Slant asymptote by division)**.** Find the slant asymptote of $f(x)=\dfrac{2x^2+3x-1}{x+2}$ and state on which side of the line the graph lies for large positive x and for large negative x. (Hint: the division leaves $f(x)=2x-1+\frac{1}{x+2}$.)

Exercise 9.6 (Full sketch)**.** Carry out the asymptote checklist for $g(x)=\dfrac{x^2-x-2}{x-3}$: holes, vertical asymptotes, slant asymptote, and all intercepts. Then sketch it, drawing the asymptotes as dashed lines first.

Exercise 9.7 (Growth-rate sorting)**.** Order the following by eventual growth, slowest to fastest, and justify the two closest calls with a limit:

$$x^{50}, \qquad (\ln x)^{1000}, \qquad 1.01^x, \qquad \sqrt{x}, \qquad x\ln x.$$

Exercise 9.8 (A crossing curve)**.** Give an explicit function with horizontal asymptote $y=2$ that *crosses* the line $y=2$ infinitely many times as $x\to\infty$. Prove your function does both. (Hint: a decaying oscillation added to 2.)

Exercise 9.9 (Prove a limit from the threshold definition)**.** Using Definition 9.1 directly, prove that $\lim_{x\to\infty} \dfrac{2x-1}{x+4} = 2$. That is, given $\varepsilon > 0$, produce an explicit M (in terms of ε) so that $x > M$ forces $\left|\frac{2x-1}{x+4} - 2\right| < \varepsilon$. (Hint: the difference simplifies to $\frac{-9}{x+4}$.)

Exercise 9.10 (Prove log loses to every power)**.** Adapt the integral bound $\ln x \le 2\sqrt{x}$ from the proof of Theorem 9.1 to show that for every fixed $p > 0$, $\lim_{x\to\infty} \dfrac{\ln x}{x^p} = 0$. (Hint: it suffices to beat x^p with $\sqrt{x}$ when $p \ge \frac{1}{2}$; for small p, bound $\ln x$ by $\frac{1}{q}x^q$ for a suitable $q < p$ using $\int t^{q-1}\,\mathrm{d}t$.)

Exercise 9.11 (Parabolic asymptote)**.** The function $f(x) = \dfrac{x^3+1}{x}$ has a top exceeding the bottom by two degrees, so no straight-line asymptote. By dividing, show that $f(x)$ becomes indistinguishable from a *parabola* for large $|x|$, and state that parabola. Confirm $f(x) - (\text{parabola}) \to 0$.

Exercise 9.12 (Conceptual)**.** True or false, with a one-line reason each. (a) A function can have at most two horizontal asymptotes. (b) A function can have infinitely many vertical asymptotes. (c) If $\lim_{x\to\infty} f(x) = L$ then f never equals L. (d) A rational function whose denominator has higher degree than its numerator always has the horizontal asymptote $y = 0$.

Part III

The Derivative

Chapter 10

The Derivative: Slope, Rate, and the Tangent

There is a single question lurking behind everything in this book, and you have been circling it for nine chapters. We learned what functions are, we tamed the exponential and the trigonometric ones, and then we spent a long time building the machinery of limits so we could speak precisely about "approaching" without ever quite arriving. All of that was preparation. The reward is the idea in this chapter, one of the great ideas in human thought:

Focus

How fast is something changing *right now* – at a single instant, not over a stretch of time? And what does it even *mean* for a quantity to have a rate of change at one frozen moment, when "rate" seems to need two moments to compare?

The answer is the **derivative**. It is, all at once, three things that turn out to be the same thing: the slope of a tangent line, an instantaneous rate of change, and the best straight-line approximation to a curve. By the end of this chapter you will see why those three are one, you will compute a few derivatives from raw first principles with nothing but algebra and a limit, and you will understand exactly what the famous symbol $\frac{\mathrm{d}y}{\mathrm{d}x}$ is really saying – and what it is *not* saying.

10.1 The problem of the instant

Let us start with the most physical version of the question, the one that drove Newton. Drop a ball. In the absence of air resistance, the distance it has fallen after t seconds is, to good accuracy,

$$s(t) = 16t^2 \quad \text{(feet, with } t \text{ in seconds)}.$$

After 1 second it has fallen 16 feet; after 2 seconds, 64 feet; after 3 seconds, 144 feet. Now I ask: *how fast is the ball moving at the instant $t = 2$?*

You know how to find an *average* speed. Average speed over a time interval is distance covered divided by time elapsed. From $t = 2$ to $t = 3$ the ball falls $144 - 64 = 80$ feet in 1 second, so its average speed over that second is 80 feet per second. But that is the average over a whole second,

during which the ball was speeding up the entire time. At the *start* of that second it was slower than 80; at the *end*, faster. The number 80 is not the speed "at $t = 2$." It is a blur of an entire second's worth of speeds.

Intuition

To get the speed *at* an instant, shrink the interval. Don't measure over a whole second – measure over a tenth of a second, then a hundredth, then a millionth. The shorter the interval, the less the speed has a chance to change across it, so the average over that tiny interval gets closer and closer to the "true" instantaneous speed. The instantaneous speed is what those averages are *heading toward*. That is a limit.

Let us actually watch them head somewhere. Take the interval from $t = 2$ to $t = 2 + h$, where h is a small time step. The distance fallen during that interval is $s(2 + h) - s(2)$, and the average speed is

$$\frac{s(2+h) - s(2)}{h} = \frac{16(2+h)^2 - 16(2)^2}{h}.$$

Expand the top:

$$16(2+h)^2 - 16(4) = 16\big(4 + 4h + h^2\big) - 64 = 64 + 64h + 16h^2 - 64 = 64h + 16h^2.$$

Divide by h (we can cancel because h, while small, is not yet zero):

$$\frac{64h + 16h^2}{h} = 64 + 16h.$$

So the average speed over the interval $[2,\ 2 + h]$ is exactly $64 + 16h$ feet per second. Watch what happens as h shrinks.

h (seconds)	average speed $64 + 16h$ (ft/s)
1	80
0.1	65.6
0.01	64.16
0.001	64.016
0.0001	64.0016

The averages are marching toward 64. And we don't have to guess – the algebra told us the average is $64 + 16h$, so as $h \to 0$ the $16h$ term vanishes and the limit is exactly 64. *The ball is moving* 64 *feet per second at the instant* $t = 2$. We never divided by zero. We divided by a small nonzero h, simplified, and only *then* let h go to zero. That two-step dance – cancel first, take the limit second – is the whole secret, and we will see it again and again.

What is really happening

Notice what the limit accomplished. The raw fraction $\dfrac{s(2+h) - s(2)}{h}$ is the indeterminate form $\frac{0}{0}$ when $h = 0$: the ball travels zero distance in zero time, and $\frac{0}{0}$ is meaningless. The instantaneous speed is *not* "$\frac{0}{0}$ evaluated." It is the *number that the fraction approaches* as h shrinks – a number the fraction itself never literally equals at $h = 0$, but circles arbitrarily

closely. The limit is precisely the tool that extracts a definite value out of a $\frac{0}{0}$ that would otherwise be nonsense. Without limits, the instant has no speed. With limits, it has exactly one.

10.2 The same picture, drawn: secants closing in on a tangent

That falling-ball calculation has a geometric twin, and once you see the picture you will never lose it. Plot the function – any function $y = f(x)$ – as a curve. Pick a point $P = (a, f(a))$ on it. We want the steepness of the curve *at* P: the slope of the line that just grazes the curve there, touching it at P and matching its direction. That grazing line is the **tangent line**.

The trouble is that slope needs *two* points – rise over run – and the tangent touches at only one. So we cheat, exactly as before. Pick a second, nearby point $Q = (a + h,\ f(a + h))$ on the curve, and draw the line through P and Q. A line through two points of a curve is called a **secant** line, and its slope we can compute directly from two points:

$$\text{slope of secant } PQ = \frac{\text{rise}}{\text{run}} = \frac{f(a+h) - f(a)}{(a+h) - a} = \frac{f(a+h) - f(a)}{h}.$$

This fraction is so important it has a name: the **difference quotient**. It is the average rate of change of f over the step from a to $a + h$ – the geometric face of the very same average speed we computed for the ball.

Now slide Q along the curve toward P. As Q approaches P, the secant line pivots about P, and its slope changes. The picture practically begs you to believe that the secants settle down toward one limiting line – the tangent – and that the secant slopes settle toward one limiting number – the slope of the tangent. Figure 10.1 shows three secants for shrinking h, each tilting closer to the dashed tangent.

Walking question

Before reading on, picture the secant pivoting and ask yourself: *why* should the secants approach a single line at all? What could go wrong? Hold a ruler against a smooth bowl and tilt it as the second contact point slides in – you can feel it lock onto one direction. Now imagine the curve has a sharp corner at P (think of the letter "V"). Slide Q in from the right and you get one limiting slope; slide it in from the left and you get a different one. The secants do *not* agree on a single line. We will meet this failure again – it is exactly where derivatives stop existing.

The two stories – falling ball, grazing line – are not analogies for each other. They are *literally the same calculation*. "Instantaneous speed" is the limit of average speeds; "slope of tangent" is the limit of secant slopes; and average speed *is* the slope of a secant on the graph of position versus time. One limit, two costumes.

10.3 The definition of the derivative

We have now done the same thing twice, so let us crystallize it once and name it.

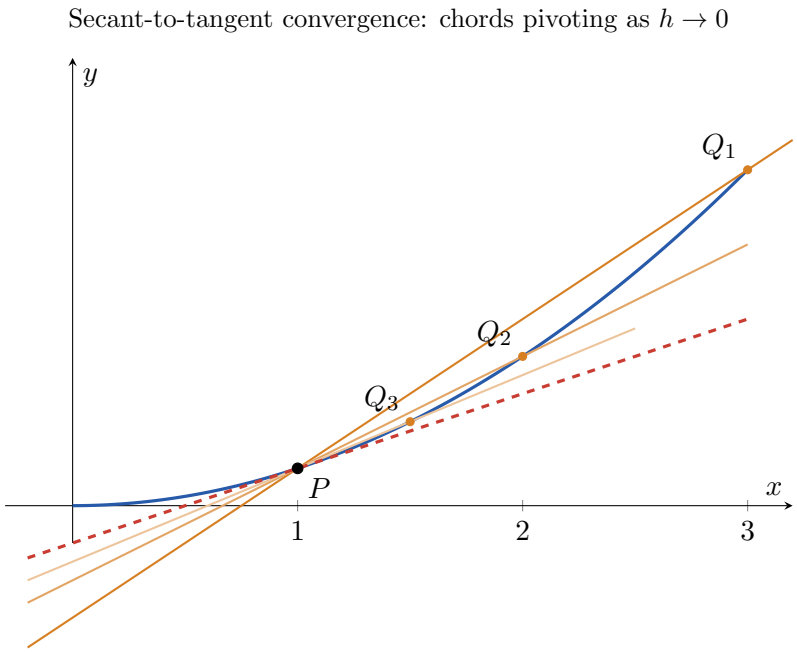


Figure 10.1: Secant lines through $P = (1, \frac{1}{2})$ and points Q_1, Q_2, Q_3 on $y = \frac{1}{2}x^2$, with Q sliding toward P. The secant slopes (2, then 1.5, then 1.25) close in on the slope of the dashed tangent line, which is 1. The tangent is the limit of the secants.

Definition 10.1 (The derivative at a point)**.** Let f be a function defined on an open interval containing the point a. The **derivative of f at a**, written $f'(a)$, is

$$f'(a) \;=\; \lim_{h \to 0} \frac{f(a+h) - f(a)}{h},$$

provided this limit exists. When it does, we say f is **differentiable at a**. The number $f'(a)$ is the slope of the tangent line to the graph $y = f(x)$ at the point $(a, f(a))$, and equivalently the instantaneous rate of change of f with respect to x at $x = a$.

Two small but important remarks before we use it.

Remark 10.1. The phrase "provided this limit exists" is doing real work. The derivative is a *limit*, and limits can fail to exist (Chapters 5–7). When the limit does not exist, f has no derivative at a – no well-defined slope, no instantaneous rate. That is not a defect in our method; it is a true fact about the function, and Section 10.8 is devoted to what such failures look like.

Remark 10.2 (A second, equivalent form). Sometimes it is cleaner to name the moving point directly. Set $x = a + h$, so $h = x - a$, and notice that $h \to 0$ is the same as $x \to a$. Then

$$f'(a) \;=\; \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

This is the identical definition with a change of variable; use whichever is more convenient for the algebra in front of you.

Focus

Read Definition 10.1 as an *instruction*, not a formula to plug into. It says: form the difference quotient (a function of h), simplify it until the offending h in the denominator is gone, and *then* see what it approaches as $h \to 0$. The art of computing derivatives from the definition is entirely the art of making that h in the denominator cancel.

10.4 Computing derivatives from first principles

Let us earn the definition by using it, with no rules yet – just algebra and a limit. (The shortcut rules of the next chapter all get *proved* this way, so this is not busywork; it is the foundation the rules stand on.)

Example 10.1 (A parabola)**.** Find the derivative of $f(x) = x^2$ at a general point x.

We build the difference quotient and chase the cancellation:

$$\begin{aligned}\frac{f(x+h)-f(x)}{h} &= \frac{(x+h)^2 - x^2}{h} \\ &= \frac{x^2 + 2xh + h^2 - x^2}{h} = \frac{2xh + h^2}{h} \\ &= \frac{h(2x+h)}{h} = 2x + h \qquad (h \neq 0).\end{aligned}$$

The h cancelled. Now take the limit:

$$f'(x) = \lim_{h \to 0} (2x + h) = 2x.$$

So the derivative of x^2 is $2x$. Sanity check against Figure 10.1, which used $\frac{1}{2}x^2$: at $x = 1$ the slope there should be $\frac{1}{2} \cdot 2 \cdot 1 = 1$, and indeed the tangent we drew had slope 1. The picture and the algebra agree.

Work it yourself

Do this one entirely by hand before reading the solution. Let $f(x) = 3x^2 - 5x + 1$. Form $\dfrac{f(x+h)-f(x)}{h}$, expand, cancel the h, and take the limit.

Solution sketch. The numerator is $3(x+h)^2 - 5(x+h) + 1 - (3x^2 - 5x + 1) = 6xh + 3h^2 - 5h$. Divide by h to get $6x + 3h - 5$, and let $h \to 0$: the derivative is $6x - 5$. Notice the constant $+1$ contributed nothing – a constant has zero rate of change, because shifting the whole graph up and down never tilts it.

Example 10.2 (A reciprocal – cancellation by a common denominator)**.** Find the derivative of $f(x) = \dfrac{1}{x}$ for $x \neq 0$.

Here the cancellation comes from combining fractions, not expanding a square:

$$\frac{f(x+h)-f(x)}{h} = \frac{1}{h}\left(\frac{1}{x+h}-\frac{1}{x}\right) = \frac{1}{h}\cdot\frac{x-(x+h)}{(x+h)\,x}$$
$$= \frac{1}{h}\cdot\frac{-h}{(x+h)\,x} = \frac{-1}{(x+h)\,x} \qquad (h\neq 0).$$

Again the h vanished, this time absorbed into the $-h$ on top. Now let $h \to 0$:

$$f'(x) = \lim_{h\to 0}\frac{-1}{(x+h)\,x} = \frac{-1}{x\cdot x} = -\frac{1}{x^2}.$$

The minus sign is the curve telling the truth: $1/x$ slopes *downward* everywhere it is defined, so its rate of change is negative, and the steeper plunge near $x=0$ shows up as the x^2 blowing up in the denominator.

Example 10.3 (A square root – cancellation by the conjugate)**.** Find the derivative of $f(x)=\sqrt{x}$ for $x>0$.

The trick when a root blocks the cancellation is to multiply by the *conjugate*, $\sqrt{x+h}+\sqrt{x}$, top and bottom. This turns a difference of roots into a difference of their squares:

$$\frac{\sqrt{x+h}-\sqrt{x}}{h} = \frac{\sqrt{x+h}-\sqrt{x}}{h}\cdot\frac{\sqrt{x+h}+\sqrt{x}}{\sqrt{x+h}+\sqrt{x}}$$
$$= \frac{(x+h)-x}{h\big(\sqrt{x+h}+\sqrt{x}\big)} = \frac{h}{h\big(\sqrt{x+h}+\sqrt{x}\big)}$$
$$= \frac{1}{\sqrt{x+h}+\sqrt{x}} \qquad (h\neq 0).$$

The lone h on top cancelled the h in the denominator. Now the limit is safe, because $\sqrt{x+h}\to\sqrt{x}$ as $h\to 0$:

$$f'(x) = \lim_{h\to 0}\frac{1}{\sqrt{x+h}+\sqrt{x}} = \frac{1}{2\sqrt{x}}.$$

Why the technique works

Why does the conjugate trick work, and why do we keep reaching for these maneuvers? Every one of them – expanding $(x+h)^2$, finding a common denominator, multiplying by the conjugate – is a way to expose a hidden factor of h in the numerator so it can cancel the h in the denominator. The difference quotient is $\frac{0}{0}$ at $h=0$ precisely *because* the numerator $f(x+h)-f(x)$ also vanishes there (it is the difference of a quantity with itself when $h=0$). A quantity that vanishes at $h=0$ has h as a factor, at least to first order. The algebra simply drags that factor out into the open where it can be cancelled. Differentiating from first principles is the systematic art of factoring out the h that *must* be there.

Common trap

The single most common error is setting $h=0$ *too early*. If you write $\dfrac{f(x+h)-f(x)}{h}$

and immediately substitute $h = 0$, you get $\frac{0}{0}$ and learn nothing – worse, you may convince yourself the derivative "doesn't exist" when it is perfectly fine. The rule is iron: *simplify until the denominator no longer contains h, and only then let $h \to 0$.* Cancellation first, limit second. Never the other way.

Example 10.4 (A constant and a line – the easy extremes)**.** If $f(x) = c$ is constant, then $f(x+h) - f(x) = c - c = 0$, so the difference quotient is $0/h = 0$ for every $h \neq 0$, and $f'(x) = 0$. A flat graph has zero slope; a quantity that never changes has zero rate of change. If $f(x) = mx + b$ is a line, then

$$\frac{m(x+h) + b - (mx+b)}{h} = \frac{mh}{h} = m,$$

so $f'(x) = m$ for every x. The derivative of a line is its slope, constant everywhere – which had better be true, since the tangent to a straight line is the line itself.

10.5 The derivative as a function

Look back at the examples. We started out asking for the slope at one point a, but the algebra naturally handed us a *formula in x*: the slope of x^2 is $2x$, valid at every x. This is a profound little shift. The derivative is not just a number attached to one point – it is a *new function* that reports the slope at every point where the original is differentiable.

Definition 10.2 (The derivative function)**.** Given f, its **derivative function** f' is defined by

$$f'(x) \;=\; \lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

on the set of all x for which the limit exists. The process of producing f' from f is called **differentiation**.

There is genuine information in graphing f and f' together. Where f rises, f' is positive; where f falls, f' is negative; where f momentarily levels off at a peak or valley, f' crosses zero. Figure 10.2 shows $f(x) = x^2$ over its derivative $f'(x) = 2x$: the parabola's slope is negative on the left, zero at the bottom, positive on the right, and the straight line $2x$ records exactly that.

Connection

This is the seed of an entire subject – the qualitative analysis of curves (Chapter 16). The sign of f' tells you whether f is increasing or decreasing; the zeros of f' locate the candidate peaks and valleys. We will eventually differentiate f' *again* to get f'', the rate at which the rate is changing – which is acceleration in the falling-ball story and concavity in the geometric one. The derivative being a function, and not merely a single number, is what makes all of that possible.

10.6 The third meaning: best linear approximation

We have met two meanings of the derivative – slope of the tangent and instantaneous rate of change. The third is subtler, and it carries most of the weight in the rest of calculus. It says: *near a, the*

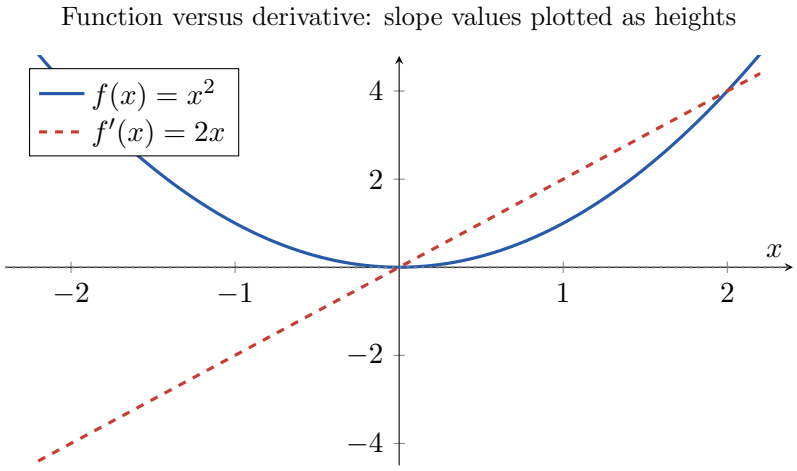


Figure 10.2: A function and its derivative. The slope of $f(x) = x^2$ at each x is read off the line $f'(x) = 2x$: negative for $x < 0$ (the parabola falls), zero at $x = 0$ (the bottom of the bowl), positive for $x > 0$ (the parabola rises). The height of the dashed line *is* the steepness of the solid curve.

function f looks like a straight line, and the derivative is the slope of that line.

Here is the idea. Unwind the definition. To say

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = f'(a)$$

is to say that for small h, the difference quotient is approximately $f'(a)$:

$$\frac{f(a+h) - f(a)}{h} \approx f'(a), \qquad \text{hence} \qquad f(a+h) \approx f(a) + f'(a)\, h.$$

Writing $x = a + h$ (so $h = x - a$), this is the **tangent line approximation**

$$\boxed{f(x) \approx f(a) + f'(a)\,(x - a)}$$

valid for x near a. The right-hand side is precisely the equation of the tangent line at a: it passes through $(a, f(a))$ and has slope $f'(a)$. So the tangent line is not merely a line that grazes the curve – it is the *best straight-line stand-in* for f near a. Among all lines through $(a, f(a))$, the tangent is the one whose error shrinks faster than h itself as $h \to 0$.

Intuition

Zoom in. Take any smooth curve, fix a point on it, and magnify the picture around that point – double the magnification, then double again, and again. The curve flattens. At high enough zoom it is visually indistinguishable from a straight line, and that line is the tangent. This is **local linearity**, and it is the real reason calculus works. Curves are hard; lines are easy. The derivative is the dictionary that translates the hard local behavior of a curve into the slope of a line. Almost every technique you will learn – linear approximation, Newton's method, the whole apparatus of differentials – is just this zoom, exploited. Figure 10.3 performs the zoom on a genuine curve.

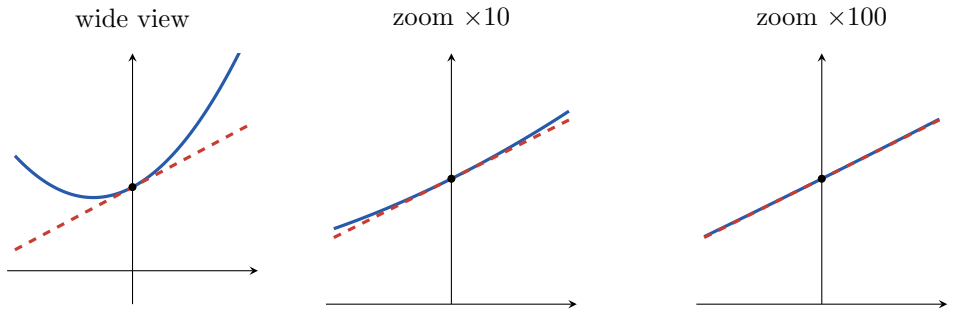


Figure 10.3: Local linearity, made literal. The same curve $f(x) = 1 + \frac{1}{2}x + \frac{1}{2}x^2$ (solid) and its tangent at $(0, 1)$ (dashed), viewed at three magnifications about that point. As we zoom, the curve becomes indistinguishable from its tangent: by $\times 100$ the two are visually one line. The derivative $f'(0) = \frac{1}{2}$ is the slope of the line the curve turns into under the microscope.

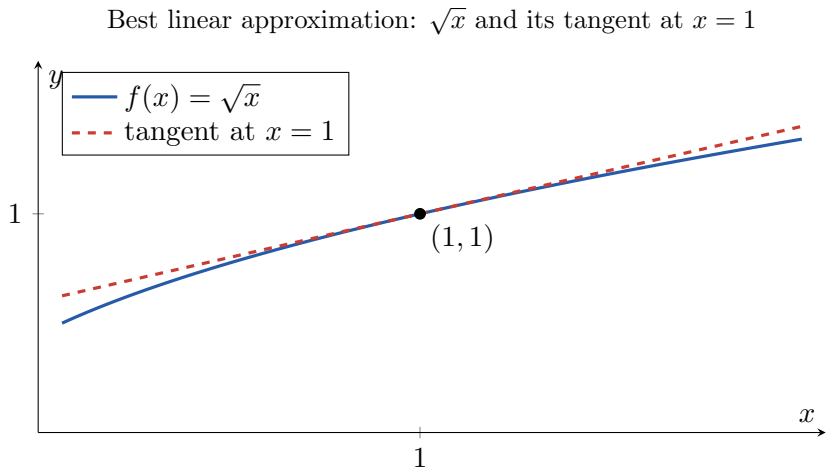


Figure 10.4: The tangent line as best linear approximation. Near $x = 1$ the curve $\sqrt{x}$ and its tangent $y = 1 + \frac{1}{2}(x - 1)$ are nearly on top of each other; they separate only as you move away from the point of contact. The approximation $\sqrt{x} \approx 1 + \frac{1}{2}(x - 1)$ gives, e.g., $\sqrt{1.04} \approx 1.02$.

Example 10.5 (Approximating a square root in your head)**.** From Example 10.3, $f(x) = \sqrt{x}$ has $f'(x) = \frac{1}{2\sqrt{x}}$. Near $a = 1$ we have $f(1) = 1$ and $f'(1) = \frac{1}{2}$, so

$$\sqrt{x} \approx 1 + \tfrac{1}{2}(x - 1).$$

Then $\sqrt{1.04} \approx 1 + \frac{1}{2}(0.04) = 1.02$. The true value is $1.0198\ldots$, so we are right to three decimal places with mental arithmetic and no calculator – the payoff of replacing a curve by its tangent line. The chapter on linear approximation and differentials develops this systematically.

Formal

Here is the precise statement that justifies calling the tangent the "best" linear approximation. Define the error of the tangent line at a as

$$E(h) = f(a + h) - \big[\, f(a) + f'(a)\, h \,\big].$$

Then differentiability at a is *equivalent* to the statement

$$\lim_{h \to 0} \frac{E(h)}{h} = 0,$$

that is, the error vanishes faster than h itself. Any *other* line through $(a, f(a))$, with slope $m \neq f'(a)$, has error $\big(f'(a) - m\big)h + E(h)$, whose ratio to h tends to the nonzero number $f'(a) - m$. Only the tangent's error is negligible compared to h. This "error smaller than first order" is the rigorous meaning of "best," and it is the doorway to Taylor's theorem later in the book.

10.7 Notation: Leibniz, Lagrange, and what $\frac{\mathrm{d}y}{\mathrm{d}x}$ really means

Calculus was invented twice, more or less simultaneously, by two people who could hardly have been more different, and we inherited *both* of their notations. It pays to know each, because each makes certain truths obvious and others awkward.

History

In the 1660s–1680s, **Isaac Newton** in England and **Gottfried Wilhelm Leibniz** in Germany independently created the calculus. Newton, thinking like a physicist, imagined quantities *flowing* in time; he called a changing quantity a *fluent* and its rate of change a *fluxion*, writing $\dot{x}$ for the fluxion of x (a dot notation still used in physics for time derivatives). Leibniz, thinking like a philosopher of symbols, sought a notation so well-designed that it would almost do the reasoning for you. He wrote the derivative as $\dfrac{dy}{dx}$ – a ratio of "infinitely small" changes dy and dx. A bitter priority dispute poisoned relations between English and Continental mathematicians for a century; the English, loyal to Newton's clumsier dot notation, fell behind the Continentals, who had Leibniz's superb symbolism. The lesson historians draw is humbling: *good notation is not decoration; it is a tool for thought.*

Lagrange's prime notation. Joseph-Louis Lagrange later introduced the clean $f'(x)$ we have been using – "f prime of x." It is compact, it makes the derivative-as-a-function idea vivid (f' is a

function just like f), and it is unbeatable for stating theorems. Its weakness: it hides *what* you are differentiating with respect to. For f' that is fine; for a quantity depending on several variables it is ambiguous.

Leibniz's ratio notation. If we name the output $y = f(x)$, then Leibniz writes

$$\frac{\mathrm{d}y}{\mathrm{d}x} \quad \text{or} \quad \frac{\mathrm{d}}{\mathrm{d}x} f(x) \quad \text{or} \quad \frac{\mathrm{d}}{\mathrm{d}x}\big(f(x)\big)\,,$$

and the value at a specific point a is written $\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=a}$. The symbol $\dfrac{\mathrm{d}}{\mathrm{d}x}$ is an *operator*: "take the derivative with respect to x of whatever follows." Leibniz notation is wonderful because it *remembers the variables*, it makes the chain rule and substitution look like fraction arithmetic (as we will see), and it makes units obvious – $\frac{\mathrm{d}s}{\mathrm{d}t}$ visibly carries units of distance over time, i.e. a speed.

What is really happening

Here is the question that trips up nearly every student, so let us answer it squarely. *Is* $\frac{\mathrm{d}y}{\mathrm{d}x}$ *a fraction?*

In the way we have built it, **no**. The symbol $\frac{\mathrm{d}y}{\mathrm{d}x}$ is a single, indivisible name for one number: the limit $\lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$. There is no genuine number called $\mathrm{d}y$ sitting on top of a genuine number $\mathrm{d}x$ that you divided. Leibniz himself thought of $\mathrm{d}x$ and $\mathrm{d}y$ as "infinitesimals" – quantities smaller than any positive number yet not zero – and treated $\frac{\mathrm{d}y}{\mathrm{d}x}$ as their literal quotient. For two centuries no one could say rigorously what an infinitesimal *was*, and the foundations were, frankly, a mess (see the History box below).

So why does the fraction notation work so flawlessly? Because $\frac{\mathrm{d}y}{\mathrm{d}x}$ is the *limit* of ordinary fractions $\frac{\Delta y}{\Delta x} = \frac{f(x+h)-f(x)}{h}$, and many of the algebraic moves you would make on the fraction $\frac{\Delta y}{\Delta x}$ *survive the limit*. The chain rule $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u}\cdot\frac{\mathrm{d}u}{\mathrm{d}x}$ looks like cancelling $\mathrm{d}u$, and it is *true* – but it is true because we will *prove* it from the definition, not because $\mathrm{d}u$ literally cancels. The notation is a brilliantly suggestive mnemonic for theorems that are established by limits. Use the fraction intuition to *guess*; lean on the limit definition to *know*.

History

The infinitesimals were more than informal – they were openly attacked. In 1734 the philosopher Bishop George Berkeley published *The Analyst*, mocking the vanishing $\mathrm{d}x$ as "the ghosts of departed quantities": in the difference quotient you first need $h \neq 0$ to divide by it, then you set $h = 0$ to finish – so which is it? It was a fair complaint, and it stood for nearly a century. The repair came in the 1820s from **Augustin-Louis Cauchy**, who based calculus on *limits* rather than infinitesimals, and was completed by **Karl Weierstrass**, whose ε–δ definition (your Chapter 6) made "limit" mean something with no ghosts in it at all. Our Definition 10.1 is the Cauchy–Weierstrass answer to Berkeley: we keep $h \neq 0$ throughout the algebra, and we never set it to zero – we take a *limit* as it approaches zero, which is a different and rigorous act. (Remarkably, in the 1960s Abraham Robinson's *nonstandard analysis* rehabilitated genuine infinitesimals on a rigorous footing, vindicating Leibniz's instincts after all. But the limit-based road is the one this book travels.)

10.8 Differentiability versus continuity

A function with a derivative at a is rather special, and it helps to know exactly *how* special. There is a clean one-way implication.

Theorem 10.1 (Differentiable $\Rightarrow$ continuous)**.** *If f is differentiable at a, then f is continuous at a.*

Intuition

Before the proof, the picture. If f has a tangent line at a, then near a the graph hugs that line – and a line has no gaps or jumps. So the graph cannot suddenly leap at a; it must arrive at $f(a)$ smoothly. Having a well-defined slope is a *stronger* demand than merely having no jump. Differentiability implies continuity, but, as we will see, continuity does not imply differentiability.

Proof. We must show $\lim_{x\to a} f(x) = f(a)$, equivalently $\lim_{x\to a}\big[f(x) - f(a)\big] = 0$. The trick is to manufacture the difference quotient by multiplying and dividing by $x - a$ (legal for $x \neq a$, which is all the limit ever sees):

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a} \cdot (x - a).$$

Now take the limit as $x \to a$. The first factor tends to $f'(a)$ by the definition of the derivative (Remark 10.2), which exists by hypothesis; the second factor tends to 0. By the product law for limits,

$$\lim_{x\to a}\big[f(x) - f(a)\big] = f'(a) \cdot 0 = 0.$$

Hence $\lim_{x\to a} f(x) = f(a)$, which is exactly continuity at a. □

The converse is false, and its failure is so instructive that we devote the rest of the section to the three classic ways a continuous function can lack a derivative. In every case the function has no jump – it is continuous – yet the limit defining $f'(a)$ does not exist. Watch *why* the limit fails each time; it is always the secant slopes refusing to settle on one number.

Failure 1: a corner – the absolute value

The cleanest example is $f(x) = |x|$ at $a = 0$. It is continuous everywhere (no jump at 0; $|x| \to 0 = |0|$). But form the difference quotient at 0:

$$\frac{f(0+h) - f(0)}{h} = \frac{|h| - 0}{h} = \frac{|h|}{h} = \begin{cases} 1 & h > 0, \\ -1 & h < 0. \end{cases}$$

As $h \to 0^+$ the quotient is $+1$; as $h \to 0^-$ it is -1. The one-sided limits disagree, so the two-sided limit does not exist, and $f'(0)$ is undefined. Geometrically the graph has a sharp **corner** at the origin: the slope is -1 coming from the left and $+1$ going to the right, and there is no single tangent line. The secants from the right insist the slope is 1; the secants from the left insist it is -1; they never agree.

Failure 2: a cusp – the cube root, sharpened

Consider $f(x) = x^{2/3}$ at $a = 0$. It is continuous ($0^{2/3} = 0$), and unlike the corner its two sides do not meet at an angle – they meet at an infinitely sharp **cusp**, both tangent directions turning vertical. The difference quotient is

$$\frac{(0+h)^{2/3} - 0}{h} = \frac{h^{2/3}}{h} = h^{-1/3} = \frac{1}{h^{1/3}},$$

and as $h \to 0$ this grows without bound (toward $+\infty$ from the right, toward $-\infty$ from the left). The limit does not exist – not even as one-sided finite limits – so $f'(0)$ does not exist. The slope is trying to be vertical on both sides at once.

Failure 3: a vertical tangent – the cube root

Now $f(x) = x^{1/3}$ at $a = 0$, again continuous. The difference quotient is

$$\frac{(0+h)^{1/3} - 0}{h} = \frac{h^{1/3}}{h} = h^{-2/3} = \frac{1}{h^{2/3}} \;\to\; +\infty \quad \text{as } h \to 0.$$

This time both sides agree – the quotient marches to $+\infty$ from either direction – so the curve genuinely has a single limiting tangent direction, but that direction is *vertical*. A vertical line has no (finite) slope, so by our definition $f'(0)$ does not exist as a number. We say the graph has a **vertical tangent** at the origin. It is differentiability's near miss: there is a tangent line, but its slope is infinite.

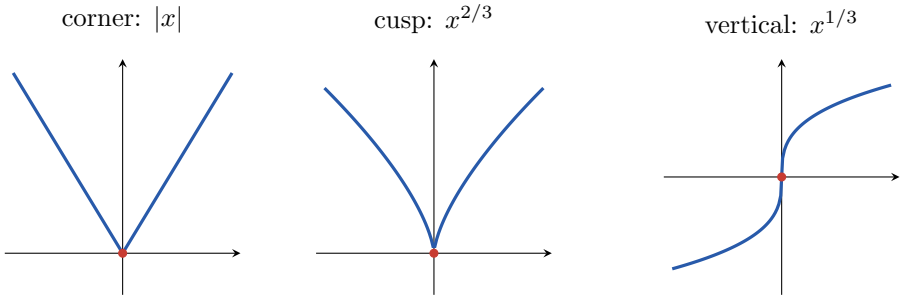


Figure 10.5: Three ways to be continuous but not differentiable at the origin. A *corner* ($|x|$): left and right slopes disagree. A *cusp* ($x^{2/3}$): the secant slopes run off to $+\infty$ on one side and $-\infty$ on the other. A *vertical tangent* ($x^{1/3}$): a genuine tangent exists, but it is vertical, so its slope is infinite.

Common trap

Do not over-learn Theorem 10.1. It runs *one way only*: differentiable $\Rightarrow$ continuous. The reverse is false, as $|x|$ shows – continuous everywhere, not differentiable at 0. So "continuous" is necessary but not sufficient for "differentiable." A handy slogan: *differentiable functions are continuous and smooth; continuous functions are merely unbroken, and may still be jagged.*

Why this matters

How bad can the gap between continuous and differentiable get? Astonishingly bad. In 1872 Weierstrass exhibited a function that is *continuous everywhere yet differentiable nowhere* –

an unbroken curve that has a corner at every single point, so jagged that no point on it admits a tangent. When it was published it scandalized mathematicians; Hermite called such functions a "lamentable plague." Today they are the cornerstone of fractal geometry and the mathematics of Brownian motion – the random path of a pollen grain in water is, with probability one, a continuous-everywhere, differentiable-nowhere curve. Continuity and differentiability are genuinely different ideas, and the chasm between them is where some of the deepest modern mathematics lives.

10.9 Rates of change everywhere

We opened with a falling ball because motion makes the derivative tangible, but the derivative is the universal language of *any* quantity changing with respect to *any* other. The pattern is always the same: a small change in the input causes a small change in the output, and the derivative is the limiting ratio.

Application

The derivative wears a different name in every field, but it is always $\lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$.

- **Physics.** If $s(t)$ is position, $s'(t)$ is velocity, and $s''(t)$ is acceleration. Newton's second law $F = ma$ is a statement about the second derivative of position.
- **Biology.** If $P(t)$ is a population, $P'(t)$ is the growth rate (births minus deaths per unit time). The model $P'(t) = kP(t)$ – growth rate proportional to current size – gives exponential growth, and is your first differential equation.
- **Economics.** If $C(q)$ is the cost of producing q items, the derivative $C'(q)$ is the *marginal cost*: the approximate cost of one more item. "Marginal" in economics almost always means "derivative."
- **Chemistry.** If $[A](t)$ is a concentration, the reaction rate is $-\frac{\mathrm{d}[A]}{\mathrm{d}t}$ – the speed at which reactant is consumed.
- **Machine learning.** Training a model means minimizing a loss function $L(\mathbf{w})$ over parameters $\mathbf{w}$; gradient descent steps *downhill* along $-L'$ (the multivariable derivative, the gradient). Every neural network is trained by computing derivatives – billions of them, by the chain rule run backwards, which is exactly "backpropagation."

The instantaneous rate of change is not a physics trick. It is the way to talk about change itself, in any domain where one quantity depends on another.

Walking question

Take a walk and find the derivatives hiding around you. The speedometer in a car literally displays $\frac{\mathrm{d}s}{\mathrm{d}t}$. The slope you feel in your legs on a hill is the derivative of elevation with respect to horizontal distance. The rate your coffee cools is the derivative of its temperature with respect to time (and, as Newton noticed, that rate is proportional to how much hotter the coffee is than the room – another differential equation). Ask of each: what is changing, with

respect to what, and is the rate positive, negative, or zero right now?

10.10 What to carry forward

Pause

The whole chapter in one breath.

- The **derivative** of f at a is $f'(a) = \lim_{h\to 0} \dfrac{f(a+h)-f(a)}{h}$, when this limit exists. It is the limit of difference quotients – of average rates – as the interval shrinks to nothing.
- It has **three faces**, all one number: the *slope of the tangent line*, the *instantaneous rate of change*, and the *slope of the best linear approximation* $f(x) \approx f(a) + f'(a)(x-a)$.
- Compute it from first principles by **cancelling the h first** (expand, common-denominator, or conjugate) and **taking the limit second**. Never set $h = 0$ before the cancellation.
- The derivative is itself a **function**, f', recording the slope at every point.
- **Notation**: Lagrange's $f'(x)$ and Leibniz's $\frac{\mathrm{d}y}{\mathrm{d}x}$ name the same thing. $\frac{\mathrm{d}y}{\mathrm{d}x}$ is *not* literally a fraction – it is the limit of ordinary fractions, which is why fraction-style manipulations keep working once we prove them.
- **Differentiable $\Rightarrow$ continuous**, but not conversely: corners ($|x|$), cusps ($x^{2/3}$), and vertical tangents ($x^{1/3}$) are continuous yet have no derivative, because the secant slopes refuse to settle on one finite number.

Next chapter we stop computing every derivative from scratch and build the *rules* – and each rule we prove will rest squarely on the limit definition you now own.

10.11 Exercises

Exercise 10.1 (Warm-ups from the definition)**.** Using only Definition 10.1 (cancel the h, then take the limit), compute $f'(x)$ for each:

(a) $f(x) = 5x - 7$.

(b) $f(x) = x^2 - 3x$.

(c) $f(x) = x^3$. *(Hint:* $(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$.*)*

(d) $f(x) = \dfrac{1}{x+1}$.

Exercise 10.2 (A root, by the conjugate)**.** Compute the derivative of $f(x) = \sqrt{2x+1}$ from first principles, using the conjugate trick. At what value of x is the slope equal to 1?

Exercise 10.3 (Instantaneous speed)**.** A particle moves so that its position at time t is $s(t) = t^2 - 4t + 5$ (in meters, t in seconds). (a) Find the average velocity over $[1, 1+h]$ and simplify.

(b) Take the limit to find the instantaneous velocity at $t = 1$. (c) At what time is the particle instantaneously at rest (velocity zero)? What is it doing at that instant – where on its path is it?

Exercise 10.4 (Tangent line and approximation)**.** Let $f(x) = x^2$. (a) Write the equation of the tangent line at $a = 3$. (b) Use it to approximate $(3.02)^2$ without squaring, and compare with the exact value. (c) Is your approximation an over- or under-estimate? Explain using the shape (concavity) of the parabola.

Exercise 10.5 (Reading a graph)**.** A function f is increasing for $x < 2$, has a peak at $x = 2$, and decreases for $x > 2$. Sketch a plausible graph of the *derivative* f'. Where is f' positive, where negative, and what happens at $x = 2$? (No formula is needed – reason from signs of slopes.)

Exercise 10.6 (Where the derivative fails)**.** For each function, decide whether it is differentiable at $a = 0$. If not, name the type of failure (corner, cusp, vertical tangent, or discontinuity) and justify with one-sided difference quotients.

(a) $f(x) = |x - 0| = |x|$ – restated for contrast with (b).

(b) $f(x) = x\,|x|$. *(Compute the left and right difference quotients at* 0*; do they agree?)*

(c) $f(x) = x^{1/3}$.

(d) $f(x) = \begin{cases} x & x \geq 0, \\ x - 1 & x < 0. \end{cases}$

Exercise 10.7 (Continuous but not differentiable – by hand)**.** Let $f(x) = |x|$. (a) Show directly from the limit definition that f is *continuous* at 0. (b) Show directly that the two one-sided limits of the difference quotient at 0 are $+1$ and -1, so $f'(0)$ does not exist. (c) In one sentence, explain why this does not contradict Theorem 10.1.

Exercise 10.8 (Prove a special case)**.** Using the binomial expansion of $(x+h)^n$, prove from Definition 10.1 that for any positive integer n,

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^n) = nx^{n-1}.$$

Hint: $(x+h)^n = x^n + nx^{n-1}h + (\textit{terms with } h^2 \textit{ or higher})$*. After cancelling* h*, every surviving term except the first still contains at least one factor of* h*.*

Exercise 10.9 (The other definition)**.** Use the equivalent form $f'(a) = \lim_{x \to a} \dfrac{f(x) - f(a)}{x - a}$ (Remark 10.2) to compute $f'(a)$ for $f(x) = x^2$ at a general a, by factoring $x^2 - a^2 = (x - a)(x + a)$. Confirm you get $2a$, matching Example 10.1.

Exercise 10.10 (Conceptual – is the fraction real?)**.** A classmate insists that $\frac{\mathrm{d}y}{\mathrm{d}x}$ *is* a fraction because the chain rule "cancels the $\mathrm{d}u$." Write a short paragraph explaining, using the $\frac{0}{0}$-and-limit idea from this chapter, in what precise sense $\frac{\mathrm{d}y}{\mathrm{d}x}$ is *not* a fraction, and why fraction-style manipulations nonetheless give correct answers.

Exercise 10.11 (A subtle differentiability question)**.** Consider

$$f(x) = \begin{cases} x^2 \sin\left(\dfrac{1}{x}\right) & x \neq 0, \\ 0 & x = 0. \end{cases}$$

(a) Using $|\sin(1/x)| \le 1$ and the squeeze theorem, show the difference quotient $\dfrac{f(h)-f(0)}{h} = h\sin(1/h)$ tends to 0, so $f'(0) = 0$ exists. (b) This function is differentiable at 0 even though it oscillates infinitely often near 0. What does this tell you about how much "wiggling" is compatible with having a derivative? *(This is a famous example; savor how the x^2 tames the oscillation.)*

Chapter 11

The Rules of Differentiation, and Why They Work

In the previous chapter you met the derivative as a *limit*: the slope of a curve at a point, the instantaneous rate of change, the number that the slopes of shrinking secant lines close in on. That definition is the bedrock. But if you had to return to the limit

$$f'(x) \;=\; \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

every single time you wanted a derivative, calculus would be unusable. Differentiating $x^7 - 4x^3 + \frac{1}{x}$ from scratch would take a page of algebra and a prayer that no h got left behind.

This chapter is about freedom from that drudgery. We will build a small toolkit of *rules* – the power rule, the constant-multiple and sum rules, the product rule, the quotient rule, and the crown jewel, the chain rule – that let you differentiate almost any function you can write down, quickly and mechanically. But this is a book about *understanding*, so we will never just hand you a rule. Every single one will be *derived* from the limit definition, and – more importantly – we will see *why* it has the shape it does. The product rule is not an arbitrary string of symbols to memorize; it is the statement that a growing rectangle gains area on two sides at once. The chain rule is not a mysterious "multiply by the derivative of the inside"; it is the plain fact that rates of change multiply when you link two processes together.

By the end you will not only be able to differentiate fluently – you will be able to *reconstruct* each rule if you ever forget it, because you will know where it comes from.

Focus

The plan: each rule gets (1) a picture or piece of intuition, (2) a careful derivation from the limit definition, (3) a *why* box exposing the mechanism, and (4) worked examples. The order is deliberate: the easy rules (power, constant multiple, sum) first, then product and quotient, then the chain rule, which secretly powers everything that comes after it in this book.

11.1 The Building Blocks: Two Derivatives We Already Know

Before any rules, we need raw material. Two derivatives are so simple we can read them straight off the limit definition, and everything else will be assembled from them.

11.1.1 The derivative of a constant is zero

Let $f(x) = c$, a constant – the graph is a flat horizontal line at height c. A flat line has slope 0 everywhere, so we already expect $f'(x) = 0$. The limit confirms it instantly:

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{c - c}{h} = \lim_{h\to 0} \frac{0}{h} = \lim_{h\to 0} 0 = 0.$$

The numerator is exactly zero for *every* h, because a constant function does not change. There is nothing accumulating, so there is no rate.

Intuition

A constant function is a thing that never moves. Its rate of change is the rate at which a parked car covers distance: zero. Keep this in your pocket – it is why additive constants (the "$+C$" you will meet again with integrals) vanish under differentiation.

11.1.2 The derivative of x is one

Let $f(x) = x$. The graph is the 45° line through the origin, slope 1 everywhere. Again the limit agrees:

$$f'(x) = \lim_{h\to 0} \frac{(x+h) - x}{h} = \lim_{h\to 0} \frac{h}{h} = \lim_{h\to 0} 1 = 1.$$

When x increases by h, the output $f(x) = x$ increases by exactly h. Output change divided by input change is $h/h = 1$. The identity function passes change through untouched, so its rate is 1.

These two facts – $\frac{\mathrm{d}}{\mathrm{d}x}(c) = 0$ and $\frac{\mathrm{d}}{\mathrm{d}x}(x) = 1$ – are the atoms. Now let us learn how to combine atoms.

11.2 Constant Multiples and Sums: Differentiation is Linear

11.2.1 The constant-multiple rule

Suppose you already know f', and you scale the whole function by a constant k, forming $g(x) = k\,f(x)$. What happens to the slope? Stretching a graph vertically by a factor of k stretches all its *rises* by k while leaving the *runs* alone – so every slope is multiplied by k.

Theorem 11.1 (Constant-multiple rule)**.** *If f is differentiable and k is a constant, then kf is differentiable and*

$$\frac{\mathrm{d}}{\mathrm{d}x}(k\,f(x)) = k\,f'(x).$$

Proof. Go to the definition and watch the constant ride along outside the difference quotient:

$$\big(kf\big)'(x) = \lim_{h\to 0} \frac{kf(x+h) - kf(x)}{h} = \lim_{h\to 0} k \cdot \frac{f(x+h) - f(x)}{h} = k \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = k\, f'(x).$$

The single nontrivial move is pulling k out of the limit, which is legal because a constant factor always survives a limit unchanged. □

11.2.2 The sum rule

Now add two functions: $g(x) = f(x) + p(x)$. If f is rising at rate f' and p is rising at rate p', then their sum rises at the combined rate $f' + p'$. Rates simply add, the way two streams pouring into one tank add their flows.

Theorem 11.2 (Sum (and difference) rule)**.** *If f and p are differentiable, then so is $f \pm p$, and*

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(f(x) \pm p(x)\big) = f'(x) \pm p'(x).$$

Proof. Form the difference quotient and regroup the four terms so the f's sit together and the p's sit together:

$$\big(f+p\big)'(x) = \lim_{h\to 0} \frac{[f(x+h) + p(x+h)] - [f(x) + p(x)]}{h} = \lim_{h\to 0} \left(\frac{f(x+h) - f(x)}{h} + \frac{p(x+h) - p(x)}{h} \right).$$

The limit of a sum is the sum of the limits (each exists, by assumption), so this is $f'(x) + p'(x)$. The subtraction case is identical with a minus sign. □

Why the technique works

Together, the constant-multiple and sum rules say the derivative is a *linear operator*: $\frac{\mathrm{d}}{\mathrm{d}x}(af + bg) = af' + bg'$. This is the single most useful structural fact about differentiation. It is what lets you differentiate a long polynomial term-by-term, never having to deal with the whole thing at once. The deep reason is that the difference quotient itself is built from subtraction and division by h – both linear operations – and the limit preserves linearity. Differentiation inherits its linearity from the linearity of "measuring change."

Common trap

Linearity covers sums and constant multiples *only*. It does *not* extend to products or quotients:

$$\frac{\mathrm{d}}{\mathrm{d}x}(f \cdot g) \;\neq\; f' \cdot g', \qquad \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{f}{g}\right) \;\neq\; \frac{f'}{g'}.$$

A great many wrong answers in calculus come from wishing products behaved as nicely as sums. They do not – and the next two sections are devoted to what really happens.

11.3 The Power Rule

Here is the workhorse. For any constant exponent n,

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}x}(x^n) = n\,x^{\,n-1}.}$$

"Bring the exponent down in front, then knock the exponent down by one." You will use this thousands of times. But where does it come from? We will prove it three ways, each of which teaches something different: first for small integers by hand, then for all positive integers by the binomial theorem, and finally we will explain why it keeps working for negative and fractional exponents.

11.3.1 Warm-up: x^2 and x^3 by hand

Example 11.1 (The derivative of x^2)**.** Let $f(x) = x^2$. Then

$$f'(x) = \lim_{h\to 0}\frac{(x+h)^2 - x^2}{h} = \lim_{h\to 0}\frac{x^2 + 2xh + h^2 - x^2}{h} = \lim_{h\to 0}\frac{2xh + h^2}{h} = \lim_{h\to 0}(2x + h) = 2x.$$

The x^2 cancels; the surviving terms all carry an h, so dividing by h leaves $2x + h$, and the leftover h dies in the limit. Result: $\frac{\mathrm{d}}{\mathrm{d}x}\big(x^2\big) = 2x = 2x^{2-1}$. The pattern holds.

Example 11.2 (The derivative of x^3)**.** Now $f(x) = x^3$. Using $(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$,

$$f'(x) = \lim_{h\to 0}\frac{3x^2h + 3xh^2 + h^3}{h} = \lim_{h\to 0}\big(3x^2 + 3xh + h^2\big) = 3x^2 = 3x^{3-1}.$$

Again the lone x^3 cancels, every surviving term has at least one h, and only the term with a *single* h — namely $3x^2h$ — contributes after we divide by h and take $h \to 0$. That single first-order term sets the instantaneous rate.

What is really happening

Look at what survives. When you expand $(x+h)^n$ you get x^n plus a term with exactly one h, plus higher-order terms in $h^2, h^3, \ldots$ The x^n cancels the $-x^n$. After dividing by h, the single-h term loses its h and becomes a clean constant in h; every other term still carries at least one factor of h and therefore vanishes as $h \to 0$. *The derivative reads off the coefficient of the linear term* — the portion of the change proportional to the nudge h. All higher powers are second-order small, vanishing at the instant. This is the governing mechanism of differential calculus: the derivative isolates the linear response of a function to a microscopic step.

11.3.2 All positive integers at once: the binomial theorem

The hand calculations practically prove themselves once you name the pattern. The binomial theorem expands

$$(x+h)^n = x^n + nx^{\,n-1}h + \binom{n}{2}x^{\,n-2}h^2 + \cdots + h^n.$$

The first term is x^n, the second is $nx^{n-1}h$ (this is the one that matters), and every later term carries h^2 or higher.

Theorem 11.3 (Power rule, positive integers)**.** *For every positive integer n, $\frac{\mathrm{d}}{\mathrm{d}x}(x^n) = nx^{n-1}$.*

Proof. Subtract x^n, divide by h:

$$\frac{(x+h)^n - x^n}{h} = \frac{nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + h^n}{h} = nx^{n-1} + \underbrace{\binom{n}{2}x^{n-2}h + \cdots + h^{n-1}}_{\text{every term has an } h}.$$

As $h \to 0$ every term in the underbrace contains at least one factor of h and so tends to 0. What remains is nx^{n-1}. □

11.3.3 Why it survives to negative and fractional exponents

The binomial proof only handles $n = 1, 2, 3, \ldots$ Yet the rule is true far more generally: $\frac{\mathrm{d}}{\mathrm{d}x}\left(x^{-2}\right) = -2x^{-3}$ and $\frac{\mathrm{d}}{\mathrm{d}x}\left(x^{1/2}\right) = \frac{1}{2}x^{-1/2}$ both obey "nx^{n-1}." We are not equipped to prove the full real-exponent case yet – the cleanest proof waits until we have the chain rule and the logarithm (next chapter), where $x^n = e^{n\ln x}$ makes it a one-liner. But two special cases are within reach right now, and they are worth doing because they showcase the algebra.

Example 11.3 (A negative exponent: $1/x$)**.** Let $f(x) = x^{-1} = 1/x$. Then

$$f'(x) = \lim_{h\to 0}\frac{1}{h}\left(\frac{1}{x+h} - \frac{1}{x}\right) = \lim_{h\to 0}\frac{1}{h}\cdot\frac{x-(x+h)}{x(x+h)} = \lim_{h\to 0}\frac{-h}{h\,x(x+h)} = \lim_{h\to 0}\frac{-1}{x(x+h)} = -\frac{1}{x^2}.$$

That is $-x^{-2} = (-1)x^{-1-1}$: the power rule with $n = -1$, exactly as advertised.

Example 11.4 (A fractional exponent: $\sqrt{x}$)**.** Let $f(x) = \sqrt{x} = x^{1/2}$ (for $x > 0$). The trick is to rationalize – multiply by the conjugate to clear the difference of square roots:

$$f'(x) = \lim_{h\to 0}\frac{\sqrt{x+h} - \sqrt{x}}{h} = \lim_{h\to 0}\frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})}.$$

The numerator collapses to $(x+h) - x = h$, which cancels the h below:

$$f'(x) = \lim_{h\to 0}\frac{1}{\sqrt{x+h} + \sqrt{x}} = \frac{1}{2\sqrt{x}} = \tfrac{1}{2}\,x^{-1/2}.$$

And $\frac{1}{2}x^{-1/2} = \frac{1}{2}x^{1/2-1}$ – the power rule again, now with $n = \frac{1}{2}$.

Why the technique works

Why does "bring down the exponent" work? Geometrically, think of x^n as the *volume of an n-dimensional cube* of side x. Nudge the side from x to $x + h$. The cube gains a thin slab on each of its n faces, each slab of area x^{n-1} and thickness h, for a leading gain of $n \cdot x^{n-1} \cdot h$. (The edges and corners where slabs overlap involve h^2 and smaller – negligible at an instant.) Divide that gain by h and you get nx^{n-1}. The n in front is literally counting the n faces of the cube that grow. We will see this same "thicken each face" picture again, in two dimensions, when we prove the product rule.

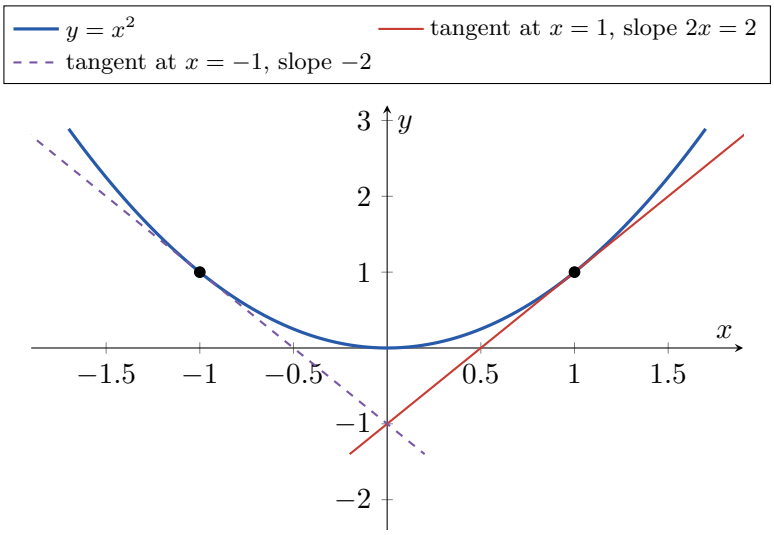


Figure 11.1: The power rule in action for $y = x^2$. At $x = 1$ the predicted slope is $2x = 2$; at $x = -1$ it is -2. The tangent lines have exactly those slopes. Differentiation reads the steepness directly off the formula $2x$.

11.3.4 Polynomials: linearity plus the power rule

The power rule plus linearity (Section 11.2) differentiate *any* polynomial on sight, term by term.

Example 11.5 (A polynomial)**.** Differentiate $p(x) = 4x^5 - 7x^3 + 2x^2 - 9x + 6$.

Work term by term. The exponent comes down, drops by one, and constants multiply through; the lone constant 6 dies:

$$p'(x) = 4 \cdot 5x^4 - 7 \cdot 3x^2 + 2 \cdot 2x - 9 \cdot 1 + 0 = 20x^4 - 21x^2 + 4x - 9.$$

Example 11.6 (Rewrite, then differentiate)**.** Differentiate $g(x) = \dfrac{3}{x^2} + 5\sqrt{x} - \dfrac{x^4}{2}$.

First rewrite every piece as a power of x so the power rule can grip it:

$$g(x) = 3x^{-2} + 5x^{1/2} - \tfrac{1}{2}x^4.$$

Now differentiate each power:

$$g'(x) = 3(-2)x^{-3} + 5 \cdot \tfrac{1}{2}x^{-1/2} - \tfrac{1}{2} \cdot 4x^3 = -6x^{-3} + \tfrac{5}{2}x^{-1/2} - 2x^3 = -\frac{6}{x^3} + \frac{5}{2\sqrt{x}} - 2x^3.$$

The lesson: rewriting roots and fractions as exponents is half the battle.

Work it yourself

Differentiate $h(x) = x^7 - 3x^4 + \dfrac{1}{x^3} - \sqrt[3]{x}$ by hand. (Hint: rewrite the last two terms as x^{-3} and $x^{1/3}$.) You should get $h'(x) = 7x^6 - 12x^3 - 3x^{-4} - \frac{1}{3}x^{-2/3}$. Then check one value, say $x = 1$, by comparing to a numerical slope $\dfrac{h(1.001) - h(0.999)}{0.002}$.

11.4 The Product Rule

We come to the first rule that genuinely surprises people. The naive guess – "the derivative of a product is the product of the derivatives" – is *wrong*, and a single example demolishes it.

Common trap

Take $f(x) = x$ and $g(x) = x$, so $f(x)g(x) = x^2$. The correct derivative is $2x$. But $f'(x)g'(x) = 1 \cdot 1 = 1 \neq 2x$. The product of the derivatives is not the derivative of the product. Something else is going on.

11.4.1 The picture: a growing rectangle

One picture makes the product rule obvious. Imagine a rectangle whose width is $f(x)$ and whose height is $g(x)$. Its area is the product $A(x) = f(x)g(x)$. Now nudge x forward by a tiny h. The width grows by $\Delta f = f(x+h) - f(x)$ and the height grows by $\Delta g = g(x+h) - g(x)$. How much new area appears?

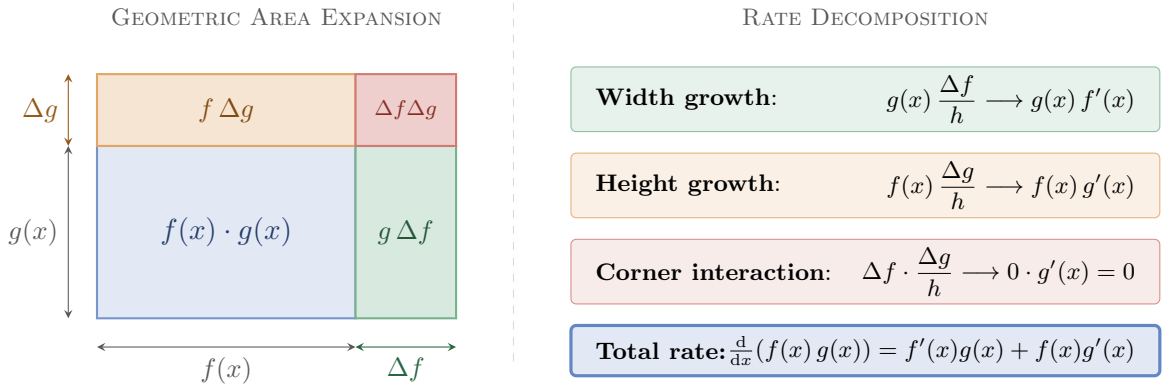


Figure 11.2: Why the product rule has two terms. Left: Growing both dimensions of a rectangle creates a tall strip $g\,\Delta f$, a wide strip $f\,\Delta g$, and a microscopic corner $\Delta f\,\Delta g$. Right: Dividing each piece by h and taking the limit $h \to 0$ reveals that the two edge strips produce finite rates gf' and fg', while the second-order corner vanishes because $\Delta f \to 0$.

The new area comes in three pieces (Figure 11.2):

- a *tall strip* along the right edge, of height g and width Δf: area $g\,\Delta f$;
- a *wide strip* along the top edge, of width f and height Δg: area $f\,\Delta g$;
- a *tiny corner* rectangle of dimensions Δf by Δg: area $\Delta f\,\Delta g$.

So the change in area is

$$\Delta A = g\,\Delta f + f\,\Delta g + \Delta f\,\Delta g.$$

Divide by h to get the rate, and watch the corner term:

$$\frac{\Delta A}{h} = g\,\frac{\Delta f}{h} + f\,\frac{\Delta g}{h} + \Delta f \cdot \frac{\Delta g}{h}.$$

As $h \to 0$, $\Delta f/h \to f'$ and $\Delta g/h \to g'$. The first two terms become gf' and fg'. The corner term is $\Delta f \cdot (\Delta g/h)$; the factor $\Delta g/h$ tends to g' (finite), but $\Delta f \to 0$, so the whole corner vanishes. The geometric insight is straightforward: *the corner is the product of two small quantities, and therefore shrinks at second order.* It contributes nothing to the instantaneous rate of change.

Theorem 11.4 (Product rule)**.** *If f and g are differentiable, then so is fg, and*

$$\frac{\mathrm{d}}{\mathrm{d}x}(f(x)\,g(x)) = f'(x)\,g(x) + f(x)\,g'(x).$$

Proof. The picture is correct, but let us pin it down with the clean algebraic trick that makes the corner term appear and disappear: *add and subtract the same mixed term.* Start from the difference quotient and insert $-f(x+h)g(x) + f(x+h)g(x)$ (a clever zero):

$$\frac{f(x+h)g(x+h) - f(x)g(x)}{h} = \frac{f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x)}{h}.$$

Group the first two and last two terms, factoring out the common piece in each:

$$= f(x+h)\,\frac{g(x+h)-g(x)}{h} \;+\; g(x)\,\frac{f(x+h)-f(x)}{h}.$$

Now take $h \to 0$ term by term. In the first term, $f(x+h) \to f(x)$ (because f, being differentiable, is continuous) and $\dfrac{g(x+h)-g(x)}{h} \to g'(x)$. In the second term, $g(x)$ is constant in h and $\dfrac{f(x+h)-f(x)}{h} \to f'(x)$. Hence the limit is

$$f(x)\,g'(x) + g(x)\,f'(x),$$

which is the product rule. □

Why the technique works

Why *two* terms? Because a product depends on *both* of its factors, and at any instant both factors are changing at once. The rectangle picture makes this vivid: area is gained on two sides simultaneously. The term $f'g$ is the contribution from the width growing while the height holds; fg' is the contribution from the height growing while the width holds. You can think of it as "hold one factor still, differentiate the other, then swap and add." The reason you *add* them (rather than, say, multiply) is that to first order the two growths are independent – their interaction, the corner $\Delta f\,\Delta g$, is second-order small and drops out. This "vary one factor at a time and sum" pattern is exactly how the product rule generalizes: for three factors,

$$(fgh)' = f'gh + fg'h + fgh',$$

one term per factor, each differentiating a single factor while freezing the others.

Example 11.7 (A first product)**.** Differentiate $y = (x^2+1)(x^3-4x)$.

Let $f = x^2+1$ and $g = x^3-4x$, so $f' = 2x$ and $g' = 3x^2-4$. Then

$$y' = f'g + fg' = (2x)(x^3-4x) + (x^2+1)(3x^2-4).$$

Expand to simplify:

$$y' = (2x^4 - 8x^2) + (3x^4 - 4x^2 + 3x^2 - 4) = 5x^4 - 9x^2 - 4.$$

Sanity check. We could instead multiply out first: $y = x^5 - 4x^3 + x^3 - 4x = x^5 - 3x^3 - 4x$, whose derivative is $5x^4 - 9x^2 - 4$. Same answer. The product rule and "expand first" must agree – and here expanding was about as fast. The product rule earns its keep when expanding is ugly or impossible, as with $x^2 \sin x$ or $e^x \ln x$, which you cannot multiply out.

Example 11.8 (When you cannot expand)**.** Differentiate $y = x^2\sqrt{x+1}$. Write $g = (x+1)^{1/2}$, so (peeking ahead to the chain rule, or just by the power rule on $x+1$) $g' = \frac{1}{2}(x+1)^{-1/2}$. With $f = x^2$, $f' = 2x$:

$$y' = 2x\sqrt{x+1} + x^2 \cdot \frac{1}{2\sqrt{x+1}} = 2x\sqrt{x+1} + \frac{x^2}{2\sqrt{x+1}}.$$

There is no way to "expand" a square root away; the product rule is essential here.

11.5 The Quotient Rule

Division is just multiplication in disguise, so we can squeeze the quotient rule out of the product rule. Suppose $q(x) = \dfrac{f(x)}{g(x)}$ (with $g(x) \neq 0$). The cleanest derivation clears the denominator first.

11.5.1 Deriving it from the product rule

Write $f = q \cdot g$. Differentiate both sides with the product rule:

$$f' = q'g + qg'.$$

Now solve for the unknown q':

$$q' = \frac{f' - qg'}{g} = \frac{f' - \dfrac{f}{g}\,g'}{g} = \frac{f'g - fg'}{g^2},$$

where in the last step we substituted $q = f/g$ and multiplied top and bottom by g.

Theorem 11.5 (Quotient rule)**.** *If f and g are differentiable and $g(x) \neq 0$, then f/g is differentiable and*

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)\,g(x) - f(x)\,g'(x)}{\big(g(x)\big)^2}.$$

Remark 11.1. The derivation above quietly assumed q was differentiable so that we could apply the product rule to $f = qg$. A fully airtight proof goes back to the limit definition (combine the fractions over the common denominator $g(x+h)g(x)$, add and subtract a mixed term just as we did for the product rule). The shortcut via $f = qg$ is the way to *remember* the rule; the limit proof is the way to *certify* it. Both give the same formula.

Why the technique works

The quotient rule's strange-looking minus sign and squared denominator are not arbitrary. Think of f/g as f times $1/g$. Increasing f pushes the quotient *up*: that is the $+f'g$ part. Increasing g pushes the quotient *down* – a bigger denominator means a smaller fraction – which is exactly why g' enters with a *minus*. The g^2 in the denominator is the price of the reciprocal: as we computed back in Example 11.3, $\frac{\mathrm{d}}{\mathrm{d}x}(1/g)$ carries a $1/g^2$, and the quotient rule inherits it. Order matters in the numerator: it is $f'g - fg'$, "derivative of top times bottom, minus top times derivative of bottom." Swapping the two terms flips the sign and gives the wrong answer.

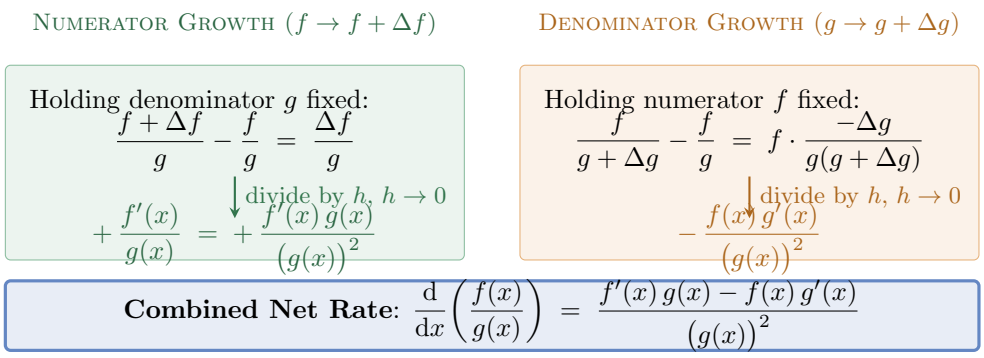


Figure 11.3: The mechanics of the quotient rule (Figure 11.3). The ratio f/g responds to two competing effects: numerator growth lifts the quotient at rate $+f'/g = +(f'g)/g^2$, while denominator growth pulls the quotient down at rate $-(fg')/g^2$. Superposing both linear responses yields the difference $f'g - fg'$ over the squared denominator g^2.

Common trap

The numerator of the quotient rule is *not* symmetric: $f'g - fg' \neq fg' - f'g$. A mnemonic that prevents the classic sign error: "low dee-high minus high dee-low, over the square of what's below." Here "high" is the numerator f, "low" is the denominator g, and "dee" means "derivative of." So: $\dfrac{\text{low} \cdot f' - \text{high} \cdot g'}{\text{low}^2} = \dfrac{gf' - fg'}{g^2}$. Same thing, written to keep the order straight.

Example 11.9 (A rational function)**.** Differentiate $q(x) = \dfrac{x^2 - 1}{x^2 + 1}$.

Top $f = x^2 - 1$, $f' = 2x$. Bottom $g = x^2 + 1$, $g' = 2x$. Then

$$q'(x) = \frac{f'g - fg'}{g^2} = \frac{2x(x^2+1) - (x^2-1)(2x)}{(x^2+1)^2}.$$

Expand the numerator:

$$2x(x^2+1) - 2x(x^2-1) = 2x\big[(x^2+1) - (x^2-1)\big] = 2x \cdot 2 = 4x.$$

So $q'(x) = \dfrac{4x}{(x^2+1)^2}$. Notice the numerator is clean while the denominator stays squared – a typical quotient-rule outcome.

Example 11.10 (Deriving the reciprocal rule)**.** Take $f = 1$ (so $f' = 0$) in the quotient rule:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{g}\right) = \frac{0 \cdot g - 1 \cdot g'}{g^2} = -\frac{g'}{g^2}.$$

This *reciprocal rule* is worth memorizing on its own. As a check, with $g = x$ it gives $\frac{\mathrm{d}}{\mathrm{d}x}(1/x) = -1/x^2$, matching Example 11.3.

History

The rules of this chapter were forged in the dramatic collision between two radically different mathematical worldviews during the 1670s and 1680s. Isaac Newton, working in Cambridge during the plague years of 1665–1666, conceived of calculus kinematically. For Newton, curves were physical trajectories traced out by moving points, fluent quantities changing continuously through time, and rates of change were *fluxions* $(\dot{x}, \dot{y})$. Newton possessed the product rule and chain rule early, but he viewed them through the mechanics of motion and guarded his manuscripts privately, sharing them only with an inner circle of English correspondents.

Gottfried Wilhelm Leibniz, working independently in Paris and Hanover, approached the problem not from kinematics, but from combinatorics and symbolic logic. In October 1684, Leibniz published the very first paper on the differential calculus in the Leipzig journal *Acta Eruditorum*: *Nova Methodus pro Maximis et Minimis, itemque Tangentibus...* ("A New Method for Maxima and Minima, as well as Tangents, which is neither impeded by fractional nor irrational quantities..."). At just six pages, Leibniz's paper changed the mathematical world because of its notation and operational perspective. Leibniz introduced the differential symbols $\mathrm{d}x$ and $\mathrm{d}y$, and treated the product and quotient rules as principles of *geometric algebra*.

To Leibniz, the product $u \cdot v$ was the area of a rectangle with sides u and v. If each side is given an infinitesimal increment $\mathrm{d}u$ and $\mathrm{d}v$, the new area is $(u+\mathrm{d}u)(v+\mathrm{d}v) = uv + u\,\mathrm{d}v + v\,\mathrm{d}u + \mathrm{d}u\,\mathrm{d}v$. By discarding the second-order corner $\mathrm{d}u\,\mathrm{d}v$ as incomparably smaller than the first-order strips, Leibniz deduced $\mathrm{d}(uv) = u\,\mathrm{d}v + v\,\mathrm{d}u$ directly. To find the differential of a quotient $q = u/v$, he inverted the geometry: writing $u = qv$, he applied the product rule to get $\mathrm{d}u = q\,\mathrm{d}v + v\,\mathrm{d}q$, and solved algebraically for $\mathrm{d}q = (\mathrm{d}u - q\,\mathrm{d}v)/v = (v\,\mathrm{d}u - u\,\mathrm{d}v)/v^2$.

This operational elegance ignited a bitter priority dispute between the Royal Society and continental mathematicians that lasted decades. It also invited fierce philosophical attack: Bishop George Berkeley, in his 1734 critique *The Analyst*, famously mocked these discarded differentials as "the ghosts of departed quantities"—vanishing increments that were treated as non-zero when dividing, and zero when discarding corners. It took until the 1820s, when Augustin-Louis Cauchy reconstructed calculus on the foundation of limits, for Leibniz's geometric algebra to gain its modern rigor. The corner $\Delta u\,\Delta v$ is not discarded by arbitrary fiat; in the limit as $\Delta x \to 0$, its contribution $(\Delta u/\Delta x)\Delta v$ vanishes because $\Delta v \to 0$. Leibniz's geometric intuition was completely vindicated; Cauchy supplied the rigorous backbone.

11.6 The Chain Rule

The chain rule is the most important differentiation rule of all. The power, product, and quotient rules handle functions *built by* arithmetic; the chain rule handles functions built by *composition* – one function fed into another, like $(3x+1)^{10}$, $\sqrt{x^2+1}$, $\sin(x^2)$, or $e^{-t/2}$. Almost every interesting function in science is a composition, so almost every derivative you ever compute will lean on this rule.

11.6.1 The intuition: rates multiply

Forget formulas for a moment and think about *linked rates of change*. Suppose a car's fuel use depends on its speed, and its speed depends on time. Ask: how fast is fuel consumption changing with time?

Suppose that for each 1 km/h of extra speed the car burns 0.2 extra litres per hour (that is $\mathrm{d}(\text{fuel})/\mathrm{d}(\text{speed}) = 0.2$), and that right now the car is accelerating at 3 km/h per second ($\mathrm{d}(\text{speed})/\mathrm{d}t = 3$). Then per second the speed rises by 3 km/h, and each of those raises fuel use by 0.2 litre/h, so fuel use climbs by $0.2 \times 3 = 0.6$ litre/h per second. *The rates multiply.*

Intuition

Gears. Picture two meshed gears. If the first turns 3 times for each turn of the input shaft, and the second turns 2 times for each turn of the first, then the second turns $3 \times 2 = 6$ times per turn of the input. Compose the linkages, multiply the gear ratios. A derivative is precisely a "how-much-output-per-unit-input ratio," and composing functions chains the linkages – so you multiply the derivatives. That is the entire content of the chain rule, before any symbols.

In Leibniz notation the chain rule is almost self-evident. If y depends on u and u depends on x, then

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u} \cdot \frac{\mathrm{d}u}{\mathrm{d}x},$$

and it *looks* as if the $\mathrm{d}u$'s simply cancel like fractions. That visual cancellation is the whole reason Leibniz's notation has survived three centuries – it makes the chain rule unforgettable. (It is not *literally* fraction cancellation, because $\mathrm{d}y/\mathrm{d}x$ is a limit, not a quotient of numbers; but it behaves enough like one that the mnemonic never lies to you here.)

11.6.2 Stating it in function notation

Let $y = f(g(x))$ – the "outer" function f applied to the "inner" function g. Setting $u = g(x)$, we have $y = f(u)$, and the chain rule reads:

Theorem 11.6 (Chain rule)**.** *If g is differentiable at x and f is differentiable at $g(x)$, then the composition $f \circ g$ is differentiable at x, and*

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(f\big(g(x)\big)\big) = f'\big(g(x)\big) \cdot g'(x).$$

In words: derivative of the outside (evaluated at the inside) times derivative of the inside.

11.6.3 Why the naive proof almost works – and how to fix it

The natural proof writes the difference quotient and inserts a clever "multiply by 1." Let $u = g(x)$ and let $\Delta u = g(x+h) - g(x)$ be the change in the inner function. Then the change in y is $\Delta y = f(g(x+h)) - f(g(x)) = f(u + \Delta u) - f(u)$, and we write

$$\frac{\Delta y}{h} = \frac{f(u+\Delta u) - f(u)}{h} = \underbrace{\frac{f(u+\Delta u) - f(u)}{\Delta u}}_{\to f'(u)} \cdot \underbrace{\frac{\Delta u}{h}}_{\to g'(x)}.$$

As $h \to 0$ we have $\Delta u \to 0$ (because g is continuous), so the first factor tends to $f'(u)$ and the second to $g'(x)$, giving $f'(u)\,g'(x)$. Beautiful – and almost correct.

Common trap

The catch: we divided by Δu, but Δu might be *zero* for some small h (imagine g momentarily flat). Then the fraction $\dfrac{f(u+\Delta u)-f(u)}{\Delta u}$ is $0/0$, undefined, and the slick cancellation is illegal. This is not pedantic fuss; it is a genuine gap that a rigorous proof must close.

11.6.4 A rigorous proof

The standard repair, due in spirit to Carathéodory, replaces the troublesome difference quotient by a single function that is well-behaved *even when* $\Delta u = 0$. Define the "error-corrected slope" of f at u:

$$\phi(t) = \begin{cases} \dfrac{f(t)-f(u)}{t-u}, & t \neq u, \\[2ex] f'(u), & t = u. \end{cases}$$

Because f is differentiable at u, the first branch tends to $f'(u)$ as $t \to u$, which is exactly the value we assigned in the second branch – so ϕ *is continuous at* u. Moreover, for *every* t (including $t = u$!), multiplying back across gives the clean identity

$$f(t) - f(u) = \phi(t)\,(t-u).$$

When $t \neq u$ this is clearing the fraction; when $t = u$ both sides are 0. By expressing the relationship as a product rather than a fraction, division by zero is bypassed cleanly.

Proof of the chain rule. Put $t = g(x+h)$ and recall $u = g(x)$, so $t - u = \Delta u = g(x+h) - g(x)$. The identity $f(t) - f(u) = \phi(t)(t-u)$ becomes

$$f\big(g(x+h)\big) - f\big(g(x)\big) = \phi\big(g(x+h)\big) \cdot \big(g(x+h) - g(x)\big).$$

Divide by h (this is now perfectly safe – no division by Δu anywhere):

$$\frac{f(g(x+h)) - f(g(x))}{h} = \phi\big(g(x+h)\big) \cdot \frac{g(x+h)-g(x)}{h}.$$

Now let $h \to 0$. Since g is differentiable, it is continuous, so $g(x+h) \to g(x) = u$; and since ϕ is continuous at u, $\phi(g(x+h)) \to \phi(u) = f'(u)$. The second factor tends to $g'(x)$ by definition. Therefore the product tends to

$$f'(u) \cdot g'(x) = f'\big(g(x)\big)\,g'(x),$$

which is the chain rule. □

Why the technique works

Both proofs agree on the punchline – "the two rates multiply" – and differ only in plumbing. The naive version divides by Δu and prays it is nonzero; the Carathéodory version replaces

"difference quotient of f" with the genuinely continuous function ϕ, which equals that quotient when $\Delta u \neq 0$ but is also defined (as $f'(u)$) when $\Delta u = 0$. The product $f(t)-f(u) = \phi(t)(t-u)$ holds with no exceptions, so we never divide by something that might vanish. The moral is general: *when a proof stumbles over a possible* $0/0$*, look for a way to write the same quantity as a product instead of a quotient.*

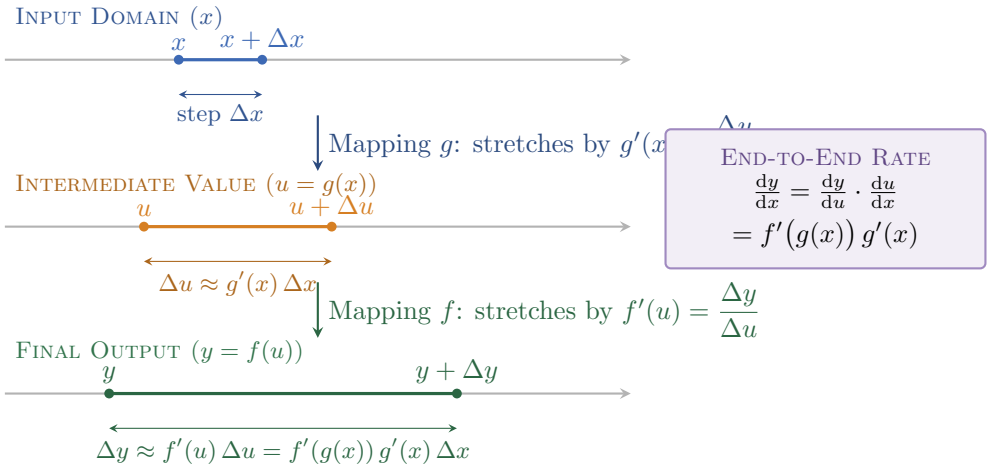


Figure 11.4: The chain rule as consecutive stretch factors across number lines. A small nudge Δx on the input axis is scaled by $g'(x)$ to produce the nudge Δu on the intermediate axis. That intermediate nudge is in turn scaled by $f'(u)$ on the output axis to produce Δy. Because the transformations act in sequence, their local scaling factors multiply.

11.6.5 Using it: peel the onion from the outside in

In practice the chain rule is a peeling procedure. Identify the outermost operation, differentiate it while leaving the inside untouched, then multiply by the derivative of the inside – and recurse if the inside is itself a composition.

Example 11.11 (A power of a polynomial)**.** Differentiate $y = (3x+1)^{10}$.

Outer function: "something to the 10th," with derivative $10(\,\cdot\,)^9$. Inner function: $3x+1$, with derivative 3. So

$$y' = 10(3x+1)^9 \cdot 3 = 30(3x+1)^9.$$

Expanding $(3x+1)^{10}$ to differentiate term by term would mean eleven terms and a binomial expansion. The chain rule does it in one line.

Example 11.12 (A square root)**.** Differentiate $y = \sqrt{x^2+1} = (x^2+1)^{1/2}$.

Outer: $(\,\cdot\,)^{1/2}$, derivative $\frac{1}{2}(\,\cdot\,)^{-1/2}$. Inner: x^2+1, derivative $2x$. Thus

$$y' = \tfrac{1}{2}(x^2+1)^{-1/2} \cdot 2x = \frac{x}{\sqrt{x^2+1}}.$$

Figure 11.5 shows this slope as an actual tangent line. This finally justifies the square-root step we borrowed in Example 11.8.

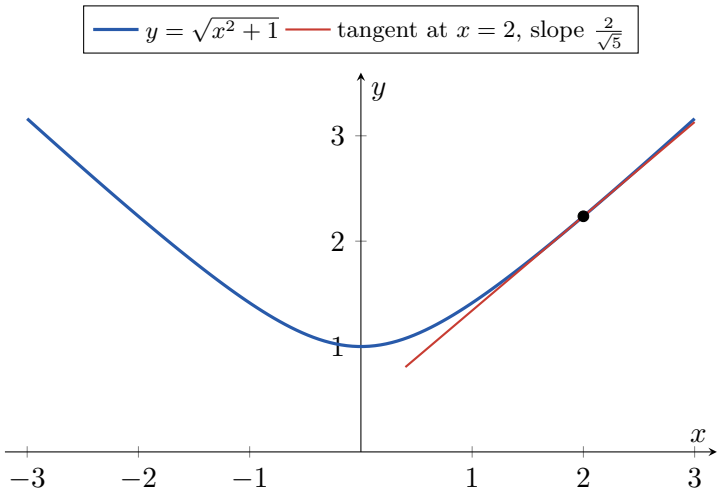


Figure 11.5: The chain rule predicts the slope of a composition. For $y = \sqrt{x^2+1}$ the chain rule gives $y' = x/\sqrt{x^2+1}$; at $x = 2$ this is $2/\sqrt{5} \approx 0.894$, the slope of the tangent line drawn here. The inner derivative $2x$ is exactly what tilts the tangent away from the slope of the bare square root.

Example 11.13 (The general power-chain shortcut)**.** Combining the power rule with the chain rule gives a pattern you will use constantly:

$$\frac{\mathrm{d}}{\mathrm{d}x}\big((g(x))^n\big) = n\big(g(x)\big)^{n-1} g'(x).$$

For instance $\frac{\mathrm{d}}{\mathrm{d}x}\big((x^3-2x)^4\big) = 4(x^3-2x)^3(3x^2-2)$.

Example 11.14 (Two layers deep)**.** Differentiate $y = \big(\sqrt{x}+1\big)^3$ – a composition inside a composition.

Peel from the outside. Outermost: $(\,\cdot\,)^3$, derivative $3(\,\cdot\,)^2$. Its inside is $\sqrt{x}+1$, whose derivative is $\frac{1}{2\sqrt{x}}$ (the $+1$ contributes nothing). So

$$y' = 3\big(\sqrt{x}+1\big)^2 \cdot \frac{1}{2\sqrt{x}} = \frac{3\big(\sqrt{x}+1\big)^2}{2\sqrt{x}}.$$

If there had been a third layer, we would simply have multiplied by one more inner derivative. The chain rule composes as many times as you need: it is gear ratios all the way down.

Common trap

The single most common chain-rule error is *forgetting the inner derivative*. People write $\frac{\mathrm{d}}{\mathrm{d}x}\big((3x+1)^{10}\big) = 10(3x+1)^9$ and stop – dropping the $\times 3$. The tell-tale check: if the inside is just x, the inner derivative is 1 and you are back to the plain power rule; the moment the inside is anything more interesting, that $\times g'(x)$ factor is doing real work. *Never let go of the inside.*

11.6.6 Combining all the rules

Real problems braid the rules together. The skill is recognizing the *outermost* structure first – is the whole expression a sum, a product, a quotient, or a composition? – and applying the matching rule, then recursing into the pieces.

Example 11.15 (Product and chain together)**.** Differentiate $y = x^2\,(2x+3)^5$.

The outermost structure is a *product*: $f = x^2$ times $g = (2x+3)^5$. Product rule first:

$$y' = f'g + fg' = 2x\,(2x+3)^5 + x^2 \cdot \frac{\mathrm{d}}{\mathrm{d}x}\big((2x+3)^5\big)\,.$$

The remaining derivative $\frac{\mathrm{d}}{\mathrm{d}x}\big((2x+3)^5\big)$ needs the *chain* rule: $5(2x+3)^4 \cdot 2 = 10(2x+3)^4$. Substitute back:

$$y' = 2x(2x+3)^5 + 10x^2(2x+3)^4.$$

You can factor out the common $2x(2x+3)^4$ to tidy up:

$$y' = 2x(2x+3)^4\big[(2x+3) + 5x\big] = 2x(2x+3)^4(7x+3).$$

Example 11.16 (Quotient and chain together)**.** Differentiate $y = \dfrac{(x+1)^3}{x^2+1}$.

Outermost structure: a *quotient*. Top $f = (x+1)^3$ with $f' = 3(x+1)^2$ (chain rule, inner derivative 1); bottom $g = x^2+1$ with $g' = 2x$. Quotient rule:

$$y' = \frac{3(x+1)^2(x^2+1) - (x+1)^3(2x)}{(x^2+1)^2}.$$

Factor $(x+1)^2$ from the numerator:

$$y' = \frac{(x+1)^2\big[3(x^2+1) - 2x(x+1)\big]}{(x^2+1)^2} = \frac{(x+1)^2\,(x^2-2x+3)}{(x^2+1)^2}.$$

(The bracket simplifies as $3x^2+3-2x^2-2x = x^2-2x+3$.)

Walking question

On your next walk, try differentiating $\big((x^2+1)^3\big)$ *two ways* in your head: once by the chain rule directly, and once by first imagining it as the product $(x^2+1)(x^2+1)(x^2+1)$ and using the three-factor product rule. Convince yourself both give $6x(x^2+1)^2$. Then ask: why *must* they agree? (Because they are the same function – and a function has only one derivative. The rules are different roads to one destination, and the destination cannot move.)

Application

Where the chain rule earns its living. In machine learning, training a neural network means minimizing a loss that is a deep composition – layer after layer after layer of functions stacked on top of one another. *Backpropagation*, the algorithm that makes modern deep learning possible, is nothing but the chain rule applied mechanically through that stack, multiplying the local rate of change at each layer to get the overall rate of change of the loss with respect to each weight. When you read that a network "propagates gradients backward," it is computing $f'(g(x)) \cdot g'(x)$ over and over, exactly the rule you just proved. Newton's gear ratios are training the chatbots.

Connection

The chain rule is the seed of multivariable calculus. When y depends on several intermediate quantities that each depend on x, the single product $f'(g(x))g'(x)$ blooms into a *sum* of such products – the multivariable chain rule – and that sum, written compactly, is a dot product of the gradient with a velocity vector. The gradient, directional derivatives, and the backpropagation above are all the chain rule wearing different clothes. We will return to this when we reach functions of several variables.

11.7 A Summary You Can Trust

You now hold the complete elementary toolkit. Every entry below was *derived*, not decreed, and each carries a one-line reason you can reconstruct.

Rule	**Formula**	**Why**
Constant	$\frac{\mathrm{d}}{\mathrm{d}x}(c) = 0$	flat line, no change
Identity	$\frac{\mathrm{d}}{\mathrm{d}x}(x) = 1$	change passes through
Power	$\frac{\mathrm{d}}{\mathrm{d}x}(x^n) = nx^{n-1}$	linear term of $(x+h)^n$ survives
Constant multiple	$(kf)' = kf'$	vertical stretch scales all slopes
Sum	$(f \pm g)' = f' \pm g'$	rates add
Product	$(fg)' = f'g + fg'$	rectangle grows on two sides
Quotient	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$	top up, bottom down
Chain	$\big(f(g(x))\big)' = f'(g(x))\, g'(x)$	rates multiply

Why this matters

Step back and admire the architecture. Two atoms ($\frac{\mathrm{d}}{\mathrm{d}x}(c) = 0$ and $\frac{\mathrm{d}}{\mathrm{d}x}(x) = 1$), one structural principle (linearity), and three combination rules (product, quotient, chain) generate the derivative of *every* function you can build from powers by the four arithmetic operations and composition. That is an enormous reach from a tiny seed. And none of it was arbitrary: each rule is forced by what "rate of change" *means* once you take the limit seriously. This is the recurring joy of calculus – a handful of plain ideas, pushed carefully, organize an entire universe of computation. In the next chapter we add the derivatives of e^x, $\ln x$, and the trig functions, and with those few new atoms the same machine differentiates essentially anything in the analyst's toolbox.

11.7.1 The Scope and Horizon of Differentiation Rules

Connection

Prior to Leibniz's 1684 operational calculus, finding the slope of a curve was an artisanal craft. Mathematicians had to devise custom geometric constructions or labor through bespoke difference-quotient expansions for every new curve. Leibniz's rules automated differentiation. By recognizing that functions are assembled algebraically from atomic pieces, calculus gained a grammar: any function constructed from powers, sums, products, quotients, and compositions could be differentiated mechanically on paper without returning to the limit definition from scratch.

Yet for all their algebraic power, differentiation rules provide purely *local* information. The derivative $f'(x_0)$ reports the instantaneous velocity or the tilt of the tangent line at one isolated point x_0. It does not, on its own, reveal the global landscape: it cannot locate the absolute peak or deepest valley on a closed domain, it cannot guarantee that a function is monotonic or invertible across an interval merely from an algebraic formula, and it gives no automatic bound on how quickly a linear approximation deteriorates as you step away from x_0.

Turning these local rates into global understanding requires three subsequent bridges across the curriculum:

1. In Chapter 14 (*Linear Approximation and Differentials*), we use the local slope to construct linear models and quantify how differentials propagate measurement errors.
2. In Chapter 15 (*The Mean Value Theorem*), we prove the foundational theoretical bridge that allows local derivative signs ($f'(x) > 0$) to guarantee global behavior (the function strictly rises across an entire interval).
3. In Chapter 16 (*Maxima, Minima, and the Shape of a Graph*), we classify critical points and inflection points, translating local derivatives into complete qualitative portraits of a function's global geometry.

11.8 Exercises

Exercise 11.1 (Power-rule drill)**.** Differentiate each, rewriting roots and reciprocals as powers first.

1. $f(x) = 6x^4 - 2x^3 + x - 11$.
2. $g(x) = \dfrac{5}{x^3} - \dfrac{2}{\sqrt{x}} + 7$.
3. $h(x) = \sqrt[3]{x^2} + 4x^{-5}$.

Exercise 11.2 (Product rule)**.** Differentiate using the product rule, then (where feasible) check by expanding first.

1. $y = (2x^2 - 1)(x^3 + x)$.
2. $y = x^3\sqrt{x+4}$.
3. $y = (x^2+1)(x^2+2)(x^2+3)$ (use the three-factor product rule).

Exercise 11.3 (Quotient rule)**.** Differentiate and simplify.

1. $y = \dfrac{x}{x^2+1}$.
2. $y = \dfrac{2x-3}{4x+5}$.
3. $y = \dfrac{x^2}{(x-1)^2}$ (quotient rule, with the chain rule on the bottom).

Exercise 11.4 (Chain rule)**.** Differentiate. Peel from the outside in, and do not drop the inner derivative.

1. $y = (5x^2-3)^7$.
2. $y = \sqrt{4-x^2}$.
3. $y = \big((x+1)^2+1\big)^3$ (two layers).

Exercise 11.5 (All rules braided together)**.** Differentiate. First identify the outermost structure (sum, product, quotient, or composition).

1. $y = x^2\sqrt{x^2+1}$.
2. $y = \dfrac{(3x-1)^4}{x^2+1}$.
3. $y = \big(x+\sqrt{x}\big)^5$.

Exercise 11.6 (Conceptual: spot the error)**.** A student writes $\frac{\mathrm{d}}{\mathrm{d}x}\left(\dfrac{x^2}{x+1}\right) = \dfrac{2x}{1} = 2x$ by "differentiating top and bottom separately." Explain in one or two sentences why this is wrong, name the rule that should have been used, and give the correct derivative.

Exercise 11.7 (Conceptual: why two terms?)**.** In your own words, explain why $\frac{\mathrm{d}}{\mathrm{d}x}(fg)$ has *two* terms rather than one, using the growing-rectangle picture. What does the third piece (the corner $\Delta f\,\Delta g$) represent, and why does it not appear in the final formula?

Exercise 11.8 (Reconstruct a rule)**.** Without looking back, derive the quotient rule from the product rule by writing $f = q\,g$ where $q = f/g$, differentiating, and solving for q'. Then check your result by computing $\frac{\mathrm{d}}{\mathrm{d}x}(1/x)$ two ways: from the quotient rule and from the limit definition.

Exercise 11.9 (Proof: the power rule for negative integers)**.** Assume the power rule holds for positive integers and that the reciprocal rule holds. Let n be a positive integer and set $x^{-n} = 1/x^n$. Use the reciprocal rule $\frac{\mathrm{d}}{\mathrm{d}x}(1/g) = -g'/g^2$ to prove

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(x^{-n}\big) = -n\,x^{-n-1},$$

thereby extending the power rule to negative integer exponents.

Exercise 11.10 (Proof: a three-factor product rule)**.** Prove that $(fgh)' = f'gh + fg'h + fgh'$ by treating fgh as the product of f with the single function (gh) and applying the two-factor product rule twice. Then state, without proof, the pattern for a product of n factors.

Exercise 11.11 (Proof: the chain rule cannot be skipped)**.** Let $F(x) = f(g(x))$ with $f(u) = u^2$ and $g(x) = x^2 + 1$.

1. Compute $F'(x)$ by the chain rule.
2. Compute $F'(x)$ instead by first expanding $F(x) = (x^2+1)^2 = x^4 + 2x^2 + 1$ and differentiating directly.
3. Verify the answers agree, and explain what would have gone wrong had you mistakenly written $F'(x) = f'(g(x)) = 2(x^2+1)$ without the factor $g'(x)$.

Exercise 11.12 (A limit that hides a derivative)**.** Recognize each limit as a derivative $f'(a)$ for a suitable f and a, and evaluate it using the rules of this chapter rather than the limit directly.

1. $\lim_{h\to 0} \dfrac{(2+h)^5 - 2^5}{h}$.
2. $\lim_{h\to 0} \dfrac{\sqrt{9+h} - 3}{h}$.

Chapter 12

Derivatives of Exponentials, Logarithms, and Trig

So far the derivative has been a machine that eats polynomials and rational functions and spits out their slopes. The power rule, the product rule, the quotient rule, the chain rule – these handle everything you can build from x by adding, multiplying, and composing. But the most *important* functions in all of science are not built that way. Populations grow like e^{kt}. Radioactive atoms decay like $e^{-\lambda t}$. A pendulum swings like $\cos(\omega t)$. The loudness you hear and the brightness you see are logarithmic. If calculus could not differentiate e^x, $\ln x$, $\sin x$, and their relatives, it would be a beautiful theory about almost nothing.

This chapter computes the derivatives of all of them. But I want to do more than hand you a table to memorize. Every one of these derivatives is *forced* by the definition of the function. The fact that $\frac{\mathrm{d}}{\mathrm{d}x}(e^x) = e^x$ is not a lucky accident – it is the whole reason the number e exists. The fact that $\frac{\mathrm{d}}{\mathrm{d}x}(\sin x) = \cos x$ is the unit circle turning. By the end you should feel that these formulas could not have come out any other way.

12.1 The Exponential: a Function Equal to Its Own Slope

Start with the most consequential single fact in differential calculus. Take any exponential function $f(x) = b^x$ with base $b > 0$ and ask for its derivative straight from the definition.

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{b^{x+h} - b^x}{h} \\ &= \lim_{h\to 0} \frac{b^x b^h - b^x}{h} \qquad \text{(law of exponents: } b^{x+h} = b^x b^h\text{)} \\ &= \lim_{h\to 0} b^x \cdot \frac{b^h - 1}{h} \\ &= b^x \cdot \underbrace{\lim_{h\to 0} \frac{b^h - 1}{h}}_{= m(b)}. \end{aligned}$$

Stop and look at what just happened, because it is remarkable. The b^x factored straight out of the limit, since it does not depend on h. What is left over is a limit that does not depend on x at all – it is just a *number*, a constant that depends only on the base b. Call it $m(b)$. So we have proven, with almost no work, the astonishing structural fact:

$$\frac{\mathrm{d}}{\mathrm{d}x}(b^x) \;=\; m(b) \cdot b^x, \qquad m(b) = \lim_{h\to 0} \frac{b^h - 1}{h}.$$

Focus

The derivative of *any* exponential function is a constant multiple of *itself*. The function b^x reproduces its own shape when differentiated; the only thing the derivative does is rescale it by the fixed number $m(b)$. No other family of functions behaves like this, and it is the source of the exponential's grip on the natural world.

Intuition

Why should the slope of b^x be proportional to its height? Picture compound interest. The more money you have, the more interest you earn per year – the growth is proportional to the current amount. The more rabbits you have, the more baby rabbits per month. Anything whose rate of change is proportional to its current size is, secretly, an exponential. The constant $m(b)$ is just the proportionality factor, the "interest rate" baked into the base.

12.1.1 What is that mysterious constant $m(b)$?

The number $m(b) = \lim_{h\to 0} \frac{b^h - 1}{h}$ has a clean geometric meaning. Set $x = 0$ in our formula: since $b^0 = 1$,

$$f'(0) = m(b) \cdot b^0 = m(b).$$

So $m(b)$ is nothing but *the slope of b^x at the point where it crosses the y-axis.* Every exponential passes through $(0, 1)$; what distinguishes them is how steeply they leave that point. And our boxed formula says something almost too good: the slope everywhere is the y-axis slope times the height. Know the starting slope, and you know every slope.

We can estimate $m(b)$ for a few bases by plugging a tiny h into $(b^h - 1)/h$.

Work it yourself

Take $h = 0.001$ and compute $(b^h - 1)/h$ with a calculator.

- $b = 2$: $(2^{0.001} - 1)/0.001 \approx 0.6934$.
- $b = 2.5$: $(2.5^{0.001} - 1)/0.001 \approx 0.9167$.
- $b = 2.718$: $(2.718^{0.001} - 1)/0.001 \approx 1.0004$.
- $b = 3$: $(3^{0.001} - 1)/0.001 \approx 1.0992$.

The slope $m(b)$ slides smoothly from below 1 (at $b = 2$) to above 1 (at $b = 3$), crossing 1 somewhere in between. At $b = 2.718$ the ratio is already a whisker over 1 – so the true crossing point, where the *limit* $m(b)$ equals exactly 1, is staring at us right around 2.718. (The finite step $h = 0.001$ overshoots the limit slightly, which is why 2.718 reads 1.0004 rather than just under 1; shrink h and the numbers settle onto $m(b) = \ln b$, the value we will prove shortly.)

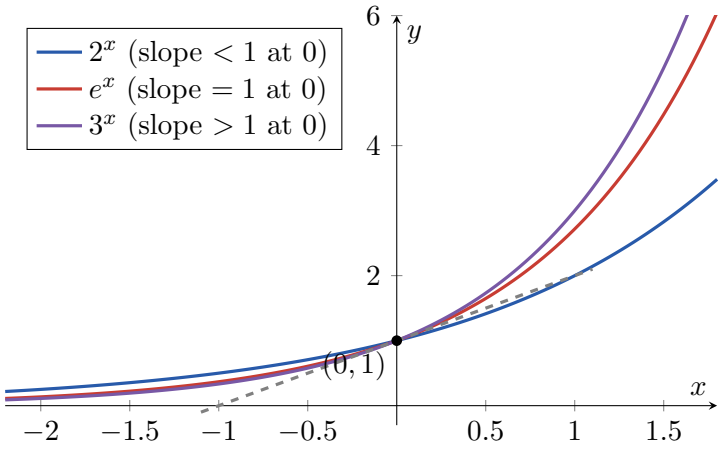


Figure 12.1: Every exponential passes through $(0, 1)$ but leaves at a different slope. The base e is the unique one whose tangent at $(0, 1)$ is the line $y = 1 + x$ (slope exactly 1, dashed).

12.1.2 Defining e as the base that makes $m(b) = 1$

We have found the cleanest possible definition of the number e. Among all exponential functions, exactly one has slope 1 at $x = 0$. Its base is what we call e.

Definition 12.1 (The number e, via slope)**.** e is the unique base $b > 0$ for which

$$\lim_{h \to 0} \frac{b^h - 1}{h} = 1, \qquad \text{equivalently} \qquad \frac{\mathrm{d}}{\mathrm{d}x} b^x \Big|_{x=0} = 1.$$

With this base the constant $m(e) = 1$, so the leftover limit disappears and we land on the jewel of the chapter.

Theorem 12.1 (The exponential is its own derivative)**.**

$$\frac{\mathrm{d}}{\mathrm{d}x}(e^x) = e^x.$$

Proof. From the general formula $\frac{\mathrm{d}}{\mathrm{d}x}(b^x) = m(b)\,b^x$ with $b = e$, and $m(e) = 1$ by Definition 12.1, we get $\frac{\mathrm{d}}{\mathrm{d}x}(e^x) = 1 \cdot e^x = e^x$. □

What is really happening

Look at what "$\frac{\mathrm{d}}{\mathrm{d}x}(e^x) = e^x$" really asserts. At every point (x, e^x) on the curve, the steepness of the graph equals its height above the axis. At height 1 the slope is 1; at height 7 the slope is 7; at height 100 the slope is 100. The curve is racing upward exactly as fast as it is tall. That self-referential loop – "my rate of growth is my current size" – is what makes e^x explode, and it is why the same function appears whenever growth feeds on itself: interest on interest, infection spreading person to person, a chain reaction. The symbol e^x is just a name for "the function that is its own rate of change and equals 1 at 0." Everything else about it follows.

History

The story of exponentials and logarithms is the story of calculus liberating itself from polynomial geometry to conquer the physical universe. In the early seventeenth century, astronomical navigation and planetary calculations (such as Johannes Kepler's calculations of planetary orbits) were paralyzed by the sheer drudgery of multiplying and dividing eight-digit numbers by hand. In 1614, John Napier published *Mirifici Logarithmorum Canonis Descriptio*, followed shortly by Henry Briggs's 1624 tables of base-10 logarithms in *Arithmetica Logarithmica*. By converting arduous multiplications into simple additions ($\log(ab) = \log a + \log b$), logarithms were celebrated as having "doubled the lives of astronomers."

Yet for more than a century, logarithms were regarded purely as computational tables and geometric areas under hyperbolas, not as functions, and certainly not as the inverses of an exponential. The decisive transformation came in 1748 with Leonhard Euler's monumental two-volume treatise *Introductio in analysin infinitorum*. Euler shifted the foundation of mathematics away from geometric curves and onto analytical *functions*. He systematically established the number $e \approx 2.718281828459$, demonstrating that the compound-interest limit $\lim_{n\to\infty}(1+1/n)^n$, the infinite power series $\sum_{k=0}^{\infty} 1/k!$, and the unique base whose exponential slope is 1 at the origin are all identical manifestations of the same constant.

This new analytical world was not without fierce controversy. A celebrated dispute erupted between Euler and his former mentor, Johann Bernoulli, over the logarithm of negative numbers. Bernoulli insisted that since $(-x)^2 = x^2$, taking logarithms must yield $2\ln(-x) = 2\ln(x)$, implying that $\ln(-x) = \ln(x)$ for all real x—so negative numbers would have ordinary, real logarithms! Euler dismantled Bernoulli's argument in 1749 by demonstrating that the logarithm is intrinsically multi-valued in the complex plane. Euler showed that $\ln(-1) = (2k+1)\pi i$ for any integer k; negative inputs rotate into the complex domain by an odd multiple of π radians. The "natural" logarithm was far deeper than either Napier or Bernoulli had dreamed: it tied real rates of change directly to rotational geometry in the complex plane ($e^{ix} = \cos x + i \sin x$).

Connection

Why does the slope-1 base equal the compound-interest limit? Here is the quick reconciliation. Suppose b is the base with $m(b) = 1$, i.e. $(b^h - 1)/h \to 1$ as $h \to 0$. For small h this says $b^h - 1 \approx h$, so $b^h \approx 1 + h$, hence $b \approx (1+h)^{1/h}$. Letting $h = 1/n$ and $n \to \infty$ turns the right side into $\lim_{n\to\infty}(1+\frac{1}{n})^n$ – exactly the compound-interest definition of e. The two definitions

are the same fact viewed from two heights.

12.2 The Natural Logarithm by Inverse Reasoning

Now that e^x is tamed, its inverse comes almost for free. Recall that $\ln x$ (the natural logarithm) is *defined* as the inverse of e^x:

$$y = \ln x \quad \Longleftrightarrow \quad e^y = x, \qquad (x > 0).$$

So ln undoes $e^{(\cdot)}$ and vice versa: $e^{\ln x} = x$ and $\ln(e^x) = x$. We differentiate it two ways – once with a slick implicit argument, and once straight from the limit – because each illuminates the other.

12.2.1 Derivative of $\ln x$ by implicit differentiation

Intuition

A function and its inverse are reflections of each other across the line $y = x$. Reflection swaps the roles of x and y, and it swaps "run" with "rise." So wherever e^x has a steep slope, its mirror image $\ln x$ has a *shallow* one — the slope of the inverse is the *reciprocal* of the slope of the original: $\big(f^{-1}\big)'(x) = 1/f'\big(f^{-1}(x)\big)$. Swapping input and output coordinates inverts the tangent slope directly; the algebraic computation below confirms this geometric reflection.

Start from the relation $e^y = x$ and differentiate both sides with respect to x. The left side needs the chain rule, since y is a function of x:

$$\frac{\mathrm{d}}{\mathrm{d}x}(e^y) = \frac{\mathrm{d}}{\mathrm{d}x}(x) \quad \Longrightarrow \quad e^y \cdot \frac{\mathrm{d}y}{\mathrm{d}x} = 1.$$

Solve for $\frac{\mathrm{d}y}{\mathrm{d}x}$ and then replace e^y by x (that is what e^y equals):

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{e^y} = \frac{1}{x}.$$

That is the result, clean and complete.

Theorem 12.2 (Derivative of the natural log)**.** *For $x > 0$,*

$$\frac{\mathrm{d}}{\mathrm{d}x}(\ln x) = \frac{1}{x}.$$

Why the technique works

The inverse-function trick works because differentiating the *identity* $e^{\ln x} = x$ forces the answer. Apply the chain rule to the left side: $\frac{\mathrm{d}}{\mathrm{d}x}\big(e^{\ln x}\big) = e^{\ln x} \cdot (\ln x)' = x \cdot (\ln x)'$. The right side is just 1. So $x \cdot (\ln x)' = 1$, giving $(\ln x)' = 1/x$ with no guessing at all. Whenever you know a function's derivative and want its inverse's derivative, differentiate the "they cancel" identity and solve. The reciprocal-of-the-slope picture and this one-line computation are the same

idea: $\big(f^{-1}\big)'(x) = 1/f'\big(f^{-1}(x)\big)$.

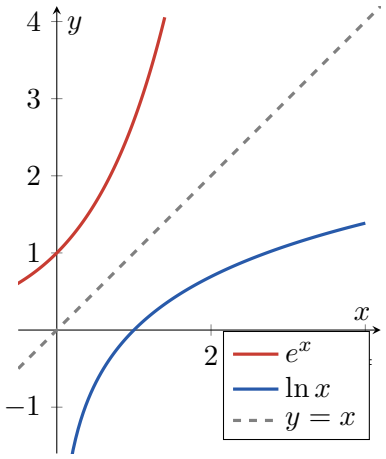


Figure 12.2: $\ln x$ is the mirror image of e^x across $y = x$. Where e^x climbs steeply, $\ln x$ rises gently: the slopes are reciprocals. As $x \to 0^+$, $\ln x$ plunges to $-\infty$ ever more steeply, matching $1/x \to +\infty$.

12.2.2 The same result straight from the limit

The same $1/x$ also drops out of the raw difference quotient, with no inverse machinery – and the derivation secretly re-derives the compound-interest formula for e along the way. We compute:

$$\frac{\mathrm{d}}{\mathrm{d}x}(\ln x) = \lim_{h\to 0} \frac{\ln(x+h) - \ln x}{h} = \lim_{h\to 0} \frac{1}{h} \ln\!\left(\frac{x+h}{x}\right) = \lim_{h\to 0} \frac{1}{h} \ln\!\left(1 + \frac{h}{x}\right).$$

Let $u = h/x$, so $h = ux$ and $u \to 0$ as $h \to 0$. Then $1/h = 1/(ux)$:

$$\frac{\mathrm{d}}{\mathrm{d}x}(\ln x) = \lim_{u\to 0} \frac{1}{ux} \ln(1+u) = \frac{1}{x} \lim_{u\to 0} \frac{\ln(1+u)}{u} = \frac{1}{x} \lim_{u\to 0} \ln\Big[(1+u)^{1/u}\Big].$$

Inside the bracket sits $(1+u)^{1/u}$, which tends to e as $u \to 0$ (the compound-interest limit again, with $u = 1/n$). Since ln is continuous we may pass the limit inside: the bracket becomes $\ln e = 1$. Therefore

$$\frac{\mathrm{d}}{\mathrm{d}x}(\ln x) = \frac{1}{x} \cdot \ln e = \frac{1}{x}.$$

Both roads lead to $1/x$, and both quietly rely on the same defining property of e. That is reassuring: the formula is not an artifact of one clever trick.

Why this matters

There is a small mystery hiding here that will pay off enormously in the integration chapters. The power rule gives $\frac{\mathrm{d}}{\mathrm{d}x}(x^n) = nx^{n-1}$ for every exponent n – *except* $n = 0$, where it produces $0 \cdot x^{-1} = 0$ instead of x^{-1}. So in the entire family $\{x^n\}$ there is no function whose derivative is $x^{-1} = 1/x$. Nature abhors the gap, and fills it with a function from a completely different family: $\ln x$. This is why $\int \frac{1}{x}\,\mathrm{d}x = \ln|x| + C$ – the logarithm is exactly the "missing" antiderivative that the power rule could never supply.

12.3 General Bases: a^x and $\log_a x$

We deliberately built everything on base e because it is the base with no clutter. Other bases inherit their derivatives from e through one rewrite.

12.3.1 Derivative of a^x

The trick is to express *any* exponential in terms of the natural one. Since $a = e^{\ln a}$, we have $a^x = \left(e^{\ln a}\right)^x = e^{(\ln a)x}$. Now it is just the chain rule with inner function $(\ln a)x$:

$$\frac{\mathrm{d}}{\mathrm{d}x}(a^x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(e^{(\ln a)x}\right) = e^{(\ln a)x} \cdot \frac{\mathrm{d}}{\mathrm{d}x}((\ln a)x) = a^x \cdot \ln a.$$

Focus

$$\frac{\mathrm{d}}{\mathrm{d}x}(a^x) = a^x \ln a.$$

Compare with our very first computation, $\frac{\mathrm{d}}{\mathrm{d}x}(a^x) = m(a)\, a^x$. We have now *identified* the mystery constant: $m(a) = \ln a$. The slope of a^x at the y-axis is the natural log of its base. So $m(2) = \ln 2 \approx 0.6931$ and $m(3) = \ln 3 \approx 1.0986$ – right where the finite-difference estimates from the lab box (0.6934 and 1.0992) were pointing. The small gap is just the overshoot from using $h = 0.001$ instead of the true limit; shrink h and the estimates march straight onto $\ln 2$ and $\ln 3$.

Work it yourself

Check the loop closes. Our finite-difference experiment gave $m(2) \approx 0.6934$. Now the theory says $m(2) = \ln 2 = 0.693147\ldots$ – spot on, the last digit's drift being only the $h = 0.001$ overshoot. And $m(e) = \ln e = 1$, recovering the defining property of e exactly. The numerical experiment and the clean theory agree as closely as you care to push the step size down.

12.3.2 Derivative of $\log_a x$

The general logarithm reduces to the natural one by the change-of-base formula $\log_a x = \dfrac{\ln x}{\ln a}$. Since $\ln a$ is a constant,

$$\frac{\mathrm{d}}{\mathrm{d}x}(\log_a x) = \frac{1}{\ln a}\,\frac{\mathrm{d}}{\mathrm{d}x}(\ln x) = \frac{1}{(\ln a)\, x}.$$

Once more e is the lucky base: with $a = e$ the factor $\ln a = 1$ vanishes and we recover $1/x$. Every stray $\ln a$ floating around in a calculus formula is the universe nagging you to switch to base e.

12.4 Logarithmic Differentiation

Here is a technique that turns the logarithm from an object of study into a *tool*. As an algebraic simplifier the natural log does two things: it converts products into sums and exponents into multi-

pliers,

$$\ln(fg) = \ln f + \ln g, \qquad \ln\big(f^{\,g}\big) = g\,\ln f.$$

Differentiation loves sums and hates products and towers of exponents. So the plan is: take ln of both sides *first* to break the mess apart, differentiate the easy pieces, then multiply back.

Why the technique works

The engine is the chain rule applied to $\ln y$. If $y = y(x)$, then

$$\frac{\mathrm{d}}{\mathrm{d}x}(\ln y) = \frac{1}{y}\,\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{y'}{y}.$$

The quantity y'/y is the *logarithmic derivative*, also known as the *relative* or *fractional* rate of change – growth measured as a percentage of current size. So whenever you compute $\frac{d}{dx}\ln y$ you are really computing y'/y; multiply by y at the end to recover y' itself. Logarithmic differentiation is just this identity used in reverse: build y'/y out of easy log-pieces, then scale up by y.

Example 12.1 (A product of three factors)**.** Differentiate $y = \dfrac{(x^2+1)^3\sqrt{x-4}}{(x+2)^5}$. Doing this with the quotient and product rules directly would be a bookkeeping nightmare. Instead take logs:

$$\ln y = 3\ln(x^2+1) + \tfrac{1}{2}\ln(x-4) - 5\ln(x+2).$$

Differentiate term by term (each is an easy chain-rule piece):

$$\frac{y'}{y} = 3\cdot\frac{2x}{x^2+1} + \frac{1}{2}\cdot\frac{1}{x-4} - 5\cdot\frac{1}{x+2} = \frac{6x}{x^2+1} + \frac{1}{2(x-4)} - \frac{5}{x+2}.$$

Finally multiply back by y:

$$y' = \frac{(x^2+1)^3\sqrt{x-4}}{(x+2)^5}\left[\frac{6x}{x^2+1} + \frac{1}{2(x-4)} - \frac{5}{x+2}\right].$$

All the structure of the original function survives out front, with a tidy sum of corrections inside the brackets.

Example 12.2 (A variable base *and* a variable exponent)**.** Differentiate $y = x^x$ for $x > 0$. Notice neither the power rule ($\frac{\mathrm{d}}{\mathrm{d}x}(x^n) = nx^{n-1}$, exponent constant) nor the exponential rule ($\frac{\mathrm{d}}{\mathrm{d}x}(a^x) = a^x \ln a$, base constant) applies – here *both* the base and the exponent vary. Logs untangle it. Take ln:

$$\ln y = x\ln x.$$

Differentiate the right side with the product rule:

$$\frac{y'}{y} = 1\cdot\ln x + x\cdot\frac{1}{x} = \ln x + 1.$$

Multiply back: $y' = x^x(\ln x + 1)$. As a sanity check, x^x has its minimum where $y' = 0$, i.e. $\ln x = -1$, i.e. $x = 1/e \approx 0.368$ – which is indeed where the graph of x^x bottoms out (Figure 12.3).

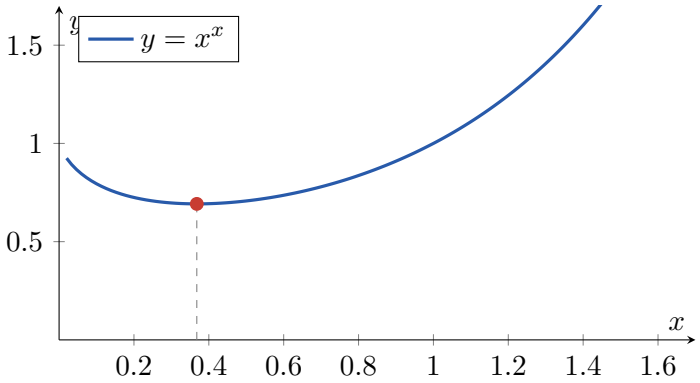


Figure 12.3: The graph of x^x for $x > 0$. Logarithmic differentiation gives $y' = x^x(\ln x + 1)$, which vanishes only at $x = 1/e \approx 0.368$, the marked low point. There $x^x = (1/e)^{1/e} \approx 0.692$.

Common trap

Do not write $\frac{\mathrm{d}}{\mathrm{d}x}(x^x) = x \cdot x^{x-1}$ (treating the exponent as constant), and do not write $\frac{\mathrm{d}}{\mathrm{d}x}(x^x) = x^x \ln x$ (treating the base as constant). *Both* are wrong because *both* the base and the exponent move. The correct derivative $x^x(\ln x + 1)$ contains a piece from each: the 1 comes from varying the base, the $\ln x$ from varying the exponent. When in doubt with anything of the form $(\text{stuff})^{(\text{stuff})}$, take logs.

12.5 Sine and Cosine: the Two Limits That Decide Everything

Now to the periodic functions. Their derivatives rest on two famous limits. Everything else is the angle-addition formula and bookkeeping. The two limits are

$$\lim_{\theta \to 0} \frac{\sin \theta}{\theta} = 1, \qquad \lim_{\theta \to 0} \frac{\cos \theta - 1}{\theta} = 0,$$

with θ measured in *radians*. We trust these from the limits chapter, but the unit-circle picture makes them believable, so look before we compute.

Intuition

On the unit circle, the *angle* θ (in radians) is the arc length swept out, and $\sin \theta$ is the vertical height of the endpoint. For a tiny angle, the arc and the vertical rise are nearly the same little segment, so $\sin \theta \approx \theta$ – their ratio heads to 1. Meanwhile $\cos \theta$ is the horizontal coordinate, which for a tiny angle is barely less than 1; it drops away only to *second* order ($\cos \theta \approx 1 - \theta^2/2$), so $(\cos \theta - 1)/\theta \approx -\theta/2 \to 0$. Height tracks angle to first order; width clings to 1.

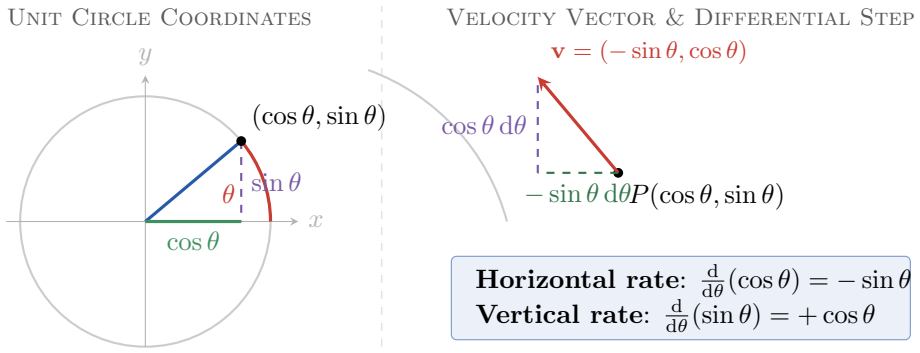


Figure 12.4: The trigonometric derivatives visualized on the unit circle. Left: The coordinates of a point at angle θ are $(\cos\theta, \sin\theta)$. Right: A counterclockwise nudge along the circle of arc length $d\theta$ moves along the tangent velocity vector $\mathbf{v} = (-\sin\theta, \cos\theta)$, perpendicular to the radius. Resolving this displacement into horizontal and vertical components reveals why $d(\cos\theta)/d\theta = -\sin\theta$ and $d(\sin\theta)/d\theta = \cos\theta$.

12.5.1 Differentiating sine

Now we run the difference quotient for $\sin x$ and use the angle-addition formula $\sin(x+h) = \sin x\cos h + \cos x \sin h$.

$$\begin{aligned}
\frac{d}{dx}(\sin x) &= \lim_{h\to 0} \frac{\sin(x+h) - \sin x}{h} \\
&= \lim_{h\to 0} \frac{\sin x\cos h + \cos x \sin h - \sin x}{h} \\
&= \lim_{h\to 0} \left[\sin x \cdot \frac{\cos h - 1}{h} + \cos x \cdot \frac{\sin h}{h}\right] \\
&= \sin x \cdot \underbrace{\lim_{h\to 0} \frac{\cos h - 1}{h}}_{=0} + \cos x \cdot \underbrace{\lim_{h\to 0} \frac{\sin h}{h}}_{=1} \\
&= \sin x \cdot 0 + \cos x \cdot 1 \;=\; \cos x.
\end{aligned}$$

Theorem 12.3 (Derivatives of sine and cosine)**.**

$$\frac{d}{dx}(\sin x) = \cos x, \qquad \frac{d}{dx}(\cos x) = -\sin x.$$

The cosine derivative falls out the same way, with $\cos(x+h) = \cos x\cos h - \sin x \sin h$. Splitting and taking limits,

$$\frac{d}{dx}(\cos x) = \cos x \cdot \left(\lim_{h\to 0} \tfrac{\cos h - 1}{h}\right) - \sin x \cdot \left(\lim_{h\to 0} \tfrac{\sin h}{h}\right) = \cos x \cdot 0 - \sin x \cdot 1 = -\sin x.$$

What is really happening

The two limits did all the work, and each played a clear role. The limit $\sin h/h \to 1$ is what makes the derivative have its full strength – it is the reason $\frac{\mathrm{d}}{\mathrm{d}x}(\sin x)$ is the *whole* cosine and not some shrunken multiple of it. That "1" is the trigonometric analogue of $m(e) = 1$ for the exponential: it is the payoff for measuring angles in radians. The other limit, $(\cos h - 1)/h \to 0$, says cosine is *flat* at its peak – it contributes nothing to first order, which is why no stray $\sin x$ term pollutes the answer. If you measured angles in degrees instead, $\sin h/h$ would tend to $\pi/180$, not 1, and every trig derivative would drag an ugly factor of $\pi/180$ forever. Radians are the natural unit for exactly the same reason e is the natural base: they make the leftover constant equal to 1.

Common trap

The radian warning is not academic. If your calculator is in degree mode, $\frac{\mathrm{d}}{\mathrm{d}x}(\sin x) \neq \cos x$ – you would need $\frac{\pi}{180}\cos x$. All of calculus assumes radians. A surprising number of wrong answers in physics and engineering trace back to a calculator left in degree mode.

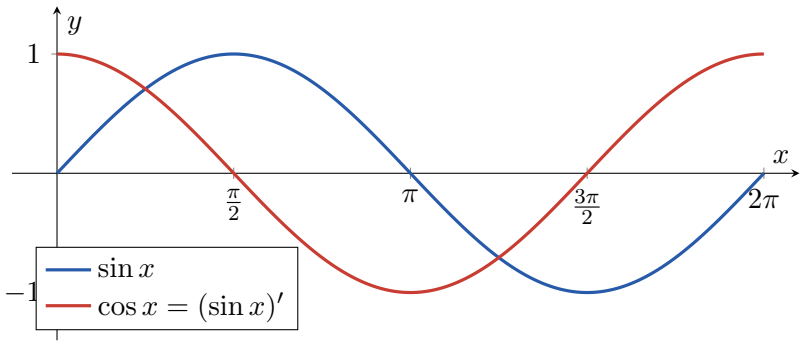


Figure 12.5: Where $\sin x$ rises most steeply (at $x = 0$, slope 1) its derivative $\cos x$ peaks at 1; where $\sin x$ flattens at its crest ($x = \pi/2$) the derivative crosses zero. The slope of sin *is* cos, point by point.

Why this matters

Differentiating sine and cosine sends you around a little cycle of length four:

$$\sin x \xrightarrow{d} \cos x \xrightarrow{d} -\sin x \xrightarrow{d} -\cos x \xrightarrow{d} \sin x.$$

Four derivatives bring you home. This is the seed of oscillation: anything satisfying $y'' = -y$ must be built from sines and cosines, and that equation governs every pendulum, spring, and wave in physics. Contrast the exponential, whose derivative cycle has length *one* ($e^x \to e^x$): exponentials grow, trig functions circle. Later, Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ will reveal these as the *same* phenomenon viewed along the real and imaginary axes – circling is just exponential growth in an imaginary direction.

12.5.2 The rest of the trig functions

With sin and cos in hand, the other four come from the quotient rule and the Pythagorean identity. Take the tangent:

$$\frac{\mathrm{d}}{\mathrm{d}x}(\tan x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\sin x}{\cos x}\right) = \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x.$$

The numerator collapsed to 1 because of $\sin^2 + \cos^2 = 1$ – a small miracle of cancellation. The same quotient-rule routine gives the rest.

Focus

$$\begin{aligned} \tfrac{\mathrm{d}}{\mathrm{d}x}(\sin x) &= \cos x, & \tfrac{\mathrm{d}}{\mathrm{d}x}(\cos x) &= -\sin x, \\ \tfrac{\mathrm{d}}{\mathrm{d}x}(\tan x) &= \sec^2 x, & \tfrac{\mathrm{d}}{\mathrm{d}x}(\cot x) &= -\csc^2 x, \\ \tfrac{\mathrm{d}}{\mathrm{d}x}(\sec x) &= \sec x \tan x, & \tfrac{\mathrm{d}}{\mathrm{d}x}(\csc x) &= -\csc x \cot x. \end{aligned}$$

Notice the pattern: every "co-" function $(\cos, \cot, \csc)$ carries a minus sign in its derivative. That bookkeeping minus is a reliable memory aid.

Example 12.3 (Using the chain rule with trig)**.** Differentiate $g(x) = \sin(3x^2+1)$ and $h(x) = \sec(5x)$.

$$g'(x) = \cos(3x^2+1) \cdot \frac{\mathrm{d}}{\mathrm{d}x}\big(3x^2+1\big) = 6x\cos(3x^2+1),$$

$$h'(x) = \sec(5x)\tan(5x) \cdot 5 = 5\sec(5x)\tan(5x).$$

The trig derivative supplies the "outer" factor; the chain rule supplies the derivative of the inside.

12.6 Derivatives of the Inverse Trig Functions

The inverse trig functions arcsin, arccos, arctan answer "which angle has this sine/cosine/tangent?" We differentiate them exactly as we did $\ln x$: implicitly, using the cancellation identity, plus one Pythagorean simplification at the end. The recurring surprise is that these derivatives are *algebraic* – no trig left in the answer at all.

12.6.1 Arcsine

Let $y = \arcsin x$, so $\sin y = x$ with $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. Differentiate $\sin y = x$ implicitly:

$$\cos y \cdot \frac{\mathrm{d}y}{\mathrm{d}x} = 1 \quad \Longrightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\cos y}.$$

Now convert $\cos y$ into something in terms of x. From $\sin^2 y + \cos^2 y = 1$ and $\sin y = x$, we get $\cos y = \sqrt{1-x^2}$. (We take the + root because on $[-\frac{\pi}{2}, \frac{\pi}{2}]$ cosine is nonnegative – the restricted range is doing real work here.) Therefore

$$\frac{\mathrm{d}}{\mathrm{d}x}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}, \qquad -1 < x < 1.$$

Why the technique works

Where did the square root come from? The implicit step gives a clean $1/\cos y$, but the answer should be a function of x, not of the hidden angle y. The right triangle with opposite side x and hypotenuse 1 has $\sin y = x$, so by Pythagoras its adjacent side is $\sqrt{1-x^2}$, and that adjacent-over-hypotenuse is $\cos y$. Drawing the triangle is the direct way to turn "$\cos(\arcsin x)$" into "$\sqrt{1-x^2}$." Every inverse-trig derivative comes out algebraic for this reason: a trig function of an inverse-trig function is always just a side ratio in a reference triangle.

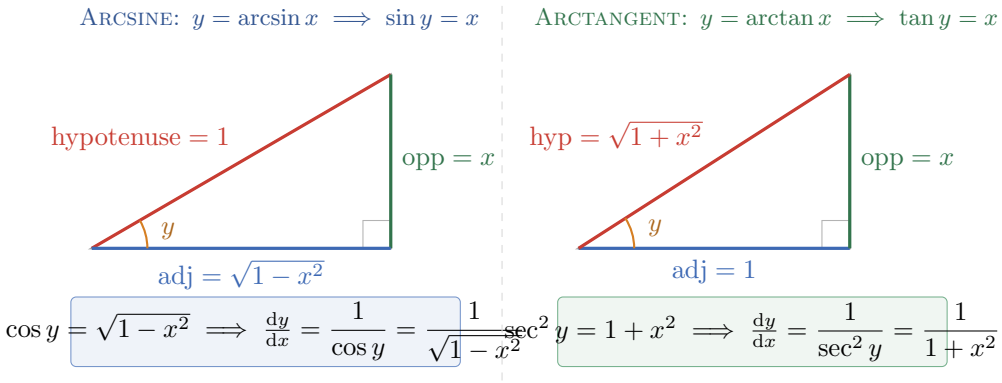


Figure 12.6: The reference triangles converting trig functions of inverse trig functions into pure algebra. Left: For $y = \arcsin x$, a right triangle with hypotenuse 1 and opposite side x forces the adjacent side to be $\sqrt{1-x^2}$ by the Pythagorean theorem, giving $\cos y = \sqrt{1-x^2}$. Right: For $y = \arctan x$, adjacent side 1 and opposite side x force the hypotenuse to be $\sqrt{1+x^2}$, giving $\sec^2 y = 1 + x^2$.

12.6.2 Arctangent

Let $y = \arctan x$, so $\tan y = x$. Differentiate:

$$\sec^2 y \cdot \frac{dy}{dx} = 1 \quad \Longrightarrow \quad \frac{dy}{dx} = \frac{1}{\sec^2 y}.$$

Use the identity $\sec^2 y = 1 + \tan^2 y = 1 + x^2$. So

$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}, \qquad x \in \mathbb{R}.$$

This one is a favorite because it is defined for all x, is beautifully simple, and – as you will see in the integration chapters – it is exactly the antiderivative needed to integrate $1/(1+x^2)$, the function whose area hides π.

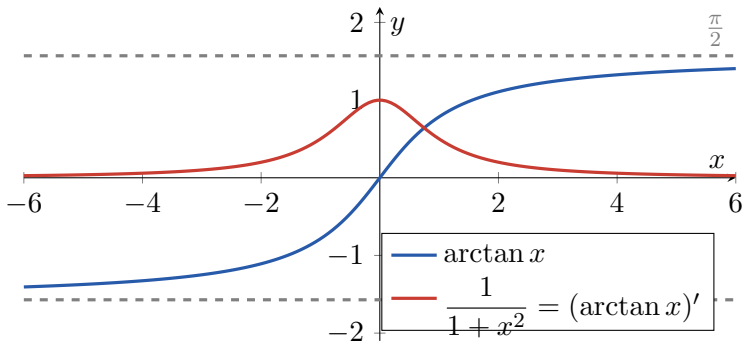


Figure 12.7: $\arctan x$ flattens toward the horizontal asymptotes $\pm\pi/2$, and exactly where it flattens its derivative $1/(1+x^2)$ dies away to 0. The derivative peaks at $x = 0$, where arctan is steepest.

Focus

The inverse-trig derivative table (each on its natural domain):

$$\frac{\mathrm{d}}{\mathrm{d}x}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}, \qquad \frac{\mathrm{d}}{\mathrm{d}x}(\arccos x) = -\frac{1}{\sqrt{1-x^2}},$$
$$\frac{\mathrm{d}}{\mathrm{d}x}(\arctan x) = \frac{1}{1+x^2}, \qquad \frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{arccot} x) = -\frac{1}{1+x^2},$$
$$\frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{arcsec} x) = \frac{1}{|x|\sqrt{x^2-1}}, \qquad \frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{arccsc} x) = -\frac{1}{|x|\sqrt{x^2-1}}.$$

Again the "co-" partners differ only by a sign. That sign comes from $\arccos x = \frac{\pi}{2} - \arcsin x$: complementary angles add to a constant, so their derivatives are negatives.

Connection

These derivatives matter most read *backwards*, as integration formulas. Differentiating the inverse trig functions is how we discover that

$$\int \frac{\mathrm{d}x}{\sqrt{1-x^2}} = \arcsin x + C, \qquad \int \frac{\mathrm{d}x}{1+x^2} = \arctan x + C.$$

Two perfectly innocent-looking algebraic integrands turn out to have *trigonometric* antiderivatives. That is the deep reason trigonometric substitution works for integrals containing $\sqrt{1-x^2}$ or $1+x^2$ – a topic we reach in the integration part. The angle is hiding inside the algebra.

12.7 A Gallery of Worked Examples

To consolidate, here are the new rules mixed with the product, quotient, and chain rules from the previous chapter.

Example 12.4 (Exponential times trig)**.** Differentiate $f(x) = e^{-2x}\cos(3x)$, the shape of a damped

oscillation. Product rule, with each factor needing the chain rule:

$$f'(x) = \underbrace{(-2)e^{-2x}}_{\frac{\mathrm{d}}{\mathrm{d}x}(e^{-2x})}\cos(3x) + e^{-2x}\underbrace{(-3\sin(3x))}_{\frac{\mathrm{d}}{\mathrm{d}x}(\cos(3x))} = -e^{-2x}\big(2\cos 3x + 3\sin 3x\big).$$

Example 12.5 (Log of a composite)**.** Differentiate $f(x) = \ln\big(\sqrt{x^2+1}\big)$. Simplify with log laws *before* differentiating: $\ln\sqrt{x^2+1} = \frac{1}{2}\ln(x^2+1)$. Then

$$f'(x) = \frac{1}{2}\cdot\frac{2x}{x^2+1} = \frac{x}{x^2+1}.$$

Always look for a log law to flatten the function first; it can turn a chain-rule slog into a one-liner.

Example 12.6 (Putting the base change to work)**.** Differentiate $f(x) = 5^{\sin x}$. Rewrite as $e^{(\ln 5)\sin x}$, or just apply $\frac{\mathrm{d}}{\mathrm{d}x}(a^u) = a^u \ln a \cdot u'$ directly with $a = 5$, $u = \sin x$:

$$f'(x) = 5^{\sin x}\,(\ln 5)\,\cos x.$$

Example 12.7 (Inverse trig inside a chain)**.** Differentiate $f(x) = \arctan(e^x)$. Outer derivative $1/(1+u^2)$ with $u = e^x$, times $u' = e^x$:

$$f'(x) = \frac{1}{1+(e^x)^2}\cdot e^x = \frac{e^x}{1+e^{2x}}.$$

Example 12.8 (Logarithmic differentiation with a root tower)**.** Differentiate $y = \dfrac{x^3\,e^{2x}}{\sqrt{1+x^2}}$. Take logs:

$$\ln y = 3\ln x + 2x - \tfrac{1}{2}\ln(1+x^2).$$

Differentiate:

$$\frac{y'}{y} = \frac{3}{x} + 2 - \frac{x}{1+x^2}.$$

Multiply back:

$$y' = \frac{x^3 e^{2x}}{\sqrt{1+x^2}}\left(\frac{3}{x} + 2 - \frac{x}{1+x^2}\right).$$

Walking question

Here is something to mull on a walk. We found three functions that are, in a sense, "eigenfunctions" of differentiation – the operation $\frac{\mathrm{d}}{\mathrm{d}x}()$ returns a simple multiple or relative of the input: $e^x \to e^x$ (returns itself), and the pair $\sin, \cos$ that rotate among themselves. Is there a function whose derivative is the function shifted, or scaled by x, in some equally clean way? What kind of equation would "$y' =$ (something simple involving y)" have to be for its solution to be elementary? You are circling the idea of a *differential equation*, the subject the whole book builds toward. Every function in this chapter is the answer to one: $y' = y$ gives e^x, $y'' = -y$ gives sin and cos, $y' = 1/x$ gives ln.

Application

These derivatives are the vocabulary of dynamics.

- **Radioactive decay / drug clearance.** If $N(t) = N_0 e^{-\lambda t}$, then $N'(t) = -\lambda N(t)$: the decay rate is proportional to how much is left. The derivative *is* the law of decay.
- **Exponential growth.** $\frac{\mathrm{d}}{\mathrm{d}t}(e^{kt}) = ke^{kt}$ is why unchecked populations, viral spread, and compound interest all share the same runaway curve.
- **Oscillation.** A mass on a spring obeys $x'' = -\omega^2 x$, solved by $x(t) = A\cos(\omega t) + B\sin(\omega t)$ precisely because of the derivative cycle of sine and cosine.
- **Machine learning.** The logistic "sigmoid" $\sigma(x) = 1/(1+e^{-x})$ has the slick derivative $\sigma'(x) = \sigma(x)\big(1-\sigma(x)\big)$ (try it!), which is why it is so cheap to train neural networks with – the gradient is built from values you already computed.

12.8 Summary

Every derivative in this chapter was forced by the *definition* of the function, not pulled from a hat.

- $\frac{\mathrm{d}}{\mathrm{d}x}(b^x) = (\ln b)\, b^x$, and $\ln b = \lim_{h\to 0}(b^h - 1)/h$ is the slope of b^x at the y-axis. The base e is defined as the one making that slope 1, which is exactly why $\frac{\mathrm{d}}{\mathrm{d}x}(e^x) = e^x$.
- $\frac{\mathrm{d}}{\mathrm{d}x}(\ln x) = 1/x$ and $\frac{\mathrm{d}}{\mathrm{d}x}(\log_a x) = 1/(x \ln a)$, by inverse reasoning or straight from the limit; $\ln x$ fills the gap the power rule leaves at exponent -1.
- Logarithmic differentiation turns products, quotients, and variable-exponent towers into sums via $y'/y = \frac{\mathrm{d}}{\mathrm{d}x}(\ln y)$.
- $\frac{\mathrm{d}}{\mathrm{d}x}(\sin x) = \cos x$ and $\frac{\mathrm{d}}{\mathrm{d}x}(\cos x) = -\sin x$, driven by $\sin h/h \to 1$ and $(\cos h - 1)/h \to 0$ from the unit circle – and radians are what make those limits clean.
- Inverse trig derivatives are *algebraic*: $(\arcsin x)' = 1/\sqrt{1-x^2}$, $(\arctan x)' = 1/(1+x^2)$, found by implicit differentiation plus a Pythagorean triangle.

The recurring theme: e and radians are the "natural" choices precisely because they make a leftover constant equal to 1. Choose them, and calculus stops nagging you with stray factors.

12.8.1 The Scope and Horizon of Transcendental Derivatives

Connection

With exponentials, logarithms, and trigonometric functions brought into differential calculus, applied mathematics exploded in scope:

- *Natural growth and decay*: Any process where rate is proportional to size ($\mathrm{d}y/\mathrm{d}t = ky$) is solved in closed form by $y(t) = y_0 e^{kt}$, modeling radioactive decay, compound capital, and population dynamics.
- *Oscillation and wave mechanics*: The derivative cycle $(\sin x)' = \cos x$ and $(\cos x)' = -\sin x$ explains why simple harmonic motion satisfies the second-order acceleration equa-

tion $\mathrm{d}^2y/\mathrm{d}t^2 = -\omega^2 y$.

- *Algebraic antiderivatives from geometry*: The remarkable discovery that inverse trigonometric derivatives are purely algebraic ($(\arctan x)' = 1/(1+x^2)$) provides the vital bridge allowing us to integrate rational expressions that would otherwise be insoluble.

Yet transcendental functions break purely algebraic solvability. In polynomial calculus, finding where a derivative vanishes or locating where two curves intersect requires solving polynomial equations, for which algebraic methods (factoring, the quadratic formula, Cardano's formula) exist. But the moment exponentials or trig functions enter, optimization and root-finding produce *transcendental equations*:

$$x\,e^x = 1, \qquad x = \cos x, \qquad e^{-x^2} = x.$$

None of these equations can be solved for x using a finite combination of elementary algebraic operations. We can write down the exact derivative formula in seconds, yet algebra cannot tell us the numerical coordinate where that rate equals zero.

Because algebra hits a firm wall when solving transcendental equations, calculus must build algorithmic tools. In Chapter 19 (*Newton's Method and Numerical Root-Finding*), we transform the derivative itself into an iterative engine that converges quadratically to the roots of transcendental equations, producing twenty decimal digits of accuracy in a handful of mechanical steps.

12.9 Exercises

Exercise 12.1 (Warm-up derivatives)**.** Differentiate each function.

(a) $f(x) = 7e^x - 4\ln x$

(b) $g(x) = e^{3x}\sin x$

(c) $h(x) = \dfrac{\cos x}{x^2}$

(d) $p(x) = \tan(2x) + \sec(2x)$

(e) $q(x) = 3^x + x^3$ (mind which rule each term needs)

(f) $r(x) = \ln(\cos x)$

Exercise 12.2 (Chain-rule combinations)**.** Differentiate.

(a) $\sin(e^x)$

(b) $e^{\sin x}$

(c) $\arctan(3x)$

(d) $\arcsin(x^2)$

(e) $\ln\big(\ln x\big)$

(f) $\cos^2(5x)$

Exercise 12.3 (Logarithmic differentiation)**.** Use logarithmic differentiation.

(a) $y = (\sin x)^x$ for $0 < x < \pi$.

(b) $y = x^{\ln x}$.

(c) $y = \dfrac{(x+1)^4(x-2)^3}{(x^2+5)^2}$.

Exercise 12.4 (Find the base)**.** Numerically estimate $m(4) = \lim_{h\to 0}(4^h - 1)/h$ with $h = 0.0001$, then confirm it equals $\ln 4$ to three decimals. Explain in one sentence why the estimate and $\ln 4$ should agree.

Exercise 12.5 (The sigmoid identity)**.** Let $\sigma(x) = \dfrac{1}{1+e^{-x}}$. Show that $\sigma'(x) = \sigma(x)\big(1 - \sigma(x)\big)$. (Hint: write $\sigma = (1+e^{-x})^{-1}$ and use the chain rule, then refactor.)

Exercise 12.6 (Tangent line)**.** Find the equation of the tangent line to $y = \ln x$ at $x = e$. Where does this tangent line cross the x-axis?

Exercise 12.7 (A radian sanity check)**.** Suppose a student insists on degrees and defines $S(x) = \sin(x^\circ)$, i.e. the sine of x degrees. Using $x^\circ = \frac{\pi}{180}x$ radians and the chain rule, show that $S'(x) = \frac{\pi}{180}\cos(x^\circ)$. Explain why this confirms that radians are the natural unit for calculus.

Exercise 12.8 (Conceptual: why proportional growth is exponential)**.** Suppose a function $y(t)$ satisfies $y'(t) = k\,y(t)$ for a constant k, and $y(0) = y_0$. Verify by differentiation that $y(t) = y_0 e^{kt}$ works. Then argue informally why *no* function of the form $y_0 b^t$ with $b \neq e^k$ can satisfy the equation. (This is the precise sense in which "rate proportional to size" means "exponential.")

Exercise 12.9 (Proof: derivative of a^x from scratch)**.** Without quoting $\frac{\mathrm{d}}{\mathrm{d}x}(a^x) = a^x \ln a$, prove it. Start from the difference quotient, factor out a^x, and identify the leftover limit $m(a)$. Then, using $a = e^{\ln a}$ and $\frac{\mathrm{d}}{\mathrm{d}x}(e^x) = e^x$ together with the chain rule, show $m(a) = \ln a$. Comment on how the two methods give the same constant.

Exercise 12.10 (Proof: arccosine and the sign)**.** Using $\arccos x = \frac{\pi}{2} - \arcsin x$, prove that $\frac{\mathrm{d}}{\mathrm{d}x}(\arccos x) = -\dfrac{1}{\sqrt{1-x^2}}$. Then re-derive the same result *independently* by implicit differentiation of $\cos y = x$, being careful about the sign of $\sin y$ on the range $y \in [0, \pi]$. Confirm the two agree.

Exercise 12.11 (Higher derivatives and the cycle)**.** Compute the first four derivatives of $f(x) = \sin x$ and confirm the period-four cycle. Then find a simple formula for the n-th derivative $f^{(n)}(x)$ in terms of n mod 4. What is $f^{(101)}(x)$?

Exercise 12.12 (Mixed challenge)**.** Differentiate $y = x^{\sin x}$ and find all x in $(0, \pi)$ where the relative rate of change y'/y equals zero. Interpret these points on the graph of $\ln y = (\sin x)(\ln x)$.

Chapter 13

Implicit Differentiation and Related Rates

So far almost every function we have differentiated arrived in a friendly form: a rule that takes an input x, does something to it, and hands back a single output $y = f(x)$. You feed in a number, you turn the crank, out comes the answer. The derivative measures how the crank's output responds when you nudge the input.

Many curves are not so accommodating. Some of the most important shapes in geometry and physics refuse to be written as $y = f(x)$. A circle is the cleanest example: the equation

$$x^2 + y^2 = 25$$

describes a perfectly definite curve, and that curve has tangent lines everywhere, with well-defined slopes. Yet there is no single function f with $y = f(x)$ whose graph is the whole circle, because most vertical lines hit the circle *twice*. The circle is a relationship between x and y, not a machine that turns x into y.

This chapter is about two closely linked ideas. The first is *implicit differentiation*: a way to find slopes on curves that are tied together by an equation, without ever solving for y. We will see that it is the chain rule operating directly on an implicit relation. The second is *related rates*: a method for problems where two or more changing quantities are locked together by a constraint, so that the rate of one drives the rate of another. A ladder slides, a balloon inflates, a shadow lengthens; in each case a single equation ties the pieces together, and differentiating that equation in time tells us how the rates relate. Both ideas spring from the same root: *when quantities are bound by an equation, their rates of change are bound too.*

13.1 Curves That Are Not Functions

Let us look hard at the circle $x^2 + y^2 = 25$. Pick a point on it, say $(3, 4)$. Geometrically the circle has an unambiguous tangent line there. We can even guess its slope: a radius runs from the center $(0, 0)$ out to $(3, 4)$ with slope $4/3$, and the tangent is perpendicular to the radius, so the tangent slope should be $-3/4$. The geometry leaves no doubt that a slope exists. The only question is how to *compute* it from the equation.

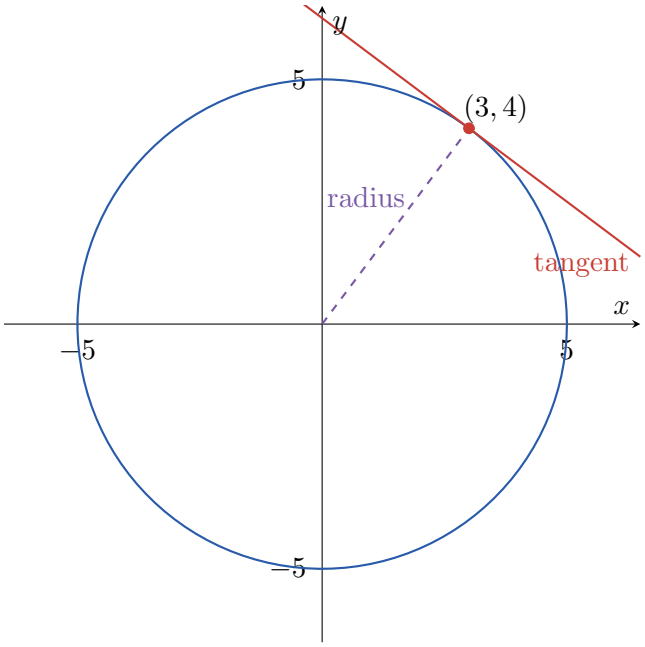


Figure 13.1: The circle $x^2 + y^2 = 25$ is not the graph of any single function of x, yet it has a definite tangent line at each point. At $(3, 4)$ the radius has slope $4/3$, so the perpendicular tangent has slope $-3/4$.

One option is brute force: solve for y. Above the x-axis the upper half of the circle is the genuine function $y = \sqrt{25 - x^2}$, and we could differentiate that. It works here, but it is already a little ugly, and for most equations it is hopeless. Try to solve

$$x^3 + y^3 = 6xy$$

for y – this is the famous *folium of Descartes* – and you will be staring at the cubic formula, producing a monstrous expression with three branches. Nobody wants to differentiate that. We need a method that respects the equation as it stands.

Intuition

The key mental shift: stop thinking of y as "the output of a formula" and start thinking of y as "whatever value (or values) the equation forces, given x." On the upper half of the circle, as x creeps from 3 toward 4, the equation *forces* y to creep downward from 4 toward 3. That forced motion of y is exactly what the derivative $\frac{dy}{dx}$ measures. The equation is the boss; y obeys.

The phrase mathematicians use is that the equation defines y *implicitly* as a function of x. "Implicit" means *implied but not spelled out*: the dependence of y on x is real and present, hiding inside the equation, even though no explicit formula $y = f(x)$ is written down. Our goal is to extract the slope of that hidden function directly.

Focus

Explicit vs. implicit.

- *Explicit:* $y = \sqrt{25 - x^2}$. The output is isolated; differentiate with the rules you already

know.

- *Implicit:* $x^2 + y^2 = 25$. The output is entangled with the input. We will differentiate *both sides of the equation*, treating y as an unknown function of x.

The method consists in differentiating both sides directly.

13.2 Implicit Differentiation: the Chain Rule on Level Curves

Here is the underlying logic. We have an equation in x and y that holds along a curve. Along that curve y is some function of x — call it $y(x)$ even if we cannot write its formula. Then *both sides of the equation are functions of* x, and they are equal for every x on the curve. If two functions of x are equal everywhere, their derivatives are equal too. So we may apply $\dfrac{\mathrm{d}}{\mathrm{d}x}$ to both sides and keep a true equation. The key step is that wherever y appears, it is $y(x)$, so differentiating it requires the chain rule.

Let us do the circle.

Example 13.1 (The circle, by implicit differentiation)**.** Differentiate both sides of $x^2 + y^2 = 25$ with respect to x:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2\right) + \frac{\mathrm{d}}{\mathrm{d}x}\left(y^2\right) = \frac{\mathrm{d}}{\mathrm{d}x}(25).$$

The first term is $2x$. The right side is 0, since 25 is constant. The middle term is the delicate one. Here y is a function of x, so y^2 is a *composition*: square-of, applied to $y(x)$. The chain rule says differentiate the outside ($u^2 \mapsto 2u$) and multiply by the derivative of the inside (y with respect to x):

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(y^2\right) = 2y \cdot \frac{\mathrm{d}y}{\mathrm{d}x}.$$

Putting it together,

$$2x + 2y\,\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \quad \Longrightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{x}{y}.$$

Now read off the slope at $(3, 4)$: $\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{3}{4}$. Exactly the geometric prediction. And the answer at $(-3, -4)$? Also $-\frac{3}{4}$ – the formula handles every branch of the curve at once, with no case-splitting.

Notice what just happened. We never solved for y. The derivative came out in terms of *both* x and y, which is perfectly fine: to use it you supply a point (x, y) that lies on the curve, and out comes the slope there. An implicit derivative is a slope *field along the curve*, expressed using the coordinates of the point.

Why the technique works

Why is differentiating both sides legitimate, and why does $\frac{\mathrm{d}y}{\mathrm{d}x}$ pop out? Two facts do all the work.

First, an equation like $x^2 + y^2 = 25$ asserts that two functions of x – the left side and the right side, once we substitute $y = y(x)$ – are equal at every point of the curve. Equal functions have equal derivatives. (This is the same principle that lets you differentiate both sides of

$\sin^2 x + \cos^2 x = 1$.) So $\dfrac{\mathrm{d}}{\mathrm{d}x}(\text{LHS}) = \dfrac{\mathrm{d}}{\mathrm{d}x}(\text{RHS})$ is a valid new equation.
Second, every term containing y is really a composition of some operation with the inner function $y(x)$. The chain rule is precisely the tool for differentiating compositions. So each y-term contributes its outer derivative times the inner derivative $\frac{\mathrm{d}y}{\mathrm{d}x}$. Implicit differentiation is nothing more than "differentiate the equation, and remember the chain rule whenever you touch y." That is why $\frac{\mathrm{d}y}{\mathrm{d}x}$ always appears *linearly* (to the first power, never squared), which is what lets us solve for it by simple algebra at the end.

What is really happening

When you write $\dfrac{\mathrm{d}}{\mathrm{d}x}(y^2) = 2y\,\frac{\mathrm{d}y}{\mathrm{d}x}$, what is really happening is a two-step relay of sensitivities. Nudge x by a tiny $\mathrm{d}x$. The curve forces y to move by $\mathrm{d}y = \frac{\mathrm{d}y}{\mathrm{d}x}\,\mathrm{d}x$. That motion of y then changes y^2 by (rate of y^2 with respect to y) times $\mathrm{d}y$, i.e. $2y\,\mathrm{d}y$. Chaining the two responses gives $2y\,\frac{\mathrm{d}y}{\mathrm{d}x}\,\mathrm{d}x$. The factor $\frac{\mathrm{d}y}{\mathrm{d}x}$ is the "exchange rate" converting a change in x into the change in y that the constraint demands. Every appearance of $\frac{\mathrm{d}y}{\mathrm{d}x}$ in implicit differentiation is one of these exchange-rate factors, inserted exactly where the chain of dependence passes through y.

13.2.1 A clean recipe

Focus

How to differentiate implicitly.

1. Apply $\dfrac{\mathrm{d}}{\mathrm{d}x}$ to *both* sides of the equation.
2. Differentiate terms in x normally. For any term involving y, use the chain rule: after differentiating with respect to y, multiply by $\frac{\mathrm{d}y}{\mathrm{d}x}$. (Products like xy also need the product rule.)
3. Collect every term containing $\frac{\mathrm{d}y}{\mathrm{d}x}$ on one side, everything else on the other.
4. Factor out $\frac{\mathrm{d}y}{\mathrm{d}x}$ and divide. The answer will usually involve both x and y.

Common trap

The single most common mistake is forgetting the $\frac{\mathrm{d}y}{\mathrm{d}x}$ factor and writing $\dfrac{\mathrm{d}}{\mathrm{d}x}(y^2) = 2y$. That is wrong: $2y$ is the derivative of y^2 *with respect to* y, not with respect to x. The whole point is that y depends on x, so the chain rule supplies the extra factor $\frac{\mathrm{d}y}{\mathrm{d}x}$. A good habit: every time your pen leaves a y-term, immediately tack on $\frac{\mathrm{d}y}{\mathrm{d}x}$ before you forget. By contrast, $\dfrac{\mathrm{d}}{\mathrm{d}x}(x^2) = 2x$ needs *no* extra factor, because $\frac{\mathrm{d}x}{\mathrm{d}x} = 1$.

Example 13.2 (A term that needs the product rule)**.** Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ for $x^2y + y^3 = 10$.

Differentiate term by term. The term x^2y is a product of x^2 and y, so use the product rule, and

remember the chain-rule factor on y:

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(x^2 y\big) = 2x \cdot y + x^2 \cdot \frac{\mathrm{d}y}{\mathrm{d}x}.$$

For y^3 the chain rule gives $3y^2 \frac{\mathrm{d}y}{\mathrm{d}x}$, and the right side is 0. So

$$2xy + x^2 \frac{\mathrm{d}y}{\mathrm{d}x} + 3y^2 \frac{\mathrm{d}y}{\mathrm{d}x} = 0.$$

Collect the $\frac{\mathrm{d}y}{\mathrm{d}x}$ terms:

$$\big(x^2 + 3y^2\big) \frac{\mathrm{d}y}{\mathrm{d}x} = -2xy \quad \Longrightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{-2xy}{x^2 + 3y^2}.$$

Work it yourself

Try one with your own pen before reading on: find $\frac{\mathrm{d}y}{\mathrm{d}x}$ for $\sin(xy) = x$. You will need the chain rule on sin, then the product rule on xy. (Solution: differentiating, $\cos(xy) \cdot \big(y + x \frac{\mathrm{d}y}{\mathrm{d}x}\big) = 1$, so $\frac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{1 - y\cos(xy)}{x\cos(xy)}$.) The structure is always the same: $\frac{\mathrm{d}y}{\mathrm{d}x}$ shows up linearly, and you solve for it.

13.2.2 The folium of Descartes

Now the curve we could not solve. The folium $x^3 + y^3 = 6xy$ is a looping cubic that crosses itself at the origin – a shape that genuinely has no single-formula description, yet implicit differentiation handles it without breaking a sweat.

Example 13.3 (Slope on the folium)**.** Differentiate $x^3 + y^3 = 6xy$ with respect to x. The right side $6xy$ is a product, so it needs the product rule (and the chain factor on y):

$$3x^2 + 3y^2 \frac{\mathrm{d}y}{\mathrm{d}x} = 6\left(y + x \frac{\mathrm{d}y}{\mathrm{d}x}\right).$$

Expand and gather $\frac{\mathrm{d}y}{\mathrm{d}x}$ terms:

$$3y^2 \frac{\mathrm{d}y}{\mathrm{d}x} - 6x \frac{\mathrm{d}y}{\mathrm{d}x} = 6y - 3x^2, \qquad \big(3y^2 - 6x\big) \frac{\mathrm{d}y}{\mathrm{d}x} = 6y - 3x^2.$$

Divide:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{6y - 3x^2}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x}.$$

The point $(3, 3)$ lies on the curve (check: $27 + 27 = 54 = 6 \cdot 9$). There the slope is

$$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{(3,3)} = \frac{6 - 9}{9 - 6} = \frac{-3}{3} = -1.$$

The tangent there is the line $y - 3 = -1(x - 3)$, i.e. $y = -x + 6$.

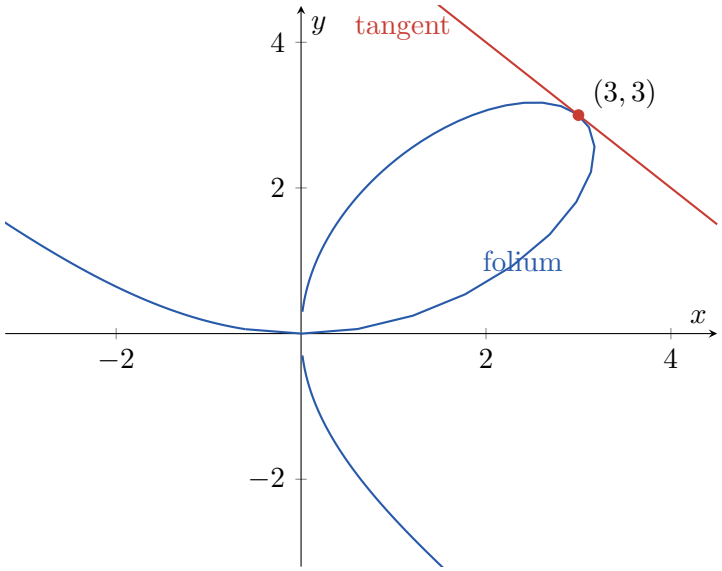


Figure 13.2: The folium of Descartes $x^3 + y^3 = 6xy$. There is no tidy $y = f(x)$ for this looping curve, but implicit differentiation gives the tangent slope at $(3, 3)$ as -1.

History

The folium of Descartes was born in 1638 out of an intellectual duel between two titans of seventeenth-century mathematics: René Descartes and Pierre de Fermat. In the 1630s, years before Newton or Leibniz were born, Fermat had devised an ingenious method for finding tangents and extrema on algebraic curves known as *adequality* (from Diophantus's *adaequare*). Fermat's technique was a conceptual forerunner to the differential: he substituted $x + e$ into the polynomial equation of a curve, treated the two sides as "approximately equal" (adequal), cancelled common terms, divided through by e, and then suppressed all remaining terms in e to read off the tangent subtangent.

Descartes, who championed his own cumbersome method of finding tangents by constructing tangent circles that intersect the curve in two coincident points, was deeply skeptical of Fermat's method. In 1638, Descartes proposed the looping curve $x^3 + y^3 = 3axy$ (now called the *folium of Descartes*) as an explicit challenge to Fermat, taunting him to find its tangent line. Descartes believed the looped geometry and the impossibility of isolating y would break Fermat's method. To Descartes' chagrin, Fermat solved it promptly and correctly.

Around the same era, the physical need for related rates arose directly from mechanics and timekeeping. Christiaan Huygens, in his celebrated work on pendulum clocks (*Horologium Oscillatorium*, 1673), sought to build marine clocks accurate enough to determine longitude at sea. Huygens discovered that a standard circular pendulum's period varies with its swing amplitude. To make the period truly constant (tautochrone), Huygens had to relate the instantaneous vertical descent rate $\frac{\mathrm{d}y}{\mathrm{d}t}$ to the horizontal rate $\frac{\mathrm{d}x}{\mathrm{d}t}$ along curved cycloidal jaws, solving an intricate related-rates problem before differential notation even existed. When Leibniz introduced his differential notation $\mathrm{d}x$ and $\mathrm{d}y$ in the 1680s, the chain rule turned these hard-won geometric geometric tangencies into fluent, three-line algebraic operations.

13.3 Tangent Lines and Higher Derivatives

Once you can find $\frac{dy}{dx}$ at a point on an implicit curve, tangent lines cost nothing extra: the tangent at (x_0, y_0) is

$$y - y_0 = m\,(x - x_0), \qquad m = \left.\frac{dy}{dx}\right|_{(x_0,y_0)}.$$

The only thing to watch is that the point must actually lie on the curve, and that you substitute *both* coordinates into the slope formula.

Example 13.4 (Tangent and a vertical tangent on an ellipse)**.** Consider the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$. Differentiate implicitly:

$$\frac{2x}{9} + \frac{2y}{4}\frac{dy}{dx} = 0 \quad\Longrightarrow\quad \frac{dy}{dx} = -\frac{4x}{9y}.$$

At the point $\left(\frac{3}{\sqrt{2}}, \sqrt{2}\right)$ (which lies on the ellipse: $\frac{9/2}{9} + \frac{2}{4} = \frac{1}{2} + \frac{1}{2} = 1$), the slope is

$$m = -\frac{4 \cdot \frac{3}{\sqrt{2}}}{9 \cdot \sqrt{2}} = -\frac{12/\sqrt{2}}{9\sqrt{2}} = -\frac{12}{9 \cdot 2} = -\frac{2}{3}.$$

Notice also *where the tangent is vertical*: the slope formula $-\frac{4x}{9y}$ blows up when $y = 0$, i.e. at $(\pm 3, 0)$, the ends of the major axis. There the tangent line is vertical – a slope our $\frac{dy}{dx}$ correctly reports as "undefined." Implicit differentiation does not hide these features; it flags them (Figure 13.3).

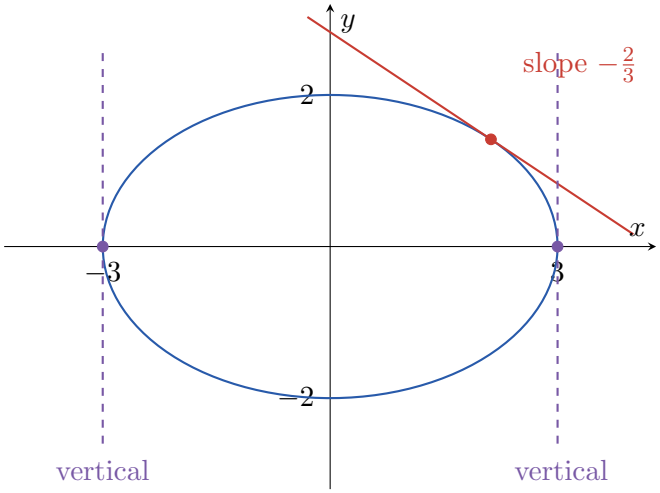


Figure 13.3: The ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$. At a typical point the implicit slope $-\frac{4x}{9y}$ gives an ordinary slanted tangent (here $-\frac{2}{3}$). At the ends of the major axis $(\pm 3, 0)$, where $y = 0$, the formula blows up and the tangent is vertical – exactly the feature that "solve for y" handles badly.

Connection

Vertical tangents are where the implicit picture genuinely beats "solve for y." If you split the ellipse into $y = \pm\frac{2}{3}\sqrt{9 - x^2}$, the derivative of each branch becomes infinite and ill-behaved

exactly at $x = \pm 3$, the seam where the two branches meet. The implicit form $\frac{dy}{dx} = -\frac{4x}{9y}$ treats the whole curve as one object and simply says: slope undefined when $y = 0$. Later, when we study curves in the plane more generally, this is the seed of the idea that a smooth curve can have a well-defined *tangent direction* even where it has no well-defined *function slope*.

13.3.1 Second derivatives implicitly

You can differentiate a second time to get $\frac{d^2y}{dx^2}$, useful for concavity. The method is the same – differentiate again, remembering the chain rule – but now you also substitute the expression for $\frac{dy}{dx}$ you already found.

Example 13.5 (Second derivative on the circle)**.** For $x^2 + y^2 = 25$ we found $\frac{dy}{dx} = -\frac{x}{y}$. Differentiate this quotient with respect to x, using the quotient rule and remembering that y depends on x:

$$\frac{d^2y}{dx^2} = -\frac{\frac{d}{dx}(x)\cdot y - x\cdot \frac{d}{dx}(y)}{y^2} = -\frac{y - x\,\frac{dy}{dx}}{y^2}.$$

Now substitute $\frac{dy}{dx} = -\frac{x}{y}$:

$$\frac{d^2y}{dx^2} = -\frac{y - x\left(-\frac{x}{y}\right)}{y^2} = -\frac{y + \frac{x^2}{y}}{y^2} = -\frac{y^2 + x^2}{y^3}.$$

But on the circle $x^2 + y^2 = 25$, so this simplifies beautifully to

$$\frac{d^2y}{dx^2} = -\frac{25}{y^3}.$$

On the upper half ($y > 0$) this is negative – the curve is concave down, as a dome should be. On the lower half ($y < 0$) it is positive, concave up. The sign of the second derivative reads off the shape of each half exactly as it should.

Walking question

Take this on a walk. The relation $x^2 + y^2 = 25$ is symmetric in x and y – swapping them leaves the circle unchanged. We found $\frac{dy}{dx} = -\frac{x}{y}$. What should $\frac{dx}{dy}$ be, and how is it related? Convince yourself geometrically that $\frac{dx}{dy} = -\frac{y}{x}$ and that $\frac{dy}{dx}\cdot\frac{dx}{dy} = 1$, then explain in words why "run over rise" is the reciprocal of "rise over run." This is the first whisper of the inverse-function relationship between derivatives.

13.4 Related Rates: When One Rate Drives Another

We now turn the same idea toward *time*. In a related-rates problem, several quantities change as time passes, and they are tied together by a geometric or physical relationship. Because the quantities

are bound by an equation, their *rates of change* are bound too – and if we know some of the rates we can solve for the others. The mathematics is implicit differentiation again, but now the hidden variable everything depends on is t, time.

Intuition

Picture a constraint as a length of inextensible string connecting two beads. If you know how fast one bead moves, the string *forces* a definite speed on the other; you do not get to choose it. A related-rates problem hands you the constraint (the string) and one or more known rates, and asks for an unknown rate. The constraint, differentiated in time, is the rule that converts the rates you know into the rate you want.

The crucial structural fact: every quantity in the problem is a function of time. So when we differentiate the constraint equation with respect to t, *every variable* gets a chain-rule factor – its own rate $\dfrac{\mathrm{d}(\cdot)}{\mathrm{d}t}$. A length x contributes $\frac{\mathrm{d}x}{\mathrm{d}t}$, an area A contributes $\frac{\mathrm{d}A}{\mathrm{d}t}$, an angle θ contributes $\frac{\mathrm{d}\theta}{\mathrm{d}t}$. There is no privileged "independent variable" among x, y, θ; they are all dependent on the one true independent variable, t.

What is really happening

Here is the real mechanism, stripped bare. A constraint $F(x, y) = c$ holds at every instant. As time ticks, the point $(x(t), y(t))$ slides along the curve $F = c$; it is not free to wander off. The condition for staying on the curve is that the *rate* at which F changes must be zero – $\frac{\mathrm{d}F}{\mathrm{d}t} = 0$ – because F is pinned at the constant c. Writing out $\frac{\mathrm{d}F}{\mathrm{d}t} = 0$ by the chain rule produces a single linear equation relating $\frac{\mathrm{d}x}{\mathrm{d}t}$ and $\frac{\mathrm{d}y}{\mathrm{d}t}$. That one equation *is* the related-rates relationship. "One rate drives another" is the statement that this equation has only one degree of freedom: fix one rate and the other is determined. Related rates is the time-derivative of a constraint, and nothing more.

13.4.1 A modeling method that always works

Related-rates problems feel hard because they come wrapped in a story. The cure is a fixed procedure that turns the story into algebra in the same way every time.

Focus

The five-step method.

1. **Draw and name.** Sketch the situation. Assign a *variable* to every quantity that changes. Use a letter, not a number, even for quantities you happen to know right now – their numerical value is only true at one instant, and plugging it in too early destroys the relationship.

2. **List knowns and the unknown rate.** Translate every "how fast" into a derivative with respect to t. Watch the signs: a shrinking quantity has a negative rate.

3. **Find the constraint.** Write a single equation connecting your variables, valid at *all* times – usually a geometric fact (Pythagoras, similar triangles, area or volume formula, trig ratio).

4. **Differentiate with respect to t.** Every variable earns its own $\dfrac{\mathrm{d}}{\mathrm{d}t}$ factor (the chain rule, as always). This gives an equation relating the rates.
5. **Substitute and solve.** *Now*, and only now, plug in the instantaneous numbers and solve for the unknown rate. Interpret the sign and units.

Common trap

Do not substitute numbers before differentiating. If you replace a changing length by its current value *before* taking $\dfrac{\mathrm{d}}{\mathrm{d}t}$, you have turned a moving quantity into a constant, and its derivative becomes a wrong 0. The numerical values describe a single frozen instant; the constraint equation and its differentiation describe *all* instants. Differentiate first (in symbols), substitute last (with numbers). This one discipline prevents the majority of related-rates errors.

13.4.2 The sliding ladder

Example 13.6 (The ladder against the wall)**.** A 10-foot ladder leans against a vertical wall. The bottom is pulled away from the wall at $2\,\mathrm{ft/s}$. How fast is the top sliding down the wall when the bottom is $6\,\mathrm{ft}$ from the wall?

Draw and name. Let x be the distance from the wall to the foot of the ladder, and y the height of the top on the wall. The ladder's length is the fixed constraint.

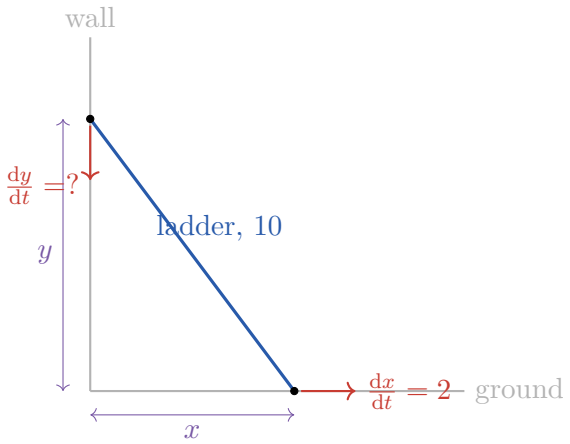


Figure 13.4: The sliding ladder. The fixed length 10 ties x and y together by $x^2 + y^2 = 100$; pulling the foot out forces the top down.

Knowns and unknown. $\frac{dx}{dt} = +2\,\mathrm{ft/s}$ (the foot moves away). We want $\frac{dy}{dt}$ at the instant $x = 6$.

Constraint. The ladder is rigid, so by the Pythagorean theorem, *at every instant*,

$$x^2 + y^2 = 100.$$

Differentiate in t. Each variable gets its rate:

$$2x\,\frac{\mathrm{d}x}{\mathrm{d}t} + 2y\,\frac{\mathrm{d}y}{\mathrm{d}t} = 0 \quad\Longrightarrow\quad \frac{\mathrm{d}y}{\mathrm{d}t} = -\frac{x}{y}\frac{\mathrm{d}x}{\mathrm{d}t}.$$

Substitute. When $x = 6$, the constraint gives $y = \sqrt{100 - 36} = 8$. So

$$\frac{\mathrm{d}y}{\mathrm{d}t} = -\frac{6}{8}\cdot 2 = -\frac{3}{2} = -1.5 \text{ ft/s}.$$

The top slides *down* at 1.5 ft/s. The negative sign is the model telling us the height is decreasing, exactly as it must.

Why this matters

There is a startling rider to the ladder problem. Keep the foot moving at a steady 2 ft/s and ask what happens as the ladder nears flat against the ground, $y \to 0$. Then $\frac{\mathrm{d}y}{\mathrm{d}t} = -\frac{x}{y}\cdot 2$ blows up to $-\infty$: the top appears to slam down infinitely fast. Of course no real ladder does this – near the end the top lifts off the wall and the model's assumptions fail. But the mathematics announces its own breakdown: the constraint $x^2 + y^2 = 100$ forces an infinite rate, which is the model's way of shouting that something physical (the ladder leaving the wall) has intervened. Good models tell you when to stop trusting them.

13.4.3 The inflating balloon

Example 13.7 (The spherical balloon)**.** Air is pumped into a spherical balloon at $100\,\mathrm{cm}^3/\mathrm{s}$. How fast is the radius growing when the radius is 5 cm? How fast is it growing when the radius is 10 cm?

Name. Let V be the volume and r the radius, both functions of t.

Knowns and unknown. $\frac{\mathrm{d}V}{\mathrm{d}t} = 100\,\mathrm{cm}^3/\mathrm{s}$. Want $\frac{\mathrm{d}r}{\mathrm{d}t}$.

Constraint. The geometry of a sphere:

$$V = \frac{4}{3}\pi r^3.$$

Differentiate in t (chain rule on r^3):

$$\frac{\mathrm{d}V}{\mathrm{d}t} = 4\pi r^2\,\frac{\mathrm{d}r}{\mathrm{d}t} \quad\Longrightarrow\quad \frac{\mathrm{d}r}{\mathrm{d}t} = \frac{1}{4\pi r^2}\frac{\mathrm{d}V}{\mathrm{d}t}.$$

Substitute. At $r = 5$:

$$\frac{\mathrm{d}r}{\mathrm{d}t} = \frac{100}{4\pi(25)} = \frac{100}{100\pi} = \frac{1}{\pi} \approx 0.318 \text{ cm/s}.$$

At $r = 10$:

$$\frac{\mathrm{d}r}{\mathrm{d}t} = \frac{100}{4\pi(100)} = \frac{1}{4\pi} \approx 0.0796 \text{ cm/s}.$$

Intuition

The factor $4\pi r^2$ in $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$ is the exact *surface area* of the sphere. The air you pump in spreads itself over the skin of the balloon as a thin shell; a thicker shell over a bigger skin means the same volume of air buys less radius. That is precisely why the radius slows as the balloon grows: at $r = 10$ the surface is four times that at $r = 5$, so the same $100\,\text{cm}^3/\text{s}$ thickens it only a quarter as fast. The derivative $\frac{dV}{dr} = 4\pi r^2 =$ (surface area) is one of the most satisfying "of course!" facts in calculus.

13.4.4 The walking shadow

Example 13.8 (The lengthening shadow)**.** A street lamp is 15 ft tall. A person 6 ft tall walks away from the base of the lamp at 5 ft/s. (a) How fast is the tip of the shadow moving? (b) How fast is the shadow's length growing?

Draw and name. Let x be the person's distance from the lamp post, and s the length of the shadow beyond the person. The tip of the shadow is then at distance $x + s$ from the post.

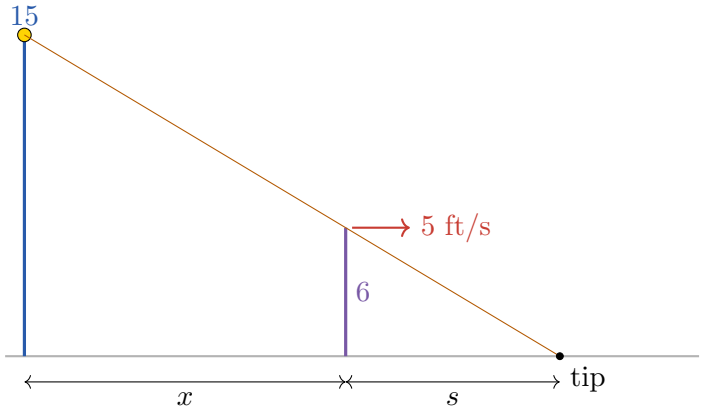


Figure 13.5: The shadow problem. Similar triangles – the big one from lamp top to shadow tip, the small one from the person's head to the tip – give the constraint.

Constraint. The lamp, the person, and the two shadow tips form *similar triangles*: the large triangle (height 15, base $x+s$) and the small triangle (height 6, base s) share the same shape. Equal ratios of height to base:

$$\frac{15}{x+s} = \frac{6}{s}.$$

Cross-multiply: $15s = 6(x+s) = 6x + 6s$, so $9s = 6x$, giving the clean linear relation

$$s = \frac{2}{3}x.$$

Differentiate in t.

$$\frac{ds}{dt} = \frac{2}{3}\frac{dx}{dt} = \frac{2}{3}(5) = \frac{10}{3}\ \text{ft/s}.$$

That answers (b): the shadow lengthens at $\frac{10}{3} \approx 3.33\,\text{ft/s}$.

For (a), the tip is at position $x + s$, so its speed is

$$\frac{\mathrm{d}}{\mathrm{d}t}(x+s) = \frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\mathrm{d}s}{\mathrm{d}t} = 5 + \frac{10}{3} = \frac{25}{3} \approx 8.33 \text{ ft/s}.$$

Common trap

Part (a) and part (b) ask different questions, and confusing them is the classic shadow trap. The *length of the shadow* (s) and the *position of its tip* ($x + s$) are not the same quantity; the tip moves because *both* the person and the shadow's far end advance. Always decide precisely which point or length the problem is asking about before differentiating. Notice too that neither rate depends on x – the answers are the same whether the person is near or far. That is special to the straight-line, point-source geometry here, and it is a nice surprise worth pausing on.

13.4.5 An angle that changes: the rising balloon

Example 13.9 (Tracking with a camera)**.** A weather balloon rises straight up at 8 m/s. A camera on the ground sits 50 m from the launch point and tilts to keep the balloon in frame. How fast is the camera's angle of elevation increasing when the balloon is 50 m high?

Name. Let h be the balloon's height and θ the camera's angle of elevation above the horizontal. The horizontal distance 50 m is constant.

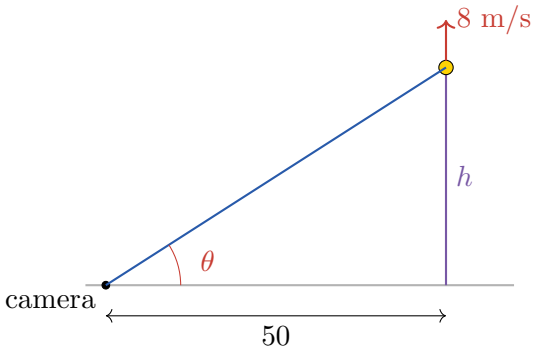


Figure 13.6: Tracking the rising balloon. The camera sits a fixed 50 m from the launch point; the line of sight makes angle θ with the ground, and $\tan\theta = h/50$. As h grows, θ opens.

Constraint. From the right triangle (Figure 13.6),

$$\tan\theta = \frac{h}{50}.$$

Differentiate in t. The left side needs the chain rule on tan:

$$\sec^2\theta\,\frac{\mathrm{d}\theta}{\mathrm{d}t} = \frac{1}{50}\frac{\mathrm{d}h}{\mathrm{d}t}.$$

Substitute. When $h = 50$, the triangle is a 50-50 right triangle, so $\theta = 45^\circ$ and the hypotenuse is $50\sqrt{2}$. Then $\sec\theta = \dfrac{50\sqrt{2}}{50} = \sqrt{2}$, so $\sec^2\theta = 2$. With $\frac{\mathrm{d}h}{\mathrm{d}t} = 8$,

$$2\,\frac{\mathrm{d}\theta}{\mathrm{d}t} = \frac{8}{50} = \frac{4}{25} \quad\Longrightarrow\quad \frac{\mathrm{d}\theta}{\mathrm{d}t} = \frac{2}{25} = 0.08 \text{ rad/s}.$$

The angle increases at 0.08 rad/s, about 4.6 degrees per second.

Common trap

The factor $\sec^2\theta$ is easy to drop. Because θ changes with time, $\tan\theta$ is a composition (tangent-of, applied to $\theta(t)$), and the chain rule supplies $\dfrac{\mathrm{d}}{\mathrm{d}t}\tan\theta = \sec^2\theta\,\dfrac{\mathrm{d}\theta}{\mathrm{d}t}$. Forgetting it is the trig version of the $\frac{\mathrm{d}y}{\mathrm{d}x}$-dropping mistake from earlier. Also: this calculation only works because θ is measured in *radians* – the derivative formula $\dfrac{\mathrm{d}}{\mathrm{d}\theta}\tan\theta = \sec^2\theta$ is a radian fact. Always work related-rates angles in radians.

13.4.6 Two moving pieces: ships on the sea

So far one rate drove one other. The method is just as happy with several changing lengths at once.

Example 13.10 (Two ships)**.** At noon, ship A is 60 km due east of ship B. Ship A sails west at 20 km/h; ship B sails north at 15 km/h. How fast is the distance between them changing at 1 p.m.?

Name. Put B's noon position at the origin. Let p be the east–west separation between the ships and q the north–south separation, and D the straight-line distance between them.

Set up. Ship A closes the east–west gap at 20 km/h, so $p = 60 - 20t$ and $\frac{\mathrm{d}p}{\mathrm{d}t} = -20$. Ship B opens a north–south gap at 15 km/h, so $q = 15t$ and $\frac{\mathrm{d}q}{\mathrm{d}t} = 15$. (Here t is hours after noon.)

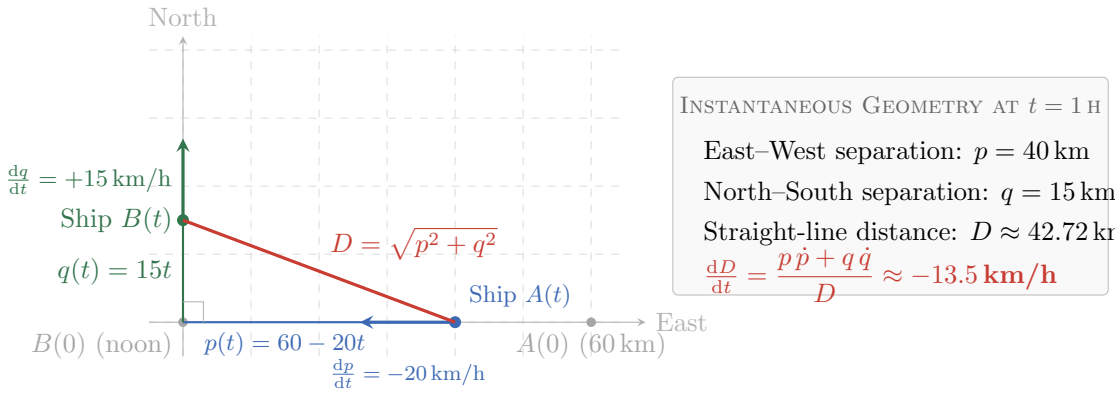


Figure 13.7: The two ships problem (Figure 13.7). Ship A sails west along the horizontal axis with $\frac{\mathrm{d}p}{\mathrm{d}t} = -20$ km/h, while ship B sails north with $\frac{\mathrm{d}q}{\mathrm{d}t} = +15$ km/h. At $t = 1$ hour, the right triangle with legs $p = 40$ and $q = 15$ gives distance $D \approx 42.72$ km; differentiating $D^2 = p^2 + q^2$ shows the distance is decreasing at approximately 13.5 km/h.

Constraint. The two ships and the right angle between their motions give

$$D^2 = p^2 + q^2.$$

Differentiate in t.

$$2D\,\frac{\mathrm{d}D}{\mathrm{d}t} = 2p\,\frac{\mathrm{d}p}{\mathrm{d}t} + 2q\,\frac{\mathrm{d}q}{\mathrm{d}t} \quad\Longrightarrow\quad \frac{\mathrm{d}D}{\mathrm{d}t} = \frac{p\,\frac{\mathrm{d}p}{\mathrm{d}t} + q\,\frac{\mathrm{d}q}{\mathrm{d}t}}{D}.$$

Substitute at $t = 1$. Then $p = 60 - 20 = 40$, $q = 15$, and $D = \sqrt{40^2 + 15^2} = \sqrt{1600 + 225} = \sqrt{1825} \approx 42.72$ km. So

$$\frac{\mathrm{d}D}{\mathrm{d}t} = \frac{40(-20) + 15(15)}{\sqrt{1825}} = \frac{-800 + 225}{42.72} = \frac{-575}{42.72} \approx -13.5 \text{ km/h}.$$

The ships are still *approaching* at about 13.5 km/h: the negative sign says the distance is shrinking, because A's rapid westward closing outweighs B's slower northward drift – for now.

Walking question

The ships' distance is decreasing at 1 p.m., but A will eventually slide past B's line and the gap must grow again. So somewhere $\frac{\mathrm{d}D}{\mathrm{d}t}$ passes through 0 – the moment of closest approach. Without solving it fully, can you see that the closest approach happens exactly when $p\,\frac{\mathrm{d}p}{\mathrm{d}t} + q\,\frac{\mathrm{d}q}{\mathrm{d}t} = 0$? That combination is a dot product in disguise: the rate of D is the component of the relative velocity along the line joining the ships. Related rates is quietly teaching you vector geometry.

13.5 What Ties It All Together

Step back and the two halves of this chapter are one idea seen from two angles.

In *implicit differentiation*, a constraint $F(x, y) = c$ binds y to x. Differentiating in x and remembering the chain rule on y yields a relationship between 1 (the rate of x with respect to itself) and $\frac{\mathrm{d}y}{\mathrm{d}x}$, which we solve for the slope.

In *related rates*, a constraint $F(x, y) = c$ binds x and y as both ride along on time. Differentiating in t and remembering the chain rule on every variable yields a relationship between $\frac{\mathrm{d}x}{\mathrm{d}t}$ and $\frac{\mathrm{d}y}{\mathrm{d}t}$, which we solve for the unknown rate.

Why this matters

Same equation, same chain rule, same algebra at the end – only the independent variable differs (x in one case, t in the other). In fact you can pass between them: if x and y both depend on t, then by the chain rule

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\mathrm{d}y}{\mathrm{d}x} \cdot \frac{\mathrm{d}x}{\mathrm{d}t}.$$

The implicit slope $\frac{\mathrm{d}y}{\mathrm{d}x}$ is exactly the exchange rate that converts the rate of x into the rate of y. Implicit differentiation finds the exchange rate; related rates uses it. The deep theme of this whole chapter is a single sentence: *when quantities obey an equation, their rates obey the differentiated equation.* Constraints on values become constraints on rates. That principle will return, in grander dress, when we reach the gradient and the method of Lagrange multipliers – where the same "the constraint pins the rate" logic governs optimization on curved surfaces.

History

This rate-chaining is the beating heart of Newtonian physics, which is why Newton's own calculus was built around it. Newton called the variables *fluents* (flowing quantities) and their time-rates *fluxions*, writing $\dot{x}$ for what we write $\frac{\mathrm{d}x}{\mathrm{d}t}$. His central problem was exactly the related-rates problem: *given a relation among fluents, find the relation among their fluxions, and vice versa.* The dot notation survives today in mechanics, where $\dot{x}$ is velocity and $\ddot{x}$ acceleration. When you differentiate $x^2 + y^2 = 100$ in time to get $x\dot{x} + y\dot{y} = 0$, you are doing precisely what Newton did in the 1660s – and the fact that it still reads cleanly three and a half centuries later is a tribute to how natural the idea is once you see the constraint as the thing that flows.

13.5.1 The Geometry and Horizons of Implicit Curves

Connection

Implicit differentiation emancipated calculus from the tyranny of explicit formulas ($y = f(x)$). In geometry, celestial mechanics, and engineering, the relations that govern variables are almost never pre-solved for y; they appear as constraint equations $F(x, y) = 0$ or geometric invariants. By differentiating both sides with respect to x or t, we can determine exact tangent slopes, normals, and velocity couplings without ever solving for y.
Yet implicit differentiation breaks down at singular points and self-intersections. When we apply the chain rule to a constraint $F(x, y) = 0$, we are secretly computing:

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \quad \Longrightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}}.$$

This relation encounters two fundamental geometric boundaries:

- Where $\frac{\partial F}{\partial y} = 0$ but $\frac{\partial F}{\partial x} \neq 0$, the tangent line is vertical; single-variable calculus cannot represent y as a function of x locally.
- Where both $\frac{\partial F}{\partial x} = 0$ and $\frac{\partial F}{\partial y} = 0$ simultaneously (as occurs at the origin $(0, 0)$ on the Folium of Descartes $x^3 + y^3 - 3axy = 0$), the formula yields an indeterminate $0/0$. At such a node or cusp, the curve crosses itself or forms a sharp point; single-variable calculus cannot determine which branch to follow, or whether a smooth tangent even exists.

To systematically handle curves and surfaces defined implicitly, we need the geometry of several variables. In Chapter 32 and Chapter 33, we develop partial derivatives, the gradient vector $\nabla F = \left(\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}\right)$, and the celebrated *Implicit Function Theorem*. That theorem precisely demarcates the boundary of solvability: whenever $\nabla F \neq \mathbf{0}$, the level set $F(x, y) = c$ is guaranteed to be a smooth, non-singular 1D curve whose normal vector is ∇F.

13.6 Exercises

Exercise 13.1. Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ by implicit differentiation for each curve.

(a) $x^2 + xy + y^2 = 7$.

(b) $\sqrt{x} + \sqrt{y} = 4$.

(c) $e^{xy} = x + y$.

(d) $y \sin x = x \cos y$.

Exercise 13.2. The curve $x^2 + y^2 = 25$ passes through $(-4, 3)$. Find the equation of the tangent line there, and confirm with the formula $\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{x}{y}$ that it is perpendicular to the radius to that point.

Exercise 13.3. For the astroid $x^{2/3} + y^{2/3} = 4$, find $\frac{\mathrm{d}y}{\mathrm{d}x}$. At which points is the tangent line horizontal, and at which is it vertical? (Hint: horizontal where $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$, vertical where it is undefined.)

Exercise 13.4. Find the second derivative $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$ for the curve $xy = 1$, first by implicit differentiation and then by solving $y = 1/x$ explicitly and differentiating twice. Confirm the two answers agree.

Exercise 13.5 (Conceptual)**.** A classmate computes $\dfrac{\mathrm{d}}{\mathrm{d}x}(y^3) = 3y^2$ while differentiating an implicit equation. Explain in one or two sentences exactly what is wrong, and what the correct expression is. Then state the analogous mistake one might make in a related-rates problem with a term r^3 when differentiating in t.

Exercise 13.6. Water drains from an inverted right-circular cone (point down) of height 12 m and top radius 3 m. The volume of water at depth h is $V = \frac{1}{3}\pi r^2 h$, where r is the surface radius at that depth. (a) Use similar triangles to express r in terms of h, and show $V = \frac{\pi}{48}h^3$. (b) If the volume decreases at $2\,\mathrm{m}^3/\mathrm{min}$, how fast is the depth dropping when $h = 4$ m?

Exercise 13.7. A kite 100 ft above the ground moves horizontally at 8 ft/s away from the person flying it. The string is taut and runs in a straight line. How fast is the string being let out when 200 ft of string is out? (Hint: let x be the horizontal distance and L the string length, with $x^2 + 100^2 = L^2$.)

Exercise 13.8. Two roads meet at a right angle. Car A is 0.3 km from the intersection driving toward it at 60 km/h; car B is 0.4 km from the intersection driving toward it at 80 km/h. How fast is the distance between the cars changing at this instant? Is it increasing or decreasing, and what does the sign mean physically?

Exercise 13.9. A trough 10 ft long has a cross-section that is an isosceles triangle, 2 ft across the top and 1 ft deep. Water is poured in at $3\,\mathrm{ft}^3/\mathrm{min}$. How fast is the water level rising when the water is 6 inches deep? (Set up similar triangles for the water's surface width in terms of depth, then relate volume to depth.)

Exercise 13.10 (Proof)**.** Suppose x and y both depend on t and satisfy the constraint $F(x, y) = c$ for all t, with y expressible as a differentiable function of x along the curve. Using only the single-variable chain rule, prove that

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\mathrm{d}y}{\mathrm{d}x} \cdot \frac{\mathrm{d}x}{\mathrm{d}t}$$

at any point where these derivatives exist. Explain in one sentence how this identity unifies implicit differentiation and related rates.

Exercise 13.11 (Conceptual)**.** In the sliding-ladder example we found $\frac{\mathrm{d}y}{\mathrm{d}t} \to -\infty$ as the top approaches the ground. Explain physically why a real ladder does not behave this way, and identify which modeling assumption ($x^2 + y^2 = 100$ holding for all time, the foot moving at constant 2 ft/s, the top staying in contact with the wall) is the first to fail. Argue that the infinite rate is the model reporting its own breakdown rather than a real physical infinity.

Chapter 14

Linear Approximation and Differentials

There is one idea hiding underneath every theorem in this book, and this chapter is where we finally say it out loud:

A smooth curve, looked at closely enough, is a straight line.

That insight forms the bedrock of differential calculus. The derivative $f'(a)$ is not merely an abstract number; it is the slope of the *tangent line that the curve approaches* under magnification near $x = a$. Once you recognize that smooth means locally linear, the surrounding ideas — tangent lines, error estimates, Newton's method, and the differentials $\mathrm{d}y$ and $\mathrm{d}x$ used throughout physics — become straightforward applications of replacing a curve by its tangent line and bounding the remainder.

This chapter makes that replacement precise. We will build the *linearization* of a function, measure the error you commit by using it, pin down what $\mathrm{d}y$ and $\mathrm{d}x$ *really mean* (they are not zero, and they are not "infinitely small"), use them to track how measurement errors propagate through a formula, and take a first look at Newton's method, which is nothing but linear approximation applied over and over. By the end the tangent line should look less like a topic and more like the seed from which all of calculus grows.

14.1 Zooming In: Why Smooth Means Straight

Take any function you like that is differentiable at a point – say $f(x) = x^2$ near $x = 1$ – and imagine doing the following experiment. Plot it. Then zoom in on the point $(1, 1)$: cut the viewing window in half, then in half again, then again. What happens to the parabola?

It flattens. The more you zoom, the straighter the visible piece of curve looks, until on a small enough window you genuinely cannot tell it from a straight line drawn with a ruler. Figure 14.1 shows three frames of this experiment.

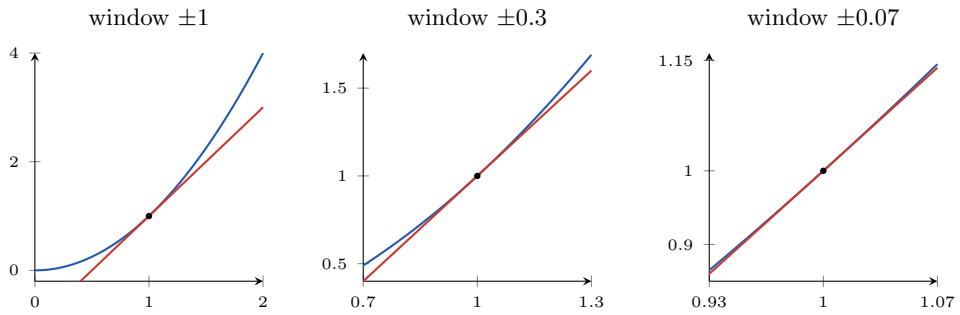


Figure 14.1: Zooming in on $y = x^2$ at $(1, 1)$. The curve (blue) and its tangent line $y = 2x - 1$ (red) become indistinguishable as the window shrinks. "Smooth" literally means "looks straight up close."

Intuition

Differentiability *is* this zoom property. To say f has a derivative at a is to say: there is a single straight line through $(a, f(a))$ that the curve melts into as you magnify. A function with a corner (like $|x|$ at 0) fails the test – no matter how far you zoom into the corner, you still see two different slopes meeting, never one line. Smoothness is the promise that the zoom converges.

Why should you care that the curve becomes straight? Because straight lines are the only functions we can compute with effortlessly. A line is just "start here, go up by the slope times how far you walk." No squaring, no exponentials, no trig tables. If we are willing to accept a small error, we can replace any hard smooth function, *near a point*, by a line and do arithmetic instead of analysis. That trade – a little error in exchange for enormous simplicity – is the engine of applied mathematics.

Walking question

Picture the surface of the Earth. Locally – in your backyard, on a soccer field – it looks flat, and flat-earth geometry (Pythagoras, straight survey lines) works beautifully. The error only shows up over hundreds of miles, where the curvature accumulates. The Earth is the function; the flat backyard is its tangent plane; the error is the curvature. Where else in life do you replace a curved thing by a flat one because the flat version is good enough up close?

14.2 The Tangent Line as the Best Linear Approximation

Let us turn the picture into algebra. We have a function f differentiable at a point a. We want the straight line that best matches f near a. A straight line through the point $(a, f(a))$ has the form

$$L(x) = f(a) + m\,(x - a),$$

and the only freedom we have is the slope m. Which slope makes L hug the curve best?

Geometrically the answer is obvious from the zoom picture: pick the slope of the curve itself at a, namely $m = f'(a)$. That gives the *tangent line*. But "looks best" is not a definition. Let us say precisely what "best" means and then prove the tangent line wins.

Definition 14.1 (Linearization)**.** Let f be differentiable at a. The **linearization** of f at a is the linear function

$$L(x) \;=\; f(a) + f'(a)\,(x - a).$$

The approximation $f(x) \approx L(x)$ for x near a is called the **linear** (or **tangent-line**) **approximation**.

Notice the anatomy of L: it carries the function's *height* $f(a)$ and its *slope* $f'(a)$ at the single point a, and nothing else. It is the most a straight line can know about f. Read it aloud: "Start at the known value $f(a)$. Walk a distance $(x - a)$. For each unit you walk, climb by the rate $f'(a)$."

14.2.1 In what sense is it "best"?

Compare the tangent line to any other line through $(a, f(a))$ with a different slope $m \neq f'(a)$. The error of a line $L_m(x) = f(a) + m(x - a)$ is

$$E_m(x) = f(x) - L_m(x).$$

At $x = a$ every such line is exact: $E_m(a) = 0$, because they all pass through the point. So the contest is about how fast the error grows as you step away. Write $x = a + h$ and divide the error by the step h:

$$\frac{E_m(a+h)}{h} = \frac{f(a+h) - f(a)}{h} - m.$$

As $h \to 0$ the difference quotient $\frac{f(a+h)-f(a)}{h}$ tends to $f'(a)$. Therefore

$$\frac{E_m(a+h)}{h} \;\longrightarrow\; f'(a) - m.$$

Here is the punchline. If $m \neq f'(a)$, this limit is a nonzero number; the error is roughly $(f'(a) - m)\cdot h$, growing *in proportion to* h. But if $m = f'(a)$, the limit is 0: the error, divided by h, dies. The error of the tangent line shrinks *faster than* h *itself*. That is the precise sense in which the tangent line is the uniquely best linear approximation.

Focus

The defining property of the tangent line. Among all lines through $(a, f(a))$, the tangent line is the only one whose error is *small compared to the step*: writing $x = a + h$,

$$\lim_{h \to 0} \frac{f(a+h) - \big[f(a) + f'(a)\,h\big]}{h} \;=\; 0.$$

Every other line leaves an error proportional to h; the tangent line leaves an error that is negligible next to h. We say the tangent line agrees with f "to first order."

Why the technique works

Why does matching the slope kill the first-order error? Because the derivative *is* the leading-order rate of change. Near a,

$$f(a + h) = f(a) + f'(a)\,h + (\text{error}),$$

and the very definition of $f'(a)$ as a limit forces the error to be small relative to h. Choosing the slope to be anything other than $f'(a)$ leaves a leftover term $(f'(a) - m)h$ that is itself proportional to h – a first-order mistake that swamps the curvature. Matching the slope is exactly the act of cancelling the largest part of the discrepancy. There is nothing magic about the tangent line; it is simply the line that has already paid off the h-sized debt, leaving only the smaller h^2-sized debt of curvature behind.

14.2.2 A first worked example

Example 14.1 (Approximating a square root)**.** Estimate $\sqrt{4.1}$ without a calculator.

Let $f(x) = \sqrt{x}$, and choose the *anchor* $a = 4$ because we know $\sqrt{4} = 2$ exactly and 4 is close to 4.1. We need the slope:

$$f'(x) = \frac{1}{2\sqrt{x}}, \qquad f'(4) = \frac{1}{2 \cdot 2} = \frac{1}{4}.$$

The linearization is

$$L(x) = f(4) + f'(4)(x - 4) = 2 + \tfrac{1}{4}(x - 4).$$

Plug in $x = 4.1$:

$$\sqrt{4.1} \approx L(4.1) = 2 + \tfrac{1}{4}(0.1) = 2 + 0.025 = 2.025.$$

The true value is $\sqrt{4.1} = 2.024845\ldots$, so our estimate is right to four decimal places. We did it with one division.

Selecting the anchor a balances proximity and evaluability. You want a point where f and f' can be computed exactly, and which sits as close to your target as possible. Here $a = 4$ serves both purposes: 4 is a perfect square and lies adjacent to 4.1.

Work it yourself

Estimate $\sqrt{4.4}$ the same way: $L(4.4) = 2 + \frac{1}{4}(0.4) = 2.1$. The true value is $2.0976\ldots$. Notice the error has grown from about 0.00015 (at the step $h = 0.1$) to about 0.0024 (at the step $h = 0.4$): the step grew by a factor of 4, and the error grew by a factor of about $16 = 4^2$. That square-law growth is the fingerprint of the curvature, and Section 14.4 explains exactly why the factor is h^2.

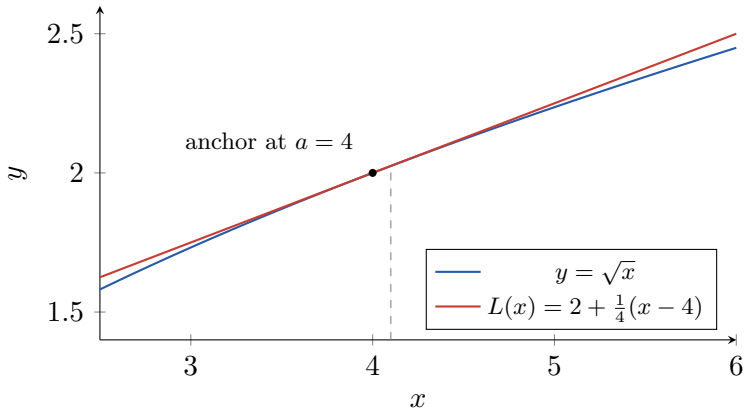


Figure 14.2: The tangent line to $\sqrt{x}$ at $a = 4$. Near the anchor the red line and blue curve are nearly identical; the gap (the error) opens slowly as you move away, and because $\sqrt{x}$ bends downward, the line sits slightly above the curve.

Common trap

Linearization is a *local* tool. It is only trustworthy near the anchor. If you used the line $L(x) = 2 + \frac{1}{4}(x - 4)$ to estimate $\sqrt{100}$ you would get $2 + \frac{1}{4}(96) = 26$, which is absurd (the truth is 10). The approximation does not "know" anything about the curve far from a – it only ever knew the height and slope at one point. Always ask: is my target genuinely close to my anchor?

14.3 What the Differentials $\mathrm{d}y$ and $\mathrm{d}x$ Really Mean

You have been writing $\frac{\mathrm{d}y}{\mathrm{d}x}$ for chapters now, defined rigorously as the limit of a difference quotient. That care was necessary while constructing the foundations. Now we can clarify the underlying geometric picture: there are two distinct quantities, $\mathrm{d}x$ and $\mathrm{d}y$, whose ratio equals the derivative when defined on the tangent line. Leibniz originally thought of $\mathrm{d}x$ as an infinitesimal change. While intuitive, that notion required two centuries of refinement to place on solid ground. Here is the modern formulation.

Definition 14.2 (The differentials $\mathrm{d}x$ and $\mathrm{d}y$)**.** Let $y = f(x)$ be differentiable at x. We declare $\mathrm{d}x$ to be an **independent real variable** – an ordinary number, which you are free to choose, representing a step in the x-direction. Then we *define* the **differential of** y by

$$\mathrm{d}y \;=\; f'(x)\,\mathrm{d}x.$$

$\mathrm{d}y$ is the rise of the *tangent line* over a run of $\mathrm{d}x$.

Read that definition slowly, because it dissolves a confusion you may have carried for years.

What is really happening

What $\mathrm{d}x$ and $\mathrm{d}y$ really are. They are not zero. They are not "infinitely small." $\mathrm{d}x$ is just a number you pick – the size of a step. And $\mathrm{d}y$ is the corresponding *rise along the tangent line*,

not along the curve. Look at Figure 14.3. Take a real step $\mathrm{d}x = \Delta x$ from the point $(a, f(a))$. Two things rise:

- the *curve* rises by $\Delta y = f(a + \Delta x) - f(a)$ – the true change;
- the *tangent line* rises by $\mathrm{d}y = f'(a)\,\mathrm{d}x$ – the linear estimate of the change.

The differential $\mathrm{d}y$ is the **approximation**; the increment Δy is the **truth**. The gap between them, $\Delta y - \mathrm{d}y$, is the error of linear approximation – exactly the quantity we studied in the last section. And now the notation $\frac{\mathrm{d}y}{\mathrm{d}x} = f'(x)$ is literally a quotient of two numbers $\mathrm{d}y$ and $\mathrm{d}x$, because that is how we defined $\mathrm{d}y$. Leibniz's fraction was real all along; we just had to say which line the rise is measured on.

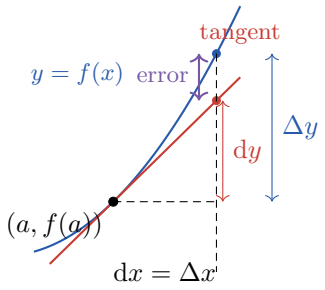


Figure 14.3: The true meaning of differentials. Over a real step $\mathrm{d}x = \Delta x$, the tangent line rises by $\mathrm{d}y = f'(a)\,\mathrm{d}x$ while the curve rises by the true amount Δy. The differential $\mathrm{d}y$ estimates Δy; their difference (purple) is the approximation error, governed by the curvature.

History

When Leibniz invented the differential notation in the 1670s and 1680s, he pictured $\mathrm{d}x$ as an *infinitesimal*: a quantity smaller than any assignable positive real number, yet not zero. With it he could write $\mathrm{d}y = f'(x)\,\mathrm{d}x$ and treat derivatives as genuine fractions, which made the chain rule $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u}\frac{\mathrm{d}u}{\mathrm{d}x}$ look like cancelling $\mathrm{d}u$'s. Newton, working concurrently, conceived of "fluxions" and evanescent increments that vanished at the moment of generation. Both pioneers obtained breathtakingly accurate physical results, but neither could logically define what an infinitely small number was. In 1734, Bishop George Berkeley published his scathing polemic *The Analyst*, asking whether these disappearing increments were "the ghosts of departed quantities"—objecting that a quantity could not logically be treated as non-zero while dividing, only to be set to zero when convenient.

The operational rescue of linear approximation occurred in the eighteenth century through Leonhard Euler. In his 1768 *Institutiones calculi differentialis*, Euler formalized the concept of using the tangent line as a computational microscope: by stepping along the tangent line, one could approximate the solutions of intractable equations and curves via discrete tangent steps ($y_{n+1} = y_n + f'(x_n)\Delta x$), creating what we now call Euler's method.

The full theoretical rigor arrived in 1823 with Augustin-Louis Cauchy's *Résumé des leçons données à l'École royale polytechnique sur le calcul infinitésimal*. Cauchy abandoned mystical infinitesimals, defining the derivative firmly as a limit of finite quotients. More importantly, Cauchy established the first rigorous *error bounds* for linear approximation: using the Mean Value Theorem, he proved that the remainder $R_1(x) = f(x) - L(x)$ is bounded precisely by

the maximum of the second derivative: $|R_1(x)| \le \frac{1}{2}M(x-a)^2$, where $|f''(t)| \le M$. Leibniz's differentials $\mathrm{d}x$ and $\mathrm{d}y$ were thereby transformed from metaphysical phantoms into rigorous, finite geometric quantities: $\mathrm{d}x = \Delta x$ is an ordinary real displacement, and $\mathrm{d}y = f'(a)\,\mathrm{d}x$ is the exact vertical rise of the tangent line.

14.3.1 Differentials as a notation for approximation

The single most useful sentence about differentials is this:

$$\boxed{\Delta y \;\approx\; \mathrm{d}y \;=\; f'(x)\,\mathrm{d}x}$$

"The true change is approximately the linear change." Everything in the rest of this chapter is an application of that boxed line. Let us redo Example 14.1 in this language to see how compact it becomes.

Example 14.2 ($\sqrt{4.1}$ via differentials)**.** We want $\sqrt{4.1}$. Anchor at $x = 4$, take the step $\mathrm{d}x = 0.1$. With $y = \sqrt{x}$ we have $\mathrm{d}y = \frac{1}{2\sqrt{x}}\,\mathrm{d}x$, so

$$\mathrm{d}y = \frac{1}{2\sqrt{4}}(0.1) = \frac{0.1}{4} = 0.025.$$

Then

$$\sqrt{4.1} = \sqrt{4} + \Delta y \approx \sqrt{4} + \mathrm{d}y = 2 + 0.025 = 2.025,$$

matching what we found before. The differential $\mathrm{d}y = 0.025$ *is* the estimated change in $\sqrt{x}$ caused by nudging x from 4 to 4.1.

Connection

The boxed approximation $\Delta y \approx f'(x)\,\mathrm{d}x$ is the one-variable seed of the *total differential* you will meet in multivariable calculus:

$$\mathrm{d}z = \frac{\partial f}{\partial x}\,\mathrm{d}x + \frac{\partial f}{\partial y}\,\mathrm{d}y.$$

There, a small move in the plane changes z by (rate in x)$\times$(step in x) plus (rate in y)$\times$(step in y): the tangent *plane* replaces the tangent line, but the idea – locally, smooth equals linear – is identical. Get it down in one variable and the multivariable version costs you almost nothing.

14.4 Linearization and Error: How Wrong Are You?

An approximation you cannot bound is a guess. The reason linear approximation is trustworthy – rather than merely convenient – is that we can say precisely how large the error is. Define the error of the linearization at a as

$$E(x) = f(x) - L(x) = f(x) - \big[f(a) + f'(a)(x-a)\big].$$

We already know (Section 14.2) that E vanishes faster than the step: $E(a) = 0$ and $E(x)/(x-a) \to 0$. That tells us the error is small, but not *how* small. The next theorem gives the size exactly, and the answer is beautiful: *the error is controlled by the second derivative* – by the curvature, which is the thing a straight line is blind to.

Theorem 14.1 (Error of linear approximation)**.** *Suppose f has a continuous second derivative on an interval containing a and x. Then there is a point c between a and x such that*

$$f(x) - \big[f(a) + f'(a)(x-a)\big] \;=\; \frac{f''(c)}{2}\,(x-a)^2.$$

Before the proof, absorb the shape of the result. Writing $h = x - a$ for the step:

$$E = \tfrac{1}{2}\, f''(c)\, h^2.$$

Three things to notice. First, the error is proportional to h^2, not h – so halving your distance from the anchor quarters the error, exactly the square-law we saw numerically in the lab box. Second, the multiplier is f'', the curvature: a flatter curve (small f'') is approximated better, and a perfectly straight f (with $f'' = 0$) has *zero* error, as it must. Third, the sign of f'' tells you which way you missed: if the curve is concave up ($f'' > 0$) the tangent line lies below it and you underestimate; concave down ($f'' < 0$) and you overestimate. That is why in Figure 14.2 the line for the downward-bending $\sqrt{x}$ sat *above* the curve, and our estimate 2.025 came out slightly high. Figure 14.4 shows this h^2 growth directly: the true error tracks a parabola, snug under the guaranteed bound.

Why the technique works

Where does the $\frac{1}{2}$ come from, and why f''? Think of building f back up from its derivatives, like reconstructing a journey from your speedometer. The value $f(a)$ is where you start. The term $f'(a)h$ is how far you would travel if your speed never changed – the linear estimate. But your speed *does* change, at the rate f'' (that is what the second derivative is: the rate the slope changes). Over a step h, a steadily changing speed contributes an *average* extra displacement. Just as distance under constant acceleration is $\frac{1}{2}at^2$ in physics, the curvature contributes $\frac{1}{2}f''h^2$. The $\frac{1}{2}$ is the same $\frac{1}{2}$ that appears in $\frac{1}{2}at^2$ – it is the average of a quantity ramping linearly from 0 to its full value. The error term is the first correction you would add to make the straight-line estimate curved.

Proof of Theorem 14.1. We use the Cauchy mean value theorem twice – but you can read the idea without the machinery: we compare the error to the function $(x-a)^2$ and apply the mean value theorem to both.

Fix a and let x vary; write $h = x - a$ and consider the two functions

$$g(t) = f(t) - \big[f(a) + f'(a)(t-a)\big], \qquad p(t) = (t-a)^2,$$

both on the interval between a and x. We want $g(x)/p(x)$. Notice $g(a) = 0$ and $p(a) = 0$, and also

$$g'(t) = f'(t) - f'(a), \qquad p'(t) = 2(t-a),$$

so $g'(a) = 0$ and $p'(a) = 0$ as well. By the Cauchy mean value theorem applied to g and p on the interval from a to x (using $g(a) = p(a) = 0$), there is a point c_1 between a and x with

$$\frac{g(x)}{p(x)} = \frac{g(x) - g(a)}{p(x) - p(a)} = \frac{g'(c_1)}{p'(c_1)} = \frac{f'(c_1) - f'(a)}{2(c_1 - a)}.$$

Now apply the ordinary mean value theorem to f' on the interval from a to c_1: there is a point c between a and c_1 (hence between a and x) with

$$\frac{f'(c_1) - f'(a)}{c_1 - a} = f''(c).$$

Substituting,

$$\frac{g(x)}{p(x)} = \frac{f''(c)}{2} \quad \Longrightarrow \quad g(x) = \frac{f''(c)}{2}\,(x-a)^2,$$

which is exactly the claim, since $g(x) = E(x)$. □

Focus

Usable error bound. You rarely know the mystery point c, but you do not need it. If $|f''(t)| \le M$ for all t between a and x, then

$$\big|f(x) - L(x)\big| \;\le\; \frac{M}{2}\,(x-a)^2.$$

Replace the unknown $f''(c)$ by an *upper bound* on the curvature and you get a guaranteed, conservative error bar.

Example 14.3 (Putting a number on the $\sqrt{4.1}$ error)**.** For $f(x) = \sqrt{x}$ we have $f''(x) = -\frac{1}{4}x^{-3/2}$. On the interval $[4, 4.1]$ the largest size of f'' occurs at the left end $x = 4$ (where $x^{-3/2}$ is biggest):

$$|f''(x)| \le \tfrac{1}{4}\cdot 4^{-3/2} = \tfrac{1}{4}\cdot\tfrac{1}{8} = \tfrac{1}{32} = M.$$

With $h = 0.1$,

$$\big|\sqrt{4.1} - 2.025\big| \le \frac{M}{2}h^2 = \frac{1}{64}\,(0.1)^2 = \frac{0.01}{64} \approx 0.000156.$$

The actual error was $0.000154\ldots$ – our bound is essentially tight, and we obtained it knowing only that $\sqrt{x}$ does not bend too sharply on $[4, 4.1]$. Because $f'' < 0$ there, we even know the *sign*: the estimate is high, as it was.

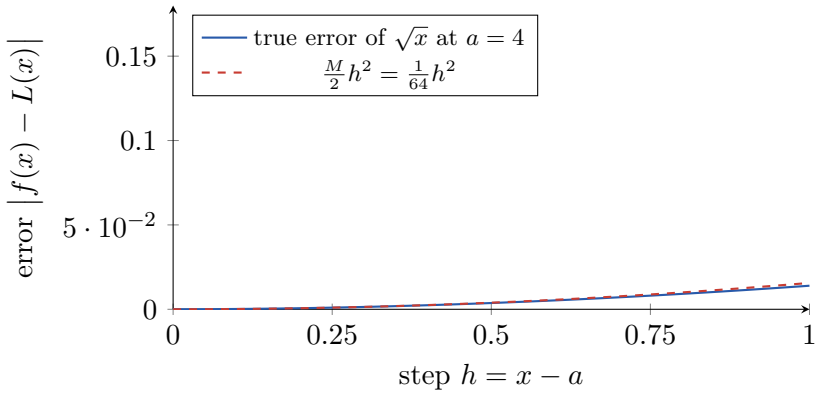


Figure 14.4: The linearization error grows like h^2. The blue curve is the actual error of approximating $\sqrt{x}$ by its tangent line at $a = 4$; the dashed red parabola is the guaranteed bound $\frac{M}{2}h^2$ from Theorem 14.1. The error hugs a parabola, not a line – doubling the step roughly quadruples the error.

Why this matters

Theorem 14.1 is the first whisper of *Taylor's theorem*, the crown jewel of approximation theory you will meet later. The pattern continues forever: the linear approximation has error $\sim f''h^2/2$; if you also match the curvature you get the quadratic approximation, with error $\sim f'''h^3/6$; and so on, each new derivative buying you another power of h. Linear approximation is Taylor's theorem stopped at the first term. Everything you learn here scales up.

14.5 Propagation of Error: Differentials Earn Their Keep

Here is where differentials stop being a notational nicety and become a working tool of every laboratory science. You measure something – a radius, a voltage, a time – and your measurement has an uncertainty. You then *compute* something from it – an area, a power, a distance. How does the uncertainty in what you measured contaminate what you computed? The answer is one application of $\mathrm{d}y = f'(x)\,\mathrm{d}x$, read in a new light: $\mathrm{d}x$ is now the *measurement error*, and $\mathrm{d}y$ is the resulting error in the computed quantity.

Example 14.4 (Error in the area of a circle)**.** You measure the radius of a circular plate as $r = 10\,\mathrm{cm}$, with your ruler good to about $\mathrm{d}r = \pm 0.05\,\mathrm{cm}$. You report the area $A = \pi r^2$. What is the uncertainty in A?

The area's differential is

$$\mathrm{d}A = \frac{\mathrm{d}A}{\mathrm{d}r}\,\mathrm{d}r = 2\pi r\,\mathrm{d}r.$$

At $r = 10$, $\mathrm{d}r = 0.05$:

$$\mathrm{d}A = 2\pi(10)(0.05) = \pi \approx 3.14\ \mathrm{cm}^2.$$

So $A = \pi(10)^2 \pm \pi \approx 314.16 \pm 3.14\ \mathrm{cm}^2$. A half-millimeter slip in the radius costs you about $3\,\mathrm{cm}^2$ of area.

Often the *relative* error – the fractional, or percentage, error – is more illuminating than the absolute error, because it is dimensionless and tells you how many significant figures you can trust. Divide:

$$\frac{\mathrm{d}A}{A} = \frac{2\pi r\,\mathrm{d}r}{\pi r^2} = 2\,\frac{\mathrm{d}r}{r}.$$

Focus

Relative error doubles for area, triples for volume. For $A = \pi r^2$,

$$\frac{\mathrm{d}A}{A} = 2\,\frac{\mathrm{d}r}{r}.$$

A 1% error in radius becomes a 2% error in area. For a volume $V = \frac{4}{3}\pi r^3$ the same calculation gives $\dfrac{\mathrm{d}V}{V} = 3\,\dfrac{\mathrm{d}r}{r}$ – a 1% radius error becomes a 3% volume error. The exponent in the power law *is* the error-amplification factor.

Why the technique works

Why does the exponent become the multiplier? Because d(power) is just the power rule turned into error language. If $y = x^n$ then $\mathrm{d}y = nx^{n-1}\mathrm{d}x$, and dividing by $y = x^n$ gives

$$\frac{\mathrm{d}y}{y} = n\,\frac{\mathrm{d}x}{x}.$$

The logarithm makes it even cleaner: $\ln y = n \ln x$, so $\frac{\mathrm{d}y}{y} = n\,\frac{\mathrm{d}x}{x}$ is exactly the differential of that identity (since $\mathrm{d}(\ln u) = \mathrm{d}u/u$). Relative error is the *differential of the logarithm*. That is why, for any product or power expression, the relative errors simply add up weighted by the exponents – taking logs turns products into sums, and differentiation turns the sums into added relative errors.

Example 14.5 (Combining several measurements)**.** A pendulum's period is $T = 2\pi\sqrt{L/g}$. Suppose you know L to within 0.5% and g to within 0.2%. How accurately do you know T?

Take logarithms first: $\ln T = \ln(2\pi) + \frac{1}{2}\ln L - \frac{1}{2}\ln g$. Now take the differential of each side, using $\mathrm{d}(\ln u) = \mathrm{d}u/u$:

$$\frac{\mathrm{d}T}{T} = \frac{1}{2}\,\frac{\mathrm{d}L}{L} - \frac{1}{2}\,\frac{\mathrm{d}g}{g}.$$

For a *worst-case* bound we add the magnitudes:

$$\left|\frac{\mathrm{d}T}{T}\right| \le \tfrac{1}{2}(0.5\%) + \tfrac{1}{2}(0.2\%) = 0.25\% + 0.1\% = 0.35\%.$$

The logarithmic trick converted a messy square-root-and-quotient formula into a one-line sum of weighted relative errors. This is the everyday error-propagation arithmetic of physics labs.

Application

Error propagation via differentials runs through all of quantitative science. A surveyor turning a measured angle into a distance, an engineer turning a measured current into dissipated power $P = I^2R$ (so $\mathrm{d}P/P = 2\,\mathrm{d}I/I$ – watch your ammeter!), an economist turning a growth rate into a doubling time, a machine learning engineer asking how a tiny perturbation $\mathrm{d}x$ to an input shifts a network's output – that last one, the *sensitivity* $\mathrm{d}y = f'(x)\,\mathrm{d}x$, is precisely a single step of backpropagation. In every case the question "how much does the output wobble when the input wobbles?" is answered by the derivative times the wobble. The differential is the universal currency of sensitivity.

14.6 The Newton Step: Linear Approximation in Reverse

So far we used the tangent line to estimate a function's *value*. Now turn the question inside out. Suppose we want to *solve* an equation $f(x) = 0$ – to find a root – but the equation is too hard to solve exactly. Examples are everywhere: $\cos x = x$, or $x^5 - x - 1 = 0$, equations with no formula for the answer. The tangent line rescues us again, and the idea is irresistibly simple:

We cannot solve $f(x) = 0$, but we can always solve $L(x) = 0$,
because L is a straight line.

Start with a guess x_0 near the root. Replace the curve by its tangent line at x_0 – the best line we know how to build. Then find where *that line* crosses zero, and call it x_1. Because the line hugs the curve, its crossing should be much closer to the true root than x_0 was. Repeat from x_1, and you spiral in. Figure 14.5 shows two steps of the dance.

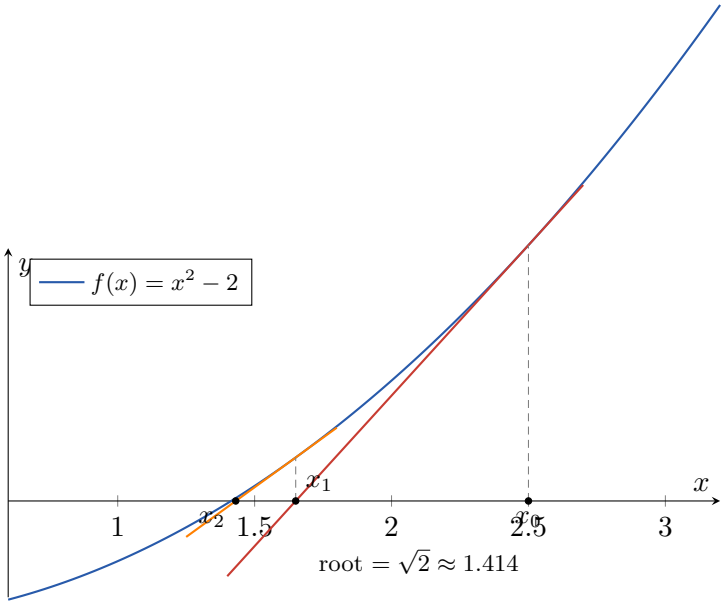


Figure 14.5: Newton's method on $f(x) = x^2 - 2$. From x_0, slide down the tangent (red) to its zero x_1; from x_1, slide down the next tangent (orange) to x_2. Each step lands far closer to the true root $\sqrt{2}$.

Let us derive the step. The tangent line at x_0 is

$$L(x) = f(x_0) + f'(x_0)(x - x_0).$$

Set $L(x_1) = 0$ and solve for x_1:

$$0 = f(x_0) + f'(x_0)(x_1 - x_0) \quad \Longrightarrow \quad x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

This single formula drives the iteration.

Focus

Newton's iteration.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

In words: from your current guess, step by "minus the value over the slope." The value $f(x_n)$ tells you how far off you are; dividing by the slope $f'(x_n)$ converts that vertical miss into the horizontal correction that a straight line of that slope would need.

Example 14.6 (Computing $\sqrt{2}$ by hand)**.** To find $\sqrt{2}$, solve $f(x) = x^2 - 2 = 0$, with $f'(x) = 2x$. The iteration becomes

$$x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n} = \frac{x_n}{2} + \frac{1}{x_n},$$

the famous "average your guess with 2/guess" rule (known to the Babylonians millennia before Newton). Start at $x_0 = 1.5$:

$$x_1 = \tfrac{1.5}{2} + \tfrac{1}{1.5} = 0.75 + 0.66667 = 1.41667,$$
$$x_2 = \tfrac{1.41667}{2} + \tfrac{1}{1.41667} \approx 0.70834 + 0.70588 = 1.41422,$$
$$x_3 = 1.414214\ldots$$

The true value is $\sqrt{2} = 1.4142136\ldots$. After just three steps we are correct to seven digits. Notice the number of correct digits roughly *doubled* each step: 1, then 3, then 6+. That explosive accuracy – *quadratic convergence* – is the reward for using the best linear approximation.

Why the technique works

Why does Newton's method converge so ferociously fast? Because the error it leaves behind is exactly the linear-approximation error of Theorem 14.1, which is of order (current error)^2. Each step uses the tangent line, whose miss is governed by $\frac{1}{2}f''h^2$; so if your current distance to the root is h, your next distance is of size h^2. Square a small number and it becomes tiny: $0.01 \to 0.0001 \to 0.00000001$. The doubling of correct digits per step is the $h \mapsto h^2$ law in disguise. Newton's method is fast for precisely the reason the tangent line is accurate: matching the slope kills the first-order error, leaving only the squared remainder.

Common trap

Newton's method is powerful but not foolproof. If $f'(x_n) = 0$ the step divides by zero – the tangent is horizontal and never reaches the axis. A bad starting guess can send the iterates flying away from the root, or trap them in a loop. And if the root is one where f' also vanishes (a double root), the fast doubling slows to a crawl. We treat all of this carefully in the dedicated Newton's-method chapter; here the point is only the *idea* – root-finding is linear approximation run backwards, solving the line instead of the curve.

History

Newton described this method around 1669 (published later) for polynomial equations, though his version did not yet use the tangent-line/derivative picture explicitly – he manipulated the polynomial algebraically. Joseph Raphson streamlined it in 1690 into an iterative form close to what we use, which is why it is often called the *Newton–Raphson method*. The clean $x_{n+1} = x_n - f(x_n)/f'(x_n)$, framed in terms of the derivative and the tangent line, is the later gift of the calculus that Newton and Leibniz had built. It is a small marvel that the same tangent line which approximates a curve's height also hands us the most powerful equation-solver in computational science.

14.7 The Seed of All of Calculus

Step back and look at what one idea – *locally, smooth is linear* – has already done in this single chapter.

- It *defined* the derivative geometrically: $f'(a)$ is the slope of the line the curve becomes under magnification.

- It gave us *linearization*: replace f by $f(a) + f'(a)(x - a)$ to compute hard values with easy arithmetic.
- It pinned down *differentials*: $\mathrm{d}y = f'(x)\,\mathrm{d}x$ is the tangent line's rise, and $\mathrm{d}y/\mathrm{d}x$ is a true ratio.
- It powered *error propagation*: the same $\mathrm{d}y = f'(x)\,\mathrm{d}x$, read as "output wobble equals slope times input wobble."
- It produced *Newton's method*: solve the tangent line instead of the curve, and iterate.

And this is only the beginning of the book. The *integral* is the same idea inverted: chop a curved region into slivers so thin that each is a straight-sided rectangle (smooth = linear on a sliver), then add the slivers up. The *Fundamental Theorem* ties the slope picture to the area picture. A *differential equation* is a recipe that gives you the local slope $\mathrm{d}y/\mathrm{d}x$ at every point and asks you to reassemble the global curve – which you do by stepping along tangent lines (that is literally Euler's method). Taylor series push linear approximation to its limit, matching infinitely many derivatives.

Why this matters

Here is the thought to carry out of this chapter. **Calculus is the systematic exploitation of one fact: that a smooth thing, examined locally, is linear – and linear things we can do anything with.** Differentiation is the act of *finding* the local line. Integration is the act of *summing* infinitely many local lines back into a whole. Every technique you will learn – the chain rule, substitution, slope fields, series – is a different way of moving between the curved world (hard, interesting, real) and the linear world (easy, computable) along the bridge that the tangent line provides. You now hold the seed. The rest of the book is watching it grow.

14.7.1 The Power and Boundaries of Linearization

Connection

Linear approximation and differentials unlocked the computational engine of applied science:

- *Instantaneous estimation*: Complicated irrational roots ($\sqrt{9.02}$), transcendental values ($\ln(1.05)$), or intractable non-linear expressions can be evaluated with elementary pencil-and-paper arithmetic: $f(a) + f'(a)\Delta x$.
- *Linear error propagation*: The differential relation $\mathrm{d}y = f'(a)\,\mathrm{d}x$ gives engineers and scientists an immediate formula for sensitivity analysis. An uncertainty in an input measurement $\mathrm{d}x$ maps directly to an uncertainty in the output $\mathrm{d}y$, quantifying tolerance stack-ups across physical assemblies.
- *The principle of linearization*: Virtually all of numerical analysis, structural mechanics, and modern physics relies on replacing formidable non-linear manifolds with their local tangent spaces.

Yet the tangent line is utterly blind to curvature. It is an affine model that assumes the derivative remains constant forever. The moment the evaluation point x moves away from the basepoint a, the approximation error begins to grow quadratically:

$$R_1(x) = f(x) - L(x) \approx \frac{1}{2} f''(a)(x - a)^2.$$

If a curve bends sharply (large $|f''|$) or if the step $\Delta x = x - a$ is not microscopic, the tangent line diverges rapidly from reality. A straight line cannot turn corners, cannot track oscillatory waves, and cannot anticipate impending inflection points or asymptotes.
To overcome the myopia of straight lines, we must allow our approximations to bend. In Chapter 30 and Chapter 31 (*Power Series and Taylor's Theorem*), we generalize the linear differential $f'(a)(x-a)$ to quadratic, cubic, and infinite-degree polynomial approximations:

$$P_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

By matching the second derivative (curvature), third derivative, and beyond, Taylor polynomials cling to curves across broad intervals, transforming linear approximation into high-precision analysis.

14.8 Exercises

Exercise 14.1 (Warm-up linearizations)**.** For each function, find the linearization $L(x)$ at the given anchor a, and use it to estimate the requested value. Compare with a calculator.

1. $f(x) = \sqrt[3]{x}$ at $a = 27$; estimate $\sqrt[3]{27.5}$.
2. $f(x) = \sin x$ at $a = 0$; estimate $\sin(0.1)$ (radians).
3. $f(x) = \ln x$ at $a = 1$; estimate $\ln(1.05)$.
4. $f(x) = (1+x)^{10}$ at $a = 0$; estimate $(1.02)^{10}$.

Exercise 14.2 (Differentials directly)**.** A cube has side length measured as $s = 5.00\,\text{cm}$ with possible error $\mathrm{d}s = \pm 0.02\,\text{cm}$. Use the differential of $V = s^3$ to estimate the largest possible error in the computed volume, and the corresponding *relative* error $\mathrm{d}V/V$. State the relative error as a percentage and confirm it equals $3\,(\mathrm{d}s/s)$.

Exercise 14.3 (Sign of the error)**.** Without computing the true value, decide whether the linear approximation of $f(x) = e^x$ at $a = 0$ *over-* or *under*-estimates $e^{0.2}$, and justify your answer using the sign of f''. Then check numerically.

Exercise 14.4 (A guaranteed error bar)**.** Let $f(x) = \sqrt{x}$ and anchor at $a = 9$. Use the linearization to estimate $\sqrt{9.3}$, then use the bound $|E| \le \frac{M}{2}h^2$ (with M an upper bound for $|f''|$ on $[9, 9.3]$) to produce a guaranteed interval that contains the true value of $\sqrt{9.3}$. Verify the true value lies inside.

Exercise 14.5 (Relative error in a power law)**.** The kinetic energy of a moving mass is $E = \frac{1}{2}mv^2$. If the speed v is known to within 1.5% and the mass m to within 0.4%, bound the relative error in E. (Hint: take logarithms first, then differentiate.)

Exercise 14.6 (Newton by hand)**.** Apply Newton's method to $f(x) = x^3 - 7$ to estimate $\sqrt[3]{7}$.

1. Write the iteration $x_{n+1} = x_n - f(x_n)/f'(x_n)$ explicitly for this f.
2. Starting from $x_0 = 2$, compute x_1 and x_2 to five decimal places.
3. How many correct digits did each step gain? Relate this to the $h \mapsto h^2$ error law.

Exercise 14.7 (When Newton stumbles)**.** Consider $f(x) = x^3 - 2x + 2$ with starting guess $x_0 = 0$.

1. Compute x_1 and x_2.
2. Show that the iteration falls into a 2-cycle ($0 \to 1 \to 0 \to \cdots$) and never approaches the real root near $x \approx -1.77$. Explain geometrically, in terms of where the tangent lines point, why the method fails to converge here.

Exercise 14.8 (Conceptual: what is dy?)**.** A student writes: "dy is the change in y when x changes by dx." Correct this sentence so that it is precise. In your corrected version, explain the difference between dy and Δy, draw the small picture that distinguishes them, and identify which one the derivative $\frac{\mathrm{d}y}{\mathrm{d}x}$ is a ratio of.

Exercise 14.9 (Prove the first-order property)**.** Let f be differentiable at a and let $L(x) = f(a) + f'(a)(x - a)$ be its linearization. Prove directly from the definition of the derivative that

$$\lim_{x \to a} \frac{f(x) - L(x)}{x - a} = 0,$$

and explain why no line of a *different* slope can satisfy this. (This is the precise statement that the tangent line is the unique best linear approximation.)

Exercise 14.10 (Proof: error formula on a polynomial)**.** Let $f(x) = x^3$ and anchor at $a = 1$.

1. Write the exact error $E(x) = f(x) - L(x)$ as a polynomial in $h = x - 1$ by expanding $(1 + h)^3$.
2. Show directly that $E(x) = \frac{f''(c)}{2}(x - 1)^2$ holds, and solve for the point c as a function of h. Confirm that c lies between 1 and x, as Theorem 14.1 promises.

Exercise 14.11 (Linearization of a product)**.** Suppose u and v are differentiable at a, with values $u(a), v(a)$ and derivatives $u'(a), v'(a)$. Using $\mathrm{d}(uv) = u\,\mathrm{d}v + v\,\mathrm{d}u$, show that the relative error of a product is the sum of the relative errors:

$$\frac{\mathrm{d}(uv)}{uv} = \frac{\mathrm{d}u}{u} + \frac{\mathrm{d}v}{v}.$$

Interpret this as the differential of the identity $\ln(uv) = \ln u + \ln v$, and state the analogous rule for a quotient u/v.

Chapter 15

The Mean Value Theorem and What It Means

A single foundational geometric principle powers much of differential calculus: the *Mean Value Theorem* (MVT). You learn that a function with positive derivative is increasing. You learn that two antiderivatives of the same function differ by a constant. You learn that you can bound the error of a linear approximation. Each of these statements feels like its own separate rule. In truth, they are all direct corollaries of this one theorem.

The Mean Value Theorem is the bridge that carries information from the derivative back to the original function. Knowing something about f' across an interval allows us to deduce rigorous facts about the values of f itself.

The progression unfolds in three stages: a horizontal baseline (Rolle's theorem), a simple tilt that yields the general MVT, and a systematic tour of its consequences.

15.1 The Picture Behind the Theorem

Take a smooth curve and draw the straight line connecting its two endpoints. That line — the *secant* — has some slope, representing the average rate of change across the interval. Now slide a line parallel to that secant until it just touches the curve. The point of tangency is a point whose instantaneous slope matches the average slope.

Intuition

The Mean Value Theorem states that somewhere on a smooth arc, the instantaneous rate of change equals the average rate of change. If you drove 120 miles in 2 hours, then at some instant your speedometer read exactly 60 mph. You cannot average 60 without matching 60 at some moment during the journey. That physical velocity observation captures the geometric content of the theorem.

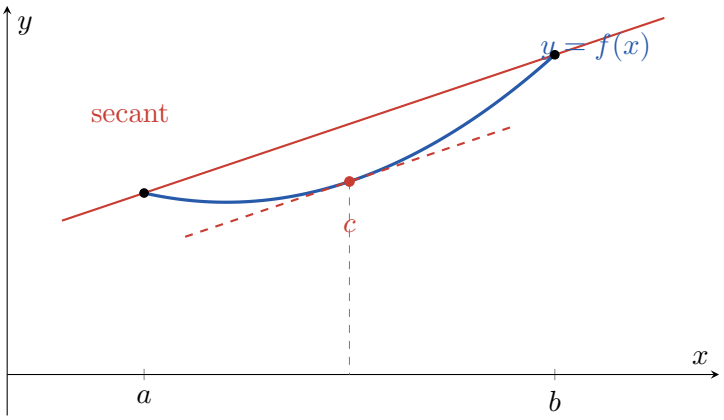


Figure 15.1: The Mean Value Theorem. The red secant joins the endpoints $(a, f(a))$ and $(b, f(b))$. Somewhere in between, at $x = c$, the tangent (dashed) is parallel to it: $f'(c)$ equals the average slope.

Notice what the picture quietly assumes. The curve has to be unbroken — if it jumped, the ruler could miss it entirely. And it has to be smooth, with a well-defined tangent everywhere inside, or there could be a sharp corner where the slope you want is never achieved. Hold onto those two requirements; they will become the hypotheses of the theorem, and we will see exactly why each one is needed.

15.2 Rolle's theorem: the flat special case

The cleanest way into the MVT is to first prove a special case so plain it almost proves itself, and then *tilt* the whole picture to get the general statement. The special case is the one where the two endpoints sit at the same height.

Intuition

If a smooth curve starts and ends at the same height, then somewhere in between it has a horizontal tangent. Throw a ball; it leaves your hand and returns to the same height; at the top of its arc, for one instant, it is moving neither up nor down. That instant is Rolle's theorem.

Theorem 15.1 (Rolle, 1691)**.** *Let f be continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b), and suppose $f(a) = f(b)$. Then there exists at least one point $c \in (a, b)$ with*

$$f'(c) = 0.$$

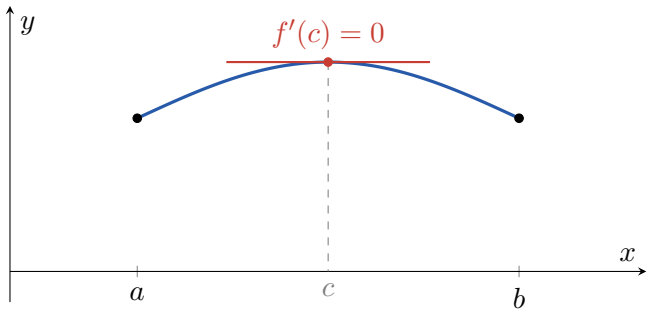


Figure 15.2: Rolle's theorem: equal endpoint heights force a horizontal tangent somewhere inside.

15.2.1 Why it has to be true

Here is the idea before the proof, because the idea is the whole point. A continuous function on a closed interval has to attain a highest value and a lowest value somewhere — it cannot run off to infinity, and it cannot sneak up to a maximum without reaching it, because there are no gaps. (This is the Extreme Value Theorem, the companion of the Intermediate Value Theorem from the continuity chapter.) Now, the endpoints are tied together at the same height. So at least one of the highest or lowest point must occur *strictly inside* the interval — unless the function is perfectly flat, in which case every interior point works anyway. And at an interior peak or valley, a differentiable function must have derivative zero. Why? Because at a peak the function isn't rising on either side, so its slope can be neither positive nor negative. The only number that is neither positive nor negative is zero.

Why the technique works

The hidden engine of Rolle's theorem is a tiny fact worth isolating, sometimes called *Fermat's theorem on interior extrema*: *if f has a local maximum or minimum at an interior point c and $f'(c)$ exists, then $f'(c) = 0$.*
Watch the mechanism. Suppose c is a local max. For $h > 0$ small, $f(c + h) \leq f(c)$, so the right-hand difference quotient $\frac{f(c+h)-f(c)}{h} \leq 0$; its limit gives $f'(c) \leq 0$. For $h < 0$ small, $f(c+h) \leq f(c)$ still, but now we divide by a *negative* h, flipping the inequality: $\frac{f(c+h)-f(c)}{h} \geq 0$, so $f'(c) \geq 0$. The derivative exists, so both one-sided limits equal it. A number that is both ≤ 0 and ≥ 0 is 0. The two sides of the peak squeeze the slope to zero from opposite directions.

Proof of Rolle's theorem. Since f is continuous on the closed, bounded interval $[a, b]$, the Extreme Value Theorem guarantees that f attains a maximum value M and a minimum value m on $[a, b]$.

Case 1: $M = m$. Then f is constant, $f'(x) = 0$ for every $x \in (a, b)$, and any interior point serves as c.

Case 2: $M > m$. At least one of these two values differs from the common endpoint height $f(a) = f(b)$. Say it is the maximum M (the minimum case is identical). Then the point where f attains M cannot be an endpoint, so it is some interior point $c \in (a, b)$. At c, f has a local maximum, f is differentiable, and by Fermat's interior-extremum fact $f'(c) = 0$. □

Common trap

Every hypothesis earns its keep. Drop any one and Rolle fails:

- *Continuity on* $[a, b]$. Let $f(x) = x$ on $[0, 1)$ and redefine $f(1) = 0$. Endpoints match ($f(0) = f(1) = 0$), but the derivative is 1 everywhere it exists — no horizontal tangent. The jump at the right end let the function "escape".
- *Differentiability on* (a, b). Let $f(x) = |x|$ on $[-1, 1]$. Endpoints match, f is continuous, but the only place the slope could be zero is the corner at 0, where f' does not exist. The peak is there, but it is a spike, not a smooth top.
- *Equal endpoints.* Without $f(a) = f(b)$ there is no reason for a flat spot — that is precisely the gap the full MVT fills next.

15.3 Tilting Rolle into the Mean Value Theorem

Rolle's theorem is the MVT for a level interval. To get the general statement, we play a trick that is one of the most useful moves in all of analysis: we *subtract off the secant line* so that the tilted problem becomes a level one.

Intuition

Imagine the MVT picture drawn on a tilted sheet of paper. Now rotate your head so the secant looks horizontal. In that tilted view, the curve starts and ends at the same height — it is exactly a Rolle picture! Rolle hands us a horizontal tangent *in the tilted frame*, which is a tangent parallel to the secant *in the original frame*. We do the "rotation" algebraically by subtracting the secant's formula from f.

Theorem 15.2 (Mean Value Theorem, Lagrange form)**.** *Let f be continuous on $[a, b]$ and differentiable on (a, b). Then there exists at least one point $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Equivalently, $f(b) - f(a) = f'(c)\,(b - a)$ for some c strictly between a and b.

Proof. The secant line through the endpoints is

$$L(x) = f(a) + \frac{f(b) - f(a)}{b - a}\,(x - a),$$

a straight line, so $L'(x) = \dfrac{f(b) - f(a)}{b - a}$ everywhere. Define the *gap function*

$$g(x) = f(x) - L(x),$$

the vertical distance from the curve to its own secant. Then:

- g is continuous on $[a, b]$ and differentiable on (a, b), being a difference of two such functions.
- $g(a) = f(a) - L(a) = f(a) - f(a) = 0$, and similarly $g(b) = f(b) - L(b) = 0$. The curve meets its secant at the endpoints, so the gap is zero there.

Thus g satisfies the hypotheses of Rolle's theorem with $g(a) = g(b) = 0$. Rolle gives a point $c \in (a, b)$ with $g'(c) = 0$. But

$$g'(x) = f'(x) - L'(x) = f'(x) - \frac{f(b) - f(a)}{b - a},$$

so $g'(c) = 0$ means exactly $f'(c) = \dfrac{f(b) - f(a)}{b - a}$. □

What is really happening

What is $g(x) = f(x) - L(x)$ really? It is the signed vertical gap between the curve and the straight line joining its ends. The proof says: the place where this gap is widest (or narrowest) — where $g' = 0$ — is the place where the curve is, for an instant, running *parallel* to the secant. That is geometrically inevitable. As you move along the curve away from the secant and then back to it, there has to be a turning point in the gap, and at that turning point the curve and the secant are momentarily heading the same direction. The MVT is not really about derivatives at all; it is about the fact that a gap that starts and ends at zero must, somewhere, stop growing.

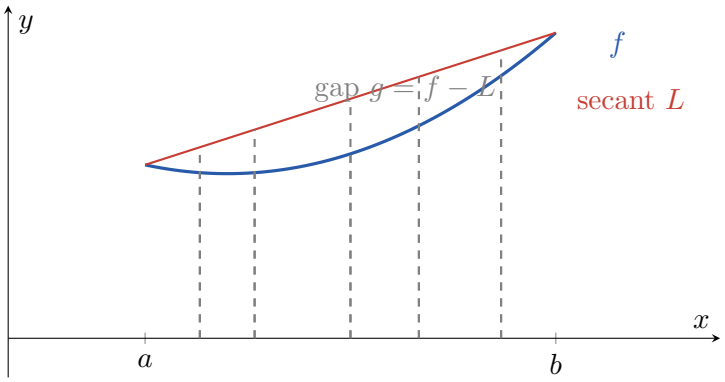


Figure 15.3: The proof in one image: $g = f - L$ is the gap between the curve and its secant. It vanishes at both ends, so by Rolle it has a horizontal tangent somewhere — and there f runs parallel to L.

History

The names attached to this theorem tell a story of fierce skepticism and slow philosophical evolution. **Michel Rolle** (1652–1719) was a conservative French algebraist and an outspoken critic of the early calculus. In the 1690s, Rolle engaged in high-profile debates at the Académie des Sciences in Paris, denouncing the infinitesimal methods of Leibniz and l'Hôpital as a collection of "ingenious fallacies" founded on unsound mysticism. In 1691, in his treatise *Méthode pour résoudre les égalitez*, Rolle proved an algebraic result purely for polynomial equations: between any two real roots of a polynomial $P(x) = 0$, there lies at least one root of what he called its derivative polynomial. Rolle never envisioned this as a statement about continuous curves or general rates of change. By a wonderful historical irony, the critic who sought to demolish calculus had his name permanently attached to its most foundational theorem.

A century later, **Joseph-Louis Lagrange** (1736–1813) realized that calculus needed a rigorous foundation free of vanishing infinitesimals. In his 1797 *Théorie des fonctions analytiques*, Lagrange attempted to rebuild the entire discipline on pure algebra, making the Mean Value Theorem the bedrock from which all of calculus should be deduced. While Lagrange's hope of bypassing limits entirely proved impossible, his intuition that the MVT was the central bridge connecting derivatives to functions was exactly right.
The modern proof and the complete conceptual structure belong to **Augustin-Louis Cauchy** (1789–1857). In his 1823 lectures at the École Polytechnique, Cauchy recognized that proving Rolle's theorem requires the Extreme Value Theorem—which in turn rests on the completeness of the real number line. Cauchy also introduced the crucial generalization known as the *Cauchy Mean Value Theorem*, which considers two functions f and g simultaneously. The MVT was used intuitively for over a century by Newton, Leibniz, and the Bernoullis before Cauchy and later Karl Weierstrass built the rigorous foundations that made its proof airtight.

15.3.1 Reading the theorem carefully

The MVT is an *existence* theorem. It promises that such a point c exists; it does not tell you where, and there may be several. For most purposes this is exactly enough, because we will use it to say "f' at some unknown point equals this average," and then bound or sign that unknown derivative.

Focus

The two faces of the MVT. Keep both forms in mind; you will reach for different ones in different situations.

- *Slope form:* $f'(c) = \dfrac{f(b) - f(a)}{b - a}$. Read it as "the average rate is hit exactly."
- *Increment form:* $f(b) - f(a) = f'(c)(b - a)$. Read it as "the total change equals some interior rate times the elapsed input." This is the workhorse form: it converts a statement about f' into a statement about $f(b) - f(a)$.

Example 15.1. Let $f(x) = x^3$ on $[0, 2]$. The average slope is $\frac{f(2)-f(0)}{2-0} = \frac{8-0}{2} = 4$. The MVT promises a $c \in (0, 2)$ with $f'(c) = 3c^2 = 4$, i.e. $c^2 = \frac{4}{3}$, $c = \frac{2}{\sqrt{3}} \approx 1.155$. Indeed 1.155 lies inside $(0, 2)$, as advertised. Notice we *found* c here only because the function is simple; in real applications we almost never solve for c — we just use the fact that some such c exists.

Work it yourself

Take $f(x) = \sqrt{x}$ on $[1, 4]$. Compute the average slope $\frac{\sqrt{4}-\sqrt{1}}{4-1} = \frac{2-1}{3} = \frac{1}{3}$. Now set $f'(c) = \frac{1}{2\sqrt{c}} = \frac{1}{3}$ and solve: $2\sqrt{c} = 3$, $\sqrt{c} = \frac{3}{2}$, $c = \frac{9}{4} = 2.25$. Check that 2.25 sits in $(1, 4)$. You have just located the exact instant where this curve's slope equals its average climb. Try the same with $f(x) = 1/x$ on $[1, 2]$ and confirm $c = \sqrt{2}$.

15.4 Why the MVT is the workhorse

Now the payoff. Everything in this section follows from the single increment form $f(b) - f(a) = f'(c)(b-a)$. The strategy is always the same: *whatever you know about f' on the interval, you know about that mystery factor $f'(c)$, and the equation passes the knowledge on to $f(b)-f(a)$.* Watch how many "obvious" facts turn out to be this one theorem in disguise.

15.4.1 Zero derivative means constant

You believe that if a function never changes (its rate is always 0) then it stays put. But why, *exactly*? "The derivative is zero" is a statement about slopes; "the function is constant" is a statement about values. The MVT is the only rigorous bridge between them.

Theorem 15.3 (Vanishing derivative)**.** *If f is continuous on $[a,b]$, differentiable on (a,b), and $f'(x) = 0$ for all $x \in (a,b)$, then f is constant on $[a,b]$.*

Proof. Pick any point $x \in (a,b]$. Apply the MVT to f on the subinterval $[a,x]$: there is a c between a and x with

$$f(x) - f(a) = f'(c)\,(x-a).$$

But $f'(c) = 0$ by hypothesis, so $f(x) - f(a) = 0$, i.e. $f(x) = f(a)$. Since x was arbitrary, f equals the constant $f(a)$ everywhere. □

Common trap

This is exactly the place where the *interval matters*. On a disconnected domain the conclusion is false. The function

$$f(x) = \begin{cases} 1, & x > 0, \\ -1, & x < 0, \end{cases}$$

has $f'(x) = 0$ everywhere it is defined, yet it is not constant. There is no contradiction: the MVT needs a single interval connecting any two points, and there is no such interval joining a negative x to a positive one inside the domain. The derivative tells you nothing about how f behaves *across a gap*.

15.4.2 Two antiderivatives differ by a constant

This is the fact that makes the "$+\,C$" in every indefinite integral legitimate. If you have found *one* antiderivative of a function, you have essentially found them all.

Corollary 15.1 (Antiderivatives up to a constant)**.** *If $F'(x) = G'(x)$ for all x in an interval I, then $F(x) = G(x) + C$ on I for some constant C.*

Proof. Let $h = F - G$. Then $h'(x) = F'(x) - G'(x) = 0$ for all $x \in I$. By the previous theorem, h is constant on I: $h(x) = C$. Hence $F(x) = G(x) + C$. □

Why this matters

This little corollary is quietly enormous. It is the reason the Fundamental Theorem of Calculus can speak of *the* antiderivative up to a constant; the reason $\int f$ is well posed; the reason solving a differential equation $y' = f$ gives a one-parameter family rather than chaos. The entire bookkeeping of integration — that mysterious $+C$ you were told to "never forget" — is the MVT speaking. You were not memorizing a rule. You were memorizing a theorem.

15.4.3 The increasing/decreasing test

Here is the workhorse role most people meet first without realizing where it comes from. We *define* "increasing" by values: f is increasing on I if $x_1 < x_2$ forces $f(x_1) < f(x_2)$. The sign of the derivative is, again, a statement about slopes. The MVT connects the two.

Theorem 15.4 (Monotonicity test)**.** *Let f be continuous on an interval I and differentiable in its interior.*

- *If $f'(x) > 0$ throughout the interior, then f is strictly increasing on I.*
- *If $f'(x) < 0$ throughout, then f is strictly decreasing on I.*
- *If $f'(x) \geq 0$ throughout, then f is nondecreasing; if $f'(x) \leq 0$, nonincreasing.*

Proof. Take any two points $x_1 < x_2$ in I. By the MVT on $[x_1, x_2]$ there is a c between them with

$$f(x_2) - f(x_1) = f'(c)\,(x_2 - x_1).$$

The factor $(x_2 - x_1)$ is positive. So the sign of $f(x_2) - f(x_1)$ is exactly the sign of $f'(c)$. If $f' > 0$ everywhere, then $f'(c) > 0$, so $f(x_2) - f(x_1) > 0$, i.e. $f(x_2) > f(x_1)$: increasing. The other three cases are read off the same equation by changing the sign or allowing equality. □

Why the technique works

Why does the *sign* of the derivative control the *direction* of the function? Because of that single MVT equation. Without it, "$f' > 0$" and "f increasing" are two unrelated sentences — one about tangent lines, one about a table of values. The MVT says the total change $f(x_2) - f(x_1)$ is built out of an actual interior slope times a positive width. You cannot accumulate a net *decrease* out of slopes that are all positive. The local information (slope at points) integrates up into global information (rise across an interval), and the MVT is the gear that meshes the two.

Common trap

A subtle but important converse warning. "Strictly increasing" does *not* require $f' > 0$ at every point. Consider $f(x) = x^3$. It is strictly increasing on all of $\mathbb{R}$, yet $f'(0) = 0$. The theorem's arrow points one way: $f' > 0 \Rightarrow$ increasing, but increasing only gives $f' \geq 0$. A single flat instant is allowed, as long as the function does not actually pause over a whole subinterval. Many students "prove" a function is increasing by checking $f' \geq 0$ and then wrongly assert it must be *strictly* increasing; that needs the extra observation that f' is not

zero on any subinterval.

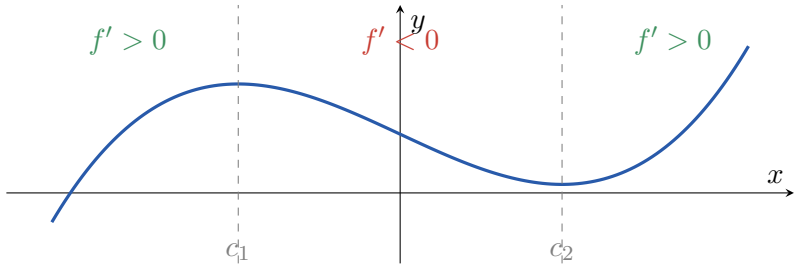


Figure 15.4: The sign of f' reads off the direction of f. Where $f' > 0$ the curve climbs; where $f' < 0$ it falls; the turning points c_1, c_2 are where $f' = 0$. The monotonicity test is just the MVT applied on each piece.

15.4.4 Error bounds: the MVT as a ruler

Perhaps the most *useful* face of the MVT in applied work is its power to *bound* how much a function can change. If you know the speedometer never exceeds 70, you know that in two hours you went at most 140 miles. That is the MVT used as an inequality.

Corollary 15.2 (Change is bounded by the steepest slope)**.** *Suppose f is differentiable on an interval and $|f'(x)| \le M$ for all x there. Then for any two points a, b in the interval,*

$$|f(b) - f(a)| \le M\,|b - a|.$$

Proof. By the MVT, $f(b) - f(a) = f'(c)(b - a)$ for some interior c. Take absolute values: $|f(b) - f(a)| = |f'(c)|\,|b - a| \le M|b - a|$, since $|f'(c)| \le M$. □

A function whose derivative is bounded by M in this way is called *Lipschitz* with constant M: it cannot change faster than M units of output per unit of input, ever. This single inequality is the backbone of error analysis, numerical methods, and the existence-and-uniqueness theory for differential equations.

Example 15.2 (Bounding a value of sine)**.** How big can $\sin(0.1)$ be, without a calculator? Apply the bound to $f = \sin$ on $[0, 0.1]$. We have $f' = \cos$, and $|\cos x| \le 1$ everywhere, so $M = 1$. Then

$$|\sin(0.1) - \sin(0)| \le 1 \cdot |0.1 - 0| = 0.1.$$

Thus $|\sin(0.1)| \le 0.1$. We just proved the famous inequality $|\sin x| \le |x|$ for all x — which is normally drawn from a picture — as a one-line consequence of the MVT, because $|(\sin)'| = |\cos| \le 1$.

Example 15.3 (A sharper estimate near a known value)**.** Estimate $\sqrt{4.1}$ and bound the error of the estimate 2.025. With $f(x) = \sqrt{x}$ on $[4, 4.1]$, the MVT gives

$$\sqrt{4.1} - \sqrt{4} = f'(c)(0.1) = \frac{1}{2\sqrt{c}}(0.1) \quad \text{for some } c \in (4, 4.1).$$

Now c is somewhere in $(4, 4.1)$, so $\frac{1}{2\sqrt{c}}$ lies between $\frac{1}{2\sqrt{4.1}}$ and $\frac{1}{2\sqrt{4}} = 0.25$. Hence

$$\sqrt{4.1} = 2 + \frac{0.1}{2\sqrt{c}}, \qquad 2 + 0.1(0.2469) \le \sqrt{4.1} \le 2 + 0.1(0.25),$$

so $2.02469 \le \sqrt{4.1} \le 2.025$. The MVT didn't just give a guess; it *trapped* the true value between two computable numbers. This is the engine behind rigorous numerical bounds.

Application

Where this bites in practice. In numerical analysis, every statement of the form "this approximation is accurate to within ϵ" is ultimately an MVT estimate: the error is some derivative (or higher derivative, via Taylor's theorem, which is the MVT applied repeatedly) evaluated at an unknown interior point, and you bound it by the worst-case size of that derivative. In machine learning, a loss function with a derivative bounded by M (an M-Lipschitz gradient) is exactly the condition that gradient descent with a small enough step size cannot overshoot — the same Corollary 15.2 controlling how far a step can move you. In control and physics, a bounded force means a bounded change in velocity over a bounded time, again the very same inequality.

15.4.5 Proving inequalities without a single graph

The MVT turns many "stare at the graph and believe it" inequalities into clean proofs. The recipe: build a function that is the difference of the two sides, show its derivative has a fixed sign, and let monotonicity finish the job.

Example 15.4 ($e^x \ge 1 + x$ for all x)**.** Let $g(x) = e^x - (1 + x)$. Then $g(0) = 0$ and $g'(x) = e^x - 1$. Now $g'(x) > 0$ for $x > 0$ and $g'(x) < 0$ for $x < 0$. By the monotonicity test, g decreases up to 0 and increases after it, so $x = 0$ is a global minimum of g, with value $g(0) = 0$. Hence $g(x) \ge 0$ everywhere, i.e. $e^x \ge 1 + x$. The tangent line $1 + x$ lies below the exponential (Figure 15.5), proved purely from the sign of a derivative.

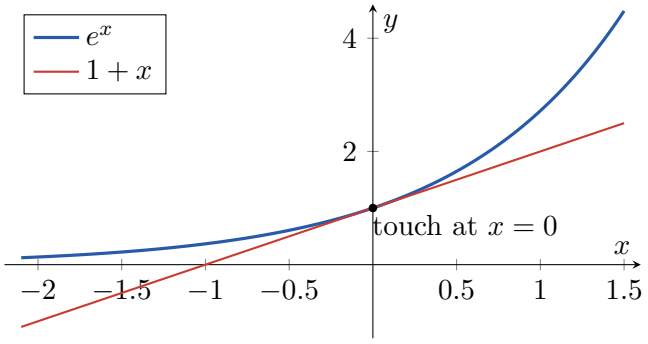


Figure 15.5: The tangent line $y = 1 + x$ at $x = 0$ lies entirely below $y = e^x$, touching only at $(0, 1)$. The single fact $g'(x) = e^x - 1$ changes sign exactly at 0 forces this: $g = e^x - (1 + x)$ falls to its minimum at 0 and climbs away on both sides.

Example 15.5 ($\ln(1 + x) < x$ for $x > 0$)**.** Let $g(x) = x - \ln(1 + x)$ on $[0, \infty)$. Then $g(0) = 0$ and $g'(x) = 1 - \frac{1}{1+x} = \frac{x}{1+x} > 0$ for $x > 0$. So g is strictly increasing on $[0, \infty)$, giving $g(x) > g(0) = 0$, that is $\ln(1 + x) < x$. Same recipe, opposite function.

Walking question

Here is a question to chew on next time you are out walking. The monotonicity test turned "slope is positive" into "function is increasing" using the MVT. But the MVT itself rested on Rolle, which rested on the Extreme Value Theorem, which rested on the *completeness* of the real numbers — the fact that there are no gaps in the number line. So when you say "a positive derivative makes a function climb," you are, four layers down, relying on there being no holes in $\mathbb{R}$. Can you feel why a function on the rationals only, with no irrational points, could fail the whole chain? (Hint: a continuous function on the rationals can fail to attain its maximum, because the max might want to live at an irrational point that isn't there.)

15.5 The Cauchy Mean Value Theorem

There is a powerful generalization that you will need the moment you reach l'Hôpital's rule, so we prove it here while the machinery is fresh. It compares the changes in *two* functions over the same interval.

Theorem 15.5 (Cauchy's Mean Value Theorem)**.** *Let f and g be continuous on $[a,b]$ and differentiable on (a,b). Then there is a point $c \in (a,b)$ with*

$$\big(f(b)-f(a)\big)\,g'(c) = \big(g(b)-g(a)\big)\,f'(c).$$

If g' is never zero on (a,b) (so $g(b) \neq g(a)$), this rearranges to

$$\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}.$$

Intuition

Think of $\big(g(t), f(t)\big)$ as a point tracing a curve in the plane as t runs from a to b. The ordinary MVT compared a slope to an average slope along the x-axis. Cauchy's version compares the *direction the curve is heading*, $f'(c)/g'(c)$, to the *straight-line direction* from start to end, $\frac{f(b)-f(a)}{g(b)-g(a)}$. It is the MVT for a parametrized path: somewhere the path points the same way as the chord joining its endpoints.

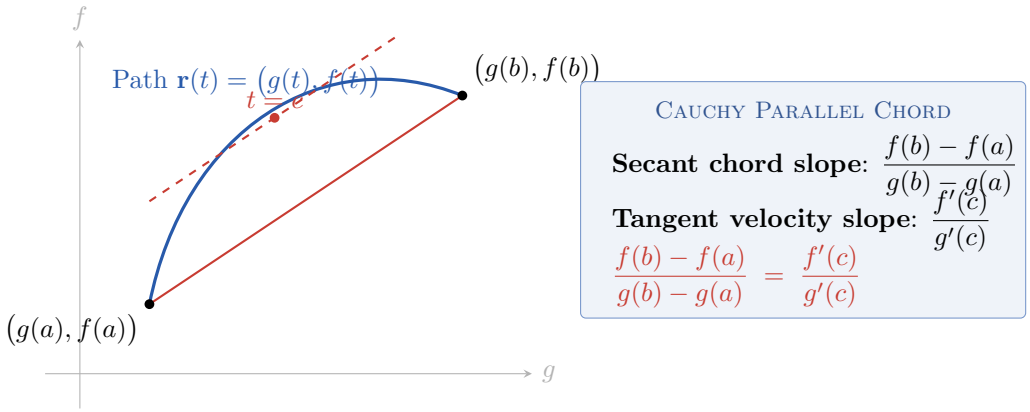


Figure 15.6: Cauchy's Mean Value Theorem viewed in the plane of the two functions (g, f) (Figure 15.6). The point $\mathbf{r}(t) = (g(t), f(t))$ traces a curve from $(g(a), f(a))$ to $(g(b), f(b))$. The secant chord connecting the endpoints has slope $\frac{f(b)-f(a)}{g(b)-g(a)}$. At some interior parameter value $t = c$, the instantaneous velocity vector $(g'(c), f'(c))$ points in the exact same direction, matching the chord's slope: $\frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}$.

Proof. We use the same subtract-a-helper trick that turned Rolle into the MVT, only now the helper is chosen to kill both endpoints at once. Define

$$h(x) = \big(f(b) - f(a)\big)\, g(x) - \big(g(b) - g(a)\big)\, f(x).$$

This h is continuous on $[a, b]$ and differentiable on (a, b). Compute the endpoint values:

$$h(a) = \big(f(b) - f(a)\big)g(a) - \big(g(b) - g(a)\big)f(a) = f(b)g(a) - g(b)f(a),$$
$$h(b) = \big(f(b) - f(a)\big)g(b) - \big(g(b) - g(a)\big)f(b) = f(b)g(a) - g(b)f(a).$$

The cross terms cancel and $h(a) = h(b)$. So Rolle applies: there is $c \in (a, b)$ with $h'(c) = 0$. But

$$h'(x) = \big(f(b) - f(a)\big)g'(x) - \big(g(b) - g(a)\big)f'(x),$$

and $h'(c) = 0$ is precisely the claimed identity. □

Connection

Notice the ordinary MVT is the special case $g(x) = x$: then $g'(c) = 1$ and $g(b) - g(a) = b - a$, and Cauchy's identity collapses to $f(b) - f(a) = f'(c)(b - a)$. So Cauchy's theorem genuinely contains the Lagrange MVT, and both descend from Rolle. Three theorems, one idea — "a difference that vanishes at both ends has a flat spot inside" — applied to three cleverly chosen helper functions. When you reach l'Hôpital's rule a couple of chapters from now, that quotient $\frac{f'(c)}{g'(c)}$ will be exactly what lets a ratio of functions be replaced by a ratio of their derivatives.

15.6 A glance ahead: the MVT seeds Taylor's theorem

The increment form $f(b) - f(a) = f'(c)(b - a)$ can be read as a *first-order* exact remainder: the change in f is captured exactly by one derivative at one mystery point. Taylor's theorem, which you

will meet in full later, is the same idea pushed to higher order. It says

$$f(b) = f(a) + f'(a)(b-a) + \tfrac{1}{2}f''(c)(b-a)^2$$

for some interior c — a "second-order MVT" whose proof is, once again, build a helper that vanishes at both ends and apply Rolle. The whole tower of polynomial approximation, and every error bound that comes with it, grows from the seed you planted in this chapter.

Why this matters

Step back and look at what one humble picture accomplished. "An average slope is attained somewhere" gave us: the meaning of the constant of integration, the test that turns derivative signs into shapes, a rigorous reason a zero rate keeps a quantity fixed, sharp numerical error bounds, a clean way to prove inequalities, the key lemma behind l'Hôpital, and the seed of Taylor's theorem. Few facts in mathematics carry so much on so slender a frame. That is what we mean by a *workhorse*: it is rarely the star of the show, but almost nothing downstream would stand without it.

15.6.1 The Scope and Horizons of the Mean Value Theorem

Connection

The Mean Value Theorem is the logical linchpin of single-variable analysis. It provides the mathematical mechanism that converts purely local information into global guarantees across an entire interval:

- *Monotonicity*: It turns local derivative signs ($f'(x) > 0$) into a global guarantee that f is strictly increasing.
- *Antiderivative uniqueness*: It proves that if $f'(x) = 0$ on an interval, f must be constant, ensuring that all antiderivatives differ by $+C$.
- *Global bounding and error control*: If $|f'(x)| \leq M$, then $|f(b) - f(a)| \leq M|b-a|$, delivering rigorous error bounds for numerical schemes.

Yet the MVT is a pure *existence theorem*. It guarantees that an intermediate point c exists where the tangent matches the secant, but it offers no algorithm whatsoever for locating c. For anything beyond basic quadratics, finding c means solving $f'(c) = \frac{f(b)-f(a)}{b-a}$, which is often a transcendental equation with no algebraic solution. The MVT guarantees that your car speedometer hit 60 mph on the highway, but tells you nothing about the mile-marker where it occurred.

From this theoretical bedrock, the curriculum branches forward:

1. In Chapter 16 (*Maxima, Minima, and the Shape of a Graph*), the MVT's monotonicity and concavity criteria become our primary blueprint for curve sketching and classifying extrema.
2. In Chapter 18 (*Indeterminate Forms and L'Hôpital's Rule*), we harness Cauchy's generalized MVT to resolve indeterminate limits of the form $0/0$ and ∞/∞ by comparing instantaneous growth rates.

15.7 Exercises

Exercise 15.1 (Finding c)**.** For each function and interval, verify the MVT hypotheses and find every c the theorem promises. (a) $f(x) = x^2 - 3x$ on $[1, 4]$. (b) $f(x) = x + \frac{1}{x}$ on $[1, 3]$. (c) $f(x) = \sin x$ on $[0, \pi]$.

Exercise 15.2 (Rolle by hand)**.** Let $f(x) = x^3 - 3x$. Find two values $a < b$ with $f(a) = f(b)$, then locate the interior point(s) where $f' = 0$ guaranteed by Rolle's theorem. Sketch the graph and mark the horizontal tangents.

Exercise 15.3 (Why a hypothesis is needed)**.** Give an explicit function on $[-1, 1]$ that is continuous, has $f(-1) = f(1)$, but has *no* point where $f' = 0$. Explain precisely which hypothesis of Rolle's theorem your example violates, and where.

Exercise 15.4 (A speed argument)**.** A car travels 90 miles in exactly 90 minutes. Use the MVT to prove that at some instant its instantaneous speed was exactly 60 mph. Then prove the weaker claim that at some instant its speed was *at least* 60 mph, using only the average, without the MVT. Which statement is stronger, and why does the MVT earn the stronger one?

Exercise 15.5 (Bounding with the derivative)**.** Suppose f is differentiable on $\mathbb{R}$ with $f(1) = 5$ and $|f'(x)| \le 2$ for all x. Find the smallest interval that must contain $f(4)$. (Use Corollary 15.2.)

Exercise 15.6 (Inequalities from monotonicity)**.** Prove each inequality by forming a difference g, signing g', and using the monotonicity test. (a) $\tan x > x$ for $0 < x < \frac{\pi}{2}$. (b) $\cos x > 1 - \frac{x^2}{2}$ for $x \neq 0$. (c) $\frac{x}{1+x} < \ln(1 + x)$ for $x > 0$.

Exercise 15.7 (Uniqueness of a solution)**.** Show that the equation $x^5 + x - 1 = 0$ has exactly one real solution. (Existence: use the Intermediate Value Theorem on $[0, 1]$. Uniqueness: if there were two solutions, Rolle's theorem would force $f' = 0$ somewhere between them — show that is impossible.)

Exercise 15.8 (Constant on an interval)**.** Suppose f is differentiable on $\mathbb{R}$ with $f'(x) = 3$ for all x. Prove, using the MVT and not antidifferentiation rules, that $f(x) = 3x + f(0)$ for every x.

Exercise 15.9 (A fixed-point bound)**.** Let g be differentiable on $\mathbb{R}$ with $|g'(x)| \le \frac{1}{2}$ for all x. Show that for any two numbers p, q, $|g(p) - g(q)| \le \frac{1}{2}|p - q|$. (Such a g is a *contraction*; this inequality is the heart of why iterating g converges to a single fixed point.)

Exercise 15.10 (Proof: a derivative that is always positive)**.** Prove carefully, citing the MVT, that if $f'(x) > 0$ for all x in an open interval I, then f is one-to-one on I. Where exactly do you use that the interval is connected?

Exercise 15.11 (Cauchy MVT in action)**.** Let $f(x) = x^2$ and $g(x) = x^3$ on $[1, 2]$. Find the point c guaranteed by Cauchy's Mean Value Theorem, i.e. solve $\frac{f(2)-f(1)}{g(2)-g(1)} = \frac{f'(c)}{g'(c)}$ for $c \in (1, 2)$. Confirm your c lies in the interval.

Exercise 15.12 (Conceptual: the role of the EVT)**.** Explain in a short paragraph why Rolle's theorem genuinely needs the Extreme Value Theorem, and why the Extreme Value Theorem in turn needs the completeness of $\mathbb{R}$. Give an example of a continuous function on the rationals in $[0, 2]$ (with the usual formula) that does *not* attain a maximum, illustrating what breaks without completeness.

Part IV

Using the Derivative

Chapter 16

Maxima, Minima, and the Shape of a Graph

Here is a question so ordinary you may never have thought to ask it carefully: how do you *know* what a graph looks like? You have surely sketched curves before — plotted a few points, connected the dots, trusted that the picture between the dots behaved itself. But that is guessing. A function can do surprising things between two sample points. It can dive down and back up. It can have a sharp bend you would never see at the resolution of your dots. To draw a graph *correctly* — to know where it rises, where it falls, where it turns around, where it bulges up and where it sags down — you need more than a handful of values. You need the derivative, and you need the derivative of the derivative.

This chapter is about reading a function the way a doctor reads an X-ray: not by staring at the surface, but by looking at the structure underneath. The first derivative f' tells you the *direction* the curve is heading — uphill or downhill. The second derivative f'' tells you how that direction is *changing* — whether the road is curving toward you or away from you. Put the two together and you can reconstruct the whole shape of a graph from almost nothing, and, far more importantly, you can find its highest and lowest points. Those highest and lowest points are the payoff. They are what optimization is made of: the cheapest design, the strongest beam, the fastest route, the maximum profit, the least error. All of it begins here, with the simple observation that *at the top of a hill, the ground is momentarily level.*

Everything in this chapter rests on the Mean Value Theorem from the previous chapter. The MVT was the bridge from f' back to f. Now we cross that bridge over and over. When we claim "$f' > 0$ means f is increasing," that is the MVT talking. When we claim a sign change in f' means a genuine peak, that is the MVT again. We will keep pointing back to it, because the real reason these tests *work* — and are not just plausible folklore — is that one theorem.

16.1 The top of the hill is level

Stand at the very top of a smooth hill. Look at the ground right at your feet. It is flat. Not flat far away — the hill falls off on both sides — but flat *right there*, at the summit. Geography forces this. If the ground at the summit were tilted even slightly uphill in some direction, then you could walk that way and go higher — so you were not at the summit after all. The only way to be at the very

top is for every direction to be level or downhill, and for a smooth curve in one variable that means the tangent line is horizontal.

Intuition

At a peak or a valley of a smooth graph, the tangent line is horizontal: the slope is zero. The reason is almost a tautology once you see it. To the left of a peak the function is climbing ($f' > 0$); to the right it is falling ($f' < 0$). A continuous slope cannot jump from positive to negative without passing through zero. So right at the top, $f' = 0$. The summit is where "going up" turns into "going down," and the turning point is level.

Let us make the words precise, because "highest point" hides a real subtlety: highest compared to *what*? Compared to the entire domain, or just compared to nearby points?

Definition 16.1 (Local and global extrema)**.** Let f be defined on a set D and let $c \in D$.

- f has a *local maximum* at c if $f(c) \geq f(x)$ for all x in some open interval around c (a local minimum if $f(c) \leq f(x)$ nearby).
- f has a *global* (or *absolute*) *maximum* at c if $f(c) \geq f(x)$ for *all* $x \in D$ (global minimum if $f(c) \leq f(x)$ for all $x \in D$).

A *local extremum* is either a local max or a local min.

A local maximum is a peak relative to its immediate surroundings — the top of *a* hill, even if a taller mountain stands a mile away. A global maximum is the tallest point in the whole landscape. Every interior global extremum is automatically a local one (being highest overall makes you highest locally), but not the reverse.

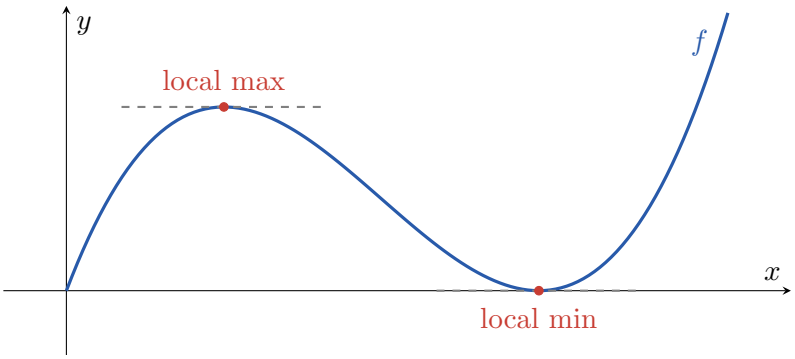


Figure 16.1: At a local maximum and a local minimum of a smooth curve, the tangent line (dashed) is horizontal. The peak is where climbing turns into falling.

16.1.1 Fermat's theorem: where to look

The flatness-at-the-summit idea has a name and a proof. It is one of the oldest results in the subject, found by Pierre de Fermat decades before Newton and Leibniz had a systematic calculus.

Theorem 16.1 (Fermat's theorem on extrema)**.** *If f has a local extremum at c, and f is differentiable at c, then $f'(c) = 0$.*

Intuition

The summit argument translates directly into algebra, exposing why differentiability is the essential hypothesis. The derivative is a single number, the limit of the difference quotients $\dfrac{f(c+h)-f(c)}{h}$ as $h \to 0$. Approaching from the right and from the left, both one-sided limits must converge to that same value. At a local maximum, the numerator $f(c+h)-f(c)$ is ≤ 0 for all sufficiently small h (no nearby point is higher). When $h > 0$, the quotient is nonpositive divided by positive, so ≤ 0. When $h < 0$, the quotient is nonpositive divided by negative, so ≥ 0. The right-hand limit forces $f'(c) \leq 0$, while the left-hand limit forces $f'(c) \geq 0$. The only real number that is simultaneously ≤ 0 and ≥ 0 is 0.

Proof. Suppose f has a local maximum at c (the minimum case is identical with inequalities flipped). Then there is an interval around c on which $f(x) \leq f(c)$, i.e. $f(c+h)-f(c) \leq 0$ for all small h.

For $h > 0$,

$$\frac{f(c+h)-f(c)}{h} \leq 0 \qquad \text{(nonpositive over positive)},$$

so the right-hand limit satisfies $f'(c) = \lim_{h\to 0^+} \dfrac{f(c+h)-f(c)}{h} \leq 0$.

For $h < 0$,

$$\frac{f(c+h)-f(c)}{h} \geq 0 \qquad \text{(nonpositive over negative)},$$

so the left-hand limit satisfies $f'(c) = \lim_{h\to 0^-} \dfrac{f(c+h)-f(c)}{h} \geq 0$.

Because f is differentiable at c, these two one-sided limits are equal and both equal $f'(c)$. The only way to have $f'(c) \leq 0$ and $f'(c) \geq 0$ at once is $f'(c) = 0$. □

History

The discovery that extrema occur where rates of change vanish predates both Newton and Leibniz. In 1636, Pierre de Fermat circulated a short manuscript, *Methodus ad disquirendam maximam et minimam*, presenting his method of *adequality*. Fermat was inspired by an observation of Pappus of Alexandria on dividing a line segment: at a maximum, a slight perturbation in the input produces virtually no change in the output. Fermat replaced x with $x+e$, set $f(x+e)$ "adequal" (approximately equal) to $f(x)$, divided through by the perturbation e, and then suppressed all terms still containing e. In modern language, Fermat was evaluating $\lim_{e\to 0} \frac{f(x+e)-f(x)}{e} = 0$, identifying peaks and troughs by finding stationary values forty years before the invention of calculus.

The second half of the story—the geometry of bending and concavity—belongs to Gottfried Wilhelm Leibniz. In his 1684 paper and subsequent memoirs in *Acta Eruditorum*, Leibniz analyzed what he called *puncta flexus contrarii*: inflection points where a curve transitions between bending upward and bending downward. Leibniz realized that just as the first differential $\mathrm{d}y$ determines the slope, the second differential d^2y determines the bending or curvature. At an inflection point, $\mathrm{d}^2y = 0$, and the tangent line actually slices through the curve rather than merely grazing one side.

Fermat and Leibniz understood the geometric intuition perfectly, but lacked the limit concept to justify why vanishing terms could be cancelled. In the nineteenth century, Cauchy and

Weierstrass provided the rigorous basis with the ε-δ limit and the Mean Value Theorem, proving that Fermat's stationary points and Leibniz's curvature transitions are exact consequences of real analysis.

Common trap

Fermat's theorem is a one-way street, and reading it backwards is the single most common mistake in this whole subject. $f'(c) = 0$ does *not* guarantee an extremum. The graph of $f(x) = x^3$ has $f'(0) = 0$, yet 0 is neither a max nor a min — the curve flattens for an instant and keeps right on rising. A zero derivative is a *candidate*, a place worth checking, nothing more. The theorem tells you where to look, not what you will find.

16.1.2 Critical points: the suspects

Fermat's theorem says smooth extrema hide where $f' = 0$. But there is a second way a function can turn around: at a *corner*, where the derivative does not exist at all. The function $f(x) = |x|$ has its minimum at 0, and $f'(0)$ simply does not exist — the slope is -1 on the left and $+1$ on the right, with no single tangent. Fermat's theorem said nothing here, because its hypothesis ("differentiable at c") failed. So our list of suspects must include both kinds of place.

Definition 16.2 (Critical point)**.** A *critical point* of f is a number c in the domain of f at which either $f'(c) = 0$ or $f'(c)$ does not exist. The value $f(c)$ is a *critical value*.

Focus

Combining Fermat's theorem with the corner case gives the rule that organizes the entire hunt for extrema:

$$\boxed{\text{Every local extremum of } f \text{ occurs at a critical point.}}$$

This does not say critical points are extrema — most are not. It says you never have to search the whole real line. The peaks and valleys can *only* sit at the finitely many places where f' is zero or undefined. Find those, and you have trapped every extremum inside a short list. That is the entire strategy.

Example 16.1 (Listing critical points)**.** Find the critical points of $f(x) = x^3 - 3x^2 - 9x + 5$.

Differentiate: $f'(x) = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x - 3)(x + 1)$. This is a polynomial, defined everywhere, so the only critical points come from $f' = 0$:

$$3(x - 3)(x + 1) = 0 \implies x = 3 \quad \text{or} \quad x = -1.$$

Two suspects: $x = -1$ and $x = 3$. We do not yet know which (if either) is a max or a min — that is the next section's job.

Example 16.2 (A critical point from a missing derivative)**.** Find the critical points of $g(x) = x^{2/3}$.

Here $g'(x) = \dfrac{2}{3}x^{-1/3} = \dfrac{2}{3\,x^{1/3}}$. There is no x making this zero (the numerator is a constant $2/3$). But at $x = 0$ the formula blows up: $g'(0)$ does not exist. Yet 0 is in the domain of g, and $g(0) = 0$ while $g(x) > 0$ for every other x. So $x = 0$ is a critical point — in fact a global minimum — found *only* because we remembered to check where the derivative fails to exist. The graph has a sharp *cusp* pointing down at the origin.

16.2 The first-derivative test: reading the sign of the slope

We have a list of suspects. Now we interrogate them. The tool is the most intuitive idea in the chapter: *the sign of f' tells you which way the function is going.* Positive slope, going up. Negative slope, going down. So to decide whether a critical point is a peak, a valley, or neither, we just watch what the slope does as we walk through it.

Why the technique works

Why does the sign of f' control whether f rises or falls? This is the Monotonicity Theorem (Theorem 15.4 in the preceding chapter), resting squarely on the Mean Value Theorem. For any two points $a < b$ in an interval, the MVT guarantees a point $\xi \in (a, b)$ satisfying

$$f(b) - f(a) = f'(\xi)\,(b - a).$$

Because $b > a$, the factor $(b - a)$ is strictly positive. Consequently, the sign of the vertical increment $f(b) - f(a)$ matches the sign of the instantaneous slope $f'(\xi)$. If $f' > 0$ across the entire interval, then $f'(\xi) > 0$ for every pair of sample points, forcing $f(b) > f(a)$ throughout: f is strictly increasing. If $f' < 0$ throughout, $f(b) < f(a)$, and f is strictly decreasing. The first-derivative test is the direct application of this relationship around a critical point.

Theorem 16.2 (First-derivative test)**.** *Suppose f is continuous at a critical point c and differentiable on an open interval around c (except possibly at c itself). Consider the sign of f' just to the left and just to the right of c:*

- *if f' changes from $+$ to $-$ at c, then f has a* local maximum *at c;*
- *if f' changes from $-$ to $+$ at c, then f has a* local minimum *at c;*
- *if f' does* not *change sign at c (stays $+$ or stays $-$), then c is* not *a local extremum.*

Proof. Suppose $f' > 0$ on an interval (a, c) to the left of c and $f' < 0$ on an interval (c, b) to the right. By the Monotonicity Theorem, f is increasing on (a, c) and decreasing on (c, b); by continuity at c these extend to f increasing on $(a, c]$ and decreasing on $[c, b)$. So for any x in (a, c) we have $f(x) < f(c)$, and for any x in (c, b) we have $f(x) < f(c)$. Thus $f(c) \geq f(x)$ for all x in (a, b): a local maximum. The minimum case reverses the inequalities. If f' keeps the same sign on both sides, f is monotone straight through c — still rising (or still falling) — so c cannot be a turning point. □

Intuition

Picture a ball rolling along the graph. Where $f' > 0$ the ground slopes up to the right, so a ball would roll *left*, downhill. Where $f' < 0$ the ground slopes down to the right, so the ball rolls *right*. A local *minimum* is a spot both neighbors roll *toward* (down from the left, down from the right) — a basin, $-$ then $+$. A local *maximum* is a spot both neighbors roll *away* from — a ridge, $+$ then $-$. And if the ball rolls the same direction on both sides, it just passes through; no resting place.

The practical method is a *sign chart*. Mark the critical points on a number line; they cut it into intervals. On each interval, f' has a constant sign (it can only change sign by passing through a zero

or a point of nondefinition — a critical point — so between consecutive critical points the sign is locked). Test one convenient point per interval and record + or −.

Example 16.3 (First-derivative test in full)**.** Classify the critical points of $f(x) = x^3 - 3x^2 - 9x + 5$ from Example 16.1.

We had $f'(x) = 3(x-3)(x+1)$, with critical points $x = -1$ and $x = 3$. These split the line into three intervals. Pick a test point in each and read the sign of f' from the factored form:

interval	$(-\infty, -1)$	$(-1, 3)$	$(3, \infty)$
test x	-2	0	4
$(x-3)$	$-$	$-$	$+$
$(x+1)$	$-$	$+$	$+$
$f' = 3(x-3)(x+1)$	$+$	$-$	$+$
f	↗	↘	↗

At $x = -1$, f' goes $+ \to -$: a **local maximum**, value $f(-1) = -1 - 3 + 9 + 5 = 10$. At $x = 3$, f' goes $- \to +$: a **local minimum**, value $f(3) = 27 - 27 - 27 + 5 = -22$. The function climbs to a peak at $(-1, 10)$, slides down into a valley at $(3, -22)$, then climbs forever.

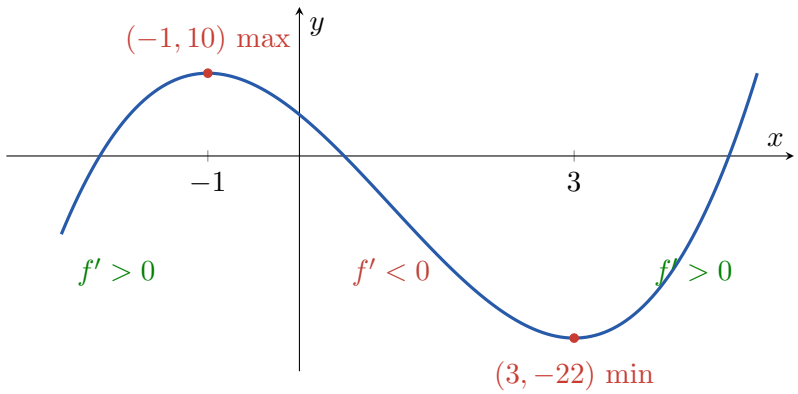


Figure 16.2: $f(x) = x^3 - 3x^2 - 9x + 5$. The slope is positive, then negative, then positive — so the curve rises, falls, rises, with a peak at $x = -1$ and a valley at $x = 3$.

Example 16.4 (A critical point that is not an extremum)**.** Classify the critical point of $f(x) = x^3$.

$f'(x) = 3x^2$, zero only at $x = 0$. But $3x^2 \geq 0$ everywhere: $f' > 0$ for $x < 0$ *and* $f' > 0$ for $x > 0$. The slope does not change sign — it dips to zero and bounces back up. By the first-derivative test, $x = 0$ is *not* an extremum. The curve flattens for one instant and keeps climbing. This is the trap from earlier made concrete: $f'(0) = 0$ did not deliver an extremum.

16.3 Concavity: which way the curve bends

Knowing where a graph rises and falls still leaves its *shape* ambiguous. A curve can climb from A to B in two very different ways: bulging up like the top of a dome, or sagging down like the inside of a bowl. Both go up; they have opposite *curvature*. To capture that, we ask not how f is changing, but how f' is changing — and the rate of change of f' is f''.

Definition 16.3 (Concavity)**.** On an interval where f is differentiable:

- f is *concave up* if f' is increasing on the interval (the slope gets ever larger — more positive or less negative as you move right);
- f is *concave down* if f' is decreasing on the interval.

Apply the Monotonicity Theorem to the function f' itself: f' is increasing exactly where *its* derivative f'' is positive. That gives the working test.

Focus

$f'' > 0 \implies$ concave up (cup-shaped, $\smile$), $\qquad f'' < 0 \implies$ concave down (cap-shaped, $\frown$).

A memory hook: $f'' > 0$ holds *water*, like a cup. $f'' < 0$ spills it, like a frown.

Intuition

There are three ways to picture concavity, and you should hold all three at once.

1. **The slope is turning.** Concave up means the tangent line is rotating counterclockwise as you move right — the slope keeps increasing. Concave down means it rotates clockwise.
2. **The curve vs. its tangent.** A concave-up curve lies *above* each of its tangent lines (the tangent underestimates it). A concave-down curve lies *below* its tangents. This is why the linear approximation from the tangent-line chapter consistently undershoots a convex function and overshoots a concave one.
3. **The chord.** A concave-up curve lies *below* the straight chord joining two of its points (think of the bottom of a valley); concave down lies above its chords. This is the geometric heart of convexity, and it is what makes inequalities like the AM–GM inequality fall out of a single picture.

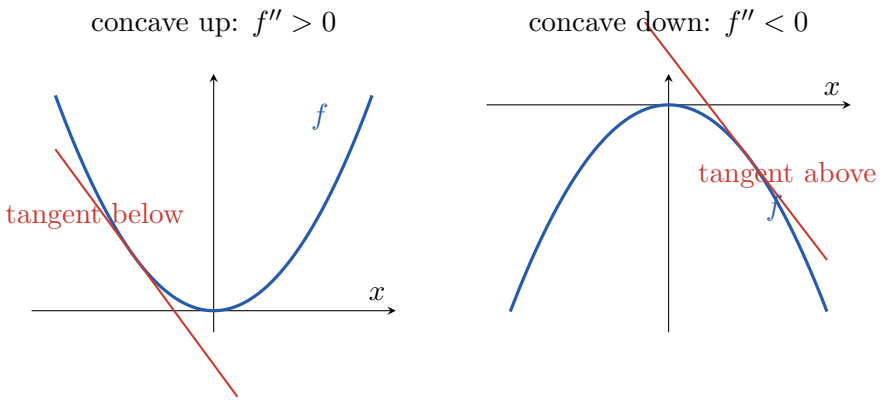


Figure 16.3: Concave up ($\smile$): the curve sits above its tangent lines. Concave down ($\frown$): the curve sits below its tangent lines.

Why the technique works

Why does a concave-up curve ($f'' > 0$) lie above its tangent lines? Fix a point a, and let

$$T(x) = f(a) + f'(a)(x - a)$$

be the tangent line at a. Define the gap $g(x) = f(x) - T(x)$, measuring how far the curve sits above the tangent line. Note that $g(a) = 0$ and $g'(x) = f'(x) - f'(a)$, so $g'(a) = 0$.
When $f'' > 0$ on the interval, f' is strictly increasing:

- For $x > a$: $f'(x) > f'(a)$, so $g'(x) > 0$. Since $g(a) = 0$ and g is increasing to the right of a, we have $g(x) > 0$.
- For $x < a$: $f'(x) < f'(a)$, so $g'(x) < 0$. Since g is decreasing toward a where $g(a) = 0$, moving leftward from a means $g(x) > 0$.

On either side of a, $g(x) \geq 0$: the curve sits strictly above its tangent line except at the point of tangency itself. The entire deduction is two successive applications of the Mean Value Theorem: first showing $f'' > 0$ makes f' increase, then showing $g' > 0$ makes g increase.

16.3.1 Inflection points: where the bend reverses

A curve does not have to keep the same concavity forever. Many of the most important shapes in science change their bending partway through. An inflection point is where that switch happens.

Definition 16.4 (Inflection point)**.** A point $\big(c, f(c)\big)$ on the graph of a continuous function is an *inflection point* if the concavity of f *changes* there — from up to down, or down to up.

Since concavity is governed by the sign of f'', an inflection point is a place where f'' changes sign. And just as f' can only change sign at a zero or a gap, f'' can only change sign where $f'' = 0$ or f'' fails to exist. Those are the *candidate* inflection points — and, exactly as with Fermat's theorem, candidacy is not guilt.

Common trap

$f''(c) = 0$ does *not* by itself make $\big(c, f(c)\big)$ an inflection point. You must check that f'' actually *changes sign*. Classic counterexample: $f(x) = x^4$ has $f''(x) = 12x^2$, which is zero at $x = 0$ but ≥ 0 on both sides — no sign change, no inflection. The curve is concave up throughout; it merely flattens momentarily at the bottom. Always confirm the sign flip, never read it off from $f'' = 0$ alone.

Example 16.5 (The S-curve and its inflection)**.** Consider $f(x) = \dfrac{1}{1 + e^{-x}}$, the logistic curve — the workhorse of population models, neural networks, and any process that grows then saturates. Find where it is concave up, concave down, and its inflection point.

Write $s = f(x)$. Differentiating with the quotient or chain rule yields $f'(x) = s(1 - s)$, and a second derivative gives

$$f''(x) = f'(x)(1 - 2s) = s(1 - s)(1 - 2s).$$

Now s ranges strictly between 0 and 1, so $s > 0$ and $(1 - s) > 0$ always; the sign of f'' is determined purely by the factor $(1 - 2s)$. That is positive when $s < \frac{1}{2}$ and negative when $s > \frac{1}{2}$. And $s = \frac{1}{2}$

happens exactly at $x = 0$. So:

$$f'' > 0 \text{ for } x < 0 \text{ (concave up)}, \qquad f'' < 0 \text{ for } x > 0 \text{ (concave down)}.$$

The concavity flips at $x = 0$: an inflection point at $\left(0, \frac{1}{2}\right)$. This is the famous "tipping point" of the S-curve — the moment of fastest growth, where the curve stops accelerating and begins to level off toward saturation.

Application

That inflection point is not a curiosity; it is often *the* quantity of interest. In an epidemic modeled by a logistic curve, the inflection point is the day new cases *peak* — after it, growth is still happening but slowing, and the worst is mathematically behind you even though the total is still rising. In a product-adoption curve it is the moment of maximum sales velocity. In both cases f (the cumulative total) is still increasing the whole time; it is f'' changing sign that signals the turn. The first derivative tells you the situation is getting worse; the second derivative tells you how fast it is getting worse, and its sign change is the first good news.

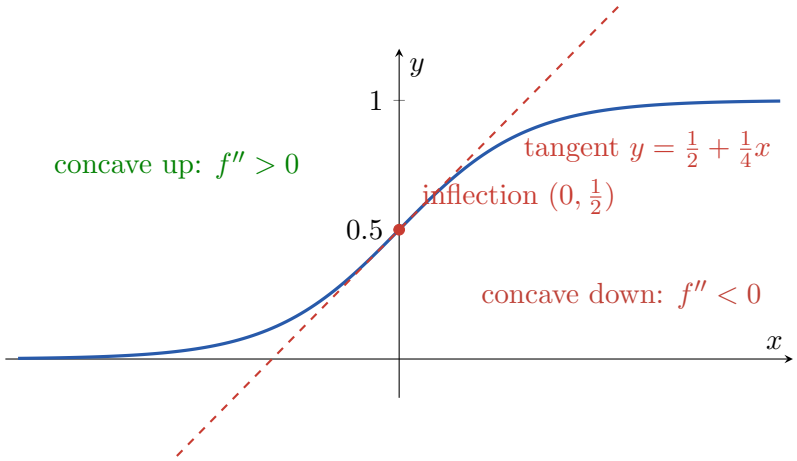


Figure 16.4: The logistic curve $f(x) = 1/(1 + e^{-x})$. For $x < 0$, $f'' > 0$ (concave up, curve above tangent); for $x > 0$, $f'' < 0$ (concave down, curve below tangent). At the inflection point $\left(0, \frac{1}{2}\right)$, the tangent line (dashed) slices directly through the curve, marking the transition from acceleration to deceleration.

What is really happening

What is the second derivative *really* measuring at an inflection point? Think of f as position, so f' is velocity and f'' is acceleration. Concave up means accelerating (speeding up in the positive direction); concave down means decelerating. The inflection point is the instant the acceleration passes through zero and reverses — the moment you stop pressing the gas and start pressing the brake. The car is still moving forward (maybe even still gaining ground for a while, because velocity has not yet hit zero), but the *trend has turned*. That is the deep content of f'' changing sign: not that motion stops, but that the *character* of the motion inverts. A peak in f is where $f' = 0$; an inflection in f is where f' is at its own peak or valley. Inflection points of f are the extrema of f'.

16.4 The second-derivative test: curvature as a shortcut

The first-derivative test always works, but it asks you to examine f' on *both* sides of a critical point. When f'' is easy to compute, there is a faster route that looks at a single point. The idea is geometric and immediate: at a critical point the tangent is horizontal, so the question "peak or valley?" becomes purely a question of *which way the curve bends there.* Bends up ($\smile$) over a horizontal tangent: a valley. Bends down ($\frown$): a peak.

Theorem 16.3 (Second-derivative test)**.** *Suppose $f'(c) = 0$ and $f''(c)$ exists.*

- *If $f''(c) > 0$, then f has a* local minimum *at c.*
- *If $f''(c) < 0$, then f has a* local maximum *at c.*
- *If $f''(c) = 0$, the test is* inconclusive *— c could be a max, a min, or neither.*

Why the technique works

Here is why the test works, tied straight back to concavity. Suppose $f''(c) > 0$. Recall (from the box above) that this means the curve lies *above* its tangent line at c in a neighborhood. But the tangent line at c is *horizontal*, because $f'(c) = 0$ — it is the constant line $y = f(c)$. So near c the curve lies above the horizontal line $y = f(c)$, i.e. $f(x) \geq f(c)$ nearby: a local minimum. If $f''(c) < 0$ the curve lies below its horizontal tangent, giving $f(x) \leq f(c)$: a local maximum. The second-derivative test is nothing but "curve-above-tangent" applied at the one place where the tangent happens to be flat. That is the whole secret.

A slightly more careful phrasing for $f''(c) > 0$: by definition $f''(c) = \lim_{x\to c} \frac{f'(x)-f'(c)}{x-c} > 0$, and since $f'(c) = 0$ this says $\frac{f'(x)}{x-c} > 0$ for x near c. So $f'(x)$ and $(x - c)$ have the same sign: $f' < 0$ just left of c and $f' > 0$ just right. That is precisely the $- \to +$ pattern the first-derivative test reads as a minimum. The two tests agree because the second is the first in disguise.

Example 16.6 (Second-derivative test, fast)**.** Reclassify the critical points of $f(x) = x^3-3x^2-9x+5$ using f''.

We have $f'(x) = 3x^2 - 6x - 9$, so $f''(x) = 6x - 6 = 6(x - 1)$. Evaluate at the two critical points:

$$f''(-1) = 6(-1 - 1) = -12 < 0 \;\Rightarrow\; \text{local max at } x = -1,$$

$$f''(3) = 6(3 - 1) = 12 > 0 \;\Rightarrow\; \text{local min at } x = 3.$$

Same verdict as the sign chart in Example 16.3, with no intervals to test — two evaluations and done. (Notice in passing that $f'' = 0$ at $x = 1$, with f'' flipping sign there: $x = 1$ is an inflection point, the midpoint between the peak and the valley, as cubics always have.)

Common trap

When $f''(c) = 0$, the second-derivative test tells you *nothing* — do not conclude "not an extremum." Compare three functions at $x = 0$, all with $f'(0) = 0$ and $f''(0) = 0$:

$$x^4 \text{ (min)}, \qquad -x^4 \text{ (max)}, \qquad x^3 \text{ (neither)}.$$

The second derivative cannot tell these apart. When it vanishes, fall back on the first-derivative test (check the sign of f' on both sides) — that one never fails to give an answer.

Why this matters

Why do we have two tests for the same thing? Because they answer different practical questions. The first-derivative test is *robust* — it works at corners and cusps where f'' does not even exist, and it is never inconclusive — but it costs you a study of f' on a whole neighborhood. The second-derivative test is *cheap* — one number per critical point — but it demands a second derivative and shrugs when that derivative is zero. Real problem-solving uses both: reach for f'' first because it is fast, and when it returns zero, retreat to the sign chart, which always closes the case. Knowing which tool to grab, and why each can fail, is the mark of understanding rather than memorizing.

16.5 Reading f, f', and f'' together: curve sketching

Now everything assembles into a single skill: drawing a graph you have never seen, accurately, from its formula alone. The discipline is to read three layers and let them constrain one another — three independent witnesses to the same shape.

Focus

The three layers of a graph:

Object	What it controls	What you read from it
f	height	intercepts, domain, asymptotes, symmetry
f'	slope	increasing/decreasing, local max/min
f''	bending	concave up/down, inflection points

f' governs whether the curve goes up or down. f'' governs whether it smiles or frowns. Together they pin down the shape between your plotted points, which is exactly the information naive dot-plotting was missing.

A reliable checklist, in order:

1. **Domain and intercepts.** Where is f defined? Where does it cross the axes ($f = 0$, and $f(0)$)?
2. **Symmetry and ends.** Even ($f(-x) = f(x)$, mirror across the y-axis)? Odd ($f(-x) = -f(x)$, rotational)? What happens as $x \to \pm\infty$ — any horizontal or slant asymptotes? Any vertical asymptotes where f blows up?
3. **First derivative.** Compute f', find critical points, build the sign chart: where increasing, where decreasing, and classify the extrema.
4. **Second derivative.** Compute f'', find where it is zero or undefined, build its sign chart: concavity intervals and inflection points.

5. **Assemble.** Plot the special points (intercepts, extrema, inflections, asymptotes), then connect them *respecting* the slope and concavity you found. There is now exactly one curve that fits all the constraints.

Example 16.7 (A full sketch: a quartic)**.** Sketch $f(x) = x^4 - 4x^3$.

1. Domain, intercepts. Defined for all x. Factor: $f(x) = x^3(x-4)$, so $f = 0$ at $x = 0$ (a triple root — the curve flattens through the axis there) and at $x = 4$. Also $f(0) = 0$.

2. Ends. As $x \to \pm\infty$, the x^4 term dominates and $f \to +\infty$ on both sides. No symmetry ($f(-x) = x^4 + 4x^3 \neq \pm f(x)$). No asymptotes.

3. First derivative. $f'(x) = 4x^3 - 12x^2 = 4x^2(x-3)$. Critical points where $f' = 0$: $x = 0$ (double) and $x = 3$. Sign chart:

	$(-\infty, 0)$	$(0, 3)$	$(3, \infty)$
$4x^2$	$+$	$+$	$+$
$(x-3)$	$-$	$-$	$+$
f'	$-$	$-$	$+$
f	$\searrow$	$\searrow$	$\nearrow$

At $x = 3$, f' goes $- \to +$: **local min**, $f(3) = 81 - 108 = -27$. At $x = 0$, f' does *not* change sign ($-$ to $-$): *not* an extremum, just a momentary flat spot on the way down.

4. Second derivative. $f''(x) = 12x^2 - 24x = 12x(x-2)$. Zero at $x = 0$ and $x = 2$. Sign chart:

	$(-\infty, 0)$	$(0, 2)$	$(2, \infty)$
$12x$	$-$	$+$	$+$
$(x-2)$	$-$	$-$	$+$
f''	$+$	$-$	$+$
shape	$\smile$	$\frown$	$\smile$

Concavity flips at *both* $x = 0$ and $x = 2$ — two inflection points: $\big(0, 0\big)$ and $\big(2, f(2)\big) = \big(2, -16\big)$.

5. Assemble. Coming from the upper left the curve falls, concave up, until $x = 0$ where it momentarily flattens (slope 0) and changes to concave down — it keeps falling but now bulging downward — through to $x = 2$, where it switches back to concave up while still descending, reaching the bottom of the valley at $(3, -27)$, then climbing away to $+\infty$. The flat spot at the origin (slope zero but *not* an extremum) and the deep minimum at $x = 3$ are the two features dot plotting would most likely get wrong.

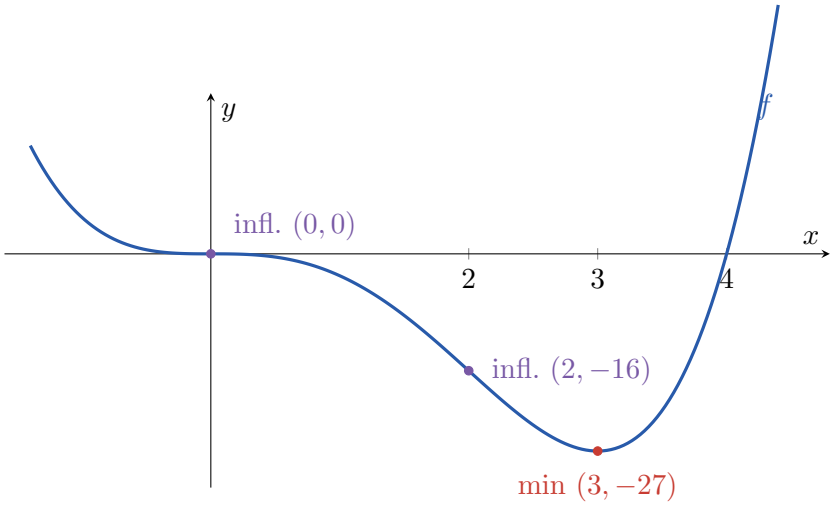


Figure 16.5: $f(x) = x^4 - 4x^3$. A flat-but-not-extreme spot at the origin, a genuine minimum at $(3, -27)$, and two inflection points (purple) at $x = 0$ and $x = 2$ where the bending reverses.

Example 16.8 (A sketch with asymptotes: a rational function)**.** Sketch $f(x) = \dfrac{x^2}{x^2 - 1}$.

1. Domain, intercepts. The denominator $x^2 - 1 = (x - 1)(x + 1)$ vanishes at $x = \pm 1$, so those are excluded: vertical asymptotes there. $f = 0$ only when the numerator $x^2 = 0$, i.e. $x = 0$, where also $f(0) = 0$. So the only intercept is the origin.

2. Symmetry, ends. $f(-x) = \dfrac{x^2}{x^2 - 1} = f(x)$: *even*, symmetric across the y-axis. As $x \to \pm\infty$, $f = \dfrac{x^2}{x^2 - 1} \to 1$: a horizontal asymptote $y = 1$. Near the vertical asymptotes the sign of the denominator tells the story: for $|x|$ slightly less than 1, $x^2 - 1 < 0$ so $f \to -\infty$; for $|x|$ slightly more than 1, $x^2 - 1 > 0$ so $f \to +\infty$.

3. First derivative. By the quotient rule,

$$f'(x) = \frac{(2x)(x^2 - 1) - x^2(2x)}{(x^2 - 1)^2} = \frac{2x^3 - 2x - 2x^3}{(x^2 - 1)^2} = \frac{-2x}{(x^2 - 1)^2}.$$

The denominator is positive wherever defined, so the sign of f' is the sign of $-2x$: positive for $x < 0$, negative for $x > 0$. The only critical point is $x = 0$, and f' goes $+ \to -$ there: a **local maximum** at $(0, 0)$. (It is a maximum of the central branch, even though far-off branches climb much higher — a vivid reminder that "local" means local.)

4. Second derivative. A computation (quotient rule again) gives

$$f''(x) = \frac{2(3x^2 + 1)}{(x^2 - 1)^3}.$$

The numerator $2(3x^2 + 1)$ is always positive, so the sign of f'' matches the sign of $(x^2 - 1)^3$, which matches the sign of $(x^2 - 1)$:

$$f'' > 0 \text{ when } |x| > 1 \text{ (concave up)}, \qquad f'' < 0 \text{ when } |x| < 1 \text{ (concave down)}.$$

There is no inflection point, because the sign change of f'' happens only at $x = \pm 1$, where f is not even defined.

5. Assemble. Three branches. The middle branch (between the asymptotes) is a downward-bulging arch peaking at the origin and diving to $-\infty$ at both $x = \pm 1$. The two outer branches descend from $+\infty$ (just outside the asymptotes) toward the horizontal line $y = 1$, concave up, mirror images of each other.

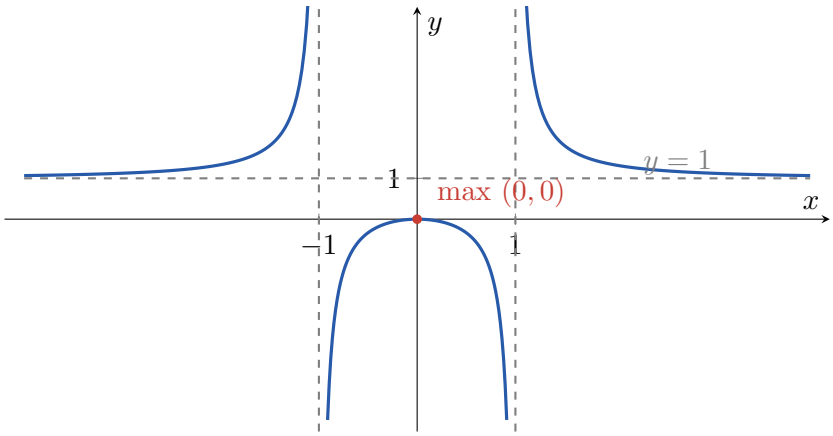


Figure 16.6: $f(x) = x^2/(x^2 - 1)$. Vertical asymptotes at $x = \pm 1$ (dashed) split the graph into three branches; horizontal asymptote $y = 1$. The central arch has a local maximum at the origin.

Walking question

Here is a graph you can build entirely in your head on a walk. Imagine a function whose *derivative* f' looks like a single hump — zero at $x = 0$, rising to a peak at $x = 2$, falling back to zero at $x = 4$, and staying near zero outside. Now ask yourself, without paper: where is f itself increasing? (Wherever $f' > 0$, which is the whole stretch from 0 to 4 — so f climbs the entire time.) Where is f *concave up*? (Wherever f' is *increasing*, i.e. from 0 to 2.) Where is f's inflection point? (At $x = 2$, the peak of f' — the steepest point of the climb.) You have just read the shape of f off the shape of f', which is the single most useful fluency in this chapter (Figure 16.7 shows a worked instance). Practice going both directions: from f to f', and from f' back up to f.

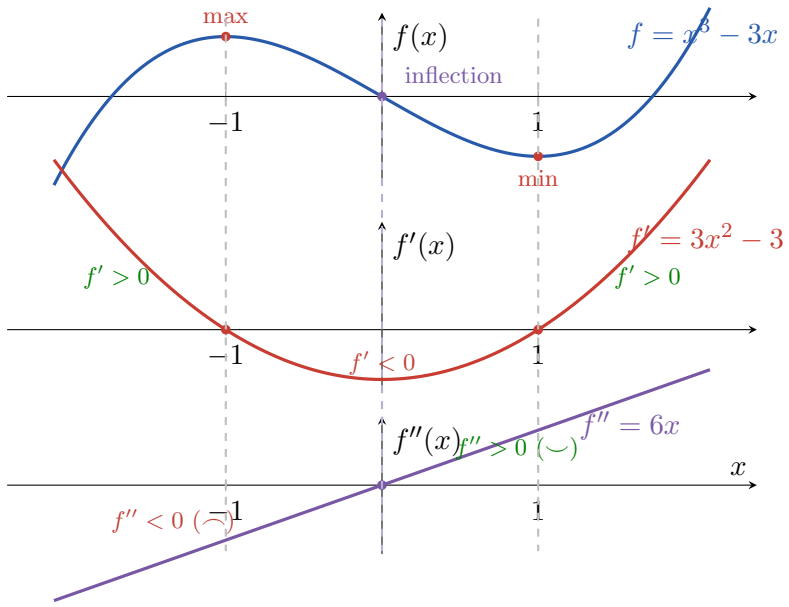


Figure 16.7: The three synchronized layers of curve analysis for $f(x) = x^3 - 3x$. Top lane: the graph of f rises where $f' > 0$ and falls where $f' < 0$. Middle lane: the zeros of f' at $x = \pm 1$ pinpoint the peak and valley of f. Bottom lane: the zero of f'' at $x = 0$ marks the inflection point, where f' attains its global minimum (steepest downhill descent) and the concavity of f flips from down (⌢) to up (⌣).

Work it yourself

Take $f(x) = xe^{-x}$ — a shape that shows up constantly in physics (radioactive decay times a linear rise), pharmacology (drug concentration over time), and probability. Work out the shape entirely by hand, then check your sketch against a plot.

1. Show $f'(x) = e^{-x}(1 - x)$. Find the one critical point and classify it with the first-derivative test. (Answer: max at $x = 1$, value $1/e \approx 0.368$.)
2. Show $f''(x) = e^{-x}(x - 2)$. Find the inflection point. (Answer: at $x = 2$, where the decay's concavity flips from down to up.)
3. As $x \to +\infty$, does $f \to 0$ or $\to \infty$? (The exponential e^{-x} beats the linear x, so $f \to 0^+$ — a horizontal asymptote $y = 0$.) As $x \to -\infty$? ($f \to -\infty$, because e^{-x} explodes.)
4. Assemble: a curve rising from below, crossing the origin, climbing to a single peak at $x = 1$, then decaying back toward zero, with the decay easing off (inflection) at $x = 2$. Sketch it, then plot it and compare.

Connection

Concavity is the one-variable shadow of a much bigger idea: *convexity*, which runs through optimization, economics, and machine learning. A function with $f'' \geq 0$ everywhere is called *convex*; the curve-above-its-chords picture is its defining property. Convex functions have a magical feature you can already anticipate from this chapter: *any local minimum is automatically a global minimum* — there are no false valleys to get stuck in, because the curve never

turns back down once it has started up. This is exactly why training a model whose loss function is convex is "easy" (gradient descent cannot be trapped) and why non-convex losses, riddled with local minima, are the hard, interesting case. The humble second-derivative test is the seed of that entire theory. We will meet the multivariable version — where f'' becomes a matrix and "concave up" becomes a statement about that matrix — when we reach functions of several variables.

16.6 Finding the global extremes on an interval

Local extrema are about neighborhoods. But the questions that matter in practice are usually *global*: over this whole range of choices, what is the absolute best? On an unbounded domain a function may have no global max at all (think $f(x) = x$, climbing forever). The clean, decidable case is a function on a *closed, bounded* interval $[a, b]$, and there the Extreme Value Theorem (from the continuity chapter) guarantees both a global max and a global min actually exist. The only question is *where*.

Focus

The closed-interval method. To find the absolute max and min of a continuous f on $[a, b]$:

1. Find all critical points of f *inside* (a, b).
2. Make a list: those critical points, plus the two endpoints a and b.
3. Evaluate f at every point on the list.
4. The largest value is the global max; the smallest is the global min.

No sign charts, no second derivatives — just evaluate and compare. The logic is airtight: a global extremum is in particular a local one *unless* it sits at an endpoint (where "local" has no left or right neighbor to compare with). Local interior extrema live at critical points (Fermat); the only other place is an endpoint. So the global champion is somewhere on the list, guaranteed.

Example 16.9 (Closed-interval optimization)**.** Find the absolute maximum and minimum of $f(x) = x^3 - 3x^2 - 9x + 5$ on $[-2, 4]$.

We already found the critical points $x = -1$ and $x = 3$ (both inside $(-2, 4)$). The list is $\{-2, -1, 3, 4\}$. Evaluate:

$$\begin{aligned} f(-2) &= -8 - 12 + 18 + 5 = 3, \\ f(-1) &= -1 - 3 + 9 + 5 = 10, \\ f(3) &= 27 - 27 - 27 + 5 = -22, \\ f(4) &= 64 - 48 - 36 + 5 = -15. \end{aligned}$$

The largest is $f(-1) = 10$: **global maximum**. The smallest is $f(3) = -22$: **global minimum**. Notice that the endpoint $x = 4$ gives -15, lower than the *other* endpoint and lower than the left critical point — a reminder that endpoints are genuine contenders and must always be on the list. Had we forgotten them, we would still have caught the true extremes here, but on a different interval an endpoint could easily be the winner.

Common trap

The closed-interval method needs both hypotheses of the Extreme Value Theorem: *continuous* function on a *closed, bounded* interval. Drop either and the guarantee evaporates. On the open interval $(0, 1)$ the function $f(x) = x$ has no maximum — it gets arbitrarily close to 1 but never reaches it, and 1 is not in the domain. On $[0, \infty)$ the function $f(x) = x$ has no maximum either. And a function with a vertical asymptote inside the interval (a discontinuity) can shoot to $+\infty$ with no max at all. When the EVT does not apply, finding global extrema is a genuinely subtler business — you must examine the behavior at the open ends or at infinity by hand.

16.7 What we built

The whole chapter is one idea unfolding: *the derivatives of f encode the shape of f*. The first derivative records direction — its zeros and nondefined points (critical points) trap every local extremum, and its sign on either side (the first-derivative test) classifies them. The second derivative records bending — its sign gives concavity, its sign *changes* give inflection points, and at a flat tangent its sign settles peak-versus-valley in one stroke (the second-derivative test). Reading both at once, you can reconstruct a graph exactly, and on a closed interval you can pin down the absolute best and worst with nothing more than a short list of values.

Underneath every one of these tests sits the Mean Value Theorem, quietly converting facts about f' into facts about f. "Positive derivative means increasing" is the MVT; the first-derivative test is that fact applied on each side of a critical point; concavity is that same fact applied to f'; the second-derivative test is concavity applied at a horizontal tangent. You are not memorizing four unrelated rules. You are watching one theorem do four jobs.

In the next chapter we put this machine to work on real questions — the largest box, the cheapest can, the shortest path, the maximum profit — where the art is no longer running the tests but *setting up* the right function to optimize. The hard part of optimization was never the calculus. It was deciding what to take the derivative of.

The Topography of Curves and the Road Ahead

Connection

This chapter transforms curve sketching from empirical guesswork into a definitive qualitative science. Armed with the first and second derivatives, you can determine:

- *Exact critical points and extrema*: Locating every candidate peak, valley, and plateau ($f' = 0$ or undefined) without plotting millions of points.
- *Global monotonicity and concavity*: Partitioning the real line into intervals where the graph rises, falls, bends upward like a cup, or sheds water like a dome.
- *Complete landscape topography*: Identifying inflections where physical quantities change acceleration, and using the closed-interval method to guarantee global optima.

Yet the critical-point framework relies on isolated critical points and well-behaved continuity of

derivatives. It falters on functions with essential singularities or infinite-frequency oscillations. Consider the canonical counterexample:

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Here $f'(0) = \lim_{h\to 0} \frac{h^2 \sin(1/h)}{h} = 0$, so $x = 0$ is a critical point. Yet for $x \neq 0$, $f'(x) = 2x\sin(1/x) - \cos(1/x)$, which oscillates between positive and negative infinitely often in every open neighborhood of 0. There are infinitely many local maxima and minima accumulating at the origin! The first derivative test cannot be applied because f' does not have constant sign on either side of 0, and $f''(0)$ does not even exist. Standard calculus tools assume the smooth, tame geometry of physics, not the pathological wildness of real analysis.

In Chapter 17 (*Optimization: Finding the Best*), we take these classification tools into the physical world. We will solve Fermat's principle of least time, derive Snell's law of optical refraction, and solve structural and economic design problems by translating real-world constraints into single-variable extrema.

Exercises

Exercise 16.1. For each function, find all critical points and classify each as a local maximum, local minimum, or neither, using the first-derivative test.

1. $f(x) = 2x^3 - 9x^2 + 12x - 3$
2. $f(x) = x^4 - 8x^2 + 3$
3. $f(x) = x^{2/3}(x - 5)$ (watch for a point where f' does not exist)

Exercise 16.2. Find the intervals of concavity and all inflection points of $f(x) = x^4 - 6x^2 + 5$. Then confirm, using the second-derivative test, the nature of each critical point.

Exercise 16.3. Find the absolute maximum and minimum of $f(x) = 2x^3 - 3x^2 - 12x + 1$ on the closed interval $[-2, 3]$. Identify whether each extreme occurs at an endpoint or at an interior critical point.

Exercise 16.4. Carry out a full curve sketch of $f(x) = \dfrac{x}{x^2 + 1}$: domain, symmetry, intercepts, end behavior and asymptotes, intervals of increase/decrease with extrema, and intervals of concavity with inflection points. Then draw the graph. (Hint: f is odd, $f \to 0$ at both ends, and you will find one max, one min, and three inflection points.)

Exercise 16.5. (Conceptual.) For each description, either sketch a function meeting it or explain briefly why none can exist.

1. $f'(2) = 0$ but f has neither a max nor a min at $x = 2$.
2. $f''(2) = 0$ but f has no inflection point at $x = 2$.
3. f has a local minimum at $x = 2$ but $f'(2)$ does not exist.
4. f has a local maximum at $x = 2$ and $f''(2) > 0$.

Exercise 16.6. (Reading f from f'.) The derivative of a function f is $f'(x) = (x-1)^2(x-4)$. Without finding f itself, determine: (a) the intervals where f is increasing and decreasing; (b) the locations and types of all local extrema; (c) whether $x = 1$ is special, and why the squared factor matters.

Exercise 16.7. (Why the test works.) Suppose f'' is continuous and $f''(c) > 0$ at a point where $f'(c) = 0$. Using the definition $f''(c) = \lim_{x\to c} \dfrac{f'(x) - f'(c)}{x - c}$, prove directly that f' is negative just to the left of c and positive just to the right, and conclude that f has a local minimum at c. (This is the second-derivative test, derived from scratch.)

Exercise 16.8. (Proof, tangent line and concavity.) Let f be twice differentiable with $f''(x) > 0$ for all x in an interval I. Prove that for any fixed $a \in I$,

$$f(x) \;\geq\; f(a) + f'(a)(x - a) \qquad \text{for all } x \in I,$$

with equality only at $x = a$. (That is: a strictly convex curve lies strictly above each of its tangent lines. Imitate the gap-function argument from the chapter: let $g(x) = f(x) - f(a) - f'(a)(x - a)$ and study the sign of g'.)

Exercise 16.9. (Inflections are extrema of f'.) Explain carefully why an inflection point of f corresponds to a local maximum or minimum of f'. Use this to find the inflection point of $f(x) = \dfrac{1}{1 + e^{-x}}$ by maximizing f' instead of analyzing f'' directly, and confirm you get $x = 0$.

Exercise 16.10. (Modeling.) A drug's concentration in the bloodstream t hours after a dose is modeled by $C(t) = 12t\, e^{-t/2}$ for $t \geq 0$. (a) When is the concentration highest, and what is that peak concentration? (b) The concentration is "falling fastest" at the inflection point of C; find that time. (c) Sketch C for $t \in [0, 12]$, marking the peak and the inflection.

Chapter 17

Optimization

Every chapter so far has been, in a quiet way, preparation for this one. We learned what a derivative is, learned to compute it, learned that it detects where a function stops rising and starts falling, and learned (through the Mean Value Theorem) why that detection is trustworthy. All of that machinery was built so that we could answer a single, very old, very human question: *what is the best you can do?*

Make the box hold the most. Use the least metal. Build the cheapest fence. Run the shortest distance. Earn the highest profit. These are not artificial textbook puzzles — they are the questions that motivated the invention of calculus in the first place. Fermat was solving max/min problems with a slope-like idea decades before Newton and Leibniz, and the whole subject of the derivative grew partly out of the desire to find peaks and valleys without guessing.

This chapter is where calculus earns its keep. The technique is the same one you already know — find where the derivative is zero, check the candidates — but the hard part, the part nobody warns you about, is everything that happens *before* you take a derivative: turning a sentence in English (or a picture of a fold-up box) into a function of one variable, on a definite domain, that you are allowed to differentiate. That translation is a craft, and most of this chapter is about learning it well enough that it stops feeling like luck.

17.1 What optimization really asks

Strip away the words and an optimization problem has exactly three ingredients.

Focus

Every applied max/min problem is built from:

1. an **objective** — the quantity you want to make as large or as small as possible (area, volume, cost, distance, time);
2. one or more **constraints** — the conditions tying your choices together (a fixed amount of fence, a fixed surface area, a point that must lie on a curve);
3. a **domain** — the set of choices that actually make sense (a length cannot be negative; a cut cannot be longer than the cardboard).

Your job is to express the objective as a function of *one* variable, find its extreme value on the domain, and confirm you really found the best.

Why aim for a single variable? We have one tool for finding peaks and valleys: set the derivative to zero. That tool needs a function of one input. So no matter how many quantities a problem mentions — length, width, height, radius, angle — our first task is to collapse them down to one, using the constraint to eliminate the rest. The constraint is not an obstacle; it is the thing that *makes the problem solvable*, because it is what lets us trade many variables for one.

Intuition

Picture the objective as a landscape and your constraint as a path drawn across it. Without the constraint you could wander the whole terrain and there might be no highest point at all. The constraint pins you to a single trail. Now "the best you can do" is just the highest point *on that trail* — a one-dimensional question, exactly the kind a derivative can answer. Much of optimization is just finding the trail.

17.1.1 A method you can trust

Here is the recipe we will follow on every single example. Write it on an index card. The discipline of doing these steps *in order* is what separates a clean solution from a tangle.

Focus

The modeling method for max/min problems.

1. **Draw a picture** and name the quantities. Introduce letters for the things that can vary.
2. **Write the objective** — the thing to be maximized or minimized — as a formula, even if it still has several variables in it.
3. **Write the constraint(s)** as equations.
4. **Reduce to one variable.** Use the constraint to eliminate every variable but one. Now the objective is a function f of a single variable.
5. **Find the domain** of that variable — the interval of values that correspond to a real, physical configuration. Note the endpoints.
6. **Differentiate and find the critical points**: solve $f'(x) = 0$ and note where f' fails to exist.
7. **Test the candidates.** Compare the values of f at the critical points *and* at the endpoints of the domain. Use a first- or second- derivative test, or the closed-interval method, to be sure which candidate wins.
8. **Answer the question that was asked** — with units, and check it is sane.

Notice that steps 1–5 contain no calculus at all. They are modeling. If you get those right, step 6 is mechanical. Most "hard" optimization problems are really just modeling problems wearing a calculus disguise.

Why the technique works

Why does "set the derivative to zero" find the best? Because at an interior point where f is largest or smallest, the curve must, for that instant, be neither rising nor falling — if it were rising, you could nudge a hair to the right and do better; if falling, a hair to the left. So the tangent there is horizontal: $f'(x) = 0$. This is *Fermat's theorem*, proved cleanly in the chapter on maxima and minima: an interior extreme of a differentiable function forces a stationary point. Optimization is just Fermat's theorem applied to a function you had to build yourself. The slope is the messenger that tells you whether moving helps; where the messenger says "moving does nothing," you may have arrived.

History

The realization that nature and mathematics optimize began in earnest with Pierre de Fermat in 1662. For centuries, optics had been stymied by the law of refraction discovered experimentally by Willebrord Snell in 1621. Fermat proposed a bold physical hypothesis: light traveling between two points does not choose the shortest geometric distance, but the path of *least time* (*Fermat's Principle*). By setting up the total transit time across an optical boundary as a single-variable function of the crossing point x,

$$T(x) = \frac{\sqrt{a^2 + x^2}}{v_1} + \frac{\sqrt{b^2 + (d - x)^2}}{v_2},$$

and setting $T'(x) = 0$ using his method of adequality, Fermat derived Snell's law $\frac{\sin\theta_1}{v_1} = \frac{\sin\theta_2}{v_2}$ from pure mathematics. Nature, it appeared, was an optimizer.

Thirty-four years later, in June 1696, Johann Bernoulli ignited a European competition by posing the celebrated *Brachistochrone problem* in *Acta Eruditorum*: given two points A and B in a vertical plane at different heights, what is the path of a frictionless bead sliding under gravity that minimizes transit time? A straight ramp is the shortest distance, but is it the fastest?

Five mathematicians solved it: Johann Bernoulli, his brother Jakob Bernoulli, Gottfried Wilhelm Leibniz, Guillaume de l'Hôpital, and Isaac Newton. Newton received the challenge upon returning from a grueling day at the Royal Mint in January 1697, worked through the night, and posted the solution anonymously the next morning. Bernoulli famously recognized the author at once, remarking: "I recognize the lion by his claw." The optimal curve was not a straight line, an arc of a circle, or a parabola—it was a *cycloid*. Bernoulli's challenge broke open the boundary of ordinary optimization: it proved that optimization applies not just to picking a single number x, but to discovering an entire unknown function or path, giving birth to the calculus of variations.

17.2 The closed-interval method: making the answer certain

Before the examples, let us nail down the one piece of theory that turns a guess into a guarantee. Setting $f'(x) = 0$ tells you where the function *might* peak. It does not by itself tell you that a peak exists, nor that the candidate you found is the global best. Two things can go wrong: the maximum might sit at an endpoint of the domain rather than at a stationary point, or the function might not have a maximum at all (think of $f(x) = x$ on all of $\mathbb{R}$ — no biggest value).

The clean fix is the **Extreme Value Theorem**, which we met in the chapter on continuity.

Focus

Extreme Value Theorem. If f is continuous on a *closed, bounded* interval $[a, b]$, then f attains both a maximum and a minimum value somewhere on $[a, b]$.

This theorem is the optimizer's safety net. As long as your objective is continuous and your domain is a closed interval $[a, b]$, a best value is *guaranteed to exist*, and it can only live in one of two kinds of places: at a critical point inside, or at an endpoint. That gives a finite checklist.

Focus

The closed-interval method. To find the absolute max and min of a continuous f on $[a, b]$:

1. Find all critical numbers in (a, b) (where $f' = 0$ or f' does not exist).
2. Evaluate f at each critical number.
3. Evaluate f at the two endpoints a and b.
4. The largest of these values is the absolute maximum; the smallest is the absolute minimum.

No derivative tests needed — just compare numbers. The theorem promises the winner is in your list.

Why the technique works

Why is comparing values *enough*? Because the Extreme Value Theorem promises the maximum is attained *somewhere*. Wherever that somewhere is, it is either an interior point or an endpoint. If interior, Fermat's theorem says it is a critical number, so it is in your list. If an endpoint, it is also in your list. There is nowhere else for it to hide. So you do not need to analyze concavity or signs of f' at all; you just need to make sure your list of suspects is complete and then read off the biggest value. The closed-interval method works precisely because the theorem rules out any fourth possibility.

Common trap

The closed-interval method requires *three* pieces, and beginners drop one:

- **Continuity**: a jump or a vertical asymptote in the domain breaks the guarantee. The function $1/x$ on $[-1, 1]$ has no maximum — it is not continuous there.
- **Closed** (endpoints included): on the open interval $(0, 1)$ the function $f(x) = x$ has no maximum, because it can get arbitrarily close to 1 without reaching it.
- **Bounded**: on $[0, \infty)$ the function $f(x) = x$ runs off to infinity.

If your domain is not a closed bounded interval, you must argue existence another way (often by the shape of the graph or a sign analysis of f').

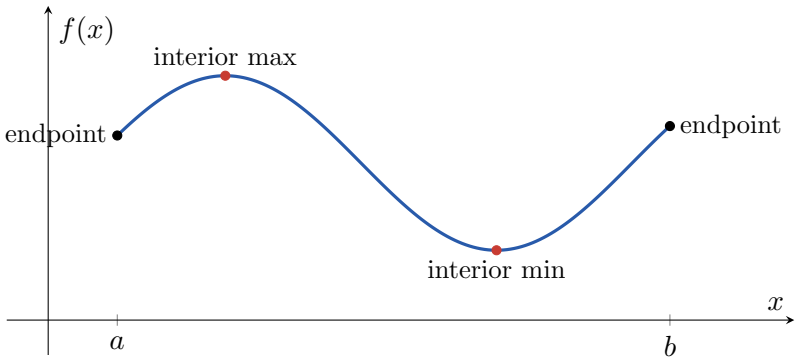


Figure 17.1: On a closed interval $[a, b]$, the absolute max and min of a continuous function can only occur at an interior critical point or at an endpoint. The closed-interval method checks all of them and compares (here both extremes happen to be interior, with the endpoints landing in between).

What is really happening

When you "set $f'(x) = 0$ and solve," what is really happening is a search for *stationary configurations* — arrangements where a tiny change in your choice produces no first-order change in the objective. Think of balancing a ball on a surface. At a peak, a valley, or a saddle, the ball is momentarily in equilibrium: the slope is zero in every direction you can move along the constraint. The derivative is measuring how sensitive your objective is to a small nudge. Setting it to zero says "find the spot where I am insensitive to nudges." That spot is a candidate for best, but you still have to check whether it is a peak (max), a valley (min), or merely a flat shelf — which is exactly why the second-derivative test and the closed-interval comparison exist. The vanishing derivative locates equilibria; the second-order information classifies them.

17.3 Worked example: the open-top box

Here is the most famous optimization problem in calculus, and deservedly so: it runs every step of the method on something you can hold in your hands.

Example 17.1 (The fold-up box)**.** You have a square sheet of cardboard 30 cm on a side. You cut a small square of side x from each corner, fold up the four flaps, and tape them to make an open-top box. What size cut x gives the box of *largest volume*?

Step 1 — picture and variables. Cutting a square of side x from each corner and folding up leaves a base that is a square of side $30 - 2x$ (you removed x from each end of every side), and the height of the box is exactly the cut x (Figure 17.2).

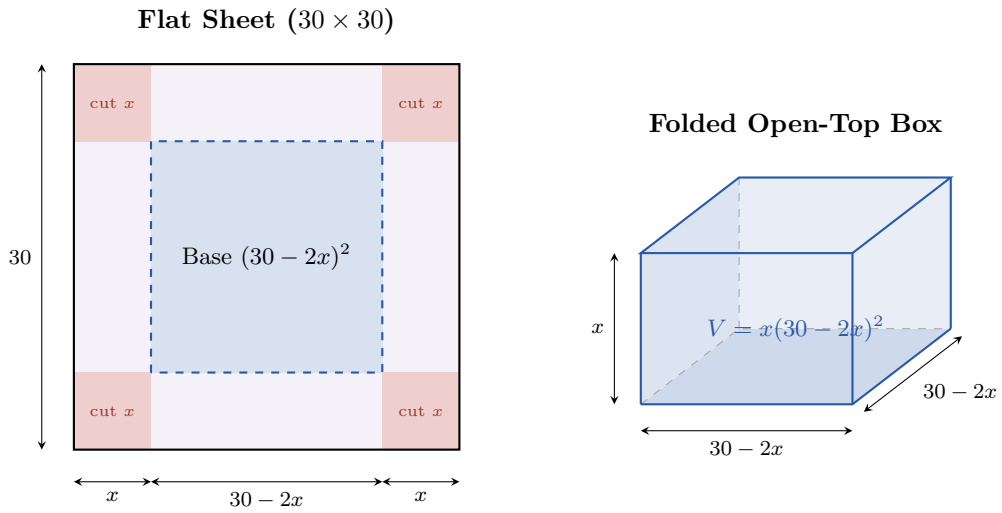


Figure 17.2: The fold-up box. Left: Removing square cutouts of side x from each corner leaves a square base of side $30 - 2x$. Right: Folding along the dashed lines creates an open-top box of height x, square base $(30 - 2x)^2$, and volume $V(x) = x(30 - 2x)^2$.

Step 2 — objective. The volume is base area times height:

$$V = (30 - 2x)^2 \cdot x.$$

This is already a function of the single variable x — the geometry handed us the constraint for free, so there is no separate elimination step here.

Step 3 — domain. What values of x make a real box? We need $x > 0$ (a genuine cut) and we need the base side $30 - 2x > 0$, i.e. $x < 15$. At $x = 0$ there is no height; at $x = 15$ there is no base. Including those degenerate endpoints (where the volume is zero) gives the closed interval

$$0 \le x \le 15.$$

We deliberately include the endpoints: now the domain is closed and bounded, V is continuous, and the Extreme Value Theorem guarantees a maximum exists. That is worth the small fiction of allowing the zero-volume cases.

Step 4 — differentiate. Expand first to make the derivative painless:

$$V = x(30 - 2x)^2 = x(900 - 120x + 4x^2) = 900x - 120x^2 + 4x^3.$$

Then

$$V'(x) = 900 - 240x + 12x^2 = 12\,(x^2 - 20x + 75) = 12\,(x - 5)(x - 15).$$

Step 5 — critical points. $V'(x) = 0$ at $x = 5$ and $x = 15$. The point $x = 15$ is an endpoint (and gives zero volume), so the only interior critical point is

$$x = 5.$$

Step 6 — test. Use the closed-interval method: evaluate V at the critical point and both endpoints.

$$V(0) = 0, \qquad V(5) = 5 \cdot 20^2 = 5 \cdot 400 = 2000, \qquad V(15) = 0.$$

The largest value is $V(5) = 2000$, so the maximum volume is $\boxed{2000 \text{ cm}^3}$, achieved by cutting squares of side $\boxed{x = 5 \text{ cm}}$.

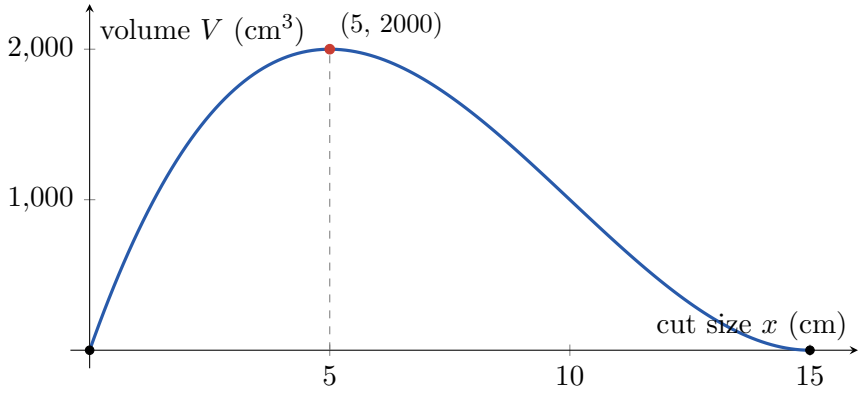


Figure 17.3: Volume of the box as a function of the corner cut x. It rises from zero, peaks at the lone interior critical point $x = 5$, and falls back to zero. The peak is the answer.

Work it yourself

Confirm the peak yourself with the second-derivative test. We have $V'(x) = 12x^2 - 240x + 900$, so $V''(x) = 24x - 240 = 24(x - 10)$. At the critical point, $V''(5) = 24(5 - 10) = -120 < 0$. A negative second derivative means concave down means a local maximum — the curve is bending downward at $x = 5$, so it is a hilltop, not a valley. This agrees with the closed-interval comparison and gives a second, independent confirmation. Then sketch the graph from $V(0) = 0$, the peak at $(5, 2000)$, and $V(15) = 0$, and watch it match Figure 17.3.

Common trap

A very common slip on the box problem is to use the side of the base, $30 - 2x$, as the height, or to forget that you remove x from *both* ends of each side, writing the base as $30 - x$ instead of $30 - 2x$. Draw the picture every time. The factor of two comes straight from the two corners on each edge, and getting it wrong quietly ruins the whole answer while leaving the algebra looking perfectly healthy.

17.4 Worked example: the cheapest can

Now a problem where the constraint really does extra work, because the objective and the constraint involve different combinations of the variables.

Example 17.2 (The minimal-surface can)**.** A cylindrical can must hold exactly 355 cm^3 (a standard soda can). What radius and height minimize the surface area — that is, use the least metal?

Step 1 — picture and variables. A cylinder has radius r and height h.

Step 2 — objective. A closed can has a top disk, a bottom disk, and a wrapped-around side (the lateral surface). The two disks contribute $2\pi r^2$; the side, if you unroll it, is a rectangle of width $2\pi r$

(the circumference) and height h, contributing $2\pi rh$. So the surface area is

$$A = 2\pi r^2 + 2\pi rh.$$

This has *two* variables. We cannot differentiate it yet.

Step 3 — constraint. The volume is fixed:

$$\pi r^2 h = 355.$$

Step 4 — reduce to one variable. Solve the constraint for the variable that is easiest to isolate. Here h comes out cleanly:

$$h = \frac{355}{\pi r^2}.$$

Substitute into the objective so that A depends on r alone:

$$A(r) = 2\pi r^2 + 2\pi r \cdot \frac{355}{\pi r^2} = 2\pi r^2 + \frac{710}{r}.$$

This single move — using the constraint to kill h — is the heart of the whole subject. We traded a two-variable problem we cannot solve for a one-variable problem we can.

Step 5 — domain. Any positive radius gives a valid can (with h adjusting to keep the volume right), so the domain is $r > 0$, the open interval $(0, \infty)$. This is *not* a closed bounded interval, so the closed-interval method does not apply directly — we will instead use the shape of $A(r)$ to confirm the minimum. Notice what happens at the ends: as $r \to 0^+$, the term $710/r \to \infty$ (a tall thin can has a huge side area); as $r \to \infty$, the term $2\pi r^2 \to \infty$ (a wide flat can has huge disks). So $A \to \infty$ at both ends, which already tells us a minimum sits somewhere in the middle.

Step 6 — differentiate.

$$A'(r) = 4\pi r - \frac{710}{r^2}.$$

Set it to zero:

$$4\pi r = \frac{710}{r^2} \quad \Longrightarrow \quad 4\pi r^3 = 710 \quad \Longrightarrow \quad r^3 = \frac{710}{4\pi} = \frac{177.5}{\pi}.$$

So

$$r = \left(\frac{177.5}{\pi}\right)^{1/3} \approx (56.50)^{1/3} \approx 3.837 \text{ cm}.$$

And the height is

$$h = \frac{355}{\pi r^2} \approx \frac{355}{\pi(3.837)^2} \approx \frac{355}{46.25} \approx 7.675 \text{ cm}.$$

Step 7 — confirm it is a minimum. Two independent checks:

- *Second derivative.* $A''(r) = 4\pi + \dfrac{1420}{r^3} > 0$ for every $r > 0$. The function is concave up everywhere, so the single critical point is a minimum — in fact the *global* minimum, since a function that is concave up on all of $(0, \infty)$ can have only one turning point.
- *End behavior.* We saw $A \to \infty$ as $r \to 0^+$ and as $r \to \infty$, with exactly one critical point in between. A continuous function that climbs to infinity at both ends and turns around once must turn at a minimum (Figure 17.4).

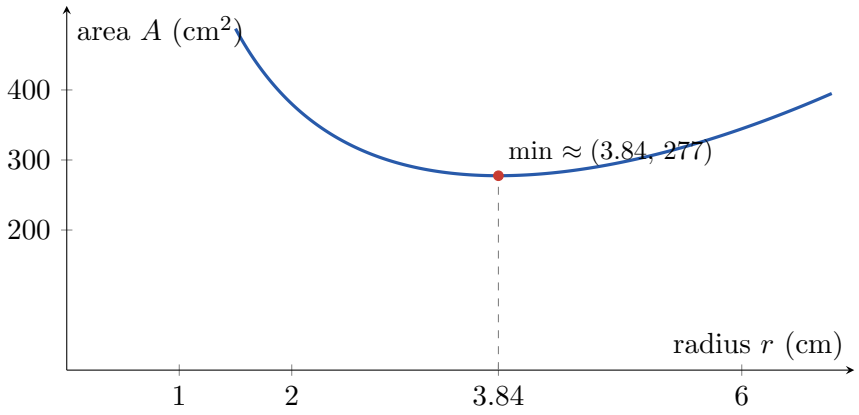


Figure 17.4: Surface area of a 355 cm^3 can as a function of radius. The two terms pull in opposite directions: $710/r$ blows up for thin cans, $2\pi r^2$ blows up for wide cans. The trade-off has a single sweet spot.

Why this matters

Look at the answer: $h \approx 7.675$ and $r \approx 3.837$, so $h = 2r$. The most efficient closed can is exactly as *tall* as it is *wide* — its height equals its diameter, so it fits snugly inside a cube. This ratio does not depend on the specific number 355; you get $h = 2r$ for *any* fixed volume. Try it symbolically: minimizing $A = 2\pi r^2 + 2V/r$ gives $4\pi r = 2V/r^2$, i.e. $2\pi r^3 = V = \pi r^2 h$, which simplifies to $h = 2r$ no matter what V is. A clean, scale-free fact sits under the arithmetic. (Real soda cans are taller and thinner than this — because the seams and the thicker top cost extra metal the simple model ignores, and because a hand likes a narrower grip. The math is right; the model is just simpler than a real can.)

Why the technique works

Why does substituting the constraint give the right answer, rather than some distortion? Because the constraint equation $\pi r^2 h = 355$ does not throw any cans away — it simply records that for each radius there is exactly one matching height. The substitution walks along the *curve of all valid cans* in the (r, h) plane, and $A(r)$ is the genuine surface area of the one valid can with that radius. We are not approximating; we are restricting attention to the feasible set and reading off its true area. Minimizing along the trail gives the minimum over all valid cans, which is exactly what was asked. This is why "reduce to one variable" loses nothing: the eliminated variable was never free — the constraint had already decided it.

17.5 Worked example: the fence with a free side

A constraint can also be the thing in short supply. This is the classic farmer problem, and it teaches a lesson about reading the geometry before reaching for algebra.

Example 17.3 (Maximum field, one free side)**.** A farmer has 200 m of fencing and wants to enclose a rectangular field. One side of the field runs along a straight river and needs no fence. What dimensions enclose the *largest area*?

Step 1 — picture and variables. Call the two sides perpendicular to the river x (there are two of them) and the side parallel to the river y (there is one of it — the opposite, river side, is free); see Figure 17.5.

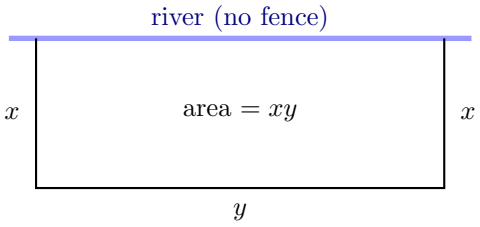


Figure 17.5: Three sides are fenced; the river bounds the fourth for free. Fencing used is $2x + y$.

Step 2 — objective. Maximize the area

$$A = xy.$$

Step 3 — constraint. Only three sides use fence, and there are 200 m of it:

$$2x + y = 200.$$

Step 4 — reduce. Solve the constraint for y: $y = 200 - 2x$. Substitute:

$$A(x) = x(200 - 2x) = 200x - 2x^2.$$

Step 5 — domain. We need $x > 0$ and $y = 200 - 2x > 0$, so $0 < x < 100$; including the degenerate endpoints (zero area) gives the closed interval $[0, 100]$, and again the Extreme Value Theorem guarantees a maximum.

Step 6 — differentiate.

$$A'(x) = 200 - 4x = 0 \quad \Longrightarrow \quad x = 50.$$

Then $y = 200 - 2(50) = 100$.

Step 7 — test. Closed-interval comparison:

$$A(0) = 0, \qquad A(50) = 50 \cdot 100 = 5000, \qquad A(100) = 0.$$

The maximum area is $\boxed{5000 \text{ m}^2}$, with the two short sides $x = 50$ m and the long side $y = 100$ m. (The second derivative $A'' = -4 < 0$ confirms concave down, a maximum.)

Intuition

Notice the answer splits the fence in a memorable way: the single parallel side y uses 100 m and the two perpendicular sides together also use $2x = 100$ m. The fence is divided *equally* between the direction along the river and the direction across it. That balance — spend half your resource each way — is the recurring signature of these rectangle problems, and once you have seen it you can often guess the answer before differentiating, then use calculus to confirm.

Common trap

The free side changes the constraint, not the area. A frequent error is to fence all four sides out of habit, writing $2x + 2y = 200$, or to optimize the perimeter instead of the area. Always ask: *which* quantity is fixed (the constraint) and *which* am I trying to make extreme (the objective)? Here the fence is fixed and the area is free to grow. Swapping their roles — maximizing the fence — is a question with no answer, and that absurdity is your signal that you mislabeled them.

17.6 Worked example: the closest point

Distance problems hide a small but important computational trick, and they show off the closed-interval reasoning on a domain that is genuinely infinite.

Example 17.4 (Nearest point on a parabola)**.** Find the point on the parabola $y = x^2$ that is closest to the point $(0, 2)$.

Step 1–2 — variables and objective. A point on the parabola is (x, x^2). Its distance to $(0, 2)$ is

$$D(x) = \sqrt{(x-0)^2 + (x^2-2)^2} = \sqrt{x^2 + (x^2-2)^2}.$$

We want to minimize this.

The trick. Minimizing the distance D is the same as minimizing the *squared* distance $S = D^2$, because squaring is an increasing function on $[0, \infty)$ — wherever D is smallest, D^2 is smallest too, and at the same x. So we minimize the square-root-free

$$S(x) = x^2 + (x^2 - 2)^2,$$

which is far easier to differentiate. This substitution costs nothing and saves a chain-rule-through-a-square-root mess.

Why the technique works

Why are you allowed to minimize D^2 instead of D? Because the squaring map $t \mapsto t^2$ is strictly increasing for $t \geq 0$, and distances are never negative. A strictly increasing function preserves order: if $D(a) < D(b)$ then $D(a)^2 < D(b)^2$, and vice versa. So D and D^2 rise and fall in lockstep; they have their minimum at the *same* x. The only thing that changes is the messiness of the derivative. This is a general and very useful move: you may replace your objective by any strictly increasing function of it (square it, take its log, etc.) without moving the location of the optimum. Optimizers reach for logs and squares constantly for exactly this reason.

Step 3 — domain. Any real x gives a point on the parabola, so the domain is all of $\mathbb{R} = (-\infty, \infty)$. As $|x| \to \infty$ the point on the parabola shoots off to infinity, so $S \to \infty$ at both ends; a minimum exists in the middle.

Step 4 — differentiate. Expand:

$$S(x) = x^2 + x^4 - 4x^2 + 4 = x^4 - 3x^2 + 4.$$

Then

$$S'(x) = 4x^3 - 6x = 2x(2x^2 - 3).$$

Step 5 — critical points. $S'(x) = 0$ gives $x = 0$ or $2x^2 = 3$, i.e. $x = \pm\sqrt{3/2} = \pm\sqrt{1.5} \approx \pm 1.225$.

Step 6 — test. Compare the squared distances at the three critical points:

$$S(0) = 0 - 0 + 4 = 4,$$

$$S\left(\pm\sqrt{1.5}\right) = (1.5)^2 - 3(1.5) + 4 = 2.25 - 4.5 + 4 = 1.75.$$

The smaller value 1.75 wins, so the closest points are $\left(\pm\sqrt{1.5},\, 1.5\right)$, at distance $D = \sqrt{1.75} \approx 1.323$ (Figure 17.6).

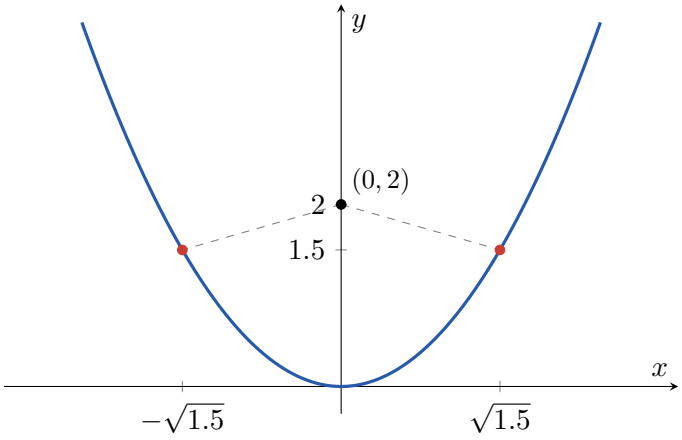


Figure 17.6: The two closest points on $y = x^2$ to the target $(0, 2)$, at $x = \pm\sqrt{1.5}$. The vertex $x = 0$ is also a critical point, but there the squared distance is *largest* among the three candidates (it is a local maximum of S, hence locally the farthest), so it loses the comparison.

Common trap

The vertex $x = 0$ is a genuine critical point — $S'(0) = 0$ — but it is *not* the answer. In fact S has a local *maximum* there: among the three critical points the vertex is locally the farthest from $(0, 2)$, not the nearest, and the comparison exposes it. This is exactly why you never stop at "I found where $S' = 0$." You must evaluate S at *all* the critical points and pick the winner. Had you found only $x = 0$ and quit, you would have reported the worst of the three candidates as the best. The derivative finds suspects; the comparison convicts.

Connection

A pretty piece of geometry sits behind the algebra. At the true closest point, the segment from $(0, 2)$ to the curve is *perpendicular* to the parabola's tangent line. To see why: if the connecting segment met the curve at any other angle, you could slide a little along the curve and get closer, because part of your motion would be "toward" the target. Only when the segment is perpendicular is there no closer direction to slide. This "shortest distance is along the normal" principle reappears throughout physics (light takes the perpendicular-incidence path, a stretched string pulls along the normal) and is the geometric soul of the Lagrange multiplier method you will meet in multivariable calculus. The condition $S'(x) = 0$ and the

condition "segment $\perp$ tangent" are the same statement in two languages.

17.7 A profit example: optimization beyond geometry

Optimization is not only about boxes and fences. The same method runs economics, biology, and machine learning. Here is an example from economics.

Example 17.5 (Maximizing profit)**.** A workshop can sell x handmade lamps per week. Market research says that to sell x lamps the price per lamp must be $p(x) = 120 - 2x$ dollars (sell more, charge less). Each lamp costs \$20 to make, plus \$300 in fixed weekly overhead. How many lamps maximize weekly profit?

Objective. Profit is revenue minus cost. Revenue is price times quantity:

$$R(x) = x\,p(x) = x(120 - 2x) = 120x - 2x^2.$$

Cost is overhead plus per-unit cost:

$$C(x) = 300 + 20x.$$

So profit is

$$P(x) = R(x) - C(x) = (120x - 2x^2) - (300 + 20x) = -2x^2 + 100x - 300.$$

Domain. We need $x \geq 0$, and the price must stay positive: $120 - 2x > 0$ means $x < 60$. So $0 \leq x \leq 60$ (a closed bounded interval — the Extreme Value Theorem applies).

Differentiate.

$$P'(x) = -4x + 100 = 0 \quad \Longrightarrow \quad x = 25.$$

Test. $P''(x) = -4 < 0$, concave down everywhere, so $x = 25$ is the global maximum. The profit is

$$P(25) = -2(625) + 100(25) - 300 = -1250 + 2500 - 300 = 950.$$

Make $\boxed{\text{25 lamps per week for \$950 profit}}$, at a price of $p(25) = 120 - 50 = \$70$ each.

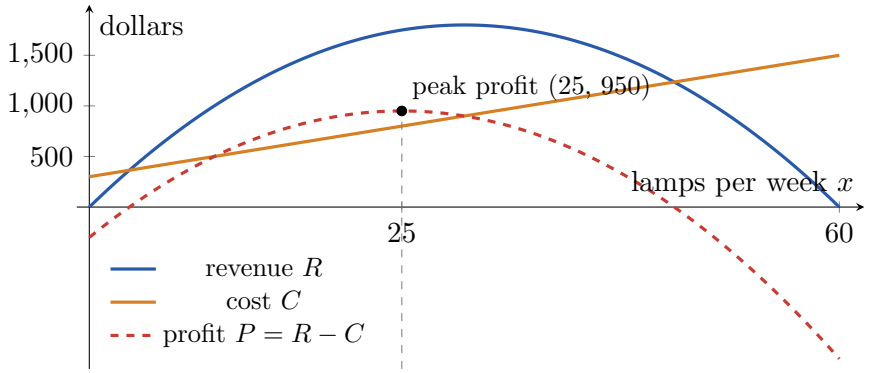


Figure 17.7: Revenue, cost, and profit for the lamp workshop. Profit peaks where its slope is zero — equivalently, where the revenue and cost curves are momentarily climbing at the *same* rate, i.e. marginal revenue equals marginal cost (at $x = 25$).

Application

Economists recognize the condition $P'(x) = 0$ as the famous rule *marginal revenue equals marginal cost*. Watch: $P' = R' - C'$, so $P' = 0$ means $R'(x) = C'(x)$. The derivative R' is the extra revenue from selling one more unit (marginal revenue) and C' is the extra cost of making one more unit (marginal cost). The rule says: keep producing as long as the next unit brings in more than it costs, and stop the instant they balance (Figure 17.7). Past that point each extra lamp costs more than it earns and profit falls. The entire theory of the profit-maximizing firm is this one derivative set to zero — the same Fermat condition as the cardboard box, in the language of money.

Application

The same skeleton drives modern machine learning. Training a model means choosing parameters to *minimize* a loss function — the objective is the error, the variables are the millions of weights, and "learning" is optimization. There is no clean constraint to substitute and no formula for the answer, so instead of solving $\nabla(\text{loss}) = 0$ exactly, the computer rolls downhill: it repeatedly steps in the direction the gradient says decreases the loss fastest (gradient descent). But the target is unchanged from this chapter — a place where the derivative vanishes, a stationary point of the objective. Every example here is a hand-solvable instance of what a neural network does numerically a billion times over.

17.8 When the candidate is not the answer: pitfalls and checks

The single most common way to lose points — and to be flatly wrong in real life — is to find a critical point and *assume* it is the extreme you want. It might be a minimum when you wanted a maximum. It might be a local extreme beaten by an endpoint. It might be a saddle-like inflection that is neither. Here is how to make sure.

Focus

Three ways to confirm you found the right extreme.

1. **Closed-interval method** (best when the domain is a closed bounded interval): evaluate the objective at every critical point and both endpoints, then read off the largest/smallest. The Extreme Value Theorem guarantees the winner is in your list. No further argument needed.

2. **First-derivative test** (works on any interval): check the sign of f' just left and just right of the critical point. Sign change $+ \to -$ means a local max; $- \to +$ means a local min; no change means neither.

3. **Second-derivative test** (quick when f'' is easy): at a critical point, $f'' < 0$ means local max (concave down), $f'' > 0$ means local min (concave up). If $f'' = 0$ the test is silent — fall back to method 1 or 2.

For an *open* or *unbounded* domain, also check the behavior at the open ends (limits as x approaches the boundary or $\pm\infty$) to be sure a max or min exists at all.

Common trap

The endpoint that wins. Suppose you must minimize $f(x) = x^3 - 3x$ on $[0, 3]$. The critical points solve $f'(x) = 3x^2 - 3 = 0$, giving $x = \pm 1$; only $x = 1$ is in the domain, and there $f(1) = 1 - 3 = -2$. If you stopped there you would report the minimum as -2 — and you would be right. But if instead the problem asked for the *maximum* on $[0, 3]$, the critical point $x = 1$ is a local *minimum*, not a maximum at all. The maximum lives at the endpoint: $f(3) = 27 - 9 = 18$ versus $f(0) = 0$. The lesson: the interesting answer is sometimes at the edge of the domain, where the derivative never vanishes (Figure 17.8). *Always check the endpoints.*

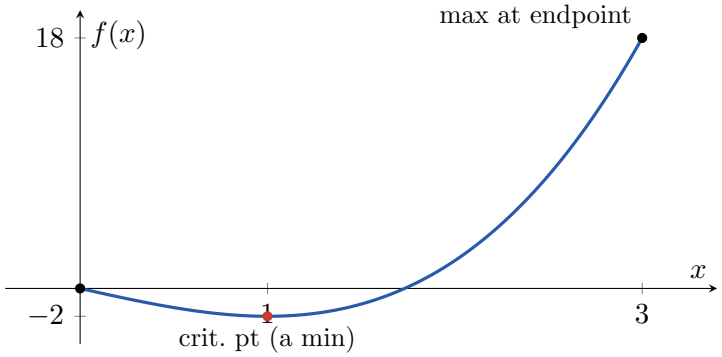


Figure 17.8: $f(x) = x^3 - 3x$ on $[0, 3]$. The only interior critical point is a *minimum*; the maximum hides at the right endpoint. Find the critical points, but let the comparison — not your hopes — pick the winner.

Walking question

Take a walk and think about *existence* versus *location*. The derivative is a tool for *location*: it points at where an extreme could be. But it says nothing about whether an extreme *exists*. What guarantees existence? On a closed bounded interval, continuity does (the Extreme Value Theorem). On an open or infinite domain you need something else — the function running off to infinity at both ends, say, or a single turning point. Ask yourself: for each worked example in this chapter, *what* guaranteed the answer existed before I trusted the critical point? You will find the closed-interval cases leaned on the theorem, while the can and the distance problems leaned on $\to \infty$ end-behavior. Existence and location are two separate questions, and a complete solution answers both.

What is really happening

When a problem has no maximum, the symptom is usually that the objective improves forever as you slide toward a boundary the domain does not include. "Find the rectangle of largest area with perimeter at most 100" has an answer; "find the rectangle of largest area" (no constraint) does not — you can always make a bigger one. The constraint is what bounds the feasible set and lets a best exist. So when a real optimization has no solution, the deepest fix is rarely more calculus; it is finding the missing constraint — the budget, the material limit, the physical wall — that the problem forgot to state. Calculus locates the optimum; modeling

guarantees there is one to locate.

17.9 A fully general template, worked symbolically

To cement that the method is one method, let us do the fence problem's cousin *with letters instead of numbers*, so you can see the structure stand on its own.

Example 17.6 (Largest rectangle inscribed under a parabola)**.** A rectangle sits with its base on the x-axis and its top two corners on the downward parabola $y = c - x^2$ (with $c > 0$), symmetric about the y-axis. What width makes its area largest?

Variables. By symmetry the top corners are at $(\pm x, c - x^2)$ for some $x > 0$. The rectangle then has width $2x$ and height $c - x^2$.

Objective.

$$A(x) = (\text{width})(\text{height}) = 2x\,(c - x^2) = 2cx - 2x^3.$$

Domain. We need the top corners above the axis, $c - x^2 \geq 0$, so $0 \leq x \leq \sqrt{c}$ — a closed bounded interval.

Differentiate.

$$A'(x) = 2c - 6x^2 = 0 \quad \Longrightarrow \quad x^2 = \frac{c}{3} \quad \Longrightarrow \quad x = \sqrt{c/3}.$$

Test. The endpoints give $A(0) = 0$ and $A(\sqrt{c}) = 0$, while the critical point gives a positive value, so the critical point is the maximum. (Or use $A''(x) = -12x < 0$ for $x > 0$.) The maximal area is

$$A\Big(\sqrt{c/3}\Big) = 2\sqrt{\tfrac{c}{3}}\,\big(c - \tfrac{c}{3}\big) = 2\sqrt{\tfrac{c}{3}} \cdot \tfrac{2c}{3} = \frac{4c}{3}\sqrt{\frac{c}{3}}.$$

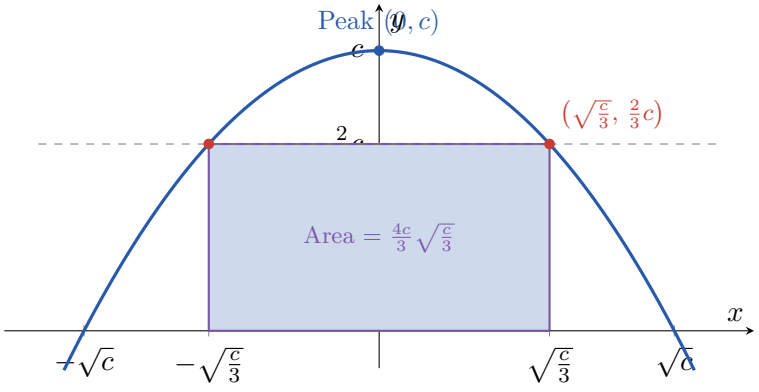


Figure 17.9: The optimal rectangle inscribed under the parabola $y = c - x^2$ (shown for $c = 3$). The upper corners meet the curve at $x = \pm\sqrt{c/3}$, fixing the optimal height at exactly two-thirds of the parabola's maximum height: $y = \frac{2}{3}c$.

Why this matters

The optimal height is $c - x^2 = c - \frac{c}{3} = \frac{2c}{3}$ — exactly *two-thirds* of the parabola's peak height c, no matter what c is. The best rectangle always reaches two-thirds of the way up. Solving the problem once, with a letter, reveals a clean ratio that a hundred numerical versions would only hint at. This is the payoff of the symbolic approach: numbers give answers, but letters give *laws*.

17.10 Summary: the shape of every solution

Focus

What to carry away.

- Optimization = objective + constraint + domain. The first and hardest job is *modeling*: turning words and pictures into a one-variable function on a definite domain.
- The constraint is your friend: it is what lets you eliminate variables and reach a single-variable function the derivative can handle.
- Critical points (where $f' = 0$ or f' is undefined) are *candidates*, not answers. Endpoints are candidates too.
- On a closed bounded interval, the Extreme Value Theorem guarantees a best exists, and the closed-interval method finds it by comparing values — no derivative test required.
- On open or infinite domains, you must separately argue that an extreme exists (end behavior, concavity, or a single turning point).
- Confirm with the closed-interval comparison, the first-derivative test, or the second-derivative test — and always sanity-check units and plausibility.
- The very same method — Fermat's vanishing slope — runs geometry, physics, economics, and machine learning. It is the most widely used idea in all of calculus.

17.10.1 Extrema in Nature and the Multivariable Frontier

Connection

Optimization gave calculus its primary mandate in the physical and commercial world. Through Fermat's stationarity principle and the closed-interval method:

- *Physical variational principles*: Light rays, gravitational orbits, and hanging chains follow trajectories that extremize fundamental physical quantities like time and action.
- *Engineering design*: Structural envelopes, beverage containers, and mechanical linkages can be designed to minimize material consumption, minimize drag, or maximize structural rigidity under physical constraints.
- *Economic equilibrium*: Monopolists and competitive firms determine profit-maximizing

production quantities by equating marginal revenue to marginal cost ($R'(q) = C'(q)$).

Yet these methods are fundamentally restricted to *single-variable* problems (or two-variable problems that can be collapsed to one variable via algebraic substitution). In the real world, optimization rarely presents itself so neatly:

- In practical engineering and machine learning, objectives depend on dozens, thousands, or billions of variables.
- Constraints are frequently non-linear or implicitly coupled equations $g_1(x, y, z) = 0, \ldots, g_k(x, y, z) = 0$ that cannot be solved algebraically for one variable in terms of the others.
- Solving single-variable equations via substitution often destroys natural physical symmetries and introduces artificial coordinate singularities.

To break through this bottleneck, we must advance to multivariable calculus. In Chapter 33 (*The Gradient, Directional Derivatives, and the Multivariable Chain Rule*), we encounter the geometric masterpiece of constrained optimization: *Lagrange multipliers*. Instead of algebraically eliminating variables, we optimize an objective $f(\mathbf{x})$ subject to $g(\mathbf{x}) = c$ by matching their normal vectors: $\nabla f = \lambda \nabla g$.

17.11 Exercises

Exercise 17.1 (Open-top box, rectangular sheet)**.** A box with an open top is made from a 16 cm $\times$ 10 cm rectangle of cardboard by cutting equal squares of side x from the corners and folding up the sides. (a) Write the volume $V(x)$. (b) State the domain of x. (c) Find the cut x that maximizes the volume, and give the maximum volume. (d) Confirm with the second-derivative test.

Exercise 17.2 (The cheapest open-top can)**.** An *open-top* cylindrical can (no lid) must hold 1000 cm^3. (a) Write the surface area as a function of the radius r alone. (b) Find the radius and height that minimize the metal used. (c) Show that for the open-top can the optimal height equals the radius, $h = r$ — contrast this with $h = 2r$ for the closed can, and explain in words why losing the lid makes the optimal can relatively shorter.

Exercise 17.3 (Fence with a divider)**.** A farmer has 240 m of fencing to build a rectangular pen *and* an internal fence parallel to one side that splits the pen into two equal halves. (a) If x is the length of the divided direction (so three segments of length x are needed) and y the other, write the constraint and the area objective. (b) Find the dimensions of maximum total area. (c) How does the answer compare to the no-divider case?

Exercise 17.4 (Closest point on a line)**.** Find the point on the line $y = 2x + 1$ closest to the origin. (a) Minimize the *squared* distance and explain why that is legitimate. (b) Verify that the segment from the origin to your point is perpendicular to the line, and explain geometrically why it must be.

Exercise 17.5 (Norman window)**.** A "Norman window" is a rectangle topped by a semicircle whose diameter equals the rectangle's width. The total perimeter is fixed at P meters. Find the dimensions admitting the most light (i.e. the largest area). Express the optimal rectangle width in terms of P.

Exercise 17.6 (Maximum-area triangle)**.** Among all right triangles with legs summing to a fixed length L (so $\text{leg}_1 + \text{leg}_2 = L$), find the one with the largest area. What is special about its shape? Prove your answer is a maximum, not a minimum.

Exercise 17.7 (An endpoint maximum)**.** Find the absolute maximum and absolute minimum of $f(x) = x^3 - 6x^2 + 9x + 2$ on the closed interval $[0, 5]$ using the closed-interval method. Identify which extreme occurs at a critical point and which (if any) at an endpoint.

Exercise 17.8 (Time, not distance: the lifeguard problem)**.** A lifeguard at point A on a straight beach must reach a swimmer at point B in the water. She runs along the sand at speed v_1 and swims at the slower speed $v_2 < v_1$. Setting up coordinates so the shoreline is the x-axis, $A = (0, a)$ with $a > 0$, and $B = (d, -h)$ with $h > 0$, she enters the water at the point $(x, 0)$. (a) Write the total travel *time* $T(x)$. (b) Show that $T'(x) = 0$ leads to the condition $\dfrac{\sin\theta_1}{\sin\theta_2} = \dfrac{v_1}{v_2}$, where θ_1, θ_2 are the angles the running and swimming paths make with the line perpendicular to the shore. (This is *Snell's law of refraction* — light minimizes travel time the same way.)

Exercise 17.9 (Why $h = 2r$, in general)**.** For a closed cylindrical can of fixed volume V, minimize the surface area *symbolically* (keep V as a letter) and prove that the optimal can always satisfies $h = 2r$, independent of V. Then prove the critical point is a minimum using the second derivative.

Exercise 17.10 (Conceptual: existence)**.** Give an example of a continuous function on an *open* interval that has a critical point which is not its absolute maximum, and explain why the closed-interval method cannot be applied. Then state precisely what additional information would let you conclude that an absolute maximum exists.

Exercise 17.11 (Proof: substitution preserves the optimum)**.** Suppose you minimize $f(x)$ over a domain D, and g is a strictly increasing function defined on the range of f. Prove that f and $g \circ f$ attain their minimum at exactly the same points of D. (This is the theorem that justifies minimizing D^2 instead of D, or minimizing a log-likelihood instead of a likelihood.)

Chapter 18

L'Hopital's Rule and Indeterminate Forms

Some limits answer themselves. If you want $\lim_{x\to 2}(x^2+1)$, you put in 2 and read off 5; continuity does the work and there is nothing to think about. The interesting limits are the ones that *refuse* to answer themselves — the ones where substituting the target value produces not a number but a piece of nonsense like $\frac{0}{0}$ or $\frac{\infty}{\infty}$. These are the *indeterminate forms*, and despite the menacing name they are the most important limits in all of calculus. The derivative itself is one of them: $f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$ is a $\frac{0}{0}$ in disguise, every single time.

This chapter is about *L'Hopital's rule*, which resolves a huge class of these forms by a single, almost shocking idea: when a ratio collapses to $\frac{0}{0}$, the limit is decided not by the values of the two functions (those are both heading to zero) but by the *rates* at which they get there. Replace the functions by their derivatives and the fog lifts. We will see why that is true, where it comes from, exactly when it is allowed, and the several ways people get it wrong. And along the way we will return to a question from much earlier in the book — which functions grow faster than which — and finally settle it cleanly.

18.1 What "indeterminate" really means

Start by getting the word right, because almost everyone misreads it. An indeterminate form is *not* an expression with no value. It is an expression whose value is *not determined by the form alone*. That is a precise and different claim, and the whole subject hinges on it.

Focus

"Indeterminate" is a statement about *information*, not about *impossibility*. The symbol $\frac{0}{0}$ tells you that both the top and the bottom are shrinking to zero — but it does not tell you *how fast*, and that is the one thing that decides the limit. The form has thrown away the information you need. Your job is to dig it back out.

To feel why $\frac{0}{0}$ genuinely carries no answer, watch four limits that all wear that same face as $x \to 0$,

yet land in four different places:

$$\frac{2x}{x} \to 2, \qquad \frac{x}{x^2} \to \infty, \qquad \frac{x^2}{x} \to 0, \qquad \frac{x}{x} \to 1.$$

In every one, plugging in $x = 0$ gives $\frac{0}{0}$. If $\frac{0}{0}$ had a fixed value, all four would have to share it. They do not: the answers are 2, ∞, 0, and 1. So $\frac{0}{0}$ cannot be a number. It is a *question*, and the answer depends entirely on the relative speed of the vanishing.

Intuition

Think of two runners both decelerating to a dead stop at the finish line. The ratio of their *remaining distances* as they crawl in is $\frac{0}{0}$ — both gaps go to zero. But whether that ratio approaches 2 or $\frac{1}{2}$ or 1 depends only on their relative *speeds* near the line. The runner who is still moving twice as fast keeps twice the gap. The form $\frac{0}{0}$ hides the speeds; L'Hopital's rule hands them back to you as derivatives.

Contrast this with a form that is *not* indeterminate. The expression $\frac{1}{0}$ (a nonzero number over something shrinking to zero) is not indeterminate: it blows up, with a sign you can usually pin down. And $\frac{0}{1}$ is not indeterminate either: it is just 0. The trouble appears only when a "smallness" or a "largeness" competes against another of the same kind, so that neither wins automatically.

Focus

The seven indeterminate forms. Up to the tricks of this chapter, all the indeterminacy in elementary calculus comes in seven flavors:

$$\frac{0}{0}, \qquad \frac{\infty}{\infty}, \qquad 0 \cdot \infty, \qquad \infty - \infty, \qquad 1^\infty, \qquad 0^0, \qquad \infty^0.$$

The first two are ratios — L'Hopital's rule attacks them directly. The middle two, $0 \cdot \infty$ and $\infty - \infty$, are *disguised* ratios; we rewrite them as $\frac{0}{0}$ or $\frac{\infty}{\infty}$. The last three are *powers*; we take a logarithm to drag them down into a product, then into a ratio. Everything reduces to the first two.

Common trap

Do not put forms like $\frac{\infty}{0}$, 0^∞, $\infty + \infty$, or $\frac{1}{0}$ on the indeterminate list. They are *determinate*: $\frac{1}{0}$ diverges, $0^\infty = 0$, $\infty + \infty = \infty$, $\frac{\infty}{0}$ diverges. The test is always the same — is there a genuine competition where the answer could come out anywhere depending on the rates? If both forces push in the same direction, the form is not indeterminate and you should just read off the answer.

18.2 The ratio-of-rates idea

Here is the insight that makes the whole chapter. Slow down for it. Suppose $f(a) = 0$ and $g(a) = 0$, and we want

$$\lim_{x \to a} \frac{f(x)}{g(x)}.$$

Near $x = a$ both functions are small. But "small" is not one number — a smooth function that passes through zero is, very close to that zero, almost exactly a straight line: its tangent line. Concretely, for x near a,

$$f(x) \approx f(a) + f'(a)(x-a) = f'(a)\,(x-a), \qquad g(x) \approx g(a) + g'(a)(x-a) = g'(a)\,(x-a),$$

because $f(a) = g(a) = 0$ kills the constant terms. Now form the ratio and watch the common factor $(x-a)$ cancel:

$$\frac{f(x)}{g(x)} \approx \frac{f'(a)\,(x-a)}{g'(a)\,(x-a)} = \frac{f'(a)}{g'(a)}.$$

The vanishing $(x-a)$ in the numerator and denominator was the entire source of the $\frac{0}{0}$ trouble — and it cancels. What survives is the ratio of the two *slopes* at a. That is L'Hopital's rule, glimpsed through the linear approximation.

What is really happening

What is really happening in $\frac{0}{0}$. Both functions are diving toward zero, and near their common zero each one is essentially its tangent line — a number (its slope) times the same small quantity $(x-a)$. The ratio of two such lines is the ratio of their slopes; the small quantity $(x-a)$ cancels top and bottom and never gets a chance to be "zero over zero." So the limit is not really $\frac{0}{0}$ at all. It is $\frac{f'(a)}{g'(a)}$ — the ratio of the rates at which the two functions leave zero. L'Hopital's rule is the act of looking past the values, which are both 0 and tell you nothing, to the slopes, which decide everything.

Look back at the four warm-up limits with this lens. For $\frac{2x}{x}$ the slopes at 0 are 2 and 1, ratio 2. For $\frac{x^2}{x}$ the numerator leaves zero with slope 0 (it is flat there) while the denominator has slope 1, ratio 0. For $\frac{x}{x^2}$ it is $\frac{1}{0}$, which is why that one blew up. The ratio of slopes predicts each answer exactly. The picture below shows the mechanism: near the shared zero, f and g are their tangent lines, and the quotient is the quotient of the lines.

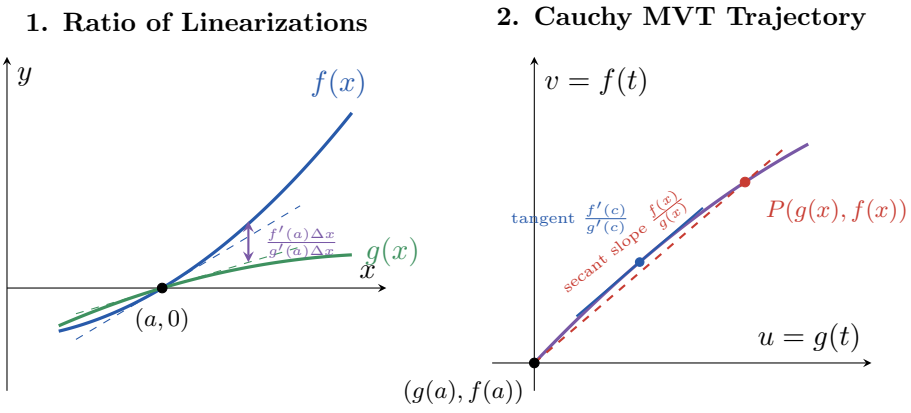


Figure 18.1: Two views of L'Hopital's ratio of rates. Left: Close to the shared root $x = a$, both curves hug their tangent lines $f(x) \approx f'(a)(x-a)$ and $g(x) \approx g'(a)(x-a)$; the $(x-a)$ factors cancel, leaving the slope ratio $f'(a)/g'(a)$. Right: In Cauchy's parametric plane $(g(t), f(t))$, the secant slope from the origin to $(g(x), f(x))$ equals the tangent slope $f'(c)/g'(c)$ at some intermediate $c \in (a, x)$. As $x \to a$, the secant swings into the initial tangent line at the origin.

Common trap

This tangent-line argument is the right *intuition*, but it is not yet a proof, and it quietly assumes more than L'Hopital's rule needs. It assumes f and g are differentiable *at* a with $g'(a) \neq 0$, and it hides what happens when both slopes are zero (giving $\frac{0}{0}$ all over again). The actual theorem, proved below with the Mean Value Theorem, is sharper: it only needs the derivatives near a, not the cancellation of tangent lines, and it lets you repeat the trick when the first attempt stalls.

18.3 L'Hopital's rule, stated precisely

Theorem 18.1 (L'Hopital's rule, the 0/0 case)**.** *Let f and g be differentiable on an open interval containing a (except possibly at a itself), with $g'(x) \neq 0$ near a. Suppose*

$$\lim_{x\to a} f(x) = 0 \qquad \textit{and} \qquad \lim_{x\to a} g(x) = 0.$$

If $\lim_{x\to a} \dfrac{f'(x)}{g'(x)}$ exists (as a finite number or as $\pm\infty$), then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}.$$

Several features of this statement are doing real work, so read them deliberately. The rule promises only one direction: *if* the limit of the derivative ratio exists, then the original limit exists and equals it. It says nothing if the derivative ratio fails to settle down — in that case the rule is simply silent, and the original limit might still exist for another reason. Notice too that we never need f or g to be defined at a; only their behavior *near* a matters, which is exactly right for a limit. And the very same statement holds verbatim for one-sided limits, for $x \to \pm\infty$, and — with one change we will note — for the $\frac{\infty}{\infty}$ case.

Why the technique works

Why the rule works: it is the Mean Value Theorem applied to two functions at once. The clean proof does not lean on tangent lines at all. It leans on *Cauchy's Mean Value Theorem*, the two-function version of the MVT you met in the chapter on the Mean Value Theorem. Cauchy's theorem says that for suitable f, g on an interval $[a, x]$ there is a point c strictly between a and x with

$$\frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c)}{g'(c)}.$$

That is the bridge. The left side is built from *values*; the right side is built from *derivatives*. If $f(a) = g(a) = 0$, the left side collapses to exactly $\frac{f(x)}{g(x)}$ — the thing we want — and it has been rewritten as a ratio of derivatives at some in-between point c. As $x \to a$, that point c is squeezed toward a as well (it is trapped between a and x), so the ratio $\frac{f'(c)}{g'(c)}$ is forced to the limit of $\frac{f'}{g'}$. The values turn into derivatives by the MVT, and the derivatives carry the answer.

Focus

Proof (the $0/0$ case, finite a). First extend f and g to be continuous at a by *defining* $f(a) = g(a) = 0$; since $f, g \to 0$ as $x \to a$, this loses nothing and makes them continuous on a closed interval with endpoint a. Fix x near a, say $x > a$ (the case $x < a$ is identical). On $[a, x]$ both functions are continuous and on (a, x) both are differentiable, and $g' \neq 0$ there. Cauchy's Mean Value Theorem then supplies a point $c \in (a, x)$ with

$$\frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c)}{g'(c)}.$$

(En route, $g(x) \neq g(a)$ because if they were equal the ordinary MVT would force g' to vanish somewhere in (a, x), which we excluded; so the left side is legal.) Using $f(a) = g(a) = 0$, the left side is just $\frac{f(x)}{g(x)}$, hence

$$\frac{f(x)}{g(x)} = \frac{f'(c)}{g'(c)}, \qquad a < c < x.$$

Now let $x \to a^+$. The point c is pinned in (a, x), so $c \to a^+$ too. By hypothesis $\frac{f'(t)}{g'(t)} \to L$ as $t \to a$, so $\frac{f'(c)}{g'(c)} \to L$, and therefore $\frac{f(x)}{g(x)} \to L$. The left-hand limit is the same argument on $[x, a]$. Combining one-sided limits gives the theorem. ■

Remark 18.1. The $\frac{\infty}{\infty}$ case is true with the identical conclusion, but its proof is fussier — you cannot simply set $f(a) = g(a) = 0$ because the functions are blowing up, not vanishing. The correct argument splits the interval and controls a tail term; it is a standard but more technical ε estimate. The good news is that you may use it with the same confidence and the same mechanics. We state it next and move on.

Theorem 18.2 (L'Hopital's rule, the ∞/∞ case)**.** *Under the same differentiability hypotheses, if $|f(x)| \to \infty$ and $|g(x)| \to \infty$ as $x \to a$ (or as $x \to \pm\infty$), and if $\lim \frac{f'(x)}{g'(x)}$ exists (finite or $\pm\infty$), then $\lim \frac{f(x)}{g(x)}$ exists and equals it.*

History

The rule carries one of the most famous and controversial backstories in the history of science. In 1691, the wealthy French aristocrat **Guillaume-François-Antoine, Marquis de l'Hôpital** (1661–1704), was fascinated by the radical new calculus invented by Leibniz. L'Hôpital hired the brilliant young Swiss mathematician **Johann Bernoulli** for private tutoring. In 1694, l'Hôpital offered Bernoulli a substantial annual retainer of 300 livres in exchange for Bernoulli sending him all his latest mathematical discoveries, with the explicit agreement that l'Hôpital could publish them as his own.

In 1696, l'Hôpital published *Analyse des Infiniment Petits pour l'Intelligence des Lignes Courbes*—the first differential calculus textbook ever printed. Embedded in Section IX was the celebrated rule for evaluating fractions that reduce to $0/0$. When l'Hôpital died in 1704, Bernoulli went public, claiming full authorship of the rule. For over two centuries, Bernoulli was widely dismissed as an envious plagiarist, until 1922, when historians discovered Bernoulli's original course notes and letters to l'Hôpital in Basel, completely corroborating Bernoulli's claim.

Beyond the priority dispute lay a deeper philosophical crisis. In the early eighteenth century,

expressions like $0/0$ drew fierce ridicule. Critics, most notably Bishop George Berkeley in *The Analyst* (1734), argued that $0/0$ was self-evident mathematical nonsense, and that dividing zero by zero to produce a finite number proved that the calculus was a web of logical fallacies. The definitive mathematical vindication came in 1823 from **Augustin-Louis Cauchy**. In his lectures at the École Polytechnique, Cauchy proved that $0/0$ is not a division of zeroes, but the limit of a ratio of two vanishing continuous curves. Deploying his newly formulated *Cauchy Mean Value Theorem*, Cauchy showed that $\frac{f(x)}{g(x)} = \frac{f(x)-f(a)}{g(x)-g(a)} = \frac{f'(c)}{g'(c)}$ for some c strictly between a and x. As $x \to a$, the squeeze forces $c \to a$, establishing the rule with airtight deductive rigor without once invoking infinitesimals.

18.4 Worked examples on the two ratio forms

Example 18.1 (The classic sin limit, again)**.** Consider $\lim_{x\to 0} \dfrac{\sin x}{x}$. Substituting gives $\frac{0}{0}$. Differentiate top and bottom *separately* (this is not the quotient rule — more on that trap shortly):

$$\lim_{x\to 0} \frac{\sin x}{x} \overset{0/0}{=} \lim_{x\to 0} \frac{\cos x}{1} = \frac{\cos 0}{1} = 1.$$

Common trap

That last example is a delicious circular trap if you are being rigorous. We computed $\frac{\mathrm{d}}{\mathrm{d}x}(\sin x) = \cos x$ in the first place by *using* $\lim_{x\to 0} \frac{\sin x}{x} = 1$. So "proving" that limit with L'Hopital is logically circular — it assumes what it sets out to show. L'Hopital gives the right *number*, but here it is not a legitimate proof. Use the geometric squeeze argument for that one. The moral: a tool that gives the right answer can still be the wrong argument.

Example 18.2 (A polynomial-flavored $0/0$)**.** $\lim_{x\to 1} \dfrac{x^3-1}{x^2-1}$ is $\frac{0}{0}$. By L'Hopital,

$$\lim_{x\to 1} \frac{x^3-1}{x^2-1} \overset{0/0}{=} \lim_{x\to 1} \frac{3x^2}{2x} = \frac{3}{2}.$$

Sanity check by factoring: $\frac{x^3-1}{x^2-1} = \frac{(x-1)(x^2+x+1)}{(x-1)(x+1)} = \frac{x^2+x+1}{x+1} \to \frac{3}{2}$. The factoring *is* the cancellation of the common $(x-1)$ that L'Hopital performs invisibly — same mechanism, two faces.

Example 18.3 (When once is not enough: applying the rule twice)**.** Take $\lim_{x\to 0} \dfrac{1-\cos x}{x^2}$, which is $\frac{0}{0}$. One application:

$$\lim_{x\to 0} \frac{1-\cos x}{x^2} \overset{0/0}{=} \lim_{x\to 0} \frac{\sin x}{2x}.$$

This new limit is *still* $\frac{0}{0}$ — the rule did not finish the job, but it left us a simpler $\frac{0}{0}$. So apply it again:

$$\lim_{x\to 0} \frac{\sin x}{2x} \overset{0/0}{=} \lim_{x\to 0} \frac{\cos x}{2} = \frac{1}{2}.$$

You may iterate L'Hopital as many times as each successive limit stays a legal $\frac{0}{0}$ or $\frac{\infty}{\infty}$. The instant a step yields a determinate form, stop and evaluate — one more differentiation past that point will give a wrong answer.

Common trap

Check the form before *every* application. The single most common L'Hopital error is to differentiate when the form is no longer indeterminate. In the example above, the moment we reached $\frac{\cos x}{2}$ the form became $\frac{1}{2}$, a perfectly ordinary number; differentiating again would give $\frac{-\sin x}{0}$, which is garbage. L'Hopital is a *conditional* rule. Verify $\frac{0}{0}$ or $\frac{\infty}{\infty}$ at each step, every time, before you reach for a derivative.

Work it yourself

Compute $\lim_{x\to 0} \dfrac{e^x - 1 - x}{x^2}$ by hand. Confirm it is $\frac{0}{0}$. Apply L'Hopital once and check you again get $\frac{0}{0}$. Apply it a second time and read off the answer. (You should land on $\frac{1}{2}$. Notice how this mirrors the previous example — both are detecting the *quadratic* coefficient in a Taylor expansion, a connection we make explicit at the end of the chapter.)

Example 18.4 (An ∞/∞ at infinity)**.** $\lim_{x\to\infty} \dfrac{\ln x}{x}$ is $\frac{\infty}{\infty}$. By Theorem 18.2,

$$\lim_{x\to\infty} \frac{\ln x}{x} \overset{\infty/\infty}{=} \lim_{x\to\infty} \frac{1/x}{1} = \lim_{x\to\infty} \frac{1}{x} = 0.$$

So $\ln x$, though it does march off to infinity, is utterly outrun by x. We will build a whole hierarchy out of facts like this in the last section.

18.5 Forcing the other forms into a fraction

L'Hopital only speaks the language of fractions, $\frac{0}{0}$ and $\frac{\infty}{\infty}$. The remaining five indeterminate forms are not fractions on their face. The art is to *algebraically rewrite* each one as a fraction without changing its value, and then let the rule loose. Each rewriting is a small, learnable move.

18.5.1 The product form $0 \cdot \infty$

A product $f \cdot g$ where $f \to 0$ and $g \to \infty$ is a tug-of-war: the 0 tries to crush the product to nothing, the ∞ tries to blow it up. Who wins is exactly the kind of question L'Hopital answers — once we make it a fraction. And there is a fraction hiding in any product, because dividing is multiplying by a reciprocal:

$$f \cdot g = \frac{f}{1/g} \quad \left(\text{form } \tfrac{0}{0}\right) \qquad \text{or} \qquad f \cdot g = \frac{g}{1/f} \quad \left(\text{form } \tfrac{\infty}{\infty}\right).$$

You get to choose which factor to flip. Flip the one whose reciprocal is friendlier to differentiate.

Example 18.5 ($x \ln x$ at the origin)**.** $\lim_{x\to 0^+} x \ln x$ is $0 \cdot (-\infty)$. Write it as a fraction by flipping the x:

$$x \ln x = \frac{\ln x}{1/x} \quad \left(\tfrac{-\infty}{\infty}\right) \quad \Longrightarrow \quad \lim_{x\to 0^+} \frac{\ln x}{1/x} \overset{\infty/\infty}{=} \lim_{x\to 0^+} \frac{1/x}{-1/x^2} = \lim_{x\to 0^+} (-x) = 0.$$

Had we flipped the other way, $\frac{x}{1/\ln x}$, the derivative of $1/\ln x$ is $-\frac{1}{x(\ln x)^2}$ and the algebra is uglier. *Choosing the right reciprocal* is most of the skill in these problems.

Walking question

Before reading on, decide on a walk: in $\lim_{x\to 0^+} x\ln x$, the function x is pulling the product to 0 and $\ln x$ is pulling it to $-\infty$. The answer came out 0. So which pull is "stronger," and what does that say about how fast $\ln x$ blows up compared to how fast x shrinks? Hold that thought until the growth-rates section.

18.5.2 The difference form $\infty - \infty$

Here two quantities both race to infinity and we subtract them; the answer depends on which is bigger and by how much. There is no universal recipe, but there are two reliable moves: *combine into a single fraction* (common denominator), or *factor out the dominant term*. Either one converts the subtraction into a ratio.

Example 18.6 (A common-denominator rescue). $\lim_{x\to 0}\left(\dfrac{1}{\sin x} - \dfrac{1}{x}\right)$ is $\infty - \infty$ (each piece blows up). Put them over a common denominator:

$$\frac{1}{\sin x} - \frac{1}{x} = \frac{x - \sin x}{x \sin x}.$$

Now it is a clean $\frac{0}{0}$. L'Hopital once:

$$\lim_{x\to 0} \frac{x - \sin x}{x \sin x} \overset{0/0}{=} \lim_{x\to 0} \frac{1 - \cos x}{\sin x + x\cos x},$$

still $\frac{0}{0}$. Again:

$$\overset{0/0}{=} \lim_{x\to 0} \frac{\sin x}{2\cos x - x\sin x} = \frac{0}{2 - 0} = 0.$$

So the two infinities cancel *exactly* in the limit; their difference settles to 0.

18.5.3 The power forms 1^∞, 0^0, ∞^0

These three look different from each other but yield to one universal trick: a power becomes a product the moment you take a logarithm. If $y = f(x)^{g(x)}$, then

$$\ln y = g(x)\,\ln f(x),$$

and each of the three power forms turns the right-hand side into a $0\cdot\infty$ product, which we already know how to fractionate. Find $\lim \ln y = L$, then exponentiate: $\lim y = e^L$. The logarithm trades an unmanageable exponent for a manageable product — that is the whole idea.

Why the technique works

Why the logarithm tames every power form. The trouble in 1^∞ and its cousins is that the *exponent* is doing the limiting, and our tools (derivatives, L'Hopital) are built for sums, products, and ratios — not for moving exponents. Taking ln is precisely the operation that *pulls the exponent down* into a coefficient: $\ln(f^g) = g\ln f$. An impossible-looking power becomes an ordinary product; the product becomes a fraction; the fraction yields to L'Hopital. Because exp is continuous, $\lim \ln y = L$ guarantees $\lim y = e^L$, so nothing is lost in the round

trip $y \to \ln y \to L \to e^L$. We are not avoiding the exponent — we are *lowering* it to where our machinery can reach.

Example 18.7 (The most famous 1^∞: compound interest revisited)**.** $\lim_{x\to\infty}\left(1+\dfrac{1}{x}\right)^x$ is 1^∞ — the base creeps to 1, the exponent runs to ∞, and they fight. Let $y = \left(1+\frac{1}{x}\right)^x$ and take logs:

$$\ln y = x \ln\left(1+\frac{1}{x}\right).$$

This is $\infty \cdot 0$. Make it a fraction by flipping the x:

$$\ln y = \frac{\ln\left(1+\frac{1}{x}\right)}{1/x} \quad \left(\tfrac{0}{0}\right).$$

Differentiate top and bottom. The top differentiates by the chain rule: $\frac{\mathrm{d}}{\mathrm{d}x}(\ln(1+1/x)) = \frac{1}{1+1/x}\cdot\left(-\frac{1}{x^2}\right)$, and the bottom gives $-\frac{1}{x^2}$:

$$\lim_{x\to\infty}\frac{\dfrac{-1/x^2}{1+1/x}}{-1/x^2} = \lim_{x\to\infty}\frac{1}{1+1/x} = 1.$$

So $\lim \ln y = 1$, hence $\lim y = e^1 = e$. This recovers, by L'Hopital, the very limit-definition of e from the early chapters — the compound-interest number falls right out. Figure 18.2 shows the approach.

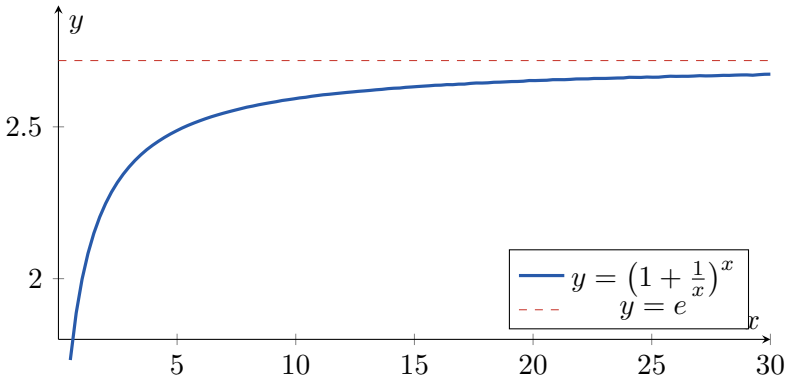


Figure 18.2: The compound-interest sequence $\left(1+\frac{1}{x}\right)^x$ climbing toward $e \approx 2.71828$ as $x \to \infty$. The form is 1^∞ — the base $1+\frac{1}{x}$ sinks to 1 while the exponent x grows — and the two effects balance at exactly e. The dashed line is the limit the curve approaches but never reaches.

Example 18.8 (An ∞^0)**.** $\lim_{x\to\infty} x^{1/x}$ is ∞^0. Logs: $\ln y = \frac{\ln x}{x}$, which we already found in Example 18.4 to be $\frac{\infty}{\infty} \to 0$. Therefore $\lim y = e^0 = 1$. A number raised to a gently vanishing power slides back down to 1.

Example 18.9 (A 0^0)**.** $\lim_{x\to 0^+} x^x$ is 0^0, perhaps the most counterintuitive of the lot. Logs: $\ln y = x \ln x$, which is the $0\cdot(-\infty)$ from Example 18.5, equal to 0. Hence $\lim_{x\to 0^+} x^x = e^0 = 1$. So x^x, despite the base sinking to 0, climbs back up to 1. The figure makes this vivid.

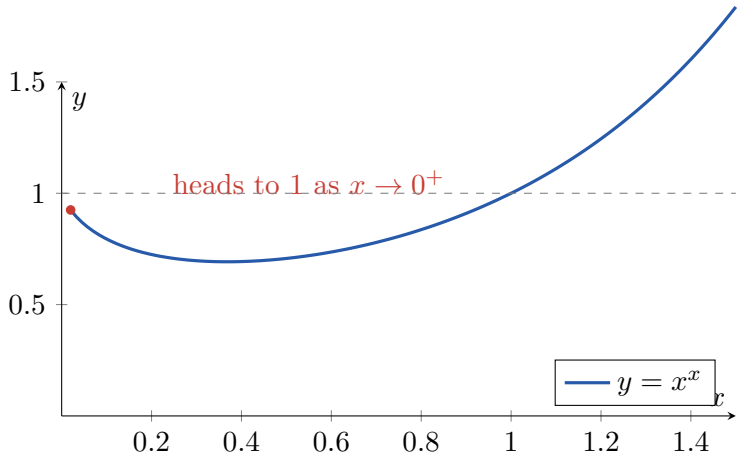


Figure 18.3: The graph of $y = x^x$ for $x > 0$. Although the base x tends to 0, the value x^x does not collapse to 0; it dips to a minimum near $x \approx 0.368$ and approaches 1 as $x \to 0^+$. The form 0^0 is genuinely indeterminate, and here it resolves to 1.

Common trap

0^0, 1^∞, and ∞^0 are *forms*, not values. People sometimes "know" that $0^0 = 1$ from algebra (where the convention is handy) and conclude that $\lim x^x$ "must" be 1 for that reason. The reasoning is wrong even though the answer is right: change the problem to $\lim_{x\to 0^+} x^{1/\ln x}$, also 0^0, and the answer is e, not 1. The form does not decide; only the logarithm-and-L'Hopital computation does.

18.6 How the rule gets misused

The rule is convenient enough that students reach for it reflexively, and that reflex causes most of the wrong answers. Here are the failure modes, each worth memorizing as a thing *not* to do.

Common trap

Trap 1: Using it on a determinate form. If the form is not $\frac{0}{0}$ or $\frac{\infty}{\infty}$, the rule simply does not apply, and using it gives nonsense. For instance $\lim_{x\to 0} \dfrac{\cos x}{x+1}$ is $\frac{1}{1} = 1$ by direct substitution. "Applying L'Hopital" would give $\lim \frac{-\sin x}{1} = 0$ — flatly wrong. *Always check the form first.*

Common trap

Trap 2: Confusing it with the quotient rule. L'Hopital says differentiate the numerator and the denominator *separately* and form a new quotient $\frac{f'}{g'}$. It does *not* say differentiate the quotient $\frac{f}{g}$ with the quotient rule $\frac{f'g - fg'}{g^2}$. These are completely different expressions. Writing the quotient rule here is one of the most frequent and most fatal slips.

Common trap

Trap 3: Continuing after the form resolves. As stressed in Example 18.3, the instant a step is no longer indeterminate you must stop. One extra differentiation past a determinate form gives a wrong number with total confidence.

Common trap

Trap 4: Treating a non-existent derivative limit as "no answer." The rule is one-directional. If $\lim \frac{f'}{g'}$ fails to exist, you may conclude *nothing* about $\lim \frac{f}{g}$; it might still exist. Take $\lim_{x\to\infty} \dfrac{x+\sin x}{x}$. This is $\frac{\infty}{\infty}$, and L'Hopital gives $\lim \frac{1+\cos x}{1} = \lim(1+\cos x)$, which *oscillates* and has no limit. A beginner concludes "the limit does not exist." But the original limit plainly *does* exist: $\frac{x+\sin x}{x} = 1 + \frac{\sin x}{x} \to 1 + 0 = 1$. L'Hopital was silent here, not authoritative.

Common trap

Trap 5: An endless loop. $\lim_{x\to\infty} \dfrac{e^x}{e^x+1}$ is $\frac{\infty}{\infty}$; L'Hopital gives $\frac{e^x}{e^x} = 1$ in one clean step, fine. But $\lim_{x\to\infty} \dfrac{\sqrt{x^2+1}}{x}$, attacked by L'Hopital, produces $\frac{x/\sqrt{x^2+1}}{1} = \frac{x}{\sqrt{x^2+1}}$, whose own L'Hopital sends you straight back to $\frac{\sqrt{x^2+1}}{x}$. You are chasing your tail. Step back and use algebra: $\frac{\sqrt{x^2+1}}{x} = \sqrt{1+\frac{1}{x^2}} \to 1$. When L'Hopital cycles, abandon it.

Pause

A healthy habit before reaching for L'Hopital: ask whether a quick algebraic move — factoring, a common denominator, dividing by the dominant power, a known limit — settles the problem first. L'Hopital is a hammer; not every limit is a nail. The examples in Traps 4 and 5 are solved faster, and more safely, by elementary algebra.

18.7 Growth rates, settled once and for all

We have brushed up against a recurring question since the chapter on limits at infinity: among the functions that all run off to infinity — powers of x, exponentials, logarithms — *which outruns which*? L'Hopital lets us answer this cleanly, because "f grows faster than g" is precisely the statement

$$\lim_{x\to\infty} \frac{g(x)}{f(x)} = 0,$$

an $\frac{\infty}{\infty}$ form tailor-made for the rule.

Theorem 18.3 (The growth hierarchy)**.** *As $x \to \infty$, for any fixed exponents $a > 0$, $p > 0$ and any base $b > 1$,*

$$(\ln x)^a \ll x^p \ll b^x \ll x! \ll x^x,$$

where $u \ll v$ means $\lim_{x\to\infty} \frac{u}{v} = 0$ (read: "v dominates u"). In words: every power of a logarithm loses to every positive power of x; every positive power of x loses to every exponential with base above 1; the exponential loses to the factorial; and the factorial loses to x^x.

Focus

Power beats logarithm. We showed $\frac{\ln x}{x} \to 0$. More generally $\frac{\ln x}{x^p}$ for any $p > 0$ is $\frac{\infty}{\infty}$, and L'Hopital gives $\frac{1/x}{p\,x^{p-1}} = \frac{1}{p\,x^p} \to 0$. So $\ln x \ll x^p$ for every $p > 0$, no matter how tiny — even $x^{0.001}$ eventually buries $\ln x$. Raising the log to a power a only delays the inevitable: a applications of L'Hopital grind $(\ln x)^a/x^p$ down to 0 as well.

Focus

Exponential beats power. Consider $\frac{x^n}{e^x}$ for a positive integer n — an $\frac{\infty}{\infty}$. Each L'Hopital step lowers the power on top by one while the e^x on the bottom regenerates unchanged:

$$\frac{x^n}{e^x} \to \frac{nx^{n-1}}{e^x} \to \frac{n(n-1)x^{n-2}}{e^x} \to \cdots \to \frac{n!}{e^x} \to 0.$$

After n steps the numerator is the constant $n!$ and the denominator still races to infinity, so the ratio dies. This is the precise sense in which exponential growth *always* overwhelms polynomial growth — the engine of why compound processes, runaway populations, and 2^n-time algorithms eventually dwarf any polynomial.

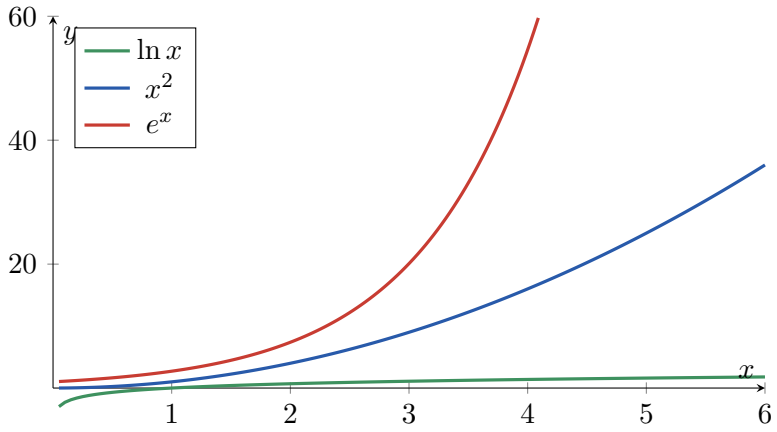


Figure 18.4: The growth hierarchy in one window. $\ln x$ barely lifts off; x^2 climbs steadily; e^x is already running away by $x \approx 4$. Pushed to infinity, the order is permanent: $\ln x \ll x^2 \ll e^x$. L'Hopital's rule is what proves the order never reverses.

Application

Why your sorting algorithm scales but your brute-force search does not. In computer science, an algorithm's running time as a function of input size n is exactly a growth-rate question. A method costing $n \log n$ steps (good sorting) versus one costing n^2 (naive sorting) versus one costing 2^n (brute-force subset search) are separated by precisely the hierarchy in Theorem 18.3: $n \log n \ll n^2 \ll 2^n$. The L'Hopital computations above are the rigorous reason a 2^n algorithm becomes hopeless near $n = 60$ while an $n \log n$ algorithm sails through n in the billions. Growth-rate order is not a heuristic; it is a theorem, and this is its proof.

Connection

The factorial and x^x ends of the hierarchy connect to *Stirling's approximation*, $n! \sim \sqrt{2\pi n}\,(n/e)^n$, which you will meet in probability and combinatorics. Stirling shows $n!$ sits, roughly, like $(n/e)^n$ — slightly below n^n, comfortably above b^n — which is exactly the placement $b^x \ll x! \ll x^x$ in our chain. The crude L'Hopital ordering and the refined Stirling estimate tell the same story at two levels of resolution.

18.8 The deeper picture: L'Hopital is Taylor's theorem in miniature

The tangent-line intuition we opened with runs even deeper. When you push it one order further, L'Hopital's rule is revealed as a low-resolution shadow of *Taylor series*, which you will meet in full later. The idea: near a zero, a function looks not just like its tangent line but like its whole power series, and the limit of a ratio is decided by the *first surviving terms* of the top and bottom.

What is really happening

Why repeated L'Hopital keeps working. Suppose near $x = 0$,

$$f(x) = \frac{f''(0)}{2}x^2 + \cdots, \qquad g(x) = g'(0)\,x + \cdots$$

Then $\frac{f}{g} \sim \frac{(f''(0)/2)x^2}{g'(0)x} = \frac{f''(0)}{2g'(0)}x \to 0$. The function whose series *starts at a higher power* vanishes faster, so it loses. Each L'Hopital step differentiates away one power of x from both series; you keep going until the surviving lowest-order terms no longer both vanish, and at that moment the ratio of their coefficients is the limit. That is why $\frac{1-\cos x}{x^2} \to \frac{1}{2}$: the series $1 - \cos x = \frac{x^2}{2} - \frac{x^4}{24} + \cdots$ starts at $\frac{x^2}{2}$, exactly matching the x^2 below, and the ratio of leading coefficients is $\frac{1/2}{1} = \frac{1}{2}$ — the same $\frac{1}{2}$ two L'Hopital steps produced. The rule is reading off the first nonzero coefficients in disguise.

Why this matters

There is something satisfying in seeing one idea reappear in three costumes. The tangent-line picture, the Cauchy MVT proof, and the Taylor-series view are not three facts about L'Hopital's rule — they are one fact at three depths. The tangent line is the first-order truth; the MVT makes it rigorous without assuming differentiability at the point; the Taylor series explains why iterating the rule must terminate and what number it terminates at. When a theorem looks the same from every angle you approach it, you have understood it.

18.8.1 The Scope, Fragilities, and Horizons of L'Hôpital's Rule

Connection

L'Hôpital's rule unlocked the direct resolution of ambiguous limiting behavior across mathematics and physics:

- *Singular ratios without factoring*: Forms like $0/0$ and ∞/∞ that defy algebraic simplification (e.g., $\frac{e^x-1}{\sin x}$ or $\frac{\ln x}{x-1}$) are evaluated purely through derivative operations.
- *Asymptotic hierarchies*: It rigorously proves the fundamental growth hierarchy: logarithms grow slower than any positive power, which grow slower than any exponential ($(\ln x)^p \ll x^q \ll e^{kx}$).
- *Indeterminate exponents*: Through logarithmic conversion ($\ln y$), tricky power limits like $\lim_{x\to\infty}(1+r/x)^x = e^r$ and $x^x \to 1$ as $x \to 0^+$ are brought into the domain of rates.

Yet l'Hôpital's rule is an incomplete and fragile tool with three structural limitations:

1. *Logical circularity*: It cannot define fundamental derivatives. Using l'Hôpital to prove $\lim_{x\to 0}\frac{\sin x}{x} = 1$ is viciously circular, because calculating $(\sin x)' = \cos x$ already requires knowing that limit.
2. *Oscillatory blind spots*: On forms like $\lim_{x\to\infty}\frac{x+\sin x}{x}$, the true limit is 1. Yet differentiating numerator and denominator yields $\frac{1+\cos x}{1}$, which oscillates forever without a limit. L'Hôpital's rule reports failure even though the original limit is completely tame!
3. *Infinite loops*: On algebraic expressions like $\lim_{x\to\infty}\frac{x}{\sqrt{x^2+1}}$, repeated differentiation flips the fraction into $\frac{\sqrt{x^2+1}}{x}$ and back again, trapping the calculation in an infinite algebraic loop.

When l'Hôpital's rule stalls, loops, or demands exhausting chains of four or five derivatives, the ultimate sovereign remedy is *Taylor series expansion*. In Chapter 30 and Chapter 31 (*Power Series and Taylor's Theorem*), we replace functions with their local polynomial expansions:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2 + \cdots$$

Comparing the first non-vanishing coefficients resolves indeterminate forms in a single inspection, exposing the exact asymptotic behavior without differentiation loops.

18.9 Exercises

Exercise 18.1 (Warm-up ratios)**.** Evaluate, checking the form before each L'Hopital step: (a) $\lim_{x\to 0}\dfrac{\tan x - x}{x^3}$; (b) $\lim_{x\to 1}\dfrac{\ln x}{x-1}$; (c) $\lim_{x\to\infty}\dfrac{x^2}{e^x}$; (d) $\lim_{x\to 0}\dfrac{e^x-1}{\sin x}$.

Exercise 18.2 (Products and differences)**.** Convert to a fraction first, then evaluate: (a) $\lim_{x\to 0^+} x^2\ln x$; (b) $\lim_{x\to(\pi/2)^-}(\sec x - \tan x)$; (c) $\lim_{x\to 0}\left(\dfrac{1}{x} - \dfrac{1}{e^x-1}\right)$.

Exercise 18.3 (Power forms)**.** Use the logarithm trick: (a) $\lim_{x\to\infty}\left(1+\dfrac{3}{x}\right)^x$; (b) $\lim_{x\to 0^+}(\sin x)^x$; (c)

$\lim_{x\to\infty}\left(\dfrac{x+1}{x-1}\right)^{x}$.

Exercise 18.4 (Spot the misuse)**.** For each, decide whether L'Hopital is legitimately applicable, and if not, find the limit by other means. (a) $\lim_{x\to 0}\dfrac{x+2}{x^2+1}$; (b) $\lim_{x\to\infty}\dfrac{2x-\cos x}{x+1}$; (c) $\lim_{x\to\infty}\dfrac{\sqrt{4x^2+1}}{x+3}$. In (b) and (c), explain in one sentence why blindly differentiating fails or stalls.

Exercise 18.5 (Three steps, then stop)**.** Evaluate $\lim_{x\to 0}\dfrac{x-\sin x}{x^3}$. State the form at each stage and identify the exact moment you must stop applying the rule. Confirm your answer against the series $\sin x = x - \frac{x^3}{6} + \cdots$.

Exercise 18.6 (Growth-rate ordering)**.** Order the following by eventual size as $x\to\infty$, smallest first, and justify each pairwise comparison with a single L'Hopital limit: x^{10}, $(\ln x)^{100}$, $1.0001^{\,x}$, $e^{\sqrt{x}}$. (The last one is subtler than it looks — compare its logarithm.)

Exercise 18.7 (A famous compound limit)**.** Show that $\lim_{x\to\infty}\left(1-\dfrac{r}{x}\right)^{x} = e^{-r}$ for any constant r, by taking logs and applying L'Hopital. Interpret the result for continuously compounded *decay* (e.g. radioactive material or a depreciating asset losing a fixed fraction per ever-shorter period).

Exercise 18.8 (Conceptual: why the form matters)**.** Construct two functions f, g with $f, g\to 0$ as $x\to 0$ for which $\lim\frac{f}{g} = 7$, and another pair for which the same form $\frac{0}{0}$ gives $\lim\frac{f}{g} = \infty$. Use your examples to explain, in your own words, the precise sense in which $\frac{0}{0}$ "contains no information."

Exercise 18.9 (Proof: the simple case from the definition)**.** Suppose f and g are differentiable *at* a with $f(a) = g(a) = 0$ and $g'(a)\neq 0$. Without invoking the full theorem, prove directly from the definition of the derivative that

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)}.$$

Hint: divide numerator and denominator by $(x-a)$ and recognize two difference quotients. Explain why this short argument needs the stronger hypothesis "differentiable *at* a," whereas Theorem 18.1 gets by with less.

Exercise 18.10 (Proof: L'Hopital can fail without the form)**.** Give an explicit pair f, g with $\lim_{x\to a}\frac{f(x)}{g(x)}$ existing but $\lim_{x\to a}\frac{f'(x)}{g'(x)}$ *not* existing, where the form is genuinely $\frac{\infty}{\infty}$. (Trap 4 contains the seed of such an example.) Explain how this is consistent with the theorem — that is, why no contradiction arises.

Chapter 19

Newton's Method

We have spent many chapters learning to differentiate. We can take a function, find its derivative, read off the slope of its tangent line at any point, and use that tangent line as the best straight-line stand-in for the curve nearby. That last idea – the tangent line as a local imitation of the curve – has been quietly doing work for us since the chapter on linear approximation. In this chapter it pays off in one of the most useful algorithms ever discovered.

Here is the problem. You have an equation. Not a friendly one like $2x - 6 = 0$, where you just rearrange and read off $x = 3$. A genuinely ugly one, like

$$x^5 - 3x + 1 = 0, \qquad \cos x = x, \qquad e^{-x} = x^2.$$

For these there is *no formula*. There is no quadratic-formula analog, no algebraic trick, that will hand you the answer. And yet these equations have real solutions – you can practically see one of them if you sketch the curves. The question is: how do you actually *find* the number, to as many decimal places as you please, when no formula exists?

Focus

Given a function f, find a number r with $f(r) = 0$ – a **root** of f – when there is no formula for r. This single problem is enormous. Almost every equation in science is "solve $f(x) = 0$" in disguise: find the equilibrium, find the break-even point, find where the orbit crosses, find the rate that makes the loan balance zero. Newton's method is the workhorse that does it, and it does it fast.

The strategy is short, and it is built entirely out of the tangent line. You guess. Your guess is wrong, but instead of throwing it away you *improve* it: you replace the hard curve with its easy tangent line, solve the easy linear problem exactly, and take that as your next guess. Then you do it again. And again. The guesses come hurtling in toward the true root – often doubling the number of correct digits at every single step. This chapter explains why it works, shows in pictures exactly what it is doing, lays out the handful of ways it can fail, and accounts for its remarkable speed.

19.1 The idea: replace the curve with its tangent

Start with a picture and a guess. We want a root of some function f – a place where its graph crosses the horizontal axis. Suppose we have a current guess, call it x_0. Almost certainly x_0 is not the root: the point $(x_0, f(x_0))$ sits above or below the axis, not on it.

Now here is the move. The curve near x_0 is hard to work with, but the *tangent line* at x_0 is trivial – it is a straight line, and finding where a straight line crosses the axis is grade-school algebra. So we make a bargain: we pretend, just for one step, that the function *is* its tangent line. We slide down (or up) that tangent line until it hits the axis, and wherever it lands we call our new, improved guess x_1.

Intuition

The whole method in one sentence: *you cannot easily find where the curve hits the axis, so instead you find where its tangent line hits the axis, and you bet that this is closer.* The tangent line hugs the curve near x_0, so if x_0 is already in the neighborhood of the root, the place the tangent crosses is a much better estimate than x_0 was. Repeat the bargain from the new point and the estimates close in.

Figure 19.1 shows a single step. The guess x_0 is off to the right of the true root r. We climb up to the point on the curve, draw the tangent there, and follow it down to the axis. It lands at x_1, noticeably closer to r than x_0 was.

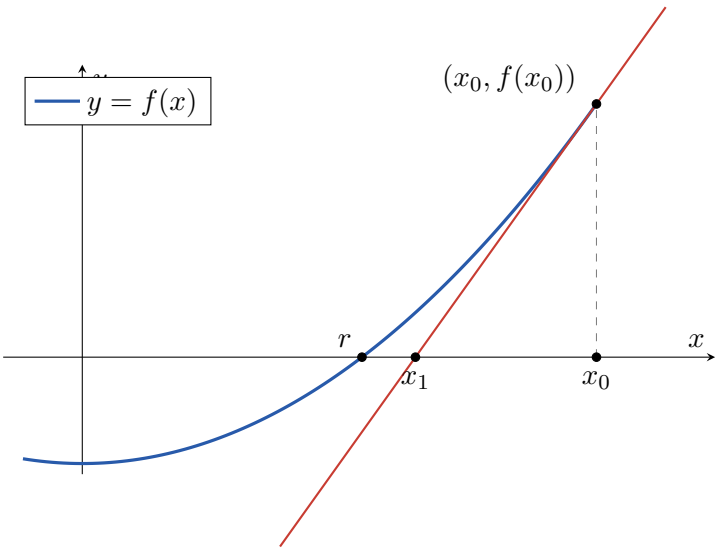


Figure 19.1: One step of Newton's method on $f(x) = x^2 - 2$, whose root is $r = \sqrt{2}$. From the guess x_0 we ride the tangent line (red) down to the axis, landing at x_1, much nearer the root r than x_0 was.

That is the entire idea. Everything else is bookkeeping: turning "follow the tangent to the axis" into a formula, and then turning the formula into a loop.

19.2 Deriving the iteration formula

Do the algebra once, carefully, because the formula is worth understanding rather than memorizing.

We are standing at x_0. The point on the curve is $(x_0, f(x_0))$, and the slope of the tangent there is $f'(x_0)$. The tangent line is therefore

$$y = f(x_0) + f'(x_0)\,(x - x_0).$$

This is nothing but the linear approximation of f at x_0 – the very same formula we built in the chapter on differentials. We want the place where this line crosses the horizontal axis, i.e. where $y = 0$. Set the right-hand side to zero and solve for x:

$$0 = f(x_0) + f'(x_0)\,(x - x_0) \quad \Longrightarrow \quad x - x_0 = -\frac{f(x_0)}{f'(x_0)} \quad \Longrightarrow \quad x = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

That landing point is our new guess x_1. Naming the general step, we get the celebrated recurrence.

Focus

Newton's iteration. Starting from a guess x_0, generate a sequence by

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}} \qquad n = 0, 1, 2, \ldots$$

Each new estimate is the old one corrected by the amount $f(x_n)/f'(x_n)$. When the x_n stop changing – when $f(x_n)$ has shrunk to essentially zero – you have your root.

Look at what the correction term $f(x_n)/f'(x_n)$ is telling you. The numerator $f(x_n)$ is how far off the axis you are – your "vertical error." The denominator $f'(x_n)$ is the slope, the rate at which the function climbs. Dividing height by slope gives you a *horizontal* distance: "if I am this high up and the ground rises at this rate, I must back up this far to reach ground level." It is exactly the run you get from a given rise and a given slope. The method is just rise over slope equals run, applied over and over.

What is really happening

What is really happening is that Newton's method *inverts the linear approximation.* Linear approximation answered the forward question: "given a small step Δx, how much does f change?" – the answer being $\Delta f \approx f'(x)\,\Delta x$. Newton's method runs that backwards. It asks: "I want f to change by a specific amount, namely by $-f(x_n)$ so as to land on zero – what step Δx achieves that?" Solving $f'(x_n)\,\Delta x = -f(x_n)$ gives $\Delta x = -f(x_n)/f'(x_n)$, and stepping by that amount is the update. So the iteration is linear approximation solved for the input instead of the output. The reason it is only *approximate* – the reason you must iterate rather than land on the root in one shot – is that the curve is not actually a straight line, so the real f is not yet zero at x_{n+1}. But it is closer, and from the closer point the linear approximation is even better, so the next correction is smaller still.

Why the technique works

Why does following the tangent get you closer to the root at all? Because near a point the tangent line and the curve agree to *first order* – they have the same value and the same slope – so over a short horizontal span they part company only slowly, by an amount governed by the curvature (the second derivative). If your guess is already reasonably near the root, the stretch between guess and root is short, the tangent and curve are nearly indistinguishable across it, and the tangent's crossing is therefore nearly the curve's crossing. The error you make by using the line instead of the curve is small, and – this is the punchline we prove later – it is small *compared to the error you started with.* Each step's leftover error is roughly the *square* of the previous error. Square a small number and you get a tiny number. That squaring is the engine of the method's speed.

19.3 A first worked example: square roots by hand

Nothing makes the method concrete like computing a number you already know. Take $\sqrt{2}$. It is the positive root of

$$f(x) = x^2 - 2, \qquad f'(x) = 2x.$$

Plug into the iteration:

$$x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n} = x_n - \frac{x_n}{2} + \frac{1}{x_n} = \frac{x_n}{2} + \frac{1}{x_n} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right).$$

That last form is famous in its own right – it is the *Babylonian method* for square roots, known thousands of years before Newton. "Average your guess with 2 divided by your guess." It is Newton's method in disguise: the ancients had the right iteration without the calculus that explains it.

Start, deliberately, with a poor guess: $x_0 = 2$.

Work it yourself

Carry these out yourself with a calculator; watch the digits lock in.

$$\begin{aligned}
x_0 &= 2 \\
x_1 &= \tfrac{1}{2}\left(2 + \tfrac{2}{2}\right) = \tfrac{1}{2}(3) = 1.5 \\
x_2 &= \tfrac{1}{2}\left(1.5 + \tfrac{2}{1.5}\right) = \tfrac{1}{2}(1.5 + 1.3333\ldots) = 1.41666\ldots \\
x_3 &= \tfrac{1}{2}\left(1.416667 + \tfrac{2}{1.416667}\right) = 1.4142156\ldots \\
x_4 &= 1.41421356237\ldots
\end{aligned}$$

The true value is $\sqrt{2} = 1.41421356237\ldots$

Stare at the errors – the distance from each guess to the true $\sqrt{2}$:

n	error $\lvert x_n - \sqrt{2}\rvert$
0	0.586
1	0.0858
2	0.00245
3	0.0000021
4	0.0000000000016

Read that column from the top down and feel what is happening. The error goes from about half, to about a tenth, to a few thousandths, to a couple millionths, to a *trillionth*. Roughly speaking, the number of correct decimal digits is *doubling* at every step: 0 correct, then 1, then 2–3, then 6, then 12. After four steps from a lazy guess of 2 we have $\sqrt{2}$ to twelve places. That doubling of correct digits is the hallmark of **quadratic convergence**, and we will explain it precisely soon.

Application

This is not a toy. The square-root key on your calculator, and the `sqrt` function buried in every programming language's math library, does not consult a table – it runs an iteration of essentially this kind, typically Newton's method or a close relative, started from a clever first guess and run just enough times to fill the available precision. Division itself, on some processors, is computed by Newton's method applied to $f(x) = 1/x - a$, which finds $1/a$ using only multiplication and subtraction – no division at all. The arithmetic happening inside your machine is shot through with this one idea.

19.4 Watching convergence in pictures

The formula is clean, but the method is at heart geometric, so watch several steps unfold on a graph. Take a genuinely non-algebraic equation:

$$f(x) = x^3 - x - 2 = 0.$$

There is no nice closed form you would want to use, but a sketch shows a single real root a bit past 1.5. Start at $x_0 = 2$.

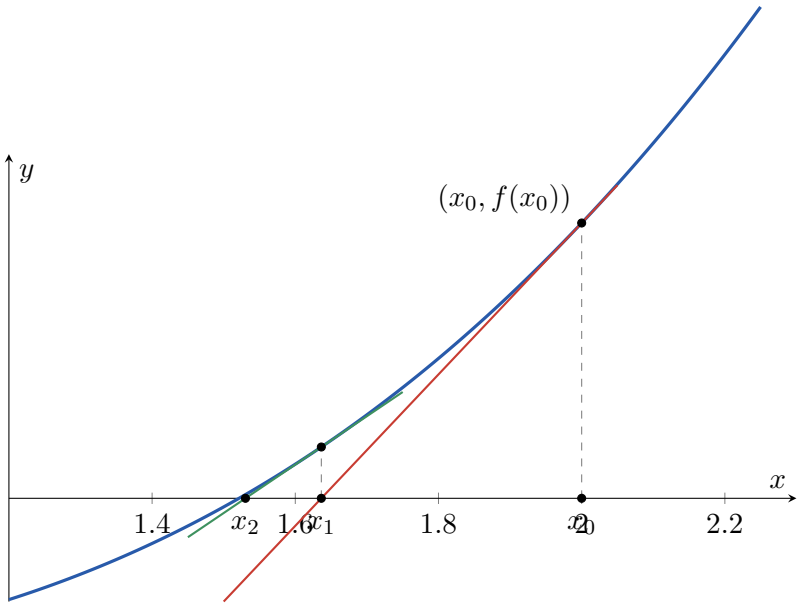


Figure 19.2: Three guesses converging on the root of $f(x) = x^3 - x - 2$. From $x_0 = 2$ the red tangent lands at x_1; from x_1 the green tangent lands at x_2, already nearly on top of the root. Each tangent is taken at a point closer to the crossing – the guesses pile up against the root.

The numbers behind the picture:

$$x_0 = 2, \quad x_1 = 1.63636\ldots, \quad x_2 = 1.53039\ldots, \quad x_3 = 1.52144\ldots, \quad x_4 = 1.52138\ldots$$

By x_4 the digits have stopped moving at the precision shown. The picture and the table tell the same story: each step the new point clings tighter to the root, because each tangent is taken at a point ever closer to the curve's actual crossing, where the line and the curve are ever harder to tell apart.

Walking question

Take this on a walk. In Figure 19.2 the curve is concave up (it bends upward) and the root is approached *from the right*, each guess landing to the left of the last but never overshooting past the root. Picture instead a curve that is concave *down* near its root, approached from the right. Sketch a tangent. Does the iterate stay on the same side of the root, or jump across it? Convince yourself that the curvature and which side you start on together decide whether Newton's method marches in steadily from one side or zig-zags across the root closing in.

19.5 When it works – and when it fails

Newton's method is fast, but it is not foolproof, and a teacher who shows you only the triumphs is doing you a disservice. The method comes with a fine-print contract, and there are four classic ways to violate it. Knowing them is what separates someone who can *use* the method from someone who merely knows the formula.

19.5.1 The happy case: a good start near a simple root

When everything goes right, two conditions hold. First, the starting guess x_0 is *close enough* to a root r. Second, the root is *simple* – meaning $f'(r) \neq 0$, so the curve crosses the axis at a genuine slant rather than grazing it tangentially. Under those conditions convergence is guaranteed to be quadratic: the count of correct digits doubles at every step. That is the regime we have been admiring.

The trouble is the phrase "close enough." Newton's method is a *local* method. It promises wonderful things once you are in the neighborhood of the root, and says almost nothing about how to get into that neighborhood in the first place. Starting far away, anything can happen.

19.5.2 Failure 1: the derivative vanishes – a flat spot

Look again at the formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

If ever $f'(x_n) = 0$, you are dividing by zero and the method blows up. Geometrically this is obvious: where the derivative is zero the tangent line is *horizontal*, and a horizontal line either never meets the axis or lies along it. There is no "where the tangent crosses" to step to. If your guess lands on or very near a flat spot of f, the tangent is nearly horizontal and it slings your next guess off to some wildly distant place – possibly farther from the root than you began.

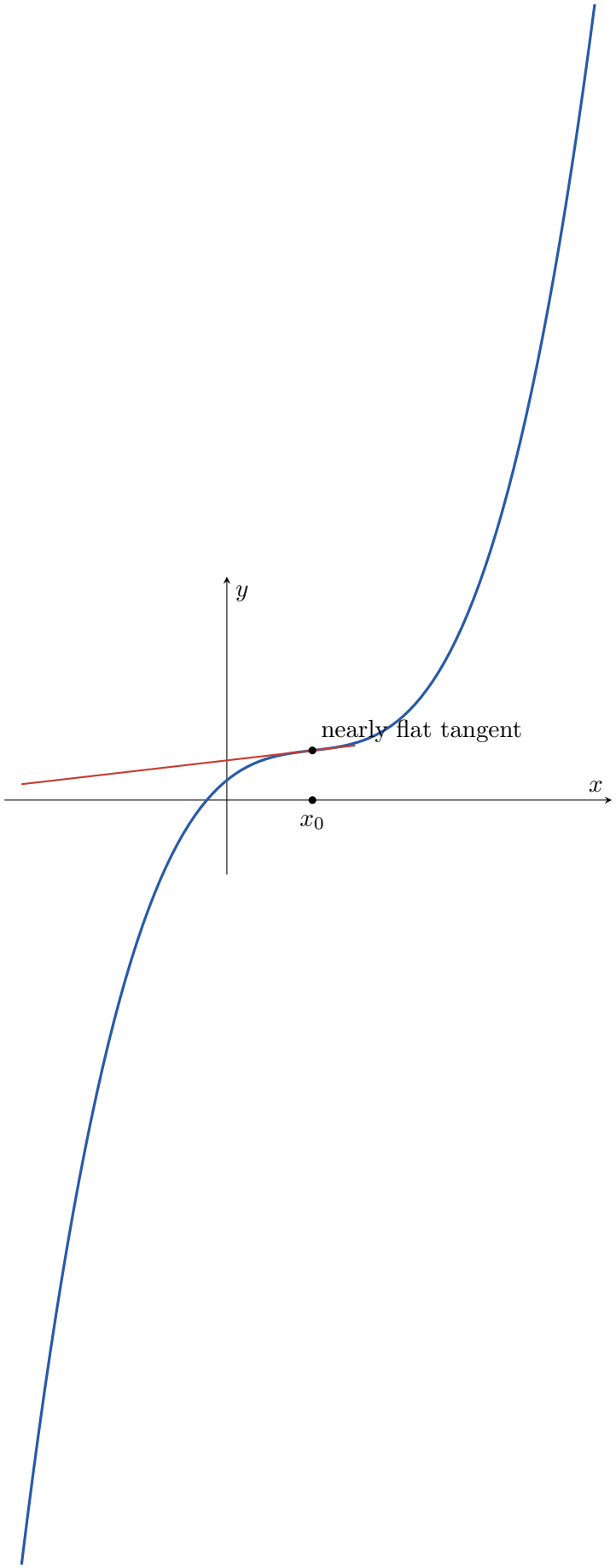


Figure 19.3: A near-horizontal tangent. The shallow slope at x_0 means the tangent travels a vast horizontal distance before it reaches the axis, flinging the next guess far away. If the slope were

19.5.3 Failure 2: a bad start – running off to infinity

Even with a perfectly nice function, a careless starting point can send the iterates marching off to infinity instead of inward to the root. The classic demonstration is

$$f(x) = \arctan x,$$

whose only root is $x = 0$. Its graph flattens out toward horizontal asymptotes $\pm\pi/2$ as $|x|$ grows, so far from the origin the slope is tiny. Start with a guess that is too large – say $x_0 = 2$ – and the tangent, being nearly flat, overshoots the root and lands *farther out* on the other side. The next step overshoots even more. The iterates spiral outward, $|x_n| \to \infty$, and the method diverges. There is a critical starting radius: inside it Newton converges to 0, outside it the method runs away. This is the price of being a local method – "close enough" was a real condition, not a formality (Figure 19.4).

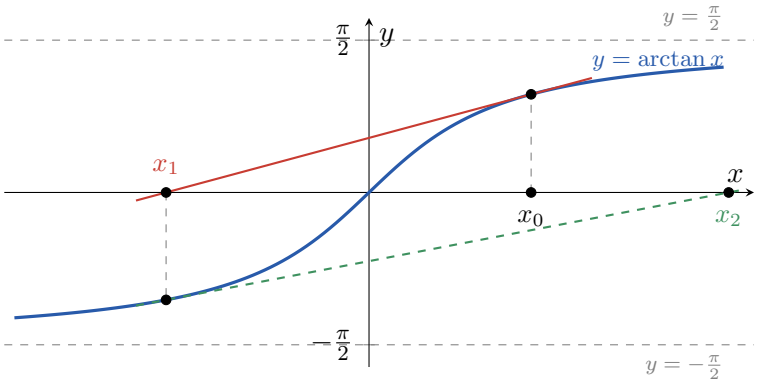


Figure 19.4: Failure through outward overshoot on $f(x) = \arctan x$. Starting from $x_0 = 1.6$, the shallow tangent (red) overshoots the root at $x = 0$, landing farther away at $x_1 \approx -2.0$. The next tangent at x_1 (dashed green) overshoots even farther to $x_2 \approx +3.55$. The iterates diverge to $\pm\infty$ in an expanding spiral.

Common trap

A common and dangerous habit is to run a fixed number of Newton iterations and *trust the result without checking*. If the start was bad, your final x_n may have nothing to do with any root – it might be drifting toward infinity, or oscillating, or have parked itself near a different root than you wanted. Always verify by plugging back in: confirm that $f(x_n)$ is genuinely small. The iteration's internal confidence is no substitute for evaluating the function.

19.5.4 Failure 3: a cycle that never lands

The most surprising failure is that the iterates can fall into a *loop*, bouncing between two values forever, never approaching any root even though one exists. The textbook example is

$$f(x) = x^3 - 2x + 2.$$

Take $x_0 = 0$. Then $f(0) = 2$ and $f'(0) = -2$, so

$$x_1 = 0 - \frac{2}{-2} = 1.$$

Now from $x_1 = 1$: $f(1) = 1 - 2 + 2 = 1$ and $f'(1) = 3 - 2 = 1$, so

$$x_2 = 1 - \frac{1}{1} = 0.$$

We are back to 0. The next step sends us to 1 again, then back to 0, forever: $0 \to 1 \to 0 \to 1 \to \cdots$. The method has entered a two-cycle and will never escape, even though f has a perfectly good real root near -1.77. Starting from the wrong place locked us into a trap with no exit.

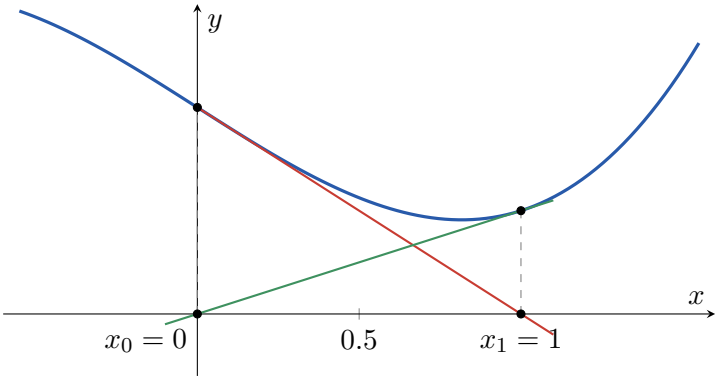


Figure 19.5: A two-cycle for $f(x) = x^3 - 2x + 2$ started at $x_0 = 0$. The red tangent at 0 crosses at 1; the green tangent at 1 crosses back at 0. Newton's method ping-pongs $0 \to 1 \to 0 \to 1 \to \cdots$ and never finds the actual root near -1.77.

19.5.5 Failure 4: a double root slows everything to a crawl

The final "failure" is gentler – the method still converges, but it loses its trademark speed. This happens at a *multiple root*, where $f(r) = 0$ and also $f'(r) = 0$, so the curve does not cross the axis cleanly but kisses it tangentially. Take

$$f(x) = (x - 1)^2, \qquad r = 1 \text{ (a double root)}.$$

The iteration becomes

$$x_{n+1} = x_n - \frac{(x_n - 1)^2}{2(x_n - 1)} = x_n - \frac{x_n - 1}{2} = \frac{x_n + 1}{2}.$$

Starting at $x_0 = 2$ this gives 1.5, 1.25, 1.125, 1.0625, ... – the error is merely *halving* each step. That is steady but slow **linear** convergence: one extra correct bit per step, not a doubling of digits. The near-flat crossing starves the denominator $f'(x_n)$, which is heading toward zero along with the numerator, and the quadratic magic is lost. (There are repaired versions of Newton's method that restore fast convergence at known multiple roots, but the plain method crawls.)

Intuition

Collect the lesson. Newton's method is a *local* method with a sharp edge to its blade: near a simple root, from a good start, it is breathtakingly fast; but a flat derivative, a careless start, an unlucky cycle, or a multiple root can stall, mislead, or diverge it. The fix in practice is humility: start from a decent guess (often found by a crude, slow, but *reliable* method first), then unleash Newton to polish that guess to full precision, and *always check the answer* by

evaluating f.

History

The algorithm bears Isaac Newton's name, but its evolution is a striking example of how modern calculus cleaned up historical algebra. In 1669, Newton wrote a manuscript titled *De analysi per aequationes numero terminorum infinitas* (circulated privately, published in 1711). In it, Newton demonstrated his method on a single cubic equation:

$$y^3 - 2y - 5 = 0.$$

Newton noted that $y = 2$ is a close guess (giving $8 - 4 - 5 = -1$). Setting $y = 2 + p$, he expanded $(2 + p)^3 - 2(2 + p) - 5 = -1 + 10p + 6p^2 + p^3 = 0$. Neglecting the nonlinear terms $6p^2$ and p^3 as small, he solved $-1 + 10p \approx 0 \implies p = 0.1$. He then set $p = 0.1 + q$, substituted again, and repeated the process. Newton's method was purely algebraic, required expanding polynomials anew at every single stage, and made zero reference to derivatives, slopes, or tangent lines!

In 1690, **Joseph Raphson** published *Analysis aequationum universalis*. Raphson recognized that Newton's tedious chain of changing variables ($y \to p \to q$) could be replaced by a clean, unified algebraic iteration: generating each new numerical approximation directly from the original polynomial. Because Raphson published this streamlined iteration fifty years before Newton's original text appeared in print, the algorithm is properly called the **Newton–Raphson method**.

The crowning step belongs to **Thomas Simpson**. In 1740, in *Essays on Several Curious and Useful Subjects in Speculative and Mixed Mathematicks*, Simpson broke free of polynomials. He was the first to frame the method using the derivative in the denominator—$x_{n+1} = x_n - f(x_n)/f'(x_n)$—and the first to recognize that the tangent line to *any* smooth geometric curve provides the path to the axis. Simpson also generalized the method to systems of simultaneous non-linear equations. It took seventy years for Newton's polynomial scratchpad to become the geometric, calculus-powered engine we use today.

19.6 Why it is so fast: quadratic convergence

We have repeatedly claimed that, in the happy case, the number of correct digits *doubles* each step. Time to prove it. The proof is short, and it pinpoints exactly which features of f make the method fly.

The tool is Taylor's theorem to second order, which you can think of for now as linear approximation *with its error term written out exactly*. It says: for a function with a continuous second derivative, expanding f about the point x_n and evaluating at the true root r,

$$f(r) = f(x_n) + f'(x_n)\,(r - x_n) + \tfrac{1}{2} f''(\xi)\,(r - x_n)^2,$$

where ξ is some point between x_n and r. The left side is $f(r) = 0$, because r is the root. So

$$0 = f(x_n) + f'(x_n)\,(r - x_n) + \tfrac{1}{2} f''(\xi)\,(r - x_n)^2.$$

Now divide the whole line by $f'(x_n)$ (legal, since we are assuming a simple root, so f' is not zero

nearby):

$$0 = \frac{f(x_n)}{f'(x_n)} + (r - x_n) + \frac{f''(\xi)}{2f'(x_n)}\,(r - x_n)^2.$$

Now rearrange to isolate the combination $x_n - f(x_n)/f'(x_n)$ – which is precisely x_{n+1}:

$$\underbrace{x_n - \frac{f(x_n)}{f'(x_n)}}_{=x_{n+1}} - r = \frac{f''(\xi)}{2f'(x_n)}\,(r - x_n)^2.$$

Define the error at step n as $e_n = x_n - r$. Then $(r - x_n)^2 = e_n^2$, and the left side is exactly $x_{n+1} - r = e_{n+1}$. We have arrived at the master equation of Newton's method:

$$\boxed{e_{n+1} = \frac{f''(\xi)}{2f'(x_n)}\,e_n^2}.$$

Focus

Read this equation slowly, for it contains the whole story. The new error e_{n+1} is proportional to the *square* of the old error e_n. Once e_n is small – say 0.01 – its square is 0.0001, which is a hundred times smaller. The next error is the square of *that*, around 10^{-8}, and the next around 10^{-16}. Each step roughly doubles the count of accurate digits. That squaring is quadratic convergence, and it is built into the geometry: it comes from the fact that the tangent line matches the curve's value *and* slope, so the leftover disagreement is governed by the next term, the curvature f'', multiplying the square of the gap.

Figure 19.6 makes the contrast between quadratic and linear convergence visible. We plot the number of correct digits against the step number for our $\sqrt{2}$ run (a simple root, quadratic) and for the double-root crawl $f(x) = (x-1)^2$ from Section 19.5 (linear). The quadratic curve bends sharply upward – each step buys roughly twice the digits of the last – while the linear one climbs at a fixed, plodding pace.

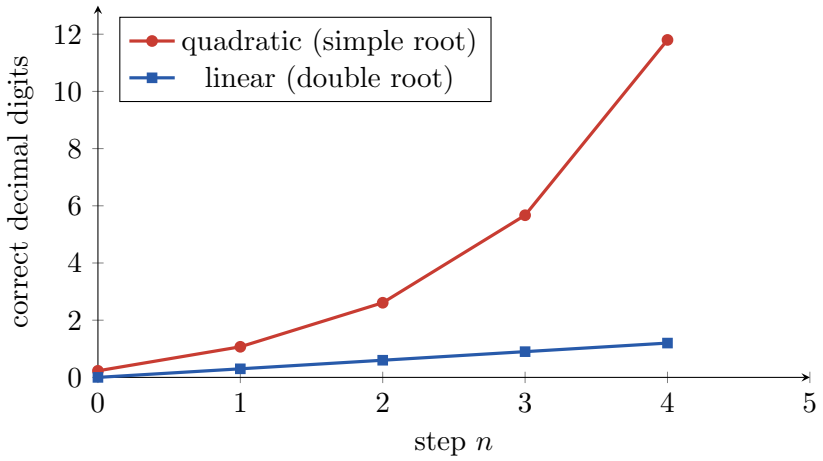


Figure 19.6: Correct digits versus step number. For a simple root (red) Newton's method gains digits at an accelerating rate – the signature of quadratic convergence. For a double root (blue) it gains a fixed amount per step, the slow march of linear convergence. Same algorithm, two radically different speeds, decided entirely by whether $f'(r) = 0$.

The master equation also explains every failure mode at a single glance:

- **The constant out front.** The error is multiplied by $C \approx |f''/2f'|$. If the curve has a clean, steep crossing – f' large, f'' modest – then C is small and convergence is ferocious. If f' is near zero (the flat-tangent failure), C blows up and a step can hurl you away. The denominator f' is the method's lifeline.
- **Why "close enough" matters.** The relation $e_{n+1} \approx Ce_n^2$ shrinks the error only if $C\,|e_n| < 1$, i.e. only once $|e_n| < 1/C$. Inside that radius each step shrinks the error and the squaring takes over; outside it, squaring a too-large number makes things *worse*. That threshold is the precise meaning of "start close enough."
- **Why double roots crawl.** At a double root $f'(r) = 0$, so the derivation above breaks – you cannot divide by $f'(x_n)$ cleanly, because it too is heading to zero. A more careful analysis (which we leave to the exercises) shows the error then merely *halves* each step: linear, not quadratic. The whole speed advantage came from $f'(r) \neq 0$.

Why this matters

There is something deeply satisfying here. The very same feature that made the tangent line a *good* approximation – that it matches the curve to first order, leaving an error of size $(\Delta x)^2$ – is what makes Newton's method *fast*. The quadratic error of linear approximation, which looked like a limitation when we first met it, turns into the quadratic *convergence* that doubles your digits. The same $(\,\cdot\,)^2$ appears in both places because it is the same fact viewed from two sides. Good approximation and fast iteration are the same coin.

19.7 Newton's method as fixed-point iteration

Newton's method belongs to a larger family of root-finding algorithms, the **fixed-point iterations**, and it is a particularly clever member of that family. Placing it there explains *why* Newton converges so much faster than its cousins, and it ties the method to a general theory worth knowing.

A fixed-point iteration is any rule of the shape

$$x_{n+1} = g(x_n),$$

for some function g. You feed in a number, the function spits out the next number, and you keep feeding the output back in. If the sequence settles down to a limit $x^\star$, then taking limits on both sides forces $x^\star = g(x^\star)$: the limit is a point that g leaves *fixed*. That is what a fixed point is – an input g returns unchanged.

Newton's method is exactly such an iteration with the particular choice

$$g(x) = x - \frac{f(x)}{f'(x)}.$$

And notice: at a root r of f we have $f(r) = 0$, so $g(r) = r - 0 = r$. The roots of f are precisely the fixed points of Newton's g. Hunting a root of f has been quietly converted into hunting a fixed point of g.

Connection

Fixed points are everywhere in mathematics, and the picture for finding them is the *cobweb diagram*: plot $y = g(x)$ and the line $y = x$ on the same axes; a fixed point is where the curve meets the diagonal. The iteration is a staircase – go vertically to the curve, horizontally to the diagonal, vertically to the curve – and whether the staircase spirals *into* the fixed point or *away* from it is governed by one number: the slope $g'(x^\star)$ at the fixed point. If $|g'(x^\star)| < 1$ the iteration is a contraction near the fixed point and homes in; if $|g'(x^\star)| > 1$ it is repelled and flies off. This is the contraction-mapping idea, and it reaches far beyond calculus – into differential equations, economics (market equilibria are fixed points), and the convergence proofs that underpin much of numerical analysis.

Now here is the punchline that elevates Newton above the ordinary fixed-point crowd. For a generic g, convergence is *linear*: the error shrinks by roughly the constant factor $|g'(x^\star)|$ each step – a fixed fraction, like steadily losing half the error every time. That is fine but slow. Newton's particular g is engineered so that its derivative is not merely less than 1 at the root – it is *exactly zero* there. Here is the verification. Differentiate $g(x) = x - f(x)/f'(x)$ using the quotient rule on the fraction:

$$g'(x) = 1 - \frac{f'(x)\,f'(x) - f(x)\,f''(x)}{\big(f'(x)\big)^2} = 1 - \frac{\big(f'(x)\big)^2 - f(x)\,f''(x)}{\big(f'(x)\big)^2} = \frac{f(x)\,f''(x)}{\big(f'(x)\big)^2}.$$

Evaluate at the root r, where $f(r) = 0$:

$$g'(r) = \frac{f(r)\,f''(r)}{\big(f'(r)\big)^2} = \frac{0 \cdot f''(r)}{\big(f'(r)\big)^2} = 0.$$

Why the technique works

Why does $g'(r) = 0$ explain the speed? In a fixed-point iteration the error multiplies by about $g'(x^\star)$ each step – so if that slope were, say, 0.5, you would lose half the error per step (linear). Newton arranges $g'(r) = 0$, which kills the linear term entirely. With the first-order term gone, the leading behavior of the error is governed by the *next* term in the Taylor expansion of g, the one carrying $g''(r)$ and a factor of e_n^2. So $e_{n+1} \approx \frac{1}{2} g''(r)\, e_n^2$ – error proportional to the square of the error, which is exactly the quadratic convergence we derived directly. Newton's method is not magic; it is the unique tuning of a fixed-point map that zeroes out the derivative at the root, and zeroing the derivative is precisely what upgrades linear convergence to quadratic.

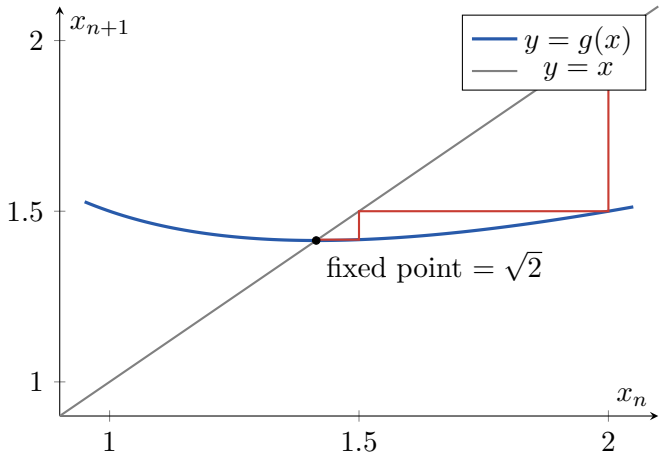


Figure 19.7: Cobweb diagram for Newton's iteration on $\sqrt{2}$, where $g(x) = \frac{x}{2} + \frac{1}{x}$. The staircase (red) descends from $x_0 = 2$ to the fixed point where $y = g(x)$ meets the diagonal $y = x$. Because the curve g is *flat* (slope 0) right at the fixed point, the staircase collapses onto it with astonishing speed.

19.8 Newton's method in real computation

Seeing the method as code is instructive, because the gap between the formula and a robust program is exactly where the failure modes of Section 19.5 demand respect. Here is the method as a small program, written in plain pseudocode that any language would render almost verbatim.

```
def newton(f, fprime, x0, tol=1e-12, max_iter=50):
    x = x0
    for n in range(max_iter):
        fx = f(x)
        if abs(fx) < tol:          # close enough: f(x) is essentially zero
            return x
        dfx = fprime(x)
        if dfx == 0:               # flat tangent: bail out, do not divide by zero
            raise ValueError("zero derivative; Newton stalled")
        x = x - fx / dfx           # the one line that is the whole method
    raise ValueError("did not converge in max_iter steps")
```

Three things in this little program encode the wisdom of the chapter. The `tol` test checks the answer by evaluating f – never trust the iterate blindly. The zero-derivative guard refuses to divide by a vanishing slope. And the `max_iter` cap stops the loop from running forever if the start was bad and the iterates are wandering off or cycling. The mathematics is one line; the engineering is the safety rail around it.

Application

Newton's method drives a great deal of modern computation, far beyond solving single equations.

- **Optimization and machine learning.** To minimize a function F, you look for where its derivative is zero – a root of F'. Applying Newton's method to F' gives the update

$x_{n+1} = x_n - F'(x_n)/F''(x_n)$, "Newton's method for optimization," which uses curvature (F'') to take far smarter steps than plain gradient descent. In many variables F'' becomes the Hessian matrix, and training large models leans on quasi-Newton methods (BFGS, L-BFGS) that approximate it.

- **Systems of equations.** Replace f by a vector of functions and f' by the Jacobian matrix; the update becomes "solve a linear system at each step." This multivariable Newton is how engineers solve the giant nonlinear systems behind circuit simulation, fluid dynamics, and structural analysis.
- **Inside the hardware.** As mentioned, reciprocals, square roots, and divisions are computed by Newton iterations chosen so that each step needs only multiplication – the operation silicon does fastest.

The reason it is chosen again and again is the quadratic convergence we proved: when each step doubles your digits, you reach full machine precision (about 16 decimal digits) in a mere handful of iterations.

19.9 A fully worked, non-algebraic example

To close, here is an equation that genuinely has no formula, the kind that proves the method earns its keep. Solve

$$\cos x = x.$$

This asks for the place where the cosine curve crosses the diagonal line $y = x$ – a number called the *Dottie number*, with no closed form whatsoever. Rewrite it as a root problem:

$$f(x) = \cos x - x = 0, \qquad f'(x) = -\sin x - 1.$$

A quick sketch (or noting $f(0) = 1 > 0$ and $f(1) = \cos 1 - 1 \approx -0.46 < 0$) shows the root lies between 0 and 1. Start at $x_0 = 1$ (radians throughout).

Work it yourself

The iteration is $x_{n+1} = x_n - \dfrac{\cos x_n - x_n}{-\sin x_n - 1}$. Step through it:

$$\begin{aligned}
x_0 &= 1 \\
x_1 &= 1 - \frac{\cos 1 - 1}{-\sin 1 - 1} = 1 - \frac{0.540302 - 1}{-0.841471 - 1} = 1 - \frac{-0.459698}{-1.841471} = 0.750364 \\
x_2 &= 0.750364 - \frac{\cos(0.750364) - 0.750364}{-\sin(0.750364) - 1} = 0.739113 \\
x_3 &= 0.739085 \\
x_4 &= 0.739085\ldots
\end{aligned}$$

The root is $0.7390851332\ldots$, and we have it to nine places after four steps.

Notice the digits settling: x_1 is right to one place, x_2 to three, x_3 to six or seven – the familiar doubling. From a single crude guess and four lines of arithmetic we have pinned down, to nine

decimal places, a number that no formula can express. This is what Newton's method buys you: it turns calculus – the tangent line, the derivative, the second-order error – into a machine that manufactures the solutions algebra cannot reach.

Pause

If you take one thing from this chapter, take the master equation $e_{n+1} \approx C\,e_n^2$ and the picture behind it: a tangent line, ridden down to the axis, again and again. The formula $x_{n+1} = x_n - f(x_n)/f'(x_n)$ is just that picture written in symbols, and the squaring of the error is just the price the curve charges for not being a straight line – a price that, near a simple root, is wonderfully small.

19.9.1 Convergence, Chaos, and Algorithmic Roots

Connection

Newton's method unlocked the computational resolution of non-linear equations across science:

- *Quadratic convergence*: When starting in the basin of attraction of a simple root, the error squares at each iteration ($e_{n+1} \approx Ce_n^2$). The number of correct decimal digits doubles with every step, reaching machine precision in 4 or 5 evaluations.
- *Solving transcendental equations*: Equations like $x = \cos x$ or $xe^x = 1$, which possess no finite algebraic formulas in radicals, are solved numerically to arbitrary precision.
- *Foundation of scientific computing*: From celestial mechanics predicting satellite orbits to GPS triangulation and circuit simulation, Newton–Raphson powers quantitative modern engineering.

Yet Newton's method is fundamentally a locally convergent nonlinear dynamical system. If its ideal conditions are breached, it displays celebrated pathologies:

1. *Zero-slope failure*: If $f'(x_n) \approx 0$, the tangent line is nearly horizontal, rocketing the next iterate x_{n+1} off toward infinity.
2. *Periodic cycle traps*: Starting at $x_0 = 0$ on $x^3 - 2x + 2 = 0$ bounces indefinitely between 0 and 1 in an unyielding 2-cycle, completely blind to the actual root at -1.769.
3. *Runaway divergence on inflection curves*: On $f(x) = \arctan x$, the curve possesses an inflection point at the root ($x = 0$). If $|x_0| > 1.3917$, the tangent line overshoots so severely that subsequent iterates diverge to $\pm\infty$.
4. *Fractal chaos*: The basins of attraction in the complex plane (and on the real line with multiple roots) have fractal boundaries with infinite detail. An arbitrarily small nudge in the initial guess can radically alter which root the method finds.

These breakdowns show why calculus alone is not enough; it must be paired with the discipline of numerical analysis. In production software, engineers use hybrid algorithms like *Brent's method*: combining the slow, foolproof reliability of the Intermediate Value Theorem's bisection with the hyper-fast quadratic acceleration of Newton–Raphson. In multivariable optimization, this logic expands into *Gauss–Newton* and *quasi-Newton methods (BFGS)*, which train modern machine learning architectures.

19.10 Exercises

Exercise 19.1. Use Newton's method to estimate $\sqrt{5}$ by finding the positive root of $f(x) = x^2 - 5$. Start at $x_0 = 2$ and carry out three iterations by hand, writing each x_n to six decimal places. Compare x_3 to the true value $2.2360679\ldots$ and count how many digits are correct at each step.

Exercise 19.2. Derive, from the general iteration, the formula for computing the cube root of a positive number a by applying Newton's method to $f(x) = x^3 - a$. Simplify your update into the form $x_{n+1} = \frac{1}{3}\left(2x_n + a/x_n^2\right)$, and use it to estimate $\sqrt[3]{2}$ starting from $x_0 = 1$ (two iterations).

Exercise 19.3. Show that applying Newton's method to $f(x) = \dfrac{1}{x} - a$ (for a positive constant a) yields the update $x_{n+1} = x_n(2 - a x_n)$, which computes $1/a$ using only multiplication and subtraction – no division. Test it with $a = 4$ and $x_0 = 0.2$, performing two steps, and check that you approach 0.25.

Exercise 19.4. The equation $x^3 - 2x + 2 = 0$ produces the two-cycle $0 \to 1 \to 0 \to \cdots$ from the start $x_0 = 0$, as shown in the chapter. (a) Verify the cycle by direct computation. (b) The equation has a real root near $x = -1.77$. Starting instead from $x_0 = -2$, perform three Newton iterations and confirm the iterates head toward that root. What does this contrast teach about the choice of starting point?

Exercise 19.5. Find the unique real root of $x^3 + x - 1 = 0$ to six decimal places using Newton's method, starting from $x_0 = 1$. Show your work for at least three iterations and verify your answer by plugging it back into the function.

Exercise 19.6 (Conceptual)**.** Explain in your own words, using a sketch, why Newton's method can fail when the starting guess lands near a point where $f'(x) = 0$. Then describe geometrically what happens to the next iterate, and why this is a genuine breakdown of the method rather than a minor inaccuracy.

Exercise 19.7 (Conceptual)**.** The function $g(x) = x - f(x)/f'(x)$ is the fixed-point map behind Newton's method. A general fixed-point iteration $x_{n+1} = g(x_n)$ converges *linearly* when $0 < |g'(x^\star)| < 1$. Using the formula $g'(x) = f(x)f''(x)/(f'(x))^2$ derived in the chapter, explain why Newton's method converges *faster* than a typical fixed-point iteration, and what specifically goes right at a simple root.

Exercise 19.8 (Multiple root)**.** For the double root of $f(x) = (x-3)^2$, show that Newton's iteration simplifies to $x_{n+1} = \frac{1}{2}(x_n + 3)$. Starting from $x_0 = 5$, compute the errors $|x_n - 3|$ for $n = 0, 1, 2, 3, 4$ and confirm that they merely *halve* each step (linear convergence) rather than the digits doubling. Explain, referring to the master equation $e_{n+1} = \frac{f''(\xi)}{2f'(x_n)} e_n^2$, why the quadratic behavior is lost at a double root.

Exercise 19.9 (Proof)**.** Let f have a continuous second derivative and a simple root r (so $f'(r) \neq 0$). Starting from the second-order Taylor expansion of f about x_n evaluated at r, derive the master equation

$$e_{n+1} = \frac{f''(\xi)}{2f'(x_n)}\, e_n^2, \qquad e_n = x_n - r,$$

for some ξ between x_n and r. Then argue that there exist a constant M and a radius $\delta > 0$ such that if $|e_0| < \delta$, the errors satisfy $|e_{n+1}| \leq M|e_n|^2$, and conclude that the convergence is quadratic.

Exercise 19.10 (Application)**.** To minimize a smooth function $F(x)$, one seeks a root of $F'(x) = 0$. Write down the Newton iteration for this root problem (it will involve F' and F''). Apply it to $F(x) = x^4 - 3x^2 + x$ starting from $x_0 = 1$, performing two iterations, and interpret geometrically what the resulting point represents.

Part V

The Integral

Chapter 20

The Integral: Area and Accumulation

For ten chapters now we have been obsessed with one operation: *taking things apart*. The derivative is a machine for slicing a curve into its instantaneous tendencies – the slope here, the slope there, the rate of change at one frozen moment. We zoomed in until the curve became a line and read off how steep that line was. That is calculus run *downhill*: from a whole thing to its local behavior.

This chapter runs calculus the other way. We are going to *add things up*. Given a rate that changes from moment to moment, we will reconstruct the total that accumulated. Given a curve, we will find the area trapped underneath it. And the astonishing thing – the thing the next chapter is entirely about – is that these two activities, taking apart and adding up, are exact inverses of one another. Differentiation undoes integration and integration undoes differentiation, the way subtraction undoes addition. But we are getting ahead of ourselves. To appreciate that miracle you first have to meet the integral on its own terms, with no derivatives in sight at all.

Focus

Two questions that look unrelated turn out to be the *same* question: (1) What is the area under a curve? (2) If I know the *rate* at which something accumulates at every instant, how much accumulates in total? This chapter builds one object – the **definite integral** – that answers both.

By the end you will know what the symbol $\int_a^b f(x)\,\mathrm{d}x$ really means: not as a rule to apply, but as a limit of a perfectly concrete sum of rectangles. You will see why that elongated S was chosen, what the $\mathrm{d}x$ is doing there, and why "area" sometimes comes out negative. We will compute an integral or two from raw first principles – no shortcuts, no Fundamental Theorem – just so you can feel in your hands what the definition is actually asking for.

20.1 Two ancient problems, one idea

20.1.1 The area problem

How much area is inside a circle? A child can answer for a rectangle (length times width) or a triangle (half base times height), because those shapes are built from straight lines and area for

straight-line shapes is elementary. But a circle is bounded by a curve, and there is no finite recipe of multiplications that gives its area exactly. The number π is, at bottom, the confession that the area of a curved region is a genuinely new kind of quantity.

The Greeks found the way through, and it is the same way we will take. Around 250 BCE, Archimedes computed the area of regions bounded by curves – a segment of a parabola, the circle – by trapping the unknown area between shapes he *could* measure. Inscribe a polygon inside the circle and circumscribe another outside it. The true area is squeezed between the two. Now add more sides. The inscribed polygon swells outward, the circumscribed one shrinks inward, and the gap between them closes. The area of the circle is the single number both sequences are converging to.

History

The dream of calculating areas under curves spans over two millennia of struggle against the paradoxes of the infinite:

- **Archimedes and Exhaustion (c. 250 BCE)**: In his masterpiece *Quadrature of the Parabola*, Archimedes calculated the exact area of a parabolic segment. He inscribed an initial triangle, then filled the remaining gaps with infinitely many smaller triangles whose areas formed a geometric series with common ratio $\frac{1}{4}$:

$$\text{Area} = T + \frac{1}{4}T + \frac{1}{16}T + \cdots = \frac{4}{3}T.$$

  Because the Greeks distrusted actual infinities, Archimedes avoided limits by using the *method of exhaustion*: a double proof by contradiction showing the area could be neither greater nor less than $\frac{4}{3}T$.

- **Cavalieri's Indivisibles (1635) and the Hobbes Controversy**: In *Geometria indivisibilibus continuorum*, the Italian mathematician Bonaventura Cavalieri took a daring leap: he imagined a 2D planar figure as composed of an infinite collection of parallel lines of zero thickness, which he called *indivisibles* (like the pages in a book, or coins in a stack). Cavalieri showed that if two regions have equal cross-sections at every height, their total areas must be equal (*Cavalieri's Principle*). But indivisibles triggered a furious backlash. Jesuit mathematicians and the English philosopher Thomas Hobbes attacked them as logically incoherent and theological heresy. Hobbes demanded: how could lines with zero width ever add up to an area with finite width? If a point has zero length, how can an infinity of points make a line?

- **Leibniz's Notation (1675)**: In October 1675, Gottfried Wilhelm Leibniz resolved the conceptual knot with one of the greatest inventions in the history of science: the integral sign $\int$. Leibniz realized that instead of zero-width indivisibles, one should sum an infinity of rectangles of height y and infinitesimal width $\mathrm{d}x$. He adopted the elongated letter "S" (for Latin *summa*) and wrote $\int y\,\mathrm{d}x$. The notation itself explained what was happening: an integral is an infinite sum of differential areas.

- **Cauchy and Riemann (1820s–1850s)**: The modern definition was completed by Augustin-Louis Cauchy and Bernhard Riemann. Riemann showed in 1854 that the integral is neither an infinite series of metaphysical indivisibles nor a mysterious sum of zeroes, but the precise limit of finite approximating sums as the mesh size shrinks to zero: $\lim_{\|\Delta\|\to 0} \sum f(x_k^*)\Delta x_k$.

20.1.2 The accumulation problem

Now a problem that seems to have nothing to do with area. A car's speedometer reads $v(t)$ at every instant t. The speed is changing – the driver accelerates, brakes, coasts. *How far did the car travel* between time $t = a$ and time $t = b$?

If the speed were *constant* at 60 miles per hour for 2 hours, distance would be easy: $60 \times 2 = 120$ miles. Rate times time. But the speed is not constant; it changes continuously, and "rate times time" has no obvious meaning when the rate is a moving target.

Here is the idea that saves us, and notice that it is *exactly* Archimedes' idea wearing different clothes. Chop the trip into tiny time intervals. Over a single hundredth of a second the speed barely changes, so on that sliver the motion is *almost* constant-speed, and distance is *almost* just "speed at that moment times the length of the sliver." Add up the slivers. The finer you chop, the better the approximation, and the true distance is the number these sums converge to.

Intuition

Both problems have the same skeleton. There is a quantity you cannot get directly because something is *curved* or *varying*. So you cut the domain into many small pieces, on each piece pretend the varying thing is constant (which is nearly true when the piece is small), compute the easy straight-line answer on each piece, add them all up, and then let the pieces shrink toward zero width. The integral is the limit of that sum.

Watch what the distance computation produces. On the time-axis, lay out the intervals. Above each one, draw a rectangle whose height is the speed during that interval. The *area* of one rectangle is (height)×(width) = (speed)×(time) = distance covered in that interval. So the total distance is the total *area* of all the rectangles – which is an approximation to the area under the speed curve $v(t)$. The accumulation problem *is* the area problem. Distance traveled is the area under the speed graph; this connection is mathematically forced.

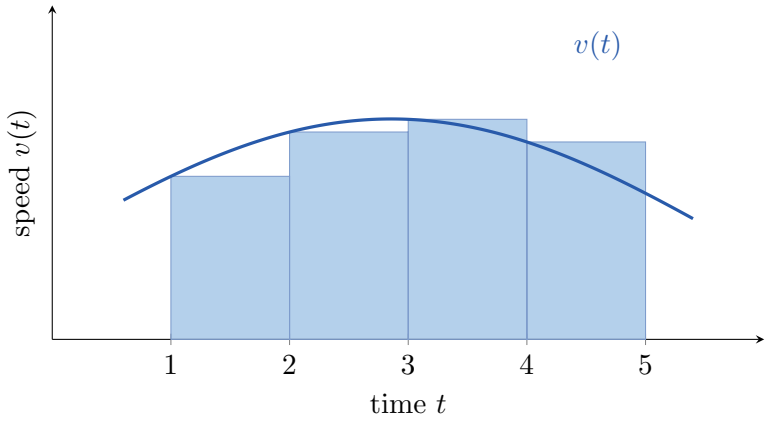


Figure 20.1: Distance traveled is the area under the speed curve. Each rectangle has height equal to the speed on that interval and width equal to the elapsed time, so its area is a distance. The sum of the rectangles approximates the area under $v(t)$; the true distance is the limit as the rectangles get thin.

20.2 Riemann sums: making the idea precise

Let us stop gesturing and pin the construction down. We have a function f on a closed interval $[a, b]$. For now, picture f as continuous and positive, so "area under the graph" is unambiguous – we will lift those assumptions soon.

20.2.1 Partitioning the interval

Cut $[a, b]$ into n subintervals. Choose points

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b.$$

This list is called a **partition** P of $[a, b]$. The k-th subinterval is $[x_{k-1}, x_k]$, and its width is

$$\Delta x_k = x_k - x_{k-1}.$$

The widths need not be equal, though most of the time we will make them equal: if we use n subintervals of the *same* width, then each has width

$$\Delta x = \frac{b-a}{n}, \qquad x_k = a + k\,\Delta x.$$

This even spacing is called a **regular** (or uniform) partition, and it is what you should picture by default.

20.2.2 Sampling a height on each piece

On each subinterval $[x_{k-1}, x_k]$ we are going to approximate the curve by a single horizontal line – a rectangle. But at what height? The curve takes many values across the subinterval; we must pick one. Choose any point $x_k^* \in [x_{k-1}, x_k]$, called a **sample point**, and use the height $f(x_k^*)$. The rectangle over the k-th subinterval then has

$$\text{area} = f(x_k^*)\,\Delta x_k.$$

Add them up over all n subintervals.

Definition 20.1 (Riemann sum)**.** Given a function f on $[a, b]$, a partition P with subintervals of width Δx_k, and a choice of sample points $x_k^* \in [x_{k-1}, x_k]$, the corresponding **Riemann sum** is

$$S(P, \{x_k^*\}) \;=\; \sum_{k=1}^{n} f(x_k^*)\,\Delta x_k.$$

This sum is just a number: a finite pile of rectangle areas, signed by the sign of f. It is named for Bernhard Riemann, who in 1854 gave the first clean general definition of the integral built exactly this way. Each Riemann sum is an *approximation* to the area we are after. To get the true area we will take a limit – but first let us look at the three most common ways to choose the sample points, because their pictures teach you what is going on.

20.2.3 Left, right, and midpoint sums

With a regular partition, three natural choices of x_k^* stand out.

- **Left sum** L_n: take $x_k^* = x_{k-1}$, the left endpoint of each subinterval. The rectangle's height is the value of f at the left edge.
- **Right sum** R_n: take $x_k^* = x_k$, the right endpoint.
- **Midpoint sum** M_n: take $x_k^* = \frac{1}{2}(x_{k-1} + x_k)$, the center of each subinterval.

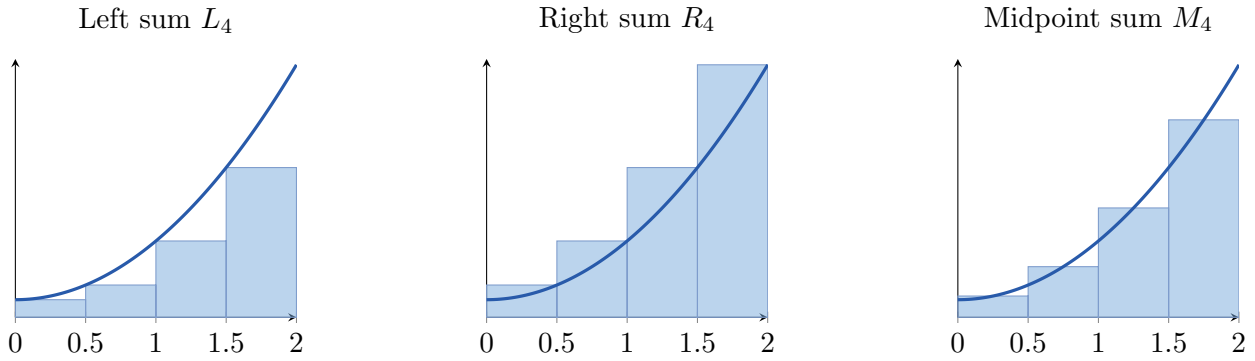


Figure 20.2: Three Riemann sums for $f(x) = x^2+0.3$ on $[0, 2]$ with four subintervals. On a rising curve the left sum *underestimates* (each rectangle ducks below the curve) and the right sum *overestimates*. The midpoint sum splits the difference and is usually closest.

Look at Figure 20.2. On a curve that is *increasing*, the left endpoint is the lowest point of f on each subinterval, so every left rectangle sits below the curve and L_n *undershoots* the true area. Symmetrically, R_n *overshoots*. The true area is sandwiched: $L_n \le \text{area} \le R_n$. (On a decreasing curve the roles swap.) That sandwich is Archimedes' inscribed-and-circumscribed idea reborn: the left sum is an inscribed staircase, the right sum a circumscribed one, and the area lives between them.

Work it yourself

Let us do one entirely by hand so the definition stops being abstract. Take $f(x) = x^2$ on $[0, 1]$ with $n = 4$ equal subintervals, so $\Delta x = \frac{1}{4}$ and the partition points are $0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$.
Left sum. Heights at $0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$:

$$L_4 = \tfrac{1}{4}\big(0^2 + \tfrac{1}{16} + \tfrac{1}{4} + \tfrac{9}{16}\big) = \tfrac{1}{4} \cdot \tfrac{14}{16} = \tfrac{14}{64} = 0.21875.$$

Right sum. Heights at $\frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$:

$$R_4 = \tfrac{1}{4}\big(\tfrac{1}{16} + \tfrac{1}{4} + \tfrac{9}{16} + 1\big) = \tfrac{1}{4} \cdot \tfrac{30}{16} = \tfrac{30}{64} = 0.46875.$$

Midpoint sum. Heights at $\frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}$:

$$M_4 = \tfrac{1}{4}\big(\tfrac{1}{64} + \tfrac{9}{64} + \tfrac{25}{64} + \tfrac{49}{64}\big) = \tfrac{1}{4} \cdot \tfrac{84}{64} = 0.328125.$$

The true area, as we will prove below, is exactly $\frac{1}{3} \approx 0.3333$. Notice $L_4 < \frac{1}{3} < R_4$ as promised, and M_4 is by far the closest – off by less than 0.006. Four rectangles already do well.

Common trap

A Riemann sum is **not** the integral. It is one finite approximation, pinned to a particular partition and a particular set of sample points. Change the sample points and you get a different number. The integral is what all these numbers *converge to* as the rectangles shrink. Students who write "$\int_a^b f = \sum f(x_k)\Delta x$" as if they were equal have skipped the only step that matters – the limit. The sum approximates; the limit *is*.

20.3 The definite integral as a limit

Now we take the limit. As we add more and more subintervals and let their widths shrink toward zero, the staircase of rectangles hugs the curve ever more tightly, and all the different Riemann sums – left, right, midpoint, any sample points at all – get squeezed toward one common value. That value is the integral.

We measure the "fineness" of a partition by its **mesh** (or norm), the width of its widest subinterval:

$$\|P\| = \max_{1\le k\le n} \Delta x_k.$$

Driving $\|P\| \to 0$ forces *every* subinterval to shrink, rather than merely the average. With that we can state the definition.

Definition 20.2 (The definite integral)**.** A function f is **integrable** on $[a, b]$ if there is a single number I with the following property: the Riemann sums approach I no matter how the sample points are chosen, as the mesh goes to zero. Precisely, for every $\varepsilon > 0$ there is a $\delta > 0$ such that for *any* partition P with $\|P\| < \delta$ and *any* choice of sample points,

$$\left|\sum_{k=1}^{n} f(x_k^*)\,\Delta x_k \;-\; I\right| < \varepsilon.$$

When this number I exists we call it the **definite integral** of f from a to b and write

$$\int_a^b f(x)\,\mathrm{d}x \;=\; \lim_{\|P\|\to 0} \sum_{k=1}^{n} f(x_k^*)\,\Delta x_k \;=\; I.$$

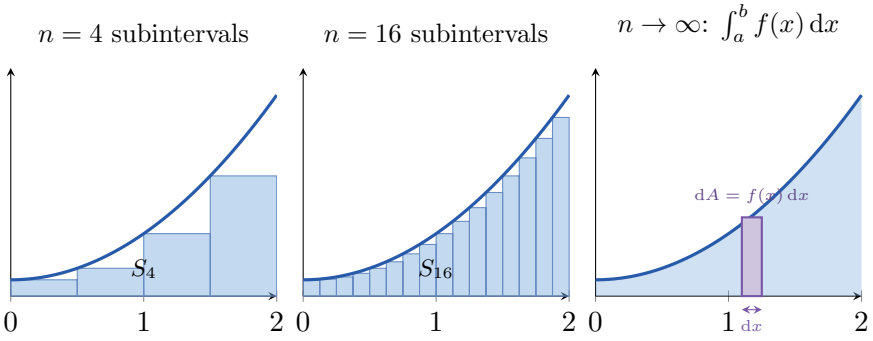


Figure 20.3: The limit of Riemann sums as the partition refines. Left: A coarse partition ($n = 4$) leaves visible staircase gaps and overshoots. Center: A finer partition ($n = 16$) hugs the curve closely as the staircase smooths out. Right: In the continuous limit $n \to \infty$ ($\Delta x \to 0$), the sum becomes the definite integral $\int_a^b f(x)\,\mathrm{d}x$, accumulating infinitely many infinitesimal strips $\mathrm{d}A = f(x)\,\mathrm{d}x$.

That "for any choice of sample points" is the whole strength of the definition. It is demanding that left sums, right sums, midpoint sums, and every wild zigzag choice in between all funnel to the *same* limit. For a badly behaved f they might not, and then f is not integrable. The good news, which we will not prove in full but which is the workhorse fact of the subject:

Theorem 20.1 (Integrability of continuous functions)**.** *If f is continuous on $[a, b]$, then f is integrable on $[a, b]$. More generally, f is integrable if it is bounded and has at most finitely many discontinuities.*

The reason continuity is enough is intuitive: on a tiny subinterval a continuous function cannot wiggle much (that is essentially *uniform* continuity on the closed interval $[a, b]$), so the gap between the tallest and shortest rectangle on that piece is small, which means left and right sums cannot disagree by much, which means everything is trapped together. Squeeze the trap shut and the limit exists.

20.3.1 Decoding the notation

The symbol $\int_a^b f(x)\,\mathrm{d}x$ is one of the best-designed notations in all of mathematics, and Leibniz designed it in 1675 to be read literally.

What is really happening

Stare at the symbol and read what each part was *built* to mean.

- The $\mathrm{d}x$ is the ghost of Δx_k – the width of a subinterval *after* it has been shrunk to nothing. It is an infinitely thin slice of the x-axis.
- $f(x)$ is the height of the curve at that slice.
- So $f(x)\,\mathrm{d}x$ is the area of one infinitely thin rectangle: height times infinitesimal width.
- The sign $\int$ is an elongated letter S, for *summa* – "sum." Leibniz literally stretched the Σ of summation into a continuous S because we are no longer adding finitely many things; we are adding a *continuum* of them.
- The $\int_a^b$ says: do this summing as x sweeps from a to b.

Read it aloud the way Leibniz meant it: *"sum, from a to b, of height times infinitesimal width."* The integral is, in one symbol, the instruction "add up infinitely many infinitely thin pieces." Everything we did with Riemann sums was the finite, limit-respecting version of exactly that sentence.

A word on the x. It is a **dummy variable**: it ranges over the subintervals and then disappears in the limit. The value of $\int_a^b f(x)\,\mathrm{d}x$ is a fixed number that does not depend on what letter you used. So $\int_a^b f(x)\,\mathrm{d}x = \int_a^b f(t)\,\mathrm{d}t = \int_a^b f(u)\,\mathrm{d}u$. The variable is scaffolding; the number is the building.

Why the technique works

Why does letting the rectangle width go to zero give the *exact* area rather than just a better guess? Because of the squeeze. On any subinterval, the true area under the curve lies between the shortest rectangle (height = minimum of f there) and the tallest (height = maximum). Summed up, the true area lies between the lower sum $\underline{S}_n$ and the upper sum $\overline{S}_n$. Each

individual gap is at most (oscillation of f on that piece)$\times$(width of that piece). As the mesh shrinks, continuity forces every oscillation toward zero, so the total gap $\overline{S}_n - \underline{S}_n$ collapses to zero. The true area is pinned between two quantities that are coming together: it can be nothing but their shared limit. No error survives the limit – that is the whole trick of calculus, and it is Archimedes' method of exhaustion made rigorous.

20.4 Computing an integral from the definition

Before we have any shortcuts, let us earn one integral the hard way – straight from Definition 20.2 – so you trust that the limit really is there and really is computable. We will compute

$$\int_0^1 x^2 \, \mathrm{d}x$$

and find it equals $\frac{1}{3}$, the number we sampled in the lab box above.

Example 20.1 ($\int_0^1 x^2 \, \mathrm{d}x$ by brute force)**.** Use a regular partition into n equal pieces, so $\Delta x = \frac{1}{n}$ and $x_k = \frac{k}{n}$. Choose right endpoints as sample points, $x_k^* = \frac{k}{n}$. The Riemann sum is

$$R_n = \sum_{k=1}^{n} \left(\frac{k}{n}\right)^2 \cdot \frac{1}{n} = \frac{1}{n^3} \sum_{k=1}^{n} k^2.$$

Now we need the sum of the first n squares – a classic closed form, provable by induction:

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

Substitute it in:

$$R_n = \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6n^2} = \frac{2n^2+3n+1}{6n^2}.$$

Divide through by n^2:

$$R_n = \frac{1}{6}\left(2 + \frac{3}{n} + \frac{1}{n^2}\right).$$

Now let $n \to \infty$ (which drives the mesh $\frac{1}{n} \to 0$). The $\frac{3}{n}$ and $\frac{1}{n^2}$ terms vanish, leaving

$$\int_0^1 x^2 \, \mathrm{d}x = \lim_{n\to\infty} R_n = \frac{1}{6} \cdot 2 = \frac{1}{3}.$$

There it is, exactly $\frac{1}{3}$, with no Fundamental Theorem, no antiderivative, nothing but a sum of squares and a limit. Notice the structure of the computation: a finite-but-growing sum, a closed form for it, then a limit that washes away every term except the one that survives. That is the integral doing precisely what its definition promised.

Work it yourself

Repeat the computation with *left* endpoints, $x_k^* = \frac{k-1}{n}$. You will get $L_n = \frac{1}{6}\left(2 - \frac{3}{n} + \frac{1}{n^2}\right)$, which is *below* $\frac{1}{3}$ for every finite n (the left sum undershoots a rising curve, as Figure 20.2 showed) – yet it still limits to the same $\frac{1}{3}$. Left and right disagree at every finite stage and agree in the limit. That agreement is integrability in action.

Why this matters

Pause on what just happened. We added up the areas of n rectangles, a quantity that grows more and more intricate as n increases, and in the limit all that intricacy boiled down to a single clean fraction, $\frac{1}{3}$. The infinite process produced something *finite and exact*. This is the recurring miracle of analysis: an infinite sum of vanishing pieces can have a perfectly ordinary number as its total. The whole edifice of integration is the disciplined harvesting of such miracles.

20.5 Signed area: when the curve dips below the axis

So far f was positive and "area under the curve" meant exactly that. But the definition of the integral never required $f \geq 0$. If $f(x_k^*)$ is *negative*, the term $f(x_k^*)\,\Delta x_k$ is negative – it *subtracts* from the sum. So what does the integral compute when the curve goes below the x-axis?

Focus

The definite integral computes **signed area**: area above the x-axis counts as positive, area below counts as negative. The integral is the area above minus the area below.

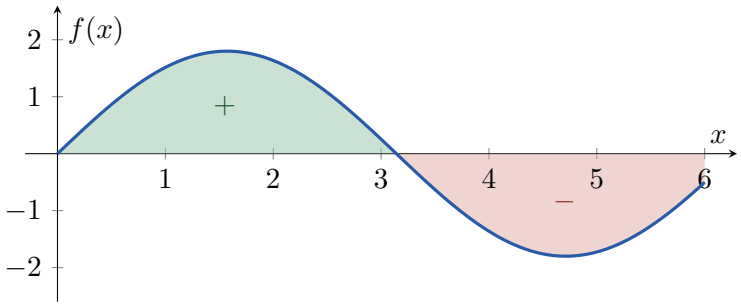


Figure 20.4: Signed area. For $f(x) = 1.8\sin x$ on $[0, 6]$, the green hump above the axis counts positively and the red dip below counts negatively. The integral $\int_0^6 1.8\sin x\,dx$ is the green area minus the red area, not the total amount of paint.

This sign convention is not an arbitrary bookkeeping rule; it is forced on us, and it is exactly what we *want*. Recall the car. If $v(t)$ is the car's *velocity* (a signed quantity – positive going forward, negative in reverse), then $\int_a^b v(t)\,dt$ is the *displacement*: net change in position. Time spent in reverse should subtract from how far you have gotten, and the negative integrand makes it do exactly that. If instead you want total distance traveled – odometer mileage, which never decreases – you integrate the *speed* $|v(t)|$, which is never negative. Displacement and distance differ precisely by the sign convention, and both are useful.

Common trap

"Find the area between the curve and the axis" and "evaluate the integral" are different requests when the curve crosses the axis. The integral gives signed area and can be zero even when there is plenty of region (a symmetric wiggle cancels itself: $\int_0^{2\pi} \sin x \, dx = 0$). For *geometric* area – total paint – you must split at the crossing points and integrate $|f|$, or equivalently add up the absolute values of the pieces. Read the question carefully.

20.6 Properties of the integral

Working from the definition every time would be unbearable. Fortunately the integral inherits a handful of clean properties straight from the Riemann sum – each one is just a fact about sums that survives the limit. Let f and g be integrable on the relevant intervals and c a constant.

Focus

(1) Linearity (sum). $\displaystyle\int_a^b \big(f(x)+g(x)\big)\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx.$

(2) Linearity (constant multiple). $\displaystyle\int_a^b c\,f(x)\,dx = c\int_a^b f(x)\,dx.$

(3) Additivity over intervals. For $a<c<b$, $\displaystyle\int_a^b f = \int_a^c f + \int_c^b f.$

(4) Orientation (reversing the limits). $\displaystyle\int_b^a f = -\int_a^b f,$ and $\displaystyle\int_a^a f = 0.$

(5) Monotonicity. If $f(x)\le g(x)$ on $[a,b]$, then $\displaystyle\int_a^b f \le \int_a^b g.$

(6) The constant. $\displaystyle\int_a^b c\,dx = c\,(b-a).$

Each of these is obvious once you see it as a statement about the rectangles.

Why linearity holds. A Riemann sum for $f+g$ is $\sum \big(f(x_k^*)+g(x_k^*)\big)\Delta x_k$. Finite sums split across addition, so this is $\sum f(x_k^*)\Delta x_k + \sum g(x_k^*)\Delta x_k$ – a Riemann sum for f plus a Riemann sum for g, on the *same* partition and *same* sample points. Take the limit of both sides; the limit of a sum is the sum of the limits. Pulling a constant c out is the same story: $\sum c\,f(x_k^*)\Delta x_k = c\sum f(x_k^*)\Delta x_k$. Linearity is nothing but the distributive law surviving a limit.

Why additivity holds. If the splitting point c is one of your partition points, then the rectangles over $[a,b]$ are literally the rectangles over $[a,c]$ followed by the rectangles over $[c,b]$. The sum over the whole is the sum over the left part plus the sum over the right part. Take the limit. This is the property that lets us handle a function defined piecewise, or split at a sign change to compute geometric area.

Why orientation flips the sign. We built the integral with a on the left and b on the right, so $\Delta x_k > 0$. If we agree to sweep the other way, every Δx_k becomes negative, flipping the sign of the

whole sum. Defining $\int_b^a f = -\int_a^b f$ makes the additivity rule $\int_a^c + \int_c^b = \int_a^b$ hold for *any* arrangement of a, b, c – even when c is outside $[a, b]$. The convention is chosen to keep the algebra seamless, and you will be grateful for it in the next chapter.

Intuition

Monotonicity is the most intuitive of all: if one curve sits above another everywhere, the area under it is larger. There is simply more region underneath. In rectangle terms, $f \le g$ makes every rectangle for f no taller than the corresponding rectangle for g, so the sums obey the same inequality, and the inequality passes to the limit. From this one fact flow all the comparison and bounding estimates you will ever make for integrals.

Example 20.2 (Properties in action)**.** Suppose someone tells you $\int_0^1 x^2\,\mathrm{d}x = \frac{1}{3}$ (we proved it) and $\int_0^1 x\,\mathrm{d}x = \frac{1}{2}$ (the area of a triangle of base and height 1). Then with no further integration:

$$\int_0^1 \left(6x^2 - 4x + 5\right)\mathrm{d}x = 6\int_0^1 x^2\,\mathrm{d}x - 4\int_0^1 x\,\mathrm{d}x + \int_0^1 5\,\mathrm{d}x = 6\cdot\tfrac{1}{3} - 4\cdot\tfrac{1}{2} + 5\cdot 1 = 2 - 2 + 5 = 5.$$

Linearity and the constant rule did all the work, dissolving a complicated integrand into simple known pieces.

20.7 The average value of a function

Here is a question the integral handles cleanly, and one that previews ideas to come. The average of finitely many numbers is their sum divided by how many there are. But what is the average *height* of a curve over an interval – the average of *infinitely many* values $f(x)$ as x ranges over $[a, b]$? You cannot divide "the sum of infinitely many values" by "infinity." The integral steps in exactly where the naive average breaks down.

Sample the curve at n evenly spaced points and average those:

$$\text{average} \approx \frac{f(x_1) + f(x_2) + \cdots + f(x_n)}{n}.$$

Multiply and divide by $\Delta x = \frac{b-a}{n}$. Since $\frac{1}{n} = \frac{\Delta x}{b-a}$,

$$\frac{1}{n}\sum_{k=1}^{n} f(x_k) = \frac{1}{b-a}\sum_{k=1}^{n} f(x_k)\,\Delta x.$$

The sum on the right is a Riemann sum for f. Let $n \to \infty$ and it becomes an integral, giving us the definition.

Definition 20.3 (Average value)**.** The **average value** of an integrable function f over $[a, b]$ is

$$\bar{f} = \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x.$$

Intuition

Rearrange it as $\bar{f} \cdot (b-a) = \int_a^b f(x)\,\mathrm{d}x$. The left side is the area of a *rectangle* of width $(b-a)$ and height $\bar{f}$. So the average value is the height of the rectangle that has the *same area* as the region under the curve – the level you would pour the area into to make it flat. If you flooded the area under the graph and let it settle, $\bar{f}$ is the water line.

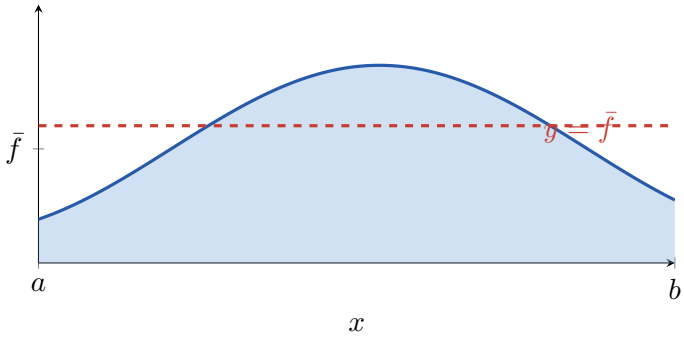


Figure 20.5: Average value as a leveling line. The dashed line $y = \bar{f}$ is set at the height that makes the rectangle over $[a, b]$ have exactly the same area as the shaded region under f. The curve's excess above the line exactly fills its deficits below.

Example 20.3 (Average value of a parabola)**.** The average value of $f(x) = x^2$ on $[0, 1]$ is

$$\bar{f} = \frac{1}{1-0}\int_0^1 x^2\,\mathrm{d}x = \frac{1}{1}\cdot\frac{1}{3} = \frac{1}{3}.$$

Sanity check: f runs from 0 up to 1 on this interval, but it spends most of the interval near the bottom (it only climbs steeply near $x = 1$). Its values near the right end, all the way up to 1, pull the average above the midpoint value $f(\frac{1}{2}) = \frac{1}{4}$, landing at $\frac{1}{3}$ – comfortably between the function's low values and its high value of 1, just as an average should be.

Connection

The average value is the bridge to the *Mean Value Theorem for integrals*, which you will meet next chapter: if f is continuous, it actually *attains* its average value at some point $c \in [a, b]$, i.e. $f(c) = \bar{f}$. Geometrically, the curve must cross its own leveling line somewhere. This is the integral's echo of the Mean Value Theorem you proved for derivatives, and it is one of the gears that drives the Fundamental Theorem of Calculus.

Application

Average value is everywhere once you look. The root-mean-square voltage of household AC power is built from the average value of $V(t)^2$ over a cycle – it is why a "120-volt" outlet has a peak voltage near 170. In probability, the *expected value* of a continuous random variable is $\int x\,p(x)\,\mathrm{d}x$, an average of outcomes weighted by the density p. In thermodynamics, the average energy of a system is an integral over all its states. Whenever you must summarize a continuously varying quantity by a single representative number, the average value – an integral divided by a length – is the right way to do it.

20.8 Accumulation: the integral as a running total

We close by returning to the theme that names this chapter, because it is the viewpoint that makes the next chapter inevitable. We have treated $\int_a^b f$ as a fixed number – one area. But let the *upper limit* move. Define a new function

$$A(x) = \int_a^x f(t)\,\mathrm{d}t,$$

the accumulated area from the fixed start a out to a variable endpoint x. (Note the dummy variable is now t, freeing x to be the endpoint.) As x slides right, $A(x)$ grows by sweeping in more area; this A is the **accumulation function**.

What is really happening

Think of f as a *rate* and A as the *running total* that rate builds up. If $f(t)$ is the rate at which water flows into a tank (liters per second), then $A(x) = \int_a^x f(t)\,\mathrm{d}t$ is the total liters that have flowed in by time x. If $f(t)$ is a country's birth rate (people per year), A is the cumulative number of births. If f is velocity, A is displacement. The integrand is always a *rate of accumulation*; the integral is always the *amount accumulated*. This single picture – rate in, total out – is the beating heart of integral calculus, and it is the exact inverse of the derivative's picture (total in, rate out).

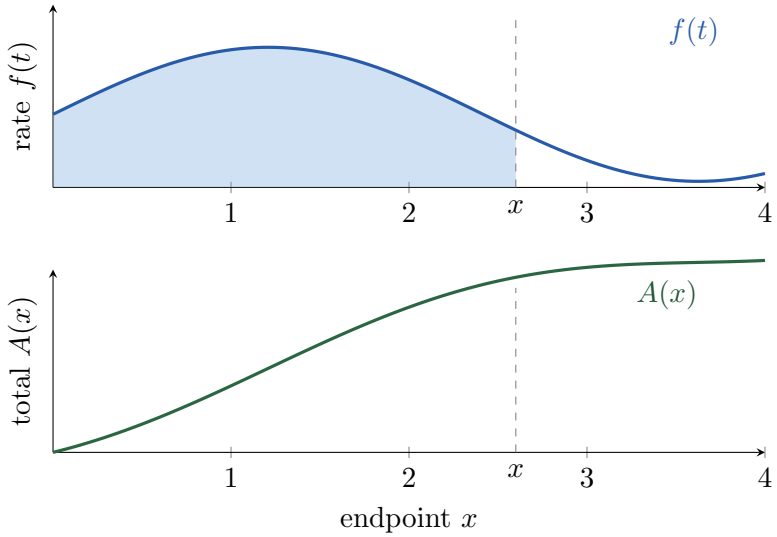


Figure 20.6: The accumulation function $A(x) = \int_a^x f(t)\,\mathrm{d}t$ (bottom) records the running area swept out under the rate f (top) as the endpoint x moves right. Where f is tall, A climbs steeply; where f is small, A nearly levels off. The *slope* of the lower curve at each x is precisely the *height* of the upper curve at that same x – a picture of $A'(x) = f(x)$ before we prove it.

Now ask the question that detonates the entire subject: *how fast does the running total $A(x)$ grow?* When the endpoint nudges forward by a tiny $\mathrm{d}x$, the area gains a thin sliver of width $\mathrm{d}x$ and height $f(x)$ – so A grows by about $f(x)\,\mathrm{d}x$, which means

$$\frac{\mathrm{d}A}{\mathrm{d}x} \approx f(x).$$

The *rate of accumulation of the total is the integrand itself.* The derivative of the area function is the curve. You can see it already in Figure 20.6: the running-total curve rises steeply exactly where the rate is tall and flattens where the rate is small, so its slope tracks the height of f point for point. Read that again: differentiation and integration are undoing each other. That equation, made exact, is the Fundamental Theorem of Calculus – and it is the whole of the next chapter. We built the integral from sums and limits alone, with no derivatives allowed; and yet at the very end the derivative has come walking back through the front door. The two halves of calculus are about to become one.

Walking question

Take this on your walk. You have known since childhood that distance is speed times time – when the speed is constant. The integral is the precise meaning of "speed times time" when the speed is *not* constant: chop time fine, on each sliver the speed is nearly constant so distance is nearly speed-times-sliver, add up the slivers, take the limit. Now turn it around: if you only had a record of total distance versus time, how would you recover the speed at an instant? You would differentiate. So the speedometer and the odometer are derivative and integral of one another. Carry that one example with you; the Fundamental Theorem is just this sentence, made airtight, for *every* pair of rate and total.

20.9 Looking back, and looking ahead

We built the definite integral from the ground up. We took two ancient, seemingly separate problems – the area under a curve and the total accumulated from a varying rate – and discovered they are the same problem. We solved it Archimedes' way: cut the domain into thin pieces, approximate the varying thing by a constant on each piece, sum the easy answers, and take a limit as the pieces vanish. That limit, when it exists, is the Riemann integral $\int_a^b f(x)\,\mathrm{d}x$, and we proved by hand that $\int_0^1 x^2\,\mathrm{d}x = \frac{1}{3}$ with nothing but a sum of squares.

We saw that the integral computes *signed* area, that it obeys clean linearity, additivity, orientation, and monotonicity rules – each one a fact about finite sums that survives the limit – and that dividing it by the interval length gives the average value of a function. Finally, by letting the upper limit roam, we built the accumulation function and glimpsed the equation $\frac{\mathrm{d}A}{\mathrm{d}x} = f(x)$ that will, next chapter, unify the two halves of calculus into a single theorem.

Why this matters

The derivative tore a curve down to its instantaneous slopes. The integral rebuilds a total from its instantaneous rates. One is analysis, the other is synthesis; one takes apart, the other puts together. That they are exact inverses is not a convenience but a deep structural truth about change itself – and it is the reason a single subject, calculus, can describe both how things vary and how much accumulates. You have now met both halves separately. Turn the page and watch them fuse.

20.9.1 The Scope of Accumulation and the Bridge Ahead

Connection

The definite integral unlocked the universal definition of physical accumulation:

- *Continuous accumulation*: Distance traveled under variable velocity ($s = \int v(t)\,\mathrm{d}t$), total work done by a variable force ($W = \int F(x)\,\mathrm{d}x$), and mass under non-uniform density are all unified under a single operation.
- *Signed net change*: By respecting the algebraic sign of $f(x)$, the integral captures net balances, cancellation between positive inflows and negative outflows, and physical displacements.
- *Rigorous measure of curved geometry*: Regions bounded by non-linear functions have unambiguous, mathematically well-defined areas as the limit of Riemann sums.

Yet Riemann sums are a magnificent *definition*, but an impractical *calculator*. To compute $\int_a^b f(x)\,\mathrm{d}x$ straight from the definition:

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k^*)\,\Delta x,$$

one needs closed-form formulas for discrete sums $\sum_{k=1}^{n} f(x_k^*)$. We know such formulas for polynomials of low degree ($\sum k, \sum k^2, \sum k^3$), but for virtually any other function—$\sin x$, e^x, $\sqrt{1-x^2}$, $\ln x$, $\frac{1}{1+x^2}$—there are no elementary summation identities. If evaluating an integral required evaluating Riemann sums from scratch, integral calculus would be an intellectual museum piece, impossible for daily scientific calculation.

The miraculous escape from Riemann sums happens in Chapter 21 (*The Fundamental Theorem of Calculus*). Isaac Newton and Gottfried Wilhelm Leibniz recognized that accumulation and differentiation are inverse operations:

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_a^x f(t)\,\mathrm{d}t = f(x).$$

Instead of constructing millions of rectangles and taking infinite limits of discrete sums, computing $\int_a^b f(x)\,\mathrm{d}x$ requires only finding an *antiderivative* F such that $F'(x) = f(x)$, evaluating $F(b) - F(a)$ in seconds. The drudgery of summation gives way to the elegance of undoing differentiation.

20.10 Exercises

Exercise 20.1 (Hand-computed Riemann sums)**.** For $f(x) = \sqrt{x}$ on $[0, 4]$ with $n = 4$ equal subintervals ($\Delta x = 1$), compute the left sum L_4, the right sum R_4, and the midpoint sum M_4. The exact value of $\int_0^4 \sqrt{x}\,\mathrm{d}x$ is $\frac{16}{3} \approx 5.333$. Which of your three sums is closest, and does $L_4 < \frac{16}{3} < R_4$ hold? Explain in one sentence why, using the fact that $\sqrt{x}$ is increasing.

Exercise 20.2 (An integral from the definition)**.** Following Example 20.1, compute $\int_0^2 x\,\mathrm{d}x$ directly from the definition using a regular partition and right endpoints. You will need $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$. Confirm that your answer equals the area of the triangle you can see in the graph.

Exercise 20.3 (Sum of cubes)**.** Using $\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$, evaluate $\int_0^1 x^3\,\mathrm{d}x$ from the definition with right endpoints. Show every step of the limit.

Exercise 20.4 (Using the properties)**.** Given $\int_1^4 f = 7$, $\int_1^4 g = -2$, and $\int_4^6 f = 5$, evaluate each of the following or state that it cannot be determined: (a) $\int_1^4 (3f - 2g)$; (b) $\int_1^6 f$; (c) $\int_6^4 f$; (d) $\int_1^4 f \cdot g$.

Exercise 20.5 (Signed versus geometric area)**.** Consider $f(x) = x^2 - 1$ on $[0, 2]$. (a) Sketch the graph and shade the regions above and below the x-axis. (b) The signed area is $\int_0^2 (x^2 - 1)\,\mathrm{d}x$; using linearity, the known integral of 1, and $\int_0^2 x^2\,\mathrm{d}x = \frac{8}{3}$ (and $\int_0^1 x^2\,\mathrm{d}x = \frac{1}{3}$), evaluate it. (c) Now find the *geometric* (total, unsigned) area by splitting at the crossing point $x = 1$ and adding absolute values. Explain why the two answers differ.

Exercise 20.6 (Average value)**.** Find the average value of $f(x) = x^3$ on $[0, 2]$ (you may use $\int_0^2 x^3\,\mathrm{d}x = 4$). Then find a point $c \in [0, 2]$ where $f(c)$ equals that average value, confirming the Mean Value Theorem for integrals in this case.

Exercise 20.7 (Accumulation reasoning)**.** Water flows into a tank at a rate $r(t) = 3 + 2t$ liters per minute, for t from 0 to 5 minutes. (a) Express the total volume that flows in as a definite integral. (b) Without yet knowing antiderivatives, estimate it with a midpoint sum using 5 subintervals of width 1. (c) The integrand is a straight line, so the exact total is the area of a trapezoid – compute it and compare to your estimate.

Exercise 20.8 (A conceptual trap)**.** A student claims: "Since $\int_0^{2\pi} \sin x\,\mathrm{d}x = 0$, the function $\sin x$ encloses no area over $[0, 2\pi]$." Explain precisely what is right and what is wrong with this statement, and write the correct integral expression for the total geometric area enclosed between $\sin x$ and the x-axis on $[0, 2\pi]$.

Exercise 20.9 (Proof: monotonicity)**.** Prove the monotonicity property: if f and g are integrable on $[a, b]$ and $f(x) \le g(x)$ for all $x \in [a, b]$, then $\int_a^b f \le \int_a^b g$. *Hint:* compare the two functions' Riemann sums on a common partition with common sample points, term by term, then pass to the limit. Note where you use $\Delta x_k > 0$.

Exercise 20.10 (Proof: a bound from monotonicity)**.** Suppose $m \le f(x) \le M$ for all $x \in [a, b]$. Prove that

$$m\,(b - a) \;\le\; \int_a^b f(x)\,\mathrm{d}x \;\le\; M\,(b - a).$$

Hint: apply the previous exercise twice, comparing f to the constant functions m and M and using $\int_a^b c\,\mathrm{d}x = c(b - a)$. Then interpret the result as a statement about the average value $\bar{f}$.

Exercise 20.11 (Order of magnitude, no computation)**.** Without evaluating any integral, rank these three numbers from smallest to largest and justify each comparison using monotonicity or a sketch: $\int_0^1 x^2\,\mathrm{d}x$, $\int_0^1 x^3\,\mathrm{d}x$, $\int_0^1 \sqrt{x}\,\mathrm{d}x$. *Hint:* on $(0, 1)$, how do x^3, x^2, and $\sqrt{x}$ compare in size?

Chapter 21

The Fundamental Theorem of Calculus

There are a handful of moments in mathematics where two ideas that grew up in separate rooms suddenly turn out to be the same idea wearing two coats. This chapter is one of those moments – arguably *the* moment of calculus.

On one side of the house we have the *derivative*: a machine for measuring instantaneous rates of change, slopes of tangent lines, how fast things are moving *right now*. On the other side we have the *integral*: a machine for adding up infinitely many infinitely thin slices, for measuring accumulated area, total distance, swept-out quantity. Slope and area. Rate and total. They look like utterly different questions. One is about a single instant; the other is about a whole interval. One is a tangent line; the other is a region of the plane.

The Fundamental Theorem of Calculus says: these two operations are inverses of each other. Differentiation and integration undo one another, the way addition undoes subtraction or squaring undoes square-rooting. The slope question and the area question are the same question read in opposite directions.

That single fact is what turns calculus from a collection of clever limit computations into a *technology*. Before the theorem, finding an area meant heroically summing infinitely many rectangles and praying the limit cooperated. After the theorem, finding an area means: guess a function whose derivative is the thing you care about, then subtract two of its values. A computation that took Archimedes a lifetime of geometric ingenuity becomes a one-line exercise a student can do in thirty seconds. That is not an exaggeration; it is the literal historical situation, and this chapter is about *why* it works.

21.1 What we are bridging

Let me set the two sides down precisely, because the whole drama depends on seeing how far apart they start.

21.1.1 The integral, briefly recalled

Given a function f on an interval $[a, b]$, the *definite integral*

$$\int_a^b f(x)\,\mathrm{d}x$$

is the limit of Riemann sums. You chop $[a, b]$ into n little pieces, build a thin rectangle on each piece whose height is a value of f there, add up the rectangle areas, and let the pieces shrink to nothing:

$$\int_a^b f(x)\,\mathrm{d}x \;=\; \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k^*)\,\Delta x_k.$$

When $f \geq 0$ this is the area under the graph. When f dips below the axis it is signed area – area above the axis counts positive, area below counts negative. The key point for us: this is a *global* object. To know it you must, in principle, account for the value of f at *every* point of $[a, b]$.

21.1.2 The derivative, briefly recalled

Given a function F, its *derivative* at a point x is the local rate of change,

$$F'(x) \;=\; \lim_{h\to 0} \frac{F(x+h) - F(x)}{h},$$

the slope of the tangent line to the graph of F at x. This is a *local* object: it cares only about the behavior of F in an infinitesimally small neighborhood of the one point x.

Intuition

Here is the puzzle in one breath. The integral needs to know about a whole interval at once. The derivative needs to know only about a single instant. How on earth could a whole-interval quantity (accumulated area) and a single-instant quantity (slope) be two faces of one coin?
The resolution is the central image of this chapter: **watch the area grow**. Don't compute the area once and for all. Instead, let the right-hand edge slide, and ask how fast the accumulated area is increasing at each moment. That *rate* of accumulation is a derivative – and it turns out to equal the height of the curve. Accumulating and rate-measuring are the same loop run forward and backward.

21.2 The area-so-far function

Everything hinges on a single new idea: instead of treating the area as a *number*, treat it as a *function*. Fix the left endpoint a and let the right endpoint vary. Define the *area-so-far* function (the standard name is the *accumulation function*):

$$A(x) \;=\; \int_a^x f(t)\,\mathrm{d}t.$$

Focus

Read $A(x)$ as "the signed area under f from the starting line a out to the moving wall at x." As you push the wall x to the right, you sweep out more area, so A grows. As you pull the wall back left, A shrinks. $A(x)$ is a running total.
Notice the bound of integration is now x, and the variable inside the integral had to be renamed t so it doesn't collide with the limit. The t is a dummy – it gets summed over and disappears. The surviving variable, the one A actually depends on, is the upper limit x.

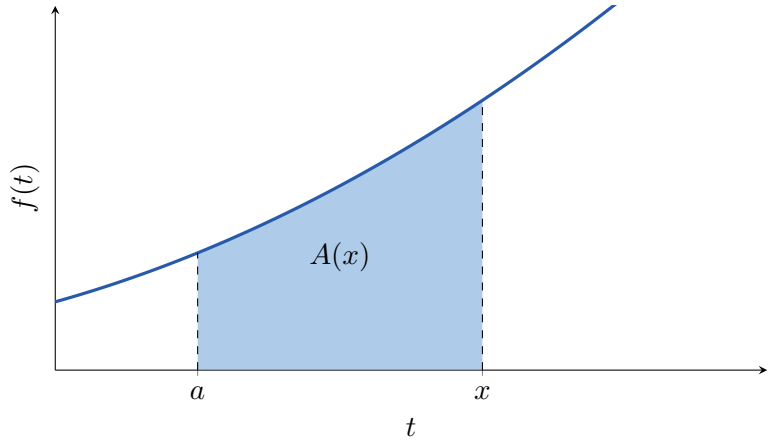


Figure 21.1: The accumulation function $A(x) = \int_a^x f(t)\,\mathrm{d}t$ is the shaded area from the fixed wall a to the sliding wall x. Sliding x to the right grows the area.

Work it yourself

Take the simplest non-trivial example: $f(t) = t$ on $[0, x]$. The region under f from 0 to x is a right triangle with both legs of length x, so its area is $\frac{1}{2}x^2$. Hence

$$A(x) = \int_0^x t\,\mathrm{d}t = \tfrac{1}{2}x^2.$$

Now do the thing this whole chapter is secretly about: differentiate the area.

$$A'(x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\tfrac{1}{2}x^2\right) = x = f(x).$$

The derivative of the area-so-far is the original height. That goes far beyond a quirk of triangles: it is the Fundamental Theorem, sneaking up on you in its smallest case. Hold onto this number; we will explain it in full next.

21.3 Part 1: differentiating the area gives back the curve

21.3.1 The infinitesimal-strip picture

Before any ε appears, let us see *why* $A'(x) = f(x)$ has to be true, using nothing but a picture and careful reasoning.

Push the right wall from x to $x + h$, where h is a tiny positive number. The area grows by a sliver: the strip sitting between x and $x + h$. Call that extra area $A(x + h) - A(x)$. What does the sliver look like? It is a thin vertical strip of width h, sitting under the curve f.

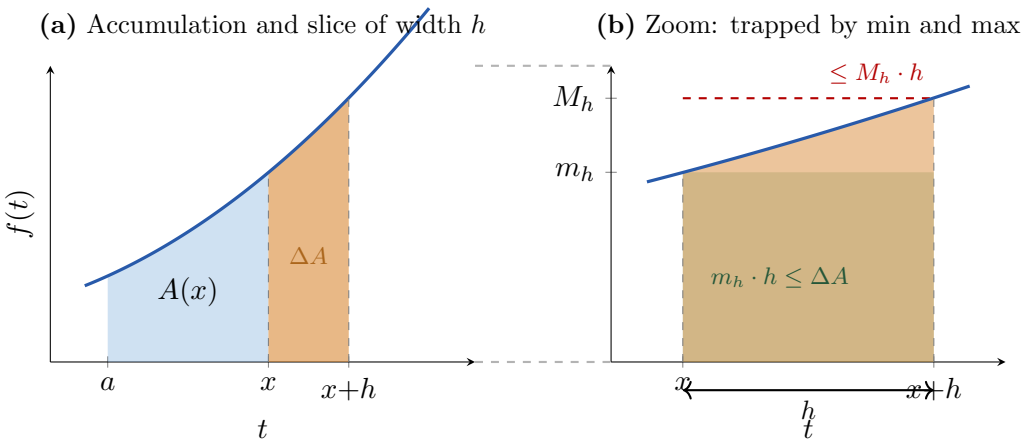


Figure 21.2: The rate of accumulation. **(a)** Growing the wall from x to $x + h$ adds the thin sliver $\Delta A = A(x+h) - A(x)$. **(b)** Zooming in on the sliver reveals it trapped between the lower rectangle of area $m_h \cdot h$ and the upper rectangle of area $M_h \cdot h$. As $h \to 0$, both bounding heights converge to $f(x)$, squeezing the average rate $\Delta A/h$ to $f(x)$.

Here is the key geometric observation: if h is small, the curve barely changes across the strip, so the strip is *almost a rectangle*: width h, height roughly $f(x)$. Its area is approximately

$$A(x+h) - A(x) \;\approx\; f(x) \cdot h.$$

Divide both sides by h:

$$\frac{A(x+h) - A(x)}{h} \;\approx\; f(x).$$

The left side is exactly the difference quotient whose limit defines $A'(x)$. As $h \to 0$ the strip gets thinner, the "almost a rectangle" becomes "a rectangle" in the limit, the approximation tightens to equality, and we land on

$$A'(x) = f(x).$$

What is really happening

What is really happening when you write $A'(x) = f(x)$? You are reading off the **instantaneous rate at which area is piling up.** The height of the curve at x *is* the speed at which the running-area total grows as the wall sweeps past x.
Think of f as a flow rate – liters per second pouring into a tank – and $A(x)$ as the total liters in the tank after time x. "How fast is the tank filling right now?" is answered by the current flow rate. That sentence is the Fundamental Theorem, Part 1, in plain language: *the rate of accumulation is the integrand itself.* The infinitesimal strip is one instant's worth of inflow; its height is the flow rate, its width $\mathrm{d}x$ is the instant, and the product $f(x)\,\mathrm{d}x$ is the dribble of area added in that instant.

21.3.2 Making it a real proof

The strip picture is convincing, but "≈" is a promise, not a proof. The real worry is this: across the strip the height is not exactly $f(x)$; the curve wobbles. We need to control that wobble and show

it vanishes in the limit. The clean tool for that is the average-value squeeze, and it works for any continuous f.

Theorem 21.1 (Fundamental Theorem of Calculus, Part 1)**.** *Let f be continuous on $[a,b]$. Define $A:[a,b]\to\mathbb{R}$ by*

$$A(x)=\int_a^x f(t)\,\mathrm{d}t.$$

Then A is differentiable on (a,b) (and one-sidedly at the endpoints), and

$$A'(x)=f(x)\qquad \text{for every } x.$$

In words: the accumulation function is an antiderivative of f.

Intuition

The idea of the proof is to trap the increment $A(x+h)-A(x)$ – the area of the strip – between the area of the shortest rectangle that fits under it and the area of the tallest rectangle that covers it. Both rectangles have width h; their heights are the minimum and maximum of f on the little interval. As $h\to 0$ that little interval collapses onto the point x, so both the minimum and the maximum of f are squeezed toward $f(x)$ (this is where continuity earns its keep). The strip's average height has nowhere to go but $f(x)$.

Proof. Fix x and take $h>0$ small enough that $x+h$ stays in $[a,b]$. Using the splitting rule for integrals (area from a to $x+h$ equals area from a to x plus area from x to $x+h$),

$$A(x+h)-A(x)=\int_a^{x+h} f-\int_a^x f=\int_x^{x+h} f(t)\,\mathrm{d}t.$$

So the increment of A is exactly the area of the strip – no approximation yet, this is an identity.

Because f is continuous on the closed interval $[x,x+h]$, the Extreme Value Theorem hands us a point where f attains its minimum value m_h and a point where it attains its maximum value M_h on that interval. Every value of f on $[x,x+h]$ lies between m_h and M_h, so the strip's area is trapped between the two rectangles of width h:

$$m_h\,h\;\le\;\int_x^{x+h} f(t)\,\mathrm{d}t\;\le\;M_h\,h.$$

Divide through by $h>0$:

$$m_h\;\le\;\frac{A(x+h)-A(x)}{h}\;\le\;M_h.$$

Now let $h\to 0^+$. The interval $[x,x+h]$ shrinks to the single point x. By continuity of f, both the minimum value m_h and the maximum value M_h converge to $f(x)$: any point achieving them is within h of x, and $f(\text{point})\to f(x)$. The middle term is squeezed between two quantities both heading to $f(x)$, so by the Squeeze Theorem

$$\lim_{h\to 0^+}\frac{A(x+h)-A(x)}{h}=f(x).$$

The case $h<0$ is identical with the roles of min and max swapped (and the inequalities flipping when you divide by a negative h), giving the same left-hand limit. The two one-sided limits agree, so $A'(x)$ exists and equals $f(x)$. □

Why the technique works

Why does the proof go through the *min* and *max* on the little interval rather than just using $f(x)$? Because we are not allowed to assume the strip is a perfect rectangle – that is the very thing we are trying to justify. The min and max rectangles are airtight bounds we can write down with no hand-waving: the true strip area genuinely sits between them. Continuity then guarantees the trap is tight, because over a shrinking interval the highest and lowest values of a continuous function cannot stay apart – they both must collapse to the value at the point. Continuity is exactly the hypothesis that makes "the curve barely changes across a thin strip" a theorem instead of a hope.

Remark 21.1. There is an equivalent and very clean way to package the squeeze, called the *Mean Value Theorem for Integrals*: for continuous f there is a point $c \in [x, x+h]$ with $\int_x^{x+h} f = f(c) \cdot h$. (The average height of the strip is actually achieved somewhere by f.) Then $\frac{A(x+h)-A(x)}{h} = f(c)$, and as $h \to 0$ the point c is trapped at x, so $f(c) \to f(x)$. Same proof, prettier costume.

Common trap

A frequent confusion: students see $\int_a^x f(t)\,\mathrm{d}t$ and try to "plug in" or differentiate with respect to t. There is no t to differentiate with respect to – t has been integrated away. The function A depends on its *upper limit* x. The statement $A'(x) = f(x)$ means: differentiate with respect to the upper limit, and you recover the integrand evaluated there.
Also resist writing $A(x) = \int_a^x f(x)\,\mathrm{d}x$ with x in both places. Using the same letter for the dummy variable and the limit is a recipe for nonsense. Always rename the inside variable.

21.3.3 The chain-rule version (variable in the limits)

Part 1 plus the chain rule lets you differentiate any accumulation whose limits are themselves functions. Suppose

$$G(x) = \int_a^{u(x)} f(t)\,\mathrm{d}t.$$

Write $G(x) = A(u(x))$ where A is the accumulation function from before, so $A' = f$. The chain rule gives

$$G'(x) = A'(u(x)) \cdot u'(x) = f\big(u(x)\big)\,u'(x).$$

If the variable sits in the lower limit instead, flip the sign (swapping limits negates the integral). The fully general rule, with both limits moving:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\int_{u(x)}^{v(x)} f(t)\,\mathrm{d}t\right) = f\big(v(x)\big)\,v'(x) - f\big(u(x)\big)\,u'(x).$$

Example 21.1. Differentiate $G(x) = \displaystyle\int_2^{x^3} \sin(t^2)\,\mathrm{d}t$.

Solution. Here $f(t) = \sin(t^2)$ and the upper limit is $u(x) = x^3$, so $u'(x) = 3x^2$. By the chain-rule version,

$$G'(x) = \sin\big((x^3)^2\big) \cdot 3x^2 = 3x^2 \sin(x^6).$$

Notice we never found a formula for G itself – there isn't an elementary one, since $\sin(t^2)$ has no elementary antiderivative. Yet we differentiated G exactly. This is what Part 1 buys you: the derivative of an integral, with no need to evaluate the integral first.

Example 21.2. Differentiate $H(x) = \displaystyle\int_x^{x^2} \frac{1}{1+\ln t}\,\mathrm{d}t$ for $x > 1$.

Solution. Both limits move: $u(x) = x$, $v(x) = x^2$. With $f(t) = \frac{1}{1+\ln t}$,

$$H'(x) = \frac{1}{1+\ln(x^2)} \cdot 2x \;-\; \frac{1}{1+\ln x} \cdot 1 = \frac{2x}{1+2\ln x} - \frac{1}{1+\ln x}.$$

21.4 Part 1 says: every continuous function has an antiderivative

Tucked inside Theorem 21.1 is a statement of pure existence that is easy to walk past but worth pausing on.

Corollary 21.1 (Existence of antiderivatives)**.** *Every function continuous on an interval has an antiderivative on that interval – namely its own accumulation function* $A(x) = \int_a^x f(t)\,\mathrm{d}t$.

Why this matters

This is more profound than it looks. "Find an antiderivative of f" usually means "find a familiar formula whose derivative is f," and for many ordinary functions – e^{-x^2}, $\frac{\sin x}{x}$, $\sqrt{1+x^3}$ – *no such formula exists.* The hunt is hopeless not because we are not clever enough but because the antiderivative is simply not expressible in elementary terms.

Part 1 reassures us that the antiderivative still *exists* as a perfectly good function; we just have to name it by its defining property – the area function – rather than by an algebraic formula. The bell-curve antiderivative, $\int_0^x e^{-t^2}\,\mathrm{d}t$, is so important to probability that it has its own name (up to a factor of $2/\sqrt{\pi}$), the error function erf. It is no less real for lacking a closed form. Part 1 is what licenses us to treat it as a genuine function with a genuine derivative (e^{-x^2}, naturally).

21.5 Part 2: evaluating an integral with any antiderivative

Part 1 tells us *a* particular antiderivative – the area function A – satisfies $A' = f$. Part 2 turns this into a computational miracle: to find the area, you may use *any* antiderivative you can get your hands on, and the answer pops out as a simple subtraction.

Theorem 21.2 (Fundamental Theorem of Calculus, Part 2)**.** *Let f be continuous on $[a,b]$ and let F be* any *antiderivative of f on $[a,b]$, meaning $F' = f$ throughout. Then*

$$\int_a^b f(x)\,\mathrm{d}x \;=\; F(b) - F(a).$$

The standard shorthand for the right side is $\Big[F(x)\Big]_a^b = F(b) - F(a)$, read "$F$ evaluated from a to b."

Intuition

Why should the area equal $F(b) - F(a)$ for *any* antiderivative, not just the area function? Because antiderivatives of the same function can differ only by a constant (this is the punchline of the Mean Value Theorem – two functions with the same derivative march in lockstep, parallel forever). The area function A and your chosen F differ by a fixed constant C. When you compute the difference $F(b)-F(a)$, the constant cancels: $(A(b)+C)-(A(a)+C) = A(b)-A(a)$. And $A(b) - A(a)$ is the total area from a to b. The $+C$ never had a chance to survive a subtraction.

Proof via Part 1. Let $A(x) = \int_a^x f(t)\,\mathrm{d}t$ be the area function. By Part 1, $A' = f$. By assumption $F' = f$ as well. Two functions with the same derivative on an interval differ by a constant (vanishing-derivative corollary of the MVT), so there is a constant C with

$$F(x) = A(x) + C \qquad \text{for all } x \in [a, b].$$

Now simply evaluate the difference:

$$F(b) - F(a) = \big(A(b) + C\big) - \big(A(a) + C\big) = A(b) - A(a).$$

But $A(a) = \int_a^a f = 0$ (an interval of zero width sweeps out no area) and $A(b) = \int_a^b f$. Therefore $F(b) - F(a) = \int_a^b f(x)\,\mathrm{d}x$. □

There is a second proof that does not lean on Part 1 at all – it grows the total out of tiny MVT-certified increments. It is worth seeing, because it makes the telescoping cancellation that powers Part 2 completely visible.

Proof via telescoping and the Mean Value Theorem. Partition $[a, b]$ into pieces $a = x_0 < x_1 < \cdots < x_n = b$. The total change in F across $[a, b]$ telescopes – each interior value is added once and subtracted once, so all that survives is the endpoints:

$$F(b) - F(a) = \sum_{k=1}^{n} \big(F(x_k) - F(x_{k-1})\big).$$

Apply the Mean Value Theorem to F on each little interval $[x_{k-1}, x_k]$: there is a sample point c_k inside with

$$F(x_k) - F(x_{k-1}) = F'(c_k)\,(x_k - x_{k-1}) = f(c_k)\,\Delta x_k.$$

(We used $F' = f$.) Substituting,

$$F(b) - F(a) = \sum_{k=1}^{n} f(c_k)\,\Delta x_k.$$

The right side is a genuine Riemann sum for f – with the MVT choosing the sample points for us. As the partition is refined and the mesh goes to 0, the left side does not move at all (it is just $F(b)-F(a)$, a fixed number), while the right side converges to $\int_a^b f$ by definition of the integral (which exists and is partition-independent because f is continuous). Therefore $F(b) - F(a) = \int_a^b f(x)\,\mathrm{d}x$. □

Why the technique works

The second proof exposes the true mechanism of Part 2: **a total change is a sum of little changes, and a little change in F is, by the MVT, exactly a height of f times a width.** Stacking those little $f(c_k)\,\Delta x_k$ bricks *is* building the area. Telescoping is what guarantees those bricks reassemble into the clean endpoint difference $F(b) - F(a)$ – every interior wall is shared by two bricks and cancels, leaving only the outer walls a and b (Figure 21.3). Integration is addition of infinitesimal increments; an antiderivative is a running total of them; so the net total is the final reading minus the initial reading. Subtraction of two antiderivative values is just the bookkeeping of "ending balance minus starting balance."

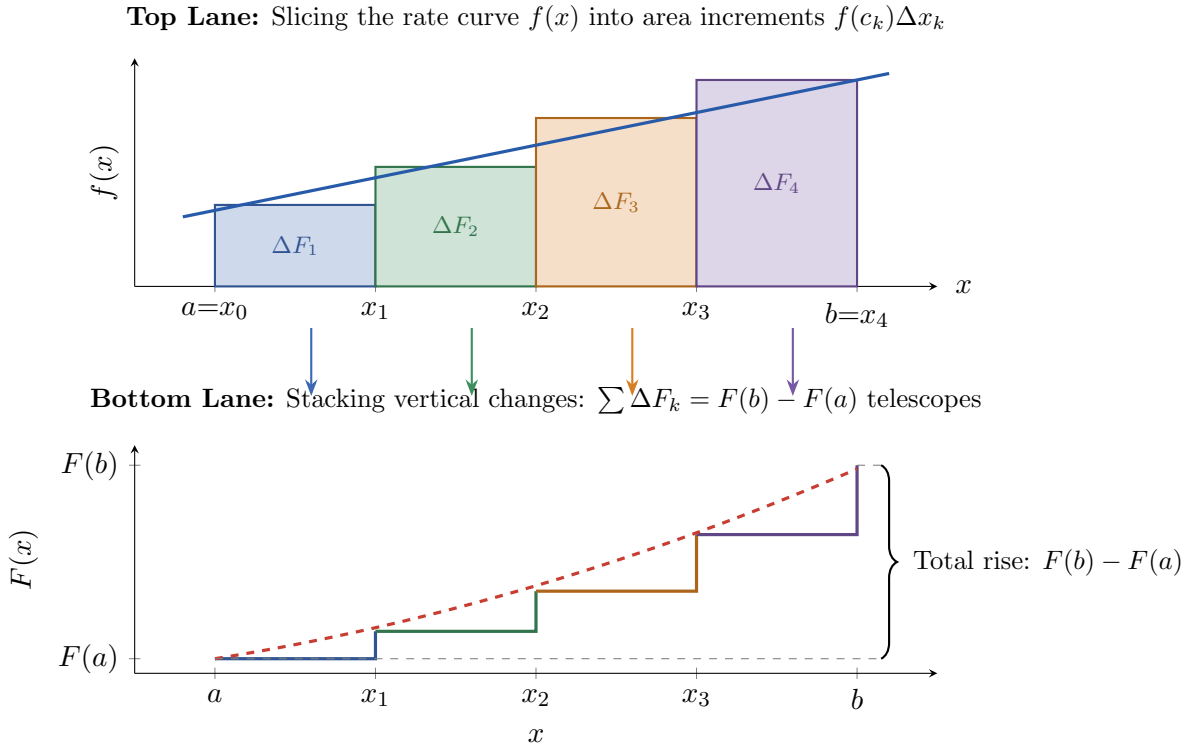


Figure 21.3: The telescoping mechanism of FTC Part 2. **Top lane:** Slices of area under the rate curve $f(x)$ have sizes $f(c_k)\Delta x_k$. **Bottom lane:** By the Mean Value Theorem, each area slice equals an exact vertical rise $\Delta F_k = F(x_k) - F(x_{k-1})$ on the antiderivative curve. Summing the slices stacks the rises, and internal heights cancel, leaving only the net rise $F(b) - F(a)$.

What is really happening

The cleanest way to feel Part 2 is the bank-account reading. Let $F(t)$ be your account balance at time t and $f(t) = F'(t)$ the instantaneous rate money flows in (deposits positive, withdrawals negative). Then $\int_a^b f(t)\,dt$ adds up all the little flows over the period – the net amount that moved – and that must equal the change in balance, $F(b) - F(a)$. You did not need to track every transaction; you only needed the balance at the start and the balance at the end. The Fundamental Theorem is the statement that *net accumulated rate equals net change in total*, and it is so intuitive in this form that it feels like it shouldn't need a proof.

The work of the chapter is showing that the intuitive banking fact and the geometric area fact are the very same theorem.

21.6 Why differentiation and integration are inverse

We can now say cleanly what "inverse operations" means here. There are two round trips, one for each order.

Focus

Integrate, then differentiate (Part 1): start with f, accumulate it into $A(x) = \int_a^x f$, then differentiate. You get f back.

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\int_a^x f(t)\,\mathrm{d}t\right) = f(x).$$

Differentiate, then integrate (Part 2): start with F, differentiate into $F' = f$, then accumulate. You recover the net change in F.

$$\int_a^b F'(x)\,\mathrm{d}x = F(b) - F(a).$$

Each operation undoes the other – with the single wrinkle that going $F \to F' \to \int F'$ recovers F only *up to its value at the starting point* (you get the change $F(b) - F(a)$, not F on the nose). That missing piece is the constant of integration, and it is the only thing the two operations cannot quite cancel.

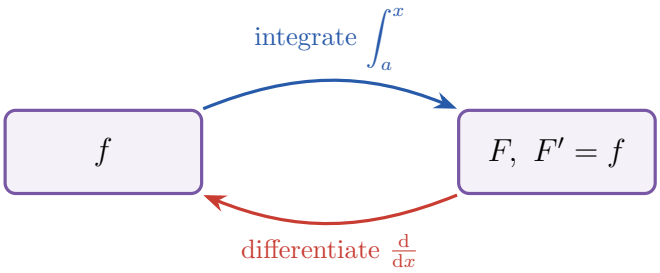


Figure 21.4: The two operations form a loop. Integrating f produces an antiderivative F; differentiating F returns f. The Fundamental Theorem says the loop closes.

Connection

This is the same pattern you meet whenever an operation has an inverse, but here it operates on whole *functions* rather than numbers. Differentiation D and integration are like a matrix and its inverse, or like a group element and its inverse: D followed by $\int$ is (almost) the identity, and $\int$ followed by D is the identity. The "almost" – the loss of the constant – is exactly why D has a one-dimensional kernel: the constant functions, whose derivative is zero, are the information differentiation throws away and integration cannot restore. You will meet this kernel-and-inverse bookkeeping again in linear algebra and again in the theory of differential

equations, where the $+C$ generalizes to the full family of solutions.

21.7 Antiderivatives and the $+C$

An *antiderivative* of f is any function F with $F' = f$. The *indefinite integral* collects them all:

$$\int f(x)\,\mathrm{d}x = F(x) + C,$$

where F is one antiderivative and C ranges over all real constants. The notation deliberately drops the limits to signal "a family of functions," as opposed to the definite integral $\int_a^b f$, which is "a single number."

Why the technique works

Why is the family *exactly* $F(x)+C$ – why no other antiderivatives lurking elsewhere? Suppose $G' = f$ too. Then $(G-F)' = f - f = 0$ on the interval. A function with zero derivative on an interval is constant (Mean Value Theorem again: if it ever changed, some secant slope would be nonzero, forcing a nonzero derivative somewhere). So $G - F = C$, i.e. $G = F + C$. The $+C$ is not sloppiness; it is the *complete* and *exact* description of the ambiguity. Geometrically: all antiderivatives are vertical shifts of one another – same shape, same slope at every x, just slid up or down. Shifting a graph vertically never changes any slope, so it never changes the derivative.

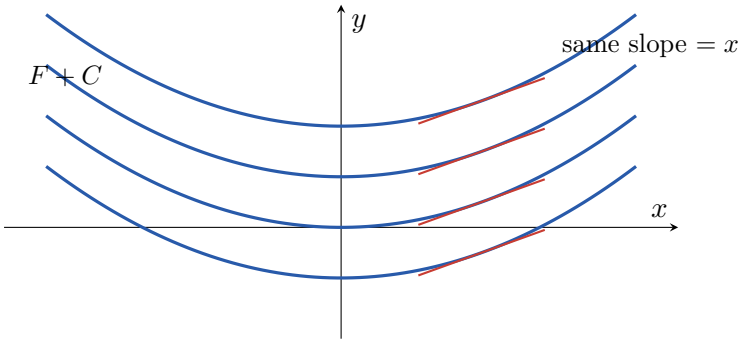


Figure 21.5: Antiderivatives of $f(x) = x$ are $\frac{1}{2}x^2 + C$, a stack of vertical shifts of one parabola. At any fixed x all of them have the identical slope x (short red tangents at $x = 1$), which is why they share the same derivative.

Common trap

Two $+C$ pitfalls.
Forgetting it on an indefinite integral. $\int 2x\,\mathrm{d}x = x^2$ is **incomplete**; the answer is $x^2 + C$. You have named one antiderivative when the question asked for all of them.
Writing it on a definite integral. $\int_0^3 2x\,\mathrm{d}x = 9$, a number – no C. If you carry the constant through, it cancels in $F(3) - F(0)$ anyway, so it is harmless but pointless. The $+C$ belongs to the indefinite world of function-families, not the definite world of numbers.

Walking question

On your walk, picture the family of all antiderivatives as a deck of identical curves stacked vertically. A definite integral asks you to read the *rise* between two x-values – and rises are unaffected by which card of the deck you read, because shifting the whole deck shifts both endpoints equally. Now ask yourself: what *one* extra piece of information would pin down a single card out of the deck? (Answer: the value of F at one point – an "initial condition." Hold that thought; it is the seed of every differential equation.)

21.8 Worked evaluations

Time to cash in. The recipe for a definite integral is now: (1) find any antiderivative F of the integrand; (2) compute $F(b) - F(a)$. Every example below is the same two moves.

Example 21.3. $\displaystyle\int_1^3 (x^2 - 2x + 5)\,\mathrm{d}x.$

Antiderivative. Reverse the power rule term by term: $F(x) = \frac{1}{3}x^3 - x^2 + 5x$. (Check: $F'(x) = x^2 - 2x + 5$. Good.)

Evaluate.

$$\Big[\tfrac{1}{3}x^3 - x^2 + 5x\Big]_1^3 = \Big(9 - 9 + 15\Big) - \Big(\tfrac{1}{3} - 1 + 5\Big) = 15 - \tfrac{13}{3} = \frac{32}{3}.$$

Example 21.4. $\displaystyle\int_0^{\pi} \sin x\,\mathrm{d}x.$

An antiderivative of $\sin x$ is $-\cos x$ (since $\frac{\mathrm{d}}{\mathrm{d}x}(-\cos x) = \sin x$). Thus

$$\Big[-\cos x\Big]_0^{\pi} = (-\cos\pi) - (-\cos 0) = -(-1) - (-1) = 2.$$

The area under one hump of the sine curve is exactly 2 – a clean number that no amount of staring at Riemann rectangles would have suggested.

Example 21.5. $\displaystyle\int_1^e \frac{1}{x}\,\mathrm{d}x.$

The antiderivative of $1/x$ (for $x > 0$) is $\ln x$. So

$$\Big[\ln x\Big]_1^e = \ln e - \ln 1 = 1 - 0 = 1.$$

The area under the hyperbola $1/x$ from 1 to e is exactly 1. This is, in fact, the defining property of e: it is the upper limit that makes this area equal to one.

Example 21.6. $\displaystyle\int_0^{\ln 2} e^x\,\mathrm{d}x.$

Since e^x is its own antiderivative,

$$\Big[e^x\Big]_0^{\ln 2} = e^{\ln 2} - e^0 = 2 - 1 = 1.$$

Example 21.7. $\displaystyle\int_0^4 \sqrt{x}\,\mathrm{d}x.$

Write $\sqrt{x} = x^{1/2}$. Reverse the power rule: raise the exponent by one to $\frac{3}{2}$ and divide, giving $F(x) = \frac{2}{3}x^{3/2}$. Then

$$\left[\tfrac{2}{3}x^{3/2}\right]_0^4 = \tfrac{2}{3}\cdot 4^{3/2} - 0 = \tfrac{2}{3}\cdot 8 = \frac{16}{3}.$$

Example 21.8 (Signed area, and why you sometimes split)**.** $\displaystyle\int_0^{2\pi} \cos x\,\mathrm{d}x$ versus the *geometric* area between $\cos x$ and the axis over the same interval.

The definite integral, using antiderivative $\sin x$:

$$\Big[\sin x\Big]_0^{2\pi} = \sin(2\pi) - \sin(0) = 0.$$

Zero – the positive area of the first hump exactly cancels the negative area of the trough (Figure 21.6). If instead you want the *total geometric area* (everything counted positive), you must split at the sign changes:

$$\int_0^{\pi/2} \cos x\,\mathrm{d}x - \int_{\pi/2}^{3\pi/2} \cos x\,\mathrm{d}x + \int_{3\pi/2}^{2\pi} \cos x\,\mathrm{d}x = 1 + 2 + 1 = 4.$$

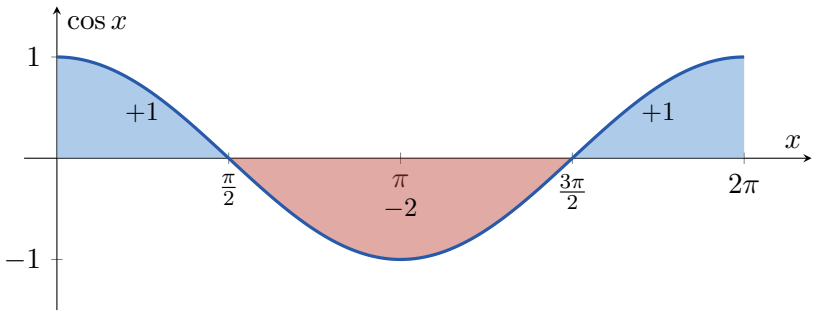


Figure 21.6: Signed area of $\cos x$ over $[0, 2\pi]$. The blue lobes above the axis contribute $+1$ each and the red lobe below contributes -2, so the definite integral is $1 - 2 + 1 = 0$. The total *geometric* area counts every lobe positively: $1 + 2 + 1 = 4$.

Common trap

The single most common Part-2 disaster is integrating across a point where the antiderivative *blows up*. Consider $\int_{-1}^{1} \frac{1}{x^2}\,\mathrm{d}x$. Blindly using $F(x) = -\frac{1}{x}$ gives $-1-(1) = -2$, a *negative* answer for the area under a strictly positive function – which contradicts the basic geometric fact that the area under an everywhere-positive curve cannot be negative. The catch: $1/x^2$ is not continuous on $[-1, 1]$; it explodes at 0. Part 2 requires f continuous on the whole interval (and F differentiable throughout). The integral here is actually improper and diverges. **Always confirm the integrand is continuous on $[a, b]$ before applying Part 2.**

Example 21.9 (Using Part 1 and Part 2 together)**.** Find the local maxima and minima of $A(x) = \displaystyle\int_0^x (t^2 - 4)\,\mathrm{d}t$ without first computing A.

Solution. By Part 1, $A'(x) = x^2 - 4$. Critical points where $A' = 0$ are $x = \pm 2$. Since $A'(x) = x^2 - 4$ is negative on $(-2, 2)$ and positive outside, A decreases then increases: $x = 2$ is a local minimum and $x = -2$ a local maximum. (If you do want a value, Part 2 gives $A(2) = \left[\frac{1}{3}t^3 - 4t\right]_0^2 = \frac{8}{3} - 8 = -\frac{16}{3}$.) The lesson: Part 1 turns questions about the *shape* of an accumulation function into ordinary derivative questions about the integrand.

21.9 A short history of the bridge

History

Pieces of both halves were known long before anyone saw they fit together. Archimedes (3rd century BCE) computed areas and volumes by his "method of exhaustion" – essentially Riemann sums done with breathtaking geometric cunning, one shape at a time, with no general machinery. Fermat, Cavalieri, and Wallis in the 1600s found area formulas for x^n. On the other side, tangent-line methods (the seeds of differentiation) were being developed by Fermat, Descartes, and especially Isaac Barrow, Newton's teacher and predecessor as Lucasian Professor of Mathematics at Cambridge.

Barrow, in his 1670 *Lectiones Geometricae* (specifically Lecture X, Proposition 11), came strikingly close to the prize. Using pure Euclidean geometry, Barrow proved that if a curve is constructed by accumulating the ordinates of another curve, its tangent at any point is determined directly by the height of the original curve. In modern terms, Barrow proved that differentiation inverts accumulation! Yet Barrow did not realize he was holding the master key to all of analysis. Working entirely with static geometric diagrams without algebraic equations or algorithmic notation, he treated the proposition as merely one clever geometric theorem among dozens, rather than a universal computational engine.

It took the algebraic and notational leap of *Isaac Newton* (who framed it through "fluxions," rates of flowing quantities, around 1666) and *Gottfried Wilhelm Leibniz* (who in 1675 independently invented the operational calculus) to recognize the Fundamental Theorem as the unifying bridge of mathematics. Armed with Cartesian coordinate geometry, power series, and algebra, Newton realized that physical motion unified rates (velocities) with accumulation (displacement). Leibniz, focusing on differential triangles and operational symbolism, engineered the supple $\int$ and $\mathrm{d}x$ symbols precisely so that the d of differentiation and the elongated-S $\int$ (for *summa*) would visibly act as inverse operators on the page: $\int \mathrm{d}F = F$ and $\mathrm{d}(\int f) = f\,\mathrm{d}x$. Together, they transformed a collection of isolated geometric triumphs into a systematic technology.

History

The bitter Newton–Leibniz priority dispute that followed – accusations of plagiarism, dueling national academies, decades of bad blood – has overshadowed the real story, which is that they discovered it independently and emphasized different things. Newton's instinct was physical: fluxions were velocities, and the theorem said position is the accumulation of velocity. Leibniz's instinct was formal: a calculus of symbols that manipulate themselves correctly if the notation is chosen well.

What both men got gloriously right was the *inverse relationship* and its power to compute. What both lacked was a rigorous account of the limiting process underneath. They reasoned with "infinitesimals" – quantities smaller than any positive number yet not zero – and Bishop Berkeley famously mocked these as "the ghosts of departed quantities." The objection was fair: nobody could say what an infinitesimal *was*. It took Cauchy (1820s) to recast everything in terms of limits, Riemann (1850s) to give the rigorous definition of the integral we used above, and Weierstrass to nail down the ε–δ bedrock. Only then did the squeeze proof of Part 1 – the proof we gave – stand on solid ground. The intuition was three centuries ahead of the rigor, and the rigor, when it finally came, confirmed the intuition exactly.

21.10 The two parts, side by side

Focus

A final consolidation. Memorize the *shapes*, not just the symbols.
Part 1 (the area-so-far has slope f):

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_a^x f(t)\,\mathrm{d}t = f(x).$$

Differentiating an accumulation returns the integrand. This is the existence half: it builds an antiderivative out of f, and it tells you the rate at which area accumulates.
Part 2 (evaluate by an antiderivative):

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a), \qquad F' = f.$$

A definite integral is a net change in any antiderivative. This is the computational half: it converts the impossible-looking infinite sum into a two-value subtraction.
They are two readings of one fact – that accumulating and rate-measuring are inverse processes – and together they are the hinge on which all of integral calculus turns. Every technique in the chapters ahead (substitution, parts, partial fractions) is, at bottom, a trick for the *first* step of Part 2: **finding an antiderivative**. Once you have F, the theorem does the rest.

Application

Physics: position from velocity. If $v(t)$ is velocity, then total displacement over $[t_0, t_1]$ is $\int_{t_0}^{t_1} v(t)\,\mathrm{d}t = s(t_1) - s(t_0)$, because position s is an antiderivative of v. The Fundamental Theorem is the statement that "displacement is the accumulation of velocity" – the area under the speedometer trace is the distance traveled. Likewise impulse is the integral of force, and work is the integral of force over distance. The theorem is why a single antiderivative replaces an infinite tally of moment-by-moment contributions.
Probability and ML. A probability density $p(x)$ has a cumulative distribution $F(x) = \int_{-\infty}^{x} p(t)\,\mathrm{d}t$; by Part 1, $F' = p$ – the density is the rate at which probability accumulates. In machine learning, the same identity underlies how a model's loss accumulates from per-example contributions, and why "integrate the rate of change back" recovers a net change.

21.11 What FTC unlocked, the Liouville barrier, and the road ahead

Connection

The Fundamental Theorem of Calculus represents the grand watershed of mathematical analysis, yet understanding its true scope requires being candid about both what it solved and the hard boundary where it halts.

Before the Fundamental Theorem, computing the area beneath a curve or the volume of a solid was an ordeal of bespoke ingenuity. A mathematician had to invent a clever summation formula for each individual curve, partition the domain into Riemann-like slices, and battle limits of algebraic series. FTC collapsed this entire landscape into a single operational pipeline: *to integrate, antidifferentiate.* Definite integrals across physics (work, impulse, center of mass), geometry (arc length, surface area), and probability (cumulative distributions) transformed overnight into simple evaluations at two boundary points, $F(b) - F(a)$.

Yet differentiation and antidifferentiation possess an asymmetric relationship: while differentiation is an algorithmic mechanical process that always succeeds on elementary formulas, antidifferentiation runs headlong into the celebrated *Liouville barrier.* FTC Part 2 requires finding an *elementary antiderivative* F such that $F' = f$. In 1835, Joseph Liouville proved that simple, smooth, elementary functions exist whose antiderivatives *cannot be expressed in terms of any finite combination of elementary functions* (powers, exponentials, logarithms, trigonometric functions, and roots). Canonical examples include:

$$\int e^{-x^2}\,\mathrm{d}x \quad \text{(the Gaussian bell curve)}, \qquad \int \frac{\sin x}{x}\,\mathrm{d}x \quad \text{(the sinc integral)}, \qquad \int \frac{e^x}{x}\,\mathrm{d}x, \qquad \int \sqrt{1 - k^2 \sin^2 x}$$

For all of these functions, FTC Part 1 guarantees that the area accumulation function $A(x) = \int_0^x f(t)\,\mathrm{d}t$ exists, is smooth, and satisfies $A'(x) = f(x)$. But FTC Part 2 is powerless to evaluate $\int_a^b f(x)\,\mathrm{d}x$ in closed elementary form, because no algebraic recipe for $A(x)$ can ever be written down.

This tension sets the roadmap for the entire remainder of our study of integration:

- *Expanding the symbolic toolkit (Chapters 22–24):* When an elementary antiderivative exists, how do we systematically find it? Chapter 22 develops *Integration by Substitution* (running the chain rule in reverse to untangle composite coordinates). Chapter 23 introduces *Integration by Parts* (inverting the product rule to disentangle products of dissimilar families). Chapter 24 builds *Partial Fractions and Trigonometric Integrals* to completely solve all rational functions.
- *When formulas fail: Numerical Quadrature (Chapter 27):* When an integrand runs into Liouville's barrier—or when we must integrate discrete empirical measurements from sensors where no formula exists at all—we return to Riemann's original insight armed with modern numerical methods: the Trapezoidal Rule and Simpson's parabolic quadrature, which achieve high precision without ever finding an antiderivative.

21.12 Exercises

Exercise 21.1. Evaluate each definite integral with Part 2.

(a) $\displaystyle\int_0^2 (3x^2 - 4x + 1)\,\mathrm{d}x$

(b) $\displaystyle\int_1^4 \frac{1}{\sqrt{x}}\,\mathrm{d}x$

(c) $\displaystyle\int_0^{\pi/4} \sec^2 x\,\mathrm{d}x$

(d) $\displaystyle\int_{-1}^1 e^{2x}\,\mathrm{d}x$

Exercise 21.2. Differentiate each accumulation function using Part 1 (and the chain rule where needed).

(a) $A(x) = \displaystyle\int_3^x \sqrt{1+t^4}\,\mathrm{d}t$

(b) $B(x) = \displaystyle\int_0^{x^2} \cos(t)\,\mathrm{d}t$

(c) $C(x) = \displaystyle\int_x^5 \frac{\mathrm{d}t}{1+t^2}$ (mind the sign)

(d) $D(x) = \displaystyle\int_{\sin x}^{\cos x} e^{t^2}\,\mathrm{d}t$

Exercise 21.3. Find the total *geometric* area enclosed between the curve $y = x^2 - 1$ and the x-axis on $[0, 2]$. (Hint: the curve crosses the axis inside the interval, so split and count each piece positively. Compare with the plain definite integral $\int_0^2 (x^2 - 1)\,\mathrm{d}x$ and explain the difference.)

Exercise 21.4. Let $G(x) = \int_0^x (t^2 - 3t + 2)\,\mathrm{d}t$. Using Part 1 only (do not first compute G), find all critical points of G and classify each as a local max or min. Then, using Part 2, find the value of G at each critical point.

Exercise 21.5 (Conceptual)**.** A student writes $\displaystyle\int_{-2}^2 \frac{1}{x^4}\,\mathrm{d}x = \Big[-\tfrac{1}{3x^3}\Big]_{-2}^2 = -\tfrac{1}{24} - \tfrac{1}{24} = -\tfrac{1}{12}$. Two things are wrong with this answer at once. Identify both: why is the numerical sign absurd, and which hypothesis of Part 2 fails? What is the correct status of this integral?

Exercise 21.6 (Conceptual)**.** Explain, in your own words and without symbols where possible, why the constant C in $\int f\,\mathrm{d}x = F + C$ cannot be detected by any definite integral $\int_a^b f$. Connect your explanation to the picture of antiderivatives as vertical shifts of one another.

Exercise 21.7. The error function is $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\,\mathrm{d}t$. Although erf has no elementary formula, compute $\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{erf}(x)$ and $\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{erf}(x^2)$ exactly. Explain why Part 1 lets you do this even though you cannot write down erf in closed form.

Exercise 21.8 (Mean Value Theorem for Integrals)**.** Suppose f is continuous on $[a, b]$. Prove there exists $c \in [a, b]$ with $\int_a^b f(x)\,\mathrm{d}x = f(c)\,(b - a)$. (Hint: apply Part 1 to get the area function A,

then apply the ordinary Mean Value Theorem to A on $[a, b]$. Interpret the result geometrically: a continuous function actually attains its average value somewhere.)

Exercise 21.9 (Proof)**.** Prove that if f is continuous on $[a, b]$ and $\int_a^x f(t)\,\mathrm{d}t = 0$ for *every* $x \in [a, b]$, then f is identically zero on $[a, b]$. (Hint: differentiate the hypothesis using Part 1.) What does this say about the only continuous function whose every accumulation vanishes?

Exercise 21.10 (Synthesis)**.** A particle moves with velocity $v(t) = t^2 - 4t + 3$ (meters per second) for $t \in [0, 4]$.

(a) Compute the net displacement $\int_0^4 v(t)\,\mathrm{d}t$.

(b) Compute the total distance traveled (integrate $|v|$, splitting at the times when $v = 0$).

(c) Explain in one sentence why the two answers differ, in terms of the particle reversing direction. Which computation used signed area, and which used geometric area?

Chapter 22

Integration by Substitution

By now you can differentiate almost anything. Hand you a function built from powers, exponentials, logarithms, and the trig family, glued together with sums, products, quotients, and compositions, and you will grind out its derivative without breaking a sweat. The chain rule, in particular, made compositions routine: the derivative of $f(g(x))$ is $f'(g(x))\,g'(x)$, and that little trailing factor $g'(x)$ – the derivative of the inside – is the whole story.

Integration is harder, and there is a real reason for it. Differentiation is a *procedure*: follow the rules and the answer falls out. Integration is a *search*. To integrate $h(x)$ is to find some F whose derivative is h, and there is no mechanical rule that always produces it. So we build a toolbox of techniques, each one a way of *recognizing* that the function in front of us is secretly the derivative of something we already know.

This chapter is about the first and most important tool in that box: **substitution**, also called u-substitution. Here is the one-sentence summary you should carry out of this chapter, because everything else is commentary on it:

Focus

Substitution is the chain rule read backwards. The chain rule tells you how the derivative of a composition picks up an extra factor from the inside; substitution teaches your eye to spot that extra factor in an integrand and to undo the composition.

We will build the idea from a picture, prove that it works (it is a one-line consequence of the chain rule and the Fundamental Theorem), learn how to choose the substitution, sort out the delicate business of what happens to $\mathrm{d}x$, and handle the definite-integral case where the limits themselves get carried along. Then we will work a long sequence of examples, from the obvious to the genuinely sneaky, and pin down what the symbol-shuffling really means.

22.1 The pattern that starts it all

We begin not with a rule but with a coincidence you can feel. Suppose I ask you to integrate

$$\int 2x\cos(x^2)\,\mathrm{d}x.$$

Stare at it. There is a composition in there, $\cos(x^2)$, whose inside is x^2. And sitting right next to it is $2x$ – which is the exact derivative of that inside. That matching factor is our signal.

Because you know the chain rule, you can run a quick experiment. What is the derivative of $\sin(x^2)$? The outside function is sin, the inside is x^2, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(\sin(x^2)\big) = \cos(x^2)\cdot \underbrace{2x}_{\text{from the inside}} = 2x\cos(x^2).$$

That is exactly our integrand. So

$$\int 2x\cos\big(x^2\big)\,\mathrm{d}x = \sin(x^2) + C,$$

and you can check it by differentiating the answer. The whole trick worked because the integrand had the shape

$$\big(\text{outside}'\text{ of the inside}\big)\times\big(\text{derivative of the inside}\big),$$

which is precisely the shape the chain rule manufactures. Substitution is the discipline of *seeing* this shape even when it is disguised, and of having a reliable bookkeeping system to exploit it.

Intuition

Think of the chain rule as a machine that, when it differentiates $F(g(x))$, spits out two pieces multiplied together: the "outer derivative" $F'(g(x))$ and the "inner derivative" $g'(x)$. If you are ever handed an integrand that looks like those two pieces multiplied together, you are looking at the output of that machine – and you can run the machine in reverse to recover $F(g(x))$.

Why the technique works

The mechanism, stated cleanly. The chain rule says

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(F(g(x))\big) = F'(g(x))\,g'(x).$$

Read that equation from right to left. It says: the antiderivative of $F'(g(x))\,g'(x)$ is $F(g(x))$. In integral notation,

$$\int F'(g(x))\,g'(x)\,\mathrm{d}x = F(g(x)) + C.$$

Substitution is nothing more than a clean way of *recognizing* an integrand as having the form $F'(g(x))\,g'(x)$ and then writing down $F(g(x))$. Every u-sub you will ever do is an instance of this single equation. The only skill is learning to see the pattern and to manage the bookkeeping when it is partly hidden.

22.1.1 Naming the inside

The mental move that turns this from a lucky guess into a method is to give the inside function a name. Call the inside u. In our example $u = x^2$. Then its derivative is $\frac{\mathrm{d}u}{\mathrm{d}x} = 2x$, which physicists and Leibniz alike happily rearrange into the suggestive form

$$\mathrm{d}u = 2x\,\mathrm{d}x.$$

Look what that does to the integral. Wherever we see x^2 we write u; wherever we see $2x\,\mathrm{d}x$ we write $\mathrm{d}u$:

$$\int \cos(\underbrace{x^2}_{u})\,\underbrace{2x\,\mathrm{d}x}_{\mathrm{d}u} = \int \cos u\,\mathrm{d}u = \sin u + C = \sin(x^2) + C.$$

The hard integral in x became a trivial integral in u. *That* is the payoff: a good substitution replaces a tangled integrand with a simple one, does the easy integral, and then translates back. We will spend the rest of the chapter making this precise and learning to do it when the pattern is buried.

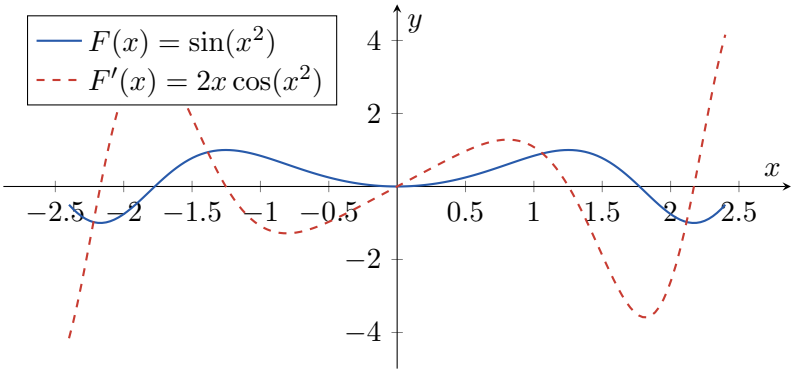


Figure 22.1: The antiderivative $F(x) = \sin(x^2)$ (solid) and its derivative $F'(x) = 2x\cos(x^2)$ (dashed). The dashed curve is the integrand we were handed; the solid curve is what substitution recovers. Notice the dashed curve reaches its peaks and troughs exactly where the solid curve is rising or falling fastest – that is the chain rule made visible.

22.2 The theorem, with proof

Intuition is wonderful, but we promised rigor. Here is substitution stated as a theorem and proved, both for indefinite and definite integrals. The proof is short because all the real work was done when we proved the chain rule and the Fundamental Theorem of Calculus; substitution just bolts them together.

Theorem 22.1 (Substitution, indefinite form)**.** *Let g be differentiable on an interval and let F be an antiderivative of a continuous function f on the range of g, so that $F' = f$. Then*

$$\int f\big(g(x)\big)\,g'(x)\,\mathrm{d}x = F\big(g(x)\big) + C.$$

Proof. We do not need to integrate anything. We only need to check that the proposed answer, $F(g(x))$, has the right derivative – because an antiderivative is, by definition, any function whose derivative is the integrand. Apply the chain rule to $F(g(x))$:

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(F\big(g(x)\big)\big) = F'\big(g(x)\big)\,g'(x) = f\big(g(x)\big)\,g'(x).$$

The first equality is the chain rule; the second uses $F' = f$. So $F(g(x))$ is indeed an antiderivative of $f(g(x))\,g'(x)$, and every antiderivative differs from it by a constant. That is exactly the claim. □

That is the whole content. The Leibniz notation $\mathrm{d}u = g'(x)\,\mathrm{d}x$ is a mnemonic device for this theorem – a way of doing the substitution on paper so that the pattern is impossible to miss. But the theorem stands on its own without any talk of "infinitesimals."

Theorem 22.2 (Substitution, definite form)**.** *With the same hypotheses, and assuming g' is continuous on $[a, b]$,*

$$\int_a^b f\big(g(x)\big)\, g'(x)\, \mathrm{d}x = \int_{g(a)}^{g(b)} f(u)\, \mathrm{d}u.$$

Proof. The Fundamental Theorem of Calculus evaluates the left-hand side using the antiderivative we just found:

$$\int_a^b f\big(g(x)\big)\, g'(x)\, \mathrm{d}x = \Big[F\big(g(x)\big)\Big]_{x=a}^{x=b} = F\big(g(b)\big) - F\big(g(a)\big).$$

Now look at the right-hand side. Since $F' = f$, the Fundamental Theorem also gives

$$\int_{g(a)}^{g(b)} f(u)\, \mathrm{d}u = \Big[F(u)\Big]_{u=g(a)}^{u=g(b)} = F\big(g(b)\big) - F\big(g(a)\big).$$

Both sides equal the same difference $F(g(b)) - F(g(a))$, so they are equal. □

Focus

The definite-form theorem contains an essential instruction hiding in the limits of integration: when you change the variable from x to $u = g(x)$, you must also change the limits. The new lower limit is $u = g(a)$ and the new upper limit is $u = g(b)$. The endpoints travel with the substitution. Forgetting this is the most common mistake in the whole chapter, and we will hammer it later.

History

The two notations in front of us belong to the two inventors of calculus, but the systematic theory of integration by substitution was forged by *Leonhard Euler* in his monumental 1768 treatise *Institutiones Calculi Integralis*. Newton had viewed calculus in terms of *fluxions* – flowing physical quantities – where changing variables was an intuitive shift of clock or ruler. Leibniz, with his unmatched notational foresight, gave us the d differentials and the integral sign $\int$.

Leibniz's notation was so cunningly engineered that $\mathrm{d}u = g'(x)\,\mathrm{d}x$ performs the coordinate change almost automatically: the differential elements scale exactly as ordinary fractions would suggest. It was Euler who elevated this formal trick into an art form, recognizing that substitution is fundamentally a geometric transformation: the factor $g'(x)$ is the one-dimensional *Jacobian stretch factor* relating the metric of the x-axis to that of the u-axis, ensuring that infinitesimal area elements remain invariant under deformation.

For over a century, mathematicians manipulated these d's as physical "infinitesimals" – quantities smaller than any positive number yet strictly nonzero. Bishop Berkeley famously lampooned them as "the ghosts of departed quantities" in 1734. The intuition was unerring, but the logical grounding was nonexistent. It was Augustin-Louis Cauchy (1823) and Karl Weierstrass who finally anchored substitution in the rigorous theory of limits, proving that the change-of-variable formula is nothing other than the Chain Rule integrated across a continuous interval.

22.3 What $\mathrm{d}x$ is really doing

You have been told to "cancel the $\mathrm{d}x$," and it works, and that should bother you a little. What is actually going on when we write $\mathrm{d}u = 2x\,\mathrm{d}x$ and then treat the d's like algebra?

What is really happening

The $\mathrm{d}x$ inside an integral is not a number being multiplied – it is a *bookkeeping tag* that records which variable you are integrating against, and, more deeply, it records how a tiny step in your variable gets stretched into a tiny step in area.
When you substitute $u = g(x)$, a small change Δx in x produces a small change $\Delta u \approx g'(x)\,\Delta x$ in u. The factor $g'(x)$ is the local *stretch factor*: it says how much the u-axis is being compressed or expanded relative to the x-axis near that point. The integral $\int f(g(x))\,g'(x)\,\mathrm{d}x$ is adding up little rectangles of height $f(g(x))$ and width $\mathrm{d}x$. To rewrite this as an integral in u, each width $\mathrm{d}x$ must be re-measured in u-units, and the conversion factor is exactly $\mathrm{d}u = g'(x)\,\mathrm{d}x$. So the "cancellation" is not a trick: $g'(x)\,\mathrm{d}x$ genuinely *is* the width of the corresponding sliver measured along the u-axis. The equation $\mathrm{d}u = g'(x)\,\mathrm{d}x$ is a statement about how widths transform, and Theorem 22.2 is the rigorous version of that statement.

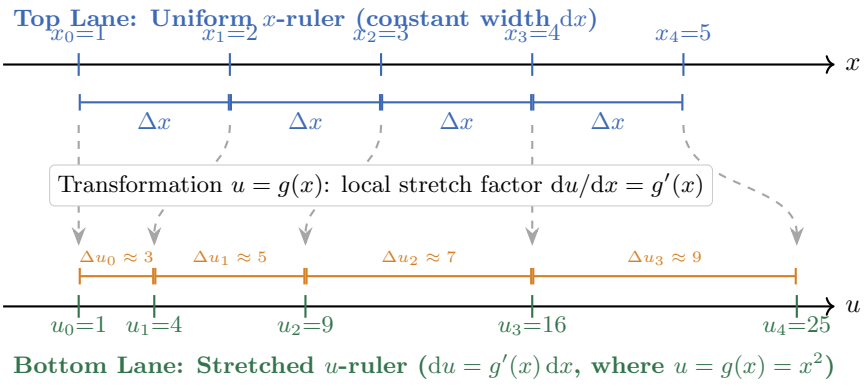


Figure 22.2: Substitution as ruler distortion (the 1D Jacobian). **Top lane:** Slices of equal width Δx along the x-axis. **Bottom lane:** Under the mapping $u = g(x)$ (here $u = x^2$), equal steps in x are non-uniformly stretched along the u-axis into increments $\Delta u \approx g'(x)\Delta x$. Slices stretch out where $g'(x)$ is large and compress where $g'(x)$ is small. The factor $g'(x)$ converts between rulers so total area is preserved.

This is why a substitution can fail or mislead if g' vanishes or changes sign on the interval: in those places the stretch factor degenerates or flips orientation, and the naive picture of "re-measuring widths" needs care. We will see a clean example of the sign issue later. For now, hold onto the geometric meaning: $\mathrm{d}u = g'(x)\,\mathrm{d}x$ is the change-of-width law, and substitution is integration performed in a rescaled coordinate.

Connection

This change-of-width factor is your first encounter with something that will return in force later in the book. When you integrate in two or three variables and change coordinates – say to polar coordinates – a similar stretch factor appears, but now it measures how *areas* (not

lengths) get rescaled, and it is called the *Jacobian*. The single $g'(x)$ here is the one-dimensional Jacobian. Master the intuition now and the multivariable change-of-variables formula will feel like an old friend wearing a bigger coat.

22.4 How to choose u

The theorem is easy; the art is picking u. What follows is a practical hierarchy of things to try. None is a guarantee – integration is a search – but in that order they resolve the large majority of substitution problems.

1. **Let u be the inside of a composition.** If the integrand contains something like $\sin(\text{stuff})$, e^{stuff}, $(\text{stuff})^n$, or $\sqrt{\text{stuff}}$, try $u = \text{stuff}$. Then check whether $du = (\text{stuff})'\,dx$ accounts for the rest of the integrand.
2. **Let u be the thing whose derivative is also present.** Scan for a chunk of the integrand that is (up to a constant) the derivative of some other chunk. Name the other chunk u. This is the same rule as above, phrased so your eye hunts for the $g'(x)$ factor.
3. **Let u be an inconvenient denominator or exponent.** A denominator $x^2 + 1$ or an exponent in e^{3x} is often the cleanest choice, especially when its derivative is lurking in the numerator.
4. **If a constant is missing, supply it.** You rarely get du exactly; usually you get a constant multiple of it. Constants are free to move across the integral sign, so you patch the discrepancy with a constant factor. (Only constants – never a variable.)

Common trap

You may multiply and divide by *constants* to make du appear, because $\int c\,h\,dx = c\int h\,dx$. You may *not* pull a variable across the integral sign to manufacture du. Writing $\int x^2 e^x\,dx = \frac{1}{?}\int \ldots$ by "factoring out an x" is illegal: $\int x\,h(x)\,dx \neq x\int h(x)\,dx$. If the leftover factor is not a constant, your chosen u is wrong, or the integral needs a different technique (such as integration by parts, coming in the next chapter).

Walking question

Before you compute anything, train the eye. Take a walk and, for each of these, just decide *what you would call* u and whether its derivative is present: $\int x\,e^{x^2}dx$; $\int \frac{\ln x}{x}\,dx$; $\int \tan x\,dx$; $\int \frac{x}{\sqrt{1-x^2}}\,dx$; $\int \cos^3 x\,dx$. Three of them are clean one-step subs, one needs a clever rewrite of $\tan x$, and one needs a trig identity before the sub appears. Sorting them in your head is the real skill.

22.5 Worked examples, from plain to subtle

What follows is a deliberately ordered sequence. Watch how the same single idea – recognize $F'(g)\,g'$ – keeps reappearing in heavier and heavier disguise.

22.5.1 The clean cases

Example 22.1 (Inside spotted directly)**.** Compute $\displaystyle\int 3x^2\,(x^3+1)^5\,\mathrm{d}x$.

The composition is $(x^3+1)^5$; its inside is x^3+1. Let $u = x^3+1$, so $\mathrm{d}u = 3x^2\,\mathrm{d}x$. The integrand contains exactly $3x^2\,\mathrm{d}x$, which is $\mathrm{d}u$. Substituting,

$$\int (x^3+1)^5 \underbrace{3x^2\,\mathrm{d}x}_{\mathrm{d}u} = \int u^5\,\mathrm{d}u = \frac{u^6}{6} + C = \frac{(x^3+1)^6}{6} + C.$$

Differentiate the answer to check: $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{(x^3+1)^6}{6}\right) = (x^3+1)^5 \cdot 3x^2$. It matches.

Example 22.2 (Patching a missing constant)**.** Compute $\displaystyle\int x\,e^{x^2}\,\mathrm{d}x$.

Let $u = x^2$, so $\mathrm{d}u = 2x\,\mathrm{d}x$. But the integrand has only $x\,\mathrm{d}x$, not $2x\,\mathrm{d}x$. No problem: $x\,\mathrm{d}x = \frac{1}{2}\,\mathrm{d}u$. Equivalently, pull the missing $\frac{1}{2}$ outside:

$$\int x\,e^{x^2}\,\mathrm{d}x = \int e^u \cdot \tfrac{1}{2}\,\mathrm{d}u = \tfrac{1}{2}\int e^u\,\mathrm{d}u = \tfrac{1}{2}e^u + C = \tfrac{1}{2}e^{x^2} + C.$$

The half is not a fudge factor. It is the exact bookkeeping the change-of-width law demands: $\mathrm{d}x = \frac{1}{2x}\,\mathrm{d}u$, and the stray x in the integrand cancels the x in $\frac{1}{2x}$, leaving the clean $\frac{1}{2}$.

Example 22.3 (The logarithm in disguise)**.** Compute $\displaystyle\int \frac{2x}{x^2+1}\,\mathrm{d}x$.

The denominator x^2+1 has derivative $2x$, which is sitting in the numerator. Let $u = x^2+1$, $\mathrm{d}u = 2x\,\mathrm{d}x$:

$$\int \frac{2x\,\mathrm{d}x}{x^2+1} = \int \frac{\mathrm{d}u}{u} = \ln|u| + C = \ln(x^2+1) + C.$$

(We may drop the absolute value because $x^2+1 > 0$ always.) This is a special case of the enormously useful pattern $\int \frac{g'(x)}{g(x)}\,\mathrm{d}x = \ln|g(x)| + C$: *a fraction whose top is the derivative of its bottom integrates to the log of the bottom.* Burn that pattern into memory; it is everywhere.

Example 22.4 (Two layers, one sub)**.** Compute $\displaystyle\int \frac{\ln x}{x}\,\mathrm{d}x$.

Here the "inside" is $\ln x$, whose derivative is $\frac{1}{x}$ – and there is a $\frac{1}{x}$ in the integrand. Let $u = \ln x$, $\mathrm{d}u = \frac{1}{x}\,\mathrm{d}x$:

$$\int \ln x \cdot \frac{\mathrm{d}x}{x} = \int u\,\mathrm{d}u = \frac{u^2}{2} + C = \frac{(\ln x)^2}{2} + C.$$

22.5.2 When the substitution must be rearranged

Sometimes after choosing u you still have leftover x's that are not part of $\mathrm{d}u$. The fix is to solve $u = g(x)$ for x and substitute those leftover x's too. This is where substitution stops being mere pattern-matching and becomes a genuine change of variable.

Example 22.5 (Solving back for the stray x)**.** Compute $\displaystyle\int x\sqrt{x+1}\,\mathrm{d}x$.

The natural choice is $u = x + 1$, so $\mathrm{d}u = \mathrm{d}x$ and $\sqrt{x+1} = \sqrt{u}$. But there is a stray x out front that is not accounted for. Since $u = x + 1$, we have $x = u - 1$. Replace it:

$$\int x\sqrt{x+1}\,\mathrm{d}x = \int (u-1)\sqrt{u}\,\mathrm{d}u = \int \left(u^{3/2} - u^{1/2}\right)\mathrm{d}u.$$

Now it is elementary:

$$= \frac{2}{5}u^{5/2} - \frac{2}{3}u^{3/2} + C = \frac{2}{5}(x+1)^{5/2} - \frac{2}{3}(x+1)^{3/2} + C.$$

The lesson: a substitution does not have to consume every x through $\mathrm{d}u$. As long as you can express *all* remaining x's in terms of u, the change of variable is legitimate and complete.

Example 22.6 (A trig identity must come first)**.** Compute $\displaystyle\int \cos^3 x\,\mathrm{d}x$.

There is no obvious inside-and-its-derivative here. The move is to spend one factor of $\cos x$ as the future $\mathrm{d}u$ and convert the rest using $\cos^2 x = 1 - \sin^2 x$:

$$\int \cos^3 x\,\mathrm{d}x = \int \cos^2 x \cdot \cos x\,\mathrm{d}x = \int (1 - \sin^2 x)\,\cos x\,\mathrm{d}x.$$

Now let $u = \sin x$, $\mathrm{d}u = \cos x\,\mathrm{d}x$:

$$= \int (1 - u^2)\,\mathrm{d}u = u - \frac{u^3}{3} + C = \sin x - \frac{\sin^3 x}{3} + C.$$

The identity created the pattern; substitution finished the job. This trick – peel off one factor to be $\mathrm{d}u$, convert the even remainder with a Pythagorean identity – is the backbone of trigonometric integrals, which get a chapter of their own.

Example 22.7 ($\tan x$, the classic rewrite)**.** Compute $\displaystyle\int \tan x\,\mathrm{d}x$.

Rewrite $\tan x = \dfrac{\sin x}{\cos x}$. Now the bottom is $\cos x$ and the top $\sin x$ is, up to a sign, its derivative. Let $u = \cos x$, $\mathrm{d}u = -\sin x\,\mathrm{d}x$, so $\sin x\,\mathrm{d}x = -\mathrm{d}u$:

$$\int \frac{\sin x}{\cos x}\,\mathrm{d}x = \int \frac{-\mathrm{d}u}{u} = -\ln|u| + C = -\ln|\cos x| + C = \ln|\sec x| + C.$$

This is again the $\int \frac{g'}{g}$ pattern wearing a trig costume.

22.5.3 The genuinely sneaky ones

Example 22.8 (When the obvious sub is not the inside)**.** Compute $\displaystyle\int \frac{1}{x\ln x}\,\mathrm{d}x$.

You might be tempted by $u = x$ or by the whole denominator. Resist, and notice the structure: $\frac{1}{x}$ is the derivative of $\ln x$. So treat $\ln x$ as the inside. Let $u = \ln x$, $\mathrm{d}u = \frac{1}{x}\,\mathrm{d}x$:

$$\int \frac{1}{\ln x}\cdot\frac{\mathrm{d}x}{x} = \int \frac{\mathrm{d}u}{u} = \ln|u| + C = \ln|\ln x| + C.$$

A logarithm of a logarithm – correct, and a small delight.

Example 22.9 (Completing the square, then substituting)**.** Compute $\displaystyle\int \frac{1}{x^2+4x+5}\,\mathrm{d}x$.

The denominator does not factor over the reals, but we can complete the square: $x^2+4x+5 = (x+2)^2+1$. Now let $u = x+2$, $\mathrm{d}u = \mathrm{d}x$:

$$\int \frac{\mathrm{d}x}{(x+2)^2+1} = \int \frac{\mathrm{d}u}{u^2+1} = \arctan u + C = \arctan(x+2) + C.$$

Here substitution did something subtler than peel off a chain rule: it *recentered* the variable so a standard form ($\int \frac{\mathrm{d}u}{u^2+1} = \arctan u$) became visible. Shifting and scaling are the humblest and most common substitutions of all.

Example 22.10 (A rationalizing substitution)**.** Compute $\displaystyle\int \frac{1}{1+\sqrt{x}}\,\mathrm{d}x$.

The square root is the obstruction. Kill it by naming it: let $u = \sqrt{x}$, so $x = u^2$ and $\mathrm{d}x = 2u\,\mathrm{d}u$. The integral becomes

$$\int \frac{2u\,\mathrm{d}u}{1+u}.$$

Now do polynomial division: $\dfrac{2u}{1+u} = 2 - \dfrac{2}{1+u}$. Hence

$$\int \left(2 - \frac{2}{1+u}\right)\mathrm{d}u = 2u - 2\ln|1+u| + C = 2\sqrt{x} - 2\ln\bigl(1+\sqrt{x}\bigr) + C.$$

Notice this substitution ran "the other way": instead of setting u equal to a chunk of the integrand and computing $\mathrm{d}u$, we set x equal to a function of u and computed $\mathrm{d}x$. Both directions are legal; Theorem 22.1 covers them as long as the change of variable is invertible on the interval in question. (Here $u = \sqrt{x} \geq 0$, which is fine for $x \geq 0$.)

Why this matters

Look back at how many different-looking integrals collapsed onto the same few shapes: $\int u^n\,\mathrm{d}u$, $\int e^u\,\mathrm{d}u$, $\int \frac{\mathrm{d}u}{u}$, and $\int \frac{\mathrm{d}u}{u^2+1}$. Substitution is, in a sense, a *universal translator*: it keeps trying to phrase the integral in one of the handful of languages you already speak fluently. The art of integration is largely the art of recognizing which translation will land you in familiar territory.

22.6 Definite integrals: carry the limits

For a definite integral you have two equally valid styles, and you should be fluent in both.

Style A – change the limits. Use Theorem 22.2 directly: when $u = g(x)$, replace the limits $x = a, x = b$ by $u = g(a), u = g(b)$ and never return to x at all.

Style B – back-substitute. Do the indefinite integral, return to x, then plug in the original limits.

Example 22.11 (The same integral both ways)**.** Compute $\displaystyle\int_0^2 x\,e^{x^2}\,\mathrm{d}x$.

Let $u = x^2$, $\mathrm{d}u = 2x\,\mathrm{d}x$, so $x\,\mathrm{d}x = \frac{1}{2}\,\mathrm{d}u$.

Style A (change the limits). When $x = 0$, $u = 0$; when $x = 2$, $u = 4$. So

$$\int_0^2 x\,e^{x^2}\,\mathrm{d}x = \frac{1}{2}\int_0^4 e^u\,\mathrm{d}u = \frac{1}{2}\Big[e^u\Big]_0^4 = \frac{e^4-1}{2}.$$

Style B (back-substitute). The antiderivative is $\frac{1}{2}e^{x^2}$, so

$$\int_0^2 x\,e^{x^2}\,\mathrm{d}x = \frac{1}{2}\Big[e^{x^2}\Big]_0^2 = \frac{1}{2}\big(e^4 - e^0\big) = \frac{e^4-1}{2}.$$

Same answer, as it must be. Style A is usually faster and cleaner once you trust it, because you never untangle u back into x.

Common trap

The signature blunder: changing the variable to u but leaving the old x-limits on the integral sign. If you write

$$\int_0^2 x\,e^{x^2}\,\mathrm{d}x \;\stackrel{?}{=}\; \frac{1}{2}\int_0^2 e^u\,\mathrm{d}u,$$

you have committed it – the "0" and "2" there are x-values, but the integral is now in u. Either change them to the u-values (0 and 4), or drop the limits entirely until you have returned to x. Mixing u-integrand with x-limits gives a wrong number every time. A safe habit: the instant you write a u on the integrand, write the u-limits on the sign, in the same stroke of the pen.

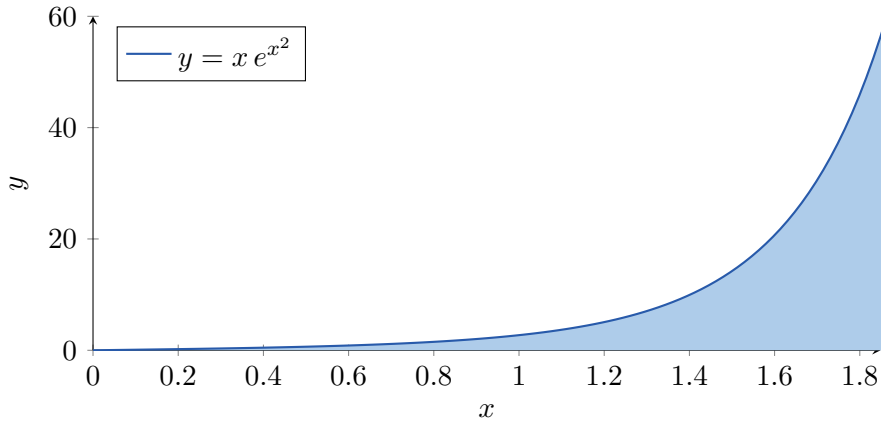


Figure 22.3: The definite integral $\int_0^2 xe^{x^2}\,\mathrm{d}x = \frac{e^4-1}{2} \approx 26.8$ is the shaded area. Substitution does not change this area; it only re-expresses it as $\frac{1}{2}\int_0^4 e^u\,\mathrm{d}u$, the same area measured along a stretched horizontal axis. The endpoints $x = 0, 2$ become $u = 0, 4$ – that is why the limits must travel.

Work it yourself

Verify the limit-changing rule with your own hands on a case where you know the answer cold. Take $\int_1^3 (2x)(x^2)\,\mathrm{d}x$, which is just $\int_1^3 2x^3\,\mathrm{d}x = \frac{1}{2}x^4\big|_1^3 = \frac{1}{2}(81-1) = 40$. Now redo it by substituting $u = x^2$, $\mathrm{d}u = 2x\,\mathrm{d}x$, with limits $u = 1$ to $u = 9$: you should get $\int_1^9 u\,\mathrm{d}u = \frac{1}{2}u^2\big|_1^9 = \frac{1}{2}(81-1) = 40$. Watch the limits change from $\{1, 3\}$ to $\{1, 9\}$ and watch the answer stay put. *That* is the theorem breathing.

22.6.1 A symmetry payoff

Changing limits is not just bookkeeping; sometimes it reveals structure you could not otherwise see.

Example 22.12 (Substitution exposes a symmetry)**.** Show that $\displaystyle\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x}\,\mathrm{d}x = \frac{\pi}{4}$.

There is no friendly antiderivative here. Instead, substitute $u = \frac{\pi}{2} - x$, the reflection that swaps the endpoints. Then $\mathrm{d}u = -\mathrm{d}x$; the limits flip ($x = 0 \to u = \frac{\pi}{2}$ and $x = \frac{\pi}{2} \to u = 0$); and $\sin x = \cos u$, $\cos x = \sin u$. Call the original integral I:

$$I = \int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x}\,\mathrm{d}x = \int_0^{\pi/2} \frac{\cos u}{\cos u + \sin u}\,\mathrm{d}u.$$

(The minus sign from $\mathrm{d}u$ was eaten by flipping the limits back to their natural order.) The last integral is the same as the original but with sin and cos traded. Add the two expressions for I:

$$2I = \int_0^{\pi/2} \frac{\sin x + \cos x}{\sin x + \cos x}\,\mathrm{d}x = \int_0^{\pi/2} 1\,\mathrm{d}x = \frac{\pi}{2},$$

so $I = \frac{\pi}{4}$. A substitution did what no antiderivative could: it turned an unsolvable-looking integral into a triviality by exploiting symmetry. Figure 22.4 shows why: the integrand and its reflection add to the constant 1, so their two equal areas together fill a rectangle of area $\frac{\pi}{2}$.

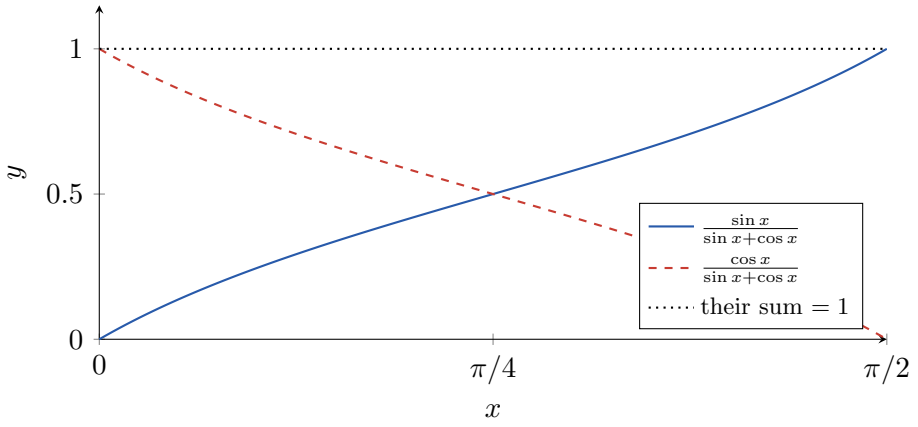


Figure 22.4: The original integrand (solid) and its reflection $x \mapsto \frac{\pi}{2} - x$ (dashed) are mirror images across $x = \pi/4$, and at every point they sum to 1 (dotted). The two shaded areas are therefore equal and together fill a $\frac{\pi}{2} \times 1$ strip, so each is $\frac{\pi}{4}$. That is the whole content of the symmetry argument, drawn.

Connection

This reflection trick generalizes to the handy fact that $\int_a^b f(x)\,\mathrm{d}x = \int_a^b f(a+b-x)\,\mathrm{d}x$ for any integrable f – a direct consequence of the substitution $u = a + b - x$. It is a workhorse in competition mathematics and in probability, where it underlies arguments about symmetric distributions.

22.7 What the symbol-shuffling really means

Time to face head-on the thing that should still nag a careful reader. When we write $\mathrm{d}u = g'(x)\,\mathrm{d}x$ and then "cancel" $\mathrm{d}x$'s, are we doing real mathematics or performing a ritual that happens to work?

What is really happening

It is real mathematics, but the symbols are doing something subtler than the ritual suggests. Strictly, $\mathrm{d}u = g'(x)\,\mathrm{d}x$ is *defined* to be shorthand for the statement of Theorem 22.2. The $\mathrm{d}x$ standing alone is not a number, so "cancelling" it is not division. What justifies the manipulation is that the shorthand was *designed* to mirror a true theorem: every legal token-move you make with the d's corresponds to a legal step in the proof.
Two things keep you safe. First, the substitution $u = g(x)$ should be *differentiable*, so that $g'(x)$ exists and the change-of-width law makes sense. Second, for definite integrals you want g' to keep a constant sign (or you must track orientation carefully), and for the reverse-direction substitutions $x = h(u)$ of Example 22.10 you want h to be *invertible* on the interval, so that travelling to u-land and back is unambiguous. When those mild conditions hold, the symbol-shuffling is guaranteed correct, because it is just a compressed transcription of the theorem we proved in three lines.

Common trap

The conditions are not decorative. Consider $\int_{-1}^{1} x^2\,\mathrm{d}x = \frac{2}{3}$, and try the substitution $u = x^2$. It is tempting to write $\mathrm{d}u = 2x\,\mathrm{d}x$, so $x\,\mathrm{d}x = \frac{1}{2}\mathrm{d}u$ – but the integrand is $x^2\,\mathrm{d}x$, not $x\,\mathrm{d}x$, and solving $x = \pm\sqrt{u}$ forces a sign *choice* that is positive on $[0,1]$ and negative on $[-1,0]$. The map $x \mapsto x^2$ is not invertible across 0 (it folds the interval in half), so a single substitution over the whole interval is illegal. The fix is to split at $x = 0$ and substitute on each piece separately. Moral: when g' changes sign – here at $x = 0$ – the change of variable can fold the domain and quietly double-count or cancel. Split first, then substitute.

Intuition

A clean way to stay safe without memorizing conditions: *always differentiate your final answer.* Substitution can be checked for free. If $\frac{\mathrm{d}}{\mathrm{d}x}$((your answer)) equals the original integrand, you are done and right, no matter how baroque the substitution was. Differentiation is the lie detector for integration, and it never sleeps.

Application

Substitution is not a classroom curiosity; it is how real models get evaluated. *Physics:* the work done by a varying force, $\int F\,\mathrm{d}x$, becomes a clean energy integral after substituting the position-dependent variable. *Probability and ML:* normalizing a Gaussian relies on the substitution $u = x/\sigma$ to reduce $\int e^{-x^2/2\sigma^2}\mathrm{d}x$ to the standard bell curve; the same change of variable is the "reparameterization trick" that lets modern generative models backpropagate through random sampling. *Biology and chemistry:* separable differential equations for population growth and reaction rates are solved by substituting to isolate each variable, then integrating. The little factor $g'(x)$ you have been tracking is, in every one of these, a unit conversion between the

natural variable of the problem and the variable that makes the integral solvable.

22.8 A compact procedure

Here is the chapter distilled into a checklist you can run on autopilot.

1. **Spot the inside.** Find a composition and let u be its inside (or: find a chunk whose derivative also appears, and call that chunk u).
2. **Differentiate:** compute $\mathrm{d}u = g'(x)\,\mathrm{d}x$.
3. **Rewrite the whole integrand** in terms of u and $\mathrm{d}u$. Patch missing *constants* freely; if a stray *variable* remains, solve $u = g(x)$ for x and substitute it too.
4. **For a definite integral, change the limits** to $u = g(a)$ and $u = g(b)$ at the same moment you change the integrand.
5. **Integrate in** u, landing in one of your known forms.
6. **Translate back to** x (indefinite) or just evaluate at the u-limits (definite).
7. **Differentiate your answer to check.** Always.

Pause

If a substitution leaves you with a leftover variable you cannot express in u, do not force it. That is the universe telling you either to pick a different u or to reach for a different tool. The very next chapter introduces *integration by parts*, the technique for integrands that are products with no inside-derivative pattern – the natural partner to substitution. Together they handle a remarkable fraction of everything you will meet.

22.9 What substitution unlocked, the exact-partner barrier, and the road ahead

Connection

Integration by substitution is the single most frequently deployed operational weapon in all of integral calculus, but like all tools, it has a precise boundary of power.
Substitution dismantled the rigid cage of Cartesian coordinates. Before Euler, computing integrals involving composite functions—such as $\int x\sqrt{x^2+1}\,\mathrm{d}x$, $\int \tan x\,\mathrm{d}x$, or Gaussian forms $\int xe^{-x^2}\,\mathrm{d}x$—required custom geometric troughs or infinite power series. Substitution transformed them into elementary polynomial or exponential primitives by stretching and reindexing the underlying variable axis. It unlocked the computation of physical work under nonlinear field potentials, probability normalizations, and the evaluation of symmetric definite integrals without finding closed-form primitives at all.

Yet because substitution is mathematically the Chain Rule run in reverse:

$$\int f(g(x))\, g'(x)\, \mathrm{d}x = F(g(x)) + C,$$

its reach encounters the *exact-partner barrier*. Its success depends upon the presence of the exact derivative partner $g'(x)\,\mathrm{d}x$ (or an easily adjusted scalar multiple $c\,g'(x)\,\mathrm{d}x$) inside the integrand.

The moment an integrand pairs a composite function with a factor that does *not* match its internal rate of change, substitution stalls completely. Compare two deceptively similar integrals:

$$\int xe^{x^2}\,\mathrm{d}x \quad \text{vs.} \quad \int xe^{x}\,\mathrm{d}x.$$

In the first, the internal function is $g(x) = x^2$, whose derivative is $g'(x) = 2x$. The factor $x\,\mathrm{d}x$ is right there waiting to be absorbed into $\frac{1}{2}\mathrm{d}u$, and the integral falls in one line to $\frac{1}{2}e^{x^2} + C$. In the second, the integrand is a product of two distinct functions (x and e^x). If you try $u = x$, you get $\mathrm{d}u = \mathrm{d}x$ and accomplish nothing. If you try $u = e^x$, you get $\mathrm{d}u = e^x\,\mathrm{d}x$ and get tangled in $\int \ln(u)\,\mathrm{d}u$. There is no inner-outer composition with an awaiting derivative factor. The same barrier halts integrals like $\int \ln x\,\mathrm{d}x$, $\int x^2 \cos x\,\mathrm{d}x$, and $\int \arctan x\,\mathrm{d}x$. Substitution cannot split a product of two independent mathematical personalities.

To break through this barrier, we must reverse a different differentiation law. Instead of inverting the Chain Rule, Chapter 23 (*Integration by Parts*) inverts the Product Rule:

$$\frac{\mathrm{d}}{\mathrm{d}x}[u(x)v(x)] = u'(x)v(x) + u(x)v'(x) \implies \int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u.$$

By trading the difficult integral $\int u\,\mathrm{d}v$ for a simpler integral $\int v\,\mathrm{d}u$, Integration by Parts unlocks precisely the products (xe^x, $x^2 \sin x$, $1 \cdot \ln x$, $1 \cdot \arctan x$) where substitution hits a brick wall.

22.10 Exercises

Exercise 22.1 (Warm-ups)**.** Evaluate each indefinite integral by a single substitution, and check by differentiating.

1. $\displaystyle\int (2x+3)^7\,\mathrm{d}x$
2. $\displaystyle\int x^2\sqrt{x^3+1}\,\mathrm{d}x$
3. $\displaystyle\int \frac{e^{1/x}}{x^2}\,\mathrm{d}x$
4. $\displaystyle\int \sin x\,\cos^4 x\,\mathrm{d}x$

Exercise 22.2 (The g'/g pattern)**.** Using $\int \frac{g'}{g}\,\mathrm{d}x = \ln|g| + C$, evaluate $\displaystyle\int \frac{3x^2-2}{x^3-2x+7}\,\mathrm{d}x$ and

$\int \cot x \, dx$. For the second, first rewrite $\cot x$ as a quotient.

Exercise 22.3 (Stray variable)**.** Evaluate $\int x(2x-1)^{10} \, dx$ by letting $u = 2x - 1$ and solving for the leftover x.

Exercise 22.4 (Definite, change the limits)**.** Evaluate $\displaystyle\int_0^1 \frac{x}{x^2+1} \, dx$ two ways: by changing the limits, and by back-substituting. Confirm the answers agree, and state the new limits explicitly.

Exercise 22.5 (Complete the square)**.** Evaluate $\displaystyle\int \frac{1}{x^2 - 6x + 13} \, dx$ by completing the square and substituting to reach an arctan.

Exercise 22.6 (Rationalize)**.** Evaluate $\displaystyle\int \frac{1}{1+\sqrt[3]{x}} \, dx$ by setting $u = \sqrt[3]{x}$. (You will need polynomial division.)

Exercise 22.7 (A trig identity first)**.** Evaluate $\int \sin^2 x \cos^3 x \, dx$. (Peel off one $\cos x$ for du and convert the rest with $\cos^2 x = 1 - \sin^2 x$.)

Exercise 22.8 (Symmetry)**.** Use the reflection substitution $u = \frac{\pi}{2} - x$ to evaluate $\displaystyle\int_0^{\pi/2} \frac{\cos^2 x}{\sin^2 x + \cos^2 x} \, dx$ without finding an antiderivative. (It should remind you of Example 22.12.)

Exercise 22.9 (Conceptual: why the constant is allowed)**.** Explain in a sentence or two why you are permitted to pull a constant factor out of an integral to manufacture du, but *not* a variable factor. Illustrate with a wrong "computation" of $\int x^2 e^x \, dx$ that mistakenly factors out an x, and point to the exact illegal step.

Exercise 22.10 (Conceptual: the folded interval)**.** Consider $\displaystyle\int_{-1}^{2} x^2 \, dx$ and the substitution $u = x^2$. Explain why a naive single substitution over $[-1, 2]$ is invalid, identify where $g'(x)$ changes sign, and describe how you would split the interval to substitute correctly. (You do not need to finish the computation.)

Exercise 22.11 (Proof)**.** Prove the reflection identity $\displaystyle\int_a^b f(x) \, dx = \int_a^b f(a+b-x) \, dx$ for continuous f, by carrying out the substitution $u = a + b - x$ carefully, including what happens to the limits and the sign of du. Then use it to evaluate $\int_0^1 \frac{1}{1+e^{x-1/2}} \, dx$ – or, more simply, re-derive Example 22.12 as a corollary.

Exercise 22.12 (Proof: substitution from the Riemann sum)**.** For a strictly increasing, continuously differentiable g on $[a, b]$, sketch why the change-of-variable formula holds at the level of Riemann sums: explain how a partition of the x-axis maps to a partition of the u-axis, why the width of the k-th subinterval transforms by approximately the factor $g'(x_k)$, and how letting the mesh go to zero recovers $\int_{g(a)}^{g(b)} f(u) \, du$. This is the "change-of-width law" of Section 22.3 made into a proof sketch.

Chapter 23

Integration by Parts

In the last chapter we learned to undo the chain rule. Substitution let us recognize when an integrand was secretly born from a composition $f(g(x))$ and to peel the composition back apart. That handles an enormous number of integrals – but it leaves a whole continent untouched. What do you do with

$$\int x\,e^x\,\mathrm{d}x, \qquad \int x\sin x\,\mathrm{d}x, \qquad \int \ln x\,\mathrm{d}x, \qquad \int e^x\cos x\,\mathrm{d}x?$$

None of these is a chain rule in disguise. Try substitution on $\int xe^x\,\mathrm{d}x$ and you will flail: setting $u = x$ does nothing, setting $u = e^x$ leaves a stray x you cannot get rid of. The trouble is that these integrands are *products* of two unrelated things – a polynomial times an exponential, a polynomial times a sine – and substitution was built to undo compositions, not products.

There is exactly one differentiation rule that creates products: the *product rule*. So if we want to integrate products, the move is to run the product rule backwards, exactly as substitution was the chain rule run backwards. The technique that comes out is called *integration by parts*, and it is the second of the two great workhorses of integration. It will not collapse an integral to nothing the way a lucky substitution sometimes does. Instead it does something subtler and, once you see it, rather beautiful: it *trades* the integral you have for a different one, in the hope that the new one is easier. Most of the art is in arranging the trade so that you come out ahead.

23.1 The Product Rule, Run Backwards

Everything in this chapter is squeezed out of one line you already know. If u and v are differentiable functions of x, the product rule says

$$\frac{\mathrm{d}}{\mathrm{d}x}(u\,v) = u\,\frac{\mathrm{d}v}{\mathrm{d}x} + v\,\frac{\mathrm{d}u}{\mathrm{d}x}.$$

Read it as a statement about how the area of a little rectangle $u \cdot v$ changes: when both sides are growing, the rectangle grows on two fronts at once, one strip of width $\mathrm{d}u$ down the side and one strip of height $\mathrm{d}v$ along the top. The product rule just adds those two strips. (We drew that picture back when we first met the product rule; hold it in mind, because it is about to pay off.)

Now integrate both sides with respect to x over the whole real line of the problem. Integration

undoes differentiation, so the left side integrates to $u\,v$ itself:

$$u\,v = \int u\,\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x + \int v\,\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x.$$

That is already the whole theorem – it just needs to be rearranged. Solve for the first integral on the right:

$$\int u\,\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = u\,v - \int v\,\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x.$$

Finally, abbreviate $\mathrm{d}v = \frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x$ and $\mathrm{d}u = \frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$ – the differentials of Chapter 14, used here as plain bookkeeping for "the slope times the step." We arrive at the formula every calculus student eventually recites in their sleep:

Focus

Integration by parts. If u and v are differentiable, then

$$\boxed{\int u\,\mathrm{d}v \;=\; u\,v \;-\; \int v\,\mathrm{d}u}$$

and, between limits a and b,

$$\int_a^b u\,\mathrm{d}v \;=\; \Big[u\,v\Big]_a^b \;-\; \int_a^b v\,\mathrm{d}u.$$

That is it. Two symbols, u and v, and the rule says: integrating $u\,\mathrm{d}v$ costs you the boundary term uv minus the price of integrating $v\,\mathrm{d}u$ instead. You have not solved the integral – you have *swapped* it for another. The bet is that $\int v\,\mathrm{d}u$ is friendlier than $\int u\,\mathrm{d}v$.

Why the technique works

Substitution is the chain rule read right to left; integration by parts is the product rule read right to left. That duality organizes the whole structure of integration technique. There are only a handful of ways to *build* a function by differentiation (sums, products, compositions), so there are only a handful of ways to *take one apart* by integration. Sums split term-by-term. Compositions are unwound by substitution. Products are unwound by parts. When you face an unfamiliar integral, the first question is always "which differentiation rule could have produced an integrand shaped like this?" – and the answer tells you which tool to reach for.

What is really happening

Look again at $\int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u$ and ask what is physically being moved around. The combination $u\,\mathrm{d}v$ is a thin horizontal slab of "stuff": a width u times an increment $\mathrm{d}v$. The formula says the total of all those strips equals the finished rectangle uv *minus* the total of the complementary strips $v\,\mathrm{d}u$. Integration by parts is thus a statement about cutting a rectangle two ways. The product uv is a region in the u–v plane; you can sweep it out in horizontal slabs ($u\,\mathrm{d}v$) or vertical slabs ($v\,\mathrm{d}u$), and the two sweeps must add up to the same rectangle. We will make this picture exact in the next section, and it explains *why* the boundary term uv is there.

23.2 The Picture: Slicing a Rectangle Two Ways

The boundary term uv can feel like it falls out of the algebra by accident. It does not. Here is a proof of integration by parts that uses no symbols at all until the very end – just the area of a rectangle.

Imagine a curve in the u–v plane traced out as x runs from a to b. At $x = a$ the point is (u_a, v_a); at $x = b$ it is (u_b, v_b). As x increases, both u and v change, and the point sweeps a path. Now look at Figure 23.1. The big rectangle reaching out to (u_b, v_b) has area $u_b v_b$. The small rectangle reaching to (u_a, v_a) has area $u_a v_a$. The L-shaped region between them – the big rectangle minus the small one – has area $u_b v_b - u_a v_a$, which is exactly the boundary term $\Big[uv\Big]_a^b$.

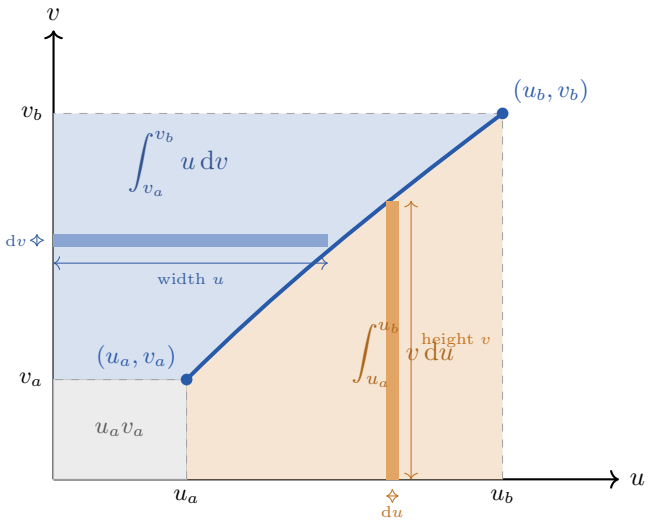


Figure 23.1: Integration by parts as bounding box geometry. The total rectangle out to (u_b, v_b) has area $u_b v_b$. Removing the bottom-left rectangle $u_a v_a$ leaves an L-shaped region of area $u_b v_b - u_a v_a = \Big[uv\Big]_a^b$. The path $(u(x), v(x))$ cuts this L-region into two complementary sweeps: horizontal strips $u\,dv$ (blue) summing to $\int u\,dv$, and vertical strips $v\,du$ (orange) summing to $\int v\,du$. Together they tile the region: $\int u\,dv + \int v\,du = \Big[uv\Big]_a^b$.

Now slice that L-shaped region two different ways. A *vertical* strip has a height v and is built from a change in u; its area is $v\,du$, and summing them gives $\int v\,du$, the orange region below the curve. A *horizontal* strip has a width u and is built from a change in v; its area is $u\,dv$, and summing gives $\int u\,dv$, the blue region to the left of the curve. The path of (u, v) is the boundary between the two regions, and together they tile the entire L. Therefore

$$\int_a^b u\,dv \;+\; \int_a^b v\,du \;=\; u_b v_b - u_a v_a = \Big[uv\Big]_a^b.$$

Move the second integral to the right and you have integration by parts, born purely from cutting a region two ways.

Intuition

This is why the boundary term is uv and not something else: uv is literally the *rectangle* whose two complementary sweeps are the two integrals. Integration by parts is conservation of area. Whatever you "give up" by not integrating $u\,dv$ directly, you get back as the rectangle minus the other sweep. Nothing is created or destroyed; the area is just being cut into different strips.

This geometric proof needs u and v to be monotonic to draw cleanly, but the algebraic derivation in Section 23.1 has no such restriction – it works for any differentiable u, v. Keep the picture for intuition and the algebra for rigor.

23.3 Choosing u and dv

The formula has a built-in decision. Faced with an integral, you must split the integrand into two pieces: one piece you will call u (and *differentiate* to get du), and one piece you will call dv (and *integrate* to get v). Every product can be split two ways, and usually only one split helps. So how do you choose?

Two principles guide every good choice.

1. u **should get simpler when you differentiate it.** The whole point is that $\int v\,du$ is easier than $\int u\,dv$. Differentiating u is your chance to make progress, so pick a u whose derivative is tamer than itself. A polynomial like x^3 is ideal: each derivative knocks the degree down, and after enough rounds it vanishes. The logarithm $\ln x$ is even better – it has no elementary antiderivative of its own, but its *derivative* $1/x$ is beautifully simple, so making $\ln x$ into u converts a hard factor into an easy one.

2. dv **must be something you can actually integrate.** You need $v = \int dv$ to be a function you know. There is no point assigning dv to a factor you cannot antidifferentiate. So $e^x\,dx$, $\sin x\,dx$, $\cos x\,dx$, and powers $x^n\,dx$ all make fine dv's; something like $\ln x\,dx$ does not (you would be stuck integrating it – which is the very problem you started with).

These two principles together produce a famous mnemonic.

Focus

LIATE – a priority order for choosing u. When a product mixes two of these types, let u be whichever appears *earlier* in the list, and let the rest be dv:

L **L**ogarithmic ($\ln x$, $\log x$)
I **I**nverse trig ($\arctan x$, $\arcsin x$)
A **A**lgebraic (powers x^n, polynomials)
T **T**rigonometric ($\sin x$, $\cos x$)
E **E**xponential (e^x, a^x)

Earlier in the list = choose as u.

Why this order? Read the list top to bottom and you are reading functions in order of "how much simpler does differentiating make them, and how easy are they to integrate." Logs and inverse trig

functions are at the top because they are miserable to integrate but pleasant to differentiate, so you always want them as u. Exponentials are at the bottom because they are trivial to integrate and never get simpler when differentiated, so you always want them as $\mathrm{d}v$. Polynomials sit in the middle: they are happy either way, but differentiating lowers their degree, so when paired with a trig or exponential they go on the u side.

Common trap

LIATE is a *heuristic, not a law*. It encodes good defaults, but it cannot think for you, and there are integrals where it points the wrong way or where *either* choice works. The genuine criterion is always the two principles above: did $\int v\,\mathrm{d}u$ come out simpler than what you started with? If a choice leads to a worse integral, you have learned something – swap your choice and try the other split. Never defend a bad choice by appeal to a mnemonic.

History

The acronym LIATE is a twentieth-century pedagogical heuristic – introduced in a 1983 paper by Herbert Kasube to assist calculus students. The foundational *method* of integration by parts has a far grander lineage: it was systematically introduced by the English mathematician *Brook Taylor* in his 1715 treatise *Methodus Incrementorum Directa et Inversa* (the same work that gave birth to Taylor series). Taylor recognized that the Product Rule of differentiation:

$$\mathrm{d}(uv) = u\,\mathrm{d}v + v\,\mathrm{d}u$$

could be integrated and rearranged to shift a derivative off an intractable factor and onto a compliant one.

In modern mathematics and physics, integration by parts is not just an integration trick; it is a structural bridge across disciplines. In nineteenth-century analysis, Pierre-Simon *Laplace* and Adrien-Marie *Legendre* recognized parts as the fundamental engine of integral transforms:

- In the **Laplace transform** $\mathcal{L}\{f'\}(s) = \int_0^\infty e^{-st} f'(t)\,\mathrm{d}t$, a single integration by parts yields $s\mathcal{L}\{f\}(s) - f(0)$, converting intractable differential equations directly into high-school algebra.
- In Hamiltonian mechanics and thermodynamics, the **Legendre transform** converts the Lagrangian $L(q, \dot{q})$ into the Hamiltonian $H(q, p) = p\dot{q} - L$ by swapping coordinates across the complementary bounding box $uv = \int u\,\mathrm{d}v + \int v\,\mathrm{d}u$.
- In quantum mechanics and modern partial differential equations, parts is how mathematicians define *weak derivatives* and define Hermitian operators: moving derivatives off un-differentiable physical wavefunctions onto smooth test functions.

Leibniz's notation $\int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u$ is what made this universal principle effortless to write and remember.

23.4 First Worked Examples

The ritual is always the same four steps: choose u and $\mathrm{d}v$; differentiate to get $\mathrm{d}u$ and integrate to get v; assemble $uv - \int v\,\mathrm{d}u$; finish the leftover integral.

Example 23.1 (The archetype: $\int xe^x\,\mathrm{d}x$)**.** Here the two factors are algebraic (x) and exponential

(e^x). LIATE says: A beats E, so $u = x$ and $\mathrm{d}v = e^x\,\mathrm{d}x$. Then

$$u = x \;\Rightarrow\; \mathrm{d}u = \mathrm{d}x, \qquad \mathrm{d}v = e^x\,\mathrm{d}x \;\Rightarrow\; v = e^x.$$

Assemble:

$$\int xe^x\,\mathrm{d}x = uv - \int v\,\mathrm{d}u = xe^x - \int e^x\,\mathrm{d}x = xe^x - e^x + C = (x-1)e^x + C.$$

Notice what happened: the troublesome factor x got differentiated down to 1, turning the leftover integral $\int e^x\,\mathrm{d}x$ into something we can do in our sleep. *That* is the payoff of choosing $u = x$.

Work it yourself

Check Example 23.1 by differentiating the answer. By the product rule, $\frac{\mathrm{d}}{\mathrm{d}x}((x-1)e^x) = 1 \cdot e^x + (x-1)e^x = e^x + xe^x - e^x = xe^x$. It matches the integrand. *Always* verify a by-parts answer this way at first – differentiation is fast and it catches sign errors instantly. Integration by parts is sign-error-prone precisely because of the minus sign in the formula.

Common trap

What if you had chosen the other way, $u = e^x$ and $\mathrm{d}v = x\,\mathrm{d}x$? Then $\mathrm{d}u = e^x\,\mathrm{d}x$ and $v = x^2/2$, and the formula gives

$$\int xe^x\,\mathrm{d}x = \frac{x^2}{2}e^x - \int \frac{x^2}{2}e^x\,\mathrm{d}x.$$

The new integral has x^2 where the old one had x – it got *worse*. The polynomial went up in degree because we integrated it instead of differentiating it. This is the classic wrong turn, and it is exactly what LIATE warns you off: never integrate the polynomial when there is an exponential or trig factor waiting to absorb the $\mathrm{d}v$ role.

Example 23.2 (The clever trick: $\int \ln x\,\mathrm{d}x$)**.** This one looks impossible – there is no second factor to play with. The trick is to *invent* one by writing $\ln x = (\ln x) \cdot 1$. Now it is a product, and LIATE is decisive: L beats everything, so $u = \ln x$ and $\mathrm{d}v = 1\,\mathrm{d}x$. Then

$$u = \ln x \;\Rightarrow\; \mathrm{d}u = \frac{1}{x}\,\mathrm{d}x, \qquad \mathrm{d}v = \mathrm{d}x \;\Rightarrow\; v = x.$$

Assemble:

$$\int \ln x\,\mathrm{d}x = (\ln x)(x) - \int x \cdot \frac{1}{x}\,\mathrm{d}x = x\ln x - \int 1\,\mathrm{d}x = x\ln x - x + C.$$

The miracle is in the cancellation $x \cdot \frac{1}{x} = 1$: differentiating $\ln x$ produced a $1/x$ that annihilated the x from v, leaving the trivial integral $\int 1\,\mathrm{d}x$. This is the standard way to integrate *any* function that is easy to differentiate but has no obvious antiderivative – log, arctan, arcsin all yield to "u = the function, $\mathrm{d}v = \mathrm{d}x$."

Example 23.3 (Inverse trig: $\int \arctan x\,\mathrm{d}x$)**.** Same trick. Write $\arctan x = (\arctan x) \cdot 1$, take $u = \arctan x$ and $\mathrm{d}v = \mathrm{d}x$, so $\mathrm{d}u = \dfrac{1}{1+x^2}\,\mathrm{d}x$ and $v = x$. Then

$$\int \arctan x\,\mathrm{d}x = x\arctan x - \int \frac{x}{1+x^2}\,\mathrm{d}x.$$

The leftover integral now *is* a substitution problem – set $w = 1 + x^2$, $\mathrm{d}w = 2x\,\mathrm{d}x$, so $\int \frac{x}{1+x^2}\,\mathrm{d}x = \frac{1}{2}\ln(1+x^2)$. Hence

$$\int \arctan x\,\mathrm{d}x = x\arctan x - \tfrac{1}{2}\ln(1+x^2) + C.$$

Notice how the two big techniques chain together: parts cracked the problem open, and substitution mopped up the remainder. Real integration is almost always a relay race between methods.

Example 23.4 (A polynomial that needs two rounds: $\int x^2 \sin x\,\mathrm{d}x$)**.** Here $u = x^2$ (algebraic, beats trig) and $\mathrm{d}v = \sin x\,\mathrm{d}x$, so $\mathrm{d}u = 2x\,\mathrm{d}x$ and $v = -\cos x$. One round:

$$\int x^2 \sin x\,\mathrm{d}x = -x^2\cos x - \int (-\cos x)(2x\,\mathrm{d}x) = -x^2\cos x + 2\int x\cos x\,\mathrm{d}x.$$

We have not finished, but we have made progress: the power dropped from x^2 to x. Attack $\int x\cos x\,\mathrm{d}x$ by parts again, with $u = x$, $\mathrm{d}v = \cos x\,\mathrm{d}x$, $\mathrm{d}u = \mathrm{d}x$, $v = \sin x$:

$$\int x\cos x\,\mathrm{d}x = x\sin x - \int \sin x\,\mathrm{d}x = x\sin x + \cos x.$$

Substitute back:

$$\int x^2 \sin x\,\mathrm{d}x = -x^2\cos x + 2\big(x\sin x + \cos x\big) + C = -x^2\cos x + 2x\sin x + 2\cos x + C.$$

Each application of parts peeled off one degree of the polynomial. A degree-n polynomial times $\sin x$ or e^x surrenders after exactly n rounds. When you see that coming, there is a shortcut – tabular integration – which we meet in Section 23.7.

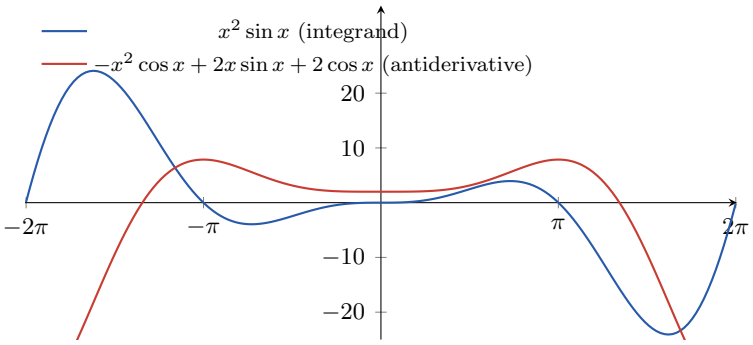


Figure 23.2: The integrand $x^2 \sin x$ (blue) and the antiderivative we computed in Example 23.4 (red). Wherever the blue curve is positive the red curve climbs, and where blue is negative red falls – the antiderivative accumulates signed area under the integrand, exactly as the Fundamental Theorem promises.

Example 23.5 (Definite integral: $\int_1^e \ln x\,\mathrm{d}x$)**.** We already found the antiderivative $x\ln x - x$ in Example 23.2, so

$$\int_1^e \ln x\,\mathrm{d}x = \Big[x\ln x - x\Big]_1^e = (e\ln e - e) - (1\cdot\ln 1 - 1) = (e-e) - (0-1) = 1.$$

Geometrically: the area under the curve $y = \ln x$ from $x = 1$ to $x = e$ is exactly 1 square unit (Figure 23.3). That clean answer is no accident – $\ln x$ and e are bound together, and the region under one natural-log arch from 1 to e has unit area.

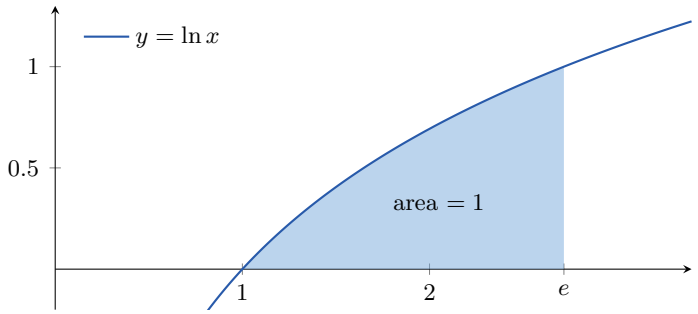


Figure 23.3: The shaded region under $y = \ln x$ from $x = 1$ to $x = e$ has area exactly 1, the value of $\int_1^e \ln x \, dx$ computed in Example 23.5. Integration by parts handed us the antiderivative $x \ln x - x$; the Fundamental Theorem turned it into this clean unit area.

Walking question

Here is a question to chew on while walking. We integrated $\ln x$ by parts and got $x \ln x - x$. But integration by parts came from the product rule, and the product rule came from differentiation. Yet $\ln x$ is not a product of anything obvious. How did a rule about products help with a single function? (Hint: the hidden product is $\ln x \cdot 1$, and the deeper truth is that "1" is the derivative of x, so we are secretly using $v = x$ as a lever. Where else might multiplying by a cleverly-chosen 1 unlock a problem?)

23.5 Reduction Formulas: Parts as a Machine

When the integrand carries a parameter – a power n, say – a single application of parts can express the integral with parameter n in terms of the *same* integral with a smaller parameter. That recursive relationship is called a *reduction formula*. It is integration by parts at its most powerful: instead of solving one integral, you solve an entire infinite family in one stroke.

Example 23.6 (The reduction formula for $\int x^n e^x \, dx$)**.** Let

$$I_n = \int x^n e^x \, dx.$$

Take $u = x^n$ and $dv = e^x \, dx$, so $du = nx^{n-1} \, dx$ and $v = e^x$. Parts gives

$$I_n = x^n e^x - \int e^x \cdot nx^{n-1} \, dx = x^n e^x - n \int x^{n-1} e^x \, dx = x^n e^x - n \, I_{n-1}.$$

So

$$\boxed{I_n = x^n e^x - n \, I_{n-1}.}$$

Each use of parts lowers the exponent by one. Starting from $I_0 = \int e^x \, dx = e^x$, we can climb the ladder to any n. For instance,

$$I_1 = xe^x - 1 \cdot I_0 = xe^x - e^x,$$
$$I_2 = x^2 e^x - 2I_1 = x^2 e^x - 2(xe^x - e^x) = x^2 e^x - 2xe^x + 2e^x,$$
$$I_3 = x^3 e^x - 3I_2 = x^3 e^x - 3x^2 e^x + 6xe^x - 6e^x,$$

and the pattern (with its alternating falling-factorial coefficients) is now unmistakable. One reduction formula generated the whole family.

Why the technique works

Why do reduction formulas exist at all? Because parts *trades a hard factor for a smaller version of itself.* When you differentiate x^n you get nx^{n-1} – structurally the same kind of object, one notch simpler. The integral on the right is therefore a clone of the integral on the left with the parameter decremented. Anytime differentiating one factor reproduces a scaled, simpler copy of that factor – powers, $\sin^n$, $\cos^n$, $(\ln x)^n$ – parts will hand you a recursion. The technique is not just for one-off integrals; it is a way of taming infinite families by induction on a parameter.

Example 23.7 (A trig reduction: $\int \sin^n x\,\mathrm{d}x$)**.** This is the classic and it is worth seeing in full, because the cleverness is real. Let

$$J_n = \int \sin^n x\,\mathrm{d}x.$$

Split off one factor of sine: write $\sin^n x = \sin^{n-1} x \cdot \sin x$, and take

$$u = \sin^{n-1} x, \qquad \mathrm{d}v = \sin x\,\mathrm{d}x,$$

so that $\mathrm{d}u = (n-1)\sin^{n-2} x\,\cos x\,\mathrm{d}x$ and $v = -\cos x$. Parts:

$$J_n = -\sin^{n-1} x\,\cos x + (n-1)\int \sin^{n-2} x\,\cos^2 x\,\mathrm{d}x.$$

Now use the Pythagorean identity $\cos^2 x = 1 - \sin^2 x$ to turn the leftover integral back into sines:

$$\int \sin^{n-2} x\,\cos^2 x\,\mathrm{d}x = \int \sin^{n-2} x\,\mathrm{d}x - \int \sin^n x\,\mathrm{d}x = J_{n-2} - J_n.$$

Substitute:

$$J_n = -\sin^{n-1} x\,\cos x + (n-1)\big(J_{n-2} - J_n\big).$$

The integral J_n has *reappeared on the right* – this is the recurring integral trick we will name in the next section. Collect the J_n terms:

$$J_n + (n-1)J_n = -\sin^{n-1} x\,\cos x + (n-1)J_{n-2},$$

$$n\,J_n = -\sin^{n-1} x\,\cos x + (n-1)J_{n-2},$$

and finally

$$\boxed{J_n = -\frac{1}{n}\sin^{n-1} x\,\cos x + \frac{n-1}{n}J_{n-2}.}$$

This drops the power by *two* each time – which is exactly right, because the base cases come in two flavors: $J_0 = \int \mathrm{d}x = x$ and $J_1 = \int \sin x\,\mathrm{d}x = -\cos x$. Every even power reduces down to J_0, every odd power down to J_1.

Connection

The reduction formula for $\int_0^{\pi/2} \sin^n x\,\mathrm{d}x$ leads to one of the most beautiful formulas in mathematics. Evaluated between 0 and $\pi/2$ the boundary term $\sin^{n-1} x\cos x$ vanishes at both ends, so the recursion becomes purely numerical: $W_n = \frac{n-1}{n}W_{n-2}$, where $W_n = \int_0^{\pi/2} \sin^n x\,\mathrm{d}x$ are

the *Wallis integrals*. Chasing the ratios W_n as $n \to \infty$ gives John Wallis's 1655 product

$$\frac{\pi}{2} = \frac{2}{1}\cdot\frac{2}{3}\cdot\frac{4}{3}\cdot\frac{4}{5}\cdot\frac{6}{5}\cdot\frac{6}{7}\cdots,$$

a product of integers that converges to half of π. That such a thing falls out of a plain reduction formula is one of the surprises that keeps the subject interesting.

23.6 The Recurring-Integral Trick

Sometimes you apply parts, then apply it again, and the integral you started with *comes back around* – the very same integral reappears on the right-hand side. A beginner panics ("I'm going in circles!"). An expert grins, because a repeated integral is something you can *solve for algebraically*, exactly the way you'd solve $x = 5 - 2x$ for x. We just saw a glimpse of this in the $\sin^n$ reduction. The cleanest example is the integral that mixes an exponential with a trig function.

Example 23.8 (Solving for the integral: $\int e^x \cos x \,\mathrm{d}x$)**.** Call it $K = \int e^x \cos x \,\mathrm{d}x$. Neither factor gets simpler when differentiated – e^x stays e^x, and $\cos x$ cycles to $-\sin x$ to $-\cos x$ – so LIATE has no traction. It does not matter which factor we pick for u; let us take $u = e^x$, $\mathrm{d}v = \cos x \,\mathrm{d}x$ consistently. Then $\mathrm{d}u = e^x \,\mathrm{d}x$, $v = \sin x$:

$$K = e^x \sin x - \int e^x \sin x \,\mathrm{d}x.$$

The new integral is no simpler, but it is the same *type*. Apply parts again to $\int e^x \sin x \,\mathrm{d}x$, keeping the same choice $u = e^x$, $\mathrm{d}v = \sin x \,\mathrm{d}x$, so $v = -\cos x$:

$$\int e^x \sin x \,\mathrm{d}x = -e^x \cos x + \int e^x \cos x \,\mathrm{d}x = -e^x \cos x + K.$$

Substitute this back into the expression for K:

$$K = e^x \sin x - \big(-e^x \cos x + K\big) = e^x \sin x + e^x \cos x - K.$$

There it is – K on both sides. Add K to both sides:

$$2K = e^x \sin x + e^x \cos x,$$

$$\boxed{K = \int e^x \cos x \,\mathrm{d}x = \frac{e^x(\sin x + \cos x)}{2} + C.}$$

We never "finished" an integral in the usual sense – we trapped the unknown integral in an equation and solved for it.

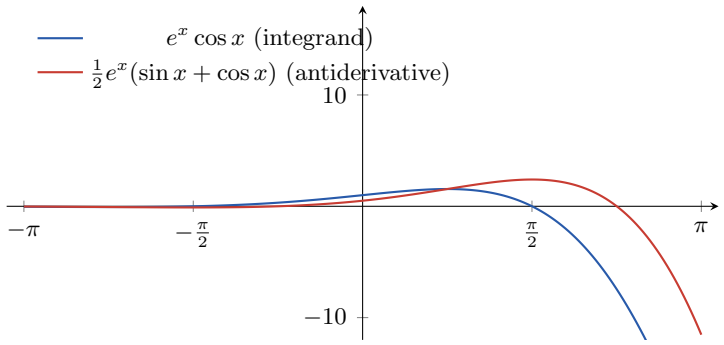


Figure 23.4: The integrand $e^x \cos x$ (blue) and the antiderivative $\frac{1}{2}e^x(\sin x + \cos x)$ (red) we trapped algebraically in Example 23.8. The exponential factor makes both curves swing ever wider, but the antiderivative still rises exactly where the integrand is positive and falls where it is negative – the recurring-integral trick produced a genuine antiderivative, not a sleight of hand.

Common trap

The fatal mistake in the recurring trick is to *switch your choice of u midstream.* If you take $u = e^x$ in the first parts and then $u = \cos x$ (i.e. $dv = e^x\,dx$) in the second, the second application simply undoes the first: you will get $K = K$, the useless identity $0 = 0$, and you'll think the method failed. It didn't – you sabotaged it. The rule is iron: *be consistent.* If u was the exponential the first time, u is the exponential every time. Consistency is what lets the integral genuinely cycle back rather than cancel.

What is really happening

Why does the integral come back rather than terminating? Because the pair $\{\sin x, \cos x\}$ is *closed* under differentiation – differentiating either one stays inside the pair – and e^x is its own derivative. So no factor ever simplifies to zero the way a polynomial eventually does; the process can only *rotate* through a finite set of shapes. After two rotations of the trig factor you are back where you started, up to a sign and the boundary terms that piled up along the way. The "recurring integral" is the visible footprint of that rotation. Linear algebra has a name for this: $\sin x$ and $\cos x$ span a two-dimensional space invariant under $\frac{d}{dx}$, and integration by parts is tracing a loop around it.

23.7 Tabular Integration: Parts on Autopilot

When one factor is a polynomial that differentiates down to zero – like x^3 or x^2 – and the other is something you can integrate over and over without effort – like e^x, $\sin x$, $\cos x$ – repeated integration by parts becomes purely mechanical. There is a bookkeeping shortcut, often called the *tabular method* or the *DI method*, that does all the rounds at once and never makes a sign error.

Here is the recipe. Make two columns. In the D column (for Differentiate) put your polynomial u and keep differentiating down to 0. In the I column (for Integrate) put the dv function and keep integrating, one row for each row of the D column. Then draw diagonal arrows from each entry in D to the entry one row *down* in I, attach alternating signs $+, -, +, -, \ldots$ starting with $+$, and read

off the products.

Example 23.9 (Tabular: $\int x^3 e^x \, dx$)**.** Differentiate x^3 down to zero; integrate e^x (it never changes):

sign	D (differentiate)		I (integrate)
$+$	x^3	$\searrow$	e^x
$-$	$3x^2$	$\searrow$	e^x
$+$	$6x$	$\searrow$	e^x
$-$	6	$\searrow$	e^x
	0		e^x

Multiply each D entry by the I entry one row down, with the sign on its left:

$$\int x^3 e^x \, dx = +x^3 e^x - 3x^2 e^x + 6x e^x - 6e^x + C = (x^3 - 3x^2 + 6x - 6)e^x + C.$$

Compare this to the laborious three-rounds-of-parts we would otherwise do, and to the reduction formula's I_3 in Example 23.6 – same answer, a fraction of the writing.

Why the technique works

Tabular integration is not a separate technique – it is ordinary parts with the bookkeeping automated. Watch the columns line up with the formula. The first diagonal $+x^3 \cdot e^x$ is the boundary term uv. The next row down is exactly the $-\int v \, du$ that parts hands you, but instead of writing the integral sign you immediately apply parts to *it*, producing the next uv term and the next leftover – and so on. The alternating signs are the chain of minus signs from each application of the formula. The table terminates precisely because differentiating the polynomial eventually gives 0, killing the final leftover integral. So the table is a flip-book of the recursion in Example 23.6, all frames drawn at once.

Example 23.10 (Tabular with a trig factor: $\int x^2 \cos x \, dx$)**.** Differentiate x^2 to zero; integrate $\cos x \to \sin x \to -\cos x \to -\sin x$:

sign	D		I
$+$	x^2	$\searrow$	$\cos x$
$-$	$2x$	$\searrow$	$\sin x$
$+$	2	$\searrow$	$-\cos x$
	0		$-\sin x$

Read the diagonals with signs:

$$\int x^2 \cos x \, dx = +x^2 \sin x - 2x(-\cos x) + 2(-\sin x) + C = x^2 \sin x + 2x \cos x - 2 \sin x + C.$$

The signs in the answer come from *two* sources – the alternating $\pm$ of the table and the minus signs that appear inside the I column as cos integrates to sin to $-\cos$. Keep them separate and the table never lies.

Common trap

Tabular integration in its simplest form requires the D column to terminate at 0 – that is, u must be a polynomial. If *neither* factor differentiates to zero (as with $\int e^x \cos x \, dx$, where both columns cycle forever), the plain table never ends. There is an extended tabular method for those cases: you stop when a row reproduces the original integrand and then solve algebraically – which is just the recurring-integral trick of Section 23.6 drawn in a table. But for that case the by-hand "solve for K" method is usually clearer, so reach for the table only when a polynomial guarantees a stop.

23.8 Putting It All Together: A Harder Example

One last integral, and this one needs taste, not just a recipe.

Example 23.11 ($\int x \arctan x \, dx$)**.** Two factors: algebraic x and inverse-trig $\arctan x$. LIATE puts I above A, so $u = \arctan x$ (differentiate the nasty one) and $dv = x \, dx$. Then

$$du = \frac{1}{1+x^2}\,dx, \qquad v = \frac{x^2}{2}.$$

Parts:

$$\int x \arctan x \, dx = \frac{x^2}{2}\arctan x - \frac{1}{2}\int \frac{x^2}{1+x^2}\,dx.$$

The leftover is a rational function. Do the polynomial-division trick: $\dfrac{x^2}{1+x^2} = 1 - \dfrac{1}{1+x^2}$. Hence

$$\int \frac{x^2}{1+x^2}\,dx = \int 1\,dx - \int \frac{1}{1+x^2}\,dx = x - \arctan x.$$

Assemble the final answer:

$$\int x \arctan x \, dx = \frac{x^2}{2}\arctan x - \frac{1}{2}\big(x - \arctan x\big) + C = \frac{x^2+1}{2}\arctan x - \frac{x}{2} + C.$$

Three techniques in one problem: parts to break it open, polynomial division to simplify the rational leftover, and the known antiderivative of $1/(1+x^2)$. This is what real integration looks like – a sequence of recognitions, each one turning the problem into a slightly easier problem until nothing is left.

Why this matters

Step back and admire what integration by parts *is*. Differentiation is a one-way street: given a formula, you can always grind out its derivative. Integration is the reverse, and the reverse of a mechanical process is generally an art – there is no algorithm that integrates every elementary function in elementary terms (some, like $\int e^{-x^2}\,dx$, have no elementary antiderivative at all). Integration by parts is one of the few genuinely *structural* tools we have: it does not compute an answer so much as *re-describe the problem*, trading one integral for another and trusting that the trade leads somewhere better. Substitution undoes composition; parts undoes multiplication. With those two reversals, plus a handful of algebraic identities, you can integrate a

startling fraction of the functions that arise in physics, probability, and engineering. The rest – the genuinely non-elementary ones – are where the next chapters, and ultimately infinite series, take over.

Application

Integration by parts is not a calculus-class curiosity; it is structural machinery throughout science. In **probability**, the expected value of a nonnegative random variable can be rewritten as $E[X] = \int_0^\infty (1 - F(x))\,\mathrm{d}x$ by integrating $\int_0^\infty x f(x)\,\mathrm{d}x$ by parts – the boundary term and the survival function $1 - F$ appear precisely because of the uv term. In **physics**, the entire framework of Lagrangian mechanics and field theory rests on "integrating by parts to move a derivative off the variation": the Euler–Lagrange equations come from doing exactly that to $\int \frac{\partial L}{\partial \dot{q}}\,\delta\dot{q}\,\mathrm{d}t$. In **signal processing**, every integration-by-parts shifts a derivative from one factor onto another, which is the seed of the theory of *distributions* (the "derivative" of a non-differentiable signal is *defined* by where parts would have moved it). In **numerical analysis**, repeated parts produces asymptotic expansions for integrals that are otherwise intractable. Learn the move well; you will meet it for the rest of your mathematical life.

23.9 Why It Works, One More Time

Here is the core idea plainly, stripped of all examples. Once you hold it you will never be confused about when to reach for parts.

Intuition

The one-sentence summary. Integration by parts moves a derivative from one factor of a product to the other, paying a boundary term uv for the move.

To see that, rewrite the formula with the differentials spelled out. Starting from $\int u\,v'\,\mathrm{d}x = uv - \int u'\,v\,\mathrm{d}x$, notice that on the left the prime (the derivative) sits on v, and on the right it sits on u. The two integrals

$$\int u\,v'\,\mathrm{d}x \quad \longleftrightarrow \quad \int u'\,v\,\mathrm{d}x$$

are the same product with the derivative *relocated* from one factor to the other, and the cost of relocation is the rectangle uv evaluated at the endpoints. That is the whole mechanism. You reach for parts whenever moving the derivative makes the integrand simpler:

- Move the derivative *onto* a polynomial repeatedly and it dies ($x^n \to \cdots \to 0$); the integral terminates. (This is tabular integration.)
- Move the derivative *off* a log or inverse-trig function and its ugly "no antiderivative" problem dissolves into a simple rational factor. (This is the $\int \ln x\,\mathrm{d}x$ trick.)
- Move the derivative around a closed family like $\{e^x, \sin x, \cos x\}$ and it cycles back, letting you solve for the integral algebraically. (This is the recurring trick.)

Every example in this chapter is one of these three motions. The formula is one line; the judgment is in seeing which direction to push the derivative so the problem gets smaller.

Formal

Statement and proof, for the record. Let $u, v : [a, b] \to \mathbb{R}$ be continuous with continuous derivatives u', v' on $[a, b]$. Then uv is differentiable and, by the product rule, $(uv)' = u'v + uv'$, which is continuous. By the Fundamental Theorem of Calculus,

$$\Big[u(x)v(x)\Big]_a^b = \int_a^b (uv)' \,\mathrm{d}x = \int_a^b u'v \,\mathrm{d}x + \int_a^b uv' \,\mathrm{d}x.$$

Rearranging gives the definite form $\int_a^b uv' \,\mathrm{d}x = \Big[uv\Big]_a^b - \int_a^b u'v \,\mathrm{d}x$; dropping the limits and writing $\mathrm{d}v = v' \,\mathrm{d}x$, $\mathrm{d}u = u' \,\mathrm{d}x$ gives the indefinite form. The only hypothesis is that u' and v' are continuous so that the Fundamental Theorem applies – everything else is algebra. □

23.10 What integration by parts unlocked, limitations, and the road ahead

Connection

Integration by parts is the calculus equivalent of a balance scale: when an integral is too heavy to lift directly, we transfer weight from one pan to the other until the problem balances.
Before Taylor's product-rule inversion, functions from different mathematical kingdoms could not be integrated together. A polynomial multiplied by an exponential ($x^n e^x$), a power of x multiplied by a trigonometric wave ($x^n \sin x$), or solitary transcendental functions lacking algebraic partners ($\ln x$, $\arctan x$, $\arcsin x$) were dead ends for substitution. By setting $\mathrm{d}v = \mathrm{d}x$, integration by parts manufactured an artificial partner from thin air, liberating inverse functions and logarithms from computational limbo. Furthermore, by generating recursive reduction formulas (such as the Wallis integrals for π), parts established a bridge between continuous integration and discrete sequence analysis.
Yet parts works only if transferring the derivative reduces the algebraic complexity of the integrand. When it does not, parts runs into two distinct failure modes:

1. *Infinite cycling and spiraling complexity:* While combinations like $e^x \sin x$ cycle cleanly and can be solved by algebraic rearrangement, other simple-looking products spiral into worse expressions. Consider $\int \frac{e^x}{x} \,\mathrm{d}x$: setting $u = 1/x$ produces $-\int \frac{e^x}{x^2} \,\mathrm{d}x$, increasing the negative power without mercy; setting $u = e^x$ produces $\int e^x \ln x \,\mathrm{d}x$. Parts cannot simplify functions where both factors resist polynomial annihilation.

2. *Rational quotients:* Integrands consisting of polynomial fractions, such as $\int \frac{x^3+5x-2}{x^4-16} \,\mathrm{d}x$ or $\int \frac{1}{x^2+1} \,\mathrm{d}x$, completely defy integration by parts. Choosing u as the numerator and $\mathrm{d}v$ as the denominator (or vice versa) leaves higher-degree rational functions or complicated logarithms that refuse to simplify.

To conquer rational quotients and intricate trigonometric combinations, Chapter 24 introduces *Partial Fractions and Trigonometric Integrals*. By deploying the Fundamental Theorem of Algebra, Chapter 24 decomposes any rational function $P(x)/Q(x)$ into simple linear and quadratic fractions whose antiderivatives are elementary logarithms and arctangents, providing a complete, deterministic algorithm for the entire universe of rational functions.

Exercises

Exercise 23.1 (Warm-ups)**.** Evaluate each indefinite integral using integration by parts. State your choice of u and $\mathrm{d}v$ each time.

(a) $\displaystyle\int x\cos x\,\mathrm{d}x$

(b) $\displaystyle\int xe^{-x}\,\mathrm{d}x$

(c) $\displaystyle\int x\,\ln x\,\mathrm{d}x$

(d) $\displaystyle\int \arcsin x\,\mathrm{d}x$

Exercise 23.2 (Two rounds and tabular)**.** (a) Evaluate $\displaystyle\int x^2e^x\,\mathrm{d}x$ by applying parts twice.

(b) Now redo it with the tabular method and confirm you get the same answer.

(c) Use the tabular method to evaluate $\displaystyle\int x^3\sin x\,\mathrm{d}x$.

Exercise 23.3 (Definite integrals)**.** Evaluate, simplifying fully:

(a) $\displaystyle\int_0^1 xe^x\,\mathrm{d}x$

(b) $\displaystyle\int_1^e x^2\ln x\,\mathrm{d}x$

(c) $\displaystyle\int_0^\pi x\sin x\,\mathrm{d}x$

Exercise 23.4 (The recurring trick)**.** (a) Evaluate $\displaystyle\int e^x\sin x\,\mathrm{d}x$ by applying parts twice and solving for the integral. (Be consistent in your choice of u.)

(b) Evaluate $\displaystyle\int e^{2x}\cos(3x)\,\mathrm{d}x$ the same way.

(c) Explain in one or two sentences why choosing u inconsistently makes the method collapse to $0=0$.

Exercise 23.5 (Reduction formulas)**.** (a) Derive the reduction formula $\displaystyle\int(\ln x)^n\,\mathrm{d}x = x(\ln x)^n - n\int(\ln x)^{n-1}\,\mathrm{d}x$.

(b) Use it to evaluate $\displaystyle\int(\ln x)^2\,\mathrm{d}x$.

(c) Derive the reduction formula $\displaystyle\int\cos^n x\,\mathrm{d}x = \frac{1}{n}\cos^{n-1}x\,\sin x + \frac{n-1}{n}\int\cos^{n-2}x\,\mathrm{d}x$, mirroring the $\sin^n$ derivation in Example 23.7.

Exercise 23.6 (Combining with substitution)**.** Each of these needs a substitution *first*, then parts. Find them.

(a) $\displaystyle\int \cos(\sqrt{x})\,\mathrm{d}x$ (try $w = \sqrt{x}$).

(b) $\displaystyle\int x^3 e^{x^2}\,\mathrm{d}x$ (try $w = x^2$, then parts).

(c) $\displaystyle\int \sin(\ln x)\,\mathrm{d}x$ (try $w = \ln x$, then the recurring trick).

Exercise 23.7 (Conceptual)**.** (a) A student integrates $\int xe^x\,\mathrm{d}x$ by choosing $u = e^x$, $\mathrm{d}v = x\,\mathrm{d}x$ and gets a *worse* integral. Explain, in terms of the two principles in Section 23.3, exactly what went wrong.

(b) Explain why $\int \sec x\,\mathrm{d}x$ cannot be cracked by the "$u = \sec x$, $\mathrm{d}v = \mathrm{d}x$" trick that worked for $\ln x$ and $\arctan x$. What is different about $\sec x$? (You do not have to evaluate it.)

(c) Why does the geometric "two ways of slicing a rectangle" proof in Section 23.2 require u and v to be increasing, while the algebraic derivation does not?

Exercise 23.8 (A proof)**.** Prove the **Wallis recursion** for the definite integrals $W_n = \displaystyle\int_0^{\pi/2} \sin^n x\,\mathrm{d}x$: show that $W_n = \dfrac{n-1}{n}\,W_{n-2}$ for $n \geq 2$. Begin from the reduction formula in Example 23.7 and explain carefully why the boundary term $-\frac{1}{n}\sin^{n-1}x\cos x$ vanishes at *both* endpoints $x = 0$ and $x = \pi/2$. Then compute W_0 and W_1, and use the recursion to find W_4 and W_5.

Exercise 23.9 (Going further)**.** Use integration by parts to establish the formula for "raising" a power of sec:

$$\int \sec^n x\,\mathrm{d}x = \frac{\sec^{n-2}x\,\tan x}{n-1} + \frac{n-2}{n-1}\int \sec^{n-2}x\,\mathrm{d}x \qquad (n \geq 2).$$

Hint: write $\sec^n x = \sec^{n-2}x \cdot \sec^2 x$, let $\mathrm{d}v = \sec^2 x\,\mathrm{d}x$ (so $v = \tan x$), and use $\tan^2 x = \sec^2 x - 1$. You will need to solve for the recurring integral, just as in Section 23.6.

Chapter 24

Partial Fractions and Trigonometric Integrals

By now you own three serious techniques of integration. Substitution runs the chain rule backwards. Integration by parts runs the product rule backwards. And underneath both sits a single quiet idea that this whole chapter is going to make loud: *we almost never integrate the integrand we are handed.* We reshape it. We push and pull on it with algebra and trigonometry until it turns into something we already recognize – a power, a logarithm, an arctangent – and then we read off the answer.

That reshaping is the real subject here. We will meet two families of integrands that look hopeless at first and become routine once you know the trick. The first is *rational functions*, ratios of polynomials like $\dfrac{3x+5}{x^2-x-2}$. The cure is *partial fractions*: a way of breaking one ugly fraction into a sum of simple ones, each of which integrates to a logarithm or an arctangent. The second is integrals built from *powers and products of sines and cosines*, and then a beautiful move called *trigonometric substitution* that turns a stubborn algebraic expression like $\sqrt{a^2-x^2}$ into a clean trig expression by drawing a right triangle and reading off its sides.

Two warnings before we start, both encouraging. Nothing in this chapter is new calculus. Every integral we "learn" to do reduces to integrals you already know; the work is algebra and trigonometry in service of calculus. And there is a single picture I want you to hold the whole way through. **Every method in this chapter reshapes the integrand into a known form.** Partial fractions reshapes a ratio into a sum of standard pieces. Trig identities reshape a product into a sum. Trig substitution reshapes a square root into a triangle. Completing the square reshapes a quadratic into a shifted square. Keep asking: *what known form am I steering toward?*

24.1 The Unifying Idea: Reshaping Into Known Forms

Let us name the small set of integrals we are allowed to treat as "known." Everything in this chapter lands on one of these.

Focus

The standard forms we steer toward:

$$\int x^n\,\mathrm{d}x = \frac{x^{n+1}}{n+1}\ (n \neq -1), \qquad \int \frac{\mathrm{d}x}{x} = \ln|x|,$$

$$\int \frac{\mathrm{d}x}{x^2+a^2} = \frac{1}{a}\arctan\frac{x}{a}, \qquad \int \frac{\mathrm{d}x}{\sqrt{a^2-x^2}} = \arcsin\frac{x}{a},$$

together with $\int e^x\,\mathrm{d}x = e^x$, $\int \sin x\,\mathrm{d}x = -\cos x$, $\int \cos x\,\mathrm{d}x = \sin x$, and whatever a single u-substitution can reach from these. *That is the entire target vocabulary.* Each technique below is a translator from a strange integrand into this small language.

Why these particular forms? Because each is a derivative you have already computed and reversed. $\ln|x|$ is there because its derivative is $1/x$; the absolute value is there because ln only eats positives, and $\dfrac{\mathrm{d}}{\mathrm{d}x}\ln|x| = 1/x$ on both sides of zero. The arctangent is there because $\dfrac{\mathrm{d}}{\mathrm{d}x}\arctan x = \dfrac{1}{1+x^2}$, and the a-flavored version is just that rule run through a substitution. The arcsine is there for the same reason with $\dfrac{1}{\sqrt{1-x^2}}$.

Intuition

Think of integration as a language-translation problem, not a calculation. You are fluent in maybe a dozen "words" (the standard forms). An arbitrary integrand is a sentence in a foreign language. You do not invent new grammar to translate it; you rewrite it, clause by clause, using identities you already know, until every clause is a word you recognize. Partial fractions, trig identities, and trig substitution are three dictionaries for three common foreign tongues.

24.2 Rational Functions and the Promise of Partial Fractions

A *rational function* is a quotient of polynomials, $R(x) = P(x)/Q(x)$. We would like to integrate any such thing. Two of them are already easy:

$$\int \frac{\mathrm{d}x}{x-3} = \ln|x-3|, \qquad \int \frac{\mathrm{d}x}{(x-3)^2} = -\frac{1}{x-3}.$$

Both are one substitution ($u = x-3$) away from a standard form. The trouble starts when the denominator factors into *several* different roots, all tangled together, as in

$$\int \frac{3x+5}{x^2-x-2}\,\mathrm{d}x.$$

The integrand as written is not any standard form. But watch what happens if we are clever. The denominator factors: $x^2-x-2 = (x-2)(x+1)$. Suppose, for a moment, that we could write

$$\frac{3x+5}{(x-2)(x+1)} = \frac{A}{x-2} + \frac{B}{x+1}$$

for some constants A and B. Then the integral would split into two pieces we *can* do:

$$\int \frac{A}{x-2}\,\mathrm{d}x + \int \frac{B}{x+1}\,\mathrm{d}x = A\ln|x-2| + B\ln|x+1|.$$

That is the whole promise. If every rational function can be torn into a sum of pieces like $\dfrac{A}{x-r}$ (and a couple of cousins we will meet), then integrating rational functions is finished: each piece is a logarithm or an arctangent, and we are done.

Focus

Partial fraction decomposition is the claim that any proper rational function $P(x)/Q(x)$ can be written as a sum of "simple" fractions, one group for each factor of the denominator. It converts an integral you cannot see into a sum of integrals you can.

Two things must be checked before we trust this. Can we always *find* the constants A, B? And, deeper, does the decomposition always *exist*? We find the constants first – it is satisfying and quick – and then come back and prove the decomposition is guaranteed, because understanding *why it exists* is more illuminating than any single computation.

24.2.1 Finding the constants

Start from the proposed identity and clear the denominator by multiplying both sides by $(x-2)(x+1)$:

$$3x+5 \;=\; A(x+1)+B(x-2).$$

This must hold for *every* x. That gives us freedom: we can plug in any x we like and squeeze out the constants. The smart choices are the values that kill a factor.

Set $x=2$: the B term dies, leaving $3(2)+5=A(2+1)$, so $11=3A$ and $A=\frac{11}{3}$.

Set $x=-1$: the A term dies, leaving $3(-1)+5=B(-1-2)$, so $2=-3B$ and $B=-\frac{2}{3}$.

Intuition

This trick of plugging in the root is sometimes called the "cover-up" method, and it has a clean meaning. To find the coefficient sitting over $(x-2)$, you cover up that factor in the original denominator and evaluate what is left at $x=2$:

$$A=\left.\frac{3x+5}{x+1}\right|_{x=2}=\frac{11}{3},\qquad B=\left.\frac{3x+5}{x-2}\right|_{x=-1}=\frac{2}{-3}=-\frac{2}{3}.$$

Why does covering up work? Multiplying by $(x-2)$ removes that factor from the denominator, and then setting $x=2$ makes every *other* term vanish, because each of them still carries a factor of $(x-2)$. You are isolating one coefficient by evaluating at the precise point where its neighbors disappear.

So the decomposition is

$$\frac{3x+5}{(x-2)(x+1)}=\frac{11/3}{x-2}-\frac{2/3}{x+1},$$

and the integral falls out:

$$\int\frac{3x+5}{x^2-x-2}\,\mathrm{d}x=\frac{11}{3}\ln|x-2|-\frac{2}{3}\ln|x+1|+C.$$

Work it yourself

Check the decomposition without integrating. Put $\dfrac{11/3}{x-2} - \dfrac{2/3}{x+1}$ over the common denominator $(x-2)(x+1)$. The numerator is $\frac{11}{3}(x+1) - \frac{2}{3}(x-2) = \frac{11}{3}x + \frac{11}{3} - \frac{2}{3}x + \frac{4}{3} = 3x+5$. It matches. The algebra of putting fractions over a common denominator is exactly what partial fractions *undoes*; that is why a check is just running the machine forward.

24.2.2 Why the decomposition must exist

It is one thing to find A and B when they are promised to be there. It is another to know they are *always* there, for any denominator. Most courses skip this part. It happens to be the best part, so we will keep it.

Why the technique works

Here is the mechanism, in the cleanest case where the denominator $Q(x) = (x-r_1)(x-r_2)\cdots(x-r_n)$ has n *distinct* real roots and the numerator P has degree less than n. We claim

$$\frac{P(x)}{Q(x)} = \frac{A_1}{x-r_1} + \frac{A_2}{x-r_2} + \cdots + \frac{A_n}{x-r_n}$$

for some constants. Why *must* such constants exist?
It comes down to counting. Clearing denominators turns the claim into

$$P(x) = A_1\,\frac{Q(x)}{x-r_1} + A_2\,\frac{Q(x)}{x-r_2} + \cdots + A_n\,\frac{Q(x)}{x-r_n}.$$

Each $\dfrac{Q(x)}{x-r_k}$ is a polynomial of degree $n-1$. The right side is a combination of n such polynomials, and as we vary the A_k it ranges over some subspace of the polynomials of degree $\le n-1$. That space has dimension n (the coefficients of $1, x, \ldots, x^{n-1}$). We have n knobs A_k and n coefficients to match – a square system. So the question is only whether those n building blocks $\dfrac{Q(x)}{x-r_k}$ are *linearly independent*; if they are, they span everything of degree $\le n-1$, and *every* valid P can be hit.

They are independent, and the cover-up trick proves it. Suppose a combination $\sum_k c_k \dfrac{Q(x)}{x-r_k}$ is the zero polynomial. Evaluate at $x = r_j$. Every term with $k \neq j$ still contains the factor $(x-r_j)$ and so vanishes there; the surviving $k=j$ term equals $c_j \prod_{m\neq j}(r_j - r_m)$, a product of nonzero numbers (the roots are distinct). For the whole thing to be zero we need $c_j = 0$. This holds for every j, so all coefficients are zero: the building blocks are independent. A square system with independent columns has exactly one solution for any right-hand side. **The decomposition exists, and it is unique.**

That argument is worth rereading, because it explains the cover-up method at the same time as it proves existence: the very evaluation that isolates A_j is the evaluation that shows the building blocks cannot cancel each other.

Connection

This is a linear-algebra fact wearing an algebra costume. "n independent vectors in an n-dimensional space span it" is the engine. Partial fractions is one of many places in calculus where a counting/dimension argument guarantees that a desired representation exists before you ever compute it – the same flavor of reasoning that guarantees a Taylor polynomial of degree n can match $n+1$ pieces of data, or that an interpolating polynomial through $n+1$ points exists.

History

The decomposition of rational functions into simple fractions is entangled with one of the most famous blunders in the history of mathematics – committed by none other than *Gottfried Wilhelm Leibniz*.

In a landmark 1702 paper in the *Acta Eruditorum*, Leibniz sought to determine whether *every* rational function could be integrated into elementary logarithms and arctangents. He believed he had found an insurmountable counterexample in the innocent-looking fraction:

$$\frac{1}{x^4+1}.$$

Leibniz factored the denominator over the complex numbers as $(x^2+i)(x^2-i)$. Because $\pm i$ are imaginary, he argued that these quadratic factors could never be recombined into real quadratic polynomials. He concluded that x^4+1 could not be factored over the real numbers, and that its integral could therefore never be evaluated using real logarithms or arctangents!

Leibniz was completely wrong. That same year, *Johann Bernoulli* recognized that complex roots always occur in conjugate pairs. Later, in 1748, *Leonhard Euler* showed the clean real factorization that Leibniz had missed by simply adding and subtracting $2x^2$:

$$x^4+1 = (x^2+1)^2 - 2x^2 = (x^2+\sqrt{2}x+1)(x^2-\sqrt{2}x+1).$$

Both quadratic factors are strictly real! The integral $\int \frac{dx}{x^4+1}$ splits neatly into partial fractions and yields elementary logarithms and arctangents.

Bernoulli boldly asserted that *every* real polynomial factors into linear and irreducible real quadratic factors. The rigorous proof of this claim had to wait for Carl Friedrich Gauss's 1799 proof of the *Fundamental Theorem of Algebra*. Thus, the universal algorithm for integrating rational functions stood as a working masterpiece for nearly a century before its algebraic foundation was fully secured.

24.3 The Four Cases of Partial Fractions

To decompose *any* proper rational function you handle the denominator's factors one type at a time. Real-coefficient polynomials factor completely into linear factors $(x-r)$ and irreducible quadratics (x^2+px+q) (those with no real root), each possibly repeated. That gives exactly four kinds of pieces.

Focus

The setup rules. First make the fraction *proper* (numerator degree < denominator degree) by long division if needed. Then for each factor of $Q(x)$, add terms:

- Distinct linear factor $(x-r)$: one term $\dfrac{A}{x-r}$.
- Repeated linear factor $(x-r)^m$: a full tower $\dfrac{A_1}{x-r}+\dfrac{A_2}{(x-r)^2}+\cdots+\dfrac{A_m}{(x-r)^m}$.
- Distinct irreducible quadratic (x^2+px+q): one term $\dfrac{Bx+C}{x^2+px+q}$ (a linear numerator, because the denominator is degree two).
- Repeated irreducible quadratic $(x^2+px+q)^m$: a tower $\dfrac{B_1x+C_1}{x^2+px+q}+\cdots+\dfrac{B_mx+C_m}{(x^2+px+q)^m}$.

The rule about repeated factors deserves a word of intuition, because students forget the lower powers.

Intuition

Why must $(x-r)^3$ contribute three terms, $\dfrac{A_1}{x-r}+\dfrac{A_2}{(x-r)^2}+\dfrac{A_3}{(x-r)^3}$, and not just the top one? Because you need enough freedom. A general numerator over $(x-r)^3$ can be *any* quadratic $a+b(x-r)+c(x-r)^2$ near $x=r$. Dividing that by $(x-r)^3$ gives exactly $\dfrac{a}{(x-r)^3}+\dfrac{b}{(x-r)^2}+\dfrac{c}{x-r}$. The three terms are the three coefficients of that local quadratic. Drop any one and you lose a degree of freedom you might need to match the original numerator.

24.3.1 Why a quadratic factor gets a linear numerator

Over an irreducible quadratic the numerator is $Bx+C$, not just a constant. The reason is the same counting principle: a piece $\dfrac{N(x)}{x^2+px+q}$ is "proper" as long as $\deg N<2$, so the most general allowed numerator has *two* free constants, B and C. Using only a constant would throw away half the freedom and the system would have no solution in general.

Common trap

The three classic errors:

1. Forgetting to do long division first when the fraction is *improper*. $\dfrac{x^3}{x^2-1}$ is not proper; you must divide to get $x+\dfrac{x}{x^2-1}$ before decomposing.
2. Writing only $\dfrac{A}{(x-r)^m}$ for a repeated factor and omitting the lower powers. The tower must be complete.
3. Putting a constant A over an irreducible quadratic instead of $Bx+C$.

Each of these makes the linear system unsolvable – a useful early warning: if your constants

come out inconsistent, suspect a malformed setup.

24.3.2 Worked example: a repeated linear factor

Example 24.1 (Repeated root)**.** Integrate $\displaystyle\int \frac{x+4}{(x-1)^2(x+2)}\,\mathrm{d}x$.

The denominator has a repeated factor $(x-1)^2$ and a single factor $(x+2)$, so the setup is

$$\frac{x+4}{(x-1)^2(x+2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{D}{x+2}.$$

Clear denominators:

$$x+4 = A(x-1)(x+2) + B(x+2) + D(x-1)^2.$$

Use the cover-up values for the easy constants. Set $x=1$: $1+4 = B(3)$, so $B = \frac{5}{3}$. Set $x=-2$: $-2+4 = D(-3)^2$, so $2 = 9D$ and $D = \frac{2}{9}$. The constant A has no root to cover up (it sits over the *first* power of a repeated factor), so match one more coefficient. The x^2 coefficient on the left is 0; on the right it is $A + D$. Hence $A = -D = -\frac{2}{9}$.

So

$$\frac{x+4}{(x-1)^2(x+2)} = \frac{-2/9}{x-1} + \frac{5/3}{(x-1)^2} + \frac{2/9}{x+2},$$

and integrating term by term,

$$\int \frac{x+4}{(x-1)^2(x+2)}\,\mathrm{d}x = -\frac{2}{9}\ln|x-1| - \frac{5/3}{x-1} + \frac{2}{9}\ln|x+2| + C.$$

Notice the middle term: the integral of $1/(x-1)^2$ is $-1/(x-1)$, a power, not a logarithm. Repeated factors produce these non-log pieces, which is exactly why their tower matters.

24.3.3 Worked example: an irreducible quadratic

Example 24.2 (Quadratic factor)**.** Integrate $\displaystyle\int \frac{2x^2 - x + 4}{x(x^2+4)}\,\mathrm{d}x$.

Set up with a linear numerator over the quadratic:

$$\frac{2x^2 - x + 4}{x(x^2+4)} = \frac{A}{x} + \frac{Bx + C}{x^2+4}.$$

Clear denominators: $2x^2 - x + 4 = A(x^2+4) + (Bx+C)x$. Set $x=0$: $4 = 4A$, so $A = 1$. Match x^2: $2 = A + B$, so $B = 1$. Match x^1: $-1 = C$. Thus

$$\frac{2x^2 - x + 4}{x(x^2+4)} = \frac{1}{x} + \frac{x-1}{x^2+4}.$$

Now integrate. The first piece is a logarithm. The second piece we *split* into a part whose numerator is the derivative of the denominator and a leftover:

$$\int \frac{x-1}{x^2+4}\,\mathrm{d}x = \int \frac{x}{x^2+4}\,\mathrm{d}x - \int \frac{1}{x^2+4}\,\mathrm{d}x.$$

For the first, $u = x^2 + 4$, $\mathrm{d}u = 2x\,\mathrm{d}x$, giving $\frac{1}{2}\ln(x^2+4)$. The second is a standard arctangent with $a = 2$: $\frac{1}{2}\arctan(x/2)$. Altogether,

$$\int \frac{2x^2 - x + 4}{x(x^2+4)}\,\mathrm{d}x = \ln|x| + \frac{1}{2}\ln(x^2+4) - \frac{1}{2}\arctan\frac{x}{2} + C.$$

What is really happening

Look at what the quadratic piece actually became: *a logarithm plus an arctangent.* This is the universal fate of an irreducible-quadratic term, and there is a reason. The Bx part of the numerator, after you arrange it to be a multiple of the denominator's derivative $2x$, integrates to $\ln(x^2+4)$ – that is substitution. The leftover constant part has no x on top, so it has nothing to substitute and instead lands on the arctangent form. Every irreducible quadratic splits its numerator into a "derivative-of-denominator" half (a log) and a "pure constant" half (an arctangent). Once you see this split you can integrate any quadratic term on autopilot.

24.3.4 Worked example: improper, needing division first

Example 24.3 (Improper fraction)**.** Integrate $\displaystyle\int \frac{x^3}{x^2-1}\,\mathrm{d}x$.

The numerator degree (3) exceeds the denominator degree (2), so do polynomial long division first: $x^3 = x(x^2-1) + x$, hence

$$\frac{x^3}{x^2-1} = x + \frac{x}{x^2-1}.$$

The leftover $\dfrac{x}{x^2-1} = \dfrac{x}{(x-1)(x+1)}$ decomposes as $\dfrac{1/2}{x-1} + \dfrac{1/2}{x+1}$ (cover-up: at $x=1$, $\frac{1}{2}$; at $x=-1$, $\frac{-1}{-2} = \frac{1}{2}$). So

$$\int \frac{x^3}{x^2-1}\,\mathrm{d}x = \frac{x^2}{2} + \frac{1}{2}\ln|x-1| + \frac{1}{2}\ln|x+1| + C = \frac{x^2}{2} + \frac{1}{2}\ln|x^2-1| + C.$$

Walking question

On your walk, picture why long division is unavoidable. A proper fraction P/Q with $\deg P < \deg Q$ shrinks toward 0 as $x \to \infty$. But $x^3/(x^2-1) \approx x$ for large x – it grows without bound. No sum of pieces like $\dfrac{A}{x-r}$ (each shrinking to zero) could ever reproduce that growth. The division peels off the polynomial part x that carries the growth, leaving a genuinely proper remainder for the fractions to handle. The methods only work on the part of the function that actually decays.

24.3.5 The shape of the picture

It helps to *see* a decomposition. Below, the rational function $\dfrac{3x+5}{(x-2)(x+1)}$ from earlier is drawn together with its two simple pieces. Near each vertical asymptote, one piece dominates and the other

is nearly flat – the decomposition is the curve's way of saying "near $x = 2$ I behave like $\frac{11/3}{x-2}$, near $x = -1$ I behave like $\frac{-2/3}{x+1}$."

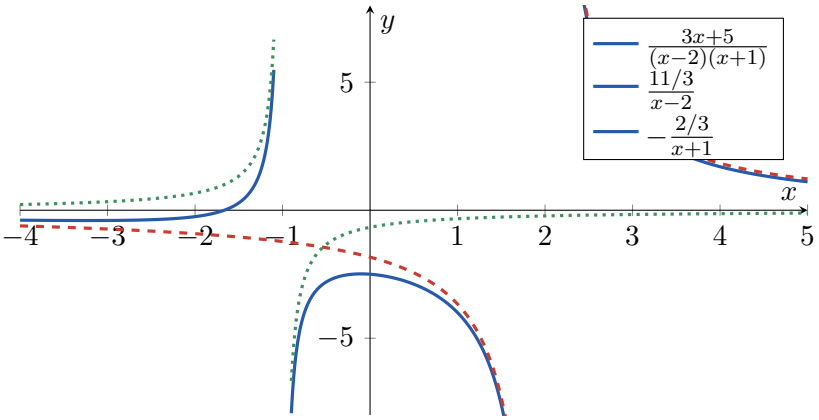


Figure 24.1: A rational function (solid) and its two partial-fraction pieces (dashed, dotted). Each piece controls the behavior near its own asymptote; the sum reproduces the original everywhere.

24.4 Trigonometric Integrals: Powers and Products

We turn now to integrals built from sines and cosines, like $\int \sin^5 x\,dx$ or $\int \sin^2 x \cos^2 x\,dx$ or $\int \sin 3x \cos 5x\,dx$. None is a standard form as written. But every one yields to the same move: *use a trig identity to reshape the integrand into something a substitution can finish.* There is no new calculus, only the Pythagorean identity, the double-angle formulas, and the product-to-sum formulas, deployed cleverly.

24.4.1 Odd powers: peel off one factor

Example 24.4 (Odd power of sine)**.** Integrate $\displaystyle\int \sin^5 x\,dx$.

There is an *odd* number of sines, so peel one off and convert the rest to cosines using $\sin^2 x = 1-\cos^2 x$. The lone sine becomes the du for $u = \cos x$.

$$\int \sin^5 x\,dx = \int (\sin^2 x)^2 \sin x\,dx = \int (1-\cos^2 x)^2 \sin x\,dx.$$

Substitute $u = \cos x$, $du = -\sin x\,dx$:

$$= -\int (1-u^2)^2\,du = -\int (1-2u^2+u^4)\,du = -\left(u - \tfrac{2}{3}u^3 + \tfrac{1}{5}u^5\right) + C.$$

Back-substitute $u = \cos x$:

$$\int \sin^5 x\,dx = -\cos x + \frac{2}{3}\cos^3 x - \frac{1}{5}\cos^5 x + C.$$

Why the technique works

Why does peeling one factor work, and why does it need an *odd* power? Substitution $u = \cos x$ needs $\mathrm{d}u = -\sin x\,\mathrm{d}x$ – it consumes exactly *one* sine. After spending that one, the remaining sines must be expressible in $u = \cos x$, and $\sin^2 x = 1 - \cos^2 x$ does that perfectly because it only ever uses sine in *even* amounts. So you need one sine to spend on $\mathrm{d}u$ and an even number left over to convert: total odd. If the power were even, you would have a leftover lone sine with nothing to pair it with, and the substitution stalls. Odd powers are easy precisely because they split as "one for $\mathrm{d}u$, the rest even."

The same trick handles odd powers of cosine (peel one cosine, set $u = \sin x$) and mixed products $\int \sin^m x \cos^n x\,\mathrm{d}x$ *whenever at least one of* m, n *is odd*: peel one factor off the odd one, convert the rest, substitute for the other function.

Example 24.5 (Mixed product, one odd power)**.** $\displaystyle\int \sin^2 x \cos^3 x\,\mathrm{d}x$. The cosine power is odd. Peel one cosine, convert the rest with $\cos^2 x = 1 - \sin^2 x$, set $u = \sin x$:

$$\int \sin^2 x(1 - \sin^2 x)\cos x\,\mathrm{d}x = \int u^2(1 - u^2)\,\mathrm{d}u = \frac{u^3}{3} - \frac{u^5}{5} = \frac{\sin^3 x}{3} - \frac{\sin^5 x}{5} + C.$$

24.4.2 Even powers: the half-angle slowdown

When *both* powers are even, the peel-one trick fails – there is no spare factor to be a $\mathrm{d}u$. Instead we lower the powers using the half-angle identities, which are the double-angle formula solved for the squares:

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \qquad \cos^2 x = \frac{1 + \cos 2x}{2}.$$

Each application trades a squared sine or cosine for a plain cosine of double the angle – lower power, in exchange for a faster oscillation. Repeat until nothing is squared.

Example 24.6 (Even power)**.** Integrate $\displaystyle\int \sin^2 x\,\mathrm{d}x$.

$$\int \sin^2 x\,\mathrm{d}x = \int \frac{1 - \cos 2x}{2}\,\mathrm{d}x = \frac{1}{2}\Big(x - \frac{\sin 2x}{2}\Big) + C = \frac{x}{2} - \frac{\sin 2x}{4} + C.$$

Intuition

There is a sanity check hiding here. The average value of $\sin^2 x$ over a full period is $\frac{1}{2}$ – because $\sin^2 + \cos^2 = 1$ and, by symmetry, $\sin^2$ and $\cos^2$ have the same average, so each must average $\frac{1}{2}$. Our antiderivative $\frac{x}{2} - \frac{\sin 2x}{4}$ grows like $\frac{x}{2}$ with a small wobble on top, so over a period it accumulates exactly $\frac{1}{2} \cdot (\text{length})$. The formula encodes the average. This "$\sin^2$ averages to a half" fact is the reason the RMS of an alternating current is the peak over $\sqrt{2}$.

Example 24.7 (Both even)**.** $\displaystyle\int \sin^2 x \cos^2 x\,\mathrm{d}x$. The slick route uses $\sin x \cos x = \frac{1}{2}\sin 2x$, so $\sin^2 x \cos^2 x = \frac{1}{4}\sin^2 2x$. Then half-angle again on $\sin^2 2x$:

$$\int \tfrac{1}{4}\sin^2 2x\,\mathrm{d}x = \frac{1}{4}\int \frac{1 - \cos 4x}{2}\,\mathrm{d}x = \frac{x}{8} - \frac{\sin 4x}{32} + C.$$

24.4.3 Products of different frequencies

For $\int \sin(mx)\cos(nx)\,dx$ with $m \neq n$ there are no powers at all, just a product of two waves at different frequencies. The reshaping tool is the *product-to-sum* identities:

$$\sin A\cos B = \tfrac{1}{2}\big[\sin(A+B) + \sin(A-B)\big],$$

and its siblings for $\cos A\cos B$ and $\sin A\sin B$. A product of two waves becomes a *sum* of two waves, and a sum is trivial to integrate.

Example 24.8. $\displaystyle\int \sin 3x\cos 5x\,dx = \frac{1}{2}\int \big[\sin 8x + \sin(-2x)\big]\,dx = -\frac{\cos 8x}{16} + \frac{\cos 2x}{4} + C.$

Connection

These product-to-sum integrals are the computational core of **Fourier series**. The reason you can extract the "amount of frequency n" inside a signal is that $\int_0^{2\pi} \sin(mx)\sin(nx)\,dx = 0$ whenever $m \neq n$ and equals π when $m = n$. That orthogonality is exactly the product-to-sum computation above: different frequencies multiply to a sum of waves that integrates to zero over a period; the same frequency multiplies to something with a nonzero average. Every time you compress audio or an image, this integral is doing the work.

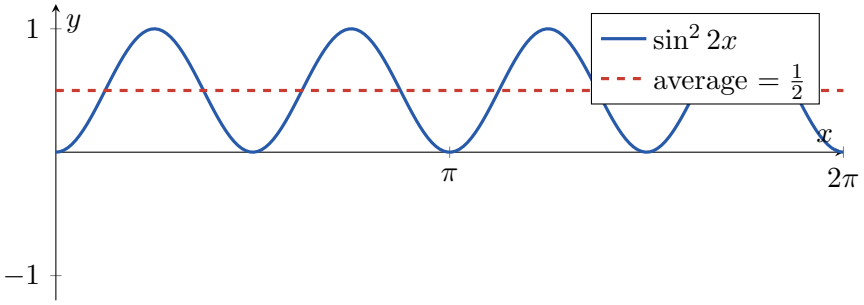


Figure 24.2: A squared sine oscillates between 0 and 1 but spends its time symmetrically, averaging $\frac{1}{2}$. The half-angle identity $\sin^2 2x = \frac{1-\cos 4x}{2}$ makes this visible: a constant $\frac{1}{2}$ plus a zero-average wobble.

24.5 Completing the Square: Taming Any Quadratic

Before trig substitution we need one small, powerful algebraic move. Many integrals hide a quadratic in the denominator or under a root that is not quite a standard form, like

$$\int \frac{dx}{x^2 + 4x + 13}.$$

The denominator is not $x^2 + a^2$, so the arctangent form does not apply directly. But *completing the square* reshapes any quadratic into a perfect square plus (or minus) a constant:

$$x^2 + 4x + 13 = (x^2 + 4x + 4) + 9 = (x+2)^2 + 9 = (x+2)^2 + 3^2.$$

Now substitute $u = x + 2$ (a pure shift, $\mathrm{d}u = \mathrm{d}x$) and it *is* the arctangent form with $a = 3$:

$$\int \frac{\mathrm{d}x}{(x+2)^2 + 3^2} = \frac{1}{3}\arctan\frac{x+2}{3} + C.$$

What is really happening

Completing the square is a *horizontal shift in disguise*. The graph of $y = x^2 + 4x + 13$ is the parabola $y = x^2 + 9$ slid two units to the left. The substitution $u = x + 2$ slides our coordinate system to sit at the bottom of the parabola, where the quadratic is symmetric and bare: $u^2 + 9$. Every quadratic is some standard parabola, just shifted; completing the square finds the shift, and the substitution undoes it. After that, the integrand is centered and the standard form snaps into place.

Example 24.9 (Linear numerator over a completed square)**.** $\displaystyle\int \frac{x+1}{x^2+4x+13}\,\mathrm{d}x$. Split the numerator into a multiple of the denominator's derivative $2x+4$ plus a leftover constant: $x+1 = \frac{1}{2}(2x+4) - 1$. Then

$$\int \frac{x+1}{x^2+4x+13}\,\mathrm{d}x = \frac{1}{2}\int \frac{2x+4}{x^2+4x+13}\,\mathrm{d}x - \int \frac{\mathrm{d}x}{x^2+4x+13}.$$

The first integral is $\frac{1}{2}\ln(x^2+4x+13)$ by substitution; the second is the arctangent we just did. So the answer is

$$\frac{1}{2}\ln(x^2+4x+13) - \frac{1}{3}\arctan\frac{x+2}{3} + C.$$

This "log half plus arctangent half" split is the same one from the partial fractions section – another sign that all these techniques are one idea.

24.6 Trigonometric Substitution: Algebra Becomes Geometry

The last technique is the most geometric one in the chapter. Integrals containing $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, or $\sqrt{x^2 - a^2}$ resist ordinary substitution because the square root has no convenient derivative hiding in the integrand. The escape is to *change variables so that the expression under the root becomes a perfect square*, using the Pythagorean identities. And the cleanest way to see which substitution to use, and how to get back, is to *draw a right triangle*.

24.6.1 The idea on a triangle

Consider $\sqrt{a^2 - x^2}$. The relation $a^2 = x^2 + (\sqrt{a^2 - x^2})^2$ is the Pythagorean theorem: a is a hypotenuse, x and $\sqrt{a^2 - x^2}$ are the two legs. So *draw* that triangle and let θ be the angle whose *opposite leg* is x:

$$\sin\theta = \frac{x}{a} \implies x = a\sin\theta, \qquad \sqrt{a^2 - x^2} = a\cos\theta.$$

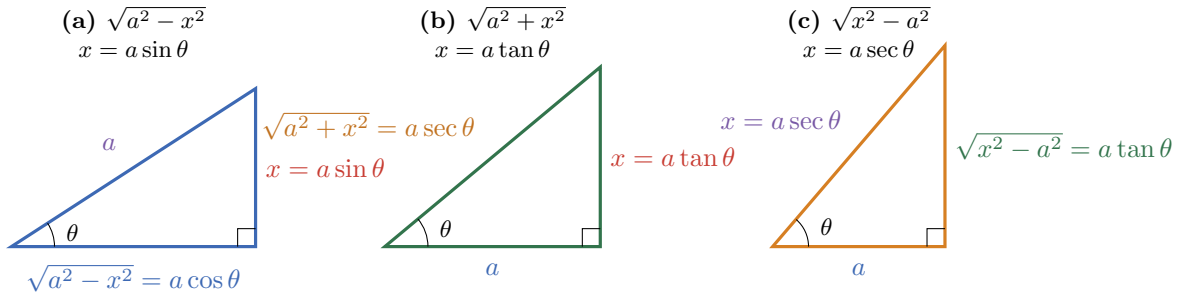


Figure 24.3: The three canonical right-triangle substitutions. Each substitution is engineered so the Pythagorean theorem collapses the square root to a single trigonometric function: **(a)** $x = a\sin\theta$ yields $a^2 - x^2 = a^2\cos^2\theta$; **(b)** $x = a\tan\theta$ yields $a^2 + x^2 = a^2\sec^2\theta$; **(c)** $x = a\sec\theta$ yields $x^2 - a^2 = a^2\tan^2\theta$. Reading the triangle's ratios translates any trig expression of θ back to x.

The triangle is not decoration; it is the whole method made visible. It tells you the substitution ($x = a\sin\theta$), it tells you $\mathrm{d}x = a\cos\theta\,\mathrm{d}\theta$, it converts the root to $a\cos\theta$, and most importantly, it tells you how to convert the *answer* back to x at the end, because every trig function of θ is just a ratio of the triangle's sides.

Focus

The three triangles.

- $\sqrt{a^2 - x^2}$: set $x = a\sin\theta$, then root $= a\cos\theta$. ($\sin^2 + \cos^2 = 1$, rearranged.)
- $\sqrt{a^2 + x^2}$: set $x = a\tan\theta$, then root $= a\sec\theta$. ($1 + \tan^2 = \sec^2$.)
- $\sqrt{x^2 - a^2}$: set $x = a\sec\theta$, then root $= a\tan\theta$. ($\sec^2 - 1 = \tan^2$.)

In each case the substitution is the one that makes a Pythagorean identity collapse the root into a single trig function. Draw the triangle and you never have to memorize which is which.

Why the technique works

Why does turning x into a trig function help at all? Because the obstacle is the square root, and a square root vanishes only when its inside is a perfect square. The Pythagorean identities are precisely the algebraic statements that $a^2 - (a\sin\theta)^2$, $a^2 + (a\tan\theta)^2$, and $(a\sec\theta)^2 - a^2$ are each a perfect square ($a^2\cos^2\theta$, $a^2\sec^2\theta$, $a^2\tan^2\theta$). Trig substitution is the deliberate choice of a new variable engineered so that the thing under the root becomes a square and the root opens up cleanly. It is substitution aimed by geometry.

24.6.2 Worked example: the area of a circle, recovered

Example 24.10 (Root of $a^2 - x^2$)**.** Integrate $\displaystyle\int \sqrt{a^2 - x^2}\,\mathrm{d}x$.

This integrand is the height of a circle of radius a, so the integral is a circular area (Figure 24.4); watch the geometry fall out of the algebra. Use the triangle: $x = a\sin\theta$, $\mathrm{d}x = a\cos\theta\,\mathrm{d}\theta$, and

$\sqrt{a^2-x^2} = a\cos\theta$. The integral becomes pure trig:

$$\int a\cos\theta \cdot a\cos\theta\,\mathrm{d}\theta = a^2\int \cos^2\theta\,\mathrm{d}\theta = a^2\Big(\frac{\theta}{2} + \frac{\sin 2\theta}{4}\Big) + C,$$

using the even-power technique from before. Now convert back via the triangle. From $\sin\theta = x/a$ we get $\theta = \arcsin(x/a)$, and $\sin 2\theta = 2\sin\theta\cos\theta = 2\cdot\frac{x}{a}\cdot\frac{\sqrt{a^2-x^2}}{a} = \frac{2x\sqrt{a^2-x^2}}{a^2}$. Therefore

$$\int \sqrt{a^2-x^2}\,\mathrm{d}x = \frac{a^2}{2}\arcsin\frac{x}{a} + \frac{x}{2}\sqrt{a^2-x^2} + C.$$

Why this matters

Test this on a whole circle. The area inside $x^2+y^2=a^2$ is four times the area under $y=\sqrt{a^2-x^2}$ from 0 to a:

$$4\int_0^a \sqrt{a^2-x^2}\,\mathrm{d}x = 4\left[\frac{a^2}{2}\arcsin\frac{x}{a} + \frac{x}{2}\sqrt{a^2-x^2}\right]_0^a.$$

At $x=a$: $\arcsin 1 = \frac{\pi}{2}$ and the square-root term is 0. At $x=0$: everything is 0. So the area is $4\cdot\frac{a^2}{2}\cdot\frac{\pi}{2} = \pi a^2$. The most famous formula in geometry falls out of a triangle substitution. The arcsin that appears is *the angle*, and integrating around the circle is literally accumulating that angle.

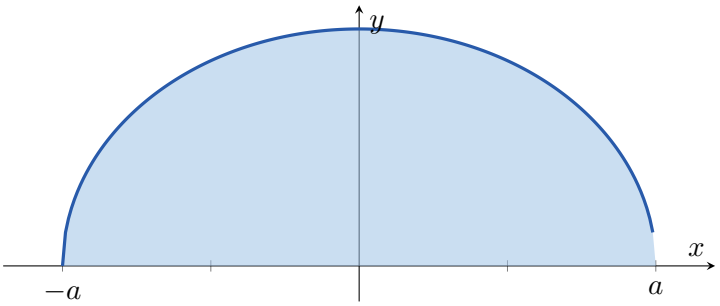


Figure 24.4: The integral $\int_{-a}^{a}\sqrt{a^2-x^2}\,\mathrm{d}x$ is the shaded area under the upper semicircle of radius a (here $a=2$). The trig-substitution antiderivative evaluates it to $\frac{\pi a^2}{2}$, the half-disc area; double it (or take four quarters, as in the box above) and the full πa^2 appears.

24.6.3 Worked example: a secant integral via the tangent triangle

Example 24.11 (Root of a^2+x^2)**.** Integrate $\displaystyle\int \frac{\mathrm{d}x}{\sqrt{x^2+9}}$.

Here $a=3$, root is $\sqrt{a^2+x^2}$, so the tangent triangle: $x = 3\tan\theta$, $\mathrm{d}x = 3\sec^2\theta\,\mathrm{d}\theta$, and $\sqrt{x^2+9} = 3\sec\theta$. Then

$$\int \frac{3\sec^2\theta}{3\sec\theta}\,\mathrm{d}\theta = \int \sec\theta\,\mathrm{d}\theta = \ln|\sec\theta+\tan\theta| + C.$$

Convert back with the triangle. With opposite $= x$, adjacent $= 3$, hypotenuse $= \sqrt{x^2+9}$, we have $\tan\theta = x/3$ and $\sec\theta = \sqrt{x^2+9}/3$. So

$$\int \frac{\mathrm{d}x}{\sqrt{x^2+9}} = \ln\left|\frac{\sqrt{x^2+9}}{3} + \frac{x}{3}\right| + C = \ln\left|x + \sqrt{x^2+9}\right| + C',$$

absorbing the constant $-\ln 3$ into C'. (This is also $\operatorname{arcsinh}(x/3)$ up to a constant – the inverse hyperbolic sine – which is no coincidence: hyperbolic functions parametrize $\sqrt{x^2+a^2}$ the way circular ones parametrize $\sqrt{a^2-x^2}$.)

Common trap

The single most forgotten step in trig substitution is **converting back to** x. Your answer must be a function of the original variable; an answer left in θ is unfinished. Always draw the triangle, label its three sides from the substitution, and read off whatever trig functions appear in your antiderivative as ratios of those sides. The triangle is your dictionary back to x.

A second trap: when the integral is *definite*, you may instead change the limits to θ-values and never convert back – often cleaner. Pick one discipline and follow it through.

24.6.4 When the root hides inside a shifted quadratic

Completing the square and trig substitution combine to handle the general quadratic-under-a-root. For $\int \frac{\mathrm{d}x}{\sqrt{x^2+4x+13}}$, first complete the square: $x^2+4x+13 = (x+2)^2+9$. Substitute $u = x+2$ to center it, giving $\int \frac{\mathrm{d}u}{\sqrt{u^2+9}}$, which is exactly the tangent-triangle integral above. Reshape (complete the square), then reshape again (trig substitution): the chapter's whole theme in one problem.

Work it yourself

Do this one by hand to feel the pipeline. Integrate $\displaystyle\int \frac{\mathrm{d}x}{\sqrt{5-4x-x^2}}$.

1. Complete the square inside the root. Write $5-4x-x^2 = -(x^2+4x)+5 = -\big((x+2)^2-4\big)+5 = 9-(x+2)^2$.
2. Substitute $u = x+2$: the integral is $\displaystyle\int \frac{\mathrm{d}u}{\sqrt{9-u^2}}$, a standard arcsine with $a = 3$.
3. Read off $\arcsin(u/3) = \arcsin\dfrac{x+2}{3} + C$.

Notice you did not even need the triangle here – once centered, it was a named standard form. Recognizing the standard form is always the fastest path; the triangle is the fallback when no name fits.

24.7 One Worldview, Several Tools

Step back and look at everything we did. One idea ran through all of it, seen from several angles.

Focus

Every technique reshapes the integrand into a known form.

- *Partial fractions*: a ratio of polynomials → a sum of $\dfrac{A}{(x-r)^k}$ and $\dfrac{Bx+C}{(x^2+px+q)^k}$, each a log, a power, or an arctangent.
- *Trig powers (odd)*: peel a factor → a polynomial in sin or cos, finished by substitution.
- *Trig powers (even)*: half-angle identities → lower powers and doubled angles, repeated to the end.
- *Products of waves*: product-to-sum → a sum of single waves.
- *Completing the square*: a general quadratic → a centered $u^2 \pm a^2$, matching the arctangent or arcsine form.
- *Trig substitution*: a square root → a triangle, where the Pythagorean identity makes the root a perfect square.

The calculus underneath never changes. What changes is your fluency at recognizing which dictionary to open.

History

By the middle of the eighteenth century, Euler had organized essentially all of these techniques into a coherent program. His *Institutiones Calculi Integralis* (1768) treats rational functions by partial fractions, trig integrals by the identities, and irrational integrands by substitutions that rationalize them – including the trigonometric and what we now call Euler substitutions for roots of quadratics. The deep message Euler made explicit is the one we have been pushing: a vast zoo of integrals is tamed not by inventing new operations but by *transforming integrands into a small canon of solvable forms*. The art of integration, as a craft, is largely the art of recognizing which transformation reveals the hidden standard form.

Why this matters

There is something quietly profound in the fact that integrating rational functions never needs anything beyond logarithms and arctangents, and that integrating $\sqrt{\text{quadratic}}$ never needs anything beyond those plus arcsines and the original roots. The "elementary functions" are closed enough to contain the answers to these whole families. This is also why it is so striking when closure *fails* – $\int e^{-x^2}\,dx$ has no elementary antiderivative at all. Knowing exactly which integrands stay inside the elementary world, and which escape it, is itself a deep theorem (Liouville's), and our reshaping techniques are the constructive half of that story: they show you the cases where the answer was always going to be a familiar friend.

24.8 What partial fractions unlocked, the Abel-Ruffini wall, and the road ahead

Connection

The partial fraction decomposition represents the pinnacle of symbolic integration: it is one of the rare domains where calculus provides a completely deterministic, algorithmic solution. Before partial fractions, integrating even simple rational functions like $\int \frac{\mathrm{d}x}{x^2-1}$ or $\int \frac{x\,\mathrm{d}x}{x^2+2x+5}$ required ad-hoc geometric tricks. Partial fractions, combined with polynomial long division, established a complete decision procedure: *every* rational function $P(x)/Q(x)$ can be integrated using exclusively rational terms, logarithms, and arctangents. Coupled with trigonometric substitutions (mapping circles and hyperbolas to trigonometric identities) and the Euler / Weierstrass half-angle substitutions (converting any rational function of sines and cosines into a rational function of $u = \tan(x/2)$), mathematicians possessed a universal solver for geometry, classical celestial mechanics, and engineering circuit models.

Yet where does this algebraic machine break down? The algorithm begins with an innocent prerequisite: *factor the denominator $Q(x)$ into linear and irreducible real quadratic factors.* For degree 1 and 2, factoring is straightforward. For degree 3 and 4, general formulas exist (Cardano's and Ferrari's formulas), though they are excruciatingly complex. But in 1824, Niels Henrik Abel and Paolo Ruffini (and later Évariste Galois) proved a landmark theorem: *there is no general algebraic formula in radicals to solve polynomial equations of degree 5 or higher.*

If your rational function has a denominator such as $Q(x) = x^5 - x - 1$, the Fundamental Theorem of Algebra guarantees that five complex roots exist, and that $Q(x)$ factors over $\mathbb{R}$ into one linear and two real quadratic factors. But Abel-Ruffini proves that you *cannot write down those factors using square roots, cube roots, or any finite algebraic radicals.* If you cannot factor $Q(x)$, you cannot set up the partial-fraction equations, and the entire deterministic integration algorithm grinds to a halt.

The study of integration now splits along two vital frontiers:

- *Singularities and Infinite Horizons (Chapter 25):* Rational functions naturally possess vertical asymptotes where denominators vanish ($x = r$), and long tails as $x \to \infty$. In Chapter 25 (*Improper Integrals*), we investigate what it means to integrate across poles and out to infinity, unlocking the mathematics of probability distributions and physical potentials.
- *Numerical Quadrature (Chapter 27):* When the Abel-Ruffini barrier prevents exact factorization of $Q(x)$, or when an algebraic antiderivative is simply too cumbersome to compute, we turn to Chapter 27 (*Numerical Integration*), where Simpson's Rule computes the definite integral to twelve decimal places in milliseconds without ever finding a single root of $Q(x)$.

24.9 Exercises

Exercise 24.1 (Distinct linear factors)**.** Decompose and integrate $\displaystyle\int \frac{5x-4}{(x-1)(x+3)}\,\mathrm{d}x$. Use the cover-up method to find both constants, and verify your decomposition by recombining over a com-

mon denominator.

Exercise 24.2 (Repeated factor)**.** Integrate $\displaystyle\int \frac{2x^2+3}{(x+1)^3}\,\mathrm{d}x$. Set up the full tower of three terms, find the constants (try the substitution $u = x+1$ as an alternative to matching coefficients), and identify which pieces become logarithms and which become powers.

Exercise 24.3 (Irreducible quadratic)**.** Integrate $\displaystyle\int \frac{x^2+x+1}{x(x^2+1)}\,\mathrm{d}x$. Decompose with a linear numerator over the quadratic, then split that piece into its "log half" and "arctangent half" as in Example 24.2.

Exercise 24.4 (Improper fraction)**.** Integrate $\displaystyle\int \frac{x^2}{x^2-4}\,\mathrm{d}x$. First explain, in one sentence, why long division is required before any decomposition, then carry it out.

Exercise 24.5 (Odd trig power)**.** Integrate $\displaystyle\int \cos^3 x\,\mathrm{d}x$ and $\displaystyle\int \sin^3 x\cos^4 x\,\mathrm{d}x$. State which factor you peel and why the substitution then closes.

Exercise 24.6 (Even trig powers)**.** Integrate $\displaystyle\int \cos^4 x\,\mathrm{d}x$ by applying the half-angle identity twice. Then, without computing, predict the average value of $\cos^4 x$ over a full period and check that your antiderivative's linear part agrees with it.

Exercise 24.7 (Product of waves)**.** Integrate $\displaystyle\int \cos 2x\cos 6x\,\mathrm{d}x$ using a product-to-sum identity. Then explain, using your answer, why $\int_0^{2\pi} \cos 2x\cos 6x\,\mathrm{d}x = 0$, and connect this to the orthogonality of distinct frequencies.

Exercise 24.8 (Completing the square)**.** Integrate $\displaystyle\int \frac{\mathrm{d}x}{x^2-6x+25}$ and $\displaystyle\int \frac{2x-1}{x^2-6x+25}\,\mathrm{d}x$. For the second, split the numerator into a multiple of the denominator's derivative plus a constant.

Exercise 24.9 (Trig substitution)**.** Integrate $\displaystyle\int \frac{x^2}{\sqrt{4-x^2}}\,\mathrm{d}x$ using the sine triangle. Draw the triangle, convert every quantity, integrate the resulting trig expression, and convert your answer back to x explicitly.

Exercise 24.10 (Trig substitution, the other two triangles)**.** Integrate $\displaystyle\int \frac{\mathrm{d}x}{x^2\sqrt{x^2-9}}$ using the secant triangle, and $\displaystyle\int \sqrt{x^2+1}\,\mathrm{d}x$ using the tangent triangle. The second will need $\int \sec^3\theta\,\mathrm{d}\theta$; derive it with one integration by parts.

Exercise 24.11 (Conceptual / proof)**.** Prove that the partial-fraction building blocks $\dfrac{Q(x)}{x-r_k}$, where $Q(x) = \prod_{i=1}^{n}(x-r_i)$ has n distinct roots, are linearly independent as polynomials. (Mimic the cover-up argument of Section 24.2.2.) Explain in a sentence why this independence is exactly what guarantees the partial-fraction decomposition exists for every proper numerator.

Exercise 24.12 (Synthesis)**.** The single integral $\displaystyle\int \frac{\mathrm{d}x}{\sqrt{x^2+2x+2}}$ uses two techniques from this chapter in sequence. Name them in order, carry the computation through to a function of x, and write one sentence explaining how this problem embodies the chapter's slogan that every method reshapes the integrand into a known form.

Chapter 25

Improper Integrals

Every integral you have met so far has been, in a quiet way, a finite object. You integrate a *bounded* function over a *bounded* interval: f does not blow up, and the interval $[a, b]$ has two concrete endpoints you can point to on the number line. Under those conditions the area under the curve is a single finite number, the Riemann sums close in on it, and the Fundamental Theorem hands you the answer through an antiderivative. Comfortable, complete, closed.

This chapter is about what happens when we break one or both of those conditions on purpose. What is the area under a curve that runs off to the right *forever*? What is the area under a curve that shoots up to infinity as you approach an endpoint? At first these questions sound either meaningless or unavoidably infinite – how could an infinitely wide region, or an infinitely tall spike, enclose any finite amount of area?

Focus

The surprise – and it genuinely surprised the people who first met it – is that sometimes an infinitely long or infinitely tall region has a perfectly finite area, and sometimes it doesn't. Whether it does depends on *how fast* the curve shrinks or grows, in a way we can make completely precise. Learning to tell the finite cases from the infinite ones is the central objective of this chapter.

The device that lets us ask these questions rigorously is the same one that made the derivative rigorous: the **limit**. We will never actually "integrate to infinity" or "integrate up to a vertical asymptote." Instead we will integrate over an ordinary, finite, well-behaved region, and then *watch what happens* as we let the edge of that region march off to infinity or creep up to the bad point. The integral over the infinite or unbounded region will be *defined* to be the limit of those ordinary integrals – if the limit exists. That single idea, applied carefully, organizes the entire chapter.

25.1 Two ways an integral can be improper

Two things can go wrong, and naming them cleanly pays off because the cures differ slightly.

Definition 25.1 (Improper integral, informally)**.** An integral $\int_a^b f(x)\,\mathrm{d}x$ is called **improper** if either

1. the interval of integration is infinite – one or both limits are $\pm\infty$ (an improper integral *of the first kind*); or
2. the integrand f is unbounded somewhere on the interval – it has a vertical asymptote at an endpoint or at an interior point (an improper integral *of the second kind*).

An integral can be improper for both reasons at once.

The ordinary Riemann integral simply has nothing to say about either case. Its whole construction – chop $[a, b]$ into finitely many pieces, build rectangles, take a limit of the sums – assumes a finite interval and a bounded function. If the interval is $[1, \infty)$ there is no finite chopping to do; if f blows up at $x = 0$, no finite collection of rectangles can pin down the area near the spike. So we are not "computing a Riemann integral" in these cases. We are *defining a new kind of object* and choosing, deliberately, to define it as a limit of ordinary integrals.

Intuition

Think of it as photographing a landscape that extends past the horizon. You cannot fit the whole thing in one frame. So you take a frame that stops at distance t, measure the area it contains, then slide the right edge of the frame out – $t = 10$, $t = 100$, $t = 10000$ – and ask whether the measured area is settling down toward a fixed number or growing without bound. The improper integral is the number it settles toward.

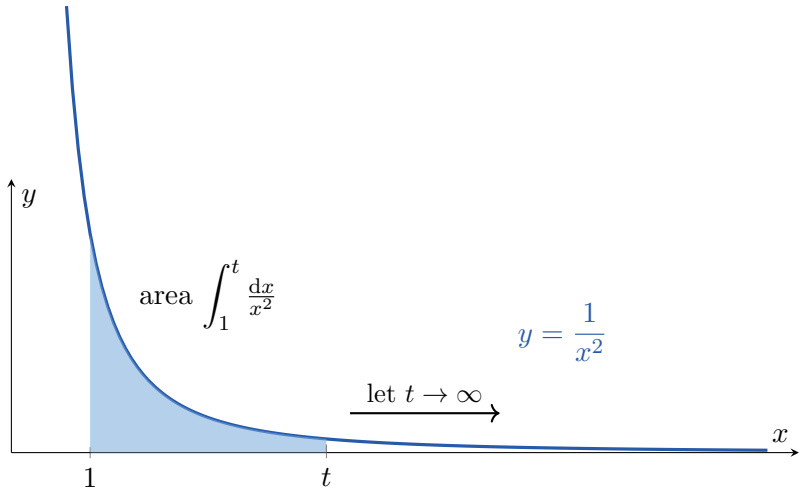


Figure 25.1: The frame that stops at $x = t$. We measure a finite area, then slide the right edge t outward and ask what number the area approaches.

25.2 Infinite intervals: the first kind

Here is the definition, stated for the most common case of a single infinite endpoint on the right.

Definition 25.2 (Improper integral over $[a, \infty)$)**.** Let f be defined on $[a, \infty)$ and integrable on every finite subinterval $[a, t]$. We define

$$\int_a^{\infty} f(x)\,dx \;=\; \lim_{t\to\infty} \int_a^t f(x)\,dx,$$

provided this limit exists as a finite number. In that case we say the improper integral **converges**, and its value is that limit. If the limit is $\pm\infty$ or fails to exist, we say the integral **diverges**.

Symmetrically, $\displaystyle\int_{-\infty}^{b} f(x)\,\mathrm{d}x = \lim_{s\to-\infty}\int_{s}^{b} f(x)\,\mathrm{d}x$. The doubly-infinite case needs a little care, and we will get to it.

What is really happening

The symbol $\int_1^\infty$ looks like it has an endpoint called "∞" that you plug into an antiderivative. It does not. There is no number ∞ to plug in, and the Riemann machinery never runs over an infinite interval. What is *really* happening is a two-step process hidden inside one piece of notation: first do an ordinary integral up to a movable finite wall t, getting a *function of* t; then take the limit of that function as $t \to \infty$. The "∞" in the upper limit is shorthand for "take the limit." Every correct calculation with these objects re-separates those two steps.

25.2.1 The canonical example: two curves that look almost identical

Run the definition on the cleanest possible pair. Consider

$$\int_1^\infty \frac{\mathrm{d}x}{x^2} \qquad \text{and} \qquad \int_1^\infty \frac{\mathrm{d}x}{x}.$$

Both integrands are positive, both decrease toward zero as $x \to \infty$, and if you sketch them they look like nearly the same gently-falling curve. Yet one encloses a finite area and the other does not. Watch.

For the first, work to the finite wall t:

$$\int_1^t \frac{\mathrm{d}x}{x^2} = \int_1^t x^{-2}\,\mathrm{d}x = \Big[-\frac{1}{x}\Big]_1^t = -\frac{1}{t} + 1 = 1 - \frac{1}{t}.$$

Now slide the wall out. As $t \to \infty$, the term $1/t$ collapses to 0, so

$$\int_1^\infty \frac{\mathrm{d}x}{x^2} = \lim_{t\to\infty}\Big(1 - \tfrac{1}{t}\Big) = 1.$$

The area under $1/x^2$ from 1 all the way out to infinity is exactly 1. A region of *infinite width* has area 1.

Now the second, the same way:

$$\int_1^t \frac{\mathrm{d}x}{x} = \Big[\ln x\Big]_1^t = \ln t - \ln 1 = \ln t.$$

And $\ln t \to \infty$ as $t \to \infty$ – slowly, but it never stops climbing. So

$$\int_1^\infty \frac{\mathrm{d}x}{x} = \lim_{t\to\infty} \ln t = \infty \quad \text{(diverges)}.$$

Intuition

What is the real difference? Both curves fall to zero, but $1/x^2$ falls *fast enough* that the leftover area out past $x = t$ shrinks to nothing, while $1/x$ falls *too slowly* – it leaves behind a thin but stubbornly non-negligible sliver of area at every stage, and those slivers add up to infinity. The whole subject is the contest between "the tail is getting thin" and "but there is infinitely much of it." Convergence means thinness wins.

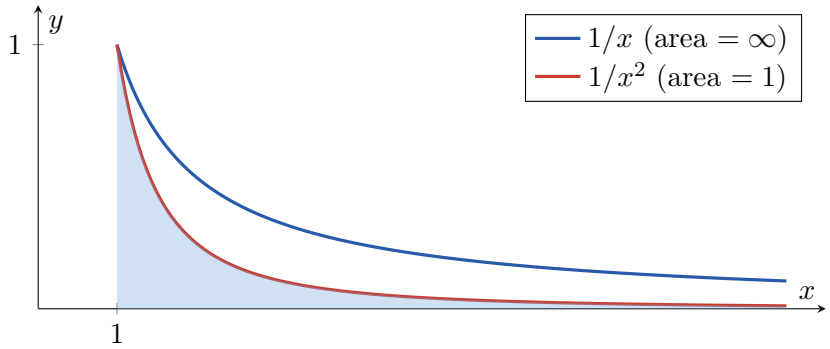


Figure 25.2: Two curves that look nearly the same near the start. The faster-falling $1/x^2$ (red, shaded) encloses finite area; the slower-falling $1/x$ (blue) encloses infinite area. The eye cannot tell them apart out in the tail – only the arithmetic of the limit can.

Work it yourself

Before reading on, compute $\displaystyle\int_1^\infty \frac{\mathrm{d}x}{x^3}$ by hand. Integrate to t, then let $t \to \infty$. (You should get the finite value $\frac{1}{2}$.) Then try $\displaystyle\int_1^\infty \frac{\mathrm{d}x}{\sqrt{x}}$ and watch it diverge. The point is to feel, in your fingers, exactly where the $1/t^{\text{something}}$ either dies or survives in the limit.

25.2.2 The p-test: the master pattern

The two examples above are the $p = 2$ and $p = 1$ cases of one general family. Settle the whole family at once, since it becomes the yardstick against which we measure everything else.

Theorem 25.1 (p-test at infinity)**.** *For a fixed real number p,*

$$\int_1^\infty \frac{\mathrm{d}x}{x^p} \quad \textit{converges if and only if} \quad p > 1,$$

and in the convergent case its value is $\dfrac{1}{p-1}$.

Proof. The only delicate part is that the antiderivative of x^{-p} has a different form when $p = 1$, so we split the argument.

Case $p \neq 1$. Then

$$\int_1^t x^{-p}\,\mathrm{d}x = \left[\frac{x^{1-p}}{1-p}\right]_1^t = \frac{t^{1-p}-1}{1-p}.$$

Everything now rides on the exponent $1-p$ of t. If $p>1$ then $1-p<0$, so $t^{1-p}=1/t^{p-1}\to 0$ as $t\to\infty$, leaving

$$\lim_{t\to\infty}\frac{t^{1-p}-1}{1-p}=\frac{0-1}{1-p}=\frac{1}{p-1},$$

a finite number: *convergence*. If instead $p<1$ then $1-p>0$, so $t^{1-p}\to\infty$ and the integral diverges.

Case $p=1$. Here the antiderivative is the logarithm, not a power:

$$\int_1^t\frac{\mathrm{d}x}{x}=\ln t\to\infty,$$

so the integral diverges. Collecting the cases: convergence happens exactly when $p>1$, with value $1/(p-1)$. □

Why the technique works

Why is $p=1$ the knife-edge – the exact boundary between win and lose? Because $1/x$ is the unique power whose antiderivative is logarithmic rather than a power. A logarithm grows slower than *any* positive power of t, yet it still grows without bound. So $1/x$ sits precisely on the border: its tail is just barely too fat to enclose finite area. Push the exponent up to 1.0000001 and the antiderivative becomes a (negative) power that converges; drop it to 0.9999999 and you still diverge. The logarithm is nature's tie-breaker, and it falls on the side of divergence. Keep the number $p=1$ in your pocket: half of all convergence questions are secretly "is my tail fatter or thinner than $1/x$?"

Common trap

"The integrand goes to zero, so the area must be finite." This is the single most common mistake, and $\int_1^\infty \mathrm{d}x/x$ is its tombstone. The integrand *does* go to zero, and the integral still diverges. Going to zero is *necessary* for convergence over an infinite interval (loosely speaking), but it is nowhere near sufficient – the tail has to go to zero *fast enough*. Never conclude convergence just from $f(x)\to 0$.

25.2.3 Exponential tails always win

Powers like $1/x^2$ converge only by a whisker. Exponentials decay so violently that they converge with room to spare, and they show up constantly in probability and physics, so here is the basic one.

Example 25.1 (The decaying exponential)**.** Evaluate $\displaystyle\int_0^\infty e^{-x}\,\mathrm{d}x$.

$$\int_0^t e^{-x}\,\mathrm{d}x=\Big[-e^{-x}\Big]_0^t=-e^{-t}+e^0=1-e^{-t}.$$

As $t\to\infty$, $e^{-t}\to 0$, so the integral converges to 1. More generally, for any rate $k>0$,

$$\int_0^\infty e^{-kx}\,\mathrm{d}x=\lim_{t\to\infty}\Big[-\tfrac{1}{k}e^{-kx}\Big]_0^t=\frac{1}{k}.$$

Application

This is the backbone of the *exponential distribution* in probability, the model for waiting times – how long until a radioactive atom decays, until the next customer arrives, until a component fails. The density is $f(x) = ke^{-kx}$ for $x \geq 0$, and the calculation above says $\int_0^\infty ke^{-kx}\,dx = 1$: total probability is one, exactly as a probability density must give. The whole edifice of continuous probability rests on improper integrals converging to 1. The mean waiting time, $\int_0^\infty x\,ke^{-kx}\,dx = 1/k$, is another improper integral – you can do it with integration by parts and the same $t \to \infty$ limit.

25.2.4 Both ends infinite: split, don't straddle

What about $\int_{-\infty}^{\infty} f(x)\,dx$, infinite at *both* ends? The correct definition splits the line at any convenient finite point c and demands that *each half* converge on its own.

Definition 25.3 (Doubly infinite integral)**.**

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{c} f(x)\,dx + \int_{c}^{\infty} f(x)\,dx,$$

and the left side converges if and only if *both* integrals on the right converge (the value is then independent of the split point c).

Common trap

You must *not* compute $\int_{-\infty}^{\infty} f$ as a single symmetric limit $\lim_{t\to\infty} \int_{-t}^{t} f$. That symmetric quantity is a different, weaker object called the *principal value*, and it can give a finite answer even when the true integral diverges. Example: $\int_{-t}^{t} x\,dx = 0$ for every t, so the symmetric limit is 0. But $\int_0^\infty x\,dx = \infty$, so the genuine improper integral $\int_{-\infty}^{\infty} x\,dx$ *diverges*. The symmetric limit was letting $+\infty$ and $-\infty$ cancel each other in lockstep, which is cheating – convergence is not allowed to depend on marching the two ends out at the same rate. Always split into two independent halves.

Example 25.2 (A doubly-infinite winner)**.** Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$. Split at $c = 0$. The antiderivative is $\arctan x$, so

$$\int_0^t \frac{dx}{1+x^2} = \arctan t - \arctan 0 \to \frac{\pi}{2},$$

and by the symmetry of the integrand the left half contributes another $\pi/2$. Both halves converge, so the whole integral converges to

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \frac{\pi}{2} + \frac{\pi}{2} = \pi.$$

A bell-shaped curve stretching to infinity in both directions, total area exactly π. The appearance of π here is a cousin of why π shows up in the Gaussian and throughout probability.

25.3 Unbounded integrands: the second kind

Now the other way an integral can be improper: the interval is finite, but the *function* runs off to infinity. Consider $\int_0^1 \mathrm{d}x/\sqrt{x}$. The interval $[0,1]$ is as tame as can be, but the integrand $1/\sqrt{x} \to \infty$ as $x \to 0^+$. There is an infinitely tall spike pinned against the left edge. Does it enclose finite area?

The cure is the same in spirit as before. We never integrate *at* the bad point. We back off to a wall a tiny distance ε away from it, integrate over the safe region, and then let the wall creep in toward the singularity.

Definition 25.4 (Singularity at an endpoint)**.** Suppose f is integrable on $[a+\varepsilon, b]$ for every small $\varepsilon > 0$ but is unbounded as $x \to a^+$. Then

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{\varepsilon \to 0^+} \int_{a+\varepsilon}^b f(x)\,\mathrm{d}x,$$

provided the limit exists (finite). Symmetrically, if the singularity is at the *right* endpoint b, back off from there: $\displaystyle\int_a^b f = \lim_{\varepsilon\to 0^+} \int_a^{b-\varepsilon} f$.

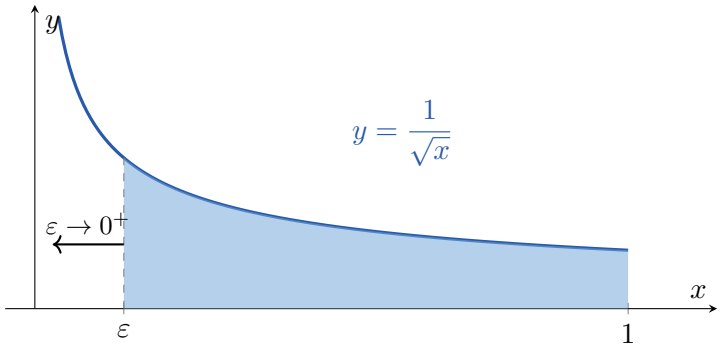


Figure 25.3: An unbounded integrand. The spike at $x = 0$ is infinitely tall, so we cut the area off at $x = \varepsilon$, measure it, and slide the wall ε toward 0. If the measured area settles, the spike has finite area despite its infinite height.

Example 25.3 (A tall spike with finite area)**.** Evaluate $\displaystyle\int_0^1 \frac{\mathrm{d}x}{\sqrt{x}}$. Back off to ε:

$$\int_\varepsilon^1 x^{-1/2}\,\mathrm{d}x = \Big[2\sqrt{x}\Big]_\varepsilon^1 = 2\sqrt{1} - 2\sqrt{\varepsilon} = 2 - 2\sqrt{\varepsilon}.$$

As $\varepsilon \to 0^+$, $\sqrt{\varepsilon} \to 0$, so the integral converges to 2. The spike is infinitely tall, and it encloses area 2.

Example 25.4 (A tall spike with infinite area)**.** Evaluate $\displaystyle\int_0^1 \frac{\mathrm{d}x}{x}$. The same procedure:

$$\int_\varepsilon^1 \frac{\mathrm{d}x}{x} = \Big[\ln x\Big]_\varepsilon^1 = \ln 1 - \ln\varepsilon = -\ln\varepsilon.$$

As $\varepsilon \to 0^+$, $\ln\varepsilon \to -\infty$, so $-\ln\varepsilon \to +\infty$: *diverges*. Again the logarithm sits on the knife-edge, again it falls on the side of divergence.

25.3.1 A p-test for spikes – and a beautiful inversion

There is a p-test here too, and comparing it to the one at infinity reveals one of the prettiest little symmetries in calculus.

Theorem 25.2 (p-test at a singularity)**.** *For a fixed real p,*

$$\int_0^1 \frac{\mathrm{d}x}{x^p} \quad \textit{converges if and only if} \quad p < 1,$$

with value $\dfrac{1}{1-p}$ *in the convergent case.*

Proof. For $p \neq 1$,

$$\int_\varepsilon^1 x^{-p}\,\mathrm{d}x = \Big[\frac{x^{1-p}}{1-p}\Big]_\varepsilon^1 = \frac{1-\varepsilon^{\,1-p}}{1-p}.$$

If $p < 1$ then $1-p > 0$ and $\varepsilon^{\,1-p} \to 0$, giving the finite limit $1/(1-p)$. If $p > 1$ then $1-p < 0$ and $\varepsilon^{\,1-p} = 1/\varepsilon^{\,p-1} \to \infty$, so it diverges. For $p = 1$ we got $-\ln\varepsilon \to \infty$ above. Hence convergence exactly when $p < 1$. □

Why this matters

Lay the two p-tests side by side and admire the reversal:

$$\int_1^\infty \frac{\mathrm{d}x}{x^p} \text{ converges} \iff p > 1, \qquad \int_0^1 \frac{\mathrm{d}x}{x^p} \text{ converges} \iff p < 1.$$

The inequality literally flips. And it *must*, because the two integrals ask opposite questions about the same exponent. Far out at infinity, you want the curve to fall *quickly* (large p) so the long tail is thin. Near zero, you want the curve to rise *slowly* (small p) so the tall spike is narrow. A large p makes a thin tail but a fat spike; a small p makes a thin spike but a fat tail. You cannot have both – which is exactly why $\int_0^\infty \mathrm{d}x/x^p$ *diverges for every* p: whatever value of p tames one end ruins the other. The single curve $1/x$ ($p = 1$) manages to diverge at both ends at once, the perfect democratic failure.

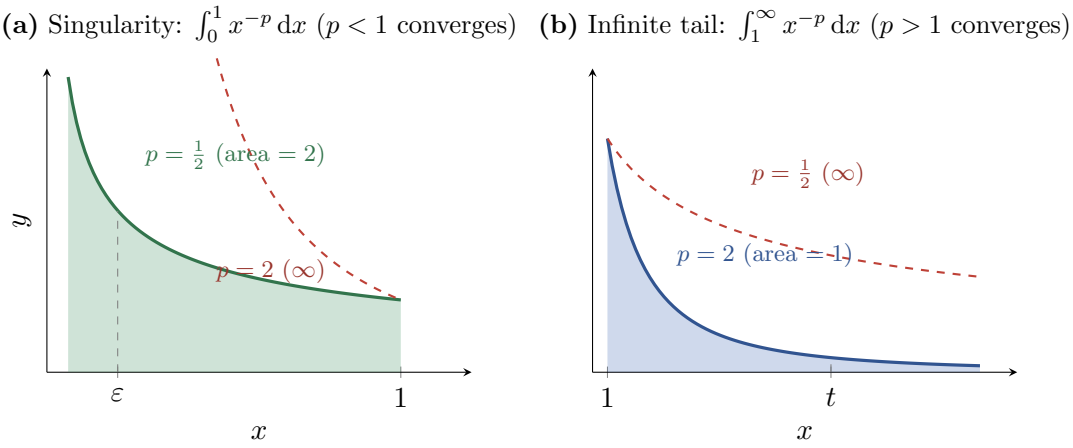


Figure 25.4: The trade-off behind the flipped p-tests. **(a)** Near a vertical singularity on $(0,1]$, small exponents $p<1$ converge (green shaded, area $=2$) because the spike is narrow, whereas large exponents $p\geq 1$ diverge (dashed red). **(b)** Out on the infinite tail $[1,\infty)$, large exponents $p>1$ converge (blue shaded, area $=1$) because the tail decays rapidly, whereas small exponents $p\leq 1$ diverge (dashed red). No single p can win at both ends – which is why $\int_0^\infty dx/x^p$ diverges for every p.

25.4 When you cannot find an antiderivative: comparison

So far every example handed us a clean antiderivative. But the questions we most want to answer often involve integrands like e^{-x^2} or $\dfrac{1}{\sqrt{x^3+1}}$ that have *no elementary antiderivative* at all. We cannot evaluate $\int_1^\infty e^{-x^2}\,dx$ by finding F and taking a limit – there is no F to find. Yet we can still answer the only question that matters for many purposes: does it *converge*?

The idea is the oldest trick in analysis: if you cannot measure a quantity, trap it between two quantities you *can* measure.

Intuition

If your messy curve sits *underneath* a curve whose area you know to be finite, then your curve – having even less area – must also have finite area. And if your messy curve sits *above* a curve whose area is infinite, then yours, having even more, must be infinite too. You never compute your own area; you sandwich it. This is the same move as the Squeeze Theorem for limits, now applied to whole accumulated areas.

Theorem 25.3 (Comparison Test for integrals)**.** *Suppose $0\leq f(x)\leq g(x)$ for all $x\geq a$, and both are integrable on every $[a,t]$. Then:*

1. *If $\displaystyle\int_a^\infty g$ converges, so does $\displaystyle\int_a^\infty f$ (the smaller one is also finite).*
2. *If $\displaystyle\int_a^\infty f$ diverges, so does $\displaystyle\int_a^\infty g$ (the bigger one is also infinite).*

The same statements hold for second-kind (unbounded-integrand) integrals.

Proof. This is where the assumption $f \geq 0$ earns its keep. Define the "area so far" function

$$A(t) = \int_a^t f(x)\,\mathrm{d}x.$$

Because $f \geq 0$, adding more interval only adds (non-negative) area, so A is a *nondecreasing* function of t. A nondecreasing function as $t \to \infty$ has only two possible fates: it climbs to a finite limit, or it runs off to $+\infty$. (There is no oscillation to worry about – it cannot come back down.) This is the Monotone Convergence principle, and it is the engine of the whole proof.

For part (1): since $f \leq g$, comparing areas over the same interval gives

$$A(t) = \int_a^t f \;\leq\; \int_a^t g \;\leq\; \int_a^\infty g =: M,$$

where M is finite by hypothesis. So A is nondecreasing *and bounded above* by M. A nondecreasing function bounded above must converge to a finite limit (it cannot escape past its ceiling). Hence $\int_a^\infty f$ converges.

Part (2) is the contrapositive of part (1): if $\int_a^\infty f$ diverged to $+\infty$ while $\int_a^\infty g$ converged, part (1) would force $\int_a^\infty f$ to converge – a contradiction. So if $\int f$ diverges, $\int g$ must diverge too. □

Why the technique works

Why does the comparison test *need* f and g to be non-negative? Because the whole argument rests on $A(t)$ being monotone – on the fact that the area "so far" can only grow as we extend the interval. If f were allowed to dip negative, $A(t)$ could rise and fall, and "bounded above" would no longer force convergence (a bounded oscillation need not settle anywhere). Sign-control is what converts "bounded" into "convergent." For integrands that change sign you need the stronger notion of *absolute convergence*, coming up at the end of the chapter.

Example 25.5 (Trapping a curve with no antiderivative)**.** Show $\displaystyle\int_1^\infty e^{-x^2}\,\mathrm{d}x$ converges. There is no elementary antiderivative, so we cannot evaluate it – but for $x \geq 1$ we have $x^2 \geq x$, hence $e^{-x^2} \leq e^{-x}$. And we computed $\int_1^\infty e^{-x}\,\mathrm{d}x = e^{-1}$, which is finite. By part (1) of the Comparison Test, our integral is squeezed under a finite area and therefore converges. We have proven it is finite without ever finding its value.

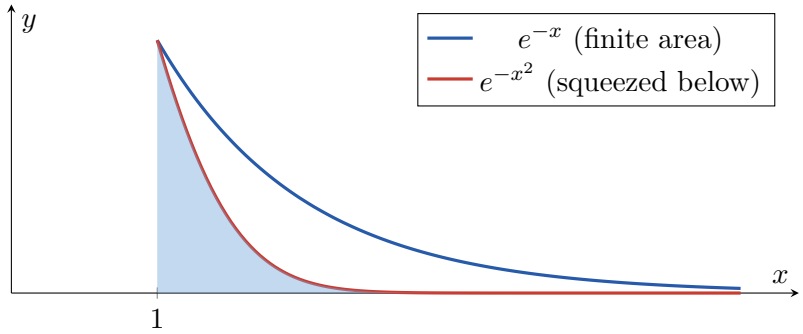


Figure 25.5: The comparison test in one picture. For $x \geq 1$ the curve e^{-x^2} (red, shaded) lies entirely under e^{-x} (blue), whose tail area is finite. A curve trapped beneath a finite area must itself have finite area – so $\int_1^\infty e^{-x^2}\,\mathrm{d}x$ converges, even though we never compute it.

Example 25.6 (Using divergence to prove divergence)**.** Show $\displaystyle\int_1^\infty \frac{\mathrm{d}x}{\sqrt{x^2+x}}$ diverges. For $x \geq 1$ we have $x^2 + x \leq x^2 + x^2 = 2x^2$, so

$$\frac{1}{\sqrt{x^2+x}} \;\geq\; \frac{1}{\sqrt{2x^2}} = \frac{1}{\sqrt{2}\,x}.$$

But $\int_1^\infty \frac{1}{\sqrt{2}\,x}\,\mathrm{d}x = \frac{1}{\sqrt{2}}\int_1^\infty \frac{\mathrm{d}x}{x}$ diverges (it is the $p = 1$ borderline). Our integrand sits *above* a divergent one, so by part (2) it diverges too.

Common trap

Comparison only works in the helpful direction. Bounding f *above* by a *divergent* g tells you nothing – a small thing under a big infinite thing can be finite or infinite. Likewise bounding f *below* by a *convergent* g tells you nothing. The useful pairings are: "smaller than convergent $\Rightarrow$ convergent" and "bigger than divergent $\Rightarrow$ divergent." Get the direction backwards and you have proven nothing. When in doubt, ask: is my known curve acting as a finite ceiling, or an infinite floor?

25.4.1 Limit comparison: comparing by behavior, not by inequality

Building an explicit inequality $f \leq g$ can be fiddly – you have to nudge the algebra into the right direction. Often you can *see* that two integrands behave the same way far out (both "act like" $1/x^2$, say) even though neither is cleanly below the other. The limit-comparison test makes that intuition rigorous.

Theorem 25.4 (Limit Comparison Test)**.** *Let $f, g > 0$ on $[a, \infty)$ and suppose*

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = L \quad \textit{with } 0 < L < \infty.$$

Then $\int_a^\infty f$ and $\int_a^\infty g$ either both converge or both diverge: they share the same fate.

The hypothesis $0 < L < \infty$ says f and g are, far out, the same size up to a bounded constant factor – and a bounded constant factor cannot turn a finite area infinite or vice versa. (We sketch the proof in an exercise; it is a clean application of the plain comparison test, bounding f between $\frac{L}{2}g$ and $2L\,g$ for large x.)

Example 25.7 (Limit comparison in action)**.** Does $\displaystyle\int_1^\infty \frac{3x^2 - x + 5}{2x^4 + 7}\,\mathrm{d}x$ converge? Forget the constants and look at the dominant powers: for large x the integrand behaves like $\dfrac{3x^2}{2x^4} = \dfrac{3}{2x^2}$. So compare against $g(x) = 1/x^2$:

$$\lim_{x\to\infty} \frac{(3x^2 - x + 5)/(2x^4 + 7)}{1/x^2} = \lim_{x\to\infty} \frac{(3x^2 - x + 5)\,x^2}{2x^4 + 7} = \lim_{x\to\infty} \frac{3x^4 - x^3 + 5x^2}{2x^4 + 7} = \frac{3}{2}.$$

The limit is finite and positive, so our integral shares the fate of $\int_1^\infty \mathrm{d}x/x^2$, which *converges* ($p = 2 > 1$). And we never had to integrate the ugly rational function.

Focus

This is the practical workhorse. For a rational integrand, compare degrees: if the denominator outranks the numerator by more than one power of x, the integrand decays faster than $1/x$ and the integral over $[1, \infty)$ converges; if it outranks by exactly one, you are on the $1/x$ borderline and it diverges; if by less, it diverges harder. "Bottom degree minus top degree" plays the role of p. One glance at the leading powers usually settles the question.

25.5 The famous surprises

We have already met one mild surprise – finite area under an infinitely long curve. The historical excitement around improper integrals came from a few results that genuinely offended common sense. They are worth meeting both for the delight and because they sharpen what "finite area" really does and does not promise.

25.5.1 Finite area, infinite length

Intuition

Take the region under $y = 1/x^2$ from 1 to ∞: area $= 1$, finite. Now ask about the *arc length* of a curve that runs to infinity horizontally – how long is the wiggly line, not the area under it. The arc length formula gives $\int_1^\infty \sqrt{1 + (y')^2}\,dx$. Since $\sqrt{1 + (y')^2} \geq 1$ for *every* curve, the arc length is at least $\int_1^\infty 1\,dx = \infty$. *Any* curve that runs to infinity horizontally has infinite length, no matter how flat it gets. So length and area are genuinely different beasts: one can be infinite while the other is finite, on the very same picture.

25.5.2 Gabriel's Horn: finite volume, infinite surface

Now the crown jewel, discovered by Evangelista Torricelli around 1641 – before calculus even had its modern notation. Take the curve $y = 1/x$ for $x \geq 1$ and spin it around the x-axis. You get an infinitely long trumpet, narrowing forever: *Gabriel's Horn* (also called Torricelli's trumpet).

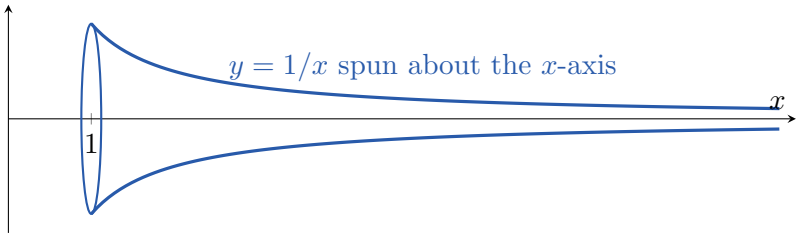


Figure 25.6: Gabriel's Horn: the curve $y = 1/x$ revolved about the x-axis, narrowing forever as $x \to \infty$. Finite volume, infinite surface area.

Two improper integrals describe the horn. Using the disk and surface-of-revolution formulas (which you will derive in the applications chapter; we just use them here):

Volume (stacking up thin disks of radius $1/x$):

$$V = \int_1^\infty \pi \left(\frac{1}{x}\right)^2 \mathrm{d}x = \pi \int_1^\infty \frac{\mathrm{d}x}{x^2} = \pi \cdot 1 = \pi.$$

A finite volume. You could fill the entire infinitely-long horn with exactly π cubic units of paint.

Surface area (wrapping a thin band of circumference $2\pi/x$ around each slice):

$$S = \int_1^\infty 2\pi \frac{1}{x}\sqrt{1 + \left(\frac{-1}{x^2}\right)^2}\,\mathrm{d}x > \int_1^\infty \frac{2\pi}{x}\,\mathrm{d}x = 2\pi \int_1^\infty \frac{\mathrm{d}x}{x} = \infty.$$

Here the surface integrand is at least $2\pi/x$ (because the square root is at least 1), and $\int_1^\infty \mathrm{d}x/x$ diverges – our old friend the $p = 1$ borderline. So the surface area is *infinite*.

Why this matters

Stand back and feel the paradox. The horn holds only π units of paint – a finite amount – yet its inner surface is *infinite*. So you could pour π units of paint in and completely fill the solid, but you could never paint the inside wall, because there is infinitely much wall to cover. "Filling with paint" and "coating with paint" sound like the same act, and the horn proves they are not. The resolution is the same lesson as finite-area / infinite-length: *a finite three-dimensional volume can be bounded by an infinite two-dimensional surface*, just as a finite area can be bounded by an infinite-length curve. The paradox is in our intuition, which secretly assumes "paint" has thickness. A real coat of paint of fixed thickness on an infinite wall needs infinite volume; it is only the idealized zero-thickness mathematical coating that the finite filling cannot supply.

History

Torricelli announced this result in 1641, and it ignited a firestorm across European philosophy and mathematics. To seventeenth-century scholars raised on Aristotle and Euclid, a geometric solid of *infinite extent* possessing a strictly *finite volume* appeared to be a self-contradiction. Galileo Galilei, Torricelli's mentor who had just grappled with the paradoxes of infinity in his 1638 *Two New Sciences*, was deeply unsettled.

The English philosopher Thomas Hobbes, an ardent critic of the emerging infinitary mathematics, vehemently attacked the result, asserting that to speak of an infinite surface enclosing a finite volume was an absurdity that destroyed the foundations of geometry. Across the English Channel, René Descartes and Marin Mersenne exchanged letters debating the horn, struggling to comprehend how a solid could be infinitely long and yet hold barely three cubic units of water.

Torricelli, working decades before Newton or Leibniz were even born, established the proof using Bonaventura Cavalieri's method of indivisibles – slicing the horn into infinitely thin cylindrical laminae. The horn was the first undeniable proof in the history of science that the infinite cannot be judged by human intuition: it must be calculated through the unyielding discipline of the limit.

Connection

The horn is the geometric twin of a fact about series you will meet later: the harmonic series $\sum 1/n$ diverges while $\sum 1/n^2$ converges. "Volume $\sim \sum 1/n^2$ converges, surface $\sim \sum 1/n$ diverges" is the same arithmetic wearing a 3D costume: the Integral Test, coming in the series chapters, says an integral $\int_1^\infty f$ and a series $\sum f(n)$ of a positive decreasing f converge or diverge *together*. Improper integrals and infinite series are two faces of one idea – adding up infinitely many shrinking contributions and asking whether the total is finite. The p-test you learned here *is* the p-series test you will learn there.

25.6 Hidden singularities and other traps

The most dangerous improper integrals are the ones that do not announce themselves. A finite interval and a formula that looks bounded can still hide a singularity in the middle, and if you integrate straight through it with the Fundamental Theorem you can get a confident, plausible, and completely wrong answer.

Common trap

Consider the innocent-looking

$$\int_{-1}^{1} \frac{\mathrm{d}x}{x^2}.$$

A careless application of the Fundamental Theorem gives $\Big[-1/x\Big]_{-1}^{1} = (-1) - (1) = -2$. This is *nonsense*: the integrand $1/x^2$ is positive everywhere it is defined, so any genuine "area" must be positive, never -2. What went wrong? The integrand blows up at $x = 0$, an *interior* point of $[-1, 1]$, so the integral is improper and the Fundamental Theorem – which assumes a continuous integrand – does not apply. The formula $-1/x$ is not a valid antiderivative *across* the discontinuity. Doing it correctly, you split at 0 and find each half is $\int_0^1 \mathrm{d}x/x^2 = \infty$, so the integral diverges. Always scan the *whole* interval, interior included, for points where the integrand is undefined before you reach for an antiderivative.

Definition 25.5 (Interior singularity)**.** If f is unbounded near an interior point c of $[a, b]$, define

$$\int_a^b f = \int_a^c f + \int_c^b f,$$

and the whole integral converges only if *both* pieces converge as improper integrals in their own right. As with two-sided infinity, you may not let the two sides cancel across the singularity.

Example 25.8 (Careful handling of a spike)**.** Evaluate $\displaystyle\int_0^8 \frac{\mathrm{d}x}{x^{1/3}}$. The integrand blows up at $x = 0$ (an endpoint here, but the moral is the same). Back off and take the limit:

$$\int_\varepsilon^8 x^{-1/3}\,\mathrm{d}x = \Big[\tfrac{3}{2}x^{2/3}\Big]_\varepsilon^8 = \tfrac{3}{2}\big(8^{2/3} - \varepsilon^{2/3}\big) = \tfrac{3}{2}\big(4 - \varepsilon^{2/3}\big).$$

As $\varepsilon \to 0^+$, $\varepsilon^{2/3} \to 0$, so the value is $\frac{3}{2} \cdot 4 = 6$. Here $p = 1/3 < 1$, so the p-test for singularities promised convergence, and indeed it converges to 6.

Walking question

Take a walk and turn this over: the integrand $1/x$ diverges at the singularity $x = 0$ (over $[0, 1]$) *and* diverges at infinity (over $[1, \infty)$). The exponent that would fix one end (p large near ∞, p small near 0) is exactly the wrong fix for the other. Convince yourself, without writing anything, that there is *no* power $1/x^p$ whose integral over the whole ray $(0, \infty)$ is finite. What does that tell you about trying to assign a finite "total area under $1/x^p$"? And what is special about $p = 1$ that makes it fail symmetrically at both ends? This is the whole chapter compressed into one curve.

25.7 When the sign keeps changing: absolute convergence

Everything in the comparison section assumed $f \geq 0$, so that the area-so-far $A(t)$ could only climb. What about an integrand that oscillates – positive, then negative, then positive – like $\dfrac{\sin x}{x}$ out to infinity? Now $A(t)$ rises and falls, and "does the area settle down?" becomes a subtler question of *cancellation*, not just of size.

Definition 25.6 (Absolute convergence)**.** The integral $\int_a^\infty f$ **converges absolutely** if $\int_a^\infty |f|$ converges. A basic theorem of analysis says: *if $\int_a^\infty |f|$ converges, then $\int_a^\infty f$ converges too.* Absolute convergence is the safe, robust kind – and what makes it so useful in practice is that it is checkable by the comparison test, because $|f| \geq 0$.

Intuition

The reason absolute convergence implies ordinary convergence: if even the *total amount of up-and-down motion* (the area of $|f|$, counting all wiggles as positive) is finite, then the actual signed area, which enjoys helpful cancellation on top of that, certainly cannot run off to infinity. Bounding the total size bounds the signed total. The converse fails: an integral can converge only *because* of cancellation, with the absolute version diverging. That fragile kind is called *conditional* convergence, and $\int_1^\infty \frac{\sin x}{x}\,dx$ is the classic example – it converges, but $\int_1^\infty \frac{|\sin x|}{x}\,dx$ diverges, because the bumps of $|\sin x|/x$ are, on average, a constant times $1/x$.

Example 25.9 (Absolute convergence via comparison)**.** Show $\displaystyle\int_1^\infty \frac{\sin x}{x^2}\,dx$ converges. We cannot sign-control $\sin x$, but we can bound its size: $|\sin x| \leq 1$, so

$$\left|\frac{\sin x}{x^2}\right| \leq \frac{1}{x^2}.$$

Since $\int_1^\infty dx/x^2$ converges, the comparison test makes $\int_1^\infty |\sin x/x^2|$ converge – that is, our integral converges *absolutely*, and hence converges. The oscillation never even had a chance to cause trouble, because the envelope $1/x^2$ already had finite area.

25.8 A gallery of fully worked examples

Here are a handful of complete computations, each touching a different technique.

Example 25.10 (Integration by parts to infinity)**.** Evaluate $\displaystyle\int_0^\infty x\,e^{-x}\,\mathrm{d}x$. Go to the wall t and integrate by parts ($u = x$, $\mathrm{d}v = e^{-x}\mathrm{d}x$, so $\mathrm{d}u = \mathrm{d}x$, $v = -e^{-x}$):

$$\int_0^t xe^{-x}\,\mathrm{d}x = \Big[-xe^{-x}\Big]_0^t + \int_0^t e^{-x}\,\mathrm{d}x = -te^{-t} + \big(1 - e^{-t}\big).$$

As $t \to \infty$, the term $te^{-t} \to 0$ (exponential decay beats the linear growth of t – you proved this in the L'Hôpital chapter), and $e^{-t} \to 0$. So the value is $0 + (1 - 0) = 1$. This is the mean of the exponential distribution with rate $k = 1$.

Example 25.11 (A logarithmic spike)**.** Evaluate $\displaystyle\int_0^1 \ln x\,\mathrm{d}x$. The integrand blows up at $x = 0^+$ (since $\ln x \to -\infty$), so this is second-kind. Back off and use parts ($u = \ln x$, $\mathrm{d}v = \mathrm{d}x$):

$$\int_\varepsilon^1 \ln x\,\mathrm{d}x = \Big[x\ln x\Big]_\varepsilon^1 - \int_\varepsilon^1 1\,\mathrm{d}x = \big(1\cdot 0 - \varepsilon\ln\varepsilon\big) - (1-\varepsilon) = -\varepsilon\ln\varepsilon - 1 + \varepsilon.$$

As $\varepsilon \to 0^+$, the product $\varepsilon\ln\varepsilon \to 0$ (again the L'Hôpital fact: a power kills a logarithm), and $\varepsilon \to 0$, leaving

$$\int_0^1 \ln x\,\mathrm{d}x = 0 - 1 + 0 = -1.$$

A finite, negative area – negative because $\ln x < 0$ on $(0, 1)$ – under a curve that dives to $-\infty$ at the left edge.

Example 25.12 (Doubly improper: both kinds at once)**.** Evaluate $\displaystyle\int_0^\infty \frac{\mathrm{d}x}{\sqrt{x}\,(1+x)}$. This is improper *twice*: the integrand blows up at $x = 0$ *and* the interval runs to ∞. The rule is to split off each bad place. Split at $x = 1$:

$$\int_0^\infty = \int_0^1 + \int_1^\infty,$$

and demand both converge. Substitute $x = u^2$, $\mathrm{d}x = 2u\,\mathrm{d}u$, so $\sqrt{x} = u$:

$$\int \frac{\mathrm{d}x}{\sqrt{x}\,(1+x)} = \int \frac{2u\,\mathrm{d}u}{u\,(1+u^2)} = 2\int \frac{\mathrm{d}u}{1+u^2} = 2\arctan u = 2\arctan\sqrt{x}.$$

Now both limits are clean. Near 0: $2\arctan\sqrt{x} \to 0$. Near ∞: $2\arctan\sqrt{x} \to 2\cdot\frac{\pi}{2} = \pi$. Each piece converges, so

$$\int_0^\infty \frac{\mathrm{d}x}{\sqrt{x}\,(1+x)} = \pi - 0 = \pi.$$

The substitution *absorbed* both singularities at once, turning a doubly improper integral into a single tidy arctangent.

Why the technique works

Why was it legitimate to use the antiderivative $2\arctan\sqrt{x}$ and just plug in the endpoints 0 and ∞, instead of writing the ε and t limits explicitly? Because once you have a single antiderivative that is *continuous all the way down to (and out to) the bad points* – and $2\arctan\sqrt{x}$ is, it has finite one-sided limits 0 and π – the limit operations reduce to evaluating those limits of the antiderivative. The ε and t are still there; you have just done them in your head. The

danger, as the $\int_{-1}^{1} dx/x^2$ trap showed, is using an antiderivative that is *not* continuous across the singularity. When the antiderivative behaves nicely at the boundary, the shortcut is safe; when it blows up there, you must keep the limits explicit and check convergence.

25.9 What we built

We extended the integral past its original home – bounded functions on bounded intervals – by the single device of *taking a limit of ordinary integrals*.

- **First kind** (infinite interval): $\int_a^\infty f = \lim_{t\to\infty} \int_a^t f$. Split doubly-infinite integrals into independent halves; never use a symmetric straddling limit.
- **Second kind** (unbounded integrand): back off from the singularity by ε and let $\varepsilon \to 0^+$. Watch for *interior* singularities that hide inside an innocent-looking interval.
- **The p-tests**, the two yardsticks: $\int_1^\infty dx/x^p$ converges iff $p > 1$; $\int_0^1 dx/x^p$ converges iff $p < 1$. The inequality flips because the two ends reward opposite behavior.
- **Comparison and limit comparison** settle convergence even with no antiderivative: trap your curve under a finite ceiling or over an infinite floor; or match its leading behavior to a known p-integral.
- **Absolute convergence** handles sign-changing integrands: if $\int |f|$ converges (checkable by comparison), so does $\int f$.
- **The surprises** – finite area under an infinite curve, finite volume inside Gabriel's Horn's infinite surface – are not paradoxes but lessons: area, length, volume, and surface are genuinely independent measurements, and the infinite must be *computed*, never guessed.

The deepest takeaway is a habit of mind: an improper integral is never evaluated "at infinity" or "at the spike." It is always an ordinary finite integral with a limit wrapped around it. Keep the two steps separate, ask whether the tail is thinner or fatter than $1/x$, and the entire subject organizes itself around that one borderline curve.

25.10 What improper integrals unlocked, conditional fragility, and the road ahead

Connection

Improper integrals brought the concept of infinity out of philosophical speculation and into computational analysis, but handling infinite boundaries requires exceptional care.
Before improper integrals were formalized by Augustin-Louis Cauchy in his 1821 *Cours d'Analyse*, calculus was restricted to compact, bounded domains. Improper integrals unlocked the mathematical foundations of modern physics, probability, and engineering:

- *Probability and Statistics:* Continuous probability distributions across infinite sample spaces depend entirely on improper integrals. The entire edifice of statistical inference

rests on the Gaussian normal distribution:

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}\,\mathrm{d}x = 1.$$

- *Field Theory and Astrophysics:* Gravitational potential energy $U = -\int_{\infty}^{r} \frac{GMm}{s^2}\,\mathrm{d}s$, electrostatic Coulomb potentials, and escape velocity calculations require integrating force over an infinite trajectory.
- *Integral Transforms:* The Laplace transform $\mathcal{L}\{f\}(s) = \int_0^{\infty} e^{-st} f(t)\,\mathrm{d}t$ and Fourier transform $\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x\xi}\,\mathrm{d}x$ convert continuous differential equations and wave signals into tractable algebraic spectra over unbounded domains.

Yet unlike finite Riemann integrals, where absolute and ordinary integrability coincide, improper integrals distinguish sharply between *absolute convergence* ($\int |f| < \infty$) and *conditional convergence*.

In 1821, Cauchy introduced the *Cauchy Principal Value*:

$$\text{P.V.} \int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = \lim_{R\to\infty} \int_{-R}^{R} f(x)\,\mathrm{d}x.$$

While the principal value can assign an answer to symmetric odd integrands (such as $\text{P.V.} \int_{-\infty}^{\infty} x\,\mathrm{d}x = 0$), it masks a profound instability: the genuine improper integral $\int_{-\infty}^{\infty} x\,\mathrm{d}x$ diverges. If one marches out to infinity at asymmetric speeds—for instance, $\lim_{R\to\infty} \int_{-R}^{2R} x\,\mathrm{d}x$—the limit shoots to $+\infty$. Even worse, conditionally convergent oscillatory integrals like $\int_0^{\infty} \frac{\sin x}{x}\,\mathrm{d}x = \frac{\pi}{2}$ do not converge absolutely ($\int_0^{\infty} \frac{|\sin x|}{x}\,\mathrm{d}x = \infty$); their finite value is a precarious artifact of positive and negative cancellation in a rigid, fixed order.

This acute difference between absolute stability and conditional fragility is the defining drama of infinite sums. In Chapter 28 (*Sequences*) and Chapter 29 (*Infinite Series Convergence*), we discover that improper integrals and infinite series are two manifestations of the same analytical structure. The *Integral Test* will prove that $\sum_{n=1}^{\infty} a_n$ converges if and only if the improper integral $\int_1^{\infty} f(x)\,\mathrm{d}x$ converges. And in Chapter 29, we will encounter the ultimate consequence of conditional fragility: Riemann's Rearrangement Theorem, showing that reordering a conditionally convergent series can alter its sum to any chosen value.

25.11 Exercises

Exercise 25.1 (Direct evaluation, first kind)**.** Evaluate each by integrating to a finite wall and taking the limit, or show it diverges:

$$\text{(a)} \int_2^{\infty} \frac{\mathrm{d}x}{x^3}, \qquad \text{(b)} \int_0^{\infty} e^{-3x}\,\mathrm{d}x, \qquad \text{(c)} \int_1^{\infty} \frac{\mathrm{d}x}{x^{4/5}}.$$

Exercise 25.2 (Direct evaluation, second kind)**.** Evaluate or show divergence, being explicit about where the integrand blows up:

$$\text{(a)} \int_0^4 \frac{\mathrm{d}x}{\sqrt{x}}, \qquad \text{(b)} \int_0^1 \frac{\mathrm{d}x}{x^{3/2}}, \qquad \text{(c)} \int_0^2 \frac{\mathrm{d}x}{(x-2)^{2/3}}.$$

Exercise 25.3 (The hidden interior singularity)**.** A student computes $\int_{-2}^{2} \frac{\mathrm{d}x}{x^4}$ by writing $\left[-\frac{1}{3x^3}\right]_{-2}^{2}$ and reports a finite number. Explain in one or two sentences why the answer is wrong, identify the singularity, and determine the correct status (converge to what, or diverge?).

Exercise 25.4 (Comparison test)**.** Use the (plain) comparison test to decide convergence. State clearly which known integral you compare to and in which direction.

$$\text{(a)} \int_1^\infty \frac{\mathrm{d}x}{x^3+5}, \qquad \text{(b)} \int_1^\infty \frac{2+\cos x}{x}\,\mathrm{d}x, \qquad \text{(c)} \int_1^\infty \frac{\mathrm{d}x}{\sqrt{x^4+1}}.$$

Exercise 25.5 (Limit comparison)**.** Use limit comparison against a suitable $1/x^p$ to decide convergence:

$$\text{(a)} \int_1^\infty \frac{x+1}{x^3-x+2}\,\mathrm{d}x, \qquad \text{(b)} \int_1^\infty \frac{\mathrm{d}x}{\sqrt{x^2+3x}}.$$

Exercise 25.6 (Absolute convergence)**.** Show that $\displaystyle\int_1^\infty \frac{\cos x}{x^2+x}\,\mathrm{d}x$ converges by proving it converges absolutely. (Bound $|\cos x|$ and compare.)

Exercise 25.7 (The p-spectrum at both ends)**.** Explain, using both p-tests, why $\displaystyle\int_0^\infty \frac{\mathrm{d}x}{x^p}$ diverges for *every* real p. (Split at $x=1$ and analyze each piece; show no single p saves both.)

Exercise 25.8 (Gabriel's Horn, your turn)**.** Repeat the Gabriel's-Horn computation for the curve $y = 1/x^2$, $x \geq 1$, revolved about the x-axis. (a) Show the volume $\int_1^\infty \pi(1/x^2)^2\,\mathrm{d}x$ is finite and find it. (b) Explain why the lower bound $\sqrt{1+(y')^2} \geq 1$ fails to decide convergence, and decide whether the surface area $\int_1^\infty 2\pi\,\frac{1}{x^2}\sqrt{1+(y')^2}\,\mathrm{d}x$ converges or diverges this time. (Hint: the square root $\sqrt{1+(2/x^3)^2}$ is bounded for $x \geq 1$, so compare to $\int_1^\infty 2\pi/x^2$.) Contrast with the $y = 1/x$ horn: why does this one *not* give the paradox?

Exercise 25.9 (Proving the Limit Comparison Test)**.** Suppose $f, g > 0$ on $[a,\infty)$ and $\lim_{x\to\infty} f(x)/g(x) = L$ with $0 < L < \infty$. Using the definition of the limit, show that there is some X beyond which $\frac{L}{2}\,g(x) \leq f(x) \leq 2L\,g(x)$. Then invoke the plain comparison test (twice) to conclude $\int f$ and $\int g$ share the same fate. (This is a clean, complete proof – write out both directions.)

Exercise 25.10 (A convergence that needs L'Hôpital)**.** Evaluate $\displaystyle\int_1^\infty \frac{\ln x}{x^2}\,\mathrm{d}x$. Integrate by parts to the wall t, then take $t \to \infty$, using $\lim_{t\to\infty} \frac{\ln t}{t} = 0$. (Answer: 1.)

Exercise 25.11 (Conceptual: why split?)**.** In two or three sentences, explain why $\int_{-\infty}^{\infty} x\,\mathrm{d}x$ is said to diverge even though $\lim_{t\to\infty}\int_{-t}^{t} x\,\mathrm{d}x = 0$. What is the difference between the principal value and genuine convergence, and why do we insist on the stricter notion?

Exercise 25.12 (A surprise of your own making)**.** Find a continuous function f on $[1,\infty)$ that does *not* tend to zero as $x \to \infty$, yet has $\int_1^\infty f\,\mathrm{d}x$ convergent. (Hint: tall thin spikes. Put a triangular spike of height 1 and base width $1/2^n$ centered at $x = n$. The integrand keeps hitting 1 – so it fails to go to zero pointwise everywhere – yet the total area is a convergent geometric series. This shows the "$f \to 0$" folklore is not even necessary in the strict pointwise sense; only the *tail area* must vanish.)

Chapter 26

Applications of Integration

By now you have built a serious machine. The definite integral $\int_a^b f(x)\,\mathrm{d}x$ is a limit of sums, the Fundamental Theorem turns that limit into an antiderivative evaluation, and you have a toolbox – substitution, parts, partial fractions – for finding antiderivatives. This chapter is where all of that pays off. We point the integral at the physical world and watch it compute areas, volumes, lengths, surface areas, work, averages, and probabilities.

But there is a single idea underneath every application in this chapter, and if you hold onto it you will never have to memorize a formula again:

To integrate something is to add up infinitely many infinitely thin slices.

Every problem here is solved the same way. You ask one question: *what am I accumulating?* You cut the thing you want into thin slices, you write down the amount in one representative slice – always of the form (something) $\mathrm{d}x$ – and then you let the integral sign add them all up. The whole art is in finding the slice. The integral does the rest.

So this is not really a chapter of formulas. It is a chapter of *one* formula, $\int(\text{slice}) = \text{total}$, applied over and over until the move becomes second nature. We begin where the integral itself began: with area.

26.1 The One Move: Slice, Measure, Sum

Before any specific application, here is what we are doing every single time, because it really is always the same.

Focus

The slicing recipe. To compute a total quantity Q by integration:

1. **Slice.** Cut the region (or solid, or curve, or process) into thin pieces indexed by a variable, say x, with thickness $\mathrm{d}x$.
2. **Measure.** Find the amount $\mathrm{d}Q$ contributed by one slice. It will look like $\mathrm{d}Q =$ (some function of x) $\mathrm{d}x$. This is the only creative step.

3. **Sum.** Add the slices: $Q = \int_a^b \mathrm{d}Q$, where a and b are the smallest and largest values of the slicing variable.

Why does this work at all? Because that is the foundational meaning of the definite integral:

$$\int_a^b g(x)\,\mathrm{d}x \;=\; \lim_{n\to\infty} \sum_{k=1}^{n} g(x_k^*)\,\Delta x,$$

a limit of Riemann sums. Each term $g(x_k^*)\,\Delta x$ is the contribution of one thin strip of width Δx. When we write "$\mathrm{d}Q = g(x)\,\mathrm{d}x$" we are naming a single term of that sum, and "$\int \mathrm{d}Q$" is the instruction to take the limit of the sum. So the slicing recipe is not a trick layered on top of integration; it is integration, told as a story about pieces.

What is really happening

When a physicist writes $\mathrm{d}Q = g(x)\,\mathrm{d}x$ and then "integrates both sides," what is really happening is *not* that $\mathrm{d}x$ is a tiny number being multiplied. The precise statement is: the total Q is the limit of sums of slice-contributions, and each slice-contribution is approximately $g(x_k)\,\Delta x$, with the approximation getting perfect as $\Delta x \to 0$. The differential notation $\mathrm{d}Q = g(x)\,\mathrm{d}x$ is just shorthand for "the rate at which Q accumulates per unit of x is $g(x)$." The error you make in one slice by treating it as a perfect rectangle is of *higher order* – it shrinks faster than Δx itself – so it washes out completely in the limit. That higher -order vanishing is the entire reason the recipe is exact rather than merely approximate.

Keep that picture in mind. Now we make a slice, eight different ways.

26.2 Area Between Curves

The integral was born to measure the area under a single curve. The first generalization is almost too easy – the area *between* two curves.

Suppose $f(x) \geq g(x)$ on $[a, b]$, so the graph of f sits above the graph of g. What is the area of the region trapped between them? Slice it vertically.

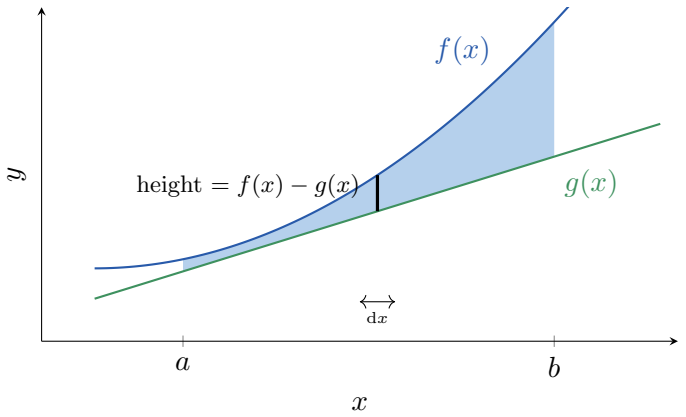


Figure 26.1: A vertical slice of the region between f (top) and g (bottom). One slice is a thin rectangle of width $\mathrm{d}x$ and height $f(x) - g(x)$.

Slice. A vertical strip at position x, thickness $\mathrm{d}x$.

Measure. The strip is (almost) a rectangle. Its width is $\mathrm{d}x$ and its height is the vertical gap between the curves, $f(x) - g(x)$. So

$$\mathrm{d}A = \big(f(x) - g(x)\big)\,\mathrm{d}x.$$

Sum.

$$\boxed{A = \int_a^b \big(f(x) - g(x)\big)\,\mathrm{d}x.}$$

That is the entire theory. We did not memorize a new formula so much as read it off the picture. The integrand is always (top minus bottom).

Common trap

Top minus bottom, not left limit minus right. The single most common error is forgetting which curve is on top, and getting a negative area. Area is positive. If you are not sure which function is larger on the interval, test one point: plug in a value of x between a and b and see which output is bigger. A second classic trap: if the curves *cross* inside $[a, b]$, the top and bottom switch. Then you must split the integral at the crossing points and use $|f - g|$ on each piece. Sketch first, always.

Example 26.1 (Area between a parabola and a line)**.** Find the area enclosed between $y = x^2$ and $y = x + 2$.

First, where do they meet? Set $x^2 = x + 2$, i.e. $x^2 - x - 2 = 0$, i.e. $(x - 2)(x + 1) = 0$. The curves cross at $x = -1$ and $x = 2$. Between those points, which is on top? Test $x = 0$: the parabola gives 0, the line gives 2. The line is on top. So

$$A = \int_{-1}^{2} \big[(x + 2) - x^2\big]\,\mathrm{d}x = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^{2}.$$

At $x = 2$: $2 + 4 - \frac{8}{3} = 6 - \frac{8}{3} = \frac{10}{3}$. At $x = -1$: $\frac{1}{2} - 2 + \frac{1}{3} = \frac{3}{6} - \frac{12}{6} + \frac{2}{6} = -\frac{7}{6}$. Subtracting,

$$A = \frac{10}{3} - \left(-\frac{7}{6}\right) = \frac{20}{6} + \frac{7}{6} = \frac{27}{6} = \frac{9}{2}.$$

The enclosed area is $\frac{9}{2}$.

26.2.1 Slicing the other way

Sometimes a region is easier to describe by horizontal strips – when the boundaries are naturally functions of y rather than x. Then you slice horizontally: a strip at height y has thickness $\mathrm{d}y$ and width (right curve) minus (left curve), both expressed as functions of y. The area is

$$A = \int_c^d \big(x_{\text{right}}(y) - x_{\text{left}}(y)\big)\,\mathrm{d}y.$$

It is the same recipe with the roles of x and y swapped. Choose whichever direction makes the slices simple.

Example 26.2 (Same region, horizontal slices)**.** Find the area enclosed by $x = y^2$ and $x = y + 2$.

Solving $y^2 = y + 2$ gives $y^2 - y - 2 = (y - 2)(y + 1) = 0$, so y runs from -1 to 2. For these y, the line $x = y + 2$ is to the right of the parabola $x = y^2$ (check $y = 0$: line gives 2, parabola gives 0). So

$$A = \int_{-1}^{2} \left[(y + 2) - y^2\right] \mathrm{d}y = \frac{9}{2},$$

identical to before – it is literally the same algebra. The lesson: pick the slicing direction that makes "right minus left" (or "top minus bottom") a single clean expression, so you never have to split the integral.

Walking question

Picture a region shaped like a sideways parabola, opening to the right, capped by a vertical line. If you slice it vertically, a single vertical strip can hit the parabola *twice* (top and bottom branches), forcing you to split the integral. Slice it horizontally and each strip hits one left edge and one right edge cleanly. Which way would you cut, and why does the geometry of the region – not any rule – decide for you?

26.3 Volumes by Slicing

Now lift the idea into three dimensions. The recipe does not change at all: cut the solid into thin slices, find the volume of one slice, integrate.

The cleanest case is slicing a solid by parallel planes. Imagine a loaf of bread. Cut it into thin slices perpendicular to the x-axis. A slice at position x is a flat slab of thickness $\mathrm{d}x$ whose flat face has some area $A(x)$. Its volume is (face area)$\times$(thickness):

$$\mathrm{d}V = A(x)\,\mathrm{d}x, \qquad \text{so} \qquad \boxed{V = \int_a^b A(x)\,\mathrm{d}x.}$$

This is the general principle of volume by slicing, and *everything* that follows – disks, washers, even shells – is a special case of it. You only ever need to know the cross-sectional area $A(x)$ as a function of where you cut.

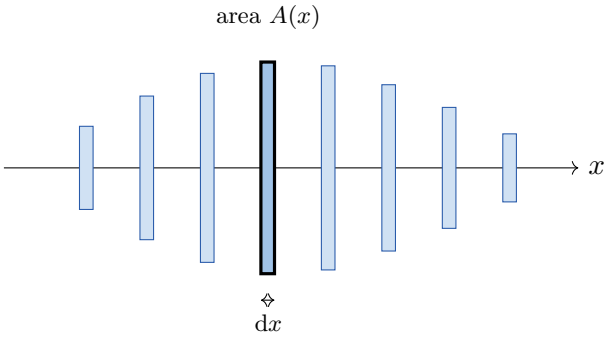


Figure 26.2: A solid cut into slabs perpendicular to the x-axis. One slab has volume $A(x)\,\mathrm{d}x$, where $A(x)$ is the area of its face.

Example 26.3 (A pyramid, derived from scratch)**.** A pyramid has a square base of side s and height h, apex directly above the center. Find its volume.

Put the apex at the origin and let x measure distance *down* from the apex, $0 \le x \le h$. At depth x the cross-section is a square. By similar triangles, the square's side grows linearly from 0 at the apex to s at the base: side $= s\,(x/h)$. So the cross-sectional area is

$$A(x) = \left(\frac{sx}{h}\right)^2 = \frac{s^2}{h^2}\,x^2.$$

Therefore

$$V = \int_0^h \frac{s^2}{h^2}\,x^2\,\mathrm{d}x = \frac{s^2}{h^2}\cdot\frac{x^3}{3}\Big|_0^h = \frac{s^2}{h^2}\cdot\frac{h^3}{3} = \frac{1}{3}s^2h.$$

There it is: the volume of a pyramid is one-third base times height. The mysterious $\frac{1}{3}$ that you memorized in geometry class is just $\int x^2 = \frac{x^3}{3}$ in disguise. The square growing like x^2 is what makes the third appear.

History

That factor of $\frac{1}{3}$ tormented the ancient world. Democritus (5th century BCE) guessed correctly that a pyramid is a third of the prism with the same base and height, but could not prove it. Eudoxus and later Archimedes proved such results with the *method of exhaustion*: trap the solid between stacks of thin prisms from below and above, and show the two stacks squeeze together. That is precisely a Riemann sum, two thousand years early. Archimedes even computed the volume of a sphere this way and was so proud of it that he asked for a sphere inscribed in a cylinder to be carved on his tombstone. What he lacked was the *Fundamental Theorem* – he had to sum each series by hand, brilliantly and laboriously. We get $\int_0^h x^2\,\mathrm{d}x = h^3/3$ for free. The whole of this chapter is Archimedes' dream automated.

26.3.1 Solids of revolution: the disk method

Now a very common special case: spin a region around an axis to sweep out a solid. If you revolve the region under $y = f(x)$ (above the x-axis, from a to b) about the x-axis, each vertical strip sweeps out a thin *disk* – a coin – of radius $f(x)$ and thickness $\mathrm{d}x$. A disk is just a cylinder, so its area is $\pi(\text{radius})^2$:

$$A(x) = \pi\,[f(x)]^2, \qquad \boxed{V = \pi\int_a^b [f(x)]^2\,\mathrm{d}x.}$$

This is not a new method – it is volume-by-slicing with a circular cross-section.

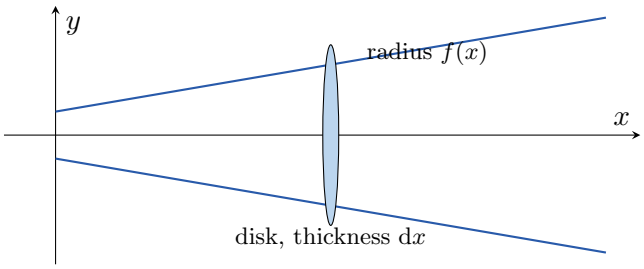


Figure 26.3: Revolving $y = f(x)$ about the x-axis. A vertical strip sweeps out a disk of radius $f(x)$; its volume is $\pi[f(x)]^2\,\mathrm{d}x$.

Example 26.4 (Volume of a sphere – Archimedes' tombstone)**.** Revolve the upper half of the circle $y = \sqrt{r^2 - x^2}$, from $x = -r$ to $x = r$, about the x-axis. You sweep out a solid ball of radius r. The disk at x has radius $\sqrt{r^2 - x^2}$, so

$$V = \pi \int_{-r}^{r} \left(\sqrt{r^2 - x^2}\right)^2 \mathrm{d}x = \pi \int_{-r}^{r} (r^2 - x^2)\,\mathrm{d}x.$$

The integrand is even, so integrate from 0 to r and double:

$$V = 2\pi \left[r^2 x - \frac{x^3}{3}\right]_0^r = 2\pi \left(r^3 - \frac{r^3}{3}\right) = 2\pi \cdot \frac{2r^3}{3} = \frac{4}{3}\pi r^3.$$

The famous $\frac{4}{3}\pi r^3$ falls out in three lines. Archimedes would weep.

26.3.2 Hollow solids: the washer method

What if the region you revolve does not touch the axis – if there is a gap? Then each slice is not a solid disk but a *washer*: a disk with a hole punched out. Revolve the region between $y = f(x)$ (outer) and $y = g(x)$ (inner), with $f \geq g \geq 0$, about the x-axis. The slice at x is an annulus with outer radius $f(x)$ and inner radius $g(x)$ (Figure 26.4). Its area is the big circle minus the little one:

$$A(x) = \pi[f(x)]^2 - \pi[g(x)]^2, \qquad \boxed{V = \pi \int_a^b \Big([f(x)]^2 - [g(x)]^2\Big)\,\mathrm{d}x.}$$

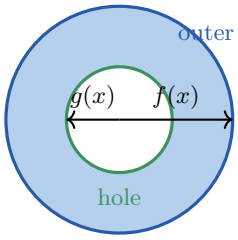


Figure 26.4: A washer cross-section: a disk of outer radius $f(x)$ with a concentric hole of inner radius $g(x)$ punched out. Its area is $\pi[f(x)]^2 - \pi[g(x)]^2$.

Common trap

Square first, then subtract – never the other way. The washer area is $\pi(R^2 - r^2)$, *not* $\pi(R - r)^2$. The annulus is the difference of two *areas*, so you square each radius and then subtract. Writing $\pi \int (f - g)^2$ is a wrong answer that looks plausible; it computes a solid of a completely different shape. Always: outer radius squared minus inner radius squared.

Example 26.5 (A washer volume)**.** Revolve the region between $y = x$ and $y = x^2$ (for $0 \le x \le 1$) about the x-axis. On this interval $x \ge x^2$, so the outer radius is $f(x) = x$ and the inner radius is $g(x) = x^2$. Then

$$V = \pi \int_0^1 \left(x^2 - x^4\right) \mathrm{d}x = \pi \left[\frac{x^3}{3} - \frac{x^5}{5}\right]_0^1 = \pi \left(\frac{1}{3} - \frac{1}{5}\right) = \pi \cdot \frac{2}{15} = \frac{2\pi}{15}.$$

26.3.3 The shell method: slicing parallel to the axis

There is a second way to slice a solid of revolution, and it is worth seeing because it pushes the slicing idea a level deeper. Instead of cutting *across* the axis into disks, cut *around* the axis into nested cylindrical shells – think of the rings of a tree, or a set of nesting tin cans.

Revolve the region under $y = f(x)$, from $x = a$ to $x = b$, about the *y-axis*. Take a thin vertical strip at position x, width $\mathrm{d}x$, height $f(x)$. When this strip revolves around the y-axis it sweeps out a thin cylindrical shell of radius x and height $f(x)$.

Now, what is the volume of a thin shell? *Unroll it.* A thin cylindrical shell of radius x, height $f(x)$, and thickness $\mathrm{d}x$, slit open and flattened, is a thin rectangular slab. Its dimensions are:

- length = circumference of the shell = $2\pi x$,
- height = $f(x)$,
- thickness = $\mathrm{d}x$.

So its volume is

$$\mathrm{d}V = \underbrace{2\pi x}_{\text{circumference}} \cdot \underbrace{f(x)}_{\text{height}} \cdot \underbrace{\mathrm{d}x}_{\text{thickness}},$$

and summing the shells,

$$\boxed{V = \int_a^b 2\pi x \, f(x) \, \mathrm{d}x.}$$

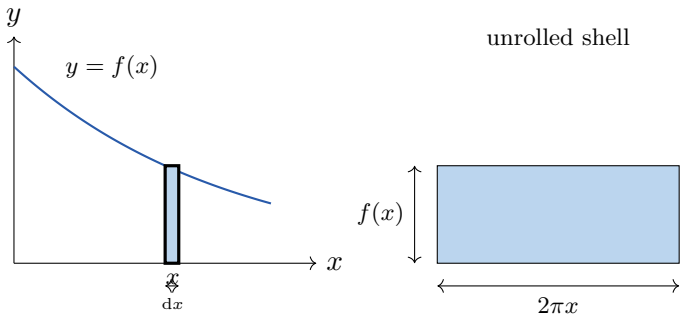


Figure 26.5: The shell method. A vertical strip at x revolves into a cylindrical shell; unrolled, the shell is a slab of dimensions $2\pi x$ by $f(x)$ by $\mathrm{d}x$.

Why the technique works

Why "unrolling" is legitimate. A real cylindrical shell is not a flat slab – its outer face ($2\pi(x+\mathrm{d}x)$ around) is slightly longer than its inner face ($2\pi x$ around). So why may we use the single circumference $2\pi x$? Write the exact shell volume as the difference of two cylinders:

$$\mathrm{d}V = \pi(x+\mathrm{d}x)^2 f(x) - \pi x^2 f(x) = \pi f(x)\big(2x\,\mathrm{d}x + (\mathrm{d}x)^2\big) = 2\pi x\, f(x)\,\mathrm{d}x + \pi f(x)(\mathrm{d}x)^2.$$

The leading term is exactly $2\pi x\, f(x)\,\mathrm{d}x$. The correction $\pi f(x)(\mathrm{d}x)^2$ involves $(\mathrm{d}x)^2$ – it is *higher order*, shrinking faster than $\mathrm{d}x$ – so when we sum and take the limit it contributes nothing. This is the same higher-order-vanishing that justified treating an area strip as a perfect rectangle. Unrolling is legitimate because the discrepancy between the inner and outer circumferences is a second-order effect.

Example 26.6 (Same solid, two methods)**.** Revolve the region under $y = x^2$, from $x = 0$ to $x = 2$, about the y-axis.

By shells (slicing parallel to the y-axis):

$$V = \int_0^2 2\pi x \cdot x^2\,\mathrm{d}x = 2\pi \int_0^2 x^3\,\mathrm{d}x = 2\pi \cdot \frac{x^4}{4}\Big|_0^2 = 2\pi \cdot 4 = 8\pi.$$

By washers, to check, we slice perpendicular to the y-axis. Here y runs from 0 to 4; at height y the solid is the region between the curve $x = \sqrt{y}$ and the line $x = 2$, revolved, giving outer radius 2 and inner radius $\sqrt{y}$:

$$V = \pi \int_0^4 \big(2^2 - (\sqrt{y})^2\big)\,\mathrm{d}y = \pi \int_0^4 (4-y)\,\mathrm{d}y = \pi\left[4y - \frac{y^2}{2}\right]_0^4 = \pi(16-8) = 8\pi.$$

The two methods agree, as they must. Notice how shells here avoided having to solve for x in terms of y. *That* is when you reach for shells: choose the method whose slices are easy to describe.

Intuition

A pocket rule for choosing. Ask: in which direction are my slices parallel to the axis of rotation? Slices *perpendicular* to the axis are disks/washers; slices *parallel* to the axis are

shells. Then ask which description of the boundary is simpler. Revolving around the y-axis a region given as $y = f(x)$? Vertical strips are parallel to the axis, so shells let you keep the natural variable x – no algebraic gymnastics. The methods never disagree; one is just less work.

26.3.4 The Pappus–Guldin Theorems: Revolution via Centroids

History

Before calculus gave us disks and shells, the ancient Greek geometer *Pappus of Alexandria* (c. 320 CE, in Book VII of his *Collection*) discovered an astonishing shortcut for solids of revolution, later rediscovered and popularized by the Swiss mathematician *Paul Guldin* in 1640.

The **Pappus–Guldin Centroid Theorems** state that when a plane region of area A is revolved about an external axis in its plane, the volume swept out is simply the area times the distance traveled by its geometric center of mass (its centroid $\bar{y}$):

$$V = 2\pi \bar{y} A.$$

Similarly, the surface area generated by revolving a curve of arc length L about an external axis is the length times the circumference traced by the curve's centroid:

$$S = 2\pi \bar{y}_{\text{curve}} L.$$

Why does this work? In the shell method, the volume integral is $\int 2\pi x f(x)\,\mathrm{d}x = 2\pi \int x f(x)\,\mathrm{d}x$. By definition, the horizontal coordinate of the centroid of the region is $\bar{x} = \frac{1}{A}\int x f(x)\,\mathrm{d}x$, so the integral is identically $2\pi \bar{x} A$! Slicing by shells reveals that Guldin's 1640 theorem is not a separate piece of magic; it is the shell integral observed from the balance point of the shape.

26.4 Arc Length: Accumulating Distance

So far we have accumulated area and volume. Now accumulate *length*. How long is a curve $y = f(x)$ from $x = a$ to $x = b$?

Slice. Cut the curve into tiny pieces. Over a sliver of horizontal width $\mathrm{d}x$, the curve rises by $\mathrm{d}y$. The piece of curve is so short that it is essentially straight – a tiny line segment (smooth means locally straight, the refrain of this whole book).

Measure. That little segment is the hypotenuse of a right triangle with legs $\mathrm{d}x$ and $\mathrm{d}y$. Pythagoras gives its length:

$$\mathrm{d}s = \sqrt{(\mathrm{d}x)^2 + (\mathrm{d}y)^2}.$$

Factor out $\mathrm{d}x$ to express everything in terms of the slope:

$$\mathrm{d}s = \sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x = \sqrt{1 + [f'(x)]^2}\,\mathrm{d}x.$$

Sum.

$$\boxed{L = \int_a^b \sqrt{1 + [f'(x)]^2}\,\mathrm{d}x.}$$

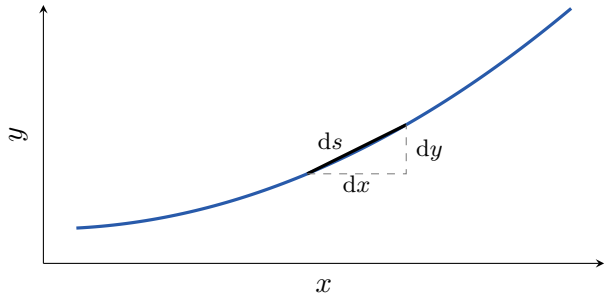


Figure 26.6: A short piece of curve is the hypotenuse of a triangle with legs $\mathrm{d}x$ and $\mathrm{d}y$. Its length is $\mathrm{d}s = \sqrt{(\mathrm{d}x)^2 + (\mathrm{d}y)^2}$.

What is really happening

The symbol $\mathrm{d}s = \sqrt{(\mathrm{d}x)^2 + (\mathrm{d}y)^2}$ is the infinitesimal version of the distance formula. What is really happening: we replace the curve by an inscribed polygon – a chain of straight chords – and add up the chord lengths. Each chord has length $\sqrt{(\Delta x)^2 + (\Delta y)^2}$. As the chords get shorter the polygon hugs the curve, and the sum of chord lengths approaches the arc length. The formula is just that limit written compactly. Arc length is the total "ground covered," integrating the local stretch factor $\sqrt{1 + (f')^2}$ that tells you how much real distance you travel per unit of horizontal travel.

Common trap

The square root usually fights back. Arc-length integrands are notoriously ugly: $\sqrt{1 + [f'(x)]^2}$ rarely has an elementary antiderivative. Even for $y = x^2$ you get $\int \sqrt{1 + 4x^2}\,\mathrm{d}x$, which needs a trig or hyperbolic substitution. Authors of textbook problems rig f so the radical collapses to a perfect square. In real life you reach for numerical integration. Do not be alarmed when an arc-length integral looks intractable – that is the normal state of affairs, not a sign you set it up wrong.

Example 26.7 (A length that actually works out)**.** Find the arc length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$. Then $f'(x) = x^{1/2}$, so $[f'(x)]^2 = x$, and

$$L = \int_0^3 \sqrt{1 + x}\,\mathrm{d}x = \frac{2}{3}(1 + x)^{3/2}\Big|_0^3 = \frac{2}{3}\left(4^{3/2} - 1\right) = \frac{2}{3}(8 - 1) = \frac{14}{3}.$$

The radical collapsed because $1 + (f')^2 = 1 + x$ is itself a clean power. That is the only reason the problem is doable by hand – and it is exactly the kind of luck textbook authors arrange.

Connection

The little relation $\mathrm{d}s = \sqrt{(\mathrm{d}x)^2 + (\mathrm{d}y)^2}$ is the seed of an enormous subject. In physics, dividing by $\mathrm{d}t$ gives speed $\frac{\mathrm{d}s}{\mathrm{d}t} = \sqrt{\dot{x}^2 + \dot{y}^2}$ – the magnitude of the velocity vector. In differential geometry, the rule for measuring $\mathrm{d}s$ at each point is called the *metric*, and replacing the flat Pythagorean $\mathrm{d}s^2 = \mathrm{d}x^2 + \mathrm{d}y^2$ with a position-dependent rule is precisely how Einstein's general relativity encodes curved spacetime. Arc length is your first encounter with a metric.

26.5 Surface Area of Revolution

Combine the last two ideas. Revolve the curve $y = f(x)$ about the x-axis and ask for the *area of the surface* it sweeps out – the skin of the solid, not its volume.

Slice. A tiny arc of the curve, of length $\mathrm{d}s$, at horizontal position x and height $f(x)$.

Measure. When that little arc revolves around the x-axis it traces a thin *band* – a ribbon (Figure 26.7). The ribbon is a tiny strip wrapped in a circle of radius $f(x)$ (the distance from the curve to the axis). Its circumference is $2\pi f(x)$ and its width is the arc length $\mathrm{d}s$. So its area is

$$\mathrm{d}S = 2\pi\, f(x)\,\mathrm{d}s = 2\pi f(x)\sqrt{1 + [f'(x)]^2}\,\mathrm{d}x.$$

Sum.

$$\boxed{S = \int_a^b 2\pi\, f(x)\sqrt{1 + [f'(x)]^2}\,\mathrm{d}x.}$$

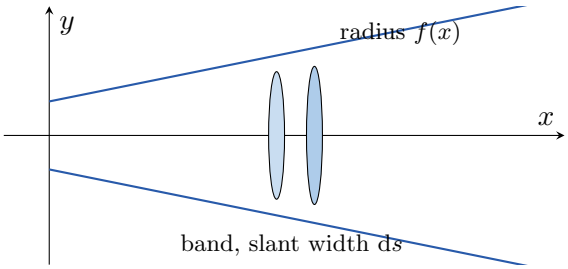


Figure 26.7: Revolving $y = f(x)$ about the x-axis. A short arc of length $\mathrm{d}s$ sweeps out a band of radius $f(x)$; its area is $2\pi f(x)\,\mathrm{d}s$.

Why the technique works

Why $2\pi f(x)\,\mathrm{d}s$ and not $2\pi f(x)\,\mathrm{d}x$? A natural instinct is to multiply the circumference $2\pi f(x)$ by the horizontal width $\mathrm{d}x$. That is wrong, and the reason is geometric. The little band is not a vertical cylinder; it is a slanted frustum of a cone, because the curve is rising. Its slant width is the *arc length* $\mathrm{d}s$, which is longer than $\mathrm{d}x$ whenever the curve has any slope. Using $\mathrm{d}x$ would systematically undercount the surface – it ignores the tilt. The correct area of a thin frustum band of radius r and slant width ℓ is $2\pi r\,\ell$; here $r = f(x)$ and $\ell = \mathrm{d}s$. So the $\mathrm{d}s$ is not a decoration; it is the whole point. (This is exactly the surface-area analogue of why arc length needs the radical.)

Example 26.8 (Surface area of a sphere)**.** Revolve $y = \sqrt{r^2 - x^2}$, from $x = -r$ to $x = r$, about the x-axis. We have $f(x) = \sqrt{r^2 - x^2}$ and

$$f'(x) = \frac{-x}{\sqrt{r^2 - x^2}}, \qquad 1 + [f'(x)]^2 = 1 + \frac{x^2}{r^2 - x^2} = \frac{r^2}{r^2 - x^2}.$$

The radical is therefore $\sqrt{1 + [f']^2} = \dfrac{r}{\sqrt{r^2 - x^2}}$, and – watch the magic –

$$\mathrm{d}S = 2\pi\sqrt{r^2 - x^2} \cdot \frac{r}{\sqrt{r^2 - x^2}}\,\mathrm{d}x = 2\pi r\,\mathrm{d}x.$$

The messy square roots cancel *exactly*, leaving a constant. Then

$$S = \int_{-r}^{r} 2\pi r\,\mathrm{d}x = 2\pi r \cdot (2r) = 4\pi r^2.$$

The surface area of a sphere is $4\pi r^2$.

Why this matters

The cancellation in that last example is not luck; it encodes a gorgeous fact known to Archimedes. Because $\mathrm{d}S = 2\pi r\,\mathrm{d}x$ is constant in x, equal slabs of a sphere – slices of equal *thickness along the axis* – have equal surface area, no matter where you cut. The skinny cap near the pole and the fat band near the equator have the same area if they span the same horizontal width. This is *Archimedes' hat-box theorem*: a sphere and its enclosing cylinder, sliced by the same parallel planes, agree band-for-band in surface area. It is why you can project the Earth onto a cylinder (the Lambert equal-area map) without distorting areas. One canceled radical, a deep theorem.

26.6 Work: Accumulating Force Over Distance

We leave geometry and integrate a physical process. In physics, *work* is force times distance – but only when the force is constant. What if the force changes as the object moves? Slice the motion.

Slice. Break the path into tiny displacements $\mathrm{d}x$.

Measure. Over a displacement so small that the force $F(x)$ barely changes, the work done is $(\text{force}) \times (\text{distance}) = F(x)\,\mathrm{d}x$.

Sum.

$$\boxed{W = \int_a^b F(x)\,\mathrm{d}x.}$$

Work is the accumulation of force over distance. This is the prototype of every "integrate a varying rate" problem in science.

Example 26.9 (Stretching a spring – Hooke's law)**.** A spring resists stretching with a force proportional to how far it is stretched: $F(x) = kx$ (Hooke's law), where k is the spring constant. How much work does it take to stretch the spring from its natural length to a displacement $x = d$?

The force is not constant – it grows as you pull – so you must integrate:

$$W = \int_0^d kx \, \mathrm{d}x = k \cdot \frac{x^2}{2}\Big|_0^d = \frac{1}{2}kd^2.$$

The $\frac{1}{2}$ appears for the same reason it always does: you are summing a linearly growing quantity, and the average of a quantity that ramps from 0 to kd is $\frac{1}{2}kd$. Work = average force times distance = $\frac{1}{2}kd \cdot d$. The integral and the averaging picture agree.

26.6.1 Pumping problems: when the slices are layers of fluid

The richest work problems involve pumping liquid out of a tank. Here the slicing is subtle and worth dwelling on, because *each slice is a different amount of stuff that has to move a different distance.* This is the problem that forces you to respect the slice.

The setup: a tank holds water; you must pump it all up and out over the top rim. Slice the water into thin horizontal layers. Consider one layer at height y, thickness $\mathrm{d}y$.

How much does this layer weigh? Its volume is (cross-sectional area at height y) times thickness, $A(y)\,\mathrm{d}y$. Multiply by the weight density of the fluid ρg (for water, about 9800 $\mathrm{N/m^3}$, or in US units the weight density is about 62.4 $\mathrm{lb/ft^3}$). So the layer's weight is $\rho g\, A(y)\,\mathrm{d}y$.

How far must this layer move? If the water exits at height H, this layer must rise a distance $(H - y)$.

Work to lift this one layer = weight × distance:

$$\mathrm{d}W = \rho g\, A(y)\,(H - y)\,\mathrm{d}y.$$

Sum over all layers:

$$\boxed{W = \rho g \int_{y_{\min}}^{y_{\max}} A(y)\,(H - y)\,\mathrm{d}y.}$$

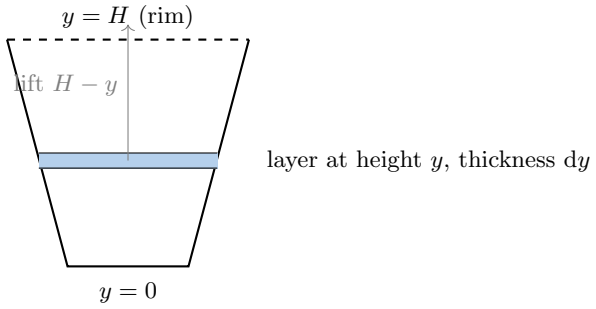


Figure 26.8: Pumping a tank. A horizontal layer at height y weighs $\rho g A(y)\,\mathrm{d}y$ and must be lifted a distance $H - y$ to clear the rim.

Common trap

The two quantities that vary are easy to confuse. In a pumping problem *two* things depend on the slice: how much water is in the layer (through $A(y)$) and how far it travels (through $H-y$). Beginners get one and forget the other – using a constant cross-section for a cone, or forgetting that lower layers must travel farther. Always write both factors explicitly and ask, for a test layer, "is this the weight? is this the distance?" Also: the lift distance is measured to the *exit point*, which may be above the tank's rim if there is a spout. Read the geometry carefully.

Example 26.10 (Pumping a cylindrical tank)**.** A cylindrical tank, radius 2 m and height 5 m, is full of water. Find the work to pump all the water out over the top rim. Use $\rho g = 9800\ \text{N/m}^3$.

Put $y=0$ at the bottom, $y=5$ at the rim, so $H=5$. The cross-section is a constant disk of area $A(y)=\pi(2)^2=4\pi$. A layer at height y must rise $(5-y)$. So

$$W = 9800\int_0^5 4\pi\,(5-y)\,\mathrm{d}y = 9800\cdot 4\pi\left[5y-\frac{y^2}{2}\right]_0^5.$$

The bracket is $25-\frac{25}{2}=\frac{25}{2}$. Hence

$$W = 9800\cdot 4\pi\cdot\frac{25}{2} = 9800\cdot 50\pi = 490000\,\pi \approx 1.54\times 10^6\ \text{J}.$$

About 1.5 million joules. Notice that even though every layer weighs the same (constant cross-section), they do *not* all do the same work, because the lower layers must be lifted farther – the $(5-y)$ factor does the bookkeeping.

Work it yourself

Redo Example 26.10 but pump the water to a spout 1 m above the rim, so $H=6$. You should only need to change one number and re-evaluate $\int_0^5(6-y)\,\mathrm{d}y = [6y-y^2/2]_0^5 = 30-12.5 = 17.5$. Confirm $W = 9800\cdot 4\pi\cdot 17.5 = 686000\,\pi \approx 2.16\times 10^6$ J. Notice how much extra work that one meter of extra lift costs – and why: every single layer pays the surcharge.

26.7 Average Value of a Function

How do you average infinitely many numbers – all the values $f(x)$ takes as x ranges over $[a,b]$? The average of finitely many numbers is their sum divided by their count. For a continuous spread of values, the sum becomes an integral and the count becomes the length of the interval:

$$\boxed{\bar{f} = \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x.}$$

Where does this come from? Sample f at n evenly spaced points and average:

$$\frac{1}{n}\sum_{k=1}^{n} f(x_k) = \frac{1}{b-a}\sum_{k=1}^{n} f(x_k)\,\frac{b-a}{n} = \frac{1}{b-a}\sum_{k=1}^{n} f(x_k)\,\Delta x.$$

We slipped $\Delta x=(b-a)/n$ in and out. As $n\to\infty$ the sum becomes the integral, giving the boxed formula. So the average value is the integral spread evenly over the interval.

Intuition

The average value $\bar{f}$ is the height of the rectangle, sitting on $[a, b]$, that has the *same area* as the region under f. You are flattening the bumpy graph into a level one without changing the total area underneath. Rearranged, $\bar{f}\,(b-a) = \int_a^b f$: average times width equals area. That is the cleanest way to remember it.

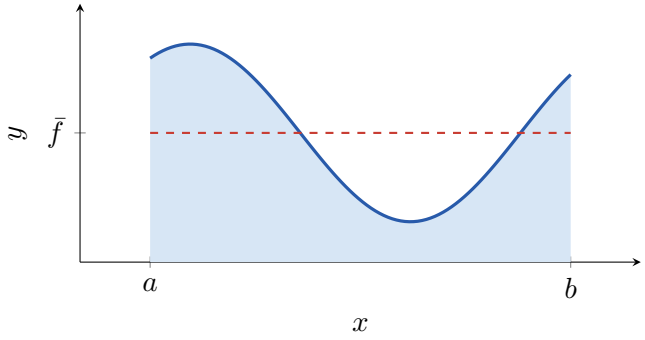


Figure 26.9: The average value $\bar{f}$ is the height of the level rectangle over $[a, b]$ whose area equals the area under f. The wiggles above the dashed line exactly fill the gaps below it.

Example 26.11 (Average value)**.** Find the average value of $f(x) = x^2$ on $[0, 3]$.

$$\bar{f} = \frac{1}{3-0}\int_0^3 x^2\,\mathrm{d}x = \frac{1}{3}\cdot\frac{x^3}{3}\Big|_0^3 = \frac{1}{3}\cdot 9 = 3.$$

The function climbs from 0 to 9, but because it spends most of the interval at smaller values, its average is only 3, well below 9/2, the naive average of the two endpoint values 0 and 9. Averages of nonlinear functions are not the average of the endpoints – the shape matters.

Connection

The *Mean Value Theorem for Integrals* says that a continuous f actually *attains* its average value: there is some $c \in [a, b]$ with $f(c) = \bar{f}$. Geometrically, the curve must cross the level $\bar{f}$ line somewhere – it cannot be entirely above its own average. This is the integral cousin of the Mean Value Theorem for derivatives you met earlier; both say a continuous quantity must, somewhere, equal its average behavior.

26.8 A Preview of Probability Density

We finish with the application that integration was, in a sense, reinvented for in the twentieth century: *continuous probability*. The slicing idea here is so natural that probability almost forces it on you.

How do you describe the probability of a quantity that can take any real value – a person's exact height, the precise time a bus arrives, the energy of a molecule? You cannot assign positive probability to each single value, because there are uncountably many and they would sum to more than 1. The probability of the height being *exactly* 1.8000… meters is zero. So instead you describe probability as something *spread out*, a density, and you accumulate it by integrating.

A **probability density function** (pdf) is a function $p(x) \geq 0$ with total integral 1:

$$\int_{-\infty}^{\infty} p(x)\,\mathrm{d}x = 1.$$

The meaning of p is exactly the slicing meaning: the probability that the quantity lands in a tiny window $[x, x+\mathrm{d}x]$ is

$$\mathrm{d}(\text{prob}) = p(x)\,\mathrm{d}x,$$

and the probability of landing in any range $[a, b]$ is the accumulation of those slices:

$$\boxed{P(a \leq X \leq b) = \int_a^b p(x)\,\mathrm{d}x.}$$

Probability is area under the density curve. That is the whole conceptual content.

What is really happening

A density is not a probability – it is a probability *per unit length*, just as mass density is mass per unit volume. By itself $p(x)$ can even exceed 1; what must stay ≤ 1 is the *area* $p(x)\,\mathrm{d}x$ accumulated over a region. This is why the units work and why "the probability of exactly x" is zero: a single point has width $\mathrm{d}x = 0$, so it carries area zero. Probability lives in the *accumulation*, never in the point. To get a real probability you must integrate the density across a window of nonzero width – you must collect a band of slices.

Example 26.12 (The uniform and exponential densities)**. Uniform on** $[0, L]$**.** A bus is equally likely to arrive at any time in $[0, L]$. The density is constant, $p(x) = 1/L$ on $[0, L]$ and 0 outside. Check: $\int_0^L \frac{1}{L}\,\mathrm{d}x = 1$. The probability the bus comes in the first third is $\int_0^{L/3} \frac{1}{L}\,\mathrm{d}x = \frac{1}{3}$ – area of a rectangle, exactly as intuition demands.

Exponential. The waiting time for a random event (a radioactive decay, a phone call) often follows $p(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. First check it is a valid density (this is an improper integral, from the previous chapter):

$$\int_0^{\infty} \lambda e^{-\lambda x}\,\mathrm{d}x = \Big[-e^{-\lambda x}\Big]_0^{\infty} = 0 - (-1) = 1.$$

The probability the event happens before time t is

$$P(X \leq t) = \int_0^t \lambda e^{-\lambda x}\,\mathrm{d}x = 1 - e^{-\lambda t}.$$

At $t = 0$ this is 0; as $t \to \infty$ it climbs to 1. The accumulation of density sweeps from impossible to certain.

Application

The **expected value** (mean) of a continuous random variable is, once again, a slicing problem. To average x over the density, weight each value x by the probability $p(x)\,\mathrm{d}x$ sitting on it, and sum:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x\,p(x)\,\mathrm{d}x.$$

This is the continuous analogue of $\sum(\text{value}) \times (\text{probability})$, and it is also exactly the formula

for the *center of mass* of a rod whose mass density is p – the balance point. Probability theory and mechanics share the integral because they share the slice: in both, you accumulate a quantity weighted by how much "stuff" (probability, or mass) sits at each location. For the exponential density a quick integration by parts gives $\mathbb{E}[X] = 1/\lambda$, the famous mean waiting time.

26.9 The Common Thread

Look back at everything we built. We solved eight very different problems – area, volume several ways, arc length, surface area, work, pumping, average value, probability – and we never once invented a genuinely new idea. Every single result came from the same three-step move:

Quantity	**One slice** $\mathrm{d}Q$	**Accumulated**
Area between curves	$(f-g)\,\mathrm{d}x$	$\int_a^b (f-g)\,\mathrm{d}x$
Volume (slicing)	$A(x)\,\mathrm{d}x$	$\int_a^b A(x)\,\mathrm{d}x$
Volume (disks)	$\pi[f]^2\,\mathrm{d}x$	$\pi\int_a^b [f]^2\,\mathrm{d}x$
Volume (washers)	$\pi([f]^2-[g]^2)\,\mathrm{d}x$	$\pi\int_a^b ([f]^2-[g]^2)\,\mathrm{d}x$
Volume (shells)	$2\pi x\,f(x)\,\mathrm{d}x$	$\int_a^b 2\pi x f(x)\,\mathrm{d}x$
Arc length	$\sqrt{1+[f']^2}\,\mathrm{d}x$	$\int_a^b \sqrt{1+[f']^2}\,\mathrm{d}x$
Surface area	$2\pi f\sqrt{1+[f']^2}\,\mathrm{d}x$	$\int_a^b 2\pi f\sqrt{1+[f']^2}\,\mathrm{d}x$
Work	$F(x)\,\mathrm{d}x$	$\int_a^b F(x)\,\mathrm{d}x$
Pumping	$\rho g A(y)(H-y)\,\mathrm{d}y$	$\rho g\int A(y)(H-y)\,\mathrm{d}y$
Average value	—	$\frac{1}{b-a}\int_a^b f\,\mathrm{d}x$
Probability	$p(x)\,\mathrm{d}x$	$\int_a^b p(x)\,\mathrm{d}x$

The middle column is the only one that ever required thought, and it always came straight from a picture: a rectangle, a coin, a shell, a triangle, a layer of water. *Find the slice, and the integral writes itself.* That is the entire craft of applied integration, and it will carry you through multiple integrals, center of mass, fluid pressure, and far beyond. Whenever you meet a new quantity to compute, do not hunt for a formula. Ask the one question this chapter is built on: *what am I accumulating, and what does one slice contribute?*

Why this matters

There is something quietly profound here. The Greeks needed a new burst of genius for each shape – a fresh exhaustion argument for the parabola, another for the sphere, another for the spiral. We needed exactly one idea, the integral, and a willingness to look closely at a single slice. That is what it means to *understand* a subject rather than to know its results: a thousand separate facts collapse into one move you can do in your sleep. The integral did not just give us answers; it gave us a way of seeing that makes the answers inevitable.

26.10 What applied integration unlocked, the elliptic barrier, and the road ahead

Connection

This chapter marks the transformation of calculus from a pure mathematical discipline into the universal engine of engineering and natural philosophy.

Armed with the slicing principle ($\mathrm{d}Q = (\text{rate or density})\,\mathrm{d}x$), centuries of fragmented Greek exhaustion proofs (Archimedes on spheres, parabolic segments, and spiral areas) and seventeenth-century mechanical theorems (Pappus and Guldin) were unified into a single standard protocol. Engineers and scientists gained the power to compute:

- Exact geometric metrics: arc length $\int \sqrt{1+[f']^2}\,\mathrm{d}x$ and surface areas of revolution;
- Mechanical dynamics: the physical work $W = \int F\,\mathrm{d}x$ required to compress springs, launch rockets against variable gravity, and pump fluids from irregularly shaped subterranean aquifers;
- Statics: hydrostatic fluid forces on submerged dams and center-of-mass centroids for non-uniform mass distributions.

Yet given how effortlessly the slicing principle sets up integrals, one might assume that any reasonably simple curve can have its physical properties evaluated in closed form. The universe delivers a swift correction: *the simplest possible curves immediately produce non-elementary integrals.*

Consider the perimeter (arc length) of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. Setting up the arc length integral using parameterization $x = a\sin\theta, y = b\cos\theta$ gives:

$$L = 4a\int_0^{\pi/2} \sqrt{1-e^2\sin^2\theta}\,\mathrm{d}\theta,$$

where $e = \sqrt{1-b^2/a^2}$ is the orbital eccentricity. Despite the ellipse being the very first conic section studied after the circle, this integral *cannot be evaluated in elementary terms.* It is the quintessential complete elliptic integral of the second kind.

The exact same non-elementary integral arises when calculating the period of a real-world pendulum. While small-angle approximations teach that $T \approx 2\pi\sqrt{L/g}$, the exact period for an initial release angle θ_0 is:

$$T = 4\sqrt{\frac{L}{g}}\int_0^{\pi/2} \frac{\mathrm{d}\phi}{\sqrt{1-\sin^2(\theta_0/2)\sin^2\phi}},$$

which is a complete elliptic integral of the first kind. The moment mechanics steps away from infinitesimal approximations, symbolic calculus hits an impasse.

How do we calculate the perimeter of an ellipse or the exact swing of a pendulum when closed-form antiderivatives are mathematically impossible?

- *High-precision numerical quadrature (Chapter 27):* Rather than abandoning the integral, we evaluate it numerically. Chapter 27 develops *Numerical Integration* (Simpson's parabolic rule and Newton-Cotes quadrature), allowing computers to evaluate elliptic integrals to arbitrary precision in microseconds.

- *Infinite polynomial expansions (Chapter 30):* Chapter 30 introduces *Taylor Series*, allowing us to expand $\sqrt{1 - e^2 \sin^2 \theta}$ into an infinite binomial series in powers of e^2, integrating term by term to yield rapidly converging series solutions.
- *Multivariable Calculus (Chapter 34):* Later, Chapter 34 extends the slicing principle to two and three dimensions, computing masses, gravitational fields, and fluxes across arbitrary spatial domains.

26.11 Exercises

Exercise 26.1 (Area, both directions)**.** Find the area of the region bounded by $y = \sqrt{x}$ and $y = x^2$. First by vertical slices, then by horizontal slices, and confirm the answers agree.

Exercise 26.2 (Crossing curves)**.** Find the total area enclosed between $y = \sin x$ and $y = \cos x$ from $x = 0$ to $x = \pi/2$. (The curves cross at $x = \pi/4$; remember to use top-minus-bottom on each piece, so the answer is positive.)

Exercise 26.3 (Volume by slicing, non-circular cross sections)**.** A solid has as its base the region between $y = 0$ and $y = 1 - x^2$ for $-1 \le x \le 1$. Every cross-section perpendicular to the x-axis is a square sitting on the base. Set up and evaluate $\int A(x)\,\mathrm{d}x$ to find the volume. (Here the side of the square at position x is $1 - x^2$, so $A(x) = (1 - x^2)^2$.)

Exercise 26.4 (Disk vs. shell)**.** Let R be the region under $y = x^2$ from $x = 0$ to $x = 1$. Find the volume when R is revolved (a) about the x-axis, and (b) about the y-axis. For each, decide whether disks/washers or shells gives the easier integral, and explain your choice in one sentence.

Exercise 26.5 (Arc length)**.** Find the arc length of $y = \frac{1}{3}\left(x^2 + 2\right)^{3/2}$ from $x = 0$ to $x = 2$. (Compute f', simplify $1 + [f']^2$ to a perfect square, and watch the radical disappear.)

Exercise 26.6 (Surface area)**.** The curve $y = \sqrt{x}$ from $x = 0$ to $x = 1$ is revolved about the x-axis. Set up the surface-area integral $\int 2\pi f \sqrt{1 + [f']^2}\,\mathrm{d}x$, simplify the integrand, and evaluate it.

Exercise 26.7 (Work, varying force)**.** A chain 10 m long weighs 5 N/m and hangs from the top of a building. How much work is required to haul the entire chain to the top? (Hint: slice the chain by length; a piece at depth x weighs $5\,\mathrm{d}x$ and must rise a distance x, so $\mathrm{d}W = 5x\,\mathrm{d}x$.)

Exercise 26.8 (Pumping a cone)**.** A tank in the shape of an inverted cone (point down) has height 4 m and top radius 2 m, and is full of water. Set up and evaluate the work to pump all the water to the top rim. (By similar triangles the radius at height y is $y/2$, so $A(y) = \pi(y/2)^2$; the lift is $4 - y$. Use $\rho g = 9800$ N/m^3.)

Exercise 26.9 (Average value, and where it is attained)**.** Find the average value of $f(x) = \sin x$ on $[0, \pi]$. Then find the value(s) of $c \in [0, \pi]$ where $f(c)$ equals that average, illustrating the Mean Value Theorem for Integrals.

Exercise 26.10 (Probability density)**.** A continuous random variable has density $p(x) = c\,x(1 - x)$ on $[0, 1]$ and zero elsewhere. (a) Find c so that $\int_0^1 p = 1$. (b) Compute $P(X \le 1/2)$. (c) Compute the mean $\mathbb{E}[X] = \int_0^1 x\,p(x)\,\mathrm{d}x$, and explain why symmetry could have told you the answer in advance.

Exercise 26.11 (Conceptual: the slice is everything)**.** In one short paragraph each, state "what one slice contributes" ($\mathrm{d}Q$) for: the area between two curves, a washer, a cylindrical shell, and the work to pump a fluid. You should be able to do this from the pictures alone, without looking up a formula – that is the point of the chapter.

Exercise 26.12 (Proof: shells and washers must agree)**.** Consider the region under $y = f(x)$ from $x = 0$ to $x = b$ (with f decreasing, $f(b) = 0$) revolved about the y-axis. Write the shell-method volume $V = \int_0^b 2\pi x f(x)\,\mathrm{d}x$. Using integration by parts with $u = f(x)$ and $\mathrm{d}v = 2\pi x\,\mathrm{d}x$, transform this into an integral over y and identify the result as the washer/disk volume. Conclude that the two methods are not two facts but one, related by integration by parts. (This is a satisfying capstone: the algebra of parts is exactly the geometry of re-slicing.)

Chapter 27

Numerical Integration

There is a quiet scandal at the heart of integral calculus, and it is time we spoke of it plainly. The Fundamental Theorem of Calculus is glorious: it tells us that to compute the definite integral $\int_a^b f(x)\,\mathrm{d}x$ we need only find an antiderivative F with $F' = f$, and then subtract: $F(b) - F(a)$. Clean. Exact. Beautiful. But it has a precondition hidden in plain sight, and the precondition fails far more often than the textbooks let on. *You have to be able to find F.*

For the polished examples we drill in a course, F is always there, waiting to be remembered. In the wild, it usually is not. The function

$$\int_0^1 e^{-x^2}\,\mathrm{d}x$$

has no antiderivative expressible in terms of the elementary functions you know – no finite combination of polynomials, roots, exponentials, logarithms, and trig functions will ever equal an antiderivative of e^{-x^2}. This is not a failure of cleverness. It is a theorem (Liouville, 1830s): the antiderivative provably does not live in that world. Yet the integral is a perfectly definite number, roughly 0.7468, and the entire field of statistics leans on it, because e^{-x^2} is the bell curve.

So what do you do when the Fundamental Theorem hands you a bill you cannot pay? You go back to what the integral *is*. It is an area. It is a limit of sums. And areas can be *estimated* – carefully, with controlled error – even when they cannot be written in closed form. That is numerical integration, and this chapter is about doing it well.

Focus

The integral $\int_a^b f\,\mathrm{d}x$ is a number: the signed area under f. The Fundamental Theorem is one road to that number, available only when an antiderivative exists in usable form. Numerical integration is a different road, always open, that returns to the original meaning of the integral as accumulated area and computes that area directly from samples of f. Our goal: get an accurate number *and* a reliable bound on how wrong it might be.

27.1 When the Antiderivative Will Not Come

First feel the size of the problem, so that numerical methods do not look like a fallback for the lazy but the main highway they actually are.

Here are integrals that arise constantly and have no elementary antiderivative:

$$\int e^{-x^2}\,\mathrm{d}x, \qquad \int \frac{\sin x}{x}\,\mathrm{d}x, \qquad \int \frac{1}{\ln x}\,\mathrm{d}x, \qquad \int \sqrt{1+\tfrac{1}{4}\cos^2 x}\,\mathrm{d}x, \qquad \int \frac{e^x}{x}\,\mathrm{d}x.$$

The first is the Gaussian, the spine of probability. The second, $\sin x/x$, is the sinc function, the heartbeat of signal processing and optics. The third, $1/\ln x$, integrated, is the logarithmic integral $\mathrm{li}(x)$ that estimates how many primes lie below x. The fourth is the arc length of an ellipse – the orbit of every planet – and the reason elliptic integrals got their name. None of them yields to substitution, parts, or partial fractions, because no elementary antiderivative exists to be found.

History

The French mathematician Joseph Liouville, in a series of papers around 1833–1841, did something that must have felt almost philosophical: he proved a *negative*. He showed precisely which functions can have elementary antiderivatives, and built the machinery to certify that e^{-x^2}, $\sin(x)/x$, and many others cannot. Before Liouville, a student who failed to integrate e^{-x^2} might reasonably suspect a gap in their own skill. After Liouville, we know the gap is in the universe of elementary functions, not in us. This is why numerical integration is not a confession of defeat. It is often the only road there is.

And even when an antiderivative *does* exist, you may not have a formula for f at all. A bridge engineer measuring the deflection of a beam, a physiologist tracking cardiac output, a data scientist with a column of numbers – they have f only as a *table of values*, sampled at points. There is no symbolic f to integrate. There is only data, and the area under it. Numerical integration is the language for turning samples into accumulated quantities.

Application

A car's speedometer reads its speed $v(t)$ each second. How far did the car travel between $t = 0$ and $t = 60$? Distance is $\int_0^{60} v(t)\,\mathrm{d}t$. You have no formula for $v(t)$ – you have sixty-one numbers. The only way to recover distance is to estimate the area under the speed curve from those samples. Every fitness tracker, every flight data recorder, every digital oscilloscope integrates this way thousands of times a second.

27.2 The Crudest Defensible Estimate: Rectangles

We already met the idea in the chapter on the integral. Chop $[a, b]$ into n equal slices of width

$$h = \frac{b-a}{n},$$

with cut points $x_0 = a,\ x_1 = a + h,\ \ldots,\ x_n = b$. On each slice approximate the area by a rectangle. If we use the left endpoint's height $f(x_{i-1})$ on each slice we get the *left Riemann sum*; the right endpoints give the right sum.

The trouble with rectangles is visible to the eye. A rectangle has a flat top, but f usually slopes. On a rising function, left rectangles sit entirely below the curve and lose area; right rectangles poke above and gain it (Figure 27.1). The error on each slice is roughly the little triangular sliver between the flat top and the sloping curve. Halve the slice width and you halve roughly each sliver's base *and*

its height – but there are twice as many of them, so the total error only halves. This is the painful arithmetic of rectangles: error $\propto h$. To get one more correct decimal digit you need ten times as many slices.

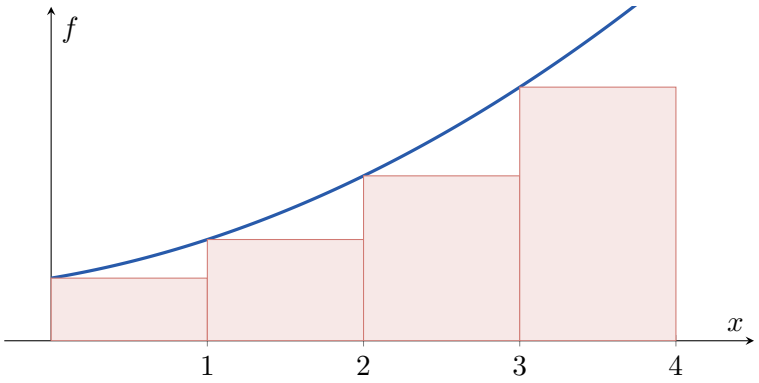


Figure 27.1: Left rectangles under a rising curve. Each rectangle's flat top falls short of the sloping curve, leaving a triangular sliver of lost area. The total error shrinks only in proportion to the slice width h. We can do much better by letting the tops slope.

Intuition

The driving principle of numerical integration is straightforward: *make the top of each piece look more like the curve.* A rectangle's top is a constant – a degree-zero polynomial. The error comes from the curve refusing to be flat. If we let the top be a sloping line (degree one), we cancel the slope error. If we let it be a parabola (degree two), we cancel the curvature error too. Each step up in the local approximation kills the leading source of error and earns us a faster-shrinking mistake. The trapezoidal rule is the line; Simpson's rule is the parabola.

27.3 The Trapezoidal Rule: Let the Tops Slope

The natural remedy for the rectangle error becomes clear once you inspect the missing sliver. Instead of a flat top, connect the two endpoints of each slice with a straight line. The flat-topped rectangle becomes a *trapezoid* whose slanted top hugs the curve far better (Figure 27.2).

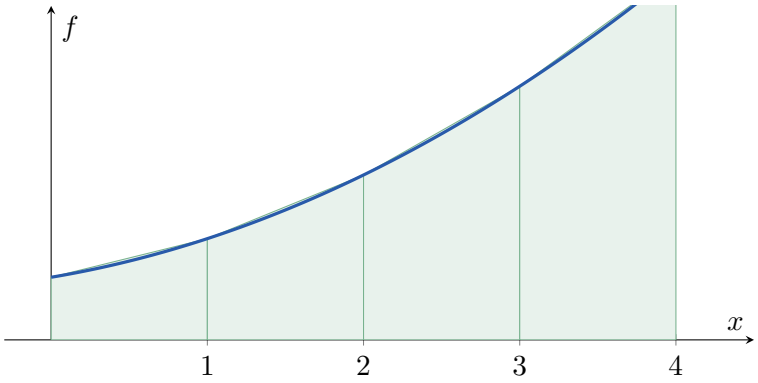


Figure 27.2: The trapezoidal rule: connect consecutive sample points with straight chords. Each trapezoid hugs the curve much more closely than a flat rectangle. The leftover error per slice now comes only from the curve's *bending*, not its sloping.

The area of a single trapezoid is its average height times its width. On the slice $[x_{i-1}, x_i]$ the two heights are $f(x_{i-1})$ and $f(x_i)$, and the width is h, so

$$\text{slice area} \approx h \cdot \frac{f(x_{i-1}) + f(x_i)}{2}.$$

Add up all n slices. Watch what happens: every interior point $x_1, \ldots, x_{n-1}$ is the right endpoint of one trapezoid and the left endpoint of the next, so its height gets counted twice. The two outer endpoints x_0 and x_n are counted once. The total telescopes into a clean formula.

Focus

Trapezoidal Rule. With $h = (b-a)/n$ and $x_i = a + ih$,

$$T_n = \frac{h}{2}\Big[\, f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n) \,\Big].$$

The pattern of weights is $1, 2, 2, \ldots, 2, 1$. Ends count once, interiors count twice, and the whole thing is scaled by $h/2$.

Why the technique works

Why is the weight pattern $1, 2, 2, \ldots, 2, 1$ and not something fancier? Because each interior sample is shared between two neighboring trapezoids – it is the right wall of one and the left wall of the next – so its height contributes to two pieces of area, hence weight 2. The two boundary samples belong to only one trapezoid each, hence weight 1. The trapezoidal rule is nothing more than "replace the curve with the connect-the-dots polygon through your samples, and take the area under the polygon." That polygon is the unique piecewise-linear function agreeing with f at every node. Integrating it exactly gives T_n. So the trapezoidal rule is exact for any function that already *is* a straight line on each slice – and approximately right to the extent that f is nearly straight at the scale of h.

Work it yourself

Estimate $\int_0^1 e^{-x^2}\,dx$ with the trapezoidal rule using $n = 4$. Then $h = 0.25$ and the nodes are $0, 0.25, 0.5, 0.75, 1$. Compute the heights $f(x) = e^{-x^2}$:

x	e^{-x^2}
0	1.000000
0.25	0.939413
0.5	0.778801
0.75	0.569783
1.0	0.367879

Now apply the weights $1, 2, 2, 2, 1$:

$$T_4 = \frac{0.25}{2}\Big[1.000000 + 2(0.939413) + 2(0.778801) + 2(0.569783) + 0.367879\Big].$$

The bracket is $1.000000 + 1.878826 + 1.557602 + 1.139566 + 0.367879 = 5.943873$. Multiply by $h/2 = 0.125$: $T_4 = 0.742984$. The true value is 0.746824. We are off by about 0.0038 – already under one percent with only five function evaluations, and we never found an antiderivative because none exists.

27.3.1 How Wrong Is the Trapezoidal Rule? The Error Comes From Curvature

Look again at Figure 27.2. Where the curve bulges *up* (concave down), the chord cuts underneath and the trapezoid undershoots. Where the curve bulges *down* (concave up), the chord rides above and the trapezoid overshoots. Either way, the error is governed by how much the curve bends – by the second derivative f''. A straight line has $f'' = 0$ and the trapezoidal rule nails it exactly. The more f curves, the more the chord misses.

Make this precise on one slice, because the mechanism is the whole story. Take a single interval of width h and ask how far the area under the chord differs from the area under f. A careful accounting (a short Taylor expansion, done in the box below) gives, for one slice,

$$(\text{true area}) - (\text{trapezoid area}) = -\frac{h^3}{12}\,f''(\xi)$$

for some point ξ inside the slice. The error on one slice is proportional to h^3 and to the curvature there.

What is really happening

Why h^3 per slice but h^2 overall. Each slice carries error of size $\sim h^3$. But to cover a fixed interval $[a, b]$ you need $n = (b - a)/h$ slices, so the errors stack up:

$$\text{total error} \sim n \cdot h^3 = \frac{b-a}{h} \cdot h^3 = (b-a)\,h^2.$$

One power of h is eaten by needing more slices as h shrinks. What survives is h^2. This is the deep pattern of the whole subject: a method with per-slice error h^{p+1} delivers *global* error h^p. The trapezoidal rule has $p = 2$. Halve h (double the work) and the error drops by a factor

of 4, not 2. That is already a world better than rectangles, where halving h only halves the error.

Why the technique works

Where the $-h^3 f''/12$ comes from. Put the slice symmetrically as $[-h/2, h/2]$ to keep the algebra clean. Taylor-expand f about the midpoint:

$$f(x) = f(0) + f'(0)\,x + \tfrac{1}{2}f''(0)\,x^2 + \cdots$$

Integrate across the slice. The odd term $f'(0)x$ integrates to zero by symmetry, leaving

$$\int_{-h/2}^{h/2} f\,\mathrm{d}x = f(0)\,h + \tfrac{1}{2}f''(0)\cdot\frac{2}{3}\Big(\tfrac{h}{2}\Big)^3 + \cdots = f(0)\,h + \frac{f''(0)\,h^3}{24} + \cdots$$

The trapezoid uses the two endpoint heights $f(\pm h/2)$. Their average is

$$\frac{f(h/2)+f(-h/2)}{2} = f(0) + \tfrac{1}{2}f''(0)\Big(\tfrac{h}{2}\Big)^2 + \cdots = f(0) + \frac{f''(0)h^2}{8} + \cdots,$$

so the trapezoid area is $h\big(f(0) + \frac{f''(0)h^2}{8}\big) = f(0)h + \frac{f''(0)h^3}{8}$. Subtract:

$$\text{true} - \text{trap} = \frac{f''(0)h^3}{24} - \frac{f''(0)h^3}{8} = f''(0)h^3\Big(\tfrac{1}{24} - \tfrac{3}{24}\Big) = -\frac{f''(0)h^3}{12}.$$

There it is. The midpoint rule (rectangle of height $f(0)$, area $f(0)\,h$) misses the truth by $+\frac{1}{24}f''(0)h^3$, while the trapezoid misses by $-\frac{1}{12}f''(0)h^3$. Both errors are a term in $f''h^3$, but the trapezoid's is exactly *twice as large, in the opposite direction.* Hold that fact – Simpson will exploit it.

Summing $-h^3 f''(\xi_i)/12$ over all n slices and replacing the average of the $f''(\xi_i)$ by a single value $f''(\xi)$ (a use of the intermediate value property) gives the standard bound.

Focus

Trapezoidal error bound. If $|f''(x)| \le M_2$ for all x in $[a,b]$, then

$$\Big|\int_a^b f\,\mathrm{d}x - T_n\Big| \le \frac{(b-a)\,h^2}{12}\,M_2 = \frac{(b-a)^3}{12\,n^2}\,M_2.$$

The error is controlled by the largest curvature on the interval and falls like $1/n^2$.

Work it yourself

Bound the error of our T_4 estimate of $\int_0^1 e^{-x^2}\,\mathrm{d}x$. We need a bound M_2 on $|f''|$ over $[0,1]$. With $f = e^{-x^2}$, $f' = -2xe^{-x^2}$ and $f'' = (4x^2-2)e^{-x^2}$. On $[0,1]$ this ranges between -2 (at $x=0$) and about 0.74 (at $x=1$), so $|f''| \le 2$; take $M_2 = 2$. With $b-a=1$, $n=4$, $h=0.25$:

$$|\text{error}| \le \frac{1\cdot(0.25)^2}{12}\cdot 2 = \frac{0.0625}{12}\cdot 2 = 0.0104.$$

Our actual error was 0.0038, comfortably inside the guaranteed bound of 0.0104. The bound is valid but conservative – it uses the worst-case curvature everywhere, while the real curvature is smaller over most of the interval. This is the typical relationship: the bound certifies, the true error beats it.

27.4 Simpson's Rule: Let the Tops Be Parabolas

The trapezoidal rule replaced the curve by straight chords and was left with an error driven by curvature, f''. The obvious next move: replace the curve not by lines but by *parabolas*, which can themselves curve. A parabola is determined by three points, so we take the samples two slices at a time – x_0, x_1, x_2 – fit the unique parabola through those three heights, and use the exact area under that parabola as our estimate for the area over $[x_0, x_2]$ (Figure 27.3).

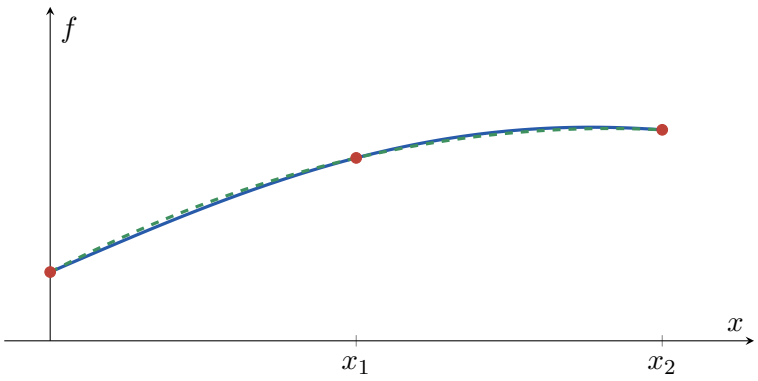


Figure 27.3: Simpson's rule fits a parabola (dashed green) through three consecutive samples and integrates the parabola exactly. A parabola can bend, so it tracks a curving function far better than a straight chord. The fit need not pass under or over the curve everywhere – it weaves through it, and the over- and under-estimates largely cancel.

The arithmetic of "integrate the parabola through three points" produces a startlingly tidy weight pattern. Over a double-slice $[x_0, x_2]$ of total width $2h$, the exact area under the interpolating parabola works out to

$$\int_{x_0}^{x_2} (\text{parabola})\,\mathrm{d}x = \frac{h}{3}\Big[\, f(x_0) + 4f(x_1) + f(x_2) \,\Big].$$

The middle point gets weight 4, the two ends weight 1, scaled by $h/3$. (The derivation is in the box below – it is just integrating a quadratic.) Now stitch these double-slices together along $[a, b]$. This requires n to be *even* so the slices pair up cleanly. Each interior odd-indexed node $x_1, x_3, x_5, \ldots$ is the apex of its own parabola and gets weight 4. Each interior even-indexed node $x_2, x_4, \ldots$ is shared as the right end of one parabola and the left end of the next, so its two weight-1 contributions add to 2. The outer ends keep weight 1.

Focus

Simpson's Rule (requires n even). With $h = (b-a)/n$,

$$S_n = \frac{h}{3}\Big[\, f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + \cdots + 4f(x_{n-1}) + f(x_n)\,\Big].$$

The weight pattern is $1, 4, 2, 4, 2, \ldots, 4, 1$: ends are 1, odd interior nodes are 4, even interior nodes are 2, all scaled by $h/3$.

Why the technique works

Deriving the $\frac{h}{3}[1,4,1]$ formula. Center the double-slice at the origin: nodes at $-h, 0, +h$ with heights y_{-1}, y_0, y_1. Any parabola is $p(x) = A + Bx + Cx^2$. Integrate it across $[-h, h]$:

$$\int_{-h}^{h} (A + Bx + Cx^2)\,\mathrm{d}x = 2Ah + \frac{2Ch^3}{3},$$

since the odd term Bx integrates to zero. Now express A and C through the sampled heights. Plugging in the nodes: $y_0 = A$, and $y_1 + y_{-1} = 2A + 2Ch^2$, so $C = (y_1 - 2y_0 + y_{-1})/(2h^2)$. Substitute:

$$2Ah + \frac{2Ch^3}{3} = 2y_0 h + \frac{h}{3}(y_1 - 2y_0 + y_{-1}) = \frac{h}{3}\big(y_{-1} + 4y_0 + y_1\big).$$

Out drops the famous $1, 4, 1$. Notice we never needed B at all – the slope term self-cancels by symmetry. This is the secret of Simpson's surprising power.

Work it yourself

Estimate $\int_0^1 e^{-x^2}\,\mathrm{d}x$ with Simpson's rule, $n = 4$, reusing the exact same five heights from the trapezoidal lab. Apply the weights $1, 4, 2, 4, 1$:

$$S_4 = \frac{0.25}{3}\Big[1.000000 + 4(0.939413) + 2(0.778801) + 4(0.569783) + 0.367879\Big].$$

The bracket: $1.000000 + 3.757652 + 1.557602 + 2.279132 + 0.367879 = 8.962265$. Multiply by $h/3 = 0.083333$: $S_4 = 0.746855$. The true value is 0.746824. The error is 0.000031 – about 31 *millionths*. Trapezoid with the very same data missed by 0.0038. Simpson, using not a single extra function evaluation, is more than a hundred times more accurate. That is the punchline of this chapter, and now we explain why.

27.5 Why Simpson Is So Astonishingly Good

Simpson's rule fits parabolas, so you would naturally guess it is exact for parabolas and merely good for everything else. It is exact for parabolas – but it is also exact for *cubics*, and that one extra degree of free accuracy is the source of its near-magical performance. The rest of this section shows why, geometrically, and what it buys.

Intuition

The cubic that Simpson sees for free. Consider $f(x) = x^3$ on the symmetric double-slice $[-h, h]$ with midpoint at 0. Its true integral is zero, because x^3 is odd: the positive bump on the right exactly cancels the negative dip on the left. Now ask what Simpson predicts. The weights $1, 4, 1$ act on the heights $(-h)^3, 0, h^3$, i.e. $-h^3, 0, h^3$. Their weighted sum is $1 \cdot (-h^3) + 4 \cdot 0 + 1 \cdot h^3 = 0$. Simpson also returns zero! The parabola Simpson fits cannot *see* the cubic wiggle – but it does not need to, because the symmetric weighting cancels the odd part of the function just as the true integral does. So Simpson gets cubics exactly right, by symmetry, even though it only ever fits quadratics. A method built to be exact on degree 2 turns out, for free, to be exact on degree 3.

Before we cash this in, meet the rule's silent partner: the *midpoint rule*. Instead of sampling at the ends of a slice (trapezoid) it samples at the *center* and uses a rectangle of that single height. You might expect a lone rectangle to be no better than the crude left/right rectangles of Section 27.2, but evaluating at the midpoint transforms the error behavior. Because the rectangle's flat top crosses the curve in the middle of the slice, it overshoots on one side and undershoots on the other, and those two triangular errors very nearly cancel (Figure 27.4). That near-cancellation is why the midpoint rule, despite using only *one* sample per slice, already matches the trapezoid's h^2 accuracy – and, as we are about to see, leans its error in the *opposite* direction, which is exactly what Simpson needs.

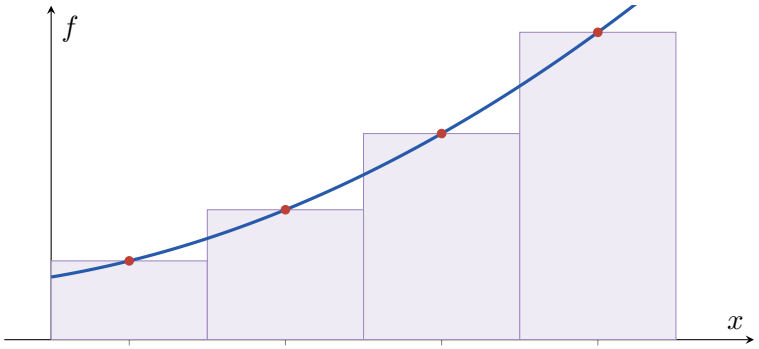


Figure 27.4: The midpoint rule: each rectangle's height is sampled at the *center* of its slice (red dots). The flat top now slices through the curve, overshooting on the low side and undershooting on the high side, so the two errors largely cancel. One sample per slice, yet accuracy $\sim h^2$ – and its leftover error tilts opposite to the trapezoid's, the fact Simpson exploits.

Why the technique works

Simpson is the trapezoid's error, cancelled. Recall the two facts we proved earlier on a single slice: the *midpoint rule* (rectangle of height f at the center) has error $+\frac{1}{24} f'' h^3$ per slice, while the *trapezoid* has error $-\frac{1}{12} f'' h^3$ – the trapezoid's leading error is exactly -2 times the midpoint's. So if we take a weighted blend that kills the f'' term,

$$\frac{2\,(\text{Midpoint}) + 1\,(\text{Trapezoid})}{3},$$

the leading curvature errors cancel: $2 \cdot (+\frac{1}{24}) + 1 \cdot (-\frac{1}{12}) = \frac{1}{12} - \frac{1}{12} = 0$. And this blend *is* Simpson's rule. On a double-slice the trapezoid uses the two endpoints and the midpoint rule

uses the center; combine them in the ratio 2:1 and you reproduce exactly the $1, 4, 1$ pattern. So Simpson is not an unrelated trick: it is precisely the linear combination of the two simple rules engineered to annihilate their common f'' error term. With the f'' term gone, the leading error is pushed all the way out to the next term, the fourth derivative $f^{(4)}$.

Because the f'' error is gone, Simpson's per-slice error is $\sim h^5 f^{(4)}$, and summed over the $n/2$ double-slices the global error is $\sim h^4 f^{(4)}$. Compare:

Rule	Exact for	Global error	Halving h divides error by
Rectangles	constants (deg 0)	$\sim h$	2
Trapezoid	lines (deg 1)	$\sim h^2$	4
Simpson	cubics (deg 3)	$\sim h^4$	16

These three growth laws are not abstractions; you can *watch* them. Figure 27.5 plots the true error of each rule against the number of slices n for our running integral $\int_0^1 e^{-x^2}\,\mathrm{d}x$, on log–log axes. On such axes a power law error $\sim n^{-p}$ becomes a straight line of slope $-p$, and the three rules sort themselves into three visibly different slopes.

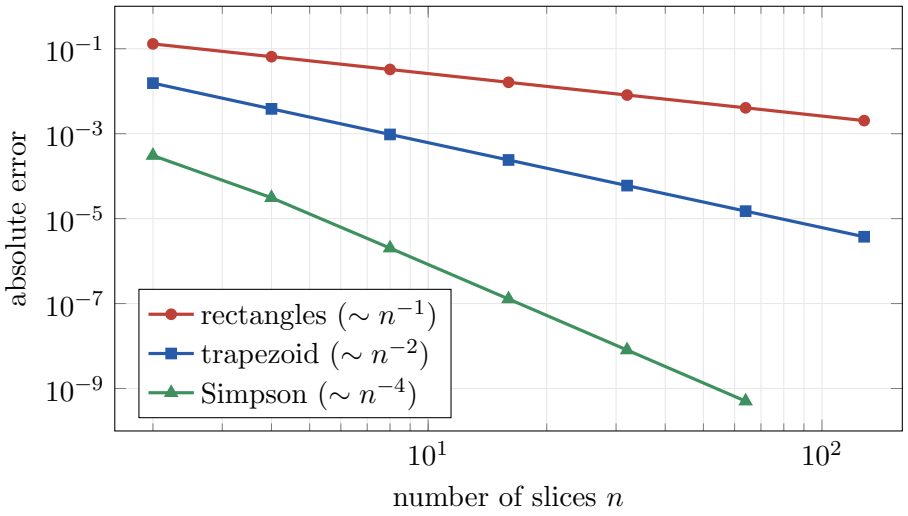


Figure 27.5: Error of each rule versus the number of slices n for $\int_0^1 e^{-x^2}\,\mathrm{d}x$, on log–log axes, where a power law appears as a straight line of slope $-p$. Rectangles fall one decade per decade ($p = 1$), the trapezoid two ($p = 2$), Simpson four ($p = 4$). The vertical gap is the whole story of the chapter: at $n = 32$, Simpson is already near the limits of single precision while the trapezoid still carries five-digit error and rectangles barely three.

Focus

Simpson error bound. If $|f^{(4)}(x)| \le M_4$ on $[a, b]$, then

$$\left| \int_a^b f\,\mathrm{d}x - S_n \right| \le \frac{(b-a)\,h^4}{180}\,M_4 = \frac{(b-a)^5}{180\,n^4}\,M_4.$$

The error falls like $1/n^4$. Doubling the number of slices divides the error by 16.

Work it yourself

Bound the error of S_4 for $\int_0^1 e^{-x^2}\,\mathrm{d}x$. We need $M_4 = \max|f^{(4)}|$ on $[0,1]$. For $f = e^{-x^2}$, $f^{(4)} = (16x^4 - 48x^2 + 12)e^{-x^2}$. At $x = 0$ this is 12; it decreases over $[0,1]$, so $|f^{(4)}| \le 12$; take $M_4 = 12$. With $b - a = 1$, $h = 0.25$:

$$|\text{error}| \le \frac{1 \cdot (0.25)^4}{180} \cdot 12 = \frac{0.00390625}{180} \cdot 12 = 0.00026.$$

Our actual error was 0.000031, well inside the certified 0.00026. Even the conservative bound promises four good digits from five function values; the truth delivers five.

Why this matters

Pause on what just happened. By choosing local pieces one degree smarter – parabolas instead of chords – and getting one more degree of exactness for free by symmetry, the error exponent jumped from h^2 to h^4. That is not a small tweak; it is a different growth law. To reach an error of 10^{-8}, the trapezoidal rule might need $n \approx 10^4$ slices while Simpson needs only $n \approx 10^2$. The same five samples, weighted $1, 4, 2, 4, 1$ instead of $1, 2, 2, 2, 1$, buy a hundredfold accuracy. This is the recurring lesson of numerical analysis: *where you put your effort matters more than how much you spend.* A cleverer formula, not a bigger computer, is what closes the gap.

27.6 A Fuller Worked Example with Both Rules

Let us integrate something with a known answer so we can check ourselves against the truth. Take

$$\int_1^2 \frac{1}{x}\,\mathrm{d}x = \ln 2 = 0.693147\ldots$$

This one *does* have an antiderivative, $\ln x$, but that makes it a perfect test bench: we can compare our estimates against the exact value to the last digit.

Use $n = 4$, so $h = 0.25$ and the nodes are $1, 1.25, 1.5, 1.75, 2$. The heights $f(x) = 1/x$:

x	$1/x$
1.00	1.000000
1.25	0.800000
1.50	0.666667
1.75	0.571429
2.00	0.500000

Trapezoidal. Weights $1, 2, 2, 2, 1$:

$$T_4 = \frac{0.25}{2}\big[1.000000 + 2(0.800000) + 2(0.666667) + 2(0.571429) + 0.500000\big].$$

Bracket: $1.000000 + 1.600000 + 1.333334 + 1.142858 + 0.500000 = 5.576192$. Times 0.125: $T_4 = 0.697024$. Error: 0.003877.

Simpson. Weights $1, 4, 2, 4, 1$:

$$S_4 = \frac{0.25}{3}\big[1.000000 + 4(0.800000) + 2(0.666667) + 4(0.571429) + 0.500000\big].$$

Bracket: $1.000000 + 3.200000 + 1.333334 + 2.285716 + 0.500000 = 8.319050$. Times 0.083333: $S_4 = 0.693254$. Error: 0.000107.

Simpson beats the trapezoidal rule by a factor of about 36 here, on identical data. Now let us check the error bounds. For $f = 1/x$ on $[1, 2]$: $f'' = 2/x^3$, maximized at $x = 1$ giving $M_2 = 2$; and $f^{(4)} = 24/x^5$, maximized at $x = 1$ giving $M_4 = 24$.

$$\text{Trap bound} = \frac{1 \cdot (0.25)^2}{12} \cdot 2 = 0.01042, \qquad \text{Simpson bound} = \frac{1 \cdot (0.25)^4}{180} \cdot 24 = 0.00052.$$

Both actual errors (0.0039 and 0.0001) sit safely below their bounds. The numbers tell a consistent story across two completely different integrands.

Work it yourself

Use the bound to choose n in advance. Suppose you want $\int_1^2 \frac{1}{x}\,dx$ to within 10^{-6} using Simpson. Solve the bound for n:

$$\frac{(b-a)^5}{180\,n^4} M_4 \le 10^{-6} \implies \frac{1 \cdot 24}{180\,n^4} \le 10^{-6} \implies n^4 \ge \frac{24}{180 \cdot 10^{-6}} = 133{,}333 \implies n \ge 19.1.$$

Round up to the next even integer: $n = 20$. With just twenty-one function values, Simpson guarantees six correct digits of $\ln 2$. The same calculation for the trapezoidal rule demands $n \ge 409$ – twenty times the work for the same accuracy. This is how production code actually decides its step size: not by trial and error, but by inverting the error bound.

Common trap

Three classic ways to wreck a numerical integral:

- **Odd n in Simpson.** Simpson pairs slices into parabolas, so n must be even. With odd n the weight pattern $1, 4, 2, \ldots$ does not close properly and your answer is silently wrong. Always check n is even before applying $h/3\,[1, 4, 2, \ldots, 4, 1]$.
- **Forgetting the rules assume smoothness.** The error bounds rely on f'' (trapezoid) or $f^{(4)}$ (Simpson) being bounded. If f has a kink, a spike, or a vertical tangent in the interval – think $\sqrt{x}$ near 0, whose derivatives blow up – the bound is infinite and the promised convergence rate evaporates. Subdivide at the bad point or change variables first.
- **Mixing up $h/2$ and $h/3$.** Trapezoid is $h/2$, Simpson is $h/3$. Swap them and every digit is off. The mnemonic: trapezoid *averages* two heights ($\div 2$); Simpson's weights $1{+}4{+}1 = 6 = 3 \times 2$ live naturally over $\div 3$.

27.7 How Computers Actually Integrate

When you type `quad(f, a, b)` in a scientific library, what runs underneath is a refined descendant of exactly what we have built. The core ideas survive; engineering wraps them.

Adaptive subdivision. A fixed, uniform h is wasteful. Where f is nearly straight, big slices suffice; where f swings violently, you want tiny slices. So real integrators are *adaptive*: they apply Simpson's rule on a region, then split the region in half and apply Simpson to each half. If the two-half estimate agrees with the whole-region estimate to within tolerance, the region is declared done; otherwise each half is recursively subdivided. Effort flows automatically to wherever the function is hardest, which is the only sensible place to spend it.

Why the technique works

How the computer estimates its own error without knowing the answer. This is the cleverest practical idea in the chapter. Run Simpson once on $[a, b]$ with step h to get $S(h)$, then again with step $h/2$ to get $S(h/2)$. We know the error scales like h^4, so $S(h)$ is off by about Ch^4 and $S(h/2)$ by about $C(h/2)^4 = Ch^4/16$. Subtract the two estimates:

$$S(h/2) - S(h) \approx \left(I - \tfrac{Ch^4}{16}\right) - \left(I - Ch^4\right) = Ch^4 - \frac{Ch^4}{16} = \frac{15}{16}\,Ch^4.$$

The thing we cannot see directly – the error $Ch^4/16$ of the better estimate – is therefore about $\frac{1}{15}$ of the *difference between the two estimates*, which we *can* see. So the computer gauges its own error by comparing two grids, with no oracle for the true value. When that estimated error drops below your tolerance, it stops. This is the engine inside every adaptive quadrature routine.

Connection

That same comparison-of-two-grids trick can be turned into *Richardson extrapolation*: form the combination $\big(16\,S(h/2) - S(h)\big)/15$, which cancels the leading h^4 error entirely and leaves an error of order h^6. Applied systematically to the trapezoidal rule, this idea generates the *Romberg integration* table – a triangular scheme that raises the accuracy order at each column. Romberg's first column is trapezoidal estimates on ever-finer grids; its second column is, beautifully, exactly Simpson's rule; later columns are higher-order rules nobody had to derive by hand. It is the same Taylor-cancellation logic we used to build Simpson, run to its logical conclusion. We meet extrapolation again, in disguise, when we study Taylor series.

Gaussian quadrature. We assumed the sample points were *equally spaced*, because that is natural and because it is forced on you when f is a table of measurements. But if you are free to choose *where* to sample f, you can do dramatically better. Gauss asked: with a fixed budget of n sample points, placed wherever we like and weighted however we like, what is the most accurate rule possible? The answer – sampling at special points called the roots of Legendre polynomials, clustered toward the ends of the interval – yields a rule exact for polynomials up to degree $2n - 1$. With the same n evaluations that give Simpson degree-3 exactness, Gaussian quadrature reaches degree $2n - 1$. For smooth functions this is the gold standard, and it is what high-precision libraries reach for first.

History

The trapezoidal rule is ancient in spirit – it is just "average the heights," a technique surveyors used for millennia. The systemic family of techniques based on fitting polynomial curves through equally spaced points is known collectively as **Newton–Cotes quadrature**, developed collaboratively by *Isaac Newton* and his Cambridge protégé *Roger Cotes* around 1700. Cotes systematically tabulated the integration weights for polynomials of degree 1 through 10. *Thomas Simpson* published the parabolic rule bearing his name in his 1743 *Mathematical Dissertations*, though James Gregory and Bonaventura Cavalieri had used geometric variants decades earlier. Simpson recognized the exceptional utility of the parabolic interpolant: by symmetry, the odd cubic error vanishes without cost, granting degree-3 accuracy for the price of degree-2. In 1814, *Carl Friedrich Gauss* revolutionized quadrature by abandoning equally spaced points entirely, showing that choosing n optimal non-uniform evaluation points (the zeros of Legendre polynomials) doubles the precision to degree $2n-1$. Today, production scientific software evaluates stubborn non-elementary integrals using adaptive Gauss–Kronrod quadrature – uniting Gauss's 1814 points with Alexander Kronrod's 1964 embedded error-estimation grid.

What is really happening

What a computer is really doing when it "integrates" is therefore never symbolic and never magic. It samples f at a handful of cleverly chosen points, multiplies those heights by precomputed weights, and adds. Every definite integral your machine evaluates – a probability, a Fourier coefficient, the energy of an orbit, the loss in a training run – comes down to a weighted sum of function values: $\sum_i w_i f(x_i)$. The trapezoidal weights are $1, 2, 2, \ldots, 2, 1$; Simpson's are $1, 4, 2, 4, \ldots, 1$; Gauss's are subtler still. But it is always the same shape. Numerical integration is the art of choosing the points x_i and the weights w_i so that this finite sum tracks an infinite, continuous accumulation as closely as the smoothness of f allows.

Walking question

On your walk, picture the speedometer-and-distance problem. You glance at the speedometer once a minute on a winding drive. The trapezoidal rule says: average each consecutive pair of readings and multiply by a minute. Simpson says: take readings three at a time and trust that the speed varied parabolically between them. Now ask yourself: on a stretch where your acceleration was steadily increasing (constant f'' but $f''' = 0$), which rule is exactly right, and why does Simpson nail it while the trapezoid does not? Then ask the harder one: if you got to choose *when* to glance, would you space your glances evenly – or look more often through the twists and less often on the straightaways?

27.8 Summary: The Logic of the Whole Subject

We began stranded – handed integrals like $\int_0^1 e^{-x^2}\,dx$ that provably have no elementary antiderivative, so the Fundamental Theorem cannot help. We returned to the integral's true meaning as area and learned to estimate that area from samples.

- **Rectangles** approximate f by constants on each slice. Global error $\sim h$. Crude.

- **Trapezoidal rule**, weights $\frac{h}{2}[1, 2, \ldots, 2, 1]$, approximates f by straight chords. Exact for lines; error driven by curvature f''; global error $\sim h^2$. Built from connect-the-dots.
- **Simpson's rule**, weights $\frac{h}{3}[1, 4, 2, 4, \ldots, 4, 1]$ with n even, approximates f by parabolas. Exact for cubics (one bonus degree, by symmetry); error driven by $f^{(4)}$; global error $\sim h^4$. Built as the curvature-cancelling blend $\frac{2\,\text{Mid}+\text{Trap}}{3}$.
- **Error bounds** let you certify accuracy and choose n in advance: trapezoid $\leq \frac{(b-a)^3}{12n^2}M_2$, Simpson $\leq \frac{(b-a)^5}{180n^4}M_4$.
- **Real software** uses these as building blocks inside adaptive subdivision (spend effort where f is wild), self-error-estimation by comparing grids, Richardson extrapolation and Romberg tables, and Gaussian quadrature (optimal sample placement). It is always, in the end, a weighted sum $\sum w_i f(x_i)$.

The throughline is a single idea, applied with increasing refinement: *replace the curve, locally, by something you can integrate exactly, and the better the local fit, the faster the error dies.* Constant, line, parabola – and the error exponent climbs h, h^2, h^4. That is calculus turning its own definition into an algorithm.

27.9 The Reach of Quadrature, the Curse of Dimensionality, and the Road Ahead

Connection

Numerical quadrature completes the transition of integration from symbolic puzzle-solving to universal numerical computation, but moving beyond one dimension introduces a fundamental barrier.

Numerical quadrature liberated mathematics, physics, and modern engineering from the constraints of Liouville's theorem. When an elementary antiderivative does not exist—as with the Gaussian error function $\text{erf}(x)$, complete elliptic integrals of the pendulum, or diffraction Fresnel integrals—Simpson's rule and Gauss–Kronrod quadrature evaluate the integral to machine precision with minimal computational cost. Furthermore, it solved the empirical problem: in real-world engineering, telemetry from accelerometers, seismographs, and medical sensors yields only discrete arrays of numbers rather than algebraic formulas. Numerical quadrature enables instantaneous recovery of velocity, position, energy, and total volume directly from sampled data.

Yet in one dimension, Simpson's rule is astonishingly efficient: halving the step size h reduces the truncation error by a factor of 16 (h^4), so a few dozen evaluation points achieve an error of 10^{-6}. The catastrophe occurs the moment we enter higher dimensions. In physics and machine learning, we routinely need to integrate over d-dimensional spaces ($d = 3$ in fluid dynamics, $d = 6$ in Hamiltonian phase space, $d = 100$ in statistical thermodynamics, or $d = 10^5$ in neural network latent spaces). If we construct a tensor-product grid using N sample points per axis, the total number of function evaluations required is:

$$N_{\text{total}} = N^d.$$

For a modest grid of $N = 100$ points:

- In 1 dimension: $10^2 = 100$ evaluations (a fraction of a millisecond);
- In 3 dimensions: $10^6 = 1{,}000{,}000$ evaluations (feasible on a laptop);
- In 10 dimensions: 10^{20} evaluations (thousands of years on the fastest supercomputer).

This exponential explosion is Richard Bellman's celebrated *curse of dimensionality*. Deterministic Newton–Cotes quadrature grids become completely paralyzed when $d > 4$.

Overcoming this barrier requires two distinct computational philosophies:

- *Multivariable Calculus (Chapter 34):* In Chapter 34 (*Multiple Integrals*), we develop analytic multi-dimensional integration using Fubini's theorem and coordinate transformations (polar, cylindrical, and spherical coordinates), exploiting physical symmetry to collapse multidimensional integrals back into nested one-dimensional problems.
- *Monte Carlo integration:* In high-dimensional physics and modern machine learning, the curse of dimensionality is bypassed by abandoning deterministic grids in favor of random sampling. Monte Carlo quadrature achieves an error rate of $O(1/\sqrt{N})$ *regardless of the dimension* d, turning an impossible 10^{20}-point grid into a manageable statistical sample.

27.10 Exercises

Exercise 27.1. Estimate $\int_0^2 \sqrt{1+x^3}\,\mathrm{d}x$ using the trapezoidal rule with $n = 4$. (Nodes $0, 0.5, 1, 1.5, 2$; compute each $\sqrt{1+x^3}$ to four decimals, then apply weights $1, 2, 2, 2, 1$ and multiply by $h/2$.)

Exercise 27.2. Estimate the same integral $\int_0^2 \sqrt{1+x^3}\,\mathrm{d}x$ with Simpson's rule, $n = 4$, reusing your five heights from Exercise 1 but with weights $1, 4, 2, 4, 1$ and factor $h/3$. By how much do the two estimates differ?

Exercise 27.3. Compute $\int_0^\pi \sin x\,\mathrm{d}x$ exactly (you know the antiderivative), then estimate it with Simpson's rule using $n = 2$ (nodes $0, \pi/2, \pi$). Find the actual error. Why is it not zero even though sin is smooth?

Exercise 27.4. For $\int_0^1 e^{-x^2}\,\mathrm{d}x$, how large must n be to guarantee the trapezoidal rule is within 10^{-4} of the truth? Use $M_2 = 2$. Then find the n needed for Simpson to reach the same tolerance using $M_4 = 12$. Comment on the ratio.

Exercise 27.5. A car's speed (in m/s) is recorded every 2 seconds: at $t = 0, 2, 4, 6, 8, 10, 12$ the readings are $0, 8, 17, 22, 24, 25, 25$. This gives $n = 6$ intervals – an even number, so Simpson applies. Estimate the distance travelled, $\int_0^{12} v\,\mathrm{d}t$, with (a) the trapezoidal rule and (b) Simpson's rule. Which would you trust more, and why?

Exercise 27.6. (Conceptual.) Explain in your own words why the trapezoidal rule *overestimates* the integral of a function that is concave up (like e^x) on the whole interval, and underestimates a concave-down function. Tie your answer to the sign of f'' in the error term $-\frac{h^3}{12}f''(\xi)$ per slice.

Exercise 27.7. (Proof.) Show directly that Simpson's rule is exact for every cubic $f(x) = ax^3 + bx^2 + cx + d$ on a single double-slice $[-h, h]$. That is, compute $\int_{-h}^{h} f\,\mathrm{d}x$ exactly and compare it to $\frac{h}{3}[f(-h)+4f(0)+f(h)]$, and verify they are equal. Identify which term of f would have to be present (degree 4) for Simpson to start being wrong.

Exercise 27.8. (Proof / derivation.) Show that for the trapezoidal rule applied to a single slice $[x_0, x_1]$ of width h, the leading error term is $-\frac{h^3}{12} f''(\xi)$ by Taylor-expanding f about the midpoint and integrating, mirroring the symmetric derivation in the chapter.

Exercise 27.9. (Self-error estimation.) Using $\int_1^2 \frac{1}{x}\,\mathrm{d}x$ with Simpson, compute S_2 (i.e. $h = 0.5$) and S_4 ($h = 0.25$). Form the estimated error of S_4 as $\frac{1}{15}(S_4 - S_2)$ and compare it to the actual error $\ln 2 - S_4$. How close is the estimate? This is exactly what adaptive software does.

Exercise 27.10. (Richardson extrapolation.) Continue the previous exercise: form the combination $S^\star = \frac{16 S_4 - S_2}{15}$, which cancels the leading h^4 error. Compute $S^\star$ and its error against $\ln 2 = 0.693147$. You should find it dramatically more accurate than either S_2 or S_4 alone – this is the first step of Romberg integration.

Part VI

Sequences and Series

Chapter 28

Sequences and Their Limits

For most of this book a limit has been a process *along a continuum*. You let x slide smoothly toward some point a and you watched what $f(x)$ did. The input could come from the left, from the right, through rationals, through irrationals – any way at all, as long as it crept up on a. That freedom was the whole content of the epsilon-delta definition.

Now we change the shape of the approach. Instead of a continuous variable gliding toward a point, picture a discrete *list* of numbers, one after another, marching out forever:

$$a_1,\ a_2,\ a_3,\ a_4,\ \ldots,\ a_n,\ \ldots$$

There is a first term, a second term, a thousandth term, but never a last term. The question is no longer "what happens as $x \to a$?" but "where is this list *headed* as we walk down it?" Does it settle on a single number? Does it wander forever? Does it blow up?

This is the idea of a *sequence* and its *limit*, and it is the gateway to the second half of calculus. Every infinite series – every Taylor expansion, every $\pi = 4(1 - \frac{1}{3} + \frac{1}{5} - \cdots)$, every decimal that never terminates – is secretly a sequence in disguise: the sequence of its partial sums. Newton's method produces a sequence of guesses. The trapezoid rule produces a sequence of approximations. Even a real number like $\sqrt{2}$ is, at bottom, the destination of a sequence of rationals. Understand how a list of numbers can *converge* and all of these come within reach at once.

The chapter proceeds in three stages. We begin by pinning down what a sequence is and what it means for one to have a limit – the same challenge-and-response game you already know, re-tuned for the integers. From there we assemble the working tools: the squeeze theorem for sequences, a short table of limits you will use constantly, and the deep fact that a sequence which climbs but stays bounded *must* converge. Last comes recursion – sequences in which each term is built from the one before – together with the idea of a *fixed point*, which is where such sequences come to rest.

28.1 What a sequence really is

Strip away the dots and a sequence is something you already understand: a function. The only twist is its domain.

Definition 28.1 (Sequence)**.** A *sequence* of real numbers is a function $a\colon \mathbb{N} \to \mathbb{R}$ whose domain is

the natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$. Instead of writing $a(n)$ we write a_n and call it the *nth term*. We denote the whole sequence by $(a_n)_{n=1}^{\infty}$, or just (a_n).

History

The intellectual drama of sequences began around 450 BCE with the Greek philosopher *Zeno of Elea*. In his famous paradoxes of motion – the *Dichotomy* and *Achilles and the Tortoise* – Zeno argued that motion across a room is impossible. Before you can reach the opposite wall, you must first cover half the distance ($a_1 = 1/2$). Before you can cover the remaining half, you must cover half of that ($a_2 = 1/2 + 1/4 = 3/4$). Then half again ($a_3 = 7/8$). You are trapped facing an infinite sequence of discrete spatial milestones:

$$a_n = 1 - \frac{1}{2^n}.$$

Zeno argued that because an infinite number of discrete acts cannot be completed in finite time, you can never reach the wall – in fact, you can never even take your first step!
For over two thousand years, Greek geometry remained paralyzed by Zeno's paradoxes because ancient thinkers lacked a formal language for the limit of an infinite sequence. They feared actual infinity, insisting that infinite processes could never terminate at a well-defined boundary. It was only with the mathematical taming of sequences that the riddle was resolved: an infinite sequence of non-zero time intervals and distances can possess a strictly finite limit. Zeno's runner arrives at $L = 1$ not by magic, but by convergence.

That is the entire definition. A sequence is a function whose inputs are $1, 2, 3, \dots$ rather than a continuous interval. Where an ordinary function f has a smooth graph you can trace with a pencil, a sequence has a graph made of *isolated dots*, one above each integer on the horizontal axis.

Intuition

Three ways to read the same object, all useful:

- **As a list.** $\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$ – the thing you write down term by term.
- **As a function.** $a_n = \frac{1}{n}$, the rule that turns an input n into an output. Plug in $n = 7$, read off $\frac{1}{7}$.
- **As a picture.** A scatter of dots at heights $1, \frac{1}{2}, \frac{1}{3}, \dots$ above the integers $1, 2, 3, \dots$, sinking toward the horizontal axis but never touching it.

Keep all three in mind. The list view is how you compute, the function view is how you reason, and the picture view is how you *see* where the sequence is going (Figure 28.1).

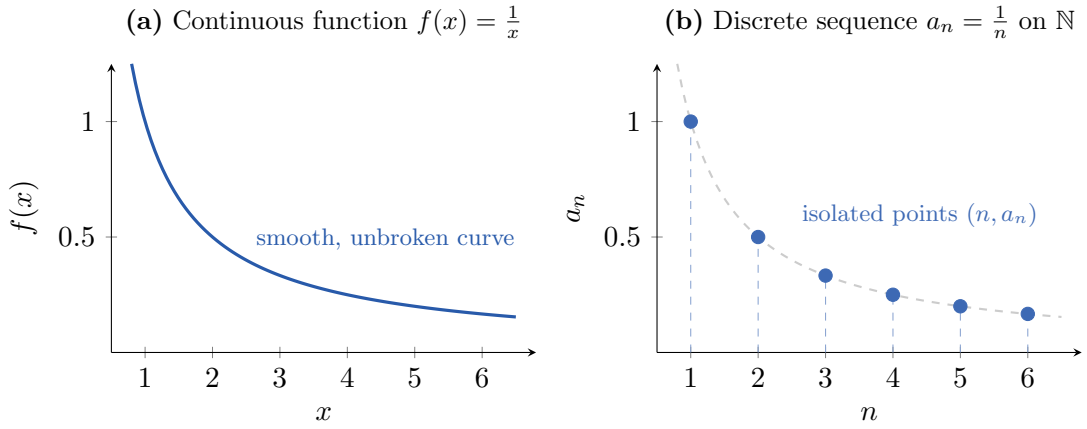


Figure 28.1: Continuous function versus discrete sequence. **(a)** A continuous function $f(x) = 1/x$ has a continuous real domain $[1, \infty)$ and a smooth, unbroken graph. **(b)** A sequence $a_n = 1/n$ restricts the domain to the positive integers $\mathbb{N} = \{1, 2, 3, \dots\}$. Its graph consists of isolated dots sampling the underlying curve at discrete integer stations.

There are three common ways to hand someone a sequence.

By a closed formula. You give a_n as an explicit expression in n. For example $a_n = \dfrac{n}{n+1}$ produces $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots$. To find the thousandth term you just substitute: $a_{1000} = \frac{1000}{1001}$. No bookkeeping required.

By recursion. You give the first term (or first few) and a rule that builds each new term from earlier ones. The classic is the Fibonacci sequence, $F_1 = F_2 = 1$ and $F_n = F_{n-1} + F_{n-2}$, which marches out as $1, 1, 2, 3, 5, 8, 13, \dots$. Recursion is powerful but slippery: to know a_{1000} you must, in principle, walk through all 999 terms before it. Section 28.7 is devoted to taming such sequences.

By description. Sometimes the rule is verbal: "let p_n be the nth prime," giving $2, 3, 5, 7, 11, \dots$. There is no tidy formula, but it is a perfectly good function $\mathbb{N} \to \mathbb{R}$.

Walking question

Write out the first five terms of $a_n = \dfrac{(-1)^n}{n}$. What is the sign doing, and what are the magnitudes doing? Where do you think this list is headed?

The terms are $-1, \frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, -\frac{1}{5}$. The sign alternates; the magnitudes $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}$ shrink toward zero. So the dots bounce above and below the axis but the bounces get ever smaller. The list is being squeezed toward 0 from both sides. Hold that image – it is exactly the case the squeeze theorem (Section 28.4) will handle.

Reading note

A word on indexing. Some sequences naturally start at $n = 0$ (powers, $2^0, 2^1, 2^2, \dots$) and some at $n = 1$. Nothing about a limit depends on where the list starts – changing or deleting the first million terms cannot change where the *tail* is headed. We will start at $n = 1$ unless a formula begs to start at 0, and we will be relaxed about it, because the mathematics is relaxed about it.

28.2 Convergence: the destination of a list

Look again at $a_n = \frac{1}{n}$. The terms are $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$. Nobody hesitates to say this sequence "goes to 0." But what, *precisely*, does that claim assert? Not that any term equals zero – none does. Not merely that the terms decrease – the sequence $1, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \dots$ also decreases (weakly) but goes to $\frac{1}{2}$, not 0. The real content is this:

> *No matter how small a margin of error you name, eventually – from some term onward – every term of the sequence lands inside that margin of* 0.

This is the challenge-and-response game again, the same one that defined $\lim_{x\to a} f(x)$, but cleaner, because there is no δ to find. The input does not slide toward a point; it just *increases*. So the response to a challenge is not "get within δ of a" but simply "go far enough out in the list."

Intuition

The game. A skeptic challenges you with a tiny positive number ε – a tolerance band of half-width ε drawn around the candidate limit L. Your job is to answer with a cutoff N: a place in the list beyond which *all* terms lie inside the band. If you can meet *every* challenge, no matter how small the ε, then L is the limit. If there is even one ε you cannot answer, L is not the limit.

The skeptic moves first and can make ε as cruelly small as desired. You move second, so you may choose N *after* seeing ε – and N is allowed to be enormous. Small ε usually forces large N; that is fine. The only thing that matters is that for each ε *some* workable N exists.

Definition 28.2 (Limit of a sequence)**.** A sequence (a_n) *converges* to the number L, written

$$\lim_{n\to\infty} a_n = L \qquad \text{or} \qquad a_n \to L,$$

if for every $\varepsilon > 0$ there exists a natural number N such that

$$n \geq N \quad \Longrightarrow \quad |a_n - L| < \varepsilon.$$

If no such L exists, the sequence *diverges*.

Read that implication out loud as a sentence in the game: "once you go past the cutoff N, every term is within ε of L." The quantity $|a_n - L|$ is the distance from the nth term to the candidate limit; demanding it be less than ε is demanding the term sit inside the tolerance band.

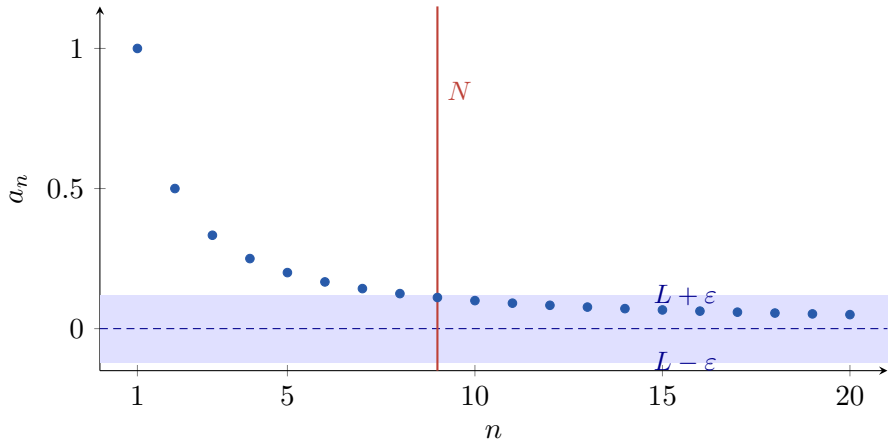


Figure 28.2: The sequence $a_n = 1/n$ converging to $L = 0$. The shaded strip is the tolerance band $|a_n - 0| < \varepsilon$ for $\varepsilon = 0.12$. To the right of the red cutoff N (here $N = 9$) every dot lies inside the band. Shrink ε and the band narrows, pushing the cutoff N rightward – but a workable N always exists.

Example 28.1 (Proving $1/n \to 0$ from the definition)**.** We claim $\lim_{n\to\infty} \frac{1}{n} = 0$. Let $\varepsilon > 0$ be given by the skeptic. We must produce N so that $n \geq N$ forces $|\frac{1}{n} - 0| = \frac{1}{n} < \varepsilon$.

Solve the target inequality for n: $\frac{1}{n} < \varepsilon$ exactly when $n > \frac{1}{\varepsilon}$. So *any* integer N larger than $\frac{1}{\varepsilon}$ will do; the existence of such an integer is guaranteed by the Archimedean property of the real numbers (no real number is larger than every integer). Concretely, take $N = \lfloor 1/\varepsilon \rfloor + 1$.

Now verify. If $n \geq N > \frac{1}{\varepsilon}$, then $n > \frac{1}{\varepsilon}$, hence $\frac{1}{n} < \varepsilon$, hence $|\frac{1}{n} - 0| < \varepsilon$. The challenge is met, and since ε was arbitrary, $\frac{1}{n} \to 0$. □

Why the technique works

Notice the *shape* of that proof, because every epsilon-N proof has it.

1. **Let $\varepsilon > 0$ be given.** You do not get to choose it; you must handle all of them at once by handling a typical one.
2. **Work backwards.** Start from the goal $|a_n - L| < \varepsilon$ and solve for what n must satisfy. This is scratch work – it tells you how big N has to be.
3. **Declare N.** Announce the cutoff your scratch work revealed.
4. **Verify forwards.** Assume $n \geq N$ and chain the inequalities to reach $|a_n - L| < \varepsilon$. This is the actual proof; steps 2–3 were the search.

The search and the proof run in opposite directions. Beginners blur them together and get confused about which inequality they are allowed to assume. Keep them separate: solve backward in the margin, then write forward on the page.

Divergence, and a sequence with no limit

A sequence diverges if no L works. There are two flavors worth naming.

Divergence to infinity. The terms $a_n = n$ grow without bound. They cannot converge to any

finite L, because once the terms exceed $L+1$ they can never return to within $\varepsilon = 1$ of L. We write $a_n \to \infty$ as shorthand, but be clear: ∞ is not a number the sequence reaches; it is a description of unbounded growth.

Oscillation. The terms $a_n = (-1)^n$ cycle $-1, 1, -1, 1, \ldots$ forever. This sequence is bounded – it never escapes $[-1, 1]$ – yet it has no limit. If we proposed $L = 1$, the term -1 keeps appearing arbitrarily far out, so the choice $\varepsilon = 1$ is never met from any cutoff on. The same defeat befalls $L = -1$, or any L in between. No single destination; no limit.

Common trap

Bounded does not mean convergent. The sequence $(-1)^n$ stays trapped in $[-1, 1]$ but never settles. Convergence requires the terms to *pile up* near one number, not merely to stay in a finite range. We will see in Section 28.6 the extra ingredient – *monotonicity* – that upgrades "bounded" into "convergent."

History

The notation lim comes from the Latin *limes*, a boundary or frontier – the Romans used it for the fortified edge of empire. The rigorous epsilon-N formulation was forged in the early nineteenth century by *Bernard Bolzano* (1817) and *Augustin-Louis Cauchy* in his landmark 1821 textbook *Cours d'Analyse*. Cauchy swept away two centuries of vague Newtonian "fluxional flow" and Leibnizian "vanishing differentials," redefining convergence strictly through inequality bounds: a sequence (a_n) converges to L if the discrepancy $|a_n - L|$ can be made smaller than any pre-assigned quantity ε for all indices past some finite threshold N.

In the 1860s, *Karl Weierstrass* perfected this framework at Berlin, showing that the real numbers possess a profound topological property: the **Bolzano–Weierstrass Theorem**. This theorem states that *every bounded sequence of real numbers contains a convergent subsequence.* It guarantees that an infinite collection of numbers trapped in a closed room cannot wander endlessly without accumulating around at least one cluster point.

Before Bolzano, Cauchy, and Weierstrass, "a_n approaches L" was an ambiguous verbal metaphor, and metaphors led even master mathematicians astray. Cauchy himself once published a false "proof" that an infinite sum of continuous functions is always continuous – an error spectacularly shattered by Joseph Fourier's heat-conduction waves, which summed infinitely many smooth sines into discontinuous square waves with vertical jumps! The cure was not sharper intuition, but Cauchy and Weierstrass's ironclad quantifier discipline: $\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N$.

28.3 The limit laws, and why they transfer

You spent a chapter proving that limits of functions respect addition, multiplication, and division. The good news: sequence limits inherit every one of those laws, with proofs that are if anything simpler, because there is no δ to juggle.

Theorem 28.1 (Algebra of sequence limits)**.** *Suppose $a_n \to A$ and $b_n \to B$. Then*

$$a_n + b_n \to A + B, \qquad a_n\, b_n \to A\, B, \qquad c\, a_n \to c\, A \quad (\text{constant } c),$$

and, provided $B \neq 0$ and $b_n \neq 0$ for all n,

$$\frac{a_n}{b_n} \to \frac{A}{B}.$$

Moreover the limit, when it exists, is unique.

We will not grind out all of these – they mirror the function case – but the sum law is short and shows the flavor, and uniqueness is worth a glance because it justifies the very phrase "the limit."

Why the technique works

Why the sum law holds. Suppose $a_n \to A$ and $b_n \to B$. We want $(a_n + b_n) \to A + B$. The distance to beat is

$$|(a_n + b_n) - (A + B)| = |(a_n - A) + (b_n - B)| \leq |a_n - A| + |b_n - B|,$$

where the last step is the triangle inequality. Each piece on the right can be made small. Given $\varepsilon > 0$, use $a_n \to A$ with tolerance $\varepsilon/2$ to get a cutoff N_1 past which $|a_n - A| < \varepsilon/2$; use $b_n \to B$ with the same tolerance to get a cutoff N_2 past which $|b_n - B| < \varepsilon/2$. Let $N = \max(N_1, N_2)$. For $n \geq N$ both bounds hold at once, so

$$|(a_n + b_n) - (A + B)| \leq |a_n - A| + |b_n - B| < \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon.$$

That is the whole proof. The two tricks – split the error with the triangle inequality, and demand $\varepsilon/2$ from each half so they recombine to ε – are the bread and butter of analysis. You will use them again and again.

Why the technique works

Why the limit is unique. Suppose, for contradiction, that $a_n \to L$ and also $a_n \to L'$ with $L \neq L'$. Let $\varepsilon = \frac{1}{2}|L - L'| > 0$, which is the half-gap between the two candidates. Far enough out, the terms are within ε of L; far enough out, they are also within ε of L'. Past both cutoffs a single term a_n would sit within ε of *both* L and L' – but the two ε-bands do not overlap, since the gap between L and L' is 2ε. A point cannot be inside both. Contradiction. So $L = L'$: a convergent sequence has exactly one limit.

Example 28.2 (Limit laws in action)**.** Find $\displaystyle\lim_{n\to\infty} \frac{3n^2 - n + 5}{2n^2 + 7n}$.

As $n \to \infty$ both top and bottom blow up; the expression is an $\frac{\infty}{\infty}$ form and we cannot read off the answer. The standard move is to divide top and bottom by the highest power present, here n^2:

$$\frac{3n^2 - n + 5}{2n^2 + 7n} = \frac{3 - \dfrac{1}{n} + \dfrac{5}{n^2}}{2 + \dfrac{7}{n}}.$$

Now each little piece is something we know. By Example 28.1, $\frac{1}{n} \to 0$; multiplying that limit by itself (the product law) gives $\frac{1}{n^2} \to 0$; scaling by constants and adding (the sum and constant-multiple laws) lets us evaluate numerator and denominator separately:

$$\text{numerator} \to 3 - 0 + 0 = 3, \qquad \text{denominator} \to 2 + 0 = 2.$$

Since the denominator's limit 2 is nonzero, the quotient law applies and

$$\lim_{n\to\infty} \frac{3n^2 - n + 5}{2n^2 + 7n} = \frac{3}{2}.$$

The leading coefficients won; everything else washed out. That is the typical fate of a ratio of polynomials of equal degree.

Connection

There is a clean bridge between sequence limits and the function limits of Chapter 9. If f is a function with $\lim_{x\to\infty} f(x) = L$, and $a_n = f(n)$, then $a_n \to L$ as well – a sequence is just the function sampled at the integers, so if the whole continuous tail heads to L, the sampled dots do too. This is what lets you import l'Hopital's rule: to find $\lim_{n\to\infty} \frac{\ln n}{n}$, study the function $\frac{\ln x}{x}$, apply l'Hopital to get $\frac{1/x}{1} \to 0$, and conclude $\frac{\ln n}{n} \to 0$.
The bridge runs only one way, though. The continuous limit existing forces the sampled limit to exist, but *not* conversely: $a_n = \sin(\pi n) = 0$ for every integer n, so $a_n \to 0$, while $\sin(\pi x)$ oscillates forever and has no limit as $x \to \infty$. Sampling can be calmer than the function it samples.

28.4 The squeeze theorem for sequences

Some sequences resist the limit laws because they are tangled with something that oscillates or otherwise misbehaves – think $\frac{\sin n}{n}$, where the numerator lurches between -1 and 1 with no pattern you can pin down. You cannot "take the limit of $\sin n$" because it has none. But you do not need to. You only need to trap the sequence between two others that you *can* handle.

Theorem 28.2 (Squeeze theorem for sequences)**.** *Suppose that, from some index onward,*

$$\ell_n \le a_n \le u_n,$$

and that both bounding sequences converge to the same limit, $\ell_n \to L$ and $u_n \to L$. Then $a_n \to L$ as well.

Intuition

The picture is exactly what the name says. Two sequences – a lower fence ℓ_n and an upper fence u_n – close in on the same height L. The sequence a_n is caught between the fences. As the gap between the fences crushes down to nothing, the trapped sequence has nowhere to go but L. It is squeezed to its destination whether it "wants" to oscillate or not.

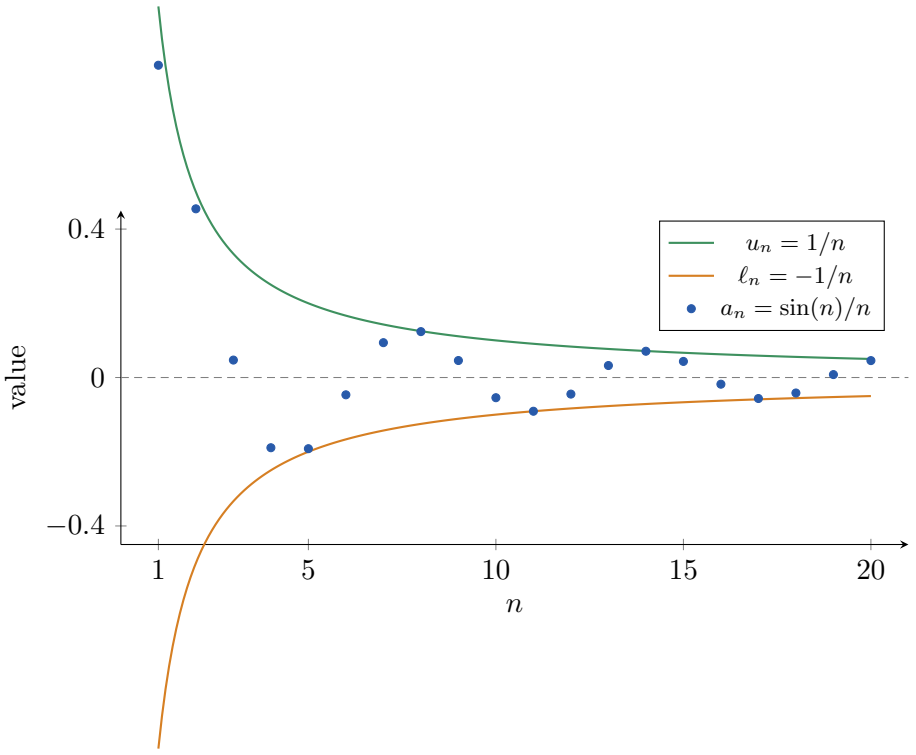


Figure 28.3: The sequence $a_n = \sin(n)/n$ trapped between the fences $\ell_n = -1/n$ and $u_n = 1/n$. Both fences converge to 0, so the pinned sequence is forced to 0 – no matter how erratically $\sin n$ jumps. (The first point $a_1 \approx 0.84$ sits on the upper fence; thereafter the gap closes fast.)

Why the technique works

Why squeezing works. Let $\varepsilon > 0$. Since $\ell_n \to L$ there is a cutoff N_1 past which $L - \varepsilon < \ell_n$. Since $u_n \to L$ there is a cutoff N_2 past which $u_n < L + \varepsilon$. Take N to be the larger of N_1, N_2 (and also large enough that the sandwich $\ell_n \le a_n \le u_n$ has kicked in). Then for $n \ge N$,

$$L - \varepsilon < \ell_n \le a_n \le u_n < L + \varepsilon,$$

so $|a_n - L| < \varepsilon$. The trapped sequence inherits the destination of its captors. Notice we never needed a formula for a_n – only the two inequalities and the two known limits.

Example 28.3 ($\sin n/n \to 0$)**.** For every n we have $-1 \le \sin n \le 1$, and dividing by the positive number n preserves the inequalities:

$$-\frac{1}{n} \le \frac{\sin n}{n} \le \frac{1}{n}.$$

Both outer sequences $\pm\frac{1}{n}$ converge to 0. By the squeeze theorem, $\dfrac{\sin n}{n} \to 0$. We tamed an untameable numerator by never touching it – we only bounded it. □

Example 28.4 (A factorial beats an exponential)**.** Show that $a_n = \dfrac{2^n}{n!} \to 0$. The factorial in the denominator eventually overwhelms the exponential on top, but let us prove it. Write out the factors for $n \ge 2$:

$$\frac{2^n}{n!} = \frac{2}{1} \cdot \frac{2}{2} \cdot \frac{2}{3} \cdot \frac{2}{4} \cdots \frac{2}{n}.$$

The first two factors multiply to 2. Every remaining factor $\frac{2}{k}$ (for $k \geq 3$) is below 1, so the product of all of them is at most the single last factor $\frac{2}{n}$ – dropping factors that are below 1 only raises the bound. Hence

$$0 \leq \frac{2^n}{n!} \leq 2 \cdot \frac{2}{n} = \frac{4}{n}.$$

Now $0 \to 0$ and $\frac{4}{n} \to 0$, so the squeeze pins $\frac{2^n}{n!}$ to 0. $\square$

28.5 A short table of limits worth memorizing

A handful of sequence limits come up so often that you should know them cold and know *why* each is true. Treat the following as a toolkit; the reasons matter more than the entries.

Limit	**Condition**	**Why**
$\frac{1}{n^p} \to 0$	$p > 0$	denominator grows without bound
$r^n \to 0$	$\lvert r\rvert < 1$	repeated shrinking factor
r^n diverges	$\lvert r\rvert > 1$	repeated growing factor
$n^{1/n} \to 1$	—	the root tames the growth
$r^{1/n} \to 1$	$r > 0$	ditto, for any positive base
$\left(1 + \frac{x}{n}\right)^n \to e^x$	$x \in \mathbb{R}$	the definition of e^x
$\frac{\ln n}{n^p} \to 0$	$p > 0$	powers beat logarithms
$\frac{n^p}{r^n} \to 0$	$r > 1$	exponentials beat powers
$\frac{r^n}{n!} \to 0$	any r	factorials beat exponentials

What is really happening

There is one organizing principle behind most of this table: a *hierarchy of growth*. Walk up the ladder and each rung utterly dwarfs the one below it, eventually and forever:

$$\ln n \ll n^p \ll r^n \ll n! \ll n^n \qquad (p > 0,\ r > 1).$$

"$\ll$" here means the ratio of the smaller to the larger goes to 0. A logarithm is slower than any positive power; any positive power is slower than any exponential with base above 1; the factorial outruns every fixed exponential; and n^n outruns even the factorial. Whenever you face a ratio of two such creatures, the answer is decided before you compute: the bigger one wins, dragging the ratio to 0 or to ∞ depending on who is on top. Figure 28.4 shows the lower rungs racing apart on a logarithmic scale. The table is just this ladder, read off rung by rung. Internalize the ladder and you can predict most sequence limits at a glance, then confirm with the squeeze theorem or l'Hopital.

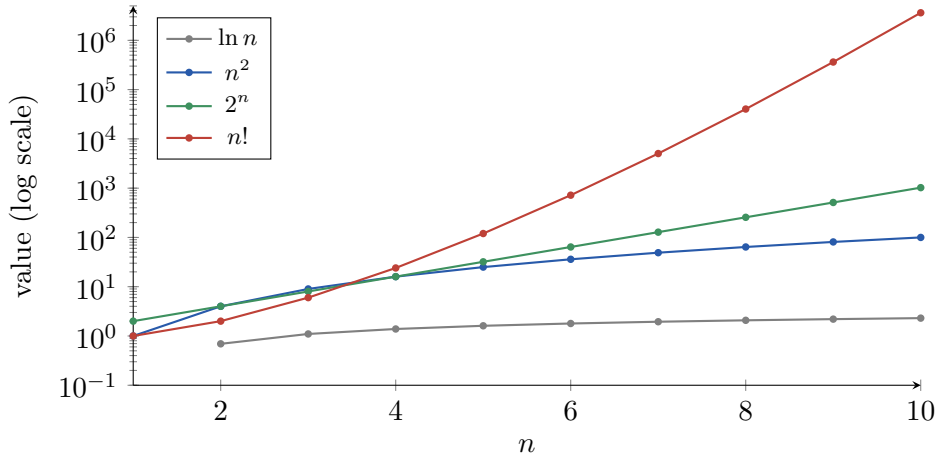


Figure 28.4: The hierarchy of growth on a logarithmic vertical scale, where each curve's height is its value and a higher curve grows faster. The logarithm $\ln n$ crawls along the bottom; the power n^2 climbs steadily; the exponential 2^n pulls ahead (note 2^n overtakes n^2 once $n \geq 5$); and the factorial $n!$ rockets past everything. This is the ladder $\ln n \ll n^p \ll r^n \ll n!$ made visible – and the reason a ratio of two such growth rates has its limit decided in advance.

Example 28.5 (The root that tames growth: $n^{1/n} \to 1$)**.** This one surprises people. The base n races to infinity, yet the exponent $\frac{1}{n}$ crashes to 0, and the two effects very nearly cancel – the limit is exactly 1. Here is a clean argument. Write $n^{1/n} = 1 + h_n$ where $h_n \geq 0$ (since $n^{1/n} \geq 1$, the excess h_n is nonnegative). Raise both sides to the nth power:

$$n = (1 + h_n)^n \geq \binom{n}{2} h_n^2 = \frac{n(n-1)}{2}\, h_n^2,$$

where we kept just the single quadratic term from the binomial expansion – every other term is nonnegative, so dropping them only weakens the right side. Solve for h_n:

$$h_n^2 \leq \frac{2}{n-1} \quad \Longrightarrow \quad 0 \leq h_n \leq \sqrt{\frac{2}{n-1}}.$$

The right side goes to 0, so by the squeeze $h_n \to 0$, hence $n^{1/n} = 1 + h_n \to 1$. The trick – keep *one* well-chosen term of a binomial expansion to get a usable inequality – is worth filing away. □

Example 28.6 (The compound-interest limit: $(1 + 1/n)^n \to e$)**.** You met e in this guise long ago: invest a dollar at 100% annual interest, compound it n times a year, and your year-end balance is $(1 + \frac{1}{n})^n$. As the compounding gets finer, the balance climbs toward a ceiling. That ceiling is Euler's number,

$$\lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n = e \approx 2.71828.$$

A few values make the climb visible: $n = 1$ gives 2; $n = 2$ gives 2.25; $n = 12$ (monthly) gives about 2.613; $n = 365$ (daily) gives about 2.7146. The sequence increases and is bounded above (by 3, say) – which, by the monotone convergence theorem of the next section, is already enough to *guarantee* a limit exists, even before we name it e. This is one of the cleanest examples of a limit that defines a number rather than merely computing one.

28.6 Monotone bounded sequences converge

Here is the theorem that makes the real numbers worth their reputation. It says that a sequence can be guaranteed to converge *without your ever knowing what it converges to* – purely from two qualitative facts about its shape.

First, two pieces of vocabulary.

Definition 28.3 (Monotone and bounded)**.** A sequence (a_n) is *increasing* if $a_n \le a_{n+1}$ for all n, and *decreasing* if $a_n \ge a_{n+1}$ for all n; it is *monotone* if it is one or the other. It is *bounded above* if there is a number M with $a_n \le M$ for all n, *bounded below* if some m has $a_n \ge m$ for all n, and *bounded* if both.

Theorem 28.3 (Monotone Convergence Theorem)**.** *An increasing sequence that is bounded above converges – and its limit is the* least upper bound *(supremum) of its terms. Symmetrically, a decreasing sequence that is bounded below converges, to the greatest lower bound of its terms.*

Intuition

Picture an increasing sequence as a runner who only ever moves forward (or stands still), but who can never cross a wall at height M. The runner's positions keep rising; they are pinned below the wall. Intuitively the runner must *pile up* against some ceiling – maybe the wall itself, maybe somewhere short of it, but *some* definite height. They cannot keep advancing forever without either crossing the wall (forbidden) or accumulating at a limit. The limit is the lowest height that the runner never exceeds: the least upper bound.

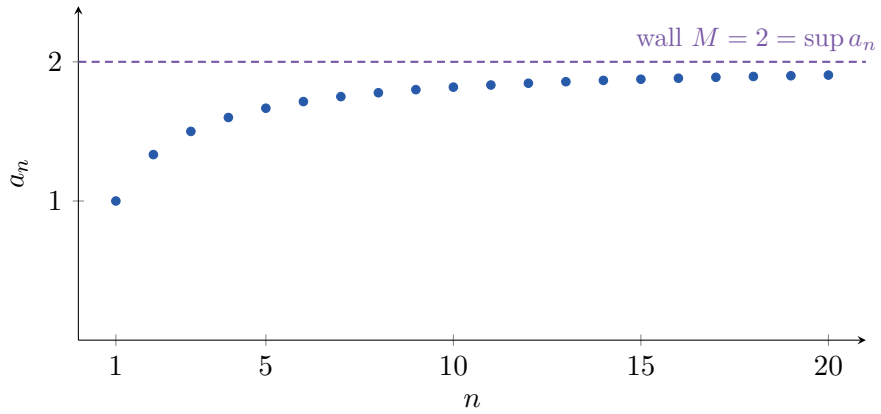


Figure 28.5: An increasing sequence ($a_n = 2 - \frac{2}{n+1}$) bounded above by the wall $M = 2$. The terms climb but never cross; they pile up against the least upper bound, which is exactly the limit. Monotone plus bounded forces a destination.

Why the technique works

Why monotone + bounded forces convergence. This is where the real numbers do their secret work. The natural numbers, the rationals – they are not enough. The decisive property is *completeness*: every nonempty set of reals that is bounded above has a *least* upper bound, a supremum, denoted sup. There are no "gaps" for a least upper bound to fall into. (The

rationals lack this: the rationals less than $\sqrt{2}$ are bounded above but have no rational least upper bound, because $\sqrt{2}$ is missing from $\mathbb{Q}$.)
Now take an increasing sequence (a_n) bounded above. Let $L = \sup\{a_n\}$, the least upper bound of the set of its terms; completeness guarantees this number exists. We claim $a_n \to L$. Let $\varepsilon > 0$. Because L is the *least* upper bound, $L - \varepsilon$ is *not* an upper bound – something smaller than L fails to bound the terms – so some term a_N exceeds $L - \varepsilon$. But the sequence is increasing, so for every $n \geq N$ we have $a_n \geq a_N > L - \varepsilon$. And every term is $\leq L$ since L bounds them all. Therefore, for $n \geq N$,

$$L - \varepsilon < a_n \leq L, \qquad \text{so} \qquad |a_n - L| < \varepsilon.$$

That is convergence to L. The engine is completeness; monotonicity is what lets a single term a_N vouch for all that follow it. □

Why this matters

Sit with what this theorem buys you. To prove a sequence converges directly from Definition 28.2, you must already *know* the limit L – you cannot check $|a_n - L| < \varepsilon$ without an L to subtract. But countless sequences in mathematics are defined in ways that hide their limit completely: the partial sums of a series, the iterates of Newton's method, the successive decimal truncations of an unknown real number. The Monotone Convergence Theorem hands you the limit's *existence* for free, in exchange for two facts you can often check by hand – "it goes up" and "it stays below something." You get the destination guaranteed without buying a map. This is why so many existence proofs in analysis end with the phrase "monotone and bounded, hence convergent."

Connection

This theorem is the precise reason an infinite decimal means anything. The string $0.333\ldots$ is the sequence of truncations $0.3,\ 0.33,\ 0.333,\ \ldots$. Each truncation is bigger than the last (increasing), and every one is below 0.4 (bounded above). Monotone plus bounded: the sequence converges, to a genuine real number – the one we cheerfully call $\frac{1}{3}$. Without the Monotone Convergence Theorem, "the number $0.333\ldots$" would be a string of digits with no guaranteed value. The same logic certifies *every* infinite decimal as a real number. Chapter 29 builds infinite series on exactly this foundation: a series converges precisely when its sequence of partial sums does.

28.7 Recursive sequences and fixed points

Many of the most important sequences are not given by a formula in n but by a *rule for getting the next term from the current one*:

$$a_{n+1} = f(a_n),$$

starting from some seed a_1. Each term is the previous term fed through the same function f. Newton's method is of this form; so is every numerical scheme that "iterates to a solution." The natural questions are: does the sequence settle down, and if so, where?

What is really happening

What is really happening: iteration and fixed points. Suppose the sequence $a_{n+1} = f(a_n)$ *does* converge, to some limit L. Then both $a_n \to L$ and $a_{n+1} \to L$ (a sequence and its own shifted copy share a limit – shifting by one place cannot change where a list is headed). If f is continuous, taking the limit of both sides of $a_{n+1} = f(a_n)$ gives

$$L = f(L).$$

A point satisfying $L = f(L)$ is called a *fixed point* of f: feeding it through f returns it unchanged. So *the only possible limits of the iteration are the fixed points of* f. This is the master key. The recursion can only come to rest where f leaves a point alone. To find the candidate limits, you do not analyze the sequence at all – you solve the equation $L = f(L)$.

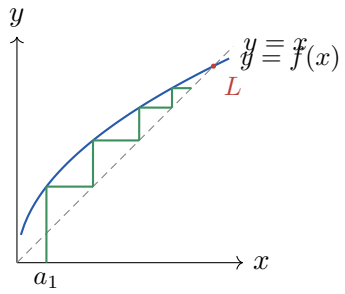


Figure 28.6: A *cobweb diagram* for the iteration $a_{n+1} = f(a_n)$ with $f(x) = \sqrt{x}$. Starting from a_1, go vertically to the curve (apply f), then horizontally to the line $y = x$ (copy the output back to the input axis), and repeat. The staircase spirals into the fixed point $L = 1$, where the curve meets the diagonal. Fixed points are exactly the crossings of $y = f(x)$ and $y = x$.

Intuition

Reading a cobweb. The line $y = x$ is the "mirror" that copies an output back into an input. A fixed point is where the curve $y = f(x)$ crosses this mirror – there, output equals input, so the iteration stops moving. Whether the staircase is drawn *toward* a fixed point (it attracts) or *away* from it (it repels) is governed by how steep f is there: a shallow curve (slope of magnitude below 1) pulls nearby points in; a steep one (slope above 1) pushes them away. The next box makes that precise.

Example 28.7 (A recursion that converges: $a_{n+1} = \sqrt{2 + a_n}$)**.** Define $a_1 = \sqrt{2}$ and $a_{n+1} = \sqrt{2 + a_n}$. The terms are the nested radicals

$$\sqrt{2}, \quad \sqrt{2 + \sqrt{2}}, \quad \sqrt{2 + \sqrt{2 + \sqrt{2}}}, \quad \dots$$

numerically 1.414, 1.848, 1.962, 1.990, 1.998, ..., visibly climbing toward 2. Let us prove they converge and find the limit, the right way.

Step 1: find the candidate limit. If $a_n \to L$, then L is a fixed point of $f(x) = \sqrt{2 + x}$, so $L = \sqrt{2 + L}$. Square: $L^2 = 2 + L$, i.e. $L^2 - L - 2 = 0$, i.e. $(L - 2)(L + 1) = 0$. The roots are $L = 2$ and $L = -1$. Since every term is positive, the only viable candidate is $L = 2$.

Step 2: show it is bounded above (by 2). Use induction. $a_1 = \sqrt{2} < 2$. If $a_n < 2$, then $a_{n+1} = \sqrt{2 + a_n} < \sqrt{2 + 2} = \sqrt{4} = 2$. So $a_n < 2$ for all n.

Step 3: show it is increasing. We check $a_{n+1} > a_n$, i.e. $\sqrt{2 + a_n} > a_n$. Squaring (both sides positive), this is $2 + a_n > a_n^2$, i.e. $a_n^2 - a_n - 2 < 0$, i.e. $(a_n - 2)(a_n + 1) < 0$. Since $0 < a_n < 2$ from Step 2, the first factor is negative and the second positive, so the product is negative. The inequality holds; the sequence increases.

Step 4: conclude. The sequence is increasing and bounded above, so by the Monotone Convergence Theorem (28.3) it converges. By Step 1 the limit can only be 2. Therefore

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + \cdots}}} = 2. \qquad \square$$

Notice the division of labor: Step 1 (fixed point) tells us *where* the limit must be; Steps 2–3 (monotone, bounded) tell us a limit *exists at all*. Neither alone is enough – the fixed-point equation also offered -1, a phantom the boundedness argument quietly rules out.

Why the technique works

Why slope decides attraction. Suppose L is a fixed point, f is differentiable, and $|f'(L)| < 1$. Near L the map moves a point at distance d from L to a point at distance roughly $|f'(L)|\, d$ – closer, because the multiplier is below 1. Each step shrinks the distance by a factor under 1, so the errors form a sequence dominated by a geometric one tending to 0: the iteration is drawn in, and we call L *attracting* (stable). If instead $|f'(L)| > 1$, each step *enlarges* the distance and the iteration is flung away (L is *repelling*). At the borderline $|f'(L)| = 1$ the linear term decides nothing and you must look closer.
In Example 28.7, $f(x) = \sqrt{2 + x}$ has $f'(x) = \frac{1}{2\sqrt{2+x}}$, so $f'(2) = \frac{1}{2 \cdot 2} = \frac{1}{4} < 1$: the fixed point 2 attracts, consistent with the staircase marching in. The phantom root -1 is not even in the domain we care about, so it never competes.

Example 28.8 (A recursion that does *not* settle)**.** Not every recursion converges. Take $a_{n+1} = 2a_n - 1$ with seed $a_1 = 2$. The fixed-point equation $L = 2L - 1$ gives $L = 1$. But $f(x) = 2x - 1$ has $f'(x) = 2 > 1$ everywhere, so the fixed point 1 repels. Starting at $a_1 = 2$ (distance 1 above the fixed point) the terms are

$$2,\ 3,\ 5,\ 9,\ 17,\ 33,\ \ldots,$$

each step doubling the distance from 1. The sequence diverges to $+\infty$. The fixed point exists, but the iteration runs from it rather than to it. This is the cautionary twin of the previous example: solving $L = f(L)$ tells you the *only possible* limit, never that a limit is *achieved*. Convergence must be earned separately – by monotone-bounded, by the slope test, or by direct estimate.

History

The fixed-point idea is one of the great unifying threads of modern mathematics. In 1922 Stefan Banach proved his *contraction mapping theorem*: if a map f shrinks distances by a fixed factor below 1 – $|f(x) - f(y)| \le k\,|x - y|$ with $k < 1$ – then iterating from *any* starting point produces a sequence converging to the one and only fixed point. The slope condition $|f'(L)| < 1$ you met above is the local, calculus-flavored shadow of Banach's global statement.

This is not abstraction for its own sake. Banach's theorem is the engine behind the existence-and-uniqueness proof for differential equations (Picard iteration, Chapter 37), behind Newton's method, behind Google's PageRank, and behind countless numerical solvers. In each case a hard problem is rephrased as "find the fixed point of this map," and a sequence of approximations is set loose to crawl toward it. The humble recursion $a_{n+1} = f(a_n)$ of this section is the seed of all of it.

Application

Babylonian square roots. To compute $\sqrt{S}$, iterate

$$a_{n+1} = \frac{1}{2}\left(a_n + \frac{S}{a_n}\right),$$

which is exactly Newton's method applied to $x^2 - S = 0$. Its fixed points solve $L = \frac{1}{2}(L + S/L)$, i.e. $2L = L + S/L$, i.e. $L^2 = S$, i.e. $L = \sqrt{S}$. For $S = 2$ starting at $a_1 = 1$:

$$1, \quad 1.5, \quad 1.41667, \quad 1.41422, \quad 1.41421356\ldots,$$

and the correct figures roughly *double* each step – a hallmark of Newton's method near a simple root. The Babylonians ran this recursion on clay tablets nearly four thousand years ago. They had no theory of limits, no ε, no N – only the observation that the iteration "stopped changing." What this chapter supplies is the why: $\frac{1}{2}(x + S/x)$ has an attracting fixed point at $\sqrt{S}$, so the clay tablets were converging all along.

Work it yourself

Pick a recursion and run it by hand for five steps, then explain its fate.

1. $a_1 = 0$, $a_{n+1} = \cos(a_n)$ (radians). Compute $a_2, \ldots, a_6$ on a calculator. The terms hover around a single number – the *Dottie number*. What equation does that number satisfy? (Answer: the fixed point $L = \cos L \approx 0.739$.) Is $|f'(L)| < 1$?
2. $a_1 = 1$, $a_{n+1} = 1 + \frac{1}{a_n}$. Compute several terms. They approach the golden ratio φ. Find the fixed-point equation $L = 1 + \frac{1}{L}$ and solve it. Which root do the terms choose, and why?

28.8 From sequences to series: a preview

Everything in this chapter was built to support one move, the move that occupies the next several chapters: making sense of an *infinite sum*.

You cannot literally add infinitely many numbers – addition is an operation on two inputs, repeated finitely. So what could

$$\tfrac{1}{2} + \tfrac{1}{4} + \tfrac{1}{8} + \tfrac{1}{16} + \cdots$$

possibly *mean*? The only sensible answer is a sequence in disguise. Form the *partial sums*, the running totals:

$$s_1 = \tfrac{1}{2}, \quad s_2 = \tfrac{1}{2} + \tfrac{1}{4} = \tfrac{3}{4}, \quad s_3 = \tfrac{7}{8}, \quad s_4 = \tfrac{15}{16}, \quad \ldots, \quad s_n = 1 - \tfrac{1}{2^n}.$$

This is a perfectly ordinary sequence (s_n). It is increasing (each new term adds a positive amount) and bounded above (by 1). By the Monotone Convergence Theorem it converges – and its limit is explicitly 1, since $1/2^n \to 0$. *That* is what we mean when we write the infinite sum and declare it equal to 1: the sequence of partial sums converges to 1.

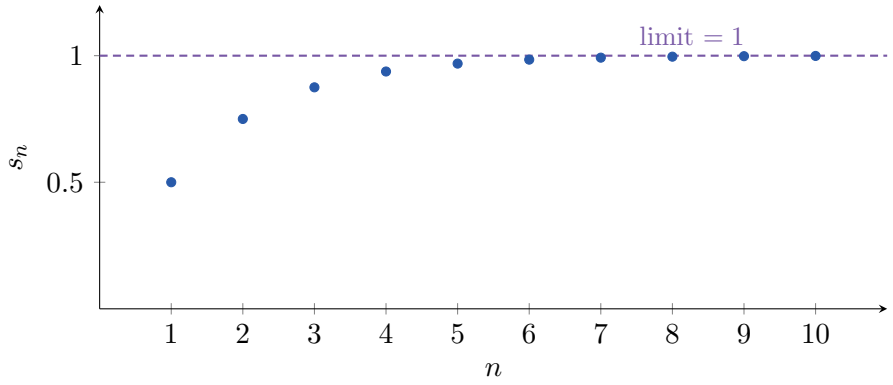


Figure 28.7: Partial sums $s_n = 1 - 2^{-n}$ of the series $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$. The running totals climb, halving the remaining gap to 1 at each step: increasing and bounded above, hence convergent. The infinite sum is *defined* to be this limit, 1.

Why this matters

This is the whole strategy of the infinite, in one sentence: **an infinite process is defined to be the limit of its finite stages.** An infinite sum is the limit of partial sums. An infinite product is the limit of partial products. A real number is the limit of its decimal truncations. A definite integral is the limit of Riemann sums. In every case the actual infinity is never performed – it is replaced by a sequence of finite, computable approximations, and the "answer" is wherever that sequence is *headed*. The theory of sequences you have just built is therefore not a side topic. It is the foundation on which the entire calculus of the infinite stands. Master where a list of numbers goes, and you have mastered, in advance, where every infinite process in this book goes.

28.9 What sequence theory unlocked, the non-constructive limit barrier, and the road ahead

Connection

Sequences are the bridge between the discrete world of arithmetic and the continuous world of analysis, transforming ancient philosophical puzzles into concrete computational machinery. Sequence theory dismantled Zeno's two-thousand-year-old paradoxes of motion. By defining limits through Cauchy's rigorous (ε, N) quantifier framework, mathematics established the algebra of limits, the Squeeze Theorem, and the Monotone Convergence Theorem. This unlocked:

- *Certification of convergence without knowing the limit:* Through the Cauchy criterion and the Bolzano–Weierstrass Theorem, mathematicians can prove that an infinite pro-

cess is guaranteed to converge purely by checking intrinsic bounds on the sequence itself, with no prior guess of the limit.

- *Universal fixed-point iteration:* Recursive sequences $a_{n+1} = f(a_n)$ and contraction mappings unlocked Newton's method for root-finding, Picard's existence proof for differential equations, and modern algorithmic optimization (such as gradient descent and PageRank).

Yet while the Monotone Convergence Theorem and Bolzano–Weierstrass certify with mathematical certainty that a bounded monotone sequence must have a limit $L \in \mathbb{R}$, they are fundamentally non-constructive: *proving that a limit exists gives no formula to evaluate it in closed form.*

A famous example is the sequence defining the Euler–Mascheroni constant:

$$\gamma_n = \left(\sum_{k=1}^{n} \frac{1}{k}\right) - \ln n.$$

By comparing the sum to the integral $\int_1^n \frac{\mathrm{d}x}{x}$, it is straightforward to prove that γ_n is strictly decreasing and bounded below by 0. By the Monotone Convergence Theorem, the sequence is guaranteed to converge to a real number $\gamma \approx 0.57721566\ldots$. Yet centuries after Euler introduced it in 1734, no one knows whether γ is rational or irrational. The sequence machinery guarantees that the number exists on the real line, but it cannot tell us its algebraic nature. Existence does not equal closed-form evaluation.

In Chapter 29 (*Infinite Series Convergence*), we put sequence theory to work on the grandest question of all: adding infinitely many numbers together. By defining an infinite series $\sum_{k=1}^{\infty} a_k$ strictly as the limit of its sequence of partial sums $S_n = \sum_{k=1}^{n} a_k$, every theorem of this chapter becomes an operational convergence test for series (the Integral Test, Comparison Test, Ratio Test, and Root Test).

Exercises

Exercise 28.1. Write out the first five terms of each sequence and predict, with a one-line reason, where it is headed.

(a) $a_n = \dfrac{2n-1}{n+3}$

(b) $a_n = \dfrac{(-1)^{n+1}}{2^n}$

(c) $a_n = \dfrac{n!}{n^n}$

(d) $a_n = \cos(\pi n)$

Exercise 28.2. Use the epsilon-N definition (28.2) directly to prove each limit. In each case, given $\varepsilon > 0$, exhibit an explicit N and verify the implication.

(a) $\lim_{n\to\infty} \dfrac{1}{n^2} = 0.$

(b) $\lim_{n\to\infty} \dfrac{n}{n+1} = 1$. *Hint: simplify* $|\frac{n}{n+1} - 1|$ *first.*

(c) $\lim_{n\to\infty} \dfrac{3n+2}{n} = 3$.

Exercise 28.3. Evaluate each limit, naming the limit laws or table entries (Section 28.5) you use.

(a) $\lim_{n\to\infty} \dfrac{4n^3 - 2n + 1}{5n^3 + n^2}$

(b) $\lim_{n\to\infty} \dfrac{\sqrt{n^2+1}}{n}$

(c) $\lim_{n\to\infty} \left(\sqrt{n+1} - \sqrt{n}\right)$ *Hint: multiply by the conjugate.*

(d) $\lim_{n\to\infty} \dfrac{2^n + 3^n}{3^n}$

Exercise 28.4. Use the squeeze theorem (28.2) to evaluate each limit. State your two bounding sequences explicitly.

(a) $\lim_{n\to\infty} \dfrac{\cos(n^2)}{n}$

(b) $\lim_{n\to\infty} \dfrac{n + \sin n}{n^2}$

(c) $\lim_{n\to\infty} \dfrac{3^n}{n!}$ *(adapt Example 28.4)*

(d) $\lim_{n\to\infty} \sqrt[n]{2^n + 3^n}$ *Hint: bound below by* 3 *and above by* $3\sqrt[n]{2}$.

Exercise 28.5. Explain carefully, using the definition of a limit, why the sequence $a_n = (-1)^n \cdot \dfrac{n}{n+1}$ diverges. (What two values do the even and odd terms approach, and why does that forbid a single limit?)

Exercise 28.6. Let $a_1 = 1$ and $a_{n+1} = \sqrt{6 + a_n}$.

(a) Find the candidate limit by solving the fixed-point equation $L = \sqrt{6 + L}$; discard the inadmissible root.

(b) Prove by induction that $a_n < 3$ for all n (bounded above).

(c) Prove the sequence is increasing.

(d) Conclude that it converges, and state the limit.

Exercise 28.7. Let $a_1 = 3$ and $a_{n+1} = \dfrac{a_n + 4}{2}$.

(a) Compute the first four terms. Is the sequence increasing or decreasing?

(b) Find the fixed point of $f(x) = \frac{x+4}{2}$ and compute f'; is the fixed point attracting or repelling?

(c) Show the sequence is bounded above by the fixed point, then conclude it converges (which form of the Monotone Convergence Theorem applies here?).

Exercise 28.8. Consider the logistic-style map $a_{n+1} = \frac{3}{2}\, a_n(1 - a_n)$ with seed $a_1 = 0.2$.

(a) Find the two fixed points of $f(x) = \frac{3}{2}x(1-x)$.

(b) Compute f' at each fixed point and classify each as attracting or repelling.

(c) Compute the first five terms numerically and say which fixed point the iteration is approaching, consistent with part (b).

Exercise 28.9. Apply the Babylonian iteration $a_{n+1} = \frac{1}{2}(a_n + \frac{5}{a_n})$ with $a_1 = 2$ to approximate $\sqrt{5}$.

(a) Compute a_2, a_3, a_4 to six decimal places.

(b) How many correct digits does each step produce? Comment on the doubling you observe and connect it to the discussion in the application box.

Exercise 28.10. (a) Using a calculator, tabulate $(1 + \frac{1}{n})^n$ for $n = 1, 10, 100, 1000, 10{,}000$ and watch it climb toward e.

(b) The sequence is increasing and bounded above. Which theorem of this chapter guarantees, before any numerical evidence, that the limit *exists*? Explain in one or two sentences what role completeness of the real numbers plays.

Exercise 28.11. Without computing exact values, use the growth hierarchy $\ln n \ll n^p \ll r^n \ll n! \ll n^n$ (Section 28.5) to decide whether each limit is 0 or $+\infty$, and justify in one line.

(a) $\displaystyle\lim_{n\to\infty} \frac{n^{10}}{2^n}$

(b) $\displaystyle\lim_{n\to\infty} \frac{n!}{100^n}$

(c) $\displaystyle\lim_{n\to\infty} \frac{(\ln n)^3}{\sqrt{n}}$

(d) $\displaystyle\lim_{n\to\infty} \frac{n^n}{n!}$

Exercise 28.12. A real number is the limit of its decimal truncations. Let a_n be π truncated to n decimal places: $a_1 = 3.1,\ a_2 = 3.14,\ a_3 = 3.141, \ldots$.

(a) Argue that (a_n) is increasing and bounded above (by 3.2, say).

(b) Which theorem certifies that this sequence has a limit – so that "the number π" is a well-defined real number rather than just an endless string of digits? Relate your answer to the connection box on infinite decimals.

Chapter 29

Infinite Series and Convergence

There is something faintly outrageous about the idea of adding up infinitely many numbers. Addition, as you first learned it, is a finite act: you line up two numbers, or ten, or a million, and you combine them. The process *stops*. But what could it possibly mean to write

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$$

and ask for "the sum"? You can never reach the end. There is no last term to add. And yet – this is the quiet miracle the chapter is built around – some of these endless sums settle down to a perfectly definite, ordinary number, while others run off to infinity or thrash around forever without ever choosing a value. Our job is to figure out *which is which*, and, more importantly, to understand *why*.

This is not an idle game. Every time you evaluate $e^{0.3}$ on a calculator, or your computer reports $\sin(1)$, or a physicist models the gravitational pull of a ring of mass, an infinite series is doing the work underneath. The functions you have grown to love – e^x, $\sin x$, $\ln(1+x)$ – *are* infinite series in disguise, and in the next chapter we will pry that disguise off. But none of it makes sense until we are clear about what it means for an infinite sum to have a value at all. So we begin where we must: with the partial sums.

29.1 What It Means to Add Infinitely Many Things

The trick – and it is the single most important idea in the chapter, the seed from which everything else grows – is that we never actually perform an infinite addition. We *can't*. Instead we perform a sequence of perfectly ordinary finite additions and watch where they are heading.

Intuition

Imagine walking toward a wall. First you cover half the remaining distance, then half of what's left, then half again. After each step you have walked a finite, knowable distance. You will never literally arrive, but it is intuitively clear *where* you are arriving: at the wall. The infinite sum is not the act of taking infinitely many steps. It is the *destination* those steps approach.

Let us make this precise. Given an infinite list of numbers $a_1, a_2, a_3, \ldots$ – which we call the *terms* –

we build a brand-new list out of them, the list of **partial sums**:

$$S_1 = a_1, \qquad S_2 = a_1 + a_2, \qquad S_3 = a_1 + a_2 + a_3, \qquad \dots, \qquad S_n = \sum_{k=1}^{n} a_k.$$

Each S_n is a finite sum – no mystery, no infinity, just arithmetic. The infinite series is then *defined* to be the limit of this sequence of partial sums, if that limit exists.

Definition 29.1 (Convergence of a series)**.** Let $a_1, a_2, a_3, \dots$ be a sequence of real numbers and let $S_n = \sum_{k=1}^{n} a_k$ be its nth partial sum. We say the series $\sum_{k=1}^{\infty} a_k$ **converges** to the number S if

$$\lim_{n\to\infty} S_n = S,$$

and we then write $\sum_{k=1}^{\infty} a_k = S$. If the sequence (S_n) has no finite limit, we say the series **diverges**.

What is really happening

The symbol $\sum_{k=1}^{\infty} a_k$ looks like an instruction to do something infinitely many times. It is not. It is shorthand for a *limit of a sequence*: the sequence of finite partial sums. When we write $\sum_{k=1}^{\infty} \frac{1}{2^k} = 1$, we are not claiming that somebody added up infinitely many fractions and got 1. We are claiming something about a sequence of ordinary numbers $\frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \frac{15}{16}, \dots$ – namely that they can be made as close to 1 as you please by going far enough out. The infinite sum is a *statement about approach*, dressed up to look like an act of addition. Once you internalize this, the paradoxes of infinity dissolve: there is no completed infinite process anywhere, only a limit.

History

Why was this definition of partial sums so desperately needed? Because without it, the greatest mathematicians in history fell into outright contradictions.

In 1703, the Italian mathematician and monk *Guido Grandi* published a study of the alternating series:

$$1 - 1 + 1 - 1 + 1 - 1 + \cdots$$

If you group adjacent pairs, you get $(1-1) + (1-1) + \cdots = 0 + 0 + \cdots = 0$. If you shift the grouping by one term, you get $1 - (1-1) - (1-1) - \cdots = 1 - 0 - 0 - \cdots = 1$. Grandi argued that since the sum was equally balanced between 0 and 1, its true value must be the arithmetic average, $1/2$. He went so far as to claim this proved God could create the universe out of nothing, since $0 = \frac{1}{2} - \frac{1}{2}$.

Even *Leonhard Euler*, whose mathematical intuition was virtually supernatural, routinely accepted that $1-1+1-1+\cdots = 1/2$. Euler derived it by expanding $\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots$ and setting $x = 1$. But Euler's bold algebraic manipulations without convergence limits soon led to absurdities, such as concluding that $1 + 2 + 4 + 8 + \cdots = -1$ (by evaluating $\frac{1}{1-x}$ at $x = 2$).

The chaos ended only in 1821, when *Augustin-Louis Cauchy* published his *Cours d'Analyse*. Cauchy insisted on Definition 29.1: a series has a sum if and only if the sequence of its finite partial sums S_n converges to a single limit. Because the partial sums of Grandi's series oscillate forever between 1 and 0 $(1, 0, 1, 0, 1, \dots)$, $\lim_{n\to\infty} S_n$ does not exist. The series has no sum; it diverges. Cauchy's partial-sum definition rescued analysis from mysticism.

So "convergence" and "divergence" are not really new ideas. They are the limit concept from earlier in the book, applied to the particular sequence (S_n). Everything we prove about series is, underneath, a fact about the convergence of these partial-sum sequences. Keep that anchor in mind whenever a test starts to feel like a magic incantation.

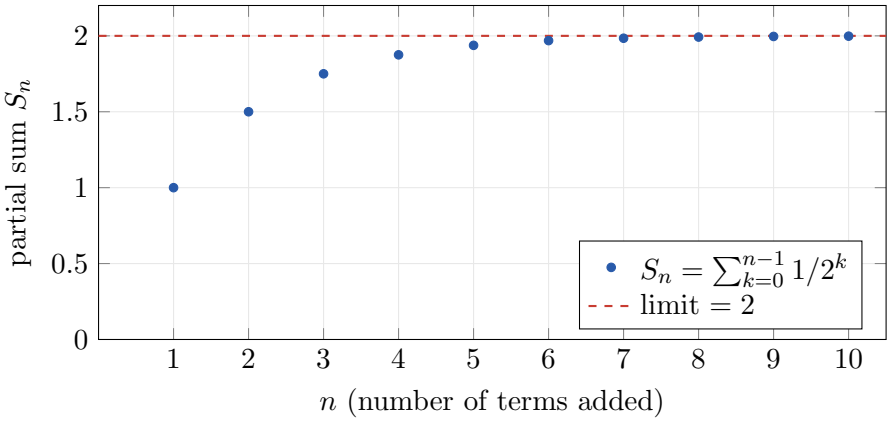


Figure 29.1: The partial sums of $1 + \frac{1}{2} + \frac{1}{4} + \cdots$ climb toward 2, halving the gap each step but never crossing. The infinite sum *is* that ceiling.

Walking question

Before reading on, settle a question in your head on a walk: the terms $\frac{1}{2^k}$ get smaller and smaller and eventually become negligibly tiny. Is "the terms shrink to zero" enough, all by itself, to guarantee the sum is finite? Hold your answer. By the end of Section 29.5 it may surprise you.

29.2 The Geometric Series: The One You Can Sum Exactly

Almost every infinite sum is impossible to evaluate in closed form – you can prove it converges, but you cannot write down a tidy formula for the total. There is one glorious exception that we can sum *exactly*, and it happens to be the most important series in all of mathematics: the **geometric series**, in which each term is a fixed multiple r of the one before.

$$a + ar + ar^2 + ar^3 + \cdots = \sum_{k=0}^{\infty} a\, r^k.$$

The number r is the *ratio*. Our walk-toward-the-wall example was the case $a = 1$, $r = \frac{1}{2}$. Let us find the total properly, by computing the partial sums and taking their limit – never by waving hands at infinity.

Focus

The summing trick. Let $S_n = a + ar + ar^2 + \cdots + ar^{n-1}$ (that's n terms). Multiply the

whole thing by r:

$$rS_n = ar + ar^2 + \cdots + ar^{n-1} + ar^n.$$

Now subtract. Look at what happens: almost everything cancels, because rS_n is just S_n shifted over by one slot.

$$S_n - rS_n = a - ar^n \quad \Longrightarrow \quad (1-r)S_n = a\,(1-r^n).$$

Provided $r \neq 1$, divide:

$$\boxed{S_n = a\,\frac{1-r^n}{1-r}}.$$

This is an *exact* formula for the finite partial sum – valuable on its own. Now we take the limit. Everything hinges on what r^n does as $n \to \infty$. If $|r| < 1$, then $r^n \to 0$ (repeatedly multiplying by something smaller than 1 in size drives you to zero). If $|r| > 1$, then $|r^n| \to \infty$ and the partial sums blow up. The boundary cases $r = 1$ and $r = -1$ misbehave in their own ways.

Theorem 29.1 (The geometric series)**.** *The geometric series $\sum_{k=0}^{\infty} a\,r^k$ converges if and only if $|r| < 1$, and in that case*

$$\sum_{k=0}^{\infty} a\,r^k = \frac{a}{1-r}.$$

If $|r| \geq 1$ (and $a \neq 0$) the series diverges.

Proof. From the boxed formula, $S_n = a\dfrac{1-r^n}{1-r}$ for $r \neq 1$. If $|r| < 1$ then $r^n \to 0$, so $S_n \to \dfrac{a(1-0)}{1-r} = \dfrac{a}{1-r}$. If $|r| > 1$ then $|r^n| \to \infty$, so S_n has no finite limit. If $r = 1$, the series is $a + a + a + \cdots$ and $S_n = na \to \pm\infty$. If $r = -1$, the partial sums bounce $a, 0, a, 0, \ldots$ and never settle. In every case with $|r| \geq 1$ the series diverges. □

Why the technique works

Why does the $|r| < 1$ condition do *all* the work? Because the geometric series carries its own fate in its terms. The "tail" that remains after n terms is itself a geometric series, $ar^n + ar^{n+1} + \cdots = r^n \cdot \frac{a}{1-r}$, a shrunken copy of the original scaled by r^n. When $|r| < 1$ that scaling factor crushes the leftover tail to nothing; the destination is in sight and the remaining journey shrinks geometrically. When $|r| \geq 1$ the tail is as big as – or bigger than – the whole, so there is never any "settling down." The self-similarity of the geometric series is exactly why it is the one we can pin down precisely, and it is why $|r|$ versus 1 is the only thing that matters.

Example 29.1 (The walk to the wall, finished)**.** With $a = 1$, $r = \frac{1}{2}$: the sum is $\dfrac{1}{1-\frac{1}{2}} = 2$. Note our list started at $k = 0$, so this is $1 + \frac{1}{2} + \frac{1}{4} + \cdots = 2$, matching Figure 29.1.

Example 29.2 (A repeating decimal is a geometric series)**.** What is $0.7777\ldots$ really? It is

$$\frac{7}{10} + \frac{7}{100} + \frac{7}{1000} + \cdots = \sum_{k=1}^{\infty} \frac{7}{10}\left(\frac{1}{10}\right)^{k-1},$$

a geometric series with first term $a = \frac{7}{10}$ and ratio $r = \frac{1}{10}$. So its value is

$$\frac{a}{1-r} = \frac{7/10}{1-1/10} = \frac{7/10}{9/10} = \frac{7}{9}.$$

Every repeating decimal is secretly a geometric series, which is why every repeating decimal is a fraction. The mysterious $0.999\ldots = 1$ is just $\dfrac{9/10}{1-1/10} = 1$ – not a trick, just the geometric formula.

History

The geometric series is ancient. Around 250 BC, Archimedes summed $1 + \frac{1}{4} + \frac{1}{16} + \cdots = \frac{4}{3}$ to find the area of a parabolic segment – a genuine infinite sum, handled with total rigor, two thousand years before "limit" had a definition. He did not say "let $n \to \infty$"; instead he proved by his *method of exhaustion* that the area could be neither more nor less than $\frac{4}{3}$, squeezing it from both sides. The intuition of convergence was available to the very best mathematicians long before the machinery. What the nineteenth century added was not the answer but the *license*: a clean definition (our partial sums) that let ordinary mathematicians sum series safely, without an Archimedes' worth of cleverness each time.

29.3 Telescoping Series: When the Middle Cancels

Besides the geometric series, there is one more family we can sum exactly, and it relies on the same essential move: arrange for the partial sum S_n to collapse to just a few surviving terms. These are the **telescoping series**, named for the way an old collapsible spyglass folds down to almost nothing.

The idea: suppose each term can be written as a *difference* of consecutive values of some sequence, $a_k = b_k - b_{k+1}$. Then

$$S_n = \sum_{k=1}^{n} (b_k - b_{k+1}) = (b_1 - b_2) + (b_2 - b_3) + (b_3 - b_4) + \cdots + (b_n - b_{n+1}).$$

Every interior term appears twice with opposite signs and cancels. Only the very first and very last survive:

$$S_n = b_1 - b_{n+1}.$$

The infinite sum is then $\lim_{n\to\infty} S_n = b_1 - \lim_{n\to\infty} b_{n+1}$, easy as soon as you know where b_k heads.

Example 29.3 (The textbook telescope)**.** Consider $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$. The terms shrink fast, but is there a sum? Use partial fractions – the key that unlocks the difference structure:

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}.$$

So with $b_k = \frac{1}{k}$ we have $a_k = b_k - b_{k+1}$, and

$$S_n = \left(1 - \tfrac{1}{2}\right) + \left(\tfrac{1}{2} - \tfrac{1}{3}\right) + \cdots + \left(\tfrac{1}{n} - \tfrac{1}{n+1}\right) = 1 - \frac{1}{n+1}.$$

As $n \to \infty$, $\frac{1}{n+1} \to 0$, so the series sums to exactly 1. Clean and complete.

Common trap

Telescoping is seductive, and the trap is assuming *everything* cancels except the first term. It does not. Always write out S_n in full and see which terms survive at *both* ends. Some telescopes leave two survivors on each side (when the difference skips a step, like $b_k - b_{k+2}$), giving $S_n = b_1 + b_2 - b_{n+1} - b_{n+2}$. And the sum exists only if the trailing survivors have a limit. If b_k does not converge, the telescope diverges no matter how prettily the middle cancels.

29.4 The First Thing to Check: The nth-Term Test

Before reaching for any heavy machinery, there is a one-second sanity check that can kill a series outright. It comes from a simple observation about *convergent* series and their terms.

Intuition

If the partial sums S_n are settling down to a fixed number S, then consecutive partial sums must be getting closer and closer to each other. But the gap between consecutive partial sums is exactly one term: $S_n - S_{n-1} = a_n$. So the terms must be shrinking to zero. A convergent sum *cannot* keep adding non-negligible chunks forever – if it did, it would never settle.

Theorem 29.2 (The nth-term test for divergence)**.** *If $\sum_{k=1}^{\infty} a_k$ converges, then $\lim_{n\to\infty} a_n = 0$. Equivalently (contrapositive): if a_n does* not *tend to 0, the series diverges.*

Proof. Suppose the series converges to S, so $S_n \to S$ and also $S_{n-1} \to S$ (same sequence, shifted). Then

$$a_n = S_n - S_{n-1} \longrightarrow S - S = 0.$$ □

Common trap

This is the single most misused fact in the chapter, so read carefully. The theorem says one thing and one thing only: *if the terms don't go to zero, the series diverges.* It does **not** say the converse. "The terms go to zero" does **not** imply the series converges. That converse is false, spectacularly so, and the harmonic series in Section 29.5 is the famous counterexample. The nth-term test is a *divergence detector only.* When the terms shrink to zero, the test is silent – it tells you nothing, and you must reach for a real test.

Example 29.4. $\sum \dfrac{n}{2n+1}$ diverges, because $\dfrac{n}{2n+1} \to \frac{1}{2} \neq 0$. We are adding chunks that stay near $\frac{1}{2}$ forever; no settling is possible. Likewise $\sum (-1)^n$ diverges since the terms don't approach 0 (they bounce between 1 and -1). One glance, done.

29.5 The Harmonic Series: A Genuine Surprise

Here is the series that should unsettle you, the one the walking question pointed toward:

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots$$

The terms shrink to zero. They shrink, in fact, quite obediently. The nth-term test says nothing against it. Surely such a sum, built of ever-tinier pieces, must add up to something finite? It does not. The harmonic series **diverges** – its partial sums crawl off to infinity, slowly but inexorably. This is one of the great surprises of mathematics, and the proof is so clever and so clean that it is worth knowing by heart.

Focus

Oresme's grouping (c. 1350). Group the terms in blocks whose lengths double, and bound each block from below:

$$\underbrace{1}_{} + \underbrace{\frac{1}{2}}_{} + \underbrace{\frac{1}{3}+\frac{1}{4}}_{>\frac{1}{4}+\frac{1}{4}=\frac{1}{2}} + \underbrace{\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}}_{>\frac{1}{8}\cdot 4=\frac{1}{2}} + \underbrace{\frac{1}{9}+\cdots+\frac{1}{16}}_{>\frac{1}{16}\cdot 8=\frac{1}{2}} + \cdots$$

In each block, replace every term by the *smallest* term in that block; the block still beats $\frac{1}{2}$. So the sum contains infinitely many chunks each exceeding $\frac{1}{2}$. Add enough blocks and you pass any number you like. The partial sums are unbounded; the series diverges.

History

Nicole Oresme, a French scholar of the fourteenth century, produced exactly this argument around 1350 – centuries before calculus existed. It is one of the earliest truly rigorous results about an infinite process. The divergence is agonizingly slow: to push the partial sum past 10 you need over twelve thousand terms; to reach 100 you need more terms than there are atoms in your body. This sluggishness is precisely why the result is shocking. Everything *looks* like it should converge. It is the slowest, gentlest possible march to infinity – but it gets there.

Why this matters

The harmonic series teaches the deepest lesson of the whole subject: *the rate at which terms shrink is everything.* The terms $\frac{1}{n}$ go to zero, but not fast enough; the terms $\frac{1}{2^n}$ go to zero fast enough. The entire theory of convergence – every test that follows – is really one question asked in many disguises: *do the terms shrink fast enough?* Geometric decay (r^n) is fast enough. Decay like $\frac{1}{n}$ is not. The borderline lies somewhere in between, and mapping that borderline precisely is what the rest of the chapter is about.

29.6 The Integral Test: Series as Areas

The harmonic series begs a follow-up: where, exactly, is the borderline? We know $\sum \frac{1}{n}$ diverges. What about $\sum \frac{1}{n^2}$, or $\sum \frac{1}{n^{1.5}}$, or $\sum \frac{1}{\sqrt{n}}$? To settle a whole family at once we connect sums to integrals – two ways of measuring the same underlying quantity, area under a decreasing curve.

Intuition

Picture the terms $a_n = f(n)$ of a positive, decreasing sequence as the heights of rectangles, each of width 1. The sum $\sum f(n)$ is the total area of an infinite staircase of rectangles.

The integral $\int_1^\infty f(x)\,dx$ is the area under the smooth curve $y = f(x)$. Because the curve is decreasing, the rectangles and the curve trap each other: the staircase is never far from the smooth area. So the sum is finite exactly when the integral is finite. We've traded a hard sum for an integral we may know how to do.

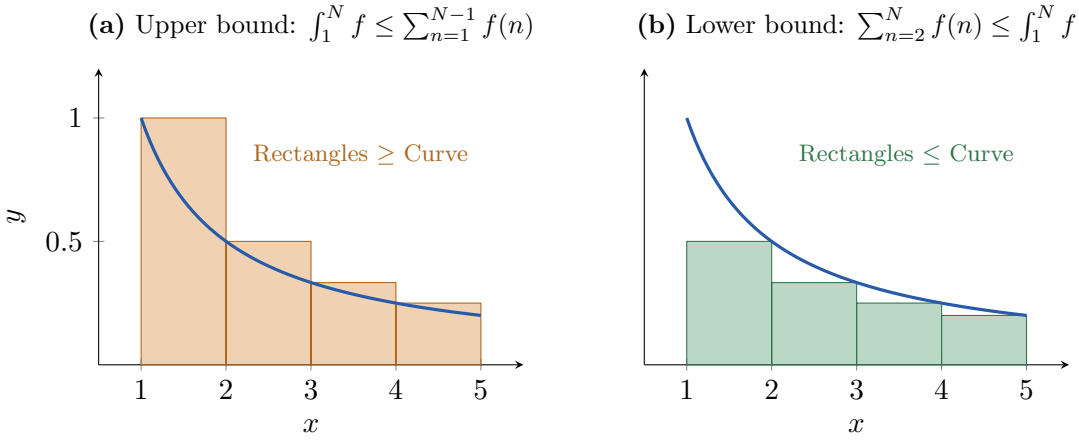


Figure 29.2: The Integral Test sandwich. **(a)** Left-endpoint rectangles sit *above* the decreasing curve $f(x)$, showing $\int_1^N f(x)\,dx \le \sum_{n=1}^{N-1} a_n$. If the integral diverges, it forces the series to diverge. **(b)** Right-endpoint rectangles sit *below* the curve on $[n-1, n]$, showing $\sum_{n=2}^{N} a_n \le \int_1^N f(x)\,dx$. If the improper integral converges, it caps the series from above, guaranteeing convergence.

Theorem 29.3 (The integral test)**.** *Let f be positive, continuous, and decreasing on $[1, \infty)$, and set $a_n = f(n)$. Then the series $\sum_{n=1}^{\infty} a_n$ and the improper integral $\int_1^\infty f(x)\,dx$ either both converge or both diverge.*

Proof. Because f decreases, on the interval $[n, n+1]$ we have $f(n+1) \le f(x) \le f(n)$. Integrating across that unit interval (width 1) gives the rectangle-vs-curve sandwich:

$$f(n+1) \;\le\; \int_n^{n+1} f(x)\,dx \;\le\; f(n).$$

Sum the right inequality for $n = 1$ to $N-1$: $\int_1^N f \le \sum_{n=1}^{N-1} f(n)$. Sum the left inequality for $n = 1$ to $N-1$: $\sum_{n=2}^{N} f(n) \le \int_1^N f$. Together,

$$\sum_{n=2}^{N} a_n \;\le\; \int_1^N f(x)\,dx \;\le\; \sum_{n=1}^{N-1} a_n.$$

If the integral converges (finite as $N \to \infty$), the left inequality bounds the partial sums above; a bounded increasing sequence converges, so the series converges. If the integral diverges, the right inequality forces the partial sums to infinity, so the series diverges. The integral and the series are locked together. □

Why the technique works

The integral test works because a sum is a *crude integral* and an integral is a *refined sum* – they are two computations of the same area, differing only by the gap between a staircase and the curve it hugs. When f decreases, that gap is itself controlled (it is squeezed between consecutive terms), so it can never accumulate enough to make the sum and the integral disagree about being finite. This is the deep reason the test needs "decreasing": without it, the rectangles could wander far from the curve and the comparison would collapse. The hypothesis is not red tape; it is the very thing that keeps staircase and curve shackled together.

The payoff is an entire family of series settled at once, and it is the convergence fact worth memorizing above all others.

Theorem 29.4 (The p-series)**.** *The series* $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^p}$ *converges if and only if* $p > 1$.

Proof. Apply the integral test to $f(x) = x^{-p}$ (positive, decreasing for $x \geq 1$ when $p > 0$; and for $p \leq 0$ the terms don't even go to zero, so it diverges by the nth-term test). For $p \neq 1$,

$$\int_1^{\infty} x^{-p}\,\mathrm{d}x = \lim_{b\to\infty} \frac{x^{1-p}}{1-p}\Big|_1^b = \lim_{b\to\infty} \frac{b^{1-p}-1}{1-p}.$$

When $p > 1$, the exponent $1-p$ is negative, so $b^{1-p} \to 0$ and the integral equals $\frac{1}{p-1}$, finite: convergence. When $p < 1$, $b^{1-p} \to \infty$: divergence. When $p = 1$, $\int_1^{\infty} \frac{1}{x}\,\mathrm{d}x = \lim_{b\to\infty} \ln b = \infty$: divergence (the harmonic series, again). □

Focus

Memorize this. $\sum \frac{1}{n^p}$ converges $\iff$ $p > 1$. So $\sum \frac{1}{n^2}, \sum \frac{1}{n^3}, \sum \frac{1}{n^{1.01}}$ converge; $\sum \frac{1}{n}, \sum \frac{1}{\sqrt{n}} = \sum n^{-1/2}, \sum \frac{1}{n^{0.99}}$ diverge. The borderline sits exactly at $p = 1$, and the harmonic series sits exactly on the borderline – on the *divergent* side. The borderline you have been hunting since Section 29.5 is now located: it is $p = 1$, and $1/n$ is the worst term that still diverges. Figure 29.3 shows the three regimes side by side.

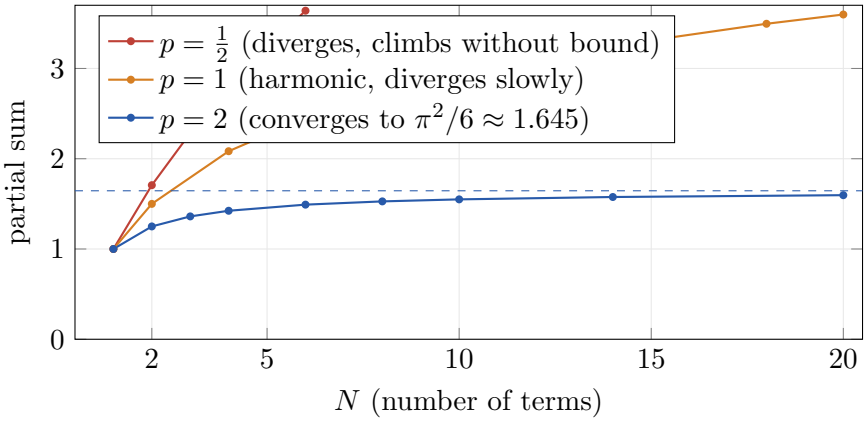


Figure 29.3: Partial sums of three p-series. For $p = 2$ they flatten toward a ceiling; for $p = 1$ and $p = \frac{1}{2}$ they keep rising, the half-power case visibly faster. Convergence is decided entirely by whether p clears 1.

Connection

The convergent p-series secretly compute famous constants. Euler stunned Europe in 1735 by proving $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ – a fact the integral test alone cannot deliver (it tells you *that* the sum is finite, not *what* it equals). As a function of p, this object is the Riemann zeta function $\zeta(p) = \sum n^{-p}$, whose behavior encodes the distribution of the prime numbers. Convergence is only the doorway; behind it lies some of the deepest mathematics there is.

29.7 Comparison Tests: Lean on a Series You Already Understand

We now have two reliable yardsticks – geometric series and p-series. The comparison tests let us measure unfamiliar series against these known ones, and the logic runs just as intuition would suggest.

Intuition

If your terms are smaller (term by term) than the terms of a series you already know converges, your sum is trapped underneath a finite total – it has nowhere to go but to a finite value. And if your terms are bigger than those of a series you know diverges, your sum is pushed up past every bound. Smaller than finite is finite; bigger than infinite is infinite. That's the whole idea.

Theorem 29.5 (Direct comparison test)**.** *Suppose $0 \le a_n \le b_n$ for all n (beyond some starting point).*

- *If $\sum b_n$ converges, then $\sum a_n$ converges.*
- *If $\sum a_n$ diverges, then $\sum b_n$ diverges.*

Proof. All terms are nonnegative, so both sequences of partial sums are increasing. Let $A_n = \sum_{k \le n} a_k$ and $B_n = \sum_{k \le n} b_k$; then $A_n \le B_n$. If $\sum b_n$ converges to B, then $A_n \le B_n \le B$, so (A_n) is increasing and bounded above, hence convergent. The second bullet is the contrapositive of the first. □

Example 29.5. Does $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2 + 3n + 1}$ converge? The denominator is bigger than n^2, so each term is *smaller* than $\frac{1}{n^2}$:

$$0 < \frac{1}{n^2 + 3n + 1} < \frac{1}{n^2}.$$

Since $\sum \frac{1}{n^2}$ converges ($p = 2 > 1$), our smaller series converges too. We never needed its value – only that it's trapped under something finite.

The direct comparison demands a clean term-by-term inequality, which can be fiddly to arrange (you may have to make the smaller series smaller, or the bigger one bigger, in just the right direction). Often it is easier to compare *growth rates* rather than the terms themselves. That is the limit comparison test, and it is the friendlier workhorse.

Theorem 29.6 (Limit comparison test)**.** *Let $a_n > 0$ and $b_n > 0$, and suppose*

$$\lim_{n \to \infty} \frac{a_n}{b_n} = L, \qquad 0 < L < \infty.$$

Then $\sum a_n$ and $\sum b_n$ either both converge or both diverge.

Why the technique works

Why does a finite, nonzero limit force the two series to share a fate? Because $\frac{a_n}{b_n} \to L$ means that eventually a_n is sandwiched between $\frac{L}{2}b_n$ and $2L\,b_n$ – the two series are, in the long run, just constant multiples of each other. And multiplying every term of a series by a fixed positive constant cannot change whether it converges (it only scales the total). Convergence is decided by the *tail*, by how fast terms shrink, and a constant factor does not change a shrink rate. So $\sum a_n$ and $\sum b_n$ live or die together. The limit comparison test is really the statement "convergence cares about growth rate, not constant multiples" made operational. You don't have to win a delicate inequality; you only have to identify which known series your terms *grow like*.

Focus

How to pick b_n in practice. Strip your term down to its dominant behavior for large n – keep only the fastest-growing parts of numerator and denominator. That skeleton is your comparison series. Example: $a_n = \dfrac{3n^2 - n}{4n^4 + 5}$ behaves like $\dfrac{3n^2}{4n^4} = \dfrac{3}{4n^2}$, so compare with $b_n = \frac{1}{n^2}$.

Example 29.6. Test $\displaystyle\sum_{n=1}^{\infty} \frac{3n^2 - n}{4n^4 + 5}$. The dominant behavior is $\frac{1}{n^2}$, so take $b_n = \frac{1}{n^2}$:

$$\frac{a_n}{b_n} = \frac{3n^2 - n}{4n^4 + 5} \cdot n^2 = \frac{3n^4 - n^3}{4n^4 + 5} \longrightarrow \frac{3}{4}.$$

The limit $\frac{3}{4}$ is finite and nonzero, and $\sum \frac{1}{n^2}$ converges, so our series converges. Notice how much cleaner this is than fighting for a direct inequality.

Example 29.7 (A divergence by limit comparison)**.** Test $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2 + 1}}$. For large n, $\sqrt{n^2 + 1} \approx n$, so compare with $b_n = \frac{1}{n}$:

$$\frac{a_n}{b_n} = \frac{n}{\sqrt{n^2 + 1}} = \frac{1}{\sqrt{1 + 1/n^2}} \longrightarrow 1.$$

Finite, nonzero. Since $\sum \frac{1}{n}$ diverges (harmonic), so does our series. The square root barely changed anything; the term still shrinks "like $1/n$," and that is fatal.

29.8 The Ratio and Root Tests: When a Series Is Secretly Geometric

The geometric series converges precisely because the ratio of consecutive terms is a constant r with $|r| < 1$. The **ratio test** takes this idea and applies it to series that are only *eventually*, *approximately* geometric – which, it turns out, is an enormous and important class, including every series built from factorials and exponentials.

Intuition

Look at the ratio $\frac{a_{n+1}}{a_n}$ of one term to the previous. For a true geometric series this ratio is always r. For a general series it varies, but if it *settles* toward some limit L, then far out in the tail the series behaves like a geometric series with ratio L. So if $L < 1$, the tail decays geometrically and the series converges; if $L > 1$, the terms eventually *grow*, dooming the sum.

Theorem 29.7 (Ratio test)**.** *For a series $\sum a_n$ with $a_n \neq 0$, suppose $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L$.*

- *If $L < 1$, the series converges (absolutely).*
- *If $L > 1$ (or $L = \infty$), the series diverges.*
- *If $L = 1$, the test is inconclusive – it tells you nothing.*

Why it works. Suppose $L < 1$. Pick any r with $L < r < 1$. Since the ratios approach L, beyond some index N we have $|a_{n+1}| < r\,|a_n|$. Then by induction $|a_{N+k}| < r^k |a_N|$, so the tail of the series is dominated, term by term, by the geometric series $|a_N| \sum r^k$, which converges because $r < 1$. By direct comparison the tail converges, hence the whole series converges absolutely. If instead $L > 1$, the terms eventually grow, so $a_n \not\to 0$ and the nth-term test kills it. When $L = 1$, the comparison geometric series sits exactly on its own boundary and tells us nothing. □

Example 29.8 (Factorials are the ratio test's home turf)**.** Test $\displaystyle\sum_{n=1}^{\infty} \frac{2^n}{n!}$. Form the ratio and watch the factorial demolish the exponential:

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n} = \frac{2}{n+1} \longrightarrow 0.$$

$L = 0 < 1$, so the series converges – and fast. (Its sum, foreshadowing the next chapter, is $e^2 - 1$.) Whenever you see factorials, reach for the ratio test first; the cancellation is almost always this clean. Figure 29.4 shows the terms diving under a geometric envelope, which is the whole mechanism behind the test.

Example 29.9 (Why $L = 1$ is genuinely useless)**.** Apply the ratio test to $\sum \frac{1}{n}$ (diverges) and to $\sum \frac{1}{n^2}$ (converges). Both give

$$\sum \tfrac{1}{n}: \ \frac{1/(n+1)}{1/n} = \frac{n}{n+1} \to 1, \qquad \sum \tfrac{1}{n^2}: \ \frac{n^2}{(n+1)^2} \to 1.$$

The ratio test returns $L = 1$ for both, yet one converges and one does not. This is concrete proof that $L = 1$ carries no information – you must switch to another test (here, the p-series rule). The ratio test is blind precisely at the borderline of polynomial decay.

The **root test** is the ratio test's close cousin, better suited when the nth term is an nth power. Instead of comparing neighbors, it extracts the "geometric ratio" directly by taking an nth root.

Theorem 29.8 (Root test)**.** *For $\sum a_n$, suppose $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L$. If $L < 1$ the series converges absolutely; if $L > 1$ (or ∞) it diverges; if $L = 1$ it is inconclusive.*

Why the technique works

The root test works for the same reason as the ratio test, even more directly. If $\sqrt[n]{|a_n|} \to L$ then for large n, $|a_n| \approx L^n$ – the term *is* essentially the nth term of a geometric series with ratio L. So $L < 1$ means geometric decay and convergence; $L > 1$ means geometric growth and divergence. The root test cuts out the middleman: rather than inferring the geometric ratio from consecutive terms, it reads it straight off the size of a_n itself. Use it whenever a_n wears an exponent of n, e.g. $a_n = \left(\frac{n}{2n+1}\right)^n$.

Example 29.10. Test $\displaystyle\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$. The nth power screams "root test":

$$\sqrt[n]{|a_n|} = \frac{n}{2n+1} \longrightarrow \frac{1}{2} < 1,$$

so the series converges. A ratio-test attack here would be a mess; the root test reads the answer off in one line.

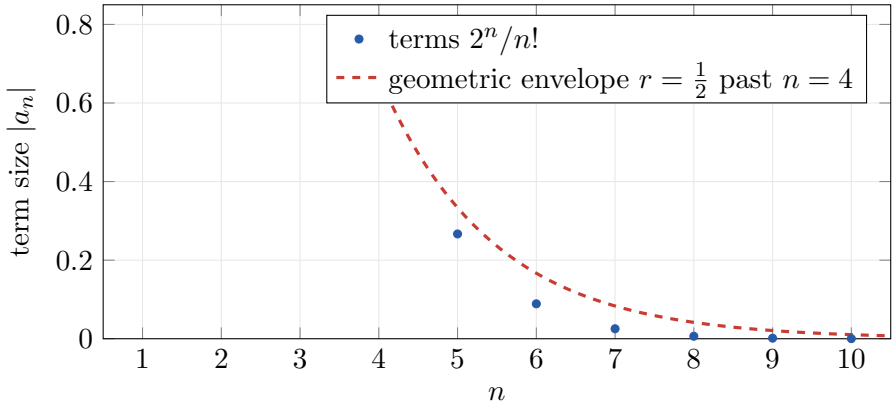


Figure 29.4: The ratio test in action on $\sum 2^n/n!$. The ratio $2/(n+1)$ drops below $\frac{1}{2}$ once $n \geq 4$, so from there on every term is bounded by a decaying geometric series (dashed). Being trapped under a convergent geometric tail is exactly what forces convergence.

29.9 Alternating Series and the Sign That Saves You

So far our terms have mostly been positive. Something remarkable happens when the signs alternate $+, -, +, -, \ldots$: the back-and-forth lets partial cancellation rescue a series that would diverge if all its terms were positive. The exemplar is the **alternating harmonic series**:

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}.$$

The positive version (the plain harmonic series) diverges to infinity. But with alternating signs, this one *converges* – to $\ln 2$, in fact. The minus signs claw back the runaway growth.

Intuition

Watch the partial sums of an alternating series whose terms shrink. You go up by 1, down by $\frac{1}{2}$, up by $\frac{1}{3}$, down by $\frac{1}{4}$. Each step overshoots the eventual limit, but by less than the last step (because the steps are shrinking). The partial sums zig-zag with ever-tightening swings – like a pendulum losing energy, homing in on a single value squeezed between every up-step and every down-step.

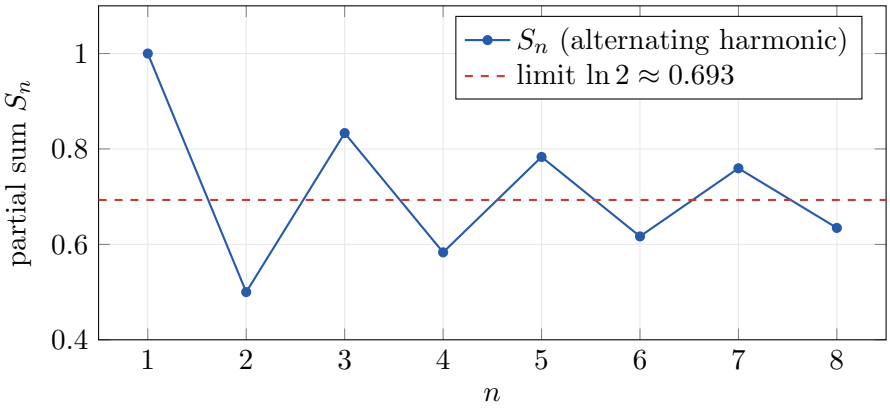


Figure 29.5: Partial sums of $1 - \frac{1}{2} + \frac{1}{3} - \cdots$ oscillate with shrinking amplitude around $\ln 2$. The odd partial sums approach from above, the even ones from below, and they trap the limit between them.

Theorem 29.9 (Alternating series test, Leibniz)**.** *If the positive numbers b_n satisfy (i) $b_{n+1} \le b_n$ (decreasing) and (ii) $b_n \to 0$, then the alternating series $\sum_{n=1}^{\infty}(-1)^{n-1}b_n$ converges. Moreover, the error from stopping at the nth term is no larger than the first omitted term: $|S - S_n| \le b_{n+1}$.*

Proof. Consider the even partial sums. Grouping in pairs, $S_{2n} = (b_1 - b_2) + (b_3 - b_4) + \cdots + (b_{2n-1} - b_{2n})$; each pair is ≥ 0 since b_k decreases, so (S_{2n}) is increasing. Regroup the other way, $S_{2n} = b_1 - (b_2 - b_3) - \cdots - b_{2n}$; each subtracted pair is ≥ 0, so $S_{2n} \le b_1$. An increasing sequence bounded above converges – call the limit S. Finally $S_{2n+1} = S_{2n} + b_{2n+1}$, and since $b_{2n+1} \to 0$ the odd sums approach the same S. So all partial sums converge to S. The error bound follows because S always lies between consecutive partial sums, which differ by exactly b_{n+1}. □

Why this matters

That error bound is a small marvel. It says: to approximate the sum to within 0.001, just keep adding terms until the terms themselves drop below 0.001, then stop – the leftover error is guaranteed smaller than the first term you threw away. No other elementary test hands you such a clean, automatic accuracy guarantee. This is why alternating series are beloved in numerical work: the error bound comes built into the series itself.

29.10 Absolute vs. Conditional Convergence: A Subtle Hierarchy

The alternating harmonic series converges, but only thanks to its signs – strip them away and it diverges. Contrast that with $\sum \frac{(-1)^{n-1}}{n^2}$, which converges even with all signs made positive (it's the

p-series $p = 2$). These two convergences are not the same *kind*, and the difference matters enormously.

Definition 29.2 (Absolute and conditional convergence)**.** A series $\sum a_n$ **converges absolutely** if $\sum|a_n|$ converges. It **converges conditionally** if $\sum a_n$ converges but $\sum|a_n|$ diverges – that is, the convergence depends on cancellation among the signs.

Theorem 29.10 (Absolute convergence implies convergence)**.** *If $\sum|a_n|$ converges, then $\sum a_n$ converges.*

Proof. The clean trick: $0 \le a_n + |a_n| \le 2|a_n|$ (the middle is either 0 or $2|a_n|$). Since $\sum 2|a_n|$ converges, by direct comparison $\sum(a_n + |a_n|)$ converges. Then $\sum a_n = \sum(a_n + |a_n|) - \sum|a_n|$ is a difference of two convergent series, hence converges. □

Focus

This theorem is what makes the ratio and root tests legitimate. When they report "$L < 1$," they are really proving $\sum|a_n|$ converges – absolute convergence – which by the theorem guarantees ordinary convergence. So those tests handle signs for free: you can throw a sign-changing series at them and they just work, because absolute convergence is the stronger, sign-blind notion.

What is really happening

What is the *real* difference between the two kinds? An absolutely convergent series converges because its terms are genuinely, sustainably small – it would converge even if you erased every minus sign. Its sum is robust: you may reorder, regroup, and rearrange the terms however you like and always reach the same total, exactly as with a finite sum. A conditionally convergent series is a far stranger animal. It converges only because of a delicate *balancing act* between positive and negative terms; the positives alone sum to $+\infty$ and the negatives alone to $-\infty$, and the finite total is just the tug-of-war between two infinities. Disturb that balance – reorder the terms – and you can make the sum come out to *any number you wish.* This is Riemann's astonishing Rearrangement Theorem. For the alternating harmonic series, one rearrangement gives $\ln 2$, another gives 5, another gives $-\pi$, all from the very same terms in a different order. Conditional convergence is convergence held together by string and good timing; absolute convergence is convergence built of stone.

History

In 1837, *Peter Gustav Lejeune Dirichlet* dropped an intellectual bombshell on the mathematical world. In a landmark paper on prime numbers in arithmetic progressions, Dirichlet showed that the alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \ln 2$ could have its terms rearranged to sum to a completely different value:

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \ln 2, \qquad \text{but} \qquad \left(1 - \frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{6} - \frac{1}{8}\right) + \cdots = \frac{1}{2}\ln 2!$$

Addition in a finite sum is commutative ($a + b = b + a$), and mathematicians had assumed without question that an infinite sum would obey the same law. Dirichlet proved that commutativity fails catastrophically for conditionally convergent series.

In 1854, *Bernhard Riemann* proved the ultimate extension of this discovery: the **Riemann Rearrangement Theorem**. Riemann proved that if a series converges conditionally, you can reorder its terms so that the newly arranged series converges to *any real number you choose*, or diverges to $+\infty$, or diverges to $-\infty$! This demonstrated that conditional convergence is not an inherent property of the set of numbers being summed, but an artifact of the rigid sequential ordering in which they are delivered. Only *absolute convergence* guarantees a rock-solid, invariant mathematical quantity.

29.11 Choosing a Test: A Strategy, Not a Lottery

With seven or eight tests in hand, the real skill is knowing which to reach for. The choice is not arbitrary – the *form* of the term usually tells you. Here is the decision flow an experienced eye runs, almost unconsciously.

Focus

A practical order of attack.

1. **Glance at the terms.** Do they fail to go to 0? Then it diverges by the nth-term test (Sec. 29.4). One look – always do this first.
2. **Recognize the species.** Is it geometric (ar^n) or telescoping (a difference $b_k - b_{k+1}$)? Then sum it *exactly* (Secs. 29.2, 29.3).
3. **Spot factorials or nth powers.** Factorials or exponentials $\Rightarrow$ ratio test. A whole term raised to the nth power $\Rightarrow$ root test (Sec. 29.8).
4. **See a rational / algebraic term** (powers of n, roots): find its dominant p-power and use limit comparison against a p-series (Secs. 29.7, 29.6).
5. **Notice alternating signs** $(-1)^n$: check absolute convergence first (it's stronger and easier); if that fails, try the alternating series test (Secs. 29.9, 29.10).
6. **If $f(n)$ has a clean antiderivative** ($\frac{1}{n \ln n}$, ne^{-n^2}): the integral test may be the most direct route (Sec. 29.6).

Common trap

The classic blunders, collected so you can avoid them:

- Concluding *convergence* from $a_n \to 0$. The nth-term test only proves divergence; it is silent on convergence. (Harmonic series.)
- Declaring victory when the ratio or root test returns $L = 1$. That is the *inconclusive* verdict – switch tests, do not guess.
- Using the integral test when f is not eventually decreasing. The hypothesis is essential; without it the rectangle-curve sandwich falls apart.
- Using limit comparison with $L = 0$ or $L = \infty$ carelessly. The clean "share a fate" conclusion needs $0 < L < \infty$; the boundary cases give only one-directional information.

- Reordering a conditionally convergent series as if order didn't matter. It does (Riemann).

Work it yourself

Classify each by hand, naming the test and the reasoning before you compute. (a) $\sum \dfrac{n}{n^2+1}$. (b) $\sum \dfrac{n!}{n^n}$. (c) $\sum \dfrac{(-1)^n}{\ln n}$ (from $n=2$). (d) $\sum \dfrac{\cos n}{n^2}$.
Hints to check yourself: (a) limit-compare with $\frac{1}{n}$, diverges. (b) ratio test, $\left(\frac{n}{n+1}\right)^n \to \frac{1}{e} < 1$, converges. (c) terms decrease to 0, alternating series test, converges – but $\sum \frac{1}{\ln n}$ diverges, so only *conditionally*. (d) $\left|\frac{\cos n}{n^2}\right| \le \frac{1}{n^2}$, comparison gives *absolute* convergence.

Application

Series are not museum pieces. In physics, the potential energy of a crystal lattice is an infinite (often conditionally convergent!) sum over ions, and getting the rearrangement right is a real computational concern. In probability, the expected value of a random variable is a series $\sum x_k p_k$, and whether it converges *absolutely* decides whether the expectation is even well-defined. In machine learning, a discounted reward $\sum_{t=0}^{\infty} \gamma^t r_t$ is a geometric-type series whose convergence (requiring $|\gamma| < 1$) is what makes reinforcement learning's value functions finite. And every special function – Bessel functions, the error function, $\zeta(s)$ – is delivered to your computer as a series whose convergence rate determines how many terms it must add to hit the precision you asked for. The tests in this chapter are, quietly, what make numerical computation possible.

29.12 What convergence theory unlocked, conditional fragility, and the road ahead

Connection

Infinite series represent one of the supreme achievements of analysis, replacing reckless algebraic manipulation with a rigorous framework for infinite accumulation.
Before Cauchy's 1821 formulation of partial sums and convergence tests, infinite series were a minefield. Brilliant mathematicians routinely derived contradictory sums from the same expression (as Grandi and Euler did with $1 - 1 + 1 - \cdots$). The convergence tests of this chapter (the nth-term test, integral test, comparison, limit comparison, ratio test, and root test) established an ironclad protocol. They unlocked:

- *Certifying convergence of non-elementary quantities:* Proving that physical constants (like the Euler–Mascheroni constant γ or $\zeta(3)$) and special functions (Bessel, Airy, Fresnel, and Riemann zeta functions) represent well-defined finite values;
- *High-precision scientific computing:* Establishing exact error bounds (via the alternating series error bound $|S - S_n| \le b_{n+1}$ and the geometric tail bound) that allow digital computers to calculate transcendental numbers to hundreds of digits with guaranteed precision.

Yet despite their immense power, the convergence tests of this chapter suffer from two fundamental boundaries:

1. *The sum-value blindspot:* The vast majority of our tests (integral test, comparison, ratio, root) are purely *qualitative*. They answer the binary question: "Does the series converge to a finite number?" They tell you virtually *nothing* about what that number is! Proving that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges takes thirty seconds using the integral test. Finding its exact numerical sum—$\frac{\pi^2}{6}$, Euler's celebrated solution to the Basel Problem in 1735—required a decade of heroic genius and techniques far beyond single-variable tests.

2. *The fragility of conditional convergence:* When a series converges only conditionally ($\sum |a_n| = \infty$), its value is mathematically fragile. By Riemann's Rearrangement Theorem, the order of terms is not an incidental bookkeeping detail; it is the entire reason the sum exists. Rearranging the terms of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ can change the sum from $\ln 2$ to 500 or $-\pi$.

In Chapter 30 (*Power and Taylor Series*), we take the decisive leap from static numerical series $\sum a_n$ to dynamic *functions of* x:

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n.$$

By treating an infinite series as an infinite polynomial, we can differentiate and integrate series term by term. This unlocks the ability to express functions like e^x, $\sin x$, and $\arctan x$ as power series, finally revealing the exact sums of countless numerical series and solving the very differential equations that govern the universe.

29.13 Exercises

Exercise 29.1 (Sum it exactly)**.** Evaluate each convergent series in closed form. (a) $\displaystyle\sum_{n=0}^{\infty} \frac{3}{4^n}$. (b) $\displaystyle\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n$. (c) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)}$ (telescope via $\frac{1}{2}\big(\frac{1}{2n-1} - \frac{1}{2n+1}\big)$). (d) Express $0.1\overline{6}$ (that is, $0.16666\ldots$) as a fraction using a geometric series.

Exercise 29.2 (The nth-term test, used correctly)**.** Determine which of these diverge by the nth-term test, and explain why the test is *silent* on the others. (a) $\sum \frac{n^2}{n^2+5}$. (b) $\sum \frac{1}{n^2+5}$. (c) $\sum \cos\big(\frac{1}{n}\big)$. (d) $\sum (-1)^n \frac{n}{n+1}$.

Exercise 29.3 (p-series and comparison)**.** Decide convergence, naming your comparison series each time. (a) $\sum \frac{1}{n^{3/2}}$. (b) $\sum \frac{1}{\sqrt[3]{n}}$. (c) $\sum \frac{n+4}{n^3-2n+7}$. (d) $\sum \frac{1}{\sqrt{n^3+1}}$.

Exercise 29.4 (Ratio and root)**.** Apply the ratio or root test, whichever is cleaner, and say what L is. (a) $\sum \frac{5^n}{n!}$. (b) $\sum \frac{n!}{10^n}$. (c) $\sum \frac{n^2}{2^n}$. (d) $\sum \left(\frac{3n+1}{4n+5}\right)^n$.

Exercise 29.5 (Integral test)**.** Use the integral test to settle each, and verify the hypotheses (positive, decreasing) hold. (a) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n \ln n}$. (b) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$. (c) $\displaystyle\sum_{n=1}^{\infty} n\, e^{-n^2}$. Comment on how (a) and (b) differ despite looking nearly identical – where is the borderline now?

Exercise 29.6 (Alternating, absolute, conditional)**.** For each, decide whether it converges absolutely, converges conditionally, or diverges. Justify all three possibilities you rule out or confirm. (a) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$. (b) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$. (c) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}\, n}{n+1}$. (d) $\displaystyle\sum_{n=2}^{\infty} \frac{(-1)^{n}}{n \ln n}$.

Exercise 29.7 (Error estimate)**.** For the alternating series $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^3}$, how many terms must you add to guarantee the partial sum is within 10^{-3} of the true value? Use the alternating-series error bound, and explain why that bound applies here.

Exercise 29.8 (Conceptual)**.** Answer in a sentence or two each, with reasons. (a) True or false: if $a_n \to 0$ then $\sum a_n$ converges. Give the canonical counterexample. (b) True or false: if $\sum a_n$ converges then $a_n \to 0$. Which theorem is this? (c) Can a series of positive terms converge conditionally? Why or why not? (d) If $\sum a_n$ and $\sum b_n$ both diverge, must $\sum(a_n + b_n)$ diverge? Find an example or prove it.

Exercise 29.9 (Prove it)**.** Prove that if $\sum a_n$ converges and $\sum b_n$ diverges, then $\sum(a_n+b_n)$ diverges. (Hint: argue by contradiction using the fact that the difference of two convergent series converges.)

Exercise 29.10 (Prove it – the heart of the matter)**.** Prove the nth-term test from scratch: if $\sum_{k=1}^{\infty} a_k = S$ (converges), then $a_n \to 0$. Your proof should use only the definition of convergence via partial sums (Definition 29.1) and the fact that if $S_n \to S$ then $S_{n-1} \to S$. Then explain in one paragraph, in your own words, why the converse fails, citing a specific series.

Exercise 29.11 (Geometric in disguise)**.** A patient takes a 100 mg dose of a drug each day. Each day, the body eliminates 25% of whatever drug is present, so 75% carries over to the next day. Model the long-run amount of drug in the body just after a dose as an infinite series, identify it as geometric, and find the steady-state amount. (This is exactly the reinforcement-learning discounted-sum structure from the application box, wearing a lab coat.)

Exercise 29.12 (Build the borderline)**.** We saw $\sum \frac{1}{n}$ diverges and $\sum \frac{1}{n^2}$ converges. Investigate the finer scale between them: (a) Argue that $\sum \frac{1}{n^{1.0001}}$ converges and $\sum \frac{1}{n^{0.9999}}$ diverges – the borderline at $p = 1$ is sharp. (b) Using the integral test, show $\sum \frac{1}{n \ln n}$ still diverges, so $\frac{1}{n}$ is not even the "last" divergent term – there is a whole ladder of ever-slower divergences below it. Reflect briefly on what this says about the idea of a single "boundary" between convergence and divergence.

Chapter 30

Power Series and Taylor Series

Somewhere in calculus the pieces you have collected fall into a single picture. Polynomials are the friendliest functions of all: you can add them, multiply them, differentiate them, integrate them, evaluate them with nothing but arithmetic. The transcendental functions e^x, $\sin x$, $\cos x$, $\ln(1+x)$ seemed to belong to a different species. You could not write $\sin x$ as any finite combination of powers of x, so to compute $\sin(0.3)$ you reached for a calculator and trusted a black box.

This chapter dissolves that wall. *Nearly every function you care about is secretly a polynomial – one of infinite degree.* The function e^x, the function $\sin x$, the function $\ln(1+x)$ – each is equal, on some interval, to a genuine sum of powers of x,

$$c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots,$$

where the coefficients c_n are specific, computable numbers. Once you know those coefficients, the mysterious transcendental function becomes something you can compute by hand, differentiate term by term, integrate term by term, and reason about with the comfortable algebra of polynomials.

These infinite polynomials are called *power series*, and the recipe that produces the coefficients is the *Taylor series*. By the end of this chapter you will see not only *how* to build them but *why* the recipe must give the coefficients it does – it is forced, with no freedom at all, by demanding that the polynomial match the function's value, slope, curvature, and every higher rate of change at a single point.

30.1 What a Power Series Is

Intuition

A power series is a polynomial that never stops. Think of it as a recipe for a function written entirely in powers of x: pick a list of numbers $c_0, c_1, c_2, \ldots$, and the series is

$$c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots$$

For most lists this sum converges only for x close to 0, and there it defines a genuine function. The family of functions you can reach this way turns out to be remarkably rich.

Definition 30.1 (Power series)**.** A *power series centered at a* is an expression of the form

$$\sum_{n=0}^{\infty} c_n(x-a)^n \;=\; c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots,$$

where the numbers c_n are the *coefficients* and the fixed number a is the *center*. For each fixed value of x this is an infinite series of real numbers in the sense of the previous chapter, and we ask the same question we always ask of an infinite sum: does it converge?

When $a = 0$ the series simplifies to $\sum c_n x^n$, and that is the case we will live in most of the time. A series centered at a is nothing more than the same idea shifted so that its natural "home base" sits at $x = a$ instead of at the origin. Everything we prove about $\sum c_n x^n$ near 0 transfers, by the substitution $x \mapsto x - a$, to $\sum c_n(x-a)^n$ near a.

A power series is a *function of* x, but only on the set of x where the series actually converges. At $x = a$ every term past the first vanishes, so the series collapses to c_0 – it always converges at its center. The interesting question is how far away from the center the convergence survives.

Example 30.1 (The geometric series is a power series)**.** The most important power series in all of mathematics is one you have already met:

$$1 + x + x^2 + x^3 + \cdots = \sum_{n=0}^{\infty} x^n.$$

Here every coefficient is $c_n = 1$. From the previous chapter you know this geometric series converges precisely when $|x| < 1$, and when it does, it sums to $\dfrac{1}{1-x}$. So we may write the exact equation

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \qquad |x| < 1.$$

This is our first function-as-infinite-polynomial. The rational function $1/(1-x)$, defined everywhere except $x = 1$, agrees with the infinite polynomial on the open interval $(-1, 1)$ and disagrees (the series diverges) everywhere else. Hold onto this example: almost every other series in this chapter is built from it by differentiating, integrating, or substituting.

30.1.1 Where does a power series live? Radius of convergence

For the geometric series the set of good x was the symmetric interval $(-1, 1)$. That symmetry holds in general: the set of points where a power series converges is *always* an interval centered at a – there is a single radius R such that the series converges inside and diverges outside.

Theorem 30.1 (Radius of convergence)**.** *For any power series $\sum_{n=0}^{\infty} c_n(x-a)^n$, exactly one of the following holds:*

1. *The series converges only at $x = a$ (we say $R = 0$).*
2. *The series converges for all real x (we say $R = \infty$).*
3. *There is a positive number R, the* radius of convergence*, such that the series converges absolutely whenever $|x - a| < R$ and diverges whenever $|x - a| > R$.*

Intuition

Picture a circle of influence around the center a. Inside the circle the powers $(x-a)^n$ shrink fast enough that the series adds up to something finite. Outside, the powers grow and overwhelm any coefficients, so the sum explodes. The boundary – the two endpoints $a \pm R$ – is a delicate place that must be checked by hand each time, because the series can go either way there.

The engine behind this theorem is a single comparison idea, resting on one lemma: if a power series converges *anywhere* away from the center, it converges *everywhere closer in.*

Lemma 30.1 (Abel's comparison)**.** *Suppose* $\sum c_n b^n$ *converges for some particular* $b \neq 0$*. Then* $\sum c_n x^n$ *converges absolutely for every* x *with* $|x| < |b|$*.*

Proof. If $\sum c_n b^n$ converges, then its terms must march to zero (the term test, from the previous chapter). A sequence that tends to zero is in particular bounded, so there is a constant M with

$$|c_n b^n| \leq M \qquad \text{for every } n.$$

Now take any x closer to the origin than b, and let $r = |x|/|b|$, a number strictly less than 1. Watch how the terms of $\sum c_n x^n$ get controlled:

$$|c_n x^n| = |c_n b^n| \cdot \left|\frac{x}{b}\right|^n \leq M\, r^n.$$

We have bounded our terms by the terms of a convergent geometric series $\sum M r^n$ (convergent because $r < 1$). By the comparison test, $\sum |c_n x^n|$ converges. That is exactly absolute convergence for $\sum c_n x^n$. □

Why the technique works

Why does the convergence set have to be an *interval* – why couldn't a power series converge at $x = 3$ and $x = 7$ but diverge at $x = 5$ in between? Abel's lemma forbids it. Convergence at the far point $x = 7$ would force convergence at every point closer to the center than 7, including $x = 5$. So the "good" set has no holes: if it contains a point, it contains everything between that point and the center. A subset of the line that contains the whole segment from the center out to each of its points, and is symmetric about the center, can only be a single interval centered at a. That is the entire content of Theorem 30.1: the radius R is simply the supremum of the distances $|b - a|$ at which convergence happens.

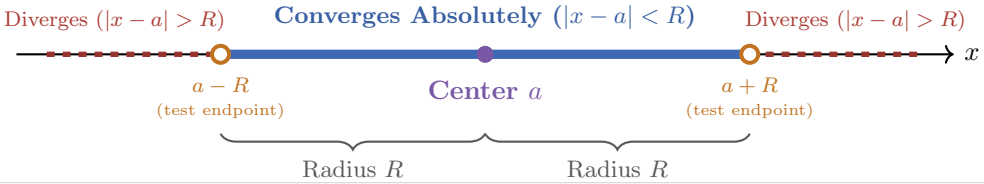
History

The Norwegian mathematical prodigy *Niels Henrik Abel* investigated the delicate boundary behavior of power series in an 1826 paper. Abel asked what happens right on the boundary circle $|x - a| = R$. He proved the celebrated **Abel's Theorem**: if a power series converges at an endpoint $x = a + R$, then the function defined by the series is continuous from the inside up to that boundary point:

$$\lim_{x \to (a+R)^-} \sum_{n=0}^{\infty} c_n (x-a)^n = \sum_{n=0}^{\infty} c_n R^n.$$

This theorem gave mathematicians the rigorous right to evaluate series at their boundary

– for example, allowing us to plug $x = 1$ into the $\ln(1+x)$ Maclaurin series to prove that $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \ln 2$, or setting $x = 1$ in the arctangent series to prove Gregory's and Leibniz's formula $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$. Without Abel's 1826 theorem, jumping from the open interval $|x| < 1$ to the boundary point $x = 1$ was an unproven leap of faith.



The 4 Possible Intervals of Convergence:

$(a-R,\, a+R)$ $\quad [a-R,\, a+R)$ $\quad (a-R,\, a+R]$ $\quad [a-R,\, a+R]$

Interior behavior is guaranteed; each endpoint must be checked independently with series tests.

Figure 30.1: The geometry of power series convergence. Inside the radius of convergence $|x-a| < R$, the series converges absolutely. Outside ($|x-a| > R$), the terms fail to go to zero and the series diverges. At the two boundary points $x = a \pm R$, the ratio test is inconclusive and the series may converge or diverge independently, leading to four possible interval types.

30.1.2 Finding the radius with the ratio test

In practice you almost always find R by feeding the series to the ratio test from the previous chapter. The ratio test asks for the limit of the absolute ratio of consecutive terms; convergence happens when that limit is below 1. For a power series the variable x rides along in that ratio, and the condition "ratio < 1" becomes a condition on $|x|$ – which is exactly the radius.

Focus

The ratio-test recipe for the radius. Apply the ratio test to $\sum c_n(x-a)^n$:

$$L(x) = \lim_{n\to\infty} \left| \frac{c_{n+1}(x-a)^{n+1}}{c_n(x-a)^n} \right| = |x-a| \cdot \lim_{n\to\infty} \left| \frac{c_{n+1}}{c_n} \right|.$$

If $\lim|c_{n+1}/c_n| = \ell$, then convergence ($L < 1$) means $|x-a| < 1/\ell$, so

$$\boxed{R = \frac{1}{\ell} = \lim_{n\to\infty} \left| \frac{c_n}{c_{n+1}} \right|}$$

(with $R = \infty$ if $\ell = 0$ and $R = 0$ if $\ell = \infty$). Always remember to test the endpoints $a \pm R$ separately.

Example 30.2 (Radius of the exponential series)**.** Consider $\displaystyle\sum_{n=0}^{\infty} \frac{x^n}{n!}$, so $c_n = 1/n!$. The ratio of consecutive coefficients is

$$\left| \frac{c_{n+1}}{c_n} \right| = \frac{1/(n+1)!}{1/n!} = \frac{n!}{(n+1)!} = \frac{1}{n+1} \to 0.$$

So $\ell = 0$ and $R = 1/\ell = \infty$. This series converges for *every* real x. The factorials in the denominator grow so violently fast that they crush any value of x^n in the numerator, no matter how large x is. (We will soon see this series is e^x.)

Example 30.3 (A series with a finite radius and interesting endpoints)**.** Consider $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{n}$, so $c_n = 1/n$. Then

$$\left|\frac{c_n}{c_{n+1}}\right| = \frac{1/n}{1/(n+1)} = \frac{n+1}{n} \to 1,$$

so $R = 1$. The series converges on $(-1, 1)$ and diverges outside $[-1, 1]$. Now the endpoints:

- At $x = 1$ the series is $\sum 1/n$, the harmonic series – diverges.
- At $x = -1$ the series is $\sum (-1)^n/n$, the alternating harmonic series – converges.

So the true interval of convergence is $[-1, 1)$: closed on the left, open on the right. This lopsidedness is exactly why the endpoints cannot be guessed and must be checked. (We will soon see this series is $-\ln(1 - x)$.)

Work it yourself

Find the radius of convergence of $\displaystyle\sum_{n=0}^{\infty} n!\, x^n$ by hand. Compute $|c_n/c_{n+1}| = n!/(n+1)! = 1/(n+1) \to 0$, so $R = 0$: this series converges *only* at $x = 0$. The coefficients grow so fast that no nonzero x can tame them. A power series really can have a vanishing radius.

Common trap

The radius is determined by the coefficients, *not* by where the function you are representing happens to misbehave – but the two are often related. Be careful, though, with two classic confusions:

- "R large means the terms are large." No: large R means the coefficients c_n *shrink* fast, so the powers can grow far before overwhelming them.
- "Convergence at one endpoint guarantees the other." No, as Example 30.3 shows. Each endpoint is its own separate question.

30.2 Building Functions from the Geometric Series

Before we develop the general Taylor machinery, let us see how much mileage we get from the single equation

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \qquad |x| < 1,$$

together with two tools we are about to justify: *you may differentiate and integrate a power series term by term inside its radius of convergence*, and *you may substitute one power series into a simple expression*. These manipulations are the bread and butter of the subject, and they let us grow a whole orchard of series from one seed.

Theorem 30.2 (Term-by-term differentiation and integration)**.** *Suppose $f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$ has radius of convergence $R > 0$. Then on the open interval $(a-R, a+R)$ the function f is differentiable (indeed infinitely differentiable), and its derivative and antiderivative are obtained term by term:*

$$f'(x) = \sum_{n=1}^{\infty} n\, c_n (x-a)^{n-1}, \qquad \int f(x)\,\mathrm{d}x = C + \sum_{n=0}^{\infty} \frac{c_n}{n+1}(x-a)^{n+1}.$$

Both new series have the same radius of convergence R.

We will not prove this in full – the careful proof belongs to the theory of uniform convergence – but the next box explains why it ought to be true, and why the radius is preserved.

Why the technique works

A finite polynomial can of course be differentiated term by term; that is just the power rule applied to each piece. The only thing in doubt for an *infinite* sum is whether the infinitely many error contributions, when you swap "differentiate" and "add up", stay under control. They do, and the reason is the absolute convergence guaranteed *strictly inside* the radius (Lemma 30.1). Inside the radius there is room to spare – the terms decay geometrically – and that slack is exactly what lets the limit defining the derivative pass through the infinite sum. As for the radius being unchanged: differentiating multiplies the coefficient c_n by n, and $\lim|nc_n/((n+1)c_{n+1})| = \lim(n/(n+1)) \cdot |c_n/c_{n+1}| = R$, since $n/(n+1) \to 1$. The factor of n is too mild to change where the series lives.

Example 30.4 (A series for $\dfrac{1}{(1-x)^2}$ by differentiating)**.** Differentiate both sides of $\dfrac{1}{1-x} = \sum_{n=0}^{\infty} x^n$. The left side differentiates to $\dfrac{1}{(1-x)^2}$. The right side differentiates term by term:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\sum_{n=0}^{\infty} x^n\right) = \sum_{n=1}^{\infty} n x^{n-1} = 1 + 2x + 3x^2 + 4x^3 + \cdots.$$

So, for $|x| < 1$,

$$\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n\, x^{n-1} = \sum_{n=0}^{\infty} (n+1)x^n.$$

We obtained a brand-new, nontrivial series for free, just by differentiating the one series we knew.

Example 30.5 (The series for $\ln(1+x)$ by integrating)**.** Start from the geometric series with x replaced by $-t$:

$$\frac{1}{1+t} = \frac{1}{1-(-t)} = \sum_{n=0}^{\infty} (-t)^n = 1 - t + t^2 - t^3 + \cdots, \qquad |t| < 1.$$

Now integrate both sides from 0 to x. On the left,

$$\int_0^x \frac{\mathrm{d}t}{1+t} = \ln(1+x).$$

On the right, integrate term by term:

$$\int_0^x \left(1 - t + t^2 - t^3 + \cdots\right)\mathrm{d}t = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots.$$

Therefore

$$\boxed{\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots}, \qquad -1 < x \le 1.$$

Notice the surprise at the endpoint: although we derived the formula assuming $|x| < 1$, at $x = 1$ the right side becomes the convergent alternating harmonic series, and one can show it does equal $\ln 2$. So

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots,$$

a strange and beautiful identity: an alternating sum of unit fractions equals the natural logarithm of 2.

Example 30.6 (The series for $\arctan x$, and a formula for π)**.** The same trick produces one of the most famous series in history. Replace x by $-t^2$ in the geometric series:

$$\frac{1}{1+t^2} = \sum_{n=0}^{\infty} (-t^2)^n = 1 - t^2 + t^4 - t^6 + \cdots, \qquad |t| < 1.$$

Integrate from 0 to x, using $\int_0^x \dfrac{\mathrm{d}t}{1+t^2} = \arctan x$:

$$\boxed{\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}}, \qquad -1 \le x \le 1.$$

Setting $x = 1$ gives Leibniz's stunning formula:

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \cdots.$$

History

The series for $\arctan x$ – and with it the formula above for $\pi/4$ – was discovered in the fourteenth century by the Indian mathematician **Madhava of Sangamagrama** and his Kerala school, roughly three hundred years before Newton and Leibniz. They had series for sine, cosine, and arctangent, and used them to compute π to eleven decimal places. In Europe the arctangent series is usually credited to James Gregory (1671) and Leibniz (1673). It is a humbling reminder that the "infinite polynomial" idea was glimpsed, with real rigor for its time, far from the usual narrative centers. The Leibniz series converges agonizingly slowly – you need hundreds of terms for a couple of digits – but it was the first time a simple, exact, infinite arithmetic recipe was given for the circle's most famous constant.

30.3 Taylor and Maclaurin Series: the Universal Recipe

So far we have manufactured series by clever manipulation of the geometric series. That is wonderful when it works, but it does not work for e^x, $\sin x$, or $\cos x$ – those are not built from $1/(1-x)$ by any obvious substitution. We need a *universal* method that takes *any* sufficiently smooth function and produces its power series directly. That method is Taylor's, and the idea behind it is one of the cleanest in all of calculus.

History

The dream of representing arbitrary curves as infinite polynomials was glimpsed in 1671 by the Scottish mathematician *James Gregory*. In correspondence with John Collins, Gregory derived Taylor-like series for $\tan x$, $\sec x$, and $\arcsin x$, discovering the expansion method more than forty years before Taylor.
In 1715, the English mathematician *Brook Taylor* published the universal formulation in his treatise *Methodus Incrementorum Directa et Inversa*. Taylor grasped the fundamental insight: the local derivatives of a function at a single basepoint $x = a$ act as a complete genetic code. Each successive derivative pins down one coefficient of the polynomial to match the curve's height, slope, curvature, and higher-order rates of change. The Scottish mathematician *Colin Maclaurin* later popularized the special case centered at $a = 0$ in his 1742 *Treatise of Fluxions*, which is why series centered at the origin bear Maclaurin's name today, though the underlying intellectual breakthrough belongs to Gregory and Taylor.

Intuition

Suppose a function f really is equal to some power series near a point a:

$$f(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots .$$

We do not yet know the coefficients. But we have a tool that extracts them one at a time: *differentiation*. Each derivative, evaluated at the center a, reaches in and pulls out exactly one coefficient, leaving the others either already gone or not yet arrived. The coefficients are not free; they are *forced* to be the derivatives of f at a.

Let us watch the extraction happen. Assume $f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$ and that we may differentiate term by term.

First, just plug in $x = a$. Every term with a factor of $(x-a)$ dies, leaving only the constant:

$$f(a) = c_0.$$

Next, differentiate once: $f'(x) = c_1 + 2c_2(x-a) + 3c_3(x-a)^2 + \cdots$, then plug in $x = a$ again. Once more all the $(x-a)$ factors vanish:

$$f'(a) = c_1.$$

Differentiate again: $f''(x) = 2c_2 + 6c_3(x-a) + 12c_4(x-a)^2 + \cdots$, and set $x = a$:

$$f''(a) = 2c_2 \quad \Longrightarrow \quad c_2 = \frac{f''(a)}{2}.$$

Once more: $f'''(x) = 6c_3 + 24c_4(x-a) + \cdots$, so $f'''(a) = 6c_3$, giving $c_3 = f'''(a)/6$. The pattern is unmistakable. Differentiating n times and evaluating at a leaves only the nth term, which has been multiplied by $n \cdot (n-1) \cdots 2 \cdot 1 = n!$. Hence

$$f^{(n)}(a) = n!\, c_n \qquad \Longrightarrow \qquad c_n = \frac{f^{(n)}(a)}{n!}.$$

Definition 30.2 (Taylor and Maclaurin series)**.** Let f be infinitely differentiable at a. The *Taylor series of f centered at a* is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots .$$

When the center is $a = 0$ this is also called the *Maclaurin series*:

$$f(0) + f'(0)\,x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots .$$

The polynomial obtained by stopping at the term of degree N is the *Taylor polynomial* $P_N(x)$ of order N.

Why the technique works

Why this exact recipe and no other? Because a power series matching f has *no freedom*. Each coefficient is pinned down by one derivative: c_n controls (and is the only coefficient controlling) the nth derivative at the center. The factor $n!$ appears because differentiating $(x-a)^n$ a total of n times produces the falling product $n(n-1)\cdots 1 = n!$. The Taylor recipe is simply the unique answer to the question: *which polynomial has the same value, the same slope, the same curvature, the same third derivative, and so on, all the way up, as f at the point a?* You are not approximating by guesswork; you are matching f to infinite order at a single point.

What is really happening

Here is what a Taylor polynomial *really* is, stripped of symbols. The order-0 polynomial $P_0 = f(a)$ is the horizontal line through the point – it gets the height right. The order-1 polynomial $P_1 = f(a) + f'(a)(x-a)$ is the tangent line – it gets the height *and* the slope right; this is exactly the linear approximation from the chapter on differentials. The order-2 polynomial adds $\frac{1}{2}f''(a)(x-a)^2$, a parabola that also matches the *curvature*. Each new term fixes the next derivative without disturbing the ones already matched, because $(x-a)^n$ and all its first $n-1$ derivatives vanish at a. So the Taylor series is a tower: each floor corrects a finer feature of the function's shape at the center, and "infinitely many floors" means matching the function's local shape to infinite precision. Whether that local matching extends to a global equality is the subtle question we take up in the next chapter.

30.3.1 The three great Maclaurin series

We now compute, from the definition, the three series that every scientist carries in their head: e^x, $\sin x$, $\cos x$. Each is centered at 0, and in each case the derivatives at 0 fall into a simple repeating pattern.

Example 30.7 (The series for e^x)**.** Let $f(x) = e^x$. The defining magic of the exponential is that it is its own derivative: $f'(x) = e^x$, $f''(x) = e^x$, and in general $f^{(n)}(x) = e^x$ for all n. Evaluated at 0, every derivative is $e^0 = 1$:

$$f^{(n)}(0) = 1 \qquad \text{for all } n.$$

So the coefficients are $c_n = f^{(n)}(0)/n! = 1/n!$, and

$$\boxed{e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots}, \qquad \text{all } x.$$

We computed the radius of this series back in Example 30.2: it is infinite. The same factorial growth that gave $R = \infty$ now reveals itself as the engine of e^x.

What is really happening

This series is arguably the *true definition* of e^x. In the chapter on the exponential we met e^x three ways: as a limit of compound interest, as the function equal to its own derivative, and as the inverse of ln. The series $\sum x^n/n!$ unifies them. It is its own derivative by term-by-term differentiation: differentiating $x^n/n!$ gives $x^{n-1}/(n-1)!$, which is just the previous term, so the whole series reproduces itself. It equals 1 at $x = 0$. Those two facts – self-derivative and value 1 at 0 – determine e^x completely. Setting $x = 1$ even gives the number e itself as a rapidly converging sum: $e = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \cdots = 2.71828\ldots$, far faster than the compound-interest limit ever did.

Example 30.8 (The series for $\sin x$ and $\cos x$)**.** Let $f(x) = \sin x$. Its derivatives cycle with period four:

$$f = \sin x, \quad f' = \cos x, \quad f'' = -\sin x, \quad f''' = -\cos x, \quad f'''' = \sin x, \ \ldots$$

Evaluating at 0, where $\sin 0 = 0$ and $\cos 0 = 1$, we get the pattern $0, 1, 0, -1, 0, 1, 0, -1, \ldots$. The even-order derivatives vanish; the odd ones alternate $+1, -1$. So only odd powers survive:

$$\boxed{\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}\, x^{2n+1}}, \qquad \text{all } x.$$

The same computation for $\cos x$ (whose derivatives at 0 run $1, 0, -1, 0, 1, \ldots$) keeps only even powers:

$$\boxed{\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!}\, x^{2n}}, \qquad \text{all } x.$$

Both have infinite radius. Notice that sin uses only odd powers (it is an odd function) and cos only even powers (it is even): the series *remembers* the symmetry of the function. And differentiating the sin series term by term gives the cos series exactly, just as it must.

Connection

Line up the three series and a miracle leaps out:

$$e^x = 1 + x + \tfrac{x^2}{2!} + \tfrac{x^3}{3!} + \cdots, \quad \cos x = 1 - \tfrac{x^2}{2!} + \tfrac{x^4}{4!} - \cdots, \quad \sin x = x - \tfrac{x^3}{3!} + \tfrac{x^5}{5!} - \cdots.$$

The terms of e^x are precisely the terms of $\cos x$ and $\sin x$ interleaved – if only we could supply the alternating signs. Feed an imaginary argument ix into the exponential series and the powers i^n cycle $1, i, -1, -i, \ldots$, producing exactly those signs. The even powers reconstruct $\cos x$, the odd powers reconstruct $i \sin x$, and you land on **Euler's formula**

$$e^{ix} = \cos x + i \sin x,$$

the deepest identity connecting exponential growth and circular motion. At $x = \pi$ it collapses to $e^{i\pi} + 1 = 0$. The power series is what makes this rigorous: it is the common language in which e^x, $\cos x$, and $\sin x$ are visibly the same kind of object.

30.4 Watching the Polynomials Close In

Definitions are one thing; *seeing* a polynomial chase down a transcendental function is another. The Taylor polynomials $P_N(x)$ are ordinary finite polynomials – you can graph them. As N grows, they hug the true function over a wider and wider stretch. Let us look.

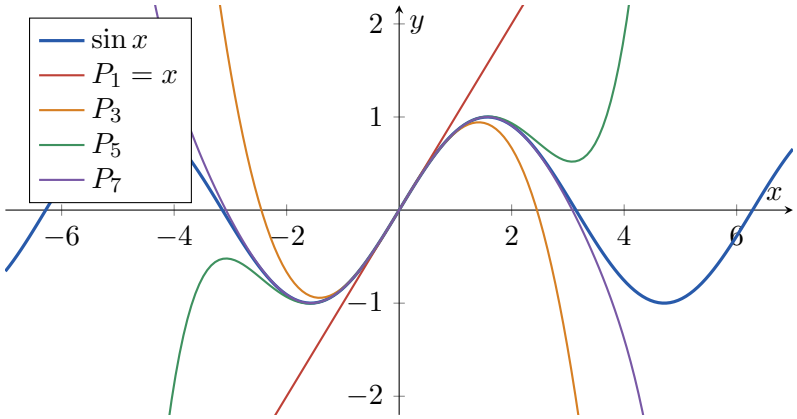


Figure 30.2: Taylor polynomials of $\sin x$ at 0. P_1 (the tangent line) is good only very near the origin; each higher-order polynomial clings to the sine curve over a longer interval before peeling away. By P_7 the match is excellent across a full period.

The picture tells the whole story. Near the center every polynomial is a good fit – that is guaranteed, because they match $\sin x$ to high order *at the origin*. The interesting drama is at the edges: a low-order polynomial cannot help shooting off to $\pm\infty$ (a degree-7 polynomial must eventually behave like $\pm x^7$), and so it eventually abandons the bounded sine wave. But it holds on longer than its predecessor, and the place where it "lets go" marches steadily outward as N increases. Take $N \to \infty$ and the letting-go point runs off to infinity: the full series equals $\sin x$ everywhere.

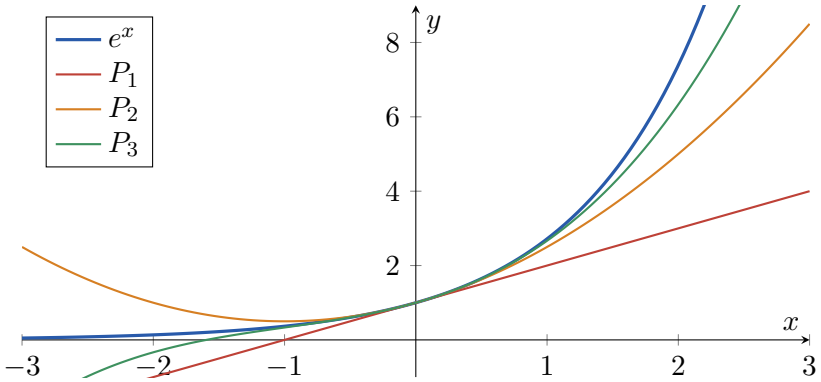


Figure 30.3: Taylor polynomials of e^x at 0. The tangent line $P_1 = 1 + x$ falls below the curve immediately; the parabola P_2 matches the upward curvature; P_3 improves the fit further. Each polynomial sits underneath e^x for $x > 0$ and creeps closer as the order rises.

Walking question

Take a walk and turn this over. The Taylor polynomial matches the function perfectly *at one point* – value, slope, curvature, and all higher derivatives, to whatever order you choose. Yet a polynomial of degree 7 must eventually run away to infinity, while $\sin x$ stays politely between -1 and 1 forever. How can these two facts coexist? The resolution: matching to high order at a *point* controls behavior *near* that point, but says nothing directly about far away. The error grows as you leave the center, slowly at first (because so many derivatives agree) and then catastrophically. The whole art of approximation theory is the contest between "how high the order" and "how far from the center", and the next chapter, on Taylor's theorem with remainder, makes that contest exact.

Work it yourself

Compute $\sin(0.5)$ by hand from the series, keeping three terms:

$$\sin(0.5) \approx 0.5 - \frac{0.5^3}{6} + \frac{0.5^5}{120} = 0.5 - \frac{0.125}{6} + \frac{0.03125}{120}.$$

That is $0.5 - 0.0208333 + 0.0002604 = 0.4794271$. The true value is $\sin(0.5) = 0.4794255\ldots$, so three terms already give six-digit accuracy. This is literally how a calculator does it: reduce the angle to a small range, then sum a handful of series terms. The black box was a power series all along.

30.5 Manipulating Series: an Algebra of Infinite Polynomials

Inside their radius of convergence, power series obey the same algebra as ordinary polynomials, and that is what makes them so useful. You may add them, subtract them, multiply them, substitute one into another, and (with care) divide them. This means you rarely have to grind out high derivatives from the Taylor definition – you start from the known series for e^x, $\sin x$, $\cos x$, $\ln(1+x)$, $1/(1-x)$ and *build*.

Focus

The toolbox of legal moves (valid wherever the series involved converge):

- **Add / subtract:** combine coefficients of like powers.
- **Multiply by a constant or a power of x:** scale or shift every term.
- **Substitute** a simple expression (like x^2 or $-x$ or $3x$) for the variable.
- **Multiply two series:** collect terms of each total degree (the Cauchy product).
- **Differentiate / integrate** term by term (Theorem 30.2).

The uniqueness of power series (each function has *one* series at a given center) is what makes this legal: whatever route you take, you must arrive at the same coefficients, so you may take the easiest route.

Example 30.9 (Substitution: the series for e^{-x^2})**.** The function e^{-x^2} – the unnormalized Gaussian bell curve – has no elementary antiderivative, so the integral $\int e^{-x^2}\mathrm{d}x$ defies the usual techniques.

But its series is effortless. Start from $e^u = \sum u^n/n!$ and substitute $u = -x^2$:

$$e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-x^2)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \cdots .$$

Now integrate term by term – something we could never do to the closed form:

$$\int_0^x e^{-t^2}\,\mathrm{d}t = x - \frac{x^3}{3} + \frac{x^5}{5\cdot 2!} - \frac{x^7}{7\cdot 3!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)\,n!}.$$

This is, up to a constant, the error function erf that governs the normal distribution in statistics. A function with no elementary antiderivative becomes completely computable through its series.

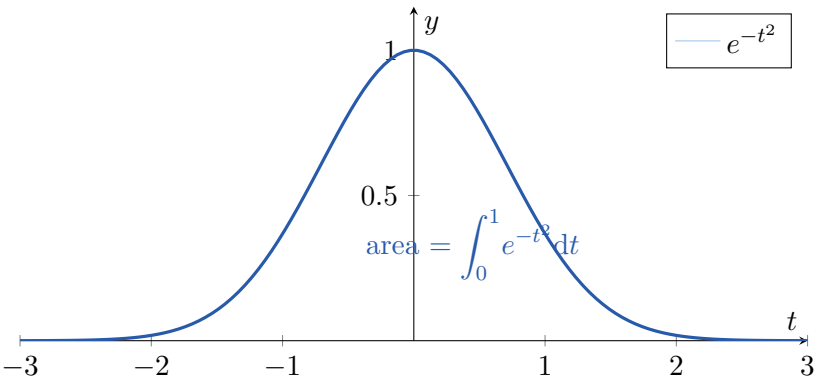


Figure 30.4: The Gaussian bell curve e^{-t^2}. The shaded area from 0 to 1 has no elementary formula, yet the term-by-term integration of Example 30.9 turns it into the convergent sum $1 - \frac{1}{3} + \frac{1}{5\cdot 2!} - \frac{1}{7\cdot 3!} + \cdots$, which any calculator can add up.

Application

The integral above is the backbone of probability and statistics. The area under the bell curve e^{-x^2} (Figure 30.4) between two points is what every "what fraction of outcomes fall within so many standard deviations" question reduces to, and no elementary formula for it exists. Statistical tables and software evaluate it by summing exactly the series of Example 30.9 (or a cleverer variant). The same situation arises in physics – Fresnel integrals for light diffraction, the integral of $\sin(x)/x$ in optics – where the only real way to get numbers is the power series. Series are not a curiosity; for a huge swath of applied mathematics they are the *only* bridge from a non-elementary function to an actual decimal.

Example 30.10 (Multiplication: the first terms of $e^x \sin x$)**.** To find the series of $e^x \sin x$ up to x^3, multiply the two series and collect by total degree:

$$e^x = 1 + x + \tfrac{x^2}{2} + \tfrac{x^3}{6} + \cdots, \qquad \sin x = x - \tfrac{x^3}{6} + \cdots .$$

Collect powers:

$$\begin{aligned} x^1 &: \quad 1\cdot x = x, \\ x^2 &: \quad x\cdot x = x^2, \\ x^3 &: \quad \tfrac{x^2}{2}\cdot x + 1\cdot\left(-\tfrac{x^3}{6}\right) = \tfrac{x^3}{2} - \tfrac{x^3}{6} = \tfrac{x^3}{3}. \end{aligned}$$

So $e^x \sin x = x + x^2 + \frac{1}{3}x^3 + \cdots$. To get the same three terms from the Taylor definition you would differentiate $e^x \sin x$ three times with the product rule – much more painful. Multiplying known series is the smart route.

Example 30.11 (Building a Taylor series at a shifted center)**.** To expand $\ln x$ around $a = 1$ (you cannot center at 0, since $\ln 0$ is undefined), write $x = 1 + (x - 1)$ and reuse the known $\ln(1 + u)$ series with $u = x - 1$:

$$\ln x = \ln\big(1 + (x-1)\big) = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \cdots, \qquad 0 < x \le 2.$$

This is the Taylor series of $\ln x$ centered at 1. The substitution did all the work the derivative computation would have done – and the radius of convergence $R = 1$ tells us the series is valid only out to $x = 0$ and $x = 2$, exactly up to the wall at $\ln 0$.

Common trap

Series manipulation is forgiving, but three mistakes recur:

- **Forgetting the new radius after substitution.** Substituting $u = -x^2$ into a series valid for $|u| < 1$ gives one valid for $|x^2| < 1$, i.e. $|x| < 1$ – you must translate the condition.
- **Mismatched starting indices when adding.** Before combining $\sum$ and $\sum$, line up the powers; a series starting at $n = 0$ and one starting at $n = 1$ cannot be merged coefficient-by-coefficient until you peel off the stray term.
- **Assuming $\ln x$ has a Maclaurin series.** It does not – $\ln 0$ is undefined, so you must shift the center, as in Example 30.11.

30.6 The Catch: When Does the Series Equal the Function?

We have been a little cavalier. The Taylor recipe produces a series from any infinitely differentiable function – but does that series actually *sum to* the function? For e^x, $\sin x$, $\cos x$, and all our examples, yes. But it is not automatic, and we should not paper over the gap.

History

For a century after Newton and Leibniz, mathematicians used power series with breathtaking freedom and almost no rigor. Euler, the supreme calculator, manipulated divergent series, summed them, and routinely got correct and astonishing results – but also occasionally absurd ones. The trouble was that "$f(x)$ equals its Taylor series" was treated as obvious. It took **Augustin-Louis Cauchy** in the 1820s to insist on a precise notion of convergence and to confront the uncomfortable question head-on. Cauchy himself produced the famous warning example below. The full clean-up – a rigorous theory of when a Taylor series converges *to its function*, via the remainder term – is the subject of the next chapter, Taylor's theorem.

Common trap

[Cauchy's cautionary function] Consider

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

This function is infinitely differentiable everywhere, and *every one* of its derivatives at 0 equals zero: $f(0) = f'(0) = f''(0) = \cdots = 0$. So its Maclaurin series is

$$0 + 0 \cdot x + 0 \cdot x^2 + \cdots = 0,$$

the zero series. But $f(x)$ is plainly *not* zero for $x \neq 0$. Here is a perfectly smooth function whose Taylor series converges everywhere – to the wrong answer. The series sees only what happens at the single point $x = 0$, and at that point f is flatter than every polynomial. The function hides all its behavior "infinitely far" from the origin in the language of derivatives.

Intuition

The moral: a Taylor series is built from purely *local* data – the derivatives at one point. Sometimes (most happy times) that local fingerprint determines the function globally, and the series reproduces it perfectly; such functions are called *analytic*. Sometimes, as with Cauchy's example, the fingerprint is misleading. The tool that decides which case you are in is the *remainder*: the difference $f(x) - P_N(x)$ between the function and its Taylor polynomial. The series equals the function exactly when this remainder shrinks to 0 as $N \to \infty$. Proving the remainder vanishes – for e^x, $\sin x$, $\cos x$, and the rest – is the precise, satisfying business of the next chapter.

Why this matters

Step back and admire what we have found. The transcendental functions – exponential growth, oscillation, the logarithm, the inverse trigonometric functions – are not a wild zoo of unrelated objects. Each is, within its interval, a single infinite polynomial with a clean, patterned list of coefficients. Differentiation, integration, multiplication, and substitution move among them as fluidly as among ordinary polynomials. The factorials, the alternating signs, the odd-and-even symmetries – all of it is bookkeeping for one idea: *match a function to a polynomial at a point, to every order at once.* That a smooth function should be "secretly a polynomial of infinite degree" is, after limits themselves, perhaps the most beautiful structural truth in calculus – and the one that makes the subject *computable*.

30.7 A Table to Carry With You

These are the series worth memorizing. Everything else is built from them.

$$\begin{aligned}
\frac{1}{1-x} &= \sum_{n=0}^{\infty} x^n &&= 1 + x + x^2 + x^3 + \cdots && |x| < 1 \\
e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} &&= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots && \text{all } x \\
\sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} &&= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots && \text{all } x \\
\cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} &&= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots && \text{all } x \\
\ln(1+x) &= \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n} &&= x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots && -1 < x \leq 1 \\
\arctan x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} &&= x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots && -1 \leq x \leq 1
\end{aligned}$$

30.8 The Reach of Power Series, the Complex Plane, and the Road Ahead

Connection

Power series and Taylor series represent the ultimate bridge between continuous analysis and discrete algebra, transforming transcendental curves into infinite polynomials that can be manipulated by high-school algebra.

Before power series, evaluating transcendental functions was an agonizing exercise in table lookups and geometric approximations. Taylor series turned every smooth function into an explicit polynomial. This unlocked:

- *Term-by-term calculus of non-elementary functions:* Functions whose antiderivatives cannot be written in closed form (by Liouville's theorem) are effortlessly integrated term by term. The Gaussian error function, central to probability and physics, is integrated in one stroke:
$$\int e^{-x^2}\,\mathrm{d}x = C + \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{n!(2n+1)},$$
yielding an alternating series that converges for *every real number* x.
- *Solving differential equations:* In mathematical physics, the fundamental equations of quantum mechanics, electromagnetism, and gravitation (the Legendre, Bessel, Hermite, and Airy equations) cannot be solved by elementary formulas. Power series expansions (the Frobenius method) provide the exact wavefunctions and fields that govern atoms and planetary orbits.

- *Euler's identity:* Comparing the series for e^x, $\cos x$, and $\sin x$ revealed Euler's formula $e^{ix} = \cos x + i \sin x$, uniting trigonometry and exponentials into a single entity.

Yet why does a power series have a finite radius of convergence R? When a function blows up on the real line—such as $f(x) = \frac{1}{1-x}$ having a vertical asymptote at $x = 1$—it is natural that its Maclaurin series cannot reach past $x = 1$. The radius $R = 1$ is enforced by an obvious physical obstacle.

Now consider the innocent-looking function:

$$f(x) = \frac{1}{1 + x^2}.$$

Its graph is a smooth, bell-shaped curve defined for *all* real numbers $x \in (-\infty, \infty)$, with no vertical asymptotes, corners, or jumps. It is infinitely differentiable everywhere. Yet its Maclaurin series is:

$$\frac{1}{1 + x^2} = 1 - x^2 + x^4 - x^6 + \cdots,$$

and its radius of convergence is strictly $R = 1$. If one plugs in $x = 1.1$, the series explodes into wild divergence $(1 - 1.21 + 1.46 - \cdots)$, even though $f(1.1) = \frac{1}{1+1.21} \approx 0.45$ is perfectly placid. Why does a completely smooth function on the real line halt its Taylor series at $R = 1$? On the real line, this question is an impenetrable mystery. The explanation exists only when you step off the real line into the *complex plane* $\mathbb{C}$. In the complex plane, the denominator $1 + z^2$ factors as $(z - i)(z + i)$, which vanishes at the imaginary points $z = \pm i$. The distance from the origin $z = 0$ to these poles is:

$$|0 - (\pm i)| = \sqrt{0^2 + (\pm 1)^2} = 1.$$

Power series do not live on a 1D line; they live in a two-dimensional disk in the complex plane. A power series' circle of convergence expands outward from the center until it strikes the *nearest singularity in the entire complex plane.* The poles at $\pm i$ are completely invisible to an observer confined to the real line, yet they cast a shadow that halts convergence at $x = \pm 1$! Real analysis cannot explain its own limits without the complex plane.

In Chapter 31 (*Taylor's Theorem and Approximation*), we turn from algebraic manipulation to rigorous quantitative error analysis. By proving Taylor's Theorem with Remainder (the Lagrange and Cauchy error bounds), we establish exactly how close a finite polynomial $P_N(x)$ is to the true function $f(x)$, providing the certified tolerances required for aerospace, scientific computing, and engineering.

30.9 Exercises

Exercise 30.1 (Radius and interval of convergence)**.** For each series, find the radius of convergence, then test the endpoints to determine the full interval of convergence.

$$\text{(a)} \sum_{n=1}^{\infty} \frac{x^n}{n^2}, \qquad \text{(b)} \sum_{n=0}^{\infty} \frac{x^n}{2^n}, \qquad \text{(c)} \sum_{n=1}^{\infty} \frac{(x-3)^n}{n\,5^n}, \qquad \text{(d)} \sum_{n=0}^{\infty} (2n)!\, x^n.$$

Exercise 30.2 (Series by manipulation)**.** Without computing a single derivative, find the Maclaurin

series (and its radius of convergence) for:

$$\text{(a) } \frac{1}{1+3x}, \qquad \text{(b) } x^2e^{-x}, \qquad \text{(c) } \cos(2x), \qquad \text{(d) } \frac{x}{1-x^2}.$$

Exercise 30.3 (Differentiate a known series)**.** Starting from $\dfrac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, differentiate twice to find a closed-form sum for $\sum_{n=2}^{\infty} n(n-1)x^{n-2}$, valid for $|x| < 1$. Then evaluate $\sum_{n=2}^{\infty} n(n-1)(1/2)^{n-2}$ as an exact number.

Exercise 30.4 (Integrate a known series)**.** Use the series for $\dfrac{1}{1+x^2}$ to write $\displaystyle\int_0^{1/2} \frac{\mathrm{d}x}{1+x^2}$ as an infinite series, and use the first three terms to estimate $\arctan(1/2)$. Compare with the true value.

Exercise 30.5 (Computing a transcendental number)**.** Use the Maclaurin series for e^x at $x = 1$ to estimate e using the terms up to $1/5!$. How many decimal places do you trust, and why does the alternating-series-style reasoning *not* directly apply here (the terms do not alternate)? Bound the tail $\sum_{n\geq 6} 1/n!$ by comparison with a geometric series.

Exercise 30.6 (A limit by series)**.** Evaluate $\displaystyle\lim_{x\to 0} \frac{\sin x - x}{x^3}$ by replacing $\sin x$ with the first few terms of its series, simplifying, and taking the limit. Then confirm with L'Hopital's rule, and reflect on which method exposed *why* the answer is what it is.

Exercise 30.7 (Product of series)**.** Find the Maclaurin series of $\dfrac{e^x}{1-x}$ up to and including the x^3 term, by multiplying the series for e^x and $\dfrac{1}{1-x}$. (You should find the coefficient of x^n is $\sum_{k=0}^{n} 1/k!$.)

Exercise 30.8 (Shifted center)**.** Find the Taylor series of $f(x) = 1/x$ centered at $a = 2$. (Hint: write $\dfrac{1}{x} = \dfrac{1}{2+(x-2)} = \dfrac{1}{2}\cdot\dfrac{1}{1+(x-2)/2}$ and use the geometric series.) State the radius of convergence and explain geometrically why it is what it is.

Exercise 30.9 (Conceptual: uniqueness)**.** Explain in your own words why a function can have only one power series at a given center. If $\sum a_n x^n = \sum b_n x^n$ for all x in some interval around 0, prove that $a_n = b_n$ for every n. (Hint: differentiate n times and set $x = 0$.)

Exercise 30.10 (Proof: term-by-term and the derivative)**.** Let $f(x) = \sum_{n=0}^{\infty} c_n x^n$ have radius $R > 0$, and let $g(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}$ be the formally differentiated series. Prove that g has the same radius of convergence R as f. (You do not need to prove $g = f'$; just the radius. Use the ratio-test characterization and the fact that $\sqrt[n]{n} \to 1$ or that $n/(n+1) \to 1$.)

Exercise 30.11 (Cauchy's function, conceptual)**.** For $f(x) = e^{-1/x^2}$ (with $f(0) = 0$), explain why every derivative at 0 is zero, at least intuitively, by arguing that e^{-1/x^2} goes to 0 as $x \to 0$ faster than any power of x. What does this example tell you about the difference between "the Taylor series converges" and "the Taylor series converges *to the function*"?

Exercise 30.12 (Euler's formula by series)**.** Substitute $x = i\theta$ into the series for e^x, group the real and imaginary terms separately, and recognize the series for $\cos\theta$ and $\sin\theta$. Conclude $e^{i\theta} = \cos\theta + i\sin\theta$, and deduce the value of $e^{i\pi}$.

Chapter 31

Taylor's Theorem and Approximation

One of the most striking facts in calculus is the subject of this chapter. If you know everything about a smooth function at a *single point* — its value, its slope, its curvature, the rate at which its curvature changes, and so on through all its derivatives — then in a great many cases you know the function *everywhere*. One point, infinitely interrogated, reconstructs the entire curve. At first encounter, this sounds almost impossible: how could the local behavior at a single point dictate the values of the function miles away?

We have already taken the first steps along this road. The linear approximation $f(x) \approx f(a) + f'(a)(x-a)$ used the value and the slope at a to build the best straight line through the point. It was faithful for x close to a, but drifted away as the curve bent. The natural question to ask is: *why stop at a line?* If matching one derivative gives the tangent line, what does matching a second derivative give? A parabola, certainly — one that not only passes through $(a, f(a))$ with the correct slope but also bends with the exact same curvature. A third derivative buys a cubic, matching the rate at which curvature changes. Each new derivative lets the polynomial hug the curve across a wider window before peeling away.

That approximating polynomial is the *Taylor polynomial*. The idea that these polynomials might close in on the function everywhere, leaving no gap at all in the limit, is the *Taylor series*. And the precise accounting of the remaining discrepancy at any finite step — the formula that transforms qualitative hope into a rigorous engineering tool — is *Taylor's theorem with remainder*. In this chapter we examine how to build these approximations, why the factorial denominators arise inevitably from differentiation, how to bound the remainder with certainty, and why the dream of infinite reconstruction succeeds for functions like e^x and $\cos x$, yet encounters fascinating barriers for others.

31.1 From a line to a polynomial: matching derivatives

Start with the linear approximation and stare at what made it good. Near a point a, the line

$$L(x) = f(a) + f'(a)(x-a)$$

agrees with f in two respects: $L(a) = f(a)$ (same height) and $L'(a) = f'(a)$ (same slope). Those two matched conditions are the whole reason the line clings to the curve near a. It is the unique line that gets both right.

Intuition

Approximation by matching is a recurring trick in all of mathematics: to mimic a complicated object near a point, build a simple object that agrees with it in as many *derivatives* as you can afford. A line matches two pieces of data (value and slope). A parabola can match three (value, slope, and curvature). A degree-n polynomial has $n+1$ free coefficients, so it can match $n+1$ pieces of data: the value and the first n derivatives. The more derivatives you pin down at a, the longer the impostor passes for the real thing as you move away from a.

So let us be greedy. We want a polynomial P_n of degree n such that

$$P_n(a) = f(a), \quad P_n'(a) = f'(a), \quad P_n''(a) = f''(a), \quad \ldots, \quad P_n^{(n)}(a) = f^{(n)}(a).$$

Match the function and its first n derivatives at the single point a. The question is what coefficients make this happen.

Here is the clean way to find them. Write the polynomial not in powers of x but in powers of $(x-a)$, because we only care about behavior near a:

$$P_n(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots + c_n(x-a)^n.$$

Now extract the coefficients one at a time by differentiating and then plugging in $x = a$, which kills every term that still has a factor of $(x-a)$ in it. Watch the pattern.

Plug in $x = a$ directly: every term but the first dies, leaving $P_n(a) = c_0$. So $c_0 = f(a)$.

Differentiate once: $P_n'(x) = c_1 + 2c_2(x-a) + 3c_3(x-a)^2 + \cdots$. Plug in $x = a$: everything but the first term dies, $P_n'(a) = c_1$. So $c_1 = f'(a)$.

Differentiate again: $P_n''(x) = 2c_2 + 3\cdot 2\, c_3(x-a) + \cdots$. Plug in $x = a$: $P_n''(a) = 2c_2$. So $c_2 = f''(a)/2$.

Once more: $P_n'''(x) = 3\cdot 2\, c_3 + 4\cdot 3\cdot 2\, c_4(x-a) + \cdots$, so $P_n'''(a) = 6c_3$, giving $c_3 = f'''(a)/6$.

The numbers $1, 2, 6, 24, \ldots$ marching down the denominators are the factorials, because differentiating $(x-a)^k$ exactly k times brings down $k\cdot(k-1)\cdots 2\cdot 1 = k!$ and turns the term into a constant, while all lower terms have already vanished and all higher terms still carry a factor of $(x-a)$. So in general

$$P_n^{(k)}(a) = k!\, c_k \quad \Longrightarrow \quad c_k = \frac{f^{(k)}(a)}{k!}.$$

Focus

The Taylor polynomial. The unique degree-n polynomial matching f and its first n derivatives at a is

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

The factorials are exactly what is needed to undo the factorials that differentiation k times produces. When $a = 0$ this is sometimes called the *Maclaurin* polynomial, but it is the same idea centered at the origin.

Why the technique works

Why factorials, really? Differentiation and the monomial $(x-a)^k$ are made for each other. Each derivative peels one factor off the front: $(x-a)^k \to k(x-a)^{k-1} \to k(k-1)(x-a)^{k-2} \to \cdots$. After exactly k steps the variable is gone and a bare $k!$ stands where the term used to be. Dividing the coefficient by $k!$ in advance is the antidote: it guarantees that when you differentiate k times and set $x = a$, the k-th term contributes precisely $f^{(k)}(a)$ and nothing else contributes at all. The factorial is not decoration; it is the gear ratio that makes the k-th coefficient control the k-th derivative cleanly and independently of all the others.

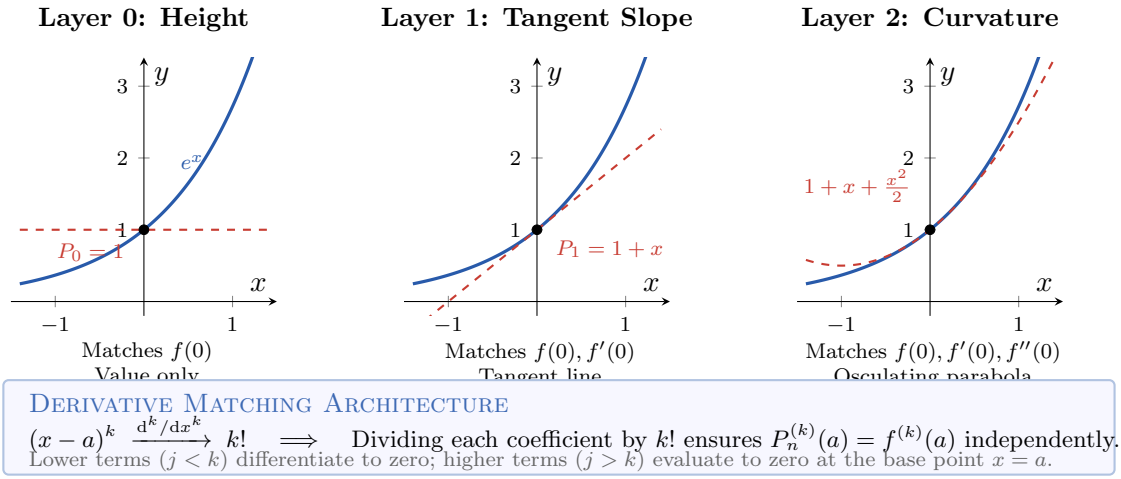


Figure 31.1: Building Taylor polynomials layer by layer at $a = 0$ for $f(x) = e^x$. Layer 0 pins the height, Layer 1 adds the tangent slope, and Layer 2 bends the polynomial to match the exact curvature. The lower lane illustrates the cascade of differentiation: each term $(x-a)^k/k!$ controls its own derivative at a without disturbing any of the previously calibrated layers.

31.1.1 A first picture: parabola beats line

Take $f(x) = \cos x$ at $a = 0$. We know $f(0) = 1$, $f'(0) = -\sin 0 = 0$, $f''(0) = -\cos 0 = -1$. So

$$P_1(x) = 1, \qquad P_2(x) = 1 - \tfrac{1}{2}x^2.$$

The linear approximation of cosine at the top of its hump is just the flat line $y = 1$ — correct in value and slope but blind to the fact that cosine curves *downward*. The quadratic $1 - \frac{1}{2}x^2$ knows about that downward bend, and you can see it follow the cosine far better.

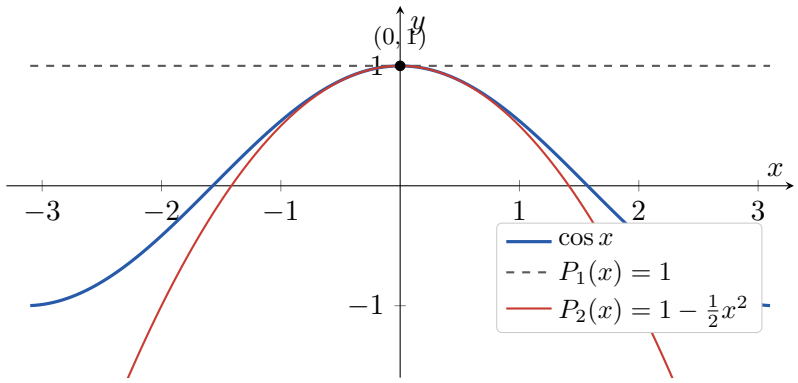


Figure 31.2: At $a = 0$, the linear approximation to $\cos x$ is the horizontal tangent line $y = 1$, which matches height and slope but ignores curvature. The quadratic $P_2(x) = 1 - \frac{1}{2}x^2$ incorporates the downward bending $f''(0) = -1$ and embraces the cosine wave over a substantial interval.

Each extra term tightens the grip. The degree-4 Taylor polynomial of cosine, $1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$, tracks the true cosine accurately well past $x = 2$. We will quantify "accurately" soon; for now, notice the qualitative law: *higher degree buys a wider window of trust.*

31.2 The remainder: naming the gap exactly

A picture that "hugs" a curve is suggestive but not yet mathematics. The exact object is the *remainder*, the difference between the function and its polynomial:

$$R_n(x) = f(x) - P_n(x).$$

This is not an approximation; it is an equation. It says $f(x) = P_n(x) + R_n(x)$ *exactly*, for every x. All the error, all the discrepancy, has been gathered up and given a name. The entire game of approximation is now the game of controlling R_n: if we can show $R_n(x)$ is small, then $P_n(x)$ is a good approximation, with a guarantee, not a hope.

What is really happening

When you write $\cos x \approx 1 - \frac{1}{2}x^2$, the "$\approx$" is hiding a precise statement. What is really there is an equation with a correction term:

$$\cos x = 1 - \tfrac{1}{2}x^2 + R_2(x),$$

and Taylor's theorem is going to hand us a *formula* for $R_2(x)$ — not its exact value (that would be asking too much), but a clean expression that lets us *bound* it. The difference between an engineer who trusts a small-angle approximation and a gambler who hopes it works is exactly this remainder term. The remainder is where approximation becomes a science.

So what should we expect of R_n? Two things, immediately. First, since P_n matches f and its first n derivatives at a, the remainder and *its* first n derivatives must all vanish at a:

$$R_n(a) = R_n'(a) = R_n''(a) = \cdots = R_n^{(n)}(a) = 0.$$

A function whose first n derivatives die at a point is extremely flat there — flatter than $(x-a)^n$. So $R_n(x)$ ought to behave like $(x-a)^{n+1}$ near a: small, and getting small *fast* as $x \to a$. That is the heart of why more terms help. Let us turn this intuition into a theorem.

31.3 Taylor's theorem with the Lagrange remainder

Here is the destination, stated cleanly, before we earn it.

Focus

Taylor's Theorem (Lagrange form of the remainder). Suppose f is $n+1$ times differentiable on an open interval containing a and x. Then there is a point c strictly between a and x such that

$$f(x) = \underbrace{\sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k}_{P_n(x)} + \underbrace{\frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}}_{R_n(x)}.$$

The remainder $R_n(x)$ looks exactly like the *next* Taylor term — the degree-$(n+1)$ term you would have written next — except that the $(n+1)$-st derivative is evaluated not at a but at some mystery point c in between.

Look at how beautifully this generalizes the Mean Value Theorem. Set $n = 0$. The theorem says $f(x) = f(a) + f'(c)(x-a)$ for some c between a and x — which, rearranged, is $\frac{f(x)-f(a)}{x-a} = f'(c)$, precisely the MVT. So Taylor's theorem is the MVT with more derivatives. The single mysterious point c where "the average is achieved" in the MVT reappears here as the point where the $(n+1)$-st derivative takes the value that makes the equation exactly balance.

Intuition

The Lagrange remainder is the optimist's dream made rigorous. You were *hoping* the error would behave like the next term $\frac{f^{(n+1)}(a)}{(n+1)!}(x-a)^{n+1}$. Taylor's theorem says: it does behave exactly like that next term — you just have to evaluate the derivative at a slightly shifted point c instead of at a. And since you usually do not need c exactly, only a *bound* on $f^{(n+1)}$ over the interval, this shifted point costs you almost nothing in practice. You trade "the exact error" (impossible to know) for "the error has this exact *form*" (immediately useful).

History

Brook Taylor published the polynomial expansion in 1715, though Newton, Gregory, and the Indian Kerala school centuries earlier already knew the series for sin, cos, and arctan. What none of them had was the *remainder* — the precise accounting of error. The early calculus treated these series as exact infinite sums and computed with them by bold instinct, usually correctly. It was Joseph-Louis Lagrange in 1797 who isolated the finite remainder term that bears his name, and Cauchy and Weierstrass in the next century who made the whole apparatus rigorous by grounding it in the limit and the Mean Value Theorem. The picture had been right for a hundred years; the guarantee came later.

31.3.1 Why it is true: the idea before the proof

There are several routes to this theorem. The cleanest conceptually is a souped-up version of Rolle's theorem, applied to an auxiliary function engineered so that its vanishing *is* the statement we want. Before the formalism, here is the idea in one breath.

We want to compare $f(x)$ with $P_n(x)$. Fix the target point $x = b$ and the center a. Build a helper function $g(t)$ of a new variable t that measures, at each t, how far f is from *its own Taylor polynomial centered at t* (evaluated at b), and rig it so that g equals 0 at the end $t = b$. Then a single mean-value step on g — this is the magic — causes almost everything to telescope away, leaving precisely the remainder formula. The telescoping is not luck; the helper is designed to telescope.

Why the technique works

Why a helper function? Because the quantity we care about, $f(b) - P_n(b)$, is a single number with no t in it — there is nothing to apply a mean-value argument to. The trick, which runs through all of analysis, is to *introduce a variable*: turn the static number into the endpoint value of a moving quantity, so that the machinery of derivatives (Rolle, MVT) has something to bite on. We promote the fixed center a to a sliding center t, watch the discrepancy change as t slides from a to b, and let the Mean Value Theorem catch the instant where the change conspires to reveal the remainder. Designing the right helper is the whole art; verifying it is then mechanical.

31.3.2 The proof

Theorem 31.1 (Taylor with Lagrange remainder)**.** *Let f be a function such that $f, f', \ldots, f^{(n)}$ are continuous on the closed interval between a and b, and $f^{(n+1)}$ exists on the open interval between them. Then there is a point c strictly between a and b with*

$$f(b) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(b-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(b-a)^{n+1}.$$

Proof. Treat a, b, n as fixed. Define a function of a sliding center t, living on the interval between a and b:

$$g(t) = f(b) - \sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!}\,(b-t)^k.$$

Read $g(t)$ out loud: it is the gap between the true value $f(b)$ and the Taylor polynomial of f *centered at t*, evaluated at b. Two endpoint values are immediate.

At $t = b$: every factor $(b-t)^k$ with $k \geq 1$ vanishes, and the $k = 0$ term is $f(b)$, so

$$g(b) = f(b) - f(b) = 0.$$

At $t = a$: the sum is exactly the Taylor polynomial $P_n(b)$, so

$$g(a) = f(b) - P_n(b) = R_n(b),$$

which is the very remainder we are chasing. Good: g carries the answer at one end and is zero at the other.

Now differentiate g with respect to t. This is the step where the design pays off. Each term of the sum is a product $\frac{f^{(k)}(t)}{k!}(b-t)^k$, so we use the product rule:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{f^{(k)}(t)}{k!}(b-t)^k\right] = \frac{f^{(k+1)}(t)}{k!}(b-t)^k \;-\; \frac{f^{(k)}(t)}{k!}\,k\,(b-t)^{k-1}.$$

The first piece is the derivative landing on $f^{(k)}$; the second is the derivative landing on $(b-t)^k$, which brings down k and lowers the power (the minus sign comes from the chain rule on $b-t$). Now sum over $k=0$ to n and watch the second pieces cancel the first pieces of the neighboring term. Writing the first-type terms as $A_k = \frac{f^{(k+1)}(t)}{k!}(b-t)^k$ and the second-type terms as $B_k = \frac{f^{(k)}(t)}{(k-1)!}(b-t)^{k-1}$ (note $\frac{k}{k!} = \frac{1}{(k-1)!}$), we have $B_k = A_{k-1}$. So

$$g'(t) = -\sum_{k=0}^{n}\big(A_k - B_k\big) = -\sum_{k=0}^{n} A_k + \sum_{k=1}^{n} A_{k-1} = -A_n,$$

because the B-sum telescopes against the A-sum, leaving only the very last first-type term (the B_0 term is zero since it has a factor $k=0$). Therefore

$$g'(t) = -\frac{f^{(n+1)}(t)}{n!}\,(b-t)^n.$$

Everything collapsed to a single clean term: this is exactly the telescoping the helper was engineered to produce.

Now we apply a mean-value argument — but cleverly, using a second helper so that we get the strong $(n+1)!$ form rather than a weaker estimate. Introduce

$$h(t) = (b-t)^{n+1}.$$

Both g and h are continuous on the closed interval and differentiable on the open one, and $h'(t) = -(n+1)(b-t)^n$ never vanishes inside. The Cauchy (generalized) Mean Value Theorem then guarantees a point c strictly between a and b with

$$\frac{g(a)-g(b)}{h(a)-h(b)} = \frac{g'(c)}{h'(c)}.$$

We computed $g(b)=0$, $g(a)=R_n(b)$, $h(b)=0$, and $h(a)=(b-a)^{n+1}$. The right side is

$$\frac{g'(c)}{h'(c)} = \frac{-\frac{f^{(n+1)}(c)}{n!}(b-c)^n}{-(n+1)(b-c)^n} = \frac{f^{(n+1)}(c)}{(n+1)!}.$$

The $(b-c)^n$ factors cancel — another piece of the design — and the $n!$ becomes $(n+1)!$. Setting the two sides equal,

$$\frac{R_n(b)}{(b-a)^{n+1}} = \frac{f^{(n+1)}(c)}{(n+1)!},$$

which rearranges to exactly

$$R_n(b) = \frac{f^{(n+1)}(c)}{(n+1)!}(b-a)^{n+1}.$$

That is the claim. □

Remark 31.1. The case $n = 0$ recovers the Mean Value Theorem itself, and the proof above, stripped to $n = 0$, *is* the standard proof of the MVT via the Cauchy form. So Taylor's theorem is not a different animal from the MVT; it is the same animal, fed more derivatives. Everything in the chapter on the Mean Value Theorem was secretly preparation for this.

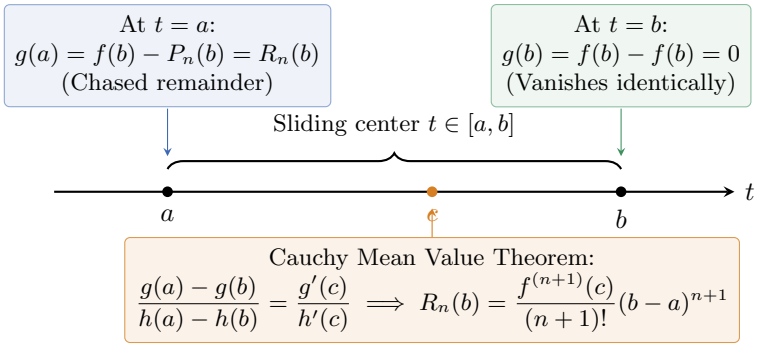


Figure 31.3: The sliding-center strategy in the proof of Taylor's theorem. The helper function $g(t)$ measures the discrepancy at b when the expansion is centered at t. Moving t from a to b connects the unknown remainder $R_n(b)$ at one boundary to zero at the other. The Cauchy Mean Value Theorem catches the intermediate point c where telescoping yields the exact Lagrange remainder.

31.4 Using the remainder to bound error

The point c is unknown, and that is fine, because we almost never want c itself. We want a guarantee. And a guarantee follows the instant we can say how big $f^{(n+1)}$ can possibly be on the interval. This is the workhorse corollary — the single most useful fact in the whole chapter.

Focus

The error bound. If $|f^{(n+1)}(t)| \le M$ for all t between a and x, then

$$|R_n(x)| \;\le\; \frac{M}{(n+1)!}\,|x-a|^{n+1}.$$

You do not need to find c. You only need a ceiling M on the next derivative.

This is the form you actually use. To approximate $f(x)$ by $P_n(x)$ and *know* how wrong you might be, you ask one question: how large can the $(n+1)$-st derivative get on the way from a to x? Whatever that ceiling M is, the error cannot exceed $\frac{M}{(n+1)!}|x-a|^{n+1}$. Two forces fight in that bound: $|x-a|^{n+1}$ grows if you reach far from a, but $(n+1)!$ in the denominator grows ferociously with n. For functions whose derivatives stay bounded (like sin, cos, e^x on any fixed interval), the factorial wins decisively, and the error collapses to zero as you take more terms.

Work it yourself

Approximate $\cos(0.5)$ using $P_2(x) = 1 - \frac{1}{2}x^2$ and bound the error *before* comparing to a calculator.
The estimate: $P_2(0.5) = 1 - \frac{1}{2}(0.25) = 1 - 0.125 = 0.875$.

The bound: here $n = 2$, so we need a ceiling on $|f^{(3)}|$. For $f = \cos$, $f'''(x) = \sin x$, and $|\sin t| \le 1$ everywhere, so $M = 1$. Then

$$|R_2(0.5)| \le \frac{1}{3!}(0.5)^3 = \frac{0.125}{6} \approx 0.0208.$$

So we may assert, with certainty, that the true $\cos(0.5)$ lies in $[0.875 - 0.021,\ 0.875 + 0.021] = [0.854,\ 0.896]$.
The check: a calculator gives $\cos(0.5) = 0.87758\ldots$, and indeed $|0.87758 - 0.875| = 0.0026$, comfortably under our promised ceiling of 0.0208. The bound held and, as usual, came out a bit pessimistic — which is exactly what you want from a guarantee.

Common trap

A bound is an upper limit, not a prediction. Beginners sometimes report "the error is 0.0208" when the theorem only promises "the error is *at most* 0.0208." The actual error here was 0.0026, almost ten times smaller. The Lagrange bound is deliberately conservative: it assumes the next derivative sits at its worst-case value M over the whole interval, which it usually does not. Never quote the bound as the error; quote it as the worst case.

Work it yourself

A harder, more satisfying one: how many terms of the Maclaurin series for e^x do we need to compute $e = e^1$ to within 10^{-6}?
For $f(x) = e^x$ centered at $a = 0$, every derivative is e^x again, and on $[0, 1]$ we have $e^t \le e < 3$, so we may take $M = 3$. With $x = 1$,

$$|R_n(1)| \le \frac{3}{(n+1)!} \cdot 1^{n+1} = \frac{3}{(n+1)!}.$$

We need $\frac{3}{(n+1)!} \le 10^{-6}$, i.e. $(n+1)! \ge 3 \times 10^6$. Now just march up the factorials: $9! = 362880$ (too small), $10! = 3628800 \ge 3 \times 10^6$. So $(n+1) = 10$, meaning $n = 9$: the polynomial through the $x^9/9!$ term suffices. Computing $\sum_{k=0}^{9} \frac{1}{k!}$ gives $2.7182815\ldots$ against the true $e = 2.7182818\ldots$, an error of about 3×10^{-7} — under our budget, as promised. The factorial in the denominator is doing extraordinary work: ten terms and we already have six digits.

Application

This is precisely how a calculator or a math library computes e^x, $\sin x$, and the rest. It does not have a transcendental oracle inside; it adds up a few Taylor terms (after first reducing the argument to a small range where convergence is fast, e.g. using $\sin(x + 2\pi) = \sin x$ and angle-halving), and the Lagrange remainder is what certifies that the result is correct to the last displayed digit. Every time you press the cosine button, a remainder estimate guaranteed the answer before it ever reached the screen.

31.5 When does the series converge to the function?

Now the bigger dream. Suppose f has *all* derivatives. We can keep building Taylor polynomials of every degree, and we may ask whether the infinite series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k$$

adds up to $f(x)$. The series equals $f(x)$ at a given point x exactly when the remainder vanishes in the limit:

$$\boxed{\lim_{n\to\infty} R_n(x) = 0}$$

This is the whole truth of convergence, and the statement repays a careful reading. The Taylor series converges *to* f at x if and only if the leftover $R_n(x)$ shrinks to zero as you include more terms. Not "if the series converges" — the series can converge to the *wrong* thing, as we will see — but if and only if the gap between f and the polynomial closes.

Why the technique works

Why is $R_n \to 0$ the exact condition, and not something about the series "converging"? Because $f(x) = P_n(x) + R_n(x)$ is an identity for every n. Rearranged, $R_n(x) = f(x) - P_n(x)$. As $n \to \infty$, $P_n(x)$ is the partial sum of the Taylor series. So $P_n(x) \to f(x)$ — the series summing to the function — is literally the same statement as $R_n(x) \to 0$. There is no daylight between them. This is why all the labor went into the remainder: it is the only thing standing between the polynomial and the function.

For the friendly functions, the Lagrange bound closes the case in one line. Take $\sin x$ centered at 0. Every derivative is one of $\pm\sin x, \pm\cos x$, so $|f^{(n+1)}(t)| \le 1$ for all t, giving

$$|R_n(x)| \le \frac{1}{(n+1)!}\,|x|^{n+1}.$$

For any fixed x, no matter how large, $\frac{|x|^{n+1}}{(n+1)!} \to 0$ as $n \to \infty$ — the factorial outruns any fixed power. Therefore the Maclaurin series for $\sin x$ converges to $\sin x$ for *every* real x. The same argument, with $M = e^{|x|}$, works for e^x. These functions are reconstructed perfectly, everywhere, from their single point of data at the origin.

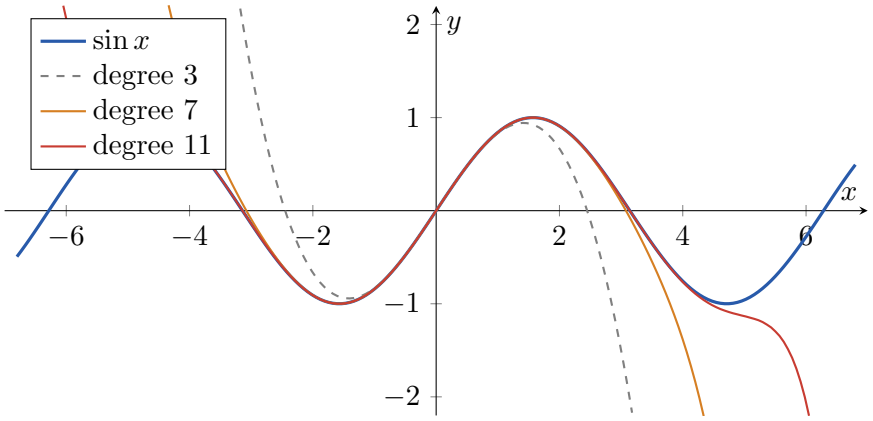


Figure 31.4: Taylor polynomials of $\sin x$ at 0 closing in on the curve. Each higher degree extends the interval over which the polynomial tracks $\sin x$ before it finally shoots off to infinity. In the limit, the series captures $\sin x$ on the entire real line.

31.5.1 Where convergence stops: the geometric warning

Not every function is captured on the whole line. Consider the homely

$$f(x) = \frac{1}{1+x^2}.$$

This function is perfectly smooth everywhere on $\mathbb{R}$ — no breaks, no corners, no vertical asymptotes, infinitely differentiable. Its Maclaurin series, obtained from the geometric series $\frac{1}{1+u} = 1-u+u^2-\cdots$ with $u = x^2$, is

$$1 - x^2 + x^4 - x^6 + \cdots,$$

which converges only for $|x| < 1$ and diverges for $|x| > 1$. Why 1? There is nothing visibly wrong with the function at $x = 1$. The graph sails smoothly through $(1, \frac{1}{2})$ without a hint of trouble.

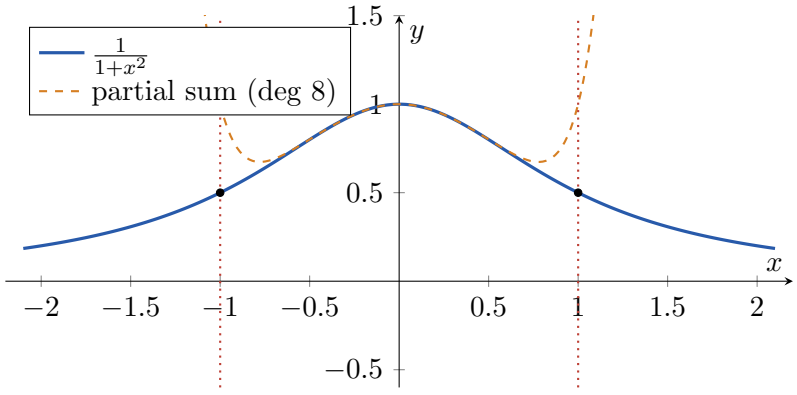


Figure 31.5: The function $\frac{1}{1+x^2}$ is smooth everywhere, yet its Maclaurin series $1-x^2+x^4-\cdots$ tracks it only on $(-1, 1)$ (dotted lines). Inside that interval the partial sums close in; just outside, they fly apart, even though the function itself is perfectly well behaved there. The wall at $|x| = 1$ is set by a singularity hiding off the real axis.

History

The mystery dissolves the moment you allow complex numbers, and this was one of the great clarifying insights of nineteenth-century analysis (Cauchy, then Weierstrass and Riemann). The function $\frac{1}{1+x^2}$, viewed over the complex plane, *blows up* at $x = \pm i$, where $1 + x^2 = 0$. Those poles sit at distance 1 from the origin. A power series centered at 0 converges inside the largest disk around 0 that reaches no singularity — and the nearest singularities, invisible on the real line, are exactly one unit away in the imaginary direction. The radius of convergence 1 was being dictated by points you could not see. The early calculus of Newton and Euler used such series freely and often correctly by instinct; it took the complex viewpoint to explain *why* the instinct sometimes failed.

Connection

This is the bridge to complex analysis. A real power series centered at a converges on an interval $(a - R, a + R)$ (the dotted wall in Figure 31.5); the number R is the *radius of convergence*, and it equals the distance from a to the nearest point — real *or complex* — where the function misbehaves. The interval on the real line is just the diameter of a hidden disk in the complex plane. Once you know this, "why does this series stop converging at $x = 1$ when the function looks fine?" is never mysterious again: look for the nearest singularity off the real axis.

31.5.2 The shocking counterexample: smooth but not analytic

There is a worse failure, and it is worth meeting because it draws the sharp line between "has a Taylor series" and "equals its Taylor series." Consider

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

This function is infinitely differentiable everywhere, including at 0. But it is so extraordinarily flat at the origin that one can show

$$f(0) = f'(0) = f''(0) = \cdots = f^{(n)}(0) = \cdots = 0$$

— *every* derivative at 0 is zero. The exponential e^{-1/x^2} decays to zero faster than any power of x can detect, so the difference quotients defining each derivative all collapse to 0.

Now build the Maclaurin series. Every coefficient $\frac{f^{(k)}(0)}{k!}$ is 0, so the Taylor series is

$$0 + 0 \cdot x + 0 \cdot x^2 + \cdots = 0.$$

The series converges beautifully — to the zero function. But $f(x)$ is *not* zero for $x \neq 0$. The Taylor series converges everywhere and yet equals f only at the single point $x = 0$. Here $R_n(x) = f(x) - 0 = f(x)$ does *not* go to zero; it just sits there equal to the whole function. The remainder never shrinks because there is nothing in the polynomial to shrink toward f.

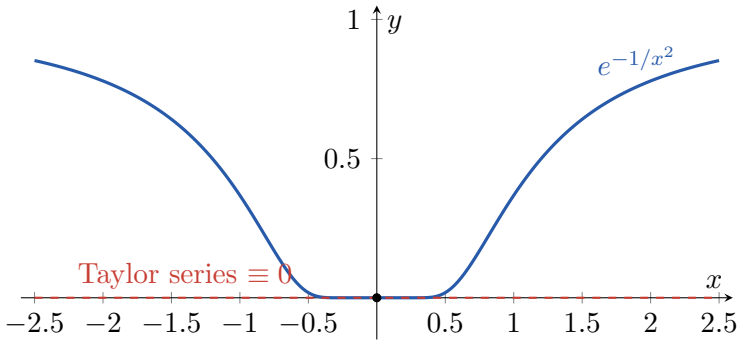


Figure 31.6: The function e^{-1/x^2} (extended by 0 at the origin) is infinitely flat at $x = 0$: every derivative there is zero. Its Taylor series is identically zero (red), and matches the function only at the single point $x = 0$. Smoothness does not guarantee that a function equals its Taylor series.

Focus

Analytic functions. A function is *analytic* at a if its Taylor series centered at a converges to the function on some interval around a — that is, if $R_n(x) \to 0$ there. Most familiar functions (e^x, sin, cos, polynomials, ln, $1/(1-x)$ away from their singularities) are analytic. The example e^{-1/x^2} shows that *infinitely differentiable is strictly weaker than analytic*: being smooth is necessary but not sufficient. Equality with the Taylor series is an extra, special grace, and it is exactly the grace that $R_n \to 0$ certifies.

Why this matters

There is something humbling and beautiful here. We began the chapter dreaming that one point, fully known, determines the whole function. For analytic functions the dream is literally true: e^x is the unique analytic function with $f(0) = 1$ and $f' = f$, and its values everywhere are encoded in its behavior at the origin. But the dream has an edge. The function e^{-1/x^2} shows that a smooth function can keep secrets — information that simply is not present in any derivative at 0. Analyticity is the property of having *no secrets*: every value is foretold by the local data. That two such different worlds live side by side among the smooth functions is one of the quietly profound facts of analysis.

31.6 Physics lives in the first two terms

A great deal of physics is secretly the statement "keep the first Taylor term that does not vanish, throw away the rest, and the world becomes solvable." Approximation is not a compromise physicists tolerate; it is often where the insight is.

31.6.1 The small-angle approximation

The pendulum equation, the bending of light, the behavior of a vibrating string — all of them simplify enormously the moment you replace $\sin\theta$ by its first Taylor term. From the series

$$\sin\theta = \theta - \frac{\theta^3}{6} + \frac{\theta^5}{120} - \cdots,$$

for small θ we keep just the leading term:

$$\sin\theta \approx \theta \qquad (\theta \text{ in radians}).$$

The remainder bound tells you exactly when this is safe. The first dropped term is $\theta^3/6$, and (using $|f^{(3)}| \le 1$) the Lagrange remainder for the degree-1 approximation is at most $\frac{|\theta|^3}{6}$. At $\theta = 0.1$ rad (about 5.7°), the error is below $\frac{(0.1)^3}{6} \approx 1.7 \times 10^{-4}$ — a relative error well under two tenths of a percent. This single approximation is why the pendulum has a period $T = 2\pi\sqrt{L/g}$ *independent of amplitude*: replacing $\sin\theta$ by θ turns the true (hard) pendulum equation into the simple harmonic oscillator, whose period does not depend on how far it swings. The famous result is an artifact of truncating a Taylor series at the first term.

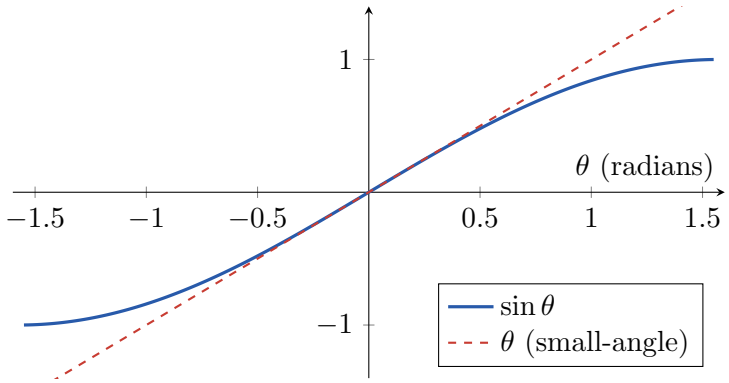


Figure 31.7: The small-angle approximation $\sin\theta \approx \theta$. Near the origin the line $y = \theta$ and the curve $y = \sin\theta$ are nearly indistinguishable; the gap grows like $\theta^3/6$, so it stays tiny while θ is small.

Application

The same "first nonvanishing term" logic runs through physics:

- **Relativistic energy.** The exact kinetic energy is $E = mc^2\big(\frac{1}{\sqrt{1-v^2/c^2}} - 1\big)$. Expanding the square root for small v/c gives $E \approx \frac{1}{2}mv^2 + \frac{3}{8}\frac{mv^4}{c^2} + \cdots$. The first term is Newton's $\frac{1}{2}mv^2$ — so classical kinetic energy is just the leading Taylor term of the relativistic one. The next term is the first relativistic correction.
- **Potential wells.** Near a stable equilibrium x_0, any smooth potential $V(x)$ has $V'(x_0) = 0$, so its Taylor expansion begins $V(x) \approx V(x_0) + \frac{1}{2}V''(x_0)(x - x_0)^2$. The quadratic term is an osculating parabola: *every* smooth potential well behaves locally like a harmonic oscillator.

In each case the Taylor remainder is the auditor that tells you when the approximation may be trusted and when the next term must be restored.

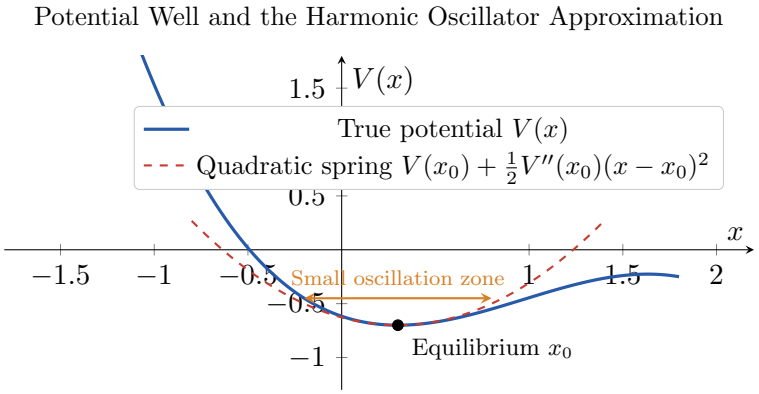


Figure 31.8: Why springs rule classical physics. Near any stable equilibrium x_0, the potential $V(x)$ has $V'(x_0) = 0$. The leading non-constant Taylor term is quadratic: $V(x) \approx V(x_0) + \frac{1}{2}V''(x_0)(x - x_0)^2$. To a particle confined to small oscillations around the trough, the true potential is indistinguishable from a parabola — an effective Hooke's-law spring with constant $k = V''(x_0)$.

Walking question

Walk and turn this over: why is it the *second* derivative that governs the behavior of a system near equilibrium, never the first? (Hint: equilibrium means the first derivative of the potential is zero, so the first surviving Taylor term is the quadratic one — and a quadratic potential is exactly a spring. Stability is $V'' > 0$; instability is $V'' < 0$.) What would a system look like if V'' also vanished and the leading term were quartic?

31.7 Worked examples, start to finish

Example 31.1 (Estimating $\sqrt{e}$ with a guarantee)**.** Approximate $\sqrt{e} = e^{0.5}$ using the degree-3 Maclaurin polynomial of e^x and bound the error.

The polynomial is $P_3(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$. At $x = 0.5$,

$$P_3(0.5) = 1 + 0.5 + \frac{0.25}{2} + \frac{0.125}{6} = 1 + 0.5 + 0.125 + 0.020\overline{83} = 1.645\overline{83}.$$

For the error, $f^{(4)}(x) = e^x$ and on $[0, 0.5]$ we have $e^t \le e^{0.5} < 1.65$, so take $M = 1.65$. Then

$$|R_3(0.5)| \le \frac{1.65}{4!}(0.5)^4 = \frac{1.65}{24} \cdot 0.0625 \approx 0.0043.$$

So $\sqrt{e} \in [1.6458 - 0.0043,\ 1.6458 + 0.0043] = [1.6415,\ 1.6501]$. The true value is $\sqrt{e} = 1.64872\ldots$, inside the interval, with actual error ≈ 0.0029 — under the promised 0.0043.

Example 31.2 (A Taylor expansion not centered at 0)**.** Approximate $\ln(1.2)$ by expanding $\ln x$ about $a = 1$.

The derivatives of $f(x) = \ln x$ are $f'(x) = x^{-1}$, $f''(x) = -x^{-2}$, $f'''(x) = 2x^{-3}$, so at $a = 1$: $f(1) = 0$, $f'(1) = 1$, $f''(1) = -1$, $f'''(1) = 2$. The degree-2 polynomial is

$$P_2(x) = 0 + 1 \cdot (x - 1) + \frac{-1}{2}(x - 1)^2 = (x - 1) - \tfrac{1}{2}(x - 1)^2.$$

At $x = 1.2$, with $(x-1) = 0.2$:

$$P_2(1.2) = 0.2 - \tfrac{1}{2}(0.04) = 0.2 - 0.02 = 0.18.$$

For the error, $f'''(x) = 2x^{-3}$ and on $[1, 1.2]$ this is largest at $x = 1$, where it equals 2, so $M = 2$. Then

$$|R_2(1.2)| \le \frac{2}{3!}(0.2)^3 = \frac{2}{6}(0.008) \approx 0.00267.$$

The true value $\ln(1.2) = 0.18232\ldots$ differs from 0.18 by 0.00232, inside the bound. Notice we expanded about 1, not 0, because ln is undefined at 0 — you center where the function is friendly and where you know the derivatives.

Example 31.3 (Reading off a limit from the first surviving term)**.** Evaluate $\lim_{x\to 0} \dfrac{x - \sin x}{x^3}$ using Taylor expansion.

Expand the numerator: $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots$, so

$$x - \sin x = \frac{x^3}{6} - \frac{x^5}{120} + \cdots.$$

Divide by x^3:

$$\frac{x - \sin x}{x^3} = \frac{1}{6} - \frac{x^2}{120} + \cdots \xrightarrow[x\to 0]{} \frac{1}{6}.$$

The limit is $\frac{1}{6}$. This is the same answer L'Hopital's rule gives after three differentiations, but Taylor expansion makes the mechanism transparent: the numerator's first nonzero term is $x^3/6$, and dividing by x^3 exposes its coefficient. Taylor series turn many delicate limits into bookkeeping of leading terms.

Connection

Example 31.3 reflects a universal computational principle. Whenever you face a 0/0 limit, expanding numerator and denominator in Taylor series and comparing their lowest surviving powers resolves it cleanly — and reveals exactly *why* L'Hopital's rule works at all. L'Hopital differentiates to peel away vanishing orders one at a time; Taylor expansion exposes all orders at once. Near a zero, every smooth function behaves like its lowest nonvanishing Taylor term.

31.8 What to carry away

- The degree-n **Taylor polynomial** $P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$ is the unique polynomial matching f and its first n derivatives at a. The factorials exactly undo the factorials that repeated differentiation produces.
- **Taylor's theorem** makes the approximation exact: $f(x) = P_n(x) + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$ for some c between a and x. The remainder is the next term with its derivative evaluated at an intermediate point. It generalizes the Mean Value Theorem (the case $n = 0$) and is proved the same way.
- The **error bound** $|R_n(x)| \le \frac{M}{(n+1)!}|x-a|^{n+1}$, where M bounds $f^{(n+1)}$, is the practical workhorse. It needs only a ceiling on the next derivative, never the mystery point c.

- The Taylor series equals $f(x)$ **exactly when** $R_n(x) \to 0$. For $e^x, \sin, \cos$ the factorial crushes the remainder everywhere. For $\frac{1}{1+x^2}$ a hidden complex singularity caps the radius of convergence. For e^{-1/x^2} the series converges to the wrong function entirely: smooth does not imply analytic.
- **Analytic** functions are those that equal their Taylor series — the ones with no local secrets. Physics lives largely in their first surviving terms: $\sin\theta \approx \theta$, $V \approx \frac{1}{2}V''(x-x_0)^2$, $E \approx \frac{1}{2}mv^2$.

31.9 Exercises

Exercise 31.1. Find the degree-3 Maclaurin polynomial of $f(x) = e^x \cos x$ by computing $f(0)$, $f'(0)$, $f''(0)$, and $f'''(0)$ directly. Confirm that the result matches $1 + x - \frac{x^3}{3}$.

Exercise 31.2. Use the degree-2 Taylor polynomial of $\sqrt{x}$ centered at $a = 4$ to estimate $\sqrt{4.2}$. Then use the Lagrange remainder to bound the error, and compare with a calculator value.

Exercise 31.3. The small-angle approximation $\sin\theta \approx \theta$ is used for $\theta = 10°$. Convert to radians, and use the remainder bound $|R_1(\theta)| \le |\theta|^3/6$ to find the maximum possible error. Is the approximation good to three decimal places?

Exercise 31.4. How many terms of the Maclaurin series for $\cos x$ are needed to compute $\cos(1)$ with error less than 10^{-8}? Justify with the Lagrange bound, then state the polynomial you would use.

Exercise 31.5. Use Taylor expansions to evaluate $\displaystyle\lim_{x\to 0} \frac{e^x - 1 - x}{x^2}$. Identify the first surviving term of the numerator and read off the answer.

Exercise 31.6 (Conceptual)**.** A classmate says, "If a function is infinitely differentiable, its Taylor series must converge to it." Give the counterexample from this chapter and explain in one or two sentences exactly which step of that reasoning fails.

Exercise 31.7 (Conceptual)**.** The Maclaurin series of $\frac{1}{1-x}$ is $1 + x + x^2 + \cdots$, with radius of convergence 1. Explain why 1 is the radius, both from the real-line behavior of the function and from the location of its singularity. Then predict the radius of convergence of the Maclaurin series of $\frac{1}{4+x^2}$ without computing the series, and justify your prediction.

Exercise 31.8 (Proof)**.** Prove the integral form of the remainder: if $f^{(n+1)}$ is continuous on the interval between a and x, then

$$R_n(x) = \frac{1}{n!}\int_a^x f^{(n+1)}(t)\,(x-t)^n\,\mathrm{d}t.$$

(Hint: start from $f(x) - f(a) = \int_a^x f'(t)\,\mathrm{d}t$ and integrate by parts repeatedly, choosing the antiderivative of 1 to be $-(x-t)$ so the powers build up. This is the cleanest remainder of all, and the Lagrange form follows from it by the Mean Value Theorem for integrals.)

Exercise 31.9 (Proof)**.** Show that for $f(x) = e^{-1/x^2}$ (with $f(0) = 0$), the first derivative at 0 is $f'(0) = 0$. (Compute $\lim_{h\to 0} \frac{e^{-1/h^2}}{h}$ by substituting $u = 1/h^2 \to \infty$ and comparing the growth of e^u against $\sqrt{u}$.) Explain in words why the same kind of argument forces *every* derivative at 0 to vanish.

Exercise 31.10. Expand $f(x) = \frac{1}{\sqrt{1+x}}$ to degree 2 about $a = 0$ to derive the approximation $\frac{1}{\sqrt{1+x}} \approx 1 - \frac{1}{2}x + \frac{3}{8}x^2$. Use it to estimate $\frac{1}{\sqrt{1.1}}$, bound the error, and explain how this is the Taylor origin of the relativistic correction term mentioned in the chapter.

Part VII

Multivariable Calculus

Chapter 32

Functions of Several Variables and Partial Derivatives

For thirty-one chapters everything has been one-dimensional. A function took a single number x and returned a single number y, and the whole drama of calculus – slope, area, accumulation – played out on a flat sheet of paper, a curve and its tangent line. That picture carried us all the way from limits to Taylor's theorem. But it is also a lie of omission. The temperature in a room is not a function of one number; it depends on *where* you are – three coordinates – and on *when* you measure it – a fourth. The cost of building something depends on dozens of quantities at once. The loss function that a neural network minimizes depends on millions of weights. Real problems are relentlessly multivariable.

This chapter is the doorway. We leave the line and step onto the plane, and eventually into space. A function will now eat *several* numbers and return one. Its graph will no longer be a curve but a *surface*. The single derivative will split into several *partial derivatives*, one for each input direction. And the simple intuition "a limit just means you can get as close as you like" will turn out to hide a genuine new subtlety, because in the plane there are infinitely many directions from which to approach a point. Almost everything you already know transfers – but it transfers with a twist, and the twist is exactly where the new geometry lives.

32.1 Functions of Two and Three Variables

A function of two variables is a rule that assigns to each ordered pair (x, y) a single real number $z = f(x, y)$. The set of pairs we are allowed to feed in is the *domain* (now a region in the plane $\mathbb{R}^2$, not an interval on the line), and the set of outputs is the *range*. Nothing about the *idea* of a function has changed: it is still a machine, still single-valued, still "one output per input." What changed is that the input slot is wider. It now takes a point of the plane.

Focus

A real-valued function of two variables is a map $f : D \to \mathbb{R}$ where $D \subseteq \mathbb{R}^2$. We write $z = f(x, y)$. For three variables, $f : D \to \mathbb{R}$ with $D \subseteq \mathbb{R}^3$, written $w = f(x, y, z)$. The output is always a *single number*; only the input has grown.

Here are functions you should hold in your head for the rest of the chapter.

Example 32.1 (A handful of surfaces)**.**

- $f(x, y) = x^2 + y^2$. The height above (x, y) is its squared distance from the origin. This is a bowl (a *paraboloid*) opening upward.
- $f(x, y) = \sqrt{1 - x^2 - y^2}$, with domain the disk $x^2 + y^2 \leq 1$. This is the upper half of the unit sphere: at the origin the height is 1, and it falls to 0 at the rim.
- $f(x, y) = x^2 - y^2$. A *saddle*: it curves up as you walk along the x-axis and down as you walk along the y-axis.
- $T(x, y, z) = \dfrac{1}{x^2 + y^2 + z^2}$, a function of three variables – think of it as the temperature near a hot point source at the origin, blazing at the center and cooling with distance.

Notice the domain is now a genuinely two-dimensional object. For $\sqrt{1 - x^2 - y^2}$ we must have $x^2 + y^2 \leq 1$, so the domain is a filled disk. For $\ln(x + y)$ we need $x + y > 0$, the open half-plane above the line $y = -x$. Sketching the domain is the first thing to do with any new multivariable function, because it tells you *where the surface lives*.

Intuition

Think of $z = f(x, y)$ as a landscape. The pair (x, y) is your position on a map – east and north – and $f(x, y)$ is the *altitude* at that spot. The graph is the terrain itself: hills, valleys, ridges, passes. Every word we are about to invent – level curve, partial derivative, tangent plane – has a hiking interpretation. When in doubt, go for a walk on the landscape.

Three variables and why you cannot draw them

A function $w = f(x, y, z)$ of three variables would have a graph living in four-dimensional space (three for input, one for output). You cannot draw that, and neither can anyone else. This is not a defect of your imagination; it is a fact about paper. The remedy is to study three-variable functions through their *level sets* (below) and through slices, holding some inputs fixed. We will lean on two-variable functions for pictures and let the three-variable case ride along on the same algebra, because every definition we give extends to n variables by simply writing more commas.

32.2 Graphs, Level Curves, and Surfaces

There are two complementary ways to see a function of two variables. The first is the *graph*: lift each domain point (x, y) to the height $z = f(x, y)$ and look at the surface that results. The second is the *contour map*: stay flat on the domain and draw the curves where the height is constant, exactly as a topographic map encodes a mountain on a flat sheet. Cartographers solved the "draw a 3D surface on 2D paper" problem centuries before calculus needed it, and their solution – contour lines – is one of the most useful pictures in all of multivariable calculus.

Definition 32.1 (Level curve, level surface)**.** For a function $f(x, y)$ and a constant c, the *level curve* (or contour) at level c is the set of domain points where the function equals c:

$$\{(x, y) : f(x, y) = c\}.$$

For $f(x, y, z)$, the analogous set $\{(x, y, z) : f(x, y, z) = c\}$ is a *level surface*.

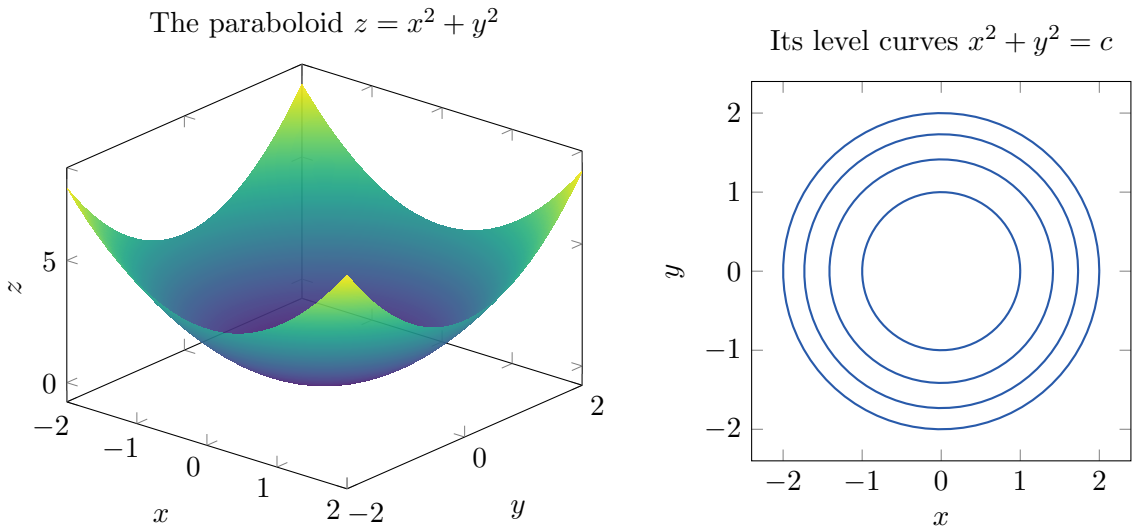


Figure 32.1: Left: the bowl $z = x^2 + y^2$ as a surface. Right: its contour map. Each circle is a level curve $x^2 + y^2 = c$; the circles are evenly spaced in height but get *closer together* as you move outward, because the bowl gets steeper. Crowded contours mean steep terrain.

Read Figure 32.1 the way a hiker reads a trail map. Each circle is a fixed altitude. Where the circles bunch up, the ground is steep; where they spread apart, it is gentle. For the bowl $x^2 + y^2 = c$, the level curve at height c is a circle of radius $\sqrt{c}$. The radii $\sqrt{1}, \sqrt{2}, \sqrt{3}, \ldots$ grow more and more slowly, so equal steps in height correspond to ever-smaller steps outward – which is precisely the contours crowding together. The contour spacing *is* the steepness, encoded in flatland.

What is really happening

A level curve is a *horizontal slice* of the graph, dropped straight down onto the domain. Slice the surface $z = f(x, y)$ with the plane $z = c$; the intersection is a curve sitting at height c in space; its shadow on the floor is the level curve. So the contour map is the collection of all these shadows, one per height. Nothing is lost: from a dense enough family of contours you can reconstruct the whole surface. This is why a topographic map, though flat, tells you everything about the mountain.

Example 32.2 (Reading a saddle from its contours)**.** Take $f(x, y) = x^2 - y^2$. The level curve at c is $x^2 - y^2 = c$. For $c > 0$ these are hyperbolas opening left-right; for $c < 0$ they open up-down; and for $c = 0$ we get $x^2 = y^2$, i.e. the two lines $y = \pm x$ crossing at the origin. That crossing pair is the signature of a saddle: the contours *intersect* themselves only at a saddle point, where the surface is going up in one direction and down in another at the very same spot.

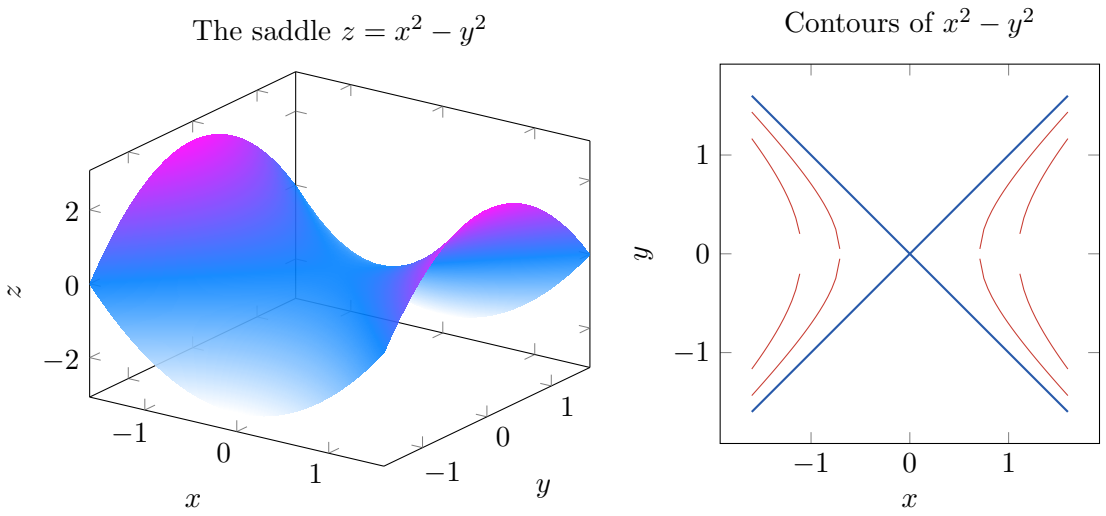


Figure 32.2: The saddle and its contours. The blue lines $y = \pm x$ are the level set $z = 0$ and cross at the saddle point; the red hyperbolas are levels $c = 0.5, 1.2$. Self-crossing contours are how a saddle announces itself on a flat map.

Walking question

On your next walk, find a mountain pass – a saddle – in your mind: the low point of a ridge between two peaks. Stand on it. Forward and backward (toward the peaks) the ground rises; left and right (down into the valleys) it falls. You are at a point that is simultaneously a *minimum* along one path and a *maximum* along another. Hold that feeling; it is the geometric heart of why finding maxima and minima in several variables is harder than in one.

Application

Level sets are everywhere once you look. Weather maps draw *isobars* (level curves of pressure) and *isotherms* (level curves of temperature). Economists draw *indifference curves*: level curves of a utility function $U(x, y)$, the combinations of two goods that leave a consumer equally happy. In machine learning, the *decision boundary* of a classifier is a level set $f(\mathbf{x}) = 0$ of the score function, separating one predicted class from another. Whenever someone says "the set of points that are equally good/hot/likely," they are naming a level set.

32.3 Limits and Continuity in Several Variables

Now the twist. We need a limit: what it means for $f(x, y)$ to approach a value L as (x, y) approaches a point (a, b). The *words* are identical to the one-variable definition. The *geometry* is profoundly different, and the difference will trip you if you are not warned.

On the line, "$x \to a$" has exactly two avenues of approach: from the left and from the right. We made peace with that using one-sided limits, and a two-sided limit existed precisely when both sides agreed. In the plane, "$(x, y) \to (a, b)$" allows approach from *every direction at once* – straight in along the x-axis, in along the y-axis, in along any line, in along a spiral, along a parabola. For the

limit to exist, *all* of these infinitely many approaches must agree on the same value L. That is a far stronger demand.

Definition 32.2 (Limit of a function of two variables)**.** We say $\lim_{(x,y)\to(a,b)} f(x,y) = L$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \quad \Longrightarrow \quad |f(x,y) - L| < \varepsilon.$$

In words: once (x, y) is within distance δ of (a, b) – inside a small disk around (a, b), in any direction – the output is within ε of L.

Compare this letter for letter with the one-variable ε–δ definition. The only change is that $|x-a| < \delta$, the length of an interval, has become $\sqrt{(x-a)^2 + (y-b)^2} < \delta$, the radius of a *disk*. The interval became a disk because the domain became two-dimensional. That single substitution carries all the new subtlety with it.

Focus

The quantity $\sqrt{(x-a)^2 + (y-b)^2}$ is just the distance from (x, y) to (a, b). The definition says: small distance forces small error, *regardless of direction*. The δ-disk has no preferred direction, which is exactly why the limit must come out the same no matter how you approach.

The two-path test: how multivariable limits fail

Here is the practical consequence, and it is the single most important computational fact in this section. If you can find *two* paths into (a, b) that give two *different* limiting values, then the limit does not exist. One disagreement among the infinitely many directions is enough to destroy the limit. Agreement along finitely many paths *never* proves a limit exists — it only fails to disprove it.

Example 32.3 (The classic non-limit)**.** Consider

$$f(x,y) = \frac{xy}{x^2+y^2}, \qquad (x,y) \neq (0,0),$$

and ask whether the limit exists as $(x, y) \to (0, 0)$.

Approach along the x-axis ($y = 0$): then $f(x, 0) = \dfrac{x \cdot 0}{x^2 + 0} = 0$ for all $x \neq 0$. So along this path $f \to 0$.

Approach along the y-axis ($x = 0$): then $f(0, y) = 0$ likewise. Still 0. So far so innocent – if these were the only two paths, we might be fooled.

Approach along the diagonal $y = x$: then

$$f(x,x) = \frac{x \cdot x}{x^2 + x^2} = \frac{x^2}{2x^2} = \frac{1}{2} \quad \text{for all } x \neq 0.$$

Along the diagonal the function is *constantly* $\frac{1}{2}$, no matter how close we get to the origin. The axes say 0; the diagonal says $\frac{1}{2}$. They disagree. Therefore

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} \quad \text{does not exist.}$$

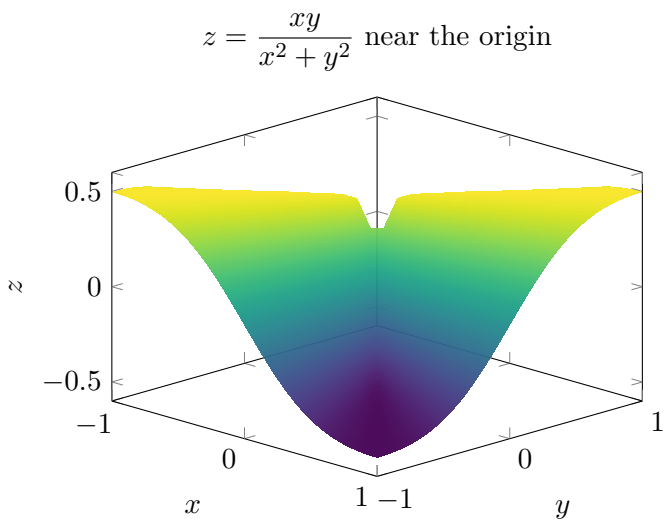


Figure 32.3: The surface $z = xy/(x^2 + y^2)$. Along the axes it sits at height 0; along the diagonals it rides at $\pm\frac{1}{2}$. The surface is pleated right at the origin – there is no single height the function settles toward, so the limit fails.

There is an even cleaner way to see the failure, straight from the contour map. Along the ray at angle θ from the origin the function $xy/(x^2 + y^2)$ is *constant*: in polar form it equals $\frac{1}{2}\sin 2\theta$, which depends on θ but not on r. So every ray into the origin carries its own fixed value, and the values disagree from one ray to the next. A function whose level sets are rays through the point you are approaching cannot have a limit there – you can read its non-existence right off Figure 32.4.

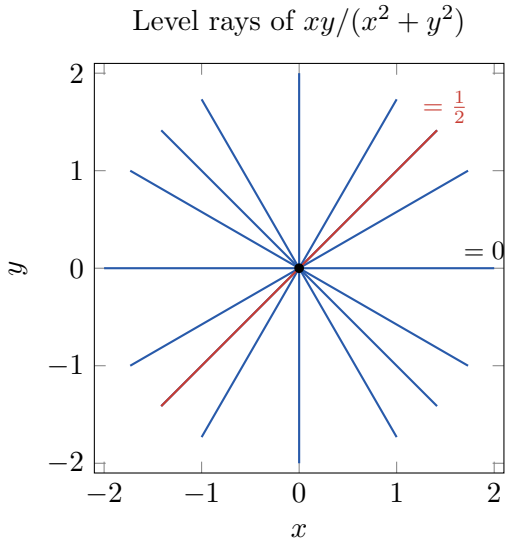


Figure 32.4: The level sets of $xy/(x^2 + y^2)$ are straight rays through the origin, because the value $\frac{1}{2}\sin 2\theta$ depends only on the direction θ. The diagonal $y = x$ (red) carries the constant value $\frac{1}{2}$; the axes (horizontal/vertical) carry 0. Different rays, different values – so no single limit can exist at the origin.

Common trap

Checking a few lines does not prove a limit exists. Worse: there are functions whose limit is the same along *every straight line* through the origin and yet still fails, because some *curved* path disagrees. The standard cautionary tale is $g(x,y) = \dfrac{x^2 y}{x^4 + y^2}$. Along every line $y = mx$ you get $g \to 0$. But along the parabola $y = x^2$,

$$g(x, x^2) = \frac{x^2 \cdot x^2}{x^4 + x^4} = \frac{x^4}{2x^4} = \frac{1}{2}.$$

So the line-limits all agree on 0, the parabola says $\frac{1}{2}$, and the limit does not exist. Moral: matching paths can only *disprove* a limit. To *prove* one exists you need an estimate that controls *all* directions at once, like the squeeze below.

Why the technique works

Why is the two-path test legitimate? Because the limit in Definition 32.2 demands a single L that works inside the *whole* disk. Any particular path of approach lives inside that disk, so if the full limit is L, the limit *restricted to any path* must also be L – the path inherits the bound. Contrapositive: if two paths produce two different restricted limits, no single L can serve both, so the full limit cannot exist. The test works because a path is a special case of "approaching," and the general statement, if true, forces every special case.

When the limit *does* exist: the squeeze

To prove a limit exists you must bound the error in a way that does not care about direction. Polar coordinates are the natural tool, because they package "distance from the center" into a single variable r.

Example 32.4 (A limit that exists)**.** Show that $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^2 y}{x^2 + y^2} = 0.$

Write $x = r\cos\theta$, $y = r\sin\theta$, so $x^2 + y^2 = r^2$ and $r \to 0$ *means* the same as $(x,y) \to (0,0)$, regardless of the angle θ. Then

$$\left|\frac{x^2 y}{x^2 + y^2}\right| = \left|\frac{r^2\cos^2\theta \cdot r\sin\theta}{r^2}\right| = r\,|\cos^2\theta\,\sin\theta| \le r.$$

The angle dropped out of the bound: whatever direction θ you came in along, the size is at most r. As $r \to 0$ this is squeezed to 0. So the limit is 0, and now we mean it for *all* approaches at once. (Concretely, given $\varepsilon > 0$, take $\delta = \varepsilon$; then distance $r < \delta$ forces the value within ε of 0.)

The contrast between Examples 32.3 and 32.4 is the whole lesson. In the bad case the angle θ survived ($f = \frac{1}{2}\sin 2\theta$ in polar form, genuinely depending on direction); in the good case the angle was crushed under a factor of r that goes to zero no matter what.

Definition 32.3 (Continuity)**.** A function f is *continuous* at (a,b) if (a,b) is in the domain, the limit there exists, and

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b).$$

The good news, which we will use without ceremony from here on: every function built from x, y, constants, and the familiar one-variable functions by adding, multiplying, dividing (where the denominator is nonzero), and composing is continuous wherever it is defined. Polynomials in x and y are continuous everywhere; $\sin(xy)$, e^{x-y}, and $\sqrt{1+x^2+y^2}$ are continuous everywhere; rational functions are continuous off their zero denominators. The pathologies above all came from a $0/0$ at a single point. Away from such points, life is calm.

History

The leap to several variables happened early and informally. Euler, in the mid-1700s, computed with functions of many variables and "partial differences" fluently, long before anyone had a careful definition of a limit even in *one* variable. The direction-dependent subtlety we just met was simply not on the radar; eighteenth-century analysts wrote down derivatives and got correct physics. The disquieting examples – functions continuous along every line yet not continuous at a point – were manufactured later, in the ε–δ era of Cauchy, Weierstrass, and especially the careful textbook tradition of the late nineteenth century, precisely to show that the naive "approach from any direction" picture needs the disk, not a bundle of lines. The counterexamples are not perversity for its own sake; they are the guardrails that tell us our definition is the *right* one.

32.4 Partial Derivatives: Slopes in the Coordinate Directions

Now we differentiate. In one variable the derivative measured one slope: how fast y changes as x changes. On a surface there is no single slope, because the answer depends on which way you walk. Walk east and the ground may rise; walk north and it may fall. So we measure slope *one direction at a time*, starting with the two coordinate directions. That is the partial derivative, and the idea behind it is almost embarrassingly simple.

Intuition

To find how the surface tilts in the x-direction at a point, *freeze* y. Holding y constant turns $f(x,y)$ back into an ordinary one-variable function of x alone – a single curve carved out of the surface by the vertical plane $y=$ const. Differentiate that curve the old way. The result is the partial derivative with respect to x. To get the y-direction, freeze x instead. The multivariable derivative is nothing but the one-variable derivative, applied along one axis with the other coordinate held still.

Definition 32.4 (Partial derivatives)**.** The *partial derivative of f with respect to x* at (a,b) is

$$\frac{\partial f}{\partial x}(a,b) \;=\; \lim_{h\to 0}\frac{f(a+h,\,b)-f(a,b)}{h},$$

and with respect to y,

$$\frac{\partial f}{\partial y}(a,b) \;=\; \lim_{h\to 0}\frac{f(a,\,b+h)-f(a,b)}{h},$$

whenever the limits exist. We also write f_x and f_y for these.

Stare at the first formula. Only the *first* slot moves: $a+h$ in the x-place, the y-place glued at b. It is exactly the one-variable difference quotient, run on the function $x \mapsto f(x,b)$. The curly ∂ (d's

rounder cousin) is the only new symbol, and it means precisely "derivative while the other variables are held fixed."

What is really happening

The notation ∂ is doing real work. The plain d in $\frac{dy}{dx}$ promised that y depended on *nothing but* x. Here f depends on both x and y, so we cannot ask for "the" rate of change of f – there isn't one. The rounded ∂ is a flag: it says *"rate of change of f as x varies, with every other variable held constant."* The symbol $\partial f/\partial x$ is a single indivisible object; the "∂f" and "∂x" on their own are not quantities you may cancel. The notation was popularized by Legendre and made standard by Jacobi in the 1840s, exactly so that nobody would mistake a partial rate for a total one.

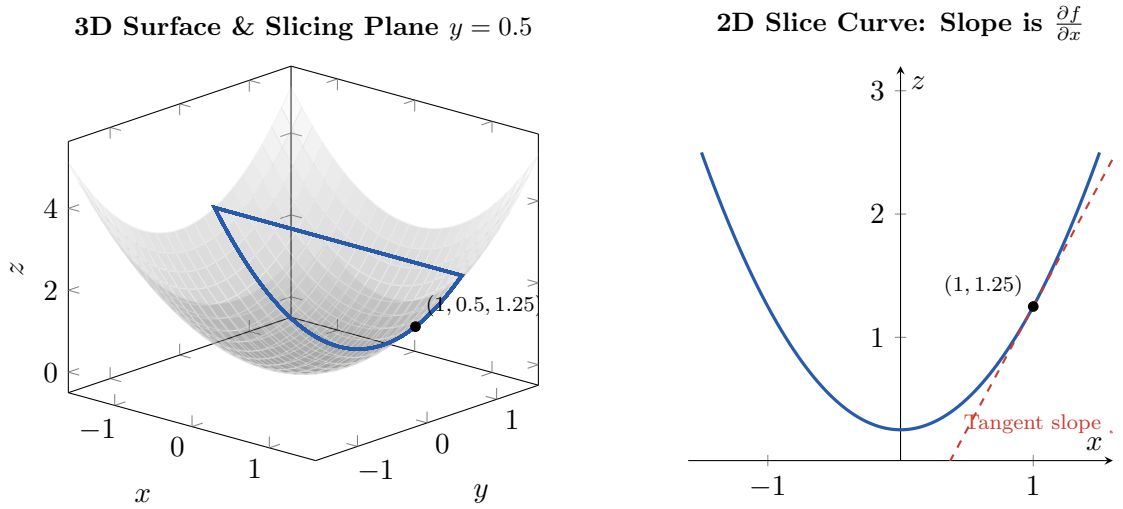


Figure 32.5: The geometric essence of a partial derivative: slicing a surface with a coordinate plane. Left: The vertical plane $y = 0.5$ slices through the paraboloid $z = x^2 + y^2$, holding y fixed and carving out a curve in space. Right: Viewed straight on in the slice plane, that curve is the ordinary single-variable parabola $z(x) = x^2 + 0.25$. The partial derivative $\frac{\partial f}{\partial x}(1, 0.5) = 2$ is simply the slope of the tangent line to this cross-sectional curve.

Work it yourself

Compute f_x and f_y for $f(x, y) = x^2+y^2$ at $(1, 0.5)$ by hand, from the definition, before reading on.
For f_x: hold $y = 0.5$, so $f(x, 0.5) = x^2 + 0.25$, whose derivative is $2x$; at $x = 1$ that is **2**.
For f_y: hold $x = 1$, so $f(1, y) = 1 + y^2$, derivative $2y$; at $y = 0.5$ that is **1**. So the bowl tilts upward twice as steeply in the x-direction as in the y-direction at that point – which matches Figure 32.5, where the blue curve climbs faster than the red.

Computing partials in practice

You almost never go back to the limit definition. The trick is the one your intuition already gave you: **to differentiate with respect to x, treat y as a constant** and apply all your one-variable

rules. Then swap roles for y.

Example 32.5 (Treat the other variable as a number)**.** Let $f(x, y) = x^3y^2 + \sin(xy) + 4y$.

With respect to x (so y is a constant):

$$f_x = 3x^2y^2 + \cos(xy) \cdot y + 0 = 3x^2y^2 + y\cos(xy).$$

The x^3 differentiated to $3x^2$ while y^2 rode along as a constant multiplier; the chain rule on $\sin(xy)$ produced $\cos(xy)$ times the x-derivative of the inside, $\partial(xy)/\partial x = y$; and the lone $4y$, a constant as far as x is concerned, died.

With respect to y (so x is a constant):

$$f_y = x^3 \cdot 2y + \cos(xy) \cdot x + 4 = 2x^3y + x\cos(xy) + 4.$$

Now x^3 was the constant multiplier, $y^2 \mapsto 2y$, the inside derivative is $\partial(xy)/\partial y = x$, and $4y \mapsto 4$. Notice how each variable's computation is a completely ordinary single-variable problem.

Example 32.6 (Three variables, same idea)**.** Let $g(x, y, z) = x^2e^{yz}$. Then, freezing the other two each time,

$$g_x = 2x\,e^{yz}, \qquad g_y = x^2z\,e^{yz}, \qquad g_z = x^2y\,e^{yz}.$$

For g_y we held x and z fixed, so e^{yz} has z as the constant multiplying y in the exponent, giving the factor z; symmetrically for g_z. Three variables, three partials, each a one-liner.

Common trap

Two slips to guard against. First, do not forget the chain-rule factor when the other variable sits *inside* a composite. In $\sin(xy)$, differentiating with respect to x gives $y\cos(xy)$, not $\cos(xy)$ – the inside xy has x-derivative y. Second, do not kill a term too eagerly: in f_x, a term like $4y$ vanishes (it is constant in x), but a term like $4xy$ does *not* – it becomes $4y$. "Constant in x" means *no x appears at all*, not merely that a y appears.

Application

Partial derivatives are the language of sensitivity. In thermodynamics, pressure, volume, and temperature are tangled together, and physicists write $(\partial P/\partial V)_T$ – "how pressure responds to volume *at fixed temperature*" – with the held variable in a subscript precisely because the answer changes if you hold something else fixed instead. In economics, $\partial(\text{output})/\partial(\text{labor})$ at fixed capital is the *marginal product of labor*. In training a neural network, every weight w_i gets its own partial $\partial(\text{loss})/\partial w_i$; the list of all of them is the gradient, and backpropagation is an algorithm for computing millions of these partials efficiently. The single most important fact about modern machine learning is that these partials can be computed at all.

32.5 Higher Partial Derivatives and Clairaut's Theorem

A partial derivative f_x is itself a function of x and y, so we can differentiate it again. This gives *second-order partial derivatives*, and now there is a genuine choice: we can differentiate first by x then by y, or first by y then by x. The notation, read inner-to-outer in the operator form, is

$$f_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2}, \qquad f_{yy} = \frac{\partial^2 f}{\partial y^2},$$

$$f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y\,\partial x}, \qquad f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x\,\partial y}.$$

Common trap

The order conventions are maddening but standard. In the *subscript* form f_{xy} you read left to right: first x, then y. In the *fraction* form $\partial^2 f/\partial y\,\partial x$ you read right to left, inside out: the ∂x next to f acts first. So f_{xy} and $\partial^2 f/(\partial y\,\partial x)$ are the *same* object. Pick one convention and translate carefully.

The two "mixed" partials f_{xy} and f_{yx} ask different-looking questions: how does the eastward slope change as you move north, versus how does the northward slope change as you move east? There is no obvious reason these should be equal. And yet – almost magically – they nearly always are.

Example 32.7 (The mixed partials match)**.** Take $f(x,y) = x^3y^2$. Then

$$f_x = 3x^2y^2, \qquad f_{xy} = \frac{\partial}{\partial y}\left(3x^2y^2\right) = 6x^2y,$$

$$f_y = 2x^3y, \qquad f_{yx} = \frac{\partial}{\partial x}\left(2x^3y\right) = 6x^2y.$$

Both routes land on $6x^2y$. Differentiating in the order x-then-y gave the same thing as y-then-x. Try it on $\sin(xy)$ or $e^x\cos y$ and the same coincidence happens. It is not a coincidence.

Theorem 32.1 (Clairaut / Schwarz)**.** *If f_{xy} and f_{yx} are both continuous on an open region containing (a,b), then*

$$f_{xy}(a,b) = f_{yx}(a,b).$$

The order of mixed partial differentiation does not matter.

Intuition

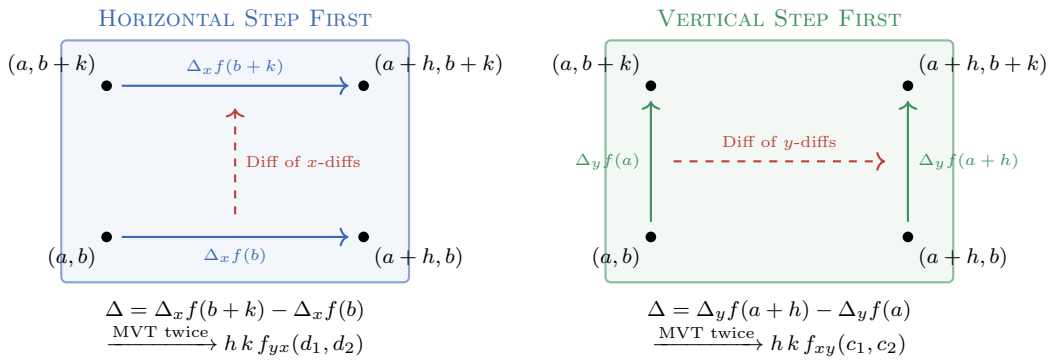
Why *should* order not matter? Picture a tiny rectangle with corners (a,b), $(a+h,b)$, $(a,b+k)$, $(a+h,b+k)$. Form the "second difference" of f around the rectangle: take the change across the top minus the change across the bottom,

$$\Delta = \big[f(a+h,b+k) - f(a,b+k)\big] - \big[f(a+h,b) - f(a,b)\big].$$

Group it the other way – change up the right side minus change up the left side – and you get *exactly the same four terms*, just reassociated:

$$\Delta = \big[f(a+h,b+k) - f(a+h,b)\big] - \big[f(a,b+k) - f(a,b)\big].$$

The first grouping, divided by hk and shrunk, becomes f_{yx}; the second becomes f_{xy}. But they were always the same number $\Delta/(hk)$. The symmetry of the mixed partials is just the symmetry of a rectangle – top-then-side equals side-then-top – pushed to the limit.



$$\Delta = f(a+h, b+k) - f(a+h, b) - f(a, b+k) + f(a, b) \text{ in both groupings } \Longrightarrow \lim_{h,k\to 0} f_{xy} = f_{yx}.$$

Figure 32.6: Why mixed partials commute: the second difference on a coordinate rectangle. Grouping the four corner values horizontally first computes how an x-step changes across a y-interval (f_{yx}); grouping vertically first computes how a y-step changes across an x-interval (f_{xy}). Because both algebraic sums are identical, the Mean Value Theorem applied twice forces $f_{xy} = f_{yx}$ as the rectangle shrinks.

Why the technique works

Here is the idea of the proof made rigorous. Define the second difference $\Delta(h, k)$ above. Apply the one-variable Mean Value Theorem twice. Group Δ as a difference in x of the y-difference: the function $\phi(x) = f(x, b+k) - f(x, b)$ satisfies $\Delta = \phi(a+h) - \phi(a) = h\,\phi'(c_1)$ for some c_1 between a and $a+h$, and $\phi'(x) = f_x(x, b+k) - f_x(x, b)$. Apply the MVT again in the y-direction to this difference: it equals $k\,f_{xy}(c_1, c_2)$ for some c_2 between b and $b+k$. So $\Delta = h\,k\,f_{xy}(c_1, c_2)$. Run the *same* argument grouping in the other order and you get $\Delta = h\,k\,f_{yx}(d_1, d_2)$. Therefore $f_{xy}(c_1, c_2) = f_{yx}(d_1, d_2)$. Now let $h, k \to 0$: all four interior points are squeezed toward (a, b), and by *continuity* of the mixed partials both sides converge to their values at (a, b), giving $f_{xy}(a, b) = f_{yx}(a, b)$. The continuity hypothesis is exactly what lets the squeeze land on a single value – without it the theorem can fail.

History

Alexis Clairaut stated the symmetry around 1740 while studying the shape of the rotating Earth, where mixed partials of a potential had to match for the physics to be consistent. Euler used it freely. But the early statements were intuitive, resting on the unexamined belief that "you can swap the limits." It was only when Weierstrass-era analysts produced functions whose mixed partials *disagree* – the standard example is $f(x, y) = xy\,\dfrac{x^2 - y^2}{x^2 + y^2}$ (with $f(0, 0) = 0$), for which $f_{xy}(0, 0) = -1$ but $f_{yx}(0, 0) = +1$ – that mathematicians saw the continuity hypothesis was not decoration. The point: the order of differentiation is *usually* irrelevant, and the theorem tells you precisely when "usually" becomes "always."

Connection

Clairaut's theorem is the seed of a deep idea. It says the mixed partials of a smooth function agree – equivalently, that a gradient vector field (f_x, f_y) has a special symmetry: $\partial_y f_x = \partial_x f_y$. In vector calculus this becomes the statement "the curl of a gradient is zero," and it is the test for whether a given vector field is *conservative* (the gradient of some potential). The harmless-looking equality $f_{xy} = f_{yx}$ grows up to become a cornerstone of the theory of force fields and potentials. You will meet it again, dressed as curl.

32.6 Tangent Planes

In one variable, the tangent line was the best straight-line approximation to a curve: zoom in on a smooth graph and it looks linear. The same zooming works on a surface, but a flat approximation in the plane is not a line – it is a *plane*. The tangent plane is the best flat approximation to the surface at a point, and the partial derivatives are precisely the two slopes that pin it down.

Intuition

Stand on the surface at the point above (a, b). You found two slopes there: $f_x(a, b)$, the tilt as you face east, and $f_y(a, b)$, the tilt as you face north. A flat plane through that point is completely determined by how steeply it rises eastward and how steeply it rises northward. Match those two slopes to the surface's two slopes and you have the unique plane that kisses the surface – the tangent plane. East-slope and north-slope are exactly enough data to fix a plane, which is why *two* partials suffice.

A plane through the point $(a, b, f(a, b))$ has the form

$$z = f(a, b) + A\,(x - a) + B\,(y - b),$$

where A is its slope in the x-direction and B its slope in the y-direction. To be tangent, those slopes must equal the surface's slopes: $A = f_x(a, b)$ and $B = f_y(a, b)$. That gives the formula.

Focus

Tangent plane. If f has continuous partials near (a, b), the tangent plane to $z = f(x, y)$ at the point $\big(a, b, f(a, b)\big)$ is

$$\boxed{z = f(a, b) + f_x(a, b)\,(x - a) + f_y(a, b)\,(y - b)}.$$

The right-hand side is the *linearization* $L(x, y)$ of f at (a, b): the best linear approximation, just as the tangent line was in one variable.

What is really happening

Compare this with the one-variable tangent line $y = f(a) + f'(a)(x - a)$. The structure is identical: value at the base point, plus slope times displacement. We simply have *two* displacement terms now, one per input direction, each weighted by its own partial. The tangent plane is two tangent lines – the slice-curves of Figure 32.5 – spanned into a flat sheet.

"Differentiable at (a, b)" means that this plane really is a good approximation: the surface peels away from the plane faster than you walk, i.e. the error is small compared to the distance $\sqrt{(x-a)^2+(y-b)^2}$. (Beware: in several variables the mere *existence* of f_x and f_y is not enough to guarantee this – continuity of the partials is the clean condition that does.)

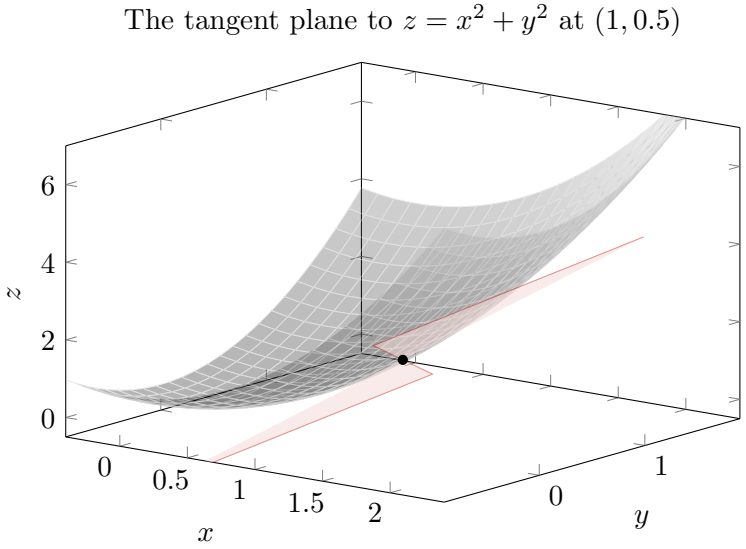


Figure 32.7: The bowl and its tangent plane at $(1, 0.5, 1.25)$. The plane has x-slope $f_x = 2$ and y-slope $f_y = 1$, matching the two slice-slopes from Figure 32.5. It touches the surface at one point and is the surface's best flat impostor nearby.

Example 32.8 (Building a tangent plane)**.** Find the tangent plane to $f(x, y) = x^2 + y^2$ at $(1, 0.5)$.

We already have $f(1, 0.5) = 1 + 0.25 = 1.25$, and from the lab box $f_x(1, 0.5) = 2$, $f_y(1, 0.5) = 1$. Plug in:

$$z = 1.25 + 2(x-1) + 1\,(y-0.5) = 2x + y - 1.25.$$

That is the plane in Figure 32.7.

Example 32.9 (Linear approximation in action)**.** Use the tangent plane to estimate $f(1.1, 0.45)$ for $f = x^2 + y^2$ without squaring anything. The linearization is $L(x, y) = 2x + y - 1.25$, so

$$L(1.1, 0.45) = 2(1.1) + 0.45 - 1.25 = 2.2 + 0.45 - 1.25 = 1.40.$$

The true value is $1.1^2 + 0.45^2 = 1.21 + 0.2025 = 1.4125$. The linear estimate 1.40 is off by only 0.0125 – excellent for a step of about a tenth, and a perfect echo of how tangent lines estimated $\sqrt{4.1}$ back in the one-variable chapter. The whole machinery of differentials and linear approximation survives the jump to several variables; it just acquires a second slope term.

Why this matters

Step back and admire the unity. One variable: derivative f', tangent *line*, linearization $f(a) + f'(a)(x-a)$. Two variables: partials f_x, f_y, tangent *plane*, linearization $f(a,b) + f_x(x-a) + f_y(y-b)$. Three variables: partials f_x, f_y, f_z, tangent *hyperplane*, the obvious sum of

three terms. The pattern never breaks – it only grows a term per dimension. In the next chapter these partials assemble into a single vector, the *gradient* $\nabla f = (f_x, f_y, f_z)$, and the linearization collapses back into the clean one-line form $f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a})$, identical in shape to the one-variable tangent line. The plurality of partials was a temporary appearance; underneath, the derivative was a single object all along.

32.7 Exercises

Exercise 32.1 (Domains and contours)**.** For each function, describe the domain as a region in the plane and sketch three level curves.

(a) $f(x, y) = \sqrt{9 - x^2 - y^2}$.

(b) $f(x, y) = \ln(x - y)$.

(c) $f(x, y) = \dfrac{1}{x^2 + y^2}$ (what happens to the contours near the origin?).

Exercise 32.2 (Reading a saddle)**.** For $f(x, y) = y^2 - x^2$, identify the level curve at $c = 0$, at $c = 1$, and at $c = -1$. Which way does the surface rise from the origin, and which way does it fall? Sketch the contour map and mark the saddle point.

Exercise 32.3 (The two-path test)**.** Show that each limit fails to exist by finding two paths giving different values.

(a) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^2 - y^2}{x^2 + y^2}$.

(b) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^2 y}{x^4 + y^2}$ (try lines, then the parabola $y = x^2$).

Exercise 32.4 (A limit that exists)**.** Using polar coordinates or a squeeze, prove that

$$\lim_{(x,y)\to(0,0)} \frac{x^3 - y^3}{x^2 + y^2} = 0.$$

Then explain in one sentence why the angle θ disappearing from your bound is exactly what ensures the limit exists.

Exercise 32.5 (Computing partials)**.** Find f_x and f_y.

(a) $f(x, y) = x^4 y - 3xy^3 + 7$.

(b) $f(x, y) = e^{xy} \sin x$.

(c) $f(x, y) = \dfrac{x}{x^2 + y^2}$.

(d) $f(x, y) = \ln\big(x^2 + y^2\big)$.

Exercise 32.6 (Three variables)**.** For $g(x, y, z) = x\, y^2\, z^3 + \dfrac{x}{y} - \cos(yz)$, compute all three first partials g_x, g_y, g_z.

Exercise 32.7 (Verifying Clairaut)**.** For $f(x, y) = x^2 e^y + \sin(xy)$, compute f_{xy} and f_{yx} separately and confirm they agree. State explicitly where in your work you used the fact that the function is built from continuous pieces.

Exercise 32.8 (Tangent plane and estimate)**.** Let $f(x, y) = \sqrt{x^2 + y^2}$ (the cone).

(a) Find the tangent plane at $(3, 4)$, where $f = 5$.

(b) Use it to estimate $f(3.1, 3.9)$, and compare with the exact value.

(c) Explain why no tangent plane exists at the origin. (Hint: walk toward $(0, 0)$ from two directions and look at the slopes – this is the cone's sharp tip.)

Exercise 32.9 (Conceptual: lines are not enough)**.** A student claims: "I checked that

$$\lim_{(x,y)\to(0,0)} \frac{x^2 y}{x^4 + y^2} = 0$$

along every straight line $y = mx$, so the multivariable limit must be 0." In a short paragraph, explain what is wrong with this conclusion, and state the curved path that refutes it.

Exercise 32.10 (Proof: Clairaut by the second difference)**.** Let $\Delta(h, k) = f(a + h, b + k) - f(a + h, b) - f(a, b + k) + f(a, b)$. Assuming f, f_x, f_y, f_{xy}, and f_{yx} are all continuous near (a, b), apply the Mean Value Theorem twice in each of the two groupings of Δ to show $f_{xy}(a, b) = f_{yx}(a, b)$. Identify the precise step where continuity of the mixed partials is needed.

Exercise 32.11 (A function continuous along every line but not at the origin)**.** Let $g(x, y) = \dfrac{x^2 y}{x^4 + y^2}$ for $(x, y) \neq (0, 0)$ and $g(0, 0) = 0$.

(a) Show g restricted to any line through the origin is continuous at the origin.

(b) Show g is nonetheless *not* continuous at the origin.

(c) In one sentence, say what this example teaches about Definition 32.3.

Chapter 33

The Gradient, Directional Derivatives, and the Chain Rule

In the last chapter we learned to take a function of several variables, freeze all but one of its inputs, and differentiate the survivor. That gave us the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$: the slopes you feel if you walk due east or due north across a landscape. They are exact answers, but to only two questions. They are also strangely timid. Landscapes are not made of north-south and east-west walks. A river runs diagonally. A ball rolls downhill in whatever direction is steepest, which is almost never one of the axis directions. If all we can ever ask is "how steep is it if I head exactly east?" we have not yet learned to read the landscape.

This chapter fixes that. We will learn to ask: *how fast does f change if I walk in any direction I choose?* The answer to that question, for *every* direction at once, turns out to be packaged inside a single vector – the **gradient** ∇f. It points in the direction of steepest ascent. Its length tells you how steep that steepest climb is. And it sits perpendicular to the level curves – the contour lines – of f, for a reason you will be able to see with your eyes by the end of Section 33.3. Once we have the gradient, the multivariable chain rule falls out as a single tidy dot product, linear approximation becomes a tangent *plane* instead of a tangent line, and the whole machinery of *gradient descent* – the algorithm that trains nearly every modern neural network – is suddenly something you understand from the inside, not as a recipe but as a walk downhill.

33.1 Walking in an Arbitrary Direction

Stand on a hillside whose height above sea level is $f(x, y)$. You are at the point (a, b). The two partial derivatives answer only two of the infinitely many walks you could take: $\frac{\partial f}{\partial x}\big|_{(a,b)}$ is your rate of climb heading east, $\frac{\partial f}{\partial y}\big|_{(a,b)}$ heading north. Now suppose you want to set off in some other direction – say northeast, or exactly 30° north of east. How steep is the ground *that* way?

Let us name the direction with a unit vector $\mathbf{u} = (u_1, u_2)$, $\|\mathbf{u}\| = 1$. Walking "in direction $\mathbf{u}$" starting from (a, b) means tracing the straight path

$$(a, b) + t\,\mathbf{u} = (a + tu_1,\ b + tu_2), \qquad t \geq 0,$$

where t is the distance you have walked (because $\mathbf{u}$ has length one, t *is* arc length). Your height

along this path is a function of the single variable t:

$$g(t) = f(a + tu_1,\ b + tu_2).$$

The slope of the ground in direction $\mathbf{u}$ is just how fast your height is changing per step taken – that is, $g'(0)$. This is exactly the rate we want.

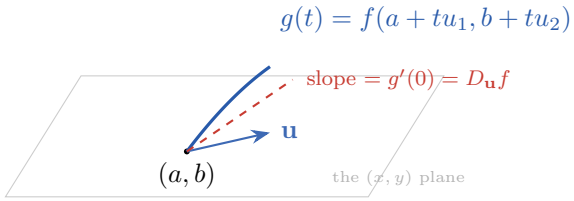


Figure 33.1: Walking from (a, b) in the unit direction $\mathbf{u}$ traces a straight line in the floor plane; the height f above that line is the single-variable function $g(t)$. Its slope at the start, $g'(0)$, is the directional derivative $D_{\mathbf{u}} f(a, b)$ – the steepness of the hillside as you set off that way.

Definition 33.1 (Directional derivative)**.** Let f be defined near (a, b) and let $\mathbf{u} = (u_1, u_2)$ be a unit vector. The **directional derivative** of f at (a, b) in the direction $\mathbf{u}$ is

$$D_{\mathbf{u}} f(a, b) \;=\; \lim_{t \to 0} \frac{f(a + tu_1,\ b + tu_2) - f(a, b)}{t},$$

provided the limit exists. It is the instantaneous rate of change of f per unit distance traveled in the direction $\mathbf{u}$.

Figure 33.1 shows the picture behind this definition: the walk in the plane, the height curve $g(t)$ above it, and the tangent slope $g'(0)$ that the limit computes. Notice the family resemblance to the ordinary derivative: numerator is "new height minus old height," denominator is "distance moved," and we let the step shrink to zero. The only new feature is that "distance moved" is now measured along a chosen compass heading.

Intuition

The partial derivatives are special directional derivatives. Take $\mathbf{u} = (1, 0)$ – walk due east – and the definition collapses to exactly the limit defining $\frac{\partial f}{\partial x}$. Take $\mathbf{u} = (0, 1)$ and you get $\frac{\partial f}{\partial y}$. So $D_{\mathbf{u}} f$ is not a new species of derivative; it is the *whole family* of slopes, one for every direction on the compass, and the two partials are the two members of that family pointing along the axes.

Common trap

The direction vector *must* be a unit vector, $\|\mathbf{u}\| = 1$. If you feed in a vector of length 3, the limit measures change per "unit of that vector," which is change over 3 actual steps – inflating your answer by a factor of 3. Whenever someone hands you a direction like $(3, 4)$, normalize it first: $\mathbf{u} = \frac{1}{5}(3, 4)$. We will see in a moment that the clean formula $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$ only measures a true slope when $\mathbf{u}$ has length one.

33.1.1 A formula instead of a limit

Computing that limit by hand for every direction would be miserable. Happily we never have to. The function $g(t) = f(a + tu_1,\, b + tu_2)$ is a composition: an outer function f of two variables, fed by two inner functions $x(t) = a + tu_1$ and $y(t) = b + tu_2$ of the single variable t. We are about to prove a chain rule for exactly this situation, but it is worth seeing where it leads first.

If f is nice enough (we make "nice enough" precise in Section 33.4), the rate $g'(0)$ turns out to be

$$g'(0) \;=\; \frac{\partial f}{\partial x}\Big|_{(a,b)} \cdot \frac{\mathrm{d}x}{\mathrm{d}t}\Big|_0 + \frac{\partial f}{\partial y}\Big|_{(a,b)} \cdot \frac{\mathrm{d}y}{\mathrm{d}t}\Big|_0 \;=\; f_x(a,b)\,u_1 + f_y(a,b)\,u_2.$$

Read that result slowly. The slope in direction $\mathbf{u}$ is a *weighted combination* of the two axis-slopes, weighted by how much of your step points east (u_1) and how much points north (u_2). If you head straight east, $u_1 = 1$, $u_2 = 0$, and you recover f_x. If you head northeast, you get a half-and-half blend (after normalizing). This is the linearity we have been promised all along: near a point, f behaves like a flat plane, and the slope of a plane in a diagonal direction is just the mix of its slopes along the two axes.

Focus

The directional derivative formula. For a differentiable f and a unit vector $\mathbf{u} = (u_1, u_2)$,

$$\boxed{D_{\mathbf{u}} f(a,b) = f_x(a,b)\,u_1 + f_y(a,b)\,u_2.}$$

The right-hand side is a dot product of two vectors: (f_x, f_y) and (u_1, u_2). The first of those vectors is so important it earns a name of its own – the gradient – which is the subject of the next section.

Example 33.1 (A first directional derivative)**.** Let $f(x,y) = x^2 + xy$ and find the rate of change at $(1,2)$ in the direction toward $(4,6)$.

First the partials: $f_x = 2x + y$ and $f_y = x$. At $(1,2)$ these are $f_x(1,2) = 2(1)+2 = 4$ and $f_y(1,2) = 1$.

Next the direction. The displacement from $(1,2)$ to $(4,6)$ is $(3,4)$, of length $\sqrt{3^2+4^2} = 5$, so the unit direction is $\mathbf{u} = \left(\frac{3}{5}, \frac{4}{5}\right)$.

Now the formula:

$$D_{\mathbf{u}} f(1,2) = f_x\,u_1 + f_y\,u_2 = 4 \cdot \tfrac{3}{5} + 1 \cdot \tfrac{4}{5} = \tfrac{12}{5} + \tfrac{4}{5} = \tfrac{16}{5} = 3.2.$$

So if you stand at $(1,2)$ and walk toward $(4,6)$, the value of f climbs at 3.2 units per unit distance. Had we forgotten to normalize and used $(3,4)$ directly, we would have gotten $4 \cdot 3 + 1 \cdot 4 = 16$ – exactly 5 times too big, the length of the vector we failed to divide out.

Work it yourself

Take $f(x,y) = x^2 + xy$ from Example 33.1 and compute $D_{\mathbf{u}} f(1,2)$ *from the raw limit*, using $\mathbf{u} = \left(\frac{3}{5}, \frac{4}{5}\right)$ and a tiny step $t = 0.001$:

$$\frac{f(1 + 0.001 \cdot 0.6,\; 2 + 0.001 \cdot 0.8) - f(1,2)}{0.001}.$$

Work out $f(1.0006,\, 2.0008)$ and $f(1,2) = 3$ on paper, subtract, divide by 0.001. You should

land very close to 3.2. Watching the limit and the formula agree – once – is worth more than ten formula-only exercises.

33.2 The Gradient Vector

We just discovered that every directional derivative is a dot product of $\mathbf{u}$ with the same fixed vector (f_x, f_y). That vector deserves its own name.

Definition 33.2 (The gradient)**.** For a function $f(x, y)$ with partial derivatives at (a, b), the **gradient** of f at (a, b) is the vector

$$\nabla f(a,b) \;=\; \Big(\frac{\partial f}{\partial x}\Big|_{(a,b)},\; \frac{\partial f}{\partial y}\Big|_{(a,b)}\Big) \;=\; \big(f_x(a,b),\, f_y(a,b)\big).$$

The symbol ∇ is read "del" or "nabla." For a function of n variables the gradient is the n-vector of all the first partials, $\nabla f = (f_{x_1}, f_{x_2}, \ldots, f_{x_n})$.

With this name in hand the directional-derivative formula becomes a one-liner:

$$\boxed{D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}.}$$

Every slope on the compass is obtained by dotting the single vector ∇f against the direction you choose to walk. The gradient is a complete summary of the local steepness of f in *all* directions at once – a remarkable compression. Two numbers, f_x and f_y, encode infinitely many slopes.

What is really happening

The gradient is not a slope; it is the machine that makes slopes. A beginner often thinks "the gradient is how steep the function is." Not quite. Steepness is a single number and it depends on which way you face. The gradient is a *vector*, and what it really is, is the linear map that eats a direction and returns the slope that way. That is the deep meaning of the dot product $\nabla f \cdot \mathbf{u}$: the gradient is the row of coefficients of the best linear approximation to f, and dotting it with $\mathbf{u}$ evaluates that linear map in the direction $\mathbf{u}$. When in later study you meet "the derivative of a multivariable function is a linear map," the gradient is that map, written as a vector.

33.2.1 The direction of steepest ascent

Now comes the payoff that makes the gradient famous. We have $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$, and we want to know: *of all the directions* $\mathbf{u}$ *I could walk, which one makes* f *increase fastest?* That is, which unit vector $\mathbf{u}$ maximizes the dot product $\nabla f \cdot \mathbf{u}$?

Recall the geometric meaning of the dot product. For any two vectors,

$$\nabla f \cdot \mathbf{u} = \|\nabla f\|\,\|\mathbf{u}\| \cos\theta = \|\nabla f\| \cos\theta,$$

since $\|\mathbf{u}\| = 1$, where θ is the angle between ∇f and your direction $\mathbf{u}$. The only thing you control here is θ, and the expression is largest exactly when $\cos\theta = 1$, i.e. when $\theta = 0$ – when $\mathbf{u}$ points the *same way* as ∇f. At that choice the climb rate is $\|\nabla f\|$. This is the whole story:

Focus

Steepest ascent. At any point where $\nabla f \neq \mathbf{0}$:

- The direction of *steepest ascent* is $\mathbf{u} = \nabla f / \|\nabla f\|$ – straight along the gradient.
- The maximum rate of increase is $\|\nabla f\|$ – the length of the gradient.
- The direction of *steepest descent* is the opposite, $-\nabla f$, with rate $-\|\nabla f\|$.
- In any direction *perpendicular* to ∇f ($\theta = 90°$, $\cos\theta = 0$), the rate of change is *zero*: you are walking along a level curve, neither climbing nor descending.

Intuition

Why should one special direction be steepest? Because slope in direction $\mathbf{u}$ is the *projection* of the gradient onto $\mathbf{u}$. Imagine the gradient as an arrow lying on your tabletop. As you swing your walking direction $\mathbf{u}$ around the compass, the slope you feel is the shadow that the gradient arrow casts onto your line of travel. The shadow is longest when you face straight along the arrow, shrinks as you turn away, and vanishes when you face perpendicular to it. There is nothing mysterious here once you see it as a shadow: a fixed arrow has its longest shadow along its own direction.

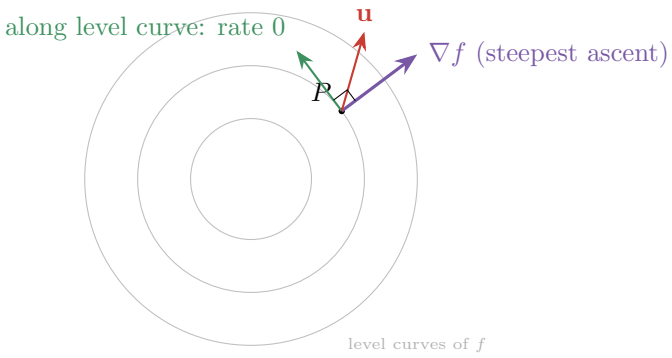


Figure 33.2: At a point P, the gradient (purple) points perpendicular to the level curve through P and aims along the steepest uphill direction. Walking along the level curve (green) keeps you at the same height: rate of change zero. Any other direction $\mathbf{u}$ (red) gives a slope between 0 and $\|\nabla f\|$.

Example 33.2 (Finding the steepest direction)**.** Let $f(x, y) = x^2 + y^2$, a round bowl, and consider the point $(3, 1)$.

The gradient is $\nabla f = (2x, 2y)$, so $\nabla f(3, 1) = (6, 2)$. The steepest uphill direction is along $(6, 2)$; normalized that is $\frac{1}{\sqrt{40}}(6, 2) = \frac{1}{2\sqrt{10}}(6, 2)$. The steepest climb rate is $\|(6, 2)\| = \sqrt{40} = 2\sqrt{10} \approx 6.32$ units per unit distance. To descend fastest – to roll the ball downhill – head in the direction $(-6, -2)$, straight back toward the bottom of the bowl at the origin. That matches intuition exactly: on a round bowl the steepest way down points at the center.

Walking question

You are hiking and the fog rolls in; you can see only the patch of ground at your feet. You want to reach the summit as fast as possible. You cannot see the peak, but you *can* feel, with each tentative step, which way the ground rises most steeply – that is the gradient at your feet. If you always step in the locally steepest uphill direction, are you guaranteed to reach the true summit? Think about a landscape with several hills. (This is exactly the promise and the peril of gradient ascent, which we revisit in Section 33.6.)

33.3 Why the Gradient Is Perpendicular to Level Curves

The single most useful geometric fact about the gradient is that it stands at a right angle to the level curves of f. We stated it above; now let us *see* why it must be true, two ways: first with a picture and a one-line argument, then with the chain rule for full rigor.

A **level curve** of f is the set of points where f takes one fixed value: $f(x, y) = c$. On a topographic map these are the contour lines; on a weather map, the isobars; on the round bowl $f = x^2 + y^2$, they are circles. Walking along a level curve, by definition, keeps f constant – your height never changes.

Why the technique works

Why $\nabla f \perp$ level curves. Walking along a level curve keeps f constant, so the rate of change of f in the direction $\mathbf{u}$ *tangent to the level curve* is zero:

$$D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u} = 0.$$

But a dot product is zero precisely when the two vectors are perpendicular. Therefore ∇f is perpendicular to the tangent of the level curve – that is, perpendicular to the level curve itself. The mechanism is exactly the steepest ascent picture run sideways: the directions of *zero* change are perpendicular to the gradient, and those are exactly the directions that keep you on a contour. Steepest ascent is therefore perpendicular to "no change," which is why the gradient crosses every contour at a right angle.

Let us make the "rate is zero along a level curve" step airtight using the chain rule, which also previews the next section. Parametrize the level curve as a path $\mathbf{r}(t) = (x(t), y(t))$ that stays on it, so $f(x(t), y(t)) = c$ for all t. Differentiate both sides with respect to t. The left side is a composition; its derivative (by the chain rule we prove in Section 33.4) is

$$\frac{\mathrm{d}}{\mathrm{d}t} f\big(x(t), y(t)\big) = f_x\, x'(t) + f_y\, y'(t) = \nabla f \cdot \mathbf{r}\,'(t).$$

The right side is the constant c, whose derivative is 0. Hence

$$\nabla f \cdot \mathbf{r}\,'(t) = 0.$$

But $\mathbf{r}\,'(t)$ is the velocity vector – it points *along* the level curve, tangent to it. A zero dot product with the tangent means ∇f is perpendicular to the curve at every point. Done.

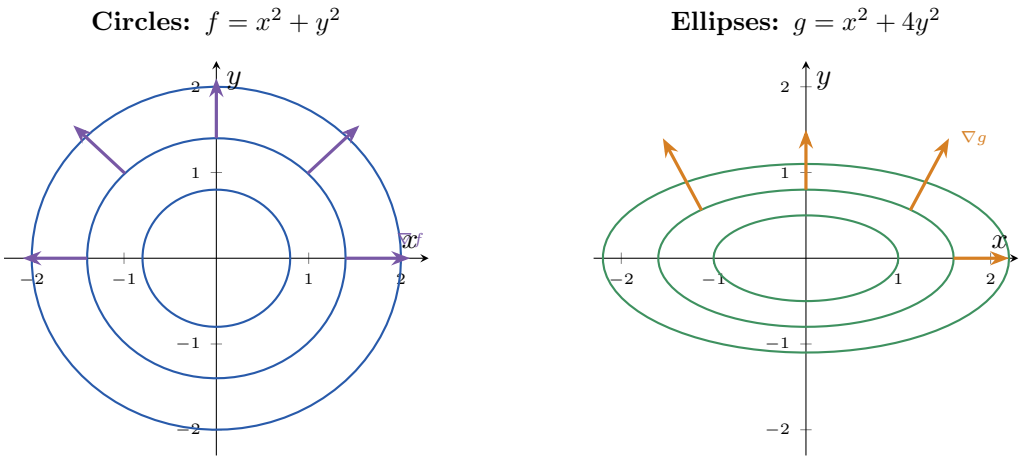


Figure 33.3: The gradient vector is strictly orthogonal to level curves everywhere. Left: For the round bowl $f(x, y) = x^2 + y^2$, level sets are circles and the gradient $\nabla f = (2x, 2y)$ radiates perpendicularly outward. Right: For the elongated elliptic paraboloid $g(x, y) = x^2 + 4y^2$, the contours are ellipses; the gradient $\nabla g = (2x, 8y)$ tilts more steeply toward the narrow y-axis, crossing the tangent to each ellipse at a precise 90° angle. Orthogonality holds universally.

Why this matters

This perpendicularity is why the gradient is the workhorse of so much applied mathematics. In physics, force is (minus) the gradient of potential energy, and so forces point "straight downhill" on the energy landscape, perpendicular to the surfaces of constant energy. In image processing, the gradient of brightness points across edges, perpendicular to them – which is exactly how an algorithm *finds* edges. In optimization, the gradient tells you the one direction that changes your objective fastest. One geometric fact – steepest change is perpendicular to no-change – echoes through all of these.

Connection

In three variables the same story holds with "curve" upgraded to "surface." The level *sets* of $g(x, y, z)$ are surfaces $g = c$, and ∇g is perpendicular to the surface – it is the surface's *normal vector*. That is the quickest way to find the tangent plane to a surface like a sphere $x^2 + y^2 + z^2 = R^2$: its gradient $(2x, 2y, 2z)$ at a point is the radial direction, so the tangent plane is perpendicular to the radius. You may have proved that in geometry the hard way; the gradient hands it to you for free.

33.4 The Multivariable Chain Rule

We have leaned on a chain rule twice now without proving it. Time to pay the debt – and the payoff is large, because the multivariable chain rule is the single computational engine behind directional derivatives, the perpendicularity proof, implicit differentiation in several variables, and (as we will see) backpropagation in neural networks.

33.4.1 One inner parameter: a path through a surface

Start with the simplest case, the one we keep using. Suppose $z = f(x, y)$, but x and y are themselves functions of a single parameter t – think of t as time, and $(x(t), y(t))$ as the position of a bug crawling across the floor while f measures the temperature of the floor. The bug's temperature over time is $F(t) = f(x(t), y(t))$. How fast is the bug heating up?

Theorem 33.1 (Chain rule, single parameter)**.** *If f is differentiable at $(x(t_0), y(t_0))$ and x, y are differentiable at t_0, then $F(t) = f(x(t), y(t))$ is differentiable at t_0 and*

$$\frac{\mathrm{d}F}{\mathrm{d}t} = \frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t} \;=\; \nabla f \cdot \mathbf{r}\,'(t),$$

where $\mathbf{r}(t) = (x(t), y(t))$.

Intuition

The formula is just bookkeeping of two contributions. The temperature changes for two reasons at once: the bug is moving east at rate $x'(t)$ into ground whose temperature rises eastward at rate f_x, contributing $f_x\, x'$; and it is moving north at rate $y'(t)$ into ground rising northward at rate f_y, contributing $f_y\, y'$. Total rate of warming = sum of the two. The single-variable chain rule was "outer slope times inner slope"; the multivariable one is "add up outer slope times inner slope over every input channel."

Why the technique works

Why the chain rule looks like this: local flatness. Near (x_0, y_0) the surface f is, to first order, a flat plane:

$$f(x_0 + \Delta x,\, y_0 + \Delta y) \approx f(x_0, y_0) + f_x\, \Delta x + f_y\, \Delta y.$$

That is the entire content of differentiability: the change in f is *linear* in the change of the inputs, with coefficients f_x and f_y. Now let the inputs change because t changed by Δt. To first order $\Delta x \approx x'(t)\, \Delta t$ and $\Delta y \approx y'(t)\, \Delta t$. Substitute:

$$\Delta f \approx f_x\, x'(t)\, \Delta t + f_y\, y'(t)\, \Delta t.$$

Divide by Δt and let $\Delta t \to 0$: the approximations become exact and you get the chain rule. The whole proof is "f is locally a plane, and the slope of a plane responds linearly to a moving input." Every chain rule in mathematics is this same idea: compose two linear approximations and the linear parts multiply (here, dot) together.

Proof of Theorem 33.1. Differentiability of f at $(x_0, y_0) = (x(t_0), y(t_0))$ means precisely that we can write the increment as its linear part plus an error that is small compared to the step:

$$f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) = f_x\, \Delta x + f_y\, \Delta y + \varepsilon_1\, \Delta x + \varepsilon_2\, \Delta y,$$

where $\varepsilon_1, \varepsilon_2 \to 0$ as $(\Delta x, \Delta y) \to (0, 0)$. Now take $\Delta x = x(t_0 + h) - x(t_0)$ and $\Delta y = y(t_0 + h) - y(t_0)$, the actual changes caused by bumping t_0 to $t_0 + h$. Divide the whole equation by h:

$$\frac{F(t_0 + h) - F(t_0)}{h} = f_x\, \frac{\Delta x}{h} + f_y\, \frac{\Delta y}{h} + \varepsilon_1\, \frac{\Delta x}{h} + \varepsilon_2\, \frac{\Delta y}{h}.$$

As $h \to 0$: differentiability of x and y gives $\Delta x/h \to x'(t_0)$ and $\Delta y/h \to y'(t_0)$, both finite. Also $(\Delta x, \Delta y) \to (0,0)$ (because x, y are continuous, being differentiable), so $\varepsilon_1, \varepsilon_2 \to 0$. The last two terms are (something $\to 0$) times (something finite), hence vanish. The survivors give

$$F'(t_0) = f_x\, x'(t_0) + f_y\, y'(t_0).$$ □

Common trap

The hypothesis "f is *differentiable*," not merely "f has partial derivatives," is essential and is a genuine pitfall in several variables. A function can have both partials existing at a point and yet not be well approximated by any plane there – in which case the chain rule formula can *fail*. (The standard cautionary example is $f(x,y) = \frac{xy}{x^2+y^2}$ at the origin, extended by $f(0,0) = 0$: both partials exist and equal 0, but f is not even continuous there.) In one variable, "derivative exists" and "differentiable" are the same thing, so this trap has no analogue; it is one of the first real surprises of higher-dimensional calculus.

Example 33.3 (Warming bug)**.** Let the floor temperature be $f(x,y) = x^2 + 3xy$ and let a bug move along $x(t) = t$, $y(t) = t^2$. Find $\frac{\mathrm{d}F}{\mathrm{d}t}$ at $t = 1$.

Partials: $f_x = 2x + 3y$, $f_y = 3x$. Inner derivatives: $x'(t) = 1$, $y'(t) = 2t$. So

$$\frac{\mathrm{d}F}{\mathrm{d}t} = (2x + 3y)(1) + (3x)(2t).$$

At $t = 1$ we have $x = 1$, $y = 1$, giving $(2+3)(1) + (3)(2) = 5 + 6 = 11$.

Sanity check by direct substitution. Along the path, $F(t) = f(t, t^2) = t^2 + 3t \cdot t^2 = t^2 + 3t^3$, so $F'(t) = 2t + 9t^2$, and $F'(1) = 2 + 9 = 11$. The chain rule and brute force agree – as they must.

33.4.2 Several inner variables: the general chain rule

Often the inner variables depend on *several* parameters. Suppose $z = f(x,y)$ but $x = x(s,t)$ and $y = y(s,t)$ depend on two parameters s and t. Then z is ultimately a function of s and t, and we want its partial derivatives. The principle is identical – add up outer-slope times inner-slope over every channel – applied one parameter at a time (hold the other fixed):

$$\frac{\partial z}{\partial s} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial s}, \qquad \frac{\partial z}{\partial t} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial t}.$$

Intuition

A *dependency tree* keeps this organized. Draw z at the top with two branches down to x and y; from each of those draw two more branches down to s and t. To find $\frac{\partial z}{\partial s}$, trace *every* path from z down to s, multiply the partial derivatives you cross along each path, and add the paths together. Two paths reach s ($z \to x \to s$ and $z \to y \to s$), giving the two terms above. "Multiply along paths, add across paths" is the chain rule for any dependency diagram, no matter how tangled – and it is exactly how automatic differentiation libraries organize backpropagation.

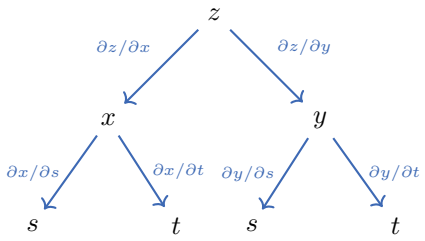


Figure 33.4: The dependency tree for $z = f(x, y)$ with x, y depending on s, t. To get $\frac{\partial z}{\partial s}$, follow each downward path that ends at s (the two leftish ones), multiply the labels along the path, and add the results.

Example 33.4 (Polar coordinates)**.** A frequent use: $f(x, y)$ with $x = r\cos\theta$, $y = r\sin\theta$. Find $\frac{\partial f}{\partial r}$.

Here the parameters are r, θ. The inner partials are $\frac{\partial x}{\partial r} = \cos\theta$ and $\frac{\partial y}{\partial r} = \sin\theta$. So

$$\frac{\partial f}{\partial r} = f_x \cos\theta + f_y \sin\theta.$$

Look closely: $(\cos\theta, \sin\theta)$ is the outward radial *unit* vector. So $\frac{\partial f}{\partial r} = \nabla f \cdot \hat{\mathbf{r}}$ is just the directional derivative of f in the radial direction – which is exactly what "rate of change as r increases" should mean. The chain rule and the directional derivative are the same machine wearing different clothes.

33.5 Linear Approximation in Several Variables: the Tangent Plane

In one variable, the derivative gave us the tangent *line*, the best straight approximation to a curve near a point. In two variables the gradient gives us the tangent *plane*, the best flat approximation to a surface near a point. The gradient is precisely the recipe for that plane.

Near (a, b), differentiability says the change in f is approximately linear in the changes of the inputs:

$$f(x, y) \approx f(a, b) + f_x(a, b)\,(x - a) + f_y(a, b)\,(y - b).$$

The right-hand side, viewed as a function of (x, y), is the equation of a plane – the **tangent plane** to the graph $z = f(x, y)$ at the point $(a, b, f(a, b))$. Using the gradient we can write it as a single dot product:

$$\boxed{L(x, y) = f(a, b) + \nabla f(a, b) \cdot \big((x, y) - (a, b)\big).}$$

What is really happening

What "differentiable" really means in several variables. It is tempting to define differentiability as "all partial derivatives exist," but that definition is too weak, as the trap above showed. The right definition is the one that matches the picture: f is differentiable at (a, b) if there is a plane L through $(a, b, f(a, b))$ such that the error $f - L$ shrinks *faster than the distance* to (a, b):

$$\lim_{(x,y)\to(a,b)} \frac{f(x, y) - L(x, y)}{\|(x, y) - (a, b)\|} = 0.$$

This is the exact two-variable echo of "smooth means locally straight": smooth now means *locally flat*, well approximated by a plane, with the error negligible compared to how far you

have stepped. When that holds, the coefficients of L are forced to be the partial derivatives, and the gradient is born. The gradient exists as a meaningful object precisely because the surface has a tangent plane.

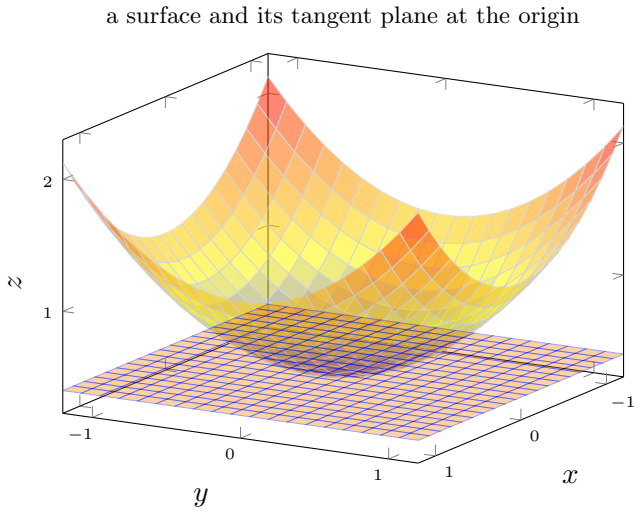


Figure 33.5: The bowl $z = 0.6x^2 + 0.6y^2 + 0.4$ and its tangent plane at the bottom point $(0, 0, 0.4)$. Because $\nabla f(0, 0) = (0, 0)$, the tangent plane is horizontal – the flat "floor" that the bowl melts into when you zoom in on its lowest point. At a slanted point the plane would tilt to match.

Example 33.5 (Estimating a value without a calculator)**.** Estimate $\sqrt{(3.02)^2 + (3.97)^2}$ using a tangent plane.

Let $f(x, y) = \sqrt{x^2 + y^2}$ and anchor at the friendly point $(a, b) = (3, 4)$, where $f(3, 4) = \sqrt{9 + 16} = 5$ exactly. The partials are

$$f_x = \frac{x}{\sqrt{x^2 + y^2}}, \qquad f_y = \frac{y}{\sqrt{x^2 + y^2}},$$

so $f_x(3, 4) = \frac{3}{5}$ and $f_y(3, 4) = \frac{4}{5}$; thus $\nabla f(3, 4) = (\frac{3}{5}, \frac{4}{5})$. The steps are $\Delta x = 3.02 - 3 = 0.02$ and $\Delta y = 3.97 - 4 = -0.03$. The linear approximation gives

$$f(3.02, 3.97) \approx 5 + \tfrac{3}{5}(0.02) + \tfrac{4}{5}(-0.03) = 5 + 0.012 - 0.024 = 4.988.$$

A calculator gives $\sqrt{3.02^2 + 3.97^2} = \sqrt{9.1204 + 15.7609} = \sqrt{24.8813} \approx 4.98812$. Our tangent-plane estimate 4.988 is correct to four significant figures (three decimal places), from arithmetic you can do in your head – because near $(3, 4)$ the surface is practically its tangent plane.

Connection

This is the engine of *error propagation* in the lab. If a quantity $f(x, y)$ is computed from two measurements x, y with small uncertainties $\Delta x, \Delta y$, then to first order the uncertainty in f is $\Delta f \approx f_x\,\Delta x + f_y\,\Delta y$. The partial derivatives are the *sensitivities*: they say how strongly each input's error feeds through to the output. The gradient is the vector of sensitivities. Scientists who have never heard the word "gradient" use this formula every time they write "$\pm$" after a result.

33.6 Gradient Descent: How Machines Learn

We have all the ingredients for one of the most consequential algorithms of our century. Here is the entire idea in one breath: *to minimize a function, keep stepping in the direction of steepest descent* – that is, in the direction of $-\nabla f$ – and you will walk downhill toward a valley.

Application

Gradient descent in machine learning. A machine-learning model has a pile of adjustable numbers called *parameters* (or weights), which we collect into a vector $\mathbf{w} = (w_1, \dots, w_n)$ – a modern network has millions or billions of them. A *loss function* $L(\mathbf{w})$ measures how badly the model does on the training data: big loss means bad predictions, small loss means good ones. Training the model means *minimizing* L over the weights.
We cannot see this n-dimensional landscape, and we certainly cannot solve $\nabla L = \mathbf{0}$ by hand for a billion variables. But we can do what the foggy hiker of the earlier walkbox does. At our current weights, we compute the gradient $\nabla L(\mathbf{w})$ – the direction in weight-space that would increase the loss fastest – and we step the *opposite* way:

$$\boxed{\mathbf{w}_{\text{new}} = \mathbf{w}_{\text{old}} - \eta \, \nabla L(\mathbf{w}_{\text{old}}).}$$

The number $\eta > 0$ is the **learning rate**, the size of the step. Repeat millions of times and the weights slide downhill into a valley of low loss – the model has *learned*. The gradient here is computed by the multivariable chain rule applied layer by layer through the network; that organized application of the chain rule is exactly what *backpropagation* is. Everything in this chapter – gradient as steepest direction, chain rule as the way to compute it – is precisely the mathematics that trains modern AI.

Intuition

Why step *against* the gradient? Because $-\nabla L$ is the direction of steepest *descent* (Section 33.2): of all directions you could step, it lowers the loss the fastest. And why multiply by a small η? Because the gradient is only the slope *at your current point*; trust it only for a short step. The linear approximation $L(\mathbf{w} - \eta \nabla L) \approx L(\mathbf{w}) - \eta \, \|\nabla L\|^2$ shows that as long as η is small and $\nabla L \neq \mathbf{0}$, the loss strictly decreases: the term $-\eta \|\nabla L\|^2$ is negative. That single inequality is the mathematical guarantee that a small gradient step makes progress.

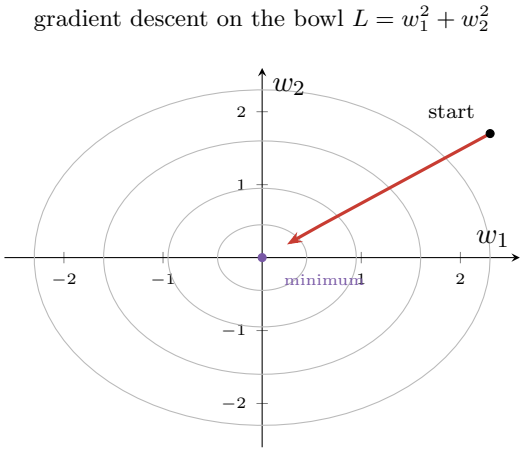


Figure 33.6: Gradient descent on $L = w_1^2 + w_2^2$. Each red step points opposite the gradient (perpendicular to the contour, straight toward the center) and is scaled by the learning rate. The iterates march into the minimum at the origin.

Example 33.6 (One descent step by hand)**.** Minimize $L(w_1, w_2) = w_1^2 + w_2^2$ starting at $\mathbf{w}_0 = (2, 3)$ with learning rate $\eta = 0.1$.

The gradient is $\nabla L = (2w_1, 2w_2)$, so $\nabla L(2, 3) = (4, 6)$. One step:

$$\mathbf{w}_1 = (2, 3) - 0.1\,(4, 6) = (2 - 0.4,\ 3 - 0.6) = (1.6,\ 2.4).$$

The loss dropped from $L(2, 3) = 4 + 9 = 13$ to $L(1.6, 2.4) = 2.56 + 5.76 = 8.32$. We moved toward the origin – the true minimum – as expected. Notice each coordinate got multiplied by $1 - 2\eta = 0.8$; iterating, the weights shrink geometrically to zero, which is gradient descent solving this bowl exactly.

Common trap

The learning rate is a genuine knife-edge. Too *small* and progress is glacial – thousands of tiny steps. Too *large* and you overshoot the valley and bounce up the far wall; the loss can *increase* and even diverge. In Example 33.6 the update multiplied each weight by $1 - 2\eta$. That factor has magnitude below 1 (so the steps converge) only when $0 < \eta < 1$; at $\eta = 1$ the weights flip sign and never shrink, and for $\eta > 1$ they blow up. Choosing η well is one of the central practical arts of training models, and it is, at heart, the trapbox warning about trusting a linear approximation too far.

Walking question

Gradient descent only ever steps downhill, so it can get *stuck* in a valley that is not the lowest valley – a *local* minimum. Picture a landscape with two basins of different depths (Figure 33.7); depending on where you start, you slide into one or the other and stop. Why, then, does gradient descent work so astonishingly well for training enormous neural networks, where the landscape has untold many local minima? (A surprising partial answer modern research offers: in very high dimensions, most of those resting points are nearly as good as the best, and true bad traps are rare. The intuition you build here is the doorway to that research.)

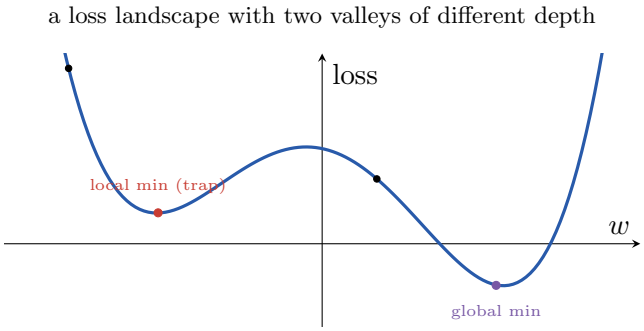


Figure 33.7: Gradient descent only steps downhill, so where you start decides where you stop. A ball released on the far left slides into the shallow left valley (a *local* minimum) and halts; one released just right of center rolls into the deeper right valley (the *global* minimum). Both are critical points where the slope is zero, but only one is the true bottom.

33.7 Finding and Classifying Critical Points: the Hessian

To find the peaks and valleys of a surface, we look for the places where it is momentarily flat – where the tangent plane is horizontal. In one variable we set $f'(x) = 0$; in several variables we set the entire gradient to zero.

Definition 33.3 (Critical point)**.** A point (a, b) is a **critical point** of f if $\nabla f(a, b) = \mathbf{0}$, that is $f_x(a, b) = 0$ and $f_y(a, b) = 0$ (or if the gradient fails to exist). At such a point the tangent plane is horizontal.

A horizontal tangent plane is necessary for a local max or min – if the ground sloped at all, you could go uphill one way and downhill the other, so it would be neither a top nor a bottom. But "flat" alone does not tell you *which*. There are three flavors of critical point, and the new one has no analogue in a single variable.

- **Local minimum:** the surface curves *up* in every direction – a bowl. You are at the bottom of a valley.
- **Local maximum:** the surface curves *down* in every direction – a dome. You are at a summit.
- **Saddle point:** the surface curves up in some directions and down in others – like a mountain pass, or the seat of a horse saddle. Up along the horse's spine, down toward the stirrups. This is genuinely new: a point can be flat yet be neither a max nor a min.

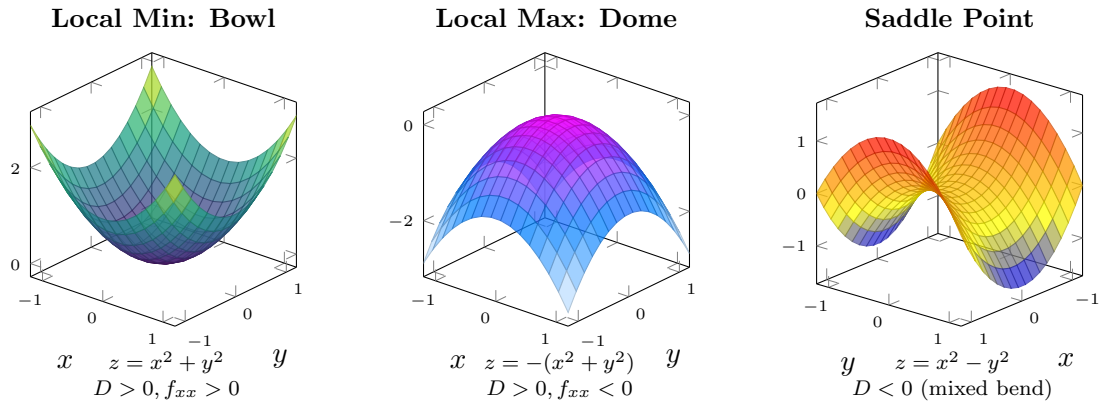


Figure 33.8: The three geometric archetypes of critical points in two variables. Left: A local minimum ($z = x^2 + y^2$) where curvature is positive in every direction ($D > 0, f_{xx} > 0$). Center: A local maximum ($z = -(x^2 + y^2)$) where curvature is negative in every direction ($D > 0, f_{xx} < 0$). Right: A saddle point ($z = x^2 - y^2$) curving upward along the x-axis and downward along the y-axis ($D < 0$), producing a horizontal tangent plane at $(0, 0)$ that is neither a peak nor a valley.

33.7.1 The second-derivative test, intuitively

How do we tell the three apart without plotting? By looking at *curvature* – the second derivatives. In one variable, after $f'(a) = 0$, the sign of $f''(a)$ decided max ($f'' < 0$, concave down) versus min ($f'' > 0$, concave up). In two variables there are *three* second partials to track,

$$f_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right), \qquad f_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right), \qquad f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right),$$

(and for the functions we meet, $f_{xy} = f_{yx}$, the mixed partials agree). We arrange them in a 2×2 grid called the **Hessian matrix**:

$$H = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{pmatrix}.$$

The Hessian is the multivariable second derivative: it records the curvature of the surface in every direction, just as the gradient recorded the slope in every direction. The quantity that decides the verdict is its **determinant**,

$$D = f_{xx} f_{yy} - (f_{xy})^2.$$

Focus

The second-derivative (Hessian) test. At a critical point (a, b), let $D = f_{xx} f_{yy} - f_{xy}^2$, all evaluated at (a, b).

- If $D > 0$ and $f_{xx} > 0$: **local minimum** (a bowl, curving up).
- If $D > 0$ and $f_{xx} < 0$: **local maximum** (a dome, curving down).
- If $D < 0$: **saddle point** (curving up one way, down another).
- If $D = 0$: the test is **inconclusive** – the curvature is too gentle to decide; look closer.

Intuition

Where does $D = f_{xx}f_{yy} - f_{xy}^2$ come from, intuitively? Two things must be true for a genuine bowl. First, it must curve up along the x-direction *and* along the y-direction: $f_{xx} > 0$ and $f_{yy} > 0$. But that is not enough – a strong mixed term f_{xy} can twist the surface so that even though both axis-directions curve up, some diagonal direction curves *down*, secretly making a saddle. The determinant D is exactly the bookkeeping that catches this twist: requiring $f_{xx}f_{yy} > f_{xy}^2$ says the genuine upward curvature beats the twisting. When the twist wins ($D < 0$), you have a saddle: up some ways, down others. The sign of f_{xx} then just picks bowl ($+$) from dome ($-$). This is the two-dimensional cousin of "concave up vs concave down," with D guarding against the diagonal sabotage that one variable cannot have.

Why the technique works

Why the determinant decides it (the deeper reason). Near a critical point the gradient is zero, so the linear part of f vanishes and the *quadratic* part takes over:

$$f(a+h, b+k) \approx f(a,b) + \tfrac{1}{2}\big(f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2\big).$$

The fate of the critical point hangs entirely on the sign of the quadratic form $Q(h,k) = f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2$. If $Q > 0$ for every direction (h,k), the surface rises every way – a min. If $Q < 0$ always – a max. If Q is positive some ways and negative others – a saddle. Completing the square (assuming $f_{xx} \neq 0$),

$$Q = f_{xx}\Big(h + \tfrac{f_{xy}}{f_{xx}}k\Big)^2 + \frac{f_{xx}f_{yy} - f_{xy}^2}{f_{xx}}\,k^2,$$

and you can read off everything: the leftover coefficient on k^2 is D/f_{xx}. For Q to keep one sign, both pieces need matching signs, which forces $D > 0$; then f_{xx} sets that sign (min if $+$, max if $-$). If $D < 0$ the two squared pieces fight, so Q takes both signs – a saddle. The determinant is not a magic recipe; it is the discriminant of a quadratic form, the exact higher-dimensional heir of the f'' sign test.

Example 33.7 (Classifying every critical point)**.** Classify the critical points of $f(x,y) = x^3 - 3x + y^2$.

Step 1: find them. $f_x = 3x^2 - 3 = 3(x^2-1)$ and $f_y = 2y$. Setting both to zero: $f_y = 0$ gives $y = 0$, and $f_x = 0$ gives $x^2 = 1$, so $x = \pm 1$. Two critical points: $(1,0)$ and $(-1,0)$.

Step 2: second partials. $f_{xx} = 6x$, $f_{yy} = 2$, $f_{xy} = 0$. So $D = f_{xx}f_{yy} - f_{xy}^2 = (6x)(2) - 0 = 12x$.

Step 3: apply the test.

- At $(1,0)$: $D = 12(1) = 12 > 0$ and $f_{xx} = 6(1) = 6 > 0$. *Local minimum.*
- At $(-1,0)$: $D = 12(-1) = -12 < 0$. *Saddle point.*

This fits the shape of f: in x it is a cubic (one local min at $x = 1$, one local max at $x = -1$), and in y it is the upward parabola y^2. At $x = 1$ both directions curve up – a true bowl. At $x = -1$ the x-cubic curves down while the y-parabola curves up – up one way, down the other, a saddle. The Hessian test read all of this off three derivatives.

Example 33.8 (A clean maximum)**.** Classify the critical point of $f(x,y) = 4 - x^2 - y^2 + 2x$.

$f_x = -2x + 2 = 0 \Rightarrow x = 1$; $f_y = -2y = 0 \Rightarrow y = 0$. One critical point $(1, 0)$. Second partials: $f_{xx} = -2$, $f_{yy} = -2$, $f_{xy} = 0$, so $D = (-2)(-2) - 0 = 4 > 0$ with $f_{xx} = -2 < 0$: a *local maximum*. Indeed $f = 5 - (x-1)^2 - y^2$ after completing the square, an upside-down bowl with summit 5 at $(1, 0)$ – exactly what the test reported.

33.8 Constrained Optimization and Lagrange Multipliers

Up to this point, our search for extreme values has operated without boundaries: we roamed across the entire xy-plane seeking summits, pits, or saddles where the gradient vanished, $\nabla f(x, y) = \mathbf{0}$. In physical, statistical, and engineering applications, however, variables are rarely free to wander without restriction. A structural engineer must maximize load-bearing capacity subject to a strict weight budget; an economist seeks to maximize utility under a fixed financial ceiling; a machine learning algorithm minimizes empirical risk subject to a spherical parameter constraint $\|\mathbf{w}\|^2 = 1$.

In single-variable calculus (Chapters 13 and 17), we attacked constrained problems through algebraic substitution: given an objective function $f(x, y)$ and a constraint equation $g(x, y) = c$, we attempted to solve the constraint explicitly for one variable in terms of the other—say, $y = h(x)$—and substitute it directly into $f(x, h(x))$, reducing the problem to single-variable critical points where $\frac{\mathrm{d}f}{\mathrm{d}x} = 0$.

Both Chapters 13 and 17 pointed out that algebraic elimination is a fragile bottleneck. If the constraint is nonlinear or implicit—such as $x^5 + x + y^3 + y = 1$ or a multivariable quadratic form—solving for one coordinate in closed form is impossible. Furthermore, eliminating variables breaks the natural geometric symmetry of the problem. As promised in Chapter 17, multivariable calculus provides the geometric breakthrough: the method of *Lagrange multipliers*.

33.8.1 The Geometry of Kissing Contours

To see the geometric core of constrained optimization, step away from formulas and picture a physical landscape.

Imagine hiking on a mountainous terrain where your elevation above sea level is described by the height function $f(x, y)$. You wish to climb as high as possible. If you were free to walk in any direction across the terrain, you would follow the gradient ∇f until you reached a mountain peak where $\nabla f = \mathbf{0}$.

Now introduce the constraint: you are confined to a narrow fenced trail painted across the mountain, defined by the equation $g(x, y) = c$. You cannot step off the path; you can only walk forward or backward along it. Which point on this fenced trail attains the highest elevation?

Picture the elevation contours (level curves) $f(x, y) = k$ on the mountain slope. As you hike along the trail $g(x, y) = c$, you cross from one elevation contour to another:

- Suppose that at your current position, the trail cuts *across* a contour line $f(x, y) = k$ at an angle. Because the trail penetrates the contour line rather than running along it, stepping forward along the trail takes you uphill toward contours of higher elevation ($f > k$), while stepping backward takes you downhill toward contours of lower elevation ($f < k$).
- Consequently, any point where the trail crosses an elevation contour at a nonzero angle cannot be an extremum on the trail. The directional derivative of f along the path's unit tangent

vector $\mathbf{T}$ is nonzero:

$$D_{\mathbf{T}} f = \nabla f \cdot \mathbf{T} \neq 0.$$

Stepping in the direction of this projection takes you higher.

Where can your climb along the trail terminate? Only when moving along the trail produces zero instantaneous change in elevation. This occurs at the exact instant where the trail no longer cuts through the contours of f, but instead runs momentarily parallel to them.

At the constrained maximum (or minimum) point (x^*, y^*), the contour line $f(x, y) = k^*$ must **kiss the constraint curve** $g(x, y) = c$ **tangentially**.

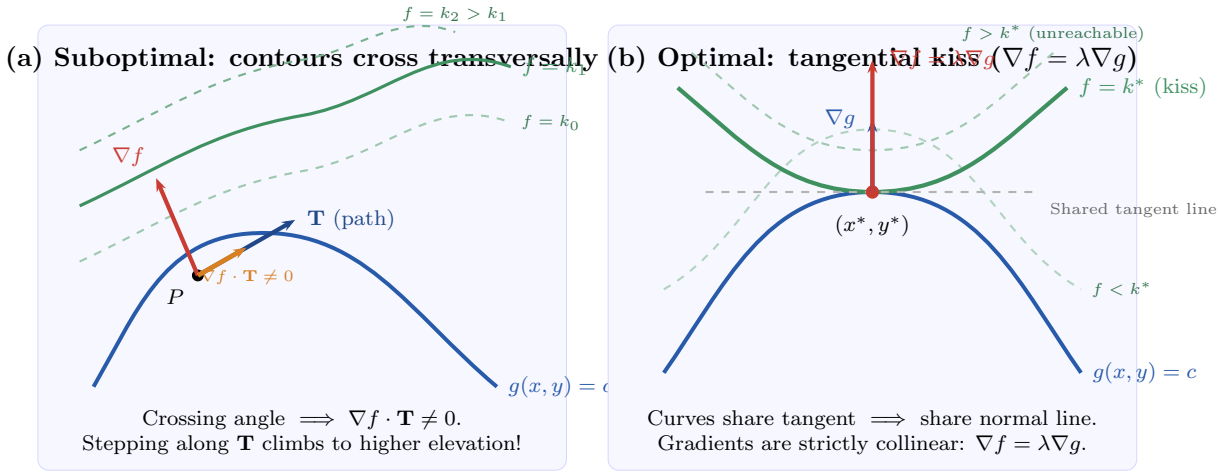


Figure 33.9: The geometric genesis of Lagrange multipliers. (a) At a suboptimal point, the elevation contour $f = k_1$ cuts across the constraint path $g(x, y) = c$; the gradient ∇f has a nonzero projection along the path, which means you can climb higher by walking along the trail. (b) At the constrained optimum (x^*, y^*), the level curve $f = k^*$ kisses the constraint path tangentially without crossing it. Because tangent lines match, their perpendicular normal vectors align along the exact same line, forcing $\nabla f(x^*, y^*) = \lambda \nabla g(x^*, y^*)$.

Now connect this geometric tangency to the property established earlier in this chapter: *the gradient vector of any differentiable function is always strictly perpendicular to its level curves* (Section 33.3).

1. The contour curve of elevation $f(x, y) = k^*$ has normal vector $\nabla f(x^*, y^*)$.
2. The constraint path $g(x, y) = c$ is itself a level curve of g, so its normal vector is $\nabla g(x^*, y^*)$.
3. Because the two curves kiss tangentially at (x^*, y^*), they share the exact same tangent line.
4. Two curves in the plane that share the same tangent line must share the same perpendicular normal line.

Therefore, the normal vectors $\nabla f(x^*, y^*)$ and $\nabla g(x^*, y^*)$ must point along the exact same line. Two vectors lie along the same line if and only if one is a scalar multiple of the other:

$$\nabla f(x^*, y^*) = \lambda \nabla g(x^*, y^*), \tag{33.1}$$

for some scalar $\lambda \in \mathbb{R}$, known as the **Lagrange multiplier** (provided $\nabla g(x^*, y^*) \neq \mathbf{0}$).

In two dimensions, expanding Equation (33.1) into components yields two scalar equations:

$$f_x(x, y) = \lambda g_x(x, y), \qquad f_y(x, y) = \lambda g_y(x, y).$$

Pairing these with the original constraint equation $g(x, y) = c$ produces a closed system of three equations in three unknowns (x, y, λ). Every candidate for a local constrained maximum or minimum must satisfy this system.

33.8.2 The Physical Meaning of λ: Shadow Price and Sensitivity

What is the physical meaning of this scalar λ? It is far more than an algebraic parameter.

Suppose the constraint budget c is relaxed by a small amount $\mathrm{d}c$—for instance, you are granted one additional unit of budget, fuel, or material. The constraint curve shifts outward, allowing you to reach a slightly higher elevation. How much does the optimal value f^* improve per unit change in the budget c?

Let $\mathbf{r}^*(c) = (x^*(c), y^*(c))$ denote the position of the constrained optimum as a function of the parameter c, and let $f^*(c) = f(\mathbf{r}^*(c))$ be the optimal value achieved. By the multivariable chain rule:

$$\frac{\mathrm{d}f^*}{\mathrm{d}c} = \nabla f(\mathbf{r}^*(c)) \cdot \frac{\mathrm{d}\mathbf{r}^*}{\mathrm{d}c}.$$

At the optimum, the gradient condition holds: $\nabla f = \lambda \nabla g$. Substituting this into the chain rule gives:

$$\frac{\mathrm{d}f^*}{\mathrm{d}c} = \lambda \nabla g(\mathbf{r}^*(c)) \cdot \frac{\mathrm{d}\mathbf{r}^*}{\mathrm{d}c} = \lambda \left(\nabla g(\mathbf{r}^*(c)) \cdot \frac{\mathrm{d}\mathbf{r}^*}{\mathrm{d}c} \right).$$

Now differentiate the constraint identity $g(\mathbf{r}^*(c)) = c$ with respect to c, again applying the multivariable chain rule:

$$\frac{\mathrm{d}}{\mathrm{d}c}\big[g(\mathbf{r}^*(c))\big] = \nabla g(\mathbf{r}^*(c)) \cdot \frac{\mathrm{d}\mathbf{r}^*}{\mathrm{d}c} = \frac{\mathrm{d}}{\mathrm{d}c}[c] = 1.$$

The dot product inside the parenthesis is identically 1. Substituting this back into our expression for $\frac{\mathrm{d}f^*}{\mathrm{d}c}$ yields:

$$\frac{\mathrm{d}f^*}{\mathrm{d}c} = \lambda. \tag{33.2}$$

The Lagrange multiplier λ is the **marginal sensitivity** or **shadow price** of the constraint:

- If $\lambda = 5$, relaxing your budget c by 0.01 will increase your optimal return by approximately $5 \times 0.01 = 0.05$.
- If $\lambda = 0$, the constraint is not actively binding at the optimum; relaxing the constraint gives no improvement because the unconstrained peak already satisfies $g(x, y) \leq c$.
- If λ is negative, increasing the budget parameter c decreases the optimal value of f.

In economics, resource allocation, and operations research, λ is often the most important quantity in the problem: it measures the marginal return of an additional unit of constrained resource.

33.8.3 Two Complete Worked Examples

Example 33.9 (Maximizing a product under a linear budget)**.** Maximize $f(x, y) = xy$ subject to the linear constraint $g(x, y) = 2x + y = 100$, where $x, y > 0$.

Step 1: Compute gradients.

$$\nabla f = \begin{pmatrix} f_x \\ f_y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}, \qquad \nabla g = \begin{pmatrix} g_x \\ g_y \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Step 2: Set up the Lagrange system $\nabla f = \lambda \nabla g$.

$$y = 2\lambda, \tag{33.3}$$
$$x = \lambda, \tag{33.4}$$
$$2x + y = 100. \tag{33.5}$$

Step 3: Eliminate λ to find the optimal ratio. From Equation (33.4), $\lambda = x$. Substituting this into Equation (33.3) gives:

$$y = 2x.$$

At the optimum, the ratio of marginal benefits equals the ratio of marginal costs ($y/x = 2/1$).

Step 4: Substitute into the constraint. Substitute $y = 2x$ into Equation (33.5):

$$2x + (2x) = 100 \implies 4x = 100 \implies x^* = 25.$$

Then $y^* = 2(25) = 50$, and the multiplier is $\lambda = x^* = 25$.

Step 5: Compute the optimal value and verify the shadow price. The maximum product is:

$$f^* = f(25, 50) = (25)(50) = 1250.$$

To verify the physical meaning of $\lambda = 25$, consider a general budget c so that $2x + y = c$. The condition $y = 2x$ remains unchanged, so $4x = c \implies x^*(c) = c/4$ and $y^*(c) = c/2$. The optimal product as a function of budget c is:

$$f^*(c) = x^*(c)\, y^*(c) = \left(\frac{c}{4}\right)\left(\frac{c}{2}\right) = \frac{c^2}{8}.$$

Differentiating with respect to the budget c:

$$\frac{\mathrm{d}f^*}{\mathrm{d}c} = \frac{\mathrm{d}}{\mathrm{d}c}\left[\frac{c^2}{8}\right] = \frac{2c}{8} = \frac{c}{4}.$$

Evaluating at our specific budget $c = 100$:

$$\left[\frac{\mathrm{d}f^*}{\mathrm{d}c}\right]_{c=100} = \frac{100}{4} = 25 = \lambda.$$

The rate at which the optimal value increases with respect to the budget is precisely the Lagrange multiplier λ.

Example 33.10 (Quadratic forms on the unit circle: The bridge to eigenvalues)**.** Let A be a real symmetric 2×2 matrix, $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$. Maximize the quadratic form:

$$f(\mathbf{x}) = \mathbf{x}^T A\mathbf{x} = ax^2 + 2bxy + cy^2$$

subject to the unit circle constraint:

$$g(\mathbf{x}) = \|\mathbf{x}\|^2 = \mathbf{x}^T\mathbf{x} = x^2 + y^2 = 1.$$

Step 1: Compute gradients in vector notation. Using the multivariable derivative of a quadratic form (or by computing partials directly):

$$\nabla f(\mathbf{x}) = \begin{pmatrix} 2ax + 2by \\ 2bx + 2cy \end{pmatrix} = 2A\mathbf{x}.$$

For the constraint $g(\mathbf{x}) = \mathbf{x}^T\mathbf{x} = x^2 + y^2$, the gradient is:

$$\nabla g(\mathbf{x}) = \begin{pmatrix} 2x \\ 2y \end{pmatrix} = 2\mathbf{x}.$$

Step 2: Apply the Lagrange condition $\nabla f = \lambda \nabla g$.

$$2A\mathbf{x} = \lambda(2\mathbf{x}) \iff A\mathbf{x} = \lambda\mathbf{x}.$$

The Lagrange multiplier condition for a quadratic form on the unit sphere is the eigenvalue equation of linear algebra:

- The critical points $\mathbf{x}^*$ of the quadratic form on the unit circle are precisely the **unit eigenvectors** of A.
- The Lagrange multipliers λ are precisely the **eigenvalues** of A.

Step 3: What value does the objective attain at an eigenvector? Multiply $A\mathbf{x} = \lambda\mathbf{x}$ on the left by $\mathbf{x}^T$:

$$\mathbf{x}^T A\mathbf{x} = \mathbf{x}^T(\lambda\mathbf{x}) = \lambda(\mathbf{x}^T\mathbf{x}).$$

Because $\mathbf{x}$ lies on the unit sphere ($\mathbf{x}^T\mathbf{x} = 1$), this simplifies to:

$$f(\mathbf{x}) = \lambda.$$

The value of the quadratic form at any critical point is the corresponding eigenvalue. Therefore:

- The **maximum** of $\mathbf{x}^T A\mathbf{x}$ on the unit sphere is the largest eigenvalue $\lambda_{\max}$.
- The **minimum** of $\mathbf{x}^T A\mathbf{x}$ on the unit sphere is the smallest eigenvalue $\lambda_{\min}$.

Step 4: A concrete numerical calculation. Let $A = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$, so $f(x, y) = 5x^2 + 4xy + 2y^2$ subject to $x^2 + y^2 = 1$. The Lagrange system $A\mathbf{x} = \lambda\mathbf{x}$ becomes:

$$\begin{aligned} (5 - \lambda)x + 2y &= 0, \\ 2x + (2 - \lambda)y &= 0. \end{aligned}$$

For a nonzero solution vector $\mathbf{x}$ to exist on the unit circle, the determinant of the system must vanish:

$$\det\begin{pmatrix} 5-\lambda & 2 \\ 2 & 2-\lambda \end{pmatrix} = (5-\lambda)(2-\lambda) - 4 = \lambda^2 - 7\lambda + 6 = 0.$$

Factoring gives $(\lambda - 6)(\lambda - 1) = 0$, yielding two eigenvalues: $\lambda_1 = 6$ and $\lambda_2 = 1$.

- For $\lambda_1 = 6$: $(5-6)x + 2y = 0 \implies -x + 2y = 0 \implies x = 2y$. Substituting into $x^2 + y^2 = 1$ gives $(2y)^2 + y^2 = 5y^2 = 1$, so $y = \pm 1/\sqrt{5}$ and $x = \pm 2/\sqrt{5}$. The maximum value of $f(x, y)$ on the circle is $f^* = \lambda_1 = 6$.
- For $\lambda_2 = 1$: $(5-1)x + 2y = 0 \implies 4x + 2y = 0 \implies y = -2x$. Substituting into $x^2 + y^2 = 1$ gives $x^2 + (-2x)^2 = 5x^2 = 1$, so $x = \pm 1/\sqrt{5}$ and $y = \mp 2/\sqrt{5}$. The minimum value of $f(x, y)$ on the circle is $f^* = \lambda_2 = 1$.

This establishes a direct bridge between multivariable calculus and Book 3 (*Linear Algebra From the Ground Up*): the spectral decomposition and principal axis theorem are constrained optimization in geometric form.

History

The word "gradient" and the ∇ symbol came of age in the nineteenth century. The inverted-delta ∇ was introduced by William Rowan Hamilton (in 1847) in his work on quaternions, and the playful name *nabla* – after an ancient Hebrew harp of similar shape – was suggested by William Robertson Smith to the physicist P. G. Tait, who developed the operator's calculus. But the *ideas* are older and grew from physics. Joseph-Louis Lagrange (1736–1813) introduced the method of multipliers in his 1788 treatise *Mécanique analytique* to solve problems in mechanics where particles are constrained to move on curved surfaces or are held together by rigid rods. Lagrange realized that the constraint force exerted by a surface acts normal to the surface, balancing the unconstrained forces along the directions of constraint. What physics called a normal reaction force, calculus crystallized as $\nabla f = \lambda \nabla g$. The Hessian is named for Otto Hesse, who studied these determinants of second derivatives in the 1840s while working on curves and surfaces. What we now package neatly as "set the gradient to zero, check the Hessian, and match normal vectors for constraints" was, for the founders, the hard-won geometry of how surfaces bend and interact.

Pause

That was a lot. Here is the spine of the chapter in six sentences. (1) The *directional derivative* $D_{\mathbf{u}} f$ is the slope of f as you walk in the unit direction $\mathbf{u}$, and it equals $\nabla f \cdot \mathbf{u}$. (2) The *gradient* $\nabla f = (f_x, f_y)$ points in the direction of steepest ascent, its length is the steepest slope, and it is perpendicular to level curves. (3) The *chain rule* "multiply along paths, add across paths" computes derivatives through any chain of dependencies and is the engine of backpropagation. (4) Linear approximation upgrades the tangent line to a tangent *plane*, $L = f(a, b) + \nabla f \cdot ((x, y) - (a, b))$. (5) Unconstrained critical points are where $\nabla f = \mathbf{0}$, classified by the *Hessian* determinant $D = f_{xx} f_{yy} - f_{xy}^2$. (6) Constrained extrema occur where the objective contour kisses the constraint path tangentially, aligning their perpendicular gradients: $\nabla f = \lambda \nabla g$, where $\lambda = \frac{\mathrm{d}f^*}{\mathrm{d}c}$ measures the shadow price of the budget. Every item on this list is one geometric idea – local flatness and normal orthogonality – seen from a different angle.

33.9 Exercises

Exercise 33.1 (Directional derivative, by formula)**.** Let $f(x, y) = x^2y + y^3$. Compute the directional derivative of f at the point $(2, 1)$ in the direction toward $(5, 5)$. (Remember to normalize the direction vector first.)

Exercise 33.2 (Steepest ascent)**.** For $f(x, y) = e^x \sin y$, find at the point $(0, \frac{\pi}{2})$: (a) the gradient, (b) the unit direction of steepest ascent, and (c) the maximum rate of increase. In what direction(s) is the directional derivative zero?

Exercise 33.3 (Chain rule, one parameter)**.** Let $z = x^2 - y^2$ with $x(t) = \cos t$ and $y(t) = \sin t$. Use the chain rule to find $\frac{\mathrm{d}z}{\mathrm{d}t}$, then verify your answer by first substituting to get z as an explicit function of t and differentiating directly. (You should find a clean trigonometric expression.)

Exercise 33.4 (Chain rule, two parameters)**.** Let $z = xy$ where $x = s + t$ and $y = s - t$. Compute $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$ using the multivariable chain rule. Then compute z directly in terms of s, t and confirm.

Exercise 33.5 (Tangent plane and estimation)**.** Let $f(x, y) = \ln(x^2 + y^2)$. (a) Write the equation of the tangent plane to the graph at $(1, 0)$. (b) Use it to estimate $f(1.1,\ 0.05)$, and compare with the true value.

Exercise 33.6 (A gradient descent run)**.** Consider $L(w_1, w_2) = w_1^2 + 4w_2^2$ with learning rate $\eta = 0.1$, starting at $(w_1, w_2) = (1, 1)$. (a) Carry out two steps of gradient descent by hand, reporting the iterates and the loss at each. (b) Each coordinate is multiplied by a constant factor every step – find the two factors. Why does w_2 converge faster than w_1, and what does that say about choosing η when the bowl is stretched?

Exercise 33.7 (Classify the critical points)**.** Find and classify all critical points of $f(x, y) = x^3 + y^3 - 3xy$ using the Hessian test. (You should find one saddle and one local extremum.)

Exercise 33.8 (The perpendicularity, concretely)**.** Let $f(x, y) = x^2 + 4y^2$, whose level curves are ellipses. At the point $(2, 1)$ (which lies on the level curve $f = 8$): (a) compute $\nabla f(2, 1)$; (b) find a vector tangent to the ellipse $f = 8$ at $(2, 1)$ (hint: parametrize, or use implicit differentiation $\frac{\mathrm{d}y}{\mathrm{d}x}$ on $x^2 + 4y^2 = 8$); (c) verify by a dot product that the gradient and the tangent are perpendicular.

Exercise 33.9 (Conceptual: partials exist but no plane)**.** Explain in your own words, with the help of the trapbox in Section 33.4, why "all partial derivatives exist at a point" is *not* enough to guarantee that the chain rule formula holds there. What stronger condition (stated in Section 33.5) is the right one, and what does it say geometrically?

Exercise 33.10 (Proof: steepest ascent)**.** Prove that among all unit vectors $\mathbf{u}$, the directional derivative $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ is maximized when $\mathbf{u}$ points along ∇f, and that the maximum value is $\|\nabla f\|$. (Use the identity $\nabla f \cdot \mathbf{u} = \|\nabla f\| \cos\theta$ and explain why $\cos\theta = 1$ is forced. State clearly where you use $\|\mathbf{u}\| = 1$.)

Exercise 33.11 (Proof: gradient perpendicular to level sets)**.** Suppose a differentiable path $\mathbf{r}(t) = (x(t), y(t))$ lies entirely on the level curve $f(x, y) = c$. Differentiate the identity $f(x(t), y(t)) = c$ with respect to t and use the chain rule to conclude that $\nabla f \cdot \mathbf{r}\,'(t) = 0$. Explain in one sentence why this proves ∇f is perpendicular to the level curve.

Chapter 34

Multiple Integrals

Back in Chapter 20 we built the single integral $\int_a^b f(x)\,\mathrm{d}x$ out of one stubbornly simple idea: chop the interval $[a, b]$ into tiny pieces, build a thin rectangle over each piece, add up their areas, and watch the sum settle onto a number as the pieces shrink. The picture was a curve over a line, and the answer was the area trapped between them.

Now the domain grows a dimension. Instead of integrating over an interval on the x-axis, we integrate over a *region* in the xy-plane – a disk, a triangle, a blob bounded by two curves. The graph of our function $f(x, y)$ is no longer a curve but a *surface* floating above that region, and the thing we add up is no longer area but *volume*: the volume of the solid trapped between the region on the floor and the surface overhead.

Astonishingly, almost nothing new has to be invented. The whole chapter is the same one idea – chop, multiply, add, take the limit – carried into two and three dimensions. The genuinely new questions are practical ones. How do you actually *compute* a sum over a two-dimensional region without drowning in it? (Answer: do it one direction at a time – that is Fubini's theorem.) And when the region is round, or the function lives most naturally in some other coordinate system, how do you change coordinates without corrupting the volume? (Answer: the Jacobian, which turns out to be a determinant, which turns out to be exactly the area-scaling factor you met in linear algebra.) Those two ideas – **iteration** and **change of variables** – are the heart of this chapter.

Focus

A single integral adds a function over an *interval* and measures *area*. A double integral adds a function over a *planar region* and measures *volume*. The definition is the same Riemann sum, just with little patches of area ΔA instead of little widths Δx. Everything hard is bookkeeping: how to organize the sum (Fubini) and how to bend the coordinate grid without losing track of area (the Jacobian).

34.1 The double integral as volume and as accumulation

34.1.1 Two readings of the same sum

When you met $\int_a^b f(x)\,\mathrm{d}x$ you learned to read it two ways at once, and both readings will carry over. The *geometric* reading: it is the signed area under the graph of f. The *accumulation* reading: if $f(x)$ is a density-per-unit-length, then the integral is the total stuff. A double integral has exactly these same two faces.

Geometric reading. Let R be a region in the xy-plane and let $f(x,y) \geq 0$ be a function defined on it. The graph $z = f(x,y)$ is a surface hovering above the plane. The double integral

$$\iint_R f(x,y)\,\mathrm{d}A$$

is the *volume* of the solid that sits over R and reaches up to the surface. When f dips below zero the contribution is negative, exactly as area went negative for the single integral when the curve dipped below the axis.

Accumulation reading. Suppose a flat metal plate occupies the region R, and $f(x,y)$ is its *density* – grams per square centimetre – which may vary from point to point. Then $\iint_R f\,\mathrm{d}A$ is the total *mass* of the plate. Here there is no surface and no volume in sight; we are simply adding up a quantity that is spread out over an area, weighting each tiny patch by how much stuff sits on it. Population over a county, rainfall over a watershed, charge over a sheet – all the same integral.

Intuition

Think of $\mathrm{d}A$ as an *infinitesimal tile* of the floor, and $f(x,y)$ as "how tall the building is" (volume reading) or "how heavy this tile is" (mass reading) at that spot. The integral lays down infinitely many infinitely small tiles over the whole region, multiplies each by its height-or-weight, and sums. The $\iint$ – a double S – is just two nested "sum" signs, one for each direction you have to sweep across the floor.

34.1.2 The Riemann sum, from scratch

Let us build the thing so the definition is not a mystery. To keep the picture clean, start with a region that is a rectangle, $R = [a,b] \times [c,d]$ – all points (x,y) with $a \leq x \leq b$ and $c \leq y \leq d$.

Cut $[a,b]$ into m equal sub-intervals of width $\Delta x = (b-a)/m$, and cut $[c,d]$ into n equal sub-intervals of height $\Delta y = (d-c)/n$. The grid lines slice R into $m \cdot n$ little rectangles, each of area

$$\Delta A = \Delta x\,\Delta y.$$

In the (i,j)-th little rectangle pick any sample point $(x_i^*,\, y_j^*)$. Over that rectangle, build a thin rectangular *box* whose base is the little rectangle and whose height is the value of the function there, $f(x_i^*, y_j^*)$. The volume of that one box is height times base area,

$$f(x_i^*, y_j^*)\,\Delta A.$$

Now add up all $m \cdot n$ boxes. You get a lumpy, blocky approximation to the solid under the surface – a city of skyscrapers approximating a smooth hill:

$$S_{m,n} = \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_i^*, y_j^*)\, \Delta A.$$

This double sum is the two-dimensional Riemann sum.

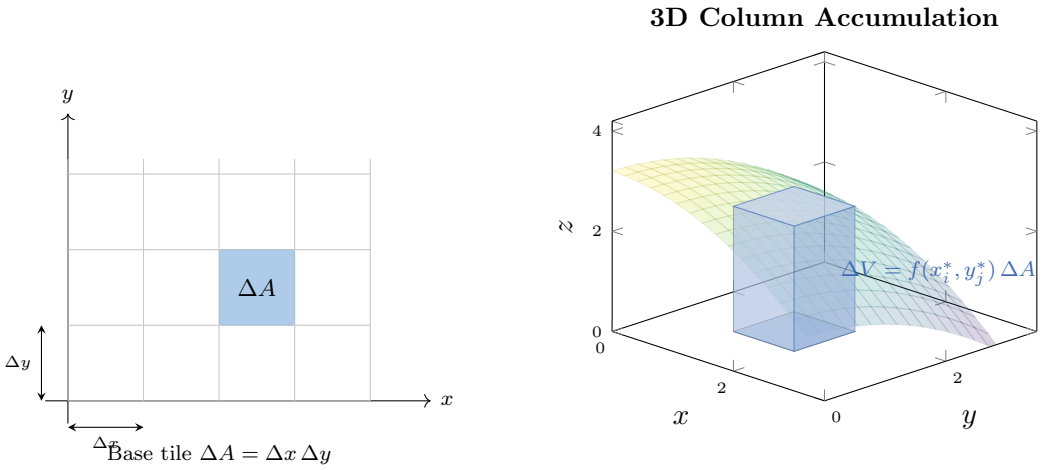


Figure 34.1: The double Riemann sum as a forest of 3D columns. Left: The domain rectangle R is partitioned into small tiles of area $\Delta A = \Delta x\, \Delta y$. Right: Over each tile, we erect a rectangular pillar of height $f(x_i^*, y_j^*)$. The volume of the pillar is $\Delta V = f(x_i^*, y_j^*)\, \Delta A$. As the grid is refined, the sum of column volumes tightly reconstructs the total volume under the surface $z = f(x, y)$.

If, as we make the grid finer and finer ($m, n \to \infty$, so $\Delta x, \Delta y \to 0$), this sum approaches one and the same number no matter how we chose the sample points, we call that number the **double integral**.

Definition 34.1 (Double integral over a rectangle)**.** Let f be defined on $R = [a, b] \times [c, d]$. If the Riemann sums $S_{m,n}$ above converge to a single limit L as $m, n \to \infty$, independent of the choice of sample points, then f is *integrable* on R and we write

$$\iint_R f(x, y)\, \mathrm{d}A = L.$$

As with one variable, every *continuous* function on a closed bounded region is integrable – the proof is the higher-dimensional twin of the one in Chapter 20 and we will not belabor it. The moral is the same: continuity is plenty to guarantee the city of skyscrapers settles onto a definite volume.

What is really happening

What is $\mathrm{d}A$ *really*? It is the ghost of $\Delta A = \Delta x\, \Delta y$ after the limit. In the sum you genuinely multiply a width by a height to get a patch of *area*, and you weight f by that patch. After the limit we write $\mathrm{d}A$ to remember "an infinitesimal patch of area." The reason $\mathrm{d}A$ will sometimes appear as $\mathrm{d}x\, \mathrm{d}y$, sometimes as $r\, \mathrm{d}r\, \mathrm{d}\theta$, and sometimes as something stranger is that those are just different ways of measuring the *area of one little patch* depending on how you have sliced the region. The entire change-of-variables story later in this chapter is one long answer to the

question: "in these coordinates, how big is one little patch?"

34.1.3 Regions that are not rectangles

Real regions are rarely rectangles. To integrate over a triangle or a disk, do the obvious thing: pretend f is zero outside the region. Enclose the region R inside a big rectangle, define a new function that equals f on R and equals 0 everywhere else in the rectangle, and integrate *that* over the rectangle. Tiles that fall outside R contribute nothing; tiles inside contribute $f\,\Delta A$ as before. The boundary tiles, the ones the curve slices through, are a vanishing fraction of the total as the grid shrinks, so they do not corrupt the limit. This little trick lets Definition 34.1 cover any reasonable region without new ideas.

Focus

Useful sanity checks you will use constantly:

- $\iint_R 1\,\mathrm{d}A =$ area of R. (Height 1 everywhere, so the "volume" is just area times height = area.)
- $\iint_R c\,f\,\mathrm{d}A = c\iint_R f\,\mathrm{d}A$ and the integral of a sum is the sum of the integrals – linearity, inherited straight from the sums.
- If R splits into two non-overlapping pieces R_1 and R_2, then $\iint_R = \iint_{R_1} + \iint_{R_2}$. Cut the region, add the volumes.

34.2 Iterated integrals and Fubini's theorem

A double sum $\sum_i \sum_j$ is intimidating only until you remember you are allowed to do the two sums *one at a time*. Fix a column, add up that column, then add up the column-totals. The continuous version of that regrouping is the workhorse of this chapter: **Fubini's theorem**, which says a double integral can be evaluated as two ordinary single integrals, nested.

34.2.1 The idea: volume by slices

Here is the picture that makes Fubini obvious. You already know one way to find a volume from Chapter 26: slice the solid into thin slabs and add the slab volumes.

Take our solid under $z = f(x, y)$ over the rectangle $[a, b] \times [c, d]$. Slice it with a plane $x =$ const, perpendicular to the x-axis. The slice is a thin slab. Its *cross-sectional area* is the area under the curve $z = f(x, y)$ as y runs from c to d – with x held fixed – which is just a single integral,

$$A(x) = \int_c^d f(x, y)\,\mathrm{d}y.$$

The slab has thickness $\mathrm{d}x$, so its volume is $A(x)\,\mathrm{d}x$. Add up the slabs from $x = a$ to $x = b$:

$$\text{Volume} = \int_a^b A(x)\,\mathrm{d}x = \int_a^b \left(\int_c^d f(x, y)\,\mathrm{d}y \right) \mathrm{d}x.$$

That nested expression is an *iterated integral*. We just sliced the solid the long way and added the slices.

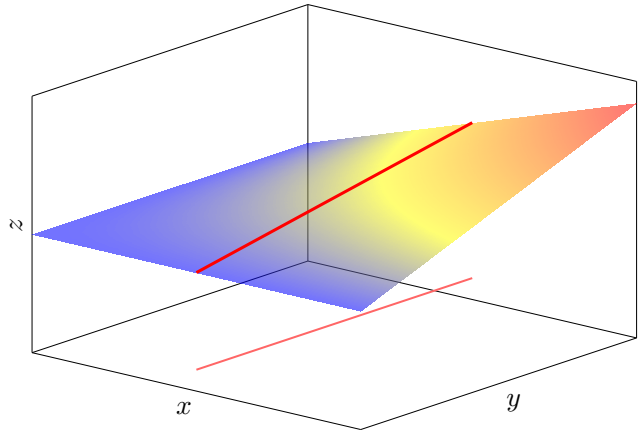


Figure 34.2: Slicing the solid at a fixed x gives a slab whose face area is $A(x) = \int_c^d f(x,y)\,\mathrm{d}y$ (the red profile). Adding the slabs $A(x)\,\mathrm{d}x$ from a to b rebuilds the volume. That is Fubini.

But we could just as well have sliced the *other* way, with planes $y = \text{const}$, giving

$$\text{Volume} = \int_c^d \left(\int_a^b f(x,y)\,\mathrm{d}x \right) \mathrm{d}y.$$

The solid does not care which way you slice it. Both iterated integrals equal the same volume, hence equal each other. That equality – and the fact that either equals the genuine double integral $\iint_R f\,\mathrm{d}A$ – is the content of Fubini.

Theorem 34.1 (Fubini, for nice functions)**.** *If f is continuous on the rectangle $R = [a,b] \times [c,d]$, then*

$$\iint_R f(x,y)\,\mathrm{d}A = \int_a^b \int_c^d f(x,y)\,\mathrm{d}y\,\mathrm{d}x = \int_c^d \int_a^b f(x,y)\,\mathrm{d}x\,\mathrm{d}y.$$

In particular both orders of integration give the same answer, and both equal the double integral.

Why the technique works

Why is the iterated integral allowed to replace the double sum at all? Because addition does not care how you group it. In the Riemann sum $\sum_i \sum_j f(x_i^*, y_j^*)\,\Delta x\,\Delta y$, factor the constant Δx out of the inner sum:

$$\sum_i \Bigg(\underbrace{\sum_j f(x_i^*, y_j^*)\,\Delta y}_{\approx A(x_i^*)} \Bigg) \Delta x.$$

The inner sum is a Riemann sum in y alone; as the grid refines it becomes $\int_c^d f(x_i^*, y)\,\mathrm{d}y = A(x_i^*)$. The outer sum is then a Riemann sum of $A(x)$ in x, becoming $\int_a^b A(x)\,\mathrm{d}x$. So Fubini is nothing but "do the y-sum first, then the x-sum," the freedom to regroup a sum, carried through the limit. The one piece of fine print – which is why Fubini is a *theorem* and not a triviality – is checking that the limits can be taken in this nested order; continuity (more generally, $\iint |f|\,\mathrm{d}A < \infty$) is what licenses it.

History

The theorem is named for Guido Fubini, who proved a sweeping version in 1907 within Lebesgue's new theory of integration. But the practice is far older: Newton, the Bernoullis, and Euler computed volumes by iterated integration as a matter of course in the 1600s and 1700s, slicing solids and adding the slices without a worry. What the twentieth century added was the precise hypothesis – that f be absolutely integrable – under which the cavalier swapping of integration order is *guaranteed* safe. It is not always safe: there are functions (non-integrable in absolute value) for which the two orders give genuinely different answers. Continuity on a bounded region spares you all such pathology, which is why every example in this book swaps freely.

34.2.2 A first computation

Example 34.1 (A double integral over a rectangle)**.** Compute $\iint_R (x+2y)\,\mathrm{d}A$ over $R=[0,2]\times[0,3]$.

Integrate in y first, treating x as a constant:

$$\int_0^3 (x+2y)\,\mathrm{d}y = \Big[xy+y^2\Big]_0^3 = 3x+9.$$

Now integrate that in x:

$$\int_0^2 (3x+9)\,\mathrm{d}x = \Big[\tfrac{3}{2}x^2+9x\Big]_0^2 = 6+18 = 24.$$

As a check, slice the other way – x first:

$$\int_0^2 (x+2y)\,\mathrm{d}x = \Big[\tfrac{1}{2}x^2+2xy\Big]_0^2 = 2+4y, \qquad \int_0^3 (2+4y)\,\mathrm{d}y = \Big[2y+2y^2\Big]_0^3 = 6+18 = 24.$$

Same number, as Fubini promised.

Common trap

The bounds and the differentials must match up *from the inside out*. In $\int_a^b\int_c^d f\,\mathrm{d}y\,\mathrm{d}x$, the inner differential $\mathrm{d}y$ pairs with the inner limits c,d, and the outer $\mathrm{d}x$ pairs with a,b. A frequent disaster is writing the limits in the wrong order, so that x ends up "integrated" over y's interval. Read the integral inside-out, like nested parentheses, and keep each variable with its own bounds.

34.2.3 Non-rectangular regions: variable bounds

For a rectangle the inner limits are constants. For a curvy region they are *functions*, because the cross-section changes length as you sweep across. There are two standard shapes.

A region is *y-simple* (vertically simple) if it sits between two curves $y=g_1(x)$ below and $y=g_2(x)$ above, for x in some interval $[a,b]$. Slice it vertically: at each fixed x, y runs from the bottom curve

to the top curve.

$$\iint_R f \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y) \, dy \, dx.$$

A region is *x-simple* (horizontally simple) if it lies between $x = h_1(y)$ on the left and $x = h_2(y)$ on the right, for y in $[c, d]$:

$$\iint_R f \, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y) \, dx \, dy.$$

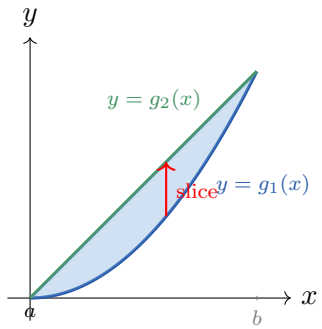


Figure 34.3: A y-simple region. A vertical slice at fixed x runs from the lower curve $y = g_1(x)$ to the upper curve $y = g_2(x)$, so the inner y-integral has those curves as its limits.

Example 34.2 (A triangular region)**.** Integrate $f(x, y) = x$ over the triangle R with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$. The hypotenuse is the line $y = x$; the bottom is $y = 0$; the right edge is $x = 1$. Treating R as y-simple: for each $x \in [0, 1]$, y runs from 0 up to x.

$$\iint_R x \, dA = \int_0^1 \int_0^x x \, dy \, dx = \int_0^1 x \cdot \Big[y\Big]_0^x \, dx = \int_0^1 x \cdot x \, dx = \int_0^1 x^2 \, dx = \frac{1}{3}.$$

34.2.4 Changing the order of integration

Here is where iterated integrals earn their keep. Sometimes one order is miserable and the reverse order is easy – or the inner integral in one order has no elementary antiderivative at all, but flips into something trivial the other way. The technique is always the same: **draw the region**, read off the *other* kind of bounds, and rewrite.

Example 34.3 (An integral you cannot do in the given order)**.** Evaluate

$$I = \int_0^1 \int_x^1 e^{y^2} \, dy \, dx.$$

The inner integral $\int e^{y^2} \, dy$ has *no* elementary antiderivative – you are stuck cold. So change the order.

First decode the region from the bounds as written: x goes from 0 to 1, and for each x, y goes from $y = x$ up to $y = 1$. That is the triangle bounded by $y = x$, $y = 1$, and $x = 0$ – the triangle with vertices $(0, 0)$, $(0, 1)$, $(1, 1)$.

Now describe the *same* triangle as x-simple. Let y run from 0 to 1; for each y, x runs from the left edge $x = 0$ to the right edge $x = y$ (the line $y = x$ solved for x). So

$$I = \int_0^1 \int_0^y e^{y^2}\,\mathrm{d}x\,\mathrm{d}y.$$

Now the inner integral is trivial – e^{y^2} is constant in x:

$$\int_0^y e^{y^2}\,\mathrm{d}x = y\,e^{y^2}, \qquad I = \int_0^1 y\,e^{y^2}\,\mathrm{d}y.$$

And *that* we can do, with the substitution $u = y^2$, $\mathrm{d}u = 2y\,\mathrm{d}y$:

$$I = \frac{1}{2}\int_0^1 e^u\,\mathrm{d}u = \frac{1}{2}(e-1).$$

The same region, swept the other direction, turned an impossible integral into a one-line substitution.

Walking question

Take Example 34.3 on your walk. The number $\frac{1}{2}(e-1)$ never needed the antiderivative of e^{y^2} – yet the inner integral as first written *was* that forbidden antiderivative. How can a quantity be perfectly computable while one route to it is blocked? (Hint: the blocked route asks for the antiderivative as a *function of* x; the open route only ever needs a *definite* integral, a single number. The double integral is a number; only our bookkeeping made it look like it needed a forbidden function.)

Common trap

When you flip the order, you must re-derive the limits from the *geometry*, not by mechanically swapping the symbols. Swapping $\int_0^1\int_x^1 \mathrm{d}y\,\mathrm{d}x$ into $\int_0^1\int_x^1 \mathrm{d}x\,\mathrm{d}y$ is nonsense – you cannot have x in the limits of an x-integral. Sketch the region every single time; let the picture tell you the new bounds.

34.3 Polar coordinates and the Gaussian integral

Some regions hate the rectangular grid. A disk, an annulus, a circular sector – their natural description is "radius from here to there, angle from here to there," not "x from here to there." For these, switch to **polar coordinates**

$$x = r\cos\theta, \qquad y = r\sin\theta,$$

where $r \geq 0$ is the distance from the origin and θ is the angle. A disk of radius a is simply $0 \leq r \leq a$, $0 \leq \theta \leq 2\pi$ – a rectangle in the (r,θ) world.

34.3.1 Why the polar area element carries an extra factor of r

Here is the subtlety that trips up everyone the first time. In rectangular coordinates, $\mathrm{d}A = \mathrm{d}x\,\mathrm{d}y$, because a little grid cell is a tiny rectangle of width $\mathrm{d}x$ and height $\mathrm{d}y$. In polar coordinates, $\mathrm{d}A$ is

$$\boxed{\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta}$$

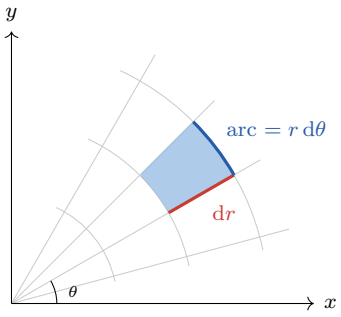
not $\mathrm{d}r\,\mathrm{d}\theta$. There is an extra factor of r, and the whole chapter's payoff is understanding where it comes from – because it is the first appearance of the change-of-variables idea that will close the chapter.

What is really happening

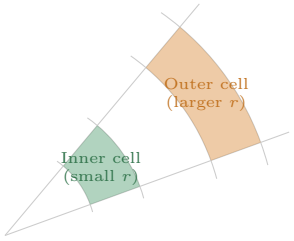
Picture the polar grid: circles of constant r and rays of constant θ. The little cell between radius r and $r + \mathrm{d}r$, and between angle θ and $\theta + \mathrm{d}\theta$, is *not* a rectangle of sides $\mathrm{d}r$ and $\mathrm{d}\theta$. Its two "radial" sides do have length $\mathrm{d}r$. But its two "arc" sides are pieces of circles, and the length of a circular arc is radius times angle: $r\,\mathrm{d}\theta$. So the cell is (very nearly) a rectangle of sides $\mathrm{d}r$ and $r\,\mathrm{d}\theta$, with area

$$\mathrm{d}A = (\mathrm{d}r)(r\,\mathrm{d}\theta) = r\,\mathrm{d}r\,\mathrm{d}\theta.$$

The factor r is the confession that polar cells *grow* as you move outward: a cell spanning one degree of angle is tiny near the origin and huge far away, in exact proportion to r. Forgetting the r is the single most common error in polar integration, and now you know it is not a rule to memorize but an arc length to remember.



$\mathrm{d}A \approx (\mathrm{d}r)(r\,\mathrm{d}\theta) = r\,\mathrm{d}r\,\mathrm{d}\theta$



Same Δr and $\Delta\theta$, but different area!
Area scales linearly with radius r.

Figure 34.4: Why the polar area element carries an extra factor of r. Left: A polar tile is bounded by two rays separated by $\mathrm{d}\theta$ and two circular arcs separated by $\mathrm{d}r$. Its circumferential width is the arc length $r\,\mathrm{d}\theta$, yielding area $\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta$. Right: Comparing two cells of identical Δr and $\Delta\theta$ at different radii demonstrates that tiles naturally widen as you move outward. The factor r accounts for this geometric expansion.

So the conversion rule is: replace $x \to r\cos\theta$, $y \to r\sin\theta$, and $\mathrm{d}x\,\mathrm{d}y \to r\,\mathrm{d}r\,\mathrm{d}\theta$, then set the limits as r- and θ-ranges.

$$\iint_R f(x,y)\,\mathrm{d}x\,\mathrm{d}y = \iint_{R'} f(r\cos\theta,\, r\sin\theta)\; r\,\mathrm{d}r\,\mathrm{d}\theta.$$

Example 34.4 (Area of a disk)**.** Take $f = 1$ over the disk of radius a. In polar:

$$\iint_R 1\,\mathrm{d}A = \int_0^{2\pi}\int_0^{a} 1\cdot r\,\mathrm{d}r\,\mathrm{d}\theta = \int_0^{2\pi}\left[\tfrac{1}{2}r^2\right]_0^{a}\mathrm{d}\theta = \int_0^{2\pi}\frac{a^2}{2}\,\mathrm{d}\theta = \pi a^2.$$

There is πa^2, falling straight out – and notice the r was essential. Without it you would have integrated $\int_0^a \mathrm{d}r = a$ and gotten the nonsensical $2\pi a$.

34.3.2 The Gaussian integral: a small miracle

We can now compute one of the most important integrals in all of mathematics – the one underneath the bell curve, statistics, the heat equation, and quantum mechanics:

$$I = \int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x.$$

The function e^{-x^2} has *no* elementary antiderivative (we noted its cousin e^{y^2} in Example 34.3). The one-variable methods of this book cannot touch it. Yet the answer is exactly $\sqrt{\pi}$, and the trick that gets it – due in spirit to Poisson and Laplace – is a jewel: *square the integral and read it as a double integral, then go polar.*

Example 34.5 ($\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x = \sqrt{\pi}$)**.** We cannot find I directly, so consider I^2. Because the integral is just a number, we may name the dummy variable of the second copy differently – call it y:

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x\right)\left(\int_{-\infty}^{\infty} e^{-y^2}\,\mathrm{d}y\right) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-x^2}e^{-y^2}\,\mathrm{d}x\,\mathrm{d}y.$$

The step that does the magic: $e^{-x^2}e^{-y^2} = e^{-(x^2+y^2)}$. The integrand depends on x and y only through $x^2+y^2 = r^2$ – it is *radially symmetric*. That is the flashing sign that polar coordinates will collapse it. The region is the whole plane, so r runs 0 to ∞ and θ runs 0 to 2π. With $x^2+y^2=r^2$ and the mandatory $\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta$:

$$I^2 = \int_0^{2\pi}\int_0^{\infty} e^{-r^2}\,r\,\mathrm{d}r\,\mathrm{d}\theta.$$

Now the factor of r – which we were forced to include – is exactly the factor that makes the radial integral elementary. Substitute $u = r^2$, $\mathrm{d}u = 2r\,\mathrm{d}r$:

$$\int_0^{\infty} e^{-r^2}\,r\,\mathrm{d}r = \frac{1}{2}\int_0^{\infty} e^{-u}\,\mathrm{d}u = \frac{1}{2}\Big[-e^{-u}\Big]_0^{\infty} = \frac{1}{2}.$$

Therefore

$$I^2 = \int_0^{2\pi}\frac{1}{2}\,\mathrm{d}\theta = \frac{1}{2}\cdot 2\pi = \pi, \qquad \Longrightarrow \qquad I = \sqrt{\pi}.$$

Why this matters

Savor what just happened. A one-dimensional integral that *provably* has no elementary antiderivative was conquered by stepping *up* into two dimensions, where its radial symmetry could be seen, and the unavoidable polar factor r – the very thing students forget – was precisely the missing piece that an antiderivative needs. The problem was unsolvable not because it was hard but because it was viewed in too few dimensions. Going up a dimension to solve a lower-dimensional problem is one of the great recurring moves in mathematics, and here you have watched it produce $\sqrt{\pi}$ out of thin air.

Application

The bell curve of statistics is $\frac{1}{\sqrt{2\pi}\,\sigma}e^{-(x-\mu)^2/2\sigma^2}$. The strange-looking constant $\frac{1}{\sqrt{2\pi}}$ is there for exactly one reason: so the total probability integrates to 1. Substituting and using $I = \sqrt{\pi}$

is what forces that normalizing constant. Every time a statistician, physicist, or machine-learning engineer writes down a Gaussian, the $\sqrt{2\pi}$ they carry around is a direct descendant of the polar computation you just did.

34.4 Triple integrals

Add one more dimension and nothing conceptually changes. A **triple integral**

$$\iiint_E f(x,y,z)\,\mathrm{d}V$$

chops a solid region $E \subset \mathbb{R}^3$ into tiny boxes of volume $\mathrm{d}V = \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$, weights each by f, and sums. Fubini still holds: evaluate it as three nested single integrals, in any of the six orders, with the inner limits possibly functions of the outer variables.

34.4.1 What it measures

If $f = 1$, the triple integral is the *volume* of E: $\iiint_E 1\,\mathrm{d}V = \mathrm{Vol}(E)$. (Just as $\iint 1\,\mathrm{d}A$ was area.) If f is a density – mass per unit volume – the integral is the total *mass* of a solid body. There is no "graph" picture this time, because the graph $w = f(x,y,z)$ would live in four dimensions; the accumulation reading is the one to hold onto. We are summing a quantity spread through a three-dimensional body.

Example 34.6 (Volume of a tetrahedron)**.** Find the volume of the solid E bounded by the coordinate planes and the plane $x+y+z=1$ (the corner of the unit cube cut by a slanted plane). Sweep z first: for fixed (x,y) in the base triangle, z runs from the floor $z=0$ up to the slanted roof $z = 1-x-y$. The base triangle (where the roof is still above the floor, i.e. $1-x-y \geq 0$) is $0 \leq y \leq 1-x$, $0 \leq x \leq 1$. So

$$\mathrm{Vol}(E) = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} 1\,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x.$$

Innermost: $\int_0^{1-x-y} \mathrm{d}z = 1-x-y$. Middle:

$$\int_0^{1-x} (1-x-y)\,\mathrm{d}y = \Big[(1-x)y - \tfrac{1}{2}y^2\Big]_0^{1-x} = (1-x)^2 - \tfrac{1}{2}(1-x)^2 = \tfrac{1}{2}(1-x)^2.$$

Outermost, with $u = 1-x$:

$$\int_0^1 \tfrac{1}{2}(1-x)^2\,\mathrm{d}x = \frac{1}{2}\int_0^1 u^2\,\mathrm{d}u = \frac{1}{2}\cdot\frac{1}{3} = \frac{1}{6}.$$

The volume is $\frac{1}{6}$, the familiar volume of that corner tetrahedron.

Work it yourself

Confirm $\frac{1}{6}$ a second way. Treat the tetrahedron as a stack of horizontal triangular slices at height z. At height z, the cross-section is the triangle $x+y \leq 1-z$, $x,y \geq 0$, a right triangle

of legs $1-z$, so of area $\frac{1}{2}(1-z)^2$. Then Vol $= \int_0^1 \frac{1}{2}(1-z)^2\,\mathrm{d}z = \frac{1}{6}$. Two slicing orders, one answer – Fubini in three dimensions, by hand.

34.4.2 Cylindrical and spherical coordinates

Just as planar disks begged for polar coordinates, round solids beg for *cylindrical* or *spherical* coordinates, and the same "how big is one little cell?" question fixes the volume element.

Cylindrical coordinates are just polar in the xy-plane with z left alone: $x = r\cos\theta$, $y = r\sin\theta$, $z = z$. A cell is a polar cell ($r\,\mathrm{d}r\,\mathrm{d}\theta$ of floor area) times a height $\mathrm{d}z$, so

$$\mathrm{d}V = r\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}z.$$

The same factor r as before, for the same arc-length reason.

Spherical coordinates measure a point by its distance ρ from the origin, the angle ϕ down from the positive z-axis (the "polar angle"), and the angle θ around the z-axis (the "azimuth"):

$$x = \rho\sin\phi\cos\theta, \quad y = \rho\sin\phi\sin\theta, \quad z = \rho\cos\phi.$$

A spherical cell has three edges: a radial edge of length $\mathrm{d}\rho$; an edge along the ϕ direction, which is an arc of a circle of radius ρ, of length $\rho\,\mathrm{d}\phi$; and an edge along the θ direction, which is an arc of a circle of radius $\rho\sin\phi$ (the distance from the z-axis), of length $\rho\sin\phi\,\mathrm{d}\theta$. Multiply the three edge lengths:

$$\boxed{\mathrm{d}V = \rho^2\sin\phi\,\mathrm{d}\rho\,\mathrm{d}\phi\,\mathrm{d}\theta}$$

What is really happening

Do not memorize $\rho^2\sin\phi$ – *rebuild* it. The cell is nearly a little box with three perpendicular edges of lengths $\mathrm{d}\rho$, $\rho\,\mathrm{d}\phi$, and $\rho\sin\phi\,\mathrm{d}\theta$. Multiplying: $\rho\cdot\rho\sin\phi = \rho^2\sin\phi$ out front, then $\mathrm{d}\rho\,\mathrm{d}\phi\,\mathrm{d}\theta$. Each factor is just an arc length, "radius times angle," for the circle that that coordinate sweeps along. The $\sin\phi$ appears because circles of constant latitude shrink toward the poles: near the north pole ($\phi \approx 0$) a full lap in θ covers almost no distance, and $\sin\phi \to 0$ records it.

Example 34.7 (Volume of a ball, finally)**.** The ball of radius a is $0 \le \rho \le a$, $0 \le \phi \le \pi$, $0 \le \theta \le 2\pi$ – a box in spherical coordinates. With $f = 1$:

$$\mathrm{Vol} = \int_0^{2\pi}\int_0^{\pi}\int_0^{a} \rho^2\sin\phi\,\mathrm{d}\rho\,\mathrm{d}\phi\,\mathrm{d}\theta.$$

The three integrals separate because each factor depends on a single variable:

$$\int_0^a \rho^2\,\mathrm{d}\rho = \frac{a^3}{3}, \qquad \int_0^{\pi}\sin\phi\,\mathrm{d}\phi = \Big[-\cos\phi\Big]_0^{\pi} = 2, \qquad \int_0^{2\pi}\mathrm{d}\theta = 2\pi.$$

Multiply: $\mathrm{Vol} = \dfrac{a^3}{3}\cdot 2\cdot 2\pi = \dfrac{4}{3}\pi a^3$. The famous formula, derived rather than recited.

34.5 The Jacobian: change of variables as area scaling

We have now twice pulled a mysterious factor out of a hat: r for polar, $\rho^2 \sin\phi$ for spherical. Each time the justification was a hand-drawn picture of "how big is one cell." It is time to see the *general* law behind these factors, and to recognize it as an old friend from linear algebra: the **determinant**.

34.5.1 The one-variable rehearsal

You already know the seed of this in one dimension; it is just u-substitution. When you substitute $x = g(u)$ in $\int f(x)\,\mathrm{d}x$, you replace $\mathrm{d}x$ by $g'(u)\,\mathrm{d}u$:

$$\int_a^b f(x)\,\mathrm{d}x = \int_{u_a}^{u_b} f(g(u))\,g'(u)\,\mathrm{d}u.$$

The factor $g'(u)$ is a *stretch factor*: near u, the map g stretches a little interval $\mathrm{d}u$ into an interval of length $|g'(u)|\,\mathrm{d}u$. (We carry an absolute value when we ignore orientation and only care about length.) The substitution rule *is* the statement "account for how much the transformation stretches length."

Why the technique works

The Jacobian is this same idea promoted to two and three dimensions. In one dimension a smooth map stretches lengths by $|g'(u)|$. In two dimensions a smooth map $(u, v) \mapsto (x, y)$ stretches *areas*, and the local stretch factor is *not* a single derivative but a determinant of partial derivatives, because near a point a smooth map looks like a *linear* map, and the factor by which a linear map scales area is exactly the absolute value of its determinant – a fact you proved in linear algebra. Change of variables in a multiple integral is u-substitution with the word "derivative" replaced by the word "determinant," for the simple reason that area-scaling in many dimensions is measured by a determinant.

34.5.2 Why a determinant: linearization

Let a transformation T carry the uv-plane to the xy-plane:

$$x = x(u, v), \qquad y = y(u, v).$$

Take a tiny rectangle in the uv-plane with corner at (u, v) and sides $\mathrm{d}u$ (along the u-axis) and $\mathrm{d}v$ (along the v-axis). Where does T send it? Each corner moves, and because the rectangle is tiny, T acts on it almost *linearly*: the image is (very nearly) a parallelogram. The side that was $\mathrm{d}u\,\mathbf{e}_u$ becomes the vector

$$\mathbf{a} = \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}\right)\mathrm{d}u,$$

because moving by $\mathrm{d}u$ in the u-direction changes x by $\frac{\partial x}{\partial u}\mathrm{d}u$ and y by $\frac{\partial y}{\partial u}\mathrm{d}u$. Likewise the $\mathrm{d}v$ side becomes

$$\mathbf{b} = \left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}\right)\mathrm{d}v.$$

The image cell is the parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$, and the area of a parallelogram spanned by two vectors is the absolute value of the 2×2 determinant of their components:

$$\mathrm{d}A_{xy} = \left| \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} \right| \mathrm{d}u\,\mathrm{d}v.$$

Intuition

Why the determinant measures area. If you have two vectors $\mathbf{a} = (a_1, a_2)$ and $\mathbf{b} = (b_1, b_2)$ in the plane, why is the area of the parallelogram they span given by the determinant $|a_1b_2 - a_2b_1|$? Enclose the parallelogram inside a rectangular bounding box of width $a_1 + b_1$ and height $a_2 + b_2$. The total area of this bounding box is $(a_1+b_1)(a_2+b_2) = a_1a_2 + a_1b_2 + a_2b_1 + b_1b_2$. To find the area of the interior parallelogram, subtract the unneeded surrounding pieces: two corner rectangles of combined area $2(a_2b_1)$, two right triangles along the horizontal sides of combined area $2 \cdot \frac{1}{2}a_1a_2 = a_1a_2$, and two right triangles along the vertical sides of combined area $2 \cdot \frac{1}{2}b_1b_2 = b_1b_2$. The subtraction leaves

$$(a_1 + b_1)(a_2 + b_2) - 2a_2b_1 - a_1a_2 - b_1b_2 = a_1b_2 - a_2b_1.$$

The determinant is the exact geometric area of the parallelogram, and its absolute value accounts for orientation.

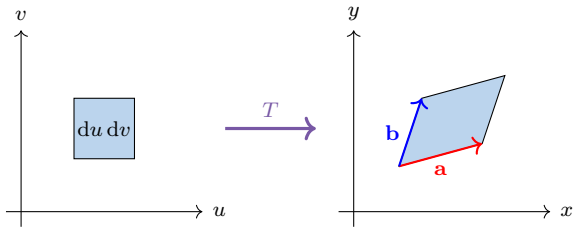


Figure 34.5: Under a smooth map T, a tiny rectangle becomes a tiny parallelogram. Its area is $|\det|\,\mathrm{d}u\,\mathrm{d}v$, where the determinant is built from the partial derivatives. That determinant is the Jacobian.

That determinant has a name.

Definition 34.2 (Jacobian)**.** For a transformation $x = x(u, v)$, $y = y(u, v)$, the *Jacobian determinant* is

$$\frac{\partial(x, y)}{\partial(u, v)} = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}.$$

Theorem 34.2 (Change of variables)**.** *If $T : (u, v) \mapsto (x, y)$ is a smooth, invertible map carrying a region S in the uv-plane onto a region R in the xy-plane, then*

$$\iint_R f(x, y)\,\mathrm{d}x\,\mathrm{d}y = \iint_S f\big(x(u, v), y(u, v)\big) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \mathrm{d}u\,\mathrm{d}v.$$

So changing variables in a double integral is: substitute the new expressions for x and y into f, replace $\mathrm{d}x\,\mathrm{d}y$ by the absolute Jacobian times $\mathrm{d}u\,\mathrm{d}v$, and re-state the limits in the new variables. It is u-substitution wearing a determinant.

Connection

This is the moment two big stories in your education shake hands. In linear algebra you learned that the determinant of a matrix is the factor by which the corresponding linear map scales area (in 2D) or volume (in 3D), and that a negative determinant means the map also flips orientation. In calculus you learned that a smooth map, viewed closely enough, *is* linear – its derivative. Put them together: the local area-scaling of a smooth map is the determinant of its derivative matrix. The Jacobian is the determinant of the derivative. Two subjects, one idea.

34.5.3 Polar coordinates, re-derived without pictures

Let us check the machine against an answer we already trust. For polar coordinates the "new variables" are (r, θ) and $x = r\cos\theta$, $y = r\sin\theta$. The partials:

$$\frac{\partial x}{\partial r} = \cos\theta, \quad \frac{\partial x}{\partial \theta} = -r\sin\theta, \quad \frac{\partial y}{\partial r} = \sin\theta, \quad \frac{\partial y}{\partial \theta} = r\cos\theta.$$

The Jacobian:

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \det\begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix} = r\cos^2\theta + r\sin^2\theta = r.$$

So $\mathrm{d}x\,\mathrm{d}y = |r|\,\mathrm{d}r\,\mathrm{d}\theta = r\,\mathrm{d}r\,\mathrm{d}\theta$ (since $r \geq 0$), exactly the factor we found by staring at one fanned-out cell. The formula and the picture agree, as they must. The spherical factor $\rho^2\sin\phi$ comes out of the analogous 3×3 Jacobian determinant in the same way; computing it is a good exercise, and the determinant earns its keep when no friendly picture is handy.

Example 34.8 (A change of variables that is not polar)**.** Evaluate $\iint_R (x+y)\,\mathrm{d}A$ where R is the parallelogram with vertices $(0,0)$, $(1,1)$, $(2,1)$, $(1,0)$ – ugly bounds in x, y. Its edges follow the lines $x - y = \text{const}$ and $y = \text{const}$, so set

$$u = x - y, \qquad v = y.$$

Then $x = u + v$, $y = v$. In the new variables R becomes the clean rectangle $0 \leq u \leq 1$, $0 \leq v \leq 1$. The Jacobian:

$$\frac{\partial(x,y)}{\partial(u,v)} = \det\begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = \det\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = 1.$$

The map is a shear, and shears preserve area – determinant 1 – so $\mathrm{d}A$ is unchanged. With $x + y = (u+v) + v = u + 2v$:

$$\iint_R (x+y)\,\mathrm{d}A = \int_0^1 \int_0^1 (u+2v)\cdot 1\,\mathrm{d}u\,\mathrm{d}v.$$

Inner: $\int_0^1 (u+2v)\mathrm{d}u = \frac{1}{2} + 2v$. Outer: $\int_0^1 (\frac{1}{2} + 2v)\mathrm{d}v = \frac{1}{2} + 1 = \frac{3}{2}$. The shear turned a parallelogram into a unit square at no cost, because its Jacobian was 1.

Common trap

Three ways people corrupt a change of variables:

- Forgetting the Jacobian entirely (the polar-r disaster, generalized).

- Forgetting the absolute value. Area is positive; if your Jacobian comes out negative it only means the map flipped orientation, and you take $|\cdot|$.
- Forgetting to also convert the *region* – substituting in f and the $\mathrm{d}A$ factor but leaving the old x, y limits. New variables, new limits; re-describe the region in u, v.

History

Carl Gustav Jacob Jacobi systematized these "functional determinants" in the 1830s and 1840s, and they have borne his name ever since – though Euler and Lagrange had used the special cases (like the polar r) long before, each time re-deriving the scaling factor by geometry. Jacobi's contribution was to see that *one* object, the determinant of the matrix of partial derivatives, governs *all* such coordinate changes at once, in any number of dimensions. It is a recurring pattern in the history of mathematics: a factor that practitioners keep rediscovering case by case finally gets unified into a single structural law – here, the determinant – and the unification is what we teach.

Why this matters

Step back and see the whole arc of this chapter as one sentence. *An integral adds up a quantity over little cells; to compute it you slice (Fubini), and to bend the coordinate grid you scale each cell by a determinant (the Jacobian).* The single integral, the double, the triple; polar, cylindrical, spherical; the $\sqrt{\pi}$ of the Gaussian – all of it is that one sentence, applied with different cells and different determinants. Calculus in many variables is not a pile of new formulas. It is the same Riemann sum you have known since Chapter 20, asked the same two questions: how shall I organize the sum, and how big is one cell?

34.6 Exercises

Exercise 34.1 (Warm-up over a rectangle)**.** Compute $\displaystyle\iint_R (4 - x - y)\,\mathrm{d}A$ over $R = [0,1] \times [0,2]$ by iterating in both orders, and confirm the two answers agree.

Exercise 34.2 (A region between curves)**.** Let R be the region bounded by $y = x^2$ and $y = 2x$. Sketch it, decide which order is convenient, and evaluate $\displaystyle\iint_R x\,\mathrm{d}A$. (First find where the two curves meet.)

Exercise 34.3 (Change the order)**.** Rewrite $\displaystyle\int_0^4 \int_{\sqrt{y}}^2 \frac{1}{x^3+1}\,\mathrm{d}x\,\mathrm{d}y$ with the order of integration reversed, then evaluate it. (As written the inner integral is unpleasant; reversed it is a clean substitution.)

Exercise 34.4 (Polar muscle)**.** Evaluate $\displaystyle\iint_R \frac{1}{1+x^2+y^2}\,\mathrm{d}A$, where R is the disk $x^2 + y^2 \le 1$, by converting to polar coordinates. Where did the factor of r make the radial integral elementary?

Exercise 34.5 (A radial Gaussian)**.** Using polar coordinates, find $\displaystyle\iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,\mathrm{d}A$. Then explain

in one sentence why this equals $\left(\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x\right)^2$, and use it to recover $\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x = \sqrt{\pi}$.

Exercise 34.6 (Triple integral over a box and a wedge)**.** (a) Compute $\iiint_E xyz\,\mathrm{d}V$ over the box $[0,1]^3$. (b) Compute the volume of the solid below $z = 1 - x$ and above the unit square $[0,1] \times [0,1]$ in the xy-plane (assume $1 - x \geq 0$ throughout).

Exercise 34.7 (Spherical coordinates)**.** Find the mass of a ball of radius a whose density at a point is proportional to its distance from the centre, $f = k\rho$. Use spherical coordinates and the volume element $\rho^2 \sin\phi\,\mathrm{d}\rho\,\mathrm{d}\phi\,\mathrm{d}\theta$. (You should find the mass is $k\pi a^4$.)

Exercise 34.8 (Compute a Jacobian)**.** For the transformation $x = u^2 - v^2$, $y = 2uv$, compute the Jacobian $\dfrac{\partial(x,y)}{\partial(u,v)}$ and simplify. (You should get $4(u^2 + v^2)$. This is the squaring map of complex numbers, $z \mapsto z^2$; the area-scaling at a point grows with distance from the origin – interpret why.)

Exercise 34.9 (Change of variables, designed)**.** Evaluate $\iint_R e^{(x-y)/(x+y)}\,\mathrm{d}A$ over the triangle with vertices $(0,0)$, $(1,0)$, $(0,1)$ using the substitution $u = x - y$, $v = x + y$. Find the Jacobian, convert the region, and reduce to a one-variable integral.

Exercise 34.10 (Conceptual: why the r?)**.** Without computing any integral, explain in your own words why the polar area element carries a factor of r but the Cartesian element $\mathrm{d}x\,\mathrm{d}y$ does not. Then explain why $\iint_R 1 \cdot \mathrm{d}r\,\mathrm{d}\theta$ (with the r wrongly omitted) does *not* give the area of a disk, and what quantity it gives instead.

Exercise 34.11 (Proof: linearity from the sum)**.** Prove that $\iint_R \big(f + g\big)\,\mathrm{d}A = \iint_R f\,\mathrm{d}A + \iint_R g\,\mathrm{d}A$ directly from the Riemann-sum definition, by writing out the sum for $f + g$ and splitting it. Identify the single property of finite sums that makes the argument work, and note that the limit step requires both integrals to exist.

Exercise 34.12 (Proof: a determinant is an area scaler)**.** Let T be the linear map sending (u, v) to $(au + bv,\ cu + dv)$. Show directly that the unit square $[0,1]^2$ maps to a parallelogram of area $|ad - bc|$, and conclude that T scales every area by the factor $|ad - bc|$. Explain how this is the constant-coefficient special case of the Jacobian theorem (Theorem 34.2).

Chapter 35

Vector Calculus: Green, Stokes, and Divergence

Every theorem in this chapter is secretly the same theorem. You already know it. Back in Chapter 21 the Fundamental Theorem of Calculus told you that to add up a rate of change over an interval, you do not have to actually add anything up – you just look at the two endpoints:

$$\int_a^b F'(x)\,\mathrm{d}x \;=\; F(b) - F(a).$$

The whole interior of $[a, b]$ – the infinitely many places where F is wiggling and changing – collapses into two numbers at the boundary. The accumulated change inside is remembered entirely by the edge.

That sentence – *the boundary remembers the inside* – is the soul of vector calculus. The great theorems of this chapter (Green's theorem, the divergence theorem, Stokes' theorem) all say the same thing in higher dimensions: to learn what is happening throughout a region, you only have to walk around its *boundary*. An integral over a solid blob equals an integral over the skin that wraps it. An integral over a flat patch equals an integral around the loop that fences it. In each case a d-dimensional accumulation is paid off by a $(d-1)$-dimensional reckoning at the edge. The Fundamental Theorem of Calculus was the one-dimensional baby version, where the "boundary" of an interval is just its two endpoints.

To say all this we need a new main character: the *vector field*. So far our functions have eaten points and spat out numbers. Now they will spit out *arrows* – a velocity at every point of a flowing river, a force at every point in a magnetic field, a push at every point of the wind. Once we can integrate arrows along curves and across surfaces, the great theorems write themselves.

Focus

One idea, four faces:

$$\begin{aligned}
&\text{FTC:} && \int_a^b F' = F(b) - F(a) && \text{(1D: endpoints)}\\
&\text{Gradient theorem:} && \int_C \nabla f \cdot d\mathbf{r} = f(\text{end}) - f(\text{start}) && \text{(curve: endpoints)}\\
&\text{Green / Stokes:} && \iint_S (\text{curl}) = \oint_{\partial S} (\text{field}) && \text{(patch: boundary loop)}\\
&\text{Divergence:} && \iiint_V (\text{div}) = \oiint_{\partial V} (\text{flux}) && \text{(solid: boundary skin)}
\end{aligned}$$

Read every line as: *accumulate a derivative inside = read the field on the boundary.*

35.1 Vector fields: a function that grows arrows

A *scalar field* assigns a number to each point: temperature in a room, altitude on a map, pressure in the ocean. We have been studying these for two chapters under the name $f(x, y)$ or $f(x, y, z)$. A *vector field* assigns an *arrow* – a vector – to each point instead.

Definition 35.1 (Vector field)**.** A vector field on a region of the plane is a function

$$\mathbf{F}(x, y) = \big(P(x, y),\, Q(x, y)\big) = P(x, y)\,\mathbf{i} + Q(x, y)\,\mathbf{j},$$

that assigns to each point (x, y) the vector $\mathbf{F}(x, y)$. In three dimensions, $\mathbf{F}(x, y, z) = (P, Q, R)$, with three component functions. We draw it by planting the arrow $\mathbf{F}(x, y)$ with its *tail* at the point (x, y).

The picture is everything. Below is the field $\mathbf{F}(x, y) = (-y, x)$. At each point we draw the arrow $(-y, x)$. Notice it always points ninety degrees counterclockwise from the position vector (x, y) – so the whole field swirls around the origin like water circling a drain, or a spinning turntable.

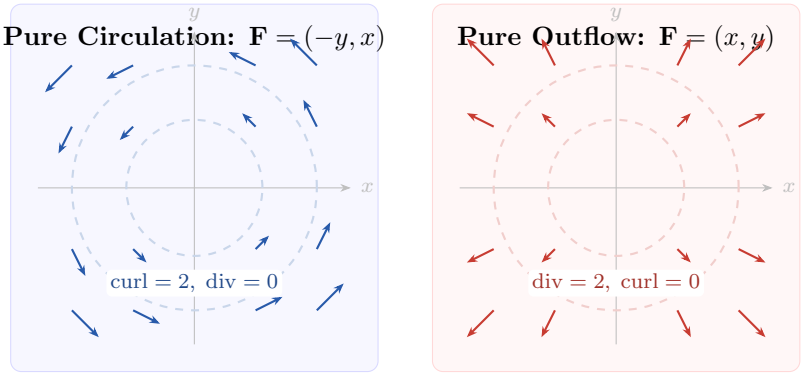


Figure 35.1: The two fundamental motions of a vector field. Left: The rotational field $\mathbf{F}(x, y) = (-y, x)$ circulates around the origin with constant scalar curl 2 and zero divergence. Right: The source field $\mathbf{F}(x, y) = (x, y)$ radiates outward from the origin with constant divergence 2 and zero curl.

Intuition

Two motions live inside every vector field, and the rest of the chapter is about measuring them. **Circulation**: does the field tend to *spin* you around a loop? (The swirl in Figure 35.1, left, does.) **Flux**: does the field tend to *carry stuff out through* a boundary? (The source in Figure 35.1, right, does.) Curl will measure local spin; divergence will measure local spreading. Everything else is bookkeeping.

Application

Vector fields are how physics stores the state of the invisible. The wind on a weather map is $\mathbf{F}(x, y) =$ (velocity of air). The gravitational pull near a planet is a field of arrows pointing inward. The electric field around a charge, the magnetic field around a wire, the flow of heat through a wall, the current in the ocean – each is a vector field, and each obeys laws written in the language of this chapter. Maxwell's four equations of electromagnetism are *nothing but* statements about the divergence and curl of two vector fields.

35.2 Line integrals: adding a field along a path

35.2.1 Work, and why we multiply force by displacement

Drag a box across a floor. The work you do is force times distance – but only the *part of the force that lies along the motion* counts. Push straight down on a box and slide it sideways and you have done no work on it at all; your downward force is perpendicular to the sideways motion. The right measure of work for a constant force $\mathbf{F}$ over a straight displacement $\mathbf{d}$ is the dot product

$$W = \mathbf{F} \cdot \mathbf{d} = |\mathbf{F}|\,|\mathbf{d}| \cos\theta,$$

which keeps exactly the component of $\mathbf{F}$ pointing along $\mathbf{d}$ and throws away the perpendicular part.

Now let the force vary from point to point – a vector field $\mathbf{F}$ – and let the path curve. We do what we always do in calculus: chop the path into tiny nearly-straight pieces, on each piece treat the force as roughly constant, compute the little bit of work $\mathbf{F} \cdot \Delta\mathbf{r}$, add them all up, and take the limit as the pieces shrink. That limit is the *line integral*.

Definition 35.2 (Line integral of a vector field)**.** Let C be a curve traced by $\mathbf{r}(t) = (x(t), y(t))$ for $t \in [a, b]$, and let $\mathbf{F}$ be a vector field along it. The line integral of $\mathbf{F}$ along C is

$$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} \;=\; \int_a^b \mathbf{F}\big(\mathbf{r}(t)\big) \cdot \mathbf{r}\,'(t)\,\mathrm{d}t.$$

Writing $\mathbf{F} = (P, Q)$ and $\mathrm{d}\mathbf{r} = (\mathrm{d}x, \mathrm{d}y)$, this is also written $\displaystyle\int_C P\,\mathrm{d}x + Q\,\mathrm{d}y$.

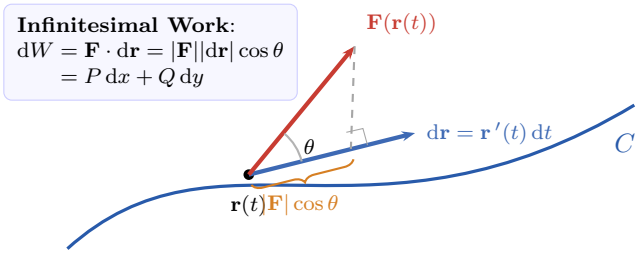


Figure 35.2: The infinitesimal work element $\mathbf{F} \cdot d\mathbf{r}$ along a curve C. At each point $\mathbf{r}(t)$, the path advances along the tangent displacement $d\mathbf{r} = \mathbf{r}'(t)\,dt$. The dot product projects $\mathbf{F}$ onto the motion: only the component $|\mathbf{F}| \cos\theta$ parallel to the curve does work. The perpendicular component produces zero motion along C.

Why the technique works

Why does $d\mathbf{r}$ become $\mathbf{r}'(t)\,dt$? Because $\mathbf{r}'(t)$ is the velocity vector – it points along the curve and its length is the speed. Over a tiny time dt the path advances by the little displacement vector $d\mathbf{r} = \mathbf{r}'(t)\,dt$. Dotting the force against this displacement gives the infinitesimal work $\mathbf{F} \cdot d\mathbf{r}$, and integrating in t adds them along the curve. The parameter t is just a clock we run along the path; the answer, reassuringly, will not depend on *how fast* we run it.

Walking question

Why doesn't the speed of traversal change the answer? If you re-parametrize and move twice as fast, then $\mathbf{r}'$ doubles but the time interval dt halves, and the product $\mathbf{r}'(t)\,dt = d\mathbf{r}$ is unchanged – it is a geometric displacement, blind to the clock. (Reversing direction, however, flips every $d\mathbf{r}$, so it flips the sign of the integral. Orientation matters; speed does not.)

35.2.2 A worked line integral, step by step

Example 35.1 (Work along a parabola)**.** Compute $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F}(x, y) = (y,\, x^2)$ along the parabola $y = x^2$ from $(0, 0)$ to $(2, 4)$.

Step 1 – parametrize. Let $x = t$, $y = t^2$, so $\mathbf{r}(t) = (t, t^2)$ for $t \in [0, 2]$. Then $\mathbf{r}'(t) = (1, 2t)$.

Step 2 – evaluate the field on the path. $\mathbf{F}(\mathbf{r}(t)) = (y,\, x^2)\big|_{x=t,\, y=t^2} = (t^2,\, t^2)$.

Step 3 – dot. $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = (t^2)(1) + (t^2)(2t) = t^2 + 2t^3$.

Step 4 – integrate.

$$\int_0^2 (t^2 + 2t^3)\,dt = \left[\frac{t^3}{3} + \frac{t^4}{2}\right]_0^2 = \frac{8}{3} + 8 = \frac{32}{3}.$$

The total work is $\frac{32}{3}$. The recipe never changes: parametrize, plug in, dot, integrate a single ordinary integral in t.

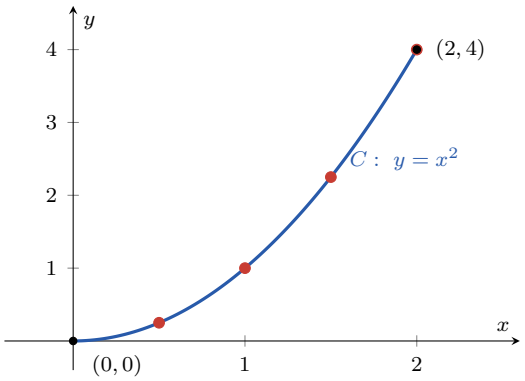


Figure 35.3: The work integral of Example 35.1: we walk the parabola C (blue) from $(0,0)$ to $(2,4)$. The line integral adds up, at each point of the path, the component of the field $\mathbf{F} = (y, x^2)$ that lies along the blue curve. The along-path total works out to $\frac{32}{3}$.

Common trap

A line integral $\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}$ is *not* the same as the arc-length integral $\int_C f \, \mathrm{d}s$ you may have seen (mass of a wire). The dot product version measures the field's component *along* the curve and is sensitive to direction; reversing the path negates it. The arc-length version uses $\mathrm{d}s = |\mathbf{r}'(t)| \, \mathrm{d}t$ (a length, always positive) and does not care about direction. They answer different questions. This chapter is almost entirely about the dot-product kind.

35.3 Conservative fields: when the path does not matter

In Example 35.1 the answer depended on the curve we chose. For some special fields it does *not*: the work depends only on where you start and where you stop, never on the route between. These are the *conservative* fields, and they are exactly the fields that are gradients.

Theorem 35.1 (Fundamental Theorem for line integrals / Gradient theorem)**.** *If $\mathbf{F} = \nabla f$ for some scalar function f (a* potential*), then for any curve C from point A to point B,*

$$\int_C \nabla f \cdot \mathrm{d}\mathbf{r} = f(B) - f(A).$$

The integral depends only on the endpoints. In particular, around any closed loop ($A = B$), $\oint_C \nabla f \cdot \mathrm{d}\mathbf{r} = 0$.

Why the technique works

This is the Fundamental Theorem of Calculus wearing a coat. Parametrize C by $\mathbf{r}(t)$, $t \in [a, b]$, and watch the chain rule do the entire job:

$$\int_C \nabla f \cdot \mathrm{d}\mathbf{r} = \int_a^b \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, \mathrm{d}t = \int_a^b \frac{\mathrm{d}}{\mathrm{d}t}\Big[f(\mathbf{r}(t))\Big] \, \mathrm{d}t = f(\mathbf{r}(b)) - f(\mathbf{r}(a)).$$

The middle equality is the multivariable chain rule (Chapter 33): $\nabla f \cdot \mathbf{r}\,' = \frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{r}(t))$. So the integrand was a perfect derivative all along, and the ordinary FTC finishes it. "The boundary remembers the inside" here means: the two endpoints remember the whole accumulated climb.

Intuition

A potential f is an *altitude*. The field ∇f points straight uphill, and its work is the height you gain. Walk any trail from the trailhead to the summit – switchbacks, scenic detours, whatever – and the elevation gained is the same, because it is just (summit altitude) minus (trailhead altitude). Walk a loop back to where you started and you gain zero net altitude. That is the whole content of conservativeness. A field is conservative precisely when it has an altitude function.

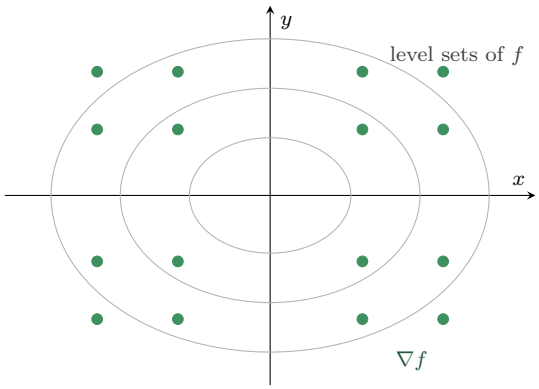


Figure 35.4: A conservative field is a gradient. Here $f(x, y) = x^2 + y^2$ is the "altitude." Its level sets (gray circles) are the contour lines, and the field $\nabla f = (2x, 2y)$ (green arrows) crosses them at right angles, always pointing uphill. Work done along any path is just the change in altitude between its endpoints; around a closed loop it is zero.

35.3.1 How to test for, and find, a potential

In the plane, write $\mathbf{F} = (P, Q)$. If $\mathbf{F} = \nabla f = (f_x, f_y)$ then $P = f_x$ and $Q = f_y$. By Clairaut's theorem (mixed partials commute, Chapter 32),

$$\frac{\partial P}{\partial y} = f_{xy} = f_{yx} = \frac{\partial Q}{\partial x}.$$

So a *necessary* condition is $P_y = Q_x$. On a region with no holes (a "simply connected" region – one where every loop can be shrunk to a point) this condition is also *sufficient*: it guarantees a potential exists.

Example 35.2 (Finding a potential)**.** Is $\mathbf{F} = (2xy + 3,\ x^2 - 1)$ conservative? Here $P = 2xy + 3$, $Q = x^2 - 1$.

Test. $P_y = 2x$ and $Q_x = 2x$. They agree, and the plane has no holes, so a potential exists.

Build it. We need f with $f_x = 2xy + 3$. Integrate in x (treating y as constant):

$$f(x, y) = x^2 y + 3x + g(y),$$

where the "constant" of integration is allowed to depend on y. Now demand $f_y = Q$:

$$f_y = x^2 + g'(y) \overset{!}{=} x^2 - 1 \;\Rightarrow\; g'(y) = -1 \;\Rightarrow\; g(y) = -y + C.$$

So $f(x,y) = x^2y + 3x - y + C$. As a check, $\nabla f = (2xy + 3,\ x^2 - 1) = \mathbf{F}$. Now any line integral of $\mathbf{F}$ is trivial: from $(0,0)$ to $(1,2)$ it is $f(1,2) - f(0,0) = (2 + 3 - 2) - 0 = 3$, no matter the path.

Common trap

The hole matters. The famous swirl $\mathbf{F} = \left(\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2}\right)$ satisfies $P_y = Q_x$ *everywhere it is defined*, yet $\oint \mathbf{F} \cdot \mathrm{d}\mathbf{r} = 2\pi \neq 0$ around the unit circle. It is *not* conservative. The catch: it is undefined at the origin, so its domain has a hole, and the loop around that hole cannot be shrunk to a point. The test $P_y = Q_x$ only promises a potential on a *simply connected* domain. Topology sneaks into calculus right here.

History

The word "conservative" comes from physics, not bookkeeping. If $\mathbf{F}$ is a force with potential energy U (so $\mathbf{F} = -\nabla U$), then the work done equals the drop in U, and the quantity (kinetic + potential energy) stays constant along any motion – energy is *conserved*. Gravity and electrostatics are conservative; friction is not (it bleeds energy as heat, and its loop integral is nonzero – that is why perpetual-motion machines fail). The nineteenth-century drive to understand which forces conserve energy, led by Helmholtz and Kelvin, is exactly what forced mathematicians to clarify when a field is a gradient.

35.4 Green's theorem: the first "boundary remembers inside"

The first of the three great theorems is also the one easiest to picture. Green's theorem relates a line integral *around* a closed loop C to a double integral *over* the flat region D that the loop encloses.

Theorem 35.2 (Green's theorem, circulation form)**.** *Let D be a region in the plane bounded by a simple closed curve C, traversed counterclockwise. If $\mathbf{F} = (P, Q)$ has continuous partials on D, then*

$$\oint_C P\,\mathrm{d}x + Q\,\mathrm{d}y \;=\; \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) \mathrm{d}A.$$

The quantity $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$ is the *microscopic circulation* – the "curl" of the field, which we meet properly in the next section. Green's theorem says: the total swirl around the boundary equals the sum of all the little swirls inside.

What is really happening

Here is the real mechanism, and it is beautiful. Tile the region D with tiny rectangles. For each tiny rectangle, run the line integral around its own little boundary, counterclockwise. Now add up all these little loop integrals. Look at any internal edge shared by two adjacent tiles: one tile traverses it left-to-right, its neighbor traverses the *same* edge right-to-left. The two contributions are equal and opposite – they **cancel**. Every interior edge cancels against

its neighbor. The *only* edges that survive the great cancellation are the ones on the outer boundary of D, where there is no neighbor to cancel with. So

$$\sum_{\text{tiles}} (\text{tiny loop integral}) = (\text{big boundary integral}).$$

Meanwhile each tiny loop integral, by a short computation, equals $\big(Q_x - P_y\big) \times (\text{area of tile})$. Summing those is the double integral. The interior cancellation is *exactly* the same telescoping that made the Fundamental Theorem of Calculus work, where the inner endpoints of adjacent subintervals cancelled and only a and b survived.

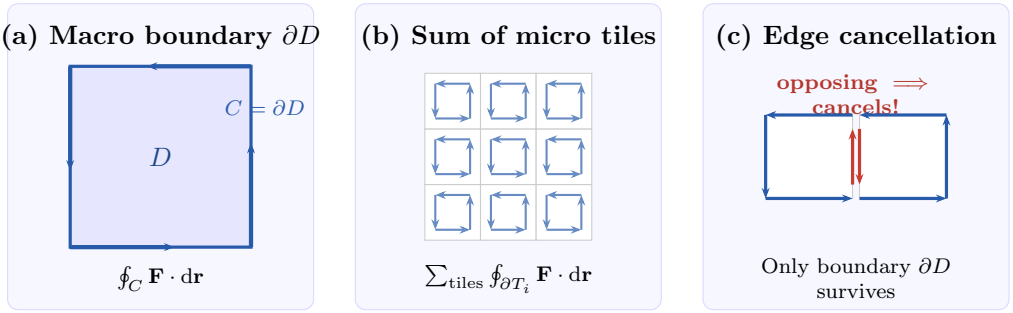


Figure 35.5: Green's theorem via microscopic edge cancellation. (a) The macro circulation $\oint_{\partial D} \mathbf{F} \cdot \mathrm{d}\mathbf{r}$ around the outer boundary loop. (b) The region D tiled into tiny sub-rectangles, each evaluated counterclockwise. (c) On every internal boundary shared by two adjacent cells, one cell traverses the segment forward while the other traverses it backward; the two opposing contributions cancel out identically. Every interior edge cancels, leaving only the outer perimeter ∂D to account for the sum of all microscopic curls.

35.4.1 Green's theorem as an area machine

A startling special case: choose $\mathbf{F}$ so that $Q_x - P_y = 1$. Then the double integral is just the *area* of D, and Green's theorem computes area from a trip around the boundary. Taking $\mathbf{F} = (-y/2,\, x/2)$ gives $Q_x - P_y = \frac{1}{2} + \frac{1}{2} = 1$, so

$$\text{Area}(D) = \frac{1}{2} \oint_C x\,\mathrm{d}y - y\,\mathrm{d}x.$$

This is the principle behind the *planimeter*, a mechanical instrument that measures the area of any shape by tracing its outline – once around the boundary and a dial reads off the enclosed area.

Example 35.3 (Area of an ellipse by its boundary)**.** Let C be the ellipse $x = a\cos t$, $y = b\sin t$, $t \in [0, 2\pi]$. Then $\mathrm{d}x = -a\sin t\,\mathrm{d}t$, $\mathrm{d}y = b\cos t\,\mathrm{d}t$, and

$$x\,\mathrm{d}y - y\,\mathrm{d}x = (a\cos t)(b\cos t)\,\mathrm{d}t - (b\sin t)(-a\sin t)\,\mathrm{d}t = ab(\cos^2 t + \sin^2 t)\,\mathrm{d}t = ab\,\mathrm{d}t.$$

Therefore

$$\text{Area} = \frac{1}{2}\int_0^{2\pi} ab\,\mathrm{d}t = \frac{1}{2}\,(ab)(2\pi) = \pi ab.$$

The familiar πab, computed without ever filling in the interior – the boundary knew the area all along.

Example 35.4 (Verifying Green's theorem directly)**.** Let $\mathbf{F} = (-y,\, x)$ (the swirl) and let D be the unit disk, C its boundary circle counterclockwise.

Right side (double integral). Here $P = -y$, $Q = x$, so $Q_x - P_y = 1 - (-1) = 2$. Thus $\iint_D 2\,\mathrm{d}A = 2 \cdot (\text{area of disk}) = 2\pi$.

Left side (line integral). Parametrize $x = \cos t$, $y = \sin t$:

$$\oint_C -y\,\mathrm{d}x + x\,\mathrm{d}y = \int_0^{2\pi} \big[(-\sin t)(-\sin t) + (\cos t)(\cos t)\big]\,\mathrm{d}t = \int_0^{2\pi} 1\,\mathrm{d}t = 2\pi.$$

Both sides give 2π. The constant microscopic swirl of 2 at every point, summed over the disk, equals the circulation you feel going once around the rim.

35.5 Curl and divergence: spin and spread, made local

Green's theorem has already handed us two local quantities worth naming. They are the derivatives the great theorems integrate, so it pays to know what each one measures physically.

35.5.1 Divergence: how much a point is a source

Definition 35.3 (Divergence)**.** For $\mathbf{F} = (P, Q, R)$, the divergence is the scalar

$$\operatorname{div}\mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}.$$

(In the plane, drop the last term.)

Intuition

Divergence measures *spreading*. Imagine $\mathbf{F}$ is the velocity of a fluid. Surround a point with a tiny balloon. If more fluid flows *out* of the balloon than in, the point is a *source* and $\operatorname{div}\mathbf{F} > 0$. If fluid piles *in*, it is a *sink* and $\operatorname{div}\mathbf{F} < 0$. If as much leaves as enters, $\operatorname{div}\mathbf{F} = 0$ and the fluid is *incompressible* there. For the source field $\mathbf{F} = (x, y)$ of Figure 35.1 (right), $\operatorname{div}\mathbf{F} = 1 + 1 = 2 > 0$ everywhere: every point is gushing. For the swirl $(-y, x)$, $\operatorname{div}\mathbf{F} = 0$: swirling moves fluid around but never makes it pile up or thin out.

35.5.2 Curl: how much a point is a whirlpool

Definition 35.4 (Curl)**.** In three dimensions, $\operatorname{curl}\mathbf{F} = \nabla \times \mathbf{F}$, the vector

$$\operatorname{curl}\mathbf{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z},\ \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x},\ \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right).$$

In the plane, only the last component survives; the *scalar curl* of (P, Q) is $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$ – exactly the integrand of Green's theorem.

Intuition

Curl measures *spin*. Drop a tiny paddlewheel into the fluid at a point. If the flow makes it rotate, the curl is nonzero, and it points along the axle (by the right-hand rule), with magnitude twice the angular speed of the spin. The swirl $(-y, x)$ has scalar curl $1-(-1)=2$ everywhere – a paddlewheel anywhere in it spins counterclockwise. The source (x, y) has curl $0-0=0$: gas rushing radially outward never spins a paddlewheel, it just carries it straight out.

Common trap

A field can *circle the origin* and still have zero curl everywhere it is defined – "curls around" (global) and "has curl" (local) are different. The hole field $\left(\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2}\right)$ has scalar curl 0 at every point of its domain, yet its loop integral around the origin is 2π. Curl is a local paddlewheel test; circulation around a big loop is a global sum. Green's theorem links them only when the field is smooth *everywhere inside* the loop – and this field is not (it blows up at the origin).

35.5.3 The microscopic geometry: where the formulas come from

Why is divergence $\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}$? Why is curl $\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}$? Neither formula is an arbitrary algebraic definition; both emerge directly from inspecting a single microscopic coordinate cell $[x, x+\Delta x] \times [y, y+\Delta y]$.

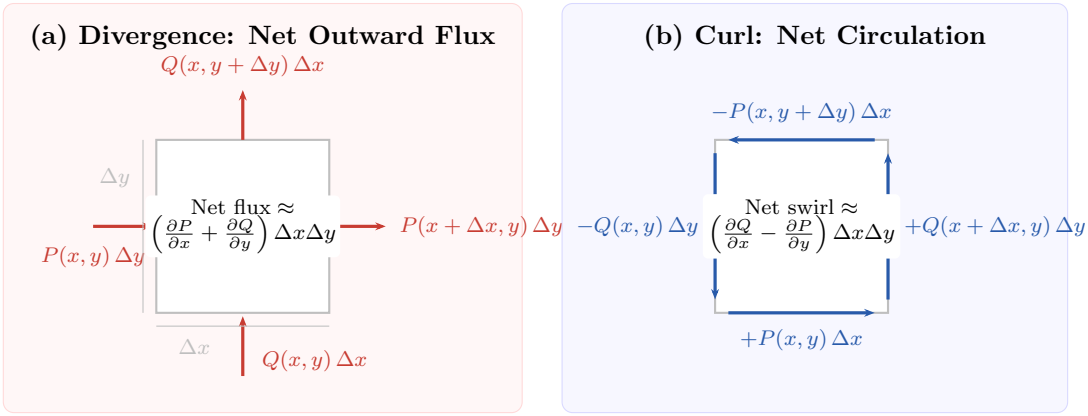


Figure 35.6: The microscopic origin of divergence and curl. (a) Divergence counts the net outward flux per unit area across the four walls of a cell: the difference in horizontal flow across Δx gives $\frac{\partial P}{\partial x}$, plus the difference in vertical flow across Δy gives $\frac{\partial Q}{\partial y}$. (b) Curl counts the net counterclockwise circulation around the four edges: the difference between the right and left edges gives $+\frac{\partial Q}{\partial x}\,\Delta x\Delta y$, while the difference between the top and bottom edges (running backward on top) gives $-\frac{\partial P}{\partial y}\,\Delta x\Delta y$.

Why this matters

Notice the symmetry. Green's theorem in *circulation form* ($\oint P\,dx + Q\,dy = \iint(Q_x - P_y)$) integrates the *curl*. There is a twin, the *flux form*, that integrates the *divergence*:

$$\oint_C \mathbf{F}\cdot\mathbf{n}\,ds = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right) dA = \iint_D \operatorname{div}\mathbf{F}\,dA,$$

where $\mathbf{n}$ is the outward unit normal. The same Green's theorem, read two ways, is the seed of *both* grand theorems to come: spin its circulation form into 3D and you get Stokes; spin its flux form into 3D and you get the divergence theorem. Two children, one parent.

35.6 Flux and the divergence theorem: the boundary skin

35.6.1 Flux across a surface

In 3D the analog of "how much field crosses the boundary loop" becomes "how much field crosses the boundary *surface*." That quantity is the *flux*.

Definition 35.5 (Flux integral)**.** Let S be a surface with a chosen unit normal direction $\mathbf{n}$. The flux of $\mathbf{F}$ through S is

$$\iint_S \mathbf{F}\cdot\mathbf{n}\,dS.$$

Only the component of $\mathbf{F}$ *piercing* the surface (along $\mathbf{n}$) contributes; field sliding *along* the surface carries nothing through.

Intuition

Hold a hoop in a river. The flux is the volume of water passing through the hoop per second. Hold it face-on to the current and the flux is large; turn it edge-on and the flux drops to zero, because now the water slides past the rim instead of through the hoop. That "how much is going through" – counting only the piercing component $\mathbf{F}\cdot\mathbf{n}$ – is flux.

35.6.2 The divergence theorem

Theorem 35.3 (Divergence theorem / Gauss's theorem)**.** *Let V be a solid region in space bounded by a closed surface $S = \partial V$ with outward unit normal $\mathbf{n}$. If $\mathbf{F}$ has continuous partials on V, then*

$$\iiint_V \operatorname{div}\mathbf{F}\,dV \;=\; \oiint_S \mathbf{F}\cdot\mathbf{n}\,dS.$$

What is really happening

The proof is the same cancellation story, one dimension up. Chop the solid V into tiny boxes. Compute the net flux out of each tiny box – its outward push minus its inward pull on all six faces. Now sum over all the boxes. Every *internal* face is shared by two adjacent boxes: what

flows *out* of one box through that face flows *into* its neighbor, so the two flux contributions are equal and opposite and **cancel**. Only the faces on the *outer skin* of V survive, with no neighbor to cancel them – and their total is the flux through $S = \partial V$. Meanwhile the net flux out of one tiny box equals $\operatorname{div}\mathbf{F} \times$ (box volume) – that is literally what divergence *is*, the outflow per unit volume. Summing gives $\iiint_V \operatorname{div}\mathbf{F}\,dV$. Interior fluxes cancel; the boundary remembers the inside.

Why this matters

The divergence theorem is a conservation law in disguise, and it is the deepest "boundary remembers inside" of them all. All the sources scattered throughout a solid – every point where fluid is being created – must account exactly for the net flow leaking out through its surface. Stuff cannot vanish: whatever is generated inside has to come out the skin. This single identity is Gauss's law of electrostatics (charge inside = field flux out), the continuity equation of fluid flow, and the heat equation's energy balance, all at once.

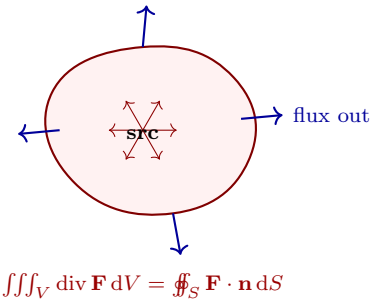


Figure 35.7: The divergence theorem. Sources sprinkled through the interior (red) must balance exactly against the total flux leaving through the outer skin (blue normals). Internal flows cancel in pairs; only the boundary is left to tell the tale.

Example 35.5 (Divergence theorem on a ball)**.** Let $\mathbf{F} = (x, y, z)$ and let V be the ball of radius a, with S its sphere.

Left side. $\operatorname{div}\mathbf{F} = 1 + 1 + 1 = 3$, so

$$\iiint_V 3\,dV = 3 \cdot \operatorname{Vol}(V) = 3 \cdot \frac{4}{3}\pi a^3 = 4\pi a^3.$$

Right side. On the sphere of radius a the outward normal is $\mathbf{n} = \frac{1}{a}(x, y, z)$, so $\mathbf{F}\cdot\mathbf{n} = \frac{1}{a}(x^2 + y^2 + z^2) = \frac{a^2}{a} = a$, a constant. Then

$$\oiint_S \mathbf{F}\cdot\mathbf{n}\,dS = a \cdot \operatorname{Area}(S) = a \cdot 4\pi a^2 = 4\pi a^3.$$

The two agree. The uniform source density 3 filling the ball is paid off exactly by the outward flux through the sphere.

35.7 Stokes' theorem: the boundary loop of a surface

Now the last of the three. Green's theorem lived in the flat plane; Stokes' theorem lifts it into 3D, where the surface S may be any curved membrane in space, like a soap film, with the line integral run around its rim ∂S.

Theorem 35.4 (Stokes' theorem)**.** *Let S be an oriented surface in space bounded by a simple closed curve $C = \partial S$, oriented compatibly (right-hand rule). If $\mathbf{F}$ has continuous partials, then*

$$\oint_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} \;=\; \iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n}\, \mathrm{d}S.$$

Intuition

The total circulation you feel marching once around the rim of a membrane equals the sum of all the little paddlewheel spins (the curl) spread across the membrane. And here is the wonder: *any* surface with the same rim gives the same answer. A flat disk, a tall dome, a crumpled sheet – if they share a boundary loop, the curl-flux through each is identical, because all that matters is the rim. (Since $\operatorname{div}(\operatorname{curl} \mathbf{F}) = 0$ always, the divergence theorem guarantees this independence: curl has no sources, so its flux depends only on the boundary.) Green's theorem is exactly Stokes' theorem for the special case where S is a flat region lying in the plane.

What is really happening

Same mechanism a third time. Tile the membrane S with tiny patches. Around each patch run the little circulation; by Green's theorem in miniature, that little circulation equals $(\operatorname{curl} \mathbf{F} \cdot \mathbf{n}) \times$ (patch area). Sum over all patches. The shared edges between neighboring patches are traversed in opposite directions and **cancel**; only the outer rim $C = \partial S$ survives. Result: rim circulation = total curl-flux. It is Green's theorem stitched over a curved surface, and the cancellation – yet again – is the FTC's telescoping in disguise.

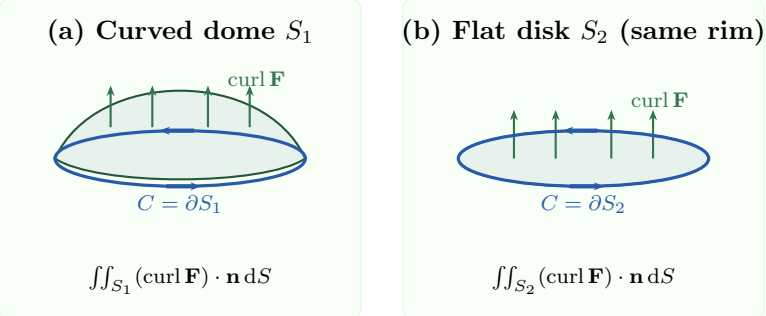


Figure 35.8: Stokes' theorem and surface independence. The circulation $\oint_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}$ around the boundary rim C equals the total flux of $\operatorname{curl} \mathbf{F}$ through *any* surface that caps C. (a) A curved dome S_1 and (b) a flat disk S_2 sharing the same rim $C = \partial S$ capture the exact same total curl flux, because any two such surfaces together form a closed container through which the divergence-free curl field has zero net flux.

Example 35.6 (Stokes on a hemisphere)**.** Let $\mathbf{F} = (-y, x, 0)$ (the swirl, lifted into 3D) and let S be the upper unit hemisphere $x^2 + y^2 + z^2 = 1$, $z \geq 0$, oriented upward. Its boundary C is the unit circle in the xy-plane.

Surface side. $\operatorname{curl}\mathbf{F} = (0, 0, Q_x - P_y) = (0, 0, 1 - (-1)) = (0, 0, 2)$. The simplest membrane with the same rim is the flat disk $z = 0$, where $\mathbf{n} = (0, 0, 1)$, so $(\operatorname{curl}\mathbf{F}) \cdot \mathbf{n} = 2$ and

$$\iint_S (\operatorname{curl}\mathbf{F}) \cdot \mathbf{n}\,\mathrm{d}S = 2 \cdot (\text{area of unit disk}) = 2\pi.$$

(We are allowed to swap the hemisphere for the flat disk precisely because Stokes says only the shared rim matters.)

Rim side. Parametrize C by $x = \cos t$, $y = \sin t$, $z = 0$:

$$\oint_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \int_0^{2\pi} \big[(-\sin t)(-\sin t) + (\cos t)(\cos t)\big]\,\mathrm{d}t = 2\pi.$$

Both give 2π. The curl pointing up through the cap is precisely the swirl you feel walking the equator.

History

"Stokes' theorem" was not proved by Stokes. In 1850 William Thomson (Lord Kelvin) wrote the result in a letter to George Stokes, who then set it as a problem on the 1854 Cambridge Smith's Prize examination – which a young James Clerk Maxwell sat. So the theorem entered the world as an *exam question*. Maxwell would later make curl and divergence the backbone of his theory of electromagnetism. The modern viewpoint – that Green, Stokes, divergence, and even the plain FTC are all the *single* generalized Stokes theorem $\int_M \mathrm{d}\omega = \int_{\partial M} \omega$ in the language of differential forms – was assembled by Poincare, Cartan, and others around 1900. One line now contains this entire chapter.

35.8 One theorem, four faces

Step back and see the family resemblance. In each statement, an integral of a *derivative* over a region equals an integral of the original *field* over the region's *boundary*:

Theorem	**Inside (a derivative)**	**Boundary (the field)**
FTC	$\displaystyle\int_a^b F'(x)\,\mathrm{d}x$	$F(b) - F(a)$
Gradient thm	$\displaystyle\int_C \nabla f \cdot \mathrm{d}\mathbf{r}$	$f(B) - f(A)$
Green	$\displaystyle\iint_D (Q_x - P_y)\,\mathrm{d}A$	$\displaystyle\oint_{\partial D} \mathbf{F} \cdot \mathrm{d}\mathbf{r}$
Stokes	$\displaystyle\iint_S \operatorname{curl}\mathbf{F} \cdot \mathbf{n}\,\mathrm{d}S$	$\displaystyle\oint_{\partial S} \mathbf{F} \cdot \mathrm{d}\mathbf{r}$
Divergence	$\displaystyle\iiint_V \operatorname{div}\mathbf{F}\,\mathrm{d}V$	$\displaystyle\oiint_{\partial V} \mathbf{F} \cdot \mathbf{n}\,\mathrm{d}S$

Why this matters

The boundary operator ∂ lowers dimension by one: the boundary of a solid is a surface, of a surface a curve, of a curve two endpoints. The derivative operator (∇, curl, div) is the perfect partner: applied inside, it produces exactly what, when summed, the boundary can account for. The two are *adjoint* – that is the one-sentence content of the generalized Stokes theorem, $\int_M \mathrm{d}\omega = \int_{\partial M} \omega$. Every theorem you have met in this book, from $\int F' = F(b) - F(a)$ onward, is one shadow of that single sentence. This is, to many mathematicians, the most beautiful theorem in calculus.

Connection

This unification is not idle elegance; it is what makes physics writable. Maxwell's equations say $\operatorname{div}\mathbf{E} = \rho/\varepsilon_0$ (charges are sources of the electric field) and $\operatorname{curl}\mathbf{B} = \mu_0\mathbf{J} + \cdots$ (currents swirl the magnetic field). Feed these into the divergence and Stokes theorems and out come Gauss's law and Ampere's law in their boundary form – the form you can actually measure with a wire and a meter. The local differential law and the global integral law are the same statement seen from inside and from the boundary.

Work it yourself

Test the unification with your hands. Take $\mathbf{F} = (-y, x)$. (1) Compute its scalar curl. (2) Compute its divergence. (3) Use Green's theorem to find its circulation around the boundary of the square $[0,1] \times [0,1]$, then check it by integrating around the four edges directly. (4) Now compute its *flux* out of that same square using the flux form of Green's theorem – and explain, before you compute, why you expect 0. (The swirl spins but does not spread.)

35.9 The Bridge to Complex Analysis: Analytic Functions, Cauchy–Riemann, and Contour Integrals

Throughout this chapter, Green's theorem, Stokes' theorem, and the divergence theorem established a single unifying principle: an integral of a derivative over an interior region is completely recorded by the field along the boundary. While historically developed to describe fluid vortices and electromagnetic fields, Green's theorem produces an even more striking payoff in two dimensions when the coordinate plane $\mathbb{R}^2$ is identified with the complex plane $\mathbb{C}$ via $z = x + iy$.

Here, Green's theorem becomes the foundation of *Complex Analysis*—a discipline where functions possessing a single complex derivative automatically possess infinitely many, and where difficult definite integrals are evaluated by counting how paths wind around isolated singularities.

35.9.1 Complex Differentiability and the Amplitwist

A complex function $f : \mathbb{C} \to \mathbb{C}$ maps an input point $z = x + iy$ to an output $w = u(x, y) + iv(x, y)$, where $u, v : \mathbb{R}^2 \to \mathbb{R}$ are its real and imaginary parts.

Just as in single-variable calculus, the complex derivative of f at a point z_0 is defined as the limit of

a difference quotient:

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}. \tag{35.1}$$

In single-variable real calculus, the increment Δx can approach zero from only two directions: left or right. In the complex plane, however, $\Delta z = \Delta x + i\Delta y$ can approach zero along an infinite family of two-dimensional trajectories. For the complex derivative $f'(z_0)$ to exist, the quotient must converge to the exact same complex number regardless of the trajectory of approach.

Compare two perpendicular directions of approach:

1. **Horizontal approach** ($\Delta z = \Delta x \in \mathbb{R}$, so $\Delta y = 0$):

$$f'(z) = \lim_{\Delta x \to 0} \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \lim_{\Delta x \to 0} \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x} = u_x + iv_x.$$

2. **Vertical approach** ($\Delta z = i\Delta y$ with $\Delta y \in \mathbb{R}$, so $\Delta x = 0$):

$$f'(z) = \lim_{\Delta y \to 0} \frac{u(x, y + \Delta y) - u(x, y)}{i\Delta y} + i \lim_{\Delta y \to 0} \frac{v(x, y + \Delta y) - v(x, y)}{i\Delta y} = \frac{u_y + iv_y}{i}.$$

   Multiplying numerator and denominator by $-i$ (recalling that $1/i = -i$):

$$\frac{u_y + iv_y}{i} = -i(u_y + iv_y) = v_y - iu_y.$$

For $f'(z)$ to exist, these two directional limits must be identical:

$$u_x + iv_x = v_y - iu_y.$$

Equating real and imaginary components yields the celebrated **Cauchy–Riemann equations**:

$$u_x = v_y \qquad \text{and} \qquad u_y = -v_x. \tag{35.2}$$

35.9.2 The Geometric View: Pure Rotation and Scaling

What do the Cauchy–Riemann equations mean geometrically?

View f as a multivariable mapping from the plane $\mathbb{R}^2$ to $\mathbb{R}^2$, sending (x, y) to $(u(x, y), v(x, y))$. Its local linear approximation is governed by the 2×2 Jacobian matrix:

$$J = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}.$$

Imposing the Cauchy–Riemann equations $v_y = u_x$ and $u_y = -v_x$, the Jacobian specializes to:

$$J = \begin{pmatrix} u_x & -v_x \\ v_x & u_x \end{pmatrix}.$$

Let $r = \sqrt{u_x^2 + v_x^2} = |f'(z)|$, and let $\theta = \operatorname{Arg}(u_x + iv_x) = \operatorname{Arg}(f'(z))$. Then this matrix factors into:

$$J = r \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

The Jacobian is not an arbitrary linear transformation: it is a pure counterclockwise rotation by angle θ accompanied by an isotropic scaling by factor r.

As Tristan Needham termed it in *Visual Complex Analysis*, a complex derivative acts locally as an **amplitwist**: it *amplifies* lengths by $r = |f'(z)|$ and *twists* directions by angle $\theta = \operatorname{Arg}(f'(z))$. Because all directions are scaled equally and angles between vectors are preserved, every complex-differentiable (holomorphic) map is *conformal*: tiny squares in the domain map to tiny, rotated squares in the range, free of any shear or directional stretching.

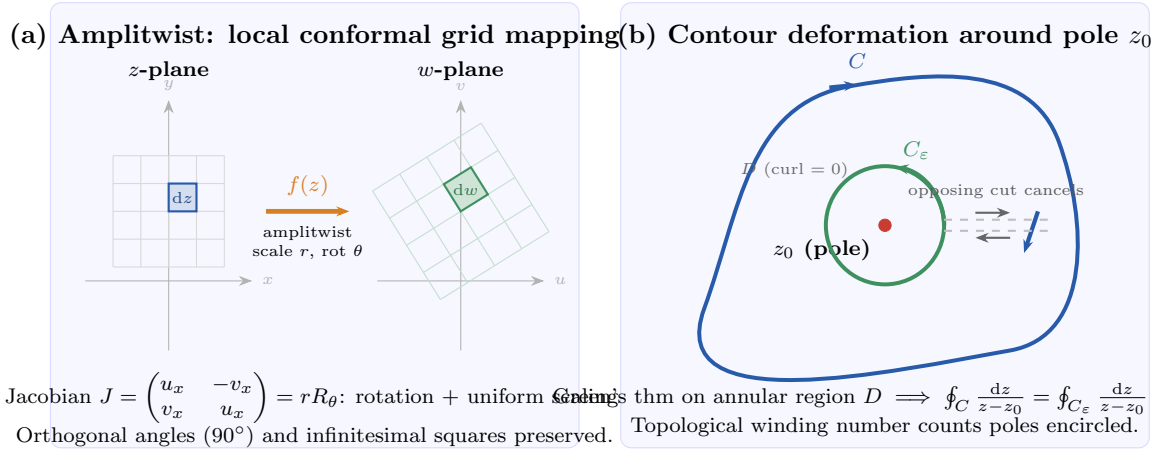


Figure 35.9: The bridge from Green's theorem to complex analysis. (a) A complex function $f(z)$ satisfies the Cauchy–Riemann equations if and only if its derivative acts locally as an amplitwist: rotating by $\theta = \operatorname{Arg}(f')$ and scaling isotropically by $r = |f'|$, preserving infinitesimal squares. (b) Green's theorem implies Cauchy's integral theorem ($\oint_C f(z)\,\mathrm{d}z = 0$ on holomorphic domains). When an isolated pole z_0 is enclosed, Green's theorem on the annular region between C and a small circle C_ε proves the contour can be deformed onto C_ε, evaluating to $2\pi i$.

35.9.3 Cauchy's Integral Theorem in Three Lines via Green's Theorem

Now consider the contour integral of a complex function $f(z)$ around a closed curve $C = \partial D$ bounding a simply connected region $D \subset \mathbb{C}$.

Expand the complex integrand and differential displacement into real and imaginary components:

$$f(z) = u + iv, \qquad \mathrm{d}z = \mathrm{d}x + i\,\mathrm{d}y.$$

Multiplying these expressions:

$$f(z)\,\mathrm{d}z = (u + iv)(\mathrm{d}x + i\,\mathrm{d}y) = (u\,\mathrm{d}x - v\,\mathrm{d}y) + i\,(v\,\mathrm{d}x + u\,\mathrm{d}y).$$

Integrating around the closed boundary loop C splits the complex contour integral into two ordinary real line integrals:

$$\oint_C f(z)\,\mathrm{d}z = \oint_C (u\,\mathrm{d}x - v\,\mathrm{d}y) + i \oint_C (v\,\mathrm{d}x + u\,\mathrm{d}y). \tag{35.3}$$

Now apply Green's Theorem, $\oint_C (P\,\mathrm{d}x + Q\,\mathrm{d}y) = \iint_D (Q_x - P_y)\,\mathrm{d}A$, directly to both integrals:

1. **Real part** ($P = u$, $Q = -v$):

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \frac{\partial(-v)}{\partial x} - \frac{\partial u}{\partial y} = -v_x - u_y.$$

   By the second Cauchy–Riemann equation, $u_y = -v_x$. Substituting this gives:

$$-v_x - (-v_x) = 0.$$

2. **Imaginary part** ($P = v$, $Q = u$):

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} = u_x - v_y.$$

   By the first Cauchy–Riemann equation, $u_x = v_y$. Substituting this gives:

$$v_y - v_y = 0.$$

Both interior curl integrands vanish identically throughout the domain D:

$$\oint_C f(z)\,\mathrm{d}z = \iint_D 0\,\mathrm{d}A + i\iint_D 0\,\mathrm{d}A = 0. \tag{35.4}$$

This is **Cauchy's Integral Theorem**:

Theorem 35.5 (Cauchy's Integral Theorem)**.** *If $f(z)$ is complex-differentiable (holomorphic) on and inside a simple closed contour C, then:*

$$\oint_C f(z)\,\mathrm{d}z = 0.$$

The central theorem of complex analysis is Green's theorem in disguise: the Cauchy–Riemann equations ensure that the vector fields $(u, -v)$ and (v, u) are identically curl-free!

35.9.4 The Single Punctured Exception: $\oint_{|z|=R} \frac{1}{z}\,\mathrm{d}z = 2\pi i$

What occurs when $f(z)$ fails to be differentiable at an isolated singularity inside C?

The standard test case is $f(z) = 1/z$, which is holomorphic everywhere except at the origin $z = 0$, where it blows up.

Let C be the circle $|z| = R$ traversed counterclockwise. Parametrize C in polar form:

$$z(\theta) = Re^{i\theta}, \qquad \mathrm{d}z = iRe^{i\theta}\,\mathrm{d}\theta, \qquad \theta \in [0, 2\pi].$$

Substitute this parametrization into the contour integral:

$$\oint_{|z|=R} \frac{1}{z}\,\mathrm{d}z = \int_0^{2\pi} \frac{1}{Re^{i\theta}}\left(iRe^{i\theta}\,\mathrm{d}\theta\right) = \int_0^{2\pi} i\,\mathrm{d}\theta = 2\pi i. \tag{35.5}$$

Observe the cancellation: the radius R cancels completely from the calculation. Whether the circular path has radius $R = 10^5$ or $R = 10^{-5}$, the line integral is invariantly $2\pi i$.

By applying Green's theorem to the annular region between an arbitrary smooth loop C encircling the origin and a small circle C_ε of radius ε, the curl is zero in the intermediate space. The connecting corridor has opposing paths that cancel (Figure 35.9b), proving that any closed contour enclosing the origin can be deformed onto C_ε without altering the integral:

$$\oint_C \frac{1}{z}\,\mathrm{d}z = \oint_{C_\varepsilon} \frac{1}{z}\,\mathrm{d}z = 2\pi i.$$

The integral $\oint_C \frac{1}{z}\,\mathrm{d}z$ acts as an integer winding counter: it evaluates to $2\pi i$ times the number of times the path C winds around the singularity.

In contrast, for every other integer power z^n with $n \neq -1$:

$$\oint_{|z|=R} z^n\,\mathrm{d}z = \int_0^{2\pi} R^n e^{in\theta}\left(iRe^{i\theta}\,\mathrm{d}\theta\right) = iR^{n+1}\int_0^{2\pi} e^{i(n+1)\theta}\,\mathrm{d}\theta = 0,$$

because integrating a periodic sinusoidal function over full periods yields zero. The term z^{-1} is the unique survivor.

35.9.5 Cauchy's Residue Theorem and Signal Processing

Any function with isolated singularities can be expanded around a pole z_0 in a Laurent series:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z-z_0)^n = \cdots + \frac{a_{-2}}{(z-z_0)^2} + \frac{a_{-1}}{z-z_0} + a_0 + a_1(z-z_0) + \ldots$$

When integrating $f(z)$ around a closed loop enclosing z_0, every term integrates to zero except $a_{-1}(z-z_0)^{-1}$, which produces $2\pi i a_{-1}$. The coefficient a_{-1} is called the **residue** of f at z_0, written $\mathrm{Res}(f, z_0)$.

Deforming a general contour around multiple enclosed isolated poles $z_1, z_2, \ldots, z_k$ yields **Cauchy's Residue Theorem**:

$$\oint_C f(z)\,\mathrm{d}z = 2\pi i \sum_k \mathrm{Res}(f, z_k). \tag{35.6}$$

In modern applied mathematics and engineering, this result powers core analysis tools:

- **The Inverse Laplace Transform**: In control systems and electrical circuit analysis, the response of a continuous-time system is converted from the frequency domain $F(s)$ back into the time domain $f(t)$ via the Bromwich contour integral:

  $$f(t) = \frac{1}{2\pi i}\int_{\gamma-i\infty}^{\gamma+i\infty} F(s)e^{st}\,\mathrm{d}s.$$

  Closing the contour into a large semicircle in the complex plane and applying the Residue Theorem transforms this continuous integral into a discrete sum of residues at the poles of $F(s)$, immediately extracting the natural frequencies, damping ratios, and resonant modes of the physical system.

- **The Inverse Z-Transform**: In digital signal processing (DSP), discrete-time signals are recovered from the Z-domain transfer function $X(z)$ by integrating around a closed contour enclosing the origin:

  $$x[n] = \frac{1}{2\pi i}\oint_C X(z)z^{n-1}\,\mathrm{d}z.$$

Residue evaluation converts this complex contour integral into closed-form filter impulse responses.

Green's theorem, originally formulated to track circulation and flux in fluid mechanics, becomes the analytical engine that drives modern signal processing and frequency-domain engineering.

35.10 Exercises

Exercise 35.1. Compute $\displaystyle\int_C \mathbf{F}\cdot\mathrm{d}\mathbf{r}$ for $\mathbf{F} = (x+y,\, y)$ along the straight segment from $(0,0)$ to $(3,1)$. Parametrize by $\mathbf{r}(t) = (3t, t)$, $t \in [0,1]$, and reduce to a single integral in t.

Exercise 35.2. Show that $\mathbf{F} = (3x^2y + e^y,\; x^3 + xe^y + 2)$ is conservative by checking $P_y = Q_x$, then find a potential f. Use it to evaluate $\int_C \mathbf{F}\cdot\mathrm{d}\mathbf{r}$ from $(0,0)$ to $(1,1)$ along *any* path.

Exercise 35.3. For $\mathbf{F} = (y,\, -x)$, compute $\oint_C \mathbf{F}\cdot\mathrm{d}\mathbf{r}$ around the unit circle counterclockwise two ways: directly by parametrization, and via Green's theorem. Confirm they agree, and explain why the nonzero answer proves $\mathbf{F}$ is not conservative.

Exercise 35.4. Use the area formula $\text{Area} = \frac{1}{2}\oint_C x\,\mathrm{d}y - y\,\mathrm{d}x$ to find the area enclosed by the astroid $x = \cos^3 t$, $y = \sin^3 t$, $t \in [0, 2\pi]$. (You should get $\frac{3}{8}\pi$.)

Exercise 35.5. For $\mathbf{F} = (x^2y,\; -xy^2,\; z)$ compute $\operatorname{div}\mathbf{F}$ and $\operatorname{curl}\mathbf{F}$ as functions of (x,y,z). Identify whether there is any point where the field is locally incompressible ($\operatorname{div} = 0$), or show that $\operatorname{div}\mathbf{F} = 1$ everywhere.

Exercise 35.6. Verify Green's theorem for $\mathbf{F} = (xy,\, x^2)$ over the triangle with vertices $(0,0)$, $(1,0)$, $(1,1)$, by computing both the boundary line integral (three edges) and the double integral of $Q_x - P_y$ over the triangle.

Exercise 35.7. Use the divergence theorem to compute the flux of $\mathbf{F} = (x^3,\, y^3,\, z^3)$ out of the unit sphere. (Hint: $\operatorname{div}\mathbf{F} = 3(x^2+y^2+z^2)$; convert the resulting triple integral to spherical coordinates.)

Exercise 35.8. Let $\mathbf{F} = (z,\, x,\, y)$ and let S be the part of the paraboloid $z = 1 - x^2 - y^2$ above the xy-plane, oriented upward, with boundary the unit circle. Use Stokes' theorem to compute $\oint_{\partial S}\mathbf{F}\cdot\mathrm{d}\mathbf{r}$ by integrating $\operatorname{curl}\mathbf{F}\cdot\mathbf{n}$ over the flat unit disk instead (legal because only the rim matters).

Exercise 35.9. Consider $\mathbf{F} = \left(\frac{-y}{x^2+y^2},\, \frac{x}{x^2+y^2}\right)$. (a) Verify that its scalar curl is 0 wherever it is defined. (b) Compute its circulation around the unit circle and get 2π. (c) Explain why parts (a) and (b) do not contradict Green's theorem. (What goes wrong at the origin?)

Exercise 35.10. In your own words, state the single sentence that contains the FTC, the gradient theorem, Green's, Stokes', and the divergence theorem at once. Identify, for each of the five, what plays the role of "the inside," "the derivative taken inside," and "the boundary."

Part VIII

Differential Equations

Chapter 36

What a Differential Equation Is

Every chapter until now has handed you a function and asked questions about it. Here is a curve; what is its slope? Here is a curve; what is the area beneath it? The function came first, fully formed, and calculus was the lens through which we studied it. This chapter turns the whole enterprise upside down, and in doing so it arrives at the reason calculus was invented in the first place.

For in the real world the function almost never comes first. What comes first is a *law* – a statement about how a quantity is changing right now, expressed in terms of how much of that quantity there is right now. A population grows faster when there are more individuals to reproduce. A hot cup of coffee cools faster when it is further above room temperature. A falling stone speeds up at a rate fixed by gravity. In each case nature does not tell you the function $y(t)$ directly. Nature tells you something about y and its rate of change $\frac{\mathrm{d}y}{\mathrm{d}t}$, tangled together in a single equation, and leaves you the job of untangling it – of finding the actual function that obeys the law.

An equation that ties a function to its own derivatives is called a *differential equation*. Solving one means finding the function hiding inside the law. This is the central problem of mathematical physics, of population biology, of engineering, of economics – of nearly every quantitative science. It is, in a real sense, the question calculus was built to answer. Newton did not invent the derivative to torment students; he invented it to write down, and then solve, the differential equations of motion that govern the planets.

This chapter is the gentle on-ramp. The clever solving techniques wait for the chapters that follow. Here we get clear on *what these equations are*, what it means to solve one, why a single law can have a whole family of solutions until an initial condition pins down just one, and how you can *see* the solutions as curves threading through a field of little slope marks, even before you have solved anything algebraically.

36.1 A law about change

Start with the simplest possible example, built from a sentence rather than from symbols.

Intuition

A quantity grows at a rate equal to its current size. That is an entire differential equation, written in English. Read it slowly. "A quantity" is some function $y(t)$ – maybe a population, maybe a bank balance. "Grows at a rate" is its derivative $\frac{\mathrm{d}y}{\mathrm{d}t}$. "Equal to its current size" means that rate equals y itself. Put the three pieces together and the sentence becomes

$$\frac{\mathrm{d}y}{\mathrm{d}t} = y.$$

That is it. That is a differential equation. It does not tell you what y is. It tells you a *relationship* that the true y – whatever it turns out to be – must satisfy at every instant.

Notice what kind of object an equation like $\frac{\mathrm{d}y}{\mathrm{d}t} = y$ is. An ordinary algebraic equation, like $x^2 = y$, relates *numbers*: plug in a number for x, get a number for y. A differential equation relates a *function* to its own derivative. Its unknown is not a number but an entire function. When we ask for a solution, we are asking: *which function, when you differentiate it, gives back a result that fits this equation?*

For $\frac{\mathrm{d}y}{\mathrm{d}t} = y$, you may already feel the answer pressing on you. We need a function that is its own derivative. The exponential $y = e^t$ is exactly that creature: $\frac{\mathrm{d}}{\mathrm{d}x}\left(e^t\right)$ – read with t in place of x – equals e^t again. So $y = e^t$ solves the equation. And it is not the only one: $y = 3e^t$ works too, because its derivative is $3e^t$, which equals the function itself. So does $y = -\frac{1}{2}e^t$, and $y = Ce^t$ for *any* constant C. We will return to this whole family shortly; for now just register the shape of things. One law, many solutions.

Definition 36.1 (Differential equation; ordinary)**.** A *differential equation* is an equation relating an unknown function to one or more of its derivatives. If the unknown function depends on a single variable – so that only ordinary derivatives appear – the equation is called an *ordinary differential equation*, abbreviated **ODE**. If the unknown depends on several variables and partial derivatives appear, it is a *partial differential equation* (PDE). This chapter, and the three that follow, are about ODEs.

The distinction in the definition matters. A PDE like the heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ involves a function $u(x, t)$ of two variables – here the temperature u at position x and time t – and is genuinely harder; PDEs are a subject of their own. By restricting to a single independent variable – usually time t or position x – we keep the unknown a curve in the plane, something we can draw. That restriction is exactly what makes the geometric picture in this chapter possible.

History

The phrase "differential equation" goes back to Leibniz, who in a 1676 manuscript was already manipulating relations between a quantity and its differential $\mathrm{d}y$. But the subject truly ignites with Newton. In the *Principia* (1687) and his earlier work on fluxions, Newton's deepest insight was that the laws of nature are most naturally stated as equations of *rates*. His second law, in the form we now write $F = m\,\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$ where $\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$ denotes the second derivative, is a differential equation for position. Newton solved the differential equation of a planet orbiting the sun and *derived* Kepler's elliptical orbits from it – a stunning demonstration that a single law of change, properly solved, reproduces the observed heavens. For the next two centuries, "doing physics" and "solving differential equations" were nearly synonymous, and the names

attached to the methods – Euler, the Bernoullis, Lagrange, Laplace – are the founders of the whole of analysis.

36.1.1 The general shape

Most of the equations we meet can be written so that the highest derivative sits alone on the left:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(t, y),$$

read aloud as "the rate of change of y is some specified function of the current time t and the current value y." The right-hand side $f(t, y)$ is the *law*: hand it where you are (t) and what you have (y), and it returns how fast y is changing. Our first example $\frac{\mathrm{d}y}{\mathrm{d}t} = y$ is the case $f(t, y) = y$ – the rate depends on y but not directly on t. The slightly different equation $\frac{\mathrm{d}y}{\mathrm{d}t} = t$ has $f(t, y) = t$ – the rate depends on the clock but not on the current value. Both are legitimate, and they behave very differently, as we will see.

What is really happening

Here is the true mechanism, stripped bare. A differential equation $\frac{\mathrm{d}y}{\mathrm{d}t} = f(t, y)$ is a *rule for the slope of a curve, given the curve's location.* It says: "If you ever find yourself at the point (t, y) in the plane, your curve must be heading in the direction with slope $f(t, y)$." That is all the equation knows. It does not see the whole curve at once; it only ever issues local instructions, slope by slope, point by point. A solution is a curve that obeys every one of those local instructions simultaneously – a curve whose slope at each of its points is exactly what the rule demands there. The entire art of this subject is reconstructing a global curve from an infinitude of local slope commands. Keep this picture in your mind; the slope field in Section 36.5 is nothing but this picture drawn explicitly.

36.2 What it means to solve one

Solving a differential equation is unlike solving the equations you are used to. When you solve $x^2 - 5x + 6 = 0$ you are hunting for *numbers* – here $x = 2$ and $x = 3$. When you solve a differential equation you are hunting for *functions*. The answer is not a value but a formula, a whole curve.

Definition 36.2 (Solution of an ODE)**.** A *solution* of the differential equation $\frac{\mathrm{d}y}{\mathrm{d}t} = f(t, y)$ on an interval I is a function $y = \varphi(t)$, differentiable on I, such that substituting φ for y makes the equation true for every t in I. That is, $\varphi'(t) = f\big(t, \varphi(t)\big)$ holds identically on I. Verifying a proposed solution is therefore a purely mechanical act: differentiate it, substitute, and check that the two sides agree.

The last sentence of the definition is a gift, and you should lean on it constantly. *Checking* a solution is always easy, even when *finding* it was hard or impossible. You differentiate, you substitute, you see whether the equation closes up. Here is that check on our running example.

Example 36.1 (Verifying a solution)**.** Claim: $y = Ce^t$ solves $\frac{\mathrm{d}y}{\mathrm{d}t} = y$ for every constant C. To check, differentiate the candidate:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(Ce^t) = Ce^t.$$

And the right-hand side of the equation is just $y = Ce^t$. The two agree – $Ce^t = Ce^t$ – for every t and every C. So the claim is verified. We have not *derived* this solution; we have only confirmed it. But confirmation is genuine knowledge: we now know with certainty that every function of the form Ce^t obeys the law.

Example 36.2 (A solution by direct integration)**.** Consider the other simple equation, $\frac{\mathrm{d}y}{\mathrm{d}t} = t$. Here the right-hand side does not mention y at all – the slope depends only on the clock. An equation of this special shape, $\frac{\mathrm{d}y}{\mathrm{d}t} = g(t)$, is solved by nothing more than integration, because it literally asks "what function has derivative $g(t)$?" – the antiderivative question from the Fundamental Theorem. So

$$y = \int t \, \mathrm{d}t = \tfrac{1}{2}t^2 + C.$$

Again a constant of integration appears, and again it is a *family* of solutions, one for each value of C. Every parabola $\frac{1}{2}t^2 + C$ is a vertical shift of every other, and each one has slope exactly t at every point, as the equation demands.

These two examples reveal something structural that you must internalize: **a first-order differential equation does not have one solution; it has a one-parameter family of solutions.** The free constant C is not a flaw in our method – it is built into the nature of the problem, and there is a clean reason why.

Why the technique works

Why must a free constant appear? Because the differential equation only constrains the *slope* of the solution curve, never its *height*. Think of $\frac{\mathrm{d}y}{\mathrm{d}t} = t$ again. The equation insists the curve have slope t at horizontal position t – but it says nothing about *where vertically* the curve sits. You are free to take any solution and slide the whole curve straight up or down by a constant; sliding does not change any slope, so the slid curve still satisfies the law. That vertical freedom is precisely the constant C. More generally, a first-order equation pins down the derivative and so pins down the solution *up to* one undetermined constant – the one piece of information ("where do we start?") that no statement purely about rates of change could ever supply. Integration always leaves a $+C$ for the same reason; a differential equation is, at bottom, a disguised integration problem.

This is the cleanest way to see why a single law yields a family. The law fixes the *shape* of motion – the slopes everywhere – but not the starting point. To select one solution from the family, we need exactly one more piece of data: an *initial condition*.

36.3 Initial conditions: pinning down one curve

An *initial condition* is a single anchoring fact of the form "at this time, the quantity had this value," written

$$y(t_0) = y_0.$$

It is the missing piece. The differential equation supplies the slopes; the initial condition supplies one point the curve must pass through. Together they typically determine exactly one solution. A differential equation packaged with an initial condition is called an *initial value problem*, or IVP.

Example 36.3 (An initial value problem)**.** Solve

$$\frac{\mathrm{d}y}{\mathrm{d}t} = y, \qquad y(0) = 5.$$

We already know the general solution is $y = Ce^t$. Now impose the initial condition: at $t = 0$, we need $y = 5$. Substitute:

$$y(0) = Ce^0 = C \cdot 1 = C = 5.$$

So $C = 5$, and the unique solution to the initial value problem is

$$y = 5e^t.$$

The family Ce^t collapsed to a single curve the instant we said where it started. Geometrically, of all the exponential curves obeying the law, exactly one passes through the point $(0, 5)$ – and that is our answer.

Focus

The two-step rhythm of solving an initial value problem is worth committing to memory, because it recurs in every chapter to come:

1. **Find the general solution** – the whole family, with its free constant(s). This is the hard, creative step, the one that needs the techniques of later chapters.
2. **Apply the initial condition(s)** to solve for the constant(s). This is the easy, mechanical step – plug in the known point and do algebra.

Mixing these up – trying to satisfy the initial condition before you have the general family – is the single most common source of confusion for beginners.

Common trap

A tempting error: thinking the initial condition has to be given "at the beginning," at $t = 0$. Not so. The condition $y(t_0) = y_0$ can anchor the curve at *any* time t_0. "Initial" is a name, not a requirement. If a problem tells you $y(3) = 7$, you use that point exactly the same way – substitute $t = 3$, set the result equal to 7, solve for the constant. The word "initial" is a historical leftover from physics, where the natural anchor is the state of a system when you start the clock.

36.3.1 Why we expect exactly one

Pause on *why* a single initial condition should select a single curve, because the picture here is so persuasive. Imagine you are a tiny walker dropped onto the plane at the starting point (t_0, y_0). The differential equation is a compass: at your current location it tells you which direction to face – it gives you a slope. You take an infinitesimal step in that direction. Now you are at a slightly new point; consult the compass again; it gives a new slope; step again. Marching forward (and backward) this way, you trace out a curve, and at every point of it your heading is exactly the slope the law prescribes – so the curve is a solution. And the curve is *forced*: at each instant you had no choice about which way to step. Start at the same point and you trace the same curve. *One starting point, one path.*

That heuristic is essentially correct, and it is the seed of both a numerical method (Euler's method, which literally takes those small steps) and a theorem (the existence and uniqueness theorem of Section 36.6). But the heuristic also quietly assumes the compass is well-behaved – that the slope does not do anything pathological. We will see in a moment that when the law misbehaves, uniqueness can genuinely fail, and the picture of the forced walker breaks down in an instructive way.

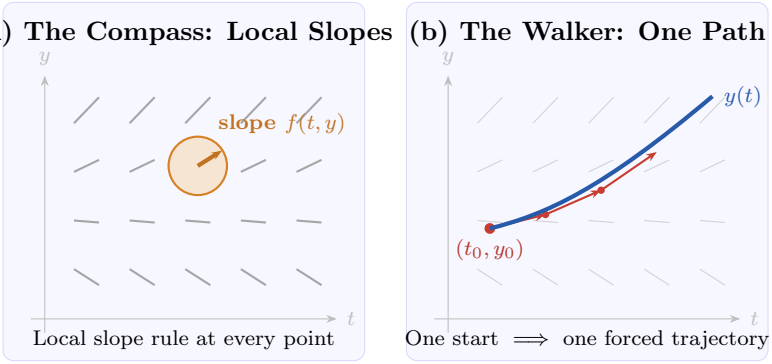


Figure 36.1: The walker and the compass. (a) A differential equation $\frac{dy}{dt} = f(t, y)$ acts as a local compass, assigning a prescribed heading to every coordinate in the plane. (b) Dropping a walker at an initial point (t_0, y_0) fixes the entire journey: stepping along the compass directions forces out a single, unique solution curve $y(t)$.

36.4 Order and linearity: classifying the zoo

Differential equations come in a vast zoo of forms, and two coarse but indispensable ways of sorting them earn their keep at once: by *order* and by *linearity*. These two words tell you, at a glance, roughly how hard an equation is and which tools stand a chance of cracking it.

Definition 36.3 (Order)**.** The *order* of a differential equation is the order of the highest derivative that appears in it.

Thus $\frac{dy}{dt} = y$ is *first order* – only the first derivative shows up. Newton's law of motion $m\frac{d^2x}{dt^2} = F$, involving the second derivative (acceleration), is *second order*. The order matters enormously: a first-order equation needs one initial condition to pin down a solution, a second-order equation needs *two* (typically an initial position *and* an initial velocity), and in general an nth-order equation carries n free constants in its general solution and needs n conditions to fix them.

Connection

The rule "order n means n free constants" is the differential-equation echo of a fact you already know from integration. Each time you integrate, you collect one constant of integration. A second-order equation is, very loosely, "two integrations deep," so it accumulates two constants; an nth-order equation, n of them. This is exactly why pinning down a thrown ball's motion ($\frac{d^2x}{dt^2} = -g$, second order) requires two facts – where you threw it from and how fast – not one. Integrating $-g$ once gives velocity with a constant; integrating again gives position with a second constant. Two integrations, two constants, two initial conditions.

The second classifier, linearity, is subtler but matters even more, because the entire theory splits

along this line. Linear equations are the ones we can largely *solve*; nonlinear ones are, in general, beyond closed-form reach and must be studied qualitatively or numerically.

Definition 36.4 (Linear ODE)**.** A differential equation is *linear* if the unknown function y and *all* of its derivatives appear only to the first power, never multiplied together, and never inside any nonlinear function. A first-order linear ODE can always be written in the standard form

$$\frac{\mathrm{d}y}{\mathrm{d}t} + p(t)\,y = q(t),$$

where p and q are given functions of t alone. If the unknown appears in any other way – squared, as y^2; multiplied by its own derivative, as $y\,\frac{\mathrm{d}y}{\mathrm{d}t}$; or inside a function, as $\sin y$ or e^y – the equation is *nonlinear*.

In plain words: in a linear equation, the unknown and its derivatives enter *gently* – each by itself, each to the first power, scaled only by coefficients that may depend on t but never on y. The functions $p(t)$ and $q(t)$ are allowed to be as wild as they like; linearity is a condition on how y enters, not on how t enters.

Example 36.4 (Sorting a handful of equations)**.** Classify a few equations to make the definitions concrete.

- $\frac{\mathrm{d}y}{\mathrm{d}t} = y$ – first order, linear ($p(t) = -1$, $q(t) = 0$).
- $\frac{\mathrm{d}y}{\mathrm{d}t} = t^2 y + \cos t$ – first order, linear; rewrite as $\frac{\mathrm{d}y}{\mathrm{d}t} - t^2 y = \cos t$, so $p = -t^2$, $q = \cos t$. The wild coefficients are fine.
- $\frac{\mathrm{d}y}{\mathrm{d}t} = y^2$ – first order, *nonlinear*: y appears squared.
- $\frac{\mathrm{d}y}{\mathrm{d}t} = \sin y$ – first order, *nonlinear*: y sits inside sin.
- $\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 3\frac{\mathrm{d}x}{\mathrm{d}t} + 2x = 0$ – second order, linear.
- $\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + \sin\theta = 0$ – second order, *nonlinear* (the pendulum equation; $\sin\theta$ is the culprit).

Why this matters

Why obsess over linearity? Because linear equations enjoy the *superposition principle*: if y_1 and y_2 both solve a linear *homogeneous* equation (one with $q = 0$), then so does every combination $c_1y_1 + c_2y_2$. Solutions add. This single algebraic fact – that the solution set is a vector space – is the foundation on which the clean, complete theory of linear ODEs is built, and it is why the chapter on second-order linear equations can give you *formulas*. Nonlinear equations have no such principle; their solutions do not add, and two simple behaviors can combine into something wildly new. Superposition is the reason linearity is the great dividing line of the whole subject – the line between equations we can conquer and equations we must respect.

36.5 The slope field: seeing solutions before solving

Here is the idea that turns an abstract equation into a picture you can read with your eyes. It rests entirely on the mechanism exposed earlier: a first-order equation $\frac{\mathrm{d}y}{\mathrm{d}t} = f(t, y)$ assigns a *slope to every point* of the plane. So *draw* those slopes.

At a grid of points (t, y), compute the number $f(t, y)$ and draw a short line segment through the point with that slope. Do this at many points and you fill the plane with a texture of tiny tilted dashes – a *slope field* (also called a direction field). Each dash is a local instruction: "a solution passing through here must be heading this way." A solution curve is then any curve that flows along the dashes, tangent to the field at every point it touches, like a leaf carried by a current.

History

The idea of reading an equation through its field of directions is older than it looks. The eighteenth-century analysts already thought of $\frac{dy}{dt} = f(t, y)$ as prescribing a direction at each point, but it was the qualitative theory of differential equations – pioneered by Henri Poincaré in the 1880s in his memoirs on curves defined by differential equations – that made the geometric viewpoint a discipline in its own right. Faced with equations that no formula could solve, Poincaré shifted the question from "what is the solution?" to "what does the family of solutions *look like*?" – where the curves go, which ones are stable, what happens in the long run. The slope field is the gateway to that whole way of thinking, and it is the seed of the phase-plane analysis you will meet at the end of the book.

Intuition

The slope field is the differential equation made *visible*. You never solve anything; you just plot the law's slope at a mesh of points. And yet, once the field is drawn, the solutions reveal themselves to the eye: trace any curve that stays tangent to the little dashes, and you are tracing a solution. Different starting heights give different curves – the family of solutions appears as a family of streamlines combing through the field.

Take the slope field for our very first equation, $\frac{dy}{dt} = y$. Here $f(t, y) = y$: the prescribed slope at a point is simply its height. Above the t-axis (where $y > 0$) slopes are positive and steepen as you rise; below it (where $y < 0$) slopes are negative and steepen as you fall; right on the axis ($y = 0$) the slope is 0, flat. Figure 36.2 shows the result, with three solution curves $y = Ce^t$ threaded through it.

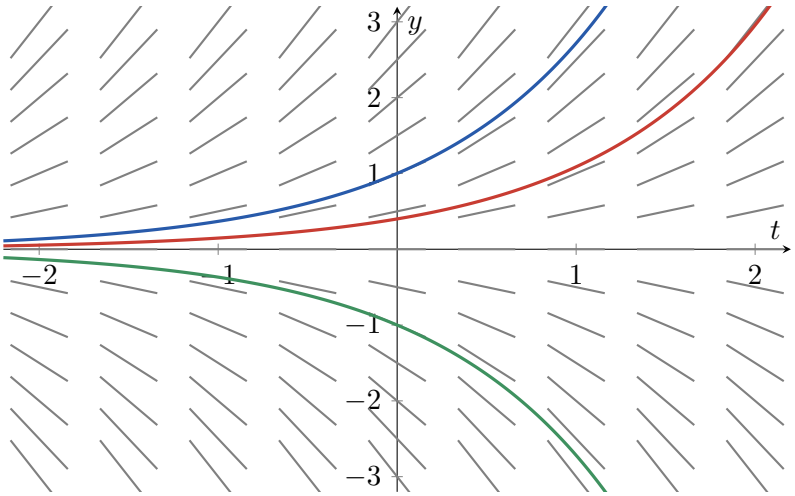


Figure 36.2: The slope field for $\frac{dy}{dt} = y$. Each gray dash has slope equal to its own height y. Three solution curves $y = Ce^t$ flow tangent to the field: $C = 1$ (blue), $C = 0.4$ (red), and $C = -1$ (green). Notice how every curve hugs the dashes everywhere it passes.

Spend a moment really looking at Figure 36.2. The dashes are flat along the t-axis and tilt ever more steeply as you move away from it – exactly because the slope equals the height. The blue curve, $y = e^t$, rises through the steepening forest of positive slopes, always tangent. The green curve, $y = -e^t$, plunges through the negative slopes below. And the t-axis itself, $y = 0$, is a perfectly flat solution – the constant zero function, with slope 0 everywhere, matching the flat dashes that lie along it. You are reading the entire solution family off a picture, having solved nothing.

Walking question

Look again at Figure 36.2 and ask yourself: through the single point $(0, 0.4)$, how many solution curves pass? Trace the dashes by eye starting there. You should find that the curve is forced – there is exactly one streamline through that point (the red one, $y = 0.4e^t$), because at every step the field dictates your heading. This is the geometric face of uniqueness: a well-behaved slope field carries exactly one solution curve through each point. Now ask the harder question: can two *different* solution curves ever cross each other? If they did, they would share a point – and at that shared point the field prescribes a single slope, so both curves would have to leave it in the same direction. Solution curves of a nice first-order equation never cross. That is uniqueness, seen.

36.5.1 Reading a field with a t-dependent slope

The previous field had slopes depending only on y, so the pattern looked the same at every horizontal position. Now take one where the clock matters too:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = t - y.$$

Now the prescribed slope at (t, y) is $t-y$, which changes as you move both horizontally and vertically. Where $t = y$ – along the diagonal line $y = t$ – the slope is zero, so the dashes lie flat there. Above that diagonal ($y > t$) slopes are negative; below it ($y < t$) slopes are positive. Everything is gently herded toward a special straight solution. Figure 36.3 shows the field with two solution curves.

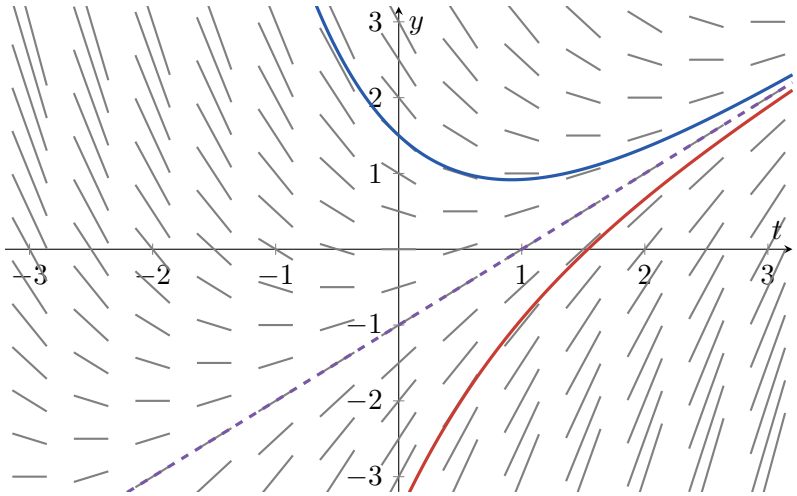


Figure 36.3: The slope field for $\frac{\mathrm{d}y}{\mathrm{d}t} = t - y$. Dashes are flat along the diagonal $y = t$ and tilt away from it. The purple dashed line $y = t - 1$ is itself a solution – an *attractor* every other solution curls toward. The blue and red curves, $y = t - 1 \pm 2.5e^{-t}$, both rush in to merge with it as t grows.

There is a striking feature in Figure 36.3 that you could not have guessed from the bare equation but that the picture announces immediately: the line $y = t - 1$ is itself a solution, and it acts as an *attractor*. Check that it solves the equation: if $y = t - 1$ then $\frac{\mathrm{d}y}{\mathrm{d}t} = 1$, while the right side is $t - y = t - (t - 1) = 1$. They match. And every other solution, no matter where it starts, bends toward this line as t increases – you can watch the blue and red curves swoop in and shadow it. This long-run behavior is exactly the kind of qualitative insight slope fields hand you for free, long before you can write the general solution $y = t - 1 + Ce^{-t}$ (which, you will be able to confirm, this is).

What is really happening

What is really happening when you "read a slope field" is that you are performing, by eye, the same construction the existence theorem performs by logic and Euler's method performs by arithmetic. All three reconstruct a global solution from purely local slope data. Your eye integrates the field into a smooth streamline; Euler's method integrates it into a polygonal approximation by taking literal small steps; the theorem proves that this integration always succeeds (under mild hypotheses) and yields exactly one curve per starting point. The slope field is therefore not a mere illustration – it is the geometric content of the equation itself. An equation you cannot solve in formulas, you can very often *understand completely* from its field.

36.5.2 A field we cannot solve, but can read: the logistic law

The two fields so far belonged to equations we could solve in closed form, so the slope field merely confirmed what algebra already knew. The real payoff of the picture comes when algebra is hard or hopeless and the field hands you the answer anyway. Here is the cleanest example in elementary modeling – the *logistic equation*, the corrected law of population growth that we will meet again in Section 36.7:

$$\frac{\mathrm{d}P}{\mathrm{d}t} = k\,P\left(1 - \frac{P}{M}\right).$$

The right-hand side is nonlinear (notice the P^2 hiding in the product), so none of the easy tricks apply; solving it takes genuine work, deferred to a later chapter. But its slope field can be drawn in a minute, and from the field the entire qualitative story leaps out.

Read the law as a rule for slope. Two heights make the right side vanish: $P = 0$ and $P = M$. At those two heights the prescribed slope is zero, so the dashes lie perfectly flat – they mark two *equilibrium* (constant) solutions, two horizontal lines that the field reveres. Between them, for $0 < P < M$, both factors P and $1 - P/M$ are positive, so the slope is positive and every solution climbs. Above the carrying capacity, $P > M$, the factor $1 - P/M$ turns negative, the slope turns negative, and solutions fall. Both regions drain toward the line $P = M$: it is an attractor, just like the line $y = t - 1$ was for the earlier field. Figure 36.4 draws it, taking $k = 1$ and $M = 3$.

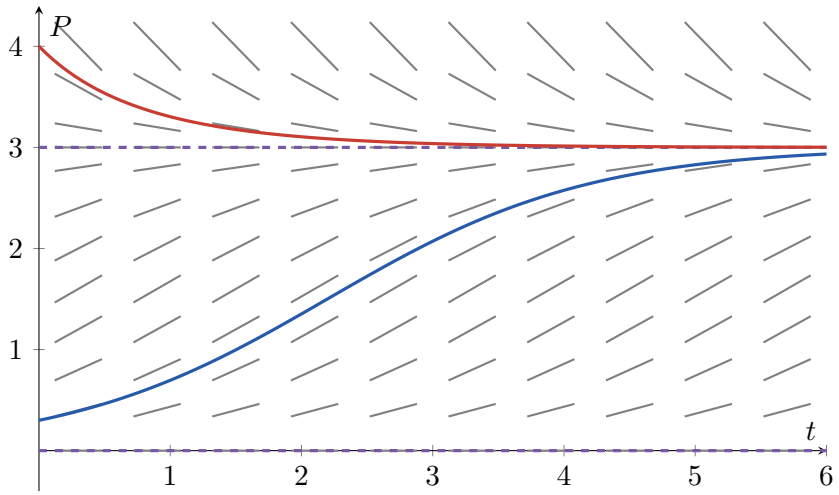


Figure 36.4: The slope field for the logistic law $\frac{dP}{dt} = P(1 - P/3)$ (so $k = 1$, $M = 3$). The dashes lie flat along the two equilibria $P = 0$ and $P = M = 3$ (purple dashed). The blue curve starts small ($P_0 = 0.3$) and traces the famous "S" shape – slow, then fast, then leveling off at the carrying capacity. The red curve starts above capacity ($P_0 = 4$) and sinks toward the same line. Every solution is herded toward $P = M$.

Walking question

Trace the blue curve in Figure 36.4 with your finger and feel the three acts of the logistic story. Near the bottom the dashes are nearly flat (small P makes the slope small), so the curve creeps – a young population grows slowly because there are few to reproduce. Through the middle, around $P = M/2 = 1.5$, the slope is steepest (the product $P(1-P/M)$ is maximal there), and the curve shoots up – this is the population explosion. Near the top the dashes flatten again as P nears M, and the curve eases into a plateau – growth stalls as the niche fills. Slow, fast, slow: that is the "S." You have just read off the qualitative life history of every self-limiting population without solving a single nonlinear equation. Now do the same for the red curve from above and watch the identical attractor $P = M$ pull it down.

This is the slope field earning its keep. The logistic equation is exactly the sort of nonlinear law that the formula-hunting methods of the next chapter *can* eventually crack – but even before they do, the picture has already told you everything a modeler usually wants: the equilibria, which one is stable, and the characteristic shape of every solution. For the many equations that resist closed-form solution entirely, this geometric reading is not a convenience but the only tool there is.

36.6 Existence and uniqueness, intuitively

We have been assuming, on the strength of the walker picture, that an initial value problem has exactly one solution. It is time to say carefully when that is true – and, just as instructively, when it fails. The governing result is one of the cornerstones of the subject.

Theorem 36.1 (Existence and Uniqueness, informal)**.** *Consider the initial value problem*

$$\frac{dy}{dt} = f(t, y), \qquad y(t_0) = y_0.$$

*If f is continuous in a rectangle around the point (t_0, y_0), then at least one solution exists through that point (*existence*). If, in addition, the partial derivative $\frac{\partial f}{\partial y}$ is continuous there, then the solution is* unique *– exactly one solution curve passes through (t_0, y_0).*

The theorem has two halves, and they require different amounts of niceness from the law f. Mere continuity of f buys you *existence*: some solution gets going. To also guarantee *uniqueness* – that the walker's path is forced, with no branching – you need the extra smoothness that $\frac{\partial f}{\partial y}$ be continuous (a slightly weaker "Lipschitz" condition is what the full proof really uses, but continuity of $\frac{\partial f}{\partial y}$ is the clean sufficient version). Intuitively, $\frac{\partial f}{\partial y}$ controls how fast the prescribed slope changes as you move up and down; keeping it under control prevents nearby curves from being able to split apart or merge.

Intuition

Read the theorem as a guarantee about the walker. *Existence* says: if the compass (the law f) points somewhere sensible at and near your starting spot, you can always start walking – a path exists. *Uniqueness* says: if the compass also turns *smoothly* as you move (no sudden infinite swings in the prescribed slope), then your path is completely determined – there is no fork in the road, no moment where the law leaves you a choice. Together they justify the whole enterprise: a sensible law plus a starting point yields one and only one future.

Why care about a theorem that merely confirms what the picture already suggested? Because without it, modeling would be built on sand. When an engineer writes down a differential equation for a bridge's vibration and a single initial state, they are implicitly claiming the bridge has *one* determined future, not several. The existence–uniqueness theorem is the license to make that claim. It promises that a well-posed law of nature, handed a present, returns a future – exactly one – and that is the bedrock assumption of deterministic science.

36.6.1 When uniqueness fails – and why that is not a paradox

Now for the cautionary tale that makes the theorem's hypotheses feel necessary rather than fussy. Consider the deceptively simple problem

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \sqrt{|y|}, \qquad y(0) = 0.$$

Here $f(t, y) = \sqrt{|y|}$ is perfectly continuous everywhere, so existence is fine – a solution exists. But watch what $\frac{\partial f}{\partial y}$ does near $y = 0$. For $y > 0$, $f = \sqrt{y}$ and $\frac{\partial f}{\partial y} = \frac{1}{2\sqrt{y}}$, which blows up to infinity as $y \to 0$. The uniqueness hypothesis fails exactly at the starting point. And uniqueness itself fails with it:

Example 36.5 (Two solutions from one start)**.** The constant function $y(t) = 0$ solves the problem: its derivative is 0, and $\sqrt{|0|} = 0$, so the equation holds, and $y(0) = 0$. But the function

$$y(t) = \tfrac{1}{4}t^2 \quad (t \geq 0)$$

also solves it. Check: $\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{1}{2}t$, while $\sqrt{|y|} = \sqrt{\frac{1}{4}t^2} = \frac{1}{2}t$ for $t \geq 0$. The two sides match, and $y(0) = 0$. So *two different* solution curves emerge from the single point $(0, 0)$ – the flat one that stays put, and the rising parabola. In fact there are infinitely many: the curve can linger at zero for any

length of time $a \geq 0$ and then peel up into the shifted parabola $y = \frac{1}{4}(t-a)^2$. Figure 36.5 shows a few of them fanning out of the single origin. Uniqueness has shattered.

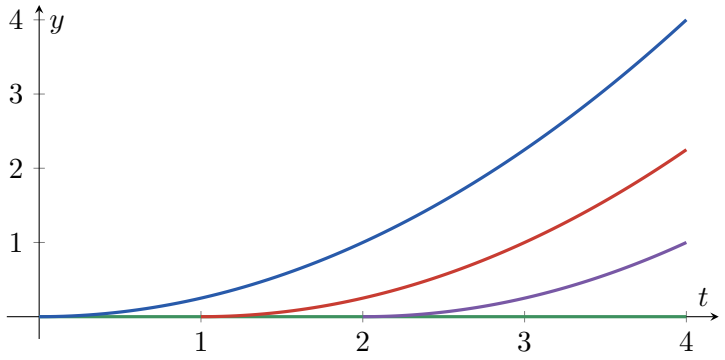


Figure 36.5: Non-uniqueness for $\frac{\mathrm{d}y}{\mathrm{d}t} = \sqrt{|y|}$, $y(0) = 0$. Infinitely many solution curves leave the single starting point $(0, 0)$: the flat solution $y = 0$ (green), and a whole fan of parabolas $y = \frac{1}{4}(t-a)^2$ that linger at zero until an arbitrary time a and then rise. The uniqueness hypothesis $\frac{\partial f}{\partial y}$ continuous fails at $y = 0$, and this is what failure looks like.

Common trap

The lesson is not that the existence–uniqueness theorem is unreliable – it is that its *hypotheses are doing real work*. When $\frac{\partial f}{\partial y}$ is continuous, uniqueness is rock-solid; when that smoothness breaks down (here, an infinite slope-sensitivity at $y = 0$), solutions really can branch. Never assume uniqueness for a nonlinear equation without checking the hypothesis. The square-root law above is not an artificial curiosity, either: it is essentially the equation governing how fast a leaking tank drains (Torricelli's law), and the non-uniqueness has a physical reading – you cannot run the drainage *backward* in time to recover when the tank was full, because many different fill-times lead to the same empty tank.

36.7 Why differential equations exist: modeling

We have treated differential equations as mathematical objects, but they earn their central place because they are the native language of *change in the world*. A modeling problem almost always arrives as a statement about rates, and translating that statement into symbols *is* writing a differential equation. To feel the equations being born, we build three of the classic models from scratch, each from a single sentence of physical reasoning.

36.7.1 Population growth

Application

The sentence: *A population grows at a rate proportional to its current size.* The reasoning is plain – twice as many individuals means twice as many births per year. Let $P(t)$ be the population. "Grows at a rate" is $\frac{\mathrm{d}P}{\mathrm{d}t}$; "proportional to its current size" means that rate is some constant k times P. So

$$\frac{\mathrm{d}P}{\mathrm{d}t} = kP.$$

This is our first equation again, wearing biological clothes. Its solution is $P(t) = P_0 e^{kt}$, where $P_0 = P(0)$ is the starting population: *exponential growth*, the model that explains why unchecked populations, compound interest, and chain reactions all explode in the same characteristic way. The constant k is the per-capita growth rate; a larger k means a steeper explosion.

Exponential growth is, of course, too optimistic forever – no real population grows without bound. The fix is itself a lesson in modeling: add a term that brakes the growth as P approaches a carrying capacity M. That gives the *logistic equation* $\frac{\mathrm{d}P}{\mathrm{d}t} = kP(1 - P/M)$, a nonlinear model whose slope field shows every solution leveling off at M. We will not solve it here, but notice how naturally a refined sentence ("growth slows as the population fills its niche") produces a refined equation.

36.7.2 Newton's law of cooling

Application

The sentence: *An object cools at a rate proportional to the gap between its temperature and the surrounding temperature.* Let $T(t)$ be the object's temperature and A the ambient temperature. The "gap" is $T - A$. The rate of cooling is $\frac{\mathrm{d}T}{\mathrm{d}t}$, and "proportional to the gap" (with cooling, so the constant is negative) gives

$$\frac{\mathrm{d}T}{\mathrm{d}t} = -k\,(T - A), \qquad k > 0.$$

When $T > A$ the right side is negative and the object cools; when $T < A$ it is positive and the object warms; when $T = A$ the rate is zero and the temperature holds. The solution is $T(t) = A + (T_0 - A)e^{-kt}$ – the temperature decays exponentially toward the ambient A, exactly the attractor behavior we saw in Figure 36.3. This is the equation forensic scientists use to estimate time of death from body temperature.

36.7.3 Motion under a force

Application

The sentence: *Force equals mass times acceleration.* Acceleration is the second derivative of position, $\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$, so Newton's second law is already a second-order differential equation:

$$m\,\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = F.$$

For a stone in free fall the force is constant, $F = -mg$, and the equation collapses to $\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -g$. Integrate once for velocity, again for position, collecting two constants – which the two initial conditions (starting height and starting velocity) fix. For a mass on a spring the force is $F = -kx$ (Hooke's law, pulling back toward rest), giving $m\,\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -kx$, whose solutions oscillate sinusoidally forever. The same skeleton, $F = ma$, with a different force law, produces falling, orbiting, oscillating, and swinging – the entire repertoire of classical motion.

Why this matters

Step back and notice the unity. Three completely different phenomena – a breeding population, a cooling cup, a falling stone – led to differential equations, and two of them led to the *same* equation, $\frac{\mathrm{d}y}{\mathrm{d}t} = ky$, up to the sign and meaning of the constant. This is what gives the subject its reach: a handful of equation-shapes recur across all of science, so that mastering them once settles problems in biology, physics, chemistry, and finance at a single stroke. The differential equation is the level of abstraction at which a population and a bank account and a radioactive sample are revealed to be the same problem. That is why this is the destination the whole book has been climbing toward: calculus exists, in the end, to write down and solve the laws of change.

Work it yourself

Before reading on, try translating each sentence into a differential equation yourself – do not solve, just write the equation.

1. A radioactive sample decays at a rate proportional to the amount remaining.
2. A skydiver's downward speed v increases due to gravity but is opposed by air resistance proportional to the speed.
3. A savings account grows by continuous interest at rate r, and money is also withdrawn at a steady w dollars per year.

Answers: (1) $\frac{\mathrm{d}N}{\mathrm{d}t} = -kN$; (2) $m\,\frac{\mathrm{d}v}{\mathrm{d}t} = mg - bv$; (3) $\frac{\mathrm{d}B}{\mathrm{d}t} = rB - w$. Notice that each is just a sentence about rates, set equal to symbols. That translation – prose to equation – is the real modeling skill, and it is almost entirely a matter of identifying "rate of change of ___" with a derivative.

36.8 Looking ahead

You now know what a differential equation is, what it means to solve one, how an initial condition selects a single solution from a family, how to classify equations by order and linearity, how to *see* solutions in a slope field, when a solution is guaranteed to exist and be unique, and why these equations are the language in which the sciences write their laws. What remains is technique – the actual machinery for producing the general solutions we have so far only verified or read off pictures.

The next three chapters supply that machinery. The chapter on first-order ODEs develops the systematic methods for first-order equations – separation of variables, which cracks $\frac{\mathrm{d}y}{\mathrm{d}t} = ky$ and the logistic equation, and the integrating factor, which conquers every first-order linear equation $\frac{\mathrm{d}y}{\mathrm{d}t} + p(t)y = q(t)$. The chapter after that tackles second-order linear equations, where superposition gives a complete and elegant theory – this is the home of springs, circuits, and resonance. And the final chapter pushes onward to systems of equations and the qualitative, geometric viewpoint, where the slope-field thinking of this chapter blossoms into the full phase-plane analysis of how coupled quantities evolve together. The pictures you have learned to read here are the ones you will be reading for the rest of the subject.

Exercises

Exercise 36.1. Verify by substitution that $y = \sin(2t)$ is a solution of the second-order equation $\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4y = 0$. Then verify that $y = \cos(2t)$ is also a solution, and that $y = 3\sin(2t) - 5\cos(2t)$ is too. What property of the equation makes this last combination work?

Exercise 36.2. For each equation state its order, and say whether it is linear or nonlinear (give the reason for any nonlinearity): (a) $\frac{\mathrm{d}y}{\mathrm{d}t} = 3y - 7$; (b) $\frac{\mathrm{d}y}{\mathrm{d}t} = y\cos t$; (c) $\frac{\mathrm{d}y}{\mathrm{d}t} = \cos(y)$; (d) $\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + x\,\frac{\mathrm{d}x}{\mathrm{d}t} + x = 0$; (e) $t^2\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - t\frac{\mathrm{d}y}{\mathrm{d}t} + y = e^t$.

Exercise 36.3. The general solution of $\frac{\mathrm{d}y}{\mathrm{d}t} = -2y$ is $y = Ce^{-2t}$. Find the particular solution satisfying $y(0) = 7$. Then find the one satisfying $y(1) = 7$ instead, and explain why the constant C comes out differently in the two cases.

Exercise 36.4. Solve the initial value problem $\frac{\mathrm{d}y}{\mathrm{d}t} = 6t^2 - 4$, $y(0) = 3$, by direct integration. (This equation's right-hand side depends only on t, so a single antiderivative does the job.)

Exercise 36.5. Consider $\frac{\mathrm{d}y}{\mathrm{d}t} = -y/t$ for $t > 0$. (a) Along which curves in the plane is the prescribed slope equal to zero? Where is it undefined? (b) Verify that $y = C/t$ is a solution for every constant C. (c) Sketch, by hand, the slope field at a few points and confirm that the curves $y = C/t$ (hyperbolas) flow tangent to it.

Exercise 36.6. For Newton's law of cooling $\frac{\mathrm{d}T}{\mathrm{d}t} = -k(T - A)$ with $k > 0$: (a) What constant function $T(t)$ is a solution? (This is the *equilibrium* temperature.) (b) Using the slope-field reasoning of Section 36.5, argue without solving that every solution approaches A as $t \to \infty$. (c) Verify that $T(t) = A + (T_0 - A)e^{-kt}$ solves the equation and satisfies $T(0) = T_0$.

Exercise 36.7. Explain, in your own words, why the general solution of an nth-order differential equation should contain n arbitrary constants. Then state how many initial conditions are needed to determine a unique solution of (a) $\frac{\mathrm{d}y}{\mathrm{d}t} = y^2$, and (b) $m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -kx$, and say physically what those conditions represent in case (b).

Exercise 36.8. Consider $\frac{\mathrm{d}y}{\mathrm{d}t} = 3y^{2/3}$ with $y(0) = 0$. (a) Verify that $y(t) = 0$ is a solution. (b) Verify that $y(t) = t^3$ is also a solution. (c) Compute $\frac{\partial f}{\partial y}$ for $f(t, y) = 3y^{2/3}$ and explain, by reference to Theorem 36.1, why the existence of two solutions does not contradict the uniqueness theorem.

Exercise 36.9. Translate each sentence into a differential equation (do not solve): (a) A tank holds V litres of water; salt water flows in carrying salt at a constant r grams per minute, and well-mixed water flows out carrying salt at a rate proportional to the current amount S of salt. (b) The velocity v of a parachutist changes under gravity minus an air-resistance force proportional to v^2. (c) A rumour spreads through a town of N people at a rate proportional to the product of those who know it, y, and those who do not, $N - y$.

Exercise 36.10. The logistic equation is $\frac{\mathrm{d}P}{\mathrm{d}t} = kP\left(1 - \dfrac{P}{M}\right)$ with $k, M > 0$. (a) Find the two constant (equilibrium) solutions by setting the right-hand side to zero. (b) For $0 < P < M$, is $\frac{\mathrm{d}P}{\mathrm{d}t}$ positive or negative? What does this say about a population that starts below carrying capacity? (c) Without solving, sketch the qualitative shape of a solution that starts at $P(0)$ slightly above zero, and explain why it has the famous "S" shape.

Exercise 36.11. The equation $\frac{\mathrm{d}y}{\mathrm{d}t} = y^2$ is nonlinear. Show that $y_1 = -1/t$ is a solution (for $t > 0$), but that $2y_1 = -2/t$ is *not* a solution. Why does this not violate any principle? Contrast with what would happen for a linear homogeneous equation, and connect your answer to the discussion of superposition in Section 36.4.

Chapter 37

First-Order Differential Equations

A differential equation is a sentence written in the language of change. Up to now we have mostly started with a function and *asked* for its derivative. A differential equation turns the question around: it hands us a relationship that the derivative must satisfy, and asks us to find the function. It says, in effect, "I will not tell you what y is. I will only tell you how fast y changes, and I will let that rate depend on where you are and what y already is. Now go find y."

This is not a niche trick. It is the way almost all of quantitative science is actually written. Newton did not hand us the position of a planet; he handed us a rule about its acceleration. A biologist does not start with a population curve; she starts with a statement that the growth rate is proportional to the population. A bank does not tell you your balance in ten years; it tells you the interest rate, which is a statement about $\frac{\mathrm{d}B}{\mathrm{d}t}$. The natural laws of the world are overwhelmingly written as differential equations, and the act of *solving* one is the act of turning a local rule about change into a global picture of behavior.

A **first-order** differential equation is one in which only the first derivative appears: no y'', no y''', just y' and y and the independent variable. The general shape is

$$\frac{\mathrm{d}y}{\mathrm{d}x} = f(x, y).$$

Read it slowly. At every point (x, y) in the plane, the right-hand side $f(x, y)$ hands you a *number* — a slope. So a first-order ODE is really a field of little slope-marks blanketing the plane, and a solution is a curve that is everywhere tangent to those marks. That picture, the *slope field*, is the single most important image in this chapter. Hold onto it.

37.1 The Picture First: Slope Fields

Before we solve anything, let us *see* what an equation like $\frac{\mathrm{d}y}{\mathrm{d}x} = f(x, y)$ is asking. The equation assigns a slope to every point. If we draw a short line segment of that slope at a grid of points, we get a slope field (also called a direction field). A solution curve is then nothing more than a path that flows along the segments, like a leaf carried on a current.

Intuition

Think of the slope field as a current in a river and the solution curves as the paths leaves take when dropped in. Drop a leaf at a point — that is choosing an initial condition — and the current carries it along a unique path. Different starting points give different paths, but they all obey the same current. Solving the ODE means *finding the shape of those paths.* You can often sketch their shape by eye, long before you can write a formula, just by following the arrows.

Consider the simplest interesting example, $\frac{dy}{dx} = y$. At a point with height $y = 2$ the slope is 2; at height $y = -1$ the slope is -1; along the x-axis the slope is 0. The slope depends only on height, not on x, so the field looks the same as you slide left or right. Figure 37.1 shows it.

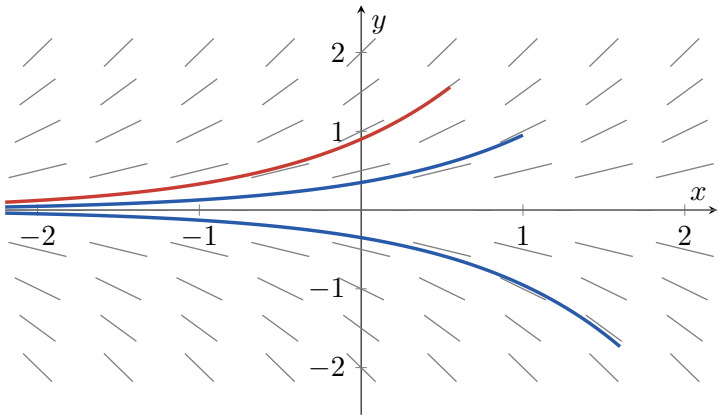


Figure 37.1: The slope field of $\frac{dy}{dx} = y$. Each gray segment has slope equal to its own height (so the marks are flat along the x-axis and steepen away from it). Three solution curves $y = Ce^x$ (blue and red) thread through the field, everywhere tangent to it. Notice how the curves never cross — one leaf, one path.

The blue and red curves are the solutions $y = Ce^x$. We have not derived that yet, but you can *believe* it from the picture: a curve hugging arrows whose slope equals their height is exactly a curve whose own slope equals its own height, and the exponential is the function that is its own derivative.

What is really happening

A first-order ODE is a *local* law: it only ever talks about one point at a time, telling you the slope there. Solving it means *integrating that local law into a global shape.* The remarkable thing — guaranteed by the Fundamental Theorem of Calculus and the existence–uniqueness theorems — is that an infinitesimal rule, repeated, knits itself into a single coherent curve. Every solution method in this chapter is a different way of doing that knitting.

Focus

Three words you must keep straight:

- A **general solution** is a whole *family* of curves, carrying one arbitrary constant C (because a first-order equation involves one integration, and integration always leaves a $+C$).

- An **initial condition** like $y(x_0) = y_0$ picks *one* curve out of the family by naming a point it must pass through. This nails down C.
- A **particular solution** is that single chosen curve.

The general solution is the river; the initial condition is where you drop the leaf; the particular solution is the leaf's path.

37.2 Separable Equations

The friendliest equations are those in which the slope $f(x, y)$ factors into a part that depends only on x times a part that depends only on y:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = g(x)\,h(y).$$

We call these **separable**, because we can herd all the y's to one side and all the x's to the other. The recipe taught in most courses is breathtakingly casual: treat $\frac{\mathrm{d}y}{\mathrm{d}x}$ as a fraction, multiply both sides by $\mathrm{d}x$, divide by $h(y)$, and integrate:

$$\frac{\mathrm{d}y}{h(y)} = g(x)\,\mathrm{d}x \quad \Longrightarrow \quad \int \frac{\mathrm{d}y}{h(y)} = \int g(x)\,\mathrm{d}x.$$

This works. But notice what we just did: we chopped the derivative $\frac{\mathrm{d}y}{\mathrm{d}x}$ in half and treated $\mathrm{d}y$ and $\mathrm{d}x$ as separate movable objects, as if the derivative were a genuine ratio of two numbers. It is not. $\frac{\mathrm{d}y}{\mathrm{d}x}$ is a *limit* of ratios, a single indivisible symbol. So why is this allowed?

Why the technique works

Separation of variables is the chain rule run backwards. Here is the rigorous derivation, with no division of $\mathrm{d}y$ by $\mathrm{d}x$.

Start from $\frac{\mathrm{d}y}{\mathrm{d}x} = g(x)\,h(y)$. Assume $h(y) \neq 0$, and divide both sides (legitimately, these are real numbers) by $h(y)$:

$$\frac{1}{h(y)}\,\frac{\mathrm{d}y}{\mathrm{d}x} = g(x).$$

Now introduce $H(y)$, an antiderivative of $1/h(y)$, so that $H'(y) = 1/h(y)$. By the *chain rule*, the left-hand side is exactly the derivative of $H(y(x))$ with respect to x:

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(H(y(x))\big) = H'(y)\,\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{h(y)}\,\frac{\mathrm{d}y}{\mathrm{d}x}.$$

So our equation says

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(H(y(x))\big) = g(x).$$

Two functions of x with the same derivative differ by a constant, so integrating both sides in x gives

$$H(y) = \int g(x)\,\mathrm{d}x + C.$$

That is *precisely* the answer the casual "multiply by $\mathrm{d}x$" recipe produces, because $\int \mathrm{d}y/h(y)$ is just $H(y)$. The fraction-juggling is a mnemonic; the chain rule is the mechanism. The $\mathrm{d}y$ and

$\mathrm{d}x$ are not really being separated — they are bookkeeping symbols that happen to track the substitution $u = y(x)$ correctly. This is the same reason u-substitution in integration works: separation of variables *is* u-substitution wearing a different hat.

Common trap

When we divided by $h(y)$ we assumed $h(y) \neq 0$. But any value $y = c$ with $h(c) = 0$ gives the *constant* solution $y \equiv c$ (its derivative is zero, and $g(x)\,h(c) = 0$ too). These constant "equilibrium" solutions are real solutions that the separation recipe can quietly lose, because you divided by zero to get them. Always check the roots of h separately. We will meet these again as equilibria in the section on autonomous equations.

37.2.1 A First Worked Example

Solve $\frac{\mathrm{d}y}{\mathrm{d}x} = xy$ with $y(0) = 3$.

Walking question

Is it separable? The right side is $x \cdot y$, a function of x times a function of y. Yes, with $g(x) = x$ and $h(y) = y$. So we expect to find $\int \mathrm{d}y/y$ on the left and $\int x\,\mathrm{d}x$ on the right.

Step 1. Separate. Divide by y (noting $y = 0$ as a possible lost solution):

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = x.$$

Step 2. Integrate both sides in x. The left integrates to $\ln|y|$ (chain rule in reverse), the right to $\frac{1}{2}x^2$:

$$\ln|y| = \tfrac{1}{2}x^2 + C.$$

Step 3. Solve for y. Exponentiate:

$$|y| = e^{C}\,e^{x^2/2}, \qquad \text{so} \qquad y = A\,e^{x^2/2},$$

where $A = \pm e^{C}$ is a new constant absorbing the sign and the e^{C}. (Letting $A = 0$ even recovers the lost solution $y \equiv 0$, so the single formula $y = Ae^{x^2/2}$ captures everything.)

Step 4. Apply the initial condition. $y(0) = 3$ means $3 = A\,e^{0} = A$, so $A = 3$. The particular solution is

$$\boxed{y = 3\,e^{x^2/2}}.$$

Check: $\frac{\mathrm{d}y}{\mathrm{d}x} = 3\,e^{x^2/2} \cdot x = x \cdot y$. It satisfies the equation, and $y(0) = 3$. Done.

Work it yourself

Solve $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x}{y}$ with $y(0) = 2$ before reading on. Separate as $y\,\mathrm{d}y = x\,\mathrm{d}x$, integrate to $\frac{1}{2}y^2 = \frac{1}{2}x^2 + C$, so $y^2 - x^2 = \text{const}$. The solution curves are *hyperbolas*. The initial condition $y(0) = 2$ gives $4 - 0 = 4$, so $y = \sqrt{x^2 + 4}$. Notice the answer is an implicit relation $y^2 - x^2 = 4$ that we then solved for y; many separable problems leave you with an implicit equation, and that is a

perfectly good answer.

37.2.2 Why an Initial Condition Pins Down One Curve

Connection

The general solution $y = Ae^{x^2/2}$ is a one-parameter family — an entire stack of curves, one for each value of A. They fill the plane without crossing, just like the leaves' paths in the river picture. The existence and uniqueness theorem (which holds whenever $f(x, y)$ and $\frac{\partial f}{\partial y}$ are continuous near the starting point) guarantees that through each point passes *exactly one* solution curve. That is why naming one point, the initial condition, selects one and only one member of the family. No two distinct solution curves of a well-behaved first-order ODE ever touch.

37.3 Exponential Growth and Decay

The most important separable equation in all of science is

$$\frac{\mathrm{d}y}{\mathrm{d}t} = k\,y,$$

which says *the rate of change is proportional to the amount present.* Read it as a sentence: "the more there is, the faster it grows" (if $k > 0$), or "the more there is, the faster it shrinks" (if $k < 0$). Money in a continuously compounded account, a colony of bacteria with unlimited food, a lump of radioactive uranium, the cooling-off of a difference in temperature — all obey this.

Separating, $\dfrac{\mathrm{d}y}{y} = k\,\mathrm{d}t$, so $\ln|y| = kt + C$, and

$$\boxed{y(t) = y_0\,e^{kt}},$$

where $y_0 = y(0)$ is the starting amount. If $k > 0$ the amount explodes; if $k < 0$ it decays toward zero. Figure 37.2 shows both.

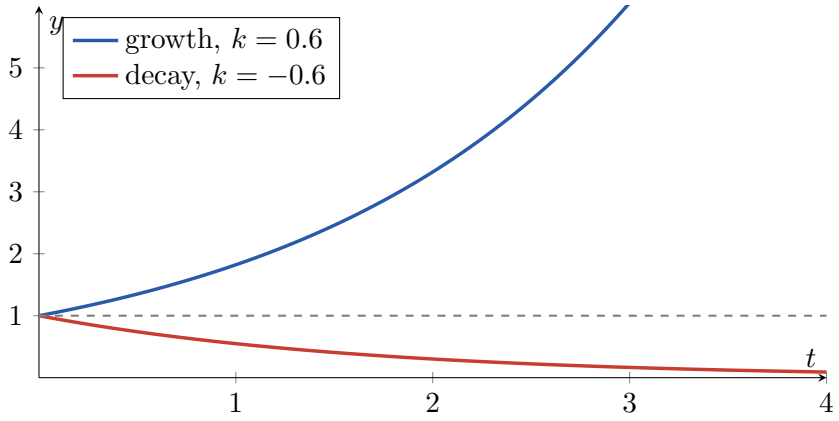


Figure 37.2: Exponential growth ($k > 0$) and decay ($k < 0$). Both start at the dashed reference and obey $y' = ky$. Growth curves upward without bound; decay falls toward the t-axis but never reaches it.

History

The radioactive decay law $N' = -\lambda N$ was pieced together by Ernest Rutherford and Frederick Soddy around 1902 from careful measurements of how thorium emanation faded. They discovered that a fixed *fraction* of atoms decayed per unit time, regardless of how many there were — the signature of $N' \propto N$. From this came the concept of *half-life*, the time $t_{1/2}$ for half the sample to vanish. Setting $\frac{1}{2} = e^{-\lambda t_{1/2}}$ gives the clean and famous relation $t_{1/2} = \dfrac{\ln 2}{\lambda}$. Half a century later this same equation, run in reverse, let Willard Libby date the dead from the carbon-14 left in their bones, an idea that won the 1960 Nobel Prize in Chemistry.

Application

Carbon dating, worked. Living tissue holds a steady ratio of carbon-14; when it dies, the ^{14}C decays with half-life $t_{1/2} = 5730$ years, so $\lambda = \dfrac{\ln 2}{5730} \approx 1.21 \times 10^{-4}$ yr^{-1}. A bone is found with 62% of the living ^{14}C level. How old is it? Solve $0.62 = e^{-\lambda t}$ for t:

$$t = -\frac{\ln 0.62}{\lambda} = -\frac{\ln 0.62}{1.21 \times 10^{-4}} \approx \frac{0.478}{1.21 \times 10^{-4}} \approx 3950 \text{ years}.$$

The bone is roughly four thousand years old. The whole calculation is one separable ODE, solved once and reused forever.

37.3.1 Newton's Law of Cooling

A hot cup of coffee in a cool room cools fast at first, then slowly, leveling off at room temperature. Newton's law of cooling captures this: the rate of cooling is proportional to the temperature *difference* between the object and its surroundings. If $T(t)$ is the object's temperature and T_a the (constant) ambient temperature,

$$\frac{\mathrm{d}T}{\mathrm{d}t} = -k\,(T - T_a), \qquad k > 0.$$

The minus sign is the physics: if $T > T_a$ the object is hotter than the room, so $\frac{dT}{dt} < 0$ and it cools; if $T < T_a$ it warms. Either way it heads toward T_a.

Why the technique works

This looks more complicated than $y' = ky$, but it is the *same* equation in disguise. Let $u = T - T_a$ be the temperature *difference*. Since T_a is constant, $\frac{du}{dt} = \frac{dT}{dt}$, and the equation becomes

$$\frac{du}{dt} = -k\,u,$$

plain exponential decay! The difference from ambient decays exponentially: $u(t) = u_0 e^{-kt}$. Translating back,

$$T(t) = T_a + (T_0 - T_a)\,e^{-kt}.$$

The substitution $u = T - T_a$ *slid the whole problem down* so that the equilibrium sat at zero, where we already know the answer, then slid it back up. A huge amount of differential-equations technique is exactly this: find the change of variable that turns your problem into one you have already solved.

Application

The coffee, worked. Coffee at $T_0 = 90°$C sits in a $20°$C room. After 5 minutes it has cooled to $70°$C. When will it reach a drinkable $50°$C?
First find k. With $T_a = 20$ and $T_0 = 90$, the difference starts at 70 and at $t = 5$ has fallen to $70 - 20 = 50$:

$$50 = 70\,e^{-5k} \;\Rightarrow\; e^{-5k} = \tfrac{5}{7} \;\Rightarrow\; k = \tfrac{1}{5}\ln\tfrac{7}{5} \approx 0.0673 \text{ min}^{-1}.$$

Now solve for the time when $T = 50$, i.e. the difference is 30:

$$30 = 70\,e^{-kt} \;\Rightarrow\; e^{-kt} = \tfrac{3}{7} \;\Rightarrow\; t = \frac{\ln(7/3)}{k} \approx \frac{0.847}{0.0673} \approx 12.6 \text{ min}.$$

So about 12.6 minutes after pouring. Notice the curve in Figure 37.3 flattening toward the room line — that flattening is the $e^{-kt} \to 0$ tail.

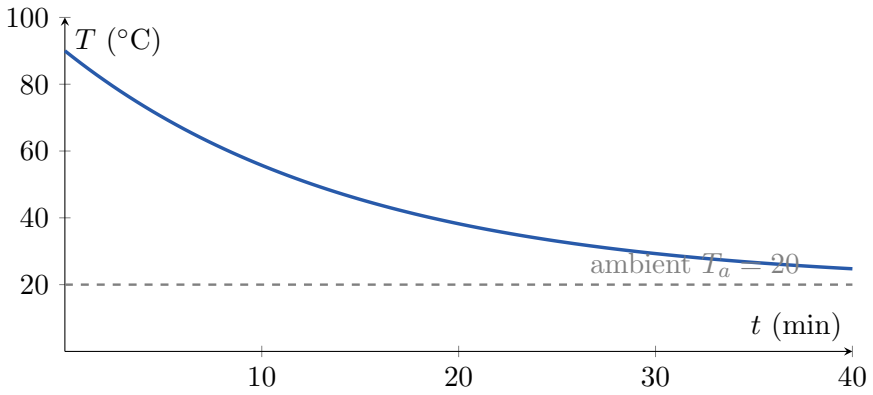


Figure 37.3: Newton cooling: the coffee's temperature decays toward the ambient line $T_a = 20°\text{C}$. Steep at first (big difference), then ever gentler. The temperature *difference* is what decays exponentially.

37.4 First-Order Linear Equations and the Integrating Factor

Not every equation separates. Consider

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = x.$$

Try to separate it: $\frac{\mathrm{d}y}{\mathrm{d}x} = x - 2y$. The right side is $x - 2y$, which does *not* factor as (function of x)(function of y). Stuck. We need a new idea.

The class of equations we can crack is the **first-order linear** equations:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)\,y = Q(x).$$

"Linear" means y and y' appear only to the first power, never multiplied together, never inside a sine or a square. The coefficients P and Q may be any functions of x. This single template covers an enormous range of models.

37.4.1 The Idea: Make the Left Side a Single Derivative

Here is the dream. The left side $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)\,y$ is *almost* the derivative of a product. Recall the product rule:

$$\frac{\mathrm{d}}{\mathrm{d}x}(\mu y) = \mu\,\frac{\mathrm{d}y}{\mathrm{d}x} + \mu' y.$$

If only we could multiply our whole equation by some cleverly chosen function $\mu(x)$ so that the left side became exactly $\frac{\mathrm{d}}{\mathrm{d}x}(\mu y)$, then the equation would collapse to

$$\frac{\mathrm{d}}{\mathrm{d}x}(\mu y) = \mu\,Q,$$

and we could just *integrate both sides* and be done. The function μ that performs this magic is called the **integrating factor**. Let us not memorize it — let us *derive* what it has to be.

Why the technique works

Deriving the integrating factor. Multiply the equation by an unknown positive function $\mu(x)$:

$$\mu \frac{\mathrm{d}y}{\mathrm{d}x} + \mu P\, y = \mu Q.$$

We *want* the left side to equal $\frac{\mathrm{d}}{\mathrm{d}x}(\mu y) = \mu \frac{\mathrm{d}y}{\mathrm{d}x} + \mu' y$. Comparing the two expressions term by term, the $\mu\, y'$ pieces already match. The remaining pieces match precisely when

$$\mu' = \mu P.$$

But that is a *separable* equation for μ! Separate it: $\dfrac{\mathrm{d}\mu}{\mu} = P\,\mathrm{d}x$, integrate to $\ln \mu = \int P\,\mathrm{d}x$, and exponentiate:

$$\boxed{\mu(x) = e^{\int P(x)\,\mathrm{d}x}}.$$

That is where the famous formula comes from. It is not pulled from a hat; it is *forced* on us by demanding that the product rule fit. The integrating factor is exactly the function whose logarithmic derivative is P, because that is what it takes to make $\mu y' + \mu P y$ collapse into one clean derivative.

What is really happening

What is really happening is a *change of viewpoint*. The combination $y' + Py$ does not look like a derivative of anything tidy. But multiply by μ and it becomes the derivative of the single quantity μy. We have re-expressed an awkward two-term expression as the rate of change of one product. Once something is written as $\frac{\mathrm{d}}{\mathrm{d}x}(\text{stuff})$, the Fundamental Theorem of Calculus lets us undo it by integration. The integrating factor is a lens that brings the left side into focus as a single derivative.

37.4.2 The Full Recipe, Derived

Putting it together: to solve $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)\, y = Q(x)$,

1. Compute the integrating factor $\mu = e^{\int P\,\mathrm{d}x}$.
2. Multiply the equation by μ. The left side is now $\frac{\mathrm{d}}{\mathrm{d}x}(\mu y)$ by construction.
3. Integrate both sides: $\mu y = \int \mu Q\,\mathrm{d}x + C$.
4. Divide by μ to get $y = \dfrac{1}{\mu}\left(\int \mu Q\,\mathrm{d}x + C\right)$.

37.4.3 Worked Example with the Integrating Factor

Solve $\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = x$ with $y(0) = 1$.

Walking question

Here $P(x) = 2$ (a constant) and $Q(x) = x$. The integrating factor will be $e^{\int 2\,dx} = e^{2x}$. We expect to multiply through by e^{2x} and watch the left side fold into $\frac{d}{dx}\big(e^{2x}y\big)$.

Step 1. Integrating factor. $\mu = e^{\int 2\,dx} = e^{2x}$.

Step 2. Multiply through.

$$e^{2x}\frac{dy}{dx} + 2e^{2x}y = x\,e^{2x}.$$

The left side is exactly $\frac{d}{dx}\big(e^{2x}y\big)$ — check it with the product rule: $\frac{d}{dx}\big(e^{2x}y\big) = e^{2x}y' + 2e^{2x}y$.

Step 3. Integrate both sides.

$$e^{2x}y = \int x\,e^{2x}\,dx.$$

The integral on the right needs integration by parts (let $u = x$, $dv = e^{2x}dx$):

$$\int x\,e^{2x}\,dx = \frac{x}{2}e^{2x} - \frac{1}{2}\int e^{2x}\,dx = \frac{x}{2}e^{2x} - \frac{1}{4}e^{2x} + C.$$

So

$$e^{2x}y = \frac{x}{2}e^{2x} - \frac{1}{4}e^{2x} + C.$$

Step 4. Solve for y. Divide by e^{2x}:

$$y = \frac{x}{2} - \frac{1}{4} + C\,e^{-2x}.$$

Step 5. Initial condition. $y(0) = 1$:

$$1 = 0 - \tfrac{1}{4} + C \;\Rightarrow\; C = \tfrac{5}{4}.$$

The particular solution is

$$\boxed{y = \frac{x}{2} - \frac{1}{4} + \frac{5}{4}\,e^{-2x}}.$$

Notice the structure: a part $\frac{x}{2} - \frac{1}{4}$ that tracks the forcing $Q = x$ (the "steady" response), plus a transient $\frac{5}{4}e^{-2x}$ that fades away. This split into a lingering particular response and a dying homogeneous transient recurs everywhere in applied mathematics.

Common trap

Two ways people wreck this method:

1. Forgetting to put the equation in *standard form* first. The coefficient of y' must be 1 before you read off P. If you start from $x\,y' + y = x^2$, divide by x to get $y' + \frac{1}{x}y = x$, so $P = 1/x$, not something else.

2. Dragging the $+C$ from the integrating-factor integral $\int P\,dx$ into μ. You do not need it: any one antiderivative works, because an extra constant in the exponent just multiplies μ by a constant, and that constant cancels when you divide by μ at the end. Use the simplest μ. The $+C$ that *does* matter is the one from Step 3, integrating μQ.

Connection

The integrating-factor trick is the same spirit as "completing the square": you massage an expression by a deliberate multiplication (or addition) until it becomes a recognizable, invertible shape. There, you force a perfect square; here, you force a perfect derivative. Recognizing the shape of a known operation, and bending your problem until it fits, is one of the great recurring moves in all of mathematics.

37.5 Autonomous Equations, Equilibria, and Stability

A first-order equation is **autonomous** when the right side does not mention the independent variable explicitly:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(y).$$

The slope depends only on *where* you are (y), never on *when* (t). Newton cooling, exponential growth, and the logistic equation we are about to meet are all autonomous. Autonomy buys us a lot: it lets us read off the long-term fate of solutions *without solving the equation at all*, just by looking at the sign of f.

37.5.1 Equilibria: Where Nothing Changes

An **equilibrium** (or steady state, or fixed point) is a value y^* where $f(y^*) = 0$. There, $\frac{\mathrm{d}y}{\mathrm{d}t} = 0$: the system is frozen. If you start exactly at y^* you stay there forever, giving the constant solution $y \equiv y^*$. These are the very "lost" constant solutions from the separable section, now promoted to center stage.

The deep question is *stability*: if you nudge the system slightly off an equilibrium, does it slide back (stable) or run away (unstable)?

Intuition

Picture a marble in a landscape. A **stable** equilibrium is the bottom of a valley: push the marble a little and it rolls back. An **unstable** equilibrium is the top of a hill: the slightest nudge and it rolls away, never to return. A **phase line** is the one-dimensional version of this picture, and the sign of $f(y)$ tells you which way the marble rolls.

37.5.2 The Phase Line

Because the equation is autonomous, we can draw everything on a single vertical y-axis, the *phase line*. Mark the equilibria (roots of f). Between them, f has a constant sign. Where $f(y) > 0$, the solution increases (draw an upward arrow); where $f(y) < 0$, it decreases (downward arrow). An equilibrium with arrows pointing *toward* it from both sides is stable; arrows pointing *away* is unstable.

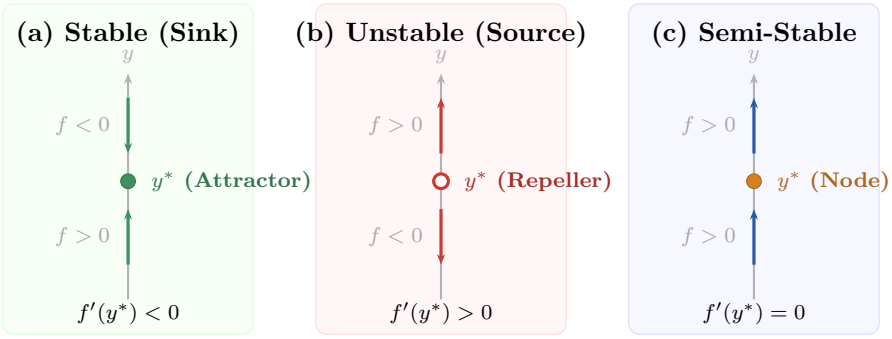


Figure 37.4: The three stability archetypes on a phase line for $\frac{dy}{dt} = f(y)$. (a) Stable Attractor (Sink): slopes point inward from both sides; small perturbations decay back to y^*. (b) Unstable Repeller (Source): slopes point outward; any perturbation drives the solution away. (c) Semi-Stable Node: slopes point in the same direction on both sides; perturbations from one side return, while perturbations to the other side run away.

Focus

The derivative test for stability. Near an equilibrium y^*, linearize: $f(y) \approx f'(y^*)(y - y^*)$ since $f(y^*) = 0$. So a small displacement $u = y - y^*$ obeys $\frac{du}{dt} \approx f'(y^*)\, u$, plain exponential behavior. Therefore:

$$f'(y^*) < 0 \implies \text{stable (perturbations decay)},$$
$$f'(y^*) > 0 \implies \text{unstable (perturbations grow)}.$$

The sign of the derivative of f at the equilibrium decides its fate. This is the one-dimensional ancestor of an enormous theory.

37.6 Logistic Growth

Exponential growth $y' = ky$ holds only while resources are unlimited. No real population grows forever; food, space, and predators eventually bite. The simplest fix, proposed by the Belgian mathematician Pierre François Verhulst in 1838, is to let the per-capita growth rate *decline* as the population approaches a ceiling M, the **carrying capacity**:

$$\frac{dP}{dt} = r\,P\left(1 - \frac{P}{M}\right).$$

Intuition

Read the two factors. When P is tiny, $1 - P/M \approx 1$, and growth is nearly exponential, $P' \approx rP$ — plenty of room, breed freely. As P climbs toward M, the factor $1 - P/M$ shrinks toward 0, throttling the growth. At $P = M$ growth stops entirely. The population is self-limiting: its own size applies the brakes.

37.6.1 The Phase Line of the Logistic

This is autonomous with $f(P) = rP(1 - P/M)$, a downward parabola in P crossing zero at $P = 0$ and $P = M$. Between them $f > 0$ (growth); below 0 and above M, $f < 0$. So:

- $P^* = 0$ is **unstable**: $f'(0) = r > 0$. A tiny population grows away from extinction.
- $P^* = M$ is **stable**: $f'(M) = -r < 0$. Populations above or below M are drawn back to it.

Every positive starting population, no matter how small or how large, ends up at the carrying capacity M. Figure 37.5 draws this verdict as a phase line, and Figure 37.6 shows the family of solution curves all funneling toward $P = M$.

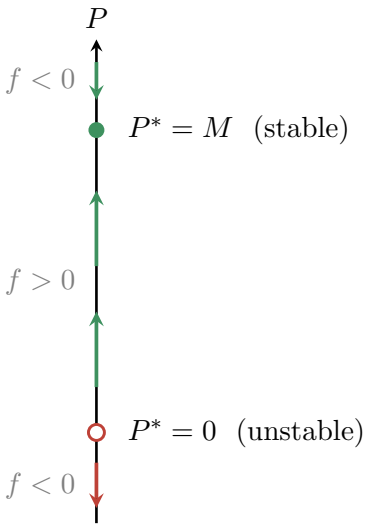


Figure 37.5: The phase line of the logistic equation $\frac{dP}{dt} = rP(1 - P/M)$. Arrows show the direction of motion: away from the open circle at $P = 0$ (unstable) and toward the filled circle at $P = M$ (stable). Reading a one-dimensional picture like this tells us the long-term fate of every solution without solving anything.

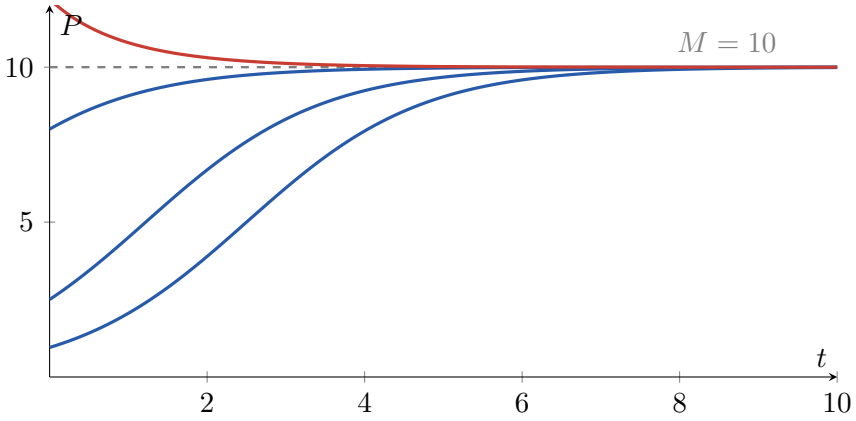


Figure 37.6: Logistic solutions for several initial populations. Small starts show the S-shaped (sigmoid) rise: nearly exponential, then bending over at the inflection near $P = M/2$, then leveling at the carrying capacity $M = 10$. A start *above* M (red) decays down to M. All roads lead to M.

37.6.2 Solving the Logistic Equation

It is separable, though the integral needs partial fractions. Separate:

$$\frac{\mathrm{d}P}{P\,(1 - P/M)} = r\,\mathrm{d}t.$$

Split the left fraction. Write $\dfrac{1}{P(1-P/M)} = \dfrac{1}{P} + \dfrac{1/M}{1-P/M}$ (you can verify by combining over a common denominator). Then

$$\int \left(\frac{1}{P} + \frac{1/M}{1 - P/M} \right) \mathrm{d}P = \ln P - \ln\!\left(1 - \frac{P}{M}\right) = \ln \frac{P}{1 - P/M}.$$

Integrating the right side gives $rt + C$, so

$$\ln \frac{P}{1 - P/M} = rt + C \quad \Longrightarrow \quad \frac{P}{1 - P/M} = C_0\, e^{rt}, \quad C_0 = e^{C}.$$

Solving algebraically for P (multiply out and collect) yields the clean **logistic function**

$$\boxed{P(t) = \frac{M}{1 + A\, e^{-rt}}}, \qquad A = \frac{M - P_0}{P_0}.$$

As $t \to \infty$, $e^{-rt} \to 0$ and $P \to M$, confirming the phase-line verdict. The curve has a single inflection point where it switches from concave-up to concave-down, located exactly at $P = M/2$, the moment of fastest growth.

Walking question

Why is the fastest growth at $P = M/2$? The growth *rate* is $f(P) = rP(1-P/M)$, a downward parabola in P. Its maximum is at the vertex, halfway between its roots 0 and M, namely $P = M/2$. So the population adds members fastest when it is exactly half-full. Before that, too few breeders; after that, too little room. The sweet spot is the middle.

Application

A yeast culture, worked. A flask of yeast follows the logistic law with carrying capacity $M = 1000$ cells and intrinsic rate $r = 0.7$ hr^{-1}, starting from $P_0 = 50$ cells. Then $A = (1000 - 50)/50 = 19$, so

$$P(t) = \frac{1000}{1 + 19\, e^{-0.7t}}.$$

How long until the culture is half full ($P = 500$)? Half-full happens when the denominator is 2, i.e. $19e^{-0.7t} = 1$:

$$t = \frac{\ln 19}{0.7} \approx \frac{2.944}{0.7} \approx 4.2 \text{ hours}.$$

That is also the moment of fastest growth. Verhulst's 1845 fit of this curve to Belgian census data was one of the first genuinely predictive uses of a differential-equation model in the social sciences.

History

Verhulst introduced the logistic curve in 1838 and named it *logistique* in 1845, but his work was largely forgotten for eighty years. It was rediscovered independently in the 1920s by Raymond Pearl and Lowell Reed, who fit it to the United States population and (over-)confidently predicted an American ceiling of about 197 million. Reality blew past that, which carries its own lesson: the logistic model is a beautiful *idealization*, exact for nothing, illuminating for almost everything. A model's job is insight, not prophecy.

37.7 Existence, Uniqueness, and the Limits of Formulas

We have leaned on the comforting belief that an initial condition picks out exactly one solution. When is that guaranteed?

Formal

Existence and Uniqueness (Picard–Lindelöf, stated plainly). Consider the initial value problem $\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$. If f and its partial derivative $\frac{\partial f}{\partial y}$ are continuous on some rectangle containing (x_0, y_0), then on some interval around x_0 there exists *one and only one* solution passing through (x_0, y_0).

This is the theorem that makes the slope-field picture trustworthy: continuous, non-crossing solution curves filling the plane, exactly one through each point. When the hypotheses fail, weird things can happen — solutions can branch or blow up in finite time.

Common trap

The equation $\frac{dy}{dx} = \sqrt{|y|}$ with $y(0) = 0$ has *two* solutions through the origin: the constant $y \equiv 0$, and the curve $y = \frac{1}{4}x^2$ (for $x \geq 0$). Uniqueness fails because $\frac{\partial f}{\partial y} = \frac{1}{2\sqrt{y}}$ blows up at $y = 0$ — the smoothness hypothesis is violated exactly there. The river forks. This is not a pathology to fear but a reminder: uniqueness is a *theorem with hypotheses*, not a law of nature.

Why this matters

Most first-order ODEs that arise in practice cannot be solved with a clean formula at all — $\frac{dy}{dx} = x^2 + y^2$ has no elementary solution. Yet the slope field still exists, the phase line still tells us about equilibria, and numerical methods (Euler and its descendants) can trace any particular solution to whatever accuracy we like. This is the quiet liberation of the subject: *understanding* a differential equation — its long-term behavior, its equilibria, its qualitative shape — does not require a formula. The geometry comes first and goes furthest. The formula, when it exists, is a bonus.

37.8 Exercises

Exercise 37.1. Solve the separable equation $\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 y$ with $y(0) = 5$. Identify the lost constant solution and check whether your general formula recovers it.

Exercise 37.2. Solve $\frac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{1+y^2}{1+x^2}$. (Hint: both sides integrate to arctangents.) Express the answer as an implicit relation, then solve for y when $y(0) = 1$.

Exercise 37.3. A sample of a radioactive isotope decays to 80% of its initial mass in 3 days. Find its half-life, and determine how long until only 10% remains.

Exercise 37.4. A turkey at 5°C is placed in a 175°C oven. After 30 minutes its internal temperature is 40°C. Using Newton's law of heating, find the constant k and predict when the turkey reaches 75°C.

Exercise 37.5. Solve the linear equation $\frac{\mathrm{d}y}{\mathrm{d}x} + \dfrac{1}{x}\,y = \cos x$ for $x > 0$ by the integrating-factor method. (Hint: $\mu = x$, and $\int x\cos x\,\mathrm{d}x$ needs parts.)

Exercise 37.6. Solve $x\,\frac{\mathrm{d}y}{\mathrm{d}x} - 2y = x^3 e^x$. First put it in standard form, then find the integrating factor. Show that $\mu = x^{-2}$ and finish the problem.

Exercise 37.7. For the autonomous equation $\frac{\mathrm{d}y}{\mathrm{d}t} = y^3 - y$, find all equilibria, draw the phase line, and classify each equilibrium as stable or unstable using the sign of f' at each.

Exercise 37.8. For the autonomous equation $\frac{\mathrm{d}y}{\mathrm{d}t} = (y-1)(y-3)^2$, find the equilibria and classify them. One of them is *semi-stable* (attracting on one side, repelling on the other) because f touches zero without changing sign there. Identify which, and explain using the phase-line arrows.

Exercise 37.9. A lake with carrying capacity $M = 20{,}000$ fish currently holds 4,000 fish and grows logistically with $r = 0.4\ \mathrm{yr}^{-1}$. Write $P(t)$ explicitly, find when the population reaches 10,000, and find when it is growing fastest.

Exercise 37.10. Mixing problem. A tank holds 100 liters of pure water. Brine containing 0.5 kg of salt per liter flows in at 4 L/min, and the well-stirred mixture flows out at the same rate. Let $S(t)$ be the kilograms of salt in the tank. Argue that $\frac{\mathrm{d}S}{\mathrm{d}t} = 2 - \dfrac{4}{100}S$, solve this first-order linear equation with $S(0) = 0$, and find the long-run amount of salt. (Hint: the equilibrium is where $S' = 0$.)

Exercise 37.11. Show by direct substitution that the logistic function $P(t) = \dfrac{M}{1 + Ae^{-rt}}$ satisfies $\frac{\mathrm{d}P}{\mathrm{d}t} = rP(1 - P/M)$. (This verifies the solution without re-deriving it, a good habit: solving is hard, checking is easy.)

Exercise 37.12. The equation $\frac{\mathrm{d}y}{\mathrm{d}x} = x^2 + y^2$ has no elementary closed-form solution. Without solving it, sketch its slope field for $-2 \le x, y \le 2$ by computing the slope at a handful of points, and describe qualitatively how a solution starting at $(0, 0)$ behaves as x increases. (This is the spirit of the final box: geometry without formulas.)

Chapter 38

Second-Order Linear Differential Equations

A first-order differential equation knows about position and velocity at the same instant: it ties the rate of change of a quantity to the quantity itself. But the physical world is full of a deeper kind of law. Newton did not write "velocity is proportional to force." He wrote "*acceleration* is proportional to force." Acceleration is the second derivative. The moment you write down $F = ma$ for almost any real system – a mass on a spring, a pendulum, a swinging bridge, a tuned radio circuit, a vibrating molecule – you get a *second-order* differential equation.

This chapter is about the most important and most tractable family of them: the *linear* ones with constant coefficients,

$$a\,y'' + b\,y' + c\,y \;=\; f(t).$$

The entire theory hangs on one lucky guess – that $y = e^{rt}$ is the natural shape of a solution – and this single guess, pushed all the way through, unfolds into oscillation, damping, resonance, and the whole vocabulary of vibration. Throughout, our running example is the *harmonic oscillator*, a mass on a spring, because nearly every second-order linear equation is secretly a harmonic oscillator wearing a disguise.

38.1 What a Second-Order Linear Equation Is

A second-order ordinary differential equation (ODE) is an equation involving an unknown function $y(t)$, its first derivative y', and its second derivative y''. It is *linear* if y, y', and y'' each appear to the first power only, never multiplied together, never inside a sine or a square root. The general second-order linear ODE with constant coefficients is

$$a\,y'' + b\,y' + c\,y = f(t), \qquad a \neq 0, \tag{38.1}$$

where a, b, c are fixed numbers and $f(t)$ is a given function, the *forcing term* or *input*. When $f(t) = 0$ the equation is called *homogeneous*; otherwise it is *inhomogeneous* (or *forced*).

Focus

The plan of the whole chapter in one breath: first solve the homogeneous equation $ay'' + by' + cy = 0$ completely (that is the "free" motion of the system, what it does when nobody pushes it), then handle the forcing $f(t)$ by adding one particular solution on top. The structure

$$y_{\text{general}} = \underbrace{y_h}_{\text{free motion}} + \underbrace{y_p}_{\text{response to forcing}}$$

is the spine of everything that follows.

Why second order means two constants

A first-order equation comes with one arbitrary constant in its general solution; you pin it down with one initial condition, usually $y(t_0)$. A second-order equation comes with *two* arbitrary constants, and you need *two* initial conditions to determine a specific solution. The natural pair is

$$y(t_0) = y_0 \quad \text{(initial position)}, \qquad y'(t_0) = v_0 \quad \text{(initial velocity)}.$$

Intuition

This matches physical reality exactly. To predict where a thrown ball will be, knowing the law of gravity is not enough – you must know where you released it *and* how fast and in what direction. Position and velocity together form the complete "state" of a Newtonian system. Two numbers in, a unique future out. The two arbitrary constants of the general solution are precisely the slots where that initial state gets plugged in.

History

The leap from first to second order is the leap from Aristotle to Newton. Aristotle's physics was, in modern language, *first order*: he believed a force produces a velocity, so remove the force and motion stops. That is the law $b\,y' = f$, with no y''. Newton's revolution in the *Principia* (1687) was to insist that force produces *acceleration*, $m\,y'' = f$. The extra derivative is the mathematical fingerprint of inertia – of the fact that things keep moving on their own. Every second-order term in this chapter is an echo of that insight.

38.2 The Harmonic Oscillator: Our Running Example

Picture a block of mass m on a frictionless table, attached to a wall by a spring (Figure 38.1). Let $y(t)$ be the displacement of the block from its rest position, positive to the right. Hooke's law says the spring pulls back with a force proportional to how far you have stretched it, and always toward rest: $F_{\text{spring}} = -k\,y$, where $k > 0$ is the spring constant. Newton's second law $F = ma$ then reads

$$m\,y'' = -k\,y, \qquad \text{i.e.} \qquad m\,y'' + k\,y = 0.$$

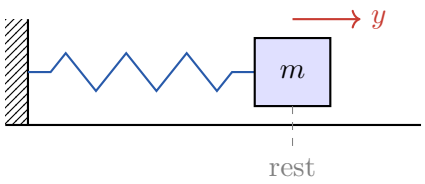


Figure 38.1: A mass on a spring. The restoring force $-ky$ always points back toward the rest position, which is what makes the motion oscillate.

Divide by m and write $\omega_0^2 = k/m$. The equation becomes the cleanest second-order ODE in all of science:

$$y'' + \omega_0^2\, y = 0. \tag{38.2}$$

The constant $\omega_0 = \sqrt{k/m}$ is the *natural frequency*. Before we solve it formally, read the equation as a sentence: *the second derivative of y is $-\omega_0^2$ times y itself.* What function is proportional to its own second derivative, with a *minus* sign? Sine and cosine. Differentiate $\cos(\omega_0 t)$ twice and you get $-\omega_0^2 \cos(\omega_0 t)$. The equation is practically telling us its own answer.

Intuition

Why *must* a spring oscillate, intuitively? When the block is displaced right, the spring pulls it left. It accelerates left, races back through the rest point with maximum speed, overshoots because of inertia, stretches the spring on the left, gets pulled back right, and so on forever. The restoring force and the inertia are locked in an eternal duel, and the trade-off between them is exactly a sine wave. The whole machinery of this chapter is just bookkeeping for that duel.

If we add friction proportional to velocity (a dashpot, air resistance, the block dragging on the table), the force law gains a term $-b\,y'$, and we get the *damped* harmonic oscillator

$$m\,y'' + b\,y' + k\,y = 0. \tag{38.3}$$

If we also push the block with an external force $F(t)$ – shaking the wall, say – we get the *forced* oscillator $my'' + by' + ky = F(t)$. These three equations, free, damped, and forced, will carry us through the entire chapter. They are special cases of the general form (38.1), and everything we prove about that form is best pictured as something happening to this block.

38.3 Superposition: Why Linearity Is Everything

The word *linear* is not decoration. It is the single property that makes these equations solvable, and it deserves to be stated as a theorem.

Theorem 38.1 (Superposition for the homogeneous equation)**.** *Suppose $y_1(t)$ and $y_2(t)$ are both solutions of the homogeneous equation $ay'' + by' + cy = 0$. Then for any constants C_1, C_2, the combination $y = C_1 y_1 + C_2 y_2$ is also a solution.*

Proof. Write $L[y] := ay'' + by' + cy$ for the operation the equation performs on a function. Differentiation is linear: the derivative of a sum is the sum of the derivatives, and constants pull out. So

$$L[C_1 y_1 + C_2 y_2] = a(C_1 y_1 + C_2 y_2)'' + b(C_1 y_1 + C_2 y_2)' + c(C_1 y_1 + C_2 y_2).$$

Distribute the derivatives and regroup by y_1 and y_2:

$$= C_1 \underbrace{(ay_1'' + by_1' + cy_1)}_{L[y_1]=0} + C_2 \underbrace{(ay_2'' + by_2' + cy_2)}_{L[y_2]=0} = C_1 \cdot 0 + C_2 \cdot 0 = 0. \qquad \square$$

What is really happening

The operator $L[y] = ay'' + by' + cy$ is a *linear machine*: feed it a scaled sum, and out comes the scaled sum of the outputs, $L[C_1y_1 + C_2y_2] = C_1L[y_1] + C_2L[y_2]$. This is the same abstract property that makes matrices, integrals, and the derivative itself behave so well. The set of all solutions of $L[y] = 0$ is therefore a *vector space* – you can add solutions and scale them and never leave the solution set. For a second-order equation this space turns out to be two-dimensional, which is the deep reason there are exactly two arbitrary constants. Finding the general solution means finding a *basis*: two independent solutions y_1, y_2 that between them span everything.

Why the technique works

Superposition is why our strategy works at all. If we can dig up just two genuinely different solutions of the homogeneous equation, linearity manufactures *all* of them for free, as combinations $C_1y_1 + C_2y_2$. We never have to search the infinite ocean of functions twice. Find two, and you are done. What "genuinely different" means precisely – linear independence – is the subject we turn to after we have found some solutions to be independent *about*.

38.4 Solving the Homogeneous Equation: Why e^{rt}?

We now want two solutions of

$$ay'' + by' + cy = 0. \tag{38.4}$$

Where do we even start? The central idea of the chapter is worth a slow read.

What is really happening

Look at what equation (38.4) asks for: a function whose second derivative, first derivative, and self can be *added up with constant weights to cancel to zero*. For that cancellation to happen at every instant t, the three functions y'', y', y had better all be the *same shape* as each other – otherwise their sum could not vanish identically. We need a function that reproduces itself under differentiation, up to a constant multiple. There is essentially only one such function in all of mathematics:

$$\frac{\mathrm{d}}{\mathrm{d}t}\, e^{rt} = r\, e^{rt}.$$

The exponential is the unique shape that differentiation cannot change, only rescale. *That* is why e^{rt} is not a trick or a lucky guess pulled from a hat – it is forced on us by the very structure of the equation. Every other method in this chapter is downstream of this one observation.

So we *try* $y = e^{rt}$, with r an unknown constant, and let the equation tell us which r works. Differentiating, $y' = re^{rt}$ and $y'' = r^2e^{rt}$. Substitute into (38.4):

$$ar^2e^{rt} + bre^{rt} + ce^{rt} = 0 \quad \Longrightarrow \quad e^{rt}\,(ar^2 + br + c) = 0.$$

Now e^{rt} is never zero, so we may divide it away. What survives is purely algebraic – the differential equation has collapsed into a quadratic:

$$\boxed{ar^2 + br + c = 0.} \tag{38.5}$$

This is the *characteristic equation* (or *auxiliary equation*) of the ODE. We have traded a calculus problem for a high-school algebra problem. Its roots r are exactly the exponents that make e^{rt} solve the differential equation.

Why the technique works

Why is collapsing a differential equation into a polynomial legitimate – and so powerful? Because differentiating e^{rt} multiplies by r, the operator $a\frac{\mathrm{d}^2}{\mathrm{d}t^2} + b\frac{\mathrm{d}}{\mathrm{d}t} + c$ acts on e^{rt} exactly the way the polynomial ar^2+br+c acts on the number r. Differentiation "becomes" multiplication by r. The hard analytic object (an operator on functions) and the easy algebraic object (a quadratic in r) are two faces of the same thing. This correspondence – calculus turning into algebra on exponentials – is the seed of the Laplace transform and of huge parts of engineering mathematics.

The quadratic (38.5) has roots given by the familiar formula

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Everything now depends on the sign of the discriminant $\Delta = b^2 - 4ac$, which splits into three cases. Each case has a completely different physical personality, and we take them one at a time.

38.5 Case 1: Two Distinct Real Roots ($\Delta > 0$)

When $b^2 - 4ac > 0$, the characteristic equation has two different real roots $r_1 \neq r_2$. Then $e^{r_1 t}$ and $e^{r_2 t}$ are both solutions, and – as we will confirm in §38.8 – they are independent. Superposition hands us the general solution

$$y(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}. \tag{38.6}$$

Physically, this case is *overdamping*: so much friction that the system oozes back to rest without ever oscillating. Both roots are typically negative (for a stable system), so both exponentials decay, just at different rates.

Example 38.1 (Overdamped motion)**.** Solve $y'' + 5y' + 6y = 0$ with $y(0) = 1$, $y'(0) = 0$.

Step 1. Characteristic equation. Replace $y'' \to r^2$, $y' \to r$, $y \to 1$:

$$r^2 + 5r + 6 = 0 \implies (r+2)(r+3) = 0 \implies r = -2,\ -3.$$

Step 2. General solution. Two distinct real roots, so

$$y(t) = C_1 e^{-2t} + C_2 e^{-3t}.$$

Step 3. Apply initial conditions. First, $y(0) = C_1 + C_2 = 1$. Differentiate: $y'(t) = -2C_1 e^{-2t} - 3C_2 e^{-3t}$, so $y'(0) = -2C_1 - 3C_2 = 0$. Solve the pair:

$$C_1 + C_2 = 1, \qquad -2C_1 - 3C_2 = 0.$$

From the first, $C_1 = 1 - C_2$; substitute: $-2(1 - C_2) - 3C_2 = 0 \Rightarrow -2 - C_2 = 0 \Rightarrow C_2 = -2$, hence $C_1 = 3$.

Step 4. Answer.

$$y(t) = 3e^{-2t} - 2e^{-3t}.$$

At $t = 0$ this is $3 - 2 = 1$ with zero slope, as required. As $t \to \infty$ both terms vanish: the block creeps back to rest and never crosses zero. That is the signature of overdamping.

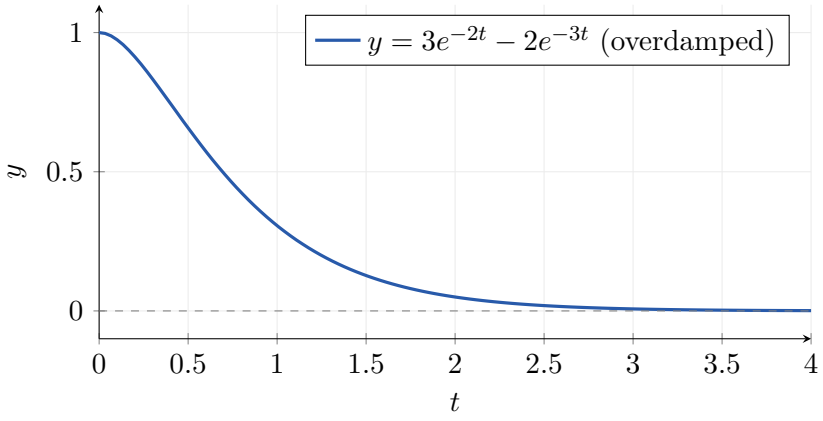


Figure 38.2: The overdamped solution of Example 38.1. It returns monotonically to rest – too much friction to overshoot, so no oscillation.

38.6 Case 3: Complex Roots ($\Delta < 0$) – Oscillation

We jump to the complex case next, out of numerical order, because it is the heart of the chapter: it is where oscillation lives. When $b^2 - 4ac < 0$, the square root of a negative number forces the roots to be a *complex-conjugate pair*

$$r = \alpha \pm i\beta, \qquad \alpha = \frac{-b}{2a}, \qquad \beta = \frac{\sqrt{4ac - b^2}}{2a} > 0.$$

Formally, superposition still applies and the general solution is $y = C_1 e^{(\alpha+i\beta)t} + C_2 e^{(\alpha-i\beta)t}$. But $y(t)$ describes the position of a real block; we want a *real* answer, with no i in sight. Euler's formula is what gets us there.

What is really happening

Recall Euler's formula, $e^{i\theta} = \cos\theta + i\sin\theta$. A complex exponential is secretly a rotation – a sine and a cosine bundled together. So when the characteristic roots go complex, the solution e^{rt} does not blow up or decay plainly; it *spins*. The real part α of the root controls a growing or shrinking envelope $e^{\alpha t}$, while the imaginary part β sets how fast the solution rotates – that is, how fast it oscillates. Complex roots are nature's way of writing "this thing vibrates." The algebra knew about oscillation before we did; Euler's formula just translates it back into the language of springs.

To extract the real solutions, factor out the common envelope:

$$e^{(\alpha\pm i\beta)t} = e^{\alpha t}\, e^{\pm i\beta t} = e^{\alpha t}\big(\cos\beta t \pm i\sin\beta t\big).$$

Two particular real combinations come straight out by adding and subtracting. Their average is the real part and (dividing by $2i$) their difference is the imaginary part:

$$\tfrac{1}{2}\big(e^{(\alpha+i\beta)t} + e^{(\alpha-i\beta)t}\big) = e^{\alpha t}\cos\beta t, \qquad \tfrac{1}{2i}\big(e^{(\alpha+i\beta)t} - e^{(\alpha-i\beta)t}\big) = e^{\alpha t}\sin\beta t.$$

By superposition (Theorem 38.1), each of these is itself a real solution. They are independent, so the real general solution is

$$\boxed{y(t) = e^{\alpha t}\big(C_1\cos\beta t + C_2\sin\beta t\big).} \tag{38.7}$$

Why the technique works

Why are we allowed to take real and imaginary parts and call them solutions? Because the coefficients a, b, c are real, the operator L maps real functions to real functions and complex to complex *without mixing*: if $L[u+iv] = 0$ with u, v real, then $L[u] + i\,L[v] = 0$, and a complex number is zero only when both its real and imaginary parts vanish. So $L[u] = 0$ and $L[v] = 0$ separately. The real and imaginary parts of a complex solution are each, on their own, real solutions. This is what lets us throw away the i at the end and keep a real-valued answer.

The undamped oscillator, finally solved

Return to the pure spring (38.2), $y'' + \omega_0^2 y = 0$. The characteristic equation is $r^2 + \omega_0^2 = 0$, giving $r = \pm i\omega_0$: pure imaginary, $\alpha = 0$ and $\beta = \omega_0$. By (38.7),

$$y(t) = C_1\cos\omega_0 t + C_2\sin\omega_0 t.$$

There is no envelope ($e^0 = 1$), so the block oscillates forever with constant amplitude – exactly the perpetual duel between spring and inertia we pictured earlier. The motion is often rewritten in *amplitude-phase form*

$$y(t) = A\cos(\omega_0 t - \varphi), \qquad A = \sqrt{C_1^2 + C_2^2}, \quad \tan\varphi = \frac{C_2}{C_1},$$

which makes the amplitude A and the natural frequency ω_0 stand out at a glance.

Example 38.2 (Undamped oscillation)**.** Solve $y'' + 9y = 0$ with $y(0) = 2$, $y'(0) = -3$.

Characteristic equation $r^2 + 9 = 0 \Rightarrow r = \pm 3i$, so $\omega_0 = 3$ and

$$y(t) = C_1\cos 3t + C_2\sin 3t.$$

Now $y(0) = C_1 = 2$. Differentiate: $y'(t) = -3C_1\sin 3t + 3C_2\cos 3t$, so $y'(0) = 3C_2 = -3 \Rightarrow C_2 = -1$. Hence

$$y(t) = 2\cos 3t - \sin 3t.$$

The amplitude is $A = \sqrt{2^2 + (-1)^2} = \sqrt{5} \approx 2.24$, and the motion repeats with period $2\pi/3$ (Figure 38.3).

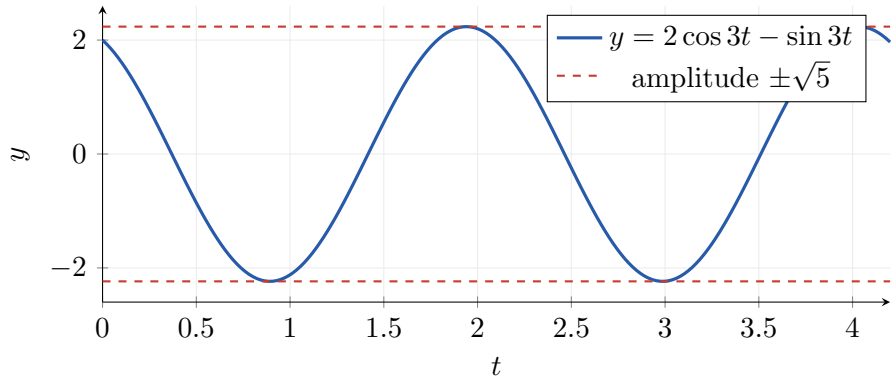


Figure 38.3: The undamped oscillation of Example 38.2: a pure sine wave of constant amplitude $\sqrt{5}$ and period $2\pi/3$, riding forever between the fixed lines $\pm\sqrt{5}$ because there is no friction to drain its energy.

Underdamping: oscillation inside a shrinking envelope

Now reinstate a *little* friction in the damped oscillator (38.3), $my'' + by' + ky = 0$, just enough that the discriminant stays negative. Then $\alpha = -b/(2m) < 0$ and the solution is

$$y(t) = e^{-(b/2m)\,t}\big(C_1 \cos \beta t + C_2 \sin \beta t\big).$$

The block still oscillates, but each swing is smaller than the last: the oscillation rides inside a decaying envelope $\pm Ae^{\alpha t}$. This is *underdamping* – a plucked guitar string, a struck tuning fork, a car on good shock absorbers nudged once. The frequency of the rings, β, is slightly lower than the undamped ω_0, because friction makes each cycle a hair slower.

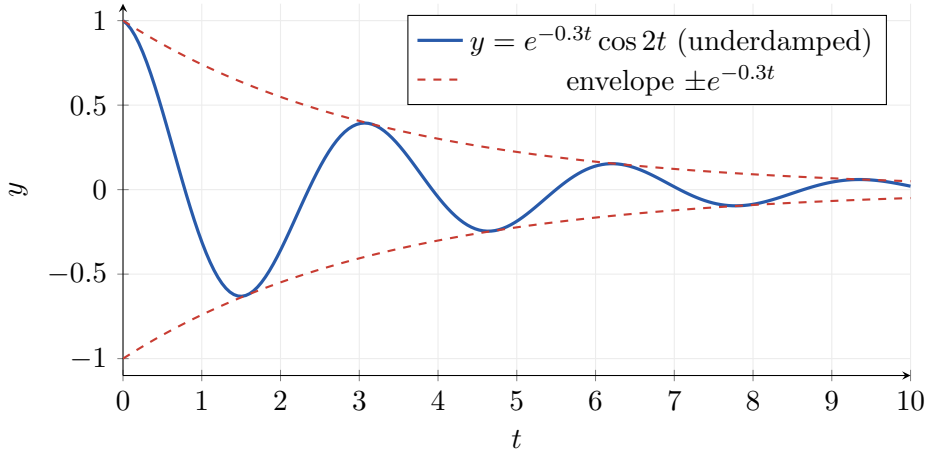


Figure 38.4: Underdamped motion: a sine wave squeezed inside the decaying envelope $\pm e^{-0.3t}$. The system oscillates but loses amplitude every cycle. This is the most common behavior of real vibrating systems.

Why this matters

Watch how three utterly different pictures – a slow monotone crawl back to rest (overdamping), an eternal pure sine wave (no damping), and a ringing wave fading inside an envelope (underdamping) – all fall out of *one* quadratic by merely changing the sign of $b^2 - 4ac$. The discriminant is a dial. Turn it through zero and the personality of the entire physical system flips from "creeps" to "rings." Few places in mathematics show so vividly how a single algebraic quantity governs qualitative behavior. The boundary itself – the dial set exactly to $\Delta = 0$ – is the repeated-root case we treat next, and it is the most delicately balanced of all.

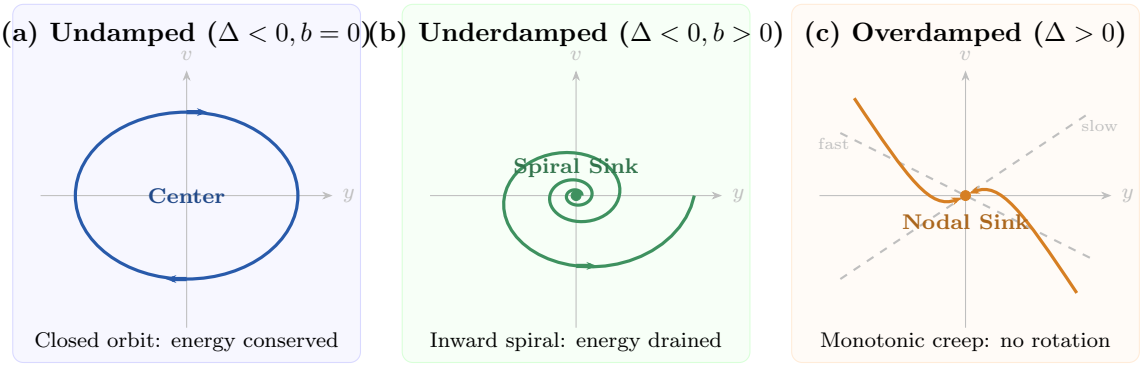


Figure 38.5: The three vibrational regimes viewed in displacement–velocity phase space (y, v) where $v = y'$. (a) Undamped motion is a closed periodic orbit (Center) cycling forever between potential and kinetic energy. (b) Underdamped motion is an inward spiral (Spiral Sink), losing amplitude on each cycle as friction extracts energy. (c) Overdamped motion is a direct approach (Nodal Sink), collapsing along the slow eigenspace into rest without a single oscillation.

38.7 Case 2: A Repeated Real Root ($\Delta = 0$)

The borderline case $b^2 - 4ac = 0$ is the strangest and most instructive. The quadratic has a single (double) root

$$r = -\frac{b}{2a},$$

and the recipe e^{rt} gives us only *one* solution, $y_1 = e^{rt}$. But a second-order equation needs *two* independent solutions to span its two-dimensional solution space. We are one solution short. Where does the second one come from?

Common trap

A tempting wrong move is to write $y = C_1 e^{rt} + C_2 e^{rt} = (C_1 + C_2)e^{rt}$. That is *not* a two-parameter family at all – it collapses to a single constant times e^{rt}. You have one solution masquerading as two. The two pieces are not independent, so the initial-value problem has no hope of matching arbitrary $y(0)$ *and* $y'(0)$. A genuinely new solution is required.

What is really happening

Here is the real reason the missing solution is $t\,e^{rt}$. Think of the repeated root as the limit of two distinct roots r and $r+\varepsilon$ that are sliding together as $\varepsilon \to 0$. For distinct roots we have the two solutions e^{rt} and $e^{(r+\varepsilon)t}$. Their difference, divided by ε, is also a solution by superposition:

$$\frac{e^{(r+\varepsilon)t} - e^{rt}}{\varepsilon}.$$

But this is exactly a difference quotient – in the limit $\varepsilon \to 0$ it becomes the *derivative of* e^{rt} *with respect to* r:

$$\frac{\partial}{\partial r}\, e^{rt} = t\, e^{rt}.$$

The second solution is the rate at which e^{rt} changes as you nudge the exponent. When two exponential solutions collide, the slot the second one vacated is filled by "t times the first." That factor of t is the ghost of the vanished second root. This is a genuine mechanism, not a memorized rule.

Let us verify directly that $y_2 = t\,e^{rt}$ works when $r = -b/(2a)$ is a double root, so that $ar^2+br+c = 0$ *and* (since r is the vertex of the parabola) $2ar + b = 0$. Compute the derivatives of $y_2 = te^{rt}$:

$$y_2' = e^{rt} + rt\,e^{rt} = (1+rt)e^{rt}, \qquad y_2'' = re^{rt} + r(1+rt)e^{rt} = (2r + r^2 t)e^{rt}.$$

Substitute into $ay_2'' + by_2' + cy_2$ and collect by powers of t:

$$e^{rt}\Big[\underbrace{(ar^2+br+c)}_{=0}\,t + \underbrace{(2ar+b)}_{=0}\Big] = 0.$$

Both bracketed quantities vanish precisely because r is a *repeated* root. So te^{rt} is a bona fide second solution. The general solution in the repeated-root case is therefore

$$\boxed{y(t) = (C_1 + C_2 t)\, e^{rt}.} \tag{38.8}$$

Physically this is *critical damping*: the exact knife's-edge amount of friction that brings the system back to rest as fast as possible *without* overshooting. A screen door closer, a sensitive galvanometer, and a well-tuned car suspension are all designed to sit at or near this critical value – any less and they would bounce, any more and they would respond sluggishly.

Example 38.3 (Critical damping)**.** Solve $y'' + 4y' + 4y = 0$ with $y(0) = 1$, $y'(0) = 1$.

Characteristic equation $r^2 + 4r + 4 = (r+2)^2 = 0$, a double root $r = -2$. So

$$y(t) = (C_1 + C_2 t)e^{-2t}.$$

Then $y(0) = C_1 = 1$. Differentiate with the product rule:

$$y'(t) = C_2 e^{-2t} - 2(C_1 + C_2 t)e^{-2t} = \big(C_2 - 2C_1 - 2C_2 t\big)e^{-2t}.$$

So $y'(0) = C_2 - 2C_1 = 1$, giving $C_2 = 1 + 2 = 3$. The solution is

$$y(t) = (1 + 3t)e^{-2t}.$$

Notice it rises briefly (the $3t$ wins early) before the exponential drags it down to rest – a single gentle bump, no oscillation. That is critical damping in one picture.

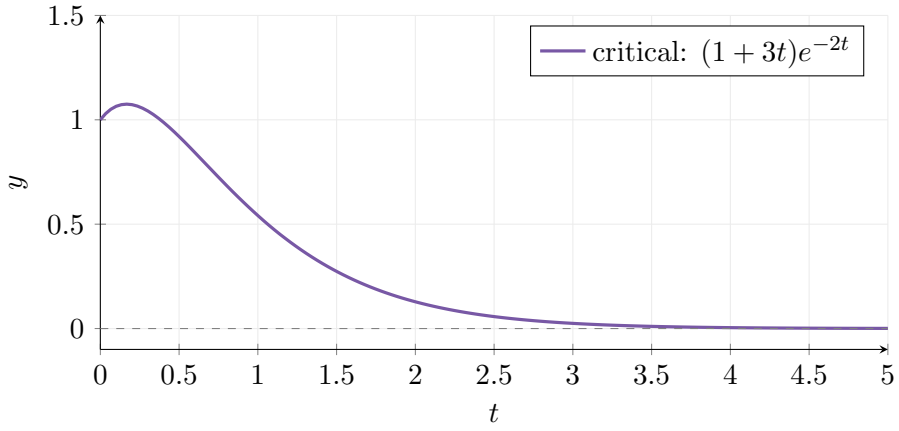


Figure 38.6: Critically damped motion from Example 38.3: one bump, then a clean return to rest with no overshoot. This is the fastest non-oscillating decay possible.

38.8 When Are Two Solutions Really Independent? The Wronskian

We have repeatedly claimed that pairs like $\{e^{r_1 t}, e^{r_2 t}\}$ or $\{e^{rt}, te^{rt}\}$ are "independent," so that $C_1 y_1 + C_2 y_2$ captures *all* solutions. Let us pin down what "independent" means, since that is exactly what guarantees we have not missed any solutions.

Two solutions y_1, y_2 are *linearly independent* if neither is a constant multiple of the other – equivalently, if $C_1 y_1 + C_2 y_2 = 0$ for all t forces $C_1 = C_2 = 0$. The clean test is the *Wronskian* determinant

$$W(y_1, y_2)(t) = \det \begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} = y_1 y_2' - y_2 y_1'.$$

Why the technique works

Why does this determinant detect independence? Suppose we want constants with $C_1 y_1 + C_2 y_2 = 0$ for all t. Differentiating gives a second equation $C_1 y_1' + C_2 y_2' = 0$. Together they are a 2×2 linear system for (C_1, C_2):

$$\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

If the determinant W is nonzero at even one point, the only solution is $C_1 = C_2 = 0$ – so y_1, y_2 are independent. The Wronskian is precisely the determinant that decides whether the "cancel to zero" system has a nontrivial solution. A nonzero Wronskian is the certificate that our two solutions genuinely span a two-dimensional space, with no overlap.

Example 38.4 (The exponential pair is independent)**.** For $y_1 = e^{r_1 t}$, $y_2 = e^{r_2 t}$ with $r_1 \neq r_2$,

$$W = e^{r_1 t} \cdot r_2 e^{r_2 t} - e^{r_2 t} \cdot r_1 e^{r_1 t} = (r_2 - r_1)\, e^{(r_1 + r_2)t}.$$

Since $r_1 \neq r_2$ and the exponential never vanishes, $W \neq 0$ everywhere. The two solutions are independent, confirming (38.6). For the repeated-root pair $\{e^{rt}, te^{rt}\}$ a similar computation gives $W = e^{2rt} \neq 0$, so that pair is independent too.

Formal

The payoff is the *Existence and Uniqueness Theorem* for constant-coefficient linear ODEs: given any initial data $y(t_0) = y_0$, $y'(t_0) = v_0$, there is one and only one solution. Because the Wronskian of our basis $\{y_1, y_2\}$ is nonzero, the system

$$C_1 y_1(t_0) + C_2 y_2(t_0) = y_0, \qquad C_1 y_1'(t_0) + C_2 y_2'(t_0) = v_0$$

always has a unique solution (C_1, C_2). So every initial-value problem is solved exactly once by our two-parameter family – nothing is missed, nothing is double-counted. This is the rigorous backbone under all the case work above.

38.9 Forcing the System: Inhomogeneous Equations

Now switch the forcing back on:

$$ay'' + by' + cy = f(t). \tag{38.9}$$

Physically, $f(t)$ is an external push: a periodic shove on the spring, a voltage applied to a circuit, gravity loading a beam. The structure theorem that organizes everything is this.

Theorem 38.2 (Structure of the general solution)**.** *Let y_p be* any one *particular solution of the inhomogeneous equation (38.9), and let $y_h = C_1 y_1 + C_2 y_2$ be the general solution of the associated homogeneous equation $ay'' + by' + cy = 0$. Then the general solution of (38.9) is*

$$y = y_h + y_p = C_1 y_1 + C_2 y_2 + y_p.$$

Proof. First, $y_h + y_p$ solves (38.9): applying the linear operator L, $L[y_h + y_p] = L[y_h] + L[y_p] = 0 + f = f$. Conversely, let Y be *any* solution. Then $L[Y - y_p] = L[Y] - L[y_p] = f - f = 0$, so $Y - y_p$ solves the homogeneous equation and must equal some $C_1 y_1 + C_2 y_2$. Hence $Y = y_h + y_p$. Every solution has this form, and every function of this form is a solution. □

Intuition

Read the structure physically. The homogeneous part y_h is the system's own free vibration – whatever it would do if you set it going and walked away. The particular part y_p is its forced response – the steady motion the external push drives it into. The true motion is the *sum*: the driven response, plus whatever leftover free vibration the initial conditions demand. In a damped system y_h decays away (it is called the *transient*), and after a while only the forced y_p survives (the *steady state*). This is why, long after you start shaking a damped spring, it settles into the rhythm of your hand and forgets how it started.

38.10 Undetermined Coefficients: Guessing y_p Intelligently

To finish the inhomogeneous problem we need just one particular solution y_p. For the common forcing functions – polynomials, exponentials, sines and cosines – there is a beautifully direct method: *guess a y_p of the same shape as f, with unknown coefficients, and let the equation solve for them.*

Why the technique works

Why should a same-shape guess work? Because differentiation keeps these special functions inside their own families. Differentiate a polynomial and you get a polynomial; differentiate e^{kt} and you get a multiple of e^{kt}; differentiate $\cos\omega t$ and you stay within $\{\cos\omega t, \sin\omega t\}$. Since $L[y]$ only differentiates and scales, if y_p is built from such a family, then $L[y_p]$ lands in the same family. To match a forcing f from that family, we only need to tune the coefficients. The method is not luck; it exploits the fact that these forcings live in finite-dimensional spaces that L maps into themselves.

The table of first guesses:

Forcing $f(t)$	Trial y_p
constant k	A
polynomial degree n	$A_n t^n + \cdots + A_1 t + A_0$
e^{kt}	Ae^{kt}
$\cos\omega t$ or $\sin\omega t$	$A\cos\omega t + B\sin\omega t$
$e^{kt}\cos\omega t$	$e^{kt}(A\cos\omega t + B\sin\omega t)$

Always include *both* a sine and a cosine when the forcing has either, because differentiation turns one into the other.

Example 38.5 (Polynomial forcing)**.** Solve $y'' - y' - 2y = 4t^2$.

Homogeneous part. $r^2 - r - 2 = (r-2)(r+1) = 0$, so $r = 2, -1$ and $y_h = C_1 e^{2t} + C_2 e^{-t}$.

Guess. The forcing is a degree-2 polynomial, so try $y_p = At^2 + Bt + C$. Then $y_p' = 2At + B$ and $y_p'' = 2A$. Substitute:

$$\underbrace{2A}_{y_p''} - \underbrace{(2At + B)}_{y_p'} - 2\underbrace{(At^2 + Bt + C)}_{y_p} = 4t^2.$$

Collect by powers of t:

$$-2A\,t^2 + (-2A - 2B)\,t + (2A - B - 2C) = 4t^2 + 0t + 0.$$

Match coefficients.

$$t^2:\ -2A = 4 \Rightarrow A = -2; \qquad t^1:\ -2A - 2B = 0 \Rightarrow B = 2; \qquad t^0:\ 2A - B - 2C = 0 \Rightarrow C = -3.$$

So $y_p = -2t^2 + 2t - 3$, and the general solution is

$$y = C_1 e^{2t} + C_2 e^{-t} - 2t^2 + 2t - 3.$$

Example 38.6 (Sinusoidal forcing – a driven oscillator)**.** Solve $y'' + 4y = 5\cos t$ (an undamped spring with $\omega_0 = 2$, pushed at frequency 1).

Homogeneous part. $r^2 + 4 = 0 \Rightarrow r = \pm 2i$, so $y_h = C_1\cos 2t + C_2 \sin 2t$.

Guess. Forcing $5\cos t$; try $y_p = A\cos t + B\sin t$. Then $y_p'' = -A\cos t - B\sin t$, and

$$y_p'' + 4y_p = (-A + 4A)\cos t + (-B + 4B)\sin t = 3A\cos t + 3B\sin t.$$

Match to $5\cos t + 0\sin t$: $3A = 5 \Rightarrow A = 5/3$, and $3B = 0 \Rightarrow B = 0$. So $y_p = \frac{5}{3}\cos t$, and

$$y = C_1 \cos 2t + C_2 \sin 2t + \tfrac{5}{3}\cos t.$$

The driving frequency 1 differs from the natural frequency 2, so the response stays bounded – the system simply hums along at the driving frequency with a modest amplitude. Watch what happens in the next section when those two frequencies coincide.

The resonance snag, and how to fix the guess

There is one trap, and it is the most physically important phenomenon in the chapter. *What if the trial y_p is already a solution of the homogeneous equation?* Then substituting it gives $L[y_p] = 0 \neq f$, and the guess cannot possibly match a nonzero forcing.

Common trap

If your trial solution duplicates a homogeneous solution, do *not* just give up – and do not try strange new shapes. The fix mirrors the repeated-root cure: multiply the trial by t (or by t^2 if the homogeneous root is doubled). For forcing e^{kt} when k is already a characteristic root, use $y_p = Ate^{kt}$. For forcing $\cos\omega_0 t$ at the natural frequency, use $y_p = t(A\cos\omega_0 t + B\sin\omega_0 t)$. The extra t is the same ghost we met with repeated roots: it appears whenever a solution slot is already occupied.

38.11 Resonance: When Forcing Hits the Natural Frequency

Take the undamped oscillator and push it at *exactly* its own natural frequency:

$$y'' + \omega_0^2 y = F_0 \cos\omega_0 t. \tag{38.10}$$

The naive guess $A\cos\omega_0 t + B\sin\omega_0 t$ is a homogeneous solution, so by the rule above we multiply by t and try

$$y_p = t\,(A\cos\omega_0 t + B\sin\omega_0 t).$$

Carrying out the differentiation (product rule twice) and substituting, the algebra delivers

$$y_p = \frac{F_0}{2\omega_0}\, t\sin\omega_0 t.$$

The crucial feature is the factor of t *outside* the sine. The amplitude of the oscillation is $\dfrac{F_0}{2\omega_0}\,t$ – it grows without bound, linearly in time. This is *resonance*.

What is really happening

Why does pushing at the natural frequency make the amplitude explode? Timing. When you drive at ω_0, every push is perfectly synchronized with the oscillator's own motion – you always shove in the direction it is already moving, like pumping a swing exactly in rhythm. Each cycle you add a little energy and *none* is wasted fighting the motion. Energy accumulates cycle after cycle, so the amplitude climbs and climbs. The factor of t in $t\sin\omega_0 t$ is the mathematical

signature of energy piling up with no escape. Off resonance, your pushes sometimes help and sometimes oppose, the books roughly balance, and the amplitude stays bounded – as we saw in Example 38.6.

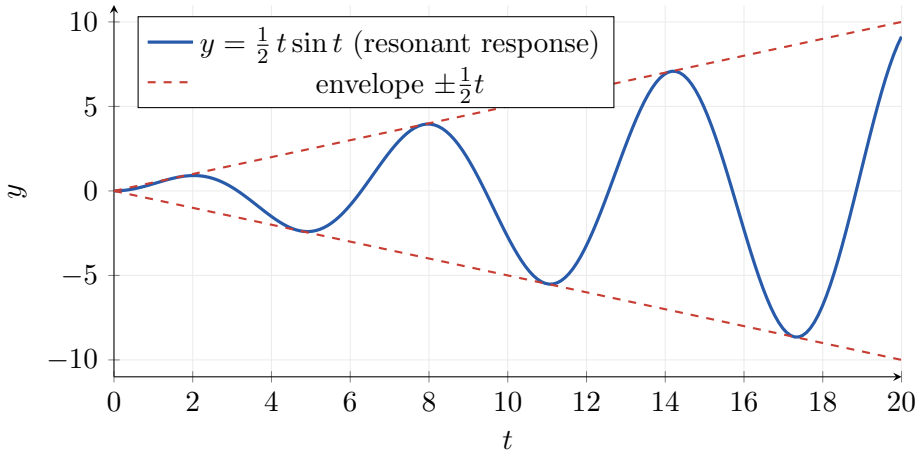


Figure 38.7: Resonance: driving at the natural frequency produces oscillations whose amplitude grows linearly in t, bounded only by the widening envelope $\pm\frac{1}{2}t$. No matter how small the push, given enough time the response becomes arbitrarily large.

History

Resonance is not a curiosity; it is a force of engineering life and death. A file of soldiers is told to break step when crossing a bridge, lest their cadence happen to match the bridge's natural frequency and pump it to destruction. The Tacoma Narrows Bridge tearing itself apart in 1940 is the textbook (if technically more subtle) cautionary image. On the constructive side, resonance is how a radio works: a tunable circuit is built to resonate at one broadcast frequency, amplifying that station hugely while ignoring all the others. A microwave oven drives water molecules near their resonance; an MRI machine resonates atomic nuclei. The growing-t solution is the equation's way of saying "this is the frequency that matters."

Application

Real systems always have a little damping, so the amplitude does not literally go to infinity – the decaying transient and the energy lost to friction eventually balance the energy you pump in, and the response levels off at a large but finite amplitude. The closer the driving frequency is to ω_0, and the smaller the damping b, the taller that resonant peak. Engineers plot the steady-state amplitude against driving frequency and get a sharp spike at ω_0 – the *resonance curve*. Designing a structure means either keeping its natural frequencies far from any expected driving frequency, or adding enough damping to tame the peak. The undamped, infinitely growing solution above is the idealized limit that warns you where the danger lies.

Example 38.7 (Pure resonance, worked)**.** Solve $y'' + y = 2\cos t$ with $y(0) = 0$, $y'(0) = 0$.

Here $\omega_0 = 1$ and the forcing frequency is also 1: resonance. The homogeneous solution is $y_h =$

$C_1 \cos t + C_2 \sin t$. Since $\cos t$ solves the homogeneous equation, multiply the trial by t:

$$y_p = t(A\cos t + B\sin t).$$

Differentiate twice:

$$y_p' = A\cos t + B\sin t + t(-A\sin t + B\cos t),$$
$$y_p'' = 2(-A\sin t + B\cos t) + t(-A\cos t - B\sin t).$$

Then $y_p'' + y_p = 2(-A\sin t + B\cos t)$ (the t-terms cancel because $\cos t, \sin t$ solve the homogeneous equation). Match to $2\cos t$: $2B = 2 \Rightarrow B = 1$ and $-2A = 0 \Rightarrow A = 0$. So $y_p = t\sin t$, and

$$y = C_1\cos t + C_2 \sin t + t\sin t.$$

Apply $y(0) = 0 \Rightarrow C_1 = 0$. Then $y' = C_2\cos t + \sin t + t\cos t$, so $y'(0) = C_2 = 0$. The clean answer is

$$y(t) = t\sin t,$$

an oscillation whose swings grow without bound – the picture of Figure 38.7 with envelope $\pm t$.

38.12 A Unified Worked Lab

Work it yourself

Take the damped, forced oscillator

$$y'' + 2y' + 5y = 10\cos t, \qquad y(0) = 0,\ y'(0) = 0,$$

and carry it through end to end. The skeleton:

1. **Homogeneous solution.** Characteristic equation $r^2+2r+5 = 0$. Discriminant $4-20 = -16 < 0$, so $r = -1 \pm 2i$. Underdamped:

$$y_h = e^{-t}(C_1\cos 2t + C_2 \sin 2t).$$

2. **Particular solution.** Forcing $10\cos t$; the frequency 1 is not a homogeneous frequency, so no resonance, no extra t. Try $y_p = A\cos t + B\sin t$ and show

$$y_p'' + 2y_p' + 5y_p = (4A + 2B)\cos t + (4B - 2A)\sin t.$$

   Match to $10\cos t + 0\sin t$: solve $4A + 2B = 10$, $-2A + 4B = 0$. Confirm $A = 2$, $B = 1$, so $y_p = 2\cos t + \sin t$.

3. **Combine.** $y = e^{-t}(C_1\cos 2t + C_2\sin 2t) + 2\cos t + \sin t$.

4. **Initial conditions.** Use $y(0) = 0$ to get $C_1 + 2 = 0$, so $C_1 = -2$. Now differentiate the full y and set $y'(0) = 0$. Since

$$y'(0) = -C_1 + 2C_2 + 1 = 0,$$

   substituting $C_1 = -2$ gives $2 + 2C_2 + 1 = 0$, so $C_2 = -\frac{3}{2}$. Final answer:

$$y(t) = -e^{-t}\left(2\cos 2t + \tfrac{3}{2}\sin 2t\right) + 2\cos t + \sin t.$$

5. **Interpret.** Identify the transient $-e^{-t}\left(2\cos 2t + \frac{3}{2}\sin 2t\right)$ (gone within a few seconds) and the steady state $2\cos t + \sin t$ (the lasting response, oscillating at the *driving* frequency 1, not the damped frequency 2). Sketch both and watch the transient fade.

Do every step with pencil before reading on; the differentiation in step 2 is the only fiddly part, and getting it by hand cements the method.

Connection

Everything here generalizes upward and sideways. *Upward in order:* an nth-order constant-coefficient linear ODE has a degree-n characteristic polynomial, and its n roots (real, repeated, or complex) build the solution by exactly the rules of this chapter – e^{rt} for each root, an extra t for each repetition, $e^{\alpha t}\cos\beta t$ pairs for complex roots. *Sideways into systems:* a single second-order equation is equivalent to a 2×2 *system* of first-order equations (set $v = y'$), and there the characteristic equation reappears as the eigenvalue equation of a matrix. The roots r become eigenvalues; oscillation becomes complex eigenvalues; damping becomes negative real parts. The next chapter follows that thread, and you will recognize every face in it.

38.13 Summary

Focus

The whole method, on one card.
To solve $ay'' + by' + cy = f(t)$:

1. Solve the homogeneous equation. Write the characteristic equation $ar^2 + br + c = 0$ and find its roots.
   - Distinct real r_1, r_2: $y_h = C_1 e^{r_1 t} + C_2 e^{r_2 t}$ (overdamped).
   - Repeated real r: $y_h = (C_1 + C_2 t)e^{rt}$ (critically damped).
   - Complex $\alpha \pm i\beta$: $y_h = e^{\alpha t}(C_1\cos\beta t + C_2\sin\beta t)$ (oscillation; underdamped if $\alpha < 0$).
2. If $f \neq 0$, find one particular y_p by undetermined coefficients – guess the shape of f, multiplying by t if the guess collides with a homogeneous solution (resonance).
3. General solution $y = y_h + y_p$. Apply the two initial conditions $y(t_0), y'(t_0)$ last, to fix C_1, C_2.

The one idea beneath it all: e^{rt} is the function differentiation cannot reshape, so it turns the differential equation into a quadratic, and the quadratic's discriminant dials the system between creeping, ringing, and – when forced at its own frequency – resonating.

38.14 Exercises

Exercise 38.1. Classify each by its discriminant as overdamped, critically damped, or underdamped, then write the general solution. (a) $y''+7y'+12y = 0$. (b) $y''+6y'+9y = 0$. (c) $y''+2y'+10y = 0$.

Exercise 38.2. Solve the initial-value problem $y'' + \omega_0^2 y = 0$, $y(0) = y_0$, $y'(0) = v_0$, for general ω_0, y_0, v_0. Then show the amplitude of the resulting oscillation is $A = \sqrt{y_0^2 + v_0^2/\omega_0^2}$. Interpret each term physically.

Exercise 38.3. For the equation $y'' + 4y' + 4y = 0$ (a double root at $r = -2$), verify by direct substitution that $y_2 = te^{-2t}$ is a solution, and compute the Wronskian $W(e^{-2t}, te^{-2t})$ to confirm the two solutions are independent.

Exercise 38.4. Solve $y'' - 3y' + 2y = 0$ with $y(0) = 0$, $y'(0) = 1$. Then describe, in one sentence each, how the solution behaves as $t \to \infty$ and as $t \to -\infty$.

Exercise 38.5. Use undetermined coefficients to find a particular solution of $y'' - y = 3e^{2t}$. Why does the plain guess Ae^{2t} work here without an extra factor of t?

Exercise 38.6. Find a particular solution of $y'' + y = 4\sin 2t$. The driving frequency is 2 and the natural frequency is 1 – explain why no resonance occurs and no extra t is needed, then solve.

Exercise 38.7. The forcing in $y'' - y = 4e^t$ matches a homogeneous solution ($r = 1$ is a root of $r^2 - 1 = 0$). Use the corrected trial $y_p = Ate^t$ to find a particular solution. This is the exponential analogue of resonance.

Exercise 38.8. Pure resonance. Solve $y'' + 9y = 6\cos 3t$ with $y(0) = 0$, $y'(0) = 0$. Identify the growing-amplitude term and state the envelope within which the solution oscillates.

Exercise 38.9. Damped driven oscillator. For $y'' + 2y' + 2y = 10\cos t$, find the steady-state (particular) solution and its amplitude. Which part of the full solution is the transient, and roughly how long until it is negligible?

Exercise 38.10. A spring-mass system has $m = 1$, spring constant $k = 4$, and damping b. (a) For which value of b is the system critically damped? (b) Sketch (qualitatively) the response to the initial data $y(0) = 1$, $y'(0) = 0$ for b slightly below, equal to, and slightly above that critical value, and explain the difference in words.

Exercise 38.11. Reduction to a system. Given $y'' + by' + cy = 0$, set $v = y'$ and write the equivalent pair of first-order equations for (y, v). Show that looking for solutions of the form $(y, v) = \mathbf{u}\, e^{rt}$ recovers exactly the characteristic equation $r^2 + br + c = 0$. (This is the bridge to the next chapter.)

Exercise 38.12. Conceptual. Explain, without solving any equation, why a second-order linear ODE needs exactly two initial conditions while a first-order one needs only one. Connect your answer to the dimension of the homogeneous solution space and to the physical idea of a system's "state."

Chapter 39

Systems of Differential Equations and Beyond

Up to now a differential equation has described a single quantity changing in time: a population, a temperature, the charge on a capacitor, the angle of a swinging pendulum. One unknown function, one equation, one story. But the world is rarely so lonely. Foxes and rabbits eat and are eaten; their fortunes are locked together, neither curve makes sense without the other. Two masses joined by springs trade energy back and forth. Three chemical species react, each feeding on and feeding the next. In every one of these settings we have not one unknown function but several, and the rate of change of each depends on *all* of them at once. We need a *system* of differential equations.

This is the chapter where two great rivers of this book finally meet. On one side flows everything we learned about derivatives, exponentials, and the single ODE $y' = ky$ whose solution is e^{kt}. On the other side flows linear algebra – vectors, matrices, and above all **eigenvalues and eigenvectors**. The astonishing payoff is this: a whole system of coupled linear equations, no matter how tangled it looks, can be *uncoupled* by choosing the right coordinates, and in those coordinates each piece is just $y' = \lambda y$ again. The eigenvectors are the right coordinates. The eigenvalues are the growth rates. Solving a linear system is, at bottom, the single equation $y' = ky$ wearing a matrix-shaped coat.

We will build the machinery, draw the pictures that make it intuitive – the *phase portraits* that turn a system into a flowing landscape – classify when equilibria attract and when they repel, and then push at the edge of the linear world to glimpse what happens when systems go nonlinear: limit cycles, sensitive dependence, chaos. We end by pointing down the roads that leave this book and head into the rest of mathematics: partial differential equations, numerical methods, dynamical systems.

39.1 What a System Is, and Why You Already Have One

A **first-order system** of differential equations is a collection of equations, one for the derivative of each unknown function, where each right-hand side may depend on all the unknowns (and possibly

on time):

$$\begin{aligned} x_1'(t) &= f_1(t, x_1, x_2, \dots, x_n), \\ x_2'(t) &= f_2(t, x_1, x_2, \dots, x_n), \\ &\vdots \\ x_n'(t) &= f_n(t, x_1, x_2, \dots, x_n). \end{aligned}$$

The word *first-order* means only first derivatives appear. The word *system* means the equations are bound together – you cannot solve one without the others, because each f_i peeks at every x_j.

It is enormously clarifying to bundle the unknowns into a single vector-valued function

$$\mathbf{x}(t) = \big(x_1(t), x_2(t), \dots, x_n(t)\big),$$

and the right-hand sides into a single vector field $\mathbf{F}(t, \mathbf{x}) = (f_1, \dots, f_n)$. Then the entire system collapses to one clean line that looks exactly like the scalar case:

$$\boxed{\mathbf{x}'(t) = \mathbf{F}\big(t, \mathbf{x}(t)\big).}$$

This is not a trick of notation; it is a genuine point of view. A single number moving on a line obeyed $x' = f(x)$. A single *point* moving through n-dimensional space obeys $\mathbf{x}' = \mathbf{F}(\mathbf{x})$. The state of the whole world is one point; the differential equation tells that point, at every moment, which direction and how fast to move.

Intuition

Picture the unknowns as the coordinates of a single dot living in a space called **phase space**. For the fox–rabbit problem that space is a plane: the horizontal axis is the rabbit population, the vertical axis the fox population. The current state of the ecosystem is *one dot* in that plane. The system of ODEs is a rule that, at every dot, plants a little arrow telling the dot where to go next. The whole job of solving the system is to follow the arrows – to let the dot drift – and watch the curve it traces. That curve is the history of the ecosystem.

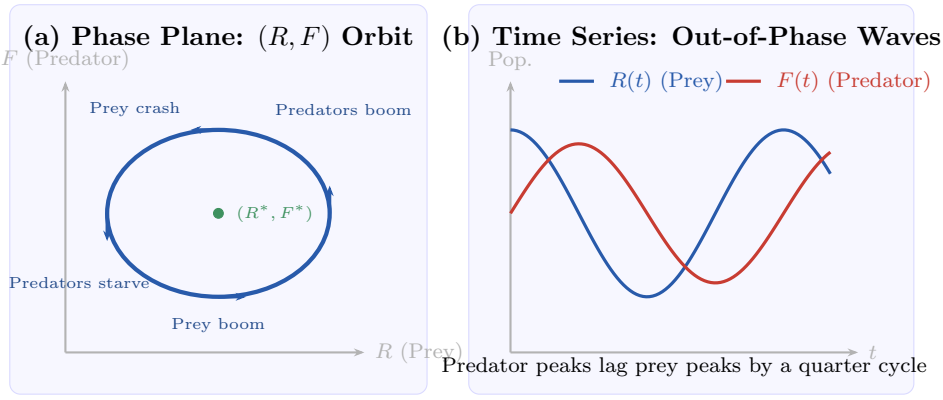


Figure 39.1: The predator–prey system in phase space and in time. (a) In phase space, the ecosystem state (R, F) traverses a closed orbit counterclockwise: abundant prey feed predators, predator overpopulation crashes the prey, food scarcity starves predators, allowing prey to recover. (b) Plotted against time, both populations oscillate periodically with identical period, with predator peaks chronically lagging behind prey peaks.

39.1.1 A first concrete example

Here is a small but real system. Let $x(t)$ and $y(t)$ be two quantities obeying

$$x' = \quad x + 2y, \qquad y' = 3x + 2y.$$

Notice the coupling: x' depends on y, and y' depends on x. You cannot integrate the first equation by itself because the $2y$ on its right is an unknown function you have not found yet. The two equations are a knot. Much of this chapter is the art of untying it.

What is really happening

What is really happening when we "solve" a system is that we replace a tangle of mutually-referencing rules with an explicit description of motion. Before solving, the system is implicit: it tells you the *velocity* of the state dot in terms of its *position*, but velocity-in-terms-of-position is a chicken-and-egg statement – where the dot goes depends on where it is, which depends on where it went. Solving turns that self-reference into a formula $\mathbf{x}(t)$ you can evaluate at any time without knowing anything else. We are trading a local rule (arrows) for a global trajectory (the curve).

39.2 Every Higher-Order Equation Is Secretly a System

The previous chapter studied second-order equations like $y'' + py' + qy = 0$, and *every* higher-order ODE turns out to be a first-order system in disguise. This is one of the most useful structural facts in all of differential equations, and it is almost trivial once you see it.

Take the second-order equation

$$y'' = -y$$

– the equation of simple harmonic motion, whose solutions you know are sines and cosines. Introduce a *second unknown* equal to the velocity:

$$x_1 = y, \qquad x_2 = y'.$$

Now ask for the derivatives of these two new unknowns. The first is immediate: $x_1' = y' = x_2$. The second uses the original equation: $x_2' = y'' = -y = -x_1$. So the single second-order equation becomes the first-order system

$$x_1' = x_2, \qquad x_2' = -x_1.$$

We have not changed the physics one bit; we have only renamed "position" and "velocity" as two coordinates and written down how each one changes. The trick generalizes word for word.

Why the technique works

Why does this always work? An nth-order equation $y^{(n)} = G(t, y, y', \ldots, y^{(n-1)})$ determines the highest derivative from all the lower ones. So if we name the lower derivatives as new variables

$$x_1 = y, \quad x_2 = y', \quad x_3 = y'', \quad \ldots, \quad x_n = y^{(n-1)},$$

then each variable's derivative is simply the *next* variable in the list ($x_1' = x_2$, $x_2' = x_3$, and

so on) until the very last, whose derivative is the only place the original equation does any work: $x_n' = G(t, x_1, \dots, x_n)$. The cascade $x_i' = x_{i+1}$ is pure bookkeeping; the genuine content of the equation lives entirely in the final slot. That is why second-order, third-order, and millionth-order equations all reduce to first-order systems: differentiation just shifts you one notch up a ladder, and the equation tells you what happens when you fall off the top.

Why this matters

This reduction is why we can afford to spend the whole chapter on first-order systems and lose nothing. A numerical solver written for $\mathbf{x}' = \mathbf{F}(\mathbf{x})$ can integrate the orbit of a planet (second order), the vibration of a bridge (fourth order), or the firing of a neuron (a tangle of several equations) without modification. First-order systems are the universal currency of dynamics. Learn them once; spend them everywhere.

Example 39.1 (Damped oscillator as a system)**.** The equation $y'' + 2y' + 5y = 0$ models a mass on a spring with friction. Set $x_1 = y$, $x_2 = y'$. Then $x_1' = x_2$, and from the equation $y'' = -2y' - 5y$, so $x_2' = -5x_1 - 2x_2$. In matrix form,

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}' = \begin{pmatrix} 0 & 1 \\ -5 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

The matrix that appears is called the **companion matrix** of the equation. We will solve systems of exactly this shape in a moment; the bottom row is literally the coefficients of the original ODE, sign-flipped.

39.3 The Matrix Form $\mathbf{u}' = A\mathbf{u}$

From here on we focus on **linear** systems with constant coefficients, because they are the ones we can solve completely and because everything we learn about them governs the local behavior of nonlinear systems too (Section 39.7). A linear constant-coefficient system in two unknowns looks like

$$\begin{aligned} u_1' &= a u_1 + b u_2, \\ u_2' &= c u_1 + d u_2, \end{aligned}$$

and packs neatly into the single matrix equation

$$\boxed{\mathbf{u}' = A\,\mathbf{u}, \qquad \mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, \quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.}$$

Here $\mathbf{u}'$ means "differentiate each component": $\mathbf{u}' = (u_1', u_2')$. The whole system is now a single object: a matrix A acting on a moving vector.

Look hard at the scalar special case. If $\mathbf{u}$ has just one component, A is just a number λ, and the equation is $u' = \lambda u$, whose solution we have known since the chapter on exponentials:

$$u(t) = u(0)\, e^{\lambda t}.$$

The entire strategy for systems is built on a single audacious hope: that the matrix equation might have solutions of the very same shape, $\mathbf{u}(t) = e^{\lambda t}\mathbf{v}$ for some fixed direction $\mathbf{v}$ and some growth rate λ. Let us test that hope.

Walking question

Suppose, hopefully, that $\mathbf{u}(t) = e^{\lambda t}\mathbf{v}$ where $\mathbf{v}$ is a constant nonzero vector. Differentiate the left side: since $\mathbf{v}$ is constant, $\mathbf{u}'(t) = \lambda e^{\lambda t}\mathbf{v}$. Now compute the right side: $A\mathbf{u}(t) = A(e^{\lambda t}\mathbf{v}) = e^{\lambda t}(A\mathbf{v})$. For these to be equal for all t, cancel the never-zero scalar $e^{\lambda t}$ and you are left staring at

$$A\mathbf{v} = \lambda\mathbf{v}.$$

That is the eigenvalue equation. Stop and feel the weight of this. The demand that $e^{\lambda t}\mathbf{v}$ solve the differential equation is *exactly* the demand that $\mathbf{v}$ be an eigenvector of A with eigenvalue λ. Calculus has handed us a pure linear-algebra problem.

Connection

This is the hinge between this book and its companion, *Linear Algebra From the Ground Up*. There, an eigenvector $\mathbf{v}$ of A is defined as a special direction that A merely *stretches* rather than rotates: $A\mathbf{v} = \lambda\mathbf{v}$, with the stretch factor λ the eigenvalue. That definition can feel abstract when you first meet it – why should anyone care about directions a matrix only stretches? Here is the answer, arriving from the calculus side: those are precisely the directions in which a linear system of ODEs behaves like a single scalar exponential. If you have not yet met eigenvalues, the chapters "Eigenvalues and Eigenvectors" and "Diagonalization" in the companion volume develop $\det(A - \lambda I) = 0$ and the solving of $(A - \lambda I)\mathbf{v} = \mathbf{0}$ from scratch; we will use those tools freely below and recall just enough to stay self-contained.

39.3.1 Recipe: eigenvalues and eigenvectors

For a 2×2 matrix $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, the eigenvalues are the roots λ of the **characteristic equation**

$$\det(A - \lambda I) = 0 \quad \Longleftrightarrow \quad \lambda^2 - \underbrace{(a+d)}_{\text{trace } \tau}\lambda + \underbrace{(ad - bc)}_{\det \Delta} = 0,$$

so $\lambda = \frac{1}{2}\left(\tau \pm \sqrt{\tau^2 - 4\Delta}\right)$. For each eigenvalue λ, the eigenvectors are the nonzero solutions $\mathbf{v}$ of the degenerate linear system $(A - \lambda I)\mathbf{v} = \mathbf{0}$. Once we have a pair of eigenvalue–eigenvector couples $(\lambda_1, \mathbf{v}_1)$ and $(\lambda_2, \mathbf{v}_2)$, each gives a solution $e^{\lambda_i t}\mathbf{v}_i$, and – because the system is linear – *any combination of solutions is again a solution.*

Theorem 39.1 (General solution by eigen-decomposition)**.** *Suppose the $n \times n$ matrix A has n linearly independent eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ with eigenvalues $\lambda_1, \dots, \lambda_n$. Then* every *solution of $\mathbf{u}' = A\mathbf{u}$ has the form*

$$\mathbf{u}(t) = c_1 e^{\lambda_1 t}\mathbf{v}_1 + c_2 e^{\lambda_2 t}\mathbf{v}_2 + \cdots + c_n e^{\lambda_n t}\mathbf{v}_n,$$

with constants $c_1, \dots, c_n$ fixed by the initial condition $\mathbf{u}(0)$.

Why the technique works

Why does the combination sweep up *every* solution, not just some? Two reasons, both worth holding onto. First, *superposition*: if $\mathbf{p}$ and $\mathbf{q}$ each solve $\mathbf{u}' = A\mathbf{u}$, then so does $c_1\mathbf{p} + c_2\mathbf{q}$, because differentiation and the matrix A are both linear – $(c_1\mathbf{p} + c_2\mathbf{q})' = c_1\mathbf{p}' + c_2\mathbf{q}' = c_1 A\mathbf{p} + c_2 A\mathbf{q} = A(c_1\mathbf{p} + c_2\mathbf{q})$. So the listed combinations really are solutions. Second, *counting*: the

existence-and-uniqueness theorem for linear systems says a solution is pinned down completely by its value at one instant, $\mathbf{u}(0) \in \mathbb{R}^n$ – that is n numbers of freedom, no more. Our family has exactly n free constants $c_1, \dots, c_n$, and because the $\mathbf{v}_i$ are independent, choosing them can hit *any* starting vector $\mathbf{u}(0) = \sum c_i \mathbf{v}_i$. A family with the right number of knobs that can be tuned to any start must, by uniqueness, contain every solution. Nothing escapes.

What is really happening

The eigenvectors are doing something profound: they are *uncoupling* the system. Write the starting state in the eigenvector basis, $\mathbf{u}(0) = \sum c_i \mathbf{v}_i$. In those coordinates the matrix A stops mixing the components – along each eigen-direction it acts as pure multiplication by λ_i. So in the right coordinates, the tangled system $\mathbf{u}' = A\mathbf{u}$ falls apart into n *independent* copies of the scalar equation $z_i' = \lambda_i z_i$, each solved by $z_i(t) = c_i e^{\lambda_i t}$. The coupling we fought in Section 39.1 was an illusion of bad coordinates. Eigenvectors are the good coordinates. This is the whole secret of linear systems: *diagonalize, and the knot is not a knot.*

39.4 A Fully Worked 2×2 System

Let us solve the system from Section 39.1 completely, step by step.

$$\mathbf{u}' = A\mathbf{u}, \qquad A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}.$$

Step 1 – eigenvalues. The trace is $\tau = 1 + 2 = 3$ and the determinant is $\Delta = (1)(2) - (2)(3) = 2 - 6 = -4$. The characteristic equation is

$$\lambda^2 - 3\lambda - 4 = 0 \implies (\lambda - 4)(\lambda + 1) = 0,$$

so $\lambda_1 = 4$ and $\lambda_2 = -1$. Two real eigenvalues of opposite sign – already this tells us (Section 39.5) that the origin will be a *saddle*.

Step 2 – eigenvector for $\lambda_1 = 4$. Solve $(A - 4I)\mathbf{v} = \mathbf{0}$:

$$A - 4I = \begin{pmatrix} 1-4 & 2 \\ 3 & 2-4 \end{pmatrix} = \begin{pmatrix} -3 & 2 \\ 3 & -2 \end{pmatrix}.$$

The top row says $-3v_1 + 2v_2 = 0$, i.e. $v_2 = \frac{3}{2}v_1$. Choosing $v_1 = 2$ to clear the fraction gives $\mathbf{v}_1 = \left(\begin{smallmatrix} 2 \\ 3 \end{smallmatrix}\right)$. (The bottom row $3v_1 - 2v_2 = 0$ is the same equation – it must be, because $A - 4I$ is singular; that redundancy is the eigenvalue working as designed.)

Step 3 – eigenvector for $\lambda_2 = -1$. Solve $(A + I)\mathbf{v} = \mathbf{0}$:

$$A + I = \begin{pmatrix} 2 & 2 \\ 3 & 3 \end{pmatrix}, \qquad 2v_1 + 2v_2 = 0 \implies v_2 = -v_1.$$

Choosing $v_1 = 1$ gives $\mathbf{v}_2 = \left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right)$.

Step 4 – assemble. By Theorem 39.1,

$$\mathbf{u}(t) = c_1\, e^{4t} \begin{pmatrix} 2 \\ 3 \end{pmatrix} + c_2\, e^{-t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Step 5 – fit an initial condition. Suppose $\mathbf{u}(0) = \binom{3}{2}$. At $t = 0$ the exponentials are 1, so we need

$$c_1 \begin{pmatrix} 2 \\ 3 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \implies \begin{cases} 2c_1 + c_2 = 3, \\ 3c_1 - c_2 = 2. \end{cases}$$

Adding the two equations: $5c_1 = 5$, so $c_1 = 1$, and then $c_2 = 3 - 2 = 1$. The trajectory through $(3, 2)$ is

$$\mathbf{u}(t) = e^{4t} \begin{pmatrix} 2 \\ 3 \end{pmatrix} + e^{-t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Intuition

Read the answer like a story of two tendencies. The e^{4t} piece grows explosively and points along $\mathbf{v}_1 = (2, 3)$: this is the *unstable* direction, the one the system is fleeing toward. The e^{-t} piece decays and points along $\mathbf{v}_2 = (1, -1)$: the *stable* direction, the one the system is collapsing onto. Almost immediately the decaying term becomes negligible and the trajectory becomes nearly parallel to $(2, 3)$, hurtling outward. The two eigenvectors are the grain of the wood; every trajectory bends to follow it.

Common trap

A frequent slip: writing $\mathbf{u}(t) = e^{\lambda_1 t}\mathbf{v}_1 + e^{\lambda_2 t}\mathbf{v}_2$ *without* the constants c_1, c_2, then being unable to match the initial condition. The constants are not optional decoration; they are the two dials you turn to aim the solution at the correct starting point. With two eigen-solutions you have two dials, exactly enough to hit any point in the plane. Drop them and you can describe only one special trajectory out of the infinitely many.

39.5 Phase Portraits: Drawing the Flow

A formula is the truth, but a picture is understanding. The **phase portrait** of a planar system is the (u_1, u_2) plane decorated with the arrows of the vector field $A\mathbf{u}$ and a representative sample of solution curves – the *trajectories* – threading through them. Reading a phase portrait is reading the destiny of every possible initial state at a single glance.

Here is the saddle we just solved. The two eigen-directions are drawn as bold lines; trajectories race away along the unstable direction $(2, 3)$ and are funneled in along the stable direction $(1, -1)$, sweeping out hyperbola-like curves between them.

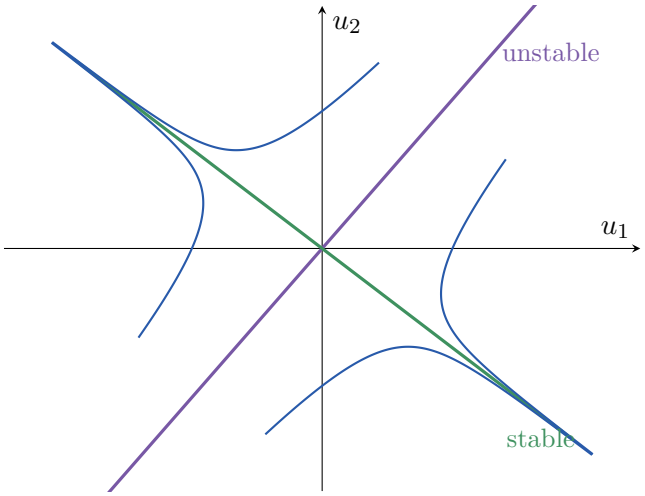


Figure 39.2: Phase portrait of the saddle $\mathbf{u}' = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 2 \end{smallmatrix}\right)\mathbf{u}$. Eigenvalues 4 and -1. Purple: the unstable eigen-direction $(2, 3)$; green: the stable eigen-direction $(1, -1)$. Trajectories flow out along purple and in along green.

39.5.1 The classification: trace and determinant decide everything

The qualitative shape of a planar phase portrait depends only on the two eigenvalues of A, and those depend only on the trace $\tau = a + d$ and determinant $\Delta = ad - bc$. Here is the full menagerie.

- **Two real eigenvalues, same sign.** If both are negative, every trajectory decays to the origin: a stable **node** (a sink), drawn in Figure 39.5. If both positive, everything flees outward: an unstable node (a source). The slower eigen-direction dominates the shape near the origin.
- **Two real eigenvalues, opposite signs.** A **saddle**, as in Figure 39.2. The origin attracts along one eigen-direction and repels along the other. Always unstable: almost every trajectory eventually leaves.
- **Complex eigenvalues $\lambda = \alpha \pm i\beta$, $\beta \neq 0$.** The solutions involve $e^{\alpha t}(\cos \beta t, \sin \beta t)$ – rotation (from the cos, sin) times scaling (from $e^{\alpha t}$). If $\alpha < 0$, trajectories spiral inward: a stable **spiral**. If $\alpha > 0$, an unstable spiral. If $\alpha = 0$ exactly, they neither grow nor shrink but circle forever: a **center**, a ring of closed orbits.

Why the technique works

Where do the rotation and the spiral come from? When the eigenvalues are complex, $\lambda = \alpha \pm i\beta$, the magic ingredient is Euler's formula, $e^{i\beta t} = \cos \beta t + i \sin \beta t$, which we proved back in the exponentials chapter. The solution $e^{\lambda t}\mathbf{v} = e^{\alpha t}e^{i\beta t}\mathbf{v}$ therefore factors into a real *growth* part $e^{\alpha t}$ and an oscillating *rotation* part $e^{i\beta t}$. The real eigenvalue gave us pure stretching; the imaginary part of a complex eigenvalue gives us turning. The imaginary axis is, quite literally, the axis of rotation in phase space. That is why an undamped spring (eigenvalues $\pm i\omega$, purely imaginary) orbits forever in a center, while a damped spring (eigenvalues with negative real part) spirals into rest.

The whole classification fits on one diagram in the (τ, Δ) plane. The parabola $\tau^2 = 4\Delta$ is the boundary where the eigenvalues switch from real to complex.

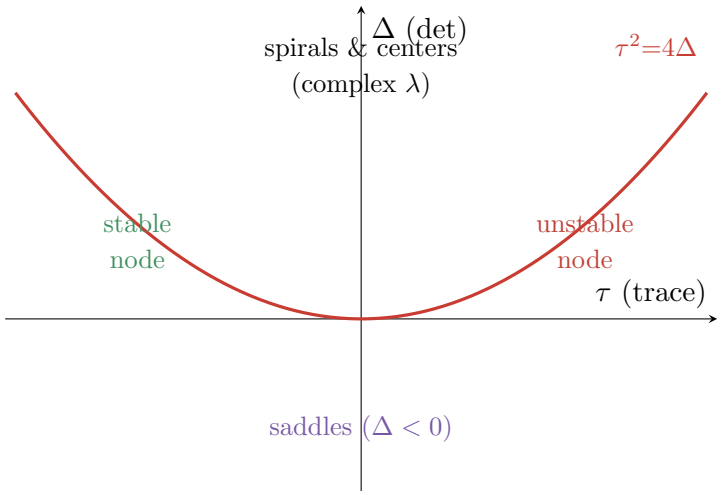


Figure 39.3: The trace–determinant plane. Below the horizontal axis ($\Delta < 0$): saddles. Above it and below the red parabola: real eigenvalues, hence nodes (stable on the left where $\tau < 0$, unstable on the right). Above the parabola: complex eigenvalues, hence spirals (stable for $\tau < 0$, centers on the line $\tau = 0$, unstable for $\tau > 0$).

Why this matters

Marvel at the economy. Two numbers – the trace and determinant of a 2×2 matrix – predict the entire qualitative fate of an infinite family of trajectories. You need not solve anything to know whether a bridge will settle or shake apart, whether a population will reach equilibrium or oscillate, whether a control system is stable. You compute τ and Δ and read off the answer from one picture. This is the gift of the eigenvalue viewpoint: it converts *dynamics* (hard, time-dependent) into *algebra* (two numbers on a chart).

39.5.2 A spiral, drawn

For concreteness, take $A = \left(\begin{smallmatrix} -0.3 & -1 \\ 1 & -0.3 \end{smallmatrix}\right)$. Its eigenvalues are $-0.3 \pm i$, complex with negative real part: a stable spiral. Every trajectory winds inward to the origin.

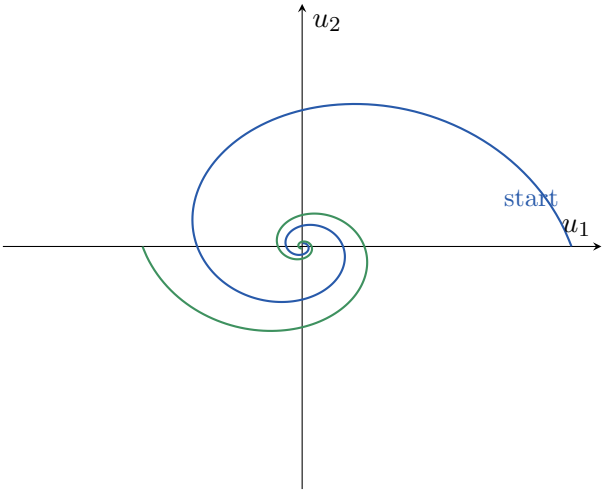


Figure 39.4: A stable spiral: $A = \left(\begin{smallmatrix} -0.3 & -1 \\ 1 & -0.3 \end{smallmatrix}\right)$, eigenvalues $-0.3 \pm i$. Rotation comes from the imaginary part $\pm i$; the inward decay comes from the real part -0.3. Two trajectories shown.

39.5.3 A stable node, drawn

For contrast, take the diagonal matrix $A = \left(\begin{smallmatrix} -1 & 0 \\ 0 & -3 \end{smallmatrix}\right)$, with eigenvalues -1 and -3 and eigenvectors along the two axes. Both eigenvalues are negative, so every trajectory decays to the origin – a stable node. Because the u_2 component dies three times faster (e^{-3t} versus e^{-t}), trajectories flatten toward the slow axis u_1 as they approach the origin, and that slow direction sets the shape you see near the origin.

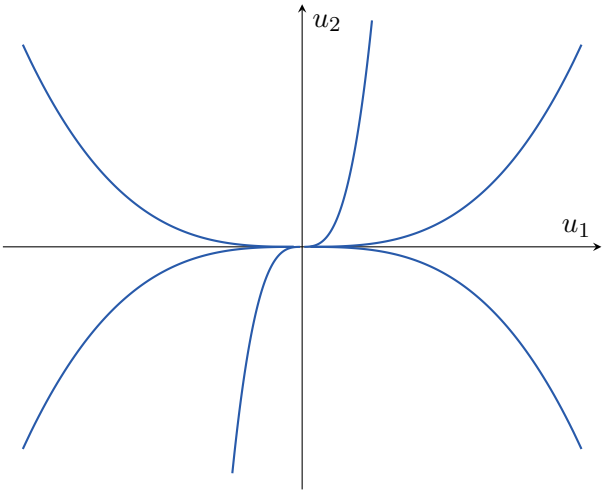


Figure 39.5: A stable node: $A = \left(\begin{smallmatrix} -1 & 0 \\ 0 & -3 \end{smallmatrix}\right)$, eigenvalues -1 and -3. Every trajectory decays to the origin. The faster-dying u_2 direction flattens each curve onto the slow u_1 axis before it arrives, so the node is "combed" along u_1.

39.6 Stability: When Does the System Settle?

A point $\mathbf{u}^*$ where $\mathbf{F}(\mathbf{u}^*) = \mathbf{0}$ is an **equilibrium** – plant the state dot there and it never moves, because its velocity is zero. For the linear system $\mathbf{u}' = A\mathbf{u}$ the origin is always an equilibrium (and the only one, if A is invertible). The practical question that haunts every engineer is: if we nudge the system slightly off equilibrium, does it return, or does it run away?

The eigenvalues answer it cleanly. A solution component behaves like $e^{\lambda t}$, which shrinks if the real part of λ is negative and grows if it is positive. Hence:

Theorem 39.2 (Linear stability)**.** *For $\mathbf{u}' = A\mathbf{u}$, the origin is*

- ***asymptotically stable*** *(every trajectory $\to \mathbf{0}$) if and only if* every *eigenvalue of A has negative real part;*
- ***unstable*** *if some eigenvalue has positive real part;*
- ***marginally stable*** *(trajectories stay bounded but need not decay) in the borderline case of purely imaginary eigenvalues, e.g. a center.*

Intuition

The rule is brutally simple: *the system is stable exactly when every growth rate is a decay rate.* One single eigenvalue poking into the right half of the complex plane – one direction in which a perturbation grows – is enough to doom stability, because a real trajectory generically has a nonzero component along every eigen-direction, and the growing one eventually swamps the rest. Stability is unanimity: *all* must decay.

Application

Why airplanes do not (usually) fall out of the sky. An aircraft in steady flight is an equilibrium of a system of differential equations governing its pitch, roll, and yaw. Engineers compute the eigenvalues of the linearized dynamics. If all have negative real part, a gust that tips the plane decays away and the plane self-corrects – the famous *phugoid* mode is a slow stable spiral you can sometimes feel as a gentle porpoising. If an eigenvalue creeps into the right half-plane, the aircraft is unstable and needs a control system (itself adding feedback terms to A, shifting the eigenvalues leftward) to fly at all. Modern fighter jets are deliberately built unstable for agility and are flyable only because a computer moves the eigenvalues thousands of times a second. Eigenvalue placement is a multi-billion-dollar industry.

History

The connection between eigenvalues and stability was forged in the late nineteenth century. *Aleksandr Lyapunov*, in his 1892 doctoral thesis *The General Problem of the Stability of Motion*, gave the first rigorous theory of stability for systems of differential equations, including methods (now called Lyapunov functions) that work even when you cannot find the eigenvalues. Around the same time the engineer-minded *Edward Routh* (1876) and the mathematician *Adolf Hurwitz* (1895) found a purely algebraic test – the Routh–Hurwitz criterion – to decide whether all roots of a polynomial have negative real part *without computing them*, which mattered enormously for designing steam-engine governors and, later, every feedback controller

ever built. Maxwell himself had opened the subject in 1868 with "On Governors," puzzling over why some steam engines hunted and surged. Stability theory was born from machines that shook themselves apart.

39.7 A Glimpse Beyond the Linear World

Everything so far rested on linearity – the right-hand side was $A\mathbf{u}$, a matrix times the state. Real systems are usually *nonlinear*: the right-hand side bends. Foxes eating rabbits, the swing of a pendulum at large angles, the weather. We cannot solve these in closed form. But two ideas carry an astonishing distance.

39.7.1 Linearization: zoom in and the curve looks straight

Near an equilibrium $\mathbf{u}^*$, expand the vector field in a Taylor series (the multivariable kind from the partial-derivatives chapters). The constant term vanishes because $\mathbf{F}(\mathbf{u}^*) = \mathbf{0}$, and to first order

$$\mathbf{F}(\mathbf{u}^* + \mathbf{h}) \approx J\,\mathbf{h}, \qquad J = \begin{pmatrix} \frac{\partial F_1}{\partial u_1} & \frac{\partial F_1}{\partial u_2} \\ \frac{\partial F_2}{\partial u_1} & \frac{\partial F_2}{\partial u_2} \end{pmatrix}_{\mathbf{u}^*},$$

where J is the **Jacobian matrix** – the matrix of partial derivatives of the components, evaluated at the equilibrium. So *near an equilibrium, a nonlinear system looks like the linear system* $\mathbf{h}' = J\mathbf{h}$, and the eigenvalues of J classify the equilibrium just as before.

Why the technique works

This is the deepest reason the linear theory was worth all our effort. A nonlinear vector field, viewed under a strong enough microscope centered on an equilibrium, is indistinguishable from its linearization – exactly as a smooth curve, magnified near a point, becomes its tangent line. The Hartman–Grobman theorem makes this precise: provided no eigenvalue of J sits on the imaginary axis, the nonlinear phase portrait near the equilibrium is a gently warped copy of the linear one. Saddles stay saddles, sinks stay sinks, spirals stay spirals. The whole trace–determinant menagerie governs nonlinear systems too – locally. Calculus's founding move, "replace the curve by its tangent," returns one last time, now in the shape of replacing a vector field by its Jacobian.

Example 39.2 (The pendulum)**.** The undamped pendulum $\theta'' = -\sin\theta$ becomes the system $x_1' = x_2$, $x_2' = -\sin x_1$ (with $x_1 = \theta$, $x_2 = \theta'$). Its Jacobian is $J = \left(\begin{smallmatrix} 0 & 1 \\ -\cos x_1 & 0 \end{smallmatrix}\right)$. At the bottom-hanging rest state $x_1 = 0$: $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, eigenvalues $\pm i$ – a *center*, the pendulum swinging forever (undamped). At the inverted balance point $x_1 = \pi$: $J = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, eigenvalues ± 1 – a *saddle*, unstable, exactly matching the physical fact that balancing a pendulum upright is hopeless: the tiniest nudge runs away along the unstable eigen-direction.

39.7.2 What linearization cannot see: limit cycles and chaos

Linearization is a local lie that happens to be locally true. Globally, nonlinear systems do things no linear system ever can.

A **limit cycle** is an isolated closed loop that nearby trajectories spiral *onto* – a self-sustaining oscillation with its own preferred amplitude, like a heartbeat or the buzz of a violin string. The Van der Pol oscillator $x'' - \mu(1 - x^2)x' + x = 0$ is the classic example; no combination of eigenvectors can produce it, because a linear center has a whole continuum of orbits with no preferred size, while a limit cycle picks out exactly one.

And in three or more dimensions, nonlinear systems can be **chaotic**: trajectories that never repeat, never settle, and depend so sensitively on their starting point that two neighboring states diverge exponentially. The most famous is the Lorenz system,

$$x' = \sigma(y - x), \qquad y' = x(\rho - z) - y, \qquad z' = xy - \beta z,$$

which Edward Lorenz extracted in 1963 from a crude model of atmospheric convection. Its trajectories are forever drawn toward a fractal object – the *Lorenz attractor*, the celebrated "butterfly" – looping around two lobes in an order no formula predicts.

History

Lorenz discovered chaos by accident. Rerunning a weather simulation on his 1960s computer, he typed in a midstream value rounded to three decimals (0.506) instead of the full six the machine carried (0.506127), expecting the same forecast. Within simulated weeks the two runs had nothing in common. That tiny truncation, amplified by the system's sensitivity, rewrote the future entirely. Lorenz named the lesson the **butterfly effect**: a butterfly flapping its wings in Brazil might set off a tornado in Texas – not because the flap is powerful, but because the atmosphere magnifies microscopic differences without bound. It meant that long-range weather prediction is impossible in principle, no matter how good the data or the model. The dream of deterministic forecasting, alive since Laplace, died on that printout.

Why this matters

Here is the philosophical jolt. The Lorenz equations are utterly *deterministic* – no randomness, no dice, just three tidy differential equations a child could write down. Yet their solutions are unpredictable in practice forever. Determinism does not imply predictability. This shattered a 300-year-old assumption, traceable to Newton and made explicit by Laplace's demon, that a universe governed by exact laws is, in principle, foreseeable. The laws can be exact and the future still unknowable, because to predict it you would need the present to *infinite* precision, and infinite precision does not exist. Chaos is where the clockwork universe meets its limits – discovered not by philosophers but by people solving differential equations.

39.8 Where to Go Next

This chapter sits at the frontier of what a first calculus book can cover. Three roads lead onward, and you are now equipped to walk all of them.

Partial differential equations. Every system here had a single independent variable, time, so the unknowns were functions of t alone – *ordinary* differential equations. But heat spreads through a *rod*, varying in both position and time; a drumhead vibrates across a *surface*; a quantum wavefunction fills *space*. These obey **partial** differential equations, where the unknown depends on several variables and *partial* derivatives appear – the heat equation $\frac{\partial u}{\partial t} = k\,\frac{\partial^2 u}{\partial x^2}$, the wave equation, Laplace's equation,

the Schrodinger equation. The partial derivatives of the multivariable chapters and the series of the Taylor chapters (Fourier series, their cousin) are exactly the tools that crack them open.

Numerical methods. Almost no differential equation arising in practice can be solved by a formula. We solve them by *marching*: from the state now, the equation gives the velocity now, so we step a tiny Δt forward and repeat. The crudest version is Euler's method, $\mathbf{u}_{n+1} = \mathbf{u}_n + \Delta t\, \mathbf{F}(\mathbf{u}_n)$; the workhorse is the fourth-order Runge–Kutta method, which samples the slope four times per step to cancel error. Every weather forecast, climate model, spacecraft trajectory, and animated-film cloth simulation is a differential equation being marched forward by a computer.

Dynamical systems. The phase-portrait, stability, and chaos ideas we touched are the doorway to a vast modern field – bifurcations (where an equilibrium suddenly changes character as a parameter crosses a threshold), strange attractors, ergodic theory, the geometry of flows. It is calculus grown up into the study of *everything that changes*.

Connection

Notice how completely linear algebra and calculus have fused by the end of this book. We began with the slope of one curve; we end with the eigenvalues of a matrix dictating the fate of a whole flowing space. The two subjects are not neighbors but the same subject seen from two sides: calculus is linear algebra done to functions and rates, linear algebra is the skeleton calculus hangs on. If this chapter sent you hunting through the companion *Linear Algebra From the Ground Up* for eigenvalues, diagonalization, and the matrix exponential e^{At} (which packages *all* solutions of $\mathbf{u}' = A\mathbf{u}$ into a single object $\mathbf{u}(t) = e^{At}\mathbf{u}(0)$), then the two books have done together what neither could alone.

Pause

Take a breath and look back at the whole arc. We turned a tangle of coupled equations into one matrix equation $\mathbf{u}' = A\mathbf{u}$; we guessed the solution $e^{\lambda t}\mathbf{v}$ and were handed the eigenvalue problem $A\mathbf{v} = \lambda\mathbf{v}$; we found that eigenvectors uncouple the system into independent scalar exponentials; we drew the flow as a phase portrait and classified it by trace and determinant; we read off stability from the sign of the eigenvalues' real parts; and we saw that even nonlinear systems bow to this analysis near their equilibria, before breaking free into limit cycles and chaos. Every step rested on something you already knew – e^{kt}, the tangent-line idea, the dot product. Nothing here was truly new; it was old ideas, recombined, reaching further than any of them could alone. That is what mathematics feels like from the inside.

39.9 Exercises

Exercise 39.1 (Reduce to a system)**.** Convert each higher-order equation to a first-order system $\mathbf{u}' = A\mathbf{u}$ (or $\mathbf{x}' = \mathbf{F}(\mathbf{x})$ if nonlinear), naming your new variables explicitly. (a) $y'' - 3y' + 2y = 0$. (b) $y''' = y$. (c) $y'' + \sin y = 0$. For (a), write down the companion matrix and state its trace and determinant.

Exercise 39.2 (Eigen-solve a system)**.** For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, find the eigenvalues and eigenvectors, write the general solution of $\mathbf{u}' = A\mathbf{u}$, and find the particular solution with $\mathbf{u}(0) = \left(\begin{smallmatrix} 1 \\ 3 \end{smallmatrix}\right)$. Classify the origin.

Exercise 39.3 (A saddle)**.** The matrix $A = \left(\begin{smallmatrix} 1 & 3 \\ 3 & 1 \end{smallmatrix}\right)$ has eigenvalues 4 and -2. Find the eigenvectors

and sketch the phase portrait by hand, marking the stable and unstable directions and the flow along each.

Exercise 39.4 (Complex eigenvalues)**.** For $A = \left(\begin{smallmatrix} 0 & -2 \\ 2 & 0 \end{smallmatrix}\right)$, show the eigenvalues are $\pm 2i$. What kind of equilibrium is the origin? Without solving, describe the shape of every trajectory. Then verify that $u_1 = \cos 2t$, $u_2 = \sin 2t$ is a solution.

Exercise 39.5 (Trace–determinant reading)**.** For each matrix, compute τ and Δ and classify the origin (stable/unstable node, saddle, stable/unstable spiral, or center) *without* finding the eigenvectors. (a) $\left(\begin{smallmatrix} -1 & 0 \\ 0 & -3 \end{smallmatrix}\right)$. (b) $\left(\begin{smallmatrix} -1 & 4 \\ -1 & -1 \end{smallmatrix}\right)$. (c) $\left(\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right)$. (d) $\left(\begin{smallmatrix} 0 & 1 \\ -4 & 0 \end{smallmatrix}\right)$.

Exercise 39.6 (Stability hinges on a parameter)**.** Consider $A = \left(\begin{smallmatrix} -1 & 1 \\ -1 & k \end{smallmatrix}\right)$ with k a real parameter. Compute the trace and determinant, and find the range of k for which the origin is asymptotically stable. What happens at the value of k where stability is lost?

Exercise 39.7 (Damped oscillator, both ways)**.** Return to $y'' + 2y' + 5y = 0$ from Example 39.1. (a) Find the eigenvalues of the companion matrix. (b) Confirm they are complex with negative real part, and say what kind of equilibrium and what physical behavior this predicts. (c) Relate the real and imaginary parts of the eigenvalues to the decay rate and oscillation frequency of the solution.

Exercise 39.8 (Linearize a nonlinear system)**.** The system $x' = x(3 - x - y)$, $y' = y(2 - x)$ models two competing species. (a) Find all equilibria (there are three, including one in the interior with both populations positive). (b) Compute the Jacobian J at the interior equilibrium. (c) Find its eigenvalues and classify that equilibrium. Is coexistence stable?

Exercise 39.9 (Pendulum equilibria)**.** Using Example 39.2, add light damping: $x_1' = x_2$, $x_2' = -\sin x_1 - 0.2\, x_2$. Recompute the Jacobian at the hanging rest state $(0, 0)$. How do its eigenvalues differ from the undamped case, and what does the change predict about the pendulum's motion?

Exercise 39.10 (Conceptual: why eigenvectors?)**.** In your own words, explain why guessing a solution of the form $\mathbf{u}(t) = e^{\lambda t}\mathbf{v}$ forces $\mathbf{v}$ to be an eigenvector of A. Then explain why a 2×2 system with two independent eigenvectors has *exactly* the right number of free constants to match any initial condition.

Exercise 39.11 (A taste of chaos, by hand)**.** You cannot solve the Lorenz system, but you can march it. With $\sigma = 10$, $\rho = 28$, $\beta = 8/3$, and step $\Delta t = 0.01$, do *three* Euler steps starting from $(x, y, z) = (1, 1, 1)$, reporting all coordinates each step. Then repeat from $(1.001, 1, 1)$. Comment on how the two runs compare after three steps – and explain why, after a few thousand steps, they would have nothing in common.

Part IX

Mathematical Background

Chapter 40

Appendix: Algebra, Functions, and Proof

Calculus does not begin from nothing. Every limit you compute, every derivative you simplify, every integral you wrestle into a closed form leans on a quieter layer of mathematics underneath: the algebra of rearranging expressions, the laws of exponents, the handling of inequalities and absolute values, a few durable facts about functions, and the small but crucial habit of *proving* that something is true rather than merely believing it. This appendix gathers that layer in one place. It is the toolbox you reach into without thinking once you are fluent — and the place to come back to when a step in a derivation suddenly feels like a magic trick.

The spirit here is the same as the rest of the book. We will not just list rules. We will ask *why* each rule holds, draw the picture that makes it obvious, and point out the traps where good students reliably slip. By the end you should feel that none of this is arbitrary: the exponent laws are forced on us once we decide what a^n should mean; the rules for inequalities follow from one plain fact about the number line; and a proof is not a ritual but an explanation that happens to be airtight.

Focus

Three things to carry away from this appendix:

1. Algebraic identities (factoring, exponent laws, the binomial pattern) are not memorized magic — each is forced by a single defining idea, and once you see the idea you can reconstruct the rule.
2. Inequalities and absolute values are the native language of calculus (limits, error bounds, convergence). The one rule everyone forgets — multiplying by a negative flips the sign — has a clean reason.
3. A proof is an explanation with no gaps. Direct proof, proof by contradiction, and induction are three reliable shapes that explanation can take.

40.1 Numbers, and the Ground We Stand On

Before algebra there are the numbers algebra rearranges. We use a nested tower of number systems, each built to fix a shortcoming of the last:

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}.$$

The naturals $\mathbb{N} = \{1, 2, 3, \ldots\}$ count things. The integers $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$ add the ability to subtract freely. The rationals $\mathbb{Q}$ — ratios p/q of integers with $q \neq 0$ — add the ability to divide. The reals $\mathbb{R}$ fill in the gaps between rationals so that the number line has no holes (a number like $\sqrt{2}$ or π lives here but not in $\mathbb{Q}$). The complex numbers $\mathbb{C}$ add a square root of -1. Calculus lives almost entirely in $\mathbb{R}$, and the "no holes" property of $\mathbb{R}$ — its *completeness* — is what makes limits behave; that story has its own appendix.

History

The tower $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$ took millennia to be accepted. For centuries, mathematicians resisted negative quantities. The third-century Greek mathematician Diophantus called the equation $4x + 20 = 4$ "absurd" because it had no positive solution. As late as the seventeenth century, René Descartes referred to negative solutions of polynomial equations as "false roots" (*racines fausses*), and Blaise Pascal claimed that subtracting 4 from 0 yields nothing because "nothing can be less than nothing." It was not until John Wallis in 1685 popularized the geometric number line—interpreting negative numbers as directed displacements to the left of the origin—that negative quantities ceased to be viewed as mathematical ghosts.

Intuition

Picture the real line as an infinitely long, perfectly smooth ruler. Every point on it is a real number, and every real number sits at exactly one point. The rationals are dense — between any two of them there is another — yet they leave gaps you cannot see but can prove are there. The reals are what you get when you plug every gap. Most of calculus is the art of saying precise things about points sitting infinitely close together on this ruler.

The arithmetic of $\mathbb{R}$ obeys a short list of field axioms that we will use without ceremony: addition and multiplication are commutative and associative, 0 and 1 are identities, every number has an additive inverse and every nonzero number a multiplicative inverse, and multiplication distributes over addition:

$$a(b + c) = ab + ac.$$

That last law, the *distributive law*, is the single most overworked fact in all of algebra. Expanding brackets, factoring, multiplying polynomials, the binomial theorem — all of it is the distributive law applied again and again.

What is really happening

When you "expand $3(x + 2)$ to get $3x + 6$" and when you "factor $3x + 6$ back to $3(x + 2)$" you are doing the exact same equation read in two directions: $a(b + c) = ab + ac$. Expanding goes left to right; factoring goes right to left. This is why factoring feels harder — running a rule *backwards* means searching for the inputs that would have produced what you see, and search is always harder than evaluation.

40.2 Algebraic Manipulation: Expanding and Factoring

40.2.1 Multiplying out

To multiply two sums, distribute every term of the first across every term of the second. For two binomials this is the familiar pattern (sometimes taught as "FOIL," though the name hides the principle):

$$(a+b)(c+d) = ac + ad + bc + bd.$$

Each term on the right is one term from the first factor times one term from the second; we collect every such pairing. There is nothing special about two terms. $(a+b)(c+d+e)$ produces six products, one for each (left term, right term) pair.

The identities worth knowing cold are the ones that appear constantly in calculus, especially when rationalizing or simplifying difference quotients:

$$(a+b)^2 = a^2 + 2ab + b^2, \tag{40.1}$$
$$(a-b)^2 = a^2 - 2ab + b^2, \tag{40.2}$$
$$(a+b)(a-b) = a^2 - b^2, \tag{40.3}$$
$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3. \tag{40.4}$$

The difference of squares (40.3) is the workhorse: it is the engine behind the trick of multiplying by a conjugate, which you meet the very first time you compute a limit like $\lim_{x\to 0}(\sqrt{x+1}-1)/x$.

Why the technique works

Why is $(a+b)^2 = a^2 + 2ab + b^2$ and *not* $a^2 + b^2$? Because squaring a sum is not the same as summing the squares — the cross terms are real. Run the distributive law all the way out:

$$(a+b)^2 = (a+b)(a+b) = a\cdot a + a\cdot b + b\cdot a + b\cdot b = a^2 + 2ab + b^2.$$

The middle $2ab$ is the two "mixed" pairings ab and ba. Forgetting it is probably the single most common algebra error in all of calculus, and the picture below shows exactly what it is.

The cross term is not abstract. It is literally the two rectangles you get when you enlarge a square.

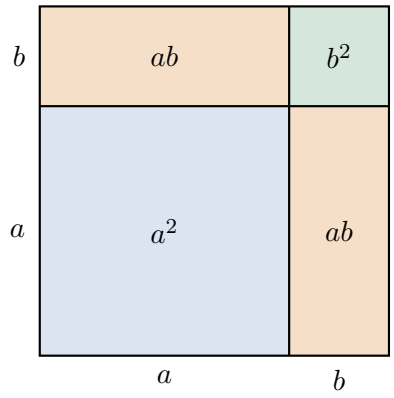


Figure 40.1: The square on side $a+b$ has area $(a+b)^2$. Cut it into four pieces: the a-square, the b-square, and *two* $a \times b$ rectangles. That is why $(a+b)^2 = a^2 + 2ab + b^2$ — the $2ab$ is the two orange rectangles, the part people forget.

40.2.2 Factoring: running the machine backwards

Factoring rewrites a sum as a product. It matters because a product is zero exactly when one of its factors is zero (the *zero-product property*), which is how we solve equations, and because a factored expression often lets a troublesome common factor cancel — the heart of computing limits of the indeterminate form $0/0$.

The basic moves, in rough order of how often you need them:

(1) Pull out a common factor.

$$6x^3 - 9x^2 = 3x^2(2x - 3).$$

Look for the largest factor shared by every term and lift it out front.

(2) Difference of squares.

$$x^2 - 25 = (x-5)(x+5), \qquad x^4 - 1 = (x^2-1)(x^2+1) = (x-1)(x+1)(x^2+1).$$

(3) Quadratic trinomials. To factor $x^2 + bx + c$, find two numbers that *multiply to* c *and add to* b. For $x^2 + 7x + 12$ we want a pair with product 12 and sum 7: that is 3 and 4, so

$$x^2 + 7x + 12 = (x+3)(x+4).$$

(4) Sum and difference of cubes.

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2), \qquad a^3 + b^3 = (a+b)(a^2 - ab + b^2).$$

Common trap

There is no real factoring of a *sum* of squares: $a^2 + b^2$ does not factor over the reals (you need complex numbers). Students who just factored $a^2 - b^2$ often write $a^2 + b^2 = (a+b)(a-b)$ out of momentum — but expanding that gives $a^2 - b^2$, the wrong sign. The difference factors; the

sum does not.

Example 40.1 (Factoring to kill a 0/0 limit)**.** Consider $\dfrac{x^2-9}{x-3}$ as $x \to 3$. Plugging in $x = 3$ gives $0/0$, which says nothing yet. Factor the top as a difference of squares:

$$\frac{x^2-9}{x-3} = \frac{(x-3)(x+3)}{x-3} = x+3 \quad (x \neq 3).$$

The offending factor $x-3$ cancels, and the simplified expression $x+3$ tends to 6 as $x \to 3$. The whole drama of the limit was a single hidden common factor; factoring exposed and removed it.

Common trap

The danger of formal cancellation without domain vigilance. Algebraic formulas manipulate symbols blindly unless you track domain restrictions. Consider the famous bogus "proof" that $1 = 2$: Start with $a = b$. Multiply by a to get $a^2 = ab$. Subtract b^2 from both sides: $a^2 - b^2 = ab - b^2$. Factor both sides: $(a-b)(a+b) = b(a-b)$. Now cancel the common factor $(a-b)$: $a + b = b$. Since $a = b$, this says $2b = b$, so dividing by b gives $2 = 1$.
Where did the reasoning derail? At the cancellation step: because $a = b$, the factor $a - b$ is identically 0. Dividing both sides by zero destroyed the equation. In calculus, when we write $\lim_{x\to 3} \frac{x^2-9}{x-3} = \lim_{x\to 3}(x+3)$, the cancellation is valid precisely because the limit process considers values of x arbitrarily close to 3, but *never* $x = 3$. Hence $x - 3 \neq 0$ throughout the approach.

Walking question

Try factoring $2x^2 + x - 6$. (Hint: look for two numbers multiplying to $2 \cdot (-6) = -12$ and adding to 1; they are 4 and -3. Split the middle term: $2x^2+4x-3x-6 = 2x(x+2)-3(x+2) = (2x-3)(x+2)$.) The technique of "splitting the middle term" turns a hard trinomial into two easy common-factor pulls.

40.2.3 The quadratic formula, and completing the square

Not every quadratic factors with nice integers. For *any* quadratic $ax^2 + bx + c = 0$ with $a \neq 0$, the solutions are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

This is not handed down from the sky — it comes from *completing the square*, a technique worth seeing in full because the same maneuver reappears in integration (trig substitution) and in finding vertices of parabolas.

Why the technique works

Start from $ax^2 + bx + c = 0$. Divide by a and move the constant:

$$x^2 + \tfrac{b}{a}x = -\tfrac{c}{a}.$$

The left side is *almost* a perfect square. The square $(x + \frac{b}{2a})^2$ expands to $x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}$, so add $\frac{b^2}{4a^2}$ to both sides to complete it:

$$\left(x + \tfrac{b}{2a}\right)^2 = \frac{b^2}{4a^2} - \frac{c}{a} = \frac{b^2 - 4ac}{4a^2}.$$

Now take square roots (this is where the $\pm$ is born) and solve for x:

$$x + \frac{b}{2a} = \pm\frac{\sqrt{b^2 - 4ac}}{2a} \implies x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The quadratic formula is just completing the square done once, symbolically, so you never have to do it again.

History

The technique of completing the square is ancient. Babylonian scribes were solving quadratics by an equivalent geometric "cut-and-paste" procedure nearly four thousand years ago. Around 825 CE the Persian mathematician al-Khwarizmi wrote *al-Kitab al-mukhtasar fi hisab al-jabr wa'l-muqabala* — "the compendious book on calculation by restoration and balancing." The word *al-jabr* ("restoration," meaning moving a subtracted term to the other side) is the root of our word *algebra*. He literally completed squares by drawing them, the same picture as Figure 40.1.

The quantity $b^2 - 4ac$ under the root is the *discriminant*. If it is positive there are two real roots; if zero, one repeated root; if negative, no real roots (the parabola misses the x-axis) and the roots are complex conjugates.

40.3 Exponents and Their Laws

40.3.1 Where the laws come from

For a positive integer n, a^n means "multiply a by itself n times." From this single definition every exponent law follows by counting factors. Take $a^m \cdot a^n$: that is m copies of a times n copies of a, which is $m + n$ copies. Hence

$$a^m \cdot a^n = a^{m+n}.$$

There is nothing to memorize; you are just counting how many a's stand in the line. The same counting gives the rest:

$$a^m \cdot a^n = a^{m+n}, \qquad \frac{a^m}{a^n} = a^{m-n}, \tag{40.5}$$

$$(a^m)^n = a^{mn}, \qquad (ab)^n = a^n b^n, \tag{40.6}$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}. \tag{40.7}$$

What is really happening

Every exponent law is really the *single* fact $a^{m+n} = a^m a^n$ wearing different clothes, because exponentiation turns addition (in the exponent) into multiplication (of the values). $(a^m)^n = a^{mn}$ is repeated addition of exponents: n copies of m. This addition-becomes-multiplication structure is exactly what makes the exponential function e^x so central in calculus — it is the function that converts the additive world of inputs into the multiplicative world of outputs, and it is the only one (up to scaling the input) that is its own derivative.

40.3.2 Extending to zero, negatives, and fractions

The definition "multiply a by itself n times" only makes sense for positive integer n. So what could a^0, a^{-3}, or $a^{1/2}$ possibly mean? One principle settles all three cases at once: *we extend the definition in the only way that keeps the law $a^m a^n = a^{m+n}$ true.* The laws are not re-proved for the new cases; they are the very thing we insist on preserving, and that insistence forces a unique answer.

Zero exponent. We need $a^0 \cdot a^n = a^{0+n} = a^n$. Dividing by a^n (for $a \neq 0$) forces

$$a^0 = 1.$$

Negative exponent. We need $a^{-n} \cdot a^n = a^0 = 1$, so a^{-n} must be the reciprocal:

$$a^{-n} = \frac{1}{a^n}.$$

Fractional exponent. We need $\left(a^{1/n}\right)^n = a^{(1/n)\cdot n} = a^1 = a$, so $a^{1/n}$ must be the number whose nth power is a — the nth root:

$$a^{1/n} = \sqrt[n]{a}, \qquad a^{m/n} = \left(\sqrt[n]{a}\right)^m = \sqrt[n]{a^m}.$$

Why the technique works

This move is worth dwelling on, because mathematics repeats it constantly. The expression $a^{1/2}$ has *no* prior meaning — you cannot multiply a by itself "half a time." We *define* it. But we do not define it by whim: we choose the one value that lets the exponent law survive. That demand pins it down uniquely to $\sqrt{a}$. Mathematics grows by exactly this method: take a rule that holds in a small world, then extend the world in whatever way keeps the rule intact. The same logic later defines a^x for irrational x, $0! = 1$, and $e^{i\theta} = \cos\theta + i\sin\theta$.

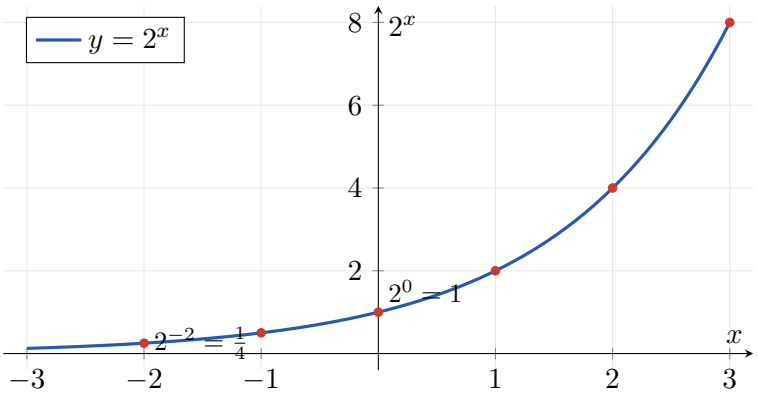


Figure 40.2: The exponential 2^x knits the integer powers (red dots) into a smooth curve. Negative exponents give reciprocals ($2^{-2} = \frac{1}{4}$), the zero exponent gives 1, and fractional exponents fill in everything between. The values for non-integer x are precisely the ones that keep $2^{m+n} = 2^m 2^n$ holding for all real m, n.

Common trap

The exponent laws govern *products with a shared base*, not sums and not mixed bases. All of the following are wrong:

$$a^m + a^n \neq a^{m+n}, \qquad (a+b)^n \neq a^n + b^n, \qquad \sqrt{a^2+b^2} \neq a+b.$$

The last is the same error as the missing cross term in $(a+b)^2$. There is no law that simplifies $a^m + a^n$ in general — the laws live on the multiplicative side only. Whenever you are tempted to "distribute" an exponent across a sum, stop: exponents distribute over products, never over sums.

40.4 Inequalities

Equations ask *where* two things are equal; inequalities ask *which is bigger*. Calculus runs on inequalities far more than on equations: the definition of a limit is an inequality ($|f(x) - L| < \varepsilon$), comparison tests for series are inequalities, error bounds are inequalities. So fluency here pays off everywhere.

The order relation $<$ on $\mathbb{R}$ obeys a handful of rules. Most are unsurprising, but one is the famous trap.

Focus

Rules for inequalities. For real numbers,

- **Add anything:** if $a < b$ then $a + c < b + c$ (for any c).
- **Multiply by a positive:** if $a < b$ and $c > 0$ then $ac < bc$.
- **Multiply by a negative — FLIP:** if $a < b$ and $c < 0$ then $ac > bc$.
- **Transitivity:** if $a < b$ and $b < c$ then $a < c$.

- **Reciprocal of positives flips:** if $0 < a < b$ then $\frac{1}{a} > \frac{1}{b}$.

Why the technique works

Why does multiplying by a negative *flip* the inequality? Picture the number line. The statement $a < b$ means a sits to the left of b. Multiplying every number by -1 is a *reflection of the line through* 0: left and right swap. So whatever was on the left (a) lands on the right, and the order reverses. More algebraically: $a < b$ means $b - a > 0$, i.e. $b - a$ is positive. Multiply a positive number by a negative c and you get a negative: $c(b - a) < 0$, that is $cb - ca < 0$, i.e. $cb < ca$, i.e. $ca > cb$. The flip is forced.

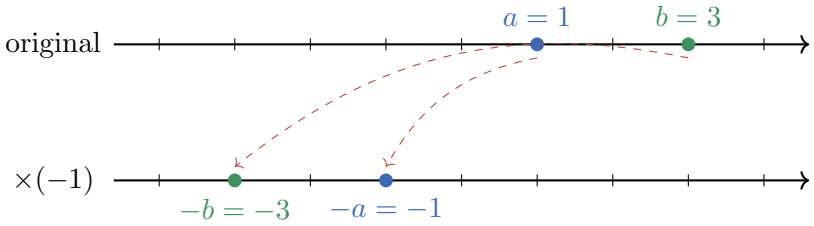


Figure 40.3: Multiplying by -1 reflects the line through 0. The blue point $a = 1$ was left of the green $b = 3$; after reflection $-a = -1$ is to the *right* of $-b = -3$. The order has flipped: $1 < 3$ but $-1 > -3$.

Common trap

The flip rule is the most common source of wrong inequality answers. When you divide both sides of $-2x < 6$ by -2, you must reverse: $x > -3$, *not* $x < -3$. A safe habit: never multiply or divide an inequality by a quantity whose sign you do not know. If you must divide by something like $x - 1$, split into cases $x > 1$ and $x < 1$ and handle each separately.

Example 40.2 (Solving a rational inequality the safe way)**.** Solve $\dfrac{x+1}{x-2} \geq 0$. Do *not* cross-multiply — you do not know the sign of $x - 2$. Instead find where each factor is zero or undefined: the numerator vanishes at $x = -1$, the denominator at $x = 2$ (where the expression is undefined). These split the line into three regions. Test one point in each:

region	$x < -1$	$-1 < x < 2$	$x > 2$
sign of $\frac{x+1}{x-2}$	$(-)/(-) = +$	$(+)/(-) = -$	$(+)/(+) = +$

So the expression is ≥ 0 on $x \leq -1$ and $x > 2$ (including $x = -1$ where it equals 0, excluding $x = 2$ where it is undefined). The solution is $(-\infty, -1] \cup (2, \infty)$.

The whole method is a *sign chart*: mark the points where the expression is zero or undefined, then record the sign in each gap. Figure 40.4 shows it at a glance.

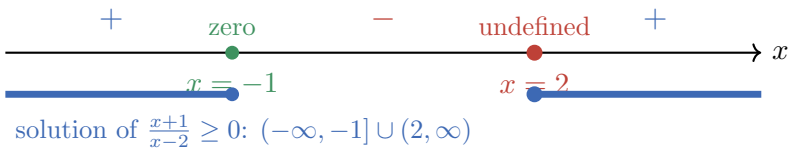


Figure 40.4: Sign chart for $\frac{x+1}{x-2}$. The two critical points — the zero at $x=-1$ (filled, included) and the undefined point at $x=2$ (open, excluded) — cut the line into three regions. One test point per region fixes the sign there. The heavy blue segments below the axis are the solution set, where the fraction is ≥ 0.

40.5 Absolute Value: Distance Made Algebraic

The absolute value $|x|$ is defined by cases,

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0, \end{cases}$$

but the case split hides the real idea, which is geometric:

Intuition

$|x|$ is the *distance from x to 0* on the number line, and more generally $|x-a|$ is the *distance between x and a*. The two cases just say "distance is never negative; if the number is already to the right of 0 its distance is itself, and if it is to the left we negate to make it positive." Read every absolute value as a distance and most manipulations become obvious.

This distance reading immediately explains how absolute-value inequalities unfold:

$$|x-a| < r \quad \Longleftrightarrow \quad a-r < x < a+r,$$

"x is within r of a," an *interval* centered at a with radius r. And the opposite,

$$|x-a| > r \quad \Longleftrightarrow \quad x < a-r \ \text{ or } \ x > a+r,$$

"x is farther than r from a," the *outside* of that interval.

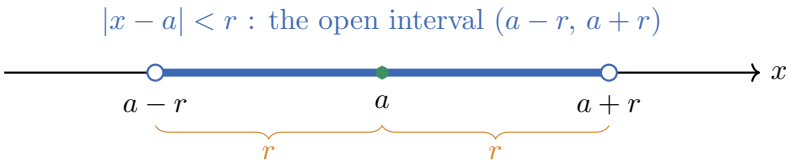


Figure 40.5: $|x-a| < r$ reads as "x is within distance r of a," so the solution set is the open interval $(a-r, a+r)$ — a fixed center a with a radius r in each direction. This exact picture is the geometry behind the ε-δ definition of a limit.

Connection

This is the whole shape of a limit, not a side topic. The statement $\lim_{x\to a} f(x) = L$ unpacks to: for every tolerance $\varepsilon > 0$ there is a $\delta > 0$ so that $|x - a| < \delta$ forces $|f(x) - L| < \varepsilon$. Both halves are absolute-value inequalities, i.e. "within distance" statements: *whenever x is within δ of a, $f(x)$ is within ε of L.* Figure 40.5 is, quite literally, the inner machinery of the central definition of calculus.

Two algebraic facts about absolute value earn their keep constantly.

Focus

Useful identities.

$$|xy| = |x|\,|y|, \qquad |x|^2 = x^2, \qquad \sqrt{x^2} = |x|.$$

The triangle inequality. For all real (indeed all complex) x, y,

$$|x + y| \le |x| + |y|.$$

The triangle inequality turns up more than any other inequality in analysis. Its name comes from the picture in the plane: the straight side of a triangle is no longer than the sum of the other two sides. On the line it says a detour through the origin is never shorter than going direct.

Why the technique works

Here is a clean proof of the triangle inequality. For any real t, $-|t| \le t \le |t|$ (a number lies between $\pm$ its size). Apply this to x and to y and add:

$$-|x| \le x \le |x|, \qquad -|y| \le y \le |y|$$

add the left and right ends:

$$-(|x| + |y|) \le x + y \le |x| + |y|.$$

A statement of the form $-c \le s \le c$ (with $c \ge 0$) is exactly $|s| \le c$. With $s = x + y$ and $c = |x| + |y|$ this reads $|x + y| \le |x| + |y|$. Done. The whole proof is the single observation that every number is sandwiched by $\pm$ its absolute value.

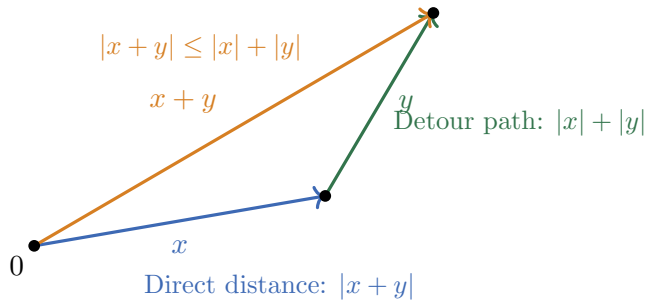


Figure 40.6: The geometric soul of the triangle inequality: the straight-line displacement $|x+y|$ from the origin to $x + y$ is never longer than the two-step detour $|x| + |y|$ that visits x along the way. Equality holds if and only if x and y point in the exact same direction.

40.6 Functions: The Facts Calculus Leans On

Functions get a full chapter early in the book; here we collect, for reference, the handful of facts the later chapters use without re-deriving.

A function $f\colon D \to \mathbb{R}$ assigns to each input x in its domain D *exactly one* output $f(x)$. The "exactly one" is the defining demand — it is what the *vertical line test* checks on a graph: a curve is the graph of a function precisely when no vertical line meets it twice.

Composition. $(g \circ f)(x) = g(f(x))$: do f first, then feed the result to g. Order matters; $g \circ f$ and $f \circ g$ are usually different functions. The chain rule is the derivative of a composition, so composition is not a curiosity — it is half of differential calculus.

Inverse functions. f^{-1} undoes f: $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$. An inverse exists exactly when f is *one-to-one* (each output comes from only one input), which the *horizontal line test* checks. The graph of f^{-1} is the graph of f reflected across the line $y = x$, because inverting swaps the roles of input and output, i.e. swaps the axes.

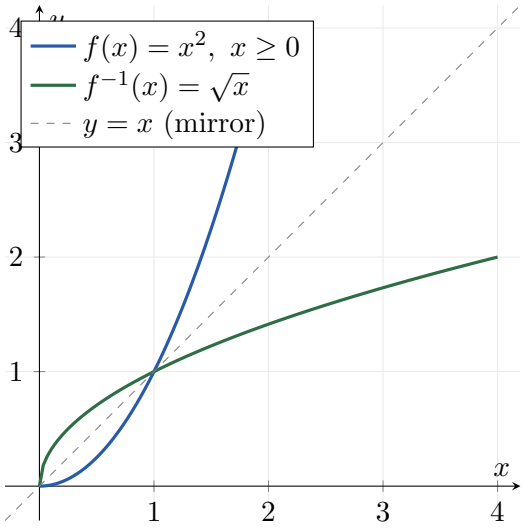


Figure 40.7: A function and its inverse are mirror images across the line $y = x$. Here $f(x) = x^2$ (restricted to $x \geq 0$ so it is one-to-one) and $f^{-1}(x) = \sqrt{x}$. Reflecting across $y = x$ swaps the x- and y-axes, which is exactly the algebra of solving $y = x^2$ for x.

Even and odd symmetry. f is *even* if $f(-x) = f(x)$ (graph symmetric across the y-axis, like x^2 or $\cos x$); *odd* if $f(-x) = -f(x)$ (graph symmetric through the origin, like x^3 or $\sin x$). These symmetries cut work in half: the integral of an odd function over a symmetric interval $[-a, a]$ is automatically 0.

Monotonicity. f is *increasing* if $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$, decreasing if the inequality reverses. A strictly monotonic function is automatically one-to-one, hence invertible. The derivative's sign detects monotonicity — this is the bridge from calculus back to this algebraic notion.

Common trap

f^{-1} means the *inverse function*, not the reciprocal. $\sin^{-1} x$ is the inverse sine (arcsine), not $1/\sin x$. The reciprocal is written $(\sin x)^{-1}$ or $1/\sin x = \csc x$. The clash of notation is unfortunate but standard; context and a sharp eye are your defense.

40.7 What a Proof Is, and Three Reliable Shapes

A theorem is a true statement; a proof is the evidence that compels belief. But "evidence" in mathematics is stricter than anywhere else: a proof is a chain of steps, each following from the previous by pure logic or from facts already granted, leaving *no* gap a skeptic could pry open. Checking a million cases is not a proof if there is a million-and-first. The reward for this strictness is total certainty: a theorem, once proved, is true forever and everywhere its hypotheses hold.

Why this matters

Why insist on proof, when experiment and pattern-spotting got us this far? Because patterns lie. Consider $f(n) = n^2 + n + 41$. For $n = 0, 1, 2, \ldots, 39$ every value is prime — forty cases in a row, overwhelming "evidence." But $f(40) = 40^2 + 40 + 41 = 41^2$ is not prime. Forty confirmations could not save a false conjecture; one proof would have settled it either way. Calculus is built on infinitely many cases (every x, every ε), so we *cannot* check them one by one. Proof is the only tool that reaches across an infinity at once.

We will meet three shapes of proof again and again. Each is a strategy for getting from hypotheses to conclusion.

40.7.1 Direct proof

The most natural shape: assume the hypothesis, and march forward by valid steps until you reach the conclusion. Most "if P then Q" theorems are proved this way — start from P, deduce, arrive at Q.

Example 40.3 (A direct proof)**.** *Claim: the sum of two even integers is even.* An even integer is one of the form $2k$ for some integer k (that is the definition we work from). So let the two even integers be $m = 2j$ and $n = 2k$. Then

$$m + n = 2j + 2k = 2(j + k).$$

Since $j + k$ is an integer, $m + n$ has the form $2 \times$ (integer), which is exactly the definition of even. Therefore $m + n$ is even. □

Notice the rhythm: unfold the definitions of the hypotheses, do careful algebra, re-fold into the definition of the conclusion. That is the whole craft of a direct proof.

40.7.2 Proof by contradiction

Sometimes the conclusion is awkward to reach head-on, but its *negation* is easy to play with. So we assume the opposite of what we want, follow the consequences, and crash into an impossibility.

Since valid reasoning from a true start can never reach a falsehood, our starting assumption (the negation) must have been false — so the original claim is true.

History

The classic example, and one of the oldest, is Euclid's proof (c. 300 BCE) that $\sqrt{2}$ is irrational — that no fraction of whole numbers equals it. The Pythagoreans are said to have found the existence of such "incommensurable" lengths so disturbing that the discovery was guarded as a secret; it shattered their creed that all of nature was ratios of whole numbers.

Example 40.4 ($\sqrt{2}$ is irrational — proof by contradiction)**.** *Suppose, for contradiction, that* $\sqrt{2}$ *is rational.* Then we can write $\sqrt{2} = p/q$ as a fraction in *lowest terms* — meaning p and q share no common factor (any common factor could be cancelled first). Square both sides:

$$2 = \frac{p^2}{q^2} \quad \Longrightarrow \quad p^2 = 2q^2.$$

So p^2 is even, which forces p itself to be even (if p were odd, p^2 would be odd). Write $p = 2r$. Substituting,

$$(2r)^2 = 2q^2 \implies 4r^2 = 2q^2 \implies q^2 = 2r^2.$$

Now q^2 is even, so q is even too. But then p and q are *both* even — they share the common factor 2 — contradicting our choice of a fraction in lowest terms. The contradiction means our supposition was impossible: $\sqrt{2}$ cannot be written as such a fraction. Hence $\sqrt{2}$ is irrational. □

What is really happening

What really powered that proof? A single squeeze: we assumed a "smallest" (lowest terms) representation and then manufactured a smaller one, which cannot be. This "no smallest counterexample" engine — formally, the well-ordering of the positive integers, the fact that any nonempty set of them has a least element — is the same engine that drives induction, the next technique. Contradiction and induction are two faces of one principle about $\mathbb{N}$.

40.7.3 Proof by induction

Induction proves a statement $P(n)$ for *every* natural number n at once, even though there are infinitely many. The image to hold is a line of dominoes.

Intuition

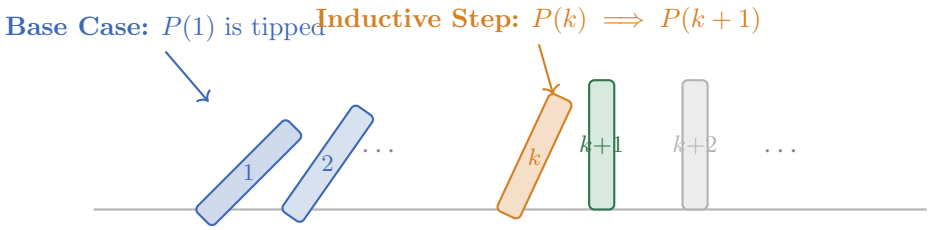
Imagine infinitely many dominoes standing in a row, one for each $n = 1, 2, 3, \ldots$. You want to know they all fall. You do not push each one — there are infinitely many. Instead you check two things: (1) the **first** domino falls, and (2) **each** domino, if it falls, knocks over the next. Granted those two facts, every domino must fall: the first falls by (1), which by (2) topples the second, which topples the third, forever. Induction is exactly this.

Focus

Principle of mathematical induction. To prove $P(n)$ holds for all integers $n \geq 1$, prove two things:

- **Base case:** $P(1)$ is true. (The first domino falls.)
- **Inductive step:** for every k, *if* $P(k)$ is true *then* $P(k+1)$ is true. (Each domino topples the next.)

The assumption "$P(k)$ is true" inside the step is called the *inductive hypothesis*. You are allowed to assume it precisely because you are about to show it would force the next case.



If the first falls, and each falling domino strikes its neighbor, the entire infinite chain must fall.

Figure 40.8: The domino model of mathematical induction. Establishing $P(1)$ guarantees the chain reaction starts. Establishing $P(k) \implies P(k+1)$ ensures the impulse propagates from each tile to the next. Together, they sweep across all integers $n \geq 1$ in a single logical stroke.

Example 40.5 (Sum of the first n integers)**.** *Claim:* for every $n \geq 1$,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Base case ($n = 1$). The left side is just 1; the right side is $\frac{1 \cdot 2}{2} = 1$. They agree, so $P(1)$ holds.

Inductive step. Assume the formula holds for $n = k$ (the inductive hypothesis):

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}.$$

We must show it then holds for $n = k + 1$. Add $(k+1)$ to both sides:

$$1 + 2 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1).$$

Factor $(k+1)$ out of the right-hand side:

$$\frac{k(k+1)}{2} + (k+1) = (k+1)\left(\frac{k}{2} + 1\right) = (k+1) \cdot \frac{k+2}{2} = \frac{(k+1)(k+2)}{2}.$$

That last expression is exactly the original formula with n replaced by $k+1$. So $P(k) \Rightarrow P(k+1)$. By induction the formula holds for all $n \geq 1$. $\square$

History

A famous (probably embroidered) story has the young Carl Friedrich Gauss, set the busywork of adding 1 through 100, answering almost instantly: 5050. His trick was to pair the sum with itself reversed — $1+100$, $2+99$, …, each pair totalling 101, and there are 100 of them, giving $100 \cdot 101 = 10100$, so the sum is half that, 5050. That pairing is the formula $n(n+1)/2$ seen geometrically, and it is a second, "direct," proof of the same fact. It is healthy to know a result two ways: induction certifies it; Gauss's pairing explains *why* the answer takes that shape.

Common trap

Induction without a base case proves nothing. "If $P(k)$ then $P(k+1)$" alone is a chain of dominoes that nobody pushed — the implications can all be valid while the statement is false for every n. Always nail down the base case. Conversely, a base case with a broken inductive step (a gap where $P(k)$ does not actually force $P(k+1)$) is equally worthless: one fallen domino, then nothing.

Connection

Induction is not confined to "sum" formulas. It underlies the power rule $\frac{d}{dx}x^n = nx^{n-1}$ for positive integer n (prove it for $n=1$, then use the product rule to climb from x^k to x^{k+1}), the binomial theorem, and the convergence arguments behind Taylor series. Whenever a statement is indexed by a whole number and each case rests on the one before, induction is the tool. It is the discrete cousin of the "accumulate one step at a time" idea that integration makes continuous.

40.8 The Binomial Theorem: Algebra and Counting Together

One identity deserves a section of its own because it threads through the entire book — it appears in the power rule's derivation, in Taylor's theorem, and in the definition of e. It answers: what is $(a+b)^n$ expanded?

We already know $(a+b)^2 = a^2+2ab+b^2$ and $(a+b)^3 = a^3+3a^2b+3ab^2+b^3$. The coefficients $1,2,1$ and $1,3,3,1$ are the rows of *Pascal's triangle*, where each entry is the sum of the two above it:

$$\begin{array}{ccccccc} & & & 1 & & & \\ & & 1 & & 1 & & \\ & 1 & & 2 & & 1 & \\ 1 & & 3 & & 3 & & 1 \\ 1 & & 4 & & 6 & & 4 & & 1 \end{array}$$

Theorem 40.1 (Binomial theorem)**.** *For any nonnegative integer n,*

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k}b^k, \qquad \binom{n}{k} = \frac{n!}{k!\,(n-k)!}.$$

What is really happening

Where do the coefficients $\binom{n}{k}$ come from? Expanding $(a+b)^n = (a+b)(a+b)\cdots(a+b)$ (n factors) means choosing, from each factor, either the a or the b, then multiplying the choices. A term $a^{n-k}b^k$ arises every time we pick b from exactly k of the n factors. The number of ways to choose which k factors contribute a b is $\binom{n}{k}$ — "n choose k." So the binomial theorem is really a *counting* statement in algebraic disguise: the coefficient is the count of ways a given mix of a's and b's can occur. This is why probability and algebra share the same triangle.

Example 40.6 (Why it matters for the power rule)**.** The derivative of x^n from the limit definition needs $(x+h)^n$. The binomial theorem gives

$$(x+h)^n = x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + h^n.$$

Subtract x^n and divide by h:

$$\frac{(x+h)^n - x^n}{h} = nx^{n-1} + \binom{n}{2}x^{n-2}h + \cdots + h^{n-1}.$$

Every term after the first carries at least one factor of h, so as $h \to 0$ they all vanish, leaving

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^n) = nx^{n-1}.$$

The binomial theorem isolated the one term that survives the limit. The power rule is the first coefficient of Pascal's triangle in motion.

40.9 Putting It Together: A Worked Gauntlet

To see these tools cooperate, run one expression through several of them.

Example 40.7 (Simplify a difference quotient)**.** Simplify $\dfrac{1}{h}\left(\dfrac{1}{x+h} - \dfrac{1}{x}\right)$ and find its limit as $h \to 0$ — this is the derivative of $1/x$ from scratch.

Step 1: combine the inner fractions over the common denominator $x(x+h)$:

$$\frac{1}{x+h} - \frac{1}{x} = \frac{x-(x+h)}{x(x+h)} = \frac{-h}{x(x+h)}.$$

Step 2: divide by h (multiply by $1/h$); the h's cancel, the move that makes the limit tractable:

$$\frac{1}{h}\cdot\frac{-h}{x(x+h)} = \frac{-1}{x(x+h)}.$$

Step 3: take the limit $h \to 0$. Now the expression is continuous at $h = 0$, so just substitute:

$$\lim_{h\to 0}\frac{-1}{x(x+h)} = \frac{-1}{x\cdot x} = -\frac{1}{x^2}.$$

So $\frac{\mathrm{d}}{\mathrm{d}x}(1/x) = -1/x^2$. Every step was pure appendix material — common denominators, sign care, cancellation — yet the result is a genuine calculus fact. This is the appendix earning its place.

Work it yourself

Run the same gauntlet on $\dfrac{\sqrt{x+h}-\sqrt{x}}{h}$. The right tool is the conjugate: multiply top and bottom by $\sqrt{x+h}+\sqrt{x}$ and use the difference of squares (40.3) to turn the top into $(x+h)-x=h$, which cancels the h below. You should land on $\dfrac{1}{2\sqrt{x}}$ in the limit — the derivative of $\sqrt{x}$. Notice it is the same three-act structure: rewrite, cancel the h, substitute.

40.10 The Scope and Limits of Finite Algebra

Every algebraic identity and proof archetype in this appendix was designed to turn intuitive steps into an airtight, repeatable discipline. Factoring identities, difference-of-squares conjugates, and rational expressions allow us to dissect difference quotients $\frac{f(x+h)-f(x)}{h}$ and cleanly isolate and cancel the vanishing factor h before passing to the limit. Extending integer exponents to negative, zero, and rational powers systematically brings reciprocal functions and roots into the single power rule $\frac{\mathrm{d}}{\mathrm{d}x}(x)^r = rx^{r-1}$. Meanwhile, inequalities and absolute values give geometric tolerances an algebraic tongue: stating that $|x-a|<\delta$ forces $|f(x)-L|<\varepsilon$ translates the idea of getting arbitrarily close into the testable contract of limits, while induction and contradiction let us verify claims across infinitely many cases at once.

Yet for all its versatility, elementary algebra possesses a fundamental boundary: *algebra is strictly finite and rational.* Addition, multiplication, factoring, and polynomial expansions are finite operations. If our universe of numbers is restricted to the rationals $\mathbb{Q}$, every field axiom and algebraic identity in this appendix holds without flaw, yet calculus completely collapses.

The rational line is perforated with gaps. As Euclid's contradiction proof established, $\sqrt{2}$ is not rational. An increasing sequence of rationals like $1, 1.4, 1.41, 1.414, \ldots$ has terms squeezing closer together, yet it possesses no landing point in $\mathbb{Q}$. On a rational domain, the Intermediate Value Theorem fails: the curve $f(x)=x^2-2$ leaps cleanly from $f(1)=-1$ to $f(2)=2$ without ever hitting zero, stepping directly through an invisible hole. Continuous bounded functions on closed rational intervals frequently fail to attain maximum values because the supremum sits in a hole.

Connection

To turn symbolic algebra into an unshakable foundation for calculus, we must move beyond finite operations into *order completeness.* This brings us to our final destination: Appendix B (*The Real Numbers and Completeness*). There we explore how Dedekind cuts and Cauchy sequences fill the gaps of $\mathbb{Q}$, establishing the Least Upper Bound Property that guarantees every bounded sequence converges, every continuous curve crossing zero strikes the axis, and every definite integral represents a genuine, well-defined real number.

Exercises

Exercise 40.1. Factor completely over the reals: (a) x^3-4x; (b) x^4-16; (c) x^2-6x+9; (d) $8x^3-27$.

For each, state how you recognized the pattern (common factor, difference of squares, perfect square, difference of cubes).

Exercise 40.2. Simplify to a single power (no negative or fractional exponents in the final answer where avoidable): (a) $\dfrac{a^5 a^{-2}}{a^{-4}}$; (b) $\left(8x^6\right)^{1/3}$; (c) $\dfrac{(2x^2)^3}{4x^4}$; (d) $\dfrac{x^{1/2}\,x^{1/3}}{x^{-1/6}}$. Explain at each step which exponent law you are invoking.

Exercise 40.3. Solve $2x^2 - 4x - 3 = 0$ two ways: once with the quadratic formula, and once by completing the square. Confirm the answers agree, and state the discriminant and what it tells you about the roots.

Exercise 40.4. Solve each inequality, writing the answer in interval notation, and say where (if anywhere) the "flip" rule was needed: (a) $3 - 2x \le 11$; (b) $\dfrac{x-1}{x+2} \le 0$; (c) $x^2 < 5x - 6$.

Exercise 40.5. Rewrite each as an interval or union of intervals on the number line, and sketch it: (a) $|x - 3| < 2$; (b) $|2x + 1| \ge 5$; (c) $|x| = |x - 4|$ (interpret this one as "equidistant from 0 and 4" before computing).

Exercise 40.6. Use the triangle inequality to prove the *reverse triangle inequality* $\big|\,|x|-|y|\,\big| \le |x-y|$. (Hint: write $x = (x - y) + y$ and apply $|x + y| \le |x| + |y|$ to get $|x| \le |x - y| + |y|$; rearrange, then do the same with the roles of x and y swapped.)

Exercise 40.7. Let $f(x) = \dfrac{2x-1}{x+3}$. (a) Find f^{-1} by solving $y = f(x)$ for x. (b) Verify $f(f^{-1}(y)) = y$. (c) State the domain of f and the domain of f^{-1}, and explain why they are what they are.

Exercise 40.8. Give a direct proof that the product of two odd integers is odd. (Start from the definition: an odd integer has the form $2k + 1$.)

Exercise 40.9. Prove by contradiction that $\sqrt{3}$ is irrational. Identify the one place where the argument differs from the $\sqrt{2}$ proof of Example 40.4 (hint: you will use that "$3 \mid p^2 \Rightarrow 3 \mid p$," which deserves a sentence of justification).

Exercise 40.10. Prove by induction that for all $n \ge 1$,

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Be explicit about your base case, your inductive hypothesis, and the algebra that turns the $n = k$ formula into the $n = k + 1$ formula.

Exercise 40.11. Prove by induction that $2^n > n$ for every integer $n \ge 1$. (This is an *inequality* induction; in the step, add to or multiply the inductive hypothesis to reach the $k + 1$ case.)

Exercise 40.12. Use the binomial theorem (Theorem 40.1) to expand $(1 + x)^4$, and verify your coefficients against the corresponding row of Pascal's triangle. Then use the same expansion to write down the first three terms of the difference quotient $\big((1 + h)^4 - 1\big)/h$ and read off the derivative of x^4 at $x = 1$.

Chapter 41

Appendix: The Real Numbers and Completeness

Every theorem in this book that begins with the comforting phrase "there exists" — a limit exists, a continuous function attains a maximum, the curve crosses zero somewhere, the bounded increasing sequence settles down to something — is, secretly, a promise about the *number line*. Not about derivatives, not about integrals, but about the raw substance the whole edifice stands on: the real numbers themselves. If that substance had even a single pinhole in it, those existence theorems would be false, and calculus would collapse into a beautiful but hollow formalism.

So this appendix takes up the question we have been politely postponing since the first chapter on limits. *What are the real numbers, really?* We all draw the number line as a smooth, unbroken ruler with no gaps. But why should it be unbroken? The rationals — fractions p/q — already seem to fill the line densely: between any two of them sits another, and another, forever. Where would a gap even hide? The startling answer, known to the Greeks and quietly terrifying to them, is that the rationals are *full of holes*, and that one of those holes sits exactly where $\sqrt{2}$ ought to be.

The cure for the holes is a single axiom — the **completeness axiom**, also called the least-upper-bound property. It is the most important sentence in analysis, and the least famous. Once you see it, you will recognise it hiding inside the Intermediate Value Theorem, the Extreme Value Theorem, the convergence of monotone sequences, the existence of the integral, and every limit you have ever computed. What follows is the story of the holes and the axiom that fills them.

41.1 A gap in the rationals

Start with the picture, where the trouble first becomes visible. Draw the rational points on a line. Between 0 and 1 you can mark $\frac{1}{2}$, and then $\frac{1}{3}, \frac{2}{3}$, and $\frac{1}{4}, \frac{3}{4}$, and on and on. The marks pile up so thickly that no matter how powerful a magnifying glass you bring, you will always find infinitely many rationals in any window, however small.

Intuition

The rationals are **dense**: between any two distinct rationals $a < b$ lies another rational, for example their average $\frac{a+b}{2}$. Repeat the averaging and you get infinitely many. Visually, the rational marks leave no *visible* space anywhere on the line. It is enormously tempting to conclude that they leave no space at all — that the rationals already *are* the continuous line. This is the trap. Density is not the same as having no holes. A sieve can be dense with metal and still be mostly holes.

To find the hole, we go hunting for the length the Greeks found: the diagonal of a unit square. By the Pythagorean theorem, a square one unit on a side has a diagonal of length d with $d^2 = 1^2 + 1^2 = 2$. So the number $\sqrt{2}$ is not an abstraction — it is a length you can draw with a ruler and compass. The question is whether that drawable length sits on a *rational* mark. It does not.

Theorem 41.1 (Irrationality of $\sqrt{2}$)**.** *There is no rational number r with $r^2 = 2$.*

Proof. Suppose, for contradiction, that such a rational exists. Write it in lowest terms as $r = p/q$, where p and q are integers with no common factor (we can always cancel until this holds). Then

$$\frac{p^2}{q^2} = 2 \quad \Longrightarrow \quad p^2 = 2q^2.$$

The right-hand side is even, so p^2 is even. But the square of an odd number is odd (an odd number is $2k+1$, and $(2k+1)^2 = 4k^2 + 4k + 1$ is odd), so p itself must be even. Write $p = 2m$. Substituting,

$$(2m)^2 = 2q^2 \quad \Longrightarrow \quad 4m^2 = 2q^2 \quad \Longrightarrow \quad q^2 = 2m^2.$$

Now the same argument applies to q: q^2 is even, so q is even. But then p and q are *both* even — they share the factor 2 — contradicting our assumption that the fraction was in lowest terms. The assumption was impossible. Hence no rational squares to 2. □

History

This argument, or something close to it, is among the oldest theorems in mathematics, attributed to the Pythagoreans of the fifth century BCE. The Pythagorean creed held that *all is number*, by which they meant whole numbers and their ratios. The discovery that the diagonal of a square is *incommensurable* with its side — that no common unit measures both — was a genuine philosophical crisis. Legend, almost certainly embroidered, says the man who revealed it, Hippasus, was drowned at sea for the impiety. What survived the crisis was the realisation that geometry knows lengths that arithmetic (of fractions) cannot name. Closing that gap between geometry and arithmetic took two and a half thousand years and was not finished until the 1870s, with the work of Dedekind and Cantor that this appendix sketches.

So there is a definite, constructible point on the line — the length $\sqrt{2}$, roughly $1.41421356\ldots$ — that no fraction lands on. The rationals, dense as they are, have a hole exactly there. And not only there: between any two rationals you can run the same trick (scale the unit square), so the holes are everywhere too. The rational line is simultaneously dense with marks and riddled with gaps. Both at once.

What is really happening

Here is the true mechanism of the gap — the very thing the completeness axiom will repair. Consider the two sets of rationals

$$A = \{x \in \mathbb{Q} : x > 0 \text{ and } x^2 < 2\}, \qquad B = \{x \in \mathbb{Q} : x > 0 \text{ and } x^2 > 2\}.$$

Every positive rational lands in exactly one of A or B (none has square equal to 2, by Theorem 41.1). Every element of A is less than every element of B. The set A has no *largest* member: given any $x \in A$, you can always nudge it up slightly and stay below $\sqrt{2}$. The set B has no *smallest* member: any $x \in B$ can be nudged down slightly and stay above $\sqrt{2}$. So the line of rationals splits cleanly into a left part with no right end and a right part with no left end — and *nothing sits in the seam between them.* That empty seam is the hole. In the reals there will always be exactly one number in such a seam; in the rationals, sometimes there is none. That single difference is completeness.

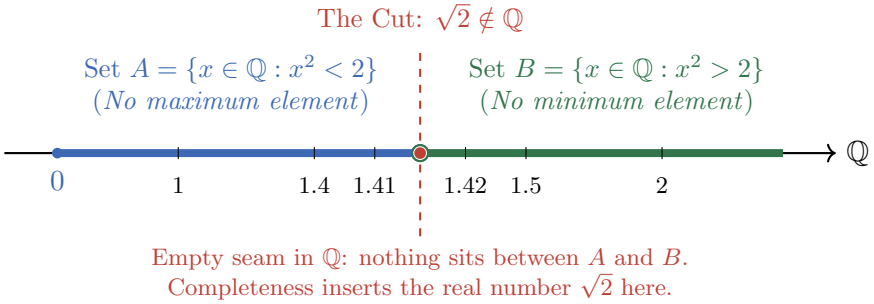


Figure 41.1: A Dedekind cut dividing the rational numbers at $\sqrt{2}$. Set A contains all positive rationals with $x^2 < 2$ (blue); set B contains all rationals with $x^2 > 2$ (green). Because $\sqrt{2}$ is irrational, set A has no largest element and set B has no smallest element. The seam between them is completely empty in $\mathbb{Q}$. Dedekind's breakthrough was defining the real number $\sqrt{2}$ to *be* this partition.

Squeezing the hole between fractions

We can watch the hole numerically. Build two sequences of rationals, one climbing toward $\sqrt{2}$ from below and one descending toward it from above, by choosing more and more decimal digits:

n	below: a_n	above: b_n
1	1.4	1.5
2	1.41	1.42
3	1.414	1.415
4	1.4142	1.4143
5	1.41421	1.41422

Each a_n is a rational with $a_n^2 < 2$ and each b_n is a rational with $b_n^2 > 2$, and the gap $b_n - a_n = 10^{-n}$ shrinks to zero. The two sequences close in on a single location on the line — you can pin it to within any tolerance you like — and yet *that location is occupied by no rational at all.* The fractions march right up to the edge of a point that, in their world, simply is not there.

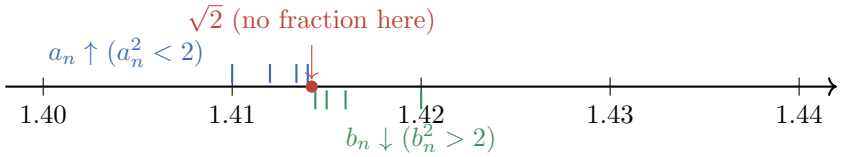


Figure 41.2: Rationals from the left (a_n, blue) and from the right (b_n, green) close in on the location of $\sqrt{2}$. The gap $b_n - a_n = 10^{-n}$ shrinks to zero, pinning a single point — but in the rational line that point is a hole. Completeness is the axiom that puts a real number there.

Why the technique works

Why should we care about one missing point when the rationals are good enough for every measurement we will ever physically make? Because calculus is not built on individual numbers; it is built on *limits*, and a limit is precisely the act of letting a sequence like a_n above "arrive somewhere." If the place it is heading does not exist, the limit does not exist, and every theorem that produces a number by a limiting process fails. The increasing sequence $1.4, 1.41, 1.414, \ldots$ is bounded above (by 2, say) and increasing, so any sane theory of limits must declare it convergent — but in $\mathbb{Q}$ it converges to nothing. The rationals are a theory of arithmetic with a fatal flaw for calculus: they let you *build* convergent processes whose *destinations* they do not contain. We need a number system with no such missing destinations.

41.2 Upper bounds, suprema, and the language of ceilings

To state the axiom that fixes everything, we first need vocabulary for "the place a set is heading toward from above." This is the notion of a *least upper bound*, and it deserves to be built slowly: it is subtle, and it underlies everything that follows.

Definition 41.1 (Bounded above; upper bound)**.** Let S be a nonempty set of real numbers. A number M is an **upper bound** for S if $x \le M$ for every $x \in S$. If S has at least one upper bound, we say S is **bounded above**. *Lower bound* and **bounded below** are defined the same way with the inequality reversed.

An upper bound is a ceiling: a number that no element of S exceeds. A set can have many ceilings. The interval $S = (0, 1)$ has 1 as a ceiling, but also 2, also 17.5, also a billion. Ceilings come in a whole upward family; once one number is a ceiling, so is every number above it.

The interesting ceiling is the *lowest* one — the tightest, the one that hugs the set from above with no slack.

Definition 41.2 (Supremum / least upper bound)**.** A number L is the **supremum** (or **least upper bound**) of a nonempty set S, written $L = \sup S$, if both:

1. L is an upper bound for S (nothing in S exceeds L), and
2. L is the *least* such bound: if M is any upper bound for S, then $L \le M$.

The mirror notion, the **infimum** or greatest lower bound, is written $\inf S$.

Intuition

Think of lowering a ceiling slowly toward a crowd of people. You keep lowering until, just before it would touch someone's head, you stop. That stopping height is the supremum: the lowest you can put the ceiling while still clearing everyone. Two flavours can happen at the moment you stop. Either the ceiling is resting on the tallest person's head (the supremum belongs to the set — it is a *maximum*), or there is no single tallest person, only an unending sequence of ever-taller people whose heights approach the ceiling without any one reaching it (the supremum is the limit of their heights but belongs to no individual). The supremum captures *both* cases in one word, which is exactly why it is more powerful than "maximum."

The second flavour is the one to internalise, because it is where the supremum earns its keep. Consider the open interval $S = (0,1) = \{x : 0 < x < 1\}$. There is no largest element: whatever $x < 1$ you name, the number $\frac{x+1}{2}$ is larger and still below 1. So S has no maximum. But it plainly has a least upper bound: $\sup S = 1$. The number 1 is an upper bound, and nothing smaller can be, because any $M < 1$ is itself beaten by an element of S (the point $\frac{M+1}{2}$ lies in S and exceeds M). The supremum exists even though the maximum does not. *This is precisely why the concept was invented.*

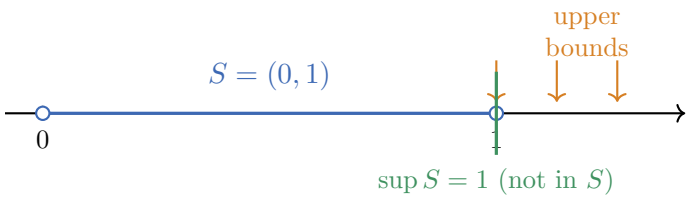


Figure 41.3: The open interval $(0,1)$ has no largest element — the right endpoint is missing (hollow circle) — yet it has a least upper bound, $\sup S = 1$. The orange arrows mark some of the upper bounds; the supremum is the leftmost of them, sitting exactly at the missing edge.

Focus

A clean way to test a candidate L for being $\sup S$ uses two complementary promises, and you will use this constantly in proofs:

1. **Ceiling:** $x \leq L$ for all $x \in S$. (Nothing pokes above L.)
2. **Tightness:** for every $\varepsilon > 0$ there is some $x \in S$ with $x > L - \varepsilon$. (You cannot lower the ceiling by any amount ε, however tiny, without cutting into the set.)

The second promise is the operational heart of the supremum. It says: elements of S come *arbitrarily close* to L from below. Whenever you see "sup" in a proof, reach for this ε characterisation — it converts the abstract "least upper bound" into a concrete sequence of elements creeping up to L, which is something you can compute with.

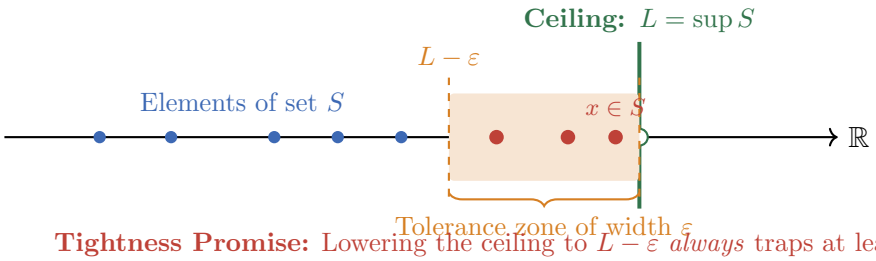


Figure 41.4: The ε-characterisation of the supremum. $L = \sup S$ acts as an impassable ceiling for S. Tightness asserts that no matter how tiny the tolerance $\varepsilon > 0$ is, lowering the ceiling to $L - \varepsilon$ breaches the set: there must exist an element $x \in S$ inside the orange corridor $(L - \varepsilon, L]$.

Common trap

Do not confuse the supremum with the maximum, and do not assume the supremum belongs to the set. For $(0, 1)$, the supremum is 1 but $1 \notin (0, 1)$. For $[0, 1]$, the supremum is 1 and $1 \in [0, 1]$, so here it is also the maximum. A maximum is a supremum that happens to be attained; a supremum need not be attained. The reverse trap also bites: a set can have a supremum without a maximum, but it can never have a maximum without that maximum being the supremum. "Max exists" is the strong statement; "sup exists" is the one the axiom will guarantee.

41.3 The completeness axiom

We have seen exactly what goes wrong in $\mathbb{Q}$: the set $A = \{x \in \mathbb{Q} : x > 0,\ x^2 < 2\}$ is bounded above (by 2, say) but has no least upper bound *inside* $\mathbb{Q}$ — its only conceivable least upper bound is $\sqrt{2}$, which $\mathbb{Q}$ does not contain. The seam between A and the larger rationals is empty. The real numbers are, by definition, the system in which this never happens.

Focus

The Completeness Axiom (least-upper-bound property). *Every nonempty set of real numbers that is bounded above has a least upper bound in* $\mathbb{R}$. In symbols: if $S \subseteq \mathbb{R}$ is nonempty and bounded above, then $\sup S$ exists and is a real number.

That is the entire axiom. One sentence. It does not mention $\sqrt{2}$, or limits, or continuity; it speaks only of sets and ceilings. Yet it is the single property that distinguishes $\mathbb{R}$ from $\mathbb{Q}$, and it is the hidden engine of every existence theorem in calculus. For all it leaves unsaid, it delivers a great deal.

What is really happening

What is the axiom *really* saying, stripped of formalism? It says **the line has no holes**. Recall the empty seam from Section 41.1: a left part A with no right end facing a right part with no left end, and nothing between. Completeness forbids exactly this. If S is bounded above, the ceilings of S form a set sitting to the right; the elements of S sit to the left; completeness demands that the boundary between "things in S" and "ceilings of S" be *an*

actual number, $\sup S$. Every potential seam is plugged by a real number. This is why the real line is drawn as an unbroken continuum: not because we can see the absence of holes (we cannot — the rationals look identical to the eye), but because we *axiomatically declare* there are none. Completeness is the mathematical content of the word "continuum."

Remark 41.1. By symmetry (replace S with $-S = \{-x : x \in S\}$ and flip every inequality), the axiom has a twin: *every nonempty set bounded below has a greatest lower bound,* $\inf S$, in $\mathbb{R}$. Suprema and infima always exist for bounded sets. You only ever need to prove the version you find convenient; the other follows for free by reflection.

History

Why is this an *axiom* and not a theorem? Because it depends on what you are willing to take for granted. For two centuries following Newton and Leibniz, calculus operated on brilliant geometric intuition, infinitesimals, and fluent quantities without any formal arithmetic definition of the continuum. Cauchy in his 1821 *Cours d'Analyse* tried to put calculus on a solid foundation of limits, but he tacitly assumed without proof that any sequence whose terms bunch together must converge to an actual number.

The crisis came to a head in the autumn of 1858. Richard Dedekind, then preparing lecture notes on differential calculus at Zurich Polytechnic, was horrified to realize that he had no rigorous way to prove that an increasing bounded sequence has a limit without appealing to the physical, geometric drawing of a line. But using geometry to justify arithmetic is circular, because analytic geometry uses numbers $(x, y) \in \mathbb{R}^2$ to define the line in the first place! Arithmetic had to stand on its own feet.

In the 1870s, four mathematicians independently constructed the reals from the rationals: Richard Dedekind, Georg Cantor, Charles Méray, and Karl Weierstrass. Dedekind's approach (published in 1872 in *Stetigkeit und irrationale Zahlen*) defined a real number to be a "cut"—a partition of $\mathbb{Q}$ into a left set and a right set, gluing a number into every empty seam. Cantor instead defined a real number as an equivalence class of Cauchy sequences of rationals. Both constructions prove completeness as a theorem.

Yet this breakthrough sparked ferocious controversy. Leopold Kronecker, a strict finitary constructivist, waged an intellectual war against Cantor and Dedekind. Kronecker famously proclaimed: *"Die ganzen Zahlen hat der liebe Gott gemacht, alles andere ist Menschenwerk"* ("God made the integers, all else is the work of man"). Kronecker argued that an irrational number like $\sqrt{2}$ has no legitimate existence unless accompanied by a finite algorithm to compute it, rejected uncompleted infinities, and attempted to suppress Cantor's publications. History ultimately sided with Dedekind and Cantor: without their rigorous model of the continuum, the foundation of modern analysis, topology, and differential equations could not stand.

Completeness builds $\sqrt{2}$

To watch the axiom fill the hole, we use it to manufacture $\sqrt{2}$ from nothing but rationals and the axiom itself.

Theorem 41.2 (Existence of square roots)**.** *There is a positive real number L with $L^2 = 2$.*

Proof sketch. Let $S = \{x \in \mathbb{R} : x > 0 \text{ and } x^2 < 2\}$. The set is nonempty (it contains 1, since

$1^2 = 1 < 2$) and bounded above (by 2, since any $x > 2$ has $x^2 > 4 > 2$ and so cannot be in S). By the completeness axiom, $L = \sup S$ exists as a real number. We claim $L^2 = 2$.

Suppose instead $L^2 < 2$. Then L is too small to be the ceiling: we can nudge it up and stay inside S. Concretely, for small $h > 0$,

$$(L+h)^2 = L^2 + 2Lh + h^2 \le L^2 + (2L+1)h \quad \text{(taking } h \le 1\text{)},$$

and choosing h small enough that $(2L+1)h < 2 - L^2$ makes $(L+h)^2 < 2$, so $L + h \in S$ — but $L + h > L$ contradicts L being an *upper* bound.

Suppose instead $L^2 > 2$. Then L is too big to be the *least* ceiling: we can shave it down and still have a ceiling. For small $h > 0$,

$$(L-h)^2 = L^2 - 2Lh + h^2 > L^2 - 2Lh,$$

and choosing h small enough that $2Lh < L^2 - 2$ makes $(L-h)^2 > 2$, so $L - h$ is still an upper bound for S — but $L - h < L$ contradicts L being the *least* upper bound.

Both inequalities are impossible, so $L^2 = 2$. The completeness axiom has produced the number the rationals were missing. □

Why this matters

Pause on what just happened. We never *constructed* $\sqrt{2}$ digit by digit. We described a set — "the positive numbers whose square is less than 2" — and the axiom handed us its ceiling, fully formed, and that ceiling turned out to square to exactly 2. The supremum is a kind of mathematical summoning: name a bounded set, and a specific number materialises at its upper edge. Nearly every existence theorem in calculus follows this pattern. You will describe a set — points where a function is still negative, heights a curve has reached, partial sums so far accumulated — and completeness will summon the number at its edge: the crossing point, the maximum, the limit, the integral. The rest of this appendix is that single move, applied over and over.

41.4 Why calculus needs completeness

We now collect the debts. Four of the load-bearing theorems of this book are, when you trace them to the foundation, restatements of the completeness axiom. We prove the first two in full and sketch the others, so you can see the axiom doing the lifting each time.

Monotone sequences converge

Theorem 41.3 (Monotone Convergence Theorem)**.** *A sequence (a_n) that is increasing ($a_1 \le a_2 \le a_3 \le \cdots$) and bounded above converges to a real limit. In fact $a_n \to \sup\{a_n : n \in \mathbb{N}\}$.*

Proof. Let $S = \{a_n : n \in \mathbb{N}\}$, the set of values the sequence takes. It is nonempty and bounded above by hypothesis, so by completeness $L = \sup S$ exists. We show $a_n \to L$.

Fix any tolerance $\varepsilon > 0$. By the tightness property of the supremum (Section 41.2), $L - \varepsilon$ is not an upper bound, so some term a_N satisfies $a_N > L - \varepsilon$. Because the sequence is increasing, every later term is at least as large: for all $n \geq N$,

$$L - \varepsilon < a_N \leq a_n \leq L < L + \varepsilon,$$

the last inequality because L is an upper bound for all the a_n. Hence $|a_n - L| < \varepsilon$ for all $n \geq N$. Since ε was arbitrary, $a_n \to L$ (Figure 41.5). □

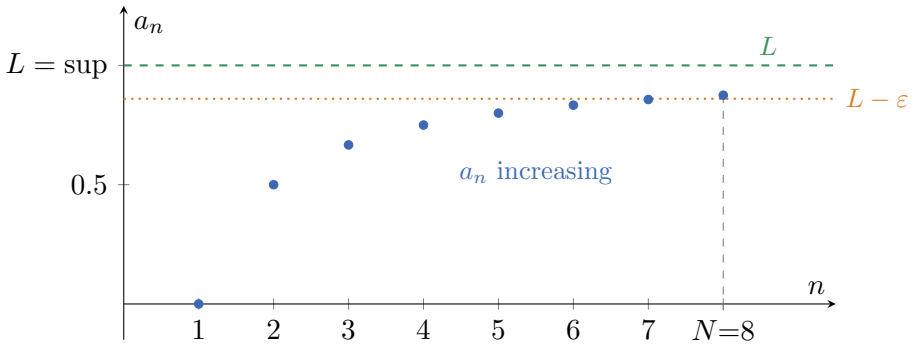


Figure 41.5: Monotone Convergence in one picture. An increasing sequence a_n (blue dots) that is bounded above presses upward against its supremum L (green dashed ceiling). For any tolerance ε, tightness of the supremum forces some term a_N above the line $L - \varepsilon$ (orange dotted); from that index on, every term is trapped in the strip $(L - \varepsilon, L]$, so $a_n \to L$.

Connection

This is the theorem that rescues the sequence $1.4, 1.41, 1.414, \ldots$ from Section 41.1. It is increasing and bounded above by 2, so by Monotone Convergence it converges — and its limit is exactly the $\sqrt{2}$ that completeness summoned in Theorem 41.2. Every decimal expansion you have ever written is secretly an appeal to this theorem: the digits $3.14159\ldots$ define an increasing, bounded sequence of rationals, and completeness is the reason the symbol "π" names an actual number rather than an unfulfilled promise. Without completeness, infinite decimals would be ink with nothing behind them.

The Intermediate Value Theorem

Theorem 41.4 (Intermediate Value Theorem)**.** *If f is continuous on $[a, b]$ and $f(a) < 0 < f(b)$, then there is a point $c \in (a, b)$ with $f(c) = 0$.*

Proof via completeness. Consider the set of inputs on which f is still negative:

$$S = \{x \in [a, b] : f(x) < 0\}.$$

It is nonempty ($a \in S$, since $f(a) < 0$) and bounded above (by b). By completeness, $c = \sup S$ exists, and $c \in [a, b]$. We claim $f(c) = 0$, and we rule out the two ways it could fail using continuity.

If $f(c) < 0$, then by continuity f stays negative on a little interval around c, so points slightly to the *right* of c also lie in S — contradicting c being an upper bound of S. If $f(c) > 0$, then by continuity

f stays positive on a little interval around c, so $c - \delta$ is already an upper bound for S for some $\delta > 0$ — contradicting c being the *least* upper bound. Both fail, so $f(c) = 0$. The crossing point is the supremum of the "still negative" set. □

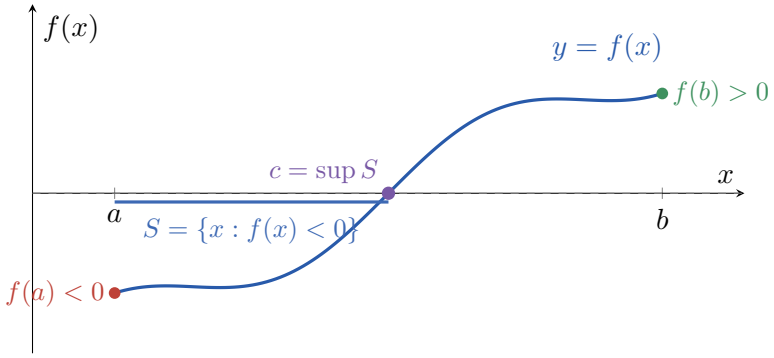


Figure 41.6: The Intermediate Value Theorem as a supremum. The curve starts below the axis and ends above it. The set S where f is still negative (blue segment) is bounded above; completeness summons $c = \sup S$, and continuity forces $f(c) = 0$ — the curve must cross exactly at the edge of S.

Why the technique works

Why does the IVT *require* completeness and not just continuity? Because the very same continuous formula can have a crossing or not, depending only on whether the line has a hole there. Take $f(x) = x^2 - 2$ on $[0, 2]$. It is continuous, $f(0) = -2 < 0$, and $f(2) = 2 > 0$. As a function of *real* inputs it crosses zero at $x = \sqrt{2}$. But as a function of *rational* inputs — exactly the same formula, restricted to $\mathbb{Q}$ — it never equals zero, because $\sqrt{2} \notin \mathbb{Q}$. The function jumps from negative to positive *without ever passing through zero*, because the zero is in a hole. Continuity is satisfied either way; what changes the conclusion is whether the underlying number system has the point $\sqrt{2}$. The IVT is false over $\mathbb{Q}$ and true over $\mathbb{R}$, and the only difference between $\mathbb{Q}$ and $\mathbb{R}$ is completeness. So completeness is not a technicality behind the IVT — it *is* the IVT.

The Extreme Value Theorem

Theorem 41.5 (Extreme Value Theorem, sketch)**.** *A function continuous on a closed bounded interval $[a, b]$ attains a maximum and a minimum value on $[a, b]$.*

The proof has two movements, and completeness drives both. First one shows a continuous function on $[a, b]$ is *bounded* — it cannot run off to $+\infty$. The cleanest argument supposes it is unbounded, extracts inputs x_n with $f(x_n) \to \infty$, and uses completeness (in the guise of the Bolzano–Weierstrass theorem, below) to find a point those inputs cluster around, where continuity then forces a contradiction. Second, knowing the value set $\{f(x) : x \in [a, b]\}$ is bounded above, completeness hands us $M = \sup\{f(x) : x \in [a, b]\}$, and a similar clustering argument produces an actual input c with $f(c) = M$. The maximum *value* is a supremum (which always exists by completeness); the content of the theorem is that this supremum is *attained* (which uses continuity on a *closed* interval). Drop completeness and the bound M might not exist; drop closedness and the supremum might not be attained, exactly as $\sup(0, 1) = 1$ is never reached inside $(0, 1)$.

Theorem 41.6 (Bolzano–Weierstrass, sketch)**.** *Every bounded sequence of real numbers has a convergent subsequence.*

The idea is a binary hunt. Enclose the bounded sequence in an interval $[A_0, B_0]$. Cut it in half; at least one half contains infinitely many terms of the sequence — keep that half and call it $[A_1, B_1]$. Halve again, keep an infinite half, and repeat forever. The left endpoints $A_0 \le A_1 \le A_2 \le \cdots$ form an increasing bounded sequence, so by Monotone Convergence (Theorem 41.3, itself completeness) they converge to a point ℓ; the interval lengths halve to zero, so the right endpoints converge to the same ℓ. Picking one sequence term from each nested interval yields a subsequence trapped in intervals shrinking to ℓ, hence converging to ℓ. The nested intervals close on a single point precisely *because the line has no holes* for them to close on emptily — completeness again.

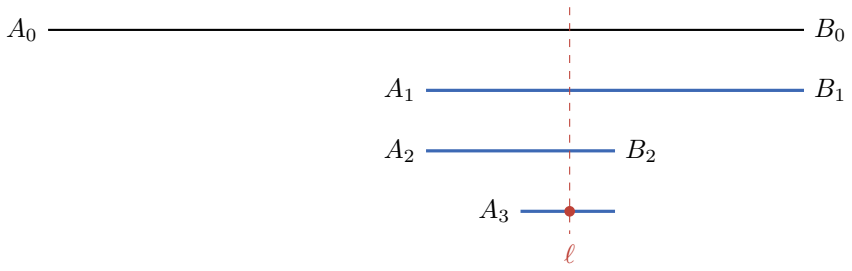


Figure 41.7: Bolzano–Weierstrass by repeated halving. Each interval keeps a half containing infinitely many sequence terms; the nested intervals shrink to a single point ℓ (dashed line), which exists because completeness leaves no empty seam for them to collapse into. A subsequence trapped in these intervals converges to ℓ.

The integral exists

When we built the definite integral as a limit of Riemann sums, we quietly used completeness twice. The lower sums (under-estimates of the area) form a set bounded above by any upper sum; completeness hands us their supremum, the *lower integral*. The upper sums form a set bounded below; completeness hands us their infimum, the *upper integral*. A function is *integrable* exactly when these two meet, and the common value — the integral — is a supremum and an infimum that completeness guaranteed existed in the first place. No completeness, no number for the area to equal.

Why this matters

Notice the unity running through all four. Limit of a monotone sequence: a supremum. Crossing point of a continuous curve: a supremum. Maximum of a continuous function: a supremum. Area under a curve: a supremum meeting an infimum. Each is the same axiom wearing a different costume. The completeness axiom is not one tool among many in analysis; it is the *ground* the tools rest on. Every time this book said "the limit exists," it was cashing a cheque written by this one sentence about ceilings. That is why an appendix on the real numbers is not an optional postscript but the foundation slipped under the whole building after the fact, so you could finally see what was holding it up.

41.5 The Archimedean property: no infinities hiding

Completeness has a quiet consequence worth stating on its own, because it is used silently in nearly every ε-argument: the real line contains no "infinitely large" or "infinitely small" numbers.

Theorem 41.7 (Archimedean property)**.** *For any real number x, there is a natural number n with $n > x$. Equivalently, for any $\varepsilon > 0$ there is a natural number n with $\frac{1}{n} < \varepsilon$.*

Proof. Suppose not: suppose some real x exceeds every natural number, so $\mathbb{N}$ is bounded above by x. Then by completeness $\mathbb{N}$ has a least upper bound $L = \sup \mathbb{N}$. Since L is the *least* upper bound, $L - 1$ is not an upper bound, so some natural number n satisfies $n > L - 1$. But then $n + 1 > L$, and $n + 1$ is also a natural number — contradicting L being an upper bound for $\mathbb{N}$. So no real number exceeds all naturals.

For the second form, given $\varepsilon > 0$ apply the first part to $x = 1/\varepsilon$ to get $n > 1/\varepsilon$, whence $1/n < \varepsilon$. □

What is really happening

The Archimedean property is what guarantees that "$\frac{1}{n}$ can be made as small as you like" — the move you make every single time you push an ε argument through. It rules out a hypothetical positive number smaller than every $\frac{1}{n}$ (an "infinitesimal"), because such a number would force $\mathbb{N}$ to be bounded. Completeness, by plugging every hole, also forbids these phantom infinities and infinitesimals: $\mathbb{R}$ is exactly Archimedean. (One *can* build consistent number systems with genuine infinitesimals — the hyperreals of nonstandard analysis — but they are not the standard reals, and they buy their infinitesimals by giving up the Archimedean property.) When this book writes "for every $\varepsilon > 0$ there is an N," it is leaning on this theorem, which leans on completeness.

Connection

The Archimedean property also explains why the rationals, holes and all, still manage to be *dense* in the reals: between any two reals $a < b$ there is a rational. Choose n with $\frac{1}{n} < b - a$ (Archimedean), so the rationals $\frac{k}{n}$ are spaced more finely than the gap $b - a$; one of them must land inside (a, b). So every real number, $\sqrt{2}$ and π included, has rationals arbitrarily close to it on both sides. The irrationals fill the holes, and the rationals crowd right up against every one of them. This is the precise sense in which the decimal expansions of $\sqrt{2}$ and π approximate them: rational truncations creeping arbitrarily close to a real destination that completeness guarantees is there.

41.6 A worked tour of suprema and infima

Abstract definitions calcify unless you compute with them. Here are four sets; for each we find sup and inf and decide whether they are attained. Work each one yourself before reading the resolution.

Example 41.1 (A set with inf attained, sup not)**.** Let $S = \{1 - \frac{1}{n} : n \in \mathbb{N}\} = \{0,\ \frac{1}{2},\ \frac{2}{3},\ \frac{3}{4},\ \ldots\}$.

Infimum. The smallest term is the first one, $1 - \frac{1}{1} = 0$. Every term is ≥ 0, and 0 is in S, so $\inf S = 0$ and it *is attained* (it is the minimum).

Supremum. The terms climb: $0, 0.5, 0.667, 0.75, 0.8, \ldots$, always below 1 since $\frac{1}{n} > 0$, yet creeping toward 1 as n grows (Archimedean: $\frac{1}{n} \to 0$). So 1 is an upper bound, and it is the least one — for any $\varepsilon > 0$, picking $n > \frac{1}{\varepsilon}$ gives a term $1 - \frac{1}{n} > 1 - \varepsilon$, so no smaller ceiling works. Thus $\sup S = 1$, and it is *not attained*: no term equals 1. Here is the open-interval phenomenon again, now in a sequence.

Example 41.2 (Sup of a set defined by an inequality)**.** Let $S = \{x \in \mathbb{R} : x^2 < 5\}$. This is the open interval $(-\sqrt{5}, \sqrt{5})$. It is bounded above by $\sqrt{5} \approx 2.236$, and $\sup S = \sqrt{5}$ (not attained, since $(\sqrt{5})^2 = 5 \not< 5$). Symmetrically $\inf S = -\sqrt{5}$ (not attained). The supremum $\sqrt{5}$ is itself an irrational number summoned by completeness — the same move as Theorem 41.2 — so this innocuous-looking set has a supremum that does not exist at all in $\mathbb{Q}$.

Example 41.3 (Attained extremes with unattained limit points)**.** Let $S = \{(-1)^n \left(1 + \frac{1}{n}\right) : n \in \mathbb{N}\}$. The even terms are $1 + \frac{1}{n}$ (just above 1, decreasing toward 1): $\frac{3}{2}, \frac{5}{4}, \frac{7}{6}, \ldots$ The odd terms are $-(1 + \frac{1}{n})$ (just below -1, increasing toward -1): $-2, -\frac{4}{3}, -\frac{6}{5}, \ldots$

Supremum. The largest single term is the first even one, $n = 2$: $1 + \frac{1}{2} = \frac{3}{2}$. All later even terms are smaller, all odd terms are negative. So $\sup S = \frac{3}{2}$, *attained.*

Infimum. The smallest single term is the first odd one, $n = 1$: -2. So $\inf S = -2$, *attained.* Both extremes here happen to be reached; the limit points ± 1 that the sequence clusters around are *not* the sup or inf — a useful reminder that "where a sequence accumulates" and "its supremum" are different questions.

Example 41.4 (An unbounded set has no supremum)**.** Let $S = \{n - \frac{1}{n} : n \in \mathbb{N}\} = \{0, \frac{3}{2}, \frac{8}{3}, \frac{15}{4}, \ldots\}$. The terms grow without bound (by the Archimedean property, $n - \frac{1}{n}$ exceeds any fixed number once n is large). So S is *not bounded above* and has *no supremum* — the completeness axiom never applies, since its hypothesis "bounded above" fails. Its infimum is $\inf S = 0$, attained at $n = 1$. The lesson: completeness guarantees a supremum *only for bounded sets*; the hypothesis is not decorative.

Work it yourself

Try these before peeking at the answers in your head. For each set, find sup and inf, and say whether each is attained.

1. $S = \{\frac{n}{n+1} : n \in \mathbb{N}\}$. *(Answer:* $\inf = \frac{1}{2}$ *attained;* $\sup = 1$ *not attained.)*
2. $S = \{x \in \mathbb{Q} : x^3 < 7\}$. *(Answer:* $\sup = \sqrt[3]{7}$*, irrational, not attained; no inf, unbounded below.)*
3. $S = \{(-1)^n + \frac{1}{n} : n \in \mathbb{N}\}$. *(Answer:* $\sup = \frac{3}{2}$ *at* $n = 2$*, attained;* $\inf = -1$*, the limit of odd terms,* not *attained.)*

Notice how often the supremum is the place the set is *heading* rather than a place it *reaches.* That gap between heading-toward and reaching is the entire reason limits — and completeness — exist.

Application

This is not idle abstraction; it is exactly how a computer finds $\sqrt{2}$. A numerical root-finder never produces $\sqrt{2}$ exactly — it produces a *bounded, monotone sequence* of rational approximations and trusts completeness to guarantee a real destination. Newton's method (Exercise 41.6) generates $1, 1.5, 1.41\overline{6}, 1.41421\ldots$, each a rational, each closer; the *number* $\sqrt{2}$ is the limit these approximations define. Every time your calculator displays 1.41421356, it is show-

ing you a truncation of a real number whose *existence* is underwritten by the completeness axiom. Floating-point arithmetic lives entirely among rationals; the meaning of its answers lives in $\mathbb{R}$.

41.7 The shape of the whole idea

The logical skeleton is more memorable than any single proof, so gather the thread.

Walking question

Walk the chain of dependence out loud:

1. The rationals are dense but *holey*: $\sqrt{2}$ names a real length that no fraction reaches (Theorem 41.1).
2. A holey number system breaks limits: a bounded increasing sequence can head toward a destination the system does not contain.
3. The cure is one axiom: *every nonempty set bounded above has a least upper bound* (Section 41.3). This declares the line hole-free.
4. From that one axiom flow all the existence theorems: square roots (Theorem 41.2), monotone convergence (Theorem 41.3), the IVT (Theorem 41.4), the EVT (Theorem 41.5), Bolzano–Weierstrass (Theorem 41.6), the Archimedean property (Theorem 41.7), and the existence of the integral.

Every "there exists" in this book traces back, through this chain, to step 3.

Why this matters

There is a deep aesthetic lesson here about how mathematics grows. The Greeks found a *problem* — the diagonal has no rational length — and for two thousand years it was patched over geometrically without being solved arithmetically. The nineteenth century finally asked the foundational question head-on: *what is a number such that limits always land somewhere?* The answer was not a clever formula but a single structural promise — completeness — chosen precisely to make the problem disappear. The reals are, in a real sense, *defined to be the number system in which calculus works*. We did not discover that limits converge and then prove it; we built the stage so that the play could run. Recognising that the foundations were laid *to order* — reverse-engineered from the theorems we wanted — is one of the quiet pleasures of understanding analysis from the ground up.

41.8 The Continuum: Triumphs, Costs, and the Horizon Ahead

Completeness is the true capstone of calculus. By sealing every hole in the rational line, the Least Upper Bound Property provides a guaranteed destination for every limiting process. In the rational numbers $\mathbb{Q}$, a Cauchy sequence whose terms crowd arbitrarily close together can aim toward an empty gap like $\sqrt{2}$ or π, having nowhere to land. Completeness guarantees that every Cauchy sequence converges to an actual real number, ensuring that $\mathbb{R}$ is a complete metric space.

Upon this single geometric rock rest all the foundational theorems of the subject. The Intermediate Value Theorem, the Extreme Value Theorem, the Monotone Convergence Theorem, and the Bolzano–Weierstrass Theorem are not separate miracles; every one of them is an immediate, inevitable consequence of order completeness. Even the definite integral $\int_a^b f(x)\,dx$ owes its life to completeness: Riemann's construction requires the supremum of lower sums to meet the infimum of upper sums, and completeness guarantees that these bounding numbers genuinely exist. The ancient dilemma of the Pythagoreans—that lengths constructible with a compass cannot be measured by rational ratios—is finally resolved: geometric continuity and numerical arithmetic become the exact same space.

Yet this profound triumph carries genuine philosophical and computational consequences. The completeness axiom is inherently non-constructive: it proves that $\sup S$ exists, but it gives no mechanical formula to generate its digits. As Alan Turing proved in 1936, the vast majority of real numbers are uncomputable—no finite algorithm running forever can compute their successive decimals. Furthermore, Georg Cantor's diagonal argument revealed that $\mathbb{R}$ is uncountably infinite, plunging mathematics into deep set-theoretic questions such as the Continuum Hypothesis, which Kurt Gödel and Paul Cohen proved to be formally undecidable within the standard axioms of set theory. Finally, while completeness rescues integrals for continuous functions, Riemann integration cannot integrate pointwise limits of rough functions, an obstacle that eventually motivated Henri Lebesgue in 1902 to invent measure theory and slice the vertical range rather than the horizontal domain.

Connection

Calculus is the gateway to the entire landscape of modern pure mathematics. With the real continuum and the calculus of one and several variables firmly in hand, three natural pathways open immediately ahead:

1. **Linear Algebra From the Ground Up (Book 3):** In multivariable calculus (Chapters 32 and 33) we saw that derivatives of multivariable maps are linear transformations represented by Jacobian matrices. Linear algebra develops vector spaces, matrix factorizations, eigenvalues, and orthogonal projections in full generality, elevating the local linear approximations of calculus into a universal language for physics, optimization, and high-dimensional geometry.

2. **Probability and Statistics From the Ground Up (Book 4):** Continuous probability density functions, the Gaussian distribution $\frac{1}{\sqrt{2\pi}}e^{-x^2/2}$, joint distributions, and the Central Limit Theorem build directly on the improper integrals, multiple integrals, and sequence limits established across this book.

3. **Real Analysis and Measure Theory:** To push beyond Riemann's integral and complete function spaces into the Hilbert space L^2, pure mathematics turns to Lebesgue integration, topology, and functional analysis—the mathematical language of modern quantum mechanics and signal analysis.

41.9 Exercises

Exercise 41.1. Adapt the proof of Theorem 41.1 to show $\sqrt{3}$ is irrational. (Hint: you need the fact that if $3 \mid p^2$ then $3 \mid p$. Show this by checking the remainder of p modulo 3.) Then explain where the analogous argument *breaks down* for $\sqrt{4}$, and why it must, since $\sqrt{4} = 2$ is rational.

Exercise 41.2. For each set, find sup and inf and state whether each is attained.

(a) $\{2 + \frac{(-1)^n}{n} : n \in \mathbb{N}\}$, (b) $(-3, 5]$, (c) $\{x \in \mathbb{R} : x^2 - x < 0\}$, (d) $\{\frac{1}{n} + \frac{1}{m} : n, m \in \mathbb{N}\}$.

Exercise 41.3. Use the ε-characterisation of the supremum (Section 41.2) to prove the following: if S is nonempty and bounded above with $L = \sup S$, then there is a sequence (x_n) of elements of S with $x_n \to L$. (Hint: for each n, apply tightness with $\varepsilon = 1/n$.)

Exercise 41.4. Let A and B be nonempty sets bounded above, and define $A + B = \{a + b : a \in A,\ b \in B\}$. Prove that $\sup(A + B) = \sup A + \sup B$. (Show $\leq$ using that $\sup A, \sup B$ are upper bounds, and $\geq$ using the ε-tightness of each, taking $\varepsilon/2$ for each set.)

Exercise 41.5. Define $f : \mathbb{Q} \to \mathbb{Q}$ by $f(x) = x^2 - 2$. Verify that f is continuous, $f(1) < 0$, and $f(2) > 0$, yet f has no zero in $\mathbb{Q}$. Explain in one or two sentences exactly which hypothesis of the Intermediate Value Theorem (Theorem 41.4) is unavailable here, and connect your answer to the completeness axiom.

Exercise 41.6. Define a sequence by $a_1 = 1$ and $a_{n+1} = \frac{1}{2}\left(a_n + \frac{2}{a_n}\right)$ (this is Newton's method applied to $x^2 - 2 = 0$). Compute a_1, a_2, a_3, a_4 as fractions and as decimals. Argue that for $n \geq 2$ the sequence is decreasing and bounded below by $\sqrt{2}$, hence convergent by the Monotone Convergence Theorem (Theorem 41.3), and show its limit L must satisfy $L = \frac{1}{2}(L + \frac{2}{L})$, so $L^2 = 2$. You have just built $\sqrt{2}$ from rationals using completeness.

Exercise 41.7. Use the Archimedean property (Theorem 41.7) to prove that $\inf\{\frac{1}{n} : n \in \mathbb{N}\} = 0$. Be explicit about both promises an infimum must satisfy: that 0 is a lower bound, and that no positive number is a lower bound.

Exercise 41.8. (Nested intervals.) Suppose $[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots$ is a shrinking chain of closed intervals with lengths $b_n - a_n \to 0$. Use the completeness axiom directly (consider $\sup\{a_n : n \in \mathbb{N}\}$) to prove there is exactly one real number belonging to *all* the intervals. Then explain why this fails over $\mathbb{Q}$ by exhibiting a nested chain of rational intervals whose only common point would be $\sqrt{2}$.

Exercise 41.9. Prove that between any two real numbers $a < b$ there is an irrational number. (Hint: between $a - \sqrt{2}$ and $b - \sqrt{2}$ there is a rational r by density; consider $r + \sqrt{2}$.) Combined with the density of the rationals, conclude that both the rationals and the irrationals are dense in $\mathbb{R}$ — every interval, however tiny, contains infinitely many of each.